Circuits, Signals and Communications

Book 3 — The Circuits, Signals & Communications Core

Electrical networks, continuous- and discrete-time signals, system behaviour, digital signal processing, information transmission, communication channels and introductory control.

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Part I — Circuit Fundamentals

Every later part of this book rests on a small set of ideas about how electrical quantities behave when components are joined by wires. Part I builds that foundation. It defines the quantities an engineer measures (charge, current, voltage, power, energy), fixes the sign conventions that keep calculations honest, and states the lumped-element assumptions under which a physical gadget may be replaced by a drawing of ideal elements. On that basis it develops Kirchhoff's laws, the systematic nodal and mesh methods that solve any linear resistive network, the theorems that compress a network into a simple equivalent at its terminals, and finally the two energy-storing elements, the capacitor and the inductor, which bring time into the picture. The examples range from medical wearables and car electrical systems to building wiring, strain-gauge bridges and amplifier models, because the same few laws govern all of them.

By the end of this part you should understand:

- what current, voltage, power and energy measure, and how reference directions and the passive sign convention decide whether an element supplies or absorbs power;
- when the lumped-element model is valid, and which physics is missing when it fails;
- how ideal, practical and dependent sources model batteries, supplies, sensors, transistors and op-amps;
- why Kirchhoff's current and voltage laws hold, and how they combine with Ohm's law to give series and parallel rules, voltage and current division, source transformations and power balance;
- how nodal, mesh and modified nodal analysis turn any linear network into a matrix equation, and when that equation has exactly one solution;
- how superposition, Thévenin, Norton and the other network theorems summarise a network at its terminals, and where they stop applying;
- how capacitors and inductors store energy, why their voltages and currents cannot jump, and how coupled coils interact.

Chapter	Central question	What it delivers
Chapter 1 — Electrical Quantities and Circuit Models	What exactly do current, voltage, power and energy measure, and when may a real device be replaced by an ideal element?	Definitions and units, reference directions, the passive sign convention, element and source models (ideal, practical, dependent), the lumped-element assumptions and their limits, schematic notation, and the circuit view of measurement (loading, burden, accuracy)
Chapter 2 — Fundamental Circuit Laws	Which physical principles fix the currents and voltages in a network, and how can a network be simplified without losing terminal information?	Ohm's law and resistivity, the topology vocabulary, KCL and KVL with their physical origin and failure modes, series–parallel and Δ–Y reduction, voltage and current division, power balance, source transformations and network reduction
Chapter 3 — Network Analysis	How can any linear network be solved by a procedure that never relies on spotting a simplification, and when is the solution unique?	Nodal and mesh analysis, supernodes and supermeshes, circuits with dependent sources, graphs, trees, incidence and loop matrices, Tellegen's theorem, modified nodal analysis as used in simulators, and the conditions for solvability and good numerical conditioning
Chapter 4 — Network Theorems	What does linearity allow us to say about a network as seen from its terminals?	Superposition, Thévenin and Norton equivalents, maximum power transfer, reciprocity, Millman's theorem, source and network equivalence, and the limits of each theorem
Chapter 5 — Capacitors and Inductors	How do elements that store energy change circuit behaviour, and what fixes their initial conditions?	Capacitance and inductance, their constitutive equations and stored energy, series and parallel combinations, continuity and initial conditions, mutual inductance and coupled coils, and the difference between ideal and practical components

How this part connects to the rest of the book

Part I treats circuits mostly at a single instant or in DC steady state. Part II (Transient and AC Analysis) lets time run: the capacitors and inductors of Chapter 5 produce the first- and second-order transients of Chapters 6 and 7, and sinusoidal sources lead to phasors, AC power and three-phase systems, all analysed with the nodal and mesh methods of Chapter 3 extended to complex impedances. Part III (Advanced Network Theory) carries the same matrices into the Laplace domain, describes networks by their terminal parameters as the theorems of Chapter 4 suggest, and replaces the lumped model by the transmission line where Chapter 1 showed it must fail. From Part IV onward, circuits become one realisation of linear time-invariant systems: the linearity that underpins superposition and Thévenin's theorem is the same linearity that makes convolution, Fourier analysis, digital filtering, modulation and feedback control possible. The mathematics of linear algebra, differential equations and transforms is drawn from *The Mathematics Universe*, the field theory behind the lumped model from *Engineering Physics*, and the device physics behind dependent-source models from *Electronics & Computing*.

Chapter 1 — Electrical Quantities and Circuit Models

Part I · Circuit Fundamentals

A wearable ECG patch is specified to record a patient's heart for seven days on a single CR2032 coin cell. In the field it shuts down after three. On the bench nothing looks wrong: a handheld multimeter in series with the cell reads an average current of about 0.9 mA, which against the cell's 225 mAh rating predicts more than ten days. The trouble lies in what the meter did not show. Every 100 ms the patch's radio wakes for 5 ms and draws 12 mA; the meter averages those bursts into a steady-looking number. The cell is not an ideal 3 V source either. It has an internal resistance that grows as it discharges, and once that resistance is large enough, each 12 mA burst pulls the terminal voltage below the microcontroller's 2.2 V brown-out threshold. The processor resets, the patch logs a fault, and two-thirds of the cell's charge is never used.

Every quantity in that story — charge, current, voltage, power, energy, the internal resistance of a source, the averaging of a meter — is a basic electrical quantity or a basic circuit model. This chapter defines them carefully, fixes the sign conventions that keep the bookkeeping honest, and states the assumptions under which a physical gadget can be replaced by a drawing of idealised elements joined by wires. Those assumptions, and their limits, carry through every later chapter of the book.

> **What exactly do current, voltage, power and energy measure, and how do we keep their signs straight?**
> **When may a real device be replaced by an ideal element or source, and when does that replacement fail?**

Concept	Engineering destination
Charge, current, voltage, power	Battery sizing, fuse and cable ratings, sensor signal levels
Reference directions, passive sign convention	Every circuit equation; deciding which elements supply and which absorb power
Energy	Battery life, electricity bills, defibrillator and pulsed-power design
Circuit elements	Resistive, capacitive and inductive models of components, cables and loads
Ideal and practical sources	Batteries, bench supplies, signal generators, solar cells, 4–20 mA transmitters
Independent and dependent sources	Transistor and op-amp small-signal models, current sensors, controlled converters
Lumped-element assumptions	Knowing when circuit theory holds and when a transmission-line or field model is needed
Circuit diagrams and notation	Reading and drawing schematics, IEC/IEEE symbols, ground conventions
Basic measurement concepts	Voltmeter and ammeter connection, loading error, burden voltage, accuracy and resolution

Roadmap. Charge, current, voltage and power (1.1) → reference directions (1.2) → the passive sign convention (1.3) → energy (1.4) → circuit elements (1.5) → ideal and practical sources (1.6) → independent and dependent sources (1.7) → the lumped-element assumptions that make all of this valid (1.8) → how circuits are drawn (1.9) → how their quantities are measured (1.10).

1.1 Charge, current, voltage and power

Charge

Electric charge is the property of matter that makes particles exert electric forces on each other. It comes in two signs and in integer multiples of the elementary charge

$$e = 1.602\,176\,634 \times 10^{-19}\ \mathrm{C}$$

(exact, since the 2019 redefinition of the SI). An electron carries $-e$, a proton $+e$. The coulomb is a large unit: one coulomb is the charge of 6.24×10^{18} electrons. Because charge is quantised in such small steps, circuit theory treats it as a continuous quantity q; the graininess becomes visible only as *shot noise* in very small currents (Chapter 29).

Charge is conserved: it can move, and positive and negative charges can be separated (as in a battery), but the net charge inside a closed surface changes only by charge crossing that surface.

> **Status of the principle: Conservation of charge**
> *Fundamental.* No experiment has ever observed net charge being created or destroyed; it is built into Maxwell's equations through the continuity equation. Kirchhoff's current law (Chapter 2) is its circuit form.

Current

Current is the rate at which charge passes through a chosen cross-section:

$$i(t) = \frac{dq}{dt}$$

measured in amperes (1 A = 1 C/s). Inverting, the charge transferred between t_0 and t is

$$q(t) - q(t_0) = \int_{t_0}^{t} i(\tau)\, d\tau .$$

A battery's capacity in ampere-hours is exactly this integral: 1 Ah = 3600 C.

By convention, current is counted as the flow of *positive* charge. In a metal the electrons drift opposite to the conventional current, which makes no difference to any circuit calculation.

It is worth knowing how slowly carriers actually move. The drift velocity is $v_d = I/(nqA)$, with n the carrier density. For copper, $n \approx 8.5 \times 10^{28}$ m^{-3}; a 16 A current in a 2.5 mm² conductor gives

$$v_d = \frac{16}{(8.5\times10^{28})(1.602\times10^{-19})(2.5\times10^{-6})} \approx 4.7\times10^{-4}\ \text{m/s},$$

about half a millimetre per second. What travels quickly along a wire is not the electrons but the electromagnetic field that pushes them, at a large fraction of the speed of light. That distinction is the root of the lumped-element assumption (Section 1.8).

Currents may be **direct** (DC, constant), **alternating** (AC, periodically reversing, as at 50 or 60 Hz in power systems) or **pulsed**, like the radio bursts of the ECG patch. Table 1.1 shows the range of magnitudes an engineer meets.

Table 1.1 — Representative current magnitudes.

Situation	Typical current
Input bias current of a CMOS op-amp	1 pA – 1 nA
Photodiode in an optical receiver	10 nA – 1 mA
Low-power microcontroller in sleep mode	0.5 – 5 μA
Indicator LED	1 – 20 mA
Industrial 4–20 mA current loop	4 – 20 mA (3.6–21 mA fault signalling)
Smartphone under load	0.1 – 3 A
Domestic socket circuit	13–16 A (Europe), 15–20 A (North America)
Electric-vehicle AC charging	16 – 32 A per phase
Car starter motor while cranking	100 – 300 A
Overhead transmission conductor	0.5 – 3 kA
Lightning return stroke (median)	about 30 kA

Voltage

The **voltage** (potential difference) between points a and b is the energy per unit charge gained by a positive test charge moved from b to a:

$$v_{ab} = \frac{dw}{dq},$$

measured in volts (1 V = 1 J/C). The double subscript is read "the voltage of a with respect to b", and $v_{ba} = -v_{ab}$. If $v_{ab} = 5$ V, then every coulomb carried from b to a gains 5 J of potential energy, and every coulomb that falls from a to b through a load gives up 5 J to that load.

Voltage is defined between *two* points. "The voltage at node 3" means with respect to a chosen reference node, called **ground** or **common**; the choice is arbitrary for calculation.

For the voltage between two points to be well defined, the energy per charge must not depend on the path taken between them. This holds when the electric field is conservative, which is true for static fields and, to excellent approximation, in circuits where no significant time-varying magnetic flux threads the paths of interest. Chapter 2 makes this the basis of Kirchhoff's voltage law and shows where it breaks.

> **Status of the principle: Path-independent voltage**
> *Approximate.* Exact for electrostatic fields; in time-varying circuits it holds when the magnetic flux linking the region between the conductors changes negligibly — the quasi-static, lumped regime of Section 1.8.

Table 1.2 collects typical voltage levels across EEE.

Table 1.2 — Representative voltage levels.

Signal or system	Typical voltage
EEG at the scalp	10 – 100 µV
ECG at the skin (R-wave)	0.5 – 3 mV
Type K thermocouple sensitivity	about 41 µV/°C
Dynamic microphone output	1 – 10 mV
Consumer line-level audio (–10 dBV)	0.316 V rms
Professional line level (+4 dBu)	1.228 V rms
Digital logic supply	0.8 – 5 V (1.8, 3.3 and 5 V common)
Lithium-ion cell	3.0 – 4.2 V (3.6–3.7 V nominal)
Automotive system	12.6 V at rest, 13.8–14.4 V charging; 48 V in mild hybrids
Mains (single-phase)	230 V rms at 50 Hz; 120 V rms at 60 Hz
Medium-voltage distribution	11, 13.8, 22, 33 kV
Transmission	132 – 765 kV ac; ±500 – ±800 kV HVDC

Power

Power is the rate at which energy is transferred. Combining the definitions of voltage and current,

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = v\, i,$$

in watts (1 W = 1 J/s = 1 V·A). It holds instant by instant for any waveform. In AC circuits $p(t)$ oscillates, and its average, reactive and apparent parts are the subject of Chapter 9.

> **Status of the principle: Power as the product of voltage and current**
> *Derived*. It follows directly from the definitions of voltage (energy per charge) and current (charge per time); it is exact at any instant for a two-terminal element in the lumped regime.

1.2 Reference directions

A circuit is analysed before its currents and voltages are known, and they may be positive or negative, constant or reversing. We therefore attach to every element an **assumed** direction for its current (an arrow) and an **assumed** polarity for its voltage (a + and a − mark). These are *reference directions*: bookkeeping labels, not predictions.

The rule that makes this work is simple. Write every equation consistently with the chosen references; solve; and interpret the sign of the answer:

- If $i = +2$ A, two amperes of positive charge flow in the direction of the arrow.
- If $i = -2$ A, two amperes flow against the arrow.
- If $v = -5$ V for an element marked + on the left, the right-hand terminal is actually 5 V more positive.

Nothing is gained by trying to guess the "true" direction before solving, and nothing is lost by guessing wrong. A negative result is not an error; it is information. What *is* an error is changing a reference halfway through, for instance writing one equation with a current arrow pointing left and another with the same current pointing right.

For time-varying quantities the reference matters even more, because the true direction changes every half-cycle. The current through a mains load is $i(t) = I_m \cos \omega t$ with respect to its arrow; it flows with the arrow for half of each cycle and against it for the other half. The reference arrow is what makes "$\cos \omega t$" mean something.

Double subscripts carry a reference with them: $v_{ab} = -v_{ba}$ and $i_{ab} = -i_{ba}$. With a ground node chosen, v_a alone means v_{a0}, and any element voltage is a difference of node voltages, $v_{ab} = v_a - v_b$ — the basis of nodal analysis (Chapter 3).

Instruments have reference directions too. A multimeter's red lead is its + reference, so a reading of −3.30 V means the red probe is at the lower potential. A clamp-on current probe carries an arrow on its jaws, and clamping one of three phase probes backwards is a classic cause of "negative power" in power-quality surveys.

1.3 Passive sign convention

The convention

Once each element has a voltage polarity and a current arrow, there are two ways the arrow can relate to the polarity: current can be referenced to enter the + terminal, or to leave it. The **passive sign convention** is the choice to always reference the current *into* the terminal marked +. With that choice,

$$p = v\,i \quad \text{is the power absorbed by the element.}$$

If $p > 0$ the element absorbs energy from the rest of the circuit (dissipating it as heat, storing it in a field, or converting it to light, sound or motion). If $p < 0$ it delivers energy to the rest of the circuit.

If the current reference instead leaves the + terminal, the same product vi is the power *delivered*, and the power absorbed is $-vi$. Figure 1.1 shows both cases.

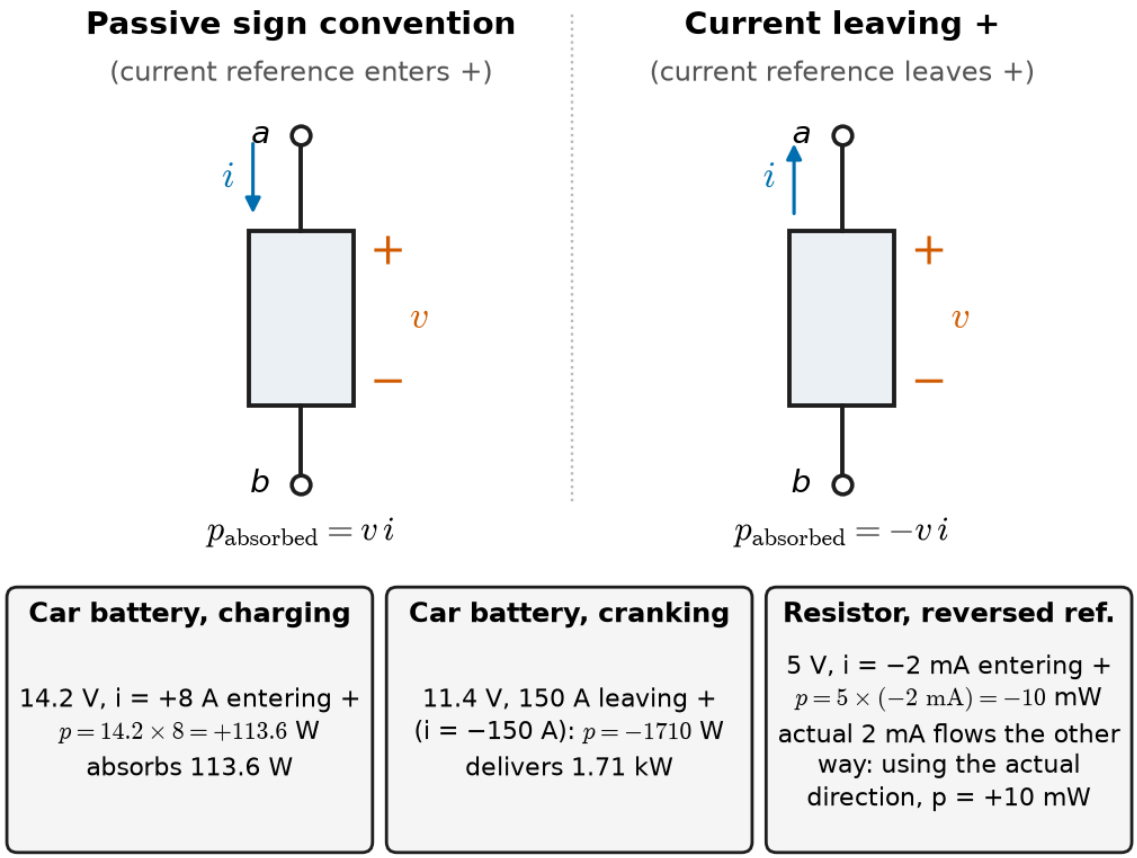


Figure 1.1 — The passive sign convention: with the current reference entering the + terminal, the product vi is the power absorbed by the element.

Every element can be labelled with the passive convention, including sources; the convention does not decide which elements are sources and which are loads — the arithmetic does. It is a convention, not a law, chosen because element laws then take their most familiar form: Ohm's law is $v = Ri$ (not $v = -Ri$), the capacitor law is $i = C\,dv/dt$, and the inductor law is $v = L\,di/dt$, all with positive coefficients. Using it everywhere means that the element equations of Chapters 2–5 can be copied without sign adjustments.

> **Boundary.**
> Electrical machines and power-system engineers often use a *generator convention* for sources and machines (current referenced out of the + terminal, so positive power means delivered). Both conventions are correct; mixing them inside a single calculation is the error. Machine conventions are treated in *Electrical Energy & Machines*.

Worked example — Who supplies and who absorbs in a car electrical system. With the engine running, a car's alternator holds the system at 14.2 V. The battery, labelled passively (+ at its positive post, current arrow entering the positive post), carries $i_B = +8.0$ A; the headlamps, ignition and electronics, labelled passively as one load, carry $i_L = +42.0$ A; the alternator, also labelled passively, carries i_A.

All three are connected between the same two conductors, so they share $v = 14.2$ V. Charge conservation at the positive bus (Chapter 2) requires the three currents entering their + terminals to sum to zero: $i_A + i_B + i_L = 0$, so $i_A = -50.0$ A.

Powers absorbed: battery $14.2 \times 8.0 = 113.6$ W (it is charging); load $14.2 \times 42.0 = 596.4$ W; alternator $14.2 \times (-50.0) = -710.0$ W. The sum is $113.6 + 596.4 - 710.0 = 0$: energy is conserved, and the negative sign identifies the alternator as the only element delivering power.

Now the engine is stopped and the starter cranks. The battery terminal falls to 11.4 V and 150 A leaves its positive post. Labelled passively, $i_B = -150$ A, so the battery absorbs $11.4 \times (-150) = -1710$ W; it delivers 1.71 kW to the starter. The same element, the same convention, and the sign of the result tells the story.

1.4 Energy

Energy is the integral of power over time:

$$w(t_0, t) = \int_{t_0}^{t} p(\tau)\, d\tau = \int_{t_0}^{t} v(\tau)\, i(\tau)\, d\tau,$$

in joules. With the passive convention, w is the energy absorbed by the element over the interval. For a constant power P over a time T, $w = PT$.

Electricity is billed in kilowatt-hours (1 kWh = 3.6 MJ) and batteries are rated in watt-hours (1 Wh = 3600 J). An ampere-hour rating is a charge, not an energy; multiplied by the average discharge voltage it gives watt-hours.

Table 1.3 — Representative energies.

Store or event	Typical energy
One radio burst of the ECG patch (2.85 V, 12 mA, 5 ms)	0.17 mJ
CR2032 coin cell	about 0.65 Wh (2.3 kJ)
Alkaline AA cell (low drain)	2.5 – 3.5 Wh
Smartphone battery	12 – 20 Wh
External defibrillator shock	120 – 360 J delivered
Electric kettle, 2 kW for 3 min	0.1 kWh (360 kJ)
Laptop battery	40 – 100 Wh
Daily household consumption	roughly 5 – 30 kWh, strongly country-dependent
Electric-vehicle battery	40 – 100 kWh

Conservation of energy in circuits

In any circuit the total power absorbed by all elements, counted with the passive convention, is zero at every instant: what the sources deliver, the other elements absorb. This was visible in the alternator example above. In Chapter 2 it appears as *power balance*, a check on every analysis, and in Chapter 3 it is proved from Kirchhoff's laws alone as Tellegen's theorem.

Worked example — Energy budget of the ECG patch. The CR2032 coin cell in the opening story is rated 225 mAh at a nominal 3.0 V.

Charge available: $Q = 0.225\ \text{Ah} \times 3600\ \text{s/h} = 810\ \text{C}$. Energy available: $W \approx QV = 810 \times 3.0 = 2430\ \text{J} = 0.675\ \text{Wh}$.

For a seven-day (168 h) life, the average current must not exceed $225/168 = 1.34\ \text{mA}$.

The current profile is a sleep current of 0.30 mA, plus a radio burst of 12 mA lasting 5 ms every 100 ms (duty cycle $D = 0.05$). The average current is

$$\bar{i} = 0.30 + (12 - 0.30) \times 0.05 = 0.885\ \text{mA},$$

so the charge budget predicts $225/0.885 = 254$ h, or 10.6 days. On charge alone the design has a margin of about 50 %.

Each burst consumes about $2.85\ \text{V} \times 12\ \text{mA} \times 5\ \text{ms} = 0.17\ \text{mJ}$. Energy is not the problem. The failure is a *power* problem: the cell cannot deliver 12 mA at a high enough voltage once its internal resistance rises. Section 1.6 completes the calculation.

1.5 Circuit elements

What an element is

A **circuit element** is an idealised two-terminal (or multi-terminal) object defined by a mathematical relation between its terminal voltage and current. It is a model, not a component. A real resistor is approximately a resistor element, but at high frequency it is better modelled as a resistor in series with a small inductance and in parallel with a small capacitance; a real cable may need resistors, inductors and capacitors per metre (Chapter 14). Modelling is the art of choosing the fewest elements that reproduce the behaviour that matters.

The basic two-terminal elements of this book are:

Table 1.4 — Basic circuit elements and their defining relations (passive sign convention).

Element	Symbol parameter	Defining relation	Energy behaviour
Resistor	R (Ω) or $G = 1/R$ (S)	$v = R\,i$	Dissipates; $p = i^2 R \geq 0$
Capacitor	C (F)	$i = C\,dv/dt$	Stores $\tfrac{1}{2}Cv^2$ in the electric field
Inductor	L (H)	$v = L\,di/dt$	Stores $\tfrac{1}{2}Li^2$ in the magnetic field
Ideal voltage source	$v_s(t)$	$v = v_s(t)$ for any i	Supplies or absorbs any power
Ideal current source	$i_s(t)$	$i = i_s(t)$ for any v	Supplies or absorbs any power
Short circuit	—	$v = 0$ for any i	No power
Open circuit	—	$i = 0$ for any v	No power
Ideal switch	state	Short when closed, open when open	No power (ideal)

Multi-terminal elements include coupled inductors and the ideal transformer (Chapter 5), the four dependent sources of Section 1.7, and the two-port networks of Chapter 12.

Classifying elements

Linear or nonlinear. An element is linear if its defining relation is linear in v and i (and their derivatives and integrals). Resistors, capacitors and inductors with constant parameters are linear; a diode, whose current rises exponentially with voltage, is not. Linearity is what makes superposition, Thévenin equivalents, phasors and transfer functions possible (Chapters 4, 8, 11).

Time-invariant or time-varying. An element whose parameters do not change with time is time-invariant. A switch, a varactor tuned by a control voltage and a condenser microphone are time-varying. Linear time-invariant (LTI) circuits are the subject of Parts II–IV.

Passive or active. An element is **passive** if the total energy it has absorbed from the start, plus whatever energy it initially held, can never become negative:

$$w(t) = w(t_0) + \int_{t_0}^{t} v\,i\,d\tau \;\geq\; 0 \quad \text{for all } t \text{ and all excitations.}$$

Resistors with $R \geq 0$, capacitors and inductors with positive values, and ideal transformers are passive. An element that can deliver net energy indefinitely — an independent source, a dependent source, a negative resistance — is **active**. A circuit made only of passive elements cannot produce growing oscillations, a fact that becomes a stability result in Chapters 13 and 34.

Lumped or distributed. A lumped element is concentrated at a point: its current is the same at both terminals and its voltage is well defined. A distributed element, such as a long cable, has voltage and

current that vary along its length. Section 1.8 states when the lumped description is allowed.

Representative component values

Table 1.5 — Typical ranges and tolerances of passive components.

Component	Typical range	Typical tolerance	Notes
Thick-film chip resistor	1 Ω – 10 MΩ	±1 % or ±5 %	Temperature coefficient 100–200 ppm/K
Thin-film / metal-film resistor	10 Ω – 1 MΩ	±0.1 % – ±1 %	5–50 ppm/K; used in precision dividers
Current-sense (shunt) resistor	0.1 mΩ – 1 Ω	±0.5 % – ±1 %	Kelvin (four-terminal) versions for accuracy
Ceramic capacitor (C0G/NP0)	0.5 pF – 0.1 µF	±1 % – ±5 %	Stable with voltage and temperature
Ceramic capacitor (X7R/X5R)	100 pF – 100 µF	±10 % – ±20 %	Capacitance falls with DC bias
Aluminium electrolytic	0.1 µF – 1 F	±20 %	ESR of mΩ to Ω; polarised
Power inductor	0.1 µH – 1 mH	±20 % – ±30 %	Saturates above rated current
RF inductor (air-core, thin-film)	1 nH – 1 µH	±2 % – ±5 %	Self-resonant frequency limits use

1.6 Ideal and practical sources

Ideal sources

An **ideal voltage source** maintains a specified voltage $v_s(t)$ across its terminals regardless of the current through it. Its current is whatever the rest of the circuit demands, positive, negative or zero. Its i–v characteristic is a vertical line at $v = v_s$ on a plot of current against voltage (Figure 1.2).

An **ideal current source** maintains a specified current $i_s(t)$ through itself regardless of the voltage across it. Its characteristic is a horizontal line at $i = i_s$.

Ideal sources admit inconsistent connections that diagrams must avoid: a shorted voltage source or two unequal voltage sources in parallel; dually, an open-circuited current source or two unequal current sources in series. Chapter 3 states these as solvability conditions.

Practical sources

Real sources lose voltage as they deliver current. The simplest model that captures this is an ideal voltage source V_s in series with an internal (source) resistance R_s:

$$v = V_s - R_s\, i,$$

where i is the current delivered out of the + terminal. Its characteristic is a straight line through the open-circuit voltage $v = V_s$ (at $i = 0$) and the short-circuit current $i_{sc} = V_s/R_s$ (at $v = 0$). An equivalent model is an ideal current source $I_s = V_s/R_s$ in parallel with the same R_s; the two describe the same terminal line and are interchangeable (source transformation, Chapter 2). A source whose R_s is small compared with its load behaves almost like an ideal voltage source; one whose R_s is large behaves almost like an ideal current source.

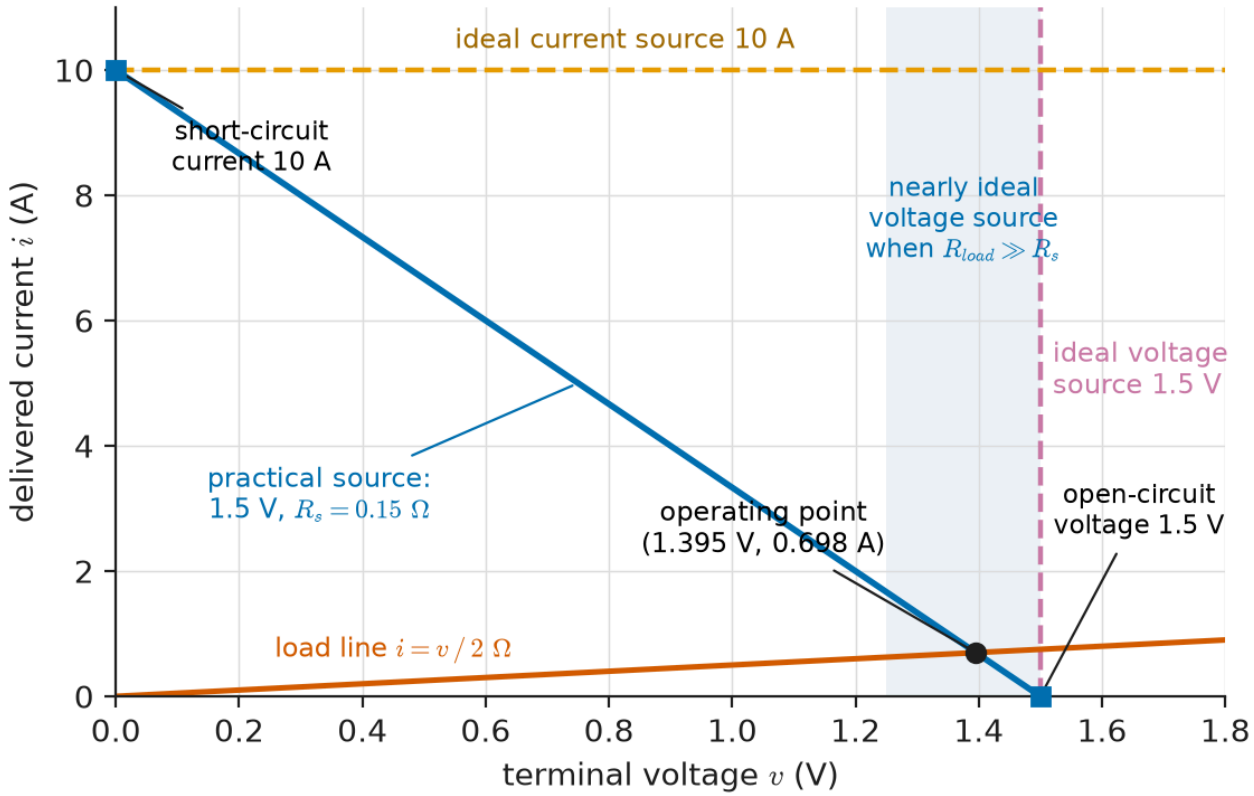


Figure 1.2 — Terminal characteristics of ideal and practical sources, with the load line of a 2 ohm resistor.

Figure 1.2 shows these lines for an alkaline AA cell. Where the source line crosses the load line $i = v/R_L$ the circuit operates; this graphical construction works for nonlinear loads too and is used for diodes and transistors in *Electronics & Computing*.

The linear model is itself an approximation. A battery's internal resistance depends on its state of charge, temperature, age and the duration of the load pulse; a solar cell's characteristic is strongly curved; a laboratory power supply is nearly ideal up to its current limit and then switches to behaving as a current source. Table 1.6 lists typical values.

Table 1.6 — Typical source parameters.

Source	Open-circuit or set value	Typical internal resistance
Alkaline AA cell (fresh)	1.5–1.6 V	0.1 – 0.3 Ω
CR2032 lithium coin cell	3.0–3.2 V	10 – 30 Ω fresh, rising to 100 Ω or more near end of life and at low temperature
18650 Li-ion cell	3.0–4.2 V	15 – 50 mΩ
12 V lead-acid car battery	12.6–12.8 V at rest	5 – 10 mΩ (healthy)
Bench power supply (CV mode)	0–30 V	a few mΩ up to its current limit
Function generator	up to 10–20 V peak-to-peak	50 Ω (by design, to match coaxial cable)
Audio power-amplifier output	—	under 0.1 Ω (damping factor above 80 into 8 Ω)
Mains socket outlet (source impedance)	230 V / 120 V rms	0.1 – 1 Ω, depending on distance from transformer
Industrial 4–20 mA transmitter	4–20 mA	very large: behaves as a current source over its compliance range
Silicon solar cell (near short circuit)	30–40 mA/cm²	large shunt resistance: behaves as a current source below its knee

> **Status of the principle: Linear (Thévenin) model of a practical source**
> *Approximate.* A first-order straight-line fit to a real source's terminal characteristic near its operating point; it ignores the dependence of internal resistance on state of charge, temperature and pulse duration, and the curvature of photovoltaic and current-limited characteristics.

Worked example — An AA cell driving a load. An alkaline cell with $V_s = 1.50$ V and $R_s = 0.15\ \Omega$ drives a 2.0 Ω load (a small motor in its running condition, approximated as a resistor).

Current: $i = V_s/(R_s + R_L) = 1.50/2.15 = 0.698$ A. Terminal voltage: $v = iR_L = 1.395$ V, a drop of 0.105 V from open circuit.

Power to the load: $i^2 R_L = 0.973$ W. Power lost inside the cell: $i^2 R_s = 0.073$ W. Efficiency $R_L/(R_L + R_s) = 93.0\ \%$.

The short-circuit current is $V_s/R_s = 10$ A — which is why AA cells can melt a stray paper clip across their terminals. The maximum power the cell could deliver to any load occurs for $R_L = R_s$ and is $V_s^2/(4R_s) = 3.75$ W, but at only 50 % efficiency (Chapter 4 proves this maximum-power-transfer result).

Worked example — Why the ECG patch dies early. Return to the CR2032. On its voltage plateau the cell's open-circuit voltage is about 2.85 V. When fresh, its internal resistance is about 15 Ω, so

during a 12 mA radio burst the terminal voltage is

$$v = 2.85 - 0.012 \times 15 = 2.67\ \text{V},$$

comfortably above the 2.2 V brown-out threshold. The burst pulls the terminal below threshold once

$$R_s > \frac{2.85 - 2.20}{0.012} = 54\ \Omega.$$

A coin cell's internal resistance climbs through this range well before its rated charge is exhausted, and climbs faster in a cold environment or when pulses are frequent. The patch fails because its designer budgeted *charge* (Section 1.4) but not *source resistance*. The engineering fixes follow from the model: a reservoir capacitor across the cell supplies the burst current and is recharged slowly between bursts, a lower brown-out threshold widens the margin, or a cell chemistry with lower internal resistance is chosen. Chapters 5 and 6 supply the tools to size the reservoir capacitor.

1.7 Independent and dependent sources

Independent sources

The sources of Section 1.6 are **independent**: their value $v_s(t)$ or $i_s(t)$ is specified from outside the circuit and does not depend on any voltage or current inside it. Independent sources represent the inputs to a circuit — batteries, mains, signal generators, microphones, antennas, sensors — and are drawn as circles (with + and – for voltage, an arrow for current).

Dependent (controlled) sources

A **dependent source** produces a voltage or current that is set by a voltage or current *elsewhere in the circuit*. There are four kinds, determined by what is controlled (voltage or current output) and what controls it (a voltage or a current):

Table 1.7 — The four dependent sources.

Name	Output	Control	Relation	Gain unit	Typical device it models
VCVS (voltage-controlled voltage source)	voltage	voltage	$v_o = \mu\, v_x$	V/V	Op-amp open-loop gain, ideal amplifier stage
CCVS (current-controlled voltage source)	voltage	current	$v_o = r_m\, i_x$	Ω (transresistance)	Transimpedance amplifier, Hall-effect or shunt current sensor
VCCS (voltage-controlled current source)	current	voltage	$i_o = g_m\, v_x$	S (transconductance)	MOSFET and BJT small-signal model, OTA
CCCS (current-controlled current source)	current	current	$i_o = \beta\, i_x$	A/A	BJT current gain, ideal current transformer, current mirror

They are drawn as diamonds, with + and − for a controlled voltage source or an arrow for a controlled current source, and the controlling variable must be labelled clearly on the diagram (Figure 1.3).

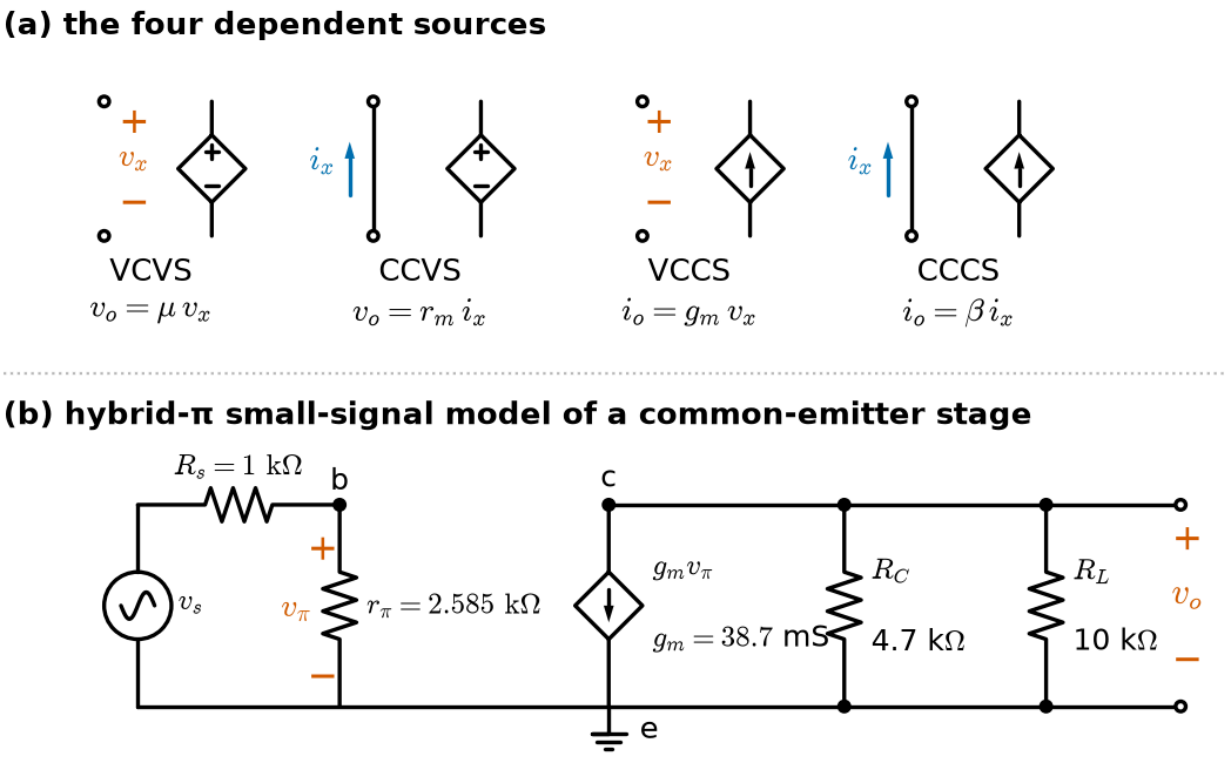


Figure 1.3 — The four dependent sources and a common use: the hybrid-pi small-signal model of a bipolar transistor.

A dependent source is **not** a source of energy in its own right. It models a device that steers power drawn from a supply left out of the diagram (an op-amp's rails, a transistor's collector supply); it is active because the signal power it delivers can exceed the power its control terminals absorb, which is what amplification means.

Dependent sources usually arise from **linearisation**. A bipolar transistor's collector current depends exponentially on its base–emitter voltage; for small variations about a bias point the slope is the transconductance $g_m = I_C/V_T$, with $V_T = kT/q \approx 25.9$ mV at 300 K. The VCCS $g_m v_{be}$ holds only while the signal is small compared with V_T.

> **Status of the principle: Linear dependent-source models of active devices**
> *Approximate.* First-order Taylor expansions of nonlinear device characteristics about a fixed operating point; accurate for small signal amplitudes and frequencies below the device's internal capacitive limits, and invalid for switching or large-signal operation.

> **Boundary.**
> How transistors and op-amps produce these characteristics, and how their bias points are set, belongs to *Electronics & Computing*. This book uses the resulting dependent-source models as network elements; Chapter 3 analyses circuits that contain them.

Worked example — Voltage gain of a transistor stage from its dependent-source model. A bipolar transistor biased at $I_C = 1.0$ mA has $g_m = I_C/V_T = 1.0\,\text{mA}/25.85\,\text{mV} = 38.7$ mS. With current gain $\beta = 100$, its input resistance is $r_\pi = \beta/g_m = 2.585$ kΩ. In the circuit of Figure 1.3, a signal source v_s with $R_s = 1$ kΩ drives the base; the collector load is $R_C = 4.7$ kΩ in parallel with an external load $R_L = 10$ kΩ.

Input side: r_π and R_s form a divider, so $v_\pi = v_s\, r_\pi/(r_\pi + R_s) = 0.721\, v_s$.

Output side: the VCCS drives $g_m v_\pi$ into $R_C \parallel R_L = 4.7 \times 10/14.7 = 3.197$ kΩ; with the current arrow pointing from collector to emitter, $v_o = -g_m v_\pi (R_C \parallel R_L)$.

Gain from the base: $v_o/v_\pi = -38.7\,\text{mS} \times 3.197\,\text{kΩ} = -123.7$. Gain from the source: $v_o/v_s = -123.7 \times 0.721 = -89.2$.

A 1 mV signal at the source therefore becomes an 89 mV inverted signal at the load. For a 1 mV-peak sinusoid the source delivers only $v_s^2/[2(R_s + r_\pi)] \approx 0.14$ nW, while the load receives $(89.2\,\text{mV})^2/(2 \cdot 10\,\text{kΩ}) \approx 0.40$ μW of signal power (peak values, sinusoidal signal): a power gain of more than a thousand, supplied by the collector power supply that the model does not show.

1.8 Lumped-element assumptions

From fields to circuits

Electromagnetism is governed by Maxwell's equations, which describe fields that vary continuously in space and propagate at a finite speed. Circuit theory replaces those fields by a handful of numbers — one voltage per node pair, one current per branch — and replaces partial differential equations by algebraic and ordinary differential equations. This is an enormous simplification. It is justified only when the following assumptions hold:

1. **Small size compared with wavelength.** The physical dimensions of the circuit are small compared with the wavelength $\lambda = v_p/f$ of the highest significant frequency, where v_p is the propagation speed in the medium. Then an electromagnetic disturbance crosses the circuit in a time much shorter than any time scale of interest, and every point of a connecting wire can be treated as responding simultaneously.
2. **Confined fields.** Electric fields are confined inside capacitors and magnetic fields inside inductors (or at least their coupling to other parts of the circuit is either negligible or modelled explicitly as mutual inductance or stray capacitance).
3. **Charge neutrality of elements.** The net charge on every element is zero: whatever current enters one terminal leaves the other at the same instant. (A capacitor holds $+q$ on one plate and $-q$ on the other, net zero.)
4. **Ideal interconnects.** Wires joining elements have negligible resistance, inductance and capacitance, or these are included as explicit lumped elements.

Under these assumptions element currents and node voltages are well defined, and the laws of Chapter 2 follow.

> **Status of the principle: Lumped-element approximation**
> *Approximate.* A quasi-static limit of Maxwell's equations, valid when circuit dimensions are much smaller than the shortest relevant wavelength (a common working rule is below $\lambda/10$, or below about one-sixth of the distance a signal edge travels in its rise time) and when radiation and stray coupling are negligible; it fails for long power lines, high-speed digital interconnects, RF and microwave circuits.

How small is small?

Table 1.8 gives wavelengths at familiar frequencies, both in free space and on a typical FR-4 printed circuit board, where a signal on a stripline travels at about $c/\sqrt{4.3} \approx 0.48c$, a delay of about 6.9 ns per metre.

Table 1.8 — Wavelength and the lumped limit at representative frequencies.

Frequency	Application	Free-space λ	λ on FR-4 stripline	$\lambda/10$ on FR-4
50 Hz	European mains	6000 km	2890 km	289 km
60 Hz	North American mains	5000 km	2410 km	241 km
20 kHz	Top of the audio band	15 km	7.2 km	720 m
1 MHz	AM radio, switching converters	300 m	145 m	14.5 m
100 MHz	FM radio, fast logic harmonics	3.0 m	1.45 m	14.5 cm
1 GHz	Cellular, DDR memory	30 cm	14.5 cm	1.45 cm
2.4 GHz	Wi-Fi, Bluetooth	12.5 cm	6.0 cm	6.0 mm
10 GHz	X-band radar, satellite	3.0 cm	1.45 cm	1.45 mm

(For power cables and overhead lines the propagation speed is close to c, so the free-space column applies; the FR-4 column is shown for board-level electronics.)

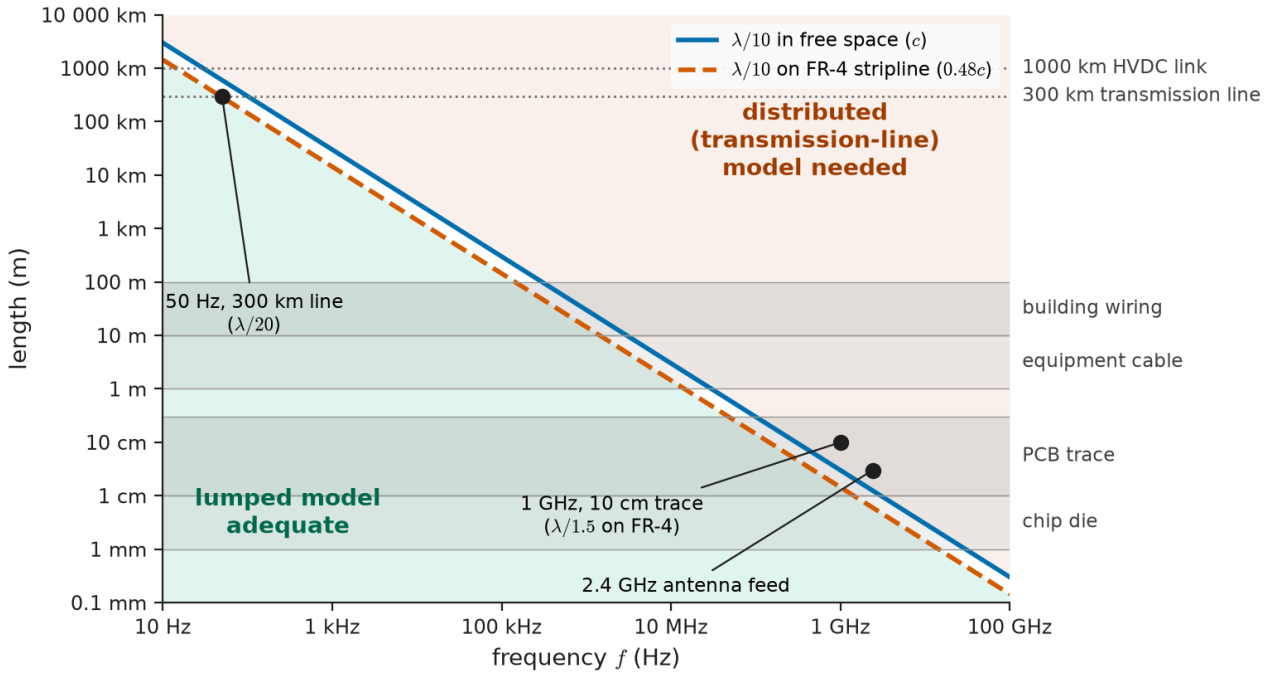


Figure 1.4 — Where the lumped model holds: one-tenth of a wavelength versus frequency, compared with typical circuit dimensions.

Figure 1.4 places typical circuit sizes against the $\lambda/10$ boundary. The conclusions are worth stating plainly:

- **Power systems.** Building wiring at 50 or 60 Hz is lumped by a factor of thousands. A 300 km transmission line at 50 Hz is about $\lambda/20$ long: lumped models with series impedance and shunt capacitance (the "nominal-π" model) are used up to roughly 250 km, but longer lines need distributed treatment (Chapter 14). Harmonics and switching transients reach far higher frequencies, so the same line can be lumped for the fundamental and distributed for a lightning surge.
- **Digital electronics.** For digital signals what matters is not the clock frequency but the edge rate: a signal edge with rise time t_r contains significant energy up to about $0.35/t_r$. A trace is commonly treated as lumped if its one-way delay is below about one-sixth of t_r.
- **RF and microwave.** Interconnects are almost always distributed; circuits are designed in terms of transmission lines and scattering parameters (Chapters 14 and 32).

Worked example — Is a PCB trace lumped? A microcontroller drives a clock line with a 1.0 ns rise time on an FR-4 stripline, delay $t_{pd} = \sqrt{4.3}/c = 6.92$ ns/m.

The significant bandwidth is about $0.35/t_r = 350$ MHz. Critical length, using the one-sixth rule: $\ell_c = t_r/(6 t_{pd}) = 1.0\text{ ns}/(6 \times 6.92\text{ ns/m}) = 0.024$ m, about 2.4 cm.

A 2 cm trace may be treated as a lumped capacitance on the driver; a 15 cm trace across the board is six times the critical length and will ring and reflect unless terminated, the "ringing bus" that Chapter 14

explains with the telegrapher's equations. The same clock with a 10 ns edge would have a 24 cm critical length: slowing edges is a legitimate way to stay in the lumped regime.

What fails, and why

When the lumped assumptions fail, the failure is not arbitrary; specific physics is missing.

- **Propagation delay.** The current entering one end of a wire no longer equals, at that instant, the current leaving the other; the distributed model of Chapter 14 is needed.
- **Radiation.** A conductor that is a significant fraction of a wavelength long radiates (an antenna, Chapter 32); energy leaves without passing through any element.
- **Induced voltage around loops.** Changing magnetic flux makes voltage path-dependent; a probe with a long ground lead "sees" voltages that are not in the circuit (Chapter 2).
- **Stray coupling.** Crosstalk and parasitic oscillation, sometimes recoverable by adding stray capacitances and mutual inductances explicitly.

> **Boundary.**
> Maxwell's equations, wave propagation and radiation are developed in *Engineering Physics*. This book uses their quasi-static limit (lumped circuits) and, from Chapter 14 on, their one-dimensional transmission-line limit.

1.9 Circuit diagrams and notation

A **circuit diagram** (schematic) records which element terminals are connected to which, nothing more. It is a topological statement: lengths and positions of the lines carry no meaning, and two drawings that connect the same terminals in the same way are the same circuit.

Nodes, branches and connections

- A **node** is a point where two or more element terminals are connected; all the wire drawn between them, however long, is one node and, under the lumped assumptions, one equipotential.
- A **branch** is a single two-terminal element (some texts use the word for a series string carrying one current).
- A **junction dot** marks a connection; lines crossing without a dot are not connected, and careful drawings avoid four-way junctions altogether.
- **Net labels** (VCC, CLK) connect every point carrying the same name, across sheets.

Reference and ground symbols

Three reference symbols are in common use: **earth (protective) ground**, three lines of decreasing length, bonded to the physical earth for safety; **chassis ground**, a hatched line, bonded to the enclosure; and **signal ground or common**, a triangle or bar, the 0 V reference of the circuit, which in a battery instrument connects to nothing outside. Analysing a circuit, all three are simply "the reference node". Designing one, the distinction matters for safety and noise: a circuit whose signal ground is

joined to earth in two places forms a ground loop, in which a changing magnetic field induces currents that appear as mains hum in audio or as baseline wander in an ECG.

Element symbols

Two symbol families coexist. **IEC 60617** (used in Europe, Asia and most of the world) draws a resistor as a rectangle; **IEEE Std 315 / ANSI Y32.2** (North America) draws it as a zigzag. Independent sources are circles; dependent sources are diamonds (Section 1.7). Both families are used in this book's figures, and engineers must read both.

Variable conventions

This book follows the widely used conventions below; later chapters add to them.

Table 1.9 — Notation conventions.

Symbol style	Meaning	Example
Lowercase, time argument	Instantaneous, possibly time-varying quantity	$v(t)$, $i_L(t)$, $p(t)$
Uppercase	DC (constant) value, or an amplitude or rms value	$V_{CC} = 5$ V, I_m, V_{rms}
Bold uppercase	Phasor (complex amplitude), from Chapter 8	$\mathbf{V} = 230\angle 0^\circ$ V
Uppercase with s argument	Laplace transform, from Chapter 11	$V(s)$, $H(s)$
Double subscript	Voltage from first to second node; current from first to second	v_{ab}, i_{ab}
Single subscript	Node voltage with respect to ground; branch or element current	v_3, i_R

Values use SI prefixes (p, n, µ, m, k, M, G); on schematics the prefix often replaces the decimal point, so "4k7" means 4.7 kΩ and "2R2" means 2.2 Ω, a habit that survives smudged printing.

Resistors and capacitors are manufactured in preferred-number series defined in **IEC 60063**, in which successive values form a geometric sequence. The number of values per decade sets the series, and the tolerance is chosen so that adjacent values roughly touch:

Table 1.10 — Preferred-number (E) series.

Series	Values per decade	Typical tolerance	Example values in one decade
E6	6	±20 %	1.0, 1.5, 2.2, 3.3, 4.7, 6.8
E12	12	±10 %	E6 plus 1.2, 1.8, 2.7, 3.9, 5.6, 8.2
E24	24	±5 %	E12 plus 1.1, 1.3, 1.6, 2.0, 2.4, 3.0, …
E96	96	±1 %	1.00, 1.02, 1.05, 1.07, …
E192	192	±0.5 % and tighter	1.00, 1.01, 1.02, 1.04, …

A calculated 300 Ω becomes 330 Ω (E12) or 300 Ω (E24); Chapter 2 checks such a choice for an LED resistor.

1.10 Basic measurement concepts

The quantities of this chapter are measured with voltmeters, ammeters, oscilloscopes and their combination in the digital multimeter (DMM). Measurement is itself a circuit problem: the instrument becomes part of the circuit, and it changes what it measures.

Connecting instruments

- A **voltmeter** goes *in parallel* with the element; an ideal one draws no current (infinite input resistance).
- An **ammeter** goes *in series*, so the circuit must be broken; an ideal one has no voltage across it (zero resistance).
- A **wattmeter** combines a series current sensor and a parallel voltage sensor and reports the average of vi (Chapter 9).
- A **clamp meter** senses the magnetic field around a conductor and measures current without breaking the circuit.

Figure 1.5 shows the models of real instruments: an ideal voltmeter in parallel with its input resistance R_{in}, and an ideal ammeter in series with its shunt and wiring resistance.

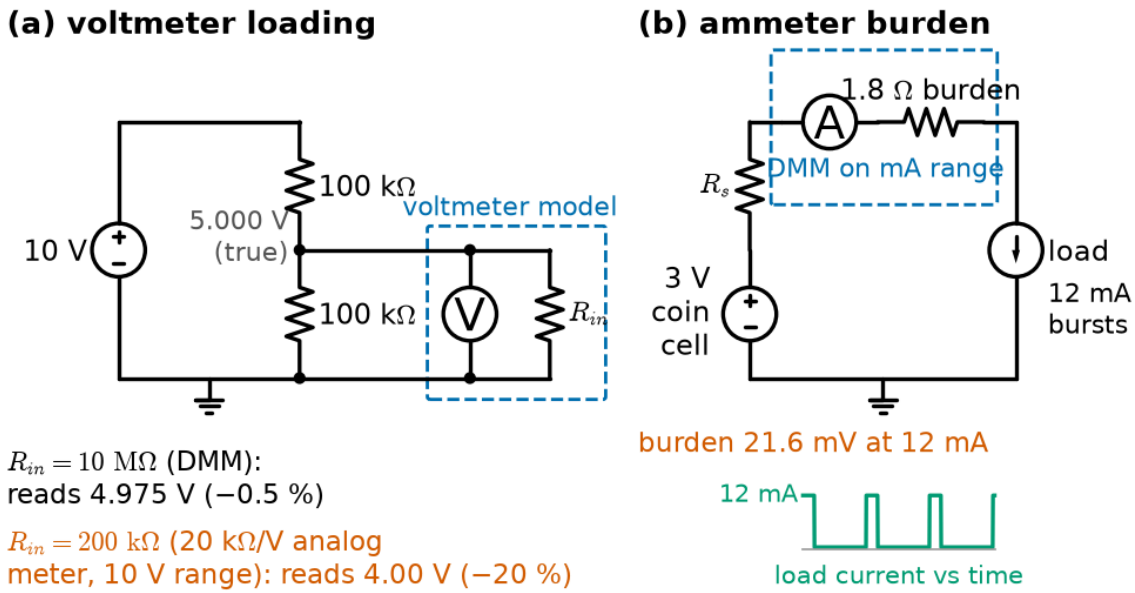


Figure 1.5 — Instruments are circuit elements: a voltmeter with finite input resistance loads a high-impedance divider, and an ammeter inserts a burden resistance in series.

Loading error

A voltmeter across a high-resistance circuit draws current that the circuit would not otherwise supply, lowering the voltage it reads.

Worked example — Voltmeter loading on a resistive divider. A 10.0 V supply drives a divider of two 100 kΩ resistors; the true midpoint voltage is 5.000 V. The lower resistor is measured with three instruments.

A DMM with $R_{in} = 10\ \text{M}\Omega$: the lower branch becomes $100\ \text{k}\Omega \parallel 10\ \text{M}\Omega = 99.01\ \text{k}\Omega$, and the reading is $10 \times 99.01/(100 + 99.01) = 4.975$ V, an error of −0.50 %.

An analogue meter rated 20 kΩ/V on its 10 V range has $R_{in} = 200\ \text{k}\Omega$: the lower branch becomes 66.67 kΩ and the reading is $10 \times 66.67/166.67 = 4.000$ V, an error of −20 %.

An oscilloscope with a 1× probe, $R_{in} = 1\ \text{M}\Omega$: lower branch 90.91 kΩ, reading 4.762 V, error −4.8 %. A 10× probe raises R_{in} to 10 MΩ and restores the DMM result, at the price of a tenfold reduction in signal.

The rule of thumb that follows — the instrument's input resistance should exceed the Thévenin resistance of the point being measured by at least a factor of 100 for 1 % accuracy — is made precise in Chapter 4. Here the Thévenin resistance of the divider midpoint is 50 kΩ, and the 10 MΩ DMM exceeds it by 200.

Burden voltage

An ammeter's series resistance develops a **burden voltage** that the circuit loses. Handheld DMMs specify it as so many millivolts per milliamp on each range; about 1–2 mV/mA on a milliamp range is typical, more on microamp ranges. In the ECG patch, a 12 mA burst through a 1.8 Ω burden drops about

22 mV — small, but enough to move a marginal brown-out event. Worse, the meter's averaging hides the bursts; a pulsed current needs a sense resistor and an oscilloscope, or an instrument that samples fast enough to resolve the pulses (Chapter 19).

Four-terminal (Kelvin) measurement

When a small resistance is measured, the resistance of the test leads and contacts adds directly to the reading: 0.1 Ω of lead resistance swamps a 10 mΩ busbar joint. The **four-terminal** or **Kelvin** method passes a known current through one pair of leads and measures the voltage with a second pair attached inside the current contacts. The sense leads carry negligible current, so V/I is the resistance between the sense points alone. Battery testers, four-terminal shunts and micro-ohmmeters all use it.

Accuracy, precision and resolution

Three terms are often confused:

- **Resolution** is the smallest change an instrument can display: one count of the least significant digit. A "6000-count" DMM on its 6 V range resolves 1 mV.
- **Accuracy** is how close the reading is to the true value. DMM specifications state it as ± (percentage of reading + number of counts), for example ±(0.5 % + 2 counts). On the 6 V range a 5.000 V reading is then uncertain by ±(25 mV + 2 mV) = ±27 mV.
- **Precision** (repeatability) is how closely repeated readings agree with each other, irrespective of whether they are correct.

A meter can be precise but inaccurate (a consistent offset), or show many digits of which none is trustworthy.

Table 1.11 — Typical instrument characteristics.

Instrument	Key parameter	Typical value
Handheld DMM, DC voltage	Input resistance; basic accuracy	10 MΩ; ±0.05 % to ±0.5 % of reading
Bench DMM (6½ digit), DC voltage	Input resistance (low ranges); accuracy	10 MΩ or above 10 GΩ selectable; about ±0.0035 %
Handheld DMM, current	Burden voltage	1–2 mV/mA (mA range), up to 100 µV/µA (µA range)
Oscilloscope input	Resistance and capacitance	1 MΩ in parallel with 10–20 pF; or 50 Ω
10:1 passive probe	Resistance and capacitance at the tip	10 MΩ in parallel with about 10 pF
Clamp meter, AC current	Accuracy	±1 % to ±3 % of reading

What a reading means for AC and pulsed signals

Average-responding meters are calibrated to read correctly only for sinusoids; **true-rms** meters compute $\sqrt{\overline{v^2}}$ and read correctly for any waveform within their bandwidth (Chapter 8). And any meter

that averages reports an average: a design that fails on *peak* current or *minimum* voltage needs an instrument that resolves the waveform in time.

> **Boundary.**
> The design of instruments themselves — analogue front ends, ADCs, probe compensation networks — belongs to *Electronics & Computing*. This chapter treats an instrument as a circuit element with known input resistance, burden and bandwidth.

What remains open

Open question	Where it is answered
How are currents and voltages constrained when elements are joined into a network?	Chapter 2 (Kirchhoff's laws, Ohm's law, power balance)
How is a network with many nodes and loops solved systematically?	Chapter 3 (nodal, mesh and matrix analysis)
Why does an instrument need input resistance far above the source's internal resistance, in general?	Chapter 4 (Thévenin's theorem, maximum power transfer)
How much energy do capacitors and inductors store, and how is a reservoir capacitor sized?	Chapter 5 (capacitors and inductors), Chapter 6 (first-order transients)
How do power and rms values work for sinusoidal and non-sinusoidal waveforms?	Chapters 8 and 9
What replaces the lumped model when a wire is a significant fraction of a wavelength?	Chapter 14 (transmission lines)
How do the device physics of transistors and op-amps give rise to dependent-source models?	*Electronics & Computing*
How does a meter that samples a pulsed signal avoid missing or misrepresenting it?	Chapter 19 (sampling and reconstruction)
What is the origin of shot noise from discrete charge carriers?	Chapter 29 (random signals and noise)

Next: Chapter 2 joins the elements of this chapter into networks and derives the two laws — Kirchhoff's current and voltage laws — that, together with Ohm's law, determine every current and voltage in a resistive circuit.

Chapter 2 — Fundamental Circuit Laws

Part I · Circuit Fundamentals

An electrician is called to a detached garage where a 3.7 kW fan heater trips nothing but "feels weak". At the consumer unit the supply reads 230 V. At the garage socket, with the heater running, it reads about 223 V; half an hour later, about 222 V. Nothing is faulty. The socket is fed by 30 m of 2.5 mm² copper cable, the heater draws about 16 A, and 60 m of copper (out and back) has a resistance of roughly 0.41 Ω, rising as the cable warms. The 7–8 V that never reach the heater are dissipated as more than 100 W of heat spread along the cable run. Three laws explain every number in that story: Ohm's law fixes the drop across each conductor, Kirchhoff's current law says the same 16 A flows in the outgoing and returning conductors, and Kirchhoff's voltage law says the drops around the loop must add up to the supply.

This chapter states those three laws, derives from them the rules that engineers use every day — series and parallel combination, voltage and current division, source transformation, network reduction — and checks every result against conservation of energy. Chapter 3 then turns the same laws into systematic methods for networks too tangled for reduction by hand.

> **Which physical principles fix the currents and voltages in a network of elements? How can a network be simplified, and how can an answer be checked, without losing any information at the terminals of interest?**

Concept	Engineering destination
Ohm's law, resistivity	Cable sizing and voltage drop, sense resistors, heater and fuse design
Circuit topology	Counting the equations a network needs; recognising series and parallel
Kirchhoff's current law	Busbar and node current checks, earth-leakage detection, current sensing
Kirchhoff's voltage law	Supply-rail budgets, loop drops in wiring, ground-loop pickup
Series and parallel combinations	Resistor networks, battery packs, parallel feeders
Voltage and current division	Sensor dividers, ADC scaling, ammeter shunts, attenuators
Equivalent resistance, Δ–Y	Input resistance of networks, bridges, R-2R converters
Power balance	Checking every analysis; efficiency and loss budgets
Source transformations	Battery-plus-charger systems, sensor front ends, Norton models
Network reduction	Hand analysis of ladders and distribution feeders

Roadmap. Ohm's law (2.1) → the topology vocabulary needed to state Kirchhoff's laws (2.2) → KCL (2.3) and KVL (2.4) → series and parallel combinations (2.5) and equivalent resistance, including Δ–Y conversion (2.6) → voltage and current division (2.7) → power balance as a check (2.8) → source transformations (2.9) → network reduction, which assembles all of these (2.10).

2.1 Ohm's law

Statement

For many conductors at constant temperature, the current is proportional to the applied voltage. With the passive sign convention (Chapter 1),

$$v = R\,i, \qquad i = G\,v, \qquad G = \frac{1}{R},$$

where R is the **resistance** in ohms (Ω) and G the **conductance** in siemens (S). A resistor obeying Ohm's law has a straight-line i–v characteristic through the origin (Figure 2.1). Its power absorption is

$$p = v\,i = i^2 R = \frac{v^2}{R} \;\geq 0,$$

so a positive resistor always absorbs power and turns it into heat.

Resistivity and geometry

For a uniform conductor of length ℓ and cross-section A,

$$R = \rho\,\frac{\ell}{A},$$

where ρ is the material's resistivity (Ω·m). The resistance grows with length and falls with area — the reason distribution cables are thick and heating elements are long thin coils. The resistivity of metals rises roughly linearly with temperature,

$$R(T) \approx R(T_0)\,[1 + \alpha\,(T - T_0)],$$

with temperature coefficient α. Table 2.1 lists values.

Table 2.1 — Resistivity and temperature coefficient at 20 °C (typical values).

Material	ρ (Ω·m)	α (1/K)	Where it is used
Silver	1.59×10^{-8}	0.0038	Contacts, RF plating
Copper (annealed)	1.72×10^{-8}	0.0039	Wiring, PCB traces, windings
Aluminium	2.82×10^{-8}	0.0040	Overhead lines, busbars
Tungsten	5.6×10^{-8}	0.0045	Lamp filaments, contacts
Constantan (Cu–Ni)	4.9×10^{-7}	about ±0.00002	Precision resistors, strain gauges
Nichrome (Ni–Cr)	1.10×10^{-6}	0.0004	Heating elements
Seawater	about 0.2	negative	Earthing, marine sensing
Intrinsic silicon	about 2.3×10^{3}	strongly negative	Semiconductor substrates
PTFE, polyethylene	above 10^{14}	—	Cable insulation

Cable resistance is standardised per kilometre; IEC 60228 limits for plain copper conductors at 20 °C are, for example, 12.1 Ω/km (1.5 mm²), 7.41 Ω/km (2.5 mm²), 4.61 Ω/km (4 mm²) and 1.83 Ω/km (10 mm²). These maxima are a little above $\rho\ell/A$ for pure copper because stranded conductors are slightly longer than the cable and the standard allows for manufacturing tolerance.

Where Ohm's law holds and where it does not

Ohm's law is not a law of nature in the sense that charge conservation is. It is an observed property of many materials — metals, carbon, metal oxides, electrolytes — within a range of conditions, and it fails in three common ways.

- **Temperature.** An incandescent lamp's tungsten filament heats from room temperature to about 2500 °C, and its resistance rises tenfold or more. A 12 V, 5 W lamp has a hot resistance of $12^2/5 = 28.8\ \Omega$ but a cold resistance of only 2–3 Ω, so it draws a large inrush current when switched on. Its i–v curve bends over (Figure 2.1). The resistor is still ohmic at each instant; its resistance simply depends on its own power history.
- **Nonlinear conduction.** A semiconductor diode conducts almost nothing below about 0.6 V and then rises exponentially; a metal-oxide varistor is nearly an open circuit until its clamping voltage. These devices have no resistance in the Ohm's-law sense, although a slope resistance dv/di can be defined at an operating point.
- **Extreme fields and frequencies.** Very high current densities (electromigration), very high fields (breakdown of insulators) and high frequencies (skin effect, which raises the AC resistance of a conductor because current crowds to its surface) all change the effective resistance.

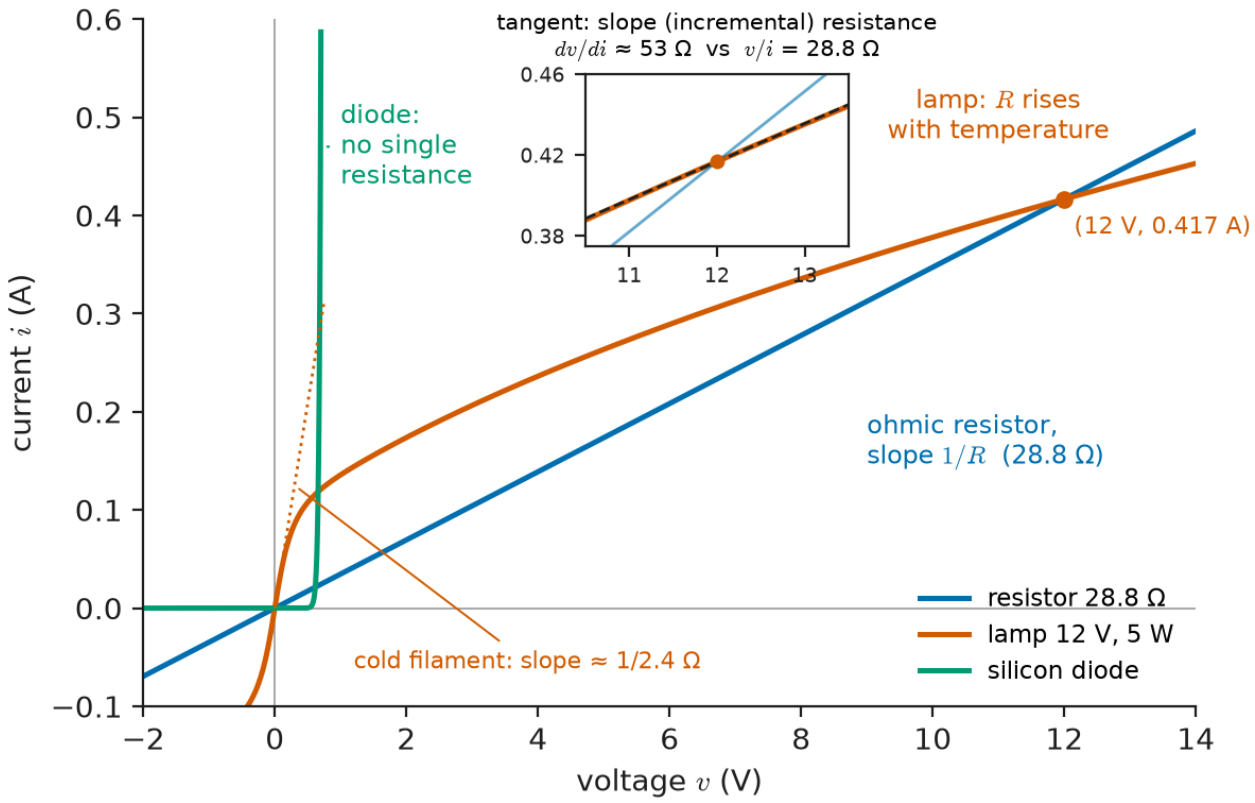


Figure 2.1 — Current–voltage characteristics: an ohmic resistor, an incandescent lamp, and a silicon diode.

> **Status of the principle: Ohm's law**
> *Empirical.* A material property observed for metals, alloys, carbon and many other conductors at fixed temperature and moderate fields and frequencies; it does not hold for semiconductor junctions, gas discharges, varistors or for conductors whose temperature changes with load.

Resistor ratings

A real resistor has a nominal value, a tolerance (Table 1.10), a temperature coefficient, a maximum working voltage and a power rating. Typical power ratings at 70 °C ambient are 0.063 W (0402 chip), 0.1 W (0603), 0.125 W (0805), 0.25 W (1206) and 0.25–1 W for leaded film resistors; wirewound and metal-clad power resistors reach hundreds of watts on a heat sink. Above the rated ambient, the permitted power is derated linearly to zero at the maximum element temperature.

Worked example — Voltage drop in a final circuit. The garage of the opening story is fed by a twin-and-earth cable with 2.5 mm² copper conductors, 30 m long. Take $\rho = 1.72 \times 10^{-8}\ \Omega\,\mathrm{m}$ at 20 °C and $\alpha = 0.00393\ /\mathrm{K}$. The heater current is 16 A.

Loop length (live out, neutral back): $\ell = 60$ m. Loop resistance at 20 °C: $R = \rho \ell / A = 1.72 \times 10^{-8} \times 60 / (2.5 \times 10^{-6}) = 0.413\ \Omega$.

Voltage drop: $\Delta v = iR = 16 \times 0.413 = 6.60$ V, which is 2.87 % of 230 V. The socket sees about 223.4 V. Power lost in the cable: $i^2 R = 16^2 \times 0.413 = 106$ W.

With the conductors at 70 °C (the rated maximum for PVC insulation), $R = 0.413 \times (1 + 0.00393 \times 50) = 0.494\ \Omega$, giving $\Delta v = 7.90$ V (3.44 %) and a loss of 126 W. The socket sees about 222.1 V.

Wiring rules commonly limit the drop from origin to load to 3–5 %, depending on country and load type. This run is within a 5 % limit but over a 3 % one at full temperature; a 4 mm² cable would cut the drop by a factor 2.5/4 to about 4.9 V. (Holding the current at exactly 16 A slightly overstates the drop: a resistive heater draws a little less current at reduced voltage. Chapter 3 solves such circuits exactly.)

2.2 Circuit topology

Kirchhoff's laws are statements about how elements are connected, so they need a vocabulary for connection. The full graph-theoretic treatment is in Chapter 3; this section introduces only what Chapter 2 uses.

- A **node** is a point at which two or more elements are joined. All the wire between element terminals that is drawn as connected is one node.
- A **branch** is a single two-terminal element (a resistor, a source).
- A **loop** is any closed path through branches that visits no node twice.
- A **mesh** is a loop that encloses no other branch when the circuit is drawn on a plane without crossings. Only **planar** circuits — those that can be drawn without crossings — have meshes.
- Two elements are in **series** if they share exactly one node and no other element connects to that node; they then carry the same current.
- Two elements are in **parallel** if they are connected between the same pair of nodes; they then have the same voltage.

If a connected circuit has b branches and n nodes, it has exactly $n - 1$ independent KCL equations and $b - n + 1$ independent KVL equations, and a planar circuit has exactly $b - n + 1$ meshes. The sum is b equations, which together with the b element laws determine the $2b$ unknown branch voltages and currents. Chapter 3 proves these counts.

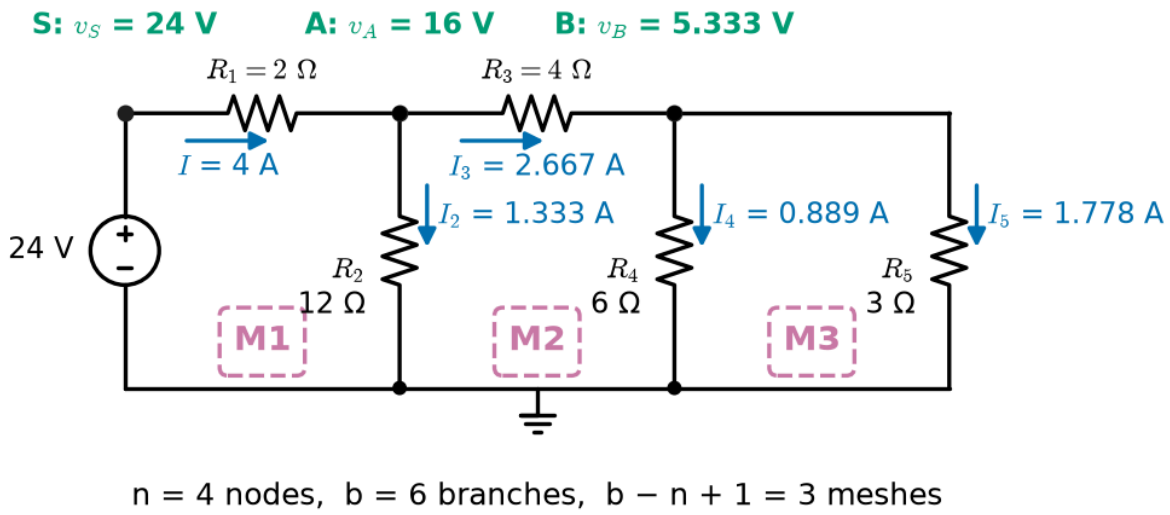


Figure 2.2 — A resistive ladder used throughout the chapter, with node names, branch currents and node voltages from the full solution.

The ladder of Figure 2.2 has $n = 4$ nodes (S, A, B and ground) and $b = 6$ branches (the source and five resistors), so it has 3 independent KCL equations and $6 - 4 + 1 = 3$ independent KVL equations, matching its three meshes. The source and R_1 are in series (node S joins only those two); R_4 and R_5 are in parallel (both run from B to ground). R_2 and R_3 are neither: node A joins three elements.

A circuit diagram can be redrawn freely without changing its topology. A common error is to see two resistors drawn side by side and assume they are in parallel; the test is not their position on the page but whether they connect the same two nodes.

2.3 Kirchhoff's current law

Statement

Kirchhoff's current law (KCL): the algebraic sum of the currents leaving any node is zero at every instant,

$$\sum_{k} i_k(t) = 0,$$

with currents entering counted negative (or, equivalently, the sum of currents entering equals the sum leaving).

KCL is charge conservation applied to a node. A node in a lumped circuit is an ideal conductor with negligible capacitance to its surroundings: it cannot accumulate charge. Whatever charge flows in per second must flow out per second.

> **Status of the principle: Kirchhoff's current law**
> *Fundamental.* It is conservation of charge applied to a region that stores no net charge; in a lumped circuit every node satisfies this exactly. It fails only when a node's own capacitance to its surroundings carries significant displacement current — at high frequency or in distributed structures.

Generalisation to closed surfaces

KCL applies not only to a node but to any closed surface drawn around part of a circuit: the net current crossing the surface is zero, because the elements inside, being lumped, hold no net charge either. The surface may cut through several wires, or enclose an entire subsystem. This is called a **cut-set** form of KCL, and Chapter 3 uses it systematically.

Two engineering consequences follow directly.

- **Residual-current devices.** A residual-current device (RCD, or GFCI in North America) passes the live and neutral conductors of a circuit together through a toroidal current transformer. If all the load current returns through the neutral, KCL for a surface around the load says the live and neutral currents sum to zero, and the transformer sees nothing. If 30 mA leaks to earth through a person or damaged insulation, the sum is 30 mA and the device trips, typically within 40 ms at five times its rated residual current. The device is a hardware implementation of KCL.
- **Current returns.** Every current must return to its source. On a printed circuit board, the return current of a fast signal flows in the ground plane directly under the trace; a gap in the plane forces it to detour, enlarging the loop and creating radiated emission. KCL guarantees there is a return path; KVL and Faraday's law (Section 2.4) decide how much trouble its shape causes.

Applying KCL

At node A of the ladder, with the currents of Figure 2.2, the current entering from R_1 is 4 A and the currents leaving through R_2 and R_3 are 1.333 A and 2.667 A: $-4 + 1.333 + 2.667 = 0$. At node B: $-2.667 + 0.889 + 1.778 = 0$. Applying KCL at the ground node gives an equation that is the negative sum of the others; that is why only $n - 1$ of the n node equations are independent.

Where KCL fails

At high frequency, a "node" — a pad, a via, a length of trace — has capacitance to nearby conductors. A stray capacitance of 1 pF at 100 MHz has a reactance of $1/(2\pi \times 10^{8} \times 10^{-12}) = 1.59\ \text{k}\Omega$, so a node at 1 V sends about 0.6 mA into its surroundings through the field. KCL still holds if that stray capacitance is drawn in as a circuit element; it fails only if the displacement current is ignored. At still higher frequencies, where the "node" is a significant fraction of a wavelength, no single lumped capacitance captures the behaviour and the distributed models of Chapter 14 take over.

2.4 Kirchhoff's voltage law

Statement

Kirchhoff's voltage law (KVL): the algebraic sum of the voltages around any closed loop is zero at every instant,

$$\sum_{k} v_k(t) = 0,$$

where each voltage is counted positive if the loop traversal goes from its + to its − mark (a drop) and negative otherwise (a rise).

KVL expresses the fact that voltage is a difference of potentials (Section 1.1). If every node has a well-defined potential, then going around any loop and returning to the starting node must bring the total change of potential back to zero. In energy terms: a charge carried around a closed loop returns to the same potential and so has gained no net energy.

> **Status of the principle: Kirchhoff's voltage law**
> *Fundamental* in the lumped regime. It follows from energy conservation for a conservative (curl-free) electric field; by Faraday's law it holds exactly when no time-varying magnetic flux links the loop and approximately when that linkage is negligible, which is part of the lumped-element assumption.

Applying KVL

Around mesh M1 of the ladder (source, R_1, R_2), traversed clockwise: a rise of 24 V through the source, a drop of $2 \times 4 = 8$ V across R_1, a drop of 16 V across R_2. Sum: $-24 + 8 + 16 = 0$. Around M2 (R_2 upward, R_3, R_4): $-16 + 4 \times 2.667 + 5.333 = 0$. Around M3 ($R_4$ upward, R_5 downward): $-5.333 + 5.333 = 0$, which is just the statement that parallel elements share a voltage.

KVL along an *open* path gives the voltage between its ends: v_{SB} equals the sum of drops from S to B along any path, $8 + 10.667 = 18.667$ V, which is also $v_S - v_B = 24 - 5.333$. This "walk the path" use of KVL is how a node voltage is read from a set of element voltages.

Where KVL fails: induced voltages

Faraday's law says that around a loop linked by a changing magnetic flux Φ,

$$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi}{dt}.$$

The voltage between two points then depends on the path between them, and KVL in its simple form is wrong. Circuit theory recovers by two devices. Where the flux is deliberate — inside an inductor or transformer — it is assigned to an element whose voltage $L\,di/dt$ or $M\,di/dt$ appears in the KVL sum (Chapter 5). Where it is not deliberate, it appears as interference.

A biomedical example shows the scale. ECG leads from the left arm and right arm to an amplifier enclose a loop area that can easily reach 0.1 m². In a magnetic field of 1 μT peak at 50 Hz, typical near mains wiring and transformers, the induced voltage is $A\,\omega B = 0.1 \times 2\pi \cdot 50 \times 10^{-6} = 31$ μV peak; at 10 μT, near a transformer, it is 0.31 mV — a third of an ECG R-wave. Twisting the lead wires together collapses the loop area and the pickup with it. The same physics explains why oscilloscope probes should be used with the shortest possible ground connection, and why ground loops in audio systems produce hum.

2.5 Series and parallel combinations

Resistors in series

Resistors $R_1, \dots, R_N$ in series carry the same current i. By KVL their voltages add, and by Ohm's law each is $R_k i$, so

$$v = (R_1 + R_2 + \cdots + R_N)\, i, \qquad R_{\mathrm{s}} = \sum_{k=1}^{N} R_k .$$

The series resistance is larger than any of its parts.

Resistors in parallel

Resistors in parallel share a voltage v. By KCL their currents add, and each is $v/R_k = G_k v$, so

$$i = (G_1 + G_2 + \cdots + G_N)\, v, \qquad G_{\mathrm{p}} = \sum_{k=1}^{N} G_k, \qquad \frac{1}{R_{\mathrm{p}}} = \sum_{k=1}^{N} \frac{1}{R_k}.$$

The parallel resistance is smaller than any of its parts. For two resistors,

$$R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2},$$

and N equal resistors R in parallel give R/N. The rule of thumb "parallel combinations are dominated by the smallest resistor, series by the largest" is a fast check on any hand calculation.

> **Status of the principle: Series and parallel combination rules**
> *Derived*. Direct consequences of KCL, KVL and the element laws; exact for linear resistors, and extended to impedances in Chapter 8 and to capacitors and inductors in Chapter 5.

Sources in series and parallel

Ideal voltage sources in series add algebraically: battery packs stack cells in series to raise voltage, 12 V lead-acid batteries contain six 2.1 V cells, and an electric-vehicle pack may contain 96 lithium-ion cells in series for a nominal 350 V. Ideal current sources in parallel add likewise; solar cells in parallel strings add their currents.

The forbidden cases are the duals noted in Chapter 1: unequal ideal voltage sources may not be paralleled, and unequal ideal current sources may not be placed in series. Real sources can be paralleled because their internal resistances absorb the difference — but the circulating current that results can be large. Two 12 V batteries at 12.6 V and 12.2 V with 10 mΩ internal resistance each, connected in parallel with no load, drive $(12.6 - 12.2)/0.02 = 20$ A from one to the other. Battery packs therefore match the state of charge of parallel strings before connecting them, and paralleled power supplies use current-sharing control.

2.6 Equivalent resistance

The idea

Any two-terminal network of resistors behaves, at its terminals, exactly like a single resistor R_{eq}: apply a voltage v, and the current drawn is v/R_{eq}. The **equivalent resistance** (or input resistance) captures everything an external circuit can learn about the network. Finding it by repeated series and parallel combination is the core skill of network reduction (Section 2.10). For the ladder of Figure 2.2, seen from the source:

$$R_{eq} = R_1 + R_2 \parallel \big(R_3 + R_4 \parallel R_5\big) = 2 + 12 \parallel \big(4 + 6 \parallel 3\big) = 2 + 12 \parallel 6 = 2 + 4 = 6\ \Omega.$$

The source therefore delivers $24/6 = 4$ A, which is the current already shown in the figure.

Networks that are neither series nor parallel

Not every network reduces by series and parallel steps. The **bridge** — four resistors in a diamond with a fifth across the middle — is the classic example: no two of its resistors are in series and no two are in parallel. Three tools handle it.

1. **Symmetry and balance.** If the bridge is balanced ($R_1/R_3 = R_2/R_4$ in the labelling of Figure 2.3), the two midpoints sit at the same potential, no current flows in the middle resistor, and it can be removed (or replaced by a short) without changing anything. This is the principle of the Wheatstone bridge used in strain-gauge and resistance-thermometer measurements (Chapter 3 analyses an unbalanced one).
2. **Δ–Y conversion.** Three resistors connected in a triangle (Δ, also called π) between three nodes can be replaced by three resistors connected in a star (Y, also called T) from the same nodes to a new internal node, without changing anything measurable at the three nodes.
3. **Systematic analysis.** Any network can be solved by nodal analysis (Chapter 3) with a test source at its terminals.

Δ–Y (π–T) transformation

Let a Δ have resistors R_{ab}, R_{bc}, R_{ca} between nodes a, b, c, and let the equivalent Y have resistors R_a, R_b, R_c from those nodes to a centre. Requiring that the resistance between every pair of nodes, with the third left open, be the same in both networks gives

$$R_a = \frac{R_{ab}R_{ca}}{R_{ab}+R_{bc}+R_{ca}}, \qquad R_b = \frac{R_{ab}R_{bc}}{R_{ab}+R_{bc}+R_{ca}}, \qquad R_c = \frac{R_{bc}R_{ca}}{R_{ab}+R_{bc}+R_{ca}}.$$

Each Y resistor is the product of the two Δ resistors touching its node, divided by the sum of all three. The inverse (Y–Δ) transformation is

$$R_{ab} = \frac{R_aR_b + R_bR_c + R_cR_a}{R_c},$$

and cyclically: each Δ resistor is the sum of the pairwise products of the Y resistors, divided by the Y resistor *opposite* it. For equal resistors, $R_Y = R_\Delta/3$. The same transformation, with impedances in place of resistances, converts star- and delta-connected three-phase loads (Chapter 10).

> **Status of the principle: Δ–Y transformation**
> *Derived*. An exact terminal equivalence for linear resistive (and, with impedances, linear AC) three-terminal networks, following from KCL, KVL and Ohm's law; it says nothing about the internal node of the Y, which does not exist in the Δ.

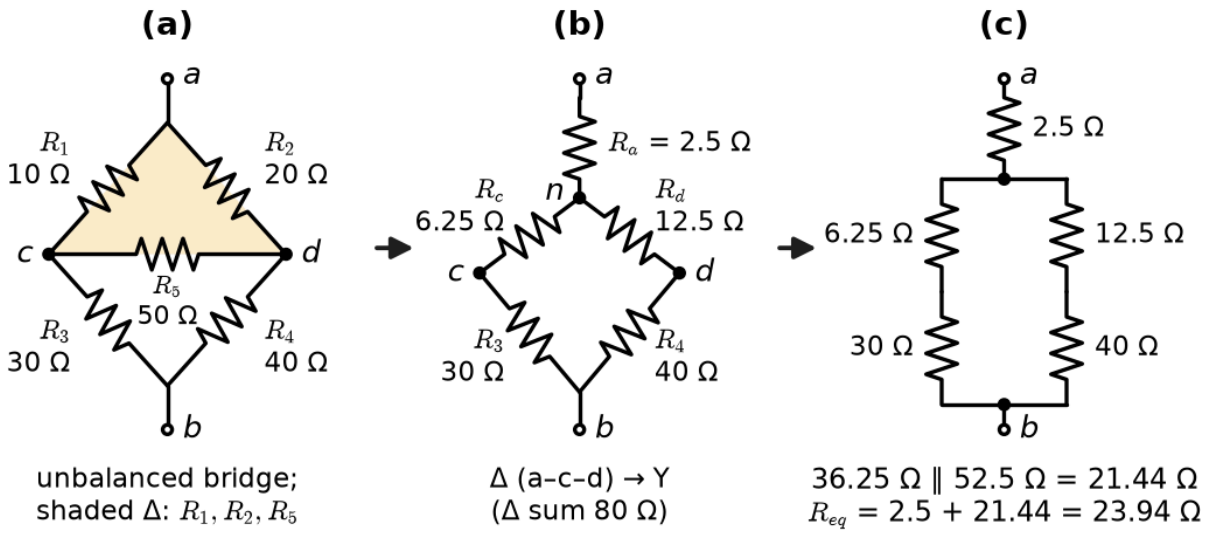


Figure 2.3 — Reducing an unbalanced bridge by a delta–wye transformation.

Worked example — Equivalent resistance of an unbalanced bridge. In Figure 2.3, $R_1 = 10\ \Omega$ (a–c), $R_2 = 20\ \Omega$ (a–d), $R_3 = 30\ \Omega$ (c–b), $R_4 = 40\ \Omega$ (d–b) and $R_5 = 50\ \Omega$ (c–d). The bridge is unbalanced: $R_1/R_3 = 0.333$ but $R_2/R_4 = 0.5$.

Convert the Δ formed by R_1, R_2, R_5 (between a, c, d) to a Y. The sum is $10 + 20 + 50 = 80\ \Omega$:

$$R_a = \frac{10 \times 20}{80} = 2.5\ \Omega, \quad R_c = \frac{10 \times 50}{80} = 6.25\ \Omega, \quad R_d = \frac{20 \times 50}{80} = 12.5\ \Omega.$$

Now R_c is in series with R_3 (36.25 Ω), R_d is in series with R_4 (52.5 Ω), and these two paths are in parallel: $36.25 \times 52.5/88.75 = 21.44\ \Omega$. Adding R_a in series:

$$R_{eq} = 2.5 + 21.44 = 23.94\ \Omega.$$

A nodal analysis with a 1 A test current injected at a (Chapter 3 method) gives $v_a = 23.94$ V, confirming the result. Had the bridge been balanced, removing R_5 would have given the answer in one line; unbalanced, the Δ–Y step was essential.

The R-2R ladder

A ladder of series resistors R and shunt resistors $2R$, terminated in $2R$, has a remarkable property: looking into any node toward the termination, the equivalent resistance is $2R \parallel 2R = R$, and adding the next series R gives $2R$ again. The structure is self-similar. As a result, a current injected at the input halves at every node — exactly the binary weighting needed by a digital-to-analogue converter, built from only two resistor values that can be matched closely on a chip. The R-2R DAC is designed in *Electronics & Computing*; its resistive core is a series–parallel network.

2.7 Voltage and current division

The voltage divider

Two resistors in series across a voltage v_s share it in proportion to their resistances. The current is $i = v_s/(R_1 + R_2)$, so the voltage across R_2 is

$$v_2 = v_s \, \frac{R_2}{R_1 + R_2}.$$

More generally, in a series string, each resistor receives the fraction R_k/R_s of the total voltage. The divider is everywhere: setting a reference voltage, scaling a 400 V DC bus down to a 3.3 V ADC input, biasing a transistor, reading a resistive sensor.

The formula holds only if nothing draws current from the junction. A load R_L across R_2 replaces R_2 by $R_2 \parallel R_L$ and lowers the output; the voltmeter-loading example of Chapter 1 was exactly this. Chapter 4 generalises the result: the unloaded divider looks to its load like a source $v_s R_2/(R_1 + R_2)$ in series with $R_1 \parallel R_2$.

The current divider

Two resistors in parallel fed by a current i_s share it in proportion to their *conductances*, that is, in inverse proportion to their resistances:

$$i_1 = i_s \, \frac{G_1}{G_1 + G_2} = i_s \, \frac{R_2}{R_1 + R_2}.$$

Note the crossed subscript: the current through R_1 involves R_2 in the numerator. The larger share takes the lower-resistance path. For N parallel branches, $i_k = i_s \, G_k / \sum G$.

> **Status of the principle: Voltage- and current-division rules**
> *Derived*. Exact consequences of KCL, KVL and Ohm's law for an unloaded series string or a set of parallel branches; any current drawn from a divider tap, or any extra branch, must be included before the rule is applied.

Sensor dividers

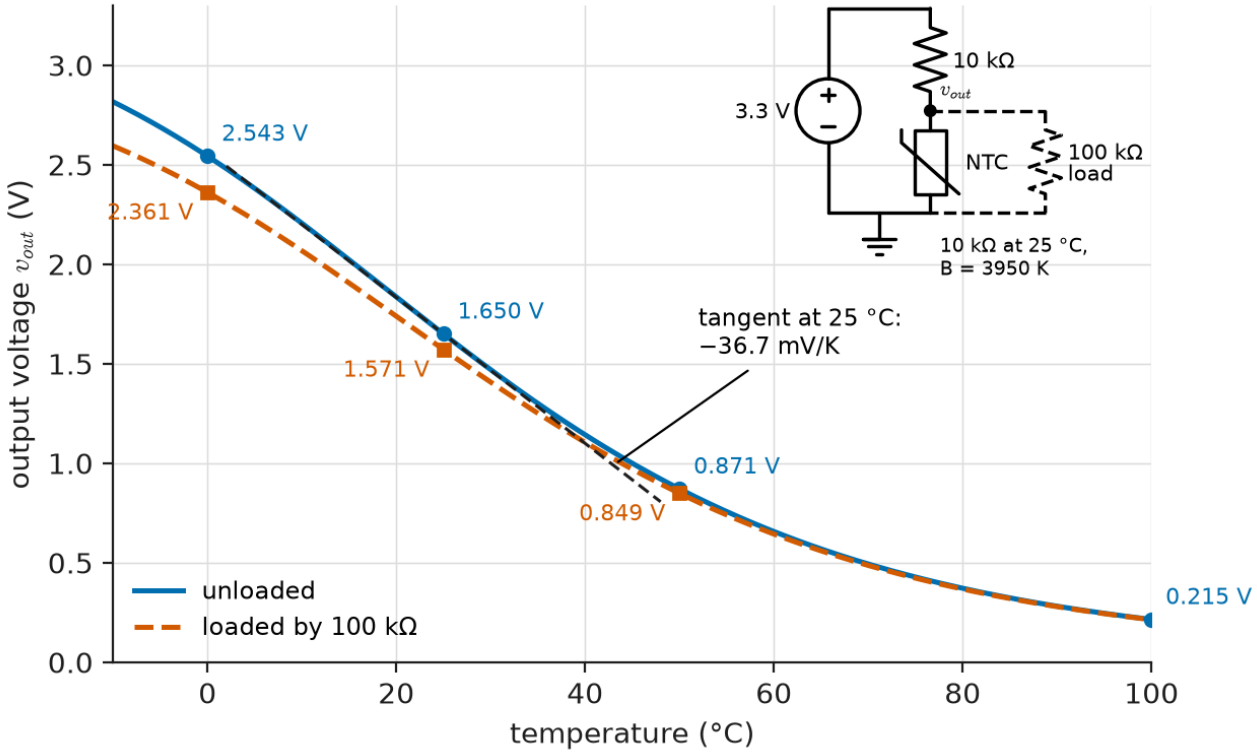


Figure 2.4 — An NTC thermistor in a voltage divider: output voltage against temperature, unloaded and loaded by a 100 kOhm input.

Worked example — An NTC thermistor divider. A temperature sensor in a battery pack uses a negative-temperature-coefficient (NTC) thermistor with $R_{25} = 10\ \text{k}\Omega$ at 25 °C and the common "B-parameter" model

$$R(T) = R_{25} \exp\left[B\left(\frac{1}{T} - \frac{1}{298.15\ \text{K}}\right)\right], \qquad B = 3950\ \text{K}.$$

It sits at the bottom of a divider with a fixed 10 kΩ resistor to a 3.3 V reference (Figure 2.4).

At 0 °C ($T = 273.15$ K): $R = 10 \times \exp[3950\,(1/273.15 - 1/298.15)] = 33.62\ \text{k}\Omega$, and $v_{out} = 3.3 \times 33.62/43.62 = 2.543$ V.

At 25 °C: $R = 10\ \text{k}\Omega$ and $v_{out} = 1.650$ V. At 50 °C: $R = 3.588\ \text{k}\Omega$ and $v_{out} = 0.871$ V. At 100 °C: $R = 0.698\ \text{k}\Omega$ and $v_{out} = 0.215$ V.

Near 25 °C the sensitivity is about −36.7 mV/K, so a 12-bit ADC with a 3.3 V range (0.81 mV per step) resolves about 0.02 K there; resolution is much coarser at the extremes where the curve flattens.

Choosing the fixed resistor equal to R_{25} places the steepest part of the curve at 25 °C.

Now connect the divider to an amplifier with a 100 kΩ input resistance. At 25 °C the bottom leg becomes $10 \parallel 100 = 9.09$ kΩ and $v_{out} = 3.3 \times 9.09/19.09 = 1.571$ V, an error of −79 mV, equivalent to more than 2 K. At 0 °C the error is −182 mV. The divider must feed a high-impedance input or a buffer, or the loading must be included in the calibration.

Shunts and current division

A moving-coil meter, an ADC current-sense input or a protection circuit often needs only a small, known fraction of a large current. A low-value **shunt** resistor in parallel with the sensing element diverts the rest.

Worked example — Extending an ammeter range with a shunt. A moving-coil movement reads full scale at 1.0 mA and has a coil resistance of 100 Ω. It must read 1.0 A full scale.

At full scale the movement carries 1.0 mA, so the shunt must carry 999 mA. The two are in parallel and share the movement's full-scale voltage $1.0\ \text{mA} \times 100\ \Omega = 0.100$ V. Hence

$$R_{sh} = \frac{0.100\ \text{V}}{0.999\ \text{A}} = 0.1001\ \Omega.$$

The current-division rule confirms it: the movement's share is $R_{sh}/(R_{sh}+100) = 0.1001/100.1001 = 0.00100$ of the total. The shunt dissipates $0.999^2 \times 0.1001 \approx 0.10$ W at full scale. Its value is so small that the resistance of connecting leads and contacts would cause serious error; this is why precision shunts have four terminals (Chapter 1). Note also the burden: the meter drops 0.1 V at full scale, the same burden-voltage issue as in Chapter 1.

A three-branch example shows the conductance form. A current of 50 mA divides among parallel resistors of 100 Ω, 220 Ω and 470 Ω, with conductances 10, 4.545 and 2.128 mS (total 16.67 mS). The branch currents are $50 \times 10/16.67 = 29.99$ mA, $50 \times 4.545/16.67 = 13.63$ mA and $50 \times 2.128/16.67 = 6.38$ mA, which sum to 50.00 mA.

Worked example — An LED current-limiting resistor. A red indicator LED with forward voltage about 2.0 V at 10 mA is driven from a 5.0 V logic rail. KVL around the loop gives the resistor voltage $5.0 - 2.0 = 3.0$ V, and Ohm's law gives $R = 3.0/0.010 = 300\ \Omega$. Choosing the E12 value 330 Ω, the current becomes $3.0/330 = 9.09$ mA, a change in brightness the eye barely notices. The resistor dissipates $3.0 \times 9.09\ \text{mA} = 27$ mW, comfortably inside the 0.1 W rating of an 0603 chip. (The LED is nonlinear; treating its voltage as constant is an approximation that works because its voltage changes only slightly with current.)

2.8 Power balance

The principle

In any circuit, at every instant, the sum of the powers absorbed by all elements — each computed with the passive sign convention — is zero:

$$\sum_{k=1}^{b} v_k \, i_k = 0.$$

Equivalently, the power delivered by the sources equals the power absorbed by everything else. This is energy conservation, but it can also be proved from KCL and KVL alone, without reference to what the elements are; that proof, known as **Tellegen's theorem**, is given in Chapter 3.

> **Status of the principle: Power balance (Tellegen's theorem)**
> *Derived.* A mathematical consequence of KCL and KVL for any lumped network, linear or not; it holds instant by instant, and with complex power in AC steady state (Chapter 9).

Power balance is the single most useful check on an analysis. A sign error in a current, a wrong reference direction or an arithmetic slip almost always breaks it.

Worked example — Power balance in the ladder. For the ladder of Figure 2.2, with all elements labelled passively:

Table 2.2 — Power balance of the ladder network.

Element	Voltage (V)	Current (A)	Power absorbed (W)
24 V source	24.000	−4.000	−96.000
$R_1 = 2\ \Omega$	8.000	4.000	32.000
$R_2 = 12\ \Omega$	16.000	1.333	21.333
$R_3 = 4\ \Omega$	10.667	2.667	28.444
$R_4 = 6\ \Omega$	5.333	0.889	4.741
$R_5 = 3\ \Omega$	5.333	1.778	9.481
Sum			**0.000**

The resistors absorb $32.000 + 21.333 + 28.444 + 4.741 + 9.481 = 96.000$ W and the source delivers 96 W. Note where the power goes: one-third is lost in R_1, the "feeder" resistance next to the source, even though it is the smallest resistor in the circuit — because it carries the full current.

Efficiency and losses

In power delivery, power balance becomes an efficiency calculation: $\eta = P_{load}/P_{source}$. In the garage example the heater receives about $16 \times 223.4 = 3.57$ kW while the cable dissipates 106 W, so 97 % of the power leaving the consumer unit reaches the heater. The same arithmetic, scaled to megawatts and kilometres, is why transmission is done at high voltage: for a fixed power $P = VI$, the line loss I^2R falls as $1/V^2$. Doubling the transmission voltage quarters the loss.

2.9 Source transformations

Equivalent practical sources

Chapter 1 modelled a practical source as an ideal voltage source V_s in series with R_s. Its terminal law is $v = V_s - R_s i$. The same law is produced by an ideal current source I_s in parallel with R_s, for which KCL at the terminal gives $i = I_s - v/R_s$, that is, $v = R_s I_s - R_s i$. The two are identical at the terminals if

$$V_s = R_s\, I_s, \qquad I_s = \frac{V_s}{R_s},$$

with the *same* resistance R_s in both. The current source arrow points toward the terminal that the voltage source's + mark identifies. This interchange is a **source transformation**; the series form is a Thévenin source and the parallel form a Norton source (Chapter 4).

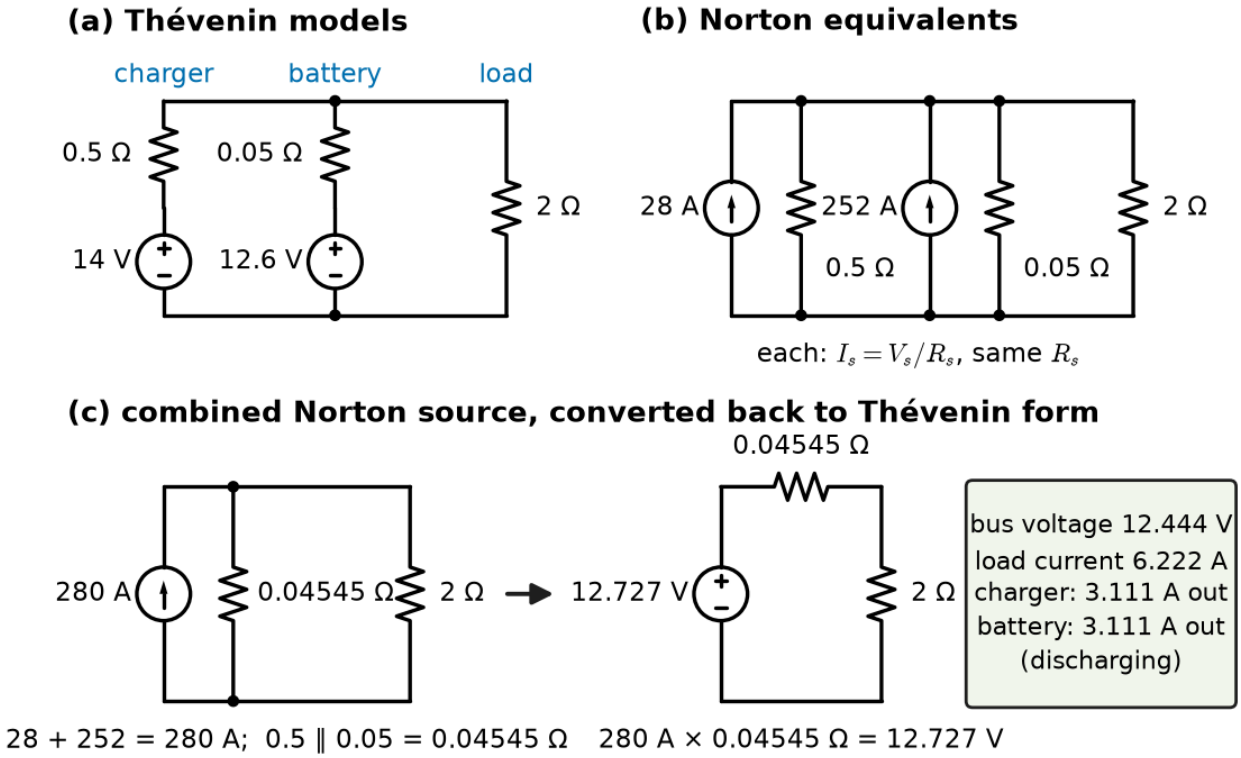


Figure 2.5 — Source transformation applied to a battery and a charger feeding a common load.

> **Status of the principle: Source transformation**
> *Derived.* Exact terminal equivalence of the series voltage-source and parallel current-source models of a linear source, from Ohm's law and KCL. It holds only at the terminals: the internal power dissipation of the two forms differs, and it does not apply to an ideal source with $R_s = 0$ or $R_s \to \infty$.

The last point deserves emphasis. Open-circuited, the Thévenin form dissipates nothing internally, while the Norton form dissipates $I_s^2 R_s$ in its parallel resistor. The two models agree about the outside world, not about what happens inside the box.

Using transformations

Source transformations are used to put sources in the form that makes the next step easiest: voltage sources in series combine by adding, current sources in parallel combine by adding. A network with several practical sources feeding a common pair of nodes reduces quickly by converting them all to Norton form, adding currents and conductances, and converting back.

Worked example — A battery and a charger in parallel. A 12 V system has a lead-acid battery (12.6 V open-circuit, internal resistance 0.05 Ω) and a small charger modelled as a 14.0 V source with 0.5 Ω output resistance. Both are connected across a 2.0 Ω load (Figure 2.5). Does the charger keep up?

Convert both to Norton form: charger $14.0/0.5 = 28$ A in parallel with 0.5 Ω; battery $12.6/0.05 = 252$ A in parallel with 0.05 Ω. The two current sources are in parallel and add to 280 A; the two resistors are in parallel, $0.5 \parallel 0.05 = 0.04545$ Ω. Converting back: a 12.727 V source in series with 0.04545 Ω.

With the 2.0 Ω load, the bus voltage is $12.727 \times 2.0/(2.0 + 0.04545) = 12.444$ V and the load current is 6.222 A. Returning to the original circuit: the charger delivers $(14.0 - 12.444)/0.5 = 3.111$ A and the battery delivers $(12.6 - 12.444)/0.05 = 3.111$ A. The battery is *discharging*: the charger cannot supply the load alone.

Power balance: the charger's ideal source delivers $14.0 \times 3.111 = 43.56$ W and the battery's $12.6 \times 3.111 = 39.20$ W, a total of 82.76 W. The load absorbs $12.444^2/2.0 = 77.43$ W and the two internal resistances absorb $3.111^2 \times 0.5 + 3.111^2 \times 0.05 = 5.32$ W. Total absorbed: 82.76 W. With the load disconnected, the bus rises to 12.727 V and the battery charges at $(12.727 - 12.6)/0.05 = 2.5$ A.

2.10 Network reduction

The procedure

Network reduction finds a desired voltage or current by repeatedly replacing parts of a network with simpler equivalents, then working back out. The standard steps are:

1. **Identify the terminals of interest** and leave them intact.
2. **Combine series and parallel resistors**, working from the end of the network farthest from the source.
3. **Transform sources** where that brings sources into series (voltage form) or parallel (current form) with each other, so they can be combined.
4. **Apply Δ–Y conversion** where a bridge or triangle blocks further series–parallel steps.
5. **Solve the reduced circuit** — usually a single loop or single node pair.
6. **Expand back**, using voltage and current division (and KCL and KVL) to recover the quantities that the reduction hid.
7. **Check** with power balance or with KCL and KVL at a few points.

Step 6 is where most errors occur, because each reduction step hides internal currents and voltages. Keeping a sketch of every intermediate circuit avoids the error.

Worked example in full

Worked example — Reducing the ladder network. Find every current and voltage in the ladder of Figure 2.2 (source 24 V; $R_1 = 2$, $R_2 = 12$, $R_3 = 4$, $R_4 = 6$, $R_5 = 3\ \Omega$).

Reduce. $R_4 \parallel R_5 = 6 \times 3/9 = 2\ \Omega$. In series with R_3: $4 + 2 = 6\ \Omega$. In parallel with R_2: $12 \times 6/18 = 4\ \Omega$. In series with R_1: $2 + 4 = 6\ \Omega$.

Solve. Source current $I = 24/6 = 4.0$ A.

Expand. By voltage division across R_1 and the $4\ \Omega$ equivalent: $v_A = 24 \times 4/6 = 16.0$ V. By Ohm's law, $I_2 = 16/12 = 1.333$ A and the current into the R_3 branch is $I_3 = 16/6 = 2.667$ A (check KCL at A: $1.333 + 2.667 = 4.0$). By voltage division in that branch, $v_B = 16 \times 2/6 = 5.333$ V. Then $I_4 = 5.333/6 = 0.889$ A and $I_5 = 5.333/3 = 1.778$ A; or by current division, $I_4 = 2.667 \times 3/9 = 0.889$ A.

Check. KVL around M2: $-16 + 4 \times 2.667 + 5.333 = 0$. Power balance (Table 2.2): 96 W delivered, 96 W absorbed.

A distribution feeder

The same procedure models a radial power-distribution feeder: a substation bus (the source), a series of cable sections (series resistances) and loads tapped off along the way (shunt resistances). Reducing from the far end gives the current drawn at the substation; expanding back gives the voltage at each tap, and the drop is largest at the far end. When loads are better described as drawing constant power than constant resistance (as many modern electronic loads do), the circuit becomes nonlinear and reduction gives only a first estimate, refined iteratively in load-flow programs.

Where reduction stops

Network reduction is fast and builds intuition, but it has limits:

- **Bridges and non-planar networks** require Δ–Y steps that quickly become tedious.
- **Multiple sources at different points** require repeated transformations or superposition (Chapter 4).
- **Dependent sources** cannot be combined with resistors in series or parallel, because their value depends on a variable that the reduction may hide. Their control variable must be kept visible, or the circuit must be solved by nodal or mesh analysis.
- **Many unknowns at once** — every node voltage in a 50-node network — are better found by a single matrix solution than by any sequence of reductions.

These limits motivate Chapter 3, which replaces ingenuity with a systematic procedure that works for every linear network and is the basis of circuit-simulation programs.

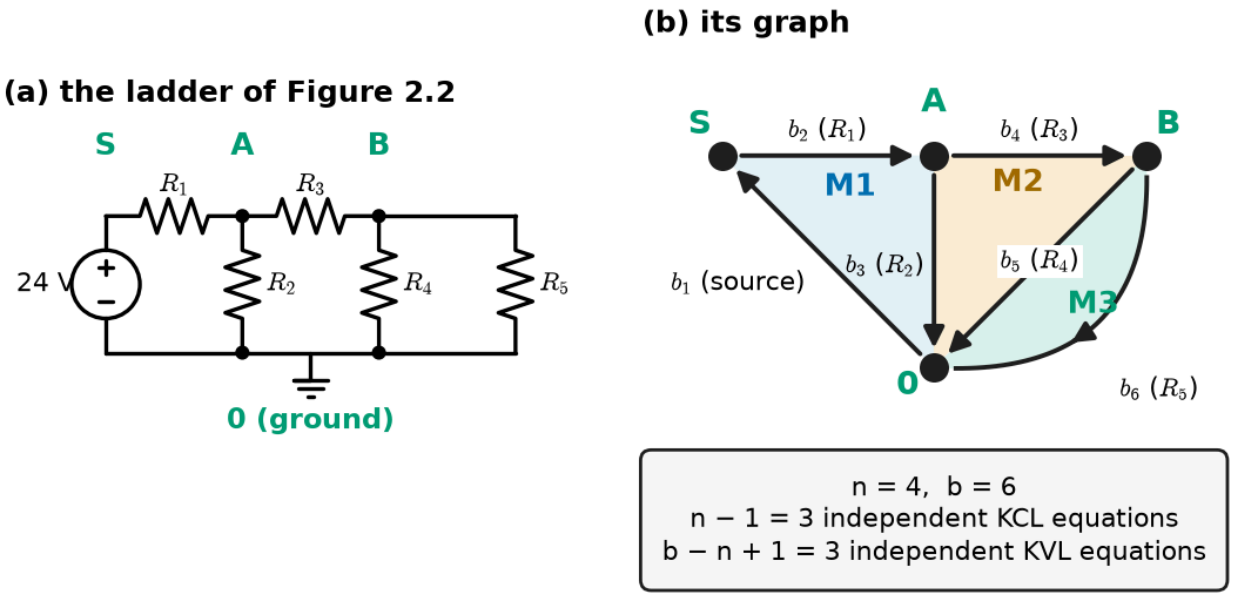


Figure 2.6 — From circuit to graph: the ladder network drawn as nodes and branches, with its three meshes.

Figure 2.6 shows the step that makes the systematic approach possible: stripping the ladder down to its **graph**, the pattern of nodes and branches with the element types forgotten. Kirchhoff's laws depend only on the graph; Ohm's law depends only on the elements. Chapter 3 exploits this separation.

What remains open

Open question	Where it is answered
How is a network with many nodes, several sources or a bridge solved systematically?	Chapter 3 (nodal and mesh analysis, matrix formulation)
Why are there exactly $n-1$ independent KCL and $b-n+1$ independent KVL equations, and how is Tellegen's theorem proved?	Chapter 3 (network topology, branch–node–loop relationships)
How are circuits with dependent sources reduced or solved?	Chapter 3 (dependent-source circuits), Chapter 4 (Thévenin with dependent sources)
How is any linear network reduced to a single source and resistance at a port, and what load takes maximum power?	Chapter 4 (Thévenin, Norton, maximum power transfer)
How do series, parallel and division rules extend to capacitors, inductors and impedances?	Chapters 5 and 8
How does KVL account for the voltage of inductors and coupled coils?	Chapter 5 (mutual inductance)
How do Δ–Y conversions apply to three-phase loads?	Chapter 10
What replaces KCL when a node is a significant fraction of a wavelength?	Chapter 14 (transmission lines)
How do electromagnetic fields give rise to Faraday induction and stray coupling?	*Engineering Physics*

Next: Chapter 3 turns Kirchhoff's laws and the element equations into nodal and mesh analysis and their matrix form, giving a procedure that solves any linear resistive network, with or without dependent sources.

Chapter 3 — Network Analysis

Part I · Circuit Fundamentals

An instrumentation engineer commissioning a weighing system expects a strain-gauge bridge to deliver 2.50 mV at full load: the textbook formula for a quarter bridge with 350 Ω gauges, 5 V excitation and a 0.2 % resistance change gives exactly that. The legacy panel meter reads 1.85 mV. The gauges are fine and the wiring is correct; what is wrong is the formula. It assumes the meter draws no current, but this meter has a 1 kΩ input, and it forms a fifth branch across the middle of the bridge. With that branch present, no two of the bridge's resistors are in series or in parallel, and the reduction tricks of Chapter 2 run out. What does not run out is Kirchhoff's laws. Written systematically — one equation per node, or one per mesh — they give the 1.85 mV reading exactly, and the same procedure scales from this five-resistor bridge to the hundred-thousand-node netlists that circuit simulators solve every day.

This chapter develops that procedure. It starts by counting how many equations a network needs, builds nodal and mesh analysis and their extensions to voltage and current sources in awkward places, adds dependent sources, and then recasts everything in matrix form with the tools of network topology. It ends with the question every simulator must answer before it solves anything: does this network have a unique solution at all?

> **How can the currents and voltages of any linear network be found by a procedure that never depends on spotting a clever simplification?**
> **When does a network have exactly one solution, and what does it mean when it does not?**

Concept	Engineering destination
Network equations	Knowing how many unknowns a problem has before solving it
Nodal analysis	Hand analysis of multi-source circuits; the core of SPICE
Mesh analysis	Planar circuits with many series sources; loop currents in power and magnetic circuits
Supernodes and supermeshes	Floating voltage sources, shared current sources, sensor excitation circuits
Dependent-source circuits	Small-signal amplifier gain, finite-gain op-amp error, feedback models
Network topology	Trees, loops and cut-sets for systematic equation writing
Branch, node and loop relationships	Incidence and loop matrices, Tellegen's theorem
Matrix formulation	Circuit simulators, power-flow programs, sensitivity analysis
Circuit solvability	"Singular matrix" errors, oscillation thresholds, ill-conditioned measurements

Roadmap. Counting the network equations (3.1) → nodal analysis (3.2) and mesh analysis (3.3) → supernodes and supermeshes for awkward sources (3.4) → dependent-source circuits (3.5) → network topology: graphs, trees and loops (3.6) → branch, node and loop relationships and Tellegen's theorem (3.7) → the matrix formulation used by simulators (3.8) → solvability: when the equations have one solution, none or infinitely many (3.9).

3.1 Network equations

Counting unknowns and equations

A network with b branches has $2b$ unknowns: one voltage and one current per branch. Three kinds of equation are available:

- b **element equations**, one per branch: Ohm's law $v_k = R_k i_k$ for a resistor, $v_k = V_s$ for a voltage source, $i_k = I_s$ for a current source, and the defining relations of dependent sources.
- $n - 1$ **independent KCL equations**, one for each node except one (the last is the sum of the others).
- $b - n + 1$ **independent KVL equations**, one per independent loop.

The total, $b + (n-1) + (b-n+1) = 2b$, matches the number of unknowns. Section 3.6 proves that the KCL and KVL counts are right for any connected network; Section 3.9 asks when the $2b$ equations are also independent, so that they have exactly one solution.

Writing all $2b$ equations at once — the **sparse tableau** formulation — is wasteful by hand. The two classical methods each use one set of Kirchhoff equations to eliminate most unknowns before writing anything down:

- **Nodal analysis** takes the $n - 1$ node voltages as unknowns. Every branch voltage is a difference of node voltages, so KVL is satisfied automatically; only the $n - 1$ KCL equations remain.
- **Mesh analysis** takes $b - n + 1$ loop currents as unknowns. Every branch current is a combination of loop currents, so KCL is satisfied automatically; only the $b - n + 1$ KVL equations remain.

The method with fewer equations is usually the better choice by hand. Table 3.1 compares some networks.

Table 3.1 — Equation counts for representative networks.

Network	Branches b	Nodes n	Nodal equations $n-1$	Mesh/loop equations $b-n+1$
Ladder of Figure 2.2	6	4	3	3
Wheatstone bridge with detector and source	6	4	3	3
Two-source bias network (Figure 3.1)	7	5	4 (2 after using the sources)	3
Cube of 12 resistors	12	8	7	5
Small-signal amplifier, 10 transistors	about 60	about 25	about 24	about 36
Transmission network, 100 buses, 150 lines	150	100	99	51

Two cautions apply to the counts. Voltage sources reduce the number of nodal unknowns (a node attached to ground through a voltage source has a known voltage), and current sources reduce the number of mesh unknowns; the "after using the sources" count is what matters. And mesh analysis applies only to planar networks, whereas nodal analysis applies to all — one reason circuit simulators are built on nodal methods.

3.2 Nodal analysis

The method

1. **Choose a reference node** (ground). Any node will do; the one with the most connections, or the one the circuit diagram already marks as ground, keeps the equations simplest.
2. **Label the remaining $n-1$ node voltages** $v_1, \dots, v_{n-1}$, each measured with respect to the reference.
3. **Write KCL at each non-reference node**, expressing every resistor current in terms of node voltages: the current leaving node j through a resistor R to node k is $(v_j - v_k)/R$.
4. **Solve** the $n-1$ linear equations.
5. **Recover** branch voltages (differences of node voltages) and currents (Ohm's law).

For a network of resistors and independent current sources, the equations take a pattern that can be written by inspection:

$$\mathbf{G}\,\mathbf{v} = \mathbf{i}_s,$$

where

- the diagonal entry G_{jj} is the **sum of conductances connected to node j**;
- the off-diagonal entry G_{jk} is **minus the sum of conductances connecting nodes j and k**;
- the right-hand entry $i_{s,j}$ is the **sum of source currents flowing into node j**.

G is the **nodal conductance matrix**. For networks of resistors it is symmetric, because the conductance between j and k appears identically in both rows. A practical voltage source is first converted to its Norton form (Chapter 2) to fit this pattern.

> **Status of the principle: Nodal analysis**
> *Derived.* KCL at $n-1$ nodes with branch currents expressed through node voltages, so that KVL holds identically; it applies to every lumped network, planar or not, and becomes the modified nodal analysis of Section 3.8 when ideal voltage sources are present.

Worked example

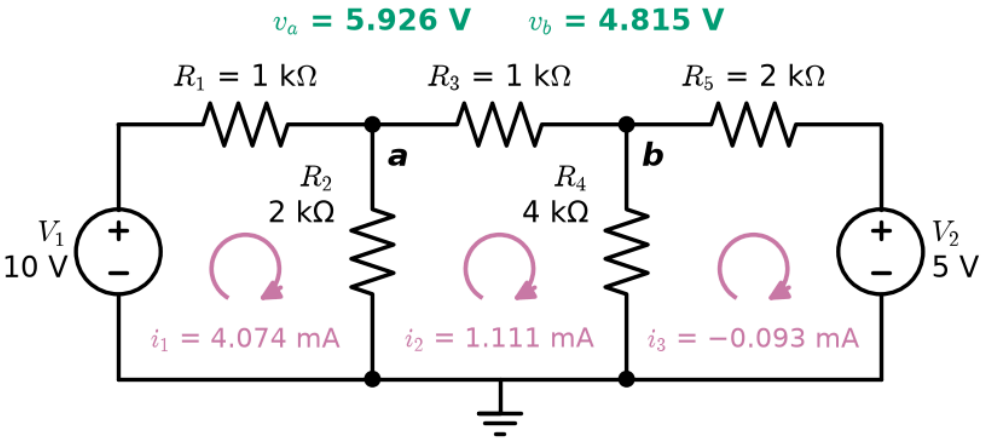


Figure 3.1 — A two-source resistive network solved by nodal and by mesh analysis.

Worked example — Nodal analysis of a two-source network. In Figure 3.1, two supplies $V_1 = 10$ V and $V_2 = 5$ V, each with a series resistor, feed a resistive network: $R_1 = 1\ \text{k}\Omega$, $R_2 = 2\ \text{k}\Omega$, $R_3 = 1\ \text{k}\Omega$, $R_4 = 4\ \text{k}\Omega$, $R_5 = 2\ \text{k}\Omega$. Find all node voltages and branch currents.

Working in kΩ, mA and V (so conductances are in mS), the network has two unknown node voltages, v_a and v_b. (The nodes at the + terminals of the sources have known voltages, 10 V and 5 V.)

KCL at a, currents leaving: $\dfrac{v_a - 10}{1} + \dfrac{v_a}{2} + \dfrac{v_a - v_b}{1} = 0.$

KCL at b, currents leaving: $\dfrac{v_b - v_a}{1} + \dfrac{v_b}{4} + \dfrac{v_b - 5}{2} = 0.$

Collected in matrix form (equivalently, by inspection after converting each source and its series resistor to a Norton source of $10/1 = 10$ mA and $5/2 = 2.5$ mA):

$$\begin{bmatrix} 2.5 & -1 \\ -1 & 1.75 \end{bmatrix} \begin{bmatrix} v_a \\ v_b \end{bmatrix} = \begin{bmatrix} 10 \\ 2.5 \end{bmatrix}.$$

The determinant is $2.5 \times 1.75 - 1 = 3.375$. By Cramer's rule,

$$v_a = \frac{10 \times 1.75 + 2.5}{3.375} = 5.926\ \text{V}, \qquad v_b = \frac{2.5 \times 2.5 + 10}{3.375} = 4.815\ \text{V}.$$

Branch currents: through R_1 from V_1 to a, $(10 - 5.926)/1 = 4.074$ mA; R_2, $5.926/2 = 2.963$ mA; R_3 from a to b, 1.111 mA; R_4, $4.815/4 = 1.204$ mA; R_5 from V_2 to b, $(5 - 4.815)/2 = 0.093$ mA.

Check KCL at b: in from R_3 (1.111) and R_5 (0.093), out through R_4 (1.204). Power: V_1 delivers $10 \times 4.074 = 40.74$ mW, V_2 delivers $5 \times 0.093 = 0.46$ mW, total 41.20 mW; the resistors absorb $16.60 + 17.56 + 1.23 + 5.80 + 0.02 = 41.20$ mW. Both supplies deliver power, but V_2 only barely: it is close to the voltage that node b would reach without it.

Practical points

- **Units.** Working in kΩ, mA and V (or Ω, A, V) keeps numbers near unity. Since $1\ \text{V}/1\ \text{k}\Omega = 1$ mA, no conversion is needed.
- **Sign discipline.** Writing every resistor current as "leaving the node" — $(v_{\text{this}} - v_{\text{other}})/R$ — produces the symmetric pattern automatically and avoids most sign errors.
- **Choice of ground.** In electronic circuits the negative supply rail or the signal common is the natural reference; in power networks the reference is the earth or a slack bus. Changing the reference shifts all node voltages by a constant and changes no element voltage or current.

3.3 Mesh analysis

The method

Mesh analysis applies to **planar** networks (Chapter 2). Each mesh — a window of the planar drawing — is assigned a **mesh current**, conventionally clockwise. A branch on the boundary of the outer region carries one mesh current; a branch shared by two meshes carries the difference of their two mesh currents. Because every branch current is constructed this way, KCL is satisfied at every node automatically: each mesh current enters and leaves every node it touches.

1. **Check the network is planar** and identify its $b - n + 1$ meshes.
2. **Assign clockwise mesh currents** $i_1, \dots, i_m$.
3. **Write KVL around each mesh**, expressing each resistor voltage as R times the net current through it in the direction of traversal.
4. **Solve** and recover branch currents as mesh currents or their differences.

For resistors and independent voltage sources the equations again follow a pattern:

$$\mathbf{R}\,\mathbf{i} = \mathbf{v}_s,$$

where R_{jj} is the **sum of resistances around mesh** j, R_{jk} is **minus the resistance shared by meshes** j **and** k (with all mesh currents clockwise), and $v_{s,j}$ is the **sum of source voltage rises around mesh** j in the clockwise direction. $\mathbf{R}$ is the **mesh resistance matrix**, symmetric for resistor networks.

> **Status of the principle: Mesh analysis**
> *Derived.* KVL around the $b-n+1$ meshes of a planar network with branch currents expressed through mesh currents, so that KCL holds identically; for non-planar networks the same idea works with the fundamental loops of Section 3.6 in place of meshes.

Worked example — Mesh analysis of the same network. The network of Figure 3.1 has $b=7$ branches (two sources and five resistors) and $n=5$ nodes, hence $7-5+1=3$ meshes: left (V_1, R_1, R_2), middle (R_2, R_3, R_4) and right (R_4, R_5, V_2). With clockwise currents i_1, i_2, i_3 (in mA):

Mesh 1: rise of 10 V through V_1; drops $1\cdot i_1$ and $2(i_1-i_2)$: $3i_1-2i_2=10$.

Mesh 2: $2(i_2-i_1)+1\cdot i_2+4(i_2-i_3)=0$: $-2i_1+7i_2-4i_3=0$.

Mesh 3: drop $2i_3$ across R_5, then traversing V_2 from + to −, a drop of 5 V, then $4(i_3-i_2)$: $-4i_2+6i_3=-5$.

$$\begin{bmatrix} 3 & -2 & 0 \\ -2 & 7 & -4 \\ 0 & -4 & 6 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix}=\begin{bmatrix} 10 \\ 0 \\ -5 \end{bmatrix} \quad\Rightarrow\quad i_1=4.074,\ i_2=1.111,\ i_3=-0.093\ \text{mA}.$$

Then $v_a=R_2(i_1-i_2)=2\times 2.963=5.926$ V and $v_b=R_4(i_2-i_3)=4\times 1.204=4.815$ V, agreeing with the nodal result. The negative i_3 means the right mesh current actually circulates counter-clockwise: 0.093 mA flows out of the + terminal of V_2 into node b, as the nodal analysis found.

Here nodal analysis needed two equations and mesh analysis three. The general guidance: nodal analysis suits networks with many parallel branches and current sources; mesh analysis suits networks with many series elements and voltage sources, such as ladder filters driven from a voltage source and the loop equations of magnetic and electromechanical models.

3.4 Supernodes and supermeshes

Voltage sources in nodal analysis

A voltage source between a non-reference node and ground simply fixes that node's voltage and removes one unknown. A voltage source between *two non-reference nodes* — a **floating** voltage source — causes a difficulty: its current is not determined by its voltage, so KCL at either node contains an unknown current that cannot be written in terms of node voltages.

The remedy is the **supernode**. Enclose the floating source and its two nodes in a closed surface. By the generalised KCL of Chapter 2, the net current leaving that surface is zero, and the source current, which is internal to the surface, does not appear. The supernode provides one KCL equation, and the source itself provides a constraint, $v_j-v_k=V_s$; together they replace the two KCL equations that could not be written.

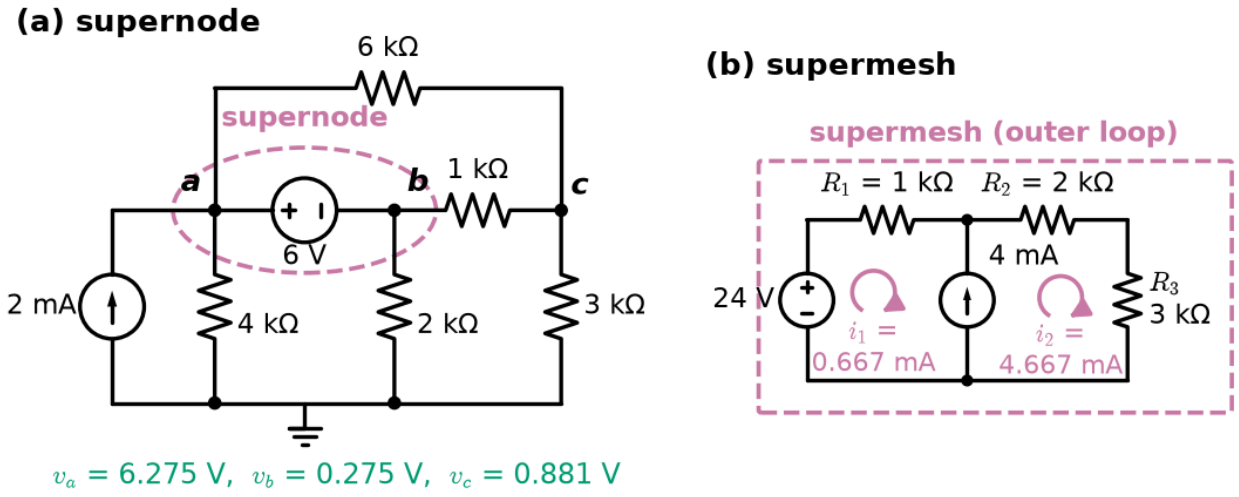


Figure 3.2 — A supernode enclosing a floating voltage source, and a supermesh around a shared current source.

Worked example — A floating voltage source. In the left panel of Figure 3.2 (values in kΩ, mA, V), a 2 mA source drives node a; a connects to ground through 4 kΩ and to node c through 6 kΩ; a 6 V source sits between a (+) and b (−); b connects to ground through 2 kΩ and to c through 1 kΩ; c connects to ground through 3 kΩ.

Constraint from the source: $v_a - v_b = 6$.

KCL for the supernode $\{a, b\}$, currents leaving through the resistors and the 2 mA entering:

$$\frac{v_a}{4} + \frac{v_a - v_c}{6} + \frac{v_b}{2} + \frac{v_b - v_c}{1} = 2.$$

KCL at c: $\dfrac{v_c - v_b}{1} + \dfrac{v_c}{3} + \dfrac{v_c - v_a}{6} = 0.$

Solving the three equations: $v_a = 6.275$ V, $v_b = 0.275$ V, $v_c = 0.881$ V.

The current through the floating source follows from KCL at b alone: the current leaving b through its resistors is $0.275/2 + (0.275 - 0.881)/1 = -0.468$ mA, so 0.468 mA flows *into* b from the resistors and out of b through the source toward a. Current therefore leaves the source's + terminal: the 6 V source delivers $6 \times 0.468 = 2.81$ mW. The 2 mA source delivers $2 \times 6.275 = 12.55$ mW, and the resistors absorb $9.844 + 4.850 + 0.038 + 0.367 + 0.259 = 15.36$ mW, matching the 15.36 mW supplied.

A floating voltage source is common in practice: the drop across a conducting diode modelled as a fixed voltage, the bias voltage generator between the bases of a push-pull output stage, the excitation source of a bridge whose negative side is not grounded, and in power systems the voltage-regulating transformer tap modelled as a series source.

Current sources in mesh analysis

The dual difficulty arises in mesh analysis. A current source on the outer boundary of a planar network simply fixes one mesh current. A current source in a branch *shared* by two meshes fixes the difference of those mesh currents, but its voltage is unknown, so neither mesh's KVL equation can be written.

The remedy is the **supermesh**: write KVL around the loop formed by the two meshes with the shared branch removed (a path that avoids the current source), and add the constraint that the difference of the two mesh currents equals the source current.

Worked example — A shared current source. In the right panel of Figure 3.2, the left mesh contains a 24 V source and $R_1 = 1\ \text{k}\Omega$; the shared branch is a 4 mA current source pointing upward; the right mesh contains $R_2 = 2\ \text{k}\Omega$ and $R_3 = 3\ \text{k}\Omega$. With clockwise mesh currents, the shared branch carries i_1 downward and i_2 upward, so

$$i_2 - i_1 = 4.$$

KVL around the supermesh (the outer loop, which avoids the current source): $24 = 1 \cdot i_1 + (2 + 3)\, i_2$.

Solving: $i_1 = 0.667$ mA and $i_2 = 4.667$ mA. The voltage across the current source equals the drop across $R_2 + R_3$, $5 \times 4.667 = 23.33$ V, so the source delivers $23.33 \times 4 = 93.3$ mW; the 24 V source delivers $24 \times 0.667 = 16.0$ mW; the resistors absorb $1 \times 0.667^2 + 5 \times 4.667^2 = 109.3$ mW. The balance holds. The voltage across the current source — 23.3 V — is something the mesh method did not need in order to find the currents, but which a real current source must be able to supply. A 4–20 mA loop transmitter, for example, has a stated *compliance* voltage range; outside it, the current-source model fails.

> **Status of the principle: Supernode and supermesh constructions**
> *Derived*. The supernode is KCL applied to a closed surface (a cut-set) instead of a single node; the supermesh is KVL around a loop that avoids the current source. Both are exact consequences of Kirchhoff's laws.

3.5 Dependent-source circuits

Procedure

Dependent sources (Chapter 1) enter nodal and mesh analysis in two steps:

1. **Treat the dependent source as if it were independent** while writing the KCL or KVL equations, using supernodes or supermeshes as needed.
2. **Add one equation expressing the source's value in terms of the circuit unknowns**: a controlling voltage as a difference of node voltages, or a controlling current as a mesh current or an Ohm's-law expression. Then substitute and collect terms.

The resulting matrix is no longer symmetric. The controlled source couples node k to node j without the reverse coupling — that one-way influence is exactly what amplification and feedback are — and the coefficient matrix can even become singular (Section 3.9). Thévenin equivalents of such circuits need a test-source method rather than resistance combination (Chapter 4).

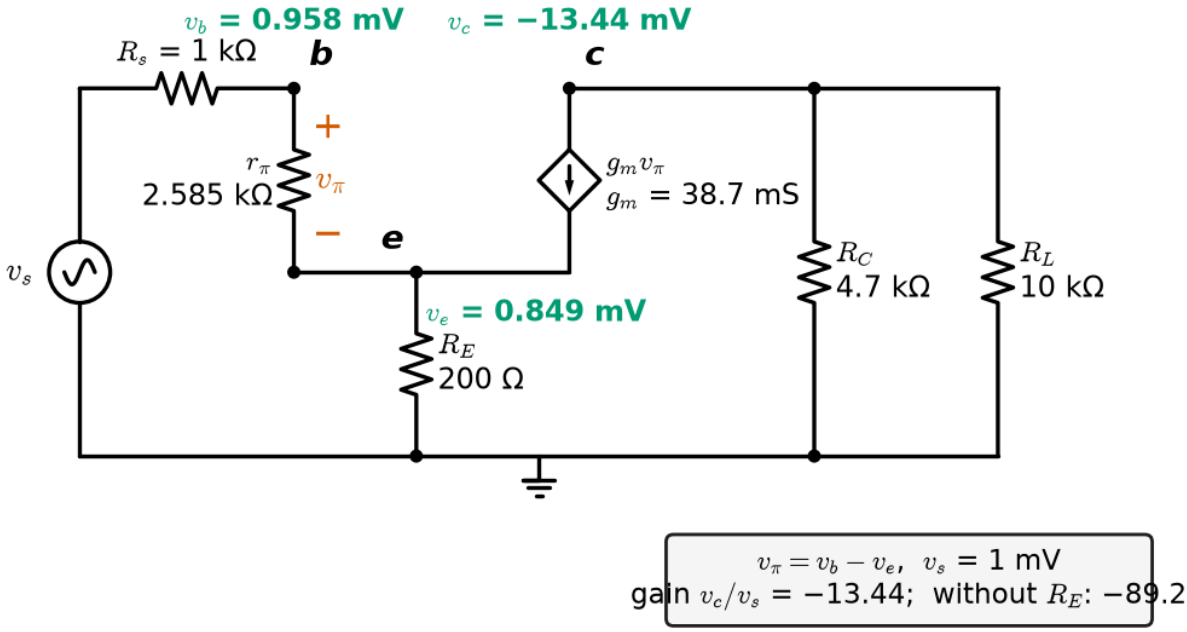


Figure 3.3 — Small-signal model of a common-emitter amplifier with emitter degeneration, prepared for nodal analysis.

Worked example — Gain of a degenerated common-emitter stage. Chapter 1 found a gain of −89.2 for a common-emitter stage ($g_m = 38.7$ mS, $r_\pi = 2.585$ kΩ, $R_s = 1$ kΩ, $R_C \parallel R_L = 3.197$ kΩ). Such a high gain depends directly on g_m, which varies with temperature and from transistor to transistor. Designers add an unbypassed emitter resistor $R_E = 200\ \Omega$ (Figure 3.3) to trade gain for predictability. The emitter is no longer grounded, the circuit no longer reduces to two independent halves, and nodal analysis is the tool.

Unknowns: v_b, v_e, v_c. Controlling variable: $v_\pi = v_b - v_e$. The VCCS carries $g_m v_\pi$ from collector to emitter. KCL, currents leaving each node:

Node b: $\dfrac{v_b - v_s}{R_s} + \dfrac{v_b - v_e}{r_\pi} = 0.$

Node e: $\dfrac{v_e - v_b}{r_\pi} + \dfrac{v_e}{R_E} - g_m(v_b - v_e) = 0$ (the source current *enters* e).

Node c: $\dfrac{v_c}{R_C} + \dfrac{v_c}{R_L} + g_m(v_b - v_e) = 0$ (the source current *leaves* c).

In matrix form, with conductances in mS ($1/R_s = 1$, $1/r_\pi = 0.3868$, $1/R_E = 5$, $1/R_C + 1/R_L = 0.3128$, $g_m = 38.68$):

$$\begin{bmatrix} 1.3868 & -0.3868 & 0 \\ -39.07 & 44.07 & 0 \\ 38.68 & -38.68 & 0.3128 \end{bmatrix} \begin{bmatrix} v_b \\ v_e \\ v_c \end{bmatrix} = \begin{bmatrix} v_s/R_s \\ 0 \\ 0 \end{bmatrix}.$$

The asymmetry is plain: g_m couples the base and emitter voltages into the collector row (row 3), but nothing couples the collector back into rows 1 and 2.

For $v_s = 1$ mV the solution is $v_b = 0.958$ mV, $v_e = 0.849$ mV, $v_c = -13.44$ mV: a voltage gain of -13.44.

The familiar design formula comes from eliminating variables symbolically: the input resistance looking into the base is $r_\pi + (\beta+1)R_E = 2.585 + 101 \times 0.2 = 22.79\ \text{k}\Omega$, and

$$\frac{v_c}{v_s} = -\frac{\beta\,(R_C \parallel R_L)}{R_s + r_\pi + (\beta+1)R_E} = -\frac{100 \times 3.197}{1 + 22.79} = -13.44,$$

matching the numerical solution exactly. The cruder estimate $-(R_C \parallel R_L)/R_E = -16.0$ ignores r_π, $1/g_m$ and R_s. The gain fell from −89 to −13, but now depends mainly on the resistor ratio rather than on g_m. That trade is the essence of negative feedback, analysed in Chapter 34.

A finite-gain op-amp

Worked example — How close is an inverting amplifier to −10? An inverting amplifier uses $R_1 = 10\ \text{k}\Omega$ and $R_2 = 100\ \text{k}\Omega$, so ideally its gain is $-R_2/R_1 = -10$. Model the op-amp as a VCVS: input resistance $R_i = 2\ \text{M}\Omega$ between its inputs, open-loop gain $A = 2 \times 10^5$, and output resistance $R_o = 75\ \Omega$ in series with a source $A(v_+ - v_-)$. The non-inverting input is grounded; a 2 kΩ load is connected to the output.

Unknowns: the inverting-input node v_n and the output v_o. With $v_s = 1\ \text{V}$,

Node n: $\dfrac{v_n - 1}{R_1} + \dfrac{v_n - v_o}{R_2} + \dfrac{v_n}{R_i} = 0.$

Node o: $\dfrac{v_o - v_n}{R_2} + \dfrac{v_o}{R_L} + \dfrac{v_o - A(0 - v_n)}{R_o} = 0.$

The second equation contains the term $A v_n / R_o$ — a coefficient of $2 \times 10^5 / 75 \approx 2667$ S in a matrix whose other entries are of order 10^{-4} to 10^{-2} S. Solving gives $v_n = 52\ \mu\text{V}$ and $v_o = -9.99943\ \text{V}$: a gain error of 57 parts per million, close to the estimate $(1 + R_2/R_1)/A = 55$ ppm. The inverting input sits at a "virtual ground" not because it is connected to ground but because the enormous gain needs only 52 µV there.

> **Boundary.**
> Why op-amps and transistors have these parameters, and how practical amplifiers are designed around them, is the subject of *Electronics & Computing*. Here they are network elements, and the methods of this chapter apply to them unchanged.

Dependent sources in mesh analysis

The same procedure works with mesh currents. A CCVS controlled by a branch current, for example $v = r_m i_x$, contributes r_m times the appropriate mesh-current combination to a KVL equation; a dependent current source shared between meshes creates a supermesh whose constraint involves the controlling variable. Because the result is again an asymmetric matrix, the choice between nodal and mesh methods rests on the same equation-count criterion as before.

3.6 Network topology

The analyses above relied on facts asserted in Chapter 2: that a network has $n-1$ independent KCL equations and $b-n+1$ independent KVL equations. Network topology — the graph theory of circuits — proves those facts and gives a systematic way to choose the equations for any network, planar or not.

Graphs

The **graph** of a network replaces every element by a line segment (an **edge**, or branch) joining two **vertices** (nodes), keeping only the pattern of connection. Assigning each branch a reference direction — normally the direction of its current reference — makes it a **directed graph**. A graph is **connected** if every node can be reached from every other along branches. Magnetically coupled but electrically separate circuits (the two windings of a transformer) form separate connected parts; they can be joined at one node each without changing any current, and this is normally done.

Trees and cotrees

A **tree** is a subset of branches that connects all n nodes without forming any loop. Every tree of a connected graph has exactly $n-1$ branches, called **twigs**. (Start from one node; each twig adds exactly one new node; after $n-1$ twigs every node is reached, and one more branch would close a loop.) The remaining $b-n+1$ branches form the **cotree**; they are called **links**. A network generally has many trees: the Wheatstone bridge with source and detector (Figure 3.4) has 16.

Fundamental loops and cut-sets

Adding any single link to a tree creates exactly one loop, consisting of that link and the unique tree path between its ends. These $b-n+1$ **fundamental loops** are independent, because each contains a link that no other contains. Hence there are at least $b-n+1$ independent KVL equations, and Section 3.7 shows there are no more.

Removing any single twig splits the tree into two parts. The set of branches joining those two parts — the twig itself and every link that crosses the gap — is a **fundamental cut-set**. There are $n-1$ of them, one per twig, and KCL applied to the surface separating the two parts gives $n-1$ independent cut-set equations. Node equations are a special case: the cut-set that isolates a single node.

> **Status of the principle: Independence counts ($n-1$ KCL, $b-n+1$ KVL)**
> *Derived*. Graph-theoretic results for any connected network: every tree has $n-1$ twigs, each fundamental loop contains its own link and each fundamental cut-set its own twig, so these sets are independent and, by the rank argument of Section 3.7, complete.

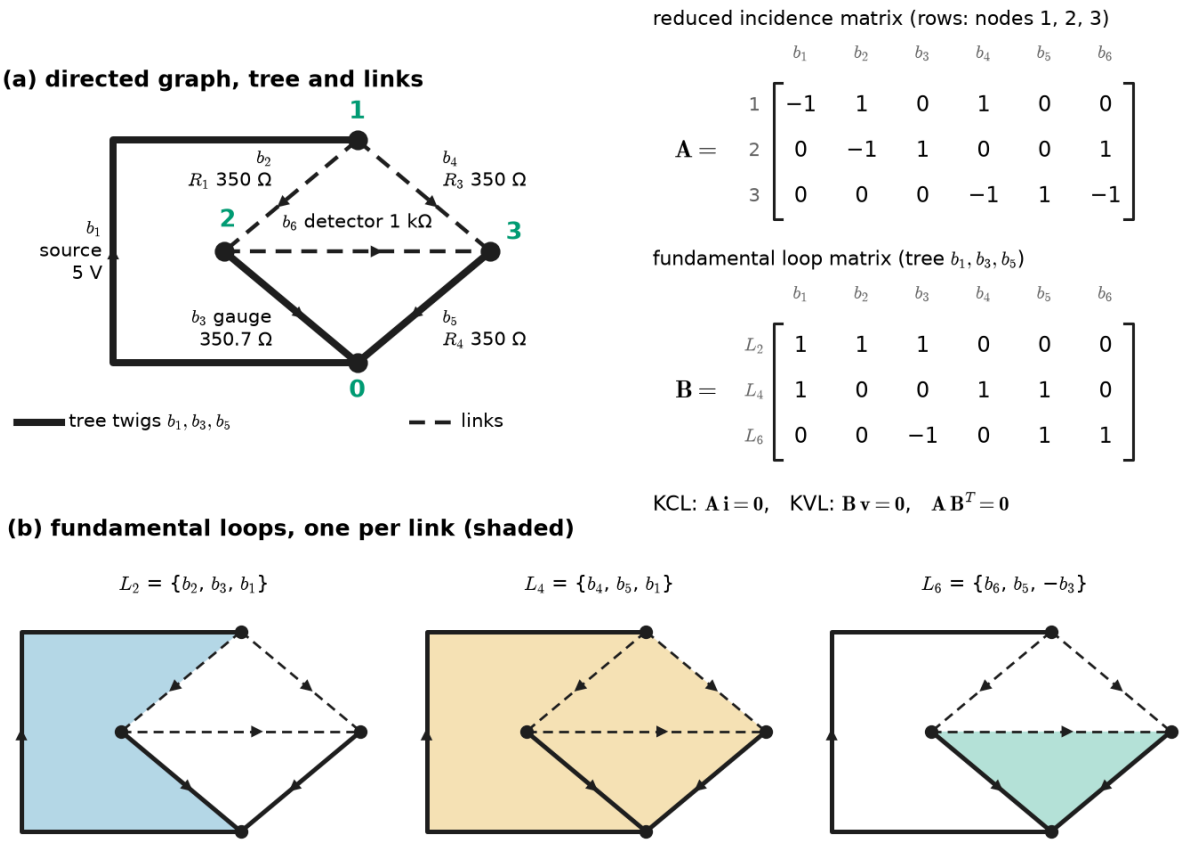


Figure 3.4 — Directed graph of the excited Wheatstone bridge, with a tree, its links and the fundamental loops.

Choosing the tree

The choice of tree is free, and a good choice simplifies analysis. Placing voltage sources in the tree (as twigs) and current sources in the cotree (as links) means each voltage source fixes one twig voltage and each current source fixes one loop current. This is possible only if the voltage sources contain no loop among themselves and the current sources contain no cut-set — the solvability conditions of Section 3.9.

3.7 Branch, node and loop relationships

The incidence matrix and KCL

Number the non-reference nodes $1, \dots, n-1$ and the branches $1, \dots, b$. The **reduced incidence matrix** $\mathbf{A}$, of size $(n-1) \times b$, has

$$a_{jk} = \begin{cases} +1 & \text{if branch } k \text{ leaves node } j, \\ -1 & \text{if branch } k \text{ enters node } j, \\ 0 & \text{otherwise.} \end{cases}$$

With $\mathbf{i}$ the vector of branch currents, KCL at all non-reference nodes is

$$\mathbf{A}\,\mathbf{i} = \mathbf{0}.$$

With $\mathbf{e}$ the vector of node voltages (relative to the reference) and $\mathbf{v}$ the vector of branch voltages (+ at the tail of each branch arrow), every branch voltage is the tail node voltage minus the head node voltage, which is

$$\mathbf{v} = \mathbf{A}^{T}\mathbf{e}.$$

This is KVL in node-voltage form: any $\mathbf{v}$ that can be written this way automatically sums to zero around every loop.

For the bridge of Figure 3.4, with branches b_1 (source, 0→1), b_2 (R_1, 1→2), b_3 (gauge, 2→0), b_4 (R_3, 1→3), b_5 (R_4, 3→0) and b_6 (detector, 2→3):

$$\mathbf{A} = \begin{bmatrix} -1 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & 1 & -1 \end{bmatrix}.$$

Its rank is $n-1 = 3$ for any connected graph: the columns of the twigs of any tree form an invertible square submatrix. That is why there are exactly $n-1$ independent KCL equations — no more, no fewer.

The loop matrix and KVL

The **fundamental loop matrix** $\mathbf{B}$, of size $(b-n+1)\times b$, has one row per fundamental loop, with $+1$ where a branch's direction agrees with the loop direction (taken as the direction of its defining link), -1 where it opposes, and 0 for branches not in the loop. KVL is

$$\mathbf{B}\,\mathbf{v} = \mathbf{0},$$

and, with $\mathbf{i}_\ell$ the vector of loop (link) currents, every branch current is a superposition of loop currents:

$$\mathbf{i} = \mathbf{B}^{T}\mathbf{i}_\ell.$$

This is KCL in loop-current form; mesh analysis is the special case in which the loops are the meshes of a planar drawing. For the tree $\{b_1, b_3, b_5\}$ of Figure 3.4:

$$\mathbf{B} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 \end{bmatrix}.$$

Orthogonality and Tellegen's theorem

Every loop crosses every cut-set, and in particular every node, an even number of times, entering as often as it leaves. In matrix form this is the orthogonality relation

$$\mathbf{A}\,\mathbf{B}^{T} = \mathbf{0},$$

which can be checked directly for the matrices above. It has a short but far-reaching consequence. Let $\mathbf{v}$ be *any* set of branch voltages satisfying KVL and $\mathbf{i}$ *any* set of branch currents satisfying KCL in the same graph — not necessarily from the same circuit, nor even at the same instant. Then $\mathbf{v} = \mathbf{A}^{T}\mathbf{e}$ for some $\mathbf{e}$, and

$$\mathbf{v}^T\mathbf{i} = \mathbf{e}^T\mathbf{A}\,\mathbf{i} = \mathbf{e}^T\mathbf{0} = 0.$$

This is **Tellegen's theorem**: $\sum_k v_k i_k = 0$. When $\mathbf{v}$ and $\mathbf{i}$ belong to the same circuit at the same instant, it is the power balance of Chapter 2. When they come from different circuits on the same graph, it underlies the reciprocity theorem (Chapter 4) and the adjoint-network method that simulators use to compute the sensitivity of an output to every component at once.

> **Status of the principle: Tellegen's theorem**
> *Derived*. It follows from KCL and KVL alone, through the orthogonality of the incidence and loop matrices; it holds for any lumped network whatever its elements — linear or nonlinear, time-invariant or not.

Because $\mathbf{A}$ has rank $n-1$ and $\mathbf{A}\mathbf{B}^T = \mathbf{0}$, the rows of $\mathbf{B}$ span the whole space of vectors orthogonal to the rows of $\mathbf{A}$, whose dimension is $b-(n-1)$. So no KVL equation exists beyond the $b-n+1$ fundamental ones: the counts of Section 3.1 are exact.

3.8 Matrix formulation

Nodal equations from the branch matrices

Collect the branch conductances of a resistive network in a diagonal matrix $\mathbf{G}_b$, and let independent current sources in parallel with the branches be collected in $\mathbf{i}_s$ (Norton form). Each branch obeys $\mathbf{i} = \mathbf{G}_b\mathbf{v} - \mathbf{i}_s$. Substituting into KCL and using $\mathbf{v} = \mathbf{A}^T\mathbf{e}$:

$$\mathbf{A}\mathbf{G}_b\mathbf{A}^T\,\mathbf{e} = \mathbf{A}\,\mathbf{i}_s \quad\Longrightarrow\quad \mathbf{Y}_n\,\mathbf{e} = \mathbf{j}_s, \qquad \mathbf{Y}_n = \mathbf{A}\mathbf{G}_b\mathbf{A}^T.$$

$\mathbf{Y}_n$ is the nodal matrix that Section 3.2 wrote by inspection; the matrix product formalises the inspection rules and is how a program assembles it. In practice simulators do not multiply matrices: they "stamp" each element directly into $\mathbf{Y}_n$, adding G to two diagonal entries and $-G$ to two off-diagonal entries for a resistor between two nodes.

The mesh counterpart is $\mathbf{B}\mathbf{R}_b\mathbf{B}^T\,\mathbf{i}_\ell = \mathbf{B}\,\mathbf{v}_s$, with $\mathbf{R}_b$ the diagonal matrix of branch resistances.

Modified nodal analysis

Ideal voltage sources have no conductance and cannot be stamped into $\mathbf{Y}_n$. **Modified nodal analysis** (MNA), introduced in the 1970s and used in SPICE and its descendants, adds the current of each ideal voltage source as an extra unknown and the source's constraint as an extra equation:

$$\begin{bmatrix} \mathbf{Y}_n & \mathbf{B}_V \\ \mathbf{C}_V & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{e} \\ \mathbf{i}_V \end{bmatrix} = \begin{bmatrix} \mathbf{j}_s \\ \mathbf{v}_s \end{bmatrix},$$

where $\mathbf{B}_V$ (and $\mathbf{C}_V = \mathbf{B}_V^T$ for independent sources) contains ± 1 entries locating each source's terminals. Inductors, VCVSs and CCVSs, and any element whose current is needed as an output, are handled the same way. MNA is in effect the supernode method made automatic: the extra rows and columns carry the information that supernodes encode by hand.

> **Status of the principle: Modified nodal analysis**
> *Derived*. Nodal analysis augmented by branch currents of voltage-defined elements; it is an exact reformulation of KCL, KVL and the element laws, and the standard formulation of general-purpose circuit simulators.

Worked example — The loaded strain-gauge bridge. Return to the opening problem (Figure 3.4). Four 350 Ω arms, one of which is a strain gauge at 350.7 Ω (a 0.2 % change, corresponding to 1000 microstrain with a gauge factor of 2); excitation 5.00 V; detector resistance 1 kΩ between the midpoints.

Using the incidence matrix of Section 3.7, restricted to the resistor branches $b_2, \dots, b_6$, and $\mathbf{G}_b = \mathrm{diag}(1/350,\ 1/350.7,\ 1/350,\ 1/350,\ 1/1000)$ S, the nodal matrix is (in mS)

$$\mathbf{Y}_n = \begin{bmatrix} 5.7143 & -2.8571 & -2.8571 \\ -2.8571 & 6.7086 & -1.0000 \\ -2.8571 & -1.0000 & 6.7143 \end{bmatrix}.$$

The excitation is an ideal voltage source from ground to node 1, so MNA adds its current i_V (leaving node 1 into the source) and the constraint $e_1 = 5$:

$$\begin{bmatrix} 5.7143 & -2.8571 & -2.8571 & 1 \\ -2.8571 & 6.7086 & -1.0000 & 0 \\ -2.8571 & -1.0000 & 6.7143 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i_V \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 5 \end{bmatrix}$$

(with e in V, i_V in mA). Solved with unrounded entries, the solution is $e_2 = 2.502173$ V, $e_3 = 2.500324$ V and $i_V = -14.28$ mA (so the source delivers 14.28 mA). The detector voltage is

$$e_2 - e_3 = 1.850 \text{ mV}.$$

The textbook formula gives the *open-circuit* output, $5 \times (350.7/700.7 - 0.5) = 2.4975$ mV, valid for an infinite detector resistance. Chapter 4 will show that the bridge looks to its detector like a 2.4975 mV source with an internal resistance of $350 \parallel 350.7 + 350 \parallel 350 = 350.17\ \Omega$; with a 1 kΩ detector, $2.4975 \times 1000/1350.17 = 1.850$ mV, confirming the matrix solution. The cure is an instrumentation amplifier with an input resistance of gigaohms, for which the loaded and open-circuit outputs agree to better than one part in a million.

Sparsity and scale

In a large circuit each node connects to only a few others, so $\mathbf{Y}_n$ is **sparse**: a 100 000-node integrated-circuit netlist might have only five or six non-zero entries per row. Simulators store only the non-zero entries and solve by sparse LU factorisation, choosing the elimination order to limit "fill-in". Nonlinear devices are handled by repeatedly linearising them about the current estimate (Newton's method), replacing each with a conductance and a current source — exactly a dependent-source model — and re-solving the linear system; transient analysis repeats that at every time step (Chapters 6 and 11). Power-system load-flow programs use the same nodal admittance matrix, called $\mathbf{Y}_{bus}$, at the scale of thousands of buses.

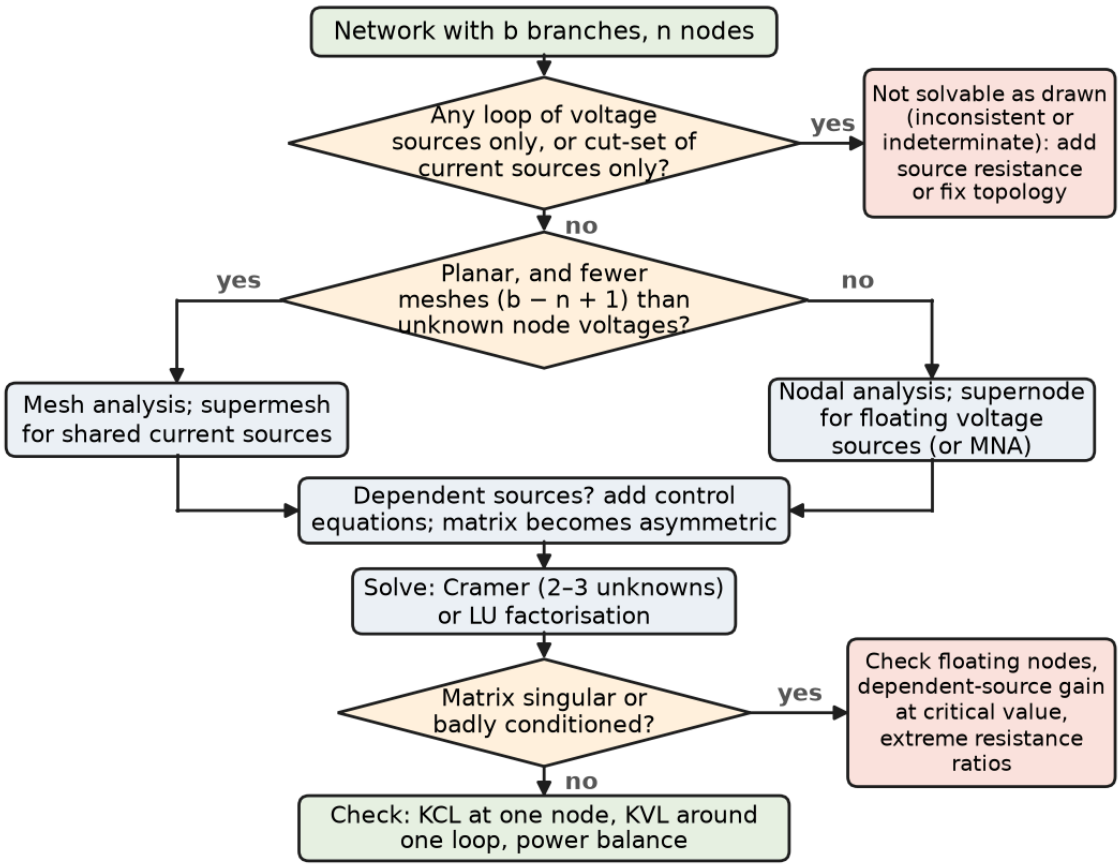


Figure 3.5 — Choosing and checking a systematic analysis method.

3.9 Circuit solvability

What can go wrong

Writing equations is mechanical; whether they have a solution is not. A linear network has exactly one solution if and only if its coefficient matrix (nodal, mesh or MNA) is non-singular. Three kinds of failure are common, and each has a physical meaning (Figure 3.5).

Inconsistent sources. A loop made only of ideal voltage sources (possibly with ideal shorts) has a KVL equation with no unknowns: either the source voltages happen to sum to zero, in which case the loop current is undetermined, or they do not, in which case there is no solution. Dually, a cut-set made only of ideal current sources (possibly with ideal opens) either over-determines or leaves undetermined the voltages across it. A charger modelled as an ideal 14 V source paralleled with a battery modelled as an ideal 12.6 V source is the textbook example; including the internal resistances, as in Chapter 2, restores a unique solution. These **topological conditions** can be checked before any arithmetic: a network of ideal sources and positive resistors is uniquely solvable exactly when it contains no loop of voltage sources and no cut-set of current sources.

Floating sub-networks. A node or group of nodes connected to the rest of the circuit only through capacitors or current sources has no DC path to the reference. In a DC analysis its potential is

undetermined: the nodal matrix has a zero eigenvalue. SPICE reports this as a "singular matrix" or "no DC path to ground" error, and the conventional remedy is a very large resistor (gigaohms) to ground. Physically, such a node really is undetermined in the ideal model; in reality leakage currents decide it, which is why an unconnected CMOS gate input drifts and must never be left floating.

Dependent sources at a critical gain. With dependent sources the matrix can become singular for particular parameter values even when the topology is fine.

Worked example — A negative conductance cancelling a resistor. A node with a 1 kΩ resistor to ground also connects to a VCCS that *injects* a current $g_m v$ proportional to the node's own voltage — the small-signal model of a cross-coupled transistor pair in an LC oscillator, which behaves as a negative conductance $-g_m$. A small test current i_t is injected into the node. KCL gives

$$\left(\frac{1}{R} - g_m\right) v = i_t, \qquad v = \frac{i_t}{1/R - g_m}.$$

With $R = 1$ kΩ and $i_t = 1$ μA: for $g_m = 0.5$ mS, $v = 2$ mV; for $g_m = 0.9$ mS, $v = 10$ mV; for $g_m = 0.99$ mS, $v = 100$ mV; at $g_m = 1.0$ mS the 1×1 nodal "matrix" is zero and no finite solution exists. The resistive network is telling the truth: at that point the circuit can sustain a voltage with no input at all. With the oscillator's LC tank added, this is precisely the start-up condition for oscillation, and for $g_m > 1/R$ the oscillation grows until device nonlinearity limits it. A singular network matrix is therefore not always a modelling error; it can mark the boundary of stability, a theme developed with characteristic equations in Chapters 7, 11 and 34.

Guaranteed solvability

For one important class the answer is always yes.

> **Status of the principle: Unique solvability of passive resistive networks**
> *Derived*. A connected network of positive resistors and independent current sources has a symmetric, positive-definite nodal matrix ($\mathbf{e}^T \mathbf{Y}_n \mathbf{e}$ equals the power the resistors would absorb at node voltages $\mathbf{e}$, which is positive for any non-zero $\mathbf{e}$), so its solution exists and is unique; adding ideal voltage sources preserves this provided they form no loop. Dependent sources and negative resistances void the guarantee.

Numerical conditioning

A non-singular matrix may still be hard to solve accurately. The **condition number** κ measures how much relative errors in the data (component tolerances, or rounding in the arithmetic) can be magnified in the solution. The bridge MNA matrix above, in the units shown, has $\kappa \approx 32$ (the value depends on how the unknowns are scaled), harmless in double-precision arithmetic, which carries about 16 significant digits. Circuits that mix very small and very large resistances are worse: a 1 mΩ current shunt and a 1 GΩ leakage path in the same network span twelve orders of magnitude, κ can approach 10^{12}, and only about four significant digits survive. Simulators mitigate this with pivoting and scaling, but the engineer should recognise the symptom: results that change in the third or fourth digit when a tolerance or solver option is changed.

Ill-conditioning also has a measurement meaning. The bridge output (1.85 mV) is the small difference of two large node voltages (2.502173 V and 2.500324 V). A 0.01 % error in one arm resistor shifts a midpoint voltage by about $5 \times 0.0001/4 = 0.125$ mV, 5 % of the 2.5 mV open-circuit signal. That is why bridge arms are matched, temperature-tracked resistors, and why the same difference-of-large-numbers problem reappears in differential signalling, in power-system state estimation and in the finite-precision arithmetic of digital filters (Chapter 23).

Checking an answer

Whatever the method, three checks catch nearly every error:

1. **KCL at a node not used in writing the equations** (for mesh analysis) or **KVL around a loop** (for nodal analysis).
2. **Power balance**: total delivered equals total absorbed (Tellegen's theorem).
3. **Limiting cases**: set a resistor to zero or infinity, or a gain to zero, and check that the answer reduces to something known. The degenerated amplifier with $R_E \to 0$ must return the −89.2 of Chapter 1; the bridge with $R_d \to \infty$ must return 2.4975 mV.

What remains open

Open question	Where it is answered
How is a whole network reduced to a single source and resistance at a pair of terminals, and how is that used for loading and maximum power?	Chapter 4 (Thévenin, Norton, maximum power transfer)
How is the reciprocity of passive networks proved and used?	Chapter 4 (reciprocity), building on Tellegen's theorem
How do nodal and mesh analysis handle capacitors, inductors and switching?	Chapters 5 and 6
How do the same matrices work with complex impedances in AC steady state?	Chapter 8 (AC nodal and mesh analysis)
How does the nodal method extend to the s-domain, and what do singular network matrices mean there?	Chapter 11 (Laplace-domain circuit analysis, poles and zeros)
How are networks described by their terminal behaviour only?	Chapter 12 (two-port networks)
Why does a singular characteristic equation mark the onset of oscillation or instability?	Chapters 7 and 34
How do finite-precision effects degrade numerical solutions and digital filters?	Chapters 21 and 23
How are transistor and op-amp parameters set by device physics and bias design?	*Electronics & Computing*

Next: Chapter 4 uses linearity to derive the network theorems — superposition, Thévenin, Norton, maximum power transfer and reciprocity — that turn the solutions of this chapter into compact models of how a network looks from its terminals.

Chapter 4 — Network Theorems

Part I · Circuit Fundamentals

A car will not start on a cold morning. The battery reads 12.6 V on a multimeter, which looks healthy, yet the starter motor barely turns. A technician clips a load tester across the terminals, draws 150 A for ten seconds, and watches the voltage collapse to 8 V. The open-circuit reading said almost nothing about the battery; the voltage *under load* said everything. What the technician measured, without naming it, was the battery's Thévenin equivalent: an ideal 12.6 V source in series with an internal resistance that had grown from about 5 mΩ when new to more than 25 mΩ after five winters. The same two numbers decide whether a USB port can charge a phone, whether a strain-gauge bridge reads correctly into a data logger, and how much fault current a substation must interrupt.

Chapter 3 gave systematic methods — nodal and mesh analysis — that solve any linear circuit completely. Completeness is not always what an engineer wants. Usually only one port matters: the terminals where a load, a sensor input or a fault is connected. The network theorems of this chapter compress everything behind that port into two or three numbers, separate the effect of each source, and expose structural properties (reciprocity, maximum power transfer) that full solutions hide.

> **How can a network of any complexity be replaced, at one pair of terminals, by a single source and a single resistance — and when does that replacement fail?**
> **Which properties of a circuit follow from linearity alone, and which need more?**

Concept	Engineering destination
Superposition	Separating DC bias from signal, noise budgets, multi-source power systems, interference analysis
Thévenin equivalent	Battery and source modelling, sensor and amplifier interfacing, fault-level calculation, loading error
Norton equivalent	Current-output sensors (photodiodes, 4–20 mA loops), transistor output models, short-circuit current
Maximum power transfer	RF and antenna matching, energy harvesting, solar maximum-power-point tracking, audio and probe design
Reciprocity	Antenna transmit/receive patterns, two-port measurements, sensitivity calculations, passive filter testing
Millman's theorem	Paralleled battery strings and rectifiers, star-point voltage of unbalanced loads, resistive summing networks
Source equivalence	Simplifying ladder networks, converting between voltage and current drive
Equivalent networks (Δ–Y, substitution)	Bridge circuits, three-phase load conversion, attenuators, reduction of meshed grids
Applications and limitations	Knowing when a linear model of a battery, diode, transformer or amplifier may be trusted

Roadmap. Linearity and superposition (4.1) → Thévenin's theorem (4.2) → Norton's theorem (4.3) → source equivalence (4.4) → maximum power transfer (4.5) → reciprocity (4.6) → Millman's theorem (4.7) → equivalent networks (4.8) → applications and limitations (4.9).

4.1 Superposition

Linearity is the root of every theorem here

A circuit is **linear** when every element obeys a linear relation between its voltage and current: resistors ($v = Ri$), the capacitors and inductors of Chapter 5 ($i = C\,dv/dt$, $v = L\,di/dt$), and linear dependent sources ($v = \mu v_x$, $i = \beta i_x$, $i = g_m v_x$, $v = r_m i_x$). Independent sources are the *inputs*; they are not themselves required to be linear functions of anything. Chapter 3 showed that the nodal equations of such a circuit take the form

$$\mathbf{G}\,\mathbf{v} = \mathbf{i}_s$$

where $\mathbf{G}$ depends only on the elements and $\mathbf{i}_s$ collects the independent sources. If $\mathbf{G}$ is invertible (the circuit is solvable, Chapter 3), then $\mathbf{v} = \mathbf{G}^{-1}\mathbf{i}_s$. Every node voltage — and therefore every branch voltage and current — is a weighted sum of the independent sources:

$$x = a_1 s_1 + a_2 s_2 + \cdots + a_N s_N$$

where x is any response, s_k are the independent sources and the coefficients a_k depend only on the network. Two properties follow directly. **Homogeneity**: scaling every source by α scales every response by α. **Additivity**: the response to several sources acting together equals the sum of the responses to each acting alone. Together they are the **superposition principle**.

The procedure

To find a response by superposition:

1. Keep one independent source; **deactivate** all the others. A deactivated voltage source is replaced by a short circuit (0 V regardless of current); a deactivated current source by an open circuit (0 A regardless of voltage).
2. Solve the simpler circuit for the response of interest.
3. Repeat for each independent source and add the partial responses, respecting reference directions.

Dependent sources are never deactivated. They are part of the network, not inputs: a current-controlled source whose controlling current happens to be zero in one sub-circuit simply produces zero there, but it must remain in place. Removing it changes $\mathbf{G}$ itself.

Sources may also be grouped: all DC sources in one sub-problem, all signal sources in another. This is exactly how a transistor amplifier is analysed — a DC bias solution plus a small-signal solution — although there the circuit is only linear about the bias point (Section 4.9).

> **Status of the principle: Superposition**
> *Derived*. A direct consequence of the linearity of the element equations and of KCL/KVL; it holds exactly for any circuit of linear elements and independent sources, and fails as soon as any element is nonlinear.

Power does not superpose

Power is a quadratic function of voltage or current. In a resistor carrying $i = i_1 + i_2$,

$$p = R(i_1 + i_2)^2 = Ri_1^2 + Ri_2^2 + 2Ri_1 i_2$$

The cross term $2Ri_1 i_2$ is missing from the sum of the individual powers. It vanishes only when the partial responses are *orthogonal* in an averaging sense — for instance, sinusoids of different frequencies averaged over a common period, or uncorrelated random noise sources averaged over time. That special case is why the average power of a signal with several harmonics is the sum of the harmonic powers (Chapter 9) and why independent noise powers add (Chapter 29). For DC sources, or for signals at the same frequency, power must be computed from the total response.

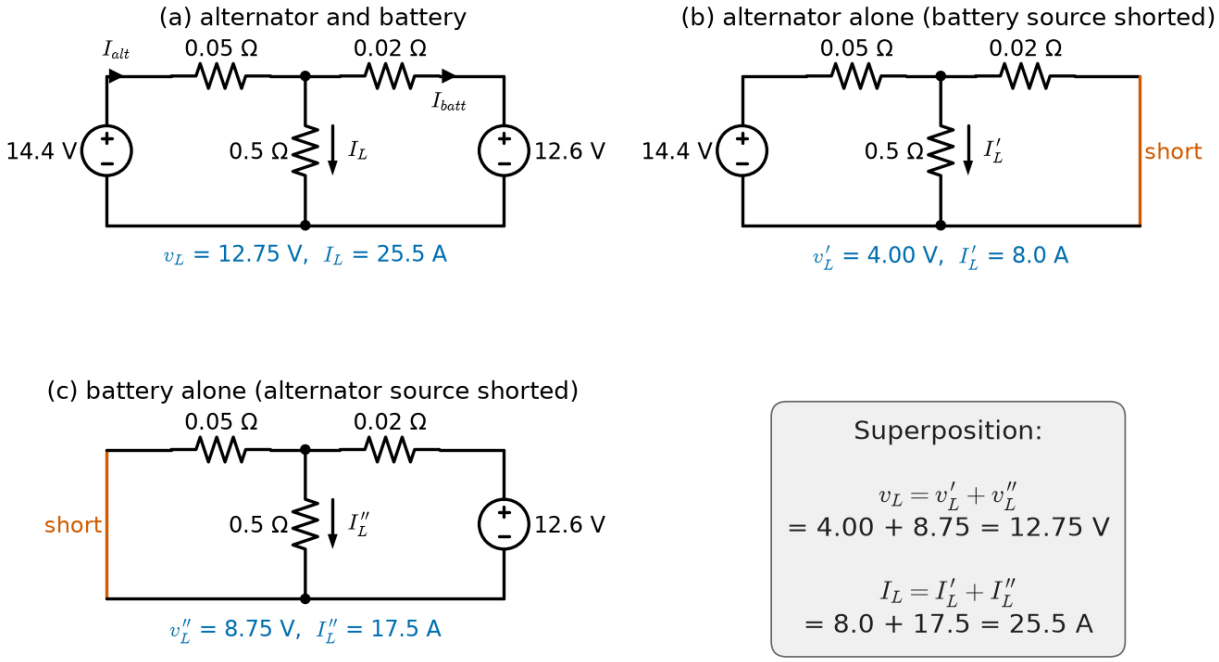


Figure 4.1 — Superposition applied to an automotive charging circuit: the full circuit and the two single-source sub-circuits.

Worked example — Alternator and battery sharing a load. A car's charging system is modelled (Figure 4.1a) as an alternator, 14.4 V behind 0.05 Ω, in parallel with a battery, 12.6 V behind 0.02 Ω, both feeding the vehicle's electrical load of 0.5 Ω (about 25 A, a typical night-time load with lights and blower).

Alternator alone (battery source shorted, Figure 4.1b). The alternator sees 0.05 Ω in series with $0.02 \parallel 0.5 = 0.019\,23$ Ω. Its current is $14.4/0.069\,23 = 208.0$ A, and the load-node voltage is $208.0 \times 0.019\,23 = 4.00$ V. The load carries $4.00/0.5 = 8.0$ A; the shorted battery branch carries $4.00/0.02 = 200$ A *into* the battery.

Battery alone (alternator source shorted, Figure 4.1c). The battery sees 0.02 Ω in series with $0.05 \parallel 0.5 = 0.045\,45$ Ω. Its current is $12.6/0.065\,45 = 192.5$ A, the node voltage is $192.5 \times 0.045\,45 = 8.75$ V, and the load carries 17.5 A while 175 A flows back into the alternator branch.

Together. Node voltage $4.00 + 8.75 = 12.75$ V; load current $8.0 + 17.5 = 25.5$ A. Alternator output current $(14.4 - 12.75)/0.05 = 33.0$ A, of which 25.5 A goes to the load and 7.5 A charges the battery: $(12.6 - 12.75)/0.02 = -7.5$ A of discharge. The partial currents in the sub-circuits are large (hundreds of amperes) because each source sees the other source's tiny internal resistance as a short; they cancel almost entirely in the sum. That is normal and harmless — sub-circuits are mathematical constructs, not operating states.

Power check. The load power is $12.75^2/0.5 = 325.1$ W. The sum of the "partial powers" would be $4.00^2/0.5 + 8.75^2/0.5 = 32.0 + 153.1 = 185.1$ W — wrong by 140 W, the cross term $2 \times 4.00 \times 8.75/0.5$. Superpose voltages and currents; compute power last.

Superposition as a design tool

Superposition is most useful not as a hand-calculation shortcut but as a way of thinking about *which source causes what*:

- **Bias and signal.** A sensor bridge excited from 5 V DC with a 1 mV, 50 Hz interference coupled in can be analysed for the DC operating point and the interference separately; the interference rejection of the instrument is a property of the second sub-circuit only.
- **Ground-loop and interference budgets.** In an audio system, each hum source (magnetic coupling, ground-potential difference, capacitive pickup) produces its own contribution at the amplifier input; superposition justifies analysing them one at a time and the linearity of the rest of the chain justifies adding them.
- **Power-system faults.** The current during a fault equals the pre-fault load current plus the current produced by a single source inserted at the fault point equal and opposite to the pre-fault voltage. This "superposition of the fault" is the basis of standard short-circuit calculation (Section 4.9).
- **Sensitivity.** Homogeneity alone tells you that doubling a bridge's excitation doubles its output, and that the output per volt of excitation (mV/V, the standard load-cell specification) is a property of the network.

4.2 Thévenin's theorem

Statement

Consider a network N of linear elements and independent sources connected at two terminals a–b to an arbitrary load (which may be nonlinear). **Thévenin's theorem** states that, as far as the load is concerned, N may be replaced by an ideal voltage source V_{Th} in series with a resistance R_{Th}, where

- V_{Th} is the **open-circuit voltage** v_{oc} at a–b (load removed);
- R_{Th} is the resistance seen looking into a–b with every *independent* source of N deactivated.

Equivalently, the terminal relation of N is the straight line

$$v = V_{Th} - R_{Th}\, i$$

where i is the current leaving terminal a into the load.

Why it is true

Drive the terminals with a test current source i (drawing current out of N) and ask for the terminal voltage v. By superposition, v is the sum of two parts: the voltage produced by the internal sources of N with the test source open — which is v_{oc} — and the voltage produced by the test source with the internal sources deactivated, which is linear in i and can be written $-R_{Th}\, i$. Hence $v = v_{oc} - R_{Th} i$ for *any* current, and therefore for whatever current the actual load draws. No assumption about the load was used; only N must be linear. The load may be a diode, a motor, a lamp or another network.

> **Status of the principle: Thévenin's theorem**
> *Derived*. Follows from superposition applied to the network seen through one port; exact for linear networks with independent and linear dependent sources, regardless of the load, provided the dependent sources are not controlled by quantities inside the load.

The proviso matters. If a dependent source inside N is controlled by a voltage or current in the load (or vice versa), the network and load are coupled by more than the two terminals and the one-port view does not apply.

Three ways to find R_{Th}

1. **Source deactivation and series–parallel reduction.** When N contains only independent sources and resistors, deactivate the sources and reduce the resistances seen from a–b (Chapter 2).
2. **Open-circuit voltage over short-circuit current.** Compute or measure v_{oc} and the short-circuit current i_{sc} (terminals shorted); then $R_{Th} = v_{oc}/i_{sc}$. This works with dependent sources and is the natural laboratory method when shorting the terminals is safe.
3. **Test source.** Deactivate the independent sources, apply a test voltage v_t (or current i_t) at the terminals, compute the resulting current (or voltage), and take $R_{Th} = v_t/i_t$. This is the only general method when dependent sources are present and $v_{oc} = 0$.

With dependent sources, R_{Th} can be smaller than any resistor in the network, larger than all of them, or even negative. Negative resistance at a port signals an active network that can deliver power into a passive load at the "wrong" sign; it is the basis of oscillator analysis (Chapter 7 shows how a negative resistance cancels circuit loss).

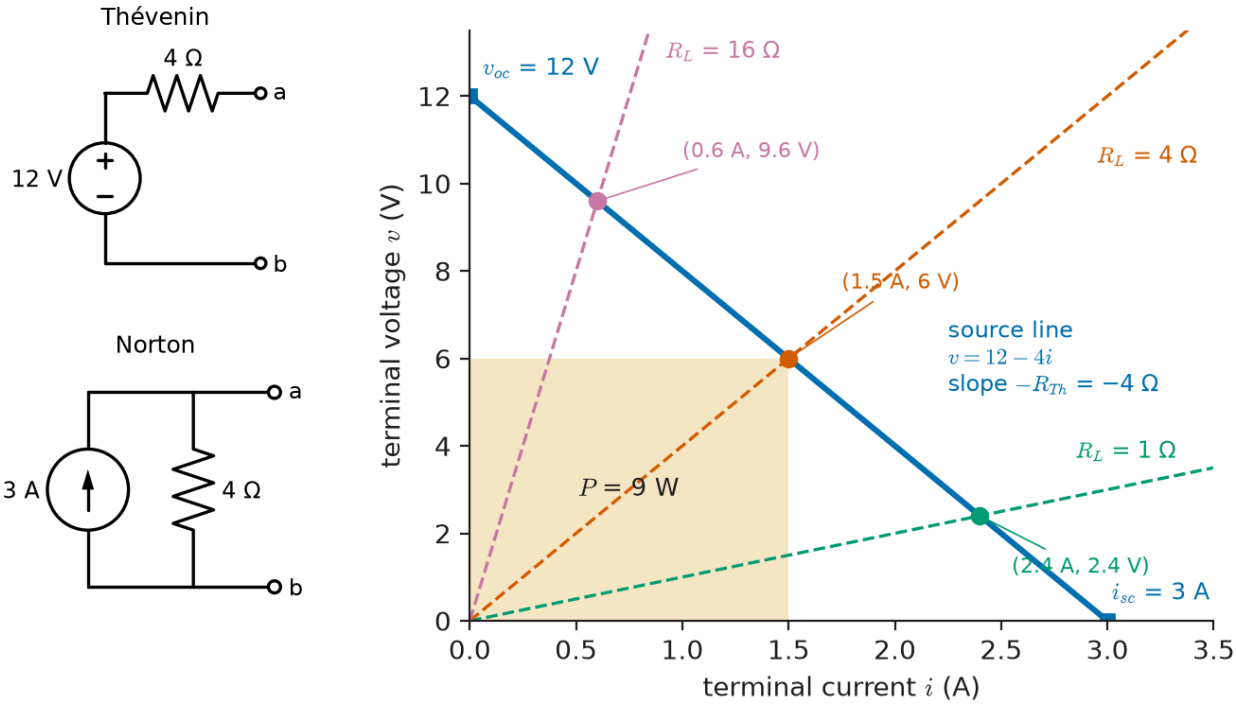


Figure 4.2 — The terminal characteristic of any linear one-port is a straight line fixed by the open-circuit voltage and the short-circuit current; Thévenin and Norton equivalents reproduce it exactly.

The terminal line and the load line

Figure 4.2 shows the geometry. A linear one-port is a straight line in the v–i plane, crossing the voltage axis at v_{oc} and the current axis at i_{sc}. A resistive load is a straight line through the origin with slope R_L. The operating point is the intersection. If the load is a nonlinear device — a diode, a lamp, a battery being charged — its characteristic is a curve, and the intersection with the Thévenin line gives the operating point graphically. This **load-line** construction is how diode and transistor bias points are found in *Electronics & Computing*; Thévenin's theorem is what reduces the rest of the circuit to a single line.

Measuring a Thévenin equivalent

Any source with accessible terminals — a battery, a bench supply, a USB port, a microphone, a sensor — can be characterised by two measurements. Measure the open-circuit voltage v_1 with a high-impedance meter; apply a known load R_L and measure v_2. Then $V_{Th} = v_1$ and

$$R_{Th} = R_L \left(\frac{v_1}{v_2} - 1 \right)$$

A USB port that reads 5.08 V open-circuit and 4.83 V while delivering 1.00 A into a 4.83 Ω load has $R_{Th} = 0.25/1.00 = 0.25\ \Omega$, most of it in the cable and connectors. The car battery of the opening reads 12.6 V open-circuit and 8.0 V at 150 A, so $R_{Th} = 4.6/150 \approx 31\ \text{m}\Omega$. Real sources are linear only over a range: the battery's internal resistance rises with discharge and falls with temperature, and a regulated supply has nearly zero R_{Th} until it reaches its current limit, where it abruptly becomes a current source. Two-point measurements should therefore be made around the intended operating point.

Worked example — A strain-gauge bridge driving a data logger. A quarter-bridge strain-gauge circuit (Figure 4.3) has four arms of nominally 350 Ω and is excited from $E = 5.000$ V. Under load the active gauge rises by 0.20 % to 350.70 Ω (with a gauge factor of 2.0, this is a strain of 1000 με). The bridge output is read by an instrument whose differential input resistance is R_m. Find the reading for $R_m = 10\ \text{k}\Omega$ and for $R_m = 1\ \text{M}\Omega$.

Label the left divider as a fixed 350 Ω on top and the gauge R_g on the bottom, so the left midpoint sits at $E\,R_g/(R + R_g)$; the right divider is two fixed 350 Ω arms, so its midpoint sits at $E/2$.

Open-circuit voltage.

$$V_{Th} = E\left(\frac{R_g}{R + R_g} - \frac{1}{2}\right) = 5.000\left(\frac{350.70}{700.70} - 0.5\right) = 2.498\ \text{mV}$$

The familiar small-strain approximation $E\,\Delta R/(4R) = 5 \times 0.70/1400 = 2.500$ mV is 0.1 % high; the bridge is slightly nonlinear in ΔR even before loading.

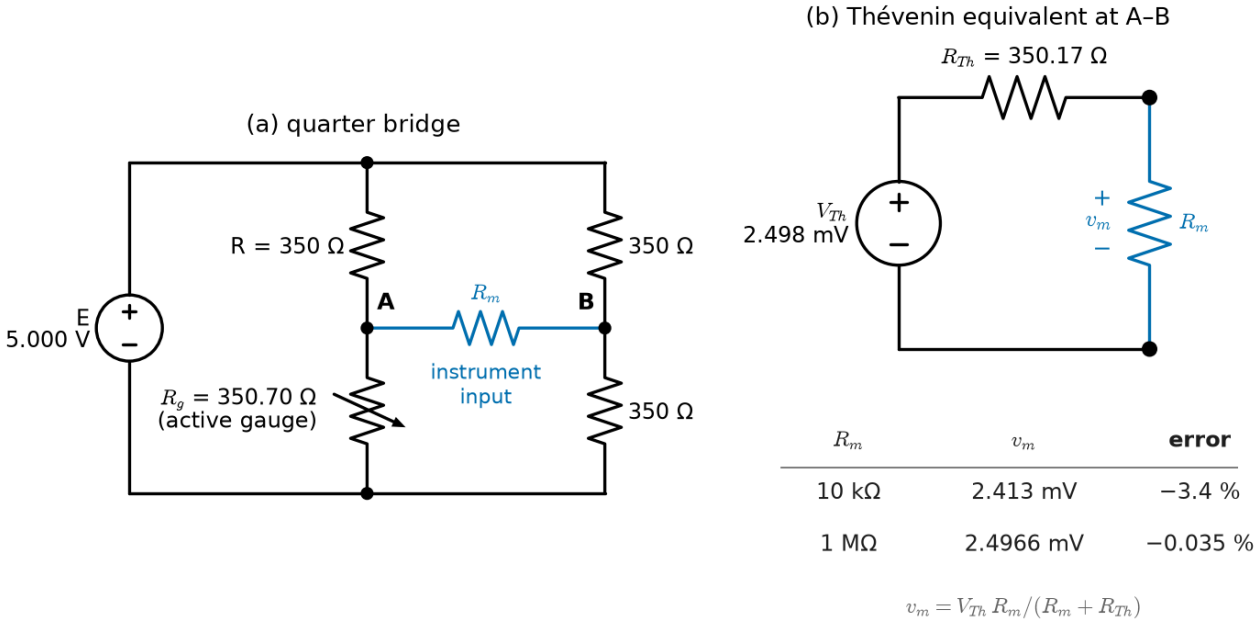
Thévenin resistance. With the excitation source shorted, the top and bottom of both dividers are tied together, so each midpoint sees its two arms in parallel, and the two midpoints are in series through the meter terminals:

$$R_{Th} = (350 \parallel 350.70) + (350 \parallel 350) = 175.17 + 175.00 = 350.17\ \Omega$$

Loaded reading.

$$v_m = V_{Th}\,\frac{R_m}{R_m + R_{Th}}$$

For $R_m = 10$ kΩ: $v_m = 2.498 \times 10\,000/10\,350.17 = 2.413$ mV, a **3.4 % loading error**, equivalent to reading 966 με instead of 1000 με. For $R_m = 1$ MΩ: $v_m = 2.4966$ mV, an error of 0.035 %. This is why strain-gauge amplifiers specify input resistances of at least 1 MΩ, and why the specification sheet of a bridge transducer always quotes its output resistance: the Thévenin resistance is the number that couples the sensor to the instrument.



R_m	v_m	error
10 kΩ	2.413 mV	−3.4 %
1 MΩ	2.4966 mV	−0.035 %

$$v_m = V_{Th}\, R_m/(R_m + R_{Th})$$

Figure 4.3 — A quarter-bridge strain-gauge circuit and its Thévenin equivalent seen by the measuring instrument.

Worked example — Thévenin equivalent of a network with a dependent source. A current-amplifying stage is modelled as follows: an independent source $V_s = 10$ V drives a resistor $R_1 = 2$ kΩ into node a; a resistor $R_2 = 8$ kΩ connects a to ground; and a current-controlled current source injects $k\,i_1$ into node a, where i_1 is the current in R_1 (flowing toward a) and $k = 3$. Find the Thévenin equivalent at node a with respect to ground.

Open-circuit voltage. KCL at a: $i_1 + k\,i_1 = v_a/R_2$, with $i_1 = (V_s - v_a)/R_1$. Substituting,

$$(1+k)\frac{V_s - v_a}{R_1} = \frac{v_a}{R_2} \;\Rightarrow\; v_a = \frac{(1+k)V_s/R_1}{1/R_2 + (1+k)/R_1} = \frac{4 \times 10/2000}{0.125\ \text{mS} + 2\ \text{mS}} = 9.412\ \text{V}$$

Short-circuit current. With a shorted to ground, $i_1 = V_s/R_1 = 5$ mA, R_2 carries nothing, and the short carries $(1+k)i_1 = 20$ mA.

Thévenin resistance. $R_{Th} = v_{oc}/i_{sc} = 9.412/0.020 = 470.6\ \Omega$.

Check with a test source. Deactivate V_s (short it) — the dependent source stays — and apply v_t at a. Then $i_1 = -v_t/R_1$, and the test source must supply $i_t = v_t/R_2 - (1+k)i_1 = v_t\,[1/R_2 + (1+k)/R_1]$, so $R_{Th} = 1/(0.125 + 2.0)\ \text{mS} = 470.6\ \ \Omega$. Note that R_{Th} is much smaller than either resistor: the dependent source multiplies the effective conductance of R_1 by $1 + k = 4$. Wrongly deactivating the dependent source would give $R_1 \parallel R_2 = 1.6$ kΩ, more than three times too large.

> **Boundary.**
> Dependent sources here stand for the linearised (small-signal) models of transistors and amplifiers. How those models are derived from device physics and bias conditions belongs to *Electronics & Computing*; this book uses them only as linear circuit elements.

4.3 Norton's theorem

Statement

Norton's theorem is the current-source dual of Thévenin's: the same linear network N may be replaced at terminals a–b by an ideal current source I_N in parallel with a resistance R_N, where

- I_N is the **short-circuit current** i_{sc} flowing from a to b through a short;
- $R_N = R_{Th}$, found exactly as in Section 4.2.

The terminal relation is the same straight line written the other way round:

$$i = I_N - \frac{v}{R_N}$$

Comparing with $v = V_{Th} - R_{Th} i$ gives the conversion between the two forms:

$$V_{Th} = I_N R_N, \qquad I_N = \frac{V_{Th}}{R_{Th}}, \qquad R_N = R_{Th}$$

> **Status of the principle: Norton's theorem**
> *Derived*. The same superposition argument as Thévenin's theorem with a test voltage instead of a test current; exact for linear networks, and equivalent to Thévenin's form whenever R_{Th} is finite and nonzero.

When the Norton form is the natural one

Mathematically the two forms are interchangeable; physically one is often more natural.

- **Photodiodes** generate a photocurrent proportional to optical power (about 0.5–0.9 A/W in silicon at its peak wavelength) in parallel with a large shunt resistance (tens of MΩ to GΩ). Their natural model is a Norton source, and a transimpedance amplifier is designed to hold their terminal voltage near zero so that the full short-circuit current is collected.
- **4–20 mA current loops** in process instrumentation use transmitters that behave as Norton sources with output resistance of megohms. The loop current is then insensitive to cable resistance and receiver resistance up to the transmitter's compliance voltage — a property easy to see in Norton form and awkward in Thévenin form, where it appears as a huge voltage behind a huge resistance.
- **Transistor output models** are Norton sources: a controlled current $g_m v_{be}$ in parallel with an output resistance r_o.
- **Power-system fault levels** are quoted as short-circuit currents (or short-circuit MVA) at a busbar — a direct statement of the Norton current of the grid behind that bus.

In Figure 4.2 the Norton equivalent of the 12 V, 4 Ω source is a 3 A source in parallel with 4 Ω. In the alternator example of Section 4.1, the alternator's Norton equivalent is $14.4/0.05 = 288$ A in parallel with 0.05 Ω, and the battery's is 630 A in parallel with 0.02 Ω; these large currents are not real fault currents (internal resistance rises and protection operates long before), but they show at once that the battery is the "stiffer" source.

Degenerate cases

If $R_{Th} = 0$, the network is an ideal voltage source at its terminals and has no Norton equivalent (the short-circuit current is infinite). If $R_{Th} \to \infty$, it is an ideal current source and has no Thévenin equivalent. Real sources always have some finite internal resistance, but *regulated* sources approximate these limits deliberately: a laboratory supply in constant-voltage mode has an output resistance of milliohms, a precision current source one of gigohms. When modelling them, use the form that remains finite.

4.4 Source equivalence

Practical sources as two-element networks

Chapter 2 introduced **source transformation**: a voltage source v_s in series with R may be exchanged for a current source v_s/R in parallel with the same R. Section 4.3 shows that this is simply the conversion between the Thévenin and Norton forms of a two-element network. Two sources are **equivalent** when they have the same terminal v–i line: the same open-circuit voltage and the same short-circuit current. Nothing more is promised.

> **Status of the principle: Source equivalence**
> *Derived*. Two linear one-ports with equal v_{oc} and i_{sc} have identical terminal characteristics; the equivalence is exact at the terminals and carries no information about what happens inside.

Equivalence is external only

The internal behaviour of equivalent sources can be completely different. With the terminals open, the Thévenin form of the 12 V, 4 Ω source dissipates nothing (no current flows), while the Norton form dissipates $3^2 \times 4 = 36$ W in its internal resistor. With the terminals shorted the situation reverses: the Thévenin resistor dissipates 36 W and the Norton resistor nothing. Neither model says anything trustworthy about the real source's internal heating; the battery's internal loss under short circuit is the Thévenin value only because the Thévenin form happens to put the resistance where the real loss mechanism is. Internal efficiency, temperature rise and component stress must be computed from a model of the actual internal structure.

Ideal sources and the rules for moving them

Some manipulations look like transformations but are not:

- An ideal voltage source with *no* series resistance has no Norton equivalent. A resistor connected *in parallel* with an ideal voltage source is irrelevant to the rest of the circuit (it changes only the source's own current) and may be removed for the external analysis.
- An ideal current source with no parallel resistance has no Thévenin equivalent. A resistor in *series* with an ideal current source is likewise irrelevant to the external circuit.

- Ideal voltage sources of different values may not be placed in parallel, and ideal current sources of different values may not be placed in series; such circuits violate KVL or KCL and have no solution (Chapter 3).
- A voltage source may be "pushed" through a node into every branch connected to that node (each branch gets a copy in series), and a current source may be split into a chain of equal sources across the elements of a loop. These **source-shifting** moves preserve all KCL and KVL equations and can turn an awkward source into one that can be transformed.

Dependent-source transformation

A dependent voltage source in series with a resistor may be transformed to a dependent current source in parallel with it, and vice versa, provided the controlling variable is *not* eliminated by the transformation. If the controlling voltage was the voltage across the very resistor being moved, it must first be re-expressed in terms of variables that survive. Losing the controlling variable is the most common error in hand analysis of amplifier models.

Source equivalence as a reduction engine

Repeated source transformation and series–parallel combination reduce ladder networks one rung at a time. Starting from the source end, each step converts a series source–resistor pair to a Norton pair, combines parallel resistors and current sources, converts back, and adds the next series resistor. The end result is the Thévenin equivalent at the far end of the ladder. This is the operation that a resistive attenuator, an R–2R digital-to-analogue converter ladder (where every node sees the same Thévenin resistance R, so each bit contributes a binary-weighted voltage) and a distribution feeder with tapped loads all invite.

4.5 Maximum power transfer

The theorem

A source with Thévenin equivalent V_{Th}, R_{Th} delivers to a resistive load R_L the power

$$P_L = \frac{V_{Th}^2 R_L}{(R_{Th} + R_L)^2}$$

Setting $dP_L/dR_L = 0$:

$$\frac{dP_L}{dR_L} = V_{Th}^2 \frac{(R_{Th} + R_L) - 2R_L}{(R_{Th} + R_L)^3} = 0 \;\Rightarrow\; R_L = R_{Th}$$

and the maximum is

$$P_{\max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{V_{Th}\, I_N}{4}$$

The maximum available power is one quarter of the product of open-circuit voltage and short-circuit current — the area of the largest rectangle that fits under the source line of Figure 4.2.

> **Status of the principle: Maximum power transfer**
> *Derived*. Follows from optimising the load power of a fixed Thévenin source; it answers "which load extracts the most power from this source", not "which source delivers the most power to this load" — if R_{Th} is adjustable, the load power is maximised by making R_{Th} zero.

Efficiency at the matched point

The efficiency of transfer — load power divided by total power taken from the ideal source — is

$$\eta = \frac{R_L}{R_{Th} + R_L}$$

At $R_L = R_{Th}$ it is exactly 50 %: half of the source's power is dissipated inside. Table 4.1 shows the trade-off for $V_{Th} = 12\ \text{V}$, $R_{Th} = 4\ \Omega$, and Figure 4.4 plots it.

Table 4.1 — Load power and efficiency for a 12 V, 4 Ω source.

R_L (Ω)	R_L / R_{Th}	I (A)	P_L (W)	η (%)
1	0.25	2.40	5.76	20.0
2	0.5	2.00	8.00	33.3
4	1	1.50	9.00	50.0
8	2	1.00	8.00	66.7
16	4	0.60	5.76	80.0

The power curve is flat near its maximum: a load mismatched by a factor of two in either direction still receives 89 % of the maximum (8.00 W of 9.00 W). The efficiency curve has no such plateau; it rises steadily with R_L.

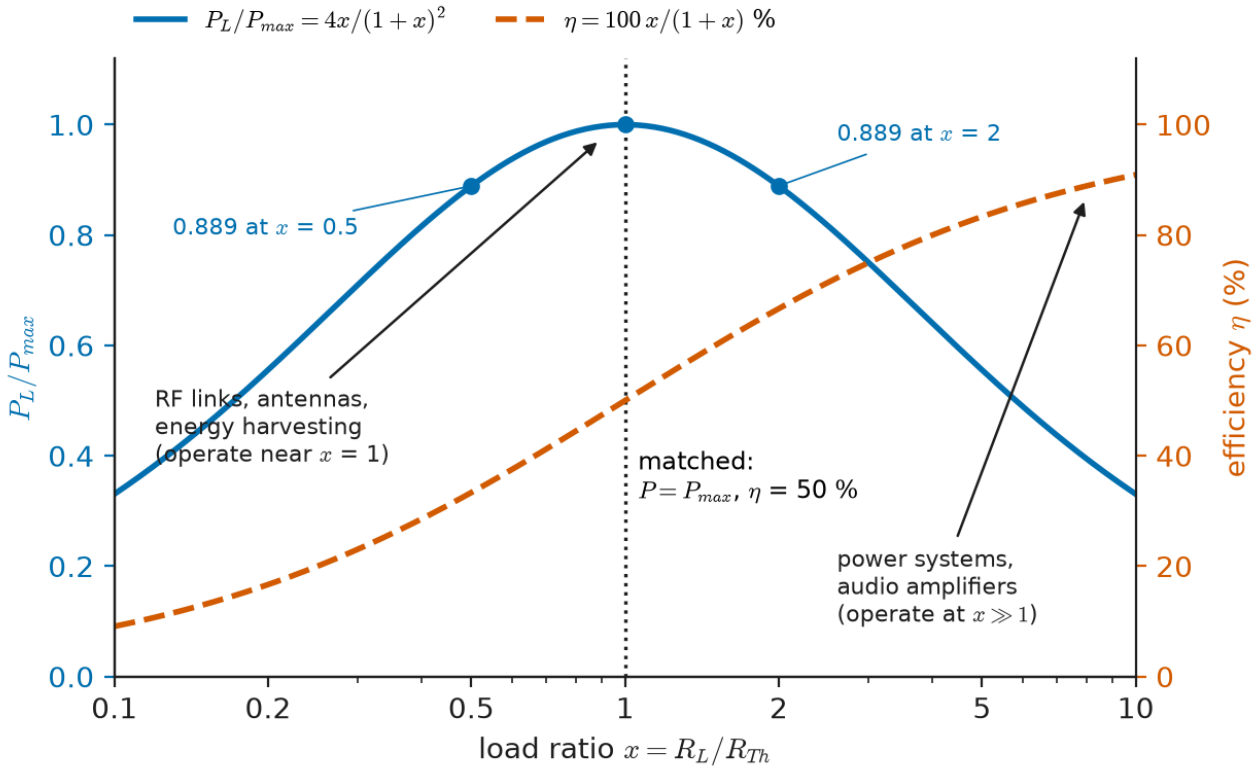


Figure 4.4 — Load power and transfer efficiency versus load resistance: power peaks at the matched load where efficiency is only 50 percent.

Two engineering regimes

Whether to match depends on what is scarce.

When the source is weak and its energy is free or already spent, match. An antenna delivers microwatts or less to a receiver; its "internal resistance" is its radiation resistance, the power dissipated there is re-radiated, not wasted heat in a battery. Extracting the maximum signal power is everything, so RF systems are built around a standard 50 Ω (or 75 Ω for video and cable television) and every source, line and load is matched to it. Energy harvesters (thermoelectric generators, piezoelectric vibration harvesters) and solar cells are run at their maximum-power point, because the energy available is fixed and limited. A photovoltaic module is nonlinear, so its maximum-power point is not simply "load = internal resistance", but the same idea — operate where $v \cdot i$ is largest on the source curve — is what an MPPT controller tracks.

When the source is strong and efficiency matters, do not match. A power system that operated at matched load would waste half its generation in its own lines and generators. A 230 V supply point with a source resistance of 0.2 Ω could in principle deliver $230^2/(4 \times 0.2) = 66$ kW to a matched load — at 50 % efficiency and with 575 A flowing. In reality loads are tens to hundreds of times the source impedance, voltage stays within a few percent of nominal, and efficiency is high. The same holds for audio amplifiers.

Worked example — Why a loudspeaker is not matched to its amplifier. A power amplifier with open-circuit output 20 V rms and output resistance 0.05 Ω drives an 8 Ω loudspeaker (treated as resistive). The current is $20/8.05 = 2.484$ A rms, the speaker power $2.484^2 \times 8 = 49.4$ W, and the

efficiency of transfer $8/8.05 = 99.4\ \%$. A matched $0.05\ \Omega$ load would in principle receive $20^2/(4 \times 0.05) = 2000\ \text{W}$ — but the amplifier would dissipate the same $2000\ \text{W}$ internally and would be destroyed long before. Audio amplifiers are designed as near-ideal voltage sources; their figure of merit is the **damping factor** R_L/R_{out}, here $8/0.05 = 160$. A high damping factor lets the amplifier's low source resistance damp the loudspeaker cone's mechanical resonance electrically (the back-EMF of the moving coil sees almost a short circuit), giving a tighter bass response. Maximum power transfer would be the wrong design goal.

Matching with reactive sources

When the source is a sinusoidal source with an internal impedance $Z_{Th} = R_{Th} + jX_{Th}$, the load that maximises average power is the **complex conjugate** $Z_L = R_{Th} - jX_{Th}$: the reactances cancel and the resistive part is matched. Chapter 8 introduces the impedances needed to state this; Chapter 9 proves it from complex power; and Chapter 14 shows how transmission-line sections and stubs realise the match at radio frequencies. The DC result of this section is the special case $X_{Th} = 0$.

4.6 Reciprocity

The theorem

Take a linear network containing only resistors (and, later, capacitors, inductors and coupled coils), with no dependent sources and no independent sources other than the one we insert. Choose two branches, call them port 1 and port 2.

Reciprocity theorem. If an ideal voltage source V inserted in branch 1 produces a current I in branch 2 (with branch 2 short-circuited or containing its normal passive element), then the same source moved to branch 2 produces the same current I in branch 1.

The dual statement holds for current sources and open-circuit voltages: a current source I injected at port 1 producing open-circuit voltage V at port 2 will, when moved to port 2, produce the same V at port 1. In both forms, the **transfer resistance** (or transfer conductance) between the two ports is the same in both directions.

Why it is true

For a network of resistors, the nodal conductance matrix $\mathbf{G}$ is symmetric: the conductance between nodes j and k appears as $-G_{jk}$ in both row j and row k. The inverse of a symmetric matrix is symmetric, so the voltage at node k due to a unit current injected at node j, $(\mathbf{G}^{-1})_{kj}$, equals the voltage at node j due to a unit current at node k, $(\mathbf{G}^{-1})_{jk}$. That is the current-source form of reciprocity. A more general proof, valid for any network whose branches obey symmetric constitutive relations, uses Tellegen's theorem — the statement that the sum of branch voltage–current products vanishes for any pair of voltage and current distributions satisfying KVL and KCL on the same graph.

> **Status of the principle: Reciprocity**
> *Derived.* Follows from the symmetry of the network matrices for bilateral linear elements (R, L, C, mutual inductance); it fails for networks with dependent sources, ferrites under static magnetic bias, and any nonlinear element.

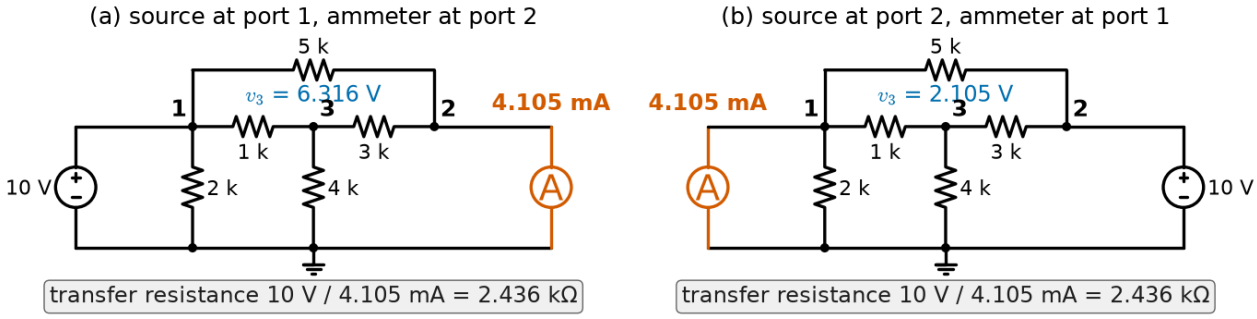


Figure 4.5 — Reciprocity: moving the source and the ammeter between two ports of a passive resistive network leaves the measured current unchanged.

Worked example — Checking reciprocity numerically. The network of Figure 4.5 has nodes 1, 2, 3 and ground, with resistors $R_{13} = 1\ \text{k}\Omega$, $R_{1g} = 2\ \text{k}\Omega$, $R_{3g} = 4\ \text{k}\Omega$, $R_{32} = 3\ \text{k}\Omega$ and $R_{12} = 5\ \text{k}\Omega$.

Source at port 1, short at port 2. Node 1 is held at 10 V and node 2 at 0 V. Node 3 is the only unknown:

$$v_3 = \frac{10/1000 + 0/4000 + 0/3000}{1/1000 + 1/4000 + 1/3000} = \frac{10\ \text{mA}}{1.5833\ \text{mS}} = 6.316\ \text{V}$$

The current into the short at node 2 is $10/5000 + 6.316/3000 = 2.000 + 2.105 = 4.105\ \text{mA}$.

Source at port 2, short at port 1. Now node 2 is at 10 V and node 1 at 0 V:

$$v_3 = \frac{10/3000}{1.5833\ \text{mS}} = 2.105\ \text{V}$$

and the short at node 1 carries $10/5000 + 2.105/1000 = 2.000 + 2.105 = 4.105\ \text{mA}$.

The internal state is completely different (node 3 sits at 6.316 V in one case and 2.105 V in the other; the 2 kΩ resistor carries current in one case and none in the other), but the transfer resistance $10\ \text{V}/4.105\ \text{mA} = 2.436\ \text{k}\Omega$ is the same in both directions.

Where reciprocity is used

- **Antennas.** An antenna's receiving pattern equals its transmitting pattern, and its gain is the same in both modes. This follows from reciprocity of Maxwell's equations in linear isotropic media, of which the circuit theorem is a special case. Radar and communication link budgets rely on it (Chapters 30 and 32).
- **Two-port parameters.** For a reciprocal two-port, $z_{12} = z_{21}$, $y_{12} = y_{21}$ and $AD - BC = 1$ (Chapter 12). This halves the measurements needed to characterise passive filters, attenuators and transmission lines.
- **Measurement convenience.** If driving port 2 is easier or safer than driving port 1 — for example, injecting a test signal at a sensor end of a cable network rather than at a high-voltage

end — reciprocity guarantees the same transfer ratio.
- **Sensitivity analysis.** The adjoint-network method for computing how every component affects one output in a single extra simulation is a direct application of reciprocity-type symmetry.

Where it fails

Amplifiers are deliberately non-reciprocal: a voltage amplifier with gain 100 in one direction has a reverse transmission near zero. Any network with dependent sources is generally non-reciprocal, because the controlled-source entries break the symmetry of $\mathbf{G}$. Non-reciprocal *passive* devices exist too — circulators and isolators use magnetised ferrites to route microwave power one way only, protecting transmitters from reflected power. Nonlinear networks (mixers, diode detectors) are not reciprocal in any useful sense.

4.7 Millman's theorem

Statement

When several branches, each consisting of a voltage source V_k in series with a resistance R_k (or simply a resistor, $V_k = 0$), are connected in parallel between two nodes, the voltage between those nodes is

$$V = \frac{\displaystyle\sum_k V_k / R_k}{\displaystyle\sum_k 1/R_k} = \frac{\sum_k G_k V_k}{\sum_k G_k}$$

It is the conductance-weighted average of the branch source voltages. Current-source branches can be included by adding their currents to the numerator (and nothing to the denominator).

Why it is true

Convert every branch to its Norton form: a current $G_k V_k$ in parallel with G_k. All the current sources are now in parallel and add; all the conductances are in parallel and add. The node voltage is total current over total conductance. Equivalently, Millman's theorem is single-node nodal analysis written in closed form.

> **Status of the principle: Millman's theorem**
> *Derived*. A one-line consequence of source transformation and KCL at a single node; exact for linear branches connected between exactly two nodes.

The result is both a shortcut and a piece of physical insight. The common voltage is pulled toward the sources with the *largest conductance* — the stiffest sources dominate. In the alternator example of Section 4.1, Millman's theorem gives the bus voltage at once:

$$V = \frac{14.4/0.05 + 12.6/0.02 + 0}{1/0.05 + 1/0.02 + 1/0.5} = \frac{288 + 630}{20 + 50 + 2} = \frac{918}{72} = 12.75\ \text{V}$$

agreeing with the superposition result.

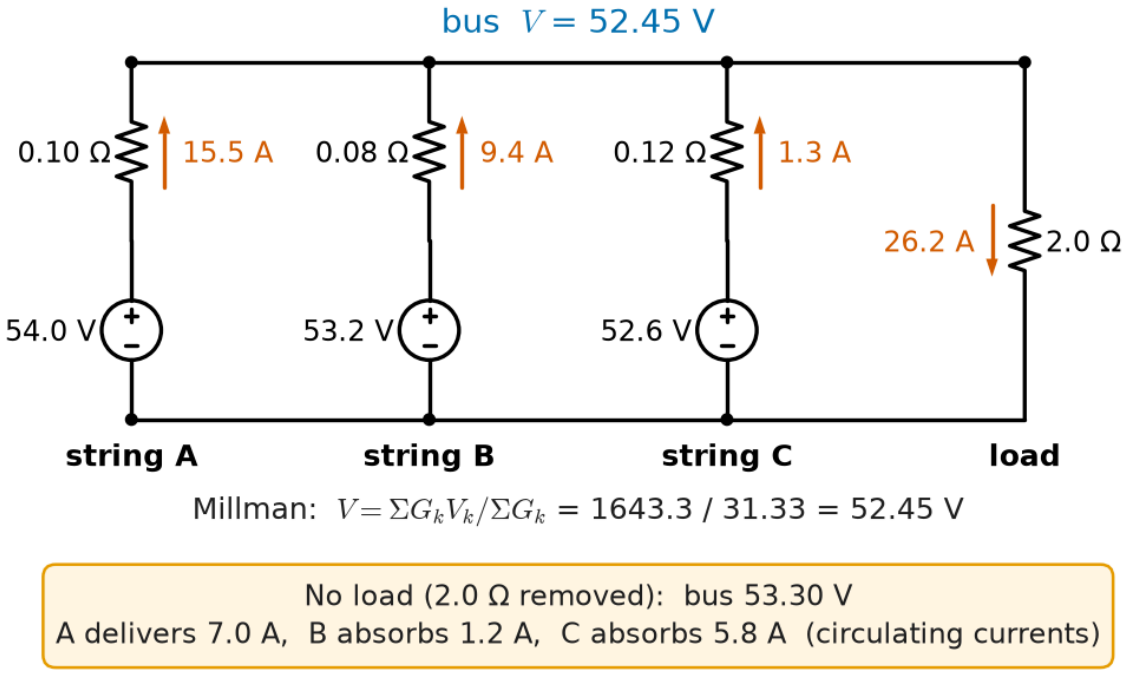


Figure 4.6 — Millman's theorem applied to three paralleled battery strings feeding a common DC bus.

Worked example — Paralleled battery strings in a telecom DC plant. A 48 V telecom power system has three valve-regulated lead-acid strings in parallel on its DC bus (Figure 4.6). Because of ageing and temperature differences, their open-circuit voltages and internal resistances differ: string A 54.0 V and 0.10 Ω, string B 53.2 V and 0.08 Ω, string C 52.6 V and 0.12 Ω (resistances include cabling and fusing). During a mains outage the strings feed a load modelled as 2.0 Ω.

Bus voltage.

$$V = \frac{54.0/0.10 + 53.2/0.08 + 52.6/0.12}{1/0.10 + 1/0.08 + 1/0.12 + 1/2.0} = \frac{540 + 665 + 438.33}{10 + 12.5 + 8.333 + 0.5} = \frac{1643.33}{31.333} = 52.45 \text{ V}$$

Branch currents. String A delivers $(54.0 - 52.45)/0.10 = 15.5$ A, string B $(53.2 - 52.45)/0.08 = 9.4$ A, string C $(52.6 - 52.45)/0.12 = 1.3$ A; the load draws $52.45/2.0 = 26.2$ A. The strings share the load very unequally: string C, which has the lowest open-circuit voltage, does almost nothing.

No load. Removing the 2.0 Ω branch, the bus floats to $1643.33/30.833 = 53.30$ V. String A then delivers 7.0 A, while B absorbs 1.2 A and C absorbs 5.8 A. These **circulating currents** flow with no external load at all. They are why paralleled strings must be matched in age and type, why each string has its own fuse, and why battery monitoring systems track string currents individually.

Millman's theorem in three-phase and signal circuits

The same formula gives the voltage of the star point of an unbalanced three-phase load relative to the supply neutral: with phase voltages V_a, V_b, V_c and load admittances Y_a, Y_b, Y_c (no neutral wire), the star-point displacement is $(Y_a V_a + Y_b V_b + Y_c V_c)/(Y_a + Y_b + Y_c)$, a phasor version of Millman (Chapter 10).

In signal circuits, a passive resistive summer — several signals each through a resistor into a common node loaded by a resistor — is a Millman network, which is why its output is a weighted average rather than a sum and why audio mixers use an active virtual-ground summing node instead.

4.8 Equivalent networks

What "equivalent" means

Two networks are **equivalent with respect to a set of terminals** when every possible set of terminal voltages produces the same set of terminal currents in both. For a two-terminal network (one-port) this reduces to the single v–i line of Section 4.2. For three or more terminals, equivalence requires the full matrix of terminal relations to agree. As with sources (Section 4.4), equivalence says nothing about internal voltages, currents or dissipation.

The theorems already met are all equivalence statements: Thévenin and Norton for one-ports with sources; series–parallel reduction for passive one-ports. Three more tools complete the set.

Delta–wye (Π–T) transformation

Some resistor arrangements are neither series nor parallel — the Wheatstone bridge with a non-zero galvanometer resistance is the classic case. Three resistors connected in a triangle between terminals a, b, c (a **delta**, Δ, or Π) are equivalent to three resistors connected from a common centre to the same terminals (a **wye**, Y, or T) when the resistance between every pair of terminals, with the third open, is the same in both. Solving the three resulting equations gives:

Δ → Y:

$$R_a = \frac{R_{ab} R_{ca}}{R_{ab} + R_{bc} + R_{ca}}, \quad R_b = \frac{R_{ab} R_{bc}}{R_{ab} + R_{bc} + R_{ca}}, \quad R_c = \frac{R_{bc} R_{ca}}{R_{ab} + R_{bc} + R_{ca}}$$

Each Y resistor is the product of the two Δ resistors adjacent to its terminal, divided by the sum of all three.

Y → Δ:

$$R_{ab} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_c}, \quad R_{bc} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_a}, \quad R_{ca} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_b}$$

Each Δ resistor is the sum of pairwise products divided by the Y resistor *opposite* it. For a balanced network (R_Δ in every arm), $R_Y = R_\Delta / 3$.

> **Status of the principle: Δ–Y transformation**
> *Derived*. Exact equivalence of two three-terminal resistive networks, obtained by matching their terminal resistances; extends to impedances in the sinusoidal steady state (Chapter 8) but not to networks containing sources inside the delta or star.

Worked example — An unbalanced bridge. A bridge network has a top node a and bottom node d, with two middle nodes b and c. The arms are $R_{ab} = 100\ \Omega$, $R_{ac} = 200\ \Omega$, $R_{bd} = 300\ \Omega$, $R_{cd} = 400\ \Omega$, and the bridging resistor $R_{bc} = 500\ \Omega$. Find the resistance between a and d, and the current drawn from a 10 V source.

No two resistors are in series or in parallel. Convert the upper delta a–b–c ($R_{ab} = 100$, $R_{bc} = 500$, $R_{ca} = 200$; sum 800 Ω) to a wye:

$$R_a = \frac{100 \times 200}{800} = 25\ \Omega, \quad R_b = \frac{100 \times 500}{800} = 62.5\ \Omega, \quad R_c = \frac{200 \times 500}{800} = 125\ \Omega$$

Now R_a leads from a to the star centre, from which two paths reach d: $R_b + R_{bd} = 362.5\ \Omega$ and $R_c + R_{cd} = 525\ \Omega$, in parallel:

$$R_{ad} = 25 + \frac{362.5 \times 525}{362.5 + 525} = 25 + 214.44 = 239.44\ \Omega$$

A 10 V source draws $10/239.44 = 41.76$ mA. (A nodal solution with 1 A injected at a gives the same 239.44 Ω, which is a useful check.) The result lies, as it must, between 240.00 Ω (bridging resistor removed) and 238.10 Ω (bridging resistor shorted); because the bridge is unbalanced ($100/300 \neq 200/400$), the 500 Ω resistor carries current and fixes where in that range the answer falls.

The Δ–Y transformation reappears in three-phase circuits, where loads connected in delta are converted to an equivalent star to compute line currents per phase (Chapter 10), and in attenuator design, where T and Π pads are two realisations of the same terminal behaviour (Chapter 12).

Substitution theorem

If the voltage across (or current through) a branch of a network is known, that branch may be replaced by an independent voltage source of that value (or an independent current source of that current) without changing any other voltage or current in the network — provided the network with the substitution still has a unique solution. The theorem is almost trivial, yet it justifies two practical moves. It lets a nonlinear element whose operating point is known (a diode conducting at 0.65 V, a Zener at 5.1 V) be replaced by a source while the rest of the circuit is analysed linearly. And it is the step in the proof of Thévenin's theorem that allows the load — whatever it is — to be replaced by a current source equal to its actual current.

Equivalence beyond resistors

All the equivalences of this chapter carry over to the sinusoidal steady state with resistances replaced by complex impedances and DC sources by phasors (Chapter 8), and to the Laplace domain with impedances $Z(s)$ and initial-condition sources (Chapter 11). There, the Thévenin impedance is a function of frequency, which is why a source can be "matched" at one frequency and badly mismatched at another, and why an oscilloscope probe loads a circuit more at high frequency than at DC.

4.9 Applications and limitations

A catalogue of applications

The theorems are used continuously, often without being named.

Table 4.2 — Typical Thévenin (source) resistances seen at practical ports.

Port	Typical open-circuit voltage	Typical source resistance	What it decides
Fresh AA alkaline cell	1.5–1.6 V	0.1–0.3 Ω	Usable current for motors and flashes
Car starter battery (new)	12.6–12.8 V	3–8 mΩ	Cranking current of several hundred amperes
USB 2.0 port plus cable	5.0–5.25 V	0.2–0.5 Ω	Voltage at the device under charging load
Laboratory supply (CV mode)	set value	1–10 mΩ	Load regulation
Domestic mains socket (230 V)	230 V	0.2–0.8 Ω (loop impedance)	Fault current, voltage drop under heavy loads
Moving-coil microphone	1–3 mV (speech)	150–600 Ω	Preamplifier input resistance (≥ 5–10× source)
Quarter-bridge strain gauge	mV range	120 or 350 Ω	Amplifier input resistance, cable effects
Thermocouple (type K)	about 41 µV/°C	1–100 Ω (wire length)	Usually negligible versus instrument input
Function generator output	set value	50 Ω	Output doubles into high-Z loads
Glass pH electrode	about 59 mV/pH at 25 °C	10–1000 MΩ	Requires electrometer-grade (> 10^{12} Ω) input
Surface ECG electrode pair	0.5–4 mV signal	1 kΩ–1 MΩ (frequency- and skin-dependent)	Amplifier input impedance and common-mode rejection

Two observations from the table recur throughout instrumentation. A measuring instrument should have an input resistance much larger than the source's Thévenin resistance — typically by a factor of 100 to 1000 for 1 % to 0.1 % loading error — and the required input resistance therefore spans from kilohms (thermocouple) to teraohms (pH electrode). And a 50 Ω function generator set to "1 V" assumes a 50 Ω load: into an oscilloscope's 1 MΩ input its open-circuit voltage is doubled, a direct consequence of $v = V_{Th} R_L / (R_{Th} + R_L)$.

Other applications follow the same pattern:

- **Fault-level calculation.** The prospective short-circuit current at a busbar is V_{Th}/Z_{Th} of the grid behind it, using the pre-fault voltage as V_{Th} (Thévenin plus superposition, Section 4.1). It sets the breaking capacity of circuit breakers and the withstand rating of busbars. A domestic socket with 0.35 Ω loop impedance has a prospective fault current of about 660 A; a large industrial switchboard may see 50 kA.
- **Voltage regulation and drop.** The voltage at the end of a feeder under load is $V_{Th} - R_{Th}I$; regulations limit the drop (typically 3–5 % for lighting circuits).
- **Interfacing logic and sensors.** Pull-up resistor sizing, the drive capability of microcontroller pins, and the input-current error of an ADC driven from a resistive divider are all Thévenin calculations.
- **Simulation.** Circuit simulators exploit superposition in small-signal (AC) analysis and Norton companion models in transient analysis: each capacitor and inductor is replaced at each time step by a resistor and a current source (Chapter 6 and Chapter 11 show where these come from).

Limitation 1 — nonlinearity

Every theorem in this chapter rests on linearity. A diode, a transistor in saturation, an iron-cored inductor near saturation, a lamp filament whose resistance rises tenfold as it heats, a battery whose internal resistance depends on state of charge: none of these is linear, and a network containing them has no Thévenin equivalent in general.

Two partial remedies are standard. If the nonlinear element is the *load*, the theorems apply to the linear rest of the network and the load-line construction (Figure 4.2) finds the operating point. If the signals are small, the nonlinear element can be **linearised** about its operating point: replaced by its incremental resistance $r = dv/di$ there. Superposition then holds for the small-signal part only, with an error that grows with signal amplitude — the origin of harmonic distortion in amplifiers. A small-signal Thévenin equivalent is valid for small deviations about one bias point and must be recomputed if the bias changes.

> **Status of the principle: Small-signal linearisation**
> *Approximate.* A first-order Taylor expansion of a nonlinear element about its operating point; accurate when signal excursions are small compared with the scale on which the element's characteristic curves, and wrong for large signals, switching and clipping.

Limitation 2 — power and energy

Superposition applies to voltages and currents, not to power or energy (Section 4.1). Thévenin and Norton equivalents reproduce terminal power exactly but internal power not at all (Section 4.4). The 50 % efficiency of maximum power transfer refers to the Thévenin model; the efficiency of a real source at matched load depends on where its internal losses actually occur.

Limitation 3 — time variation and switching

The theorems apply to linear *time-invariant* networks in the form stated, and to each interval of a switched network separately. When a switch operates, the network changes and its Thévenin equivalent

changes with it; Chapter 6 handles this by solving interval by interval with continuity conditions on capacitor voltages and inductor currents. Superposition remains valid for linear time-varying networks, but the convenient "one Thévenin equivalent for all time" does not.

Limitation 4 — the lumped-element assumption

All of the above assumes lumped elements connected by ideal wires, so that KCL and KVL hold (Chapter 1). When interconnections become a significant fraction of a wavelength, the "terminals" of a network are no longer well defined: the voltage between two points depends on the path, and the Thévenin impedance seen at the end of a cable depends on the cable's length and termination. At that point the port description moves to transmission-line theory and scattering parameters (Chapters 14 and 32), which are the high-frequency generalisation of the same idea — characterise a network only by what it does at its ports.

Limitation 5 — what the equivalent hides

An equivalent is a black box by design. It cannot say which internal component is overheating, how close an internal node is to breakdown, or what happens if the internal structure changes (a fuse blowing, a cell failing). Equivalents are tools for interfacing; internal design and protection need the full model.

What remains open

Open question	Where it is answered
How do capacitors and inductors, whose equations involve derivatives, fit into these linear theorems?	Chapter 5 (element equations, initial conditions); Chapter 6 (first-order circuits solved via Thévenin equivalents)
How do Thévenin equivalents and superposition handle switching transients?	Chapters 6 and 7
What becomes of Thévenin, Norton, Millman and maximum power transfer with sinusoidal sources and complex impedances?	Chapter 8 (phasor equivalents); Chapter 9 (conjugate matching from complex power)
How does Millman's theorem give the neutral shift of an unbalanced three-phase load?	Chapter 10
How are equivalents with initial conditions written in the s-domain?	Chapter 11
How are networks with two ports characterised, and what does reciprocity imply for their parameters?	Chapter 12
What replaces the port idea when interconnects are electrically long?	Chapter 14 (transmission lines); Chapter 32 (S-parameters)
Why do independent noise powers add even though power does not superpose in general?	Chapter 29
How are transistor and op-amp small-signal models derived?	*Electronics & Computing*

Next: Chapter 5 introduces the two energy-storing elements, the capacitor and the inductor, whose memory of past voltage and current turns the algebraic circuits of Part I into the differential equations of Part II.

Chapter 5 — Capacitors and Inductors

Part I · Circuit Fundamentals

A technician switches off a variable-speed drive, waits for the display to go dark, opens the cabinet — and receives a painful shock from the DC bus. The mains has been disconnected for thirty seconds, yet the 470 μF bus capacitors still hold several hundred volts. A few metres away, a maintenance engineer opens a contactor feeding a large solenoid valve; the contact draws a bright arc, and over months the contact surfaces erode until the contactor welds shut. Neither event can be explained by the resistive circuits of Chapters 2–4, in which every voltage and current is fixed instantly by the present values of the sources. The capacitor remembers its voltage after the source has gone; the inductor insists on keeping its current flowing after the switch has opened. Both store energy, and both release it on their own terms.

This chapter introduces the two energy-storage elements. Their defining equations involve time derivatives, so circuits containing them obey differential rather than algebraic equations — the subject of all of Part II. Before solving those equations we need the elements themselves: where capacitance and inductance come from, how they combine, how much energy they hold, what they force on a circuit at the instant of switching, how two inductors couple through a shared magnetic field, and how far real components depart from the ideal.

> **Why does a capacitor hold its voltage and an inductor its current, and what does that imply at the instant a switch operates?**
> **How much energy do they store, and what do real capacitors and inductors look like once their parasitic resistance, inductance and capacitance are included?**

Concept	Engineering destination
Capacitance	Decoupling, filtering, timing, sensors (touch, humidity, level), memory cells, cable and PCB parasitics
Inductance	Switching converters, filters and chokes, relays and solenoids, current sensing, wiring parasitics
Constitutive equations	Ripple calculation, slew-rate limits, inductive kick, numerical simulation
Energy storage	Defibrillators, flash units, hold-up time, supercapacitor buffers, ignition coils, safety discharge
Series and parallel combinations	Voltage rating of capacitor banks, inductor arrays, charge sharing and inrush
Initial conditions	Every switching transient of Chapters 6 and 7
Mutual inductance and coupled coils	Transformers, current transformers, Rogowski coils, wireless charging, crosstalk
Ideal versus practical components	ESR heating, self-resonance, component selection, EMC

Roadmap. Capacitance (5.1) → inductance (5.2) → constitutive equations and the duality between the two (5.3) → energy storage (5.4) → series and parallel combinations (5.5) → initial conditions at switching instants (5.6) → mutual inductance (5.7) → coupled coils and the transformer limit (5.8) → ideal versus practical components (5.9).

5.1 Capacitance

Charge and voltage on two conductors

Any two conductors separated by an insulator form a capacitor. When a charge $+q$ is moved onto one conductor and $-q$ onto the other, an electric field appears in the insulator and a voltage v appears between the conductors. For linear insulating materials the field, and therefore the voltage, is proportional to the charge. The constant of proportionality is the **capacitance**:

$$C = \frac{q}{v}$$

measured in farads (F = C/V). One farad is an enormous capacitance for a conventional component; practical values run from picofarads (pF) to millifarads (mF), while electrochemical supercapacitors reach thousands of farads.

Capacitance depends only on geometry and on the **permittivity** $\varepsilon = \varepsilon_r \varepsilon_0$ of the insulator (the **dielectric**), where $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m. The simplest case is two parallel plates of area A separated by d, with d small compared with the plate dimensions so that fringing at the edges can be neglected:

$$C = \frac{\varepsilon_r \varepsilon_0 A}{d}$$

Two plates of 10 cm × 10 cm, 1 mm apart in air, give only $8.854 \times 10^{-12} \times 0.01/0.001 = 88.5$ pF. To make large capacitances in small volumes, manufacturers increase A (rolling or stacking long electrodes), reduce d (films of a few micrometres or less) and raise ε_r (ceramics with ε_r of thousands).

> **Status of the principle: Linear capacitance** $q = Cv$
> *Approximate.* Follows from electrostatics for a linear dielectric; holds well for air, glass, polymer films and C0G ceramics, but high-permittivity ceramics (X7R, X5R) lose a large fraction of their capacitance under DC bias and with temperature, and electrolytic capacitors depend on frequency and age.

> **Boundary.**
> How permittivity arises from atomic and molecular polarisation, and why ferroelectric ceramics have such large and nonlinear ε_r, belongs to *The Materials Core*; the field theory of capacitance for arbitrary geometries belongs to *Engineering Physics*. This chapter uses C as a circuit parameter.

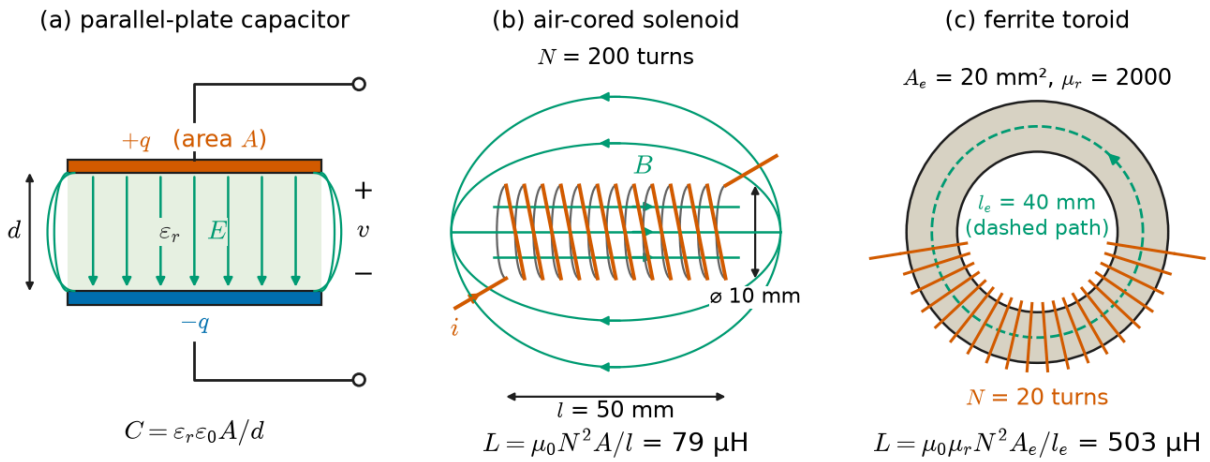


Figure 5.1 — Geometry of the two storage elements: a parallel-plate capacitor stores energy in its electric field, a solenoid and a toroid store energy in their magnetic fields.

Capacitor technologies

Table 5.1 — Capacitor families (typical values).

Family	Dielectric ε_r	Typical range	Typical voltage	Characteristics and uses
C0G/NP0 ceramic (MLCC)	20–100	0.1 pF–0.1 µF	16–1000 V	Stable (±30 ppm/°C), low loss; RF, timing, filters
X7R/X5R ceramic (MLCC)	2000–4000	100 pF–100 µF	4–100 V	High density; lose 30–80 % of C at rated DC bias; decoupling
Polypropylene film	2.2	1 nF–100 µF	63 V–2 kV	Very low loss, self-healing; snubbers, PFC, audio, pulse
Polyester (PET) film	3.3	1 nF–10 µF	50–630 V	General purpose coupling and bypass
Aluminium electrolytic	about 8–10 (Al_2O_3)	1 µF–1 F	6.3–550 V	Polarised, high C, high ESR, limited life; bulk storage
Tantalum / polymer	about 27 (Ta_2O_5)	0.1 µF–1 mF	2.5–50 V	Compact, low ESR (polymer); polarised
Electric double-layer (supercapacitor)	—	0.1–3000 F	2.5–3.0 V per cell	Very high C, low voltage; hold-up and buffering

Worked example — Capacitance of a multilayer ceramic chip. A small MLCC has $n = 200$ active dielectric layers, each $d = 1.0$ µm thick, with electrode overlap of 0.8 mm × 0.4 mm and $\varepsilon_r = 3000$ (X5R). The layers are electrically in parallel, so

$$C = n\frac{\varepsilon_r \varepsilon_0 A}{d} = 200 \times \frac{3000 \times 8.854 \times 10^{-12} \times (0.8 \times 10^{-3})(0.4 \times 10^{-3})}{1.0 \times 10^{-6}} = 1.70\ \mu\text{F}$$

in a package roughly 1.0 mm × 0.5 mm. At its 6.3 V rating, the field in each layer is $6.3/1.0 \times 10^{-6} = 6.3$ MV/m. That strong field is precisely what drives the X5R ceramic toward saturation of its polarisation: such parts commonly deliver only 40–60 % of their nominal capacitance at rated voltage. A designer who needs 1 µF at 5 V across such a part may need a nominal 2.2 µF or a larger case size.

Capacitance is everywhere

Capacitance is not only a component. Every pair of conductors has it: a PCB trace over a ground plane (typically 0.5–1.5 pF/cm), a coaxial cable (about 100 pF/m for 50 Ω cable, 67 pF/m for 75 Ω), a transistor's junctions, a person's finger near a touch-screen electrode, the windings of a transformer. This **stray** or **parasitic** capacitance limits the speed of logic, couples interference between circuits, and sets the bandwidth of sensors. Capacitive sensors exploit the same physics deliberately: a change in d (pressure diaphragms, MEMS accelerometers), in A (rotary encoders, liquid-level probes) or in ε_r (humidity sensors, proximity detectors) changes C and is read out electronically.

5.2 Inductance

Flux linkage and current

A current i in a coil produces a magnetic field; the magnetic flux ϕ through each turn, summed over all N turns, is the **flux linkage** $\lambda = N\phi$. For coils in linear magnetic media, λ is proportional to i, and the constant is the **self-inductance**:

$$L = \frac{\lambda}{i} = \frac{N\phi}{i}$$

measured in henries (H = Wb/A = V·s/A). Practical inductors range from nanohenries (RF coils, bond wires, PCB vias) through microhenries (switching converters) and millihenries (audio crossovers, line filters) to henries (relay coils, old-style chokes, large reactors).

Inductance depends on geometry and on the **permeability** $\mu = \mu_r \mu_0$ of the medium, with $\mu_0 = 4\pi \times 10^{-7}$ H/m. The flux in a magnetic path of length l and cross-section A driven by Ni ampere-turns is $\phi = Ni/\mathcal{R}$, where $\mathcal{R} = l/(\mu A)$ is the path's **reluctance**. Hence

$$L = \frac{N^2}{\mathcal{R}} = \frac{\mu_r \mu_0 N^2 A}{l}$$

The N^2 dependence is essential: one factor of N comes from the ampere-turns that produce the flux, and the second from the turns that link it. For a long air-cored solenoid the same formula applies with $\mu_r = 1$, approximately (it overestimates L when the coil is not much longer than its diameter). Manufacturers of cores quote the **inductance factor** $A_L = L/N^2$ (nH per turn squared), which contains all the geometry and material information.

> **Status of the principle: Linear inductance $\lambda = Li$**
> *Approximate.* Holds exactly for air-cored coils and well for gapped cores and ferrites below saturation; iron and ferrite cores saturate above a material-dependent flux density (about 0.3–0.5 T for ferrites, 1.5–2.0 T for silicon steel), beyond which L falls sharply, and they also show hysteresis.

Worked example — Two inductors from first principles. (a) An air-cored solenoid of $N = 200$ turns, length 50 mm and diameter 10 mm (Figure 5.1b) has $A = \pi(5 \times 10^{-3})^2 = 7.854 \times 10^{-5}$ m² and

$$L = \frac{4\pi \times 10^{-7} \times 200^2 \times 7.854 \times 10^{-5}}{0.050} = 79.0\ \mu\text{H}$$

(The long-solenoid formula is a few percent high for a coil only five diameters long.) (b) A ferrite toroid with $\mu_r = 2000$, magnetic path length $l_e = 40$ mm and cross-section $A_e = 20$ mm² (Figure 5.1c) has

$$A_L = \frac{\mu_r \mu_0 A_e}{l_e} = \frac{2000 \times 4\pi \times 10^{-7} \times 20 \times 10^{-6}}{0.040} = 1257\ \text{nH/turn}^2$$

so just 20 turns give $L = 1257 \times 400$ nH $= 503$ µH — six times the solenoid's inductance with one tenth of the turns. The price is saturation. The flux density in the core is $B = \mu_r \mu_0 N i / l_e$; with a ferrite limit of about 0.4 T, the current at which this core saturates is

$$I_{sat} = \frac{B_{sat}\, l_e}{\mu_r \mu_0 N} = \frac{0.4 \times 0.040}{2000 \times 4\pi \times 10^{-7} \times 20} = 0.32 \text{ A}$$

Power inductors that must carry amperes therefore use cores with a small **air gap**, which adds reluctance, lowers A_L and raises I_{sat}; most of the energy is then stored in the gap (Section 5.4).

> **Boundary.**
> Magnetic materials, hysteresis loops and core losses belong to *The Materials Core*; magnetic circuits of machines and transformers in depth belong to *Electrical Energy & Machines*. Here L is a circuit parameter.

Inductance is everywhere too

Every current loop has inductance. A straight wire or PCB trace has roughly 0.5–1 nH per millimetre (depending on how far away its return path is), a surface-mount capacitor's mounting adds 0.3–2 nH, and a power cable a few hundred nanohenries per metre. In fast digital and power circuits, this **parasitic inductance** produces voltage spikes $L\,di/dt$ whenever current changes rapidly: 10 nH carrying a current that changes by 10 A in 10 ns generates 10 V. Layout for switching converters is largely the art of minimising the area of loops that carry fast-changing current.

5.3 Constitutive equations

The capacitor

Differentiate $q = Cv$ with respect to time, using $i = dq/dt$ (Chapter 1), with the passive sign convention (current entering the positive terminal):

$$i = C\frac{dv}{dt}$$

Integrating from a reference time t_0:

$$v(t) = v(t_0) + \frac{1}{C}\int_{t_0}^{t} i(\tau)\, d\tau$$

The voltage at any time depends on the *entire history* of the current, summarised by the initial value $v(t_0)$. The capacitor is an element with **memory**.

The inductor

Differentiate $\lambda = Li$ and use Faraday's law $v = d\lambda/dt$:

$$v = L\frac{di}{dt}$$

$$i(t) = i(t_0) + \frac{1}{L}\int_{t_0}^{t} v(\tau)\, d\tau$$

The inductor's current is the running integral of its voltage; the initial current $i(t_0)$ carries its memory.

> **Status of the principle: Capacitor and inductor laws $i = C\,dv/dt$, $v = L\,di/dt$**
> *Derived.* From charge conservation and Faraday's law applied to linear, time-invariant elements under the lumped assumption; if C or L varies with time (a capacitive microphone, a moving relay armature) the correct forms are $i = d(Cv)/dt$ and $v = d(Li)/dt$, which contain extra terms.

Four consequences

1. **DC behaviour.** If the voltage across a capacitor is constant, its current is zero: in DC steady state **a capacitor is an open circuit**. If the current through an inductor is constant, its voltage is zero: in DC steady state **an inductor is a short circuit**. These two rules give the "before" and "after" states of every switching problem in Chapter 6.
2. **Continuity.** A finite current can change a capacitor's voltage only at a finite rate, $dv/dt = i/C$. Unless an infinite (impulsive) current flows, **capacitor voltage cannot jump**. Dually, unless an impulsive voltage is applied, **inductor current cannot jump**. Section 5.6 turns this into the rules for initial conditions.
3. **Waveform transformation.** A capacitor converts a voltage ramp into a constant current; an inductor converts a constant voltage into a current ramp. A triangular voltage applied to a capacitor produces a square-wave current (Figure 5.2), and a square voltage applied to an inductor produces a triangular current — the exact operating waveform of every switching power converter.
4. **Sinusoids stay sinusoids.** If $v = V_m \sin \omega t$ across a capacitor, $i = \omega C V_m \cos \omega t$: the current has the same frequency, amplitude multiplied by ωC, and leads the voltage by 90°. For an inductor, the current lags the voltage by 90°, and the amplitude ratio is ωL. These ratios, $1/(\omega C)$ and ωL, are the **reactances** that Chapter 8 turns into complex impedances. A 10 µF capacitor across 230 V, 50 Hz mains has reactance $1/(2\pi \times 50 \times 10 \times 10^{-6}) = 318\ \Omega$ and draws 0.72 A rms while dissipating (ideally) nothing.

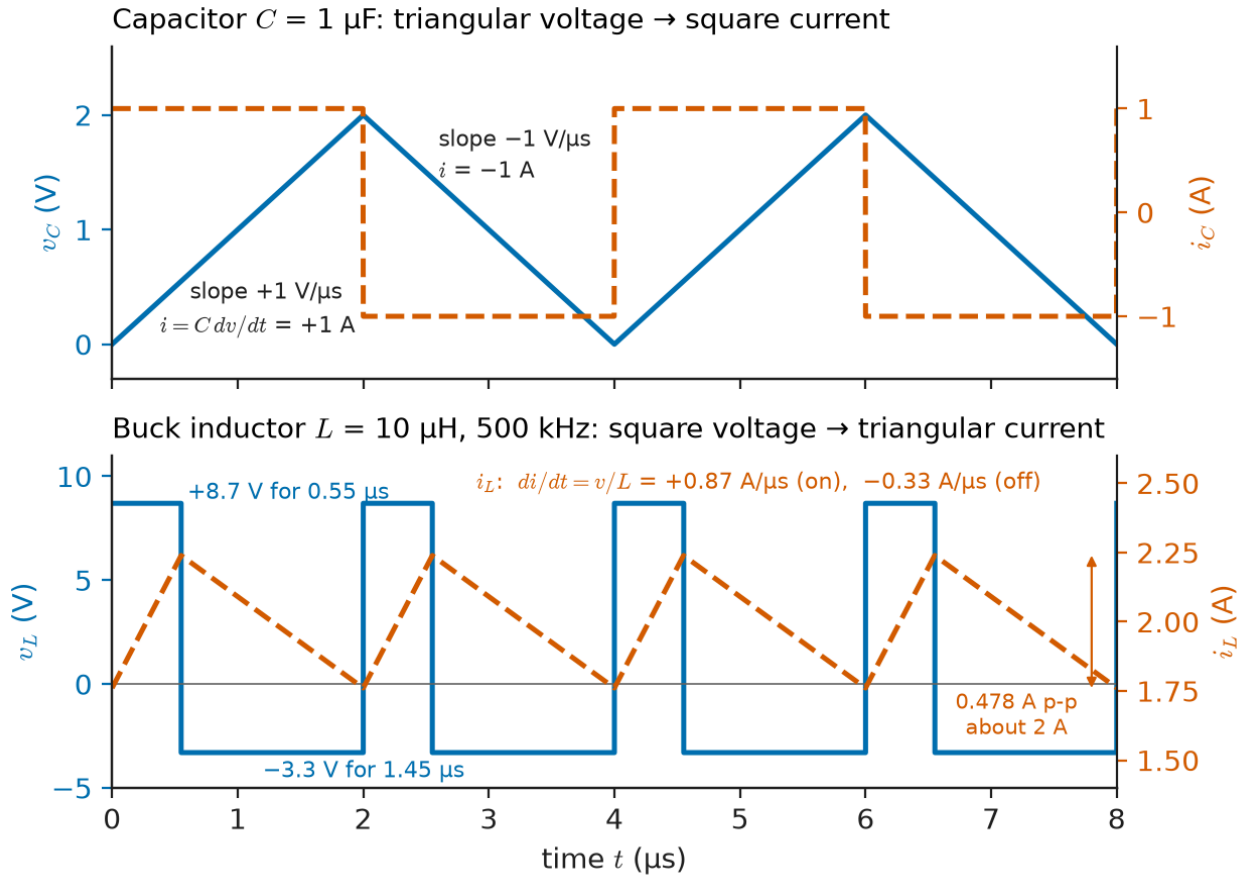


Figure 5.2 — Constitutive equations at work: a triangular voltage across a capacitor gives a square current, and a square voltage across an inductor gives a triangular current.

Worked example — Inductor ripple in a buck converter. A buck converter steps 12 V down to 3.3 V, switching at 500 kHz through a 10 µH inductor (Figure 5.2, bottom). While the switch is on, the inductor sees $12 - 3.3 = 8.7$ V and its current rises at

$$\frac{di}{dt} = \frac{v}{L} = \frac{8.7}{10 \times 10^{-6}} = 8.7 \times 10^{5}\ \text{A/s}$$

The switch is on for a fraction $D = 3.3/12 = 0.275$ of the 2 µs period, i.e. 0.55 µs, so the current rises by $8.7 \times 10^{5} \times 0.55 \times 10^{-6} = 0.478$ A. While the switch is off, the inductor sees -3.3 V for 1.45 µs and falls by $3.3/10^{-5} \times 1.45 \times 10^{-6} = 0.478$ A — the same amount, as it must be in steady state (zero average inductor voltage, or **volt-second balance**). The ripple current, 0.48 A peak-to-peak, flows into the output capacitor; its size sets the output voltage ripple and the inductor's RMS current, both design constraints met in *Electronics & Computing*.

Duality

The capacitor and inductor equations are identical under the exchange

$$v \leftrightarrow i, \qquad C \leftrightarrow L, \qquad q \leftrightarrow \lambda, \qquad \text{series} \leftrightarrow \text{parallel}, \qquad \text{open} \leftrightarrow \text{short}$$

Every statement about one element has a dual statement about the other. This **duality** halves the work of Chapters 6 and 7: an RC circuit and its dual RL circuit have the same equations with the variables

swapped, and a series RLC circuit is the dual of a parallel RLC circuit. Duality extends to whole planar networks (meshes ↔ nodes, KVL ↔ KCL), a correspondence first noticed in Chapter 3.

Table 5.2 — Summary of element relations.

Property	Resistor	Capacitor	Inductor
Defining relation	$v = Ri$	$q = Cv$	$\lambda = Li$
v–i law	$v = Ri$	$i = C\,dv/dt$	$v = L\,di/dt$
Continuous variable	—	voltage	current
DC steady state	resistor	open circuit	short circuit
Energy	dissipated, $\int Ri^2 dt$	stored, $\tfrac{1}{2}Cv^2$	stored, $\tfrac{1}{2}Li^2$
Series combination	$R_1 + R_2$	$C_1C_2/(C_1 + C_2)$	$L_1 + L_2$
Parallel combination	$R_1R_2/(R_1 + R_2)$	$C_1 + C_2$	$L_1L_2/(L_1 + L_2)$
Sinusoidal amplitude ratio V_m/I_m	R	$1/(\omega C)$	ωL

5.4 Energy storage

Energy in a capacitor

The power into a capacitor is $p = vi = Cv\,dv/dt$. Integrating from an uncharged state ($v = 0$) to voltage v:

$$w_C = \int_0^t p\,d\tau = \int_0^v Cv'\,dv' = \tfrac{1}{2}Cv^2 = \frac{q^2}{2C}$$

The energy depends only on the present voltage, not on how it was reached. It is stored in the electric field of the dielectric, with energy density $\tfrac{1}{2}\varepsilon E^2$ (J/m³).

Energy in an inductor

Similarly, $p = vi = Li\,di/dt$, so

$$w_L = \tfrac{1}{2}Li^2 = \frac{\lambda^2}{2L}$$

stored in the magnetic field with energy density $B^2/(2\mu)$. In a gapped core, μ in the gap is μ_0, thousands of times smaller than in the ferrite, so for the same B the gap stores thousands of times more energy per unit volume. That is why the gap, not the ferrite, is where a power inductor keeps its energy.

> **Status of the principle: Stored energy $\frac{1}{2}Cv^2$, $\frac{1}{2}Li^2$**
> *Derived*. From integrating the power vi into a linear lossless element; for nonlinear elements (saturating cores, ferroelectric ceramics) the stored energy must be computed as $\int v\,dq$ or $\int i\,d\lambda$ along the actual characteristic.

Both stored energies are non-negative, and both are **state functions**: an ideal capacitor or inductor returns exactly the energy it received. An ideal capacitor in a circuit can absorb energy for a while and deliver it later, but over a full cycle of periodic operation its net energy intake is zero. This is the circuit statement of the fact that reactive elements exchange but do not consume energy — the origin of reactive power (Chapter 9).

Energy and continuity

Energy gives a physical reason for the continuity rules of Section 5.3. If a capacitor's voltage jumped instantaneously from v_1 to v_2, its stored energy would change by a finite amount in zero time, which requires infinite power. Real sources cannot supply infinite power; real circuits therefore round off any attempted jump over a time set by their resistance and stray inductance. When a circuit is *forced* to approximate a jump — a charged capacitor connected abruptly to another, or an inductor's current interrupted by an opening switch — the energy has to go somewhere, and it goes into heat, arcs, radiation or breakdown.

Table 5.3 — Representative stored energies.

Application	Element	Values	Stored energy
MCU decoupling capacitor	100 nF at 3.3 V	—	0.54 µJ
Camera flash capacitor	120 µF at 330 V	—	6.5 J
Drive DC-link bank	470 µF at 560 V	—	74 J
External defibrillator	32 µF at 5.0 kV	monophasic design	400 J
Supercapacitor cell	3000 F at 2.7 V	—	10.9 kJ (3.0 Wh)
Relay coil	0.5 H at 50 mA	—	0.63 mJ
Automotive ignition coil (primary)	4 mH at 8 A	—	0.13 J
Buck-converter inductor	10 µH at 2.2 A peak	—	24 µJ
Superconducting magnetic energy storage	multi-H at kA	research and utility units	MJ range

Worked example — Defibrillator energy and the inductive kick of a relay. (a) A monophasic external defibrillator stores energy in a 32 µF capacitor. Charged to 5.0 kV it holds $\frac{1}{2} \times 32 \times 10^{-6} \times 5000^2 = 400$ J; to store a 360 J setting it must be charged to $\sqrt{2 \times 360/32 \times 10^{-6}} = 4.74$ kV. Delivered to a thorax of roughly 50 Ω, this energy drives currents of tens of amperes for a few milliseconds; the

waveform-shaping inductor and the patient resistance form exactly the second-order circuits of Chapter 7.

(b) A relay coil of 0.5 H carries 50 mA. Its stored energy is only $\tfrac{1}{2} \times 0.5 \times 0.05^2 = 0.625$ mJ. But if a transistor switching the coil turns off in 1 µs, the coil tries to maintain its current by producing $v = L\,di/dt \approx 0.5 \times 0.05/10^{-6} = 25$ kV. Long before that, the transistor breaks down or an arc forms, and the 0.625 mJ is dumped into it — on every operation. The universal remedy is a **freewheeling diode** across the coil, which gives the current a path and limits the voltage to about 0.7 V above the supply; Chapter 6 analyses the decay that follows and why it slows the relay's release.

Safety: stored energy outlives the source

The drive cabinet of the opening holds $\tfrac{1}{2} \times 470 \times 10^{-6} \times 560^2 = 74$ J at 560 V — enough to be dangerous. Standards for electrical equipment therefore require stored voltages to fall below a safe level (for machinery, typically 60 V within 5 s) after disconnection, achieved with **bleeder resistors** whose design is a first-order discharge problem (Chapter 6). The rule for maintenance is to measure, not to assume, that capacitors are discharged.

5.5 Series and parallel combinations

Capacitors

Capacitors in **parallel** share the same voltage; their charges add, $q = (C_1 + C_2 + \dots)v$, so

$$C_{par} = C_1 + C_2 + \cdots + C_n$$

Capacitors in **series** carry the same current, so (starting uncharged) they acquire the same charge; their voltages add, $v = q(1/C_1 + 1/C_2 + \dots)$, so

$$\frac{1}{C_{ser}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$$

Capacitors combine like conductances, the opposite of resistors. Two equal capacitors in series give half the capacitance at twice the voltage rating.

Inductors

Uncoupled inductors in **series** carry the same current and their voltages add: $L_{ser} = L_1 + L_2 + \dots$. In **parallel** they share the same voltage and their currents add: $1/L_{par} = 1/L_1 + 1/L_2 + \dots$. Inductors combine like resistors. These rules hold only if the inductors' magnetic fields do not link each other; when they do, mutual inductance adds or subtracts (Section 5.8).

> **Status of the principle: Series and parallel combination of C and L**
> *Derived*. From KCL, KVL and the element laws; exact for ideal elements, with the proviso that the initial voltages (for capacitors) or currents (for inductors) of the individual elements must also be combined, and that inductors must be magnetically uncoupled.

Initial conditions in combinations

A series string of capacitors with different initial voltages behaves externally as C_{ser} with an initial voltage equal to the sum of the individual initial voltages — but the individual voltages are no longer determined by the external behaviour alone. Parallel inductors with initial currents behave as L_{par} with the sum of the currents, while a current may circulate around the loop they form without appearing outside. The equivalent captures the terminal behaviour; the internal distribution must be tracked separately when it matters (for voltage ratings, for example).

Voltage sharing in series capacitor banks

Series capacitors share voltage according to their capacitances only for fast transients. Over seconds and longer, the DC voltage divides according to their **leakage resistances**, which can differ by factors of several between nominally identical electrolytic capacitors. Series banks on DC buses therefore include **balancing resistors**, chosen so that the current through them is large compared with the leakage current mismatch.

Worked example — Balancing two electrolytics on a 400 V bus. Two 100 µF, 250 V aluminium electrolytic capacitors are placed in series on a 400 V DC link, giving 50 µF rated for 500 V. Their leakage currents at operating voltage are 20 µA and 60 µA. Without balancing, the capacitor with lower leakage would charge toward a higher share of the voltage until the leakages equalise, and could exceed its 250 V rating. With a 47 kΩ resistor across each capacitor, KCL at the midpoint (top-capacitor leakage $I_1 = 20$ µA, bottom $I_2 = 60$ µA) gives

$$\frac{400 - v_m}{R} + I_1 = \frac{v_m}{R} + I_2 \;\Rightarrow\; v_m = \frac{400}{2} - \frac{R(I_2 - I_1)}{2} = 200 - \frac{47 \times 10^{3} \times 40 \times 10^{-6}}{2} = 199.06\ \text{V}$$

so the bottom capacitor sees 199.1 V and the top one 200.9 V — well balanced. The price is $400^2/94\ \text{k}\Omega = 1.7$ W of continuous dissipation; the resistors double as bleeders.

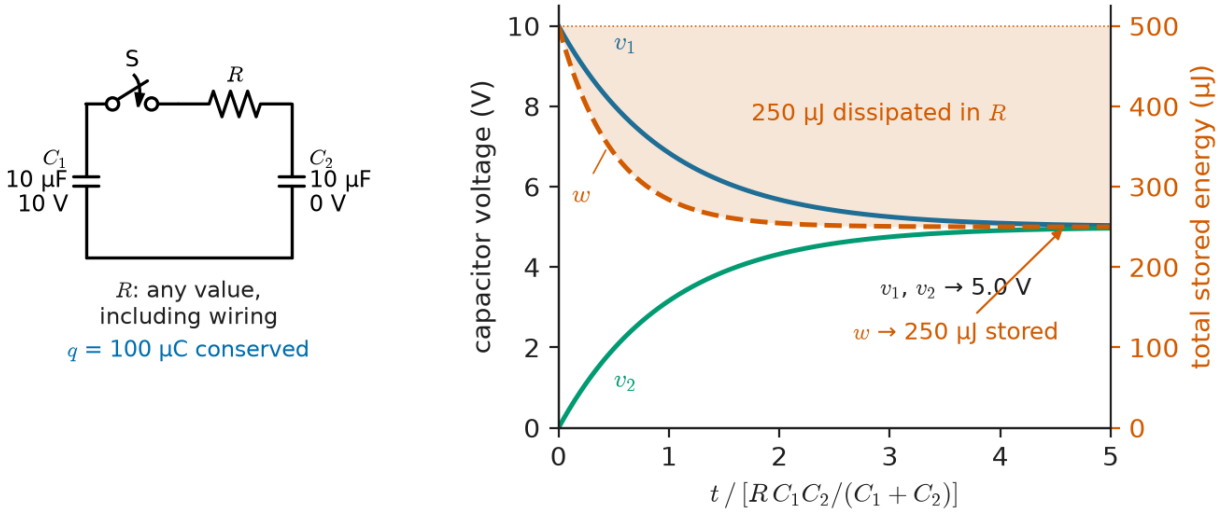


Figure 5.3 — Charge sharing between two capacitors: closing the switch conserves charge but loses half the energy, whatever the resistance in the loop.

Charge sharing and the "lost" energy

Worked example — Connecting a charged capacitor to an uncharged one. A 10 μF capacitor charged to 10 V is connected through a switch to an identical uncharged capacitor (Figure 5.3). Charge is conserved: the initial charge $q = 10 \times 10^{-6} \times 10 = 100$ μC spreads over 20 μF, so both end at

$$V_f = \frac{C_1 V_1}{C_1 + C_2} = 5.0 \text{ V}$$

The stored energy before is $\frac{1}{2} \times 10 \times 10^{-6} \times 10^2 = 500$ μJ; after it is $\frac{1}{2} \times 20 \times 10^{-6} \times 5^2 = 250$ μJ. Half the energy has disappeared. It was dissipated in the resistance of the connecting path — and remarkably, the amount, 250 μJ, does not depend on that resistance. A large resistance gives a small current for a long time, a small one a large current for a short time; $\int i^2 R\,dt$ comes out the same (Chapter 6 confirms this by integrating the exponential current). With an ideal zero-resistance wire the circuit has no consistent solution; in practice the energy goes into wiring resistance, the capacitors' ESR, a spark, and radiation from the loop inductance.

The same physics appears in **hot-plugging**: inserting a board with 1000 μF of uncharged bulk capacitance onto a live 12 V backplane through 20 mΩ of connector and trace resistance would draw a peak of $12/0.020 = 600$ A and dissipate $\frac{1}{2} \times 1000 \times 10^{-6} \times 12^2 = 72$ mJ in the contacts, enough to pit them. Hot-swap controllers limit the inrush by charging the capacitance slowly. Switched-capacitor circuits — sample-and-hold stages, charge pumps, switched-capacitor filters — pay a related energy price on every clock cycle, which sets their power consumption.

5.6 Initial conditions

The continuity rules

From Section 5.3, in the absence of impulsive currents or voltages:

$$v_C(0^+) = v_C(0^-), \qquad i_L(0^+) = i_L(0^-)$$

where 0^- is the instant just before a switching event at $t = 0$ and 0^+ the instant just after. Capacitor voltages and inductor currents are the **state variables** of a circuit: together with the sources, they determine everything else. All other quantities — capacitor currents, inductor voltages, resistor voltages and currents — may jump.

> **Status of the principle: Continuity of capacitor voltage and inductor current**
> *Derived*. From the integral forms of the element laws, provided all currents into capacitors and all voltages across inductors remain finite; violated in idealised circuits that connect capacitors directly to voltage sources or other capacitors, or that interrupt inductor currents abruptly.

The procedure at a switching instant

1. **Before switching** ($t = 0^-$): the circuit is usually in DC steady state. Replace capacitors by open circuits and inductors by short circuits; solve the resistive circuit for every $v_C(0^-)$ and $i_L(0^-)$.

2. **At $t = 0^+$**: in the post-switching circuit, replace each capacitor by a voltage source of value $v_C(0^+) = v_C(0^-)$ and each inductor by a current source of value $i_L(0^+) = i_L(0^-)$. This is a purely resistive circuit — solve it for every other quantity at 0^+.
3. **Initial derivatives**: from the 0^+ solution, $dv_C/dt(0^+) = i_C(0^+)/C$ and $di_L/dt(0^+) = v_L(0^+)/L$. These derivatives are the second initial condition needed by second-order circuits (Chapter 7).
4. **Final values** ($t \to \infty$): if the sources are DC, the circuit settles to a new DC steady state; again treat capacitors as open and inductors as short.

The substitution in step 2 is an application of the substitution theorem of Chapter 4: at 0^+ the capacitor voltage is known, so the capacitor may be replaced by a source of that voltage.

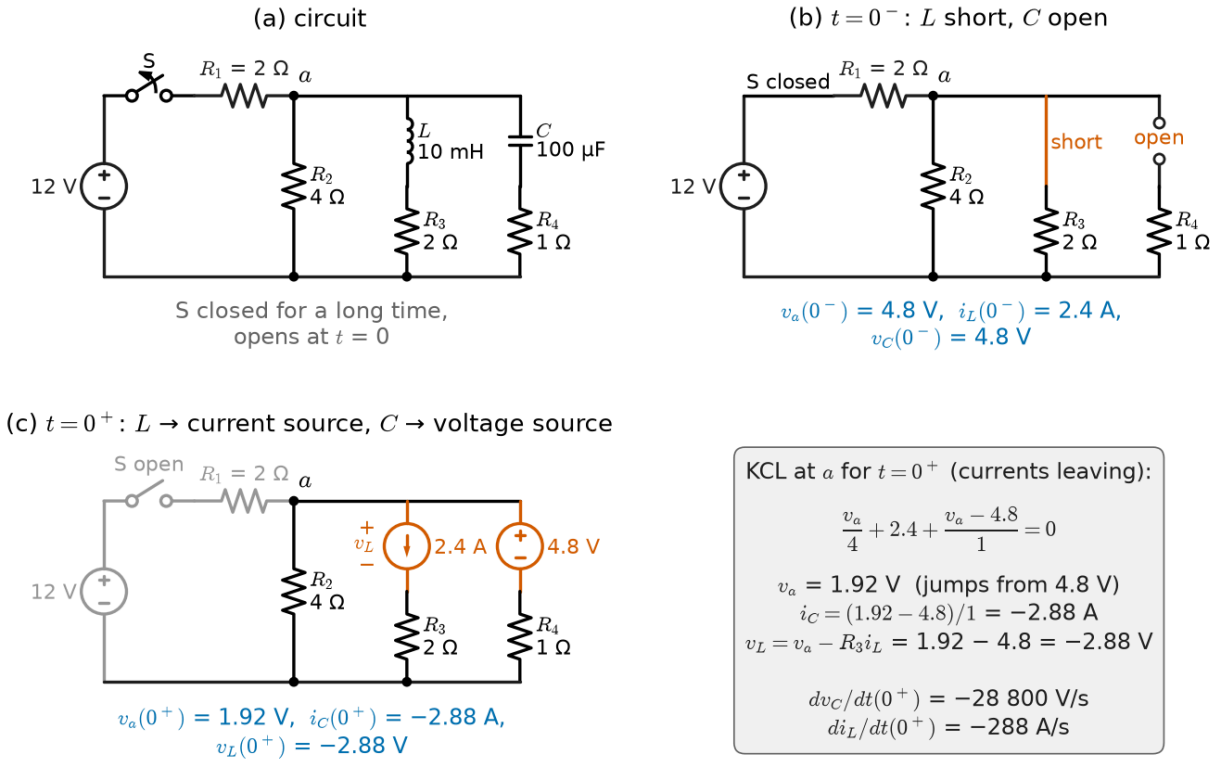


Figure 5.4 — Finding initial conditions: the circuit before switching (DC steady state), and the equivalent resistive circuit at t = 0+ with the capacitor and inductor replaced by sources.

Worked example — Initial values and initial slopes. In Figure 5.4a, a 12 V source feeds node a through $R_1 = 2\ \Omega$ and a switch. From a to ground are three branches: $R_2 = 4\ \Omega$; $L = 10$ mH in series with $R_3 = 2\ \Omega$; and $C = 100$ µF in series with $R_4 = 1\ \Omega$. The switch has been closed for a long time and opens at $t = 0$.

At 0^-. The capacitor is open, the inductor shorted. Node a sees $R_2 \parallel R_3 = 1.333\ \Omega$, so $v_a = 12 \times 1.333/(2 + 1.333) = 4.80$ V. Then $i_L(0^-) = 4.80/2 = 2.40$ A, and since no current flows in R_4, $v_C(0^-) = 4.80$ V.

At 0^+. With the switch open, KCL at a (currents leaving):

$$\frac{v_a}{R_2} + i_L + \frac{v_a - v_C}{R_4} = 0 \ \Rightarrow\ v_a\left(\frac{1}{4} + 1\right) = -2.40 + 4.80 \ \Rightarrow\ v_a(0^+) = 1.92\ \text{V}$$

So v_a jumps from 4.80 V to 1.92 V — allowed, because it is not a state variable. The capacitor current is $i_C = (1.92 - 4.80)/1 = -2.88$ A (the capacitor discharges), R_2 carries 0.48 A, and the inductor's 2.40 A continues; check: $0.48 + 2.40 - 2.88 = 0$. The inductor voltage is $v_L = v_a - R_3 i_L = 1.92 - 4.80 = -2.88$ V.

Initial slopes.

$$\frac{dv_C}{dt}(0^+) = \frac{-2.88}{100 \times 10^{-6}} = -28\,800 \text{ V/s}, \qquad \frac{di_L}{dt}(0^+) = \frac{-2.88}{10 \times 10^{-3}} = -288 \text{ A/s}$$

Final values. With no source, everything decays to zero: $v_C(\infty) = 0$, $i_L(\infty) = 0$. Chapter 7 shows that this circuit, with one capacitor and one inductor, is second order, and that $v_C(0^+)$, $i_L(0^+)$ and these slopes are exactly the data its solution needs.

When continuity is violated

Two circuit configurations force jumps and therefore impulses:

- A **loop consisting only of capacitors and ideal voltage sources** (a "capacitor loop"). KVL around the loop may demand that capacitor voltages change instantly — as in the charge-sharing example with zero resistance — and an impulse of current flows. The charge, not the voltage, is conserved across the event.
- A **cut-set consisting only of inductors and ideal current sources** (an "inductor cut-set"), such as two inductors carrying different currents suddenly placed in series. KCL demands the currents become equal instantly; an impulse of voltage appears, and flux linkage, not current, is conserved.

Real circuits contain resistance and stray inductance or capacitance, so the "impulse" becomes a short, large pulse, which can be destructive (arcing contacts, inrush currents). The ideal impulse model, made precise with the unit impulse $\delta(t)$ of Chapter 6 and Chapter 15, predicts the total charge or flux transferred correctly even though it hides the detail.

5.7 Mutual inductance

Flux shared between two coils

Place two coils close together. Current i_1 in coil 1 produces flux, part of which links coil 2. The flux linkage of coil 2 due to i_1 is proportional to i_1; the constant is the **mutual inductance** M_{21}. With both coils carrying current, and linear media,

$$\lambda_1 = L_1 i_1 + M_{12} i_2, \qquad \lambda_2 = M_{21} i_1 + L_2 i_2$$

Energy considerations (below) show that $M_{12} = M_{21} = M$ for any linear magnetic medium — a form of reciprocity (Chapter 4). Differentiating with Faraday's law:

$$v_1 = L_1 \frac{di_1}{dt} \pm M \frac{di_2}{dt}, \qquad v_2 = \pm M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$

The sign of the mutual term depends on how the coils are wound relative to each other — whether their fluxes aid or oppose.

The dot convention

Circuit diagrams cannot show winding directions, so a dot is placed at one end of each coupled coil. The rule:

> If both currents enter (or both leave) the dotted terminals, the mutual voltage terms have the same sign as the self-inductance terms (+M). If one current enters a dotted terminal and the other leaves one, the mutual terms are negative (−M).

Equivalently: a current entering a dotted terminal of one coil induces, in the other coil, a voltage that is positive at that coil's dotted terminal. Dots are found experimentally by applying a step of current to one coil and observing the polarity of the voltage spike on the other, or by series aiding and opposing measurements (Section 5.8).

Coupling coefficient

The fraction of flux shared is expressed by the **coupling coefficient**

$$k = \frac{M}{\sqrt{L_1 L_2}}, \qquad 0 \le k \le 1$$

The upper bound follows from energy. The total energy stored in two coupled coils is

$$w = \tfrac{1}{2} L_1 i_1^2 + \tfrac{1}{2} L_2 i_2^2 \pm M i_1 i_2$$

This must be non-negative for every pair of currents, since a passive magnetic field cannot store negative energy. The worst case is $i_2/i_1 = \mp M/L_2$, which gives $w = \tfrac{1}{2} i_1^2 (L_1 - M^2/L_2)$; non-negativity requires $M^2 \le L_1 L_2$, i.e. $k \le 1$.

> **Status of the principle: Mutual inductance and $k \le 1$**
> *Derived*. The coupled-coil equations follow from Faraday's law and linear flux superposition; $M_{12} = M_{21}$ and $k \le 1$ follow from energy conservation and the non-negativity of magnetic energy. Valid for linear, non-saturating media.

Table 5.4 — Typical coupling coefficients.

Arrangement	Typical k
Two coils on a closed ferrite or laminated core (power transformer)	0.98–0.999
Bifilar-wound pulse transformer	above 0.99
Gapped-core flyback transformer	0.95–0.99
Wireless (inductive) charging pads, aligned, few mm gap	0.3–0.7
Near-field RFID / NFC reader and card	0.01–0.3
Adjacent PCB traces (crosstalk)	0.05–0.5
Air-cored coils, coaxial, separated by one diameter	below 0.1

Where mutual inductance matters

Mutual inductance is the operating principle of transformers, current transformers and Rogowski coils, of resonant wireless power and of near-field communication. It is also the mechanism of **inductive crosstalk**: a fast-changing current in one PCB trace induces a voltage $M\,di/dt$ in its neighbour. And it is a hazard in power systems, where a transmission line running parallel to a telephone line or a pipeline induces voltages in it during earth faults.

Worked example — A Rogowski coil measuring a 1 kA current. A Rogowski coil is an air-cored toroidal winding placed around a conductor; it has no magnetic core to saturate, so its mutual inductance M with the conductor is constant and its output is $v = M\,di/dt$. A coil with $M = 200$ nH encircles a busbar carrying a 50 Hz current of 1000 A rms, i.e. $i = 1414\sin(2\pi\cdot 50\,t)$ A. The output is

$$v = M\frac{di}{dt} = 200\times10^{-9}\times 2\pi\times 50\times 1414\cos(2\pi\cdot 50\,t) = 88.9\cos(2\pi\cdot 50\,t)\ \text{mV}$$

— 88.9 mV peak, 62.8 mV rms. The output is proportional to the *derivative* of the current, so an integrator (analog or digital) follows the coil to recover i. Because there is no iron, the same coil measures 100 kA fault currents or kiloampere pulses in pulsed-power and welding equipment without saturating — the advantage over a conventional current transformer.

5.8 Coupled coils

Series connection: aiding and opposing

Two coupled coils in series carry the same current. If it enters both dotted terminals (series **aiding**), the mutual flux adds; if it enters one dotted terminal and leaves the other (series **opposing**), it subtracts:

$$L_{aid} = L_1 + L_2 + 2M, \qquad L_{opp} = L_1 + L_2 - 2M$$

Measuring both gives $M = (L_{aid} - L_{opp})/4$, the standard bench method for mutual inductance and dot polarity: the connection that gives the larger inductance is the aiding one.

Worked example — Two coupled coils. Coils with $L_1 = 2$ mH and $L_2 = 8$ mH are wound on a common former with $k = 0.9$. Then $M = 0.9\sqrt{2 \times 8}\,\text{mH} = 3.6$ mH, and

$$L_{aid} = 2 + 8 + 7.2 = 17.2\ \text{mH}, \qquad L_{opp} = 2 + 8 - 7.2 = 2.8\ \text{mH}$$

Recovering M from the measurements: $(17.2 - 2.8)/4 = 3.6$ mH. The ratio $\sqrt{L_2/L_1} = 2$ is the effective turns ratio (inductance scales as N^2), so these coils approximate a 1:2 transformer with 10 % of the flux not shared.

Parallel connection

Two coupled coils in parallel (both dotted terminals joined) have equivalent inductance

$$L_{par} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M}$$

and with the opposite polarity the sign of $2M$ in the denominator reverses. For $k = 1$ the numerator vanishes, so if $L_1 \neq L_2$ (unequal turns) the parallel pair is a short circuit: the two coils try to impose different voltages on each other, and large circulating currents flow. Paralleled transformer windings must therefore have equal turns and correct polarity.

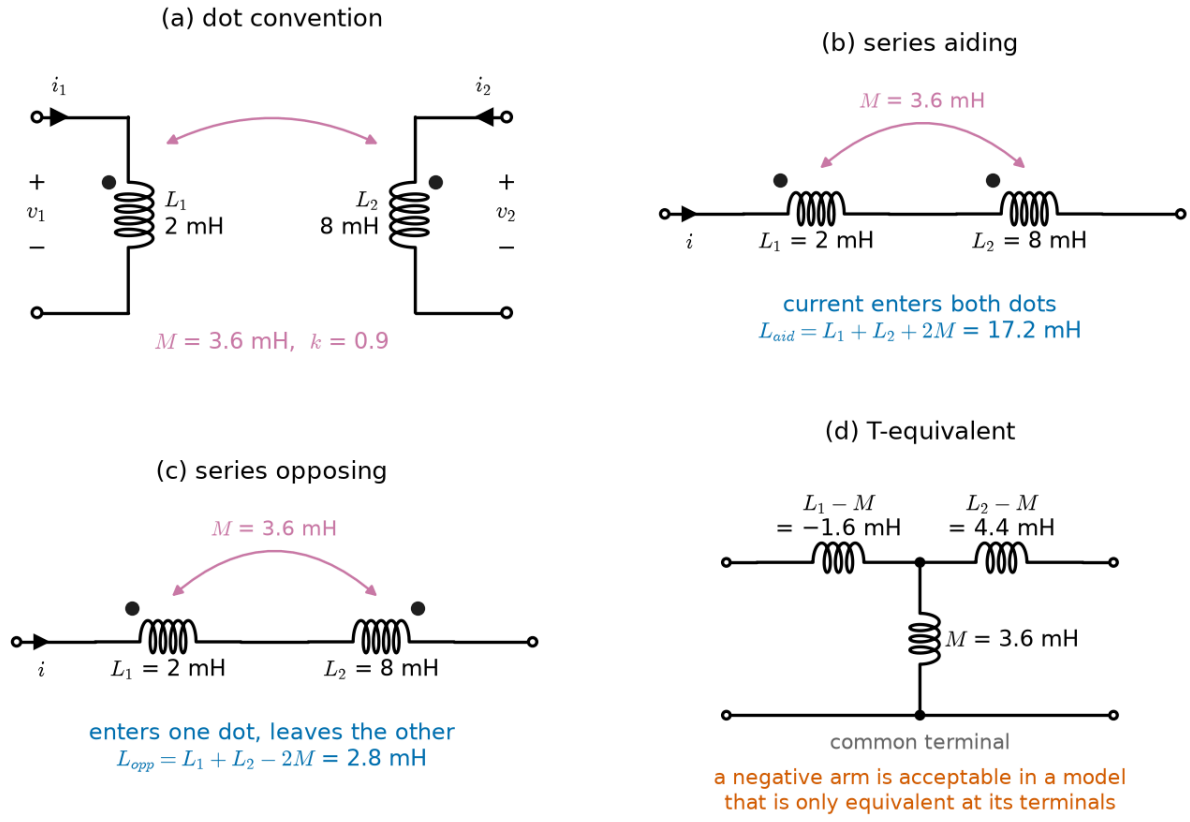


Figure 5.5 — Coupled coils: dot convention, series aiding and opposing connections, and the T-equivalent circuit.

The T-equivalent

When two coupled coils share a common terminal (or can be given one without changing the circuit — the bottom terminals of a transformer connected to a common ground), they can be replaced by three

uncoupled inductors in a T (Figure 5.5d): $L_1 - M$ and $L_2 - M$ in the series arms and M in the shunt arm. This uses the equivalent-network idea of Chapter 4 and turns a coupled-coil problem into an ordinary circuit problem. One series arm can be negative — here $L_1 - M = -1.6$ mH — which is unphysical for a single component but perfectly valid in a model that is only claimed to be equivalent at its terminals.

Analysing circuits with coupled coils

In mesh analysis, coupled coils add mutual-voltage terms to each mesh equation: the voltage across coil 1 in mesh 1 contains $L_1\, di_1/dt$ plus or minus $M\, di_2/dt$, with the sign from the dot rule. Nodal analysis is awkward because the coupled-coil equations give voltages in terms of current derivatives; either use the T-equivalent or invert the inductance matrix. In the sinusoidal steady state (Chapter 8) the mutual terms become $\pm j\omega M I$, and a load Z_L on coil 2 appears in coil 1 as a **reflected impedance** $(\omega M)^2/(j\omega L_2 + Z_L)$, the basis of wireless-power and transformer analysis.

The ideal-transformer limit

Let $k \to 1$ (no leakage flux) and let $L_1, L_2 \to \infty$ with the ratio $L_2/L_1 = n^2$ fixed (infinite core permeability, so no magnetising current is needed). The coupled coils then become an **ideal transformer**:

$$\frac{v_2}{v_1} = n = \frac{N_2}{N_1}, \qquad \frac{i_2}{i_1} = -\frac{1}{n}$$

with the currents defined entering both dotted terminals. The ideal transformer stores no energy and dissipates none; it simply scales voltage up and current down (or vice versa) and transforms a load resistance R_L on the secondary into R_L/n^2 seen at the primary. A real transformer is modelled as an ideal transformer plus a magnetising inductance (finite L_1), leakage inductances (from $k < 1$), winding resistances and a core-loss resistance.

> **Status of the principle: Ideal transformer**
> *Approximate.* The limit of coupled coils with $k \to 1$ and infinite self-inductances; accurate for well-designed iron- or ferrite-cored transformers within their rated frequency band and flux, and it fails at DC (no transformer action, since $v = d\lambda/dt$ requires changing flux), at low frequency (magnetising current), at high frequency (leakage and winding capacitance) and in saturation.

> **Boundary.**
> Transformer construction, equivalent circuits with core loss, efficiency, regulation and three-phase transformer connections belong to *Electrical Energy & Machines*. This book uses the coupled-coil model and the ideal transformer as circuit elements, and returns to them in two-port form in Chapter 12.

5.9 Ideal versus practical components

Why a capacitor is not a capacitor

Every real capacitor has resistance in its electrodes, leads and (for electrolytics) electrolyte; inductance in its current path; and a dielectric that leaks a little. Its simplest useful model (Figure 5.6a) is

- **ESR** (equivalent series resistance) in series: from milliohms for MLCCs to ohms for small electrolytics; it dissipates I_{rms}^2 ESR when ripple current flows, and it limits how well the capacitor can filter;
- **ESL** (equivalent series inductance) in series: 0.2–1 nH for MLCCs, 5–20 nH for leaded electrolytics, set mainly by package geometry and mounting;
- **leakage resistance** R_p in parallel: gigohms to teraohms for film and C0G parts, much less for electrolytics (whose leakage is specified as a current, e.g. $0.01CV$ or 3 µA, whichever is greater).

Above its **self-resonant frequency** $f_{SR} = 1/(2\pi\sqrt{\mathrm{ESL}\cdot C})$, a capacitor behaves as an inductor. Its impedance magnitude falls as $1/(\omega C)$ below f_{SR}, bottoms out at ESR near f_{SR}, and rises as $\omega\,\mathrm{ESL}$ above it (Figure 5.6c).

Other non-idealities matter in specific applications: **dielectric absorption** (a capacitor discharged briefly recovers part of its voltage, up to a few percent for some dielectrics — a hazard in high-voltage capacitors and an error source in sample-and-hold circuits), **voltage and temperature coefficients** (Table 5.1), **piezoelectric noise** (class-2 ceramics "sing" audibly under ripple voltage and generate microphonic noise in audio circuits), and **ageing** (electrolytic capacitors lose electrolyte; their life roughly halves for every 10 °C rise in temperature).

Why an inductor is not an inductor

A real inductor (Figure 5.6b) has

- **DCR**, the DC resistance of its winding, in series: milliohms for power inductors, ohms for small chokes; at high frequency, skin and proximity effects raise the effective resistance well above DCR;
- **core loss**, modelled as a resistance in parallel or folded into a frequency-dependent series resistance;
- **interwinding capacitance** C_p in parallel, typically a few pF, which resonates with L at the inductor's self-resonant frequency, above which it behaves as a capacitor;
- **saturation** (Section 5.2): the inductance falls as the current approaches I_{sat}; datasheets give the current at which L has dropped by 10–30 %;
- **stray field**: unshielded inductors radiate magnetic field and couple to nearby circuits.

The **quality factor** $Q = \omega L/R$ measures how close to ideal an inductor is at a given frequency. A 10 µH inductor with 50 mΩ effective series resistance has $Q = 2\pi \times 10^5 \times 10 \times 10^{-6}/0.05 = 126$ at 100 kHz; a good capacitor has a much larger Q, which is why resonant-circuit losses are usually dominated by the inductor (Chapter 13).

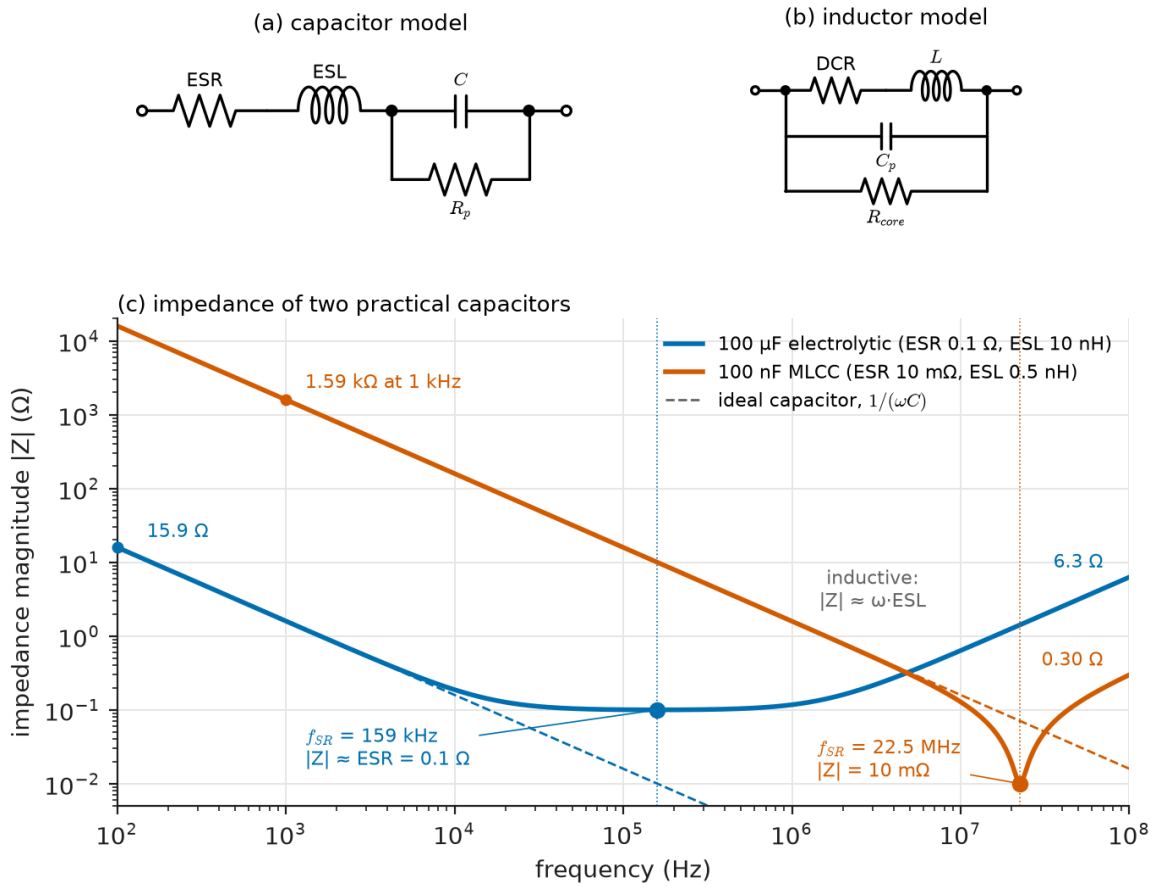


Figure 5.6 — Practical capacitor and inductor models, and the impedance of real capacitors versus frequency showing self-resonance.

Worked example — Self-resonance and ripple heating of decoupling capacitors. A 100 µF aluminium electrolytic (ESR 0.1 Ω, ESL 10 nH) and a 100 nF MLCC (ESR 10 mΩ, ESL 0.5 nH) sit on the same supply rail.

Self-resonant frequencies.

$$f_{SR,el} = \frac{1}{2\pi\sqrt{10\times10^{-9}\times100\times10^{-6}}} = 159\ \text{kHz}, \qquad f_{SR,MLCC} = \frac{1}{2\pi\sqrt{0.5\times10^{-9}\times100\times10^{-9}}} = 22.$$

Impedance at selected frequencies (Table 5.5, computed from $|\text{ESR} + j\omega\,\text{ESL} + 1/(j\omega C)|$):

Table 5.5 — Impedance magnitude of two practical capacitors.

Frequency	100 µF electrolytic	100 nF MLCC
1 kHz	1.59 Ω	1.59 kΩ
100 kHz	0.100 Ω	15.9 Ω
1 MHz	0.117 Ω	1.59 Ω
10 MHz	0.636 Ω	0.128 Ω
100 MHz	6.28 Ω	0.298 Ω

The electrolytic is the better capacitor below about 5 MHz, the small MLCC above. At 100 MHz the "100 µF" part behaves as a 10 nH inductor with impedance 6.3 Ω; no amount of extra electrolytic capacitance will filter noise at that frequency. This is why decoupling networks combine bulk electrolytics, mid-value ceramics and small ceramics placed as close as possible to IC pins, and why ESL — mostly mounting inductance — is the decisive parameter at high frequency.

Ripple heating. If the electrolytic carries 2 A rms of ripple current at 100 kHz (where its impedance is essentially its ESR), it dissipates $2^2 \times 0.1 = 0.4$ W. Its datasheet ripple-current rating exists to keep the resulting core temperature rise, and so the electrolyte loss rate, within limits.

Table 5.6 — Typical parasitics and tolerances of practical components.

Component	Tolerance	ESR or DCR	ESL or C_p	Self-resonance (typical)
100 nF X7R MLCC, 0603	±10 % (plus DC-bias loss)	10–30 mΩ	0.5–0.8 nH	15–25 MHz
1 µF X7R MLCC, 0805	±10 %	5–10 mΩ	0.5–1 nH	5–8 MHz
10 pF C0G MLCC, 0402	±0.25–0.5 pF	low (high Q)	0.3–0.5 nH	1.5–3 GHz
100 µF / 25 V aluminium electrolytic	±20 %	0.05–0.5 Ω	5–20 nH	100–500 kHz
1 µF polypropylene film	±5 %	5–20 mΩ	5–15 nH	1–2 MHz
10 µH shielded power inductor	±20 %	20–60 mΩ DCR	2–10 pF	15–40 MHz
100 nH wire-wound RF inductor	±2–5 %	0.1–0.5 Ω	0.1–0.3 pF	1–1.5 GHz
0.2–1 H relay coil	±10 %	50–2000 Ω	tens of pF	kHz range

Choosing the model

How much of the practical model to keep depends on the frequency range and the question. At mains frequency a 100 µF electrolytic is a capacitor with a little ESR; at 100 kHz it is mostly ESR; at 10 MHz it is an inductor. A relay coil is an inductor with large series resistance — in fact, at DC it is just a resistor that sets the holding current, and its inductance appears only during switching. A component model is good when every element that affects the answer at the frequencies and amplitudes of interest is present, and no more; adding all parasitics everywhere hides the physics without improving the answer.

> **Status of the principle: Lumped RLC models of practical components**
> *Approximate.* Valid when the component is small compared with the wavelength at the frequencies of interest and when the element values (ESR, ESL, core loss) are taken at those frequencies; ESR and core loss vary strongly with frequency and temperature, and at microwave frequencies the components themselves become distributed structures (Chapters 14 and 32).

What remains open

Open question	Where it is answered
How do voltages and currents evolve after switching in circuits with one capacitor or one inductor?	Chapter 6
What happens when a capacitor and an inductor exchange energy — ringing, damping, resonance?	Chapter 7
How do the reactances $1/(\omega C)$ and ωL become complex impedances in AC analysis?	Chapter 8
Why do reactive elements give rise to reactive power and power-factor correction?	Chapter 9
How are initial conditions written as sources in the s-domain?	Chapter 11
How are transformers and coupled coils described as two-ports?	Chapter 12
How do resonant circuits and the Q of real inductors set filter bandwidth?	Chapter 13
What happens when the "lumped" capacitance and inductance are distributed along a line?	Chapter 14
How are dielectrics, magnetic cores and their losses engineered?	*The Materials Core*
How are power transformers designed and rated?	*Electrical Energy & Machines*

Next: Chapter 6 opens Part II by solving the simplest circuits with memory — one resistor network and one capacitor or one inductor — and shows that every such circuit relaxes exponentially with a single time constant.

Part II — Transient and AC Analysis

Part I treated circuits whose voltages and currents are fixed, instant by instant, by the sources acting at that instant. Capacitors and inductors break that rule: they remember. A circuit containing them responds to a switching event with a transient that may take microseconds or minutes to die away, and responds to a sinusoidal source with currents that lead or lag the voltage. This part develops the two complementary ways of handling such circuits. The first is the time domain: write the differential equation, find its natural and forced responses, and match initial conditions — the method of switching transients, relay drive, power-on reset, ringing and damping. The second is the sinusoidal steady state: replace the differential equation by complex algebra with phasors and impedances — the method of mains power, audio and radio-frequency circuits, filters and three-phase systems. Between them they cover nearly every analysis an engineer performs on a linear circuit before reaching for transforms.

By the end of this part you should understand:

- why one storage element gives a single exponential with one time constant, and how the initial value, final value and time constant determine any first-order step response;
- how two storage elements exchange energy, and how the damping ratio and natural frequency decide between overdamped, critically damped and underdamped (ringing) responses;
- how phasors and complex impedance turn the sinusoidal steady state into the resistive circuit methods of Part I, including nodal and mesh analysis and the network theorems;
- what RMS values, active, reactive, apparent and complex power mean, why power factor matters on an electricity bill, and how it is corrected;
- how balanced and unbalanced three-phase systems are analysed, why the neutral carries current, and how three-phase power is measured.

Chapter	Central question	What it delivers
Chapter 6 — First-Order Circuits	How does a circuit with one capacitor or inductor move from one state to another after switching?	Natural and forced response, time constants, the three-number method, inductive switching and suppression, pulse and square-wave response, step and impulse responses
Chapter 7 — Second-Order Circuits	What happens when energy can swing between a capacitor and an inductor?	Characteristic equations, overdamped, critically damped and underdamped responses, natural frequency, damping ratio, resonance and transient energy exchange
Chapter 8 — Sinusoidal Steady-State Analysis	How can circuits driven by sinusoids be solved without differential equations?	RMS values, phasors, impedance and admittance, AC nodal and mesh analysis, series and parallel AC circuits, resonance, frequency response and power calculations
Chapter 9 — AC Power	Where does power go in an AC circuit, and why is some of it "reactive"?	Instantaneous, active, reactive, apparent and complex power, power factor and its correction, energy consumption, non-sinusoidal effects and practical metering
Chapter 10 — Three-Phase Circuits	Why is bulk power generated and distributed in three phases, and how are such systems analysed?	Three-phase generation and phase sequence, star and delta connections, line and phase quantities, balanced and unbalanced loads, three-phase power, measurement and neutral currents

The order is deliberate. Chapters 6 and 7 work entirely in the time domain and establish the vocabulary — natural frequency, time constant, damping, steady state versus transient — that the rest of the book reuses under other names: the natural frequencies become poles in Chapter 11, the step and impulse responses become the system descriptions of Chapters 15–17, and the damping ratio becomes a control specification in Chapters 34 and 35. Chapters 8–10 then specialise to sinusoidal excitation, the case on which the electricity supply, most measurement and all of radio rest. The phasor methods of Chapter 8 are the first appearance of frequency-domain thinking; Chapter 13 extends them into Bode plots and filters, Chapter 14 into transmission lines, and Chapter 18 into the Fourier transform. The power concepts of Chapters 9 and 10 are the circuit-theory foundation that *Electrical Energy & Machines* builds on for generators, transformers and power systems. Part II thus stands between the static circuit laws of Part I and the general network and signal theory of Parts III and IV: it is where circuits first acquire time and frequency.

Chapter 6 — First-Order Circuits

Part II · Transient and AC Analysis

A microcontroller board works perfectly on the bench supply but fails to start one time in twenty when powered from a cheap wall adapter. The adapter's output rises slowly, over tens of milliseconds, and the processor's reset line — held low by a resistor and a capacitor — releases before the supply has settled. Elsewhere, an ECG monitor in an ambulance shows a flat line for several seconds every time a patient is moved, then the trace slowly drifts back to the centre of the screen. And in a factory a 24 V solenoid valve is driven by a transistor that fails within weeks, while an identical transistor driving a resistive heater lasts for years. Three unrelated failures, one piece of mathematics: each involves a single energy-storage element, a resistance network and a switching event, and each is governed by an exponential with one time constant.

Chapter 5 gave the element laws $i = C\,dv/dt$ and $v = L\,di/dt$ and the rules for initial conditions. This chapter puts them to work in the simplest dynamic circuits: those whose behaviour is described by a single first-order differential equation. Despite their simplicity, first-order circuits account for a large share of everyday engineering — timing, filtering, debouncing, power-on reset, relay and solenoid drive, sensor response, bleeder discharge and the settling of every measurement front end.

> **How does a circuit with one capacitor or one inductor move from one state to another after a switch operates, and how long does it take?**
> **How do steps, pulses, ramps, sinusoids and impulses each shape the response, and why do all the answers share one exponential?**

Concept	Engineering destination
RC circuits	Timing and reset circuits, low-pass and high-pass filters, ADC input settling, sensor front ends
RL circuits	Relays, solenoids, contactors, motor windings, current ramp in converters, fault-current asymmetry
Natural response	Capacitor discharge, bleeder resistors, inductive kick and snubbing
Forced response	Tracking of ramps and sinusoids by sensors and filters, steady-state ripple
Time constants	Settling-time specifications, rise time and bandwidth, thermal and sensor lag
Switching behavior	Contact arcing, freewheeling diodes, sequential switching, debouncing
Initial and final values	The three-number method for any first-order step problem
Piecewise excitation	PWM filtering, square-wave testing, pulse trains
Step and impulse inputs	System characterisation, ECG baseline recovery, charge-sensitive detectors

Roadmap. RC circuits (6.1) → RL circuits (6.2) → natural response (6.3) → forced response and the complete response (6.4) → time constants (6.5) → initial and final values (6.6) → switching behavior (6.7) → piecewise excitation (6.8) → step and impulse inputs (6.9).

6.1 RC circuits

Reducing any first-order circuit to one loop

A **first-order circuit** contains resistors, sources and exactly one energy-storage element (or several that can be combined into one, Chapter 5). The key simplification comes from Chapter 4: everything connected to the capacitor, however complicated, is a linear one-port and can be replaced by its Thévenin equivalent V_{Th}, R_{Th}. The circuit then becomes a single loop: a source V_{Th}, a resistor R_{Th} and the capacitor C (Figure 6.1a).

KVL around the loop, with capacitor current $i = C\,dv/dt$:

$$V_{Th} = R_{Th} C \frac{dv}{dt} + v$$

or, defining the **time constant** $\tau = R_{Th} C$,

$$\tau \frac{dv}{dt} + v = V_{Th}(t)$$

This is a linear, first-order ordinary differential equation with constant coefficients. Its state variable is the capacitor voltage v — the quantity that cannot jump (Chapter 5). Once $v(t)$ is known, every other voltage and current in the original circuit follows from the resistive network by ordinary analysis, with the capacitor replaced by a voltage source of value $v(t)$ (substitution theorem, Chapter 4).

The resistance that matters is the **Thévenin resistance seen by the capacitor**: deactivate all independent sources and find the resistance between the capacitor's terminals. It is not, in general, any single resistor in the circuit, and it must be recomputed whenever a switch changes the network.

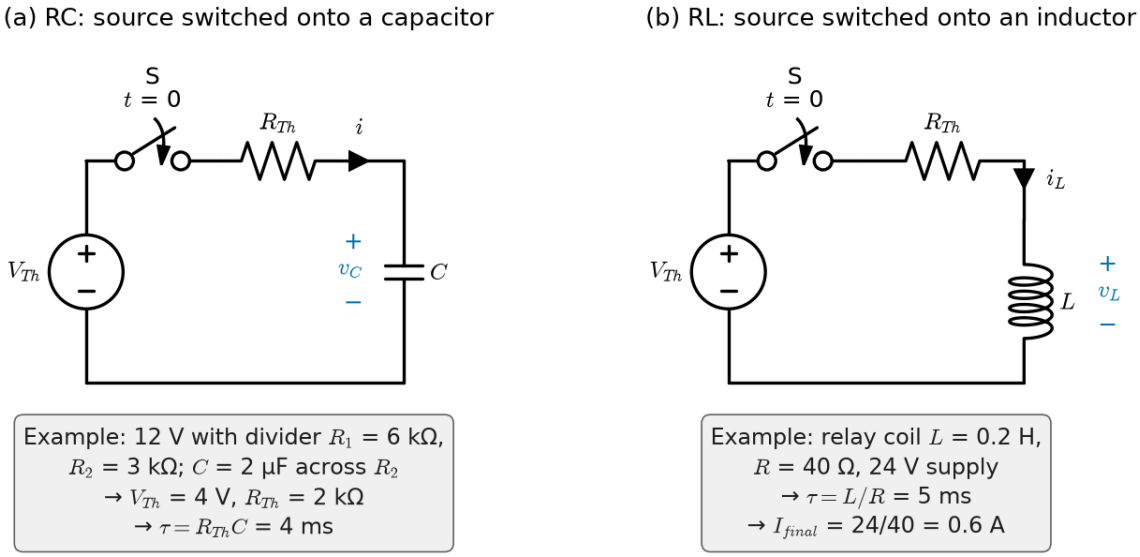


Figure 6.1 — The generic first-order circuits: a Thévenin source switched onto a capacitor, and onto an inductor.

Worked example — Reducing a circuit to its Thévenin form. A 12 V source feeds a divider of $R_1 = 6$ kΩ (top) and $R_2 = 3$ kΩ (bottom); a 2 μF capacitor, initially uncharged, is connected across R_2 by a switch closing at $t = 0$ (Figure 6.1a). Seen from the capacitor, the divider has

$$V_{Th} = 12 \times \frac{3}{6+3} = 4.0\ \text{V}, \qquad R_{Th} = 6 \parallel 3 = 2.0\ \text{k}\Omega$$

so $\tau = 2 \times 10^{3} \times 2 \times 10^{-6} = 4.0$ ms. The capacitor charges toward 4.0 V, not 12 V, and does so with a 4 ms time constant, not $R_1 C = 12$ ms or $R_2 C = 6$ ms. As Section 6.4 shows, $v(t) = 4.0(1 - e^{-t/4\,\text{ms}})$ V; it reaches 3.0 V at $t = 4 \ln 4 = 5.55$ ms.

RC circuits as filters

The same equation, read with an arbitrary input $V_{Th}(t) = v_{in}(t)$, describes an **RC low-pass filter**: the output v_C follows slow input changes and smooths fast ones. Taking the output across the resistor instead, $v_R = v_{in} - v_C$, gives an **RC high-pass filter** (also called a coupling or AC-coupling network), which passes changes and blocks steady levels. Both are first-order systems governed by the same τ; Sections 6.8 and 6.9 show their time-domain behaviour, and Chapters 8 and 13 their frequency response.

6.2 RL circuits

The dual equation

Replace the capacitor by an inductor. The rest of the circuit is now best represented by its **Norton** equivalent (I_N, R_N), the dual of the Thévenin form, because the inductor current is the state variable. KCL at the inductor node, with inductor voltage $v = L\,di/dt$ appearing across R_N:

$$I_N = \frac{L}{R_N}\frac{di}{dt} + i \quad \Rightarrow \quad \tau\frac{di}{dt} + i = I_N(t), \qquad \tau = \frac{L}{R_N}$$

Equivalently, with a Thévenin source and series resistance (Figure 6.1b), KVL gives $V_{Th} = R_{Th}i + L\,di/dt$, the same equation with $I_N = V_{Th}/R_{Th}$. The RL time constant is L/R: a *larger* resistance gives a *faster* response, the opposite of the RC case. This is duality (Chapter 5) at work: $C \leftrightarrow L$, $R \leftrightarrow G = 1/R$.

> **Status of the principle: The first-order circuit equation $\tau\,dx/dt + x = x_{in}(t)$**
> *Derived*. KCL/KVL plus the linear element laws, after Thévenin or Norton reduction of the resistive network; exact for linear time-invariant circuits with one storage element, and approximate for circuits with several storage elements when one time constant dominates.

Where RL circuits appear

Every electromagnetic actuator is an RL circuit: relays, contactors, solenoid valves, fuel injectors, loudspeaker voice coils at low frequency, the windings of DC motors with the rotor stalled. A coil of inductance L and winding resistance R driven from a voltage V cannot reach its final current V/R instantly; the current rises with time constant L/R. That delay is the pull-in time of a relay and the opening delay of a fuel injector. Power systems are RL circuits on a large scale: a feeder and transformer have X/R ratios from about 2 (distribution) to 30 or more (near large generators), and the time constant L/R governs how fault currents behave (Section 6.4).

Worked example — Pull-in time of a relay. A 24 V relay coil has $L = 0.2$ H and $R = 40\ \Omega$ (Figure 6.1b). Its time constant is $\tau = 0.2/40 = 5$ ms and its final current $24/40 = 0.6$ A. The armature pulls in when the coil current reaches 0.40 A. The current is $i(t) = 0.6(1 - e^{-t/5\,\text{ms}})$ A, so pull-in occurs at

$$t = -\tau \ln\left(1 - \frac{0.40}{0.60}\right) = 5\ln 3 = 5.49\ \text{ms}$$

(In reality the armature's motion changes the magnetic gap and therefore L while it moves, which adds a characteristic dip in the current waveform; the linear model gives the time to the start of motion.) Driving the coil from 48 V through an extra 40 Ω series resistor keeps the final current at 0.6 A but halves the time constant to 2.5 ms and halves the pull-in time to 2.75 ms — the classic **forcing** technique for fast actuators, later replaced by current-controlled drivers.

6.3 Natural response

Source-free decay

The **natural response** (also called the source-free or zero-input response) is what a circuit does when the sources are removed and only the stored energy drives it. With $V_{Th} = 0$:

$$\tau\frac{dv}{dt} + v = 0$$

Try $v = Ae^{st}$: substitution gives $(\tau s + 1)Ae^{st} = 0$, so the **characteristic equation** is $\tau s + 1 = 0$ with root $s = -1/\tau$. The initial condition $v(0^+) = V_0$ fixes $A = V_0$:

$$v(t) = V_0\, e^{-t/\tau}, \qquad t \geq 0$$

For the inductor, identically, $i(t) = I_0\, e^{-t/\tau}$ with $\tau = L/R$. All other quantities decay with the same exponential: in the RC case the discharge current is $i = -v/R = -(V_0/R)e^{-t/\tau}$, jumping from zero to $-V_0/R$ at the switching instant and then decaying.

The root $s = -1/\tau$ is the circuit's **natural frequency** — a single, real, negative number. It does not depend on what started the decay or on which variable is observed; it is a property of the circuit alone. Chapter 7 shows that second-order circuits have two natural frequencies, which may be complex, and Chapter 11 identifies them as the poles of the circuit's transfer function.

Energy in the natural response

All of the stored energy ends up in the resistance. For the RC discharge,

$$w_R = \int_0^\infty \frac{v^2}{R}\, dt = \frac{V_0^2}{R}\int_0^\infty e^{-2t/\tau} dt = \frac{V_0^2}{R}\cdot\frac{\tau}{2} = \tfrac{1}{2}CV_0^2$$

The value of R cancels: a large resistance dissipates slowly, a small one quickly, but the total is the stored energy. This confirms the charge-sharing result of Chapter 5. When a 10 μF capacitor at 10 V is connected to an uncharged 10 μF capacitor through any resistance, the loop behaves as a single capacitor $C_1C_2/(C_1+C_2) = 5$ μF holding the initial voltage difference of 10 V, and its natural response dissipates $\tfrac{1}{2}\times 5\times 10^{-6}\times 10^2 = 250$ μJ in the resistance — independent of R, exactly as found there.

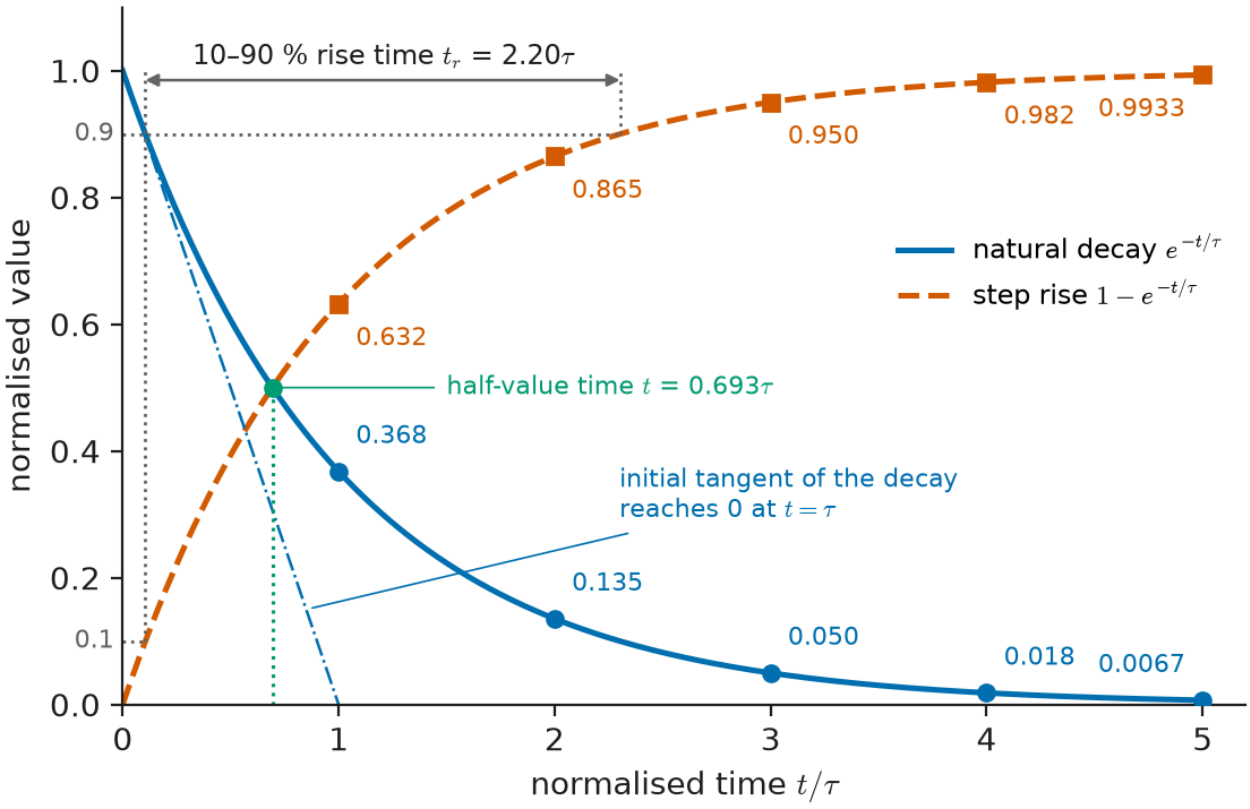


Figure 6.2 — Normalised first-order responses: the natural decay and the step rise, with the initial tangent and the values at one to five time constants.

Worked example — Sizing a bleeder resistor for a drive DC link. A motor drive's DC link has 470 µF charged to 400 V. Machinery safety standards commonly require stored voltage to fall below 60 V within 5 s of disconnection. With a bleeder resistor R across the capacitor,

$$60 = 400\,e^{-5/(RC)} \;\Rightarrow\; RC = \frac{5}{\ln(400/60)} = \frac{5}{1.897} = 2.636\ \text{s} \;\Rightarrow\; R \le \frac{2.636}{470 \times 10^{-6}} = 5.61\ \text{k}\Omega$$

But a 5.61 kΩ resistor across 400 V dissipates $400^2/5610 = 28.5$ W *continuously* while the drive runs. That is why drives either accept longer discharge times with warning labels, use several resistors in series to share voltage and heat, or switch in a discharge resistor only when the mains is removed. The stored energy, $\tfrac{1}{2} \times 470 \times 10^{-6} \times 400^2 = 37.6$ J, is the same whichever resistor dissipates it.

The inductive kick

The natural response of an RL circuit hides a hazard. When the switch feeding an inductor opens, the inductor current must continue at its initial value I_0 (Chapter 5). If the only path left is through a resistance R_s, the inductor voltage jumps to $I_0 R_s$; if no path is provided, R_s is effectively the open switch — megohms or more — and the voltage rises until something breaks down. Section 6.7 analyses the practical suppression circuits.

6.4 Forced response

The complete response as natural plus forced

With a source present, the differential equation $\tau\,dx/dt + x = x_{in}(t)$ is inhomogeneous. Its general solution is the sum of

- a **forced response** (particular solution) $x_f(t)$ — any one function that satisfies the equation with the actual input; for the common inputs it has the same form as the input;
- a **natural response** $Ae^{-t/\tau}$ (homogeneous solution) — the same exponential as in Section 6.3, with a constant A now fixed by the initial condition of the *complete* response.

$$x(t) = x_f(t) + \left[x(0^+) - x_f(0^+)\right] e^{-t/\tau}$$

The natural part is a correction that bridges the gap between where the circuit actually starts and where the forced solution "would like" it to be. Since $\tau > 0$, the natural part decays, and after a few time constants only the forced response remains. In this sense the forced response is the **steady-state** response and the natural response is the **transient**. Superposition (Chapter 4) offers a second decomposition of the same answer: the **zero-input response** (initial energy, no source) plus the **zero-state response** (source, no initial energy). Both decompositions are correct; they divide the same total differently.

> **Status of the principle: Complete response = natural + forced**
> *Derived*. A theorem of linear ordinary differential equations: the general solution of a linear inhomogeneous equation is any particular solution plus the general homogeneous solution. It relies on linearity and fails for circuits containing nonlinear elements.

Forced responses for standard inputs

Table 6.1 — Forced response of $\tau\,dx/dt + x = x_{in}(t)$ for standard inputs.

Input $x_{in}(t)$	Forced response $x_f(t)$	Interpretation
Constant X	X	Settles to the input level
Ramp αt	$\alpha(t - \tau)$	Follows the ramp with a delay of τ, error $\alpha\tau$
Exponential Xe^{-at}, $a \neq 1/\tau$	$\dfrac{X}{1 - a\tau}e^{-at}$	Same exponential, rescaled
Sinusoid $X_m \sin \omega t$	$\dfrac{X_m}{\sqrt{1 + (\omega\tau)^2}} \sin(\omega t - \arctan \omega\tau)$	Attenuated and phase-lagged sinusoid

Each entry can be verified by substitution. The ramp row says something useful about sensors: a first-order sensor tracking a steadily changing quantity reads it with a constant lag of one time constant. A sheathed thermocouple with $\tau = 8$ s, measuring a furnace whose temperature ramps at 3 °C/s, reads

$3 \times 8 = 24$ °C low for as long as the ramp continues — not a calibration error but a dynamic one, which disappears when the ramp stops (after a few further time constants).

The sinusoid row is the time-domain origin of everything in Chapter 8. For $\omega\tau \ll 1$ the output follows the input; at $\omega\tau = 1$ the amplitude is $1/\sqrt{2}$ (−3 dB) and the lag is 45°; for $\omega\tau \gg 1$ the amplitude falls as $1/(\omega\tau)$ and the lag approaches 90°. The frequency $\omega = 1/\tau$ is the **corner** or **cutoff** frequency of the RC or RL filter.

Worked example — Switching a sinusoid onto an RC filter. A sinusoid $v_{in} = V_m \sin \omega t$ is switched at $t = 0$ onto an RC low-pass with $R = 1$ kΩ, $C = 1$ µF ($\tau = 1$ ms) at $\omega = 1000$ rad/s (159.2 Hz), so $\omega\tau = 1$. The capacitor is initially uncharged. From Table 6.1 the forced response is

$$v_f(t) = \frac{V_m}{\sqrt{2}} \sin\left(\omega t - 45^\circ\right)$$

whose value at $t = 0$ is $\frac{V_m}{\sqrt{2}} \sin(-45^\circ) = -0.5 V_m$. The capacitor starts at zero, so the natural part must supply $0 - (-0.5V_m) = +0.5V_m$:

$$v_C(t) = \frac{V_m}{\sqrt{2}} \sin(\omega t - 45^\circ) + 0.5 V_m \, e^{-t/1\,\text{ms}}$$

The transient raises the first peak to $0.756 V_m$ (at $t = 2.28$ ms), about 7 % above the steady-state amplitude $0.707 V_m$, and has died to below 1 % of V_m after 5 ms. Switching at a different point of the sine changes the size and sign of the transient; switching at the instant when $v_f(0) = 0$ produces no transient at all.

Fault-current asymmetry: the RL version

The same analysis applied to an RL circuit explains one of the central facts of power-system protection. A short circuit on a feeder makes the network an RL circuit driven by the sinusoidal source voltage. The forced current is a sinusoid lagging the voltage by $\varphi = \arctan(X/R)$, but the inductor current must start from its pre-fault value (approximately zero for this purpose). The natural response therefore contains a decaying **DC offset**, whose size depends on the instant in the voltage cycle at which the fault occurs. In the worst case the forced current is at its negative peak $-I_m$ at $t = 0$, the offset starts at $+I_m$, and

$$i(t) = I_m \left[e^{-t/\tau} - \cos \omega t \right], \qquad \tau = \frac{L}{R} = \frac{X/R}{\omega}$$

For $X/R = 10$ at 50 Hz, $\tau = 31.8$ ms and the first peak, at $t = 9.77$ ms, is $1.73 I_m$. Circuit breakers must withstand and interrupt this **asymmetrical** current, not just the symmetrical value; the "peak factor" κ used in fault-current standards is exactly this ratio, tabulated against X/R.

6.5 Time constants

Reading the exponential

The time constant is the single number that characterises a first-order circuit's speed. Its meanings, all equivalent:

1. The time for the natural response to fall to $1/e = 36.8\ \%$ of its initial value (or for a step response to cover 63.2 % of the change).
2. The time at which the **initial tangent** of the response reaches the final value (Figure 6.2) — if the circuit kept changing at its initial rate, it would finish in exactly one time constant.
3. The ratio of stored energy capacity to the rate at which the resistance can move it: $\tau = RC$ or L/R.
4. The reciprocal of the magnitude of the natural frequency, $\tau = 1/|s|$.

Table 6.2 — Remaining fraction of the transient after n time constants.

Elapsed time	Remaining e^{-n}	Completed $1 - e^{-n}$	Typical use
0.693τ	50.0 %	50.0 %	Half-life of decay
1τ	36.8 %	63.2 %	Definition
2τ	13.5 %	86.5 %	—
2.303τ	10.0 %	90.0 %	—
3τ	4.98 %	95.0 %	"Settled" for rough work
4τ	1.83 %	98.2 %	—
4.6τ	1.0 %	99.0 %	1 % settling
5τ	0.67 %	99.3 %	"Settled" in common practice
6.9τ	0.1 %	99.9 %	0.1 % settling (about 10 bits)
$(N+1)\ln 2 \cdot \tau$	$2^{-(N+1)}$	—	Settling to ½ LSB of an N-bit converter

The last row matters for data acquisition: an ADC input or a sample-and-hold capacitor must settle to within half a least-significant bit, which for a 16-bit converter takes $17 \ln 2 = 11.8$ time constants. Fast, high-resolution converters therefore need very low source resistance, which is why they are driven by dedicated amplifiers.

Rise time and bandwidth

The **10–90 % rise time** of a first-order step response is $t_r = \tau \ln 9 = 2.20\tau$. Since the -3 dB bandwidth of the same circuit is $f_{3dB} = 1/(2\pi\tau)$,

$$t_r = \frac{2.2}{2\pi f_{3dB}} \approx \frac{0.35}{f_{3dB}}$$

This rule links the time and frequency domains for any system that behaves approximately as first order. An oscilloscope with 100 MHz bandwidth has an intrinsic rise time of about 3.5 ns and cannot faithfully display edges much faster than that. Chapter 18 generalises the relation (time–bandwidth product) to other systems.

> **Status of the principle:** $t_r \approx 0.35/f_{3dB}$
> *Approximate.* Exact for a first-order system; a good estimate (within about 10 %) for many well-damped amplifiers and instruments, but not for systems with overshoot or sharp-cutoff filters, where the constant ranges from about 0.35 to 0.45.

Representative time constants

Table 6.3 — Time constants across EEE (typical values).

System	Elements	Typical τ
On-chip logic gate driving local wire	1 kΩ, 10 fF	10 ps
50 Ω line driving 10 pF probe	50 Ω, 10 pF	0.5 ns
10× oscilloscope probe tip	9 MΩ, about 1.1 pF (compensated)	about 10 µs
Switch debounce filter	10 kΩ, 100 nF	1 ms
Power-on reset	100 kΩ, 1 µF	100 ms
Relay or solenoid coil	0.2 H, 40 Ω	5 ms
Distribution feeder fault offset ($X/R \approx 3$–10)	—	10–30 ms at 50 Hz
Generator-near fault offset ($X/R \approx 30$–50)	—	100–160 ms at 50 Hz
ECG amplifier AC coupling (0.05 Hz corner)	—	3.2 s
Sheathed thermocouple in air	thermal RC	1–30 s
Large power transformer thermal	thermal RC	1–3 h

The last entries are not electrical circuits at all: heat flow through a thermal resistance into a heat capacity obeys the same first-order equation, which is why "thermal time constant" appears on every power-semiconductor and transformer datasheet.

Measuring a time constant

On an oscilloscope, a single exponential is recognised by its constant ratio over equal time intervals: each interval of τ divides the remaining distance to the final value by e. To measure τ, read the final value, find the time at which 63.2 % (or 36.8 %) of the change has occurred, or fit a straight line to the logarithm of the remaining distance — its slope is $-1/\tau$. The log plot is also the best test of first-order behaviour: a second time constant appears as a bend in the line.

6.6 Initial and final values

The three-number method

When all sources are constant after the switching event, the forced response is a constant — the final value $x(\infty)$ — and the complete response of *any* voltage or current in a first-order circuit is

$$x(t) = x(\infty) + \left[x(0^+) - x(\infty)\right] e^{-(t-t_0)/\tau}, \qquad t \geq t_0$$

Three numbers determine everything: the initial value $x(0^+)$, the final value $x(\infty)$ and the time constant τ. Each is found from a resistive circuit:

1. **Initial value.** Find $v_C(0^-)$ or $i_L(0^-)$ from the circuit before switching (usually in DC steady state: capacitor open, inductor short). Continuity gives $v_C(0^+)$ or $i_L(0^+)$. For any *other* variable x, solve the post-switching circuit with the capacitor replaced by a voltage source $v_C(0^+)$ or the inductor by a current source $i_L(0^+)$ (Chapter 5).
2. **Final value.** Solve the post-switching circuit with the capacitor open or the inductor shorted.
3. **Time constant.** Deactivate the independent sources of the post-switching circuit and find the Thévenin resistance seen by the storage element; $\tau = R_{Th}C$ or L/R_{Th}.

The method works for any variable, not just the state variable, because every voltage and current in a linear first-order circuit is a linear combination of the source and the state variable, and so shares its exponential.

> **Status of the principle: Three-number formula** $x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$
> *Derived*. The complete response of a linear first-order circuit with constant sources after the switching instant; it fails for time-varying sources (use Section 6.4), for circuits with more than one independent storage element (Chapter 7), and if $R_{Th} \leq 0$, in which case the circuit does not settle.

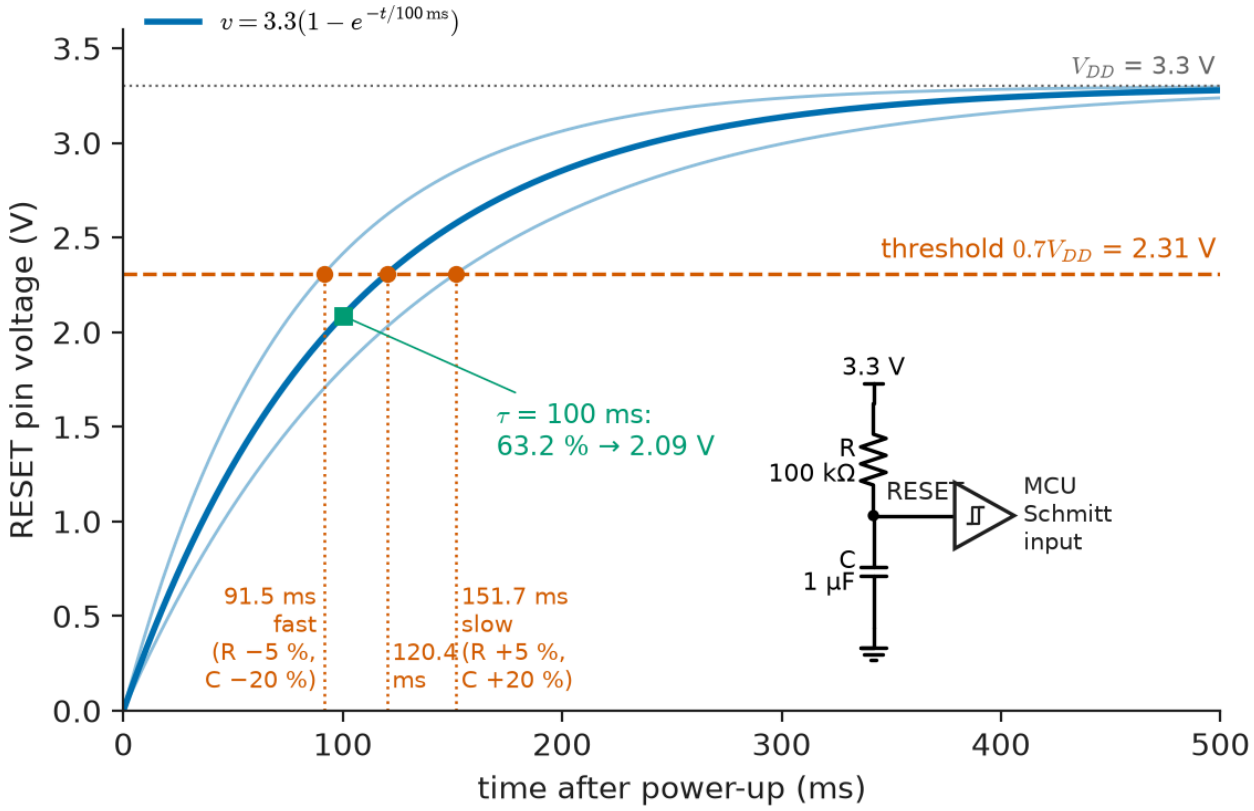


Figure 6.3 — Power-on reset: the RC voltage rises toward the supply and releases the processor when it crosses the logic threshold; component tolerances spread the release time.

Worked example — Power-on reset timing. A microcontroller's reset input is held low by a capacitor $C = 1$ µF charged through $R = 100$ kΩ from the 3.3 V supply (Figure 6.3). The reset is released when the input crosses $0.7V_{DD} = 2.31$ V. Assume the supply rises instantly.

Initial value $v(0^+) = 0$; final value $v(\infty) = 3.3$ V; time constant $\tau = 100 \times 10^3 \times 1 \times 10^{-6} = 0.1$ s. Then $v(t) = 3.3(1 - e^{-t/0.1})$, and the threshold is crossed when $e^{-t/\tau} = 0.3$:

$$t = \tau \ln \frac{1}{0.3} = 0.1 \times 1.204 = 120.4 \text{ ms}$$

With a ±5 % resistor and a ±20 % capacitor, the release time spans $0.95 \times 0.80 \times 120.4 = 91.5$ ms to $1.05 \times 1.20 \times 120.4 = 151.7$ ms. The reset is useful only if the supply has settled before the shortest of these. The opening's failure — a slowly rising adapter — breaks the "supply rises instantly" assumption: the capacitor's final value is itself moving, the RC voltage can cross the threshold while V_{DD} is still low, and the processor starts with an unstable supply. Dedicated supervisor ICs, which compare V_{DD} against a reference and only then start a timer, exist because the simple RC reset depends on how the supply ramps.

Worked example — Any variable, not only the state. Return to the circuit of Section 6.1 (12 V, $R_1 = 6$ kΩ, $R_2 = 3$ kΩ, $C = 2$ µF switched across R_2 at $t = 0$). What is the current in R_1? Before switching, R_1 and R_2 form a simple divider: $i_{R1}(0^-) = 12/9 = 1.333$ mA. At 0^+, the capacitor (still at 0 V) shorts R_2, so $i_{R1}(0^+) = 12/6 = 2.0$ mA — a jump, allowed because i_{R1} is not a state variable. Finally $i_{R1}(\infty) = 1.333$ mA again (capacitor open). The time constant is the same 4 ms. Hence

$$i_{R1}(t) = 1.333 + (2.0 - 1.333)e^{-t/4\,\text{ms}}\ \text{mA} = 1.333 + 0.667\,e^{-t/4\,\text{ms}}\ \text{mA}$$

No differential equation was solved; three resistive circuits gave the answer.

6.7 Switching behavior

Real switches and the ideal model

The analysis so far treats a switch as ideal: zero resistance when closed, infinite when open, transition in zero time. Real switches differ in ways that matter precisely in first-order circuits:

- **Mechanical contacts bounce.** A pushbutton or relay contact makes and breaks several times over 0.1–10 ms before settling. A logic input sees a burst of pulses; an RC filter followed by a Schmitt trigger (a **debounce** circuit, τ of a few milliseconds) turns the burst into one clean edge.
- **Opening contacts arc.** If the circuit is inductive, the current continues across the opening gap as an arc until the stored energy is dissipated or the voltage across the gap can no longer sustain it. DC arcs are hard to extinguish because the current never passes through zero; AC arcs usually extinguish at the next current zero.
- **Semiconductor switches have finite speed and voltage ratings.** A transistor turns off in nanoseconds to microseconds, so $L\,di/dt$ can be enormous, and the device fails if the resulting voltage exceeds its breakdown rating.

Turning off an inductive load

The opening example — a transistor that fails driving a solenoid but not a heater — is an RL switching problem. When the transistor turns off, the coil current I_0 must continue. Four common arrangements decide where it goes (Figure 6.4).

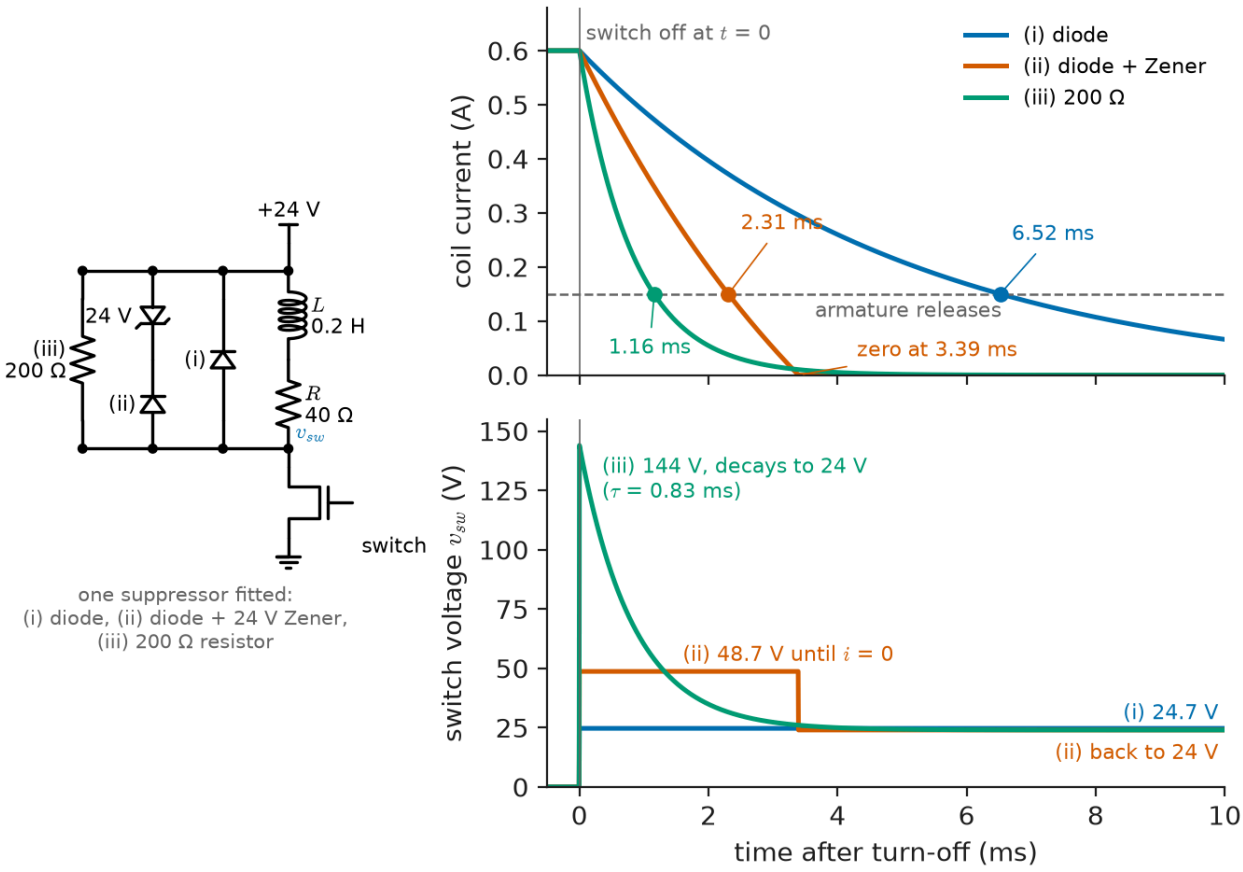


Figure 6.4 — Turning off a 0.2 H, 40 ohm relay coil from 0.6 A: coil current and switch voltage for three suppression circuits.

Worked example — Suppressing the turn-off of a relay coil. The relay of Section 6.2 ($L = 0.2$ H, $R = 40\ \Omega$, 24 V, $I_0 = 0.6$ A) is switched off by a transistor at $t = 0$. The armature releases when the current falls to 0.15 A.

No suppression. The coil's stored energy, $\tfrac{1}{2} \times 0.2 \times 0.6^2 = 36$ mJ, has no path. The voltage rises until the transistor breaks down (avalanche) and absorbs the energy — 36 mJ per operation, repeated every time. That is the transistor that fails within weeks.

Freewheeling diode across the coil. The current circulates through the diode (forward drop $V_D \approx 0.7$ V). The coil equation is $L\,di/dt = -(Ri + V_D)$, a first-order equation with constant input whose final value would be $-V_D/R = -17.5$ mA (it stops at zero, when the diode turns off):

$$i(t) = \left(I_0 + \frac{V_D}{R}\right) e^{-t/\tau} - \frac{V_D}{R}, \qquad \tau = \frac{L}{R} = 5\ \text{ms}$$

The current reaches 0.15 A at $t = -\tau \ln[(0.15 + 0.0175)/(0.6 + 0.0175)] = 6.52$ ms. The transistor sees only $24 + 0.7 = 24.7$ V. Safe, but the release is slow: the diode lets the coil keep its current, so the relay lingers closed.

Diode plus 24 V Zener in series. Now $L\,di/dt = -(Ri + V_D + V_Z)$, and the effective final value is $-(0.7 + 24)/40 = -0.6175$ A. The current reaches 0.15 A at

$$t = -\tau \ln \frac{0.15 + 0.6175}{0.6 + 0.6175} = 2.31\ \text{ms}$$

almost three times faster. The transistor sees $24 + 0.7 + 24 = 48.7$ V, which must be within its rating. The Zener absorbs most of the energy at a high voltage, speeding the decay.

Resistor $R_s = 200\ \Omega$ across the coil. The current transfers into R_s, so the coil voltage jumps to $0.6 \times 200 = 120$ V and the transistor sees $24 + 120 = 144$ V at the instant of turn-off. The time constant becomes $L/(R + R_s) = 0.2/240 = 0.833$ ms and the release occurs at $0.833 \ln 4 = 1.16$ ms. Fast, but with a high peak voltage, and the resistor wastes $24^2/200 = 2.9$ W whenever the relay is energised.

The trade-off — release speed against switch voltage — is general: the faster the stored energy is removed, the higher the voltage at which it must be removed, because the energy $\tfrac{1}{2}LI_0^2$ is fixed and the time to remove it scales roughly as LI_0/V. RC **snubbers** across contacts and transistors, varistors across mains-connected coils, and the active clamp circuits of integrated low-side drivers are all variations on these four arrangements.

Turning off a capacitive circuit, turning on into a capacitor

The dual problem arises when a switch *closes* onto a capacitor: the capacitor's voltage cannot jump, so the switch carries a current limited only by the circuit resistance — the inrush current of Chapter 5. A rectifier charging a large reservoir capacitor at switch-on, a hot-plugged board and a power-factor-correction capacitor bank switched onto a live bus all exhibit it. The remedies are dual to those for inductors: add series resistance temporarily (an **NTC thermistor** or a precharge resistor bypassed by a relay after a few time constants), or ramp the voltage slowly with a controlled transistor.

Sequential switching

When switches operate at several instants, the circuit is solved interval by interval. At each switching instant, the state variable carries over continuously; the time constant and final value are recomputed for the new circuit; and time is measured from the new instant.

Worked example — Charge then discharge. A 10 μF capacitor, initially uncharged, is charged from 10 V through $R_1 = 1$ kΩ for 5 ms; at $t = 5$ ms the switch transfers the capacitor to a discharge resistor $R_2 = 4$ kΩ.

Interval 1 ($0 \le t \le 5$ ms): $\tau_1 = 10$ ms, final value 10 V: $v(t) = 10(1 - e^{-t/10\,\text{ms}})$, so $v(5\,\text{ms}) = 10(1 - e^{-0.5}) = 3.935$ V. The capacitor is far from fully charged.

Interval 2 ($t \ge 5$ ms): $\tau_2 = 40$ ms, final value 0: $v(t) = 3.935\, e^{-(t-5\,\text{ms})/40\,\text{ms}}$. At $t = 25$ ms, $v = 3.935\, e^{-0.5} = 2.387$ V.

The same bookkeeping, repeated periodically, gives the response to pulse trains (Section 6.8).

6.8 Piecewise excitation

Inputs built from steps

Many practical inputs are **piecewise constant**: pulses, square waves, PWM signals, staircases from a DAC. Two equivalent methods handle them.

1. **Interval by interval.** As in Section 6.7: in each interval the input is constant, the three-number formula applies, and the state variable at the end of one interval is the initial value for the next.
2. **Superposition of shifted steps.** Write the input as a sum of steps, e.g. a rectangular pulse of height V and width T as $V[u(t) - u(t-T)]$, where $u(t)$ is the unit step. If $s(t)$ is the response to $u(t)$, linearity and time invariance give the response $V[s(t) - s(t-T)]$.

For a pulse into an RC low-pass from rest, both methods give

$$v(t) = \begin{cases} V\left(1 - e^{-t/\tau}\right), & 0 \le t \le T \\ V\left(1 - e^{-T/\tau}\right) e^{-(t-T)/\tau}, & t \ge T \end{cases}$$

If $T \gg \tau$ the output is a faithful pulse with rounded edges; if $T \ll \tau$, it barely rises — to about $V\,T/\tau$ — before decaying. The second regime, in which the output is proportional to the *area* VT of the pulse, leads to the impulse idea of Section 6.9.

Periodic steady state under a square wave

When a square wave of period T and duty cycle D drives a first-order circuit, the response settles after a few time constants into a periodic waveform, rising toward the high level during the high interval and falling toward the low level during the low interval. In steady state the waveform must repeat, so the voltage at the start of each period equals that at its end. For a 0-to-V square wave into an RC low-pass, with $a = e^{-DT/\tau}$ and $b = e^{-(1-D)T/\tau}$:

$$v_{\max} = V\frac{1-a}{1-ab}, \qquad v_{\min} = v_{\max}\,b$$

For a 50 % duty cycle, the shape depends on τ/T (Figure 6.5): for $\tau \ll T$ the output reproduces the square wave with exponential edges; for $\tau \approx T$ it becomes a "shark-fin"; for $\tau \gg T$ it is a small triangular ripple around the average value DV. The high-pass output (across the resistor) shows the complementary picture: for $\tau \ll T$, narrow spikes at each edge (a differentiator); for $\tau \gg T$, a square wave with slight **tilt** (sag), which is how AC-coupled oscilloscope inputs and audio coupling capacitors distort low-frequency square waves.

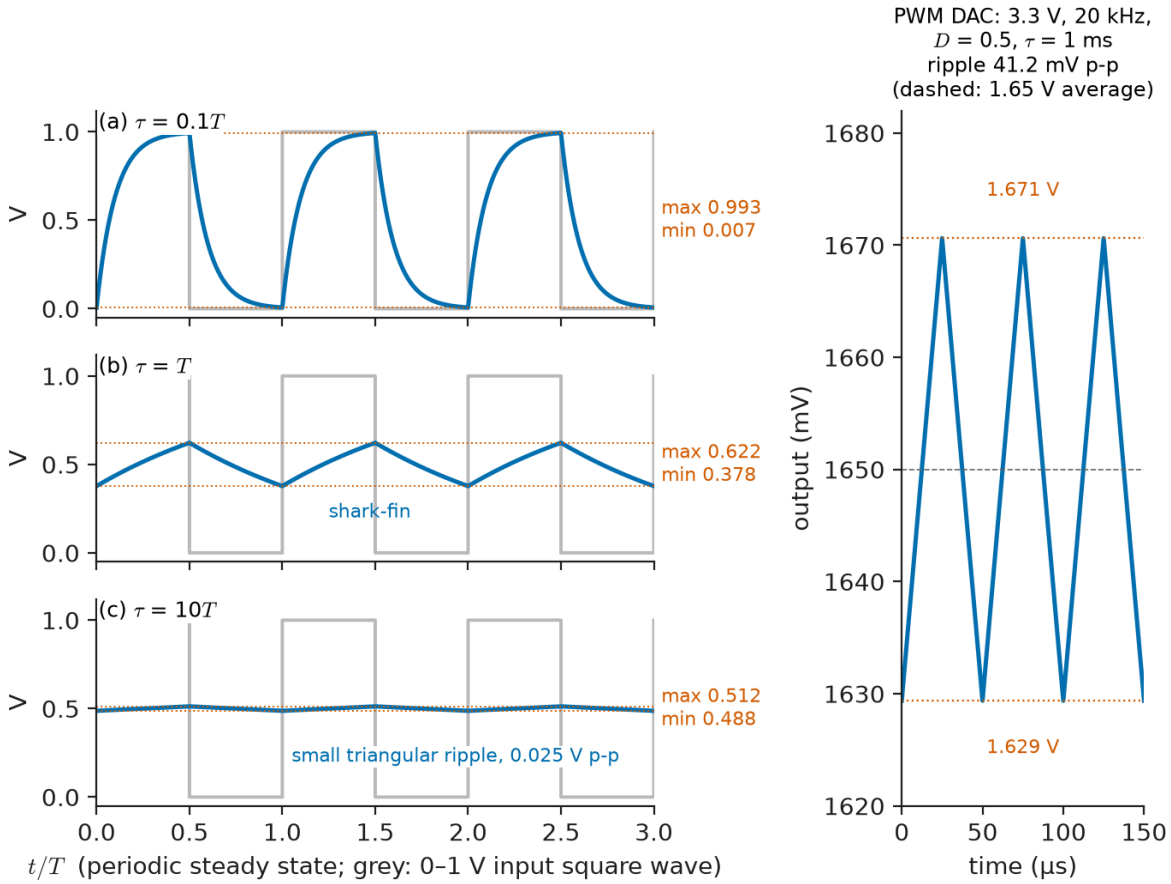


Figure 6.5 — Periodic steady-state response of an RC low-pass filter to a 0 to 1 V square wave of 50 percent duty cycle for three ratios of time constant to period.

Table 6.4 — Steady-state extremes for a 0–1 V, 50 % duty square wave into an RC low-pass.

τ/T	$v_{\max}$ (V)	$v_{\min}$ (V)	Peak-to-peak ripple (V)
0.1	0.993	0.007	0.987
1	0.622	0.378	0.245
10	0.512	0.488	0.025

Worked example — A PWM digital-to-analogue converter. A microcontroller produces an analogue voltage by driving a 3.3 V PWM signal at 20 kHz ($T = 50$ µs) into an RC low-pass with $R = 10$ kΩ and $C = 100$ nF ($\tau = 1$ ms, so $\tau/T = 20$). At $D = 0.5$:

Exact steady-state ripple. $a = b = e^{-25\,\mu\text{s}/1\,\text{ms}} = 0.9753$, giving $v_{\max} = 3.3 \times (1 - 0.9753)/(1 - 0.9753^2) = 1.6706$ V and $v_{\min} = 1.6294$ V: ripple 41.2 mV peak-to-peak about the average 1.65 V.

Approximation for $\tau \gg T$. The capacitor voltage hardly changes over a period, so the capacitor current is nearly constant in each interval: $(V - DV)/R$ while high, $-DV/R$ while low. The ripple is then

$$\Delta v \approx \frac{V\,D(1-D)\,T}{\tau} = \frac{3.3 \times 0.25 \times 50 \times 10^{-6}}{10^{-3}} = 41.25\ \text{mV}$$

in excellent agreement. The ripple is largest at $D = 0.5$.

Settling. When the duty cycle changes, the average output moves with the circuit's time constant. To settle within half an LSB of an 8-bit output (2^{-9} of full scale) takes $\tau \ln 512 = 6.24$ ms. Reducing the ripple below ½ LSB (6.4 mV at 3.3 V) needs a larger τ — and makes settling slower still. This ripple–speed conflict of a single pole is why practical PWM DACs use second-order or higher filters (Chapter 13) or a higher PWM frequency.

6.9 Step and impulse inputs

The unit step and the step response

The **unit step** $u(t)$ is 0 for $t < 0$ and 1 for $t > 0$; it models a switch closing onto a 1 V source at $t = 0$. The **step response** $s(t)$ is the zero-state response (all initial conditions zero) to $u(t)$. For the RC low-pass with output across C:

$$s(t) = \left(1 - e^{-t/\tau}\right) u(t)$$

and for the high-pass with output across R:

$$s_{HP}(t) = e^{-t/\tau} u(t)$$

The step response is the most widely used characterisation of a dynamic system in the time domain: rise time, settling time, overshoot (absent in first order) and steady-state error are all read from it (Chapter 35 uses the same quantities as control specifications). It is also directly measurable: apply a fast edge and observe the output.

The unit impulse

Consider a rectangular pulse of width ϵ and height $1/\epsilon$, so that its area is 1. As $\epsilon \to 0$ the pulse becomes infinitely tall and narrow with unit area: the **unit impulse** or Dirac delta $\delta(t)$, defined by its effect under an integral,

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1, \qquad \int_{-\infty}^{\infty} f(t)\, \delta(t - t_0)\, dt = f(t_0)$$

and related to the step by $\delta(t) = du/dt$. It models any event that is much shorter than the circuit's time constant but delivers a finite total: a charge packet from a particle detector, a lightning-induced current pulse, a hammer tap on an accelerometer, a short current spike delivered into a capacitor.

> **Status of the principle: Impulse idealisation**
> *Approximate.* A short pulse of area A produces essentially the same response as $A\,\delta(t)$ when its duration is much shorter than the circuit's shortest time constant; a pulse of width 0.1τ gives an output 4.8 % below the impulse prediction, a pulse of width 0.01τ about 0.5 % below.

The rigorous theory of $\delta(t)$ as a generalised function belongs to *The Mathematics Universe*; Chapter 15 introduces it as a signal in its own right. Here it is enough to know how a circuit responds.

The impulse response

The **impulse response** $h(t)$ is the zero-state response to $\delta(t)$. Because δ is the derivative of u, and differentiation commutes with a linear time-invariant system, the impulse response is the derivative of the step response:

$$h(t) = \frac{ds}{dt}$$

For the RC low-pass,

$$h(t) = \frac{1}{\tau} e^{-t/\tau} u(t)$$

Physically, the impulse delivers a finite charge to the capacitor in zero time — an **instantaneous jump** of capacitor voltage, which is permitted precisely because the input is impulsive (the exception to continuity noted in Chapter 5). An impulse of voltage $A\delta(t)$ through R deposits charge A/R, raising the capacitor voltage to $A/(RC) = A/\tau$; thereafter the circuit is source-free and decays with its natural response. **The impulse response is the natural response started by an impulse.**

> **Status of the principle:** $h(t) = ds/dt$
> *Derived.* Holds for every linear time-invariant system, because the impulse is the derivative of the step; it fails for time-varying or nonlinear systems.

The impulse response contains everything about a linear time-invariant circuit: by superposition, any input can be built from shifted, scaled impulses, and the output is the corresponding sum of shifted, scaled impulse responses — the **convolution integral** of Chapter 17. Its Laplace transform is the transfer function of Chapter 11. First-order circuits are where this idea first becomes concrete.

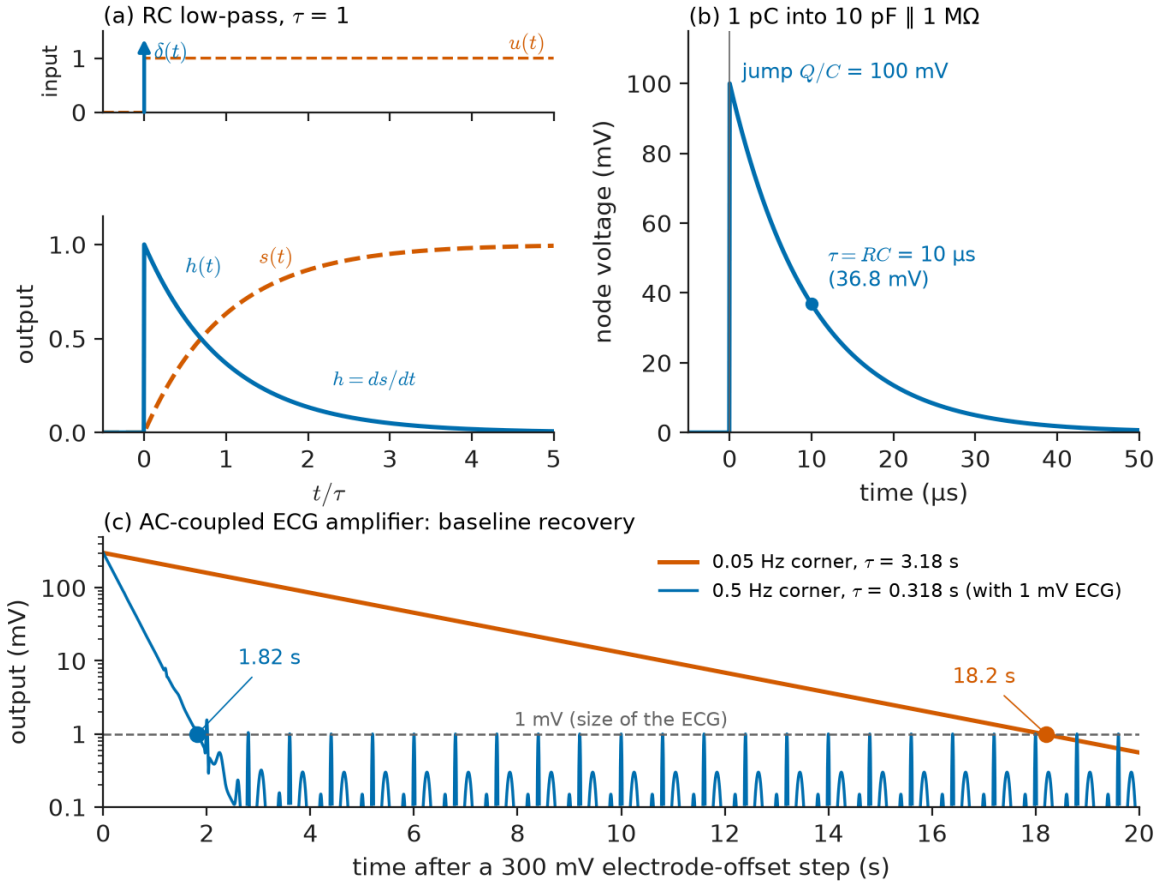


Figure 6.6 — Step and impulse responses of first-order circuits, and the baseline recovery of an AC-coupled ECG amplifier after an electrode-offset step.

Worked example — A charge-sensitive radiation detector. A semiconductor radiation detector deposits a charge of $Q = 1$ pC in about 10 ns onto the input node of its preamplifier, which presents $C = 10$ pF in parallel with a $R = 1$ MΩ bias resistor ($\tau = 10$ µs). Since 10 ns is a thousandth of τ, the charge pulse is effectively an impulse of current $Q\,\delta(t)$. It raises the node voltage instantly by

$$\Delta v = \frac{Q}{C} = \frac{10^{-12}}{10 \times 10^{-12}} = 0.10 \text{ V}$$

after which $v(t) = 0.10\, e^{-t/10\,\mu\text{s}}$ V (Figure 6.6b). The peak is proportional to the deposited charge and therefore to the particle's energy, which is the basis of pulse-height spectroscopy. The time constant sets the maximum count rate: pulses arriving less than a few τ apart pile up on each other's tails.

Worked example — ECG baseline recovery. Diagnostic ECG amplifiers are AC-coupled with a high-pass corner of 0.05 Hz, so as not to distort the slow ST segment of the heartbeat. The corresponding time constant is

$$\tau = \frac{1}{2\pi \times 0.05} = 3.18 \text{ s}$$

When the patient moves, the electrode–skin half-cell potential can shift abruptly by hundreds of millivolts — far larger than the 1 mV heart signal. Treat it as a 300 mV step into the high-pass: the output jumps by 300 mV (driving the display off-scale or the amplifier into saturation), then decays as $300 e^{-t/\tau}$ mV. It returns to within 1 mV of baseline after

$$t = \tau \ln \frac{300}{1} = 3.18 \times 5.70 = 18.2 \text{ s}$$

This is the ambulance monitor's flat line and slow drift from the opening. Monitoring-mode ECGs therefore use a 0.5 Hz corner ($\tau = 0.318$ s, recovery in 1.82 s), accepting some distortion of the ST segment, and many instruments add a "baseline reset" that momentarily shorts the coupling capacitor — a deliberate switching event that collapses the time constant. The trade-off between low-frequency fidelity and recovery time is a property of a single time constant and cannot be escaped without a different architecture (such as DC-coupled, high-resolution front ends that remove the offset digitally).

Probe compensation: a step response that must be flat

A 10× oscilloscope probe is a resistive divider (9 MΩ in the probe, 1 MΩ in the scope) with capacitance across each part (the probe's tip capacitance and the scope's input plus cable capacitance). A step applied to such a divider initially divides according to the *capacitances* (Chapter 5: series capacitors share a fast transient by charge) and finally according to the *resistances*. The two ratios agree, and the step response is flat with no exponential at all, exactly when $R_1 C_1 = R_2 C_2$. The adjustable capacitor in every passive probe sets this condition; the square-wave calibrator on the scope front panel exists to check it, displaying overshoot (overcompensation) or a rounded leading edge (undercompensation) when the condition is not met. The condition $R_1 C_1 = R_2 C_2$ is the case in which the circuit's natural response is not excited at all, because the initial and final values coincide.

What remains open

Open question	Where it is answered
What happens with two storage elements — oscillation, overshoot, damping?	Chapter 7
How are steady-state sinusoidal responses found directly, without solving the differential equation?	Chapter 8
What is the frequency response of RC and RL filters, and how is it plotted?	Chapter 8; Chapter 13 (Bode plots)
How do initial conditions and switching appear in the s-domain, and why are $-1/\tau$ the poles?	Chapter 11
How is the step response of a long cable or a fast interconnect different from a lumped RC?	Chapter 14
How is $\delta(t)$ defined as a signal, and what are its properties?	Chapter 15
How does the impulse response determine the output for any input (convolution)?	Chapter 17
How is the time–bandwidth relation $t_r \approx 0.35/f_{3dB}$ generalised?	Chapter 18
How do first-order lags appear in feedback loops and control specifications?	Chapters 33–35
How are freewheeling diodes, snubbers and gate drivers designed in practice?	*Electronics & Computing*

Next: Chapter 7 adds a second storage element, letting energy swing between a capacitor and an inductor, and shows how damping decides whether the result is a slow approach, a fastest-possible settle or a ringing oscillation.

Chapter 7 — Second-Order Circuits

Part II · Transient and AC Analysis

A technician probes a 3.3 V clock line on a new board and sees every rising edge overshoot to almost 5.7 V, then ring down in a decaying sinusoid at about 130 MHz. The logic family is rated for 3.6 V absolute maximum, and a design review is called. Before anyone redesigns the clock driver, a senior engineer swaps the 15 cm ground lead on the probe for a 1 cm spring clip. The overshoot drops to about a quarter of the step and the ringing frequency jumps to about 500 MHz. Most of what the technician saw was the probe itself: the inductance of the ground lead and the capacitance of the probe tip formed a resonant circuit, and the fast clock edge struck it like a bell. In a substation, the same mathematics appears at a very different scale: closing a breaker onto a discharged 3 Mvar capacitor bank draws a current spike of two kiloamperes oscillating at a few hundred hertz, and the bus voltage rings above its normal peak. And in an emergency room, a defibrillator delivers a 400 J pulse through a series inductor whose shape is chosen so the current through the patient's heart rises smoothly and does not swing strongly negative.

All three are **second-order circuits**: circuits whose behaviour is governed by a second-order differential equation, because they contain two independent energy-storage elements. Chapter 6 showed that one storage element gives a single exponential. Two storage elements can do something one cannot: pass energy back and forth between them, so the response can oscillate.

> **Why do circuits with an inductor and a capacitor ring, and what decides whether they ring, creep or settle as fast as possible?**
> **How are the frequency, the decay rate and the overshoot of that ringing predicted from component values, and what happens to the energy?**

Concept	Engineering destination
Series and parallel RLC circuits	Probe and interconnect ringing, LC filters in converters, capacitor-bank switching
Characteristic equation and its roots	Every linear transient; the bridge to Laplace poles (Chapter 11) and control (Chapter 34)
Overdamped response	Snubbers, slow but monotone sensor and supply responses
Critically damped response	Meter movements, defibrillator pulse shaping, fastest non-overshooting designs
Underdamped response	Ringing on digital edges, power-supply load-step overshoot, tuned sensors
Natural frequency and damping ratio	Specifications of filters, servos, accelerometers and control loops
Resonance	Radio tuning, harmonic resonance in power networks, wireless power
Transient energy exchange	Tank circuits, inductive proximity sensors, quality factor, losses

Roadmap. Building the circuits and their equations (7.1) → the characteristic equation (7.2) → natural frequency and damping ratio as the two numbers that classify everything (7.3) → the three natural responses: overdamped (7.4), critically damped (7.5), underdamped (7.6) → complete responses to steps, with probe ringing and capacitor-bank inrush as cases (7.7) → resonance as the sinusoidally driven view (7.8) → energy exchange and quality factor (7.9) → where the second-order model stops being valid (7.10).

7.1 RLC circuits

Why two storage elements make a second-order equation

In Chapter 6 each circuit had one capacitor or one inductor, and its state at any instant was fully described by one number: the capacitor voltage or the inductor current. Kirchhoff's laws plus one constitutive equation gave a first-order differential equation, and the natural response was $e^{-t/\tau}$.

Put an inductor and a capacitor in the same circuit, and two numbers are needed to describe the state: v_C and i_L. Each obeys its own first-order constitutive law,

$$i_C = C\frac{dv_C}{dt}, \qquad v_L = L\frac{di_L}{dt}$$

and the circuit laws couple them. Eliminating one variable in favour of the other produces a single **second-order** linear differential equation with constant coefficients. Equivalently, the circuit has a two-dimensional state vector $(v_C,\, i_L)$; that state-space viewpoint returns in Chapter 35.

Two storage elements do not always give an oscillating circuit. Two capacitors (or two inductors) in an RC ladder give a second-order equation too, but as Section 7.2 shows, its roots are always real. Ringing needs one element that stores energy in an electric field and one that stores it in a magnetic field, so that the energy can slosh between them.

There is also a counting subtlety. Two capacitors directly in parallel, or two inductors in series, are really one element (Chapter 5) and add only one order. The order of a circuit equals the number of *independent* storage elements: capacitors that cannot be combined and do not form a loop with only voltage sources and other capacitors, inductors that do not form a cutset with only current sources and other inductors.

The series RLC circuit

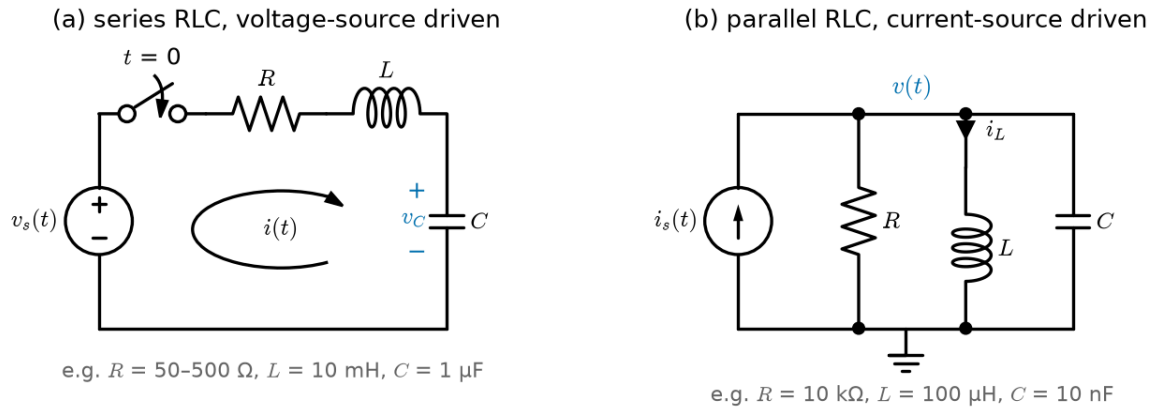


Figure 7.1 — The two canonical second-order circuits: series RLC driven by a voltage source and parallel RLC driven by a current source.

Figure 7.1 shows the two canonical circuits. In the series circuit (left), one current i flows through all three elements. KVL around the loop gives

$$Ri + L\frac{di}{dt} + v_C = v_s(t), \qquad i = C\frac{dv_C}{dt}$$

Substituting i and dividing by LC:

$$\frac{d^2 v_C}{dt^2} + \frac{R}{L}\frac{dv_C}{dt} + \frac{1}{LC}v_C = \frac{1}{LC}v_s(t)$$

Differentiating the KVL equation instead and eliminating v_C gives the same left-hand side for the current:

$$\frac{d^2 i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{1}{LC}i = \frac{1}{L}\frac{dv_s}{dt}$$

The left-hand side, which fixes the natural response, is identical. That is general: every voltage and current in a linear circuit shares the same characteristic equation, because they are all linear combinations of the same two state variables. Only the forcing terms and initial conditions differ.

The parallel RLC circuit

In the parallel circuit (Figure 7.1, right), one voltage v appears across all three elements. KCL at the top node:

$$\frac{v}{R} + i_L + C\frac{dv}{dt} = i_s(t), \qquad v = L\frac{di_L}{dt}$$

Eliminating v in favour of i_L:

$$\frac{d^2 i_L}{dt^2} + \frac{1}{RC}\frac{di_L}{dt} + \frac{1}{LC}i_L = \frac{1}{LC}i_s(t)$$

The parallel circuit is the **dual** of the series one: swap voltage and current, L and C, and R and $G = 1/R$. In particular, the damping term R/L of the series circuit becomes $1/(RC)$ in the parallel one. A large series resistance damps a series circuit; a *small* parallel resistance damps a parallel one. Keeping this duality in mind halves the formulas to remember.

> **Status of the principle: Second-order circuit equation**
> *Derived.* It follows from KCL and KVL (Chapter 2) and the linear constitutive laws of R, L and C (Chapters 1 and 5); it is exact for ideal lumped, linear, time-invariant elements.

Other second-order circuits

Real circuits rarely look exactly like Figure 7.1. A step-down (buck) converter's output filter is an inductor feeding a capacitor in parallel with a load resistor; a power-distribution network on a board is a decoupling capacitor in series with its mounting inductance and ESR; a capacitor bank on a bus sees the bus's Thévenin inductance. The method is the same in every case:

1. Reduce the resistive part of the network around the two storage elements to a Thévenin or Norton equivalent (Chapter 4).
2. Write KCL/KVL with the two state variables v_C and i_L.
3. Eliminate one variable to obtain $\ddot{x} + a_1\dot{x} + a_0 x = f(t)$.

The buck filter with a resistive load, for example, gives $\ddot{v} + \frac{1}{RC}\dot{v} + \frac{1}{LC}v = \frac{1}{LC}v_{in}$: it is a parallel-RLC damping term (R across C) with a series-RLC forcing (the source drives through L). The coefficients a_1 and a_0 are what matter, and Section 7.3 extracts two physically meaningful numbers from them.

The mechanical analogy

The series RLC equation has the same form as the equation of a mass on a spring with a viscous damper:

$$m\ddot{x} + b\dot{x} + kx = F(t)$$

with $L \leftrightarrow m$ (inertia), $R \leftrightarrow b$ (friction), $1/C \leftrightarrow k$ (stiffness), charge $q \leftrightarrow x$ and current $\leftrightarrow$ velocity. The analogy is exact, which is why MEMS accelerometers, loudspeaker cones, galvanometer movements and vehicle suspensions are all described with the same vocabulary of natural frequency and damping ratio that this chapter develops. Chapter 33 uses it systematically to model mixed electrical–mechanical systems.

7.2 Characteristic equations

From the differential equation to an algebraic one

Write the homogeneous equation (no source, which gives the natural response) in the general form

$$\frac{d^2x}{dt^2} + a_1\frac{dx}{dt} + a_0 x = 0$$

where x is any circuit variable. Constant-coefficient linear equations are solved by exponentials, because differentiating e^{st} only multiplies it by s. Substituting $x = Ae^{st}$:

$$\left(s^2 + a_1 s + a_0\right) Ae^{st} = 0$$

A nontrivial solution requires

$$s^2 + a_1 s + a_0 = 0$$

This is the **characteristic equation**. For the two canonical circuits:

$$\text{series RLC:}\quad s^2 + \frac{R}{L}s + \frac{1}{LC} = 0, \qquad \text{parallel RLC:}\quad s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0$$

Its two roots are

$$s_{1,2} = -\frac{a_1}{2} \pm \sqrt{\left(\frac{a_1}{2}\right)^2 - a_0}$$

and the general natural response is $x_n(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ when the roots are distinct. The constants A_1 and A_2 come from two initial conditions. The mathematics of linear ODEs (existence, uniqueness and the repeated-root case) is developed in *The Mathematics Universe*; here it is used.

> **Status of the principle: Characteristic-equation solution**
> *Derived*. A theorem of linear constant-coefficient ODEs: every solution of the homogeneous equation is a combination of $e^{s_k t}$ (times powers of t for repeated roots), where s_k are the characteristic roots.

Three kinds of roots

The discriminant $(a_1/2)^2 - a_0$ sorts every second-order circuit into one of three cases:

- **Positive discriminant:** two distinct real negative roots. The response is a sum of two decaying exponentials — **overdamped**.
- **Zero discriminant:** one repeated real root. The response is $(A_1 + A_2 t)e^{st}$ — **critically damped**.
- **Negative discriminant:** a complex-conjugate pair $s = -\alpha \pm j\omega_d$. The response is a decaying sinusoid — **underdamped**.

For passive circuits $a_1 \geq 0$ and $a_0 > 0$, so the real parts of the roots are never positive: the natural response always decays (or, with zero resistance, persists without decay). Active circuits — an oscillator, an amplifier with too much feedback — can make a_1 negative, and then the roots move into the right half of the s-plane and the response grows. That is the subject of Chapter 34.

Roots in the complex plane

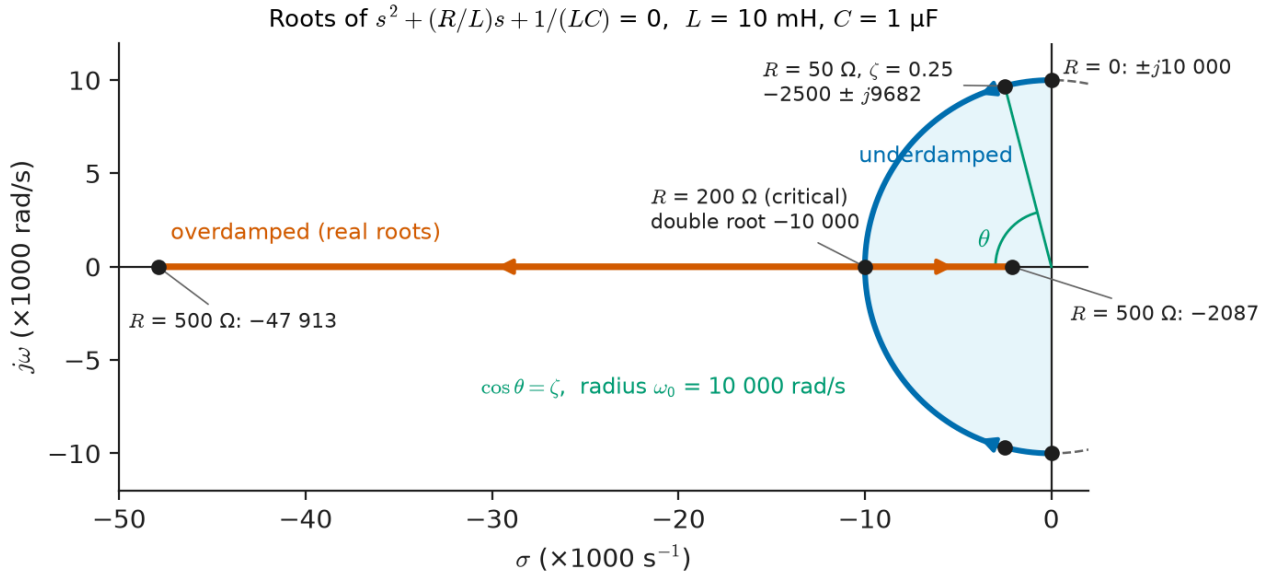


Figure 7.2 — Characteristic roots of the series RLC circuit in the s-plane as the series resistance increases from zero.

Figure 7.2 traces the roots of a series RLC with $L = 10$ mH and $C = 1$ µF as R increases. At $R = 0$ they sit on the imaginary axis at $\pm j\omega_0$; as R rises they move along a circle of radius ω_0 into the left half-plane, collide on the negative real axis at the critical resistance, and then split, one root heading toward the origin and the other toward $-\infty$. The geometry is worth memorising: for complex roots, the **distance from the origin** is ω_0, the **real part** is the decay rate, the **imaginary part** is the ringing frequency, and the **angle** from the negative real axis has cosine equal to the damping ratio ζ (Section 7.3). In Chapter 11 these roots reappear as the poles of a transfer function, and in Chapter 34 the path in Figure 7.2 is recognised as a simple root locus.

Why two RC sections never ring

For two cascaded RC sections with equal R and C, nodal analysis gives

$$\tau^2 s^2 + 3\tau s + 1 = 0, \qquad \tau = RC$$

whose discriminant $9\tau^2 - 4\tau^2 = 5\tau^2$ is always positive. With $R = 1\,\text{k}\Omega$ and $C = 1\,\mu\text{F}$ the roots are -382 and $-2618\ \text{s}^{-1}$. This is not an accident of values: a network of resistors and capacitors only (or resistors and inductors only) always has real, negative natural frequencies, because it can store energy in only one form and has no mechanism to swap it. Oscillation in a passive circuit requires both L and C.

Initial conditions

The natural response has two unknown constants, so two initial conditions are needed: $x(0^+)$ and $\dot{x}(0^+)$. Circuits supply them through the continuity rules of Chapter 5: capacitor voltage and inductor current cannot jump (unless driven by an impulse). So the procedure is:

1. Find $v_C(0^-)$ and $i_L(0^-)$ from the circuit before switching (usually a DC steady state, with C open and L shorted).

2. Set $v_C(0^+) = v_C(0^-)$, $i_L(0^+) = i_L(0^-)$.
3. Get the derivative from the constitutive law: $\dot{v}_C(0^+) = i_C(0^+)/C$ or $\dot{i}_L(0^+) = v_L(0^+)/L$, where $i_C(0^+)$ or $v_L(0^+)$ follows from KCL/KVL applied to the circuit at $t = 0^+$ with the storage elements replaced by sources of their known values.

For the series circuit with a charged capacitor V_0 and no initial current, for instance, $v_C(0^+) = V_0$ and $\dot{v}_C(0^+) = i(0^+)/C = 0$. Section 7.4 uses exactly these values.

7.3 Natural frequency and damping ratio

Two numbers that classify every second-order system

The coefficients a_1 and a_0 have units of s^{-1} and s^{-2}, and their raw values do not reveal much. Engineers rewrite the characteristic equation in the **standard form**

$$s^2 + 2\zeta\omega_0 s + \omega_0^2 = 0$$

which defines two quantities:

- the **undamped natural frequency** $\omega_0 = \sqrt{a_0}$ (rad/s), the frequency at which the circuit would oscillate with no loss;
- the **damping ratio** $\zeta = a_1/(2\omega_0)$ (dimensionless), the ratio of the actual damping to the damping that would make the circuit critically damped.

It is also convenient to define the **neper frequency** or decay rate $\alpha = \zeta\omega_0 = a_1/2$ and, for $\zeta < 1$, the **damped natural frequency** $\omega_d = \omega_0\sqrt{1-\zeta^2}$. In this notation the roots are

$$s_{1,2} = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

and the three cases of Section 7.2 become $\zeta > 1$ (overdamped), $\zeta = 1$ (critically damped), and $0 < \zeta < 1$ (underdamped). The undamped case $\zeta = 0$ gives a pure oscillation.

For the two canonical circuits:

$$\text{both:}\quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\text{series:}\quad \alpha = \frac{R}{2L},\quad \zeta = \frac{R}{2}\sqrt{\frac{C}{L}}; \qquad \text{parallel:}\quad \alpha = \frac{1}{2RC},\quad \zeta = \frac{1}{2R}\sqrt{\frac{L}{C}}$$

The quantity $Z_0 = \sqrt{L/C}$, with units of ohms, is the **characteristic impedance** of the LC pair. It is the ratio of peak voltage to peak current when energy swings between them (Section 7.9). With it the damping ratios become $\zeta = R/(2Z_0)$ for the series circuit and $\zeta = Z_0/(2R)$ for the parallel one: the series circuit is critically damped when $R = 2Z_0$ and the parallel one when $R = Z_0/2$. Comparing a resistor with Z_0 is the fastest way to judge damping by eye.

Quality factor

Filter and resonator designers prefer the **quality factor**

$$Q = \frac{1}{2\zeta}$$

so that $Q = Z_0/R$ for a series circuit and $Q = R/Z_0$ for a parallel one. A lightly damped circuit has large Q. Section 7.9 gives Q its energy meaning, and Chapters 8 and 13 its meaning as the sharpness of a resonance peak. Critical damping corresponds to $Q = 0.5$; the "maximally flat" Butterworth second-order filter to $\zeta = 0.707$, $Q = 0.707$.

> **Status of the principle: Natural frequency and damping ratio**
> *Derived.* A re-parameterisation of the characteristic equation; ω_0 and ζ carry exactly the same information as a_0 and a_1, but in a form that maps directly onto observable ringing frequency, decay and overshoot.

Representative values

Table 7.1 — Second-order circuits across EEE (typical values).

System	L	C	$f_0 = \omega_0/2\pi$	$Z_0 = \sqrt{L/C}$	Typical ζ
Scope probe, 15 cm ground lead	150 nH	10 pF	130 MHz	122 Ω	0.1 (with 25 Ω source)
Scope probe, 1 cm spring tip	10 nH	10 pF	500 MHz	32 Ω	0.4
Board decoupling capacitor + mounting	1 nH	100 nF	16 MHz	0.1 Ω	0.1–1 (set by ESR)
Buck-converter output filter, 1 Ω load	10 µH	100 µF	5.0 kHz	0.32 Ω	0.16 from load, plus ESR
Inductive proximity-sensor tank	100 µH	10 nF	159 kHz	100 Ω	0.005 ($Q = 100$)
AM radio tuned circuit	200 µH	127 pF	1 MHz	1.26 kΩ	0.005 ($Q \approx 100$)
Monophasic defibrillator, 50 Ω patient	50 mH	32 µF	126 Hz	40 Ω	0.63
11 kV capacitor bank on 250 MVA bus	1.54 mH	78.9 µF	456 Hz	4.4 Ω	0.01–0.1

The table makes one point forcefully: the same equation spans nine decades of frequency. Everything below is written in terms of ω_0 and ζ, so it applies to every row.

7.4 Overdamped response

Form of the response

For $\zeta > 1$ the roots are real, distinct and negative:

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}, \qquad s_2 < s_1 < 0$$

and the natural response is

$$x(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

From the initial conditions $x(0) = X_0$ and $\dot{x}(0) = D_0$:

$$A_1 + A_2 = X_0, \qquad s_1 A_1 + s_2 A_2 = D_0 \quad \Rightarrow \quad A_1 = \frac{D_0 - s_2 X_0}{s_1 - s_2}, \quad A_2 = \frac{s_1 X_0 - D_0}{s_1 - s_2}$$

The response is a sum of a **slow** exponential (time constant $1/|s_1|$) and a **fast** one ($1/|s_2|$). The fast term dies quickly and mainly shapes the start of the transient — it makes the slope continuous, which a single exponential could not — while the slow term dominates the tail. As ζ grows the roots separate further and the slow root approaches $-\omega_0/(2\zeta)$, which is why heavily overdamped circuits are sluggish: adding damping beyond critical makes the response *slower*, not faster.

A heavily overdamped second-order circuit behaves almost like a first-order one with time constant $1/|s_1|$. This **dominant-pole** approximation is the reason so many real systems that contain several storage elements can be treated with Chapter 6 methods.

Worked example — Overdamped discharge of a series RLC. A series RLC circuit has $L = 10$ mH and $C = 1\ \mu$F; the capacitor is charged to $V_0 = 10$ V and the inductor current is zero when a switch closes the loop through $R = 500\ \Omega$. Find $v_C(t)$.

The LC parameters are $\omega_0 = 1/\sqrt{(10^{-2})(10^{-6})} = 10^4$ rad/s ($f_0 = 1.59$ kHz) and $Z_0 = \sqrt{L/C} = 100\ \Omega$, so critical damping would need $R = 2Z_0 = 200\ \Omega$. With $R = 500\ \Omega$: $\alpha = R/(2L) = 25\,000\ \text{s}^{-1}$ and $\zeta = \alpha/\omega_0 = 2.5$ — overdamped. The roots are

$$s_{1,2} = -25\,000 \pm \sqrt{25\,000^2 - 10^8} = -25\,000 \pm 22\,913$$

so $s_1 = -2087\ \text{s}^{-1}$ and $s_2 = -47\,913\ \text{s}^{-1}$. Initial conditions: $v_C(0) = 10$ V and $\dot{v}_C(0) = i(0)/C = 0$. Then

$$A_1 = \frac{0 - (-47\,913)(10)}{-2087 + 47\,913} = 10.455\ \text{V}, \qquad A_2 = 10 - 10.455 = -0.455\ \text{V}$$

$$v_C(t) = 10.455\,e^{-2087t} - 0.455\,e^{-47\,913t}\ \text{V}$$

The fast term is only 4.6 % of the response and is gone in about 0.1 ms; the slow term, with time constant 0.48 ms, governs the rest. The capacitor voltage falls below 1 % of its initial value (0.1 V) after 2.23 ms. The discharge current $i = -C\,dv_C/dt$ peaks at 18.1 mA after 68 µs. Note that the large resistor limits the current and *slows* the discharge.

7.5 Critically damped response

Form of the response

At $\zeta = 1$ the two roots merge at $s = -\alpha = -\omega_0$. A single exponential cannot satisfy two initial conditions, and the second independent solution of the ODE is $te^{-\alpha t}$:

$$x(t) = (A_1 + A_2 t)\, e^{-\alpha t}, \qquad A_1 = X_0, \quad A_2 = D_0 + \alpha X_0$$

The critically damped response is the boundary between the two other cases. Among all responses that do not overshoot the final value, it is the one that gets there fastest for a given ω_0 — which is why it is a classic design target for indicating instruments (a moving-coil meter needle should settle quickly without wobbling past the reading), for some actuators and for certain pulse-shaping networks.

In practice "fastest without overshoot" is a fragile specification. Component tolerances move ζ by ±10–20 %, and a circuit tuned to exactly $\zeta = 1$ may end up at 0.85 (a small overshoot) or 1.2 (somewhat slower). Many designers therefore aim for $\zeta \approx 0.7$–0.8, accepting an overshoot of a few percent in exchange for a faster rise and lower sensitivity; Chapter 35 treats this trade-off as a control-design specification.

Worked example — Critically damped discharge. In the circuit of the previous example, set $R = 200\ \Omega$. Then $\alpha = 200/(2 \times 0.01) = 10^4\ \text{s}^{-1} = \omega_0$, so $\zeta = 1$. With $X_0 = 10$ V and $D_0 = 0$: $A_1 = 10$ V, $A_2 = \alpha X_0 = 10^5$ V/s, and

$$v_C(t) = 10\left(1 + 10^4 t\right) e^{-10^4 t}\ \text{V}$$

The capacitor voltage drops below 0.1 V after 0.66 ms — 3.4 times faster than with $R = 500\ \Omega$. The current

$$i(t) = -C\frac{dv_C}{dt} = \frac{V_0}{L}\, t\, e^{-\alpha t}$$

peaks at $t = 1/\alpha = 100\ \mu$s with $i_{\max} = V_0/(L\alpha)\, e^{-1} = 36.8$ mA, about twice the overdamped peak. Faster discharge costs a larger current; the energy delivered to the resistor is the same 50 µJ ($\tfrac{1}{2}CV_0^2$) in both cases.

> **Status of the principle: Critical damping as the fastest non-overshooting response**
> *Derived*. For a pure second-order system with fixed ω_0, the critically damped response reaches any band around the final value faster than any overdamped one and is the limiting case with no overshoot; for higher-order systems the statement holds only approximately, for the dominant pair.

7.6 Underdamped response

Form of the response

For $0 < \zeta < 1$ the roots are complex conjugates, $s_{1,2} = -\alpha \pm j\omega_d$. Using Euler's formula, the pair of complex exponentials combines into a real decaying sinusoid:

$$x(t) = e^{-\alpha t}\left(B_1 \cos \omega_d t + B_2 \sin \omega_d t\right), \qquad B_1 = X_0, \quad B_2 = \frac{D_0 + \alpha X_0}{\omega_d}$$

or, equivalently, $x(t) = K e^{-\alpha t}\cos(\omega_d t - \theta)$ with $K = \sqrt{B_1^2 + B_2^2}$. Three features dominate:

- **Ringing frequency.** The oscillation is at $\omega_d = \omega_0\sqrt{1-\zeta^2}$, slightly below ω_0. For $\zeta = 0.25$ the difference is only 3 %; for $\zeta = 0.1$ it is 0.5 %. In practice the frequency one reads off an oscilloscope is almost the undamped ω_0.
- **Envelope.** The oscillation is bounded by $\pm K e^{-\alpha t}$, an exponential with time constant $1/\alpha = 1/(\zeta\omega_0)$. A practical settling time to within 2 % is about $4/\alpha$.
- **Logarithmic decrement.** The ratio of successive peaks of the same sign is constant:

$$\delta = \ln\frac{x_n}{x_{n+1}} = \alpha T_d = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}}$$

Measuring two successive peaks on an oscilloscope therefore gives ζ directly — a standard way to identify the damping of a cable, a PCB power plane, a structure or a MEMS resonator from a recorded transient.

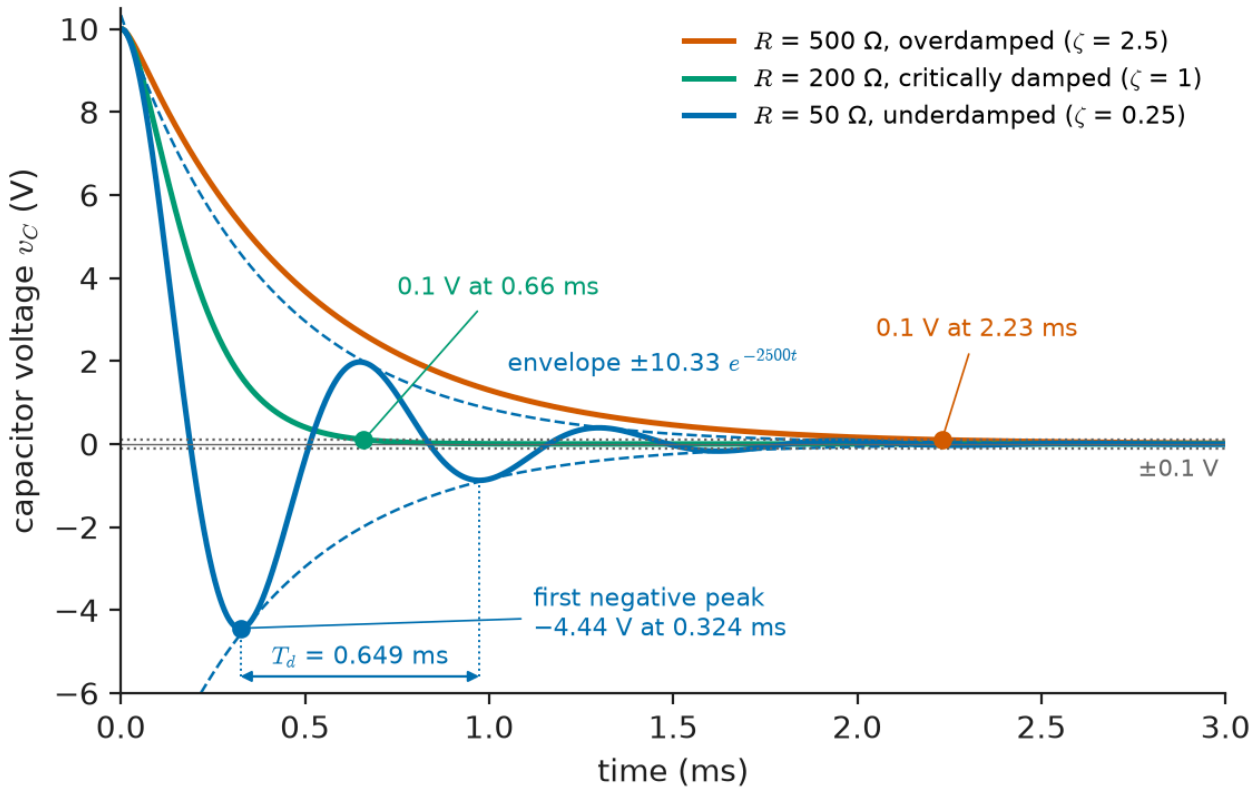


Figure 7.3 — Natural response of a series RLC circuit (L = 10 mH, C = 1 uF, capacitor initially at 10 V) for three resistances.

Worked example — Underdamped discharge. Reduce the resistance in the same circuit to $R = 50\ \Omega$. Now $\alpha = 2500\ \text{s}^{-1}$ and $\zeta = 0.25$. The damped frequency is $\omega_d = \sqrt{10^8 - 2500^2} = 9682$ rad/s ($f_d = 1.541$ kHz, period $T_d = 0.649$ ms). With $X_0 = 10$ V and $D_0 = 0$:

$$B_1 = 10\ \text{V}, \quad B_2 = \frac{2500 \times 10}{9682} = 2.582\ \text{V}$$

$$v_C(t) = e^{-2500t}\left(10\cos 9682t + 2.582\sin 9682t\right)\ \text{V}$$

The capacitor voltage first swings to a negative peak at $t = \pi/\omega_d = 0.324$ ms, where $v_C = -10\,e^{-\alpha\pi/\omega_d} = -4.44$ V: the inductor has driven the capacitor through zero and recharged it with the opposite polarity. The logarithmic decrement is $\delta = 1.62$, so each full cycle multiplies the amplitude by $e^{-1.62} = 0.197$. The envelope $10.33\,e^{-2500t}$ falls below 0.1 V after 1.85 ms, so this underdamped circuit takes longer to settle to 1 % than the critically damped one (0.66 ms), even though it is the only one of the three that crosses zero at all, at 0.19 ms. Figure 7.3 compares all three cases.

The comparison in Figure 7.3 summarises Sections 7.4–7.6: the overdamped circuit is slow because the damping resistor throttles the energy flow; the underdamped one is fast to *reach* the final value but slow to *stay* there; the critically damped one is the compromise.

> **Status of the principle: Damped-sinusoid response**
> *Derived*. Exact for an ideal linear second-order circuit with $0 < \zeta < 1$; for real components L, C and especially R vary with frequency (skin effect, dielectric loss, core loss), so the decay measured in a real circuit is often faster than the DC resistance predicts.

7.7 Complete response: forced plus natural

Structure of the complete response

With a source present the solution has two parts:

$$x(t) = x_f(t) + x_n(t)$$

where x_f is any particular solution of the full equation (the **forced response**, with the same form as the forcing: a constant for a DC step, a sinusoid at the source frequency for an AC source), and x_n is the natural response of Sections 7.4–7.6, whose two constants are now fixed by applying the initial conditions to the **sum**. This is exactly the procedure of Chapter 6, with one extra constant. For a DC step the forced response is the final DC value, found from the circuit with capacitors open and inductors shorted.

For a unit step applied to a system in standard form with zero initial conditions and unity DC gain ($\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = \omega_0^2 u(t)$), the underdamped result is

$$x(t) = 1 - \frac{e^{-\zeta\omega_0 t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi)\,, \qquad \phi = \arccos\zeta$$

This is the capacitor voltage of a series RLC when a DC step is applied through R and L, and it is the curve that power-supply engineers, oscilloscope designers and control engineers all describe with the same handful of metrics.

Step-response metrics

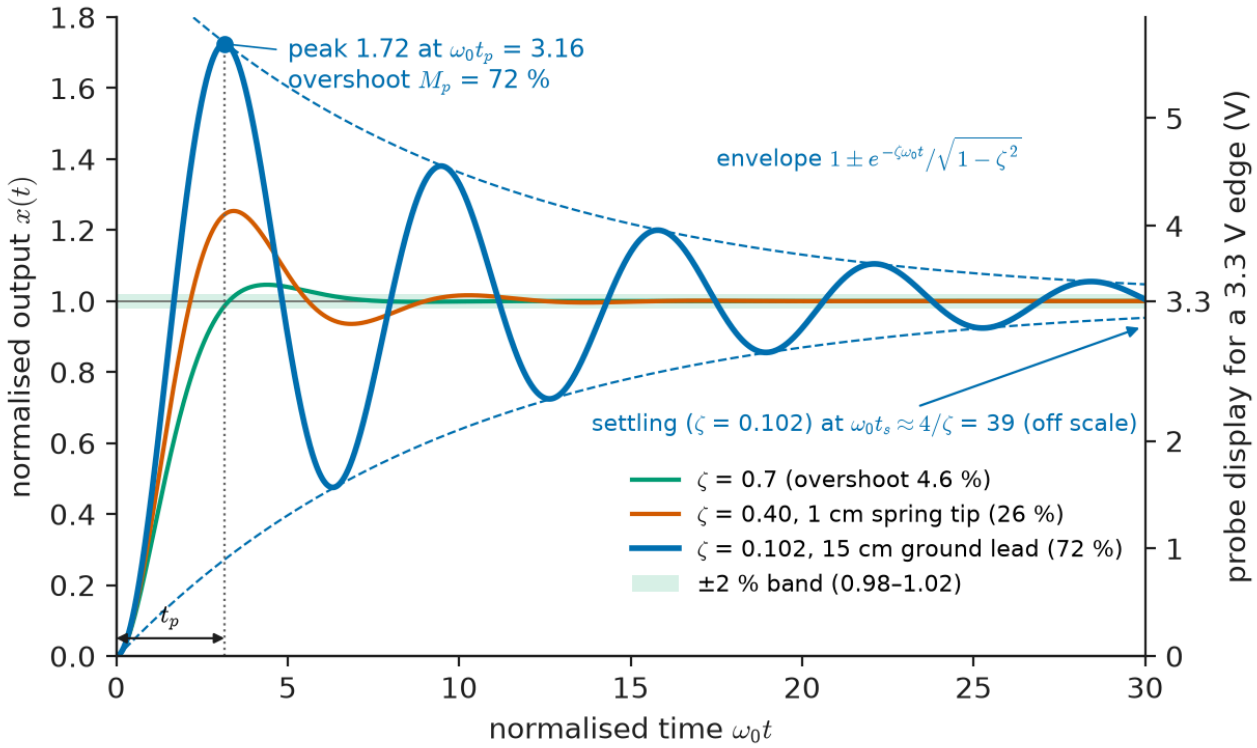


Figure 7.4 — Underdamped step response of a second-order system with the standard metrics marked, illustrated by probe ringing on a 0 to 3.3 V edge.

Figure 7.4 marks the standard quantities on an underdamped step response:

- **Peak time** $t_p = \pi/\omega_d$, the time of the first maximum.
- **Percent overshoot**

$$M_p = \exp\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$

which depends *only* on ζ: 72 % at $\zeta = 0.1$, 44 % at 0.25, 16 % at 0.5, 4.6 % at 0.7, 0 at 1.

- **Settling time** $t_s \approx 4/(\zeta\omega_0)$ to stay within ±2 % of the final value.
- **Rise time** (10–90 %), which for moderate damping is roughly $1.8/\omega_0$.

> **Status of the principle: Step-response formulas for overshoot, peak and settling time**
> *Derived* for an ideal second-order system with no zeros, driven by an ideal step. *Approximate* when applied to higher-order circuits through a dominant complex pole pair, or when the input edge has a rise time comparable to the ringing period, which reduces the overshoot.

Worked example — Oscilloscope-probe ringing. A passive 10:1 probe presents about $C = 10$ pF at its tip, and its 15 cm ground lead has an inductance of about $L = 150$ nH (roughly 1 nH per mm for a loop of this size). It probes a CMOS driver with an output resistance of $R = 25\ \Omega$ producing a 3.3 V edge with 0.5 ns rise time. Model the arrangement as a series RLC driven by a step.

$$\omega_0 = \frac{1}{\sqrt{150\times10^{-9}\times 10\times10^{-12}}} = 8.165\times10^{8}\ \text{rad/s}, \quad f_0 = 130\ \text{MHz}$$

$$Z_0 = \sqrt{L/C} = 122.5\ \Omega, \qquad \zeta = \frac{R}{2Z_0} = 0.102$$

The overshoot is $M_p = \exp(-\pi\times0.102/\sqrt{1-0.0104}) = 0.724$, so the displayed edge peaks at $3.3\times1.724 = 5.69$ V, at $t_p = \pi/\omega_d = 3.9$ ns after the edge, and rings at 129 MHz. Settling within 2 % takes $4/(\zeta\omega_0) = 48$ ns — nearly a whole period of a 20 MHz clock. The edge rise time (0.5 ns) is much shorter than the ringing period (7.7 ns), so the step model is a good one and nearly the full overshoot is excited.

What can be done? Damping to $\zeta = 0.7$ would need a total series resistance $R = 2\zeta Z_0 = 171\ \Omega$; probe manufacturers indeed build damping resistors into probe tips, but adding 146 Ω also slows the edge. The better fix is to reduce L: with a 1 cm spring tip ($L \approx 10$ nH), f_0 rises to 503 MHz, Z_0 falls to 32 Ω, $\zeta = 0.40$ and the overshoot drops to 26 % — and, because 503 MHz is above the bandwidth of a typical 200–500 MHz oscilloscope, much of the remaining ringing is filtered by the instrument. The lesson is general: when a measurement rings at a frequency set by the measurement setup, it is the setup that is being measured.

Worked example — Energising a capacitor bank. A 3 Mvar, 11 kV (line-to-line) capacitor bank is switched onto a bus whose three-phase fault level is 250 MVA. Estimate the peak inrush current and its frequency. Use a single-phase equivalent (star, line-to-neutral); the three-phase background is in Chapter 10.

The bus Thévenin reactance per phase is $X_s = V^2/S_{sc} = 11\,000^2/250\times10^6 = 0.484\ \Omega$, so $L_s = X_s/(2\pi\cdot50) = 1.54$ mH. The bank's capacitance per phase is $C = Q/(\omega V^2) = 3\times10^6/(314.16\times11\,000^2) = 78.9\ \mu$F. The inrush is the response of a series LC (the resistance is small, ζ of order 0.01–0.1) to a step equal to the instantaneous source voltage at the closing instant. The worst case is closing at the voltage peak, $V_{pk} = 11\,000\sqrt{2/3} = 8.98$ kV:

$$f_0 = \frac{1}{2\pi\sqrt{L_sC}} = 456\ \text{Hz}, \qquad Z_0 = \sqrt{L_s/C} = 4.42\ \Omega$$

$$\hat{I}_{inrush} \approx \frac{V_{pk}}{Z_0} = \frac{8980}{4.42} = 2.03\ \text{kA}$$

The bank's rated peak current is $\sqrt{2}\times3\times10^6/(\sqrt{3}\times11\,000) = 223$ A, so the inrush is 9.1 times rated, oscillating at 456 Hz and decaying over a few cycles of the mains. A neat result falls out of the algebra: the ratio of inrush to rated peak current, and the ratio $f_0/50$ Hz, are both $\sqrt{S_{sc}/Q_c} = \sqrt{250/3} = 9.13$. Simultaneously, the capacitor voltage, starting at zero and driven by an almost constant 8.98 kV over the short oscillation, overshoots toward twice the peak, almost 18 kV. In practice damping, the instant of closing and the network beyond the bus all reduce this, but switching overvoltages of 1.5–2 per unit are routinely seen, and they are why capacitor banks are fitted with inrush-limiting reactors, pre-insertion resistors or point-on-wave controlled switching. The equipment side belongs to *Electrical Energy & Machines*.

> **Boundary.**
> The single-phase equivalent captures the dominant oscillation but not the coupling between phases, the travelling-wave behaviour on long cables, or the arc physics in the breaker. Those are treated with three-phase (Chapter 10), transmission-line (Chapter 14) and power-system switching models.

7.8 Resonance

The driven view of the same circuit

So far the circuits have been left to themselves after a switching event. Drive them continuously with a sinusoid and the natural frequency appears in a different guise: **resonance**. If a series RLC is driven at frequency ω, the inductive and capacitive voltages are opposite in sign and their magnitudes are $\omega L I$ and $I/(\omega C)$. At

$$\omega = \omega_0 = \frac{1}{\sqrt{LC}}$$

they cancel exactly, the source sees only R, and the current is maximum. Meanwhile the voltages across L and C individually are Q times the source voltage. A 1 mV signal at the antenna of an AM receiver with $Q = 100$ produces 100 mV across its tuning capacitor; a 230 V supply driving an accidentally resonant series LC with $Q = 20$ could put 4.6 kV across the capacitor.

The full frequency-domain treatment — impedance, the shape of the resonance curve, bandwidth ω_0/Q — needs phasors and is developed in Chapter 8 and, with Bode plots and filters, in Chapter 13. What the time-domain view adds is the answer to *how resonance builds up*.

Build-up and ring-down

Switch on a sinusoid at exactly ω_0 to a series RLC at rest. The complete response is the forced sinusoid (amplitude V/R in current) plus a natural response that must cancel it at $t = 0$. Because the natural response oscillates at almost the same frequency and decays at rate α, the sum is a sinusoid whose amplitude grows as

$$\hat{i}(t) \approx \frac{V}{R}\left(1 - e^{-\alpha t}\right)$$

The time constant of build-up, $1/\alpha = 2L/R$, is the same as the time constant of ring-down after the source is removed. Expressed in cycles, it is Q/π cycles. A high-Q circuit therefore responds strongly but slowly: an AM tuned circuit with $Q = 100$ at 1 MHz takes about 32 µs (32 cycles) to build up to 63 % of its final amplitude, and the same time to forget a signal. This is the time-domain face of the bandwidth–selectivity trade-off: a resonator that rejects neighbouring frequencies must integrate over many cycles to distinguish them, and so cannot follow fast amplitude changes. It reappears in Chapter 18 as the time–frequency relationship.

Harmonic resonance in power systems

Resonance is not always wanted. The capacitor bank of the previous section forms, with the bus inductance, a resonant circuit at 456 Hz — the 9.1th harmonic of 50 Hz. If a nearby variable-speed drive injects 9th- or 11th-harmonic current, that current sees a high-impedance parallel resonance near 450 Hz and produces a large harmonic voltage on the bus, overheating capacitors and disturbing other loads. Power engineers therefore check the ratio $\sqrt{S_{sc}/Q_c}$ when siting banks and, when it lands near a characteristic harmonic (5th, 7th, 11th, 13th), they add a series reactor to *detune* the bank, typically to 3.8–4.2 times the fundamental (below the 5th harmonic). Chapter 9 returns to harmonics as a power-quality problem.

> **Status of the principle: Resonance at $\omega_0 = 1/\sqrt{LC}$**
> *Derived.* Exact for ideal lumped L and C; the frequency of maximum response shifts slightly from ω_0 for voltages measured across L or C and for low-Q circuits, and stray capacitance and inductance shift ω_0 in real components (self-resonance, Chapter 5).

7.9 Transient energy exchange

The lossless LC: energy sloshing back and forth

The physical reason a second-order circuit can ring is that energy can move between two different stores. In the inductor it is magnetic, in the capacitor electric:

$$w_L = \tfrac{1}{2}Li_L^2, \qquad w_C = \tfrac{1}{2}Cv_C^2$$

Consider a lossless LC loop ($R = 0$) with the capacitor initially charged to V_0. The solution is $v_C = V_0\cos\omega_0 t$, $i = V_0\sqrt{C/L}\,\sin\omega_0 t = (V_0/Z_0)\sin\omega_0 t$. Then

$$w_C = \tfrac{1}{2}CV_0^2\cos^2\omega_0 t, \qquad w_L = \tfrac{1}{2}L\frac{V_0^2 C}{L}\sin^2\omega_0 t = \tfrac{1}{2}CV_0^2\sin^2\omega_0 t$$

and $w_C + w_L = \tfrac{1}{2}CV_0^2$ at every instant. The energy oscillates between the two stores at **twice** the circuit frequency, $2\omega_0$, while the total stays constant. When all the energy is in the capacitor, the current is zero; a quarter period later the capacitor is empty and the current is at its peak V_0/Z_0. This gives the characteristic impedance its physical meaning: Z_0 is the ratio of peak voltage to peak current when a fixed amount of energy sloshes between L and C. A low-Z_0 circuit swings large currents at modest voltage (a capacitor bank on a stiff bus: 4.4 Ω), a high-Z_0 circuit swings large voltages at small current (a radio tank: 1.26 kΩ).

> **Status of the principle: Energy conservation in the LC circuit**
> *Fundamental.* The sum of stored electric and magnetic energy plus energy dissipated equals energy supplied; in a lossless lumped LC circuit the stored total is constant, as energy conservation requires.

Damping as dissipation

Add resistance and the total stored energy decays. Differentiating $w = w_L + w_C$ along the series-circuit equation:

$$\frac{dw}{dt} = Li\frac{di}{dt} + Cv_C\frac{dv_C}{dt} = i\left(L\frac{di}{dt} + v_C\right) = -Ri^2$$

The stored energy falls at exactly the rate the resistor dissipates heat. In the underdamped case the energy decays, on average, as $e^{-2\alpha t}$ (the amplitude decays as $e^{-\alpha t}$), with ripples at $2\omega_d$ as the energy moves between a store that is lossy (the inductor carries the current that heats R) and one that is not.

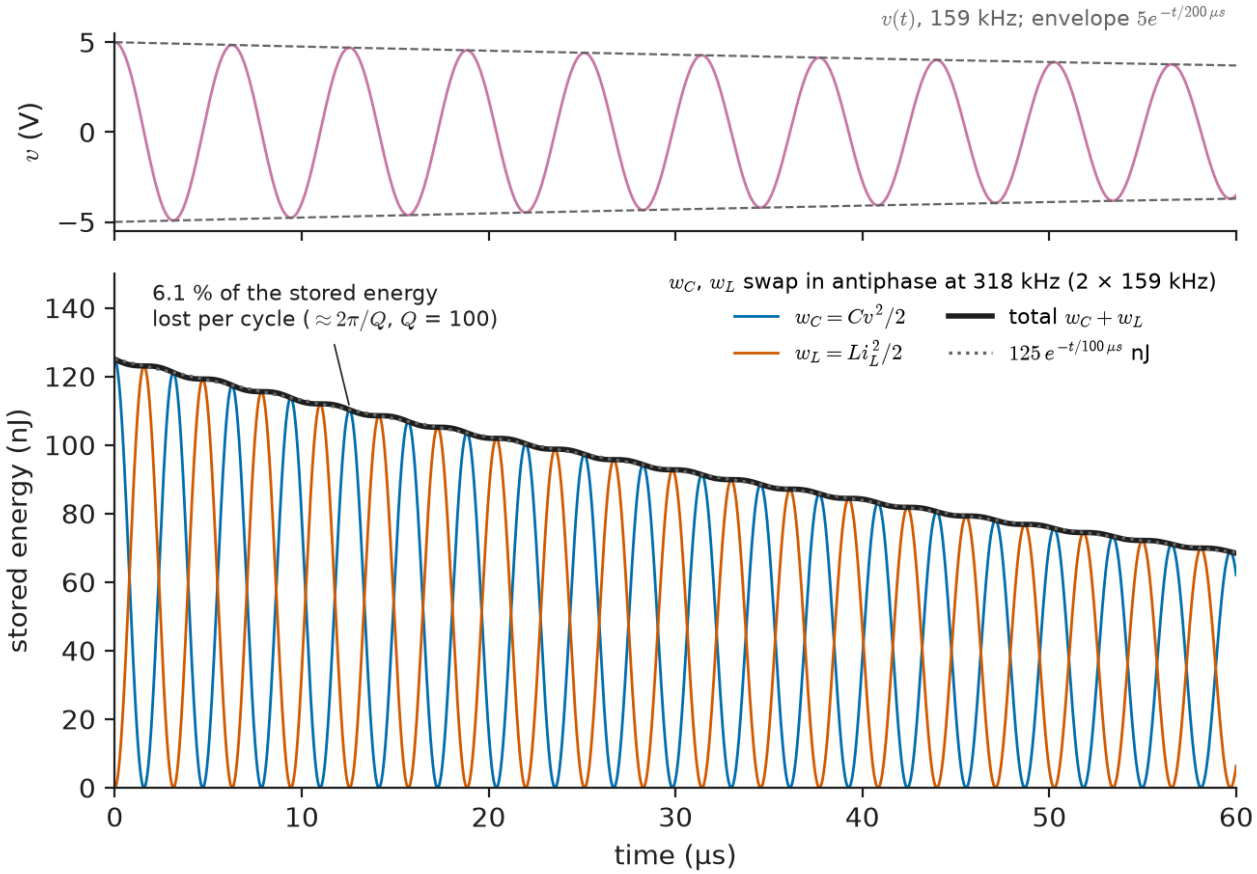


Figure 7.5 — Energy exchange in a lightly damped parallel LC tank (L = 100 uH, C = 10 nF, R = 10 kohm, Q = 100) after release with 5 V on the capacitor.

Quality factor as an energy ratio

The energy view gives the most general definition of Q:

$$Q = 2\pi\,\frac{\text{peak energy stored}}{\text{energy dissipated per cycle}}$$

For a lightly damped circuit the energy lost per cycle is $1 - e^{-2\alpha T} \approx 4\pi\alpha/\omega_0 = 2\pi/Q$ of the stored energy, consistent with $Q = \omega_0/(2\alpha) = 1/(2\zeta)$. This definition applies to any resonator — a quartz crystal ($Q \sim 10^4$–10^6), a microwave cavity (10^3–10^4), a tuning fork, a building swaying in the wind — and is how Q is measured from a ring-down when no circuit model is available.

Worked example — Inductive proximity-sensor tank. An inductive proximity sensor uses a parallel tank with $L = 100\ \mu\text{H}$ (a ferrite-cored coil facing the target), $C = 10$ nF and an equivalent parallel loss resistance $R = 10\ \text{k}\Omega$. A metal target nearby absorbs energy through eddy currents, which lowers R; the electronics detects the faster decay.

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = 159.2\ \text{kHz}, \quad Z_0 = \sqrt{L/C} = 100\ \Omega, \quad Q = \frac{R}{Z_0} = 100, \quad \zeta = 0.005$$

With 5 V on the capacitor at release, the stored energy is $\tfrac{1}{2}CV^2 = 125$ nJ and the peak inductor current is $V/Z_0 = 50$ mA. The voltage envelope decays with time constant $2RC = 200\ \mu$s — about 32 cycles — and the stored energy with time constant $RC = 100\ \mu$s. Each cycle loses $1 - e^{-2\pi/Q} = 6.1$ % of the stored energy ($2\pi/Q = 6.3$ % in the small-loss approximation), so after 10 cycles the amplitude is $e^{-10\pi/Q} = 73$ % of its starting value. If a steel target halves R to 5 kΩ, Q falls to 50 and the amplitude after 10 cycles drops to 53 %: a change easily detected by a comparator. Figure 7.5 shows the energy sloshing between w_L and w_C while the total decays.

Worked example — Defibrillator pulse. A classic monophasic damped-sine defibrillator stores energy in a capacitor $C = 32\ \mu$F charged to 5 kV and discharges it through a series inductor $L = 50$ mH into the patient, modelled as a resistance $R = 50\ \Omega$ (transthoracic impedance is typically 25–150 Ω). The stored energy is $\tfrac{1}{2}CV_0^2 = 400$ J.

$$\omega_0 = \frac{1}{\sqrt{0.05 \times 32 \times 10^{-6}}} = 790.6\ \text{rad/s}\ (f_0 = 126\ \text{Hz}), \quad \alpha = \frac{R}{2L} = 500\ \text{s}^{-1}, \quad \zeta = 0.632$$

The circuit is underdamped. With $v_C(0) = V_0$ and $i(0) = 0$, the patient current is

$$i(t) = \frac{V_0}{\omega_d L}\, e^{-\alpha t} \sin \omega_d t, \qquad \omega_d = 612.4\ \text{rad/s}$$

It peaks at 61.4 A after 1.45 ms and crosses zero at $\pi/\omega_d = 5.13$ ms, after which it reverses with a peak of only 4.7 A — the ratio is $e^{-\alpha\pi/\omega_d} = 7.7$ %, the same overshoot formula as in Section 7.7. Integrating Ri^2 over the pulse returns the full 400 J (in practice internal resistance of the inductor and electrodes takes a share). The inductor's purpose is visible here: without it, the capacitor would dump its charge as a single exponential with a peak of 100 A and a sharp leading edge; with it, the current rises smoothly over about 1.5 ms. The damping, however, depends on the patient. At 25 Ω, $\zeta = 0.32$ and the reverse swing is larger; at 100 Ω, $\zeta = 1.26$ and the pulse is overdamped and longer. The critical resistance is $2Z_0 = 79\ \Omega$. Modern biphasic defibrillators control the waveform electronically instead, partly to remove this dependence; the circuit theory above is the reason that was necessary.

7.10 Where the second-order model ends

The second-order model is powerful because so many real situations are dominated by one L and one C. It is still a model, and it fails in recognisable ways.

More than two storage elements. A real buck converter has input and output filters; a probe has a cable, a compensation network and an input amplifier. The full circuit is of higher order, and its response is a sum of several exponentials and damped sinusoids. Often one complex pair dominates — it is closest to the imaginary axis and decays slowest — and the second-order formulas apply to it

approximately. When two pairs have similar frequencies, the response beats and the simple overshoot formula fails. Chapter 11 handles arbitrary order with Laplace transforms and partial fractions.

Frequency-dependent losses. The resistance that damps a ringing PCB trace is not its DC resistance: skin effect raises it roughly as $\sqrt{f}$, and dielectric loss adds a term proportional to f. Damping inferred from a measured decrement is therefore often several times what the DC values suggest. Capacitor ESR and inductor core loss behave similarly.

Nonlinearity. A ferrite inductor saturates at high current, so L falls as the current rises; ceramic capacitors (class II dielectrics) lose capacitance with DC bias. Large transients in power circuits therefore ring at a frequency that changes during the transient. An extreme case is **ferroresonance**, where a saturable transformer inductance and cable or grading capacitance produce sustained, erratic overvoltages that no linear model predicts.

Distributed behaviour. The lumped model assumes the circuit is small compared with a wavelength (Chapter 1). At 500 MHz the free-space wavelength is 60 cm, and on FR-4 about 30 cm; a 5 cm trace is already a sixth of a wavelength. Beyond this point the "ringing" is really reflections on a transmission line, with a rich set of resonances instead of one. Chapter 14 provides that model.

> **Boundary.**
> The derivation of solutions to linear ODEs, including repeated roots and the theory of complex exponentials, belongs to *The Mathematics Universe*. Ringing in transistor drivers, snubber design for switching devices and the circuitry of oscilloscope probes belong to *Electronics & Computing*; capacitor-bank switching equipment and power-system protection to *Electrical Energy & Machines*.

What remains open

Open question	Where it is answered
How does a circuit respond to a sinusoid that has been applied for a long time, without solving the differential equation?	Chapter 8 (phasors and impedance)
How sharp is a resonance in the frequency domain, and what are its bandwidth and power?	Chapter 8 (resonance, frequency response); Chapter 13 (quality factor, bandwidth)
How are circuits with many storage elements and arbitrary inputs solved systematically?	Chapter 11 (Laplace-domain analysis, poles, partial fractions)
Why do higher-frequency interconnects ring with many resonances rather than one?	Chapter 14 (transmission lines)
How do overshoot and settling-time specifications become controller design targets?	Chapter 34 (characteristic equations, root locus); Chapter 35 (transient-response specifications)
What happens when the damping term becomes negative and a circuit oscillates on purpose?	Chapter 34 (stability); oscillator circuits in *Electronics & Computing*
How are electrical and mechanical second-order systems combined into one model?	Chapter 33 (dynamic-system modeling)

Next: Chapter 8 drives these circuits with sinusoids and replaces the differential equation with complex-number algebra, so that resonance, impedance and frequency response can be calculated directly.

Chapter 8 — Sinusoidal Steady-State Analysis

Part II · Transient and AC Analysis

A maintenance electrician clamps a meter on the supply to a workshop air compressor and reads 230 V and 23 A. Multiplying, she expects the motor to be drawing 5.29 kW, yet the plug-in energy monitor in the circuit says 4.23 kW, and the motor nameplate says nothing about either number. No instrument is faulty. The current is not in step with the voltage: it reaches its peak about 2 ms after the voltage does, and for part of every cycle energy flows back from the motor's magnetic field into the supply. A cardiologist meets the same mathematics from the other side. On a bedside monitor an ECG shows the ST segment displaced, suggesting ischaemia, but the same patient's diagnostic 12-lead recording looks normal. The monitor's input filter, set to remove baseline wander, passes the slow components of the heartbeat with the wrong amplitude and a phase shift of tens of degrees, and that distorts the waveform shape.

Both problems involve circuits driven by sinusoids long enough for every transient of Chapters 6 and 7 to have died away: the **sinusoidal steady state**. In this regime the differential equations of the last two chapters collapse into algebra with complex numbers, and the whole of DC circuit analysis (Chapters 2–4) can be reused with resistances replaced by complex impedances.

> **Why can a linear circuit driven by a sinusoid be analysed with complex numbers instead of differential equations?**
> **How are currents, voltages, resonances, frequency responses and power found in AC circuits, and what does "230 V" actually mean?**

Concept	Engineering destination
Sinusoidal excitation	Mains supplies, test tones, carriers, sensor excitation
RMS and average values	Mains ratings, meter readings, heating of cables and resistors, PWM drives
Phasors	Every AC calculation in power, electronics and telecom
Complex impedance	Motor and transformer equivalent circuits, filter design, cable loading
Admittance	Parallel loads, power-factor correction, network admittance matrices
AC nodal and mesh analysis	Multi-source AC networks, bridge circuits, power-flow equations
Series and parallel AC circuits	Load modelling, crossover networks, LC filters
Resonance	Radio tuning, tank circuits, harmonic filters, wireless charging
Frequency response	Audio filters, ECG and EEG front ends, anti-alias filters, instrument bandwidth
Power calculations	Load ratings, efficiency, meter readings (developed in Chapter 9)

Roadmap. Sinusoids and why they matter (8.1) → measuring their size with RMS and average values (8.2) → phasors (8.3) → impedance (8.4) and admittance (8.5) → series and parallel AC circuits (8.6) → nodal and mesh analysis in the phasor domain (8.7) → resonance (8.8) → frequency response (8.9) → average power from phasors (8.10) → the limits of the method (8.11).

8.1 Sinusoidal excitation

Why sinusoids

Sinusoids deserve a chapter of their own for three reasons.

1. **They are what generators produce.** A coil rotating at constant speed in a uniform magnetic field produces a sinusoidal EMF; every synchronous generator in a power system is designed to approximate this. Oscillators in radios, function generators and sensor excitation circuits produce sinusoids because a lightly damped resonator (Chapter 7) naturally rings sinusoidally.
2. **They pass through linear circuits unchanged in shape.** Differentiating or integrating a sinusoid gives a sinusoid of the same frequency. A linear, time-invariant circuit driven by a sinusoid therefore responds, in steady state, with a sinusoid of the same frequency; only amplitude and phase change. In the language of Chapter 17, complex exponentials are the **eigenfunctions** of linear time-invariant systems.
3. **Everything else can be built from them.** Fourier's theorem (Chapter 18) represents periodic waveforms as sums of sinusoids, and aperiodic ones as integrals over sinusoids. Knowing a circuit's response to every sinusoid is knowing its response to everything.

Parameters of a sinusoid

A sinusoidal voltage is written

$$v(t) = V_m \cos(\omega t + \theta)$$

where V_m is the **amplitude** (peak value), $\omega = 2\pi f$ the **angular frequency** in rad/s, $f = 1/T$ the frequency in hertz, T the period and θ the **phase angle**. This book uses the cosine as the reference; a sine is a cosine delayed by 90°: $\sin \omega t = \cos(\omega t - 90^\circ)$. Mixing degrees and radians inside $\omega t + \theta$ is common in engineering practice and harmless as long as the conversion is made before evaluating.

Two sinusoids of the same frequency differ only in amplitude and phase. If $v = V_m \cos(\omega t + \theta_v)$ and $i = I_m \cos(\omega t + \theta_i)$, the voltage **leads** the current by $\theta_v - \theta_i$ when that difference is positive: its peaks occur earlier. A phase difference ϕ corresponds to a time shift $\Delta t = \phi / \omega$; at 50 Hz, 36.87° is 2.05 ms, the lag the electrician's compressor current showed.

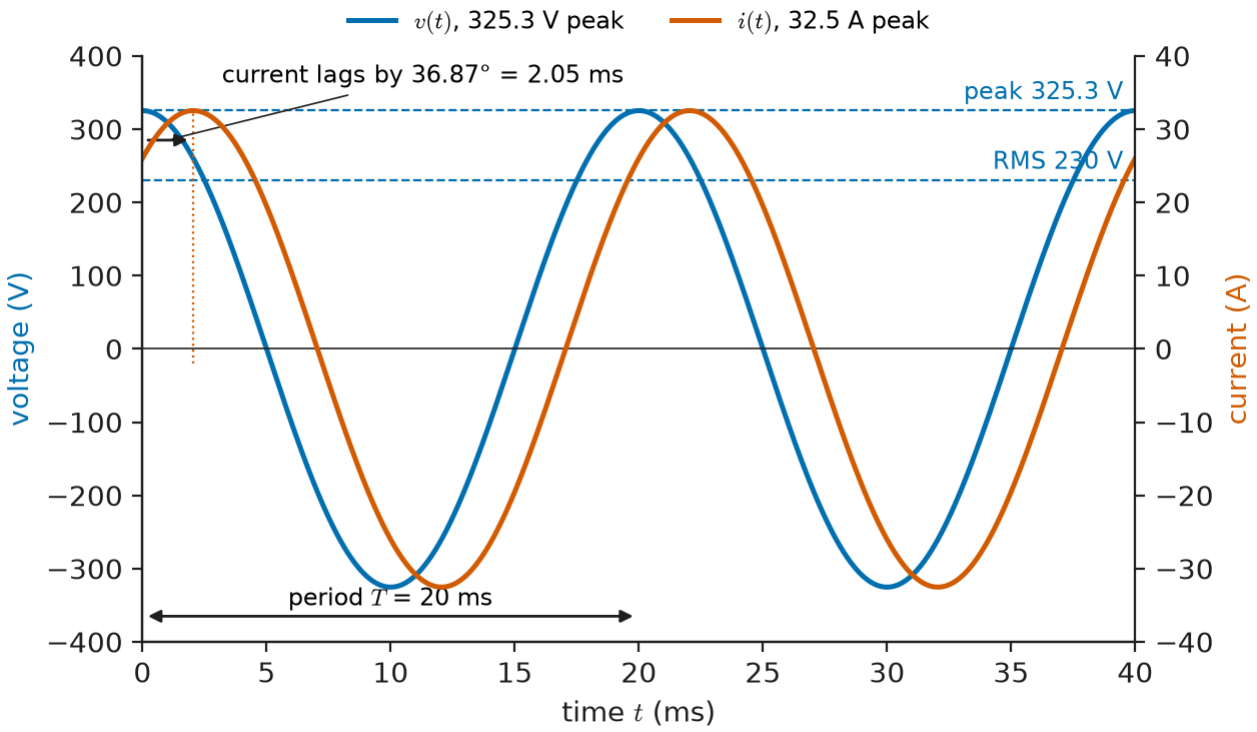


Figure 8.1 — A 230 V, 50 Hz mains voltage and the 23 A current of an inductive load lagging it by 36.87 degrees.

Frequencies across EEE

Table 8.1 — Representative sinusoidal frequencies.

Application	Frequency	Notes
Public mains supply	50 Hz (Europe, Asia, Africa, Australia), 60 Hz (Americas, parts of Asia)	Held within about ±0.2 Hz in large grids
Railway traction (some European systems)	16.7 Hz	Legacy low-frequency AC
Aircraft electrical systems	400 Hz	Lighter transformers and generators
Audio test tone	1 kHz	Audio band 20 Hz–20 kHz
Induction heating, wireless EV charging	10–100 kHz (EV charging about 85 kHz)	Resonant converters
Medical ultrasound	2–15 MHz	Piezoelectric transducers
AM broadcast (medium wave)	530–1700 kHz	Channel spacing 9 or 10 kHz
FM broadcast	88–108 MHz	
Cellular, Wi-Fi	0.7–6 GHz	e.g. Wi-Fi 2.4 GHz and 5 GHz bands
Automotive radar	76–81 GHz	

When is a circuit in steady state?

Switching a sinusoidal source onto a circuit produces, as Chapters 6 and 7 showed, a forced response plus a natural response. The natural response decays with the circuit's time constants (or as $e^{-\alpha t}$ for second-order circuits), and once it has gone only the forced sinusoid remains. "Steady state" therefore means "several of the slowest time constants after the last disturbance". For a mains-frequency motor circuit with L/R of a few milliseconds that is a few cycles; for a high-Q crystal it may be milliseconds to seconds. Everything in this chapter assumes that this has happened. Switching transients, inrush and start-up are the province of Chapters 6, 7 and 11.

8.2 RMS and average values

Average value

The **average** (mean, DC) value of a periodic waveform is

$$V_{avg} = \frac{1}{T}\int_{0}^{T} v(t)\,dt$$

A pure sinusoid has zero average. Rectified waveforms do not: a half-wave rectified sine of peak V_m has average V_m/π, a full-wave rectified sine $2V_m/\pi$. The average is what a moving-coil (d'Arsonval) meter

responds to, what charges a battery, what electroplates a metal, and what an ideal low-pass filter extracts.

RMS value

For heating, the average is the wrong measure: a resistor heats equally for positive and negative current. The **root-mean-square** (effective) value is defined as the DC value that would dissipate the same average power in a resistor:

$$\frac{V_{rms}^2}{R} = \frac{1}{T}\int_0^T \frac{v^2(t)}{R}\,dt \quad \Rightarrow \quad V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt}$$

For a sinusoid, the average of $\cos^2$ over a period is $\tfrac{1}{2}$, so

$$V_{rms} = \frac{V_m}{\sqrt{2}} \approx 0.707\,V_m$$

Mains voltages are always quoted as RMS: "230 V" means a peak of $230\sqrt{2} = 325.3$ V and a peak-to-peak swing of 650.5 V; "120 V" means a peak of 169.7 V. Insulation, capacitors and semiconductors on mains equipment must therefore be rated for the peak, not the RMS value, plus transients.

> **Status of the principle: RMS as the heating-equivalent value**
> *Derived*. It follows from Joule heating $p = v^2/R$ and the definition of average power; it is exact for any waveform in a linear resistor whose resistance does not change during the cycle.

If a waveform is a sum of components at different frequencies (including DC), the cross terms average to zero over a common period and the RMS values add in quadrature:

$$V_{rms} = \sqrt{V_0^2 + V_{1,rms}^2 + V_{2,rms}^2 + \cdots}$$

This result underpins the treatment of harmonics in Chapter 9 and of noise power in Chapter 29.

Form factor and crest factor

Two ratios describe the shape of a waveform:

$$\text{form factor} = \frac{V_{rms}}{|V|_{avg}}, \qquad \text{crest factor} = \frac{V_{peak}}{V_{rms}}$$

where $|V|_{avg}$ is the average of the rectified waveform.

Table 8.2 — RMS, rectified average, form factor and crest factor of common waveforms (peak V_m).

Waveform	RMS	Rectified average	Form factor	Crest factor
Sine	$V_m/\sqrt{2} = 0.707V_m$	$2V_m/\pi = 0.637V_m$	1.111	1.414
Full-wave rectified sine	$0.707V_m$	$0.637V_m$	1.111	1.414
Half-wave rectified sine	$V_m/2$	$V_m/\pi = 0.318V_m$	1.571	2.0
Square ($\pm V_m$)	V_m	V_m	1.0	1.0
Triangle or sawtooth	$V_m/\sqrt{3} = 0.577V_m$	$V_m/2$	1.155	1.732
Unipolar pulse train, duty D	$V_m\sqrt{D}$	DV_m	$1/\sqrt{D}$	$1/\sqrt{D}$
Rectifier–capacitor (SMPS) input current	—	—	typically 1.5–2.5	typically 2.5–3.5

Worked example — True RMS versus average-responding meters. Many low-cost multimeters measure the rectified average and multiply by 1.111 — the form factor of a sine — to display "RMS". What do they show for other waveforms?

For a square wave of peak 1 V, the rectified average is 1 V, so the meter shows 1.111 V; the true RMS is 1 V, an error of +11.1 %. For a triangle wave of peak 1 V, the average is 0.5 V, so the meter shows 0.555 V against a true 0.577 V, −3.8 %. For a half-wave rectified sine of peak 1 V (AC coupled readings aside), the average is 0.318 V, the display 0.354 V, the true RMS 0.5 V: −29 %. For a 12 V PWM drive at 25 % duty, a DC meter reads the average, 3 V, while the heating in a resistive load corresponds to $12\sqrt{0.25} = 6$ V RMS — four times the power the average would suggest. This is why electricians checking variable-speed drives, dimmers and switch-mode supplies use **true-RMS** instruments, which compute $\sqrt{\overline{v^2}}$ from samples or with a thermal or analog-computing converter. Even these are specified only up to a stated crest factor (typically 3 at full scale) and bandwidth.

8.3 Phasors

The idea

Solving a circuit with sinusoidal sources directly in the time domain means substituting $A\cos(\omega t + \phi)$ into the differential equations and grinding through trigonometric identities. The phasor method replaces this with complex arithmetic, based on Euler's formula

$$e^{j\phi} = \cos\phi + j\sin\phi$$

A sinusoid can be written as the real part of a rotating complex number:

$$v(t) = V_m\cos(\omega t + \theta) = \mathrm{Re}\left\{V_m e^{j\theta} e^{j\omega t}\right\}$$

All voltages and currents in a linear circuit in sinusoidal steady state share the same factor $e^{j\omega t}$. Only the complex constant in front — amplitude and phase — differs from one variable to another. That constant is the **phasor**.

RMS phasors

This book uses **RMS phasors**, written in bold, because they make the power formulas of Section 8.10 and Chapter 9 free of factors of $\frac{1}{2}$:

$$v(t) = \sqrt{2}\,V\cos(\omega t + \theta) \quad \longleftrightarrow \quad \mathbf{V} = V e^{j\theta} = V\angle\theta$$

where V is the RMS value. The reverse step is $v(t) = \mathrm{Re}\{\sqrt{2}\,\mathbf{V}e^{j\omega t}\}$. Many circuit-theory texts use amplitude (peak) phasors instead; the analysis is identical, only power formulas pick up a factor $\frac{1}{2}$. Whichever convention is used, it must be used consistently in a calculation.

Phasors can be written in polar form $V\angle\theta$ or rectangular form $a + jb$, with $V = \sqrt{a^2+b^2}$, $\theta = \arctan(b/a)$ (in the correct quadrant). Addition is easiest in rectangular form, multiplication and division in polar form. The algebra of complex numbers itself is covered in *The Mathematics Universe*.

Two properties that make phasors work

Linearity. The phasor of a sum of same-frequency sinusoids is the sum of their phasors. KCL and KVL, which are sums of currents and voltages, therefore hold for phasors exactly as for instantaneous values:

$$\sum_k \mathbf{I}_k = 0 \text{ at a node}, \qquad \sum_k \mathbf{V}_k = 0 \text{ around a loop}$$

Differentiation becomes multiplication by $j\omega$. Since $\frac{d}{dt}e^{j\omega t} = j\omega e^{j\omega t}$,

$$\frac{dv}{dt} \;\longleftrightarrow\; j\omega\mathbf{V}, \qquad \int v\,dt \;\longleftrightarrow\; \frac{\mathbf{V}}{j\omega}$$

Multiplying by j rotates a phasor by +90°; dividing by j rotates it by −90°. A differential equation in the time domain thus becomes an algebraic equation in the phasor domain.

> **Status of the principle: Phasor transformation**
> *Derived.* It follows from linearity and time invariance of the circuit and from the eigenfunction property of $e^{j\omega t}$; it is exact for linear time-invariant circuits in sinusoidal steady state at a single frequency, and gives nothing about transients.

Phasor diagrams

A **phasor diagram** draws the phasors as arrows in the complex plane. It shows at a glance which quantity leads and which lags, and it makes KVL and KCL visible as closed polygons of arrows. Power engineers draw them for every machine and line; instrumentation engineers use them for bridges and lock-in amplifiers. Figure 8.3 shows the diagram for the compressor of the opening, solved in the next section.

8.4 Complex impedance

Impedance of R, L and C

Apply the phasor rules to the three constitutive laws:

$$v = Ri \;\to\; \mathbf{V} = R\,\mathbf{I}, \qquad v = L\frac{di}{dt} \;\to\; \mathbf{V} = j\omega L\,\mathbf{I}, \qquad i = C\frac{dv}{dt} \;\to\; \mathbf{V} = \frac{1}{j\omega C}\,\mathbf{I}$$

Each is an "Ohm's law" $\mathbf{V} = Z\mathbf{I}$ with a complex **impedance** Z:

$$Z_R = R, \qquad Z_L = j\omega L, \qquad Z_C = \frac{1}{j\omega C} = -\frac{j}{\omega C}$$

In a resistor, voltage and current are in phase. In an inductor, the voltage leads the current by 90° (the current builds up only after voltage has been applied). In a capacitor, the current leads the voltage by 90° (charge must flow before voltage appears). The mnemonic "ELI the ICE man" — E leads I in an L, I leads E in a C — has served generations of electricians.

A general two-terminal impedance is written

$$Z = R + jX = |Z|\angle\theta_Z$$

where R is the **resistance** and X the **reactance**, both in ohms. Inductive reactance $X_L = \omega L$ is positive; capacitive reactance $X_C = -1/(\omega C)$ is negative. The impedance angle θ_Z is the angle by which voltage leads current. Figure 8.2 places these impedances in the complex plane: inductive impedances lie in the upper half, capacitive ones in the lower. The resistance $R = \mathrm{Re}\{Z\}$ of a composite impedance is in general *not* the resistance of any single resistor; it is whatever real part results from the combination.

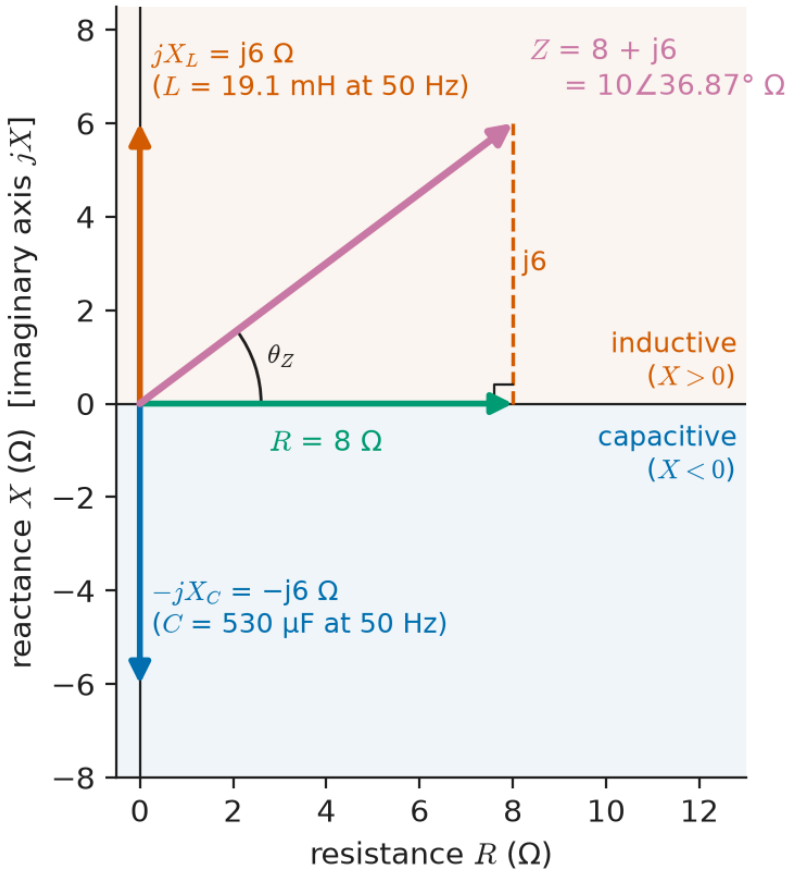


Figure 8.2 — Impedances in the complex plane: resistor, inductor, capacitor and the 8 + j6 ohm compressor load.

Table 8.3 — Magnitude of reactance at representative frequencies.

Component	50 Hz	1 kHz	1 MHz
1 µH (a few cm of wire or a small SMD inductor)	0.31 mΩ	6.3 mΩ	6.3 Ω
10 mH (audio or mains choke)	3.14 Ω	62.8 Ω	62.8 kΩ
100 nF (decoupling capacitor)	31.8 kΩ	1.59 kΩ	1.59 Ω
1 µF	3.18 kΩ	159 Ω	0.16 Ω
100 µF (electrolytic)	31.8 Ω	1.59 Ω	1.6 mΩ (ESR and ESL dominate)

Table 8.3 explains much practical folklore: a 100 nF capacitor is practically an open circuit at mains frequency but a short at radio frequencies, which is why it decouples digital supplies; a few centimetres of wire are irrelevant at 50 Hz and decisive at 100 MHz (Chapter 7's probe). At high frequency every real capacitor has series inductance and every real inductor parallel capacitance (Chapter 5), and the simple impedances above apply only below self-resonance.

> **Status of the principle: Impedance of ideal R, L and C**
> *Derived*. Exact for ideal lumped linear elements in sinusoidal steady state; *approximate* for real components, whose parasitic elements, skin effect and dielectric loss make R and X depend on frequency in more complex ways.

Impedances combine like resistances

Because KVL and KCL hold for phasors and every element obeys $\mathbf{V} = Z\mathbf{I}$, all the DC network rules of Chapters 2–4 carry over with R replaced by Z: series impedances add, parallel admittances add, voltage and current dividers, source transformations, Thévenin and Norton equivalents, superposition (for sources at the same frequency), nodal and mesh analysis. The only change is that the arithmetic is complex.

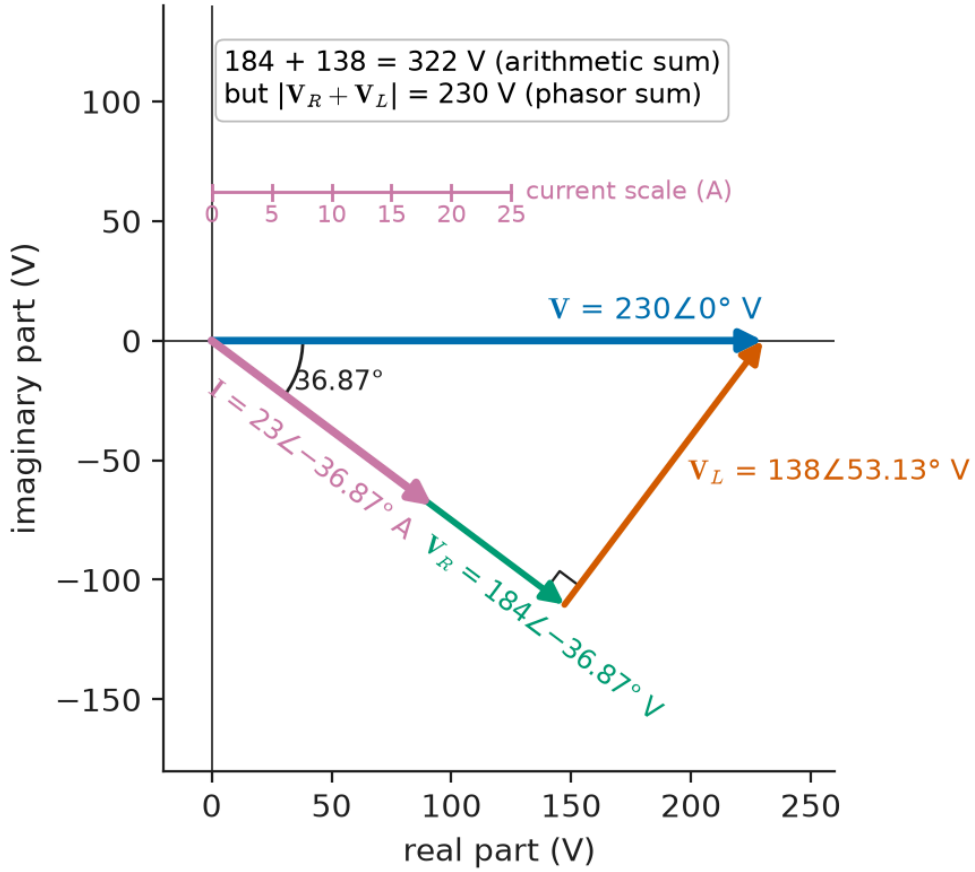


Figure 8.3 — Phasor diagram of the compressor modelled as R = 8 ohm in series with X_L = 6 ohm on a 230 V, 50 Hz supply.

Worked example — The compressor as a series RL load. Model the compressor of the opening as $R = 8\ \Omega$ in series with $L = 19.1$ mH on a 230 V, 50 Hz supply. Find the current and the element voltages.

The reactance is $X_L = \omega L = 2\pi \times 50 \times 0.0191 = 6.00\ \Omega$, so

$$Z = 8 + j6 = 10\angle 36.87^\circ\ \Omega$$

Taking the supply as reference, $\mathbf{V} = 230\angle 0^\circ$ V:

$$\mathbf{I} = \frac{\mathbf{V}}{Z} = \frac{230\angle 0^\circ}{10\angle 36.87^\circ} = 23\angle -36.87^\circ \text{ A} = 18.4 - j13.8 \text{ A}$$

$$\mathbf{V}_R = R\mathbf{I} = 184\angle -36.87^\circ \text{ V}, \qquad \mathbf{V}_L = jX_L\mathbf{I} = 138\angle 53.13^\circ \text{ V}$$

The two element voltages add to 322 V arithmetically but to 230 V as phasors, because they are 90° apart: $\sqrt{184^2 + 138^2} = 230$. A voltmeter placed across each in turn would read 184 V and 138 V, which confuses anyone expecting KVL to hold for RMS readings. KVL holds for phasors and instantaneous values, never for RMS magnitudes. Figure 8.3 shows the closed triangle. The 36.87° lag (2.05 ms at 50 Hz) is the one in Figure 8.1, and the reason the meter product $230 \times 23 = 5290$ disagrees with the 4232 W energy monitor is the subject of Section 8.10.

8.5 Admittance

Definition

The reciprocal of impedance is the **admittance**

$$Y = \frac{1}{Z} = \frac{\mathbf{I}}{\mathbf{V}} = G + jB$$

measured in siemens (S). Its real part G is the **conductance** and its imaginary part B the **susceptance**. For the ideal elements:

$$Y_R = \frac{1}{R} = G, \qquad Y_L = \frac{1}{j\omega L} = -\frac{j}{\omega L}, \qquad Y_C = j\omega C$$

An inductor has negative susceptance, a capacitor positive — the opposite signs to their reactances.

Converting between impedance and admittance

For $Z = R + jX$,

$$Y = \frac{1}{R + jX} = \frac{R}{R^2 + X^2} - j\frac{X}{R^2 + X^2}$$

so $G = R/(R^2 + X^2)$ and $B = -X/(R^2 + X^2)$. The conductance of a series RL branch is **not** $1/R$ unless $X = 0$. For the compressor, $Z = 8 + j6\ \Omega$ gives

$$Y = \frac{8 - j6}{100} = 0.08 - j0.06 \text{ S}$$

Why admittance is useful

Parallel admittances add, exactly as parallel conductances do. Power systems are full of loads in parallel across a common supply, so power engineers reason with admittances. The network **admittance matrix** $\mathbf{Y}_{bus}$ of a power system — the AC version of the nodal conductance matrix of Chapter 3 — is the starting point of every load-flow study.

Admittance also makes one fact plain: the compressor's admittance $0.08 - j0.06$ S is the same as a conductance of 0.08 S ($R_p = 12.5\ \Omega$) in **parallel** with an inductive susceptance of −0.06 S ($X_p = 16.67\ \Omega$). At a single frequency, any series RX branch has an equivalent parallel RX pair, and vice versa. The equivalence holds only at that frequency: change ω and the two models diverge.

> **Status of the principle: Series–parallel equivalence at one frequency**
> *Derived*. Two impedances with equal complex values are indistinguishable at that frequency; the equivalence breaks at other frequencies because the element values scale differently with ω.

8.6 Series and parallel AC circuits

Series circuits and the voltage divider

Impedances in series add:

$$Z_{eq} = Z_1 + Z_2 + \cdots + Z_n$$

and the voltage divides in proportion to impedance:

$$\mathbf{V}_k = \frac{Z_k}{Z_{eq}}\mathbf{V}$$

Because the impedances are complex, the divided voltages can have different phases from the source, and an individual element voltage can exceed the source voltage when inductive and capacitive reactances partly cancel in Z_{eq} — the voltage magnification of a series resonance (Section 8.8).

Parallel circuits and the current divider

Admittances in parallel add, $Y_{eq} = Y_1 + Y_2 + \cdots$, and currents divide in proportion to admittance:

$$\mathbf{I}_k = \frac{Y_k}{Y_{eq}}\mathbf{I}$$

For two branches, $Z_{eq} = Z_1 Z_2/(Z_1 + Z_2)$ and $\mathbf{I}_1 = \mathbf{I}\, Z_2/(Z_1 + Z_2)$, as for resistors.

Worked example — A capacitor across the compressor. A capacitor is connected across the compressor terminals so that the supply current is in phase with the voltage. What capacitance is needed, and what does the supply current become?

The compressor admittance is $0.08 - j0.06$ S. Adding a capacitor $Y_C = j\omega C$ in parallel cancels the susceptance if $\omega C = 0.06$ S:

$$C = \frac{0.06}{2\pi \times 50} = 191\ \mu\text{F}$$

Then $Y_{eq} = 0.08$ S, purely real, and $\mathbf{I}_{supply} = 230 \times 0.08 = 18.4\angle 0^\circ$ A. The compressor still draws 23 A at −36.87°, and the capacitor draws $230 \times 0.06 = 13.8\angle 90^\circ$ A; these add as phasors to 18.4 A. The supply current has dropped from 23 A to 18.4 A with no change in what the motor does — the essence of

power-factor correction, which Chapter 9 develops properly, including why a capacitor this large is rarely fitted to a single small motor.

Series–parallel reduction and AC Thévenin equivalents

Any ladder of impedances can be reduced step by step, exactly as resistive networks were in Chapter 2. Likewise, any linear two-terminal AC network at a single frequency has a Thévenin equivalent $\mathbf{V}_{th}$ in series with Z_{th} (or a Norton equivalent $\mathbf{I}_N$ in parallel with Y_N), found by the methods of Chapter 4 with complex arithmetic. The power-system "fault level" used in Chapter 7's capacitor-bank example is an AC Thévenin impedance in disguise: $Z_{th} \approx jV^2/S_{sc}$.

8.7 AC nodal and mesh analysis

The method

The systematic methods of Chapter 3 apply unchanged:

- **Nodal analysis.** Choose a reference node, write KCL at each other node in terms of node-voltage phasors and branch admittances, and solve $\mathbf{Y}\mathbf{V} = \mathbf{I}_s$, where $\mathbf{Y}$ is the complex nodal admittance matrix.
- **Mesh analysis.** Assign mesh-current phasors, write KVL around each mesh with branch impedances, and solve $\mathbf{Z}\mathbf{I} = \mathbf{V}_s$.

Supernodes, supermeshes and dependent sources are handled as in Chapter 3. For passive reciprocal networks the matrices are symmetric, as in the DC case; they are now complex. A computer solves them with the same linear-algebra routines, in complex arithmetic.

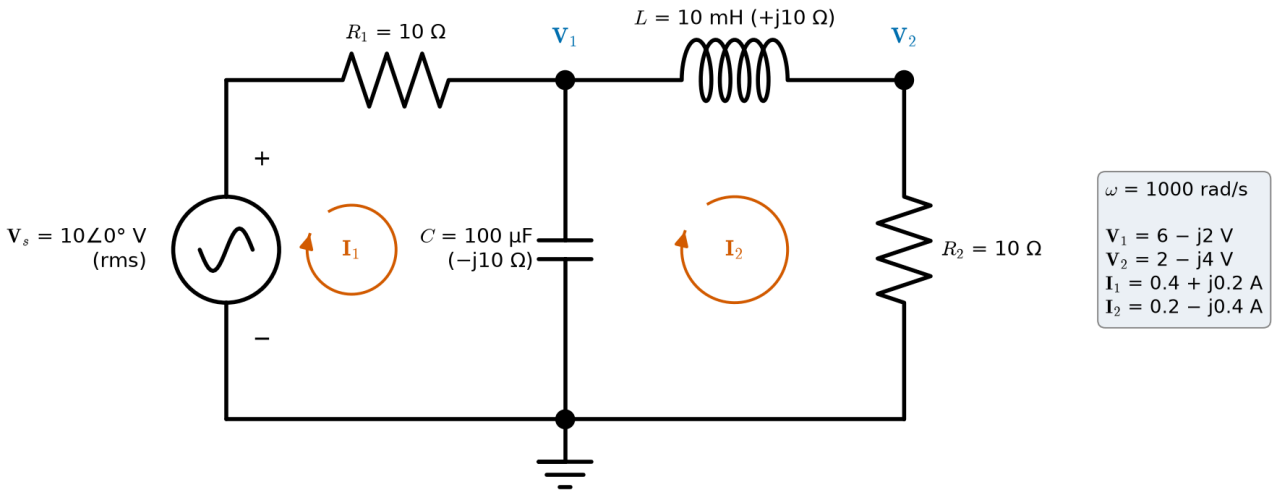


Figure 8.4 — Two-node AC circuit used for nodal and mesh analysis at omega = 1000 rad/s.

Worked example — Nodal and mesh analysis of a two-node circuit. In Figure 8.4 a source $\mathbf{V}_s = 10\angle 0^\circ$ V (RMS) at $\omega = 1000$ rad/s drives $R_1 = 10\ \Omega$ into node 1; node 1 has $C = 100\ \mu$F to

ground and $L = 10$ mH to node 2; node 2 has $R_2 = 10\ \Omega$ to ground. Find all node voltages and branch currents.

Impedances: $Z_C = 1/(j \times 1000 \times 10^{-4}) = -j10\ \Omega$; $Z_L = j \times 1000 \times 0.01 = j10\ \Omega$.

Nodal equations. KCL at node 1 (currents leaving):

$$\frac{\mathbf{V}_1 - 10}{10} + \frac{\mathbf{V}_1}{-j10} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j10} = 0$$

KCL at node 2:

$$\frac{\mathbf{V}_2 - \mathbf{V}_1}{j10} + \frac{\mathbf{V}_2}{10} = 0$$

Multiplying both by 10 and using $1/j = -j$:

$$\begin{aligned}(1 + j - j)\mathbf{V}_1 + j\mathbf{V}_2 &= 10\\ j\mathbf{V}_1 + (1 - j)\mathbf{V}_2 &= 0\end{aligned}$$

The capacitor and inductor admittances at node 1 cancel, which leaves $\mathbf{V}_1 = 10 - j\mathbf{V}_2$. From the second equation, $\mathbf{V}_1 = -(1 - j)\mathbf{V}_2/j = (1 + j)\mathbf{V}_2$. Equating, $(1 + j)\mathbf{V}_2 = 10 - j\mathbf{V}_2$, so $(1 + 2j)\mathbf{V}_2 = 10$ and

$$\mathbf{V}_2 = \frac{10}{1 + 2j} = 2 - j4 = 4.47\angle -63.43^\circ\ \text{V}, \qquad \mathbf{V}_1 = (1 + j)(2 - j4) = 6 - j2 = 6.32\angle -18.43^\circ\ \text{V}$$

Branch currents: source current $(10 - \mathbf{V}_1)/10 = 0.4 + j0.2$ A; capacitor current $\mathbf{V}_1/(-j10) = 0.2 + j0.6$ A; inductor and R_2 current $\mathbf{V}_2/10 = 0.2 - j0.4$ A. KCL at node 1 checks: $(0.2 + j0.6) + (0.2 - j0.4) = 0.4 + j0.2$.

Mesh check. With $\mathbf{I}_1$ in the left mesh and $\mathbf{I}_2$ in the right:

$$\begin{bmatrix} 10 - j10 & j10 \\ j10 & 10 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \end{bmatrix}$$

(the right-mesh self-impedance is $-j10 + j10 + 10 = 10\ \Omega$). Solving gives $\mathbf{I}_1 = 0.4 + j0.2$ A and $\mathbf{I}_2 = 0.2 - j0.4$ A, and the capacitor current $\mathbf{I}_1 - \mathbf{I}_2 = 0.2 + j0.6$ A, matching the nodal result. The input impedance seen by the source is $10/(0.4 + j0.2) = 20 - j10\ \Omega$: net capacitive, even though the circuit contains equal reactances, because of where they sit.

Sources at different frequencies

Phasors are tied to one frequency. If a circuit contains sources at different frequencies — a 50 Hz supply and a 150 Hz harmonic current, or an audio signal and a DC bias — each frequency must be solved separately with its own impedances, and the **time-domain** results added (superposition). Adding phasors of different frequencies is meaningless. DC is just the case $\omega = 0$, where inductors are shorts and capacitors opens.

> **Status of the principle: AC nodal and mesh analysis**
> *Derived*. KCL and KVL hold for phasors, and every element obeys $\mathbf{V} = Z\mathbf{I}$; the methods are exact for linear time-invariant circuits at a single frequency.

8.8 Resonance

Series resonance

The impedance of a series RLC branch is

$$Z(j\omega) = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

The reactance vanishes at

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

the natural frequency of Chapter 7. At this **series resonance** the impedance is a minimum, equal to R, the current is maximum and in phase with the source, and the voltages across L and C are equal and opposite, each of magnitude

$$V_L = V_C = \frac{\omega_0 L}{R} V = QV, \qquad Q = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

Below ω_0 the branch is capacitive; above it, inductive.

Bandwidth

As the frequency moves away from ω_0 the reactance grows and the current falls. The **half-power frequencies** ω_1 and ω_2 are where $|X| = R$, the current is $1/\sqrt{2}$ of its peak, and the power in R is half its peak. Their separation is the **bandwidth**

$$B = \omega_2 - \omega_1 = \frac{R}{L} = \frac{\omega_0}{Q}$$

and $\omega_0 = \sqrt{\omega_1 \omega_2}$ (the resonance is at the geometric mean). Exactly, $\omega_{1,2} = \mp\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \omega_0^2}$; for $Q \gtrsim 10$ the curve is nearly symmetric and $\omega_{1,2} \approx \omega_0 \mp B/2$.

> **Status of the principle: Bandwidth–Q relation $B = \omega_0/Q$**
> *Derived*. Exact for the half-power bandwidth of an ideal series or parallel RLC; the symmetric approximation $\omega_0 \pm B/2$ is *approximate*, good for Q above about 10.

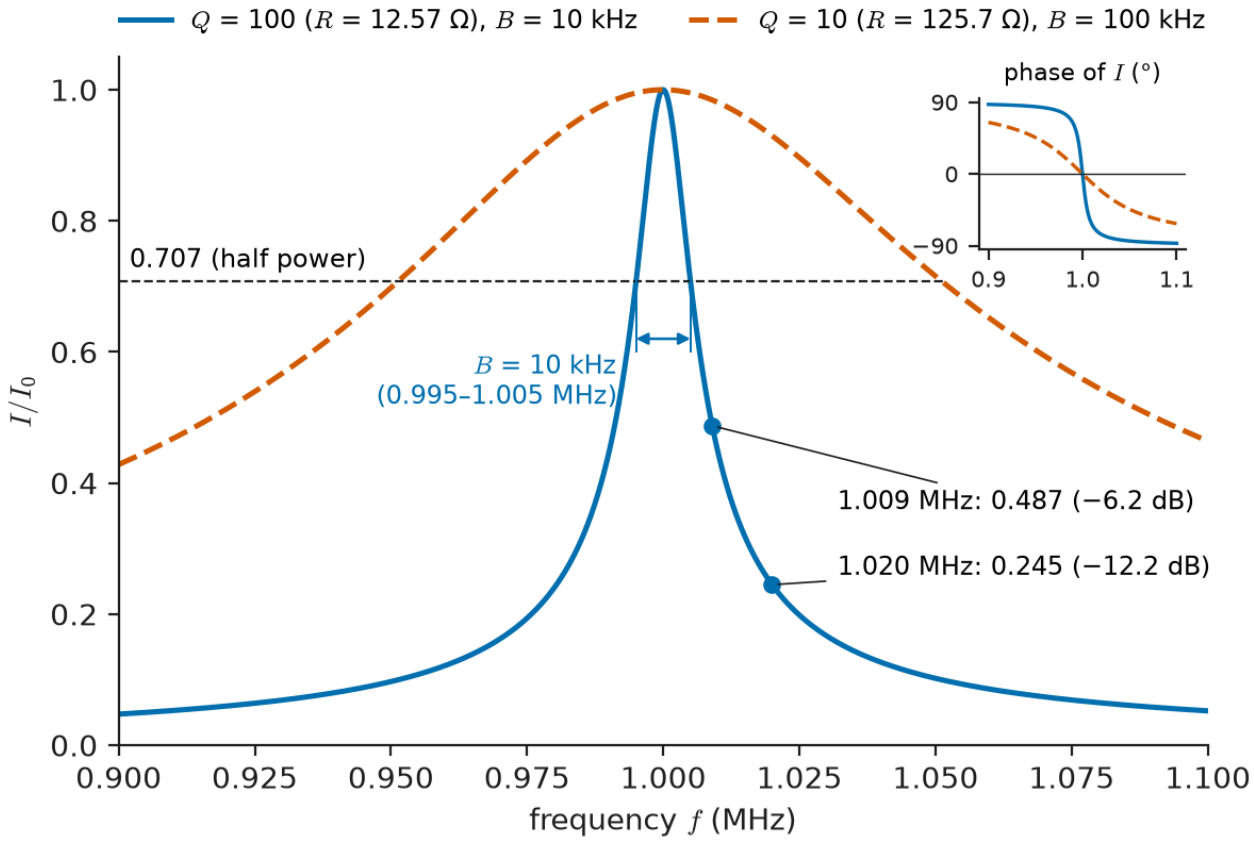


Figure 8.5 — Normalised current of a series RLC versus frequency for Q = 10 and Q = 100, centred on the 1 MHz AM tuner example.

Worked example — AM receiver tuning circuit. An AM receiver's antenna circuit uses $L = 200\ \mu$H with an effective series resistance of $R = 12.57\ \Omega$. What capacitance tunes it to a station at 1 MHz, what are Q and the bandwidth, and how well does it reject a station 9 kHz away?

$$C = \frac{1}{(2\pi f_0)^2 L} = \frac{1}{(2\pi \times 10^6)^2 \times 200 \times 10^{-6}} = 126.7\ \text{pF}$$

$$X_L = 2\pi f_0 L = 1257\ \Omega, \qquad Q = \frac{X_L}{R} = 100, \qquad B = \frac{f_0}{Q} = 10\ \text{kHz}$$

Figure 8.5 plots the resonance curve. At the adjacent European channel, 1.009 MHz, the current ratio is $R/|Z| = 0.487$ (−6.2 dB); at 1.02 MHz it is 0.245 (−12.2 dB); at 1.1 MHz, 0.052 (−25.6 dB). A single tuned circuit therefore gives only modest adjacent-channel rejection — which is exactly why practical receivers use the superheterodyne principle (Chapter 26) to do their selectivity at a fixed intermediate frequency with several cascaded resonators. A 1 mV signal at resonance appears as $Q \times 1$ mV $= 100$ mV across the capacitor: the tuned circuit is also a passive voltage amplifier. Covering the medium-wave band 530–1700 kHz with this inductor needs a variable capacitor from about 44 pF to 451 pF.

Parallel resonance

Dually, a parallel RLC has admittance $Y = G + j(\omega C - 1/(\omega L))$, which is minimum (impedance maximum, equal to R) at the same ω_0. At **parallel resonance** the currents in L and C are Q times the

source current and circulate between them, while the line current is small; $Q = R/(\omega_0 L) = R\sqrt{C/L}$ and again $B = \omega_0/Q$.

A real tank is usually a capacitor in parallel with a coil that has series resistance r. For $Q_L = \omega_0 L/r \gg 1$ it behaves like an ideal parallel tank with a **dynamic resistance**

$$R_D \approx \frac{L}{Cr} = Q_L^2 r$$

For example, a 10 kHz tank with $L = 1$ mH, $r = 5\ \Omega$ and $C = 253$ nF has $Q_L = 12.6$ and $R_D = 790\ \Omega$: the resonant impedance is 158 times the coil's resistance. Parallel resonance is used in tuned amplifiers and oscillator tanks; it is also what makes harmonic resonance between capacitor banks and network inductance dangerous (Chapter 7) — at resonance a small harmonic current produces a large voltage.

> **Boundary.**
> Quality factor, bandwidth and the design of passive and active filters around resonant circuits are developed in Chapter 13. Microwave resonators, where the lumped model is replaced by cavities and lines, are in Chapter 32.

8.9 Frequency response

Transfer function in the sinusoidal steady state

Most circuits are used as signal processors: an input sinusoid produces an output sinusoid of the same frequency, scaled and shifted. The ratio of output phasor to input phasor, as a function of frequency, is the **frequency response**

$$H(j\omega) = \frac{\mathbf{V}_{out}}{\mathbf{V}_{in}} = |H(j\omega)|\,\angle\phi(\omega)$$

$|H|$ is the **magnitude response** (gain) and ϕ the **phase response**. Magnitudes are usually expressed in decibels, $20\log_{10}|H|$: -3 dB is a factor 0.707 (half power), -20 dB a factor 0.1. The frequency response is computed by solving the circuit with phasors at a general ω, which is exactly the work of Sections 8.4–8.7 with ω left as a symbol.

First-order low-pass and high-pass sections

The RC low-pass (series R, output across C) is a voltage divider:

$$H_{LP}(j\omega) = \frac{1/(j\omega C)}{R + 1/(j\omega C)} = \frac{1}{1 + j\omega RC} = \frac{1}{1 + j\,f/f_c}, \qquad f_c = \frac{1}{2\pi RC}$$

At the **corner (cut-off) frequency** f_c the magnitude is $1/\sqrt{2}$ (-3 dB) and the phase $-45°$. Far above f_c the magnitude falls as f_c/f: -20 dB per decade. Exchanging R and C gives the high-pass

$$H_{HP}(j\omega) = \frac{j\,f/f_c}{1 + j\,f/f_c}$$

with −3 dB and +45° at f_c, and a phase lead that approaches 90° well below it.

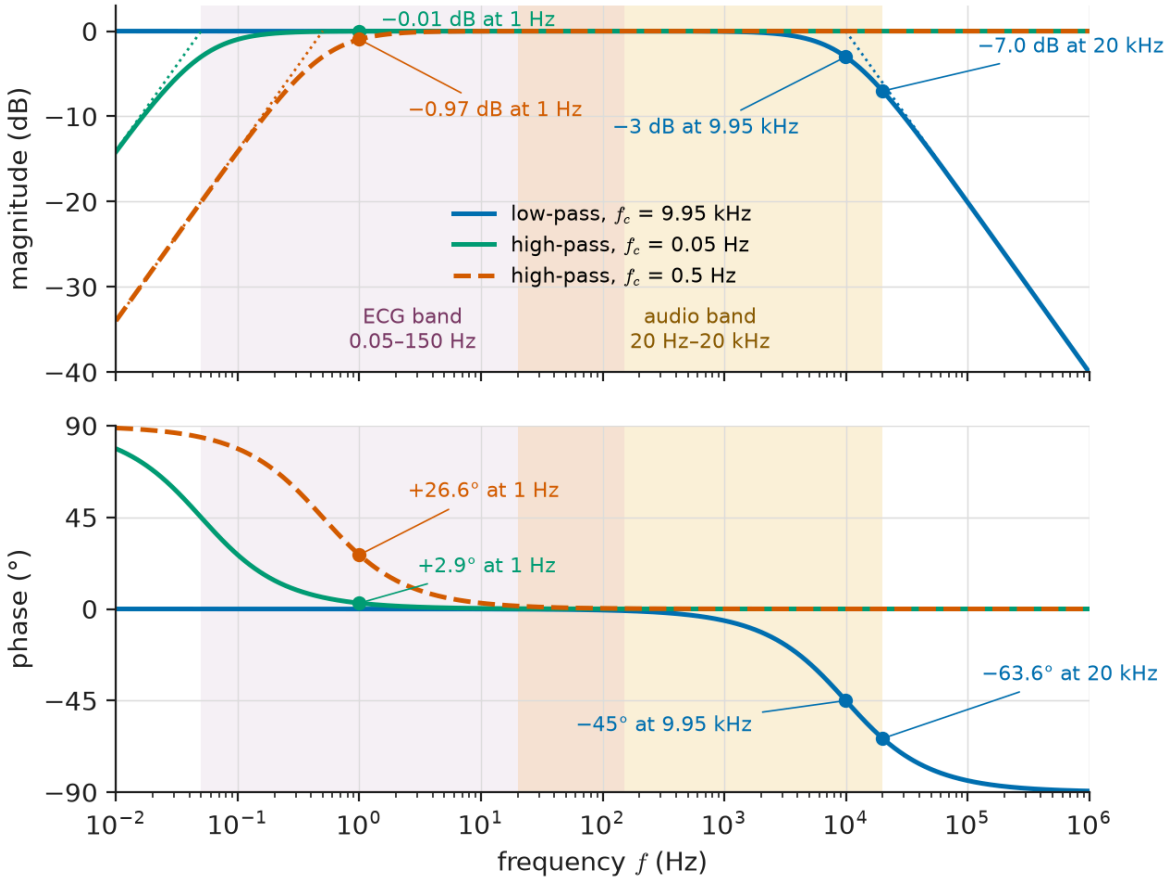


Figure 8.6 — Magnitude and phase of first-order RC sections: an audio low-pass with fc = 9.95 kHz and two ECG high-pass settings with fc = 0.05 Hz and 0.5 Hz.

Worked example — Audio low-pass and ECG high-pass. (a) An audio line input uses $R = 1.6\ \text{k}\Omega$ and $C = 10$ nF to remove radio-frequency pickup. (b) An ECG amplifier blocks electrode DC offset and baseline wander with a high-pass RC section, set either to 0.05 Hz (diagnostic) or 0.5 Hz (monitoring). Find the response at the frequencies that matter.

(a) $f_c = 1/(2\pi \times 1600 \times 10^{-8}) = 9.95$ kHz. At 1 kHz the gain is 0.995 (−0.04 dB), phase −5.7°; at 20 kHz, the top of the audio band, 0.445 (−7.0 dB), phase −63.6°; at 100 kHz, 0.099 (−20.1 dB). The filter removes RF pickup but is too aggressive for high-fidelity audio: it cuts the top octave by several decibels. Moving f_c to about 50 kHz (e.g. $C = 2$ nF) would be the usual compromise.

(b) The fundamental of a heartbeat at 60 beats/min is 1 Hz, and the ST segment and T wave contain significant energy down to about 0.5 Hz. With $f_c = 0.05$ Hz the gain at 1 Hz is 0.999 and the phase lead 2.9°; at 0.5 Hz, 0.995 and 5.7° — negligible distortion. With $f_c = 0.5$ Hz the gain at 1 Hz is 0.894 (−0.97 dB) with a 26.6° phase lead, and at 0.5 Hz it is 0.707 with 45° lead. Different harmonics of the beat are shifted by different amounts, which reshapes the slow parts of the waveform and can depress or elevate the ST segment. This is why diagnostic ECG standards specify a low-frequency limit of 0.05 Hz (or an equivalent phase-linear digital filter), while monitoring equipment accepts 0.5 Hz in exchange for a steadier baseline. With a 1 µF coupling capacitor, the 0.05 Hz corner needs $R = 3.18\ \text{M}\Omega$, the 0.5 Hz corner 318 kΩ.

Figure 8.6 shows both responses. The ECG case illustrates a point developed fully in Chapter 18: preserving a waveform's *shape* requires not just flat magnitude but a phase that is linear in frequency (constant group delay) across the band of interest.

> **Status of the principle: Frequency response of a linear circuit**
> *Derived*. The sinusoidal-steady-state ratio of output to input phasors; it fully characterises a stable linear time-invariant circuit, and through Fourier analysis determines its response to any input (Chapter 18). It says nothing about unstable circuits, whose response does not reach a steady state.

8.10 Power calculations

Average power from phasors

In the sinusoidal steady state with $v = \sqrt{2}V\cos(\omega t + \theta_v)$ and $i = \sqrt{2}I\cos(\omega t + \theta_i)$, the instantaneous power $p = vi$ oscillates at 2ω about an average value (derived in Chapter 9):

$$P = VI\cos(\theta_v - \theta_i)$$

The product of RMS values, VI, is the **apparent power** in volt-amperes; the factor $\cos(\theta_v - \theta_i)$ is the **power factor**. For the compressor, $230 \times 23 = 5290$ VA and $\cos 36.87^\circ = 0.8$, so $P = 4232$ W — the energy monitor's reading. The difference is not lost power: it is energy that flows into the motor's magnetic field and back each half-cycle.

With RMS phasors, power is most conveniently computed as the **complex power**

$$S = \mathbf{V}\mathbf{I}^{*} = VI\angle(\theta_v - \theta_i) = P + jQ$$

whose real part is the average power P and whose imaginary part Q is the **reactive power** in volt-amperes reactive (var). For an impedance $Z = R + jX$ carrying current $\mathbf{I}$: $P = I^2R$ and $Q = I^2X$. Only resistances absorb average power; reactances exchange energy but absorb none on average. Chapter 9 develops these quantities, their measurement and their economic consequences.

Worked example — Power in the two-node circuit. In the circuit of Figure 8.4, the source current is $\mathbf{I}_s = 0.4 + j0.2$ A ($|\mathbf{I}_s|^2 = 0.2$ A^2) and the R_2 current is $0.2 - j0.4$ A (also 0.2 A^2). The resistors absorb $P_{R1} = 0.2 \times 10 = 2$ W and $P_{R2} = 0.2 \times 10 = 2$ W. The capacitor current magnitude squared is $0.2^2 + 0.6^2 = 0.4$ A^2, so $Q_C = 0.4 \times (-10) = -4$ var; the inductor carries 0.2 A^2, so $Q_L = 0.2 \times 10 = +2$ var. The source delivers

$$S = \mathbf{V}_s\mathbf{I}_s^{*} = 10 \times (0.4 - j0.2) = 4 - j2 \text{ VA}$$

matching $P = 2 + 2 = 4$ W and $Q = -4 + 2 = -2$ var: complex power is conserved, as Chapter 9 proves in general. The negative Q confirms that the circuit is net capacitive, as its input impedance $20 - j10\ \Omega$ said.

Maximum average power transfer

Chapter 4 showed that a resistive load extracts maximum power from a resistive Thévenin source when $R_L = R_{th}$. With a complex source impedance $Z_{th} = R_{th} + jX_{th}$, the average power in a load $Z_L = R_L + jX_L$ is

$$P_L = \frac{V_{th}^2 R_L}{(R_{th} + R_L)^2 + (X_{th} + X_L)^2}$$

which is maximised by cancelling the reactance, $X_L = -X_{th}$, and then matching the resistance, $R_L = R_{th}$. The condition is the **conjugate match** $Z_L = Z_{th}^*$, and the maximum power is $V_{th}^2/(4R_{th})$. Conjugate matching is how an antenna is matched to a receiver front end and how a wireless-power coil is tuned to its load; Chapters 14 and 32 refine it for lines and RF networks. As in Chapter 4, it is a condition for maximum power *transfer*, not for efficiency, and power systems never operate there.

8.11 Limits of phasor analysis

The phasor method is exact under precise assumptions, and each one fails somewhere in practice.

Steady state only. Phasors describe the forced response after transients have decayed. Inrush current, switching overvoltages and start-up behaviour need the time-domain methods of Chapters 6 and 7 or the Laplace transform (Chapter 11), which handles transient and steady state together.

A single frequency. Non-sinusoidal periodic sources must be decomposed into harmonics (Chapter 18) and each harmonic solved separately. Mains currents drawn by rectifiers, drives and LED drivers are strongly non-sinusoidal; Chapter 9 shows what that does to power and power factor.

Linearity. A saturating transformer, a diode, a varistor or an arc does not have an impedance: its current is not proportional to its voltage, and a sinusoidal voltage produces a distorted current. Phasor analysis can only be applied to such elements after linearising them about an operating point (the small-signal model of *Electronics & Computing*).

Lumped elements. At frequencies where the circuit dimensions approach a tenth of a wavelength the impedance of a "wire" is no longer negligible or even simple. Transmission-line phasor analysis (Chapter 14) extends the method to distributed circuits.

Time invariance. A switched circuit — a chopper, a mixer, a PWM inverter — changes its topology every cycle. It may be analysed with phasors only through averaged or harmonic-balance models.

> **Boundary.**
> Complex-number algebra and Euler's formula belong to *The Mathematics Universe*; small-signal linearisation of transistors and diodes belongs to *Electronics & Computing*; the internal equivalent circuits of motors and transformers, used here only as R–L loads, belong to *Electrical Energy & Machines*.

What remains open

Open question	Where it is answered
What exactly is reactive power, how is it billed and corrected, and how is power measured when waveforms are distorted?	Chapter 9 (AC power)
How are three sinusoidal sources 120° apart combined to supply power efficiently?	Chapter 10 (three-phase circuits)
How are transients and steady state handled together for arbitrary inputs?	Chapter 11 (Laplace-domain analysis)
How are frequency responses plotted and approximated, and how are filters designed?	Chapter 13 (Bode plots, quality factor, passive and active filters)
What happens to phasor analysis when the circuit is a wavelength long?	Chapter 14 (transmission lines)
How are non-sinusoidal signals decomposed into sinusoids, and what does phase linearity mean for waveform shape?	Chapter 18 (Fourier series, transforms, group delay, distortion)
How do receivers achieve the selectivity a single tuned circuit cannot?	Chapter 26 (superheterodyne receivers)

Next: Chapter 9 takes the power formula introduced in Section 8.10 and builds from it the complete theory of active, reactive, apparent and complex power, power-factor correction, energy billing and power measurement with distorted waveforms.

Chapter 9 — AC Power

Part II · Transient and AC Analysis

A plastics-moulding plant receives its monthly electricity bill. Beside the expected charge for 208 000 kWh of energy there are two lines the production manager has never looked at: a "reactive energy" charge of almost a thousand US dollars, and a "maximum demand" charge billed not in kilowatts but in kilovolt-amperes. The same week, the site electrician reports that the 630 kVA supply transformer runs at 85 % of its rating although the plant draws only 400 kW, and that adding the planned new extruder line would overload it. A few kilometres away, an office building that has replaced all its desktop computers finds that the neutral conductors of its distribution boards are running hot, and an audit shows a power factor of 0.67 even though the current drawn by each computer is almost exactly in phase with the voltage.

All of these are questions about **AC power**: how energy flows in circuits where voltage and current are sinusoidal (or nearly so), what is billed and why, which part of the current does useful work and which merely loads the conductors, and how any of it is measured. Chapter 8 introduced the formula $P = VI\cos\phi$. This chapter builds the full theory on it.

> **Why does an AC load draw more current than its useful power requires, and what does that cost?**
> **How are active, reactive, apparent and complex power defined, corrected and measured, and what happens to these definitions when the current is distorted?**

Concept	Engineering destination
Instantaneous power	Torque pulsation, DC-link capacitor sizing in single-phase converters, thermal cycling
Active power	Energy bills, efficiency, heating, mechanical output
Reactive power	Voltage regulation in grids, magnetising of motors and transformers, capacitor banks
Apparent power	Ratings of transformers, generators, UPS units, cables and switchgear
Complex power	Load-flow studies, summing plant loads, power balance checks
Power factor	Tariff penalties, conductor and transformer loading, supply standards
Power-factor correction	Capacitor banks, active PFC in power supplies, STATCOMs
Energy consumption	Metering, demand charges, appliance energy labels, efficiency projects
Non-sinusoidal considerations	Harmonics from drives, LED and IT loads; neutral overheating; THD limits
Practical power measurements	Wattmeters, power analysers, revenue meters, instrument-transformer errors

Roadmap. Instantaneous power (9.1) → its average (active) and oscillating (reactive) parts (9.2) → apparent power and equipment ratings (9.3) → complex power and its conservation (9.4) → power factor (9.5) and its correction (9.6) → energy and how it is billed (9.7) → what changes when waveforms are not sinusoidal (9.8) → how power is actually measured (9.9).

9.1 Instantaneous power

Power at every instant

The power delivered to any two-terminal element at each instant is the product of its voltage and current (with the passive sign convention of Chapter 1):

$$p(t) = v(t)\, i(t)$$

This holds for any waveforms. For the sinusoidal steady state, take the voltage as reference and let the current lag by the angle $\phi = \theta_v - \theta_i$:

$$v(t) = \sqrt{2}\, V \cos \omega t, \qquad i(t) = \sqrt{2}\, I \cos(\omega t - \phi)$$

where V and I are RMS values. Using $2 \cos a \cos b = \cos(a-b) + \cos(a+b)$:

$$p(t) = V I \cos \phi + V I \cos(2\omega t - \phi)$$

The instantaneous power is a constant plus a sinusoid at **twice** the supply frequency. Expanding the second term separates two physically different flows:

$$p(t) = \underbrace{V I \cos \phi \; (1 + \cos 2\omega t)}_{\text{always } \geq 0} + \underbrace{V I \sin \phi \, \sin 2\omega t}_{\text{average zero}}$$

The first term pulses between 0 and $2VI\cos\phi$ but never reverses: it is energy flowing one way, from source to load, and its average is $VI\cos\phi$. The second term alternates symmetrically: energy flows into the load for a quarter cycle and back to the source for the next, with zero net transfer. Its amplitude is $VI\sin\phi$.

> **Status of the principle: Instantaneous power $p = vi$**
> *Fundamental.* It follows from the definitions of voltage (energy per charge) and current (charge per time) and from energy conservation; it holds for any waveform and any lumped element.

The three elementary cases

- **Resistor** ($\phi = 0$): $p = VI(1 + \cos 2\omega t)$, always positive. A resistor only absorbs energy.
- **Inductor** ($\phi = +90^\circ$): $p = VI \sin 2\omega t$. The inductor absorbs energy as its current builds up, storing it in its magnetic field ($\frac{1}{2}Li^2$), and returns it all as the current falls. Average zero.
- **Capacitor** ($\phi = -90^\circ$): $p = -VI \sin 2\omega t$. The same exchange, with the electric field, in the opposite phase. Average zero.

A capacitor and an inductor on the same supply exchange energy in antiphase: while one absorbs, the other releases. This is why a capacitor can "supply" the reactive needs of a motor locally (Section 9.6), and it is the steady-state version of the energy sloshing studied in Section 7.9.

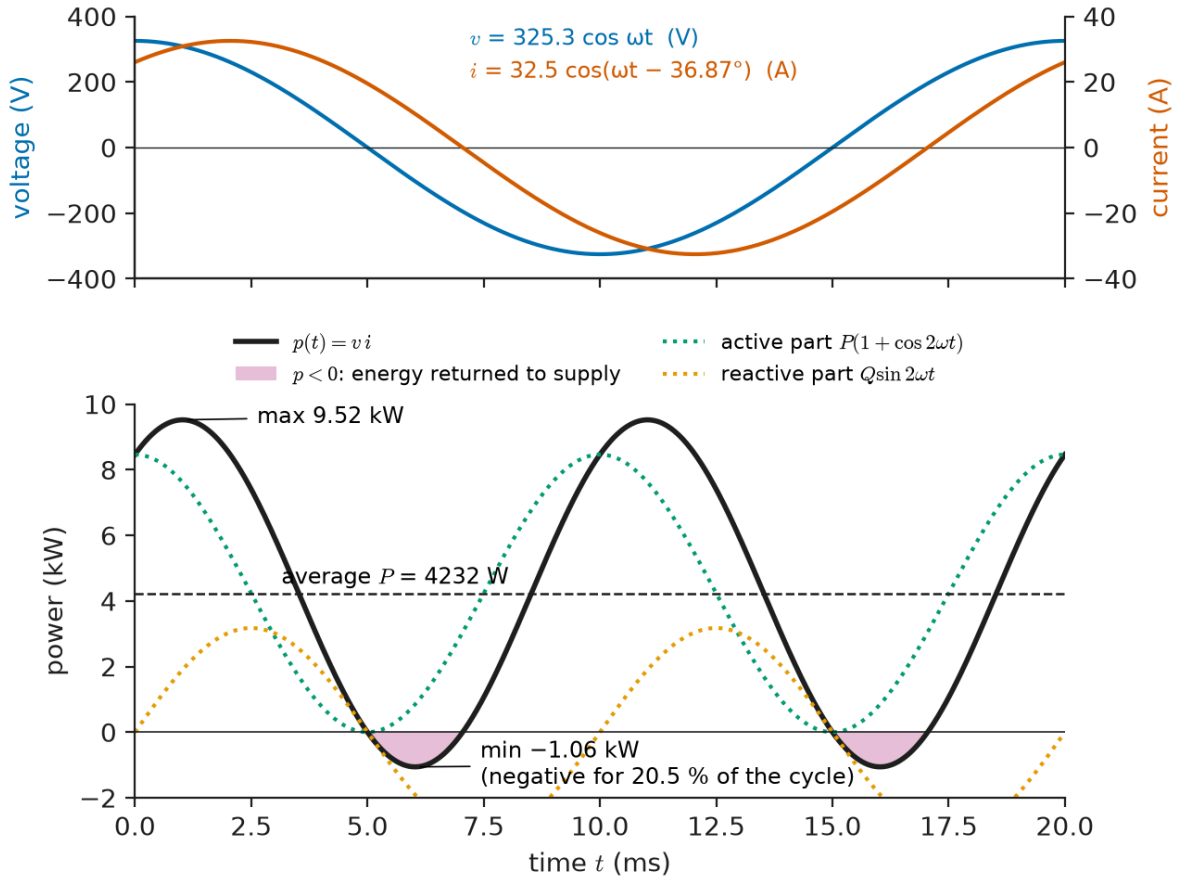


Figure 9.1 — Voltage, current and instantaneous power of the 230 V, 23 A compressor load with power factor 0.8 lagging over one 50 Hz cycle.

Worked example — Instantaneous power of the compressor. The compressor of Chapter 8 draws 23 A at 230 V with the current lagging by 36.87° ($\cos\phi = 0.8$, $\sin\phi = 0.6$). Describe $p(t)$.

$$VI = 230 \times 23 = 5290 \text{ VA}, \qquad p(t) = 4232 + 5290\cos(2\omega t - 36.87^\circ) \text{ W}$$

The power oscillates at 100 Hz between $5290(0.8+1) = 9522$ W and $5290(0.8-1) = -1058$ W. For $2\phi/2\pi = 20.5$ % of each cycle the power is negative: the motor returns energy from its magnetic field to the supply. The average, 4232 W, is the useful power. The decomposition gives an active component swinging between 0 and 8464 W and a reactive component of amplitude 3174 W. Figure 9.1 shows the curves.

Why the double-frequency pulsation matters

The average is what the energy meter integrates, but the pulsation is real and has engineering consequences. A single-phase motor's electrical input pulsates at 100 Hz, so its torque does too, which is one reason single-phase motors hum and vibrate more than three-phase ones. A single-phase inverter feeding the grid from a solar panel or battery must buffer the 100 Hz pulsation in its DC-link capacitor, and that capacitor is often the component that limits the inverter's lifetime. A balanced three-phase

system, remarkably, delivers *constant* total instantaneous power, because the pulsations of the three phases cancel — one of the main reasons power is generated and transmitted three-phase (Chapter 10).

9.2 Active and reactive power

Active power

The **active power** (real power, average power) is the average of $p(t)$ over a period:

$$P = \frac{1}{T}\int_0^T p(t)\,dt = V I \cos\phi \qquad [\mathrm{W}]$$

It is the rate at which energy is converted irreversibly into heat, mechanical work, light, chemical energy or radiation. Only resistive parts of impedances absorb it: for a branch $Z = R + jX$ carrying RMS current I, $P = I^2 R$.

Reactive power

The **reactive power** is the amplitude of the oscillating, zero-average component:

$$Q = V I \sin\phi \qquad [\mathrm{var}]$$

measured in volt-amperes reactive (var, kvar, Mvar). With the convention $\phi = \theta_v - \theta_i$, an inductive load (current lagging) has $Q > 0$ and is said to **absorb** reactive power; a capacitive load has $Q < 0$ and is said to **generate** or supply it. For a branch $Z = R + jX$: $Q = I^2 X$. In particular

$$Q_L = I^2 \omega L = \frac{V^2}{\omega L}, \qquad Q_C = -I^2 \frac{1}{\omega C} = -\omega C V^2$$

Reactive power is *not* power in the energy-conversion sense: no fuel is burned to supply it. It is a measure of the energy that sloshes back and forth each cycle between the source and the load's magnetic and electric fields. The peak energy stored in an inductor is $L I_{pk}^2/2 = L I^2$, and $Q_L = \omega L I^2 = \omega \times$(peak stored energy) — reactive power is the stored-energy exchange rate.

> **Status of the principle: Active and reactive power for sinusoidal waveforms**
> *Derived.* P and Q follow from averaging and decomposing $p(t) = vi$ for same-frequency sinusoids; the definitions are exact for sinusoidal voltage and current and require generalisation for distorted waveforms (Section 9.8).

Why reactive power matters

If reactive power transfers no net energy, why does anyone care? Three reasons.

1. **It costs current.** Delivering a given P at a lower power factor needs a larger current, $I = P/(V\cos\phi)$. Every conductor, transformer winding and switch in the path carries that current.
2. **Current costs losses.** Losses in the supply path are $I^2 R_{line}$, so they scale as $1/\cos^2\phi$. At $\cos\phi = 0.7$ the losses for the same useful power are twice those at unity power factor.

3. **Reactive power moves voltage.** For a load at the end of a line with impedance $R + jX$, the voltage drop is approximately

$$\Delta V \approx \frac{RP + XQ}{V}$$

On high-voltage lines $X \gg R$, so the voltage at a substation is governed mainly by reactive power flow: grid operators control voltage by injecting or absorbing vars with generators, capacitor banks, reactors and static compensators. On a low-voltage feeder with $R = 0.4\ \Omega$ and $X = 0.3\ \Omega$ supplying 10 kW at 230 V, the drop is $4000/230 = 17.4$ V at unity power factor but $(4000 + 0.3 \times 7500)/230 = 27.2$ V if the load also absorbs 7.5 kvar ($\cos\phi = 0.8$).

> **Status of the principle: Voltage-drop approximation $\Delta V \approx (RP + XQ)/V$**
> *Approximate.* It neglects the quadrature component of the drop and is accurate when the drop is a small fraction (below about 10 %) of the voltage, as in normal power-system operation.

9.3 Apparent power

Definition

The product of RMS voltage and RMS current is the **apparent power**

$$S = VI \qquad [\text{VA}]$$

It is the power a load *would* absorb if voltage and current were in phase, and it is the quantity that directly loads the supply equipment: transformer and generator windings heat according to current, insulation is stressed according to voltage, so their capability is the product VI regardless of phase. For sinusoids,

$$S^2 = P^2 + Q^2$$

Equipment rated in VA

Table 9.1 — Typical apparent-power ratings.

Equipment	Typical rating	Why VA, not W
Distribution transformer (urban)	250–1600 kVA at 11 kV/400 V	Winding heating set by current
Large generator	100–1500 MVA	Stator current and rotor field limits
Office UPS	1–20 kVA, often with a W rating at 0.8–1.0 power factor	Inverter switches limited by current
Wind turbine converter	2–15 MVA	Converter current
Domestic consumer unit (main fuse)	60–100 A at 230 V, i.e. 14–23 kVA	Fuse responds to current
Portable generator	2–8 kVA	Alternator current

The plant transformer of the opening is a case in point. Rated at 630 kVA, it supplies 400 kW at power factor 0.75, i.e. $S = 400/0.75 = 533$ kVA, or 85 % of its rating. The same 400 kW at power factor 0.95 would be 421 kVA, 67 %. At 0.95 the transformer could deliver up to $0.95 \times 630 = 599$ kW — enough for the planned extruder line without replacing the transformer. Raising power factor buys capacity.

9.4 Complex power

Definition

The three quantities fit into one complex number. With RMS phasors $\mathbf{V} = V\angle\theta_v$ and $\mathbf{I} = I\angle\theta_i$, the **complex power** is

$$S = \mathbf{V}\mathbf{I}^{*} = VI\angle(\theta_v - \theta_i) = P + jQ$$

where the conjugate of the current makes the angle the difference $\theta_v - \theta_i = \phi$. Then $|S|$ is the apparent power, $\operatorname{Re} S = P$ and $\operatorname{Im} S = Q$. (With amplitude phasors, $S = \tfrac{1}{2}\mathbf{V}_m\mathbf{I}_m^{*}$.) For an impedance Z:

$$S = I^2 Z = \frac{V^2}{Z^{*}}$$

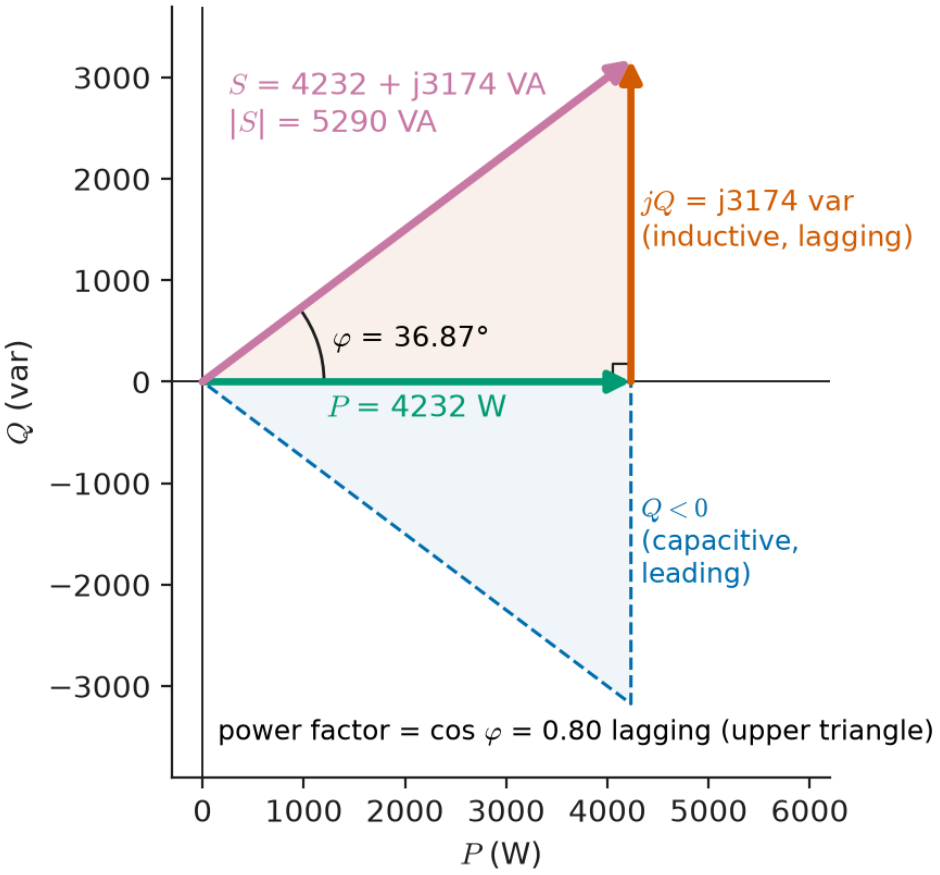


Figure 9.2 — The power triangle of the compressor load: complex power S = 4232 + j3174 VA.

The geometric picture is the **power triangle** of Figure 9.2: P along the real axis, Q vertical, S the hypotenuse, at the angle ϕ. Inductive loads point upward, capacitive loads downward.

Conservation of complex power

In any circuit, the complex power delivered by all sources equals the sum of the complex powers absorbed by all elements:

$$\sum_{\text{sources}} S_k = \sum_{\text{elements}} S_k$$

This is a consequence of KCL and KVL alone (it is a form of **Tellegen's theorem**), so it holds for any lumped network in sinusoidal steady state. Its real part says active power is conserved — energy conservation. Its imaginary part says the same for reactive power, which is less obvious but equally exact. Section 8.10 checked it in a small circuit. Its practical value is that the total load on a supply is found by **adding complex powers** of the individual loads, without solving any circuit.

> **Status of the principle: Conservation of complex power**
> *Derived*. It follows from KCL and KVL through Tellegen's theorem; it holds for any lumped network in sinusoidal steady state, and separately for each frequency when several are present.

Apparent powers do **not** add: $|S_1 + S_2| \le |S_1| + |S_2|$, with equality only when the loads have the same power factor.

Worked example — Total load of a plant. A plant has four groups of load on its 400 V supply: resistive heating and lighting 60 kW at unity power factor; induction motors 180 kW at 0.80 lagging; variable-speed drives with front-end rectifiers 90 kW at 0.92 lagging (fundamental displacement power factor, see Section 9.8); and welding sets 40 kW at 0.60 lagging. Find the total complex power and power factor.

The reactive power of each group is $Q = P\tan(\arccos \mathrm{pf})$:

Load	P (kW)	pf	Q (kvar)	$\lvert S\rvert$ (kVA)
Heating, lighting	60	1.00	0	60.0
Motors	180	0.80 lag	135.0	225.0
Drives	90	0.92 lag	38.3	97.8
Welding	40	0.60 lag	53.3	66.7
Total	**370**	**0.853 lag**	**226.7**	**433.9**

$$S_{tot} = 370 + j226.7 \ \mathrm{kVA}, \qquad |S_{tot}| = 433.9 \ \mathrm{kVA}, \qquad \mathrm{pf} = \frac{370}{433.9} = 0.853$$

The arithmetic sum of the apparent powers would be 449.5 kVA, 3.6 % too high. The motors dominate the reactive demand: they draw 60 % of the total kvar.

9.5 Power factor

Definition

The **power factor** is the ratio of active to apparent power:

$$\mathrm{PF} = \frac{P}{S}$$

For sinusoidal voltage and current it equals $\cos\phi$, and is described as **lagging** (inductive load, current lags) or **leading** (capacitive load). It is a number between 0 and 1 that says what fraction of the current-carrying capacity of the supply path is doing useful work. The definition P/S is the general one; $\cos\phi$ is only the sinusoidal special case, and Section 9.8 shows how badly the two can disagree for distorted currents.

Typical values

Table 9.2 — Representative power factors (typical values).

Load	Power factor	Remarks
Resistive heater, incandescent lamp	1.0	
Induction motor, full load	0.80–0.90 lagging	Magnetising current roughly constant
Induction motor, 25 % load	0.40–0.60 lagging	Lightly loaded motors are the worst offenders
Fluorescent lamp, magnetic ballast, uncorrected	0.5 lagging	Usually corrected to 0.9 with a capacitor
Arc welder, arc furnace	0.6–0.8 lagging	Strongly varying
Variable-speed drive (diode rectifier input)	0.95–0.98 displacement, 0.65–0.9 total	Distorted current
PC or LED driver without PFC	0.5–0.7 total	Displacement close to 1; distortion dominates
Power supply with active PFC	0.95–0.99	Required above 75 W in the EU (IEC 61000-3-2 harmonic limits)
Underground cable at light load	leading	Cable capacitance generates vars

Consequences

For fixed active power P at voltage V, the current and the supply-path losses are

$$I = \frac{P}{V \cdot \mathrm{PF}}, \qquad P_{loss} = I^2 R_{line} \propto \frac{1}{\mathrm{PF}^2}$$

A plant at PF 0.7 draws 43 % more current and causes twice the line losses of one at unity. Utilities therefore require large consumers to keep their power factor above a threshold (commonly 0.9–0.95) and charge for reactive energy or for kVA demand when they do not (Section 9.7). Grid codes impose similar requirements on generators and renewable plants, which must be able to supply or absorb reactive power on demand.

9.6 Power-factor correction

The principle

Most poor power factors are inductive: motors and transformers need magnetising current that lags the voltage by 90°. A capacitor in parallel with the load draws a current that *leads* by 90°. If the two reactive currents are equal, they circulate between the capacitor and the load, and the supply provides only the active current. In terms of power, the capacitor's negative Q cancels the load's positive Q:

$$Q_{supply} = Q_{load} + Q_C, \qquad Q_C = -\omega C V^2$$

The load itself is unchanged — it still draws its full current — but the supply, cables and transformer upstream of the capacitor carry less. Correction should therefore be placed as close to the load as practical: correcting at the main switchboard relieves the transformer and the utility, but not the internal cables to the motors.

Sizing the capacitor

To move a load of active power P from power factor $\cos\phi_1$ to $\cos\phi_2$, the reactive power must fall from $P\tan\phi_1$ to $P\tan\phi_2$, so the capacitor must supply

$$Q_C = P\left(\tan\phi_1 - \tan\phi_2\right)$$

and, for a single-phase connection at RMS voltage V, $C = Q_C/(\omega V^2)$.

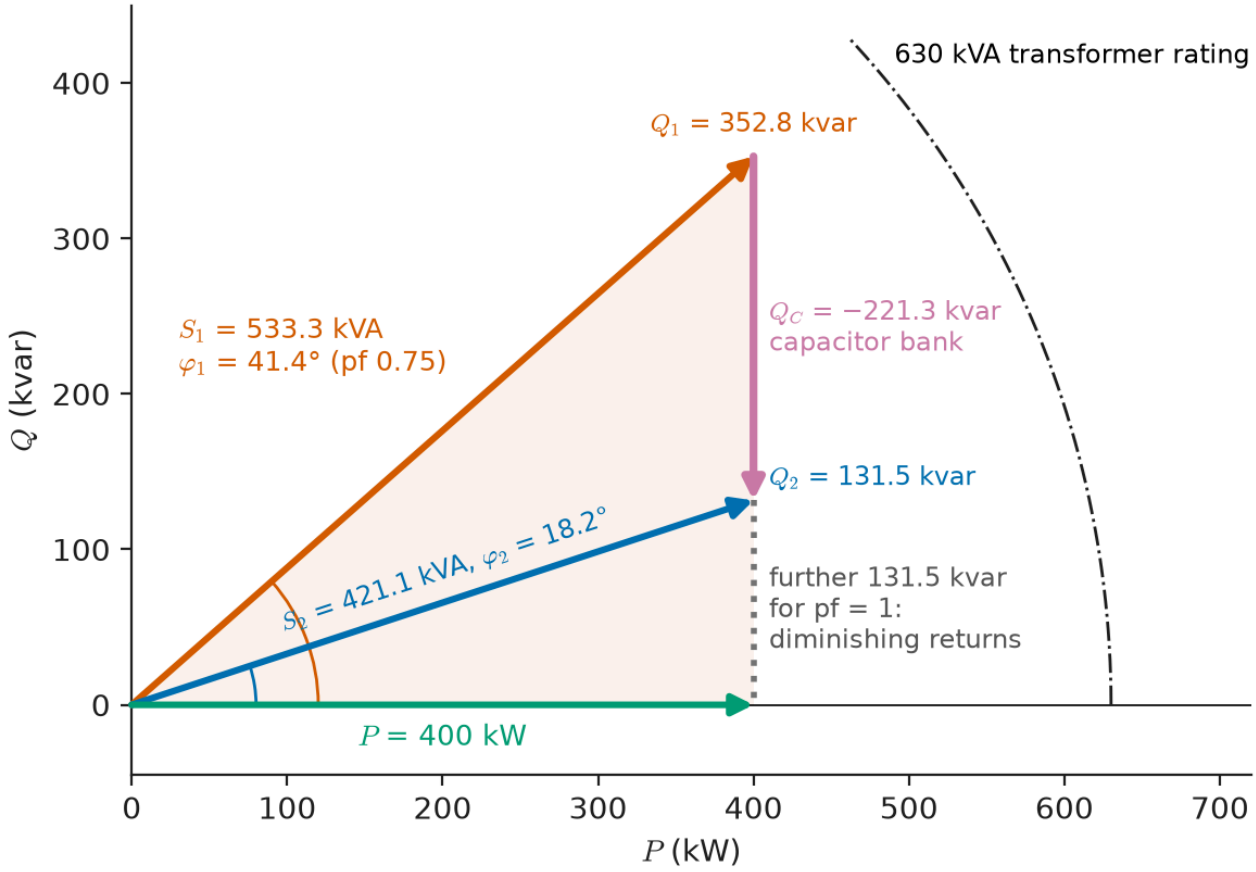


Figure 9.3 — Power-factor correction of a 400 kW plant from 0.75 to 0.95 lagging, shown as power triangles.

Worked example — Correcting a plant from 0.75 to 0.95. The plant of the opening draws 400 kW at power factor 0.75 lagging from a 400 V, 50 Hz three-phase supply. Size a capacitor bank to raise the power factor to 0.95, and find the effect on current, losses and transformer loading.

$$\tan\phi_1 = \tan(\arccos 0.75) = 0.8819, \qquad \tan\phi_2 = \tan(\arccos 0.95) = 0.3287$$

$$Q_1 = 400 \times 0.8819 = 352.8\ \text{kvar}, \quad Q_2 = 400 \times 0.3287 = 131.5\ \text{kvar}, \quad Q_C = 221.3\ \text{kvar}$$

The apparent power falls from 533.3 kVA to 421.1 kVA. The line current, $I = S/(\sqrt{3}\,V_{LL})$ for a three-phase supply (Chapter 10), falls from 769.8 A to 607.7 A. Losses in the supply cables and transformer

windings scale as I^2, so they fall to $(607.7/769.8)^2 = 0.623$ of their former value — a 37.7 % reduction. If those losses were 20 kW, they become 12.5 kW. The transformer loading falls from 85 % to 67 % of its 630 kVA rating, freeing the capacity noted in Section 9.3.

For the bank itself, with three capacitors connected in delta, each sees the 400 V line voltage and supplies one third of Q_C:

$$C_\Delta = \frac{Q_C}{3\omega V_{LL}^2} = \frac{221.3 \times 10^3}{3 \times 314.16 \times 400^2} = 1468\ \mu\text{F per phase}$$

A star connection would need three times that capacitance at $1/\sqrt{3}$ of the voltage, which is why low-voltage banks are almost always delta-connected. In practice the bank would be built from standard 25 or 50 kvar steps switched by a controller.

> **Status of the principle: Capacitive power-factor correction**
> *Approximate*. The sizing formula is exact for sinusoidal voltage and current at a fixed load; real loads vary, the capacitor's kvar varies with V^2, and harmonics can interact with the capacitor's resonance (Section 9.8).

How far to correct

Correcting to unity would need a further 131.5 kvar — 59 % more capacitor — to remove the last 5 % of current. The return diminishes because $\cos\phi$ is flat near $\phi = 0$. Targets of 0.95–0.98 are normal. There are also hazards on the far side of unity:

- **Over-correction.** When the plant's load drops at night but a fixed bank stays connected, the plant becomes capacitive. Leading power factor raises the voltage on the feeder (the XQ term of Section 9.2 changes sign) and may also be penalised. Automatic banks switch capacitor steps in and out with the load.
- **Self-excitation.** A capacitor left across an induction motor that is disconnected from the supply while still spinning can keep it excited as a generator, producing dangerous voltages. Capacitors fitted directly at motors are therefore limited to about 90 % of the motor's no-load magnetising kvar.
- **Harmonic resonance.** A capacitor bank and the supply inductance form a parallel resonance at $f_0 = f_1\sqrt{S_{sc}/Q_C}$ (Chapter 7). If that falls near a harmonic produced by drives or rectifiers, harmonic voltages are amplified. Banks in plants with significant non-linear load are **detuned** with series reactors so that the combination is inductive at all harmonic frequencies.
- **Switching transients.** Energising a bank produces the inrush current and overvoltage computed in Section 7.7.

Other correction methods

Capacitor banks are the cheapest source of reactive power, but not the only one. **Synchronous condensers** — synchronous machines running unloaded with adjustable field — generate or absorb vars continuously and add inertia to the grid. **Static var compensators** (thyristor-switched capacitors and thyristor-controlled reactors) and **STATCOMs** (voltage-source converters) provide fast,

continuous control for grids and arc furnaces. Inside equipment, **active PFC** stages in switch-mode power supplies shape the input current into a sinusoid in phase with the voltage, correcting both displacement and distortion; the circuits are in *Electronics & Computing*.

> **Boundary.**
> The construction, protection and switching of capacitor banks, synchronous condensers and static compensators belong to *Electrical Energy & Machines*. This chapter supplies the circuit theory that sizes them.

9.7 Energy consumption

Energy from power

Energy is the time integral of instantaneous power:

$$W = \int_{t_1}^{t_2} p(t)\,dt$$

Over many cycles of a steady load the double-frequency pulsation integrates to nothing and $W = P\,\Delta t$. The billing unit is the **kilowatt-hour**: 1 kWh = 3.6 MJ. A 2 kW kettle running for 3 minutes uses 0.1 kWh; a refrigerator averaging 150 W at a 35 % compressor duty cycle uses $0.15 \times 0.35 \times 8760 = 460$ kWh per year; a device with a 5 W standby draw uses 43.8 kWh per year doing nothing.

Table 9.3 — Representative annual energy consumption (typical values).

Load	Annual energy
Household (Europe, without electric heating)	2500–4500 kWh
Household (North America)	10 000–11 000 kWh
Electric car, 15 000 km at 0.17 kWh/km	2500 kWh
1 MW data-centre IT load at PUE 1.4	12.3 GWh
400 kW plant, two shifts (about 6200 h/yr)	2.5 GWh

Reactive energy and demand

Revenue meters for commercial and industrial customers record more than kWh:

- **Reactive energy** (kvarh), $\int Q\,dt$, measured separately for lagging and sometimes leading operation. Tariffs typically allow a free reactive energy equal to $\tan\phi_{lim}$ times the active energy (for a limit of 0.9, $\tan\phi = 0.484$) and charge for the excess.
- **Maximum demand**, the highest average power (kW) or apparent power (kVA) over any metering interval (typically 15 or 30 minutes) in the billing period. A kVA demand charge

penalises poor power factor directly, because it bills the capacity of the network the customer occupies.

> **Boundary.**
> Tariff structures, thresholds and prices differ between countries and utilities and change over time. The figures in the next example are illustrative, chosen to show the mechanism, not quoted from any specific tariff.

Worked example — The plant's bill before and after correction. The 400 kW plant of Section 9.6 operates 520 h in a billing month. Its tariff charges reactive energy above $0.484\times$kWh at 0.012 USD per kvarh and has a demand charge of 8 USD per kVA of monthly maximum demand. The capacitor bank costs about 25 USD per kvar installed. Estimate the monthly savings and the payback time.

Active energy: $400 \times 520 = 208\,000$ kWh. Reactive energy before correction: $352.8 \times 520 = 183\,439$ kvarh. The free allowance is $0.484 \times 208\,000 = 100\,739$ kvarh, leaving 82 700 kvarh chargeable: 992 USD. After correction to 0.95 the reactive energy is $131.5 \times 520 = 68\,366$ kvarh, within the allowance: no charge.

The kVA demand falls from 533.3 to 421.1 kVA, saving $112.3 \times 8 = 898$ USD per month. Together with the reduced losses (about 7.5 kW less, or roughly 3900 kWh per month), the savings exceed 2000 USD per month at an energy price of about 0.12 USD per kWh, against a bank cost of $221.3 \times 25 \approx 5500$ USD: a payback of two to three months. Few efficiency investments in an industrial plant are as quick, which is why power-factor correction is usually the first item on an energy audit.

Energy efficiency through the circuit lens

Every efficiency calculation is a ratio of active powers: output power divided by input power, or useful energy over metered energy. Reactive power does not appear in the efficiency of a load, but it appears in the losses of the *supply*, so the system efficiency depends on power factor even when the load's does not. The same applies inside equipment: a power converter's input-filter capacitors circulate reactive current that heats their ESR and the converter's wiring without contributing to the output.

9.8 Non-sinusoidal considerations

Distorted currents

The office building of the opening is full of switch-mode power supplies whose input stage is a diode bridge feeding a smoothing capacitor. The capacitor charges only near the peaks of the supply voltage, so the current flows in short pulses, one per half-cycle. The voltage stays nearly sinusoidal (the supply is stiff), but the current is far from it. Variable-speed drives, LED drivers, battery chargers, arc furnaces and saturated transformers all distort current in similar ways. The sinusoidal theory of Sections 9.1–9.6 must be generalised.

Any periodic current can be expanded as a Fourier series (Chapter 18) of harmonics at integer multiples hf_1 of the fundamental:

$$i(t) = I_0 + \sum_{h=1}^{\infty} \sqrt{2}\, I_h \cos\left(h\omega_1 t - \phi_h\right)$$

with I_h the RMS value of the hth harmonic. Half-wave-symmetric waveforms (as from a bridge rectifier) contain only odd harmonics. The RMS value adds in quadrature (Section 8.2):

$$I_{rms} = \sqrt{I_0^2 + I_1^2 + I_2^2 + \cdots}$$

The **total harmonic distortion** of the current is

$$\mathrm{THD}_I = \frac{\sqrt{\sum_{h\geq 2} I_h^2}}{I_1}$$

Typical values: below 5 % for a power supply with active PFC; 30–50 % for a six-pulse drive with a DC choke; 80–150 % for a small single-phase rectifier–capacitor supply.

Active power with harmonics

When voltage and current both contain harmonics, the average of their product contains only products of components at the **same** frequency, because products of different harmonics average to zero over a period (orthogonality of sinusoids):

$$P = V_0 I_0 + \sum_{h=1}^{\infty} V_h I_h \cos\phi_h$$

If the supply voltage is a pure sinusoid ($V_h = 0$ for $h \geq 2$), **only the fundamental current carries active power**: $P = V_1 I_1 \cos\phi_1$. The harmonic currents add to the RMS current, and so to losses and apparent power, without delivering any energy to the load.

> **Status of the principle: Active power as a sum over harmonics**
> *Derived.* A consequence of the orthogonality of sinusoids of different frequencies over a common period; it is exact for periodic waveforms in steady state and fails for non-periodic (interharmonic or fluctuating) loads unless averaged over a suitable long window.

Power factor with harmonics

With sinusoidal voltage, $S = V_1 I_{rms}$, and the true power factor factors into two parts:

$$\mathrm{PF} = \frac{P}{S} = \frac{V_1 I_1 \cos\phi_1}{V_1 I_{rms}} = \underbrace{\cos\phi_1}_{\text{displacement}} \times \underbrace{\frac{I_1}{I_{rms}}}_{\text{distortion}} = \frac{\cos\phi_1}{\sqrt{1 + \mathrm{THD}_I^2}}$$

The **displacement power factor** $\cos\phi_1$ is what a capacitor can correct. The **distortion factor** I_1/I_{rms} is not: a capacitor does nothing to the harmonic currents except possibly resonate with them.

Distortion is reduced only by changing the load (active PFC, multi-pulse rectifiers, chokes) or by filtering (passive tuned filters, active harmonic filters).

The apparent power no longer satisfies $S^2 = P^2 + Q^2$. With sinusoidal voltage one writes

$$S^2 = P^2 + Q_1^2 + D^2$$

where $Q_1 = V_1 I_1 \sin\phi_1$ is the fundamental reactive power and $D = V_1\sqrt{\sum_{h\geq 2} I_h^2}$ is called the **distortion power**. When the voltage is also distorted, there is no unique accepted way to split S into components: Budeanu's, Fryze's and the IEEE 1459 definitions of "reactive power" disagree, and instruments that implement different ones give different var readings on the same load. Active power and RMS values remain unambiguous; reactive power with distorted waveforms is a definition, not a measurement, and the definition should be stated.

> **Status of the principle: Power-factor decomposition into displacement and distortion factors**
> *Derived* for sinusoidal voltage. For distorted voltage the decomposition of apparent power into reactive and distortion parts is a matter of convention, with several standards in use.

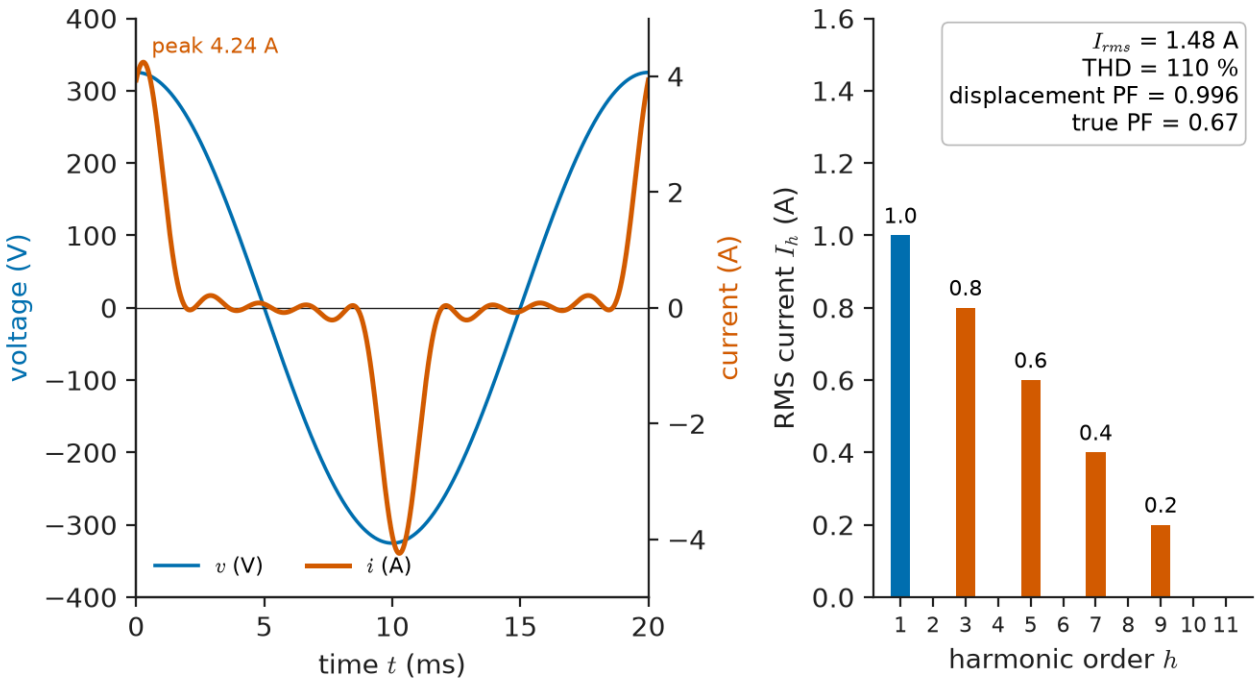


Figure 9.4 — Current drawn by a rectifier-capacitor power supply on a 230 V, 50 Hz supply: waveform and harmonic spectrum.

Worked example — A computer power supply. A desktop computer's power supply (without active PFC) on a 230 V sinusoidal supply draws a current with RMS harmonic components $I_1 = 1.0$ A (lagging the voltage by 5°), $I_3 = 0.8$ A, $I_5 = 0.6$ A, $I_7 = 0.4$ A and $I_9 = 0.2$ A. Find the RMS current, THD, active power, apparent power and power factor.

$$I_{rms} = \sqrt{1.0^2 + 0.8^2 + 0.6^2 + 0.4^2 + 0.2^2} = \sqrt{2.20} = 1.483\ \text{A}$$

$$\mathrm{THD}_I = \frac{\sqrt{0.64 + 0.36 + 0.16 + 0.04}}{1.0} = \sqrt{1.20} = 1.095 = 110\ \%$$

$$P = 230 \times 1.0 \times \cos 5^\circ = 229.1\ \mathrm{W}, \qquad Q_1 = 230 \times 1.0 \times \sin 5^\circ = 20.0\ \mathrm{var}$$

$$S = 230 \times 1.483 = 341.1\ \mathrm{VA}, \qquad \mathrm{PF} = \frac{229.1}{341.1} = 0.672$$

The displacement power factor is $\cos 5^\circ = 0.996$ — an electrician measuring the phase between voltage zero-crossing and the fundamental current would think the load nearly ideal — but the distortion factor is $1/\sqrt{1 + 1.20} = 0.674$, and that sets the true power factor. The distortion power is $D = \sqrt{341.1^2 - 229.1^2 - 20.0^2} = 252$ VA, much larger than Q_1. Correcting with a capacitor would remove at most the 20 var of Q_1 and leave the power factor near 0.67. Figure 9.4 shows the waveform and its spectrum.

Triplen harmonics and the neutral

The third harmonic and its odd multiples (9th, 15th, …) — the **triplen** harmonics — have a property that only shows in three-phase systems: the triplen currents of the three phases are in phase with each other, so instead of cancelling in the neutral as balanced fundamental currents do, they add. A building full of single-phase IT loads, each drawing 0.8 A of third harmonic per ampere of fundamental, can carry a neutral current larger than the phase currents. That is what overheated the office neutrals in the opening; Chapter 10 shows the arithmetic. Standards such as IEC 61000-3-2 (equipment up to 16 A per phase) and IEEE 519 (system level) limit harmonic emissions for this reason.

Effects of harmonics on equipment

Harmonic currents produce extra losses out of proportion to their RMS value: eddy-current losses in transformer windings grow roughly as $h^2 I_h^2$, so transformers serving non-linear loads are derated or specified with a "K-factor". Skin effect raises the resistance of conductors at harmonic frequencies. Capacitor banks, whose admittance rises with h, attract harmonic current and may fail. Harmonic voltage drops across the supply impedance distort the voltage seen by all other customers, which is why utilities limit voltage THD (typically to 5–8 % at low voltage).

9.9 Practical power measurements

The measurement problem

Measuring power means measuring the average of a product of two signals that may be large (kilovolts, kiloamperes), distorted and noisy, with an accuracy of 0.1–1 %. There are three steps: scale the voltage and current to instrument levels, form the product, and average it.

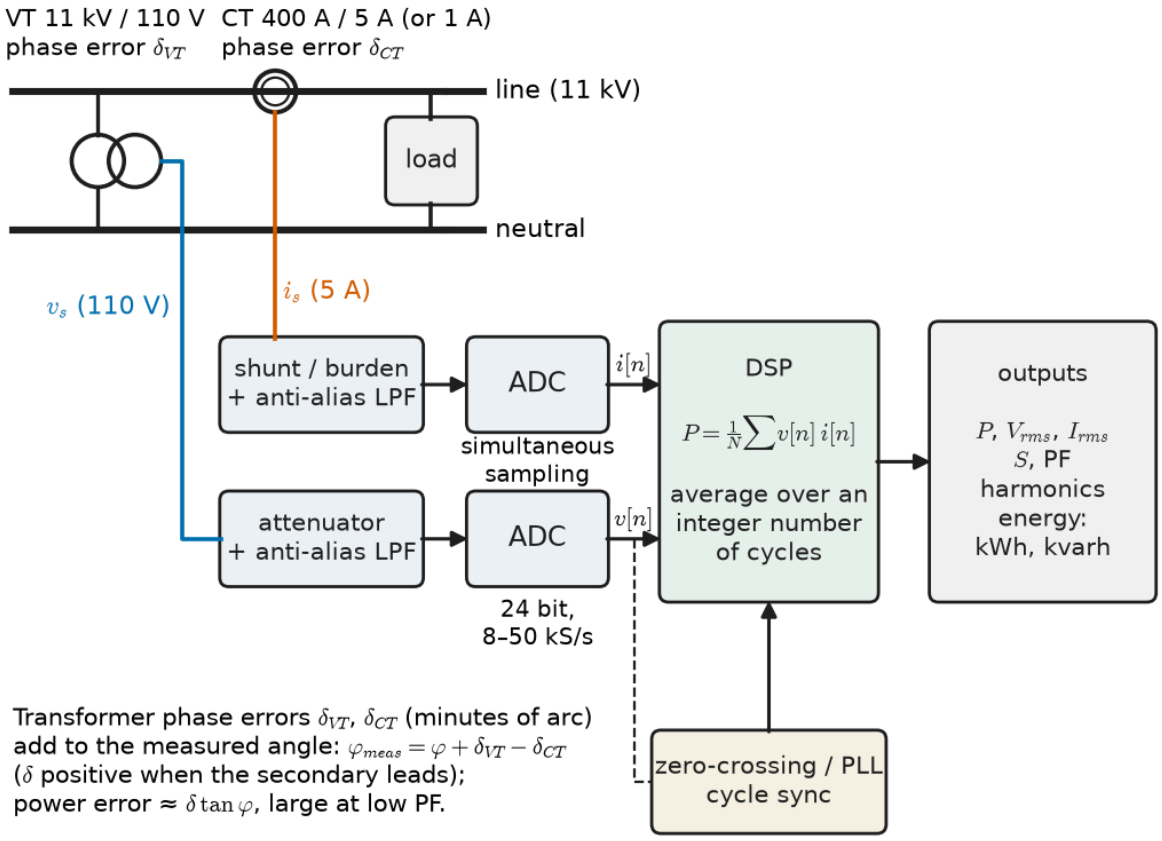


Figure 9.5 — Signal chain of a digital power analyser or revenue meter connected through instrument transformers.

Analog wattmeters

The classical **electrodynamometer wattmeter** has a fixed current coil carrying the load current and a moving voltage coil connected (through a series resistor) across the supply. The torque on the moving coil is proportional to the instantaneous product $v\,i$, and the inertia of the movement averages it. It measures true active power for any waveform within its bandwidth, which is why it was the reference instrument for a century. Induction-disc energy meters, still in service in some places, integrate power over time by driving an aluminium disc whose speed is proportional to P.

Digital power analysers and electronic meters

Modern instruments and revenue meters (Figure 9.5) sample voltage and current simultaneously with a pair of ADCs and compute

$$P = \frac{1}{N}\sum_{n=0}^{N-1} v[n]\,i[n], \qquad V_{rms} = \sqrt{\frac{1}{N}\sum v[n]^2}, \qquad I_{rms} = \sqrt{\frac{1}{N}\sum i[n]^2}$$

over a window that contains an integer number of cycles (synchronised to the measured frequency); otherwise the incomplete cycle leaks into the average. From P, V_{rms} and I_{rms}, the instrument derives $S = V_{rms}I_{rms}$ and $\mathrm{PF} = P/S$. Reactive power is computed by one of the conventions of Section 9.8, often by multiplying the voltage samples with current samples delayed by a quarter cycle, or from the fundamental components obtained by a DFT (Chapter 22). Harmonic measurement requires a sampling

rate high enough to capture the harmonics of interest without aliasing: to measure up to the 50th harmonic of 50 Hz (2.5 kHz), the sampling rate must exceed 5 kHz and the anti-alias filter must reject everything above it (Chapter 19). Power-quality standards (IEC 61000-4-7) specify 10-cycle (50 Hz) or 12-cycle (60 Hz) windows for this reason.

> **Status of the principle: Active power as the average of sampled $v \cdot i$**
> *Approximate.* It equals the true average power when the signals are sampled fast enough to avoid aliasing and averaged over an integer number of periods; errors arise from sampling skew between channels, quantisation, and non-integer windows.

Instrument transformers and accuracy

At medium and high voltage, and at currents above about 100 A, the signals reach the meter through **current transformers** (CTs, e.g. 400/5 A or 400/1 A) and **voltage transformers** (VTs, e.g. 11 kV/110 V). Each introduces a **ratio error** (a gain error) and a **phase displacement** δ between primary and secondary. Accuracy classes (IEC 61869) specify both: a class 0.5 CT has a ratio error within ±0.5 % and a phase displacement within ±30 minutes of arc (0.5°) at rated current. Revenue metering uses class 0.2 or 0.2S transformers and class 0.5S or 0.2S meters; protection CTs are designed for a different job (faithful reproduction of fault currents) and are less accurate at normal currents.

Phase displacement matters far more than its small size suggests. If the measured phase angle is $\phi - \delta$ instead of ϕ, the measured power is $VI\cos(\phi - \delta)$, a relative error of

$$\frac{\cos(\phi - \delta) - \cos\phi}{\cos\phi} \approx \delta \tan\phi$$

which grows without bound as the power factor falls.

Worked example — Phase error at low power factor. A test on a lightly loaded motor at power factor 0.2 is made through a CT with 0.5° phase displacement. Estimate the error in measured power, and compare with power factors 0.8 and 1.0.

At PF 0.2, $\phi = 78.46^\circ$ and $\tan\phi = 4.90$. With $\delta = 0.5^\circ = 0.00873$ rad, the relative error is $4.90 \times 0.00873 = 0.0428$: **4.3 %**. The exact formula gives 4.27 %. At PF 0.8 ($\tan\phi = 0.75$) the error is 0.65 %, and at PF 1.0 it is only -0.004 % (second order in δ). Measuring no-load or short-circuit losses of transformers and motors, where the power factor may be 0.1–0.2, therefore requires class 0.1–0.2 transformers or phase-compensated instruments; a class 0.5 CT that is perfectly adequate for billing a factory can give a no-load loss figure that is several percent wrong.

Clamp meters and portable analysers

For field work, current is measured with clamp-on probes: a split-core CT for AC, or a Hall-effect or fluxgate clamp for AC with DC. A Rogowski coil — an air-cored flexible loop whose output is proportional to di/dt and is integrated electronically — handles very large and harmonic-rich currents without saturating, at the cost of lower accuracy at low current. Portable power-quality analysers combine voltage leads and clamps with the sampling chain of Figure 9.5, and log P, Q, PF, THD and individual harmonics over days to diagnose problems like those in the opening.

Measuring three-phase and DC power

Three-phase power needs either three single-phase measurements or, on a three-wire system, two wattmeters whose readings sum to the total power (Blondel's theorem): the subject of Chapter 10. DC power in converters and batteries is simply VI, but measured on top of large ripple, it needs the same true-average computation; the interface with sampling and quantisation is in Chapters 19 and 21.

> **Boundary.**
> The design of instrument transformers and revenue-meter hardware belongs to *Electrical Energy & Machines*, and that of ADCs and signal-conditioning circuits to *Electronics & Computing*. This chapter supplies what they must compute.

What remains open

Open question	Where it is answered
How is power delivered and measured in balanced and unbalanced three-phase systems, and why is balanced three-phase power constant?	Chapter 10 (three-phase power, two-wattmeter method, neutral currents)
How large are triplen-harmonic neutral currents in a building, and why don't they cancel?	Chapter 10 (neutral currents)
How is a distorted waveform decomposed into harmonics, and what do magnitude and phase mean for each?	Chapter 18 (Fourier series and transforms)
How fast must a power analyser sample, and what does its anti-alias filter have to do?	Chapter 19 (sampling, aliasing, anti-aliasing filters)
How are harmonics computed from samples, and what errors does windowing introduce?	Chapter 22 (DFT, spectral leakage, windowing)
How are the resonances of capacitor banks and supply inductance analysed across frequency?	Chapter 7 (resonance); Chapter 13 (resonant circuits, passive filters)
How are active PFC circuits, rectifiers and inverters built?	*Electronics & Computing*; machines and power hardware in *Electrical Energy & Machines*

Next: Chapter 10 extends single-phase AC power to the three-phase systems that generate, transmit and distribute almost all electrical energy, where balanced power is constant and neutral currents reveal the harmonics of Section 9.8.

Chapter 10 — Three-Phase Circuits

Part II · Transient and AC Analysis

A facilities engineer is called to an office building because the neutral conductor feeding the third floor is running hot. A clamp meter shows about 80 A in each of the three phase conductors, comfortably inside their rating, yet the neutral, which the original designers sized at half the phase conductor because "a balanced load returns nothing in the neutral", is carrying more than 120 A. Across town, at a small dairy, a different fault: a loose neutral connection at the service entrance. Within minutes the lights on one phase go dim, a refrigerator on another phase hums and fails, and a television on the third phase is destroyed by an overvoltage, even though the utility's supply never left its tolerance band. Both incidents are three-phase circuit problems. Neither can be understood with the single-phase ideas of Chapters 8 and 9 alone, and both are understood completely with the phasor tools those chapters built, once we take seriously that three sinusoids 120° apart are not three separate circuits but one.

> **Why is electrical power generated, transmitted and used in three phases, and how are line and phase quantities related in star and delta connections?**
> **What happens when the load is not balanced, how is three-phase power measured, and why can a neutral conductor carry more current than any phase?**

Concept	Engineering destination
Three-phase generation	Synchronous generators, inverters, wind and solar grid interfaces
Phase sequence	Motor direction, synchronising a generator to the grid, protection relays
Star and delta connections	Distribution transformers, motor windings, capacitor banks, star–delta starters
Line and phase quantities	Nameplate ratings, cable sizing, instrument transformer selection
Balanced loads	Per-phase equivalent circuits for motors, drives and transmission studies
Unbalanced loads	Residential feeders, single-phase traction loads, broken-neutral faults
Three-phase power	Constant torque in machines, generator and transformer ratings
Power measurement	Two-wattmeter method, revenue meters, power analysers
Neutral currents	Neutral sizing, triplen harmonics from electronic loads, transformer heating
Practical applications	Industrial plants, data centres, traction, rectifiers, HVDC converters

Roadmap. Generating three phases (10.1) → which phase leads (10.2) → star and delta connections (10.3) and the $\sqrt{3}$ relations between line and phase quantities (10.4) → balanced loads and the per-phase equivalent (10.5) → unbalanced loads and the neutral-displacement problem (10.6) → total power and its remarkable constancy (10.7) → how it is measured (10.8) → the neutral conductor and triplen harmonics (10.9) → where all of this appears in practice (10.10).

10.1 Three-phase generation

From one coil to three

A single coil rotating at angular speed ω in a uniform magnetic field produces an EMF $e(t) = E_m \sin \omega t$. That is the single-phase source of Chapter 8. Its instantaneous power into a resistive load, $p(t) = E_m I_m \sin^2 \omega t$, pulsates at twice the supply frequency between zero and twice its average. For a lamp the pulsation is invisible; for a 5 MW generator the pulsating electrical power appears as a pulsating torque on the shaft, which shakes the machine and its foundation at 100 Hz.

Place three identical coils on the same rotor (or, as in practice, on the stator with a rotating field), displaced mechanically so that their EMFs are displaced in time by one third of a cycle. For a two-pole machine the coils are 120° apart in space. The three EMFs are

$$e_a(t) = E_m \sin \omega t, \qquad e_b(t) = E_m \sin(\omega t - 120^\circ), \qquad e_c(t) = E_m \sin(\omega t - 240^\circ)$$

and they are shown in Figure 10.1. At every instant they sum to zero:

$$e_a + e_b + e_c = E_m \left[\sin \omega t + \sin(\omega t - 120^\circ) + \sin(\omega t + 120^\circ) \right] = 0$$

This identity is the source of almost every advantage of three-phase systems. Its phasor form is even shorter. With the phasor operator

$$a = 1 \angle 120^\circ = -\tfrac{1}{2} + j\tfrac{\sqrt{3}}{2}, \qquad a^2 = 1 \angle 240^\circ, \qquad a^3 = 1, \qquad 1 + a + a^2 = 0$$

a balanced set of phase voltages is $\mathbf{V}_a = V \angle 0^\circ$, $\mathbf{V}_b = a^2 \mathbf{V}_a$, $\mathbf{V}_c = a \mathbf{V}_a$, and $\mathbf{V}_a + \mathbf{V}_b + \mathbf{V}_c = \mathbf{V}_a (1 + a^2 + a) = 0$.

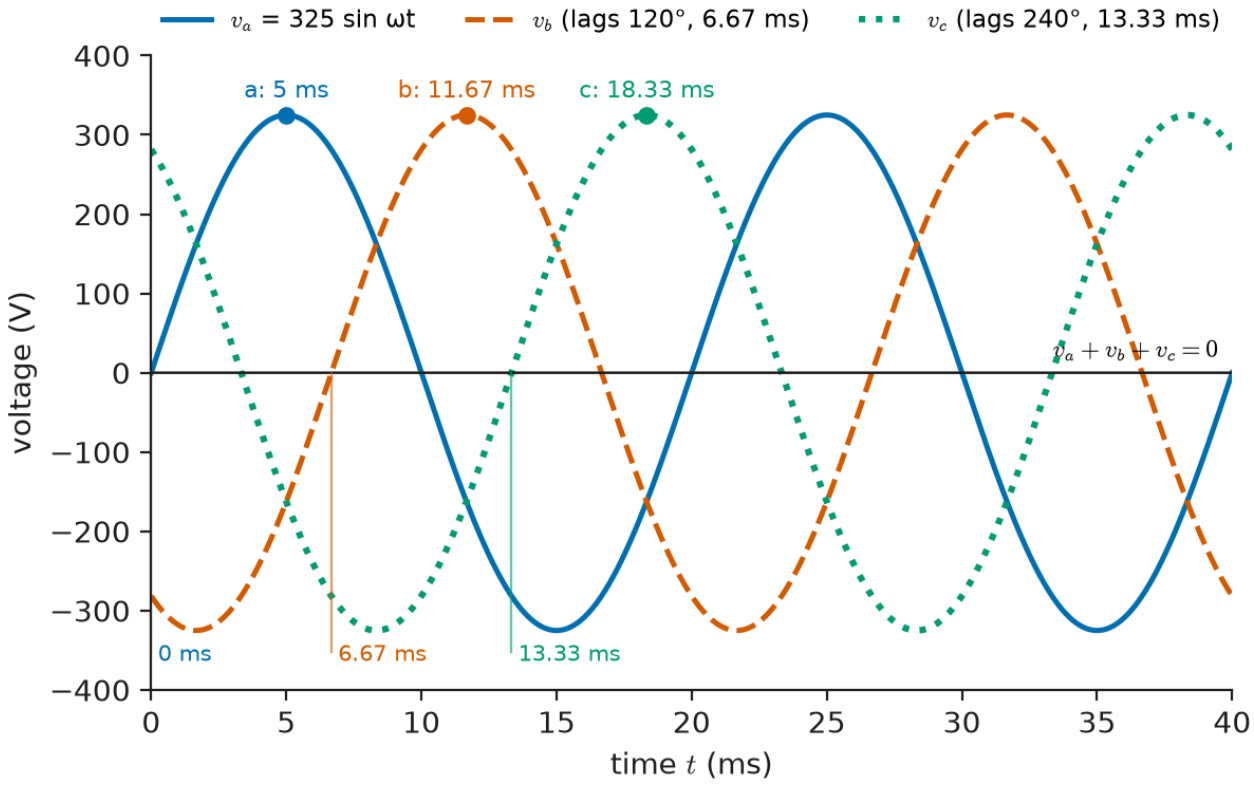


Figure 10.1 — The three phase voltages of a 230 V, 50 Hz supply; at every instant their sum is zero.

Frequency, speed and poles

A machine with p poles produces one electrical cycle for every pair of poles that passes a coil, so the electrical frequency and the rotor speed n in revolutions per minute are linked by

$$f = \frac{p\,n}{120}$$

A two-pole turbo-generator in a 50 Hz grid turns at 3000 rpm (3600 rpm in a 60 Hz grid); a four-pole machine at 1500 rpm; a hydro generator with 48 poles at 125 rpm. The construction of these machines, the shaping of the air-gap field to make the EMF sinusoidal and the excitation control that sets E_m belong to *Electrical Energy & Machines*. For circuit analysis the generator is simply three sinusoidal voltage sources of equal amplitude, 120° apart, each with a small internal impedance.

> **Boundary.**
> How the rotating field is produced, why the EMF is sinusoidal and how generators are excited and synchronised are machine topics treated in *Electrical Energy & Machines*. Power-electronic inverters that synthesise three-phase voltages from DC (in solar plants, drives and battery systems) are treated in *Electronics & Computing*. Here a three-phase source is a set of ideal or near-ideal phasor sources.

Why three phases and not two, four or six

Three is the smallest number of equally spaced phases that gives (i) instantaneous power that is constant in time for a balanced load (Section 10.7), (ii) a rotating magnetic field from stationary windings, which makes the induction motor possible, and (iii) a return current that cancels, so that the "return conductor" can be omitted or reduced. A two-phase system (90° apart) also gives constant power and a rotating field but needs a return conductor carrying $\sqrt{2}$ times the phase current. Six- and twelve-phase systems exist inside rectifiers to reduce ripple, but each extra phase adds a conductor without improving power constancy.

The conductor saving is easy to quantify. Deliver power P at line voltage V over a given distance with given losses. Single-phase two-wire: current P/V in each of two conductors, loss $2P^2R_1/V^2$. Three-phase three-wire at unity power factor: line current $P/(\sqrt{3}V)$ in each of three conductors, loss $3 \cdot P^2R_3/(3V^2) = P^2R_3/V^2$. Equal losses require $R_3 = 2R_1$, i.e. each three-phase conductor has half the cross-section of a single-phase conductor. Total conductor volume: three half-sized conductors against two full ones, $1.5/2 = 75\,\%$. Three-phase transmission uses three quarters of the copper or aluminium of single-phase transmission for the same power, voltage and loss.

Table 10.1 gives the three-phase supply voltages an engineer meets most often.

Table 10.1 — Representative three-phase system voltages (line-to-line / line-to-neutral, nominal).

System	Line-to-line	Line-to-neutral	Frequency	Typical use
IEC low voltage (Europe, South Asia, Africa, Australia)	400 V	230 V	50 Hz	Homes, offices, small industry
North American low voltage	208 V	120 V	60 Hz	Commercial buildings
North American industrial	480 V	277 V	60 Hz	Motors, lighting in industry
Industrial low voltage	690 V	400 V	50 Hz	Large drives, wind turbines
Medium-voltage distribution	11 kV (also 6.6, 13.8, 20, 33 kV)	6.35 kV	50/60 Hz	Feeders, large motors
Subtransmission	66–132 kV	38–76 kV	50/60 Hz	Regional networks
Transmission	220–400 kV (up to 765 kV and more)	127–231 kV	50/60 Hz	Bulk power transfer
Generator terminals	11–25 kV	6.4–14.4 kV	50/60 Hz	Step-up transformer input

Statutory tolerances are typically ±10 % at the low-voltage service point (230 V +10 %/−10 % in IEC countries, 120 V ±5 % in North America).

10.2 Phase sequence

Positive and negative sequence

The order in which the three phase voltages reach their positive peaks is the **phase sequence** (or phase rotation). In Figure 10.1, v_a peaks first, then v_b a third of a cycle later, then v_c: the sequence is abc, also called positive sequence. Reversing the direction of rotation of the generator, or swapping any two supply leads, gives acb, the negative sequence.

In phasor form, with $\mathbf{V}_a$ as reference:

Sequence	$\mathbf{V}_a$	$\mathbf{V}_b$	$\mathbf{V}_c$
Positive (abc)	$V\angle 0^\circ$	$V\angle -120^\circ$	$V\angle +120^\circ$
Negative (acb)	$V\angle 0^\circ$	$V\angle +120^\circ$	$V\angle -120^\circ$

Phasors rotate counter-clockwise, so in a positive-sequence diagram an observer at a fixed point sees a, then b, then c pass.

Why sequence matters

The sequence has no effect on a balanced static load: the magnitudes of currents and powers are the same either way. It matters whenever something rotates or whenever phases are compared:

- **Motor direction.** A three-phase induction motor turns in the direction of the rotating field, which is set by the sequence. Swapping two supply leads reverses a conveyor, a pump or a hoist. Pumps run backwards deliver little flow and can damage seals, so commissioning procedures include a sequence check.
- **Generator synchronisation.** A generator may be connected to the grid only when its voltage, frequency, phase angle *and* sequence match. Closing a breaker with the wrong sequence is a three-phase short circuit through two machines.
- **Two-wattmeter readings and power-factor meters.** The assignment of the "high" and "low" readings in Section 10.8 depends on sequence.
- **Unbalanced loads.** In Section 10.6 the voltages on the loads of an unbalanced, ungrounded star depend on the sequence: swapping two phases changes which load sees the overvoltage.

A phase-sequence indicator is a tiny unbalanced circuit used on purpose: a capacitor and two lamps in a star with no neutral. The lamp connected to the phase that follows the capacitor's phase glows brightly, the other dimly. The principle is the neutral-displacement calculation of Section 10.6.

10.3 Star and delta connections

Star (wye, Y)

In a **star** connection one end of each of the three source windings (or load impedances) is joined at a common point, the **neutral** n, and the other ends are brought out to the three **lines** a, b, c (Figure 10.2a). The neutral may be brought out as a fourth conductor (three-phase four-wire, the normal low-voltage distribution system) or left unconnected (three-wire). In distribution systems the neutral is usually earthed at the transformer.

The voltages from line to neutral, $\mathbf{V}_{an}, \mathbf{V}_{bn}, \mathbf{V}_{cn}$, are the **phase voltages** of a star. The voltages between lines, $\mathbf{V}_{ab} = \mathbf{V}_{an} - \mathbf{V}_{bn}$ and so on, are the **line voltages**. The current in each line is the same as the current in the corresponding phase winding.

Delta (mesh, Δ)

In a **delta** connection the three windings or impedances are joined end to end in a closed loop, and the lines are taken from the three junctions (Figure 10.2b). There is no neutral point. Each phase element is connected directly between two lines, so the phase voltage *is* a line voltage. Each line current is the difference of two phase currents, by KCL at the junction.

A delta of sources is a closed loop of EMFs. It is permissible only because a balanced set sums to zero, so no circulating current flows (KVL around the loop gives $\mathbf{E}_{ab} + \mathbf{E}_{bc} + \mathbf{E}_{ca} = 0$). Any imbalance or any third-harmonic content in the EMFs drives a circulating current limited only by the winding impedances. This is one reason why generators are almost always star-connected, while delta windings appear in transformers where the circulating third-harmonic current is wanted (Section 10.9).

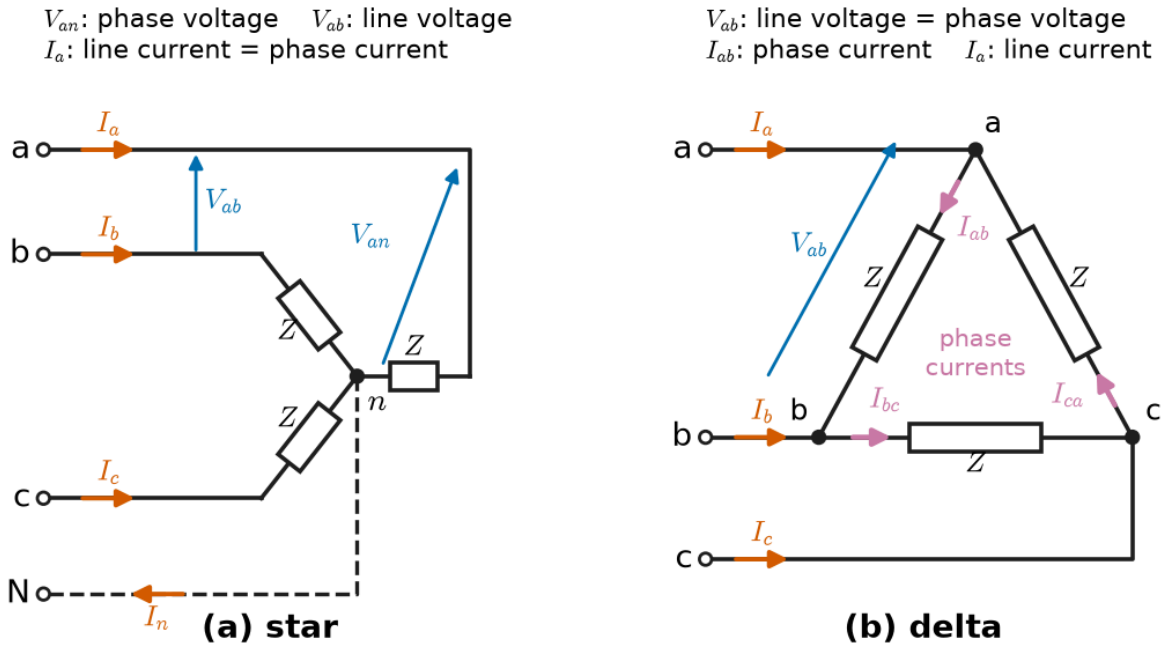


Figure 10.2 — Star (a) and delta (b) connections of a three-phase load, with line and phase quantities marked.

Source and load combinations

Source and load may each be star or delta, giving four combinations: Y–Y, Y–Δ, Δ–Y and Δ–Δ. Practical systems contain all four, often in a single path: a star-connected generator feeds a delta–star step-up transformer, which feeds a star–star or delta–star transmission network, a delta–star distribution transformer and finally star-connected household loads and delta-connected motors. For analysis, any balanced system can be reduced to an equivalent Y–Y system (Section 10.5), and unbalanced systems are solved directly with nodal or mesh analysis.

Star–delta equivalence for balanced loads

The general Y–Δ (or T–π) transformation of Chapter 4 applies to phasor impedances. For a balanced load with Z_Δ in each delta branch, the equivalent star has

$$Z_Y = \frac{Z_\Delta \, Z_\Delta}{3 Z_\Delta} = \frac{Z_\Delta}{3}$$

and conversely $Z_\Delta = 3 Z_Y$. A delta of 30 Ω resistors draws the same line currents as a star of 10 Ω resistors on the same supply.

> **Status of the principle: Star–delta equivalence**
> *Derived*. Follows from KCL and KVL applied to the three terminals; exact for linear impedances at a single frequency, and the balanced form $Z_Y = Z_\Delta/3$ holds at every frequency only when each branch has the same impedance function.

10.4 Line and phase quantities

Voltages in a star

For a positive-sequence star with $\mathbf{V}_{an} = V_p \angle 0^\circ$:

$$\mathbf{V}_{ab} = \mathbf{V}_{an} - \mathbf{V}_{bn} = V_p \left(1 - a^2\right) = V_p \left(\tfrac{3}{2} + j\tfrac{\sqrt{3}}{2}\right) = \sqrt{3}\, V_p \angle 30^\circ$$

and similarly $\mathbf{V}_{bc} = \sqrt{3} V_p \angle -90^\circ$, $\mathbf{V}_{ca} = \sqrt{3} V_p \angle 150^\circ$. The line voltages form a balanced set of their own, larger by $\sqrt{3}$ and leading the corresponding phase voltages by 30° (Figure 10.3):

$$V_L = \sqrt{3}\, V_p, \qquad \mathbf{V}_{ab} = \sqrt{3}\, \mathbf{V}_{an} \angle 30^\circ \quad \text{(positive sequence)}$$

For negative sequence the magnitude relation is the same but the line voltage *lags* by 30°. A 230 V phase voltage gives $\sqrt{3} \times 230 = 398.4$ V, which is why the IEC system is described as 230/400 V. North American 120 V phases give $\sqrt{3} \times 120 = 207.8$ V, the "208 V" of Table 10.1.

The line current in a star equals the phase current: $I_L = I_p$.

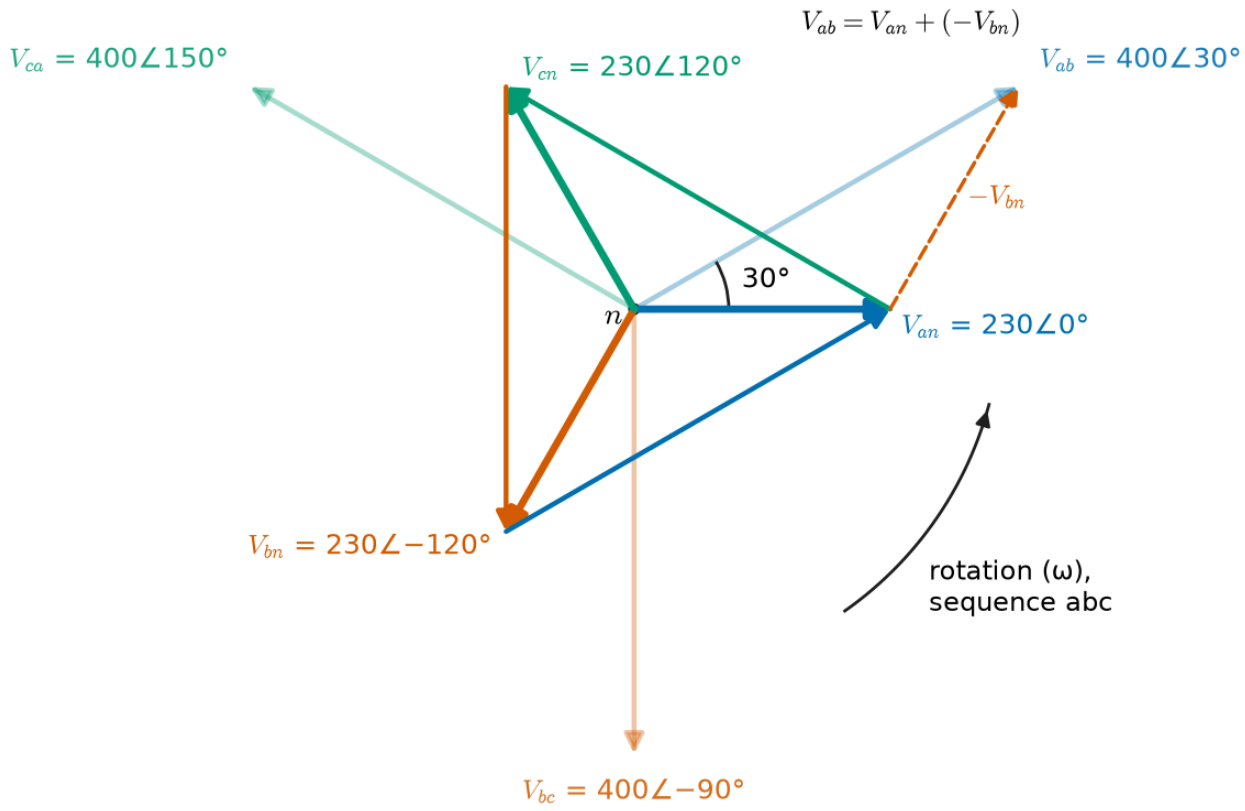


Figure 10.3 — Phase and line voltage phasors of a positive-sequence star: line voltages are larger by the square root of 3 and lead by 30 degrees.

Currents in a delta

For a balanced delta with phase currents $\mathbf{I}_{ab} = I_p\angle\theta$, $\mathbf{I}_{bc} = a^2\mathbf{I}_{ab}$, $\mathbf{I}_{ca} = a\mathbf{I}_{ab}$, KCL at node a gives the line current:

$$\mathbf{I}_a = \mathbf{I}_{ab} - \mathbf{I}_{ca} = \mathbf{I}_{ab}\,(1 - a) = \sqrt{3}\,\mathbf{I}_{ab}\angle -30^\circ$$

The line currents are larger than the phase currents by $\sqrt{3}$ and lag them by 30° (positive sequence). The line voltage of a delta equals its phase voltage: $V_L = V_p$.

Summary

Quantity	Star (Y)	Delta (Δ)
Line voltage vs phase voltage	$V_L = \sqrt{3}\,V_p$, leads by 30°	$V_L = V_p$
Line current vs phase current	$I_L = I_p$	$I_L = \sqrt{3}\,I_p$, lags by 30°
Total apparent power	$3V_pI_p = \sqrt{3}V_LI_L$	$3V_pI_p = \sqrt{3}V_LI_L$
Neutral available	Yes	No
Phase element voltage stress	$V_L/\sqrt{3}$	V_L

The angles in the table are for positive sequence; for negative sequence the signs of the 30° shifts reverse.

> **Status of the principle: The √3 relations**
> *Derived*. Consequences of KVL (star) and KCL (delta) applied to a perfectly balanced, sinusoidal, positive- or negative-sequence set; they do not hold for unbalanced voltages or currents, nor for harmonic components that are not themselves balanced positive- or negative-sequence sets (triplens, Section 10.9).

Ratings and nameplates

By convention a three-phase equipment rating quotes the *line* voltage and the *line* current: "400 V, 22 A, 11 kW" on a motor nameplate means 400 V between any two terminals and 22 A in each supply cable, whatever the internal connection. Many motors are marked "Δ 400 V / Y 690 V": the same winding, each phase designed for 400 V, can be run in delta on a 400 V supply or in star on a 690 V supply, where each phase again sees $690/\sqrt{3} = 398$ V. The star–delta starter exploits the same fact: starting a 400 V delta motor in star puts $400/\sqrt{3}$ V on each phase, so each phase current falls by $\sqrt{3}$ and the line current (no longer multiplied by $\sqrt{3}$) falls to one third of the direct-on-line value, along with the starting torque. The motor is switched to delta once it is near speed.

10.5 Balanced loads

The per-phase equivalent

A **balanced** three-phase system has balanced source voltages and identical impedances in each phase of every load and line. For such a system, the neutral points of all star-connected parts are at the same potential, whether or not they are joined by a wire. To see this, take a Y–Y system without a neutral wire and write nodal analysis at the load neutral N, with the source neutral n as reference:

$$\frac{\mathbf{V}_{an}-\mathbf{V}_{Nn}}{Z}+\frac{\mathbf{V}_{bn}-\mathbf{V}_{Nn}}{Z}+\frac{\mathbf{V}_{cn}-\mathbf{V}_{Nn}}{Z}=0 \quad\Rightarrow\quad \mathbf{V}_{Nn}=\frac{\mathbf{V}_{an}+\mathbf{V}_{bn}+\mathbf{V}_{cn}}{3}=0$$

Because $\mathbf{V}_{Nn}=0$, a neutral wire (of any impedance) would carry no current, and each phase can be solved alone, as if it were a single-phase circuit between line and neutral. This is the **per-phase equivalent**: convert every delta to its equivalent star, draw phase a from source neutral to load neutral, solve it, and obtain phases b and c by rotating the results by $-120°$ and $+120°$. The method reduces a balanced three-phase power system of any size to a single-phase problem; it is the basis of load-flow studies, transmission-line models (Chapter 12) and machine equivalent circuits.

> **Status of the principle: Per-phase equivalent**
> *Derived*. Exact for linear, perfectly balanced systems with balanced sources, by the symmetry argument above; it fails whenever any element is unbalanced (a single-phase load, an open conductor, a single-line-to-ground fault), and the system must then be solved as a whole or by symmetrical components.

Worked example — Balanced star load on a 400 V supply. A three-phase load with $Z = 8 + j6\ \Omega$ per phase is connected in star to a 400 V, 50 Hz, positive-sequence supply. Take $\mathbf{V}_{an}$ as reference.

Phase voltage: $V_p = 400/\sqrt{3} = 230.94$ V, so $\mathbf{V}_{an} = 230.94\angle 0^\circ$ V.

Impedance: $|Z| = \sqrt{8^2 + 6^2} = 10\ \Omega$, $\varphi = \arctan(6/8) = 36.87^\circ$.

Line (= phase) current: $\mathbf{I}_a = 230.94\angle 0^\circ / 10\angle 36.87^\circ = 23.094\angle -36.87^\circ$ A; $\mathbf{I}_b = 23.094\angle -156.87^\circ$ A; $\mathbf{I}_c = 23.094\angle 83.13^\circ$ A.

Power per phase: $P_p = I_p^2 R = 23.094^2 \times 8 = 4266.7$ W; $Q_p = I_p^2 X = 3200$ var.

Total: $P = 12.8$ kW, $Q = 9.6$ kvar, $S = \sqrt{12.8^2 + 9.6^2} = 16.0$ kVA, power factor $\cos 36.87^\circ = 0.80$ lagging. Check with line quantities: $\sqrt{3} V_L I_L = \sqrt{3} \times 400 \times 23.094 = 16\,000$ VA.

Worked example — The same impedances in delta. Reconnect the three $8 + j6\ \Omega$ impedances in delta on the same supply.

Each impedance now sees the full line voltage: $\mathbf{I}_{ab} = 400\angle 30^\circ / 10\angle 36.87^\circ = 40\angle -6.87^\circ$ A.

Line current: $I_L = \sqrt{3} \times 40 = 69.28$ A, at $\mathbf{I}_a = 69.28\angle -36.87^\circ$ A (30° behind $\mathbf{I}_{ab}$).

Power: $P = 3 \times 40^2 \times 8 = 38.4$ kW, $Q = 28.8$ kvar, $S = 48$ kVA.

The delta connection draws exactly three times the power and three times the line current of the star connection. This factor of 3 is the star–delta starter of Section 10.4 seen from the other side; it is also the reason a delta-connected heater accidentally wired for a star supply overheats.

Power-factor correction in three-phase plants

Chapter 9 sized a capacitor to correct a single-phase load. In a three-phase plant the capacitor bank is a three-phase load of its own, connected in star or delta, and its total reactive power is what matters. For the star load of the first example (12.8 kW, 9.6 kvar), raising the power factor to 0.95 lagging requires the total reactive power to fall to $12.8 \tan(\arccos 0.95) = 4.207$ kvar, so the bank must supply $Q_C = 9.6 - 4.207 = 5.393$ kvar, or 1.798 kvar per phase.

In delta, each capacitor sees 400 V: $C_\Delta = Q_{C,p} / (\omega V_L^2) = 1798 / (2\pi \cdot 50 \cdot 400^2) = 35.8\ \mu$F. In star, each sees 230.9 V: $C_Y = 1798 / (2\pi \cdot 50 \cdot 230.94^2) = 107.3\ \mu$F.

The delta bank needs one third of the capacitance, but each capacitor must be rated for line voltage. At low voltage, where 440 V or 480 V capacitors are cheap, delta is the usual choice; at medium voltage, where insulation dominates the cost, star-connected (often ungrounded) banks are common. After correction the line current falls from 23.09 A to $12\,800 / 0.95 / (\sqrt{3} \times 400) = 19.45$ A.

10.6 Unbalanced loads

Where unbalance comes from

Most loads in a building or a village are single-phase, connected between one line and neutral. Distribution engineers spread them across the three phases, but at any instant the three phases carry different loads: one floor's air conditioners are on, a kitchen's ovens are off. Railways supplied from the public grid draw large single-phase loads between two lines. A blown fuse in one phase of a capacitor bank, a broken conductor or a single-line-to-ground fault also unbalance the system. Even generator and transformer impedances are not perfectly equal. Unbalance is the normal condition of a low-voltage network; balance is an approximation that improves as more loads are aggregated.

Four-wire star with a solid neutral

When a star load is connected to a star source through a neutral conductor of negligible impedance, the load neutral is held at the source neutral potential. Each load impedance then sees its own phase voltage, independently of the others, and the phases are solved separately. The neutral carries the phasor sum of the line currents:

$$\mathbf{I}_N = \mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c$$

Worked example — A residential feeder, with and without its neutral. A 230/400 V four-wire supply feeds three groups of resistive load (heaters, kettles, ovens): 9.2 kW on phase a, 4.6 kW on phase b and 2.3 kW on phase c, all rated at 230 V.

Load resistances: $R_a = 230^2/9200 = 5.75\ \Omega$, $R_b = 11.5\ \Omega$, $R_c = 23\ \Omega$.

With the neutral intact. Each load sees 230 V: $\mathbf{I}_a = 40\angle 0^\circ$ A, $\mathbf{I}_b = 20\angle -120^\circ$ A, $\mathbf{I}_c = 10\angle 120^\circ$ A. The neutral current is

$$\mathbf{I}_N = 40 + 20\angle -120^\circ + 10\angle 120^\circ = 26.46\angle -19.1^\circ\ \text{A}$$

Smaller than the largest phase current, as is usual for fundamental-frequency unbalance, but far from zero.

With the neutral broken at the service head. The load neutral N is now a floating node. Nodal analysis with the source neutral as reference gives the **neutral displacement voltage**

$$\mathbf{V}_{Nn} = \frac{\mathbf{V}_{an}Y_a + \mathbf{V}_{bn}Y_b + \mathbf{V}_{cn}Y_c}{Y_a + Y_b + Y_c}$$

With $Y_a = 0.1739$ S, $Y_b = 0.0870$ S, $Y_c = 0.0435$ S: $\mathbf{V}_{Nn} = 86.93\angle -19.1^\circ$ V. The voltages across the loads are $\mathbf{V}_{aN} = \mathbf{V}_{an} - \mathbf{V}_{Nn}$ and so on:

Phase	Load (rated)	Voltage with neutral	Voltage with broken neutral	Power with broken neutral
a	9.2 kW	230 V	150.6 V	3.94 kW
b	4.6 kW	230 V	260.8 V	5.91 kW
c	2.3 kW	230 V	301.1 V	3.94 kW

The heavily loaded phase drops to 65 % of its rated voltage; the lightly loaded phase rises to 131 %. Electronic equipment on phase c, designed for 230 V +10 %, sees 301 V and fails. This is the dairy of the chapter opening. Figure 10.4 shows the geometry: the load neutral is pulled towards the phase with the largest admittance, away from the lightly loaded phases.

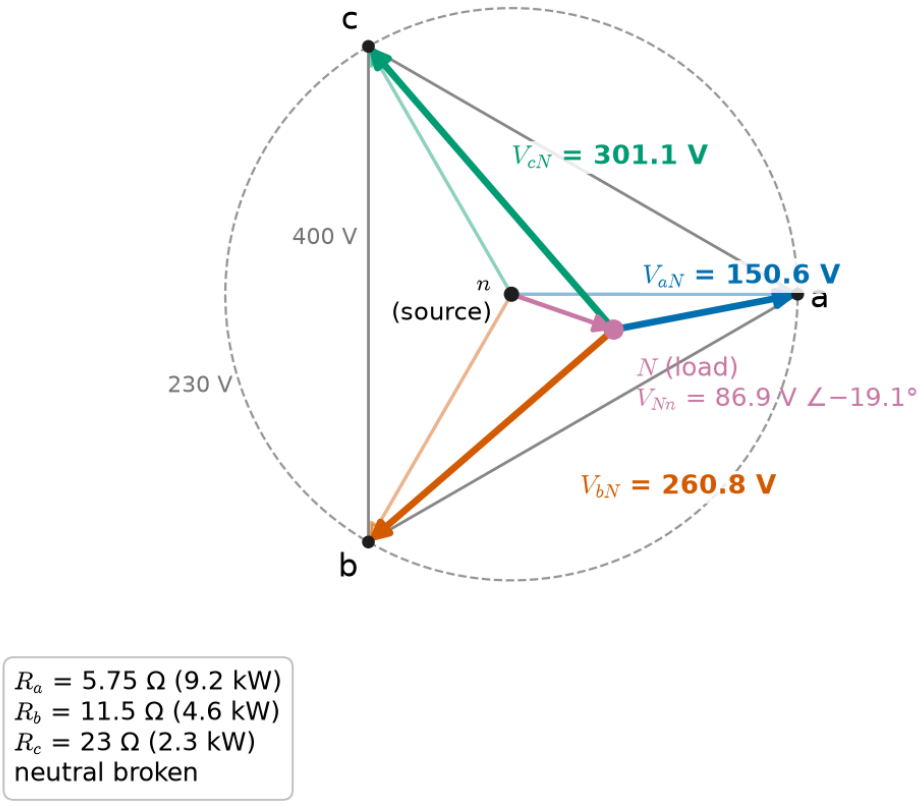


Figure 10.4 — Neutral displacement in an unbalanced star load with a broken neutral: the load neutral moves towards the heavily loaded phase.

The same calculation explains the two-lamp phase-sequence indicator of Section 10.2 and the danger of a high-impedance neutral joint: it is not necessary for the neutral to be completely open for significant displacement to occur. Distribution codes therefore forbid fuses and switches in the neutral alone, require neutral conductors to be mechanically robust, and recommend protection that disconnects on overvoltage.

Three-wire unbalanced loads

A three-wire supply (no neutral available) feeding an unbalanced star behaves exactly like the broken-neutral case: the line currents must sum to zero, so the load neutral floats to $\mathbf{V}_{Nn}$ above. An unbalanced delta load is simpler: each element is connected between two lines and sees its own line voltage regardless of the others, so the phase currents follow directly from $\mathbf{I}_{ab} = \mathbf{V}_{ab}/Z_{ab}$, and the line currents from KCL. A 20 kVA single-phase welder connected between lines a and b of a 400 V supply, for example, draws $20\,000/400 = 50$ A in lines a and b and nothing in line c. Large single-phase loads such as traction substations are connected line-to-line for this reason and rotated among the three line pairs from one substation to the next.

For general unbalanced networks, nodal or mesh analysis (Chapter 3) in phasor form handles any configuration; there is nothing special about three phases except the bookkeeping.

Symmetrical components: a first look

An unbalanced set of three phasors can be written as the sum of three balanced sets: a **positive-sequence** set (abc), a **negative-sequence** set (acb) and a **zero-sequence** set (three equal phasors in phase). For currents,

$$\begin{aligned}
\mathbf{I}_0 &= \tfrac{1}{3}\left(\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c\right) \\
\mathbf{I}_1 &= \tfrac{1}{3}\left(\mathbf{I}_a + a\,\mathbf{I}_b + a^2\mathbf{I}_c\right) \\
\mathbf{I}_2 &= \tfrac{1}{3}\left(\mathbf{I}_a + a^2\mathbf{I}_b + a\,\mathbf{I}_c\right)
\end{aligned}$$

For the residential feeder with its neutral intact, $\mathbf{I}_1 = 23.33\angle 0^\circ$ A, $\mathbf{I}_2 = 8.82\angle 19.1^\circ$ A and $\mathbf{I}_0 = 8.82\angle -19.1^\circ$ A. The neutral current is $3\mathbf{I}_0 = 26.46$ A, as found directly. The ratio $|\mathbf{I}_2|/|\mathbf{I}_1| = 37.8\,\%$ is the current unbalance factor.

The value of the decomposition is that, in a network whose elements are themselves symmetrical (lines, transformers, machines), each sequence set flows in its own uncoupled per-phase network with its own impedance. Negative-sequence currents produce a field rotating backwards in a motor, causing rotor heating; zero-sequence currents flow only where there is a neutral or earth path. This is the standard tool of fault analysis and protection.

> **Status of the principle: Symmetrical components**
> *Derived*. A linear change of variables that is exact for any set of three phasors; it simplifies analysis only when the network itself is symmetrical, so that the sequence networks decouple.

> **Boundary.**
> Sequence impedances of machines and transformers, fault calculations and protection settings are power-system topics treated in *Electrical Energy & Machines*.

Voltage unbalance in practice

Supply standards limit voltage unbalance rather than current unbalance, typically to 2 % negative-sequence voltage at the point of supply (EN 50160, IEC 61000-2-2). A quick estimate uses line voltages:

the NEMA definition is the maximum deviation from the average, divided by the average. For measured line voltages of 400, 396 and 388 V, the average is 394.7 V, the maximum deviation 6.67 V, and the unbalance 1.69 %. Motors are sensitive to this: a rule of thumb is that the extra winding temperature rise in per cent is roughly twice the square of the percentage voltage unbalance, so 1.7 % unbalance gives about 6 % more temperature rise, and motor derating curves begin at about 1 %.

10.7 Three-phase power

Instantaneous power is constant

For a balanced load with phase voltage $v_a = \sqrt{2}V_p\cos\omega t$ and current $i_a = \sqrt{2}I_p\cos(\omega t - \varphi)$, the instantaneous power in phase a is (Chapter 9)

$$p_a(t) = V_p I_p\left[\cos\varphi + \cos(2\omega t - \varphi)\right]$$

The other phases have the same form with ωt replaced by $\omega t - 120^\circ$ and $\omega t + 120^\circ$. Their double-frequency terms are three cosines at $2\omega t - \varphi$, $2\omega t - \varphi - 240^\circ$ and $2\omega t - \varphi + 240^\circ$, again a balanced set, and they sum to zero. What remains is

$$p(t) = p_a + p_b + p_c = 3V_pI_p\cos\varphi$$

constant at every instant. For the star load of Section 10.5, a time-domain evaluation gives exactly 12 800 W at every instant over two cycles, although each individual phase power swings between $4266.7 - 5333.3 = -1066.7$ W and $4266.7 + 5333.3 = 9600$ W at 100 Hz (the per-phase apparent power is 5333.3 VA).

> **Status of the principle: Constant instantaneous power of a balanced three-phase system**
> *Derived.* Follows from the identity $1 + a + a^2 = 0$ applied to the double-frequency terms; it holds for balanced sinusoidal operation at any power factor and fails under unbalance (a negative-sequence component produces a 2ω power pulsation) and with harmonics.

A constant electrical power implies a constant electromagnetic torque in a machine running at constant speed, which is why three-phase generators and motors run smoothly while large single-phase machines vibrate at twice the line frequency. It also means a three-phase rectifier draws power from the grid with little ripple and needs a much smaller smoothing capacitor than a single-phase one.

Total active, reactive and apparent power

With V_L and I_L the line quantities and φ the angle of the per-phase load impedance (the angle between *phase* voltage and *phase* current, not between line voltage and line current):

$$P = \sqrt{3}\,V_L I_L\cos\varphi, \qquad Q = \sqrt{3}\,V_L I_L\sin\varphi, \qquad S = \sqrt{3}\,V_L I_L$$

and the complex power of the whole load is $\mathbf{S} = \mathbf{V}_a\mathbf{I}_a^* + \mathbf{V}_b\mathbf{I}_b^* + \mathbf{V}_c\mathbf{I}_c^*$, valid for balanced and unbalanced loads alike. The same formula holds for star and delta. For an unbalanced load the per-

phase powers are added; the scalar formula $\sqrt{3}V_L I_L$ no longer gives the apparent power, and different definitions of three-phase apparent power (arithmetic, vector, effective) give different numbers. Revenue meters and power-quality standards (IEEE 1459) specify which is used.

Representative ratings

Table 10.2 — Typical three-phase equipment ratings.

Equipment	Rating	Line voltage	Full-load line current	Typical power factor
Small induction motor	2.2 kW	400 V	4.7 A	0.80–0.83
Medium induction motor	11 kW	400 V	21–22 A	0.84–0.86
Large induction motor	110 kW	400 V	190–200 A	0.86–0.89
Distribution transformer	630 kVA	400 V (secondary)	909 A	Load dependent
Distribution transformer	630 kVA	11 kV (primary)	33 A	Load dependent
Data-centre UPS	500 kVA	400 V	722 A	0.9–1.0
Electric-vehicle fast charger	150 kW	400 V	220 A	>0.98
Turbo-generator	600 MVA	22 kV	15.7 kA	0.85–0.9 rated

The transformer currents follow from $I_L = S/(\sqrt{3}V_L)$: $630\,000/(\sqrt{3}\times 400) = 909$ A and $630\,000/(\sqrt{3}\times 11\,000) = 33.1$ A.

10.8 Power measurement

The three-wattmeter method

A wattmeter (electrodynamic in the past, digital sampling today) measures the average of the product of the voltage across its voltage element and the current through its current element: $W = \frac{1}{T}\int_0^T v\,i\,dt = \mathrm{Re}\{\mathbf{V}\mathbf{I}^{*}\}$. In a four-wire system, three wattmeters, each with its current element in one line and its voltage element between that line and the neutral, read the three phase powers directly; their sum is the total power, balanced or not. Modern revenue meters for four-wire supplies are three-element meters of exactly this kind.

Blondel's theorem and the two-wattmeter method

When there is no neutral, the three-wattmeter method still works if the voltage elements are joined to *any* common point, because the sum of the line currents is zero:

$$\sum_k \mathrm{Re}\{(\mathbf{V}_k - \mathbf{V}_x)\mathbf{I}_k^*\} = \sum_k \mathrm{Re}\{\mathbf{V}_k\mathbf{I}_k^*\} - \mathrm{Re}\left\{\mathbf{V}_x \sum_k \mathbf{I}_k^*\right\} = P$$

for any potential $\mathbf{V}_x$. Choose the common point to be one of the lines, say b: the wattmeter in line b then has zero voltage across it and reads zero, and can be removed. This is **Blondel's theorem**: in a system of n conductors, $n-1$ wattmeters suffice to measure total power. For a three-wire system, two wattmeters suffice (Figure 10.5).

> **Status of the principle: Blondel's theorem**
> *Derived.* A direct consequence of KCL ($\sum \mathbf{I}_k = 0$ in an n-wire system with no other return path); it holds for unbalanced loads and non-sinusoidal waveforms, but fails if current returns through an uninstrumented path such as earth or an unmetered neutral.

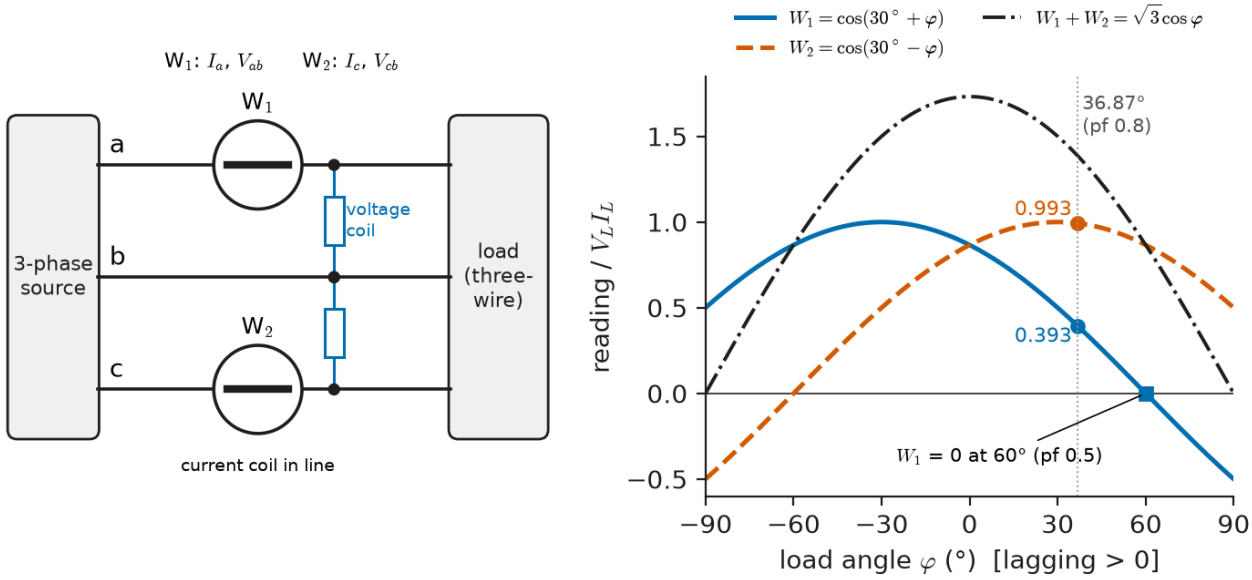


Figure 10.5 — Two-wattmeter connection on a three-wire supply, and the readings as functions of load power-factor angle.

For a balanced load and positive sequence, the meter with its current element in line a and voltage element across $\mathbf{V}_{ab}$ reads, and the meter in line c across $\mathbf{V}_{cb}$ reads,

$$W_1 = V_L I_L \cos(30^\circ + \varphi), \qquad W_2 = V_L I_L \cos(30^\circ - \varphi)$$

with sum $W_1 + W_2 = 2V_L I_L \cos 30^\circ \cos\varphi = \sqrt{3} V_L I_L \cos\varphi = P$ and difference $W_2 - W_1 = V_L I_L \sin\varphi$, so that

$$Q = \sqrt{3}\,(W_2 - W_1), \qquad \tan\varphi = \sqrt{3}\,\frac{W_2 - W_1}{W_2 + W_1}$$

The two readings therefore give the power factor of a balanced load as well as its power. Several special cases are useful checks:

Load angle φ	Power factor	W_1	W_2
0°	1.0	Equal to W_2	$P/2$
36.87°	0.8 lagging	$0.393\,V_L I_L$	$0.993\,V_L I_L$
60°	0.5 lagging	0	P
75°	0.26 lagging	Negative	Larger than P
90°	0	$-V_L I_L/2$	$+V_L I_L/2$

Below a power factor of 0.5 one meter reads negative, and with analogue instruments its current or voltage connections had to be reversed and the reading subtracted. A lightly loaded induction motor, whose power factor can be 0.2–0.3, is the classic case.

Worked example — Two-wattmeter readings for the balanced star load. For the 400 V load of Section 10.5 ($I_L = 23.094$ A, $\varphi = 36.87^\circ$ lagging):

$W_1 = 400 \times 23.094 \times \cos 66.87^\circ = 3628.7$ W, $W_2 = 400 \times 23.094 \times \cos 6.87^\circ = 9171.3$ W.

Sum: 12 800.0 W, the load power. $\sqrt{3}(W_2 - W_1) = \sqrt{3} \times 5542.6 = 9600$ var, the load reactive power. $\tan\varphi = 9600/12\,800 = 0.75$, $\varphi = 36.87^\circ$. A direct phasor calculation of $\mathrm{Re}\{\mathbf{V}_{ab}\mathbf{I}_a^*\}$ and $\mathrm{Re}\{\mathbf{V}_{cb}\mathbf{I}_c^*\}$ gives the same two numbers, which confirms the assignment of W_1 to line a for positive sequence. With negative sequence the two readings swap meters, which is why a sequence check precedes the interpretation of the readings.

For an *unbalanced* load the sum $W_1 + W_2$ is still the total power (Blondel), but the reactive-power and power-factor formulas no longer apply.

Other methods and modern instruments

- **One wattmeter, balanced load.** With a balanced four-wire load, one line-to-neutral wattmeter times three gives the total. With a balanced three-wire load, an artificial star point made of three equal resistors provides the neutral.
- **Reactive power by one wattmeter.** A wattmeter with its current element in line a and its voltage element across $\mathbf{V}_{bc}$ (which is in quadrature with $\mathbf{V}_{an}$) reads $V_L I_L \sin\varphi = Q/\sqrt{3}$ for a balanced load. Many analogue varmeters were built on this principle.
- **Instrument transformers.** At medium and high voltage, wattmeters and meters are fed from voltage transformers (typically 110 V secondary line-to-line) and current transformers (1 A or 5 A secondary). Accuracy classes such as 0.2 and 0.5 (IEC 61869) specify ratio and phase-angle errors; a CT phase error of 10 minutes of arc at low power factor produces a noticeable power error, because the error is proportional to $\tan\varphi$.
- **Digital power analysers and smart meters.** These sample all voltages and currents at kHz to MHz rates and compute P as the average of $\sum_k v_k i_k$, so they automatically include harmonics. They report the total, per-phase and sequence quantities, harmonic spectra (Chapter 22) and unbalance factors. A three-wire meter uses the two-element (Blondel) connection; a four-wire meter uses three elements.

10.9 Neutral currents

Fundamental-frequency neutral current

In a four-wire star, the neutral carries $\mathbf{I}_N = \mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 3\mathbf{I}_0$, the zero-sequence current. For linear loads drawing sinusoidal current, the neutral current is zero when the loads are balanced and, as the residential example showed, generally smaller than the largest phase current when they are not: its maximum for loads that never exceed a given phase current $I_{\max}$ equals $I_{\max}$, reached when one phase carries $I_{\max}$ and the other two carry nothing. This is the reasoning behind the old practice of using a reduced neutral, half or two thirds the phase conductor size, on large feeders.

Triplen harmonics add in the neutral

Modern loads are not linear. A switched-mode power supply with a capacitor-input rectifier (in every computer, LED driver, television and phone charger) draws current in short pulses near the voltage peaks. Its current waveform contains odd harmonics; typical single-phase electronic loads have a third-harmonic current of 60–90 % of the fundamental, fifth 40–60 %, seventh 20–40 %.

Consider the h-th harmonic of the three phase currents of a balanced set of such loads. Phase b is phase a delayed by one third of a fundamental period, so its h-th harmonic is delayed by $h \times 120^\circ$:

Harmonic order h	Shift of phase b relative to a	Sequence
1, 7, 13, …	-120°	Positive
5, 11, 17, …	$-600^\circ \equiv +120^\circ$	Negative
3, 9, 15, … (triplens)	$-360^\circ \equiv 0^\circ$	Zero

The third, ninth and fifteenth harmonics of the three phases are *in phase*. They do not cancel at the neutral point; they add. The neutral of a balanced set of electronic loads carries three times the third-harmonic phase current, at 150 Hz (180 Hz in a 60 Hz system).

> **Status of the principle: Triplen harmonics are zero-sequence**
> *Derived*. Follows from the time shift of $T/3$ between phases of a balanced set of identical periodic loads; it holds exactly only for identical loads on a balanced supply, and approximately for the aggregate of many similar electronic loads.

Worked example — Neutral current from electronic loads. Each phase of an office floor carries a fundamental current of 100 A rms plus a third-harmonic current of 70 A rms (70 %), from balanced single-phase computer and lighting loads; ignore higher harmonics.

Phase current rms: $I_p = \sqrt{100^2 + 70^2} = 122.1$ A.

Neutral current: the fundamentals cancel and the third harmonics add: $I_N = 3 \times 70 = 210$ A rms, at 150 Hz.

Ratio: $I_N / I_p = 210 / 122.1 = 1.72$.

A neutral sized at half the phase conductor would carry 3.4 times its intended current. This is the overheated neutral of the chapter opening. It is why IEC 60364-5-52 forbids a reduced neutral once the third harmonic exceeds 15 %, derates the cable when it lies between 15 and 33 %, and requires the cable to be sized for the neutral current above 33 %; data centres and office buildings often use a double-size neutral. Figure 10.6 shows the waveforms.

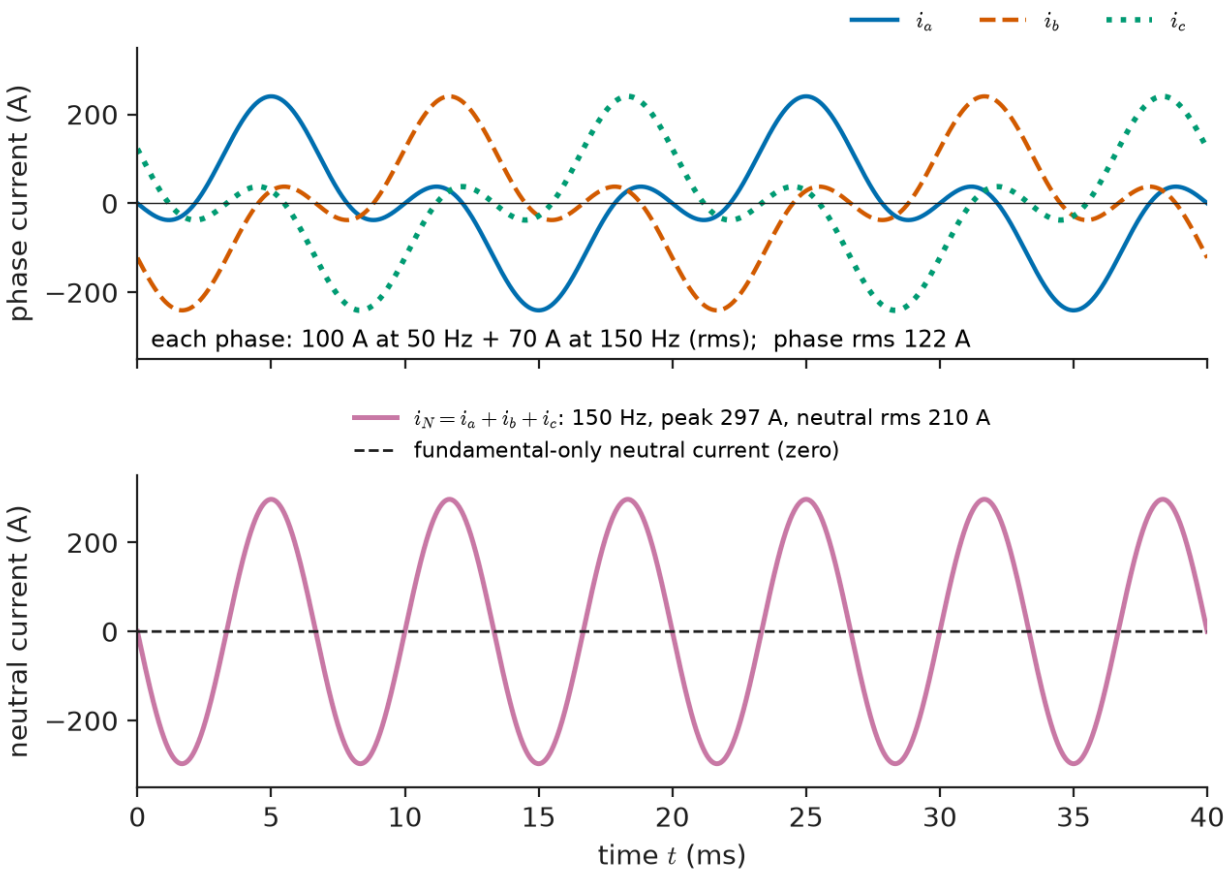


Figure 10.6 — Phase and neutral currents for balanced electronic loads: fundamentals cancel in the neutral, third harmonics add.

Where the triplen currents go

At a delta–star distribution transformer, the zero-sequence (triplen) currents in the star-side windings induce corresponding currents in the delta winding, where they circulate around the closed delta without appearing in the primary lines. The delta therefore "traps" the triplens: the medium-voltage network does not see them, but the transformer does, as extra copper and eddy-current loss. Transformers for heavily nonlinear loads are derated by a **K-factor** or specified by harmonic loss factor (IEEE C57.110). The same zero-sequence path is why generator windings are star-connected with a high-impedance earth: a delta would let third-harmonic EMFs circulate.

Other neutral issues

- **Neutral-to-earth voltage.** Current in the neutral produces a voltage drop along it, so the neutral at a remote socket can be several volts above earth. Sensitive instrumentation sees this as common-mode noise; with triplen currents it is a 150 Hz hum.

- **Residual-current devices.** An RCD compares the phase and neutral currents of a circuit and trips when their sum exceeds typically 30 mA, the signature of current leaking to earth through a person or faulty insulation. Three-phase RCDs sum all four conductors.
- **Harmonic mitigation.** Zigzag transformers provide a low-impedance zero-sequence path near the loads; active harmonic filters inject cancelling currents; power-factor-corrected power supplies draw nearly sinusoidal current and remove the problem at its source.

10.10 Practical applications

Generation, transmission and distribution

Every large power system is three-phase from generator to distribution transformer. Transmission uses three-wire lines (often with overhead earth wires for lightning protection, not as a return), because the balanced load of thousands of customers makes a neutral unnecessary and the 75 % conductor saving matters at hundreds of kilometres. Per-phase equivalents with ABCD parameters (Chapter 12) and distributed-parameter models (Chapter 14) are the analytical tools for these lines. Distribution transformers are commonly delta–star (Dyn11 in IEC notation: delta primary, star secondary with neutral brought out, secondary leading primary by 30°), giving a four-wire 230/400 V system and trapping triplen currents.

Motors and drives

Roughly half of all electrical energy in industrialised countries drives three-phase motors. The rotating field from three stationary windings makes the induction motor self-starting, rugged and cheap. Motor connection (star or delta), starting (direct-on-line, star–delta, soft starter, variable-frequency drive) and the effect of supply unbalance on motor heating all follow from the relations of Sections 10.4–10.6. A variable-frequency drive rectifies the three-phase supply to DC and inverts it back to three-phase at variable frequency; its input stage is a six-pulse rectifier that draws characteristic 5th and 7th harmonic currents (negative and positive sequence, so they do not appear in the neutral but do distort the supply).

Rectifiers and DC systems

A three-phase six-pulse bridge rectifier produces a DC output whose ripple is at six times the supply frequency (300 Hz at 50 Hz) with a peak-to-peak ripple of only about 13.4 % of the peak line voltage, against 100 % at twice the supply frequency for a single-phase bridge without smoothing. Twelve-pulse rectifiers, built from two six-pulse bridges fed by star and delta transformer windings with their 30° phase shift, cancel the 5th and 7th harmonics; they are used in aluminium smelters, traction substations and HVDC converter stations.

Data centres, buildings and EV charging

A data centre is a large, nearly balanced, highly nonlinear three-phase load. Its design involves the neutral-sizing rules of Section 10.9, power measurement per phase and per rack (Section 10.8), and three-phase UPS systems. Electric-vehicle chargers above about 7 kW use three-phase supplies (11 kW

and 22 kW AC chargers at 400 V draw 16 A and 32 A per phase). Heat pumps and lifts in buildings are three-phase; lighting and sockets are distributed across the phases to keep unbalance low.

Railways and other single-phase loads on three-phase grids

AC electric railways (25 kV, 50 Hz in many countries) draw single-phase power from two lines of the three-phase grid. A 10 MW train load on one line pair produces a large negative-sequence current in the grid; railway substations therefore alternate their connections among the three line pairs and are separated by neutral sections in the overhead line. Where the grid is weak, static balancers or Scott-connected transformers convert the single-phase load into a nearly balanced three-phase demand.

Instrumentation and signal circuits

Three-phase ideas reach beyond power. Resolvers and synchros transmit shaft angles with three-wire signals. Three-phase sinusoidal encoders and the space-vector (Clarke and Park) transformations used in motor control are applications of the same a-operator algebra. In biomedical and audio instrumentation, the 50/60 Hz hum and its triplen harmonics that couple from building wiring are often the dominant interference, and identifying them begins with the neutral-current picture of Section 10.9.

What remains open

Open question	Where it is answered
How do switching events such as energising a three-phase motor or transformer produce transient currents beyond the phasor steady state?	Chapter 11 (Laplace-domain analysis of switching transients)
How is a three-phase transmission line represented as a per-phase network with input and output ports?	Chapter 12 (two-port and ABCD models)
How do long lines, with distributed inductance and capacitance, behave at power and at high frequency?	Chapter 14 (transmission lines)
How are the harmonic spectra of rectifier and electronic-load currents computed and measured?	Chapters 18 and 22 (Fourier series and DFT)
What filters suppress harmonics and interference from three-phase systems?	Chapters 13 and 23 (passive and digital filters)
Sequence networks, fault currents, protection and the machines themselves	*Electrical Energy & Machines*
Rectifier, inverter and motor-drive circuits	*Electronics & Computing*
How is the frequency of a three-phase grid regulated by feedback?	Chapters 33–35 (control fundamentals); grid control in *Electrical Energy & Machines*

Next: Chapter 11 opens Part III by moving circuit analysis into the complex-frequency domain, where switching transients, initial conditions and steady state are handled by a single algebraic method.

Part III — Advanced Network Theory

Parts I and II analysed circuits one situation at a time: resistive networks with the laws and theorems of Chapters 1–5, switching transients with differential equations in Chapters 6 and 7, and sinusoidal steady state with phasors in Chapters 8–10. Each method worked, but each needed its own setup, and none of them said much about how one network behaves when it is connected to another, or what happens when a wire becomes long compared with a wavelength. Part III joins these methods into one framework. The Laplace transform turns every lumped linear circuit, with its stored energy and its sources, into algebra in the complex frequency s. A circuit is then summed up by a few rational functions: a transfer function, a driving-point impedance, a matrix of two-port parameters. Their poles and zeros predict ringing, overshoot, settling, selectivity and stability. Evaluated on the $j\omega$ axis, the same functions give the frequency response and the filter. When interconnects grow long, the lumped model gives way to distributed lines, which bring characteristic impedance, reflections and matching. This is the network theory behind filter design, amplifier interfacing, power-line modelling, high-speed digital interconnects and the signals-and-systems view of Part IV.

By the end of this part you should understand

- how to transform a circuit, including initial capacitor voltages and inductor currents, into the s-domain and solve it with the nodal, mesh and Thévenin methods of Part I;
- what a transfer function is, and how its poles and zeros set the transient response, the steady-state response and stability;
- how partial-fraction expansion recovers time responses, and how it separates transient from steady-state components;
- how a linear network is described at its ports by Z, Y, hybrid or ABCD parameters, how to convert between them, and how cascaded, terminated and interconnected two-ports behave;
- what reciprocity and symmetry imply for passive networks, and why amplifiers are non-reciprocal;
- how magnitude and phase responses are drawn as Bode plots, and how resonance, Q and bandwidth shape passive and active filters;
- why a lumped circuit fails at high frequency or over long distances, and how the telegrapher's equations, characteristic impedance, reflection coefficient, standing-wave ratio and the Smith

chart describe transmission lines and their matching.

Chapter	Central question	What it delivers
11. Laplace-Domain Circuit Analysis	How can a circuit with stored energy and arbitrary inputs be solved by algebra alone?	s-domain element models with initial-condition sources, transfer functions, pole-zero maps, partial fractions, and the split of a response into transient and steady-state parts
12. Two-Port Networks	How can a network be described by its terminal behaviour so that it can be terminated and cascaded without re-solving its interior?	Z, Y, h and ABCD parameters, conversion between them, terminated and cascaded two-ports, reciprocity and symmetry, amplifier and transmission-line models
13. Network Frequency Response	How does a network treat each frequency, and how are its magnitude and phase designed?	Bode plots from poles and zeros, resonant circuits, Q and bandwidth, passive filters, the interface to active filters, and frequency-domain signs of stability
14. Transmission Lines	What happens when the connections themselves are long enough to have travel time?	Distributed parameters, telegrapher's equations, characteristic impedance, propagation constant, reflection, standing waves, matching, the Smith chart and high-frequency interconnects

How this part connects to the rest of the book. Part III takes the phasor methods of Part II (Chapters 8–10) and the per-phase models of three-phase systems and widens them into a general network theory. Its main results are used throughout the rest of the book. The transfer function and the s-plane of Chapter 11 come back in Part IV as the system function of continuous-time LTI systems (Chapters 16–18). Their discrete-time counterparts, the z-plane and digital filters, appear in Chapters 20 and 23. Pole locations and frequency response are the language of feedback and stability in Part VII (Chapters 33–35). Two-port and transmission-line models lead into RF and microwave networks and S-parameters (Chapter 32), and into the channels of the communication chapters of Part VI. The mathematics of the Laplace transform itself is in *The Mathematics Universe*. The field theory behind transmission lines is in *Engineering Physics*. The transistor circuits that these chapters model as two-ports belong to *Electronics & Computing*.

Chapter 11 — Laplace-Domain Circuit Analysis

Part III · Advanced Network Theory

An inverter for a small battery system has a 48 V DC link. To avoid a destructive inrush when the battery contactor closes, the link is first energised through a 10 Ω precharge resistor and a 10 mH filter choke into the 100 µF link capacitor. On the bench everything works. In the field, a technician cycles the system quickly, so the capacitor still holds 12 V from the previous run when the precharge contactor closes. A week later a 50 V-rated electrolytic capacitor in the link has vented. The oscilloscope trace that finally explains it shows the link voltage overshooting to almost 54 V and ringing at about 138 Hz before settling at 48 V. Every ingredient of the failure was in Chapter 7: a series RLC circuit with an initial condition and a step input. But solving it by the classical method (write the differential equation, find the natural response, find a particular solution, fit two constants to the initial conditions) is laborious, error-prone, and hides the structure that tells an engineer how to fix it. This chapter replaces that procedure with algebra. In the Laplace domain the circuit, its initial conditions and its source become one rational function; the overshoot, the ringing frequency, the settling time and the final value are read off from the locations of its poles.

> **How can a circuit with stored energy and arbitrary sources be analysed with the same algebra as a resistive network?**
> **What do the poles and zeros of a network tell us about its transient and steady-state behaviour, and how is the time response recovered from them?**

Concept	Engineering destination
s-domain circuit equations	Switching transients in power converters, relays and breakers
s-domain impedance and admittance	Filter design, matching networks, driving-point models of sensors and loudspeakers
Initial conditions	Precharge circuits, capacitor banks, pulsed-power and defibrillator discharge
Transfer functions	Amplifier and filter specifications, control-loop plant models, instrument front ends
Pole-zero analysis	Ringing and overshoot prediction, stability checks, notch filters
Partial-fraction expansion	Closed-form time responses for design equations and simulation checks
Network response	Step and impulse testing of instruments, settling time of ADC drivers
Transient and steady-state separation	Inrush and DC offset in power systems, settling of measurement channels

Roadmap. Why the s-domain (11.1) → transforming the circuit laws (11.2) and the elements (11.3) → stored energy as initial-condition sources (11.4) → transfer functions (11.5) and their poles and zeros

(11.6) → inverting by partial fractions (11.7) → complete network response (11.8) → separating transient from steady state, with the initial- and final-value theorems (11.9).

11.1 Why move to the s-domain

What the classical method costs

Chapters 6 and 7 solved first- and second-order circuits by writing a differential equation, solving its homogeneous part, guessing a particular solution and fitting constants to the initial conditions. That method has three weaknesses. It requires a differential equation to be written first, which for a network with five energy-storage elements means eliminating variables among coupled ODEs. It treats the initial conditions separately, after the solution form has been chosen, so it is easy to fit the wrong initial value (the inductor current at 0^+ instead of the inductor voltage, for example). And it handles each kind of input (step, ramp, sinusoid, pulse) with a separate guess.

Chapter 8 showed a better way for one special case: in sinusoidal steady state, impedances $j\omega L$ and $1/(j\omega C)$ turn every network into an algebraic problem. The Laplace transform generalises that idea from $j\omega$ to the complex frequency $s = \sigma + j\omega$, and from sinusoidal steady state to arbitrary inputs that start at $t = 0$, with initial conditions built in.

The transform in one paragraph

The (unilateral) Laplace transform of a signal $f(t)$ defined for $t \geq 0^-$ is

$$F(s) = \mathcal{L}\{f(t)\} = \int_{0^-}^{\infty} f(t)\, e^{-st}\, dt$$

which converges for $\mathrm{Re}\{s\}$ larger than some abscissa set by the growth of f. The lower limit 0^- (just before any switching at $t = 0$) is deliberate: it lets the transform include impulses at $t = 0$ and makes initial conditions mean "the state of the circuit before the switch operates", which is the state an engineer actually knows. The transform is linear, it turns differentiation into multiplication by s minus the initial value, and integration into division by s. These two properties are the whole reason it is useful for circuits. The theory of the transform, including existence, uniqueness and the inversion integral, belongs to *The Mathematics Universe*; Table 11.1 lists the results this chapter uses.

Table 11.1 — Laplace-transform pairs and properties used in circuit analysis.

Time function or operation ($t \geq 0$)	Transform
Unit impulse $\delta(t)$	1
Unit step $u(t)$	$1/s$
Ramp $t\,u(t)$	$1/s^2$
$t^n/n!$	$1/s^{n+1}$
e^{-at}	$1/(s+a)$
$t\,e^{-at}$	$1/(s+a)^2$
$\sin \omega t$	$\omega/(s^2+\omega^2)$
$\cos \omega t$	$s/(s^2+\omega^2)$
$e^{-at}\sin \omega t$	$\omega/[(s+a)^2+\omega^2]$
$e^{-at}\cos \omega t$	$(s+a)/[(s+a)^2+\omega^2]$
Linearity $af(t)+bg(t)$	$aF(s)+bG(s)$
Derivative df/dt	$sF(s)-f(0^-)$
Integral $\int_{0^-}^{t} f(\tau)d\tau$	$F(s)/s$
Time delay $f(t-t_0)u(t-t_0)$	$e^{-st_0}F(s)$
Exponential shift $e^{-at}f(t)$	$F(s+a)$
Convolution $\int_0^t f(\tau)g(t-\tau)d\tau$	$F(s)G(s)$
Initial value $f(0^+)$	$\lim_{s\to\infty} sF(s)$
Final value $f(\infty)$ (if it exists)	$\lim_{s\to 0} sF(s)$

> **Boundary.**
> Convergence, the inversion integral and the two-sided transform are mathematics treated in *The Mathematics Universe*. This chapter uses the table above as established and concentrates on how a circuit is put into a form the table can invert.

What s means physically

A signal e^{st} with $s=\sigma+j\omega$ is an exponentially growing or decaying sinusoid: $e^{\sigma t}(\cos\omega t + j\sin\omega t)$. The real part σ (in nepers per second) is the decay or growth rate; the imaginary part ω (in radians per second) is the oscillation frequency. Constant signals have $s=0$; pure sinusoids have $s=\pm j\omega$; decaying transients have $\sigma<0$. Every signal a lumped linear circuit produces after switching is a sum of such terms, which is why the complex plane of s is the natural map of circuit behaviour (Section 11.6).

11.2 Circuit equations in the s-domain

Kirchhoff's laws transform directly

KCL at a node says $\sum_k i_k(t) = 0$ at every instant. Because the transform is linear, the transformed currents also sum to zero: $\sum_k I_k(s) = 0$. The same holds for KVL around any loop: $\sum_k V_k(s) = 0$. The topology of the network, and therefore every nodal, mesh and matrix formulation of Chapter 3, carries over unchanged to the s-domain.

> **Status of the principle: Kirchhoff's laws in the s-domain**
> *Derived*. Linearity of the Laplace transform applied to KCL and KVL, which are themselves fundamental (charge and energy conservation in lumped circuits); valid wherever the lumped-element assumption holds.

Element laws transform into algebra

The three passive element laws, with the passive sign convention, transform as follows.

- **Resistor.** $v = Ri$ becomes $V(s) = R\,I(s)$.
- **Inductor.** $v = L\,di/dt$ becomes $V(s) = sL\,I(s) - L\,i_L(0^-)$.
- **Capacitor.** $i = C\,dv/dt$ becomes $I(s) = sC\,V(s) - C\,v_C(0^-)$, or equivalently $V(s) = I(s)/(sC) + v_C(0^-)/s$.

Every derivative has become a multiplication by s, and every initial condition has become a known term on the right-hand side. A linear circuit with any number of energy-storage elements therefore produces, in the s-domain, a set of *linear algebraic* equations in the unknown node voltages or mesh currents, with coefficients that are polynomials in s. Solving them is Cramer's rule or Gaussian elimination, exactly as in Chapter 3.

Coupled coils transform the same way. The equations $v_1 = L_1 di_1/dt + M\,di_2/dt$ and $v_2 = M\,di_1/dt + L_2 di_2/dt$ become

$$V_1 = sL_1 I_1 + sMI_2 - L_1 i_1(0^-) - M i_2(0^-), \qquad V_2 = sMI_1 + sL_2 I_2 - M i_1(0^-) - L_2 i_2(0^-)$$

so a transformer or coupled-inductor filter is handled without any change of method. Dependent sources keep their algebraic definitions: a VCCS $i = g_m v_x$ becomes $I = g_m V_x(s)$.

The general procedure

1. Determine the state of every capacitor voltage and inductor current at $t = 0^-$, from the circuit before switching (usually a DC steady state, in which capacitors are open circuits and inductors short circuits).
2. Redraw the circuit for $t \geq 0$ with every element replaced by its s-domain model (Section 11.3) including initial-condition sources (Section 11.4), and every source replaced by its transform.
3. Solve the resulting algebraic network for the wanted quantity by any method: nodal, mesh, division, Thévenin.
4. Simplify to a ratio of polynomials in s and invert by partial fractions (Section 11.7).

5. Check the result at $t = 0^+$ and $t \to \infty$ with the initial- and final-value theorems (Section 11.9).

11.3 Impedance and admittance in the s-domain

Generalised impedance

With zero initial conditions, each element law reduces to $V(s) = Z(s)I(s)$:

Element	Impedance $Z(s)$	Admittance $Y(s)$	Value at $s = j\omega$
Resistor	R	$G = 1/R$	R
Inductor	sL	$1/(sL)$	$j\omega L$
Capacitor	$1/(sC)$	sC	$1/(j\omega C)$

Impedances in series add; admittances in parallel add; voltage and current division, source transformation, Thévenin and Norton equivalents, star–delta transformation and every other result of Chapters 2–4 hold for $Z(s)$ and $Y(s)$, because they depend only on linearity and Kirchhoff's laws. Setting $s = j\omega$ recovers the phasor impedances of Chapter 8, so the s-domain contains AC analysis as a special case.

> **Status of the principle: s-domain impedance**
> *Derived.* Follows from the transformed element laws with zero initial state; it holds for linear time-invariant lumped elements and fails for nonlinear elements (saturating inductors, varactor capacitance, diodes), which must be linearised about an operating point first.

Impedance functions of simple networks

Combining elements gives rational functions of s. Three examples recur throughout this book.

A series RLC branch:

$$Z(s) = R + sL + \frac{1}{sC} = \frac{LCs^2 + RCs + 1}{sC}$$

A parallel RLC tank:

$$Z(s) = \frac{1}{1/R + sC + 1/(sL)} = \frac{1}{C}\,\frac{s}{s^2 + s/(RC) + 1/(LC)}$$

An RC low-pass section, as a voltage ratio:

$$\frac{V_o(s)}{V_i(s)} = \frac{1/(sC)}{R + 1/(sC)} = \frac{1}{1 + sRC}$$

Impedances of passive RLC networks are always ratios of polynomials with real, non-negative coefficients; they belong to the class of **positive-real** functions, whose real part is non-negative for

$\mathrm{Re}\{s\} \geq 0$. This property, which expresses the fact that a passive network cannot deliver net energy, is the starting point of network synthesis: given a positive-real function, a passive network that realises it can always be found.

Thévenin and Norton in the s-domain

An s-domain Thévenin equivalent consists of a source $V_{Th}(s)$, the transform of the open-circuit voltage including any contribution from initial conditions, in series with $Z_{Th}(s)$, the impedance seen with all independent sources (and initial-condition sources) set to zero. This is often the fastest route to a transient response. For example, a sensor with source impedance $Z_s(s)$ driving a cable modelled as a capacitance is a first-order problem once Z_s is known, regardless of how complicated the sensor's internal network is.

Representative s-domain impedances

Table 11.2 — Representative networks and their pole locations.

Network	Element values (typical)	Characteristic poles	Behaviour
Anti-aliasing RC (audio ADC)	1 kΩ, 100 nF	-1.0×10^4 rad/s	Corner 1.59 kHz
ECG amplifier AC-coupling high-pass	Corner 0.05 Hz	-0.314 rad/s	Time constant 3.2 s
Oscilloscope probe with 10 cm ground lead	~100 nH, ~12 pF	$\sigma \pm j\,9.1 \times 10^8$ rad/s	Rings near 145 MHz
Loudspeaker (closed box, $Q_{tc} = 0.4$)	$f_s = 40$ Hz	-126 and -503 rad/s	Overdamped bass roll-off
Inverter DC-link precharge	10 Ω, 10 mH, 100 µF	$-500 \pm j866$ rad/s	Overshoot 16 %
Transmission network fault, X/R = 10–30, 50 Hz	—	-10.5 to -31.4 rad/s	DC offset decays in 32–95 ms
32.768 kHz watch crystal	$Q \approx 5 \times 10^4$	$-2.06 \pm j2.06 \times 10^5$ rad/s	Ring-down time constant 0.49 s

The table spans more than nine decades of $|s|$, yet every entry is analysed with the same algebra.

11.4 Initial conditions

The state of a circuit

The energy stored in a circuit at the switching instant is described completely by the capacitor voltages and inductor currents at $t = 0^-$. These are the **state variables**: they are continuous in time whenever

the currents into capacitors and the voltages across inductors are finite, because changing them instantaneously would require infinite power. Chapter 5 established this continuity; Chapter 6 used it to find 0^+ values. In the s-domain we do not need 0^+ values at all: the transformed element laws use the 0^- values, and the transform itself produces whatever happens at 0^+, including any jumps.

Initial-condition source models

The transformed element laws can be read as small circuits (Figure 11.1).

Inductor with initial current $I_0 = i_L(0^-)$. Series form: $V(s) = sL\,I(s) - LI_0$, an impedance sL in series with a voltage source LI_0 (a constant times the transform of an impulse) oriented to oppose the reference direction of V. Parallel form: $I(s) = V(s)/(sL) + I_0/s$, an impedance sL in parallel with a current source I_0/s (a step) in the direction of the reference current.

Capacitor with initial voltage $V_0 = v_C(0^-)$. Series form: $V(s) = I(s)/(sC) + V_0/s$, an impedance $1/(sC)$ in series with a voltage source V_0/s (a step). Parallel form: $I(s) = sCV(s) - CV_0$, an admittance sC in parallel with a current source CV_0 (an impulse of charge) directed opposite to the reference current.

The series forms are convenient in mesh analysis, the parallel forms in nodal analysis. Either form, used consistently, gives the same answer; they are source transformations of each other.

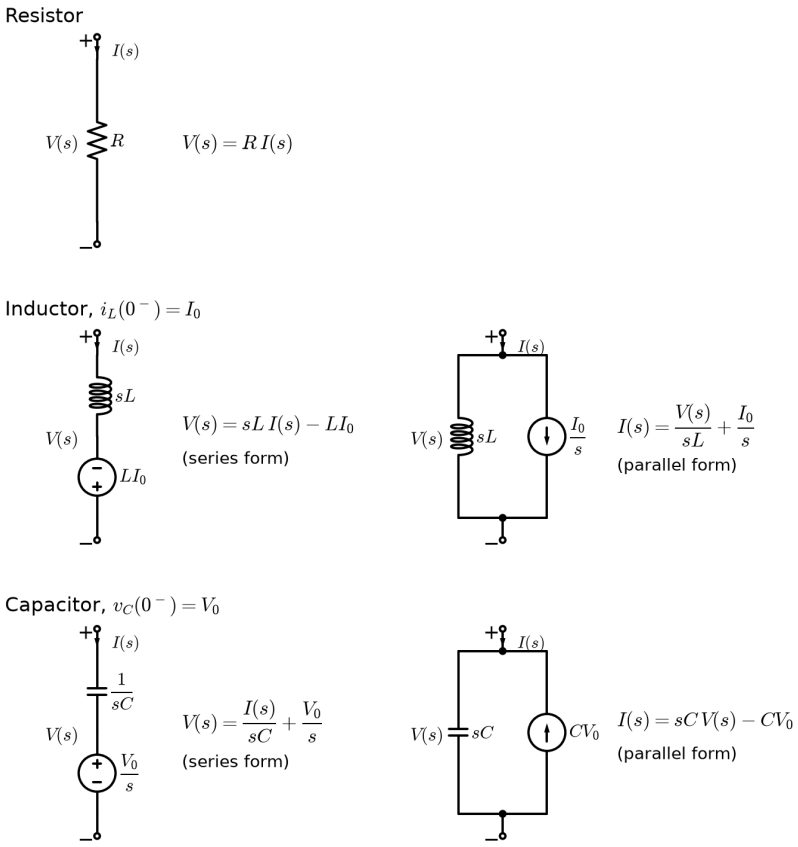


Figure 11.1 — s-domain models of the resistor, inductor and capacitor, with series and parallel initial-condition sources.

> **Status of the principle: Initial-condition source models**
> *Derived*. Direct readings of the transformed constitutive laws with the 0^- convention; exact for linear time-invariant capacitors and inductors.

Worked example — DC-link precharge with a partly charged capacitor

The inverter of the chapter opening: a 48 V source is connected at $t = 0$ through $R = 10\ \Omega$ and $L = 10$ mH to $C = 100\ \mu$F. Before the switch closes, $i_L(0^-) = 0$ and $v_C(0^-) = 12$ V.

Step 1: s-domain circuit. The source becomes $48/s$. The capacitor becomes $1/(sC)$ in series with a source $12/s$. The inductor carries no initial current, so it is just sL.

Step 2: mesh equation. KVL around the single loop:

$$\frac{48}{s} = \left(R + sL + \frac{1}{sC}\right) I(s) + \frac{12}{s}$$

so

$$I(s) = \frac{36/s}{R + sL + 1/(sC)} = \frac{36/L}{s^2 + (R/L)s + 1/(LC)} = \frac{3600}{s^2 + 1000s + 10^6}$$

The initial voltage has simply been subtracted from the source: only the 36 V difference drives current.

Step 3: poles. The denominator has roots $s = -500 \pm j866.0$ rad/s. In the notation of Chapter 7, $\alpha = R/(2L) = 500\ \text{s}^{-1}$, $\omega_0 = 1/\sqrt{LC} = 1000$ rad/s, $\zeta = \alpha/\omega_0 = 0.5$ and $\omega_d = \sqrt{\omega_0^2 - \alpha^2} = 866.0$ rad/s (137.8 Hz). The circuit is underdamped.

Step 4: invert. Writing the denominator as $(s + 500)^2 + 866.0^2$ and using Table 11.1:

$$i(t) = \frac{3600}{866.0}\, e^{-500t} \sin(866.0\, t) = 4.157\, e^{-500t} \sin(866.0\, t)\ \text{A}$$

The current peaks when $\tan(\omega_d t) = \omega_d/\alpha$, at $t = 1.209$ ms, where $i = 1.967$ A.

Step 5: capacitor voltage. $V_C(s) = I(s)/(sC) + 12/s = 12/s + 3.6 \times 10^7/[s(s^2 + 1000s + 10^6)]$. Partial fractions (Section 11.7) give

$$V_C(s) = \frac{48}{s} - \frac{36\,(s + 1000)}{(s + 500)^2 + 866.0^2}$$

$$v_C(t) = 48 - 36\, e^{-500t} \left[\cos(866.0\, t) + 0.5774\, \sin(866.0\, t)\right]\ \text{V}$$

The first peak occurs at $t = \pi/\omega_d = 3.628$ ms, where $v_C = 48 + 36\, e^{-500 \times 3.628 \times 10^{-3}} = 53.87$ V. The overshoot is 16.3 % of the 36 V step, the standard $e^{-\pi\zeta/\sqrt{1-\zeta^2}}$ of a second-order system with $\zeta = 0.5$ (Figure 11.2). A numerical integration of the circuit ODE gives 53.869 V at the same instant.

Consequence. The 50 V capacitor is overstressed by almost 4 V whenever the link is re-energised from a partly charged state. Notice that the overshoot is 16 % of the *difference* between source and initial voltage, so it is worst from a fully discharged capacitor (48 V step, peak $48 + 0.163 \times 48 = 55.8$

V). The design fix is to raise the damping: at $R = 2\sqrt{L/C} = 20\ \Omega$ the circuit is critically damped (worked in Section 11.7) and the capacitor voltage never exceeds 48 V.

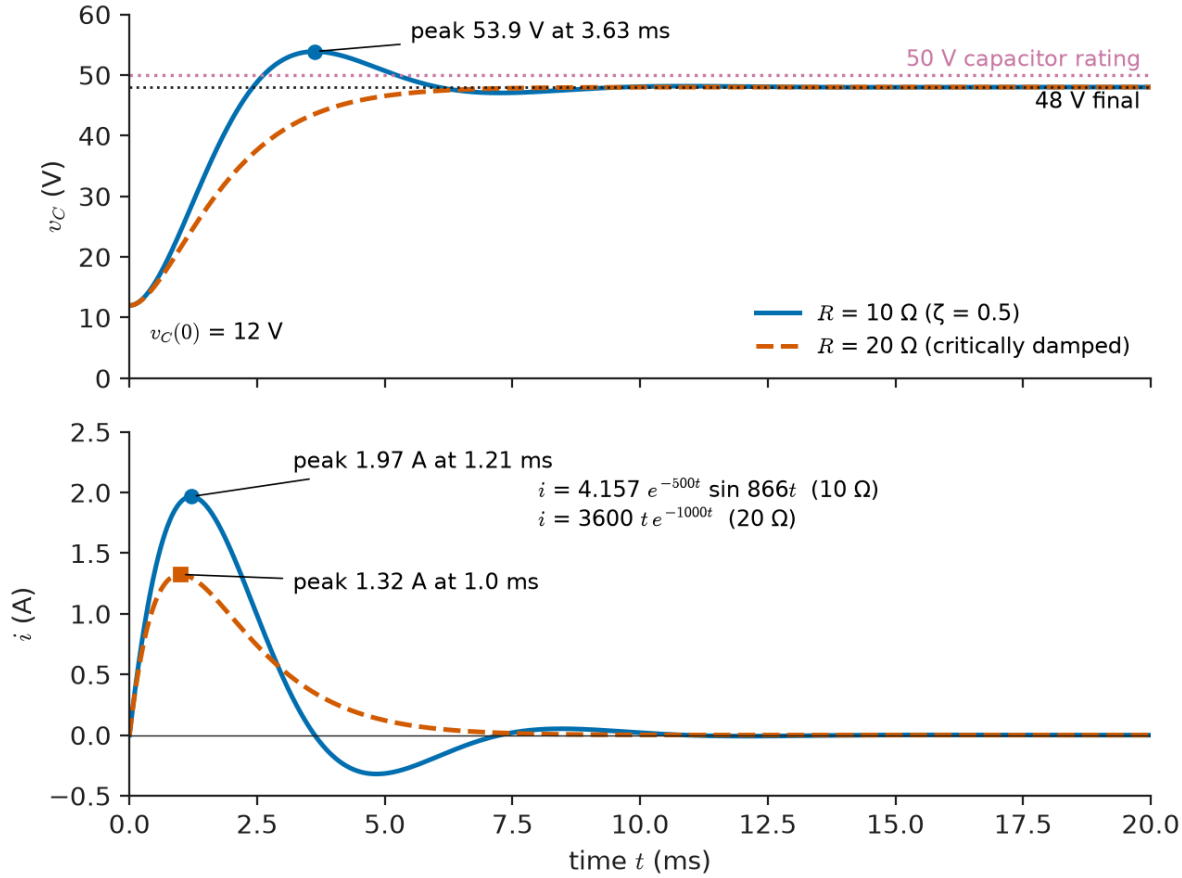


Figure 11.2 — DC-link precharge with a capacitor initially at 12 V: underdamped (10 ohm) and critically damped (20 ohm) responses.

Impulsive cases: when the state must jump

Continuity of capacitor voltage and inductor current fails in idealised circuits that force a jump: closing a switch that connects two capacitors charged to different voltages, or opening a switch in series with an inductor carrying current. Classical methods struggle here because the 0^+ values are not given by continuity. The s-domain with the 0^- convention handles them automatically: the transform of the response contains a constant term (an impulse), and the jump appears in the inverse transform.

Consider two 100 µF capacitors, one charged to 48 V and one discharged, joined by an ideal switch at $t = 0$ with no resistance. In the s-domain the loop contains the two capacitor impedances and the step source $48/s$. The current is $I(s) = (48/s)/(2/(sC)) = 24C = 2.4 \times 10^{-3}$, a constant, whose inverse is $2.4 \times 10^{-3}\,\delta(t)$: an impulse carrying 2.4 mC from one capacitor to the other in zero time. Both capacitors end at 24 V. The stored energy falls from 0.1152 J to 0.0576 J; the missing half is dissipated in whatever resistance, arc or radiation the real circuit has, independently of its value. Real circuits always have some resistance and inductance, so the "impulse" is a large, short current pulse; the s-domain idealisation gives its charge exactly and tells the designer that a current-limiting element is needed. The same analysis governs paralleling capacitor banks in substations (inrush currents of several

kA), hot-plugging boards onto a powered backplane, and the opening of an inductive circuit, where the impulse of voltage is what destroys relay contacts and transistors without a flyback diode or snubber.

11.5 Transfer functions

Definition

For a linear time-invariant network with a single input $x(t)$ and a single output $y(t)$, and *zero initial conditions*, the ratio of the output transform to the input transform does not depend on the input:

$$H(s) = \left.\frac{Y(s)}{X(s)}\right|_{\text{zero initial state}}$$

$H(s)$ is the **transfer function** (or network function). Depending on what x and y are, it is a voltage gain, a current gain, a transimpedance (output voltage per input current) or a transadmittance. When input and output are at the same port, it is a **driving-point** function: the input impedance $Z(s) = V(s)/I(s)$ or admittance $Y(s)$.

Because the input-output relation is algebraic in the s-domain, the transfer function of a lumped network is a ratio of polynomials in s with real coefficients:

$$H(s) = \frac{b_m s^m + \cdots + b_1 s + b_0}{a_n s^n + \cdots + a_1 s + a_0}$$

The order n of the denominator equals at most the number of independent energy-storage elements. For physical networks driven by finite sources, $m \le n$ (the function is *proper*); a transfer function with $m > n$ would amplify high frequencies without bound and corresponds to an ideal differentiator, which no real network achieves over all frequencies.

> **Status of the principle: Transfer function**
> *Derived.* Exists for linear time-invariant systems because the s-domain input-output relation is multiplication; it describes only the zero-state response and does not exist for nonlinear or time-varying circuits (mixers, switched-capacitor circuits analysed in continuous time, saturating magnetics).

Impulse response

If the input is a unit impulse, $X(s) = 1$ and $Y(s) = H(s)$. The transfer function is therefore the transform of the **impulse response** $h(t)$, and the zero-state response to any input is the convolution $y(t) = \int_0^t h(\tau)x(t-\tau)\,d\tau$, whose transform is $Y(s) = H(s)X(s)$. The time-domain view of this result is the subject of Chapter 17; here it means that one function, $H(s)$, describes the network's response to every input.

Frequency response is $H(j\omega)$

Evaluating the transfer function on the imaginary axis gives the sinusoidal steady-state gain and phase: $H(j\omega) = |H(j\omega)|e^{j\angle H(j\omega)}$. This is how a network's transient behaviour and its frequency response are tied together: they are the same function evaluated in different parts of the s-plane. Chapter 13 develops the frequency-response view (Bode plots, bandwidth, filters); this chapter concentrates on time response.

Worked example — A two-section RC anti-aliasing filter

An audio data-acquisition front end uses two identical RC sections in cascade before the ADC, with $R = 1\ \text{k}\Omega$ and $C = 100$ nF, driven from a low-impedance source and feeding a high-impedance ADC buffer. Find the transfer function.

Let V_1 be the voltage at the middle node and $\tau = RC = 10^{-4}$ s. Nodal analysis at the middle node and at the output node:

$$\frac{V_1 - V_i}{R} + sCV_1 + \frac{V_1 - V_o}{R} = 0, \qquad \frac{V_o - V_1}{R} + sCV_o = 0$$

The second equation gives $V_1 = (1 + s\tau)V_o$. Substituting in the first:

$$(2 + s\tau)(1 + s\tau)V_o - V_o = V_i \quad \Rightarrow \quad H(s) = \frac{V_o}{V_i} = \frac{1}{\tau^2 s^2 + 3\tau s + 1}$$

The denominator is *not* $(1 + s\tau)^2$, which would be the product of two isolated sections. The second section loads the first. Its poles are

$$s = \frac{-3 \pm \sqrt{5}}{2\tau} = -3819.7\ \text{rad/s} \ \text{ and } \ -26\,180\ \text{rad/s}$$

that is, at 607.9 Hz and 4166.7 Hz, rather than a double pole at 1591.5 Hz. The −3 dB frequency is 595.6 Hz, much lower than the 1591.5 Hz of a single section. The loading effect between cascaded sections is precisely what the two-port methods of Chapter 12 are designed to handle: there the same $H(s)$ falls out of a product of two ABCD matrices.

11.6 Pole-zero analysis

Factored form

Factoring numerator and denominator of a rational transfer function gives

$$H(s) = K\,\frac{(s - z_1)(s - z_2)\cdots(s - z_m)}{(s - p_1)(s - p_2)\cdots(s - p_n)}$$

The roots z_i of the numerator are the **zeros**, the roots p_i of the denominator the **poles**, and K the gain constant. Because the coefficients are real, complex poles and zeros occur in conjugate pairs. Apart from K, the function is completely determined by its poles and zeros, so a plot of them in the s-plane (crosses for poles, circles for zeros) is a complete picture of the network.

Poles are the natural frequencies

The poles of a transfer function are the roots of the circuit's characteristic equation: they are the complex frequencies s at which the network can sustain a response with no input. Each pole contributes a term $e^{p_i t}$ to the impulse response. Their location therefore determines the character of every transient (Figure 11.3):

Pole location	Contribution to $h(t)$	Circuit example
Negative real axis, $p = -\alpha$	$e^{-\alpha t}$, decays with time constant $1/\alpha$	RC low-pass, RL winding
Complex pair, $-\alpha \pm j\omega_d$	$e^{-\alpha t}\cos(\omega_d t + \theta)$, damped ringing	Underdamped RLC, probe ringing
Imaginary axis, $\pm j\omega_0$	Sustained oscillation	Lossless LC tank, oscillator at threshold
At the origin	Constant (a step in h)	Ideal integrator, capacitor fed by current
Right half-plane	Growing exponential or oscillation	Unstable amplifier, oscillator start-up
Repeated pole	Multiplied by t, t^2, …	Critically damped RLC

For a complex pair $p = -\alpha \pm j\omega_d$, the distance from the origin is the undamped natural frequency $\omega_0 = \sqrt{\alpha^2 + \omega_d^2}$, and the cosine of the angle from the negative real axis is the damping ratio $\zeta = \alpha/\omega_0$. Poles far from the imaginary axis die out quickly; poles close to it ring for a long time. The quality factor of a pole pair is $Q = \omega_0/(2\alpha) = 1/(2\zeta)$.

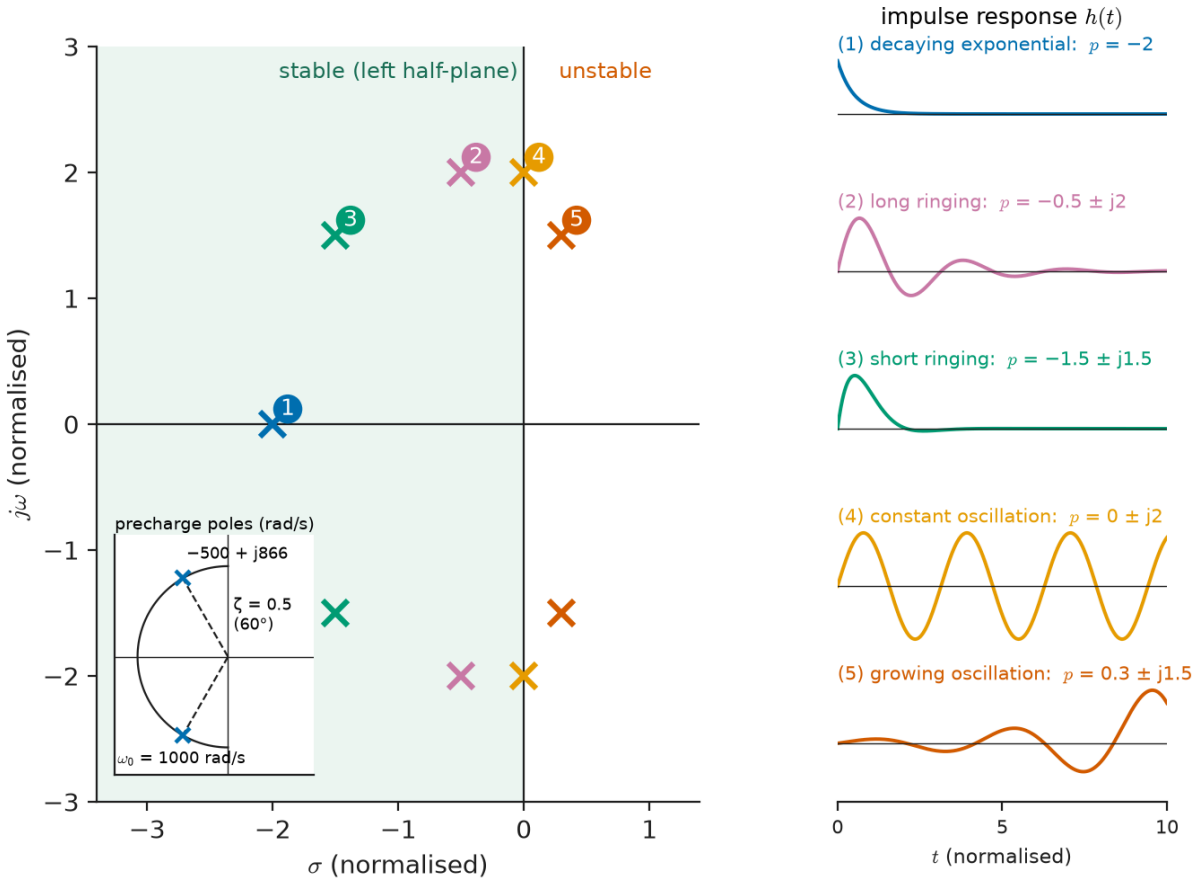


Figure 11.3 — The s-plane as a map of transient behaviour: pole positions and the impulse-response shapes they produce.

> **Status of the principle: Pole location and stability**
> *Derived.* A response term e^{pt} decays if and only if $\mathrm{Re}\{p\} < 0$, so a network is stable (every bounded input gives a bounded output) if and only if all poles of its transfer function lie in the open left half-plane; this holds for any lumped LTI system, with the proviso that a pole cancelled by a zero can still be excited by initial conditions.

Passive networks are stable

A network made of positive resistors, capacitors, inductors and coupled coils cannot generate energy. Its poles therefore lie in the closed left half-plane, and on the imaginary axis only if they are simple and correspond to lossless resonances. Instability requires an active element: a transistor, an op-amp, a negative-resistance device, or feedback. This is the reason the stability tests of Chapter 34 are needed for amplifiers and control loops but not for passive filters.

Zeros are transmission blocks

A zero of a transfer function is a complex frequency at which the input produces no output. Zeros on the imaginary axis are frequencies that the network blocks completely: the twin-T notch at 50 Hz used to reject mains hum in an ECG or strain-gauge amplifier has a pair of zeros at $\pm j2\pi \cdot 50$ rad/s, and the LC trap in a television IF strip a pair at the trapped frequency. A zero at the origin means DC is blocked (a

coupling capacitor). Zeros at infinity (when $m < n$) mean the response falls at high frequency, at $20(n-m)$ dB per decade. Zeros in the right half-plane, which arise in bridged networks, lattice all-pass filters and some amplifiers, give a response that initially moves the wrong way and an extra phase lag; they matter greatly in control (Chapter 34).

Zeros do not affect *which* exponentials appear in a response; that is fixed by the poles. They determine *how much* of each exponential appears, through the residues of Section 11.7.

Worked example — Pole-zero map of a parallel resonant tank

A parallel tank with $R = 1\ \text{k}\Omega$, $L = 1\ \text{mH}$ and $C = 1\ \mu\text{F}$ (the kind of circuit used as a tuned load or as a model of a piezoelectric transducer near resonance) has the driving-point impedance

$$Z(s) = \frac{1}{C}\,\frac{s}{s^2 + s/(RC) + 1/(LC)} = 10^6\,\frac{s}{s^2 + 1000s + 10^9}$$

Zeros. One at $s = 0$ (the inductor shorts DC) and one at $s = \infty$ (the capacitor shorts high frequencies).

Poles. $s = -500 \pm j31\,619$ rad/s. So $\alpha = 500\ \text{s}^{-1}$, $\omega_0 = 31\,623$ rad/s ($f_0 = 5.033$ kHz), $Q = \omega_0/(2\alpha) = 31.6$, which agrees with $Q = R\sqrt{C/L}$ for a parallel tank.

Consequences. On the imaginary axis at $s = j\omega_0$ the impedance equals $R = 1\ \text{k}\Omega$, purely resistive. The poles are close to the axis ($\zeta = 0.016$), so a current impulse (a sudden charge dump into the tank) produces a voltage that rings at 5.03 kHz and decays with time constant $1/\alpha = 2$ ms, about 10 cycles per time constant. The half-power bandwidth of the impedance is $2\alpha = 1000$ rad/s (159 Hz). The pole-zero map alone gives frequency, selectivity and ring-down time, without solving any differential equation.

11.7 Partial-fraction expansion

The idea

The transform of any response computed by the s-domain method is a rational function $F(s) = N(s)/D(s)$. If it is strictly proper ($\deg N < \deg D$), it can be written as a sum of simple terms, one for each pole, each of which is in Table 11.1. The coefficients of those terms are the **residues**. Inverting is then a matter of reading the table term by term.

If $F(s)$ is not strictly proper ($\deg N \ge \deg D$), first divide: $F(s) = Q(s) + R(s)/D(s)$. The polynomial part $Q(s)$ inverts to impulses and their derivatives at $t = 0$, which appear, for example, in the current of a capacitor driven by an ideal voltage step.

Distinct real poles

For $F(s) = N(s)/[(s-p_1)(s-p_2)\cdots(s-p_n)]$ with distinct p_i:

$$F(s) = \sum_{i=1}^{n} \frac{k_i}{s - p_i}, \qquad k_i = \left[(s-p_i)F(s)\right]_{s=p_i}, \qquad f(t) = \sum_{i=1}^{n} k_i e^{p_i t}$$

The formula for k_i is Heaviside's "cover-up" rule: cover the factor $(s-p_i)$ and evaluate what remains at $s=p_i$. Equivalently, $k_i = N(p_i)/D'(p_i)$.

Complex-conjugate poles

A complex pair $p, p^* = -\alpha \pm j\omega_d$ gives conjugate residues k and k^*, and their two terms combine into a real damped sinusoid:

$$\frac{k}{s-p} + \frac{k^*}{s-p^*} \;\longrightarrow\; 2|k|\,e^{-\alpha t}\cos(\omega_d t + \angle k)$$

In hand work it is usually quicker to avoid complex arithmetic: keep the quadratic factor, write the term as $(As+B)/[(s+\alpha)^2+\omega_d^2]$, find A and B by matching coefficients, and split it into the $e^{-\alpha t}\cos$ and $e^{-\alpha t}\sin$ pairs of Table 11.1. The capacitor voltage of the precharge example was found this way.

Repeated poles

A pole of multiplicity r at p contributes r terms:

$$\frac{k_r}{(s-p)^r} + \frac{k_{r-1}}{(s-p)^{r-1}} + \cdots + \frac{k_1}{s-p}, \qquad k_{r-j} = \frac{1}{j!}\,\frac{d^j}{ds^j}\left[(s-p)^r F(s)\right]_{s=p}$$

which invert to $k_r\, t^{r-1}e^{pt}/(r-1)!$ and so on. The factor t^{r-1} is the signature of a repeated pole.

> **Status of the principle: Partial-fraction expansion**
> *Derived.* An algebraic identity for rational functions, combined with the uniqueness of the inverse Laplace transform; it applies to every lumped LTI network response and is replaced by numerical root-finding when the denominator is of high order.

Worked example — Critically damped precharge

Repeat the precharge of Section 11.4 with $R = 20\ \Omega$, so that $R^2 = 400 = 4L/C$ and the circuit is critically damped. The mesh equation now gives

$$I(s) = \frac{3600}{s^2 + 2000s + 10^6} = \frac{3600}{(s+1000)^2} \quad\Rightarrow\quad i(t) = 3600\,t\,e^{-1000t}\ \text{A}$$

which peaks at $t = 1/1000$ s $= 1$ ms with $i = 3600 \times 10^{-3} \times e^{-1} = 1.324$ A. For the capacitor voltage,

$$V_C(s) = \frac{12}{s} + \frac{3.6\times10^7}{s(s+1000)^2} = \frac{12}{s} + \frac{k_0}{s} + \frac{k_2}{(s+1000)^2} + \frac{k_1}{s+1000}$$

Cover-up at $s=0$: $k_0 = 3.6\times10^7/10^6 = 36$. Cover-up of the square at $s=-1000$: $k_2 = 3.6\times10^7/(-1000) = -36\,000$. Derivative rule: $k_1 = \frac{d}{ds}\left[3.6\times10^7/s\right]_{s=-1000} = -3.6\times10^7/10^6 = -36$. Hence

$$v_C(t) = 48 - 36\,e^{-1000t} - 36\,000\,t\,e^{-1000t} = 48 - 36\,(1+1000t)\,e^{-1000t}\ \text{V}$$

which rises monotonically from 12 V to 48 V with no overshoot (dashed curves in Figure 11.2). The price is a slower approach: the critically damped voltage reaches 47 V (within 1 V of final) at about 5.4 ms, against a first crossing of 47 V at about 2.3 ms for the underdamped circuit, which then overshoots. In a

precharge circuit the slower, safe response is the right choice; in a measurement channel the trade between speed and overshoot is the subject of the filter and control chapters.

Worked example — Step response of the two-section RC filter

Apply a 1 V step to the filter of Section 11.5. With $p_1 = -3819.7$ and $p_2 = -26\,180$ rad/s,

$$V_o(s) = \frac{1}{s}\,\frac{1}{\tau^2 (s - p_1)(s - p_2)} = \frac{k_0}{s} + \frac{k_1}{s - p_1} + \frac{k_2}{s - p_2}$$

$k_0 = 1/(\tau^2 p_1 p_2) = 1$ (since $\tau^2 p_1 p_2 = 1$); $k_1 = 1/[\tau^2 p_1 (p_1 - p_2)] = -1.1708$; $k_2 = 1/[\tau^2 p_2 (p_2 - p_1)] = +0.1708$. Thus

$$v_o(t) = 1 - 1.1708\, e^{-3819.7 t} + 0.1708\, e^{-26\,180 t} \ \text{V}$$

Checks: $v_o(0) = 1 - 1.1708 + 0.1708 = 0$, as it must be with both capacitors initially discharged; the slope at $t = 0$ is $1.1708 \times 3819.7 - 0.1708 \times 26\,180 = 0$, as it must be because the output capacitor's current comes through the middle node, which itself starts at zero. Values: 0.213 V at 0.1 ms, 0.456 V at 0.2 ms, 0.827 V at 0.5 ms, 0.974 V at 1 ms. The 10–90 % rise time is 0.586 ms, against $2.2\tau = 0.22$ ms for one section alone. The slow pole p_1 dominates: its residue is seven times that of the fast pole and its time constant seven times longer. Figure 11.4 shows the pole positions and the step response.

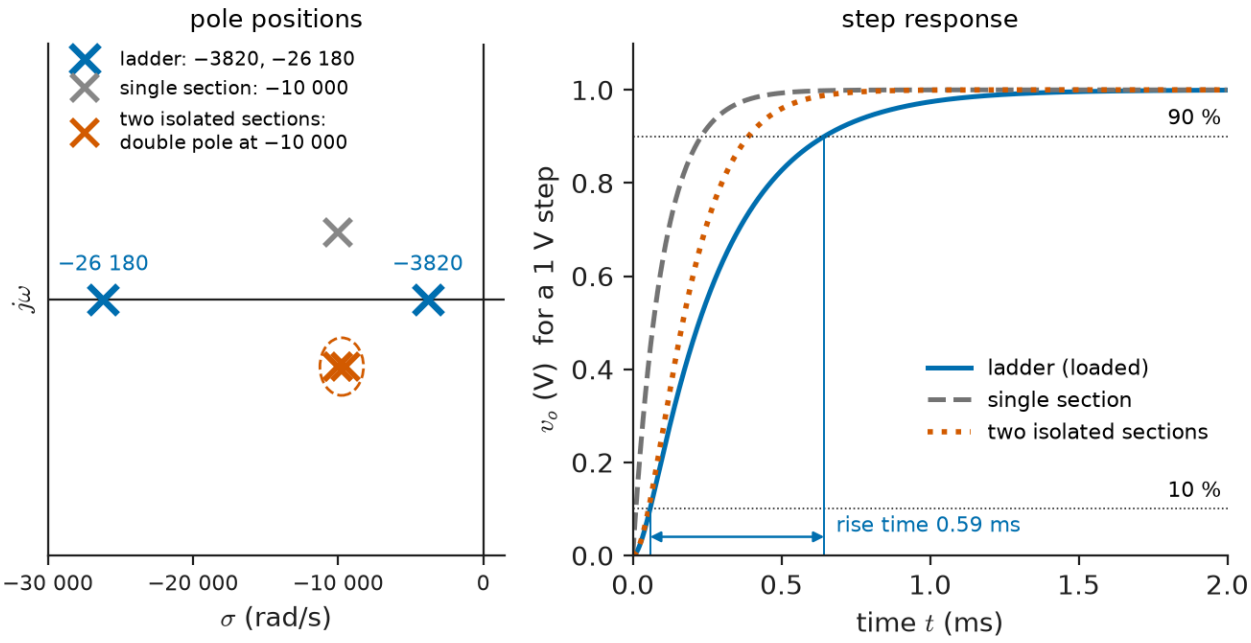


Figure 11.4 — Two-section RC filter (R = 1 kilohm, C = 100 nF): pole positions and step response compared with a single section.

11.8 Network response

Zero-input and zero-state responses

Because the s-domain equations are linear in both the sources and the initial-condition terms, any response splits into two parts:

$$Y(s) = \underbrace{H(s)X(s)}_{\text{zero-state}} + \underbrace{\frac{N_0(s)}{D(s)}}_{\text{zero-input}}$$

The **zero-state response** is what the inputs produce from a de-energised circuit; it is the transfer function times the input transform. The **zero-input response** is what the stored energy produces with all independent sources set to zero; its numerator depends on the initial conditions, but its denominator is the same characteristic polynomial $D(s)$, so it contains the same natural frequencies. The complete response is their sum.

In the precharge example, the zero-input part is the response to $v_C(0^-) = 12$ V with the 48 V source shorted, and the zero-state part is the response to the 48 V step with the capacitor discharged. For the current, the two parts are $-12/L$ and $+48/L$ over the same quadratic, giving the net $36/L$ found directly.

> **Status of the principle: Decomposition into zero-input and zero-state responses**
> *Derived*. Superposition applied to the s-domain equations, treating initial-condition sources as additional inputs; exact for linear networks, invalid for nonlinear ones, where the response to stored energy and the response to inputs interact.

Response to standard inputs

Input	$X(s)$	Zero-state response transform	Use
Unit impulse	1	$H(s)$	Characterises the network; hammer test, short pulse test
Unit step	$1/s$	$H(s)/s$	Settling time, overshoot, rise time; switch-on behaviour
Ramp	$1/s^2$	$H(s)/s^2$	Tracking error of sweeps and ramps
Sinusoid $\cos \omega t$	$s/(s^2+\omega^2)$	$H(s)\, s/(s^2+\omega^2)$	Switch-on transient plus AC steady state
Delayed pulse	$(1-e^{-sT})/s$	$H(s)(1-e^{-sT})/s$	Response to digital pulses and bursts

The step response is the integral of the impulse response, $Y(s) = H(s)/s$, which is why a laboratory step test, easier to generate than an impulse, characterises a network just as completely. For the two-section filter the impulse response is

$$h(t) = \frac{1}{\tau^2 (p_1 - p_2)} \left(e^{p_1 t} - e^{p_2 t} \right) = 4472 \left(e^{-3819.7 t} - e^{-26\,180 t} \right) \ \text{s}^{-1}$$

and its integral reproduces the step response of Section 11.7.

Pulses and piecewise inputs

The time-delay property turns piecewise inputs into sums of shifted steps. A rectangular pulse of height V and width T is $V[u(t) - u(t - T)]$, with transform $V(1 - e^{-sT})/s$. The response is the step response minus the same response delayed by T. This is how the effect of a finite-width digital pulse passing through an RC interconnect, or of a 1 ms defibrillator-test pulse through a biomedical amplifier's input network, is computed: one step response, superposed twice.

Multi-input networks

When a network has several independent sources and stored energy, the s-domain solution by nodal or mesh analysis gives the complete response at once. For hand work, superposition is often clearer: compute the response to each source and to each initial condition separately, using the transfer function from that source to the output, and add. This is particularly useful in instrumentation, where a sensor signal, an interference source and an offset all drive the same front end and their contributions to the output must be ranked.

11.9 Transient and steady-state separation

Which poles produce what

The response transform $Y(s) = H(s)X(s)$ (plus any zero-input part) has two kinds of poles:

- the poles of $H(s)$, the network's **natural frequencies**, which in a stable network all lie in the left half-plane and give terms that decay to zero: the **transient response**;
- the poles of $X(s)$, the **forced frequencies** of the input, which give terms of the same form as the input: the **steady-state** (or forced) response. A step has a pole at $s = 0$ and gives a constant; a sinusoid has poles at $\pm j\omega$ and gives a sinusoid at the same frequency.

The partial-fraction expansion separates the two automatically: the terms belonging to the network poles are the transient, the terms belonging to the input poles are the steady state. The residues at the input poles are the steady-state amplitudes; for a sinusoidal input they equal $|H(j\omega)|$ and $\angle H(j\omega)$, which shows that phasor analysis (Chapter 8) is the steady-state part of the Laplace solution.

> **Status of the principle: Transient–steady-state separation**
> *Derived*. Follows from partial-fraction expansion of $H(s)X(s)$ when the poles of H and X are distinct; it requires a stable network (otherwise the "transient" never dies), and when an input pole coincides with a network pole (resonant excitation) the terms merge into a $t\,e^{pt}$ response that is neither purely transient nor steady.

Worked example — Energising an inductive load on the mains

A large inductive load (a transformer's short-circuit impedance, a reactor or a motor at standstill) with $R = 1\ \Omega$ and $X = \omega L = 10\ \Omega$ at 50 Hz (so $L = 31.83$ mH and $\tau = L/R = 31.83$ ms) is switched onto a 230 V, 50 Hz supply, $v(t) = 325.3 \sin(\omega t + \theta)$, at $t = 0$. The current is initially zero.

Steady state by phasors. $|Z| = \sqrt{1 + 100} = 10.05\ \Omega$, $\varphi = \arctan 10 = 84.29^\circ$, so the steady-state current is $i_{ss}(t) = 32.37 \sin(\omega t + \theta - 84.29^\circ)$ A.

Complete solution by Laplace. For switching at a voltage zero ($\theta = 0$), $V(s) = 325.3\,\omega/(s^2 + \omega^2)$ and

$$I(s) = \frac{325.3\,\omega}{(s^2 + \omega^2)\,(Ls + R)} = \frac{k}{s + R/L} + \frac{As + B}{s^2 + \omega^2}$$

The quadratic term inverts to the steady-state sinusoid above. The cover-up rule at the network pole $s = -R/L = -31.42$ rad/s gives the transient residue

$$k = \frac{325.3\,\omega/L}{(R/L)^2 + \omega^2} = \frac{325.3 \times X}{R^2 + X^2} = \frac{3253}{101} = 32.21\ \text{A}$$

so that

$$i(t) = 32.37 \sin(\omega t - 84.29^\circ) + 32.21\, e^{-t/31.83\,\text{ms}}\ \text{A}$$

At $t = 0$ the two terms cancel ($32.37 \sin(-84.29^\circ) = -32.21$), as the continuity of inductor current requires. The second term is a **DC offset**: a decaying unidirectional component. Because τ is longer than a cycle, it has barely decayed half a cycle later, when the sinusoid reaches its own positive peak. The total current peaks at 56.2 A at $t = 9.45$ ms, 1.74 times the steady-state peak (Figure 11.5).

If instead the switch closes at $\theta = \varphi = 84.29^\circ$ (close to a voltage peak), the transient residue is zero and the current starts directly in its steady state. Switching at a voltage zero is the worst case for an inductive circuit. This is why circuit breakers must be rated for an asymmetrical peak current of up to about 2.5 times the symmetrical rms value (IEC 62271-100 uses 2.5 for 50 Hz systems with $X/R = 14$), why controlled ("point-on-wave") switching closes reactors near voltage peaks, and why, in a transformer with a saturable core, the same DC offset of flux drives the core into saturation and produces inrush currents of several times rated current. The saturating case is nonlinear and outside the Laplace method, but the linear analysis shows where the offset comes from.

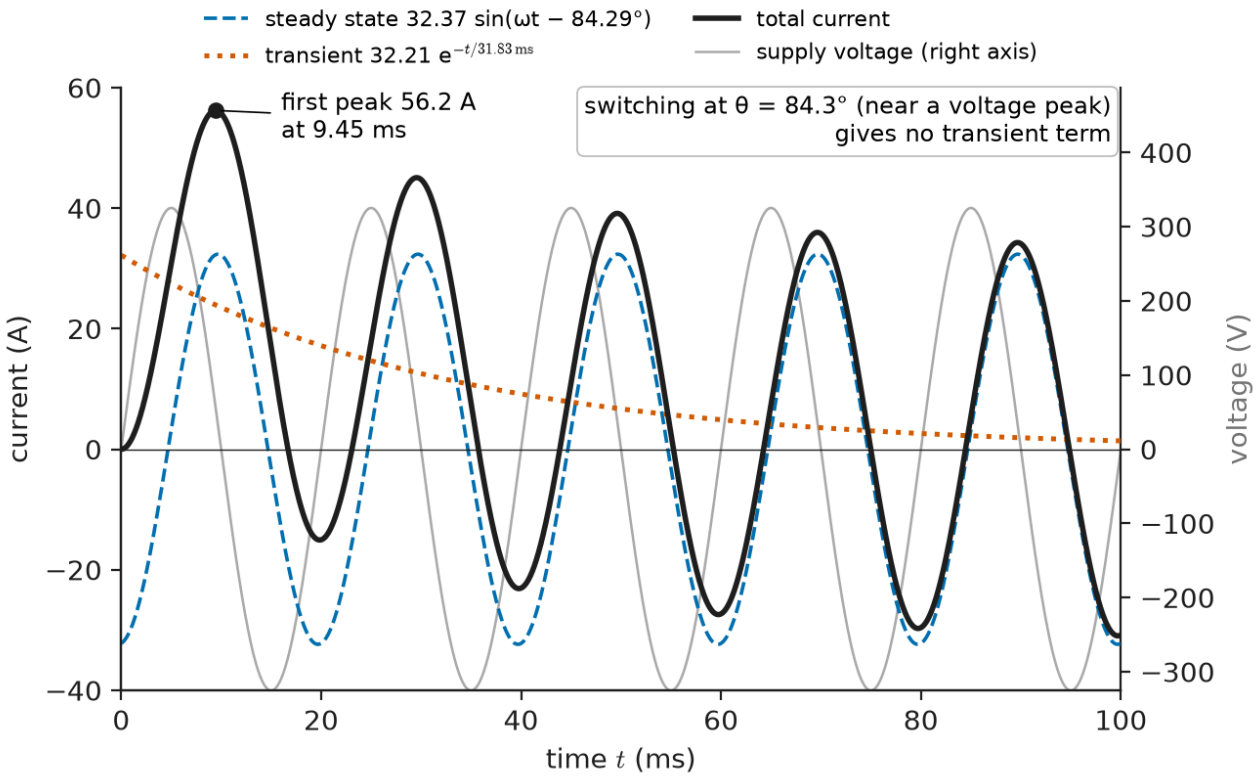


Figure 11.5 — Energising an R-L load (1 ohm, 31.8 mH) at a voltage zero on a 230 V 50 Hz supply: steady-state and transient components add to an asymmetrical current.

Initial- and final-value theorems

Two limits of the transform give values of the time response without inverting it:

$$f(0^+) = \lim_{s\to\infty} sF(s), \qquad f(\infty) = \lim_{s\to 0} sF(s)$$

The final-value theorem is valid only when $f(t)$ actually settles: all poles of $sF(s)$ must lie in the open left half-plane. It gives nonsense for a sinusoidal steady state (poles on the imaginary axis) or an unstable network.

Applied to the precharge example: $\lim_{s\to\infty} sV_C(s) = 12$ V, the initial capacitor voltage; $\lim_{s\to 0} sV_C(s) = 12 + 3.6\times 10^7/10^6 = 48$ V, the final value. For the current, $\lim_{s\to\infty} sI(s) = 0$ (the inductor prevents a jump) and $\lim_{s\to 0} sI(s) = 0$ (the capacitor blocks DC). For the RL energisation, the final-value theorem must not be used: $sI(s)$ has poles at $\pm j\omega$.

> **Status of the principle: Initial- and final-value theorems**
> *Derived*. Limits of the Laplace integral; the initial-value theorem requires $F(s)$ strictly proper (otherwise f contains an impulse at $t = 0$), and the final-value theorem requires all poles of $sF(s)$ in the open left half-plane.

Settling time and the dominant pole

The transient dies away at the rate of the slowest network pole, the one nearest the imaginary axis. A practical measure is the settling time to within 2 % of the final value, approximately $4/\alpha_{\min}$ where $\alpha_{\min}$ is the smallest decay rate. For the two-section filter, $4/3819.7 = 1.05$ ms; for the underdamped precharge, $4/500 = 8$ ms; for the RL energisation, $4\tau = 127$ ms, more than six cycles of the mains. An ADC driver that must settle to half an LSB of a 16-bit converter needs $\ln(2^{17}) \approx 11.8$ time constants of its dominant pole, which sets the minimum bandwidth of the driver for a given sample rate.

> **Status of the principle: Dominant-pole approximation**
> *Approximate.* When one real pole (or pair) is much closer to the imaginary axis than the others, typically by a factor of 5 or more, and its residue is not small, the response after the first few fast time constants is well described by that pole alone; it fails when poles are comparably spaced or when a nearby zero nearly cancels the dominant pole.

Where the Laplace method stops

The method requires linear, time-invariant, lumped elements. It does not handle saturation, hysteresis, diodes and switches that change state during the transient (each interval must be solved separately with the end state of one as the initial state of the next), or distributed elements whose impedance is not rational in s. Transmission lines (Chapter 14) produce transforms containing $e^{-s\ell/v}$ and $\sqrt{s}$-type functions; they are still handled in the s-domain, but partial fractions no longer suffice. For large networks, the s-domain formulation is the basis of state-space and circuit-simulator methods (Chapters 33 and 35), which replace hand factoring by numerical eigenvalue computation.

What remains open

Open question	Where it is answered
How does a network with an input port and an output port compose with others, so that loading is handled systematically?	Chapter 12 (two-port networks, ABCD cascades)
How are magnitude, phase, bandwidth and Q displayed and designed from poles and zeros?	Chapter 13 (Bode plots, resonance, passive filters)
What happens when interconnects are long enough that the lumped model fails?	Chapter 14 (transmission lines)
How do impulse response, convolution and transfer function relate for general signals and systems?	Chapters 16–18 (systems, LTI theory, Fourier analysis)
What is the discrete-time counterpart of the s-plane?	Chapter 20 (Z-transform, poles and zeros in the z-plane)
How is stability tested and controlled when feedback can move poles into the right half-plane?	Chapter 34 (Routh–Hurwitz, root locus, Nyquist)
State-space form of network equations	Chapters 33 and 35
The Laplace transform itself: existence, inversion integral, proofs	*The Mathematics Universe*
Transformer inrush and breaker asymmetry in power systems	*Electrical Energy & Machines*

Next: Chapter 12 treats a network as a box with two ports and characterises it by a small matrix of s-domain functions, so that amplifiers, filters and transmission lines can be cascaded and terminated without re-solving them.

Chapter 12 — Two-Port Networks

Part III · Advanced Network Theory

A technician needs 20 dB of attenuation in front of a spectrum analyser and finds two 10 dB coaxial pads in a drawer. Screwed together, they give 20.0 dB, exactly, as the label promises. The same technician then builds a crude anti-aliasing filter by putting two identical RC sections in series, each with a corner frequency of 1.59 kHz and each, on its own, 3 dB down at that frequency. The pair is not 6 dB down at 1.59 kHz but 9.5 dB down, and its corner has moved to 600 Hz. A transmission engineer has a related problem at a hundred thousand times the power: a 150 km, 220 kV line must deliver 100 MW at 0.9 power factor to a city substation, and the operator needs to know what voltage to hold at the power station, and what the substation voltage will do at night when the load drops away. In each case a network sits between a source and a load, and the question is what it does to the signal or the power passing through it, and what happens when it is connected to something else. Re-solving the whole circuit every time a load or source changes is wasteful and hides the answer's structure. This chapter treats each network as a box with an input port and an output port, characterised by four numbers (or four functions of s), and develops the algebra for terminating, converting and combining such boxes.

> **How can a linear network be described by its terminal behaviour alone, so that it can be connected to any source and load without re-solving its interior?**
> **Why do some networks cascade by simple multiplication and others not, and what do reciprocity and symmetry buy us?**

Concept	Engineering destination
Z parameters	Transformer open-circuit tests, sensor and antenna coupling, series-connected networks
Y parameters	Transistor models at high frequency, nodal simulators, parallel-connected networks
Hybrid parameters	Bipolar-transistor small-signal models, series-shunt feedback amplifiers
ABCD parameters	Transmission lines, cascaded filters and attenuators, power-system line models
Parameter conversion	Combining datasheet models given in different forms, simulator interfaces
Cascaded networks	Filter chains, attenuator pads, multi-section lines, amplifier chains
Reciprocity and symmetry	Passive filters and pads, test-fixture de-embedding, measurement checks
Amplifier and transmission applications	Loaded gain and impedance of amplifiers; sending-end voltage, regulation and Ferranti rise on lines

Roadmap. The two-port model and its assumptions (12.1) → the four principal parameter sets: Z (12.2), Y (12.3), hybrid (12.4), ABCD (12.5) → converting between them (12.6) → terminating a two-port

with source and load (12.7) → combining two-ports, above all in cascade (12.8) → reciprocity and symmetry (12.9) → amplifiers and transmission systems (12.10).

12.1 The two-port model

Ports and the port condition

A **port** is a pair of terminals at which the current entering one terminal equals the current leaving the other. A **two-port** is a network with two such pairs, conventionally an input port 1 and an output port 2 (Figure 12.1). Each port has a voltage V_k and a current I_k, with I_k defined as flowing *into* the network at the positive terminal. Four quantities V_1, I_1, V_2, I_2 describe everything the outside world can observe.

The port condition is not automatic. Inside a larger circuit, the current into terminal 1 need not return through terminal 1′; a path through some other connection can carry part of it. The two-port description is valid only when the external connections enforce the port condition, for example because the source and load each connect to one port only. Section 12.8 shows where this matters.

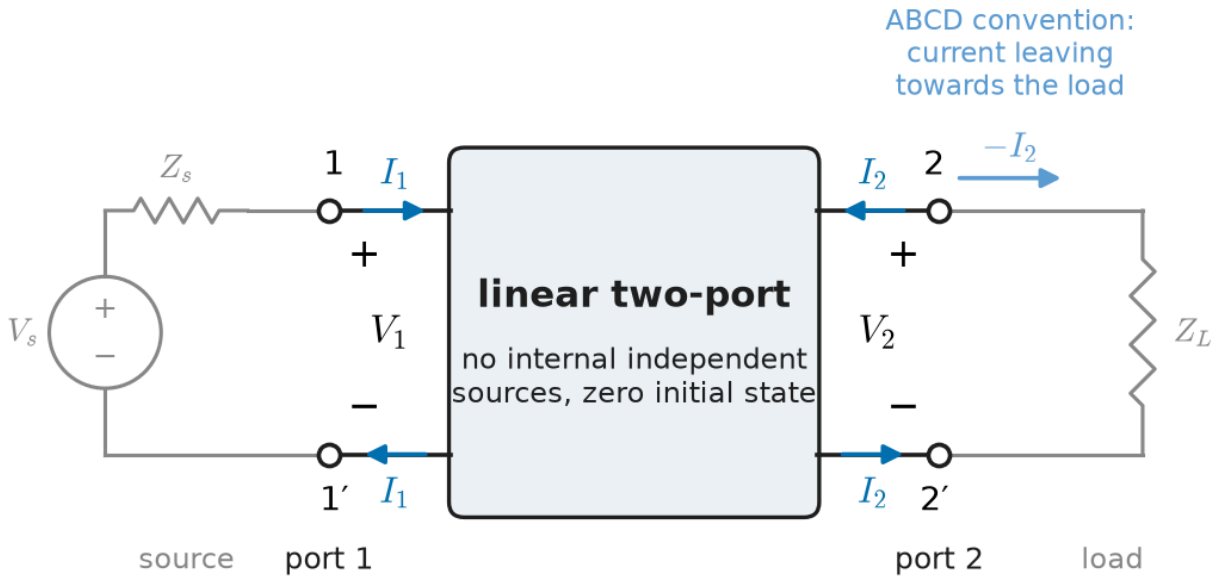


Figure 12.1 — A two-port network with port voltages and currents, and the alternative current direction used by ABCD parameters.

Assumptions

The two-port descriptions of this chapter assume that the network is

- **linear** and **time-invariant**, so that its port quantities are related by linear equations with constant coefficients in the s-domain or in phasor form;
- free of **independent sources** inside (dependent sources are allowed, and are what make amplifiers possible);

- at **zero initial state** when described in the s-domain (stored energy would add source-like terms, Chapter 11).

Under these assumptions, any two of the four port quantities can be chosen as independent variables, and the other two are linear functions of them. There are $\binom{4}{2} = 6$ ways to choose, so six parameter sets exist. Four are in common use and are developed here: impedance (Z), admittance (Y), hybrid (h, with its inverse g) and transmission (ABCD). The parameters are functions of s (or of $j\omega$ in sinusoidal steady state); for purely resistive networks they are constants.

> **Status of the principle: Two-port representation**
> *Derived*. Linearity and the absence of internal independent sources make two of the port variables linear functions of the other two (by superposition); the description fails for nonlinear networks (large-signal transistors, saturating transformers), for networks with internal sources unless an offset term is added, and wherever the port condition is violated.

What the model buys

Once a network has been reduced to four parameters, its interior can be forgotten. Its input impedance with any load, its gain from any source, its output impedance, and its behaviour when connected to another two-port all follow from those four numbers by formulas that never change. This is the same economy as Thévenin's theorem (Chapter 4), extended from one port to two. It is how a transistor datasheet describes a device without a circuit diagram, how a power-system engineer represents 150 km of line in a load-flow program, and how a vector network analyser characterises a filter.

12.2 Z parameters

Definition

Choose the two currents as independent variables. Superposition gives

$$\begin{aligned} V_1 &= Z_{11} I_1 + Z_{12} I_2 \\ V_2 &= Z_{21} I_1 + Z_{22} I_2 \end{aligned} \qquad \text{or} \qquad \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \mathbf{Z} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

Each parameter is measured with one port open-circuited (zero current), which is why they are called **open-circuit impedance parameters**:

$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}, \quad Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0}, \quad Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}, \quad Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0}$$

Z_{11} and Z_{22} are the driving-point impedances with the other port open; Z_{12} and Z_{21} are transfer impedances. All four are in ohms.

The T equivalent

Any reciprocal two-port ($Z_{12} = Z_{21}$) can be represented by a T network: series arm $Z_{11} - Z_{12}$ at port 1, series arm $Z_{22} - Z_{12}$ at port 2, and shunt arm Z_{12} (Figure 12.2a). Conversely, a T network with series arms Z_a (port 1) and Z_b (port 2) and shunt arm Z_c has

$$Z_{11} = Z_a + Z_c, \qquad Z_{12} = Z_{21} = Z_c, \qquad Z_{22} = Z_b + Z_c$$

A non-reciprocal network adds a current-controlled voltage source $(Z_{21} - Z_{12})I_1$ in series with port 2.

Worked example — Parameters of a resistive T network. Take $Z_a = 10\ \Omega$, $Z_b = 20\ \Omega$, $Z_c = 30\ \Omega$. Then

$$\mathbf{Z} = \begin{bmatrix} 40 & 30 \\ 30 & 50 \end{bmatrix}\ \Omega, \qquad \Delta Z = Z_{11}Z_{22} - Z_{12}Z_{21} = 2000 - 900 = 1100\ \Omega^2$$

Check by inspection: with port 2 open, port 1 sees $10 + 30 = 40\ \Omega$, and the open-circuit output voltage is the voltage across the 30 Ω shunt, $30I_1$. This network is used again in the following sections, so that every parameter set can be checked against the same circuit.

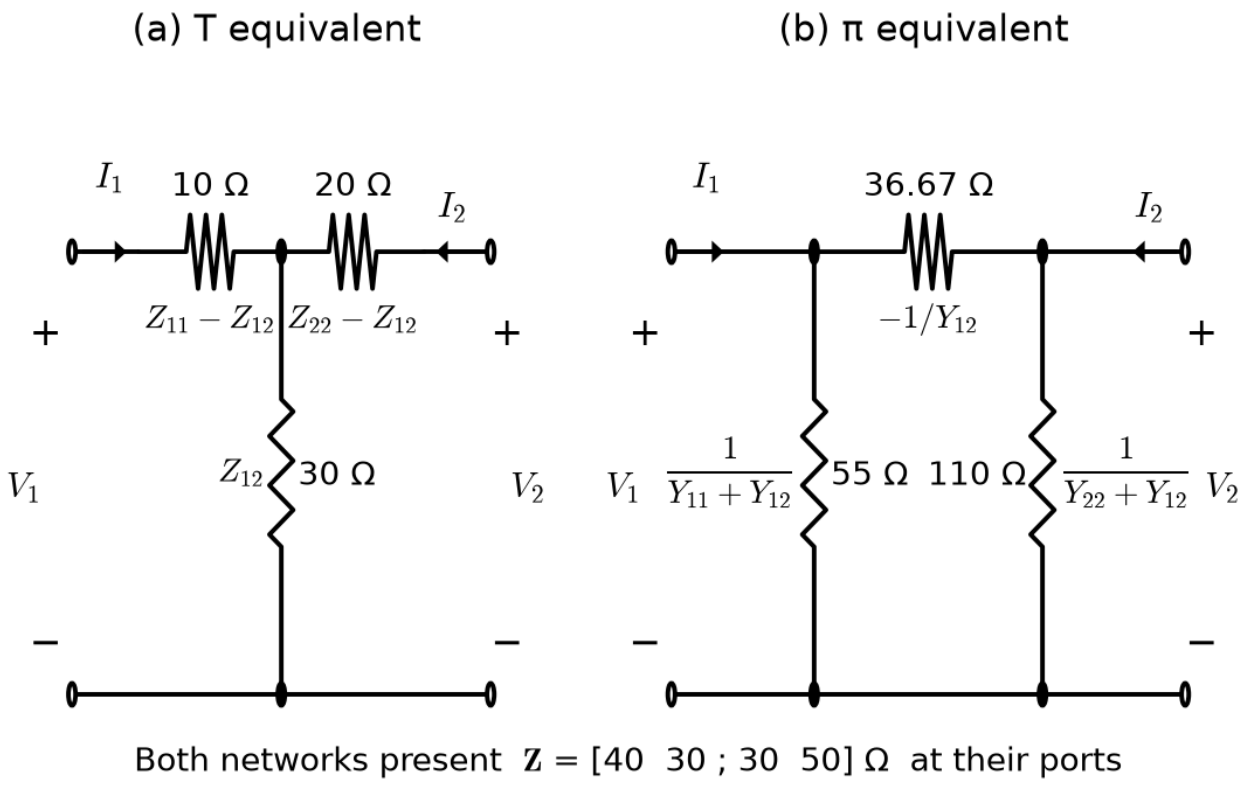


Figure 12.2 — The T and pi equivalents of the same reciprocal two-port, using the worked-example values.

Measurement and physical meaning

Z parameters are natural when open circuits are easy to make. The **open-circuit test of a power transformer** is a Z-parameter measurement: rated voltage is applied to one winding with the other open, and the measured current and power give Z_{11} (dominated by the magnetising branch) and, from the open-circuit secondary voltage, Z_{21}. Z parameters also describe the mutual coupling of two antennas

or two coils: for coupled inductors, $Z_{11} = R_1 + sL_1$, $Z_{22} = R_2 + sL_2$ and $Z_{12} = Z_{21} = sM$, which is exactly the s-domain coupled-coil law of Chapter 11.

Z parameters do not exist for every network. An ideal transformer has no finite open-circuit impedance, and a network consisting of a single series impedance between the ports has Z_{11} undefined (with port 2 open no current flows, whatever the voltage). In such cases another parameter set is used.

12.3 Y parameters

Definition

Choose the two voltages as independent variables:

$$\begin{aligned} I_1 &= Y_{11}V_1 + Y_{12}V_2 \\ I_2 &= Y_{21}V_1 + Y_{22}V_2 \end{aligned} \qquad Y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2=0},\; Y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1=0},\; Y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2=0},\; Y_{22} = \left.\frac{I_2}{V_2}\right|_{V_1=0}$$

These are the **short-circuit admittance parameters**, in siemens. When both exist, $\mathbf{Y} = \mathbf{Z}^{-1}$.

The π equivalent

A reciprocal two-port ($Y_{12} = Y_{21}$) is represented by a π network: shunt admittance $Y_{11} + Y_{12}$ at port 1, series admittance $-Y_{12}$ between the ports, shunt admittance $Y_{22} + Y_{12}$ at port 2. For a π network of shunt admittances Y_A and Y_B and series admittance Y_C: $Y_{11} = Y_A + Y_C$, $Y_{22} = Y_B + Y_C$, $Y_{12} = Y_{21} = -Y_C$. A non-reciprocal network adds a voltage-controlled current source $(Y_{21} - Y_{12})V_1$ across port 2; with $Y_{12} = 0$ and $Y_{21} = g_m$, this is the familiar transconductance model of a field-effect transistor.

Worked example — Y parameters of the T network. Inverting $\mathbf{Z}$ of Section 12.2:

$$\mathbf{Y} = \frac{1}{1100}\begin{bmatrix} 50 & -30 \\ -30 & 40 \end{bmatrix} = \begin{bmatrix} 45.45 & -27.27 \\ -27.27 & 36.36 \end{bmatrix}\ \text{mS}$$

Check Y_{11} by inspection: with port 2 shorted, port 1 sees $10 + (30 \parallel 20) = 10 + 12 = 22\ \Omega$, and $1/22 = 45.45$ mS. The π equivalent has shunt arms $1/(45.45 - 27.27)\ \text{mS} = 55\ \Omega$ at port 1 and $1/(36.36 - 27.27)\ \text{mS} = 110\ \Omega$ at port 2, and a series arm $1/27.27\ \text{mS} = 36.67\ \Omega$ (Figure 12.2b). These are the values given by the star–delta transformation of Chapter 4: the sum of pairwise products of the T arms is $10 \cdot 20 + 20 \cdot 30 + 30 \cdot 10 = 1100$, and dividing by the opposite arm gives 36.67, 55 and 110 Ω.

Where Y parameters are natural

Y parameters are measured with short circuits, which are easy to provide at high frequency with a large capacitor, so they were the classical way to characterise RF transistors before S-parameters (Chapter 32). They are the natural form for **parallel-connected** networks (Section 12.8) and for nodal analysis: a circuit simulator stamps each element's Y-matrix into the global nodal admittance matrix. A FET at moderate frequency is well described by $Y_{11} = s(C_{gs} + C_{gd})$, $Y_{12} = -sC_{gd}$, $Y_{21} = g_m - sC_{gd}$, $Y_{22} = g_{ds} + sC_{gd}$; the small feedback term Y_{12} is what makes a high-gain stage prone to oscillation and gives the Miller effect.

12.4 Hybrid parameters

The h parameters

Choose I_1 and V_2 as independent variables:

$$V_1 = h_{11} I_1 + h_{12} V_2$$
$$I_2 = h_{21} I_1 + h_{22} V_2$$

$$h_{11} = \left.\frac{V_1}{I_1}\right|_{V_2=0} (\Omega), \quad h_{12} = \left.\frac{V_1}{V_2}\right|_{I_1=0}, \quad h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0}, \quad h_{22} = \left.\frac{I_2}{V_2}\right|_{I_1=0} (\mathrm{S})$$

The four parameters have mixed dimensions, hence "hybrid": h_{11} is the input impedance with the output shorted, h_{12} the reverse voltage ratio with the input open, h_{21} the forward current ratio with the output shorted, and h_{22} the output admittance with the input open. The equivalent circuit (drawn for a transistor stage in Figure 12.5) is a Thévenin form at the input (h_{11} in series with a controlled voltage source $h_{12}V_2$) and a Norton form at the output (a controlled current source $h_{21}I_1$ in parallel with h_{22}).

Why the bipolar transistor uses h parameters

At low frequency a bipolar transistor in common-emitter connection has a low input impedance (hundreds of ohms to tens of kilohms) and a high output impedance (tens of kilohms or more). Shorting the output (for h_{11}, h_{21}) is easy to do with a large capacitor across a high impedance; opening the input (for h_{12}, h_{22}) is easy because the source can be an inductor or large resistor compared with a low input impedance. The measurement conditions match the device, so the parameters can be measured accurately. The h parameters also have direct physical meaning: h_{fe} (the common-emitter h_{21}) is the small-signal current gain β, h_{ie} is the base input resistance r_π, and h_{oe} is the inverse of the collector output resistance.

The subscripts i, r, f, o (input, reverse, forward, output) followed by e, b, c (the common terminal) are the standard datasheet notation: $h_{ie} = h_{11}$, $h_{re} = h_{12}$, $h_{fe} = h_{21}$, $h_{oe} = h_{22}$ for the common-emitter connection.

Table 12.1 — Representative h parameters of small-signal bipolar transistors (common emitter, room temperature, audio frequencies).

Parameter	Symbol	Typical range	Example (general-purpose NPN, $I_C \approx 2$ mA)
Input impedance	h_{ie}	0.5–10 kΩ (falls as I_C rises)	1.2 kΩ
Reverse voltage ratio	h_{re}	$0.5–8 \times 10^{-4}$	2.5×10^{-4}
Forward current gain	h_{fe}	50–500	100
Output admittance	h_{oe}	1–50 µS	25 µS (40 kΩ)

The spread within one transistor type is large (datasheets for common types give h_{fe} ranges of 3:1 or more), which is the reason amplifier designs use feedback to make the gain depend on resistor ratios rather than on h_{fe}.

> **Boundary.**
> Biasing, the physics behind these parameters, the hybrid-π model and transistor amplifier design belong to *Electronics & Computing*. Here a transistor is a two-port whose small-signal parameters are given.

The g parameters

The inverse hybrid set takes V_1 and I_2 as independent variables:

$$I_1 = g_{11}V_1 + g_{12}I_2, \qquad V_2 = g_{21}V_1 + g_{22}I_2$$

with $\mathbf{g} = \mathbf{h}^{-1}$. The g parameters suit devices with high input impedance and low output impedance, such as a voltage amplifier or an op-amp follower, whose ideal model is $g_{11} = 0$, $g_{12} = 0$, $g_{21} = A$, $g_{22} = 0$.

Worked example — h parameters of the T network. For the T network ($\mathbf{Z} = [40, 30; 30, 50]\ \Omega$), with port 2 shorted, the input impedance is 22 Ω and the short-circuit current is divided between the 30 Ω and 20 Ω arms so that $I_2 = -\frac{30}{50}I_1$:

$$h_{11} = 22\ \Omega, \quad h_{21} = -0.6, \quad h_{12} = \frac{Z_{12}}{Z_{22}} = 0.6, \quad h_{22} = \frac{1}{Z_{22}} = 0.02\ \text{S}$$

Note $h_{12} = -h_{21}$; this is the h-parameter form of reciprocity (Section 12.9).

12.5 ABCD parameters

Definition

For cascading, the natural independent variables are the *output* quantities, so that the output of one network becomes the input of the next. The **transmission** or **ABCD parameters** express the input port in terms of the output port:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

The output current is taken as $-I_2$, the current *leaving* port 2 towards the load. With this sign, the current leaving one network is the current entering the next, and the matrices multiply without sign changes. The parameters are

$$A = \left.\frac{V_1}{V_2}\right|_{I_2=0}, \quad B = \left.\frac{V_1}{-I_2}\right|_{V_2=0}\ (\Omega), \quad C = \left.\frac{I_1}{V_2}\right|_{I_2=0}\ (\text{S}), \quad D = \left.\frac{I_1}{-I_2}\right|_{V_2=0}$$

A is the reciprocal of the open-circuit voltage gain, D the reciprocal of the short-circuit current gain, B the negative of the reciprocal short-circuit transfer admittance, C the open-circuit transfer admittance. A good network has small A for voltage transmission: $A = 1$ means no voltage change with the output open.

> **Status of the principle: ABCD (chain) parameters**
> *Derived.* Equivalent to any other two-port description when it exists; they exist whenever the network transmits (when V_2 and I_2 can be chosen independently), and fail for networks with no transmission between the ports, such as two isolated one-ports.

Elementary sections

The ABCD matrices of a few elementary sections allow almost any ladder network to be assembled by multiplication.

Table 12.2 — ABCD matrices of elementary two-ports.

Two-port	A	B	C	D	$AD - BC$
Series impedance Z	1	Z	0	1	1
Shunt admittance Y	1	0	Y	1	1
Ideal transformer $n:1$	n	0	0	$1/n$	1
T network (Z_a, Z_c shunt, Z_b)	$1 + Z_a/Z_c$	$Z_a + Z_b + Z_aZ_b/Z_c$	$1/Z_c$	$1 + Z_b/Z_c$	1
π network (Y_A, Z series, Y_B)	$1 + ZY_B$	Z	$Y_A + Y_B + ZY_AY_B$	$1 + ZY_A$	1
RC low-pass section (series R, shunt C)	$1 + sRC$	R	sC	1	1
Lossless line, length ℓ, impedance Z_0	$\cos\beta\ell$	$jZ_0\sin\beta\ell$	$j\sin(\beta\ell)/Z_0$	$\cos\beta\ell$	1
Ideal voltage amplifier, gain μ	$1/\mu$	0	0	0	0

For the T network of the running example: $A = 1 + 10/30 = 1.333$, $B = 10 + 20 + 200/30 = 36.67\ \Omega$, $C = 1/30 = 33.33$ mS, $D = 1 + 20/30 = 1.667$, and $AD - BC = 2.222 - 1.222 = 1$.

The ideal transformer has no Z or Y parameters but a simple ABCD matrix. Its $n:1$ ratio scales voltage by n and current by $1/n$, so it transforms a load impedance Z_L at port 2 into $n^2 Z_L$ at port 1, the impedance-transformation rule used to match a loudspeaker to an output stage or a 50 Ω line to an antenna.

Transmission lines in ABCD form

ABCD parameters were developed for power-line engineering, where they are still called the "generalised circuit constants". A line of any length is represented in each phase (using the per-phase equivalent of Chapter 10) by a two-port relating sending-end voltage and current to receiving-end voltage and current. For a short line (below roughly 80 km at 50 Hz), the shunt capacitance is neglected and the line is a series impedance Z: $A = D = 1$, $B = Z$, $C = 0$. For a medium line (roughly 80–250 km), the **nominal π** model places half the total shunt admittance Y at each end: $A = D = 1 + ZY/2$, $B = Z$, $C = Y(1 + ZY/4)$. For long lines the exact distributed solution gives $A = D = \cosh \gamma\ell$, $B = Z_c \sinh \gamma\ell$, $C = \sinh(\gamma\ell)/Z_c$ (Chapter 14). Section 12.10 works a medium-line example.

Table 12.3 — Representative per-phase parameters of 50 Hz lines and cables (typical values per km).

Line or cable	r (Ω/km)	x (Ω/km)	b (µS/km)	Surge impedance (Ω)
11 kV overhead, 100 mm² Al	0.3	0.35–0.4	3	~350–400
11 kV XLPE cable, 185 mm² Al	0.16–0.21	0.09–0.11	60–100	~30–40
132 kV overhead, single conductor	0.1–0.2	0.38–0.42	2.7–3.0	~380–400
220 kV overhead, single conductor	0.06–0.1	0.38–0.42	2.8–3.0	~370–390
400 kV overhead, quad bundle	0.02–0.03	0.26–0.33	3.5–4.3	~250–300

Cables have much lower series reactance and fifteen to thirty times the shunt susceptance of overhead lines, which is why long AC cables are limited by their charging current and why long submarine links use DC.

12.6 Parameter conversion

Why convert

Datasheets, measurements and models arrive in different forms: an RF transistor in Y or S parameters, a bipolar transistor in h parameters, a line in ABCD form, a transformer from open- and short-circuit tests. Combining them requires a common form, and each interconnection (Section 12.8) has a preferred one. Since each parameter set is a rearrangement of the same two linear equations, conversion is algebra: solve the defining equations of one set for the variables of another.

The conversion table

Write $\Delta Z = Z_{11}Z_{22} - Z_{12}Z_{21}$, and similarly ΔY, Δh and $\Delta T = AD - BC$.

Table 12.4 — Conversions between two-port parameter sets.

To ↓ / from →	Z	Y	h	ABCD
Z_{11}	Z_{11}	$Y_{22}/\Delta Y$	$\Delta h/h_{22}$	A/C
Z_{12}	Z_{12}	$-Y_{12}/\Delta Y$	h_{12}/h_{22}	$\Delta T/C$
Z_{21}	Z_{21}	$-Y_{21}/\Delta Y$	$-h_{21}/h_{22}$	$1/C$
Z_{22}	Z_{22}	$Y_{11}/\Delta Y$	$1/h_{22}$	D/C
Y_{11}	$Z_{22}/\Delta Z$	Y_{11}	$1/h_{11}$	D/B
Y_{12}	$-Z_{12}/\Delta Z$	Y_{12}	$-h_{12}/h_{11}$	$-\Delta T/B$
Y_{21}	$-Z_{21}/\Delta Z$	Y_{21}	h_{21}/h_{11}	$-1/B$
Y_{22}	$Z_{11}/\Delta Z$	Y_{22}	$\Delta h/h_{11}$	A/B
h_{11}	$\Delta Z/Z_{22}$	$1/Y_{11}$	h_{11}	B/D
h_{12}	Z_{12}/Z_{22}	$-Y_{12}/Y_{11}$	h_{12}	$\Delta T/D$
h_{21}	$-Z_{21}/Z_{22}$	Y_{21}/Y_{11}	h_{21}	$-1/D$
h_{22}	$1/Z_{22}$	$\Delta Y/Y_{11}$	h_{22}	C/D
A	Z_{11}/Z_{21}	$-Y_{22}/Y_{21}$	$-\Delta h/h_{21}$	A
B	$\Delta Z/Z_{21}$	$-1/Y_{21}$	$-h_{11}/h_{21}$	B
C	$1/Z_{21}$	$-\Delta Y/Y_{21}$	$-h_{22}/h_{21}$	C
D	Z_{22}/Z_{21}	$-Y_{11}/Y_{21}$	$-1/h_{21}$	D

Each conversion exists only when its denominator is nonzero. That is the algebraic form of the existence remarks made earlier: a series impedance has $C = 0$ and therefore no Z parameters; a shunt admittance has $B = 0$ and no Y parameters; a network with no transmission has $Z_{21} = 0$ (or $Y_{21} = 0$) and no ABCD parameters.

> **Status of the principle: Parameter conversion**
> *Derived*. Algebraic rearrangement of the same pair of linear equations; exact wherever the relevant denominator is nonzero, and numerically ill-conditioned when it is close to zero (for example, converting a nearly unilateral transistor's h parameters to Z parameters at very low output conductance).

Worked example — Converting the T network both ways. Starting from the ABCD values of Section 12.5 ($A = 4/3$, $B = 36.67\ \Omega$, $C = 1/30$ S, $D = 5/3$, $\Delta T = 1$):

$h_{11} = B/D = 36.67/1.667 = 22\ \Omega$, $h_{12} = \Delta T/D = 0.6$, $h_{21} = -1/D = -0.6$, $h_{22} = C/D = 0.02$ S, in agreement with Section 12.4.

$Z_{11} = A/C = 40\ \Omega$, $Z_{12} = Z_{21} = 1/C = 30\ \Omega$, $Z_{22} = D/C = 50\ \Omega$, in agreement with Section 12.2.

$Y_{11} = D/B = 45.45$ mS, $Y_{12} = Y_{21} = -1/B = -27.27$ mS, $Y_{22} = A/B = 36.36$ mS, in agreement with Section 12.3.

Every set describes the same three resistors; the choice is a matter of convenience.

12.7 Terminated two-ports

Input impedance, gain and output impedance

In use, a two-port sits between a source (Thévenin voltage V_s, impedance Z_s) at port 1 and a load Z_L at port 2, which enforces $V_2 = -I_2 Z_L$. Substituting this into any parameter set gives the quantities a designer needs. In ABCD form, with $-I_2 = V_2/Z_L$:

$$Z_{in} = \frac{V_1}{I_1} = \frac{AZ_L + B}{CZ_L + D}, \qquad \frac{V_2}{V_1} = \frac{Z_L}{AZ_L + B}, \qquad \frac{-I_2}{I_1} = \frac{1}{CZ_L + D}$$

Looking back into port 2 with the source connected, the Thévenin equivalent at the output is

$$Z_{out} = \frac{DZ_s + B}{CZ_s + A}, \qquad V_{oc} = \frac{V_s}{A + CZ_s}$$

In h-parameter form, which is the usual one for transistor stages:

$$Z_{in} = h_{11} - \frac{h_{12}h_{21}Z_L}{1 + h_{22}Z_L}, \quad A_I = \frac{I_2}{I_1} = \frac{h_{21}}{1 + h_{22}Z_L}, \quad A_V = \frac{V_2}{V_1} = \frac{-h_{21}Z_L}{h_{11} + \Delta h\, Z_L}, \quad Z_{out} = \frac{h_{11} + Z_s}{\Delta h + h_{22}Z_s}$$

These formulas show the central message of two-port theory: input impedance depends on the load, and output impedance depends on the source, *through the reverse transmission* (B, C, or h_{12}). A **unilateral** network, with no reverse transmission ($h_{12} = 0$, $Y_{12} = 0$, $Z_{12} = 0$), has an input impedance independent of the load and an output impedance independent of the source. Ideal amplifiers are unilateral; real ones, and all passive networks, are not.

Worked example — Loading the T network. Terminate the T network with $Z_L = 100\ \Omega$. From the ABCD form:

$$Z_{in} = \frac{(4/3)(100) + 36.67}{(1/30)(100) + 5/3} = \frac{170}{5} = 34\ \Omega, \qquad \frac{V_2}{V_1} = \frac{100}{133.33 + 36.67} = 0.588$$

Direct check: port 1 sees $10 + 30 \parallel (20 + 100) = 10 + 24 = 34\ \Omega$; the voltage across the shunt arm is $V_1 \times 24/34$, and the load receives $100/120$ of it, $V_1 \times 0.588$. With port 2 open, by contrast, $Z_{in} = Z_{11} = 40\ \Omega$; with port 2 shorted, $22\ \Omega$. The load is visible from the input.

Image impedance and matched termination

For a symmetrical network ($A = D$), there is one load impedance which, when connected to port 2, makes the input impedance equal to itself. Setting $Z_{in} = Z_L = Z_0$ in the ABCD formula gives

$$Z_0 = \sqrt{\frac{B}{C}} = \sqrt{Z_{oc}\,Z_{sc}}$$

where $Z_{oc} = A/C$ and $Z_{sc} = B/D$ are the input impedances with the output open and shorted. This **image** (or iterative) **impedance** is the impedance level of the network. A chain of identical symmetrical sections, each terminated in Z_0, presents Z_0 at every junction; each section then multiplies the signal by the same factor. This is why matched attenuators and lines cascade by simple multiplication of their gains. For a line, Z_0 is the characteristic impedance of Chapter 14.

12.8 Cascaded networks and other interconnections

Cascade: multiply the ABCD matrices

When the output port of network a is connected to the input port of network b, $V_2^{(a)} = V_1^{(b)}$ and $-I_2^{(a)} = I_1^{(b)}$. Then

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \mathbf{T}_a \begin{bmatrix} V_2^{(a)} \\ -I_2^{(a)} \end{bmatrix} = \mathbf{T}_a \mathbf{T}_b \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} \qquad \Rightarrow \qquad \mathbf{T} = \mathbf{T}_a \mathbf{T}_b$$

The ABCD matrix of a cascade is the product of the individual matrices, in the order the signal passes through them. Matrix multiplication is not commutative: an RC low-pass section followed by a transformer is not the same network as the transformer followed by the section, and the products differ.

> **Status of the principle: Cascade rule**
> *Derived.* Follows from KCL and KVL at the junction between the two networks; it is exact for any linear two-ports with defined ABCD parameters, and automatically includes the loading of each network by the next.

The cascade rule explains both of the technician's results from the chapter opening.

Worked example — Two RC sections in cascade. Each section, series $R = 1\ \text{k}\Omega$ and shunt $C = 100$ nF, has $\mathbf{T}_1 = \begin{bmatrix} 1 + s\tau & R \\ sC & 1 \end{bmatrix}$ with $\tau = RC = 0.1$ ms. The cascade:

$$\mathbf{T} = \mathbf{T}_1 \mathbf{T}_1 = \begin{bmatrix} (1 + s\tau)^2 + s\tau & R(2 + s\tau) \\ sC(2 + s\tau) & 1 + s\tau \end{bmatrix}$$

With the output open (high-impedance ADC input), $V_2/V_1 = 1/A = 1/(\tau^2 s^2 + 3\tau s + 1)$, exactly the transfer function obtained by nodal analysis in Chapter 11. The extra $s\tau$ in A, beyond the $(1 + s\tau)^2$ that would describe two isolated sections, is the loading of the first section by the second.

At $\omega = 1/\tau = 10^4$ rad/s (1591.5 Hz), one section alone gives $|H| = 1/|1 + j| = 0.707$ (−3.01 dB). Two isolated sections would give $0.707^2 = 0.5$ (−6.02 dB). The cascade gives $A = (1 + j)^2 + j = 3j$, so $|H| = 1/3 = 0.333$ (−9.54 dB). The −3 dB frequency of the cascade is 595.6 Hz. Inserting a unity-gain buffer between the sections (a unilateral two-port with $A = 1$, $B = C = D = 0$ in the ideal case) removes the loading, and the cascade then gives the product response, $1/(1 + s\tau)^2$. Figure 12.3 compares the three responses.

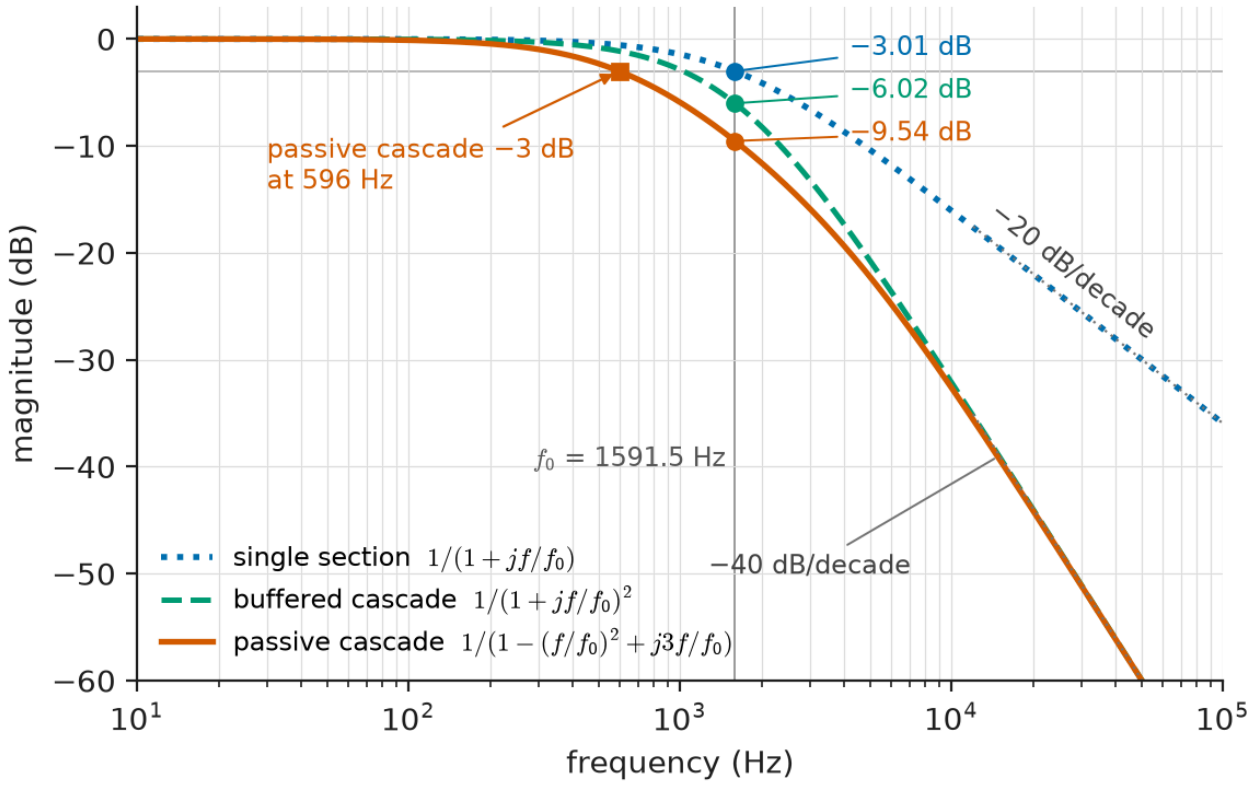


Figure 12.3 — Magnitude response of two RC sections (1 kilohm, 100 nF): single section, ideal product of two buffered sections, and the loaded passive cascade.

Worked example — Two matched 10 dB attenuators. A symmetrical π attenuator for $Z_0 = 50\ \Omega$ and attenuation $K = 10^{10/20} = 3.162$ has shunt resistors $R_{sh} = Z_0(K+1)/(K-1) = 96.25\ \Omega$ and a series resistor $R_{se} = Z_0(K^2-1)/(2K) = 71.15\ \Omega$. Its ABCD matrix is the product shunt × series × shunt:

$$\mathbf{T}_{pad} = \begin{bmatrix} 1.7393 & 71.15\ \Omega \\ 28.46\ \text{mS} & 1.7393 \end{bmatrix}, \qquad AD - BC = 1.000$$

Image impedance: $\sqrt{B/C} = \sqrt{71.15/0.02846} = 50.0\ \Omega$. Gain into 50 Ω: $50/(1.7393 \times 50 + 71.15) = 0.3162$, i.e. −10.00 dB. Cascading two pads: $\mathbf{T} = \mathbf{T}_{pad}^2 = \begin{bmatrix} 5.05 & 247.5\ \Omega \\ 99.0\ \text{mS} & 5.05 \end{bmatrix}$, and the gain into 50 Ω is $50/(5.05 \times 50 + 247.5) = 0.1$, exactly −20.00 dB.

The pads cascade by addition of decibels because each is terminated in its own image impedance: the second pad presents 50 Ω to the first, which is what the first was designed to see. The RC sections do not, because the second section's input impedance at 1.59 kHz is not the open circuit the first section's −3 dB figure assumed. The matrix product handles both cases without distinction; the "dB add" shortcut is valid only for image-matched chains.

Series, parallel and hybrid connections

Two-ports can also be interconnected port by port in four other ways (Figure 12.4). Each adds the matrices of one parameter set:

Connection	Input ports	Output ports	Rule
Series–series	In series	In series	$\mathbf{Z} = \mathbf{Z}_a + \mathbf{Z}_b$
Parallel–parallel (shunt–shunt)	In parallel	In parallel	$\mathbf{Y} = \mathbf{Y}_a + \mathbf{Y}_b$
Series–parallel (series–shunt)	In series	In parallel	$\mathbf{h} = \mathbf{h}_a + \mathbf{h}_b$
Parallel–series (shunt–series)	In parallel	In series	$\mathbf{g} = \mathbf{g}_a + \mathbf{g}_b$
Cascade	Output of a to input of b	—	$\mathbf{T} = \mathbf{T}_a \mathbf{T}_b$

These rules are the basis of the classical analysis of feedback amplifiers. An amplifier with a feedback network whose input is in parallel with the amplifier's output and whose output is in series with the amplifier's input (voltage-sensing, voltage-mixing, or series–shunt feedback, as in the non-inverting op-amp stage) is a series–parallel connection, and the h parameters of amplifier and feedback network add. The other three feedback topologies map onto Z, Y and g additions in the same way.

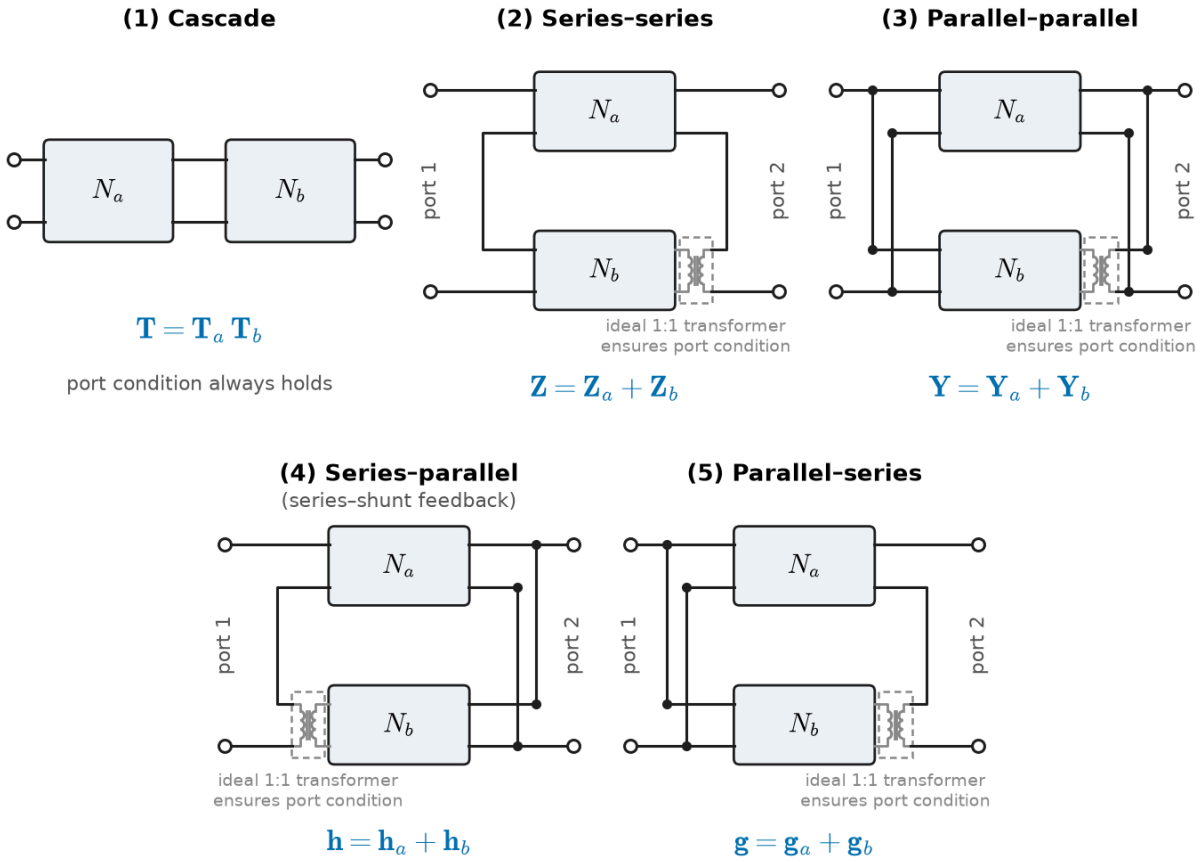


Figure 12.4 — The five standard interconnections of two two-ports and the parameter set that adds or multiplies in each.

When the addition rules fail: the port condition

The addition rules assume that each two-port still satisfies the port condition after the connection. Connecting two three-terminal networks (both with a common bottom rail) in series–series, for example, shorts part of one network through the common connection, and the current entering a port

no longer equals the current leaving it. The test for validity, known as the **Brune test**, checks that no voltage appears across the open connection points when the ports are excited as in the final interconnection. Where the test fails, an ideal 1:1 isolating transformer at one port of one network restores the port condition. The cascade connection never has this problem, because each network's ports remain separately connected; this is another reason the ABCD form dominates in practice.

> **Status of the principle: Parameter addition for series and parallel interconnections**
> *Derived.* KVL and KCL at the interconnected ports give the sums directly; valid only if the port condition holds for each constituent network after connection (Brune test), and invalid otherwise unless isolating transformers are used.

12.9 Reciprocity and symmetry

Reciprocity

Chapter 4 stated the reciprocity theorem: in a network of linear passive bilateral elements, the transfer impedance from one port to another is the same in both directions. For a two-port this becomes a constraint on the parameters:

Parameter set	Reciprocity condition
Z	$Z_{12} = Z_{21}$
Y	$Y_{12} = Y_{21}$
h	$h_{12} = -h_{21}$
g	$g_{12} = -g_{21}$
ABCD	$AD - BC = 1$

A reciprocal two-port therefore has only three independent parameters, and is fully described by a three-element T or π network. Every network of resistors, capacitors, inductors, coupled coils and ideal transformers is reciprocal. Every passive network in this chapter has satisfied $AD - BC = 1$: the T network, the RC sections, the attenuator, and, below, the transmission line.

> **Status of the principle: Two-port reciprocity**
> *Derived.* A consequence of the symmetry of the network's nodal admittance matrix, which follows from the bilateral nature of R, L, C, coupled coils and ideal transformers (ultimately from the time-reversal symmetry of Maxwell's equations in such media); it fails for networks containing dependent sources (transistors, amplifiers), ferrites in a magnetic bias field (isolators, circulators) and other non-reciprocal media.

Non-reciprocity is what makes amplification and isolation possible. A transistor with $h_{fe} = 100$ and $h_{re} = 2.5 \times 10^{-4}$ is very far from reciprocal ($h_{12} \neq -h_{21}$), and its usefulness comes from that

asymmetry: signals pass forward with gain and backward with heavy attenuation. A microwave isolator uses a magnetised ferrite to pass power one way and absorb it the other.

Reciprocity is also a practical check. A measured passive network whose ΔT is not 1 within the measurement uncertainty indicates a fixture problem, a calibration error or an active element somewhere in the path.

Symmetry

A two-port is **symmetrical** if its ports can be interchanged without changing the external behaviour. The conditions are:

Parameter set	Symmetry condition (in addition to reciprocity)
Z	$Z_{11} = Z_{22}$
Y	$Y_{11} = Y_{22}$
h	$\Delta h = h_{11}h_{22} - h_{12}h_{21} = 1$
ABCD	$A = D$

A symmetrical reciprocal two-port has only two independent parameters. The attenuator pad ($A = D = 1.7393$) and the transmission line ($A = D$ in the nominal π model and in the exact model) are symmetrical; the T network of the running example ($Z_{11} = 40$, $Z_{22} = 50$) and the single RC section ($A = 1 + s\tau$, $D = 1$) are not. Physically symmetrical networks are electrically symmetrical, but the converse need not hold: networks with different internal structure can have equal A and D.

For a symmetrical reciprocal network the two remaining parameters can be taken as the image impedance $Z_0 = \sqrt{B/C}$ and the **image propagation constant** γ, defined by $\cosh\gamma = A$. Then

$$\mathbf{T} = \begin{bmatrix} \cosh\gamma & Z_0 \sinh\gamma \\ \sinh(\gamma)/Z_0 & \cosh\gamma \end{bmatrix}$$

For the 10 dB pad, $\cosh\gamma = 1.7393$ gives $\gamma = 1.1513$ Np, which is $1.1513 \times 8.686 = 10.0$ dB, as it must be. For a uniform transmission line, γ becomes the propagation constant times the length (Chapter 14). This image-parameter form is the bridge between lumped filter sections and distributed lines, and it is the classical basis of filter design by constant-k and m-derived sections.

12.10 Applications in amplifiers and transmission systems

Amplifier stages as two-ports

An amplifier stage is a non-reciprocal two-port. Its h, y or g parameters at the operating point are given by the device datasheet or a simulator; its behaviour with a given source and load follows from the terminated-two-port formulas of Section 12.7. Figure 12.5 shows the common-emitter stage in h-parameter form.

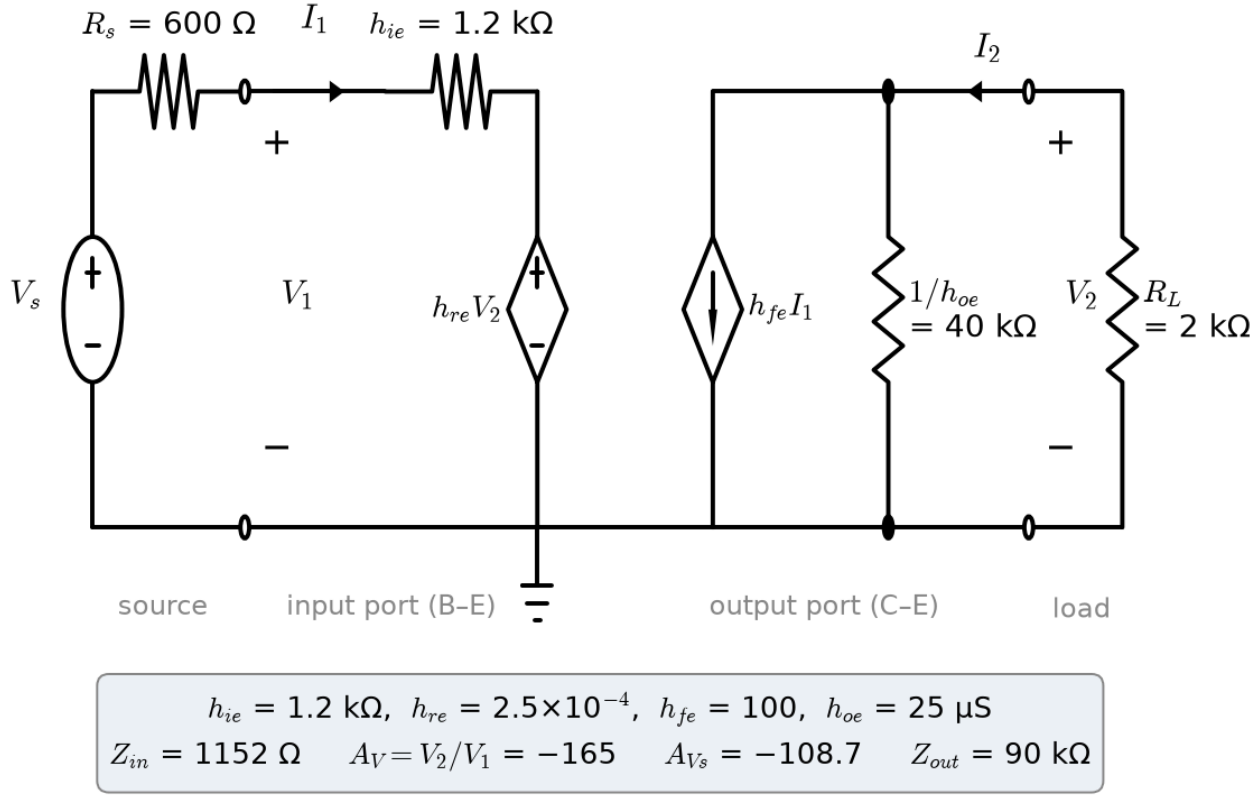


Figure 12.5 — h-parameter small-signal model of a common-emitter stage with source and load, with the worked-example values.

Worked example — Loaded gain of a common-emitter stage. A transistor biased at about 2 mA has $h_{ie} = 1.2\ \text{k}\Omega$, $h_{re} = 2.5 \times 10^{-4}$, $h_{fe} = 100$, $h_{oe} = 25\ \mu\text{S}$ (Table 12.1). It is driven from a source with $R_s = 600\ \Omega$ and drives a load $R_L = 2\ \text{k}\Omega$ (the collector resistor in parallel with the next stage's input, as seen by the signal).

$\Delta h = h_{ie}h_{oe} - h_{re}h_{fe} = 0.030 - 0.025 = 0.005.$

Current gain: $A_I = h_{fe}/(1 + h_{oe}R_L) = 100/1.05 = 95.24.$

Input impedance: $Z_{in} = h_{ie} - h_{re}h_{fe}R_L/(1 + h_{oe}R_L) = 1200 - 47.6 = 1152.4\ \Omega.$

Voltage gain: $A_V = -h_{fe}R_L/(h_{ie} + \Delta h\, R_L) = -200\,000/1210 = -165.3.$

Overall gain from the source: $A_{Vs} = A_V\, Z_{in}/(Z_{in} + R_s) = -165.3 \times 1152.4/1752.4 = -108.7.$

Output impedance: $Z_{out} = (h_{ie} + R_s)/(\Delta h + h_{oe}R_s) = 1800/(0.005 + 0.015) = 90\ \text{k}\Omega.$

Compare the simplified, unilateral model ($h_{re} = 0$, $h_{oe} = 0$): $Z_{in} = 1.2\ \text{k}\Omega$, $A_V = -h_{fe}R_L/h_{ie} = -166.7$, $Z_{out} = \infty$. The gain and input impedance change by less than 4 %, which justifies the simplified model for quick design at audio frequencies. The output impedance, however, is not infinite but 90 kΩ, more than twice $1/h_{oe} = 40\ \text{k}\Omega$: through the source resistance, the reverse transmission h_{re} partly cancels the output conductance, so that Z_{out} depends on R_s. At high frequencies, where the base–collector capacitance makes the reverse transmission much larger (the Miller effect), the unilateral approximation fails, and the full two-port description, usually in Y or S parameters, is required.

Three further amplifier applications rest on the same algebra:

- **Multistage amplifiers.** Converting each stage to ABCD form and multiplying gives the overall gain with all interstage loading included; this is how an audio preamplifier chain or an RF receiver front end is analysed before simulation.
- **Feedback amplifiers.** The series–shunt connection of Section 12.8 adds the h parameters of the amplifier and the feedback network; from the sum, the closed-loop gain and the raised input and lowered output impedances follow. Chapter 34 treats the stability side of feedback.
- **Stability of RF stages.** A two-port with reverse transmission can present a negative-resistance input when loaded with certain impedances, and oscillate. Stability criteria for two-ports (Rollett's K-factor) are written in Y or S parameters (Chapter 32).

Transmission systems as two-ports

Worked example — A 150 km, 220 kV medium-length line. A three-phase overhead line has, per phase, $z = 0.08 + j0.40\ \Omega$/km and $y = j2.8\ \mu$S/km, and is 150 km long. It delivers 100 MW at 0.9 power factor lagging to a substation held at 220 kV line-to-line. Use the nominal π model (Figure 12.6) and per-phase (line-to-neutral) quantities.

Totals: $Z = z\ell = 12 + j60\ \Omega$; $Y = y\ell = j420\ \mu$S.

ABCD constants:

$$A = D = 1 + \frac{ZY}{2} = 1 + \frac{(12 + j60)(j4.2 \times 10^{-4})}{2} = 0.9874 + j0.00252 = 0.9874\angle 0.146^\circ$$

$$B = 12 + j60\ \Omega, \qquad C = Y\left(1 + \frac{ZY}{4}\right) = -5.3 \times 10^{-7} + j4.174 \times 10^{-4}\ \text{S}$$

Check: $AD - BC = 1.0000$, as reciprocity requires.

Receiving end: $V_R = 220/\sqrt{3} = 127.02$ kV (reference). $I_R = 100 \times 10^6/(\sqrt{3} \times 220 \times 10^3 \times 0.9) = 291.6$ A at -25.84°.

Sending end:

$$V_S = AV_R + BI_R = 136.97\angle 6.09^\circ\ \text{kV}\ (237.2\ \text{kV line-to-line}), \qquad I_S = CV_R + DI_R = 269.1\angle -15.48^\circ$$

The sending-end current is *smaller* than the receiving-end current, because the line's shunt capacitance supplies part of the load's reactive power.

Sending-end power: $S_S = 3V_S I_S^* = 102.84 + j40.66$ MVA, so the sending-end power factor is 0.930 and the line loss is 2.84 MW. Transmission efficiency $= 100/102.84 = 97.2\,\%$.

Voltage regulation: if the load is removed with V_S unchanged, $I_R = 0$ and $V_R = V_S/A$, which is 240.3 kV line-to-line. The regulation is $(240.3 - 220)/220 = 9.2\,\%$.

Comparison with the exact distributed model (Chapter 14): $\cosh\gamma\ell = 0.98743 + j0.00251$ and $Z_c \sinh\gamma\ell = 11.90 + j59.76\ \Omega$, within 0.5 % of the nominal π values. At 150 km the lumped model is adequate; at 400 km it is not.

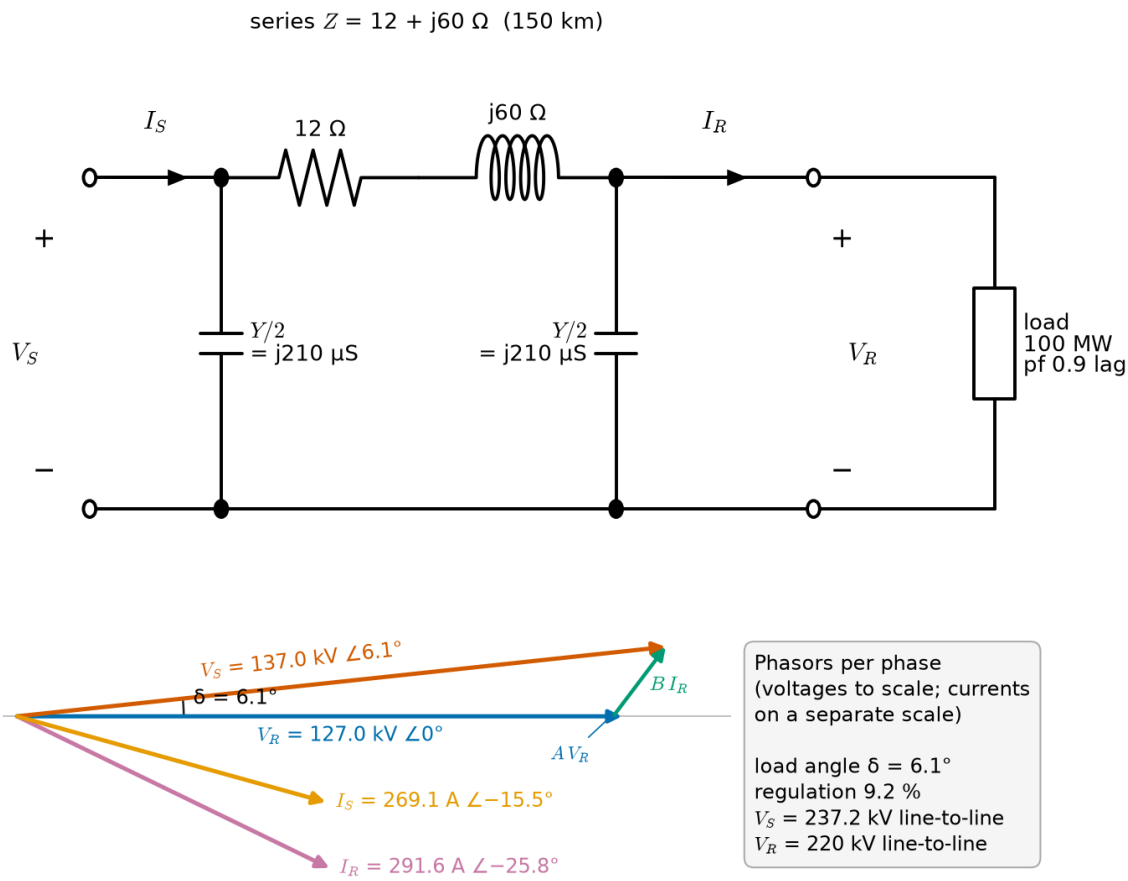


Figure 12.6 — Nominal pi model of the 150 km, 220 kV line and its sending- and receiving-end phasors at full load.

The Ferranti effect

At light load the receiving-end voltage of a long line rises above the sending-end voltage. With V_S held at 220 kV and the receiving end open, $V_R = V_S/|A| = 220/0.9874 = 222.8$ kV. The cause is the line's charging current flowing through its series inductance: the capacitive current produces a voltage rise along the inductive reactance. The rise grows roughly with the square of line length (since $1 - A \approx -ZY/2 \propto \ell^2$); on a 400 km, 400 kV line it is around 10 %, and long cables show it even more strongly because of their large capacitance. Grid operators control it at night by switching in shunt reactors, which, in two-port terms, cascade a shunt admittance $-jB_L$ at the receiving end to cancel part of C.

Other transmission applications

- **Compensation.** A series capacitor halfway along a line, which reduces the effective B and raises transfer capability, is analysed as a cascade: line half × series capacitor × line half.
- **Transformers in the chain.** The step-up transformer at a power station, the line and the step-down transformer at the substation form a cascade of three ABCD matrices, from which the overall regulation and losses follow.
- **Telephone and DSL loops.** Twisted-pair subscriber loops of a few kilometres are cascades of line sections, bridged taps and loading coils; DSL modems estimate the loop's ABCD (chain) matrix to predict attenuation at each tone frequency.
- **Instrumentation cables and sensor interfaces.** A thermocouple or strain-gauge sensor, its cable, a protection network and an amplifier input form a cascade whose overall transfer function

and noise depend on every loading interaction, which the ABCD product captures.

- **High frequencies.** Above a few hundred megahertz, open and short circuits cannot be realised accurately, and the parameters that are measured instead are the scattering (S) parameters, defined with respect to matched 50 Ω terminations; they convert to ABCD form for cascading. They are treated in Chapter 32.

> **Boundary.**
> The design of line conductors, insulation, compensation equipment and the power-flow and stability studies that use these line models belong to *Electrical Energy & Machines*. This chapter supplies the network model on which those studies rest.

What remains open

Open question	Where it is answered
How do two-port parameters vary with frequency, and how are they displayed?	Chapter 13 (frequency response, Bode plots, filters)
What is the exact two-port model of a line with distributed parameters, and what happens at high frequency?	Chapter 14 (telegrapher's equations, $\cosh \gamma\ell$ form, reflection, Smith chart)
How are two-ports characterised at RF and microwave frequencies, where open and short circuits are impractical?	Chapter 32 (S-parameters, microwave networks)
Back: the s-domain transfer functions that two-port parameters contain	Chapter 11
Back: reciprocity, Thévenin equivalents and star–delta transformation	Chapter 4
How do feedback connections of two-ports affect stability?	Chapter 34 (feedback and stability)
How are block diagrams and signal-flow graphs used to combine subsystems more generally?	Chapter 33 (dynamic-system modelling)
Transistor biasing and amplifier circuit design	*Electronics & Computing*
Line design, compensation and power-flow studies	*Electrical Energy & Machines*

Next: Chapter 13 takes the transfer functions and two-port gains of this part onto the $j\omega$ axis, and develops magnitude and phase response, Bode plots, resonance, Q and passive filters.

Chapter 13 — Network Frequency Response

Part III · Advanced Network Theory

A clinical ECG recorder is returned with a complaint: the ST segment of every heartbeat looks depressed, and a cardiologist nearly diagnoses ischaemia in a healthy patient. The amplifier is not broken. Its gain at 10 Hz is exactly as specified, its noise is low and its input impedance is high. The fault is a single capacitor in the input coupling network, chosen so that the low-frequency corner sits at 0.5 Hz instead of 0.05 Hz. At 0.5 Hz the network does more than lose 3 dB of amplitude: it shifts the phase of the slow components of the waveform by 45°, and the slow, flat ST segment is distorted into a slope. Nothing in a single-frequency gain test reveals the problem. The amplifier has to be described by how *both* its magnitude and its phase change across the whole band the signal occupies.

That description is the network's frequency response. Chapter 8 introduced it as the phasor ratio $\mathbf{V}_{\text{out}}/\mathbf{V}_{\text{in}}$ of a circuit driven by one sinusoid; Chapter 11 showed that the same ratio is the transfer function $H(s)$ evaluated on the imaginary axis, $s = j\omega$. This chapter turns those two ideas into an engineering tool: reading a response off a pole-zero map, sketching it in minutes with Bode asymptotes, sizing resonant circuits by their quality factor and bandwidth, designing passive filters, connecting them to active stages, and recognising in a frequency plot the signs of a circuit that is about to oscillate.

> **How does the location of poles and zeros decide the gain and phase a network applies at each frequency?**
> **How are resonance, quality factor and bandwidth related, and how do they set the selectivity of filters and tuned circuits? What can a frequency plot say about whether a feedback circuit will be stable?**

Concept	Engineering destination
Magnitude and phase response	Amplifier specifications, sensor front ends, waveform fidelity in ECG and audio
Poles and zeros	Fast mental evaluation of filters, compensation networks, notch placement
Bode plots	Datasheet frequency plots, loop-gain sketches, compensation design
Resonant circuits	Radio tuning, IF filters, wireless-power coils, harmonic filters in power systems
Quality factor	Selectivity of receivers, crystal and ceramic resonators, losses in reactive components
Bandwidth	Channel spacing, measurement bandwidth, rise time of instruments
Passive filters	Anti-aliasing, EMI filters, RF band selection, loudspeaker crossovers
Active-filter interface	Source and load effects when passive networks meet op-amp stages
Stability-related frequency behaviour	Phase margin of regulators and amplifiers, ringing, oscillation

Roadmap. Magnitude and phase as two views of $H(j\omega)$ (13.1) → poles and zeros as the geometry behind them (13.2) → Bode asymptotes for fast sketching (13.3) → resonant circuits (13.4), quality factor (13.5) and bandwidth (13.6) → passive filters built from these pieces (13.7) → what happens when a passive network meets an active stage (13.8) → frequency-domain signs of instability (13.9).

13.1 Magnitude and phase response

The frequency response as a complex function

Let a linear time-invariant network be described by its transfer function $H(s) = Y(s)/X(s)$ (Chapter 11), where X and Y are the Laplace transforms of input and output with zero initial conditions. If the network is stable and is driven by $x(t) = A\cos(\omega t + \theta)$, then once the transients have died away the output is

$$y_{\text{ss}}(t) = A\,|H(j\omega)|\cos\big(\omega t + \theta + \angle H(j\omega)\big)$$

This is the whole content of the **frequency response**: a sinusoid in gives a sinusoid of the same frequency out, scaled by the **magnitude response** $|H(j\omega)|$ and shifted by the **phase response** $\angle H(j\omega)$. The result rests on the fact that complex exponentials e^{st} pass through an LTI system unchanged in form (they are its eigenfunctions, Chapter 17), and on stability, which guarantees that the natural response decays and leaves only the forced part.

> **Status of the principle: Sinusoidal steady-state response**
> *Derived.* Follows from linearity, time invariance and the eigenfunction property of $e^{j\omega t}$; it holds only for stable networks and only after the transient has decayed.

Writing $H(j\omega) = |H|e^{j\phi}$ separates two pieces of engineering information. The magnitude says which frequencies are passed and which are suppressed. The phase says how much each frequency is delayed. A pure time delay T has $H(j\omega) = e^{-j\omega T}$: unity magnitude, phase $-\omega T$ falling linearly with frequency. Any network whose phase in the band of interest is approximately linear therefore delays the signal without distorting its shape; departures from linear phase reshape the waveform even when the magnitude is perfectly flat. This was the ECG fault: the magnitude error at the 1 Hz components was only a fraction of a decibel, but their phase was shifted by tens of degrees, and the waveform changed. Group delay, $\tau_g = -d\phi/d\omega$, quantifies this and is developed in Chapter 18.

Decibels and logarithmic frequency

Gains that span many orders of magnitude are expressed in decibels,

$$|H|_{\text{dB}} = 20\log_{10}|H(j\omega)|$$

The factor 20 (rather than 10) is used because $|H|$ is a ratio of amplitudes and power is proportional to amplitude squared; when input and output share the same impedance, the decibel gain equals the power gain in decibels. Useful anchors are $\sqrt{2} \to 3.01$ dB, $2 \to 6.02$ dB, $10 \to 20$ dB and $1000 \to 60$ dB. Decibels turn products of cascaded gains into sums, which is why Bode plots (Section 13.3) can be built by adding curves. Frequency is plotted on a logarithmic axis, where a **decade** is a factor of 10 and an **octave** a factor of 2.

The first-order low-pass as the reference case

The RC voltage divider with output across the capacitor is the simplest filter and the building block of the rest of the chapter:

$$H(s) = \frac{1}{1 + sRC}, \qquad H(j\omega) = \frac{1}{1 + j\omega/\omega_c}, \qquad \omega_c = \frac{1}{RC}$$

so that

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega/\omega_c)^2}}, \qquad \angle H(j\omega) = -\arctan\frac{\omega}{\omega_c}$$

At the **corner frequency** ω_c the magnitude is $1/\sqrt{2}$ (−3.01 dB) and the phase is −45°. Well below the corner the network is transparent; well above it the magnitude falls in proportion to $1/\omega$ (−20 dB per decade) and the phase approaches −90°.

Worked example — audio anti-aliasing RC. A first-order RC filter ahead of an audio converter uses $R = 1.6$ kΩ and $C = 10$ nF. The time constant is $RC = 16$ μs, so

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(16 \times 10^{-6})} = 9.95 \text{ kHz}$$

At 1 kHz, $f/f_c = 0.1005$, $|H| = 0.995$ (−0.04 dB) and the phase is −5.7°. At 20 kHz, the top of the audio band, $f/f_c = 2.01$, $|H| = 1/\sqrt{1+4.04} = 0.445$ (−7.0 dB) and the phase is −63.6°. At 100 kHz, $|H| = 0.099$ (−20.1 dB) and the phase is −84.3°. Two lessons follow. The filter already removes 7 dB at the top of the audio band, which is audible; and a decade above the corner it attenuates by only 20 dB, far less than the 80–100 dB a 16-bit or 24-bit converter needs against aliasing. A single pole is not a usable anti-aliasing filter on its own; Section 13.7 and Chapter 19 show what is.

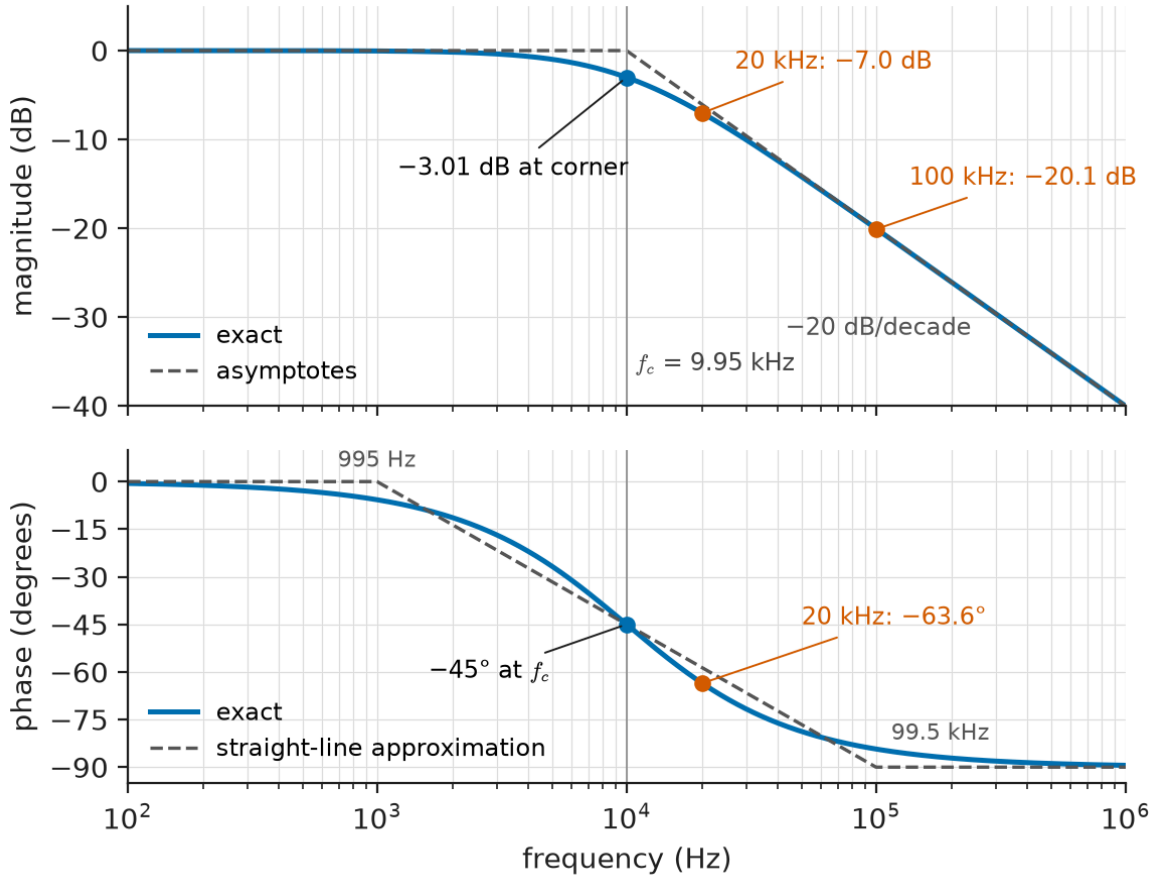


Figure 13.1 — Magnitude and phase of the first-order RC low-pass with R = 1.6 kOhm and C = 10 nF, exact curves with straight-line asymptotes.

Figure 13.1 shows both curves on logarithmic axes. The magnitude curve is close to two straight lines that meet at the corner; the phase curve is an S-shape that does most of its turning within one decade on either side of the corner. These two shapes are the alphabet from which Bode plots are written.

Magnitude and phase are not independent

For a large and important class of networks, the magnitude curve determines the phase curve. A network whose poles and zeros all lie in the left half of the s-plane is called **minimum-phase**. For such a network, Bode's gain–phase relation states that the phase at any frequency is approximately proportional to the local slope of the magnitude curve on log–log axes: a slope of $-20n$ dB/decade, sustained over a wide band, corresponds to a phase near $-90n$ degrees. A flat region has little phase shift; a region falling at −40 dB/decade has close to −180°.

> **Status of the principle: Bode gain–phase relation**
> *Derived.* A consequence of the analyticity of $\ln H(s)$ in the right half-plane for stable minimum-phase networks; it fails for networks with right-half-plane zeros or pure delays, which add phase lag without changing magnitude.

The practical value is enormous. A feedback designer who sees a loop gain falling at −40 dB/decade through 0 dB knows, without computing anything, that the phase there is close to −180° and the loop will be on the edge of instability (Section 13.9). A delay or a right-half-plane zero, by contrast, adds lag the magnitude curve does not reveal.

13.2 Poles and zeros

The factored transfer function

Every lumped linear network with constant elements has a rational transfer function, the ratio of two polynomials in s with real coefficients. Factoring numerator and denominator gives

$$H(s) = K\,\frac{(s - z_1)(s - z_2)\cdots(s - z_m)}{(s - p_1)(s - p_2)\cdots(s - p_n)}$$

The roots z_k of the numerator are the **zeros**; the roots p_k of the denominator are the **poles**. Real coefficients force complex poles and zeros to come in conjugate pairs. For a physical network with energy-storing elements, n equals the number of independent capacitor voltages and inductor currents (the order of the network), and $m \le n$ for any network that does not have infinite gain at infinite frequency.

The poles are the natural frequencies of the network: they are the exponents in its unforced response (Chapter 11). A pole at $s = -\alpha$ produces a term $e^{-\alpha t}$; a conjugate pair $-\alpha \pm j\omega_d$ produces a decaying oscillation $e^{-\alpha t}\cos(\omega_d t + \theta)$. The zeros are the complex frequencies at which the network transmits nothing: an input $e^{z_k t}$ produces no forced output. A zero on the $j\omega$ axis at $\pm j\omega_z$ is a **notch**: a sinusoid at ω_z is blocked completely.

Geometric evaluation of the frequency response

Setting $s = j\omega$, each factor $(j\omega - z_k)$ or $(j\omega - p_k)$ is a complex number: the vector drawn from the zero or pole to the point $j\omega$ on the imaginary axis. Hence

$$|H(j\omega)| = |K|\frac{\prod_k |j\omega - z_k|}{\prod_k |j\omega - p_k|}, \qquad \angle H(j\omega) = \angle K + \sum_k \angle(j\omega - z_k) - \sum_k \angle(j\omega - p_k)$$

The magnitude is the product of the distances from the zeros divided by the product of the distances from the poles; the phase is the sum of the zero angles minus the sum of the pole angles. As ω sweeps up the imaginary axis, the point $j\omega$ moves, the vectors change length and direction, and the response is traced out (Figure 13.2).

> **Status of the principle: Pole-zero geometry of the frequency response**
> *Derived*. Exact algebra on the factored rational transfer function; it applies to any lumped LTI network, but the frequency response it computes is physically observable only when all poles are in the left half-plane.

This geometric view explains the shapes of all frequency responses at a glance:

- A pole close to the $j\omega$ axis makes one distance very small as $j\omega$ passes it, so the magnitude rises to a sharp peak there: **resonance** is a pole pair near the axis.
- A zero on or near the axis makes a distance vanish, so the magnitude falls to a sharp dip: a **notch**.
- Far above all poles and zeros, every distance is approximately ω, so $|H| \propto \omega^{m-n}$: the high-frequency roll-off is $20(n-m)$ dB per decade and the phase tends to $-90(n-m)$ degrees.
- At $\omega = 0$ the distances are the magnitudes of the poles and zeros themselves, giving the DC gain $|K| \prod |z_k| / \prod |p_k|$.

Worked example — a lightly damped resonance read from the pole-zero map. A vibration-sensor conditioning network has

$$H(s) = \frac{10(s+2)}{s^2 + 2s + 101}$$

with s in rad/s. The poles are the roots of $s^2 + 2s + 101$, namely $p_{1,2} = -1 \pm j10$, and the zero is at $z = -2$. At DC the gain is $10 \cdot 2/101 = 0.198$ (−14.1 dB). At $\omega = 10$ rad/s the point $j10$ sits directly opposite the upper pole. The distances are $|j10 - z| = \sqrt{2^2 + 10^2} = 10.20$, $|j10 - p_1| = |j10 - (-1 + j10)| = 1.00$ and $|j10 - p_2| = |1 + j20| = 20.02$. So

$$|H(j10)| = \frac{10 \times 10.20}{1.00 \times 20.02} = 5.09 \quad (+14.1 \text{ dB})$$

The angles are 78.7° from the zero, 0° from the upper pole and 87.1° from the lower pole, giving $\angle H = 78.7^\circ - 0^\circ - 87.1^\circ = -8.4^\circ$. The response at 10 rad/s is 25.7 times larger than at DC, a 28.2 dB resonant peak produced entirely by the upper pole lying only 1 rad/s from the axis. Moving that pole further left (more damping) would lengthen the short vector and flatten the peak; this is exactly what a damping resistor does in a real circuit.

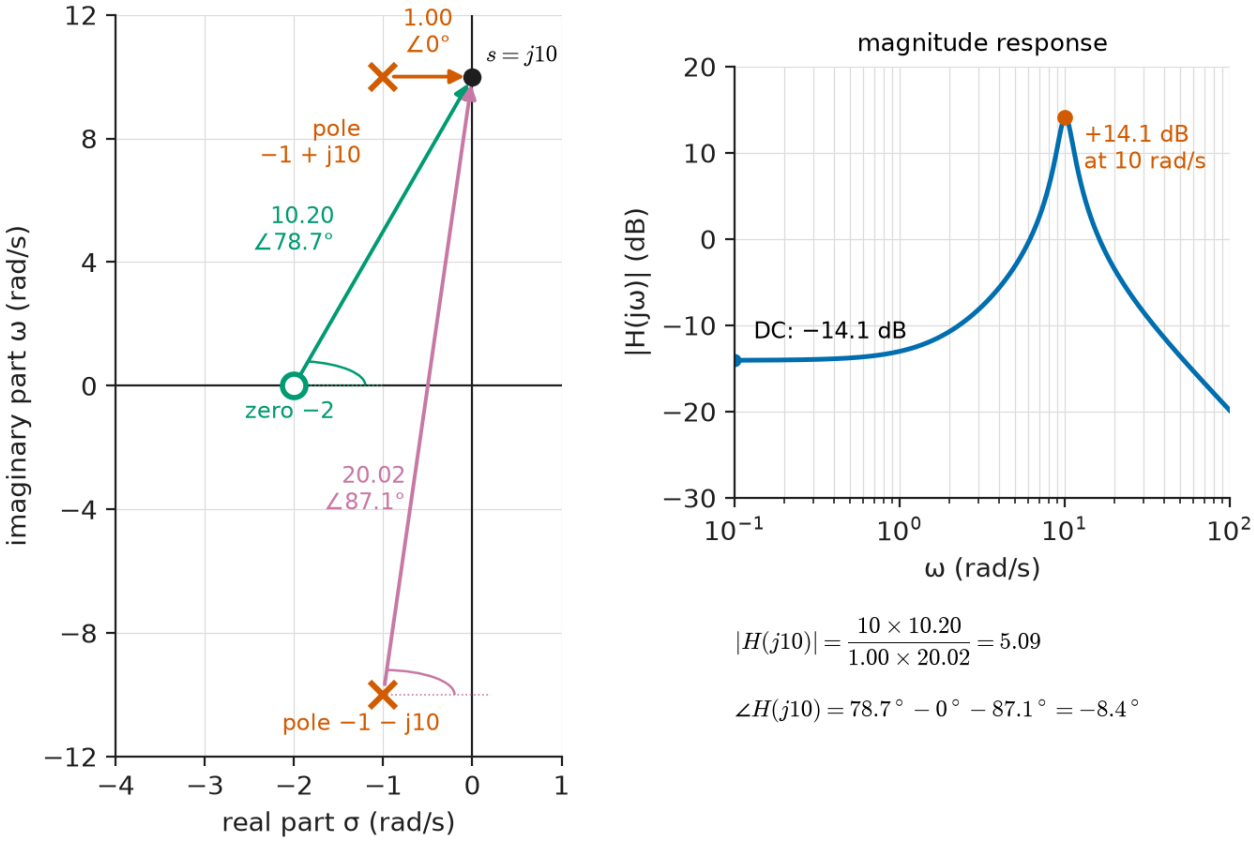


Figure 13.2 — Pole-zero map of H(s) = 10(s+2)/(s^2+2s+101) with the vectors used to evaluate H at s = j10.

Pole and zero locations and what they mean in a circuit

The table collects the common locations and their meaning.

Location	Time-domain signature	Frequency-domain signature	Typical circuit
Real pole at $-\omega_p$	Decaying exponential, time constant $1/\omega_p$	-3 dB corner at ω_p, then -20 dB/decade	RC or RL low-pass
Pole at origin	Step becomes ramp (integration)	-20 dB/decade everywhere, $-90°$	Ideal integrator, capacitor charged by a current
Real zero at $-\omega_z$	Derivative-like term	$+20$ dB/decade above ω_z, $+90°$	Lead network, RC high-pass corner
Zero at origin	Blocks DC	$+20$ dB/decade from DC	Coupling capacitor, AC-coupled input
Complex pole pair near axis	Ringing at ω_d, slow decay	Resonant peak near ω_n	Tuned circuit, under-damped LC filter
Zero pair on $j\omega$ axis	Input at ω_z produces no output	Notch at ω_z	Twin-T or bridged-T notch, LC trap
Right-half-plane pole	Growing response	Frequency response not observable	Oscillator, unstable loop
Right-half-plane zero	Initial response in the wrong direction	Phase lag with rising magnitude	Boost converter, some feedback paths

13.3 Bode plots

Why straight lines work

Hendrik Bode's insight was that on log–log axes the factors of $H(j\omega)$ become additive, and each factor is very close to a pair of straight lines. Write the transfer function in **time-constant form**, with every real pole and zero normalised to unity at DC:

$$H(j\omega) = K_0\,(j\omega)^{\ell}\,\frac{\prod_k (1 + j\omega/\omega_{zk})}{\prod_k (1 + j\omega/\omega_{pk})}\cdot\frac{1}{\prod_i \left[1 - (\omega/\omega_{ni})^2 + j2\zeta_i\omega/\omega_{ni}\right]}$$

where ℓ counts zeros ($\ell > 0$) or poles ($\ell < 0$) at the origin and the last factor collects complex pole pairs with natural frequency ω_n and damping ratio ζ (Chapter 7). Then

$$|H|_{\text{dB}} = 20\log K_0 + 20\ell\log\omega + \sum_k 20\log|1 + j\omega/\omega_{zk}| - \sum_k 20\log|1 + j\omega/\omega_{pk}| - \cdots$$

and the phase is likewise a sum. Each term can be sketched separately and added.

The building blocks

Factor	Magnitude asymptotes	Exact correction at corner	Phase
Constant K_0	Flat at $20\log K_0$	—	0° (or 180° if $K_0<0$)
$(j\omega)^{\pm 1}$	±20 dB/decade through 0 dB at $\omega=1$	—	±90°
$(1+j\omega/\omega_p)^{-1}$	0 dB, then −20 dB/decade from ω_p	−3.0 dB	0 to −90°, −45° at ω_p
$(1+j\omega/\omega_z)$	0 dB, then +20 dB/decade from ω_z	+3.0 dB	0 to +90°, +45° at ω_z
Complex pole pair	0 dB, then −40 dB/decade from ω_n	$-20\log(2\zeta)$ dB	0 to −180°, −90° at ω_n; steepness grows as ζ falls

For the phase of a real pole, the standard straight-line approximation is 0° up to $\omega_p/10$, a slope of −45° per decade to $10\omega_p$, then −90°. Its worst error is 5.7°, at $0.1\omega_p$ and $10\omega_p$. The magnitude asymptote errs by at most 3 dB, at the corner, and by 1 dB an octave either side. For complex poles the asymptote can be badly wrong near ω_n: at $\zeta=0.1$ the true response at ω_n is +14 dB above the asymptote. Lightly damped pairs must always be corrected by hand (Section 13.5).

> **Status of the principle: Bode asymptotes**
> *Approximate.* The straight lines are the exact limits of each factor for $\omega \ll$ and $\omega \gg$ the corner; the error is bounded (3 dB and 5.7° for real factors) when corners are separated by a decade or more, and grows when corners crowd together or complex pairs are lightly damped.

Procedure

1. Put H into time-constant form and list the corner frequencies in ascending order.
2. Find the low-frequency asymptote: the constant and any $(j\omega)^{\ell}$ factor.
3. Move up in frequency; at each corner change the slope by +20 dB/decade for a zero, −20 for a pole, ±40 for a complex pair.
4. Add the corner corrections where accuracy matters.
5. Sketch the phase by summing the straight-line phase of each factor.

Worked example — Bode plot of an ECG front end. A diagnostic ECG amplifier needs a gain of 1000, a high-pass corner to remove electrode offset drift and a low-pass corner to limit noise. Suppose a poorly designed version uses corners at $f_1=0.5$ Hz and $f_2=150$ Hz:

$$H(jf)=1000\cdot\frac{jf/f_1}{1+jf/f_1}\cdot\frac{1}{1+jf/f_2}$$

The asymptotes are: a +20 dB/decade line rising to 60 dB at 0.5 Hz, a flat 60 dB plateau to 150 Hz, and −20 dB/decade above. Exact values computed from the expression:

Frequency	Magnitude (exact)	Phase (exact)	Comment
0.05 Hz	40.0 dB	+84.3°	On the rising asymptote
0.5 Hz	57.0 dB	+44.8°	High-pass corner: −3 dB
5 Hz	60.0 dB	+3.8°	Phase error still almost 4°
8.66 Hz	60.0 dB	0°	Geometric centre $\sqrt{f_1 f_2}$
50 Hz	59.5 dB	−17.9°	Mains frequency still passed at full gain
150 Hz	57.0 dB	−44.8°	Low-pass corner
1 kHz	43.4 dB	−81.4°	Falling at −20 dB/decade

The magnitude looks excellent: within 0.5 dB of 60 dB from about 2 Hz to 50 Hz. The phase tells the real story. The ST segment and T wave contain energy between roughly 0.05 Hz and 1 Hz, where this network applies roughly 26° (at 1 Hz) to 84° (at 0.05 Hz) of phase lead; the slow segments are differentiated rather than reproduced, which is the distortion described at the opening. Diagnostic ECG standards therefore place the high-pass corner at 0.05 Hz (a time constant of 3.2 s), or require linear-phase filtering, precisely because of the phase response. The 50 Hz mains interference, meanwhile, passes at almost full gain; it must be removed separately with a notch (Section 13.7).

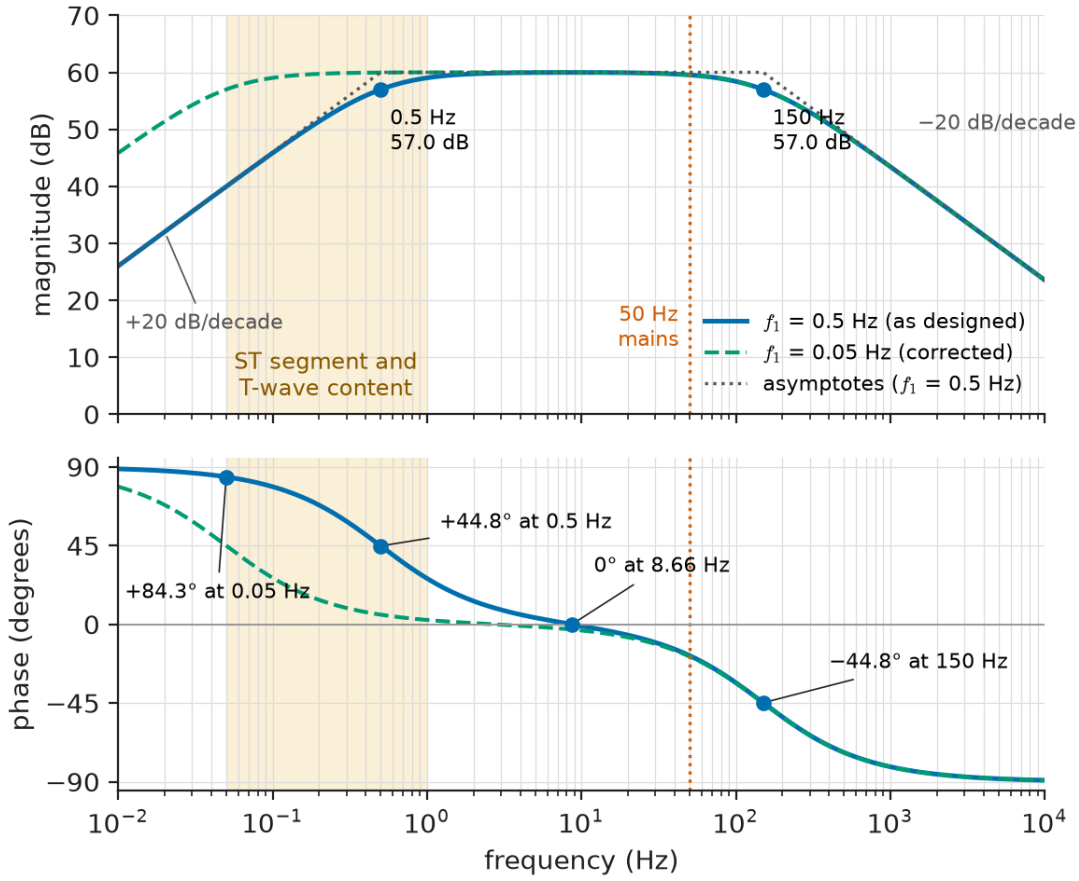


Figure 13.3 — Bode plot of an ECG front end with a 0.5 Hz high-pass and 150 Hz low-pass corner and mid-band gain 1000.

Figure 13.3 shows the plot with the diagnostically important band shaded. The general lesson is that every corner affects the phase for a decade on either side of it: to keep phase distortion below about 6° over a band, both corners must lie at least a decade outside it.

13.4 Resonant circuits

The series RLC circuit

Resonance appeared in Chapter 7 as the natural oscillation of an RLC circuit and in Chapter 8 as the frequency at which reactances cancel. Here it is treated as a frequency-selective network. A series RLC circuit driven by a voltage source has impedance

$$Z(j\omega) = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

which is purely resistive and minimum at the **resonant frequency**

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

The current, and therefore the voltage across R, peaks there. Taking the resistor voltage as the output,

$$H_R(s) = \frac{V_R}{V_s} = \frac{sR/L}{s^2 + sR/L + 1/(LC)}$$

a second-order **band-pass** response whose poles are

$$p_{1,2} = -\frac{R}{2L} \pm j\sqrt{\omega_0^2 - \left(\frac{R}{2L}\right)^2}$$

For small R the poles sit just to the left of $\pm j\omega_0$, and the geometric picture of Section 13.2 predicts a sharp peak at ω_0. The output across C gives a low-pass response, across L a high-pass response, and across the series LC pair a notch: the same circuit supplies all four basic filter types, depending on where the output is taken.

The parallel RLC circuit

The dual circuit, R, L and C in parallel fed by a current source, has admittance

$$Y(j\omega) = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$

which is minimum at the same $\omega_0 = 1/\sqrt{LC}$. The impedance therefore *peaks* at resonance, reaching R, and the voltage across the tank is maximum. At resonance the inductor and capacitor currents are each Q times the source current but cancel each other, circulating inside the tank. This is the **tank circuit** of oscillators, the IF filter of radio receivers and the tuned load of RF amplifiers.

Practical tanks: the lossy inductor

Real inductors are not ideal. Their winding resistance, skin effect and core loss are often summarised by a series resistance r in the inductor branch, so a practical parallel tank is L in series with r, in parallel with C. Near resonance this branch can be replaced by an equivalent parallel combination: with the coil quality factor $Q_L = \omega L / r$,

$$R_p = r(1 + Q_L^2) \approx Q_L^2 r, \qquad L_p = L\left(1 + \frac{1}{Q_L^2}\right) \approx L$$

The series-to-parallel conversion is exact at one frequency and very accurate over the narrow band of a high-Q tank.

> **Status of the principle: Series-to-parallel equivalence of a lossy reactance**
> *Approximate.* Exact at the frequency of conversion; over a band it holds while $Q_L \gg 1$ and the fractional bandwidth is small, because both R_p and L_p depend on frequency through Q_L.

Worked example — a 10.7 MHz IF tank. An FM receiver's intermediate-frequency stage uses a tank tuned to 10.7 MHz with $L = 1.0\ \mu$H of series resistance $r = 0.5\ \Omega$. The capacitor is

$$C = \frac{1}{\omega_0^2 L} = \frac{1}{(2\pi \cdot 10.7 \times 10^6)^2 (1.0 \times 10^{-6})} = 221\ \text{pF}$$

The inductive reactance is $\omega_0 L = 67.2\ \Omega$, so $Q_L = 67.2/0.5 = 134$ and the equivalent parallel resistance is $R_p = 0.5(1 + 134.5^2) = 9.04\ \text{k}\Omega$. Unloaded, the tank would have a bandwidth $f_0/Q_L = 79.6$ kHz. The next stage presents a load of $10\ \text{k}\Omega$ in parallel with the tank, so the total resistance is $9.04 \parallel 10 = 4.75\ \text{k}\Omega$ and the loaded quality factor is $Q = 4750/67.2 = 70.6$. The bandwidth widens to $10.7\ \text{MHz}/70.6 = 152$ kHz, which happens to suit the 150–200 kHz occupied by a broadcast FM channel. The example shows a central design fact: the bandwidth of a tuned circuit is set as much by what it is connected to as by its own components.

Resonance across EEE

Application	Typical resonant frequency	Typical loaded Q	What resonance does
AM broadcast tuner	0.53–1.7 MHz	50–150	Selects one station, magnifies antenna voltage
FM IF transformer	10.7 MHz	30–100	Sets the IF bandwidth
Quartz crystal	32.768 kHz to tens of MHz	10^4–10^6	Frequency reference for clocks and radios
Wireless-power (Qi) coil	87–205 kHz	20–100	Increases coupling efficiency between coils
Power-system harmonic filter	250 Hz (5th harmonic on 50 Hz)	20–100	Low-impedance shunt path for a harmonic current
Accelerometer or MEMS gyroscope	1–50 kHz	10 (damped) to 10^4 (vacuum)	Sets sensing bandwidth or sensitivity

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13.5 Quality factor

The energy definition

"Quality factor" has several working definitions; the one that applies to every resonator, electrical or not, is based on energy:

$$Q = 2\pi \left. \frac{\text{peak energy stored}}{\text{energy dissipated per cycle}} \right|_{\omega=\omega_0} = \omega_0 \frac{W_{\text{stored}}}{P_{\text{loss}}}$$

A resonator with $Q = 100$ loses $2\pi/100 \approx 6\%$ of its stored energy per cycle. In a series RLC circuit the stored energy at resonance is $\tfrac{1}{2}LI_m^2$ (it moves between L and C but its total is constant) and the average loss is $\tfrac{1}{2}I_m^2 R$, so

$$Q_{\text{series}} = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

For the parallel circuit the roles of R invert:

$$Q_{\text{parallel}} = \frac{R}{\omega_0 L} = \omega_0 C R = R\sqrt{\frac{C}{L}}$$

A small series resistance and a large parallel resistance both mean small losses and high Q. The quantity $\sqrt{L/C}$ is the **characteristic impedance** of the resonator, the impedance level at which energy sloshes between the two reactances; it reappears as the characteristic impedance of a transmission line in Chapter 14, for the same physical reason.

> **Status of the principle: Energy definition of quality factor**
> *Fundamental.* Follows from energy conservation in a resonator with small losses; it applies equally to LC tanks, cavities, quartz crystals and mechanical resonators, which is why their Q values can be compared directly.

Q, damping ratio and pole position

The second-order denominator can be written in three equivalent ways:

$$s^2 + \frac{\omega_0}{Q}s + \omega_0^2 = s^2 + 2\zeta\omega_n s + \omega_n^2, \qquad Q = \frac{1}{2\zeta}, \quad \omega_n = \omega_0$$

The poles are at $-\omega_0/(2Q) \pm j\omega_0\sqrt{1 - 1/(4Q^2)}$. They lie on a circle of radius ω_0 in the s-plane, and Q sets the angle: the real part is $\omega_0/(2Q)$, so high Q puts the poles close to the $j\omega$ axis. In the time domain, the envelope of the natural response decays as $e^{-\omega_0 t/(2Q)}$, so the oscillation rings for about Q/π cycles before its amplitude falls to e^{-1}. A quartz crystal with $Q = 10^5$ at 10 MHz rings for 3 ms after excitation stops; an LC tank with $Q = 50$ dies in about 16 cycles.

Voltage and current magnification

At resonance in the series circuit the current is V_s/R, and the capacitor voltage is

$$|V_C| = \frac{I}{\omega_0 C} = \frac{V_s}{\omega_0 C R} = Q\,V_s$$

The same holds for the inductor voltage. In the parallel circuit the reactive branch currents are Q times the source current. Magnification is useful at the input of a radio, where a 10 µV antenna signal appears as 1 mV across the tuning capacitor of a $Q = 100$ circuit. It is a hazard in power systems: a capacitor bank that resonates with the supply inductance near a harmonic frequency can amplify a modest harmonic current into a large one, overheating the capacitors and blowing fuses. The same arithmetic explains both.

Representative quality factors

Resonator or component	Typical Q	Main loss mechanism
Ferrite-core power inductor at 100 kHz	10–50	Core loss and winding resistance
Air-core RF inductor, 1–100 MHz	50–250	Skin and proximity effect in the wire
Multilayer ceramic capacitor (C0G), 100 MHz	500–2000	Electrode resistance (ESR)
Microstrip resonator on FR-4, 2 GHz	50–150	Dielectric loss ($\tan\delta \approx 0.02$) and conductor loss
Metal waveguide cavity, 10 GHz	10^3–10^4	Wall currents
Quartz crystal	10^4–10^6	Mounting and internal friction
Tuning-fork watch crystal (32.768 kHz)	50 000–100 000	Acoustic loss, air damping

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13.6 Bandwidth

Half-power bandwidth of a resonator

The **bandwidth** of a band-pass response is the width of the band over which the output power is at least half its peak value, so that $|H| \geq |H|_{\max}/\sqrt{2}$. For the series RLC band-pass, normalised so that the peak gain is 1,

$$|H(j\omega)| = \frac{1}{\sqrt{1 + Q^2\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)^2}}$$

The half-power points occur where $Q(\omega/\omega_0 - \omega_0/\omega) = \pm 1$, which gives

$$\omega_{1,2} = \omega_0\left(\sqrt{1 + \frac{1}{4Q^2}} \mp \frac{1}{2Q}\right), \qquad \omega_2 - \omega_1 = \frac{\omega_0}{Q}, \qquad \omega_1\omega_2 = \omega_0^2$$

The bandwidth is exactly ω_0/Q for any Q, and the centre frequency is the *geometric* mean of the half-power frequencies. For $Q \gtrsim 10$ the response is also nearly symmetric on a linear frequency axis. The relation

$$B = \frac{f_0}{Q}$$

is the single most useful formula in tuned-circuit design: it ties selectivity (narrow B) to low loss (high Q).

Status of the principle: Bandwidth–Q relation
Derived. Exact for the second-order band-pass $s\,\omega_0/Q\,/(s^2 + s\omega_0/Q + \omega_0^2)$; for other responses (higher order, or low-pass rather than band-pass) the bandwidth must be computed from the actual transfer function.

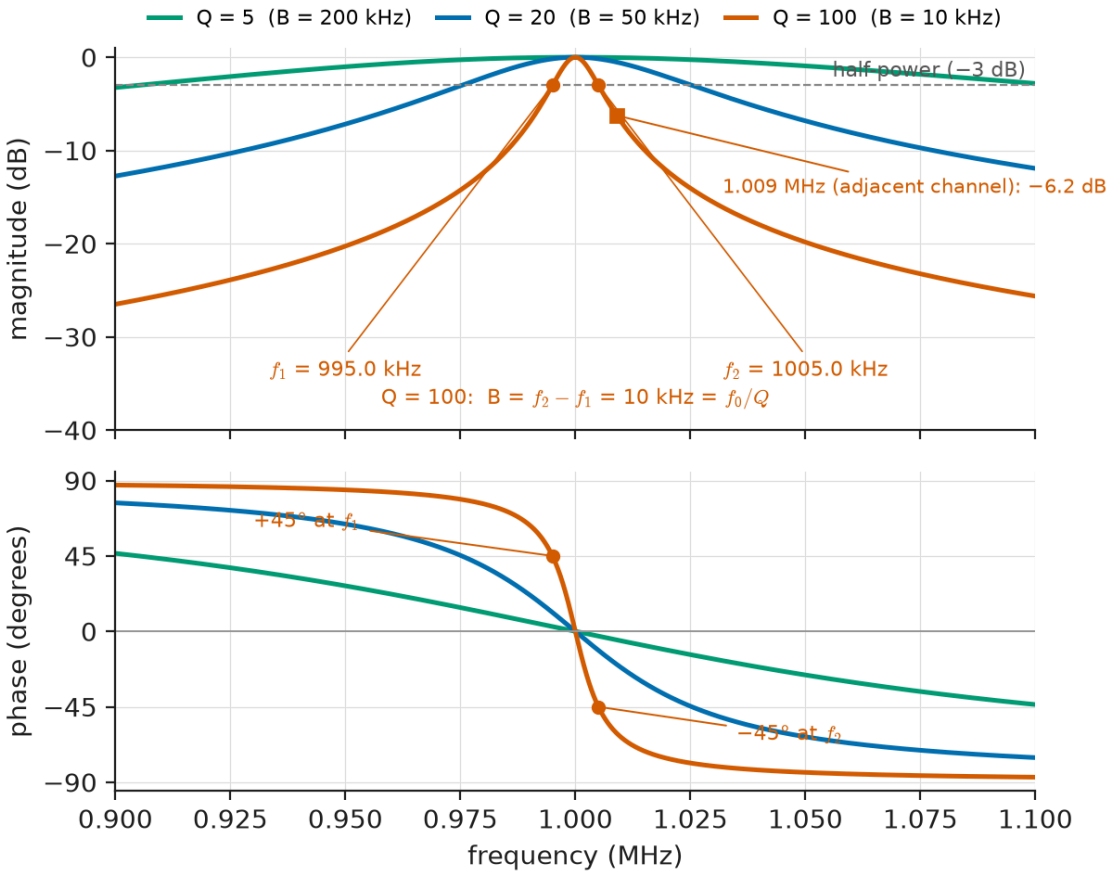

Figure 13.4 — Normalised series-RLC band-pass response for Q = 5, 20 and 100 around 1 MHz, showing that bandwidth equals fo/Q.

Worked example — selectivity of an AM tuner. A medium-wave receiver uses a series tuned circuit with $L = 200\ \mu\text{H}$ to receive a station at 1.000 MHz. The required capacitor is

$$C = \frac{1}{(2\pi \times 10^6)^2 (200 \times 10^{-6})} = 126.7\ \text{pF}$$

The inductive reactance at resonance is $\omega_0 L = 1257\ \Omega$. If the total series resistance (coil loss plus the antenna's radiation and loss resistance as reflected into the circuit) is $R = 12.6\ \Omega$, then $Q = 1257/12.6 = 100$ and $B = 1\ \text{MHz}/100 = 10\ \text{kHz}$. The exact half-power frequencies are $f_1 = 995.0$ kHz and $f_2 = 1005.0$ kHz. In regions using 9 kHz channel spacing, the adjacent station at 1.009 MHz sees

$$|H| = \frac{1}{\sqrt{1 + 100^2\left(1.009 - 1/1.009\right)^2}} = 0.487 \quad (-6.2\ \text{dB})$$

and a station two channels away, at 1.018 MHz, is attenuated by only 11.4 dB. A single tuned circuit cannot separate adjacent AM stations: suppressing a neighbour by 40–60 dB requires a much steeper

skirt than a second-order resonance provides. This is the historical reason for the superheterodyne receiver (Chapter 26), which moves every station to a fixed intermediate frequency where several coupled resonators or a ceramic filter can provide steep skirts once, for all stations. Figure 13.4 shows how the resonance narrows as Q grows while the skirts far from f_0 still fall only at −20 dB/decade.

The example also illustrates voltage magnification: a 10 µV antenna voltage appears as $QV_s = 1$ mV across the tuning capacitor.

Bandwidth of low-pass systems and rise time

For low-pass systems the bandwidth is the frequency where the gain has fallen by 3 dB from its DC value. For a first-order low-pass, $B = f_c = 1/(2\pi\tau)$, and the 10–90 % rise time of its step response is $t_r = \tau \ln 9 = 2.2\tau$. Eliminating τ,

$$t_r \approx \frac{0.35}{B}$$

This relation, exact for a single pole, holds within about 10–20 % for many well-damped higher-order low-pass systems and is used routinely to relate oscilloscope bandwidth to the fastest edge it can display: a 100 MHz oscilloscope has an intrinsic rise time near 3.5 ns.

> **Status of the principle: Rise-time–bandwidth product**
> *Approximate.* Exact (0.35) for a single real pole; between about 0.33 and 0.45 for typical well-damped responses; meaningless for strongly resonant systems whose step response rings.

Representative bandwidths

System	Bandwidth	Comment
Diagnostic ECG	0.05–150 Hz	High-pass corner set low for phase fidelity
EEG	0.5–70 Hz typical	Notch at 50 or 60 Hz common
Telephone voice channel	300–3400 Hz	Historical analog channel, 8 kHz sampling
High-fidelity audio	20 Hz–20 kHz	Limit of human hearing
AM broadcast channel	9 or 10 kHz	Depends on region
FM broadcast channel	≈200 kHz	Carson's-rule estimate for 75 kHz deviation
General-purpose oscilloscope	50 MHz–1 GHz	Rise time 7 ns to 0.35 ns

13.7 Passive filters

Specifications

A filter is a network designed to pass a range of frequencies (the **passband**) and reject another (the **stopband**). Its specification states the passband edge f_p and the maximum passband ripple or loss, the stopband edge f_s and the minimum stopband attenuation, and often limits on phase or group-delay variation. The region between f_p and f_s is the **transition band**. The four classic types are low-pass, high-pass, band-pass and band-stop (notch); all-pass networks, which have flat magnitude and shape only the phase, are used for delay equalisation.

The steepness of the transition is set by the filter order n. Far into the stopband, an n-th-order low-pass falls at $20n$ dB/decade (6n dB/octave). The required order can be estimated from the specification before any circuit is drawn.

Classic approximations

Real filters approximate the ideal "brick wall" with rational functions chosen to optimise one property:

Approximation	Passband	Transition	Phase/delay	Typical use
Butterworth	Maximally flat, no ripple	Moderate	Moderate nonlinearity	General purpose, audio, anti-aliasing
Chebyshev type I	Equiripple	Steeper than Butterworth for same order	Worse	RF band filters, where ripple is acceptable
Inverse Chebyshev	Flat	Steep, with stopband zeros	Moderate	Anti-aliasing with notches
Elliptic (Cauer)	Equiripple	Steepest for a given order	Worst	Channel filters, tight specifications
Bessel	Gently rolling	Slow	Nearly linear phase	Pulse and waveform fidelity, ECG, data acquisition

The Butterworth low-pass of order n has

$$|H(j\omega)|^2 = \frac{1}{1 + (\omega/\omega_c)^{2n}}$$

whose poles lie equally spaced on a semicircle of radius ω_c in the left half-plane. Every n gives -3 dB at ω_c, but higher n keeps the passband flatter and the transition steeper.

> **Status of the principle: Classic filter approximations**
> *Derived*. Each is the solution of a stated optimisation problem (maximal flatness, equiripple, linear phase) over rational functions; they say nothing about component tolerance, parasitics or loss, which set how closely a built filter follows the ideal.

The LC ladder

At radio frequencies, and wherever power must pass with little loss, filters are built as **ladders** of series inductors and shunt capacitors between equal source and load resistances. Normalised element values ("g-values") are tabulated for each approximation; for Butterworth filters

$$g_k = 2\sin\frac{(2k-1)\pi}{2n}, \qquad k = 1, \dots, n$$

giving $1, 2, 1$ for $n = 3$ and $0.765, 1.848, 1.848, 0.765$ for $n = 4$. Scaling to a cut-off ω_c and impedance level R_0 gives shunt capacitors $C_k = g_k/(\omega_c R_0)$ and series inductors $L_k = g_k R_0/\omega_c$.

Doubly terminated LC ladders have a remarkable property: at frequencies of maximum power transfer the sensitivity of the response to every element value is zero to first order. A 5 % error in an inductor changes the passband response far less than 5 %. This low sensitivity is why LC ladders remain the prototype for active and digital filter designs (Chapter 23).

Worked example — a 10 MHz Butterworth low-pass for a 50 Ω system. A third-order Butterworth low-pass is to remove harmonics from a signal generator output, with cut-off $f_c = 10$ MHz between 50 Ω terminations. With $\omega_c = 2\pi \times 10^7 = 6.283 \times 10^7$ rad/s,

$$C_1 = C_3 = \frac{1}{\omega_c R_0} = \frac{1}{(6.283\times10^7)(50)} = 318\ \text{pF}, \qquad L_2 = \frac{2R_0}{\omega_c} = \frac{100}{6.283\times10^7} = 1.59\ \mu\text{H}$$

Including the 6 dB loss of the 50 Ω divider (so that the reference is the voltage delivered to a matched load), the response is $-10\log_{10}(1 + (f/f_c)^6)$ dB. A circuit simulation of the ladder confirms the formula: −0.07 dB at 5 MHz, −3.01 dB at 10 MHz (phase −135°), −18.1 dB at 20 MHz, and −28.6 dB at 30 MHz, the third harmonic of a 10 MHz signal. A first-order RC with the same cut-off would give only −10.0 dB at 30 MHz. In practice standard values of 330 pF and 1.5 µH would be used, moving the corner by a few percent.

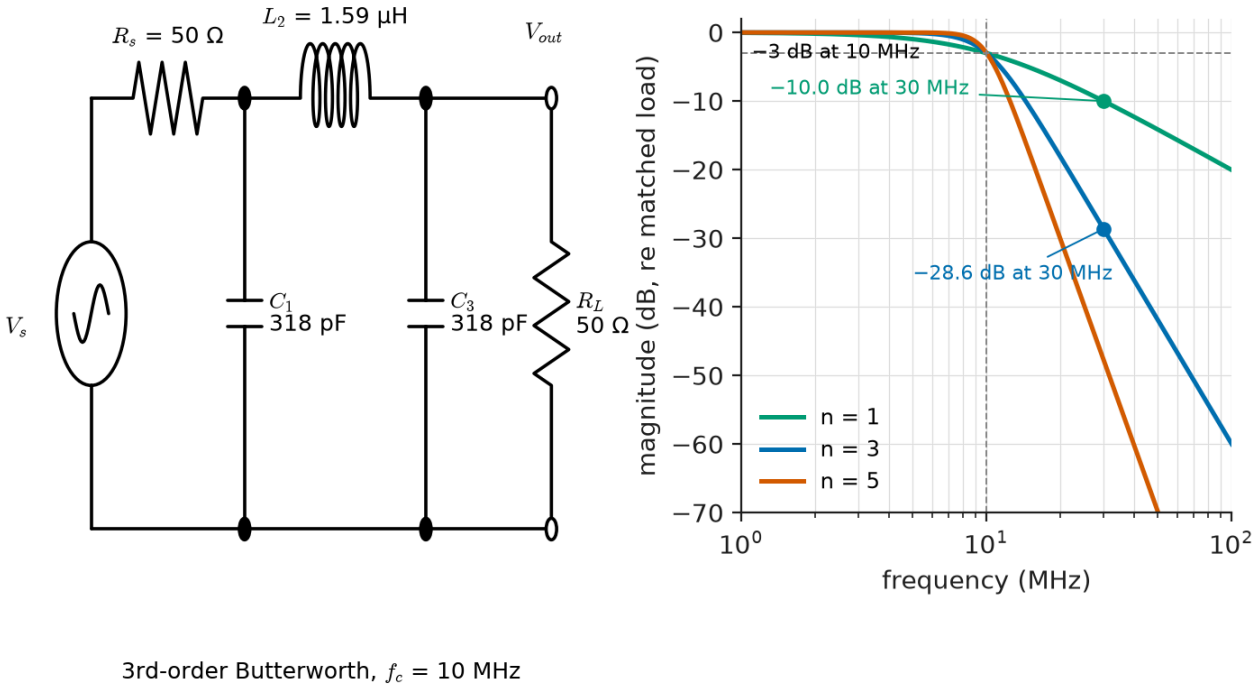

Figure 13.5 — Third-order Butterworth LC low-pass ladder for 10 MHz in a 50-ohm system, with its response compared with first-order and higher-order Butterworth filters.

Band-pass, band-stop and notch filters

A low-pass prototype becomes a band-pass filter by replacing each series inductor with a series LC resonator and each shunt capacitor with a parallel LC resonator, all tuned to the centre frequency ω_0; the bandwidth of the result equals the prototype cut-off. The same transformation family gives high-pass and band-stop filters. These frequency transformations are also used, in discrete-time form, for digital filters (Chapter 23).

A **notch** suppresses one frequency. The canonical passive notch is a series LC trap across the signal path: at its resonance it short-circuits the line. In biomedical and instrumentation amplifiers the twin-T RC network places a zero pair on the $j\omega$ axis at $\omega_z = 1/(RC)$ without inductors, removing 50 or 60 Hz mains pickup. Its passive Q is only 0.25, so the notch is broad and attenuates neighbouring frequencies; feedback around an amplifier sharpens it. In power systems, the "single-tuned harmonic filter" is the same idea at high power: a series RLC branch tuned slightly below the 5th, 7th or 11th harmonic provides a low-impedance path that diverts harmonic current away from the supply.

Limitations of passive filters

Passive filters need no power supply, can handle large signals and add no active noise. Their limits are practical. At audio and lower frequencies the inductors become large, heavy, lossy and susceptible to magnetic pickup; a 1 H inductor for a 100 Hz filter is not a component anyone wants. Passive networks cannot give gain, and their response depends on the source and load impedances. Above a few hundred megahertz, the "lumped" components themselves become transmission-line sections, and filters are built from lines (Chapter 14) or cavities (Chapter 32). Between these limits, below about 1 MHz, the inductor is replaced by an active circuit, which is the subject of the next section.

13.8 Active-filter interface

Why active filters

An active filter replaces inductors with combinations of resistors, capacitors and an amplifier (usually an op-amp) whose feedback synthesises the required complex poles. It offers gain, isolation from the load and practical component values at low frequencies. The design of op-amp circuits themselves belongs to *Electronics & Computing*; what this book owns is the network interface: the transfer function the active stage realises, and what happens when it is connected to real sources and loads.

> **Boundary.**
> The internal design of op-amps and of particular active-filter topologies (Sallen–Key, multiple-feedback, state-variable, biquad) is treated in *Electronics & Computing*. Here an op-amp is modelled at the network level: a voltage-controlled voltage source with very high gain, high input impedance and low output impedance, valid well below its gain-bandwidth limit.

The second-order section as a building block

Almost all active filters are cascades of second-order sections, each realising

$$H(s) = K\frac{\omega_0^2}{s^2 + (\omega_0/Q)s + \omega_0^2}$$

(for a low-pass) with one conjugate pole pair. An n-th-order filter is factored into its pole pairs (plus one real pole if n is odd), and each pair is assigned to one section. The fourth-order Butterworth, for example, needs two sections with $Q = 0.541$ and $Q = 1.307$, both at $\omega_0 = \omega_c$. Because each op-amp output is a near-ideal voltage source, sections can be cascaded without loading one another, and the overall response is simply the product of the section responses. This is the main network advantage of active filters over passive ladders, where every element interacts with every other.

A widely used unity-gain section (the Sallen–Key low-pass) uses two resistors R_1, R_2, a capacitor C_1 from the middle node back to the output and a capacitor C_2 from the op-amp input to ground. Its parameters are

$$\omega_0 = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}, \qquad Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{C_2(R_1 + R_2)}$$

Source impedance

The equations above assume the section is driven from an ideal voltage source. In practice the source has an output resistance R_s, and since it is in series with the input resistor, the section actually sees $R_1 + R_s$.

Worked example — a Sallen–Key filter driven from a sensor. A unity-gain Sallen–Key low-pass with Butterworth response at 1 kHz is designed with $R_1 = R_2 = 10\ \text{k}\Omega$. With equal resistors, $Q = \tfrac{1}{2}\sqrt{C_1/C_2}$, so $Q = 0.707$ needs $C_1 = 2C_2$, and $\omega_0 = 1/(R\sqrt{C_1 C_2}) = 1/(R\sqrt{2}\,C_2)$ gives

$$C_2 = \frac{1}{2\pi (1000)(10^4)\sqrt{2}} = 11.25\ \text{nF}, \qquad C_1 = 22.5\ \text{nF}$$

Driven from an ideal source it meets its specification exactly: $f_0 = 1000$ Hz, $Q = 0.707$. Suppose instead it is driven directly from a sensor with $1\ \text{k}\Omega$ output resistance. The effective R_1 becomes $11\ \text{k}\Omega$; recomputing, $f_0 = 953$ Hz and $Q = 0.706$. The corner has moved down 4.7 %, and the response at 1 kHz is now −3.45 dB instead of −3.01 dB. If the source resistance is $10\ \text{k}\Omega$, comparable to the filter resistors, f_0 falls to 707 Hz and Q to 0.667: the filter is no longer the one designed. The rule is that the source impedance must be either much smaller than the filter's input impedance level, or absorbed into the design (here by reducing R_1 by R_s), or isolated by a buffer.

Load impedance and the passive–active boundary

At the output, the op-amp's low output impedance makes the response almost independent of load, which is why active filters are usually placed last before a heavy load or a cable. A passive network driving an active stage sees the stage's input impedance, which for most op-amp inputs is high and can be ignored; but a passive filter that drives an ADC input directly sees the ADC's switched sampling capacitor, which draws current pulses at each sample. This is a frequent source of errors in data-acquisition designs and is why a small RC "charge reservoir" is placed at the converter input (Chapter 19).

The general principle is the two-port view of Chapter 12: a filter's transfer function is defined *between specified terminations*, and any change in those terminations changes the transfer function. Passive filters are designed for their terminations; active stages are used to make terminations irrelevant.

Where active filters stop working

The op-amp model holds only while the op-amp has plenty of loop gain at the frequencies involved. As f_0 approaches a few percent of the op-amp's gain-bandwidth product, the realised Q rises above its design value, and high-Q sections can even oscillate; the effect is a stability phenomenon of the type described in the next section. Typical practice keeps the gain-bandwidth product at least $100\,Q\,f_0$. Above about 1 MHz, passive LC, ceramic, SAW or BAW and transmission-line filters take over again; switched-capacitor filters, whose corner is set by a clock, cover roughly 1 Hz to 100 kHz.

13.9 Stability-related frequency behaviour

Poles, peaking and ringing

A passive RLC network is always stable: its poles lie in the left half-plane because its resistors dissipate energy. Networks with active elements (amplifiers, dependent sources, negative resistances) can place poles on or to the right of the $j\omega$ axis, and then the network oscillates or its output grows until something saturates. Long before that point, the frequency response shows warning signs.

The clearest is **resonant peaking**. For a second-order low-pass with damping ratio ζ below $1/\sqrt{2}$, the magnitude peaks above its DC value by

$$M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

at $\omega_r = \omega_n\sqrt{1-2\zeta^2}$. The same poles make the step response overshoot by $e^{-\pi\zeta/\sqrt{1-\zeta^2}}$. A frequency plot with a peak is therefore a time response that rings, and the peak height is a quantitative measure of how close the poles are to the axis.

Damping ratio ζ	$Q = 1/(2\zeta)$	**Peak** M_r	**Peak (dB)**	**Step overshoot**	**Equivalent phase margin**
0.1	5.0	5.03	14.0	73 %	11°
0.2	2.5	2.55	8.1	53 %	23°
0.3	1.67	1.75	4.8	37 %	33°
0.5	1.0	1.15	1.2	16 %	52°
0.707	0.707	1.00	0.0	4.3 %	66°
1.0	0.5	no peak	—	0 %	76°

The last column anticipates the feedback view developed next: for a standard second-order closed loop, the phase margin of the corresponding open loop is tied one-to-one to ζ, and a rule of thumb follows that $\zeta \approx \mathrm{PM}/100^\circ$ for phase margins below about 60°.

Loop gain and the frequency-domain view of feedback

When a network contains a feedback loop (an amplifier with a feedback network, a voltage regulator, a phase-locked loop) its closed-loop transfer function has the form

$$T(s) = \frac{G(s)}{1 + L(s)}$$

where $L(s)$ is the **loop gain**, the product of all gains around the loop. The closed-loop poles are the roots of $1 + L(s) = 0$. On the imaginary axis this equation says that the network is on the edge of oscillation at any frequency where $L(j\omega) = -1$: a gain of exactly 1 and a phase of exactly −180°. A signal travelling around the loop then returns with the same amplitude and inverted, and negative feedback has turned into positive feedback.

Two frequencies on the Bode plot of L therefore matter:

- the **gain crossover** ω_{gc}, where $|L| = 1$ (0 dB), at which the **phase margin** is $\mathrm{PM} = 180^\circ + \angle L(j\omega_{gc})$;
- the **phase crossover** ω_{pc}, where $\angle L = -180^\circ$, at which the **gain margin** is $\mathrm{GM} = -20\log|L(j\omega_{pc})|$ dB.

For loop gains that are stable on their own and whose magnitude crosses 0 dB once, positive margins mean a stable closed loop. The margins quantify how much extra gain or phase lag the loop can tolerate, from component tolerances, temperature, ageing or an unexpected load, before it oscillates.

> **Status of the principle: Gain and phase margins as stability indicators**
> *Derived*. Follows from the Nyquist criterion (Chapter 34) for open-loop-stable systems with a single gain crossover; for loops with several crossings, unstable open-loop poles or conditionally stable behaviour, the full Nyquist plot is needed.

Worked example — a three-pole loop. A feedback loop contains three identical first-order stages, each a buffered RC section with corner ω_1, and a gain A_0:

$$L(j\omega) = \frac{A_0}{(1 + j\omega/\omega_1)^3}$$

Each pole contributes $-60°$ when $\omega/\omega_1 = \tan 60^\circ = \sqrt{3}$, so the phase crossover is at $\omega_{pc} = 1.732\,\omega_1$. There the magnitude of each factor is $\sqrt{1+3} = 2$, so $|L| = A_0/8$. The loop oscillates if $A_0 \geq 8$ (18.1 dB). Take $A_0 = 4$. The gain crossover satisfies $(1+x^2)^{3/2} = 4$, so $x = \sqrt{4^{2/3} - 1} = 1.233$. The phase there is $-3\arctan(1.233) = -152.9^\circ$, giving

$$\text{PM} = 180^\circ - 152.9^\circ = 27.1^\circ, \qquad \text{GM} = 20\log_{10}\frac{8}{4} = 6.0\ \text{dB}$$

The loop is stable but poorly damped. Computing the closed-loop response $T = L/(1+L)$ confirms this: the DC gain is $4/5 = 0.8$, and the response peaks 9.6 dB above it at $1.35\,\omega_1$. The closed-loop poles are at $-2.59\,\omega_1$ and $(-0.21 \pm j1.37)\,\omega_1$, a complex pair very close to the axis. With a $27°$ phase margin, the table above predicts $\zeta \approx 0.25$ and overshoot near 45 %, which is what such a loop delivers. Raising A_0 to 7.9 moves the pair to $(-0.004 \pm j1.72)\,\omega_1$, essentially on the axis at the predicted phase-crossover frequency $\sqrt{3}\,\omega_1$.

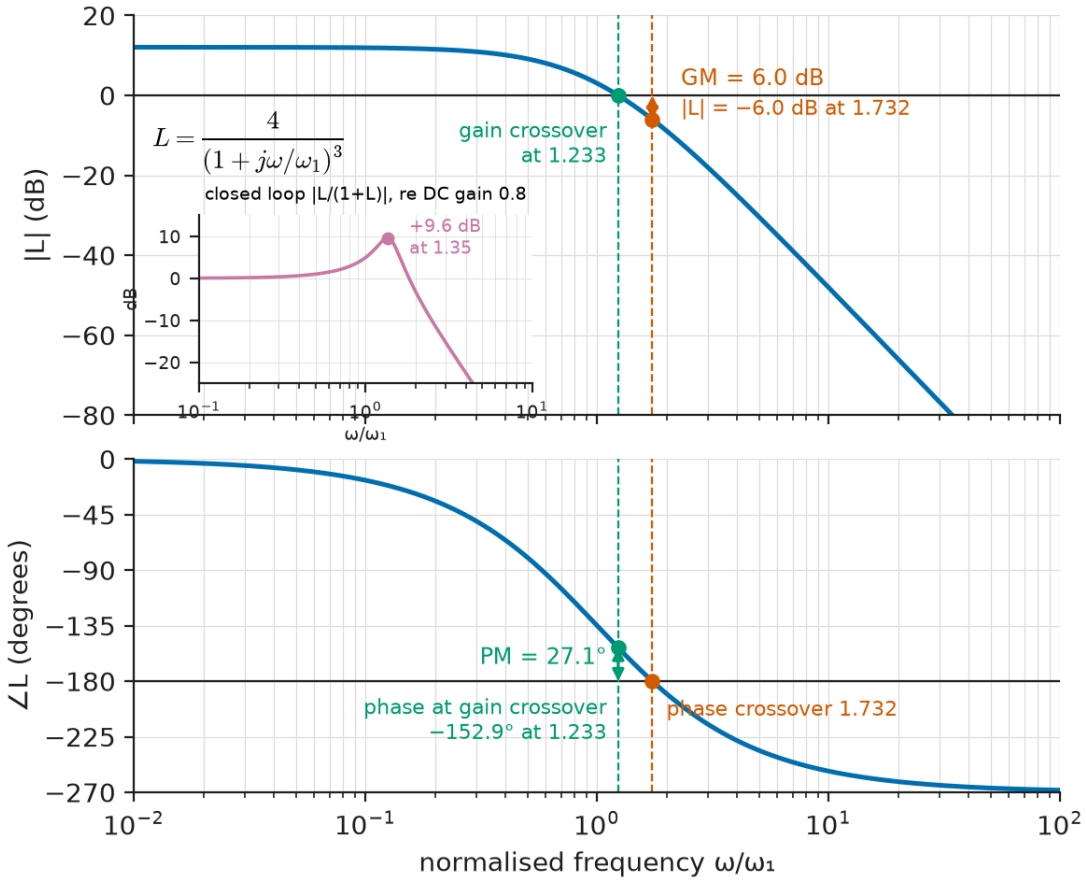

Figure 13.6 — Loop-gain Bode plot of L = 4/(1 + j omega/omega1)^3 showing gain crossover, phase crossover, phase margin and gain margin.

Figure 13.6 shows the margins on the loop-gain plot. The same arithmetic, with a loading correction, explains the classic RC **phase-shift oscillator**: three RC sections in a passive ladder (not buffered) produce 180° at a frequency where their attenuation is 1/29, so an inverting amplifier with gain at least 29 oscillates there. An oscillator is a feedback loop designed to fail the stability test at exactly one frequency.

Signs of marginal stability in measured responses

In practice, a circuit approaching instability shows itself in measurements before it oscillates: a closed-loop gain peak of several decibels, a sharp peak in a regulator's output impedance, or a peak that grows when a capacitive load is added or the temperature changes. A familiar example is an op-amp driving a long cable. The cable's capacitance, together with the op-amp's output resistance, adds a pole inside the feedback loop; the phase margin shrinks, the closed-loop response peaks at high frequency and square waves ring. The cure (a small series resistor at the output, or a compensation capacitor) is a frequency-response fix. Similar effects appear in switch-mode power supplies whose output capacitor ESR moves a zero, in power systems where long lines and generator controls interact to produce lightly damped electromechanical oscillation modes near 0.1–2 Hz, and in drones whose attitude loop rings as the gain is raised.

> **Boundary.**
> Systematic stability analysis (Routh–Hurwitz, root locus, the full Nyquist criterion) and the design of compensators to achieve specified margins are developed in Part VII (Chapters 34 and 35). This section introduces only the frequency-domain signatures that a network analyst encounters.

What remains open

Open question	Where it is answered
How do magnitude and phase extend to non-periodic signals, and what is group delay in general?	Chapter 18 (Frequency-Domain System Analysis)
What happens to frequency response when wires are a sizeable fraction of a wavelength?	Chapter 14 (Transmission Lines)
How is an anti-aliasing filter specified from converter resolution and sample rate?	Chapter 19 (Sampling and Reconstruction)
How are Butterworth, Chebyshev and elliptic designs carried into discrete time?	Chapter 23 (Digital Filters)
How do resonators and filters behave at microwave frequencies and in S-parameter terms?	Chapter 32 (RF and Microwave Fundamentals)
How does a superheterodyne receiver use fixed IF filters to obtain selectivity?	Chapter 26 (Analog Communication)
How is stability decided rigorously from the Nyquist plot, and how are margins designed?	Chapters 34 and 35 (Feedback and Stability; Control Design Fundamentals)
How are op-amp active-filter circuits designed and built?	*Electronics & Computing*
How are the transfer functions recapped here derived from Laplace transforms?	Chapter 11 and *The Mathematics Universe*

Next: Chapter 14 follows signals onto cables, PCB traces and power lines whose length is comparable to a wavelength, where the lumped circuits of this chapter give way to distributed transmission lines.

Chapter 14 — Transmission Lines

Part III · Advanced Network Theory

A new digital board works on the bench with short test leads and fails in the enclosure. The clock output of a processor drives a memory chip 15 cm away along a printed trace. On an oscilloscope the clock at the memory input no longer looks like a square wave: each rising edge overshoots to 6.7 V on a 5 V logic level, falls back to 4.4 V, rises again, and takes several nanoseconds to settle. Occasionally the memory sees two clock edges where the processor sent one. Every component is within specification, and Kirchhoff's laws, applied to the trace as a wire, predict a clean waveform. The prediction fails because the trace is not a wire: the edge takes almost a nanosecond to travel its length, and in that time the voltage at the two ends is not the same.

The same physics appears at every scale. A 300 km, 400 kV overhead line carries 50 Hz power, but its voltage at the open receiving end rises above the sending-end voltage. A technician finds a fault in a buried cable by sending a pulse and timing its echo. A radio transmitter shuts down its power amplifier because the antenna cable reflects power back. In each case the interconnect must be treated as a **transmission line**: a structure with inductance, capacitance and loss distributed along its length, on which voltage and current travel as waves.

> **When does a connection stop behaving as a wire and start behaving as a transmission line?**
> **What decides how much of a signal is reflected at a load, and how can reflections be removed? How do loss and frequency change what a long line delivers?**

Concept	Engineering destination
Distributed parameters	Cable and PCB-trace models, overhead-line constants in power systems
Telegrapher's equations	Wave propagation on any two-conductor line, from power lines to coax
Characteristic impedance	50 Ω RF systems, 75 Ω video, 100 Ω differential pairs, surge impedance of power lines
Propagation constant	Delay and attenuation budgets for cables, DSL reach, line loadability
Reflection coefficient	Signal integrity, TDR fault location, antenna matching
Standing-wave ratio	Transmitter protection, antenna tuning, RF power measurement
Impedance matching	Quarter-wave transformers, stub tuners, termination of digital buses
Smith chart	Graphical design of matching networks, reading VNA measurements
Lossy and lossless lines	Cable selection, long-distance power transmission, telephone and DSL loops
High-frequency interconnects	PCB layout rules, differential signalling, termination strategies

Roadmap. Distributed parameters replace lumped elements (14.1) → the telegrapher's equations (14.2) → their wave solution defines characteristic impedance (14.3) and the propagation constant (14.4) → a load sends part of the wave back, measured by the reflection coefficient (14.5) and the standing-wave ratio (14.6) → reflections are removed by matching (14.7), designed on the Smith chart (14.8) → loss changes the picture (14.9) → all of it applied to PCB and cable interconnects (14.10).

14.1 Distributed parameters

When the lumped model fails

Every circuit in Chapters 1–13 rested on the **lumped-element assumption**: the circuit is small enough that a change of current at one end of a wire is felt instantly at the other end, so that one voltage describes each node and one current each branch. Electromagnetic disturbances travel at a finite speed v, close to the speed of light $c = 3.00 \times 10^8$ m/s in air and slower in dielectrics. A signal of frequency f has wavelength $\lambda = v/f$. The lumped model is accurate when the physical length ℓ of the circuit is small compared with λ; a common working threshold is

$$\ell < \frac{\lambda}{10}$$

For signals with fast edges rather than a single frequency, the equivalent statement compares the propagation delay $t_d = \ell/v$ with the rise time t_r (Section 14.10).

Signal	Frequency or edge	Wavelength (in the medium)	Length where line effects matter
Mains power, overhead line	50 Hz	≈5900 km	≳ 100–300 km
Telephone voice	3.4 kHz	≈50 km on a loaded pair, less unloaded	Every subscriber loop
AM radio	1 MHz	300 m in air	Antenna feeders, towers
FM radio and VHF	100 MHz	2 m in coax (VF 0.66)	≳ 20 cm
Wi-Fi	2.4 GHz	6.2 cm on FR-4	≳ a few millimetres
Digital logic, 1 ns edge	knee ≈ 350 MHz	≈ 42 cm on FR-4	≳ 3–7 cm

The last column does not mark a sharp boundary: line effects grow continuously with ℓ/λ. The table shows why the same 50 Hz power that is lumped inside a building becomes distributed across a continent, and why every trace on a multi-gigahertz board is a transmission line.

The per-unit-length model

A **transmission line** is any pair of conductors (two wires, a coaxial cable, a trace over a ground plane, a twisted pair, a phase conductor over earth) whose cross-section is uniform along its length. A short

piece of length Δz has:

- **series resistance** $R\,\Delta z$, from the finite conductivity of both conductors (Ω/m);
- **series inductance** $L\,\Delta z$, from the magnetic flux linking the current loop (H/m);
- **shunt conductance** $G\,\Delta z$, from leakage through the dielectric (S/m);
- **shunt capacitance** $C\,\Delta z$, from the electric field between the conductors (F/m).

R, L, G and C are the **primary line constants** or distributed parameters. They are fixed by geometry and materials. For a coaxial cable with inner radius a, outer radius b and dielectric permittivity ε_r,

$$L = \frac{\mu_0}{2\pi}\ln\frac{b}{a}, \qquad C = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(b/a)}$$

and for most lines $LC = \mu\varepsilon$, a relation that ties the propagation speed to the dielectric alone. R grows with frequency through the skin effect (Section 14.9), and G grows in proportion to frequency for most dielectrics.

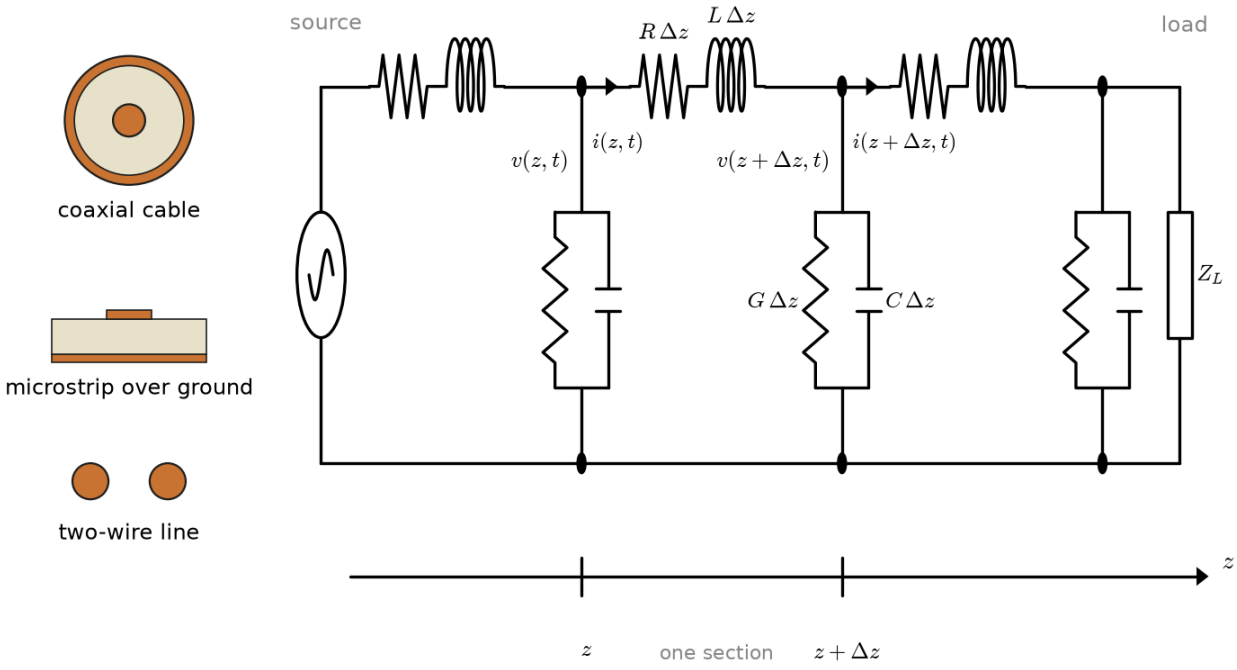

Figure 14.1 — Lumped equivalent of a short section of transmission line of length delta z, with series R and L and shunt G and C per unit length.

Figure 14.1 shows the model as a ladder of infinitesimal sections. It is still a circuit, and Kirchhoff's laws still apply to each section, because each section is short compared with a wavelength. The new physics enters only through the limit $\Delta z \to 0$.

> **Boundary.**
> The derivation of L and C for arbitrary geometries, and the field picture of a TEM wave guided between two conductors, belong to *Engineering Physics*. This chapter takes the per-unit-length constants as given and builds the circuit theory of lines from them.

Representative line constants

Line	R	L	C	G	Use
RG-58 coax (50 Ω, VF 0.66)	≈ 0.03 Ω/m at 1 MHz, rises as $\sqrt{f}$	250 nH/m	100 pF/m	negligible below 100 MHz	Lab RF, instrumentation
RG-6 coax (75 Ω, VF 0.85, foam)	≈ 0.02 Ω/m at 1 MHz	295 nH/m	52 pF/m	small	TV, satellite
0.4 mm telephone pair	280 Ω/km (loop)	0.6 mH/km	50 nF/km	≈ 1 µS/km	Subscriber loops, DSL
400 kV overhead line, bundled	≈ 0.03 Ω/km	≈ 1.0 mH/km	≈ 11.5 nF/km	≈ 0	Bulk power transmission
50 Ω microstrip on FR-4	≈ 5–10 Ω/m at 1 GHz	≈ 300 nH/m	≈ 120 pF/m	from $\tan\delta \approx 0.02$	PCB signals

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14.2 Telegrapher's equations

Derivation from the section model

Apply KVL around one section of Figure 14.1. The voltage drop along the section is caused by the series resistance and inductance:

$$v(z,t) - v(z+\Delta z,t) = R\,\Delta z\, i(z,t) + L\,\Delta z\,\frac{\partial i(z,t)}{\partial t}$$

Apply KCL at the node $z+\Delta z$. The current lost along the section flows through the shunt conductance and capacitance:

$$i(z,t) - i(z+\Delta z,t) = G\,\Delta z\, v(z+\Delta z,t) + C\,\Delta z\,\frac{\partial v(z+\Delta z,t)}{\partial t}$$

Dividing by Δz and letting $\Delta z \to 0$ gives the **telegrapher's equations**:

$$\frac{\partial v}{\partial z} = -R\,i - L\frac{\partial i}{\partial t}, \qquad \frac{\partial i}{\partial z} = -G\,v - C\frac{\partial v}{\partial t}$$

They were derived by Oliver Heaviside in the 1880s to explain why telegraph pulses on long submarine cables smeared out, and they describe every uniform two-conductor line.

> **Status of the principle: Telegrapher's equations**
> *Fundamental.* They follow from Maxwell's equations for a transverse electromagnetic (TEM) wave between two conductors, with R, L, G, C capturing the geometry; they are exact for TEM lines with uniform cross-section and an excellent approximation (quasi-TEM) for microstrip and twisted pairs below the frequencies at which higher-order modes or radiation appear.

The lossless wave equation

For a lossless line ($R = G = 0$), differentiating one equation with respect to z and substituting the other gives

$$\frac{\partial^2 v}{\partial z^2} = LC\,\frac{\partial^2 v}{\partial t^2}$$

which is the one-dimensional wave equation. Its general solution, found by d'Alembert, is the sum of two arbitrary waveforms travelling in opposite directions:

$$v(z,t) = v^{+}\!\left(t - \frac{z}{v}\right) + v^{-}\!\left(t + \frac{z}{v}\right), \qquad v = \frac{1}{\sqrt{LC}}$$

A forward wave v^{+} moves toward increasing z at speed v without changing shape; a backward wave v^{-} moves the other way. For RG-58 with $L = 252.5$ nH/m and $C = 101$ pF/m, $v = 1/\sqrt{LC} = 1.98 \times 10^{8}$ m/s, which is 0.66 of the speed of light. This **velocity factor** equals $1/\sqrt{\varepsilon_r}$ for a TEM line filled with a dielectric of relative permittivity ε_r; for solid polyethylene ($\varepsilon_r = 2.25$) it is 0.667. The delay per metre is 5.05 ns/m, a figure every engineer who has synchronised instruments with cables learns to use.

The phasor form

For sinusoidal steady state at angular frequency ω, the time derivatives become $j\omega$ and the equations become ordinary differential equations in the phasors $V(z)$ and $I(z)$:

$$\frac{dV}{dz} = -(R + j\omega L)I = -Z'I, \qquad \frac{dI}{dz} = -(G + j\omega C)V = -Y'V$$

with $Z' = R + j\omega L$ the series impedance per metre and $Y' = G + j\omega C$ the shunt admittance per metre. Eliminating I gives

$$\frac{d^2V}{dz^2} = \gamma^2 V, \qquad \gamma = \sqrt{Z'Y'} = \sqrt{(R + j\omega L)(G + j\omega C)}$$

whose solution is

$$V(z) = V^{+}e^{-\gamma z} + V^{-}e^{+\gamma z}, \qquad I(z) = \frac{1}{Z_0}\left(V^{+}e^{-\gamma z} - V^{-}e^{+\gamma z}\right)$$

The two terms are again forward and backward waves; γ and Z_0 are the subject of the next two sections. These equations, with the two constants V^{+} and V^{-} fixed by the source and the load, are the entire sinusoidal theory of transmission lines.

14.3 Characteristic impedance

Definition

Substituting the forward wave alone into $dV/dz = -Z'I$ shows that its voltage and current are in a fixed ratio at every point:

$$Z_0 = \frac{V^+}{I^+} = \sqrt{\frac{Z'}{Y'}} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

This is the **characteristic impedance** of the line. It is not the impedance of any component and cannot be measured with an ohmmeter. It is the ratio of voltage to current in a single travelling wave: a source launching a wave onto a long line sees Z_0, regardless of what is connected at the far end, until a reflection returns. For a lossless line, or any line at frequencies where $\omega L \gg R$ and $\omega C \gg G$,

$$Z_0 = \sqrt{\frac{L}{C}}$$

which is real and independent of frequency. For RG-58, $\sqrt{252.5 \times 10^{-9} / 101 \times 10^{-12}} = 50.0\ \Omega$.

> **Status of the principle: Characteristic impedance**
> *Derived*. A direct consequence of the telegrapher's equations; the lossless form $\sqrt{L/C}$ is approximate, valid when $\omega L \gg R$ and $\omega C \gg G$, which for cables means above roughly 100 kHz.

Physical interpretation

A useful way to see Z_0 is to imagine an infinitely long line. A source connected to it sees a resistance Z_0 forever, because the energy it sends travels away and never comes back; a lossless line of infinite length absorbs power exactly like a resistor. A finite line terminated in a resistor equal to Z_0 is indistinguishable from an infinite line: this is **matched termination**, and it is the goal of most interconnect design.

The value of Z_0 depends only on the cross-section geometry, logarithmically for coax:

$$Z_0 = \frac{60}{\sqrt{\varepsilon_r}} \ln \frac{b}{a}\ \Omega$$

so doubling the diameter ratio changes Z_0 by only $41.6/\sqrt{\varepsilon_r}\ \Omega$. Practical characteristic impedances therefore span a modest range, roughly 20 Ω to 600 Ω.

Line or system	Typical Z_0	Reason for the value
RF instrumentation coax	50 Ω	Compromise between minimum loss (≈77 Ω) and maximum power handling (≈30 Ω) in air-dielectric coax
Video, cable TV, satellite coax	75 Ω	Close to minimum attenuation
Twisted pair (Ethernet, Cat 5e/6)	100 Ω differential	Wire gauge and twist geometry
USB 2.0 and HDMI differential pairs	90 Ω and 100 Ω differential	Interface standards
Single-ended PCB trace (DDR, general logic)	40–60 Ω	Trace width and dielectric height
Telephone pair at voice frequencies	≈ 500–1700 Ω, angle ≈ −45°	Loss-dominated regime (Section 14.9)
Twin-lead ribbon feeder	300 Ω	Wide spacing in air
Overhead power line (surge impedance)	≈ 250–400 Ω	Conductor height and bundle geometry
Underground power cable	≈ 20–60 Ω	Thin insulation, high capacitance

Surge impedance loading of power lines

Power engineers call Z_0 the **surge impedance** of a line, because it governs the travelling waves set up by lightning and switching surges. It also defines a natural power level. If a lossless line is terminated in its surge impedance, voltage and current keep a constant ratio along the whole line, and the reactive power generated by the line capacitance exactly balances that absorbed by its inductance. The real power delivered at that condition is the **surge impedance loading**:

$$\mathrm{SIL} = \frac{V_{\mathrm{LL}}^2}{Z_0}$$

Worked example — surge impedance of a 400 kV line. A 400 kV line has $L = 1.0$ mH/km and $C = 11.5$ nF/km per phase. The lossless surge impedance is $\sqrt{1.0 \times 10^{-3} / 11.5 \times 10^{-9}} = 295\ \Omega$. Including $R = 0.03$ Ω/km at 50 Hz gives $Z_0 = 295.2 - j14.1\ \Omega$, magnitude 295.6 Ω and angle −2.7°: the loss changes Z_0 by less than 0.3 % in magnitude. The SIL is $(400 \times 10^3)^2 / 295 = 543$ MW. Below SIL the line generates net reactive power and the receiving-end voltage tends to rise; above SIL it absorbs reactive power and the voltage sags. The wave speed is $1/\sqrt{LC} = 2.95 \times 10^5$ km/s, close to the speed of light, so the 50 Hz wavelength is 5900 km. A 300 km line is $300/5900 \times 360^\circ = 18.3^\circ$ long, and its open-circuit receiving-end voltage is $1/\cos 18.3^\circ = 1.053$ times the sending-end voltage: the **Ferranti effect**, a 5.3 % rise produced entirely by distributed capacitance. Long lines are therefore operated near SIL or compensated with shunt reactors, a topic of *Electrical Energy & Machines*.

14.4 Propagation constant

Attenuation and phase constants

The **propagation constant** is complex:

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

A forward wave $V^{+}e^{-\gamma z} = V^{+}e^{-\alpha z}e^{-j\beta z}$ decays in amplitude at the rate set by the **attenuation constant** α (in nepers per metre) and rotates in phase at the rate set by the **phase constant** β (in radians per metre). One neper is a factor e in amplitude, equal to 8.686 dB, so

$$\text{attenuation in dB/m} = 8.686\,\alpha$$

The phase constant determines the wavelength on the line, $\lambda = 2\pi/\beta$, and the **phase velocity**, $v_p = \omega/\beta$.

> **Status of the principle: Propagation constant**
> *Derived*. Follows from the phasor telegrapher's equations; α and β inherit the frequency dependence of R, L, G, C, so a measured $\gamma(\omega)$ also characterises the line's materials.

The lossless case

When $R = G = 0$, $\gamma = j\omega\sqrt{LC}$, so $\alpha = 0$ and

$$\beta = \omega\sqrt{LC}, \qquad v_p = \frac{1}{\sqrt{LC}}, \qquad \lambda = \frac{v_p}{f}$$

The phase velocity is independent of frequency: all frequency components of a signal travel together and the waveform arrives undistorted, only delayed. This is the ideal that real interconnects approach at high frequency.

Electrical length

What matters for the behaviour of a line is not its physical length but its length in wavelengths, the **electrical length**

$$\theta = \beta\ell = 2\pi\frac{\ell}{\lambda}$$

usually quoted in degrees. A line with θ of a few degrees behaves almost like a lumped connection; at $\theta = 90^\circ$ (a quarter wavelength) it inverts impedances (Section 14.7); at $\theta = 180^\circ$ it reproduces the load impedance at its input. For the 400 kV example, 300 km at 50 Hz is 18.3°; for a 10 cm FR-4 trace at 2.4 GHz with $v_p = 1.45 \times 10^8$ m/s, $\lambda = 6.0$ cm, and the trace is about 600° long.

14.5 Reflection coefficient

Boundary condition at the load

Let a line of characteristic impedance Z_0 be terminated at $z = \ell$ by a load Z_L. At the load the line voltage and current must satisfy Ohm's law for the load, $V(\ell) = Z_L I(\ell)$. A single forward wave has $V/I = Z_0$, so unless $Z_L = Z_0$ it cannot satisfy the boundary condition alone. A reflected wave must appear, with just the amplitude needed to make the total voltage and current match the load. Writing $V_L = V^+ + V^-$ and $I_L = (V^+ - V^-)/Z_0$ (with phasors referred to the load position) and setting $V_L/I_L = Z_L$ gives the **reflection coefficient**

$$\Gamma_L = \frac{V^-}{V^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

> **Status of the principle: Reflection coefficient**
> *Derived.* Follows from continuity of voltage and current at the line–load junction and the fixed V/I ratio of each travelling wave; it applies to any linear load and, at an instant, to any junction between lines of different Z_0.

The three limiting cases are worth memorising:

Load	Γ_L	What happens
Matched, $Z_L = Z_0$	0	No reflection; all incident power absorbed
Open circuit, $Z_L \to \infty$	+1	Voltage doubles at the load; current is zero
Short circuit, $Z_L = 0$	−1	Voltage is zero; current doubles
Pure reactance, $Z_L = jX$	$\lvert\Gamma\rvert = 1$, angle varies	All power reflected, with a phase shift
Resistive, $R_L > Z_0$	Real, between 0 and +1	Partial reflection, same polarity
Resistive, $R_L < Z_0$	Real, between −1 and 0	Partial reflection, inverted polarity

For a passive load, $|\Gamma_L| \le 1$. The fraction of incident power reflected is $|\Gamma_L|^2$; the fraction delivered to the load is $1 - |\Gamma_L|^2$. RF engineers express these as the **return loss**, $\mathrm{RL} = -20\log_{10}|\Gamma_L|$ dB, and the **mismatch loss**, $-10\log_{10}(1 - |\Gamma_L|^2)$ dB. In S-parameter language (Chapter 32), Γ measured at a port is S_{11}.

Worked example — a 75 Ω camera on a 50 Ω cable, and a dipole on a 50 Ω feed. A $75\ \Omega$ video load at the end of a $50\ \Omega$ cable gives $\Gamma_L = (75 - 50)/(75 + 50) = 0.20$. The return loss is $-20\log 0.20 = 14.0$ dB, 4 % of the incident power is reflected, and the mismatch loss is only 0.18 dB. The more serious consequence on a video line is the echo: a reflection of 20 % in amplitude, delayed by the cable round trip, appears as a faint "ghost" image. A half-wave dipole antenna slightly longer than resonant presents roughly $Z_L = 73 + j42.5\ \Omega$ to a $50\ \Omega$ feed. Then

$$\Gamma_L = \frac{23 + j42.5}{123 + j42.5} = 0.274 + j0.251 = 0.371\angle 42.5^\circ$$

The return loss is 8.6 dB and 13.8 % of the power is reflected (mismatch loss 0.64 dB). Trimming the dipole to resonance removes the reactance and brings Γ down to $(73-50)/123 = 0.187$.

Reflections in the time domain

The reflection coefficient applies equally to fast edges, which makes it the main tool of **signal integrity** and **time-domain reflectometry**. When a step travels down a line and meets a load, the reflected step has amplitude Γ_L times the incident step and travels back toward the source, where it reflects again with the source reflection coefficient $\Gamma_S = (Z_S - Z_0)/(Z_S + Z_0)$. The voltage at any point is the running sum of all the waves that have passed it. A **bounce** (lattice) **diagram** tracks the waves graphically: position along the horizontal axis, time downward, each wave a diagonal line.

Worked example — ringing on an unterminated clock trace. A logic driver with 25 Ω output resistance switches from 0 to 5 V into a 50 Ω PCB trace 15 cm long, ending at a CMOS input that behaves as an open circuit. With a delay of 6.0 ps/mm the one-way delay is $t_d = 0.90$ ns. The initial wave is set by the divider between source resistance and Z_0:

$$V_1 = 5 \times \frac{50}{25+50} = 3.33\ \text{V}, \qquad \Gamma_S = \frac{25-50}{25+50} = -\frac{1}{3}, \qquad \Gamma_L = +1$$

At $t = t_d$ the 3.33 V wave reaches the open end, reflects with $\Gamma_L = +1$, and the load voltage jumps to $3.33 + 3.33 = 6.67$ V, the overshoot seen in the opening. The reflected 3.33 V wave returns to the source, reflects as -1.11 V, and arrives at the load at $3t_d$, where it doubles again, bringing the load to $6.67 - 2.22 = 4.44$ V. Continuing, the load voltage at $t_d, 3t_d, 5t_d, 7t_d, 9t_d$ is 6.67, 4.44, 5.19, 4.94 and 5.02 V. It converges to 5 V, the DC value Kirchhoff's laws predict, but only after several round trips of 1.8 ns each. A 25 Ω series resistor at the driver, making $Z_S = 50\ \Omega$, would give $\Gamma_S = 0$: the load would jump cleanly from 0 to 5 V at t_d and stay there.

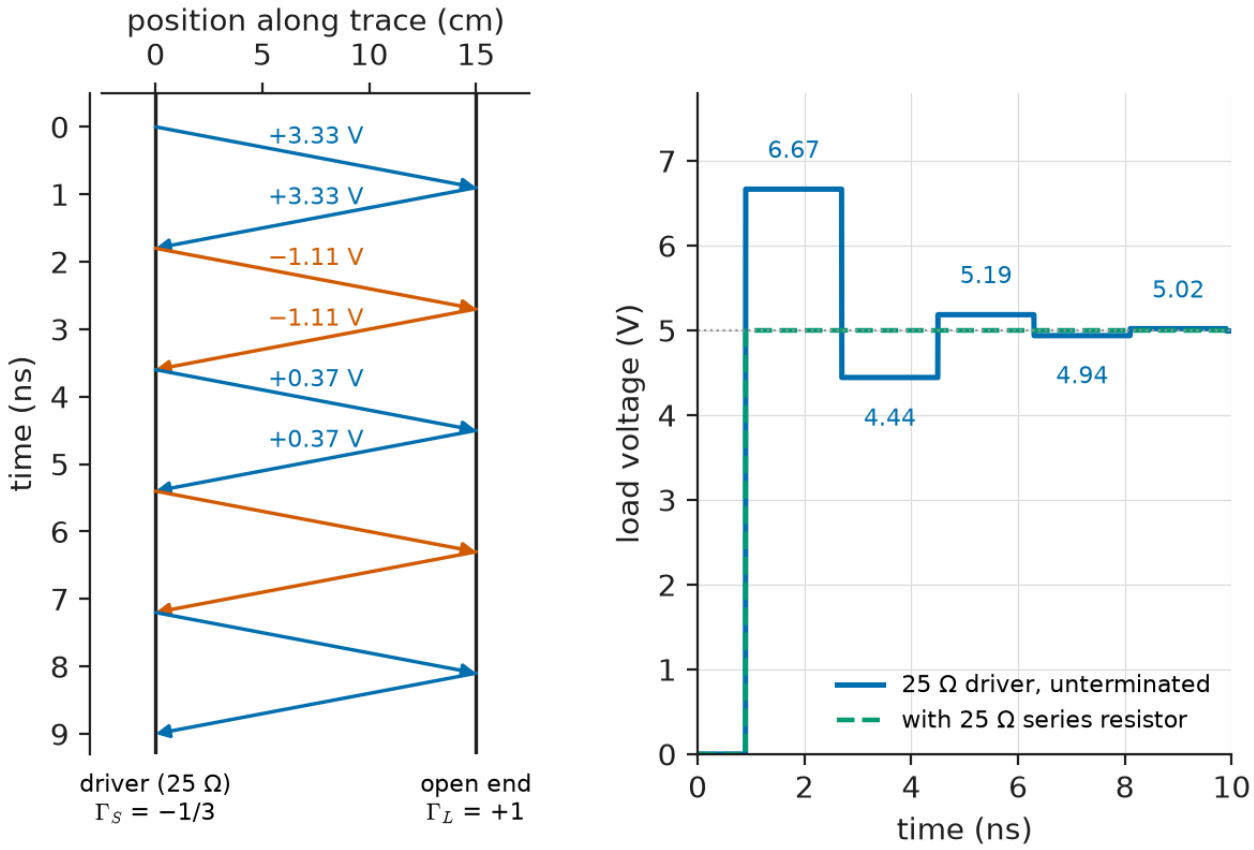

Figure 14.2 — Bounce diagram and load voltage for a 5 V step from a 25-ohm driver into an open-ended 50-ohm, 0.90 ns trace.

Figure 14.2 shows the lattice and the staircase waveform. On a real board the edges are not instantaneous and the input has a few picofarads of capacitance, so the staircase becomes a damped ringing waveform, but the levels follow the bounce diagram closely.

Time-domain reflectometry

The same mechanism locates faults. A **time-domain reflectometer** (TDR) launches a fast step or pulse into a cable and records the echoes. The delay of an echo gives the distance to the discontinuity, and the sign and size of the echo give its nature: an open circuit returns a positive step, a short a negative one, a crushed or water-logged section a smaller dip. If an echo returns $1.2\ \mu\text{s}$ after launch on a cable with velocity factor 0.66, the fault lies at

$$d = \frac{v\,t}{2} = \frac{0.66 \times (3.00 \times 10^{8}) \times (1.2 \times 10^{-6})}{2} = 119\ \text{m}$$

TDR is used on telephone and data cables, on underground power cables (where the high-voltage version is called surge or impulse-current fault location), on PCB traces to measure Z_0 along their length, and in soil-moisture probes, where the delay measures the soil's permittivity.

14.6 Standing-wave ratio

Standing waves

With a sinusoidal source, the forward and reflected waves coexist on the line and interfere. Measured from the load ($d = \ell - z$ is the distance toward the source), the voltage magnitude is

$$|V(d)| = |V^+|\,\left|1 + \Gamma_L e^{-j2\beta d}\right|$$

As d increases, the reflected term rotates twice as fast as βd, and the magnitude oscillates between a maximum $|V^+|(1 + |\Gamma_L|)$ where the two waves add in phase and a minimum $|V^+|(1 - |\Gamma_L|)$ where they oppose. Maxima and minima repeat every half wavelength and are a quarter wavelength apart. This stationary pattern of maxima and minima is a **standing wave** (Figure 14.3).

The **voltage standing-wave ratio** is the ratio of maximum to minimum:

$$\mathrm{VSWR} = \frac{V_{\max}}{V_{\min}} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}, \qquad |\Gamma_L| = \frac{\mathrm{VSWR} - 1}{\mathrm{VSWR} + 1}$$

It runs from 1 (perfect match) to infinity (total reflection). Before vector network analysers became common, VSWR was the easiest quantity to measure on a line: a slotted line with a sliding probe recorded $V_{\max}$ and $V_{\min}$ directly, and the position of the minimum gave the phase of Γ_L.

> **Status of the principle: Standing-wave ratio**
> *Derived*. An algebraic consequence of the superposition of forward and reflected waves on a lossless line; on a lossy line the ratio varies along the line and is lowest near the source, which can make a badly mismatched antenna look acceptable from the transmitter end.

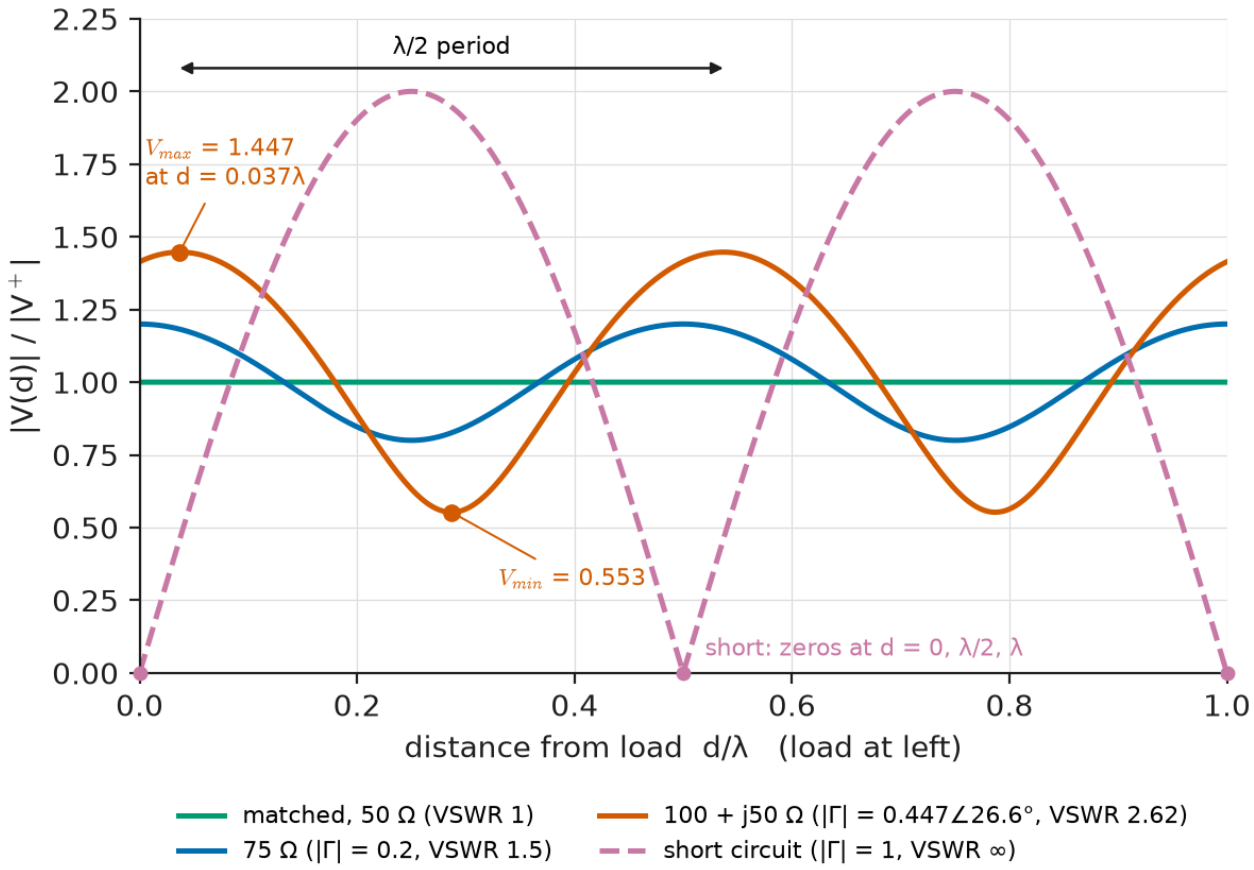

Figure 14.3 — Voltage magnitude along a lossless 50-ohm line for four loads, showing standing waves and VSWR.

What VSWR means in practice

VSWR	|Γ|	Return loss	Power reflected	Typical interpretation
1.0	0	∞	0 %	Ideal termination
1.2	0.091	20.8 dB	0.8 %	Good test-equipment port
1.5	0.200	14.0 dB	4.0 %	Typical antenna specification limit
2.0	0.333	9.5 dB	11.1 %	Marginal; many transmitters begin to reduce power
3.0	0.500	6.0 dB	25.0 %	Poor; protection circuits often trip
∞	1	0 dB	100 %	Open, short or pure reactance

High VSWR harms a transmitter in two ways: reflected power is not radiated, and the voltage maxima along the line and at the amplifier output can exceed device ratings. Transmitters measure forward and reflected power with a directional coupler and fold back their output when the reflection exceeds a threshold.

Input impedance of a terminated line

The ratio $V(d)/I(d)$ at a distance d from the load is the impedance seen looking toward the load. For a lossless line,

$$Z_{\text{in}}(d) = Z_0 \frac{Z_L + jZ_0 \tan \beta d}{Z_0 + jZ_L \tan \beta d}$$

This **impedance transformation** formula shows that a line changes the impedance its load presents, periodically with period $\lambda/2$. Special cases include a short-circuited line, $Z_{\text{in}} = jZ_0 \tan \beta d$, and an open-circuited line, $Z_{\text{in}} = -jZ_0 \cot \beta d$: lossless line stubs behave as pure inductors or capacitors of any value, depending on length. A shorted 50 Ω line one-eighth of a wavelength long has $\tan(\pi/4) = 1$ and presents $+j50\ \Omega$, an inductor; an open one of the same length presents $-j50\ \Omega$, a capacitor. Above about 1 GHz, stubs replace lumped inductors and capacitors in filters and matching networks.

14.7 Impedance matching

Why match

Matching serves three different purposes, which sometimes conflict:

1. **Maximum power transfer** from a source to a load (Chapter 4), which for a lossless line requires the load to present the conjugate of the source impedance at the input.
2. **Absence of reflections** on the line, which requires $Z_L = Z_0$ at the load; this protects transmitters, avoids video ghosts, and keeps digital edges clean.
3. **Low noise** in receivers, where the source impedance that minimises the amplifier's noise figure is generally not the conjugate match (Chapter 29).

In most RF and high-speed digital systems the second requirement dominates, and everything is designed to the same reference impedance, usually 50 Ω, so that any component can be connected to any other.

Termination in digital interconnects

Scheme	Arrangement	Advantage	Cost
Series (source) termination	Resistor $Z_0 - R_{\text{out}}$ at driver	No DC power; one resistor	Half-amplitude wave travels down the line; only the far end sees a clean edge
Parallel termination	$R = Z_0$ from far end to ground or V_{TT}	Clean edge everywhere	DC power dissipation
Thévenin termination	Two resistors to supply and ground	Sets a mid-rail bias	Continuous current
AC termination	$R = Z_0$ in series with C at far end	No DC power	Needs $RC \gg$ bit time
Differential termination	$R = Z_{\text{diff}}$ (100 Ω) across the pair	Standard for LVDS, Ethernet, USB	Requires differential routing

The quarter-wave transformer

At $\beta d = \pi/2$ (a quarter wavelength) the transformation formula reduces to

$$Z_{\text{in}} = \frac{Z_T^2}{Z_L}$$

for a line section of characteristic impedance Z_T. To match a real load R_L to a line Z_0, choose

$$Z_T = \sqrt{Z_0 R_L}$$

To match a 100 Ω load to 50 Ω, a quarter-wave section of $\sqrt{50 \times 100} = 70.7\ \Omega$ is needed. It works exactly only at the design frequency. At 10 % above or below that frequency, the section is 99° or 81° long and the input impedance becomes $50.6 \pm j5.5\ \Omega$, a VSWR of 1.12, which is still good; wider bandwidths are obtained with several cascaded sections of stepped impedance. Quarter-wave transformers are used in antenna feeds, power dividers, and the impedance-inverting sections of RF amplifiers.

> **Status of the principle: Quarter-wave impedance inversion**
> *Derived*. Exact for a lossless line at the frequency where it is a quarter wavelength long; the match degrades smoothly away from that frequency, and loss in the section adds a resistive error.

Lumped and stub matching

A general complex load can be matched by moving along the line until the real part of the admittance equals $1/Z_0$, then cancelling the remaining susceptance with a reactive element in parallel. At low RF that element is a capacitor or inductor (an L-network); above about 1 GHz it is a short- or open-circuited stub. This **single-stub match** is worked through on the Smith chart in the next section.

14.8 Smith chart

Mapping impedance to reflection coefficient

The impedance transformation formula is awkward to apply by hand, especially for complex loads. Phillip Smith's 1939 chart makes it geometric. Normalise every impedance to the line, $z = Z/Z_0 = r + jx$. The reflection coefficient is

$$\Gamma = \frac{z - 1}{z + 1}$$

and moving a distance d toward the source multiplies it by $e^{-j2\beta d}$: on a lossless line, $|\Gamma|$ stays the same and its angle decreases by $2\beta d$. On the complex Γ-plane, then, moving along a lossless line is simply **rotation about the centre**, clockwise toward the generator, by 720° per wavelength.

The Smith chart is the Γ-plane with the grid of constant r and constant x drawn on it. The bilinear map $z \mapsto \Gamma$ sends lines of constant resistance to circles tangent at $\Gamma = 1$, and lines of constant reactance to arcs through the same point. The whole right half of the impedance plane (all passive loads) fills the unit circle.

> **Status of the principle: Smith chart**
> *Derived.* A graphical form of the bilinear transformation between normalised impedance and reflection coefficient; it is exact for lossless lines, and loss can be included by spiralling inward at the rate $e^{-2\alpha d}$.

Key features:

Point or curve	Meaning
Centre, $\Gamma = 0$	Matched, $z = 1$
Right edge, $\Gamma = +1$	Open circuit
Left edge, $\Gamma = -1$	Short circuit
Upper half	Inductive impedances ($x > 0$)
Lower half	Capacitive impedances ($x < 0$)
Circle about the centre through the load	Constant $\|\Gamma\|$ and constant VSWR
Where that circle crosses the right horizontal axis	Value of r equals the VSWR
Rotating 180° about the centre	Quarter-wave move; turns impedance into admittance

The last property lets the same chart be read as an admittance chart, which is convenient for shunt elements.

Worked single-stub match

Worked example — matching $100 + j50\ \Omega$ with a shunt stub. A load $Z_L = 100 + j50\ \Omega$ terminates a 50 Ω line. Normalising, $z_L = 2 + j1$, so

$$\Gamma_L = \frac{1 + j1}{3 + j1} = 0.40 + j0.20 = 0.447\angle 26.6^\circ$$

The VSWR is $1.447/0.553 = 2.62$ and 20 % of the power is reflected (mismatch loss 0.97 dB). On the chart the load lies in the upper half on the circle $|\Gamma| = 0.447$. To use a shunt stub, work in admittance: the normalised load admittance is $y_L = 1/z_L = 0.4 - j0.2$. Rotating toward the generator on the constant-$|\Gamma|$ circle, the admittance first meets the unit-conductance circle $g = 1$ at a distance $d_1 = 0.199\lambda$ from the load, where $y = 1 + j1.0$. A shunt element with normalised susceptance $-j1.0$ placed there cancels the imaginary part and leaves $y = 1$: matched. A short-circuited stub has normalised admittance $-j\cot\beta\ell_s$, so $\cot\beta\ell_s = 1$ and $\ell_s = \lambda/8 = 0.125\lambda$. The second solution is $d_2 = 0.375\lambda$, where $y = 1 - j1.0$, with a shorted stub of 0.375λ. The shorter combination is normally chosen because it has the wider bandwidth. At 2.4 GHz on a microstrip with $\lambda = 7.3$ cm (effective permittivity 2.9), the stub would sit 14.6 mm from the load and be 9.2 mm long.

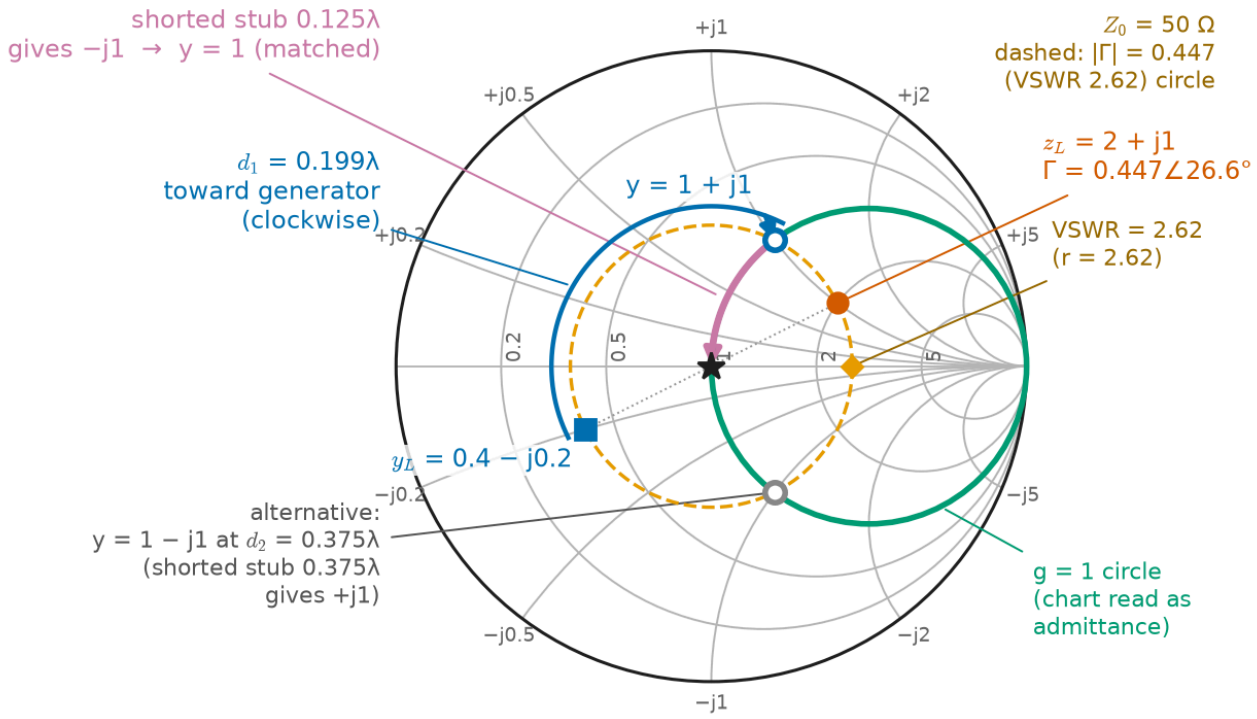

Figure 14.4 — Single shunt-stub match of the load 100 + j50 ohm on a 50-ohm line, plotted on the Smith chart.

Figure 14.4 shows the construction. Network analysers display measured S_{11} directly on a Smith chart for the same reason: a single glance shows whether a load is inductive or capacitive, how far it is from the match, and how it moves with frequency.

14.9 Lossy and lossless lines

Sources of loss

Two loss mechanisms dominate real lines:

- **Conductor loss**, through R. At high frequency current crowds into a thin layer at the conductor surfaces, the **skin depth** $\delta = \sqrt{\rho/(\pi f \mu)}$. For copper, δ is 9.3 mm at 50 Hz, 66 µm at 1 MHz, 6.6 µm at 100 MHz, 2.1 µm at 1 GHz and 0.66 µm at 10 GHz. Since the effective conducting area shrinks with δ, R grows as $\sqrt{f}$, and surface roughness of PCB copper adds further loss at gigahertz frequencies.
- **Dielectric loss**, through $G = \omega C \tan\delta_d$, where $\tan\delta_d$ is the loss tangent of the insulation. It grows in proportion to f, and dominates above a few gigahertz on FR-4 ($\tan\delta_d \approx 0.02$); low-loss laminates ($\tan\delta_d \approx 0.002$–$0.004$) and PTFE-based materials (≈ 0.001) are used where this matters.

> **Boundary.**
> The conductivity, permittivity and loss tangent of cable and board materials, and their dependence on frequency and temperature, are properties covered in *The Materials Core*.

The low-loss approximation

For most cables at radio frequencies, $R \ll \omega L$ and $G \ll \omega C$. Expanding γ to first order in the small ratios gives

$$\alpha \approx \frac{R}{2Z_0} + \frac{G Z_0}{2}, \qquad \beta \approx \omega\sqrt{LC}, \qquad Z_0 \approx \sqrt{\frac{L}{C}}$$

The first term of α is conductor loss and the second dielectric loss. The phase behaviour remains that of the lossless line, and only the amplitude decays. This is why the lossless results of Sections 14.5–14.8 can be used for RF design with a separate attenuation budget.

> **Status of the principle: Low-loss line approximation**
> *Approximate.* Valid when $R/(\omega L)$ and $G/(\omega C)$ are both well below about 0.1; it fails at audio frequencies on telephone pairs and at very low frequencies on any cable, where R dominates ωL.

Cable or line	Loss at 10 MHz	Loss at 100 MHz	Loss at 1 GHz
RG-58 (50 Ω, solid PE)	≈ 4.5 dB/100 m	≈ 16 dB/100 m	≈ 60–70 dB/100 m
RG-213 (50 Ω, solid PE)	≈ 2 dB/100 m	≈ 7 dB/100 m	≈ 25–30 dB/100 m
LMR-400 (50 Ω, foam)	≈ 1.3 dB/100 m	≈ 4 dB/100 m	≈ 13 dB/100 m
Cat 6 twisted pair (100 Ω)	≈ 6 dB/100 m	≈ 20 dB/100 m	not specified
50 Ω microstrip on FR-4	≈ 0.01 dB/cm	≈ 0.03 dB/cm	≈ 0.1 dB/cm (≈ 0.5–1 dB/cm at 10 GHz)

The values are typical manufacturer figures; they rise roughly as $\sqrt{f}$ where conductor loss dominates and faster where dielectric loss takes over.

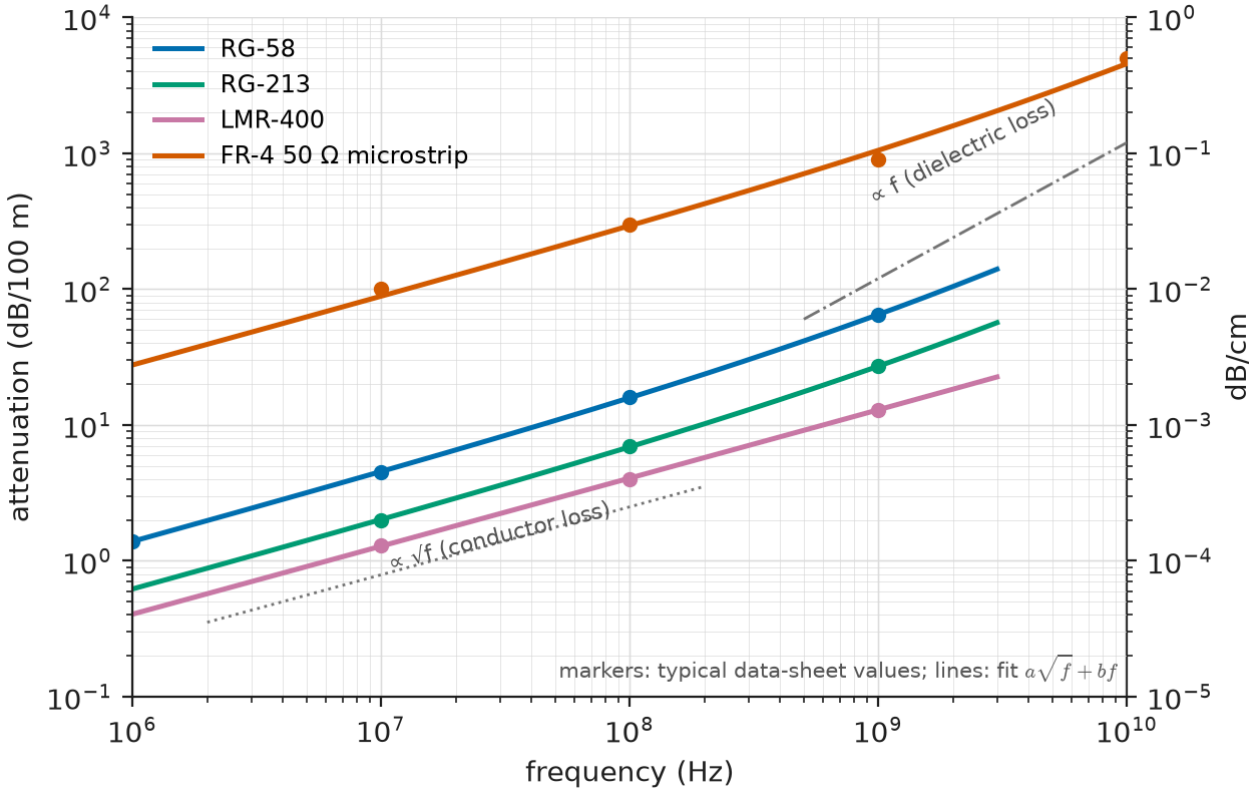

Figure 14.5 — Attenuation versus frequency for common cables and for an FR-4 microstrip, showing the square-root-of-frequency conductor-loss region and the linear dielectric-loss region.

Distortionless lines and the telephone loop

When loss is not small, Z_0 and γ depend strongly on frequency, so different frequency components of a signal travel at different speeds and are attenuated differently: the line distorts. Heaviside showed that if

$$\frac{R}{L} = \frac{G}{C}$$

then $\gamma = \sqrt{RG} + j\omega\sqrt{LC}$ and $Z_0 = \sqrt{L/C}$ exactly, at all frequencies: the line attenuates but does not distort. Real cables have $R/L \gg G/C$, so the condition was approached by *adding* inductance, either by winding iron tape around the conductors or by inserting loading coils every 1.8 km or so. Loaded telephone cables carried voice across continents for most of the twentieth century.

Worked example — a telephone subscriber pair. A 0.4 mm pair has $R = 280\ \Omega$/km (loop), $L = 0.6$ mH/km, $C = 50$ nF/km and negligible G. At 1 kHz, $\omega L = 3.8\ \Omega$/km is only 1.3 % of R, so the line is resistance-dominated. Computing exactly,

$$Z_0 = \sqrt{\frac{280 + j3.77}{j3.14\times10^{-4}}} = 944\angle -44.6^\circ\ \Omega$$

with attenuation 1.81 dB/km and phase velocity 2.98×10^{4} km/s, one-tenth of the speed in air. At 300 Hz the attenuation is 1.00 dB/km and $|Z_0| = 1720\ \Omega$; at 3.4 kHz it is 3.28 dB/km and $|Z_0| = 512\ \Omega$. A 5 km loop therefore attenuates the top of the voice band by 16 dB but the bottom by only 5 dB, and the waveform is distorted. At 1 MHz, where DSL operates, ωL dominates, Z_0 approaches $\sqrt{L/C} = 110\ \Omega$ and is almost real, but the skin-effect resistance has risen sharply, and real 0.4 mm pairs lose 20–30 dB/km at 1 MHz (the constant-R model used here underestimates this). This is why DSL rates fall so steeply with loop length, and why loading coils, good for voice, had to be removed before DSL could be deployed: they make the line a low-pass filter above about 3.5 kHz.

Long power lines

At 50 or 60 Hz an overhead line is close to lossless in the sense of Section 14.4: from the 400 kV example, $\alpha = 4.4\times10^{-4}$ dB/km, or 0.13 dB of wave attenuation over 300 km. Its line effects are therefore phase effects: the Ferranti rise, the dependence of receiving-end voltage on loading relative to SIL, and the limit on transmittable power set by the angle across the line. Lines shorter than about 80 km are modelled as a series impedance, medium lines (80–250 km) as a lumped π-network, and long lines with the exact distributed equations in ABCD form (Chapter 12), with $A = D = \cosh\gamma\ell$, $B = Z_0\sinh\gamma\ell$ and $C = \sinh(\gamma\ell)/Z_0$.

14.10 High-frequency interconnects

When is a trace a line?

On a digital board the relevant signal is not a sinusoid but an edge with rise time t_r. Its spectrum extends to roughly the **knee frequency** $f_{\text{knee}} \approx 0.35/t_r$ to $0.5/t_r$. A trace must be treated as a transmission line when its one-way delay is a significant fraction of the rise time. Common rules of thumb are

$$t_d > \frac{t_r}{6}\quad\text{(conservative)}\qquad\text{or}\qquad 2t_d > t_r\quad\text{(reflections clearly visible)}$$

> **Status of the principle: Critical length for interconnects**
> *Empirical.* The fractions $1/6$ to $1/2$ come from simulation and measurement experience, not from a sharp physical boundary; reflections grow continuously with t_d/t_r, and the acceptable level depends on the logic family's noise margin.

Worked example — critical trace length. A logic family with 1 ns edges drives traces on FR-4 ($\varepsilon_r = 4.3$). In stripline (trace buried between planes) the delay per length is $\sqrt{\varepsilon_r}/c = 6.9$ ps/mm; in microstrip (surface trace with field partly in air, effective permittivity about 3.3) it is about 6.0 ps/mm. The conservative criterion $t_d > t_r/6 = 167$ ps gives a critical length of about 24 mm in stripline and 28 mm in microstrip; the looser criterion $2t_d > t_r$ gives 72 mm and 83 mm. The knee frequency is 350 MHz, where the wavelength in stripline is $145\ \text{mm/ns} \div 0.35\ \text{GHz} = 414$ mm, so $\lambda/10 = 41$ mm: the different rules agree that traces longer than a few centimetres need controlled impedance and termination. The 15 cm clock trace of the opening example is well beyond all of them. With 100 ps edges, as in modern serial links, the critical length shrinks to a few millimetres and every trace is a line.

Controlled-impedance traces

A PCB trace becomes a controlled transmission line when its width, the dielectric height and permittivity, and its distance to the reference plane are held to specified tolerances. For a surface microstrip, a widely used curve fit (IPC-2141) is

$$Z_0 \approx \frac{87}{\sqrt{\varepsilon_r + 1.41}} \ln\frac{5.98h}{0.8w + t} \quad \Omega$$

for trace width w, copper thickness t and dielectric height h. On 0.2 mm FR-4 ($\varepsilon_r = 4.3$) with 35 µm copper, a 0.35 mm wide trace gives $Z_0 \approx 48.6\ \Omega$. The formula is accurate to a few percent in its range of validity (roughly $0.1 < w/h < 2$); field solvers are used for production designs, and fabricators typically hold Z_0 to ±10 %.

> **Status of the principle: Closed-form microstrip impedance**
> *Empirical.* A curve fit to field-solver results and measurements over a limited range of geometries; it ignores frequency dependence of ε_r, solder mask and conductor roughness.

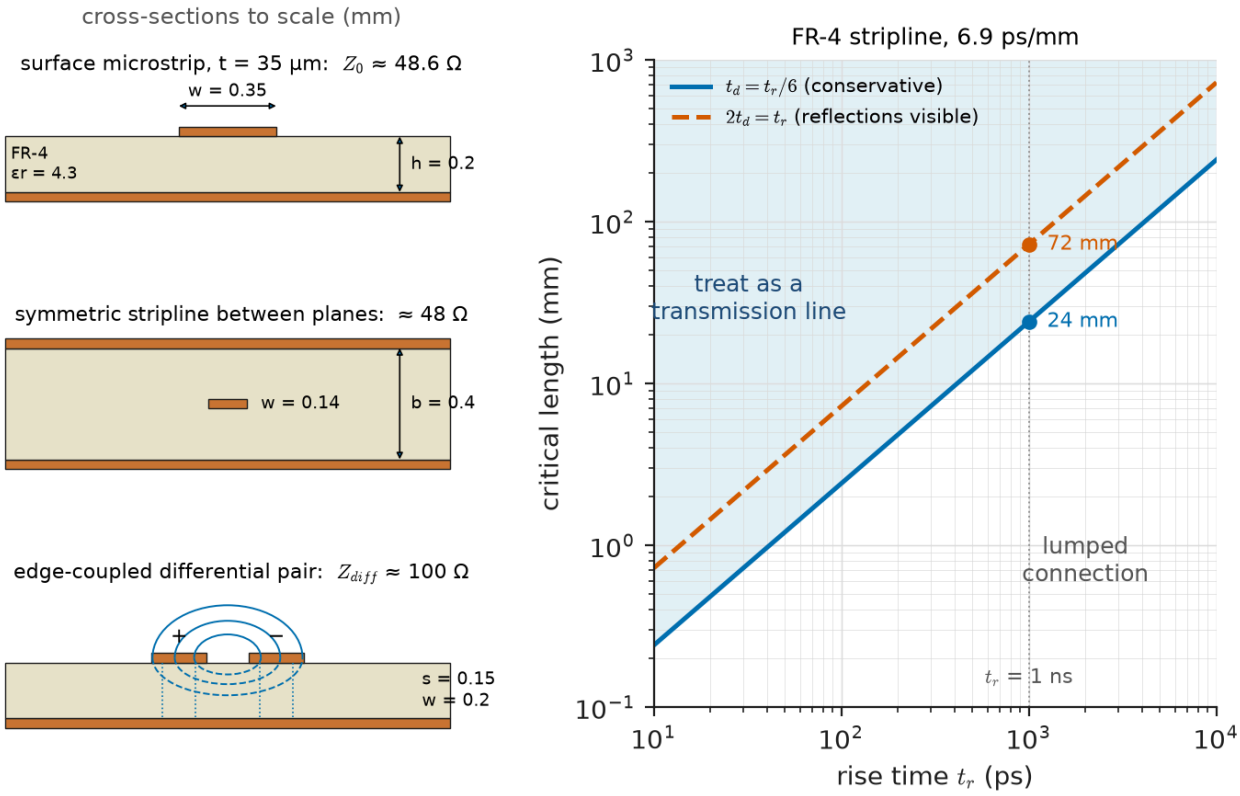

Figure 14.6 — Cross-sections of common PCB transmission lines and the critical-length criterion versus rise time.

Differential signalling

Most fast serial interfaces (USB, PCI Express, Ethernet, HDMI, LVDS) use **differential pairs**: two coupled lines carrying equal and opposite signals, received by a circuit that responds only to the difference. The pair has a **differential impedance**, typically 85–100 Ω, set by the trace geometry and the coupling between the two traces. Because the two currents are equal and opposite, the pair radiates little and rejects noise coupled equally onto both lines. A differential termination resistor equal to the differential impedance absorbs the incoming wave at the receiver.

Other discontinuities

Any change in cross-section is a change in Z_0, and therefore a reflection. Common ones on boards and in cables are:

Discontinuity	Effect	Typical mitigation
Via between layers	Capacitive dip plus inductive bump; stub resonance at high GHz	Back-drilling, short stubs, ground vias alongside
Connector	Impedance step of 5–20 %	Controlled-impedance connectors
Component pad or IC input	Capacitive load of 1–5 pF	Minimise stubs; series termination
Break in reference plane	Return current detours; inductive bump and radiation	Never route over plane splits
Branch (stub) to a second receiver	Reflections from the stub end	Daisy-chain (fly-by) topology, as in DDR memory
Bend in a trace	Small capacitive discontinuity	Mitered or curved corners

Above roughly 10 Gb/s the cumulative effect of loss, reflections and crosstalk is assessed with S-parameters (Chapter 32) and compensated at the receiver with equalisation, a digital signal-processing problem (Chapter 23). The physics underneath is still that of this chapter: travelling waves on distributed RLGC lines, and reflections wherever the impedance changes.

What remains open

Open question	Where it is answered
How are L and C derived from Maxwell's equations, and what is a TEM wave?	*Engineering Physics*
How are matching networks, couplers and stubs described with S-parameters?	Chapter 32 (RF and Microwave Fundamentals)
How does an antenna turn a guided wave into a radiated one?	Chapter 32 (RF and Microwave Fundamentals)
How does dispersion limit data rate on an optical fibre, the optical analogue of a lossy line?	Chapter 31 (Optical Communications)
How is the received noise and link budget of a cable or radio link computed?	Chapters 25, 29 and 30
How is the frequency-dependent loss of a channel equalised digitally?	Chapters 18 and 23
How are long power lines compensated and operated near their stability limit?	*Electrical Energy & Machines*
What do the ABCD parameters of a distributed line mean for cascading?	Chapter 12 (Two-Port Networks)
How do conductor and dielectric materials set R and G?	*The Materials Core*

Next: Chapter 15 opens Part IV by stepping back from circuits to the signals they carry, defining the continuous- and discrete-time signals and the elementary operations used in the rest of the book.

Part IV — Signals and Systems

Parts I–III treated circuits as the object of study: given a network, find its voltages and currents. Part IV turns the question around. The voltage on a busbar, the output of an ECG electrode, the echo received by a radar and the samples stored by an audio converter are *signals*, functions that carry information. The amplifier, filter, cable, sensor, digital algorithm or control loop that acts on them is a *system*. This part develops the general theory of both, independent of the hardware that realises them, so that the same ideas serve a power-quality analyser, a hearing aid, a seismometer and a mobile-phone modem. Its central result is that a linear time-invariant system is completely described by one function, its impulse response or equivalently its frequency response, and that continuous signals can be replaced by discrete samples without loss under conditions that can be stated exactly.

By the end of this part you should understand:

- how signals are classified (continuous or discrete, periodic or aperiodic, even or odd, energy or power) and built from steps, impulses, ramps and exponentials;
- what linearity, time invariance, causality, stability and memory mean, and how to test a system for each;
- why the impulse response and convolution determine the output of any linear time-invariant system, and how differential and difference equations describe the same systems;
- how Fourier series and transforms turn convolution into multiplication, and what magnitude, phase, bandwidth, group delay and distortion mean for a signal passing through a system;
- when a continuous signal can be recovered from its samples, what aliasing is, and how anti-aliasing and reconstruction filters work in practice;
- how the Z-transform, poles and zeros describe discrete-time systems and their stability, and how a digital filter is defined.

Chapter	Central question	What it delivers
15 — Signal Fundamentals	How are signals described, classified and manipulated?	Continuous- and discrete-time signals, periodicity, symmetry, energy and power, the step, impulse, ramp and exponential, time shifting, scaling and amplitude operations
16 — System Fundamentals	What properties of a system decide how it can be analysed?	Classification by linearity, time invariance, causality, stability and memory; continuous and discrete systems; input–output representations
17 — Linear Time-Invariant Systems	How does one response determine all responses?	Impulse response, convolution, differential and difference equations, eigenfunctions, causal and stable LTI systems, interconnections
18 — Frequency-Domain System Analysis	What does a system do to each frequency in a signal?	Fourier series and transforms, frequency response and transfer functions, filtering, bandwidth, group delay, distortion, time–frequency relationships
19 — Sampling and Reconstruction	When do samples capture a continuous signal completely?	The sampling theorem, Nyquist frequency, aliasing, ideal and practical sampling, sample-and-hold, reconstruction, the quantization interface, anti-aliasing filters
20 — Discrete-Time Signals and Systems	How are sampled signals and digital systems analysed?	Sequences, discrete convolution, difference equations, the Z-transform, system functions, poles and zeros, discrete-time stability, digital-filter fundamentals

How this part connects to the rest of the book. Part IV generalises what Parts II and III did for circuits. The first- and second-order responses of Chapters 6–7, the phasors of Chapter 8, the Laplace-domain transfer functions of Chapter 11 and the frequency responses of Chapter 13 reappear here as special cases of LTI-system theory, now applied to any system and to discrete as well as continuous time; transmission lines (Chapter 14) become an example of a system with delay and dispersion. Part IV in turn supplies the foundations for everything that follows. Part V (Digital Signal Processing) builds the DFT, digital filters and multirate and adaptive methods directly on Chapters 19 and 20. Part VI (Communication Theory) uses spectra, bandwidth, filtering and sampling to describe modulation, channels and noise, and adds the random-signal models only touched on here. Part VII (Control-System Fundamentals) reuses transfer functions, stability and frequency response to design feedback. The mathematics of the transforms themselves (Fourier, Laplace and Z, and generalised functions such as the impulse) is recapped where needed and developed in full in *The Mathematics Universe*.

Chapter 15 — Signal Fundamentals

Part IV · Signals and Systems

A power-quality analyser is clipped onto the supply of a factory that keeps tripping its variable-speed drives. For a week it records the line voltage. Most of the time the record is a slightly flattened 50 Hz sine wave, 230 V rms, repeating every 20 ms with a few percent of fifth and seventh harmonic. Twice a day, when the utility switches a capacitor bank on a nearby substation, a burst of 500–800 Hz oscillation lasting a few milliseconds rides on top of the sine wave, its peak reaching 1.6 times the normal value. The drives trip on those bursts. To diagnose the problem the engineer must describe the recorded signal precisely: is it periodic or not, how much power does each part carry, when does the transient start and how quickly does it decay, how does the analyser's sampled record relate to the continuous voltage on the busbar, and how would the waveform look if it were shifted, scaled, filtered or combined with the drive's own current?

Those questions are not specific to power quality. An ECG, a radar echo, a seismic trace, a sampled audio track and the bit stream in a fibre all raise them. This chapter builds the vocabulary of signals: the classifications (continuous or discrete, periodic or not, even or odd, energy or power), the elementary building-block signals (steps, impulses, ramps and exponentials), and the operations (shifting, scaling, reflection, addition, multiplication, differentiation and integration) from which the rest of Parts IV–VII are constructed.

> **What distinguishes a continuous-time signal from a discrete-time one, and how are they related?**
> **How are signals classified by symmetry, repetition and energy content, and why do those classes matter for analysis? What are the elementary signals and operations from which any engineering signal can be modelled?**

Concept	Engineering destination
Continuous-time signals	Sensor outputs, circuit voltages and currents, received radio waveforms
Discrete-time signals	ADC records, digital audio, sampled control loops, digital communication symbols
Periodic and aperiodic signals	Mains and harmonics, clocks, rotating-machine vibration, transients and pulses
Even and odd signals	Symmetry shortcuts in Fourier analysis, filter design, correlation
Energy and power signals	Pulse energy in radar and defibrillators, rms ratings, SNR and link budgets
Unit step and impulse	Switching events, test inputs, impulse response of systems
Ramp and exponential signals	Charging transients, sweep generators, damped oscillations, phasors
Time shifting and scaling	Propagation delay, Doppler, playback speed, decimation
Signal operations	Modulation, mixing, averaging, integration in sensors and controllers

Roadmap. Two kinds of independent variable: continuous time (15.1) and discrete time (15.2) → classification by repetition (15.3), symmetry (15.4) and energy (15.5) → the elementary signals: step and impulse (15.6), ramp and exponential (15.7) → operations on the time axis (15.8) and on amplitude (15.9).

15.1 Continuous-time signals

What a signal is

A **signal** is a function that carries information about the behaviour of a physical system. In this book the information is almost always carried as a voltage, a current or a quantity converted into one (pressure by a microphone, acceleration by an accelerometer, light intensity by a photodiode, the heart's electrical activity by ECG electrodes). The independent variable is usually time; in images it is position, and in some sensors (a line-scan camera, an ultrasound array) both.

A **continuous-time** (CT) signal $x(t)$ is defined for every real value of t in some interval. The voltage on the busbar, the current in an inductor and the pressure at a microphone diaphragm are continuous-time signals: nature does not skip instants. The amplitude may also take any real value, in which case the signal is called **analog**. Continuous-time signals are what circuits produce and process; Parts I–III were, in effect, the theory of continuous-time signals in electrical networks.

Real and complex signals

Physical signals are real. It is nevertheless very convenient to work with complex-valued signals, above all the complex exponential

$$x(t) = Ae^{j(\omega t+\phi)} = A\cos(\omega t+\phi) + jA\sin(\omega t+\phi)$$

The phasors of Chapter 8 were a disguised use of this signal: a real sinusoid is the real part of a rotating complex exponential, or equivalently the sum of two counter-rotating ones,

$$A\cos(\omega t+\phi) = \frac{A}{2}e^{j(\omega t+\phi)} + \frac{A}{2}e^{-j(\omega t+\phi)}$$

Complex signals also appear directly in communications, where a receiver's in-phase and quadrature channels are combined into one complex **baseband** signal $x_I(t) + jx_Q(t)$ (Chapter 27).

Deterministic and random signals

A signal whose value at every instant is fixed by a formula or a known record is **deterministic**: a 50 Hz sine wave, a clock, a radar chirp. A signal that can be described only statistically is **random**: thermal noise, speech before it is spoken, the arrival pattern of data packets. Every real measurement is a deterministic part plus a random part. This chapter and Chapters 16–20 treat deterministic signals; random signals and noise are the subject of Chapter 29.

Representative engineering signals

Signal	Type	Typical amplitude	Bandwidth or rate	Comment
Mains voltage (Europe, Asia, Africa)	CT, periodic	230 V rms, 325 V peak	50 Hz plus harmonics to ≈ 2.5 kHz	120 V, 60 Hz in North America
Surface ECG	CT, quasi-periodic	0.5–4 mV	0.05–150 Hz	Sampled at 250–1000 Hz
Scalp EEG	CT, random-like	10–100 µV	0.5–70 Hz	Very low amplitude
Microphone output	CT	1–50 mV	20 Hz–20 kHz	Speech or music
CD audio	DT	16-bit samples	44.1 kHz sampling	Studio: 24-bit, 48–192 kHz
Telephone speech	DT	8-bit companded	8 kHz sampling	300–3400 Hz voice band
Seismic accelerometer	CT, then DT	µg to g	0.01–50 Hz, sampled at 100–200 Hz	Earthquake and structural monitoring
Radar pulse	CT, aperiodic burst	kW–MW peak	1–100 MHz instantaneous	Repeats at the pulse repetition frequency
GPS signal at the antenna	CT, random-like spread spectrum	≈ −130 dBm	2 MHz (L1 C/A)	Below the thermal noise floor

15.2 Discrete-time signals

Sequences

A **discrete-time** (DT) signal $x[n]$ is defined only at integer values of an index n; it is a **sequence** of numbers. Square brackets distinguish it from a continuous-time signal. Discrete-time signals arise in two ways:

1. **By sampling** a continuous-time signal at regular intervals T_s: $x[n] = x_c(nT_s)$, where $f_s = 1/T_s$ is the **sampling rate**. The power-quality analyser of the opening samples the busbar voltage, typically at 10–25 kHz; a digital ECG machine at 500 Hz; a CD at 44.1 kHz.
2. **Naturally**, when the quantity is defined only at discrete instants or positions: the daily closing value of an energy price, the number of packets arriving per millisecond, the pixel values along a line of an image, the bits of a digital message.

A discrete-time signal whose amplitude is also restricted to a finite set of levels is a **digital** signal. An ADC produces digital signals; the step from discrete-time to digital, **quantization**, adds an error that is analysed in Chapters 19 and 21. In the theory of Parts IV and V, discrete-time signals are usually treated as having real or complex amplitudes and quantization is considered separately.

Notation and the role of the sampling interval

Once a sequence has been formed, the index n is dimensionless: the sequence does not "know" its sampling rate. A 50 Hz sine wave sampled at 1 kHz gives

$$x[n] = \sin\left(2\pi \frac{50}{1000} n\right) = \sin(0.1\pi n)$$

and so does a 5 Hz sine wave sampled at 100 Hz. The quantity that matters inside the sequence is the **normalised (digital) frequency** $\Omega = \omega T_s = 2\pi f / f_s$, measured in radians per sample. Physical frequency is restored only when the sequence is related back to a sampling rate. This scale-free property is what lets one digital filter design serve many sampling rates.

> **Status of the principle: Discrete-time model of a sampled signal**
> *Approximate.* The sequence $x[n] = x_c(nT_s)$ represents the continuous signal without loss only if x_c is bandlimited below $f_s/2$ and the samples are exact; aliasing, sampling jitter and quantization are the three ways the model departs from the physical record (Chapters 19 and 21).

Worked example — what a sampled record contains. A diagnostic ECG is recorded for 10 s at 500 samples per second with a 12-bit ADC. The record contains $10 \times 500 = 5000$ samples, and at 12 bits each it occupies $5000 \times 12/8 = 7500$ bytes per lead; a standard 12-lead record is therefore about 90 kB. The sampling interval is 2 ms. A QRS complex lasting 80–100 ms is described by 40–50 samples, and the R-wave peak, which rises in about 20 ms, by about 10 samples. Whether that is enough to measure the amplitude of the peak accurately is a question about the signal's bandwidth (the diagnostic band extends to 150 Hz, below the 250 Hz half-sampling rate), answered in Chapter 19.

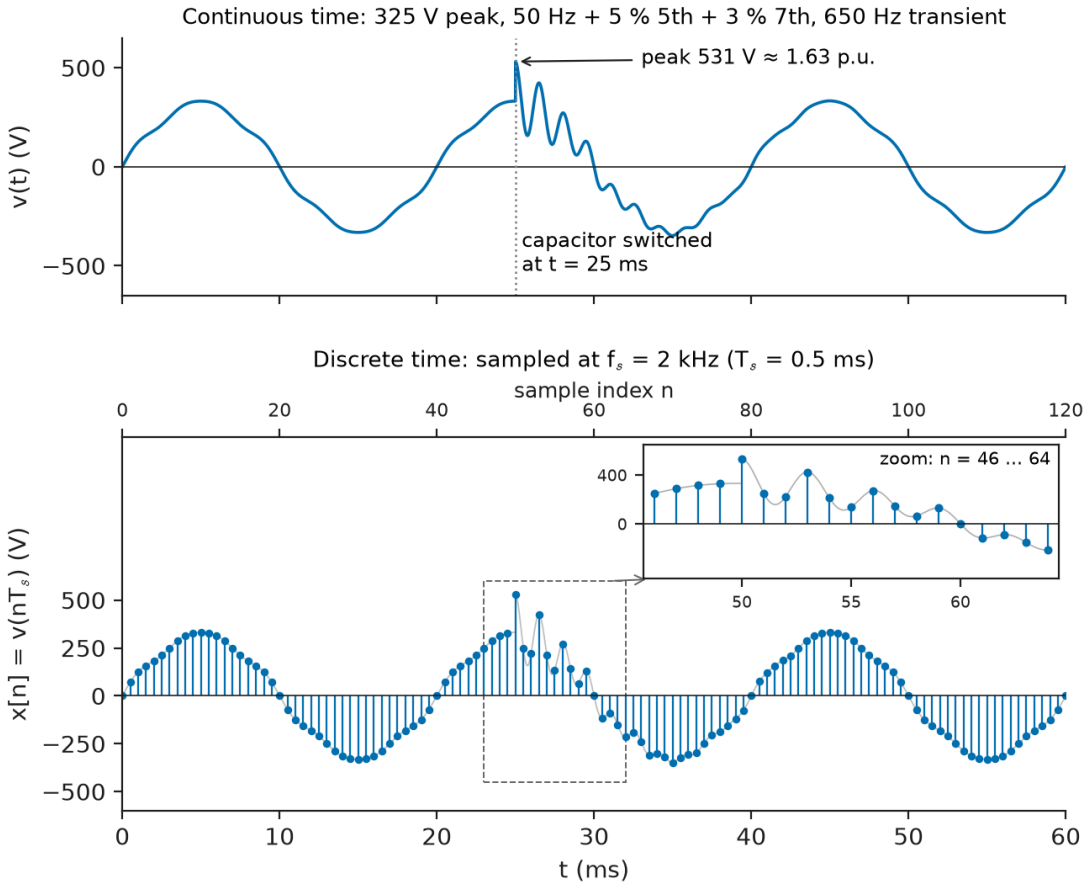

Figure 15.1 — A continuous-time busbar voltage with a capacitor-switching transient, and the discrete-time record made by a power-quality analyser sampling it.

Figure 15.1 shows the two descriptions of the same voltage. The continuous signal exists at every instant; the analyser knows it only at the stems. Everything in the analyser's report (rms value, harmonic content, transient detection) is computed from those stems.

Continuous versus discrete: a comparison

Property	Continuous time	Discrete time
Independent variable	$t \in \mathbb{R}$	$n \in \mathbb{Z}$
Typical origin	Physical quantity, circuit variable	Sampled measurement, digital computation
Elementary impulse	Dirac $\delta(t)$, a generalised function	Kronecker $\delta[n]$, an ordinary sequence
Sinusoid	Periodic for every ω	Periodic only for rational $\Omega/2\pi$
Distinct frequencies	$-\infty < \omega < \infty$	$-\pi < \Omega \leq \pi$ (frequencies repeat every 2π)
Energy	$\int \lvert x(t)\rvert^2 dt$	$\sum \lvert x[n]\rvert^2$
Calculus	Derivative, integral	Difference, running sum
System model	Differential equations	Difference equations

15.3 Periodic and aperiodic signals

Continuous-time periodicity

A continuous-time signal is **periodic** if there is a positive T such that

$$x(t+T) = x(t) \quad \text{for all } t$$

The smallest such T is the **fundamental period** T_0, and $f_0 = 1/T_0$ is the fundamental frequency. A signal that is not periodic is **aperiodic**. The mains voltage, a crystal-oscillator clock and the vibration of a machine running at constant speed are periodic (to good approximation); a lightning surge, a radar echo and a spoken word are aperiodic. Strictly, no physical signal is periodic, since a periodic signal must have existed forever; "periodic" always means "periodic over the observation interval", and the error made by that idealisation reappears as spectral leakage in Chapter 22.

A sum of periodic signals is periodic if and only if the ratio of every pair of periods is a rational number; the fundamental period of the sum is then the least common multiple of the individual periods. The distorted mains voltage

$$v(t) = 325\sin(2\pi\,50t) + 16.3\sin(2\pi\,250t) + 9.8\sin(2\pi\,350t)\ \text{V}$$

has components with periods 20 ms, 4 ms and 2.86 ms, all of which fit an integer number of times into 20 ms, so $T_0 = 20$ ms. In contrast, $\cos(2\pi t) + \cos(2\sqrt{2}\,\pi t)$ is not periodic: the period ratio $\sqrt{2}$ is irrational, and the waveform never repeats exactly. Such **almost-periodic** signals occur when two unsynchronised sources are combined, for example a 50 Hz grid and a 49.8 Hz islanded generator, whose sum beats at 0.2 Hz; because both frequencies drift and their ratio is not a ratio of small integers, the sum does not repeat over any practical record.

Discrete-time periodicity

A sequence is periodic with period N (a positive integer) if $x[n+N] = x[n]$ for all n. This differs from the continuous case in an important way. The discrete sinusoid $\cos(\Omega_0 n)$ is periodic only if there is an integer N with $\Omega_0 N = 2\pi k$ for some integer k, that is, only if

$$\frac{\Omega_0}{2\pi} = \frac{k}{N} \quad \text{is a rational number}$$

The fundamental period is the smallest such N, obtained by writing k/N in lowest terms.

> **Status of the principle: Periodicity of discrete-time sinusoids**
> *Derived.* Follows directly from the requirement that $\Omega_0 N$ be an integer multiple of 2π with N an integer; it has no continuous-time counterpart, where every sinusoid is periodic.

Worked example — periodic and non-periodic sequences. The sequence $\cos(0.3\pi n)$ has $\Omega_0/2\pi = 0.15 = 3/20$, so it is periodic with $N = 20$: twenty samples cover three cycles of the underlying cosine. The sequence $\cos(0.3n)$ has $\Omega_0/2\pi = 0.3/2\pi$, which is irrational, so the samples never repeat, although they lie on a cosine envelope with a "period" of $2\pi/0.3 = 20.9$ samples. For a sum, $x[n] = \cos(2\pi n/12) + \sin(2\pi n/18)$ has periods 12 and 18, and the sum repeats after

$\mathrm{lcm}(12, 18) = 36$ samples. In practice: a 50 Hz signal sampled at 1 kHz repeats every 20 samples, and at 48 kHz every 960; a 60 Hz signal sampled at 48 kHz repeats every 800 samples; but a 50 Hz signal sampled at 44.1 kHz repeats every 882 samples only because 44 100/50 happens to be an integer. If the grid drifts to 50.01 Hz, the sampled record is no longer periodic over any practical length, and a power-quality analyser must synchronise its sampling to the measured grid frequency to avoid leakage errors in its harmonic measurements.

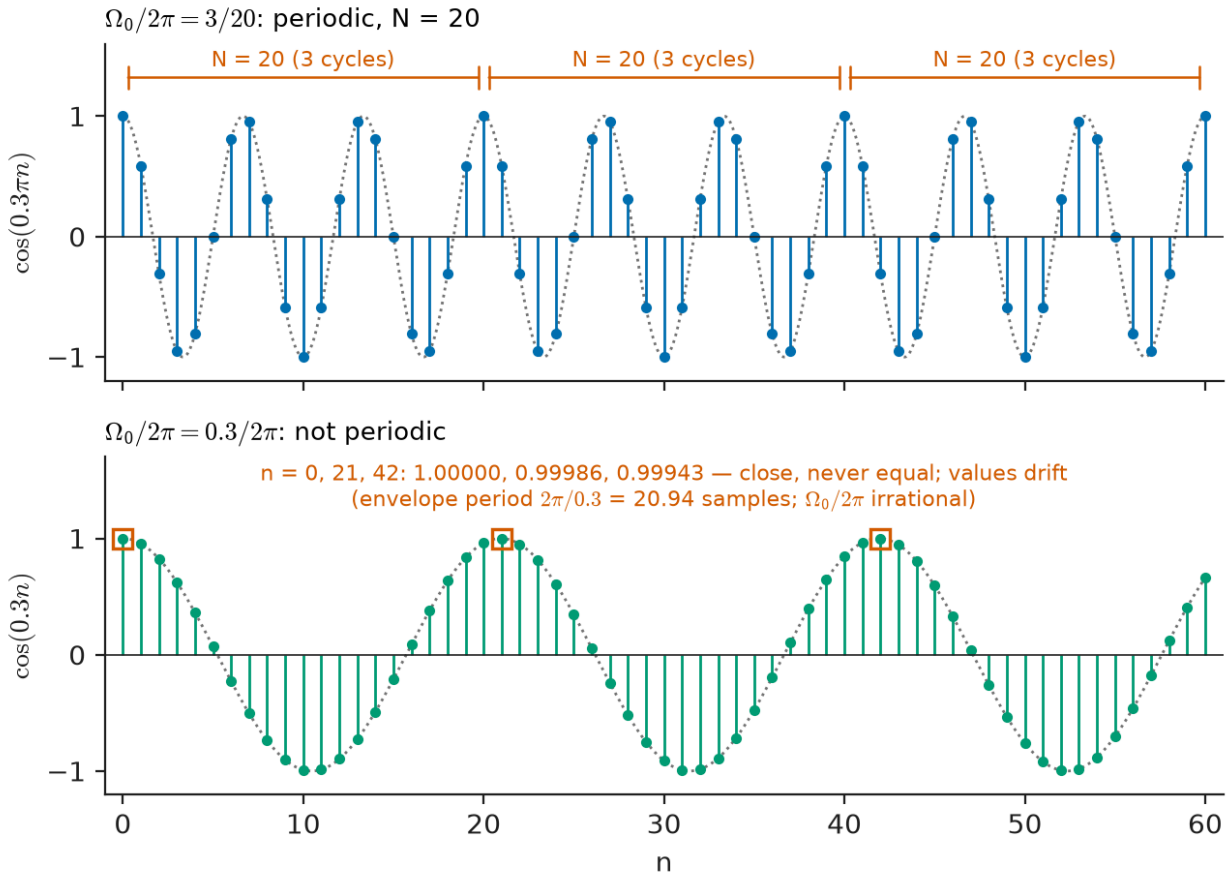

Figure 15.2 — The discrete sinusoids cos(0.3 pi n), which repeats every 20 samples, and cos(0.3 n), which never repeats.

Figure 15.2 contrasts the two. A second difference from continuous time is that discrete frequencies separated by 2π are identical: $\cos((\Omega_0 + 2\pi)n) = \cos(\Omega_0 n)$ for every integer n. Only frequencies in a band of width 2π, conventionally $-\pi < \Omega \le \pi$, are distinct, and the highest frequency a sequence can represent is $\Omega = \pi$, alternating $+1, -1, +1, \ldots$. This is the discrete-time face of aliasing (Chapter 19).

Quasi-periodic and transient signals

Many signals of engineering interest are neither strictly periodic nor arbitrary. A heartbeat repeats with a period that varies by several percent from beat to beat (heart-rate variability is itself a diagnostic quantity); the vibration of a gearbox repeats at each shaft revolution while the speed drifts; speech vowels are periodic for tens of milliseconds at a time. These **quasi-periodic** signals are analysed over windows short enough that the period is nearly constant, which leads to the time-frequency methods of Chapter 24. A **transient** is an aperiodic signal that is significant only for a finite time, such as the capacitor-switching burst of the opening or the step response of a circuit.

15.4 Even and odd signals

Definitions

A signal is **even** if it is symmetric about $t = 0$,

$$x(-t) = x(t)$$

and **odd** if it is antisymmetric,

$$x(-t) = -x(t)$$

The same definitions hold for sequences with n in place of t. The cosine is even, the sine is odd, and an odd signal must be zero at the origin. Many signals are neither, but every signal can be split uniquely into an even and an odd part:

$$x_e(t) = \frac{x(t) + x(-t)}{2}, \qquad x_o(t) = \frac{x(t) - x(-t)}{2}, \qquad x(t) = x_e(t) + x_o(t)$$

> **Status of the principle: Even–odd decomposition**
> *Derived*. An algebraic identity valid for any signal; uniqueness follows because a signal that is both even and odd must be zero.

Products follow sign rules: even × even and odd × odd are even, even × odd is odd. The integral of an odd signal over a symmetric interval is zero. These two facts are the source of most symmetry shortcuts in signal analysis: the Fourier series of an even periodic signal contains only cosines, that of an odd one only sines (Chapter 18), and the Fourier transform of a real even signal is real.

For complex signals the useful symmetry is **conjugate symmetry**, $x(-t) = x^*(t)$. Real signals have conjugate-symmetric Fourier transforms, which is why the spectrum of any real signal is fully described by its positive-frequency half.

Symmetry and the choice of time origin

Symmetry depends on where $t = 0$ is placed. A square wave that switches at $t = 0$ is odd; shifted by a quarter period it becomes even. Choosing the origin to make a signal even or odd often halves the work of an analysis. In a three-phase power system, choosing the time origin at the peak of the phase-a voltage makes it a pure cosine (even), while the phase-b and phase-c voltages, delayed by 120° and 240°, are neither. Half-wave symmetry, $x(t + T_0/2) = -x(t)$, is a related property: a waveform with it contains only odd harmonics. Most mains-supplied loads draw currents with half-wave symmetry, which is why power-system harmonic spectra are dominated by the 3rd, 5th, 7th, … and even harmonics indicate a DC component or asymmetric rectification.

Worked example — even and odd parts of a decaying exponential. The voltage on a discharging capacitor, normalised, is $x(t) = e^{-t}u(t)$: zero before $t = 0$, then decaying with time constant 1. Its reflection is $x(-t) = e^{t}u(-t)$. So

$$x_e(t) = \tfrac{1}{2}e^{-|t|}, \qquad x_o(t) = \tfrac{1}{2}\,\mathrm{sgn}(t)\,e^{-|t|}$$

The even part is a symmetric two-sided exponential with peak 0.5; the odd part is a step-like signal jumping from −0.5 to +0.5 at the origin and decaying both ways. Their energies (Section 15.5) are each $2\int_0^\infty \tfrac{1}{4}e^{-2t}dt = \tfrac{1}{4}$, and they add to the energy of $x(t)$, $\int_0^\infty e^{-2t}dt = \tfrac{1}{2}$, as they must: the even and odd parts of any real energy signal are orthogonal, so their energies add. For a sequence, $x[n] = \{1, 2, 3\}$ at $n = 0, 1, 2$ (zero elsewhere) has $x_e[n] = \{1.5, 1, 1, 1, 1.5\}$ and $x_o[n] = \{-1.5, -1, 0, 1, 1.5\}$ for $n = -2, \ldots, 2$.

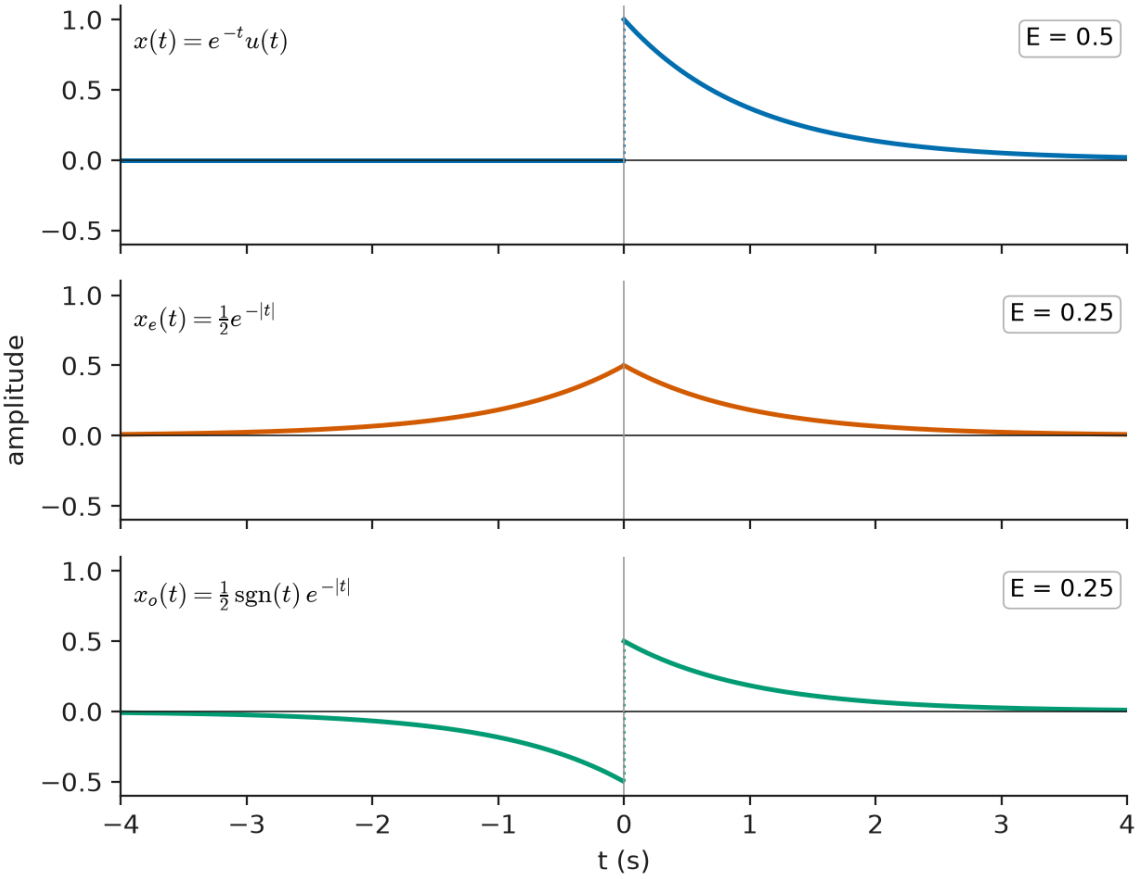

Figure 15.3 — Decomposition of the causal exponential e^(-t)u(t) into its even and odd parts.

Figure 15.3 shows the decomposition. The even part of a causal signal mirrors the future into the past, and the odd part makes the two cancel for $t < 0$; this pairing is the time-domain root of the relation between the real and imaginary parts of a causal system's frequency response.

15.5 Energy and power signals

Signal energy and power

The terms energy and power are borrowed from circuits. If $x(t)$ is the voltage across, or the current through, a 1 Ω resistor, the instantaneous power is $x^2(t)$ and the energy dissipated is its integral. For signals in general, the **energy** and the **average power** are defined as

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2\,dt, \qquad P_x = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} |x(t)|^2\,dt$$

and for sequences

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2, \qquad P_x = \lim_{N\to\infty} \frac{1}{2N+1}\sum_{n=-N}^{N} |x[n]|^2$$

These "normalised" quantities have units of V²·s and V² (or A²·s and A²) rather than joules and watts; the physical energy and power follow by dividing by the actual resistance for a voltage, or multiplying for a current. The ratio of two signal powers, such as signal-to-noise ratio, does not depend on the normalisation.

A signal is an **energy signal** if $0 < E_x < \infty$; then $P_x = 0$. It is a **power signal** if $0 < P_x < \infty$; then $E_x = \infty$. Transients, pulses and anything of finite duration and finite amplitude are energy signals. Periodic signals and stationary random signals are power signals. Some signals are neither: the ramp $t\,u(t)$ grows without bound and has infinite power.

> **Status of the principle: Energy–power classification**
> *Derived*. A mathematical consequence of the definitions; it is a modelling choice for physical signals, which always have finite energy over any finite observation, and the class chosen decides whether a Fourier transform (energy) or a power spectrum (power) is the appropriate description.

For a periodic signal the limit reduces to an average over one period,

$$P_x = \frac{1}{T_0}\int_{T_0} |x(t)|^2 dt, \qquad X_{\text{rms}} = \sqrt{P_x}$$

so the power is the square of the rms value of Chapter 8. For a sinusoid of peak A, $P = A^2/2$; for a sum of sinusoids at different frequencies the powers add, because the cross-terms average to zero.

Worked examples

Worked example — the mains as a power signal. The mains voltage $v(t) = 325.3\sin(2\pi\,50t)$ V has normalised power $325.3^2/2 = 52\,900$ V², so $V_{\text{rms}} = 230$ V. Across a $52.9\ \Omega$ kettle element it delivers $52\,900/52.9 = 1000$ W. Its energy over all time is infinite, as for any periodic signal. Adding 5 % fifth and 3 % seventh harmonic (16.3 V and 9.8 V peak) raises the power by the sum of their individual powers, giving $V_{\text{rms}} = 230\sqrt{1 + 0.05^2 + 0.03^2} = 230.4$ V; the total harmonic distortion is $\sqrt{0.05^2 + 0.03^2} = 5.8\ \%$. The rms value barely changes, but the waveform's shape and peak do, which is why power-quality standards limit harmonic content directly rather than through rms.

Worked example — the energy of a defibrillator pulse. A simple monophasic defibrillator discharges a $C = 100\ \mu$F capacitor charged to $V_0 = 2000$ V into a patient whose chest presents about 50 Ω. The voltage is $v(t) = V_0 e^{-t/\tau}u(t)$ with $\tau = RC = 5$ ms. This is an energy signal. The energy delivered to the resistance is

$$E = \frac{1}{R}\int_0^{\infty} V_0^2 e^{-2t/\tau}dt = \frac{V_0^2\tau}{2R} = \frac{(2000)^2(5\times10^{-3})}{2\times 50} = 200\ \text{J}$$

exactly the $\frac{1}{2}CV_0^2 = 200$ J stored in the capacitor, as energy conservation requires. The peak current is 40 A. Since the energy density decays with time constant $\tau/2$, 86 % of the energy is delivered in the first 5 ms and 98 % in the first 10 ms. Modern defibrillators shape the pulse as a truncated biphasic waveform precisely to control how energy and current are distributed in time, which is a signal-design problem.

Worked example — a radar pulse train. A surveillance radar transmits rectangular pulses of 1 µs duration at 1 MW peak power, repeated 1000 times per second. One pulse is an energy signal carrying $10^6 \times 10^{-6} = 1$ J. The pulse train is a power signal: its average power is $1\ \text{J} \times 1000\ \text{s}^{-1} = 1$ kW, the peak power times the duty cycle of 0.001. The transmitter must handle 1 MW peaks but only 1 kW of average heat, and the detection range depends on the energy per pulse (Chapter 29), which is why long pulses compressed after reception (chirp radar) are used to obtain energy without extreme peak power.

Crest factor and peak-to-average ratio

Power and peak are different things, and circuits care about both: heating depends on rms, while breakdown, clipping and ADC overload depend on peak. The **crest factor** is the ratio of peak to rms value; its square, expressed in decibels, is the **peak-to-average power ratio** (PAPR).

Signal	Crest factor	PAPR	Consequence
Square wave	1	0 dB	Most power per peak volt
Sinusoid	$\sqrt{2} = 1.414$	3.0 dB	Basis of rms ratings
Triangle wave	$\sqrt{3} = 1.732$	4.8 dB	Sweep generators
PWM, 12 V at 25 % duty	2.0	6.0 dB	3 V average, 6 V rms
Speech	≈ 4–10	≈ 12–20 dB	Audio headroom in amplifiers and ADCs
OFDM (Wi-Fi, LTE) before clipping	≈ 3–4	≈ 10–12 dB	Power-amplifier back-off, low efficiency
Rectifier input current	≈ 2–3	6–10 dB	Extra heating in transformers and cables

The PWM entry is a short calculation: a 12 V pulse train with duty cycle $D = 0.25$ has average $12D = 3$ V, power $12^2 D = 36\ \text{V}^2$ and rms $12\sqrt{D} = 6$ V, so a voltmeter that reads average and one that reads true rms disagree by a factor of 2.

15.6 Unit step and impulse

The unit step

The **unit step** is the idealisation of a switch closing at $t = 0$:

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

Its value at $t = 0$ is usually left undefined or set to $1/2$; it never matters in integrals. The discrete-time step is $u[n] = 1$ for $n \geq 0$ and 0 otherwise, with no ambiguity. Multiplying a signal by $u(t)$ makes it **causal** (zero for $t < 0$): $e^{-t/\tau}u(t)$ is a decay that starts at the switching instant. A rectangular pulse of width T is a difference of steps, $u(t) - u(t - T)$; a gated burst of a carrier, as in a radar pulse, is a carrier multiplied by such a pulse.

The unit impulse

Consider a rectangular pulse of width ε and height $1/\varepsilon$, so that its area is always 1. As $\varepsilon \to 0$ the pulse becomes infinitely tall and narrow with unchanged area. The limit is the **unit impulse** or **Dirac delta** $\delta(t)$, defined not by its values but by what it does inside an integral:

$$\int_{-\infty}^{\infty} \delta(t)\,dt = 1, \qquad \int_{-\infty}^{\infty} x(t)\,\delta(t - t_0)\,dt = x(t_0)$$

The second relation is the **sifting property**: integrating against a shifted impulse extracts the signal's value at that instant. A related identity is $x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$. The step and impulse are related by

$$\delta(t) = \frac{du(t)}{dt}, \qquad u(t) = \int_{-\infty}^{t} \delta(\tau)\,d\tau$$

The impulse is the mathematical model of anything that happens much faster than the system observing it can respond: a lightning stroke seen by a power system, a hammer tap on a structure under modal test, a short laser pulse on a photodetector, the charge delivered by an ionising particle in a radiation detector.

> **Status of the principle: The impulse as a physical model**
> *Approximate.* $\delta(t)$ is a generalised function, not an ordinary one; as a model of a physical pulse it is valid when the pulse is much shorter than the fastest time constant of the system it drives, so that only the pulse's area matters.

Worked example — impulsive charging of a capacitor. A radiation detector's charge-sensitive input is modelled as a 10 nF capacitor. A particle deposits 1 μC of charge in a few nanoseconds, much faster than any other time constant in the circuit. The current is modelled as $i(t) = Q\,\delta(t)$ with $Q = 1\ \mu\text{C}$, and the capacitor voltage is

$$v(t) = \frac{1}{C}\int_{-\infty}^{t} Q\,\delta(\tau)\,d\tau = \frac{Q}{C}\,u(t) = \frac{10^{-6}}{10^{-8}}u(t) = 100\,u(t)\ \text{V}$$

An impulse of current produces a step of voltage whose height depends only on the charge, not on the details of the pulse shape. This is why such amplifiers are called charge-sensitive, and why the impulse model is the right one: the measured quantity is the area.

The impulse has a scaling property that is easily overlooked:

$$\delta(at) = \frac{1}{|a|}\delta(t)$$

because compressing the narrow pulse by a also divides its area by $|a|$.

The discrete-time impulse

In discrete time there is no limiting process. The **unit sample** (Kronecker delta) is simply

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

with $\delta[n] = u[n] - u[n-1]$ and $u[n] = \sum_{k=-\infty}^{n} \delta[k]$. Any sequence can be written as a weighted sum of shifted unit samples,

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

which is trivial-looking but becomes the basis of convolution in Chapter 17: if a system's response to $\delta[n]$ is known, its response to any input follows. The continuous-time analogue, $x(t) = \int x(\tau)\delta(t-\tau)\,d\tau$, plays the same role.

> **Boundary.**
> The rigorous theory of generalised functions (distributions), which makes statements such as $du/dt = \delta$ precise, belongs to *The Mathematics Universe*. This book uses the impulse through its sifting and integration properties only.

15.7 Ramp and exponential signals

The ramp

The **unit ramp** is the integral of the step:

$$r(t) = t\,u(t) = \int_{-\infty}^{t} u(\tau)\,d\tau$$

with discrete counterpart $r[n] = n\,u[n]$. Ramps model a capacitor charged by a constant current ($v = It/C$), the sweep of an oscilloscope timebase or a radar's frequency chirp, the position of a vehicle at constant speed, and the slowly increasing setpoint used to test how well a control system tracks (Chapter 34). A ramp is neither an energy nor a power signal. The sequence of increasingly smooth signals, impulse → step → ramp → parabola, each the integral of the previous, is used to probe systems with inputs of increasing order.

Steps, ramps and impulses combine into piecewise-linear signals. A trapezoidal logic edge with rise time t_r is $\frac{1}{t_r}[r(t) - r(t-t_r)]$; a triangular pulse of half-width T is $\frac{1}{T}[r(t+T) - 2r(t) + r(t-T)]$. Writing

waveforms this way makes their derivatives easy: the derivative of the trapezoidal edge is a rectangular pulse of height $1/t_r$, the shape of the current a logic edge drives into a capacitive load.

Real exponentials

The **real exponential** $x(t) = Ae^{\sigma t}$ grows if $\sigma > 0$ and decays if $\sigma < 0$. Decaying causal exponentials $Ae^{-t/\tau}u(t)$ are the natural response of every first-order circuit (Chapter 6), with time constant τ; a decay to 1 % takes 4.6τ. Growing exponentials describe unstable systems and avalanche processes.

In discrete time the corresponding signal is the **geometric sequence** $x[n] = Aa^n$. It decays if $|a| < 1$ and grows if $|a| > 1$; if a is negative it alternates in sign. A sampled continuous exponential gives $a = e^{\sigma T_s}$. The causal sequence $0.9^n u[n]$ is an energy signal with $E = \sum_{n\ge 0} 0.81^n = 1/(1-0.81) = 5.26$; it is the impulse response of the simplest recursive digital filter, $y[n] = 0.9y[n-1] + x[n]$, the digital analogue of an RC low-pass (Chapter 20).

Complex exponentials and damped sinusoids

The most important signal in this book is the **complex exponential**

$$x(t) = Ae^{st}, \qquad s = \sigma + j\omega$$

Using Euler's formula,

$$Ae^{st} = Ae^{\sigma t}\big(\cos\omega t + j\sin\omega t\big)$$

so its real part is a sinusoid with an exponential envelope. Four cases cover most of engineering:

s	Signal	Example
$\sigma < 0$, $\omega = 0$	Decaying exponential	RC discharge, thermal cooling
$\sigma = 0$, $\omega \neq 0$	Constant-amplitude sinusoid (phasor)	Mains, carriers, clocks
$\sigma < 0$, $\omega \neq 0$	Damped sinusoid	Ringing RLC circuit, struck bell, capacitor-switching transient
$\sigma > 0$, $\omega \neq 0$	Growing sinusoid	Oscillator start-up, unstable control loop, flutter

The complex frequency s is the variable of the Laplace transform (Chapter 11), and the locations of a circuit's poles in the s-plane are precisely the complex frequencies of the exponentials in its natural response. The capacitor-switching transient of the opening is a damped sinusoid,

$$v_{\text{tr}}(t) = 200\,e^{-(t-t_0)/4\,\text{ms}}\cos\big(2\pi\,650\,(t-t_0)\big)\,u(t-t_0)\ \ \text{V}$$

corresponding to $s = -250 \pm j4084$ rad/s. Reading σ and ω off a recorded transient identifies the network that produced it: the oscillation frequency points to the resonance between the capacitor bank and the supply inductance, and the decay rate to the damping provided by the loads.

Status of the principle: Complex exponentials as the basis of signal analysis
Derived. Every signal of practical interest can be built from complex exponentials (Fourier and Laplace analysis), and linear time-invariant systems map each exponential to a scaled copy of itself (Chapter 17); together these make e^{st} the natural building block.

In discrete time the analogue is $x[n] = Az^n$ with complex $z = re^{j\Omega}$: the envelope grows or decays as r^n and the phase rotates by Ω per sample. The Z-transform (Chapter 20) is built on this signal as the Laplace transform is built on e^{st}.

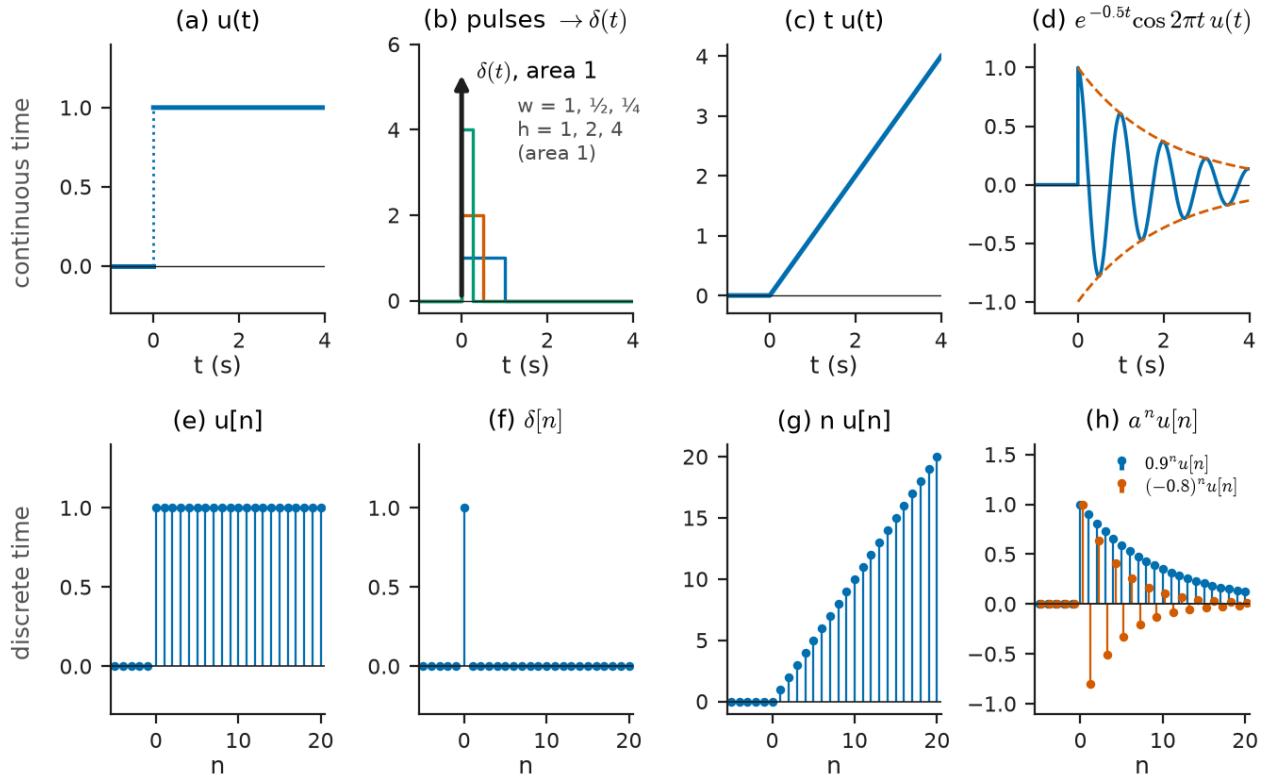

Figure 15.4 — The elementary signals: step, approximations to the impulse, ramp, decaying exponential and damped sinusoid, in continuous and discrete time.

Figure 15.4 collects the elementary signals. Each later chapter uses them as test inputs or as building blocks: steps for transient response (Chapters 6–7, 34), impulses for system characterisation (Chapter 17), exponentials for frequency analysis (Chapters 18 and 20).

15.8 Time shifting and scaling

Time shifting

The signal $x(t - t_0)$ is $x(t)$ delayed by t_0 if $t_0 > 0$ (moved to the right on the time axis) or advanced if $t_0 < 0$. Whatever x did at $t = 0$, $x(t - t_0)$ does at $t = t_0$. In discrete time, $x[n - n_0]$ is a delay by n_0 samples, with n_0 an integer.

Delays are everywhere: propagation along a cable (5 ns per metre in coax, Chapter 14), the travel time of sound (2.9 ms per metre in air), the round trip of a radar pulse (6.67 μs per kilometre of range), the latency of a digital filter, and the transport delay of material on a conveyor in a process plant. A pure delay changes no property of the signal except its timing, and for a sinusoid it is equivalent to a phase shift $-\omega t_0$, a phase proportional to frequency (Chapter 13).

Time scaling

The signal $x(at)$ with $a > 1$ is $x(t)$ **compressed** in time by the factor a: events happen a times faster. With $0 < a < 1$ it is **expanded**. Playing an audio recording at double speed is $x(2t)$; every frequency doubles and the pitch rises by an octave. Time scaling also appears physically. The Doppler effect compresses the echo from an approaching target. For sound from a source approaching at 30 m/s, the received signal is the transmitted one scaled by $343/(343 - 30) = 1.096$, so a 1 kHz tone is heard at 1096 Hz. For electromagnetic waves the scaling is tiny, and it is usually described as a frequency shift: a 10 GHz radar sees a 30 m/s target shifted by $2vf/c = 2.0$ kHz.

In discrete time, scaling is not a simple change of variable because an must be an integer. **Decimation** $x[Mn]$ keeps every M-th sample and discards the rest, losing information unless the signal was oversampled; **expansion** inserts zeros between samples. Both are central to multirate processing (Chapter 24).

Reflection

With $a = -1$, $x(-t)$ is the **time reversal** of $x(t)$, its mirror image about $t = 0$. Time reversal appears in convolution (Chapter 17), in matched filters, whose impulse response is the time-reversed expected pulse (Chapter 27), and in the even–odd decomposition of Section 15.4.

Combined operations

The general transformation of the independent variable is $x(at - b)$. Because it is easy to apply the operations in the wrong order, the safest method is to substitute: the value $x(\tau)$ appears in the new signal at the time t for which $at - b = \tau$, that is $t = (\tau + b)/a$. Equivalently, shift first, then scale: form $v(t) = x(t - b)$, then $v(at) = x(at - b)$.

> **Status of the principle: Transformation of the independent variable**
> *Derived*. The rule $t = (\tau + b)/a$ is direct substitution; it holds for any signal and any real $a \neq 0$, while in discrete time only integer shifts and integer decimation or expansion factors are defined.

Worked example — shifting and scaling a pulse. Let $x(t)$ be a rectangular pulse of height 1 between $t = 0$ and $t = 2$ s, followed by a linear fall to zero at $t = 3$ s. Find $y(t) = x(2t - 3)$. The pulse's start, $\tau = 0$, maps to $t = (0 + 3)/2 = 1.5$ s; its top's end, $\tau = 2$, maps to $t = 2.5$ s; the end of the fall, $\tau = 3$, maps to $t = 3$ s. The result is a pulse of height 1 from 1.5 s to 2.5 s with a fall to zero at 3 s: half as long and centred later. The wrong order (scale first, giving $x(2t)$, then shift by 3 to give $x(2(t - 3)) = x(2t - 6)$) would place it at 3 to 4.5 s. Figure 15.5 shows each step.

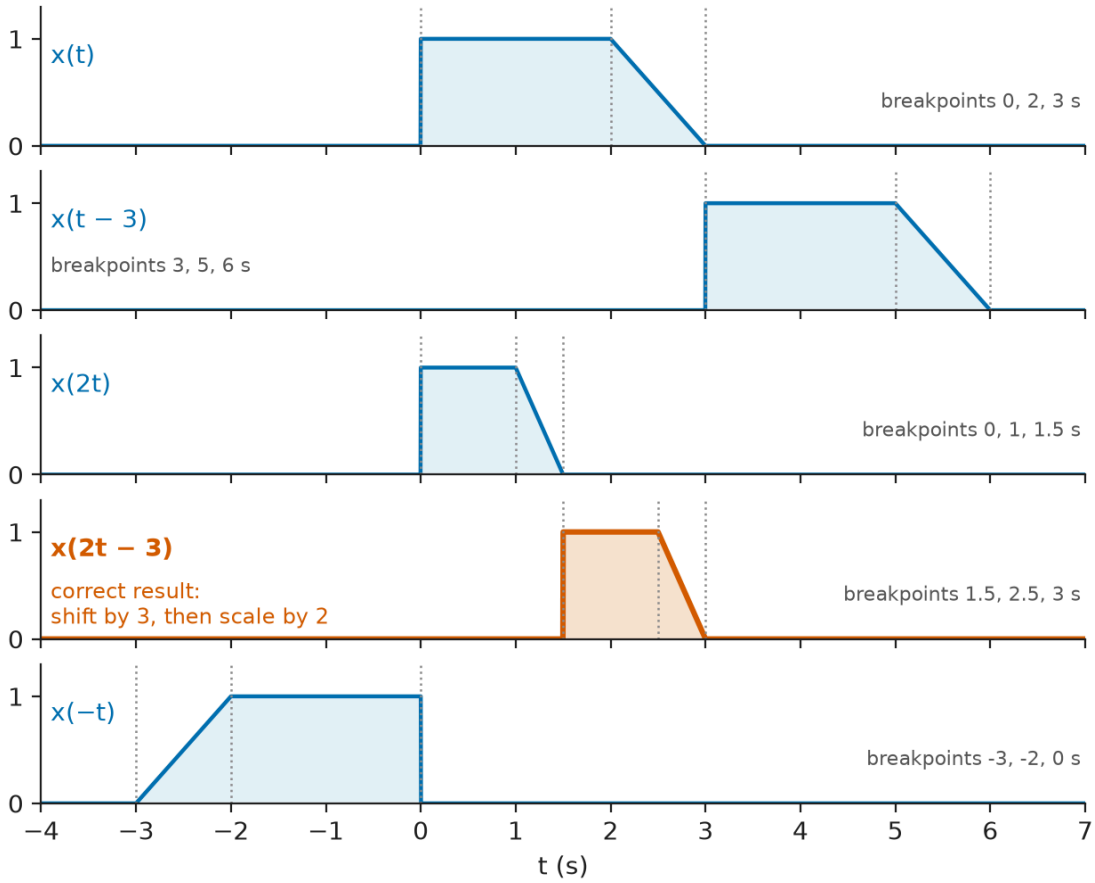

Figure 15.5 — Time shifting, scaling and reflection of a pulse: x(t), x(t-3), x(2t), x(2t-3) and x(-t).

Time shifts in systems: a preview

Whether a system treats a delayed input by producing an equally delayed output is the property of **time invariance**, defined in Chapter 16. For time-invariant systems, the response to $x(t - t_0)$ is $y(t - t_0)$; for time-varying systems, such as a sampled-data system observed between samples or a mixer driven by a local oscillator, it is not. The operations of this section are therefore the language in which system properties are stated.

15.9 Signal operations

Amplitude operations

Operations on the dependent variable are as important as those on time. The basic ones, and where each appears in hardware, are:

Operation	Definition	Typical realisation
Amplitude scaling	$y(t) = a\,x(t)$	Amplifier, attenuator, sensor gain
Offset	$y(t) = x(t) + c$	Bias network, DC level shift for an ADC
Addition	$y(t) = x_1(t) + x_2(t)$	Summing amplifier, superposition of signal and noise, mixing desk
Multiplication	$y(t) = x_1(t)\,x_2(t)$	Mixer, modulator, gating, windowing, power measurement $v \cdot i$
Differentiation	$y(t) = dx/dt$	Capacitor current, inductor voltage, edge detection
Integration	$y(t) = \int_{-\infty}^{t} x(\tau)\,d\tau$	Capacitor voltage from current, energy meter, integral controller
Nonlinear mapping	$y(t) = g(x(t))$	Rectifier, limiter, quantizer, square-law detector

In discrete time, differentiation becomes the **first difference** $y[n] = x[n] - x[n-1]$ and integration the **running sum** $y[n] = \sum_{k \le n} x[k]$, the digital analogues used in every software controller and digital meter.

Addition and multiplication

Addition is the operation behind superposition: a measured signal is the sum of the wanted signal, interference and noise, and much of signal processing consists of undoing that sum. Multiplication is different in kind. When a signal is multiplied by a sinusoid, its spectrum is shifted in frequency; this is **modulation**, the basis of radio (Chapters 26–27). Amplitude modulation of a 1 MHz carrier by an audio tone,

$$y(t) = \bigl[1 + m\cos(2\pi f_m t)\bigr]\cos(2\pi f_c t)$$

is an offset, a scaling and a multiplication combined. Multiplication by a rectangular pulse **gates** or **windows** a signal, keeping only a segment of it; every finite measurement is a window applied to an infinitely long signal, with consequences for spectral analysis (Chapter 22). Multiplication of voltage by current gives instantaneous power, which an electronic energy meter computes sample by sample.

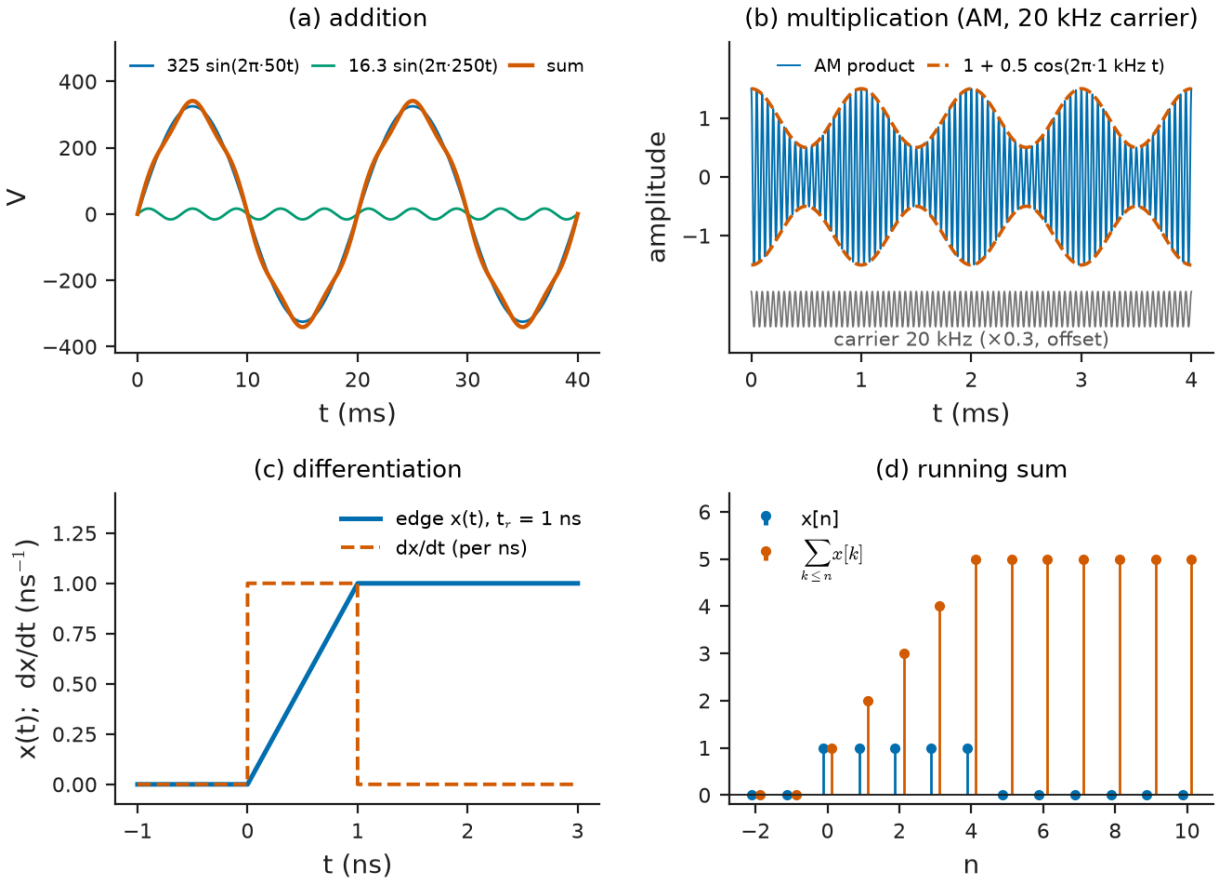

Figure 15.6 — Signal operations illustrated: addition of a 50 Hz signal and its fifth harmonic, multiplication of a carrier by an offset message (AM), differentiation of a trapezoidal edge, and the running sum of a sampled pulse.

Differentiation and integration

Differentiation emphasises rapid changes and suppresses slow ones; integration does the opposite. These are the time-domain views of high-pass and low-pass behaviour. A capacitor's current is $C\,dv/dt$, so a fast logic edge produces a current spike; an inductor's voltage is $L\,di/dt$, so interrupting current in a relay coil produces a high voltage that must be clamped. Integration recovers velocity from an accelerometer and displacement from velocity, which is why inertial navigation accumulates drift: any small offset is integrated into a growing ramp, and twice-integrated into a parabola. Differentiation amplifies noise, whose rapid fluctuations have large derivatives; practical differentiators are always band-limited.

Signal decomposition into elementary pieces

The operations and elementary signals combine to build complex waveforms. A pulse-width-modulated motor drive voltage is a sum of shifted, scaled steps. A sampled signal is a sum of scaled, shifted impulses. A periodic signal is a sum of scaled, shifted complex exponentials (its Fourier series), and an aperiodic one an integral of them (its Fourier transform). The general idea is to express an arbitrary signal as a combination of simple ones whose effect on a system is known, and then use linearity to find the system's response to the whole. That programme, which runs through Chapters 16–20, depends only on the vocabulary of this chapter.

> **Status of the principle: Decomposition of signals into elementary components**
> *Derived*. The representations as sums of shifted impulses, steps or complex exponentials are exact for well-behaved signals (with convergence conditions recapped in Chapter 18); their usefulness for system analysis depends on the system being linear, which is the subject of Chapter 16.

Operations on discrete-time signals in practice

Worked example — a moving-average smoother. A process sensor sampled at 10 Hz gives temperature readings corrupted by fast fluctuations. A five-point moving average,

$$y[n] = \frac{1}{5}\big(x[n] + x[n-1] + x[n-2] + x[n-3] + x[n-4]\big)$$

combines delay, scaling and addition. Fed with the unit sample $\delta[n]$, it outputs five samples of 0.2 at $n = 0, \dots, 4$ and zero otherwise: its impulse response is a short rectangular pulse. Fed with a unit step, it outputs 0.2, 0.4, 0.6, 0.8 and then 1.0 from $n = 4$ onward: the running sum of the impulse response, a ramp that rises over 0.5 s (five samples at 0.1 s). The step response shows that the smoother delays a sudden temperature change by an average of 2 samples (0.2 s) and spreads it over 0.5 s. That trade between smoothing and delay is the most basic fact of filtering, and the steps of this example (impulse response, step response as its running sum) are developed generally as convolution in Chapter 17.

What remains open

Open question	Where it is answered
What properties (linearity, time invariance, causality, stability) make a system tractable?	Chapter 16 (System Fundamentals)
How does the impulse response determine a system's response to any input?	Chapter 17 (Linear Time-Invariant Systems)
How is a signal represented as a sum of complex exponentials, and what is its spectrum?	Chapter 18 (Frequency-Domain System Analysis)
When does a sequence of samples represent a continuous signal exactly, and what is aliasing?	Chapter 19 (Sampling and Reconstruction)
How are sequences and difference equations analysed with the Z-transform?	Chapter 20 (Discrete-Time Signals and Systems)
What does quantization do to a discrete-time signal?	Chapters 19 and 21
Why does a finite record of a periodic signal show leakage in its spectrum?	Chapter 22 (Discrete Fourier Transform)
How are random signals and noise described by power and correlation?	Chapter 29 (Random Signals and Communication Noise)
How are generalised functions such as $\delta(t)$ defined rigorously?	*The Mathematics Universe*

Next: Chapter 16 turns from signals to the systems that act on them, and defines the properties that decide which analysis methods apply.

Chapter 16 — System Fundamentals

Part IV · Signals and Systems

A biomedical engineer is commissioning a new bedside ECG monitor. On the bench, fed from a calibrated simulator, the trace is perfect. On the ward it is not. With the patient lying still, the baseline wanders slowly up and down with breathing; when the patient lifts an arm, the trace jumps off the screen and takes several seconds to come back; a fine 50 Hz ripple rides on every beat; and when the engineer doubles the simulator amplitude to check the gain, the recorded R-wave grows by slightly less than a factor of two. Each of these complaints is a statement about the *system* between the electrodes and the screen, not about the heart. The baseline wander tests whether the filters respond the same way at every moment. The slow recovery after a large artefact tests whether the amplifier stays in its linear range and how long its memory of the past is. The ripple asks what the filter does to a 50 Hz input. The missing gain asks whether doubling the input really doubles the output.

Chapter 15 described signals. This chapter describes what acts on them. It introduces a small set of yes-or-no properties — linear or not, time-invariant or not, causal or not, stable or not, with memory or without — that classify a system before any detailed analysis is attempted, and it shows the main ways an input–output relationship can be written down. These properties decide which tools are allowed later: convolution and the frequency response (Chapters 17 and 18) are valid *only* for systems that pass the linearity and time-invariance tests.

> **What exactly is a "system", and which of its properties can be decided from its input–output description alone?**
> **Why do linearity and time invariance matter so much, and how do real circuits approximate them? What do causality and stability mean precisely, and how are they tested?**

Concept	Engineering destination
System as an input–output map	Amplifiers, filters, sensors, channels, controllers, power networks
Continuous and discrete systems	Analog front ends, ADC/DSP chains, sampled-data control
Memory	Capacitors and inductors, filters, integrators, echo paths
Linearity, superposition	Small-signal amplifiers, harmonic analysis in power systems, distortion budgets
Time invariance	Filters and channels with fixed parameters; failure in modulators, switched circuits, fading radio channels
Causality	Real-time filtering, control loops, latency budgets; non-causal offline processing
BIBO stability	Filter design, integrator drift, feedback amplifiers, power-system oscillations
Input–output representations	Differential and difference equations, block diagrams, state-space models, impulse and frequency responses

Roadmap. What a system is and how systems are classified (16.1) → continuous, discrete and mixed systems (16.2) → memory (16.3) → linearity and its approximations (16.4) → time invariance (16.5) → causality (16.6) → stability (16.7) → ways of writing the input–output relationship (16.8).

16.1 Systems and system classification

A system as an operator

A **system** is any process that takes an input signal and produces an output signal. The word covers an RC low-pass filter, a power transformer, a loudspeaker, a radio channel between a base station and a phone, a digital filter running on a microcontroller, the arterial tree between heart and fingertip, and a building swaying under wind load. What all of these share, for the purposes of this part of the book, is that we can describe them by a rule that maps an input signal to an output signal:

$$y(t) = \mathcal{T}\{x(t)\} \qquad \text{or} \qquad y[n] = \mathcal{T}\{x[n]\}$$

The operator $\mathcal{T}$ acts on the *whole* input signal, not on a single value. This point matters: $y(t_0)$ may depend on x at times other than t_0. A capacitor voltage at noon depends on all the current that flowed into it before noon.

Many real systems have several inputs and outputs — a three-phase motor drive, a MIMO radio link, a stereo amplifier. The properties below extend to vector signals without change, and most of Part IV is written for the single-input single-output case.

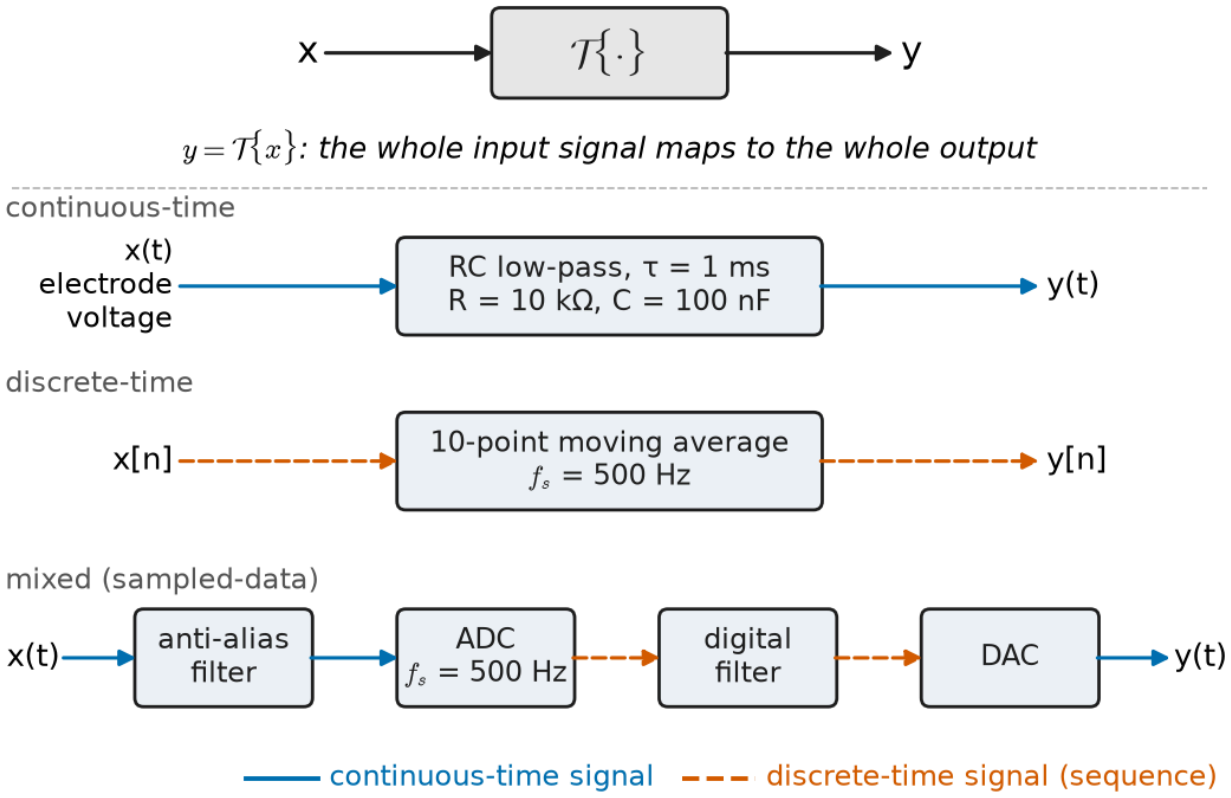

Figure 16.1 — A system maps a whole input signal to an output signal; the same abstraction covers analog, digital and mixed chains.

Figure 16.1 shows the abstraction for three chains that appear in the ECG monitor of the opening: a continuous-time RC filter, a discrete-time moving average, and a mixed chain that converts between the two.

Why classify

Detailed models — differential equations, transfer functions, circuit simulations — are expensive. Before building one, an engineer wants to know which mathematics is even legitimate. Five properties do most of that work:

Property	Question it answers	If yes, it allows	Typical violator
Memoryless	Does y at time t depend only on x at time t?	Static transfer-characteristic analysis	Any circuit with C or L
Linear	Does superposition hold?	Decomposition of inputs, phasors, Fourier methods	Diode, saturating amplifier, squarer
Time-invariant	Does a delayed input give an equally delayed output?	Impulse response, convolution, transfer functions	Mixer, chopper, fading channel
Causal	Does the output depend only on present and past inputs?	Real-time implementation	Zero-phase offline filter
BIBO stable	Does every bounded input give a bounded output?	Safe operation, meaningful frequency response	Integrator, oscillator, unstable loop

Two further properties are occasionally needed. A system is **invertible** if distinct inputs always produce distinct outputs, so that the input can be recovered; a channel equaliser or an echo canceller tries to build the inverse of a channel, which is possible only if the channel is invertible (a squarer $y = x^2$ is not, because x and $-x$ give the same output). A system is **real** if real inputs give real outputs; almost all physical systems are, while the complex-baseband models used in communication (Chapter 27) are not.

Each property is tested against the *definition*, applied to arbitrary inputs. A single counter-example is enough to show a property fails; showing that it holds requires an argument for all inputs. The following sections give each definition, the standard tests and the engineering systems that pass or fail.

16.2 Continuous and discrete systems

Three kinds of system

A **continuous-time system** maps a continuous-time input to a continuous-time output: every passive circuit, analog amplifier, mechanical structure and propagation channel. Its natural mathematical description is a differential equation.

A **discrete-time system** maps a sequence $x[n]$ to a sequence $y[n]$. Discrete-time systems exist as algorithms — a digital filter, a data-compression stage, a control law executed every millisecond — but also arise naturally where data are counted per period: monthly energy consumption, daily peak load, per-revolution measurements of a rotating machine. The natural description is a difference equation.

A **hybrid** or **sampled-data system** contains both. Every digital instrument, digital audio chain, software-defined radio and digital controller is hybrid: a continuous-time plant or sensor, an ADC, a discrete-time algorithm, a DAC and usually a reconstruction filter. The interfaces are covered in Chapter 19 (sampling and reconstruction) and Chapter 21 (DSP workflow); within each section of the chain the classification below applies separately.

System	Domain	Typical rate or bandwidth
RC anti-alias filter in an audio ADC	continuous	cut-off around 20–24 kHz
Digital audio equaliser	discrete	44.1 or 48 kHz sample rate
ECG acquisition and filtering	hybrid	250–1000 Hz sample rate typical
Protection relay (digital)	hybrid	16–96 samples per mains cycle typical
Power-system electromechanical dynamics	continuous	oscillation modes about 0.1–2 Hz
Digital motor current loop	hybrid	8–20 kHz control rate typical
LTE baseband receiver	discrete	30.72 MHz sample rate for a 20 MHz carrier

The two are not separate subjects

The same physical system often has both descriptions. An RC filter with time constant $\tau = RC$ obeys

$$\tau \frac{dy}{dt} + y(t) = x(t)$$

If the input is held constant over each interval of length T (as a DAC does) and the output is read every T seconds, the exact relationship between samples is

$$y[n] = a\,y[n-1] + (1-a)\,x[n-1], \qquad a = e^{-T/\tau}$$

which is a discrete-time system. Section 16.8 returns to this example. The properties of this chapter are defined in parallel for both domains; wherever the text writes a statement in t, the same statement holds with n in place of t and integer shifts in place of real ones.

16.3 Memory and memoryless systems

Definition

A system is **memoryless** (static, instantaneous) if the output at each instant depends only on the input at that same instant:

$$y(t) = f\big(x(t), t\big)$$

Otherwise it has **memory** (it is dynamic). A resistor, $v(t) = R\,i(t)$, is memoryless. An ideal transformer, $v_2 = (N_2/N_1)v_1$, is memoryless. So is an ideal diode, an ideal comparator, a quantiser in an ADC and the static characteristic $y = a_1 x + a_2 x^2$ of an amplifier at low frequency. A capacitor is not:

$$v(t) = v(t_0) + \frac{1}{C}\int_{t_0}^{t} i(\lambda)\,d\lambda$$

The capacitor voltage summarises the entire past current. The state variable $v(t_0)$ is exactly the "memory" — a single number that carries everything about the past that the future needs (Chapter 35 builds state-space models on this idea).

Worked example — the memory of a capacitor. A 10 µF hold capacitor in a sample-and-hold circuit is at 0 V. A constant current of 2 mA charges it for 5 ms and then stops. While current flows, $dv/dt = i/C = 2 \times 10^{-3}/10^{-5} = 200$ V/s, so after 5 ms the voltage is $200 \times 0.005 = 1.0$ V. At $t = 8$ ms the input current is zero, yet the output is still 1.0 V. A memoryless system with zero input would have to give zero output; the capacitor does not, because its output records the integral of the past. The same arithmetic in reverse explains droop: a leakage of 10 nA discharges the capacitor at $10^{-8}/10^{-5} = 1$ mV/s, which is why hold times in precision sample-and-hold circuits are kept short.

Degrees of memory

Memory can be short or long, finite or infinite:

- A **pure delay**, $y(t) = x(t - t_0)$, remembers exactly t_0 seconds of past input. A 1 km length of coaxial cable with velocity factor 0.66 is a delay of about 5.05 µs; an acoustic echo path in a room is a delay of tens of milliseconds.
- An **N-point moving average**, $y[n] = \frac{1}{N}\sum_{k=0}^{N-1} x[n-k]$, has finite memory of N samples.
- An **RC filter** has infinite memory that fades: the influence of an input t seconds ago decays as $e^{-t/\tau}$. After 5τ it is below 1 %.
- An **ideal integrator** or accumulator has infinite memory that never fades. This is useful (a charge amplifier, an energy meter, the integral term of a PID controller) and dangerous (any offset accumulates without limit, Section 16.7).

A practical rule: a system's memory length is the time over which a disturbance continues to affect the output. For the ECG monitor of the opening, the slow recovery after an arm movement is the memory of the 0.05 Hz high-pass filter (time constant $1/(2\pi \cdot 0.05) \approx 3.2$ s) — clinical monitors often reduce it by switching temporarily to a shorter time constant after a large artefact, which deliberately makes the system time-varying for a moment.

Differentiators

A differentiator $y = dx/dt$ — the inductor law $v = L\,di/dt$ is one — is usually classified as having memory, because the derivative at t needs the input in an arbitrarily small neighbourhood around t, not only the value at t. The memory is infinitesimally short, but it is enough to make the response depend on how the input is changing, and it is what lets an inductor produce a large voltage when a current is interrupted.

16.4 Linearity

Definition: superposition

A system is **linear** if, for any inputs x_1, x_2 and any constants a, b,

$$\mathcal{T}\{a\,x_1 + b\,x_2\} = a\,\mathcal{T}\{x_1\} + b\,\mathcal{T}\{x_2\}$$

The single statement contains two properties. **Additivity**: the response to a sum is the sum of the responses. **Homogeneity** (scaling): doubling the input doubles the output. Neither alone is sufficient

in general, although for physical systems with continuous characteristics additivity usually implies homogeneity.

The engineering value of linearity is decomposition. If a system is linear, a complicated input can be broken into simple pieces — impulses, steps, sinusoids — the response to each piece found once, and the answers added. Fourier analysis of harmonics in a power network, the phasor method of Chapter 8, the superposition theorem of Chapter 4 and convolution in Chapter 17 are all this idea.

> **Status of the principle: Superposition**
> *Derived.* For a system that satisfies the linearity definition, superposition is an exact mathematical consequence; whether a physical circuit satisfies the definition is a separate, approximate question about its components and signal levels.

Quick tests

Three consequences give fast tests:

1. **Zero in, zero out.** Setting $a = 0$ shows that a linear system must give $y = 0$ for $x = 0$ (with zero initial conditions). Any system with an output offset fails.
2. **Scaling.** Apply x and $2x$; the output must exactly double.
3. **Two-tone test.** Apply x_1, then x_2, then $x_1 + x_2$; compare.

Examples, checked against the definition:

System	Linear?	Reason
$y = Kx$ (ideal amplifier)	yes	direct
$y(t) = \int_{-\infty}^{t} x(\lambda)\,d\lambda$	yes	integral is linear
$y(t) = x(t)\cos\omega_c t$ (mixer)	yes	scaling and addition pass through the product with a fixed function
$y = 2x + 1$	no	fails zero-in, zero-out
$y = x^2$	no	$(x_1 + x_2)^2 \neq x_1^2 + x_2^2$
$y = \lvert x\rvert$ (rectifier)	no	fails for $a = -1$
$y[n] = \text{round}(x[n])$ (quantiser)	no	fails scaling
RC circuit with nonzero initial voltage	no (strictly)	output nonzero for zero input

The mixer row surprises many students: multiplying by a *known* time function is linear. Linearity concerns how the output depends on the input, not whether the formula contains a product. The mixer fails a different test (Section 16.5).

Incrementally linear systems and initial conditions

The system $y = 2x + 1$ is not linear, but its *change* in output is linear in the change in input. Such a system is called **incrementally linear**: it is a linear system plus a fixed response that does not depend on the input. A circuit with an initial capacitor voltage is exactly this:

$$y(t) = \underbrace{y_{zi}(t)}_{\text{zero-input}} + \underbrace{y_{zs}(t)}_{\text{zero-state}}$$

The zero-input response comes from stored energy; the zero-state response is linear in the input. Chapter 17 uses this split systematically. In practice, an amplifier with a DC offset, a sensor with a zero error or a circuit started from a nonzero state is analysed by removing the fixed part and treating the remainder as linear.

Linear models of nonlinear devices

No physical device is exactly linear. Resistors heat up and change value; capacitor dielectrics have voltage-dependent permittivity (class-2 ceramic capacitors can lose more than half their capacitance at rated DC bias); iron cores saturate; every amplifier clips at its supply rails. Linearity is an approximation that becomes good when signals are small enough. The standard way to obtain it is **linearisation** about an operating point: write the device law $y = f(x)$, choose a bias point X_0, and keep the first term of the Taylor series,

$$y \approx f(X_0) + f'(X_0)\,(x - X_0)$$

The constant term is the operating point; the slope $f'(X_0)$ is the small-signal gain. This is how transistor amplifiers are analysed (in *Electronics & Computing*), how power-flow models are linearised for stability studies, and how a strain gauge bridge is treated as linear in strain.

Worked example — how small is "small signal" for a diode? A silicon diode biased at $I_D = 1$ mA follows $i = I_S(e^{v/V_T} - 1)$ with $V_T \approx 25.85$ mV at 300 K. The small-signal resistance is $r_d = V_T/I_D = 25.85\ \Omega$, so the linear model predicts a current change $\Delta i = \Delta v / r_d$. The exact change for a voltage step Δv above the bias point is $\Delta i = I_D(e^{\Delta v/V_T} - 1)$. For $\Delta v = 1$ mV: linear 0.0387 mA, exact 0.0394 mA, error 1.9 %. For $\Delta v = 5$ mV: linear 0.193 mA, exact 0.213 mA, error 9.4 %. For $\Delta v = 10$ mV: linear 0.387 mA, exact 0.472 mA, error 18 %. The diode is "linear" to within a few percent only for excursions of a few millivolts — a small fraction of V_T. This is the quantitative content of the phrase "small-signal".

> **Status of the principle: Small-signal linearisation**
> *Approximate.* A smooth nonlinear device behaves linearly about its operating point when the signal excursion is small compared with the scale over which its slope changes (for a diode, a few millivolts compared with $V_T \approx 26$ mV); it fails for large signals, at clipping and at discontinuities such as switching.

What nonlinearity does to signals

The signature of nonlinearity is **new frequencies**. A linear system can change the amplitude and phase of each sinusoid it is given, but never creates a frequency that was not in the input (Chapter 17 proves this using eigenfunctions). A nonlinear system does. Consider an amplifier whose static characteristic is

$$y = a_1 x + a_2 x^2$$

driven by $x = A\cos\omega t$. Using $\cos^2\theta = \tfrac{1}{2}(1+\cos 2\theta)$,

$$y = \frac{a_2 A^2}{2} + a_1 A\cos\omega t + \frac{a_2 A^2}{2}\cos 2\omega t$$

The squared term creates a DC shift and a **second harmonic**. With two input tones ω_1 and ω_2, it also creates sum and difference frequencies $\omega_1 \pm \omega_2$ — **intermodulation**.

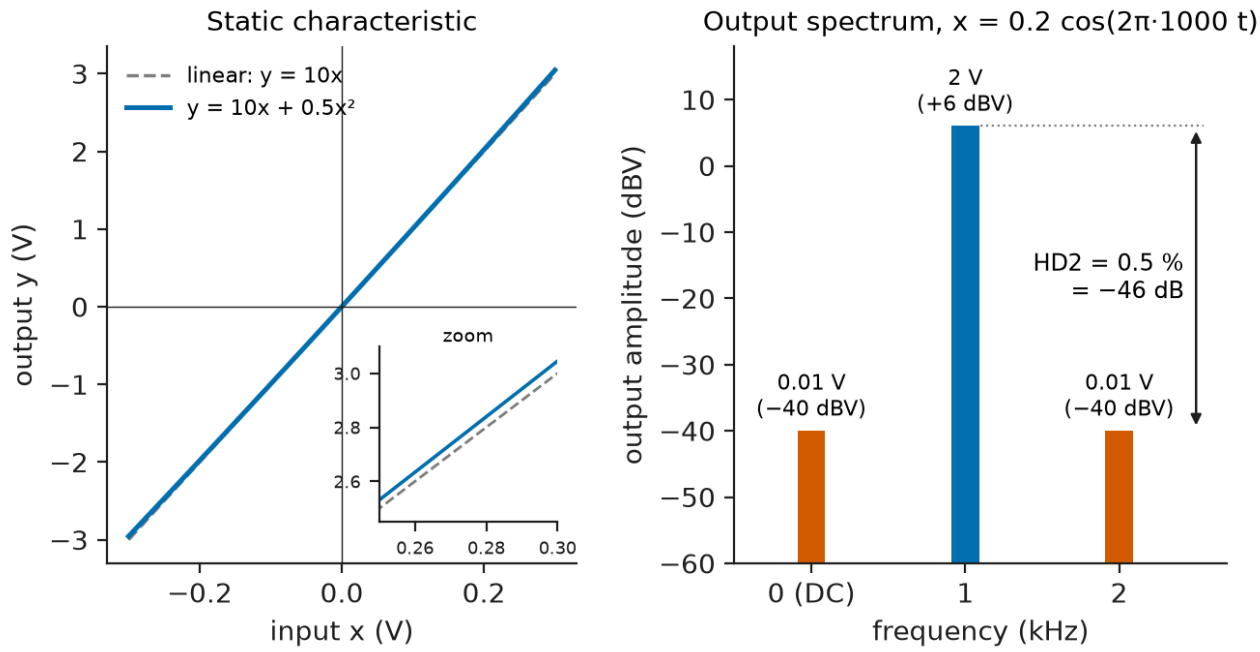

Figure 16.2 — A weak quadratic term in an amplifier characteristic creates a DC shift and a second harmonic that a linear system could never produce.

Worked example — second-harmonic distortion of an amplifier. An audio preamplifier has $a_1 = 10$ and a weak curvature $a_2 = 0.5\ \text{V}^{-1}$ (Figure 16.2). The input is a 1 kHz tone of amplitude $A = 0.2$ V. The fundamental output is $a_1 A = 2.0$ V. The second-harmonic amplitude at 2 kHz is $a_2 A^2/2 = 0.5 \times 0.04/2 = 0.010$ V, and there is a DC shift of the same 0.010 V. The second-harmonic distortion is $0.010/2.0 = 0.5\ \%$, or $20\log_{10}(0.005) = -46$ dB. Because the ratio is $a_2 A/(2a_1)$, it is proportional to amplitude: halving the input to 0.1 V halves the distortion to 0.25 %. This proportionality is a practical diagnostic. Distortion that falls as the signal falls is curvature of the characteristic; distortion that stays roughly constant at low level (crossover distortion in a class-B output stage, quantisation in an ADC) comes from a nonlinearity near zero, where small-signal linearisation does not help.

The same mathematics describes harmonic currents drawn from the mains by rectifier loads (Chapter 9 and Chapter 18), the third-order intercept point of a radio receiver's low-noise amplifier (Chapter 32),

and the spurious tones in a digitised signal.

Linearity in practice

Device or system	Typical linearity figure
Metal-film resistor	voltage coefficient below about 1 ppm/V
C0G/NP0 ceramic capacitor	essentially no voltage coefficient
X7R ceramic capacitor at rated DC bias	capacitance loss often 30–70 %
Good audio power amplifier, rated power	THD about 0.001–0.05 %
Instrumentation amplifier	gain nonlinearity about 1–10 ppm of full scale
16-bit precision ADC	integral nonlinearity about ±1–2 LSB
Current transformer, protection class 5P	composite error 5 % at the accuracy-limit current
Mobile-phone power amplifier	operated near compression; linearised by digital predistortion

The last row shows that linearity is also a design trade-off. A transmitter amplifier is most efficient near saturation, where it is least linear; communication standards specify how much spectral regrowth is allowed into the next channel, and the engineer chooses the operating point accordingly.

16.5 Time invariance

Definition

A system is **time-invariant** (shift-invariant) if a time shift of the input produces the same time shift of the output and nothing else. If $y(t) = \mathcal{T}\{x(t)\}$, then for every t_0

$$\mathcal{T}\{x(t - t_0)\} = y(t - t_0)$$

and in discrete time, for every integer n_0, $\mathcal{T}\{x[n - n_0]\} = y[n - n_0]$. Informally: the system behaves the same today as it did yesterday. An RC filter built from stable components is time-invariant; the same filter with a resistor that warms up during operation is, strictly, not, though the variation may be negligible over the time of interest.

The test is mechanical. Compute the response to the shifted input; separately, shift the original output; compare. If the system's formula contains t explicitly (other than through x), or contains time scaling such as $x(2t)$, it usually fails.

Examples

System	Time-invariant?	Comment
$y(t) = x(t - t_0)$ (delay)	yes	
$y(t) = \int_{-\infty}^{t} x(\lambda)\, d\lambda$	yes	
$\tau\, dy/dt + y = x$ with constant τ	yes	coefficients constant
$y(t) = x(t)\cos \omega_c t$ (mixer, AM modulator)	no	gain depends on absolute time
$y(t) = x(2t)$ (playback at double speed)	no	shifts are compressed
$y[n] = x[2n]$ (downsampler)	no	a shift of 1 at the input is not a shift at the output
Mobile radio channel with moving user	no	multipath gains change with position
Switched-capacitor filter	no, strictly	periodically time-varying; nearly LTI for signals well below the clock
Battery with state-of-charge-dependent resistance	no, slowly	treated as LTI over short intervals

Worked example — a mixer is linear but not time-invariant. An AM modulator multiplies its input by a 1 MHz carrier, $y(t) = x(t)\cos(2\pi \cdot 10^6\, t)$. Feed it a short rectangular pulse of amplitude 1 V and width 0.2 µs starting at $t = 0$. During the pulse the output is $\cos(2\pi 10^6 t)$, which starts at its positive peak, $+1$ V. Now delay the input by $t_0 = 0.5$ µs. The output during the delayed pulse is $\cos(2\pi 10^6 t)$ for $0.5 \le t \le 0.7$ µs, which starts at $\cos(\pi) = -1$ V. Shifting the *original* output by 0.5 µs would instead give $\cos\left(2\pi 10^6 (t - 0.5\times 10^{-6})\right)$, which starts at $+1$ V. The two differ in sign, so the system is time-varying. Linearity still holds: doubling the pulse doubles the output at every instant. This combination — linear, time-varying — is exactly what a modulator needs to be, because its purpose is to move a signal to a new frequency, something no LTI system can do (Chapter 26).

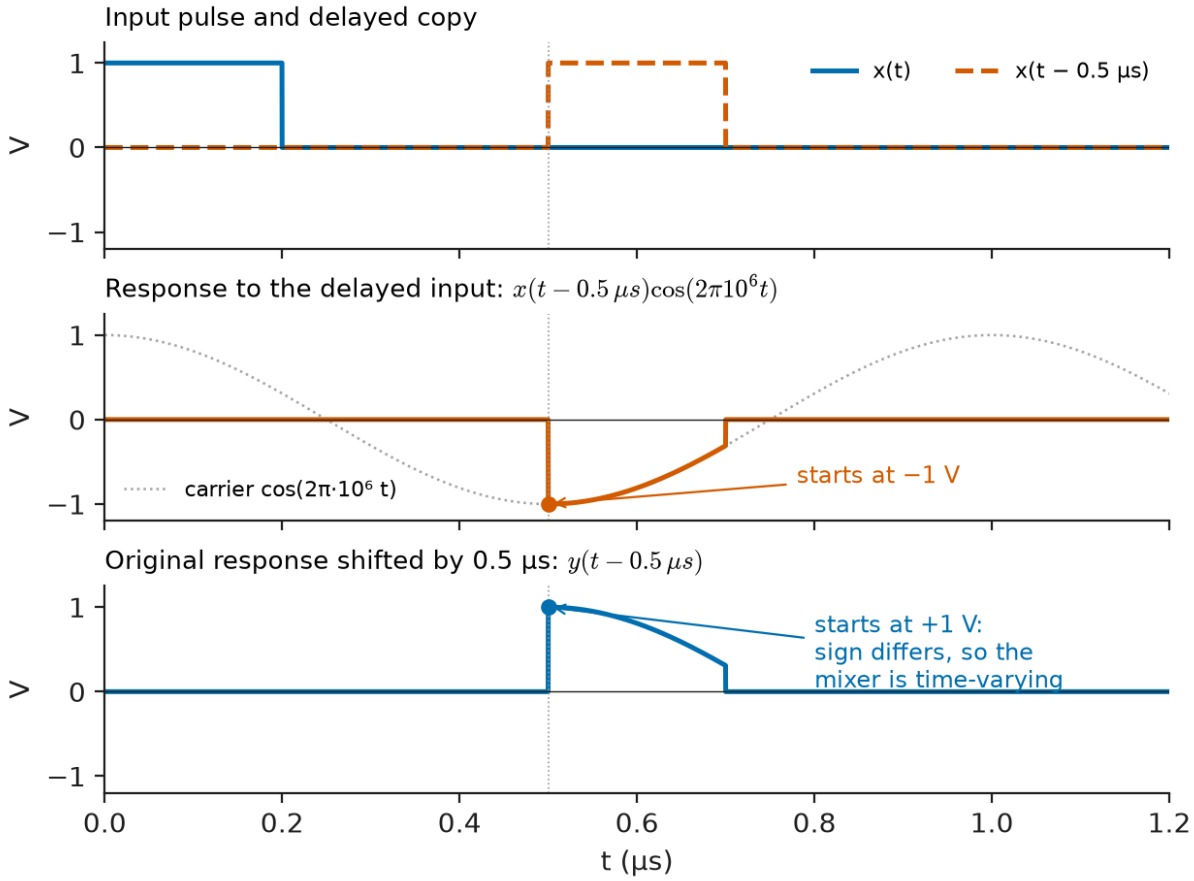

Figure 16.3 — Time-invariance test on a 1 MHz mixer: delaying the input does not simply delay the output.

Figure 16.3 shows the test graphically. A related family, **periodically time-varying** systems — mixers, choppers, switched-mode converters, sample-and-hold circuits — are time-varying but in a regular way, and they have their own analysis methods (averaging for converters, conversion matrices for mixers).

Linear time-invariant systems

The systems that satisfy both linearity and time invariance form the class **LTI** (in discrete time sometimes LSI, linear shift-invariant). They are the central object of Part IV because, as Chapter 17 shows, an LTI system is completely described by one signal, its impulse response, and its output is the convolution of input with impulse response. Every lumped circuit of resistors, capacitors, inductors, coupled coils and linear dependent sources with constant values is LTI once the initial conditions are separated off. Many systems that are not exactly LTI are treated as LTI over a limited range of amplitude and time: a small-signal amplifier over a frequency band, a radio channel over a few milliseconds (its coherence time), a power network around a load-flow operating point.

16.6 Causality

Definition

A system is **causal** if its output at any time depends only on the present and past values of the input:

$$y(t_1) \text{ depends only on } x(t) \text{ for } t \le t_1$$

Equivalently, if two inputs agree up to time t_1, their outputs agree up to time t_1. For a linear system with zero initial conditions this reduces to: if $x(t) = 0$ for $t < t_0$, then $y(t) = 0$ for $t < t_0$. A causal system cannot respond before it is excited.

> **Status of the principle: Causality of physical systems**
> *Fundamental.* A physical system operating in real time cannot respond to an input that has not yet arrived; this follows from the ordering of cause and effect and, for electromagnetic systems, from the finite speed of propagation.

Every circuit, mechanical structure and propagation path is causal when its input and output are physical quantities measured on the same clock. The interest of the definition lies in the systems that are *not* causal, all of which are algorithms working on stored data:

- A **centred moving average**, $y[n] = \tfrac{1}{3}\big(x[n-1] + x[n] + x[n+1]\big)$, needs the next sample.
- A **zero-phase filter**, made by filtering a record forwards and then backwards, removes phase distortion completely (Chapter 18) but uses the whole record.
- **Interpolation** that reconstructs a value between samples using samples on both sides.
- **Smoothing** of a recorded seismogram, biomedical record or vibration log.

These are perfectly legitimate offline. In real time, a non-causal operation is made causal by **delaying** its output: a centred N-point average becomes causal if its output is labelled $(N-1)/2$ samples later. The price is **latency**.

Worked example — the latency of a mains-hum filter. An ECG front end samples at $f_s = 500$ Hz. A 10-point moving average,

$$y[n] = \frac{1}{10}\sum_{k=0}^{9} x[n-k]$$

is causal. Averaging over exactly ten samples spans 20 ms, one full cycle of 50 Hz mains, so 50 Hz interference averages to zero; its gain is exactly zero at 50 Hz and also at 100 Hz, 150 Hz and so on. The gain at 10 Hz is 0.936 and at 25 Hz is 0.639, so it also smooths the QRS complex, whose content extends well above 25 Hz — Chapter 23 shows how to do better. The filter's output corresponds to the input $(N-1)/2 = 4.5$ samples earlier, a delay of $4.5/500 = 9$ ms. For a monitor display, 9 ms is irrelevant. For a pacemaker that must sense an R-wave and respond, or for a protection relay that must trip within one cycle, every millisecond of filter latency is budgeted. Making the filter centred (non-causal) would remove the label offset in an offline plot, but cannot remove the physical waiting time.

Causality and time variable

Causality refers to time. When the independent variable is not time — position along an image row, distance along a pipeline in a leak-detection survey, frequency in a spectrum — "causal" has no physical meaning, and image filters are routinely symmetric (non-causal). The concept also needs care with **predictors**: a system that estimates $x(t+\Delta)$ from past data is causal (it uses the past), even though its output refers to the future.

16.7 Stability

BIBO stability

Several definitions of stability exist. The one used for input–output descriptions is **bounded-input bounded-output (BIBO) stability**: a system is BIBO stable if every bounded input produces a bounded output,

$$|x(t)| \le B_x < \infty \text{ for all } t \quad \Longrightarrow \quad |y(t)| \le B_y < \infty \text{ for all } t$$

The bound B_y may depend on B_x but must be finite for every bounded input. A single bounded input that produces an unbounded output — even one carefully chosen — shows the system is unstable.

Stability is the property engineers most need and most often lose. An amplifier with too much feedback oscillates; a digital filter with a coefficient quantised the wrong way grows without bound; a power system after a fault can swing apart; an integrator fed with a small offset drifts to its rail. For LTI systems the test reduces to a condition on the impulse response (absolutely integrable or summable) and, equivalently, on pole locations — all poles in the left half of the s-plane or inside the unit circle of the z-plane. Chapter 17 derives this; Chapters 11, 20 and 34 use it.

> **Status of the principle: BIBO stability**
> *Derived*. BIBO stability is a mathematical property of the input–output map; for LTI systems it is equivalent to absolute integrability of the impulse response, a result proved from the convolution integral in Chapter 17.

Examples

Moving average. For the 10-point average, $|y[n]| \le \frac{1}{10}\sum |x[n-k]| \le B_x$. Every bounded input gives an output bounded by the same value: stable.

RC low-pass. The output is a weighted average of the past input with weights $e^{-t/\tau}/\tau$ that integrate to 1; the output can never exceed the largest input magnitude. Stable.

Integrator and accumulator. The accumulator $y[n] = y[n-1] + x[n]$ is the discrete integrator. A bounded constant input $x[n] = c$ gives $y[n] = (n+1)c$, which grows without bound. Unstable — even though no individual sample is large and nothing in the system "explodes".

Worked example — integrator drift. A vibration-monitoring system integrates an accelerometer signal to obtain velocity, using an accumulator at $f_s = 1$ kHz with scaling $T = 1$ ms per sample. The ADC has an uncorrected offset equivalent to 1 m/s² (1 % of a 100 m/s² full scale). After one second the integrated velocity error is $1 \times 1 = 1$ m/s; after one minute, 60 m/s. The same mechanism makes a MEMS rate gyroscope with a 0.01 °/s bias accumulate a heading error of $0.01 \times 3600 = 36°$ per hour. The physical sensor is fine; the integrator is not BIBO stable. Practical systems replace the pure integrator by a "leaky" one — a low-pass filter with a long but finite time constant, or a high-pass filter before the integrator — which is stable, at the price of losing the slowest components of the true signal.

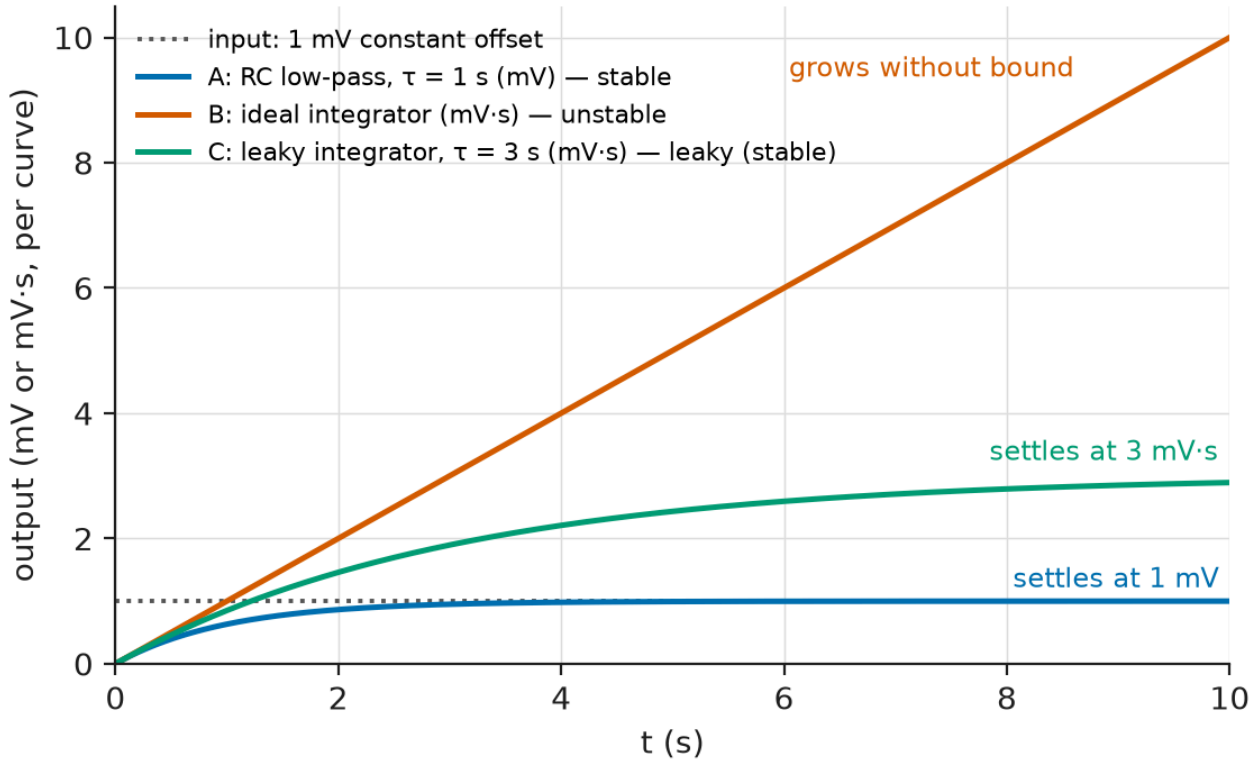

Figure 16.4 — A bounded input gives a bounded output from an RC filter but an unbounded drift from an integrator.

Figure 16.4 compares the three behaviours for the same bounded input.

Marginal stability. An ideal LC tank, or a system with a pair of poles exactly on the imaginary axis, responds to a brief kick with a sinusoid that neither grows nor decays. Its response to some bounded inputs is bounded, but a sinusoidal input at exactly its resonant frequency produces an output whose amplitude grows linearly in time. It is therefore **not** BIBO stable. Oscillators are deliberately built near this condition, with amplitude limited by a nonlinearity; filters must keep well away from it.

Other notions of stability

BIBO stability is about forced response. **Internal (Lyapunov) stability** concerns the natural response: a system is asymptotically stable if every initial state decays to zero with zero input. For LTI systems without hidden pole–zero cancellations the two coincide; when a cancellation hides an unstable mode, the input–output map may look stable while an internal variable grows — a capacitor voltage drifting to the supply in a circuit whose output looks fine. Chapter 35 returns to this through controllability and observability. Power-system engineers additionally distinguish **small-signal**

stability (the linearised model about an operating point) from **transient stability** (whether the nonlinear system recovers after a large disturbance such as a fault); only the first is a property of an LTI model.

System	BIBO stable?	Remark
Resistive divider	yes	memoryless with finite gain
Passive RLC filter with $R > 0$	yes	energy dissipated
Ideal integrator, accumulator	no	constant input → ramp
Lossless LC tank	no	resonant input → growing response
Op-amp with too much phase lag in feedback	no	oscillates
FIR digital filter	always yes	finite sum of bounded terms
IIR digital filter	depends	poles must lie inside the unit circle
Synchronous generator, lightly damped mode	yes, marginally	poorly damped electromechanical oscillation

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16.8 Input–output representations

The properties above describe *what kind* of system we have. To compute responses we need a concrete description. There are several, each suited to different tasks, and they are interchangeable for LTI systems.

Explicit input–output formula

The simplest description gives y directly in terms of x: $y = Kx$, $y = x^2$, $y(t) = x(t - t_0)$, $y[n] = \frac{1}{N}\sum x[n-k]$. Memoryless systems and FIR filters have such formulas. Many systems do not, because the output depends on its own past.

Differential equations

Continuous-time lumped systems are described by ordinary differential equations relating input and output, obtained from KCL, KVL and the element laws (Chapters 5–7) or from Newton's laws for mechanical systems (Chapter 33):

$$a_N \frac{d^N y}{dt^N} + \cdots + a_1 \frac{dy}{dt} + a_0 y = b_M \frac{d^M x}{dt^M} + \cdots + b_1 \frac{dx}{dt} + b_0 x$$

Constant coefficients make the system time-invariant; the absence of products and powers of x and y makes it linear (with zero initial conditions). The order N equals the number of independent energy-storage elements. The solution needs N initial conditions, which carry the system's memory of the past.

Difference equations

The discrete-time counterpart is the linear constant-coefficient difference equation,

$$\sum_{k=0}^{N} a_k \, y[n-k] = \sum_{k=0}^{M} b_k \, x[n-k]$$

which can be solved recursively: with $a_0 = 1$, $y[n]$ is computed from the current and past inputs and the past outputs. This is how every IIR digital filter, digital controller and recursive estimator is executed. If all a_k for $k \geq 1$ are zero the equation is an explicit FIR formula.

Block diagrams

A block diagram draws the system as an interconnection of a few primitive operations: gains, adders, integrators (continuous time) or unit delays (discrete time). Any linear constant-coefficient differential or difference equation can be drawn this way, and the diagram is directly a recipe for implementation — an analog computer, an active filter or a DSP program. Chapter 33 develops block-diagram algebra and signal-flow graphs.

State-space form

A state-space model writes an Nth-order equation as N coupled first-order equations in a state vector $\mathbf{q}$:

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}x(t), \qquad y(t) = \mathbf{C}\mathbf{q}(t) + D\,x(t)$$

The natural states of a circuit are capacitor voltages and inductor currents. State-space handles multiple inputs and outputs, nonzero initial conditions and time-varying systems naturally, and is the basis of modern control (Chapter 35) and of circuit simulators.

Impulse response and frequency response

For LTI systems two further descriptions are complete on their own: the **impulse response** $h(t)$ or $h[n]$ — the output when the input is a unit impulse — and the **frequency response** $H(j\omega)$, which gives the gain and phase shift at each frequency, along with its generalisation the **transfer function** $H(s)$ or $H(z)$. Chapter 17 shows how h determines the output for any input; Chapter 18 develops $H(j\omega)$; Chapters 11 and 20 use $H(s)$ and $H(z)$. These are the representations of choice for filters, channels and amplifiers because they can be measured directly: an impulse or step test for h, a swept sine or a noise excitation for $H(j\omega)$.

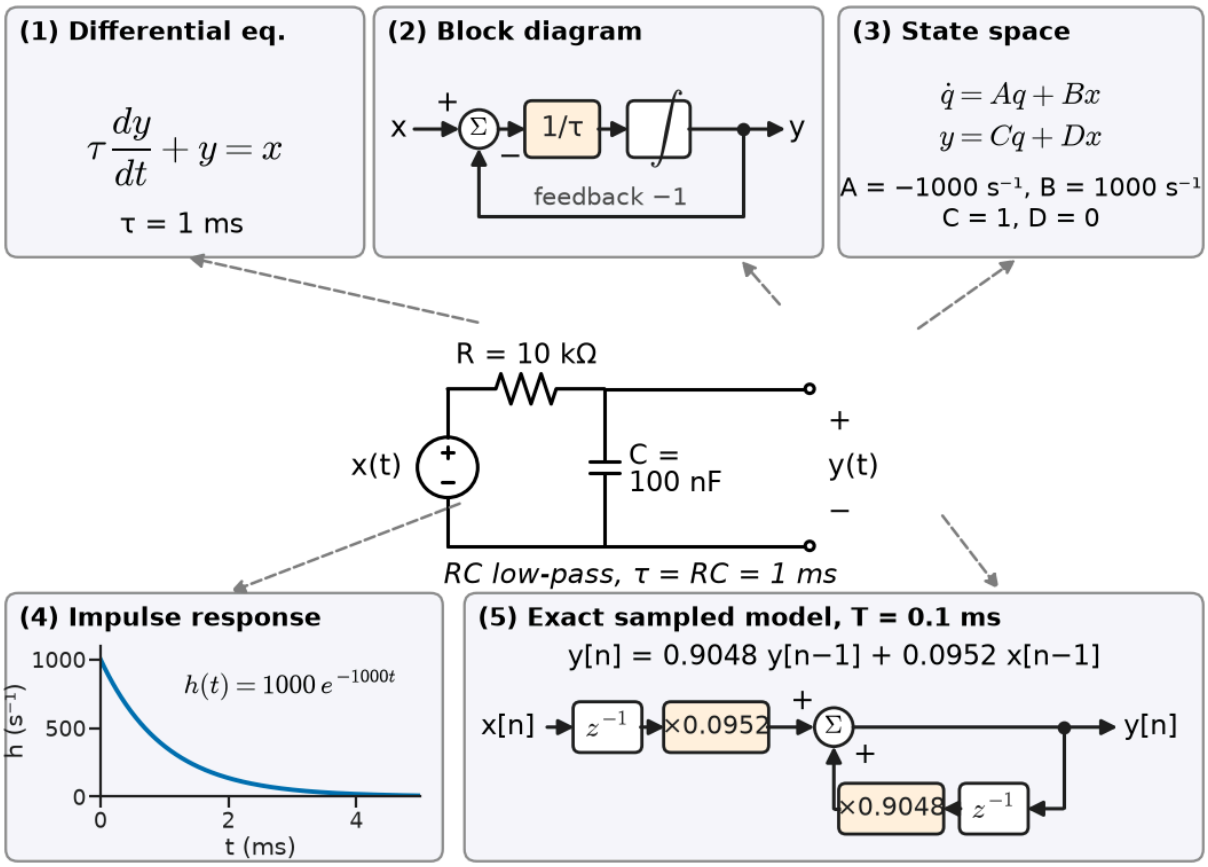

Figure 16.5 — One RC low-pass filter written in five equivalent input-output representations.

One system, five descriptions

Figure 16.5 collects the descriptions for the RC low-pass filter with $R = 10\ \text{k}\Omega$, $C = 100\ \text{nF}$, $\tau = RC = 1\ \text{ms}$:

Representation	RC low-pass, $\tau = 1$ ms
Differential equation	$\tau\, dy/dt + y = x$
State-space	$\dot{q} = -1000\, q + 1000\, x$, $y = q$ (units s^{-1})
Impulse response	$h(t) = 1000\, e^{-1000t}$ for $t \geq 0$
Frequency response	$H(j\omega) = 1/(1 + j\omega\tau)$, cut-off 159 Hz
Exact sampled model, $T = 0.1$ ms	$y[n] = 0.9048\, y[n-1] + 0.0952\, x[n-1]$

Worked example — discretising the RC filter. A digital controller must simulate the RC filter at a sample interval $T = 0.1$ ms. Two discrete models are common. The exact one for a piecewise-constant input uses $a = e^{-T/\tau} = e^{-0.1} = 0.9048$, giving $y[n] = 0.9048\, y[n-1] + 0.0952\, x[n-1]$. The simpler forward-Euler approximation replaces dy/dt by $(y[n] - y[n-1])/T$ evaluated one step back, giving $y[n] = (1 - T/\tau)\, y[n-1] + (T/\tau)\, x[n-1] = 0.9\, y[n-1] + 0.1\, x[n-1]$. For a unit step at $t = 0$, the continuous system reaches $1 - e^{-1} = 0.6321$ at $t = 1$ ms. The exact discrete model reaches $1 - 0.9048^{10} = 0.6321$ after 10 steps — identical, by construction. The Euler model reaches $1 - 0.9^{10} =$

0.6513, about 3 % high. Both discrete models are linear, time-invariant, causal and stable, as the continuous filter is; but they are different systems, and a careless discretisation can even destroy stability (forward Euler becomes unstable if $T > 2\tau$). Chapter 23 treats discretisation methods properly.

Choosing a representation

Task	Preferred representation
Deriving a model from a circuit or mechanism	differential equation, state-space
Implementing a digital filter or controller	difference equation, block diagram
Predicting the response to a measured transient	impulse response and convolution
Specifying or measuring a filter, amplifier or channel	frequency response
Locating instability and resonances	transfer function poles and zeros
Nonlinear or time-varying systems	state-space, simulation

> **Boundary.**
> The mathematical theory of differential and difference equations, and of the transforms that solve them, belongs to *The Mathematics Universe*; this book uses the results. The internal design of amplifiers, ADCs and DACs that implement these systems belongs to *Electronics & Computing*.

What remains open

Open question	Where it is answered
How is an LTI system completely characterised by one signal, and how is its output computed?	Chapter 17 (impulse response, convolution)
Why are complex exponentials special for LTI systems, and what is the frequency response?	Chapter 17 (eigenfunctions), Chapter 18
How are BIBO stability and causality read from pole locations?	Chapter 17, Chapter 11 (s-domain), Chapter 20 (z-domain)
What happens at the boundary between continuous and discrete systems?	Chapter 19 (sampling and reconstruction)
How are digital filters designed to meet specifications without losing stability?	Chapter 23
How are nonlinear distortion and noise quantified in communication systems?	Chapter 18 (distortion), Chapter 29 (noise)
How does feedback change linearity, sensitivity and stability?	Chapter 17 (interconnection), Chapter 34
How are state-space models used for design, and what are controllability and observability?	Chapter 35
How are transistor amplifiers linearised about an operating point?	*Electronics & Computing*

Next: Chapter 17 takes the systems that pass both the linearity and time-invariance tests and shows that a single response — to an impulse — determines everything they do.

Chapter 17 — Linear Time-Invariant Systems

Part IV · Signals and Systems

An acoustic consultant is asked why speech is hard to understand in a newly built lecture hall. She does not play speech into the room. She stands at the lectern, bursts a balloon, and records the result at a seat in row twenty: a sharp click, a cluster of strong early reflections over the first 50 ms, then a dense decaying tail that takes about 1.8 s to fall by 60 dB. From that single recording she can compute what *any* sound produced at the lectern will sound like at that seat — a lecturer's voice, a piano, a fire alarm — without ever playing them. Audio engineers use the same trick in reverse to put a dry studio recording "into" a cathedral. A radio engineer does the equivalent with a channel sounder to predict how a 5G signal will be smeared by multipath; a cardiologist's ECG system designer does it with a step input to characterise the amplifier; a seismologist does it with a controlled explosion.

All of this rests on one fact: for a system that is linear and time-invariant, the response to a brief impulse contains everything there is to know about the system. This chapter establishes that fact, derives the operation — convolution — that turns the impulse response into the output for any input, connects it to the differential and difference equations that physical systems actually obey, and shows why sinusoids and complex exponentials pass through LTI systems unchanged in shape. These results are the foundation for the frequency-domain view of Chapter 18 and for everything in Parts V–VII.

> **Why does one measured response — to an impulse — determine the output of an LTI system for every input?**
> **How do differential and difference equations, impulse responses and convolution describe the same system? Why do complex exponentials pass through LTI systems unchanged in shape, and what does that imply for stability and interconnection?**

Concept	Engineering destination
Impulse response	Room acoustics, channel sounding, loudspeaker and sensor characterisation
Convolution	Filtering, reverberation, multipath channels, blurring and deblurring
Differential-equation models	RC, RL and RLC circuits, sensors, mechanical structures, plant models
Difference equations	Digital filters, smoothers, digital controllers, recursive estimators
Zero-input and zero-state response	Switching transients, start-up of filters and converters
Eigenfunctions	Phasor analysis, frequency response, modulation and demodulation
Causality and stability from h	Real-time filters, feedback amplifiers, IIR filter design
Interconnection	Cascaded filter stages, parallel channels, feedback amplifiers, echo cancellation

Roadmap. Impulse response and the decomposition of inputs (17.1) → convolution, its mechanics and properties (17.2) → differential-equation systems (17.3) → difference-equation systems (17.4) → the structure of the system response (17.5) → eigenfunctions (17.6) → causality and stability read from h (17.7) → series, parallel and feedback interconnections (17.8).

17.1 Impulse response

Discrete time first

The idea is cleanest in discrete time. Any sequence can be written as a sum of scaled, shifted unit impulses (Chapter 15):

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

Each term is a single sample: the value $x[k]$ placed at position k. Define the **impulse response** $h[n]$ as the output of the system when the input is $\delta[n]$ and the system starts at rest:

$$h[n] = \mathcal{T}\{\delta[n]\}$$

Now apply the two LTI properties in turn. Time invariance says the response to $\delta[n-k]$ is $h[n-k]$. Linearity says the response to $x[k]\,\delta[n-k]$ is $x[k]\,h[n-k]$, and the response to the sum is the sum of the responses. Therefore

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$

This is the **convolution sum**, written $y = x * h$. Nothing else about the system was used. Its internal structure — whether it is a digital filter with ten coefficients, a cable, an amplifier or a room — is irrelevant once h is known.

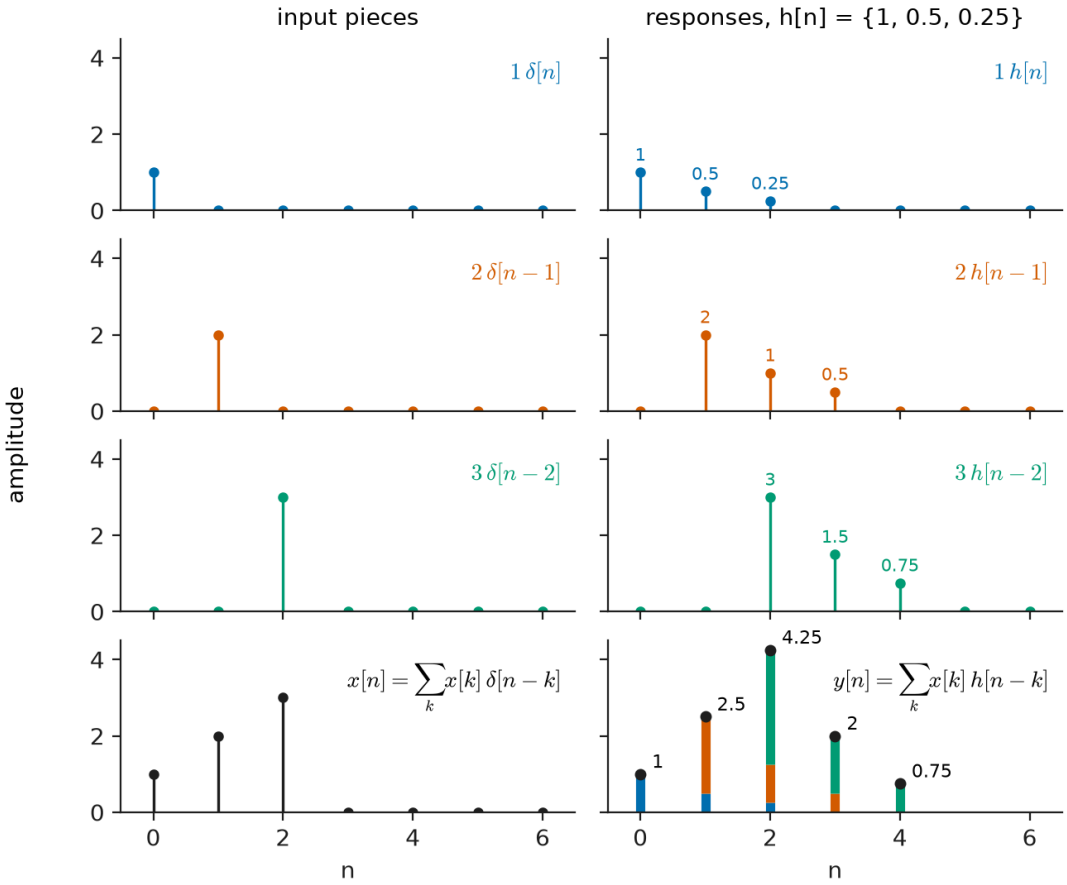

Figure 17.1 — An input sequence is a sum of shifted impulses; for an LTI system the output is the same sum of shifted, scaled impulse responses.

Continuous time

In continuous time the same argument uses the sifting property of the Dirac impulse,

$$x(t) = \int_{-\infty}^{\infty} x(\lambda)\,\delta(t-\lambda)\,d\lambda$$

which expresses x as a continuum of impulses of strength $x(\lambda)\,d\lambda$. With $h(t) = \mathcal{T}\{\delta(t)\}$, linearity and time invariance give the **convolution integral**

$$y(t) = \int_{-\infty}^{\infty} x(\lambda)\,h(t-\lambda)\,d\lambda = (x * h)(t)$$

A more physical route reaches the same result: approximate $x(t)$ by a staircase of narrow rectangular pulses of width Δ; each pulse is close to an impulse of area $x(k\Delta)\Delta$ when Δ is short compared with the system's response time; add the responses; let $\Delta \to 0$. The approximation explains how impulse responses are measured. No real input is a Dirac delta, but any pulse much shorter than the fastest feature of $h(t)$ acts as one. A balloon burst lasting a millisecond is "an impulse" to a room whose response lasts seconds; a 10 ns pulse is an impulse to a 100 kHz amplifier but not to a 1 GHz one.

> **Status of the principle: Convolution representation of LTI systems**
> *Derived*. It follows exactly from linearity, time invariance and the impulse decomposition of the input; it holds for any system with those two properties and for no other.

Step response

The impulse is an idealisation that is often hard to apply in practice — a large, short pulse may overload an amplifier. The **unit step response** $s(t)$, the output for input $u(t)$, is easier to measure and contains the same information. Since $u(t) = \int_{-\infty}^{t} \delta(\lambda)\,d\lambda$ and the system is LTI,

$$s(t) = \int_{-\infty}^{t} h(\lambda)\,d\lambda, \qquad h(t) = \frac{ds}{dt}$$

and in discrete time $s[n] = \sum_{k \le n} h[k]$, $h[n] = s[n] - s[n-1]$. Oscilloscope probe compensation, the specification of amplifier settling time, and the calibration of ECG amplifiers with a 1 mV step all use step responses.

Measured impulse responses

System	Typical length of impulse response
RC anti-alias filter, $f_c = 20$ kHz	about 40 µs (to 1 %)
ECG high-pass filter, 0.05 Hz	about 15 s (to 1 %)
Living room, reverberation time T_{60}	about 0.3–0.6 s
Concert hall, T_{60}	about 1.8–2.2 s
Indoor radio channel, RMS delay spread	about 10–100 ns
Urban cellular channel, RMS delay spread	about 0.5–5 µs
Hybrid echo on a long-distance telephone circuit	tens of ms
Broadband seismometer	tens to hundreds of seconds

The lengths span thirteen orders of magnitude, but the mathematics is the same in every row.

17.2 Convolution

Mechanics: flip, shift, multiply, integrate

To evaluate $y(t) = \int x(\lambda) h(t-\lambda)\,d\lambda$ at one particular time t:

1. Write both signals as functions of the dummy variable λ.
2. **Flip** h: $h(-\lambda)$.
3. **Shift** it right by t: $h(t-\lambda)$.

4. **Multiply** pointwise by $x(\lambda)$.
5. **Integrate** (or sum) the product over λ.

Repeating for each t traces out $y(t)$. The flipped impulse response acts as a weighting window that slides along the input; $h(t-\lambda)$ says how much the input at time λ contributes to the output at time t. For a causal system the window only looks backwards, and its shape is the system's memory.

Worked example — discrete convolution by hand. A three-tap FIR filter has $h[n] = \{1,\ 0.5,\ 0.25\}$ for $n = 0, 1, 2$. The input is $x[n] = \{1,\ 2,\ 3\}$ for $n = 0, 1, 2$. Using $y[n] = \sum_{k} x[k]\,h[n-k]$:

- $y[0] = x[0]h[0] = 1$
- $y[1] = x[0]h[1] + x[1]h[0] = 0.5 + 2 = 2.5$
- $y[2] = x[0]h[2] + x[1]h[1] + x[2]h[0] = 0.25 + 1 + 3 = 4.25$
- $y[3] = x[1]h[2] + x[2]h[1] = 0.5 + 1.5 = 2.0$
- $y[4] = x[2]h[2] = 0.75$

The output has $3 + 3 - 1 = 5$ samples: convolving sequences of lengths L and M gives length $L + M - 1$. Figure 17.1 shows the same calculation as a sum of three shifted, scaled copies of h. A useful check: the sum of the output samples, 10.5, equals the product of the input sum (6) and the impulse-response sum (1.75), because convolution multiplies DC gains.

Worked example — a rectangular pulse through an RC filter. An RC low-pass filter with $\tau = RC = 1$ ms has $h(t) = \frac{1}{\tau} e^{-t/\tau} u(t)$. The input is a 5 V pulse lasting 2 ms, from $t = 0$ to $t = 2$ ms — for instance a gating pulse on a long line. For $0 \le t \le 2$ ms the flipped, shifted impulse response overlaps the pulse from $\lambda = 0$ to $\lambda = t$:

$$y(t) = \int_{0}^{t} 5 \cdot \frac{1}{\tau} e^{-(t-\lambda)/\tau}\, d\lambda = 5\bigl(1 - e^{-t/\tau}\bigr)$$

At $t = 1$ ms, $y = 5(1 - e^{-1}) = 3.16$ V; at the end of the pulse, $y(2\text{ ms}) = 5(1 - e^{-2}) = 4.32$ V. For $t > 2$ ms the overlap is $\lambda \in [0, 2\text{ ms}]$, and the result is $y(t) = 4.32\, e^{-(t-2\,\text{ms})/\tau}$; at $t = 3$ ms, $y = 4.32 e^{-1} = 1.59$ V. This is the familiar charge–discharge waveform of Chapter 6, now obtained without writing a differential equation. Figure 17.2 shows the sliding window at three instants.

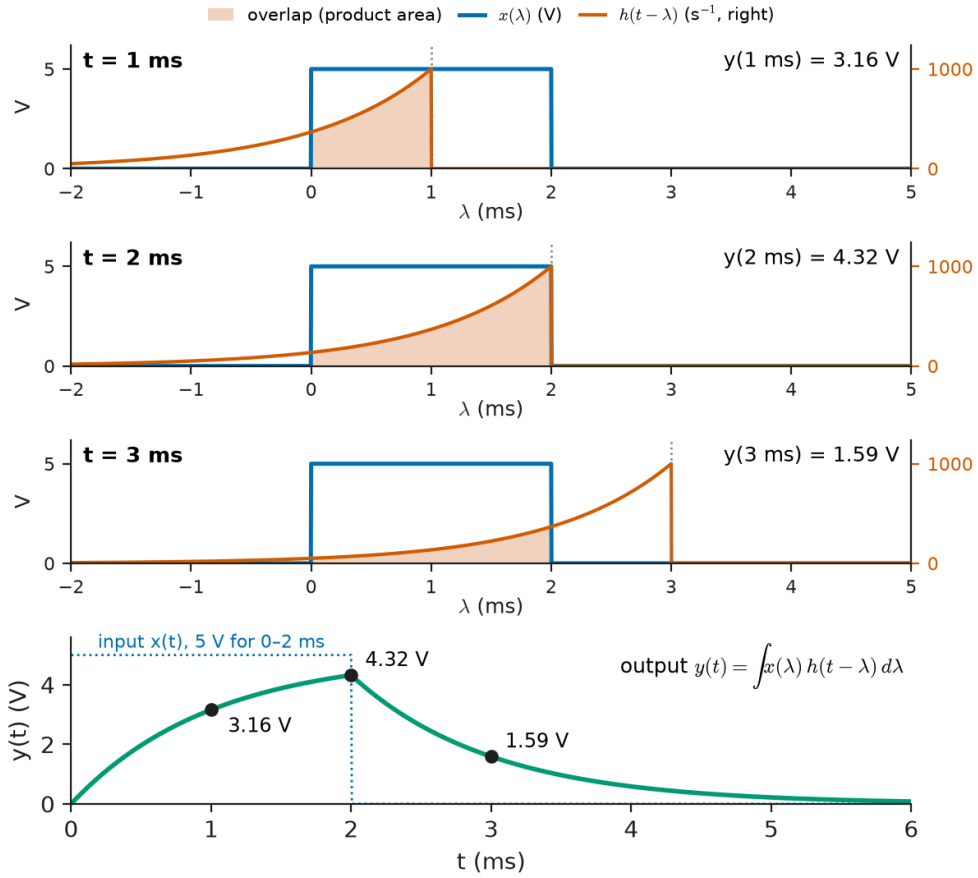

Figure 17.2 — Graphical convolution: the flipped RC impulse response slides across a 2 ms rectangular pulse, and the area of the product gives the output.

Properties

Convolution behaves like multiplication, and each algebraic property has an engineering reading:

Property	Statement	Engineering meaning
Commutative	$x * h = h * x$	input and system are interchangeable mathematically
Associative	$(x * h_1) * h_2 = x * (h_1 * h_2)$	a cascade is one system with impulse response $h_1 * h_2$
Distributive	$x * (h_1 + h_2) = x * h_1 + x * h_2$	parallel paths add
Identity	$x * \delta = x$	a wire has impulse response δ
Shift	$x(t) * \delta(t - t_0) = x(t - t_0)$	a pure delay has impulse response $\delta(t - t_0)$
Width	widths add (supports add in length)	filters smear signals; cascading smears more
Area	$\int y = \int x \cdot \int h$	DC gains multiply

The width property explains intersymbol interference in digital communication. A transmitted symbol of duration T_s convolved with a channel whose delay spread is T_d occupies about $T_s + T_d$: when T_d is comparable to T_s, successive symbols overlap. In an urban cellular channel with a 5 μs delay spread,

symbols of 1 µs would overlap badly; this is one reason LTE and 5G use OFDM symbols of about 67 µs (at 15 kHz subcarrier spacing) with a cyclic prefix of about 4.7 µs (Chapter 30).

The cost of convolution

Evaluating the convolution sum directly for an FIR impulse response of M samples costs M multiply–accumulate operations per output sample. For a short filter this is trivial: a 10-tap mains-hum average at 500 Hz needs 5000 operations per second. For a measured room response it is not. Convolution reverb for a concert hall with a 2 s impulse response sampled at 48 kHz uses $M = 96\,000$ taps; at 48 000 output samples per second, direct convolution needs $96\,000 \times 48\,000 \approx 4.6 \times 10^{9}$ multiply–accumulates per second per channel. Real-time convolution reverbs therefore compute the convolution in blocks using the FFT, where the cost grows roughly as $\log M$ per sample rather than M (Chapter 22). Recursive (IIR) systems are the other escape: when h has the form of decaying exponentials, a few feedback coefficients generate an arbitrarily long response at fixed cost (Section 17.4).

Convolution in the wider world

Convolution is not only a filtering operation in circuits. A camera's optical blur is the convolution of the scene with the lens's point-spread function; a radar return is the transmitted pulse convolved with the target's reflectivity profile; a seismogram is the source wavelet convolved with the earth's layered reflectivity; drug concentration in blood after repeated doses is the dose sequence convolved with the body's clearance response. In each case, undoing the convolution — **deconvolution** — is valuable and difficult, a theme revisited in Section 17.8.

17.3 Differential-equation representation

Where the equations come from

Lumped LTI systems obey linear differential equations with constant coefficients. A series RC circuit with output across the capacitor gives

$$RC\,\frac{dy}{dt} + y = x$$

A series RLC circuit with output across the capacitor gives

$$LC\,\frac{d^2y}{dt^2} + RC\,\frac{dy}{dt} + y = x$$

Mass–spring–damper systems, accelerometers, DC motors, thermal systems and many process plants give equations of the same form (Chapter 33). It is convenient to write the second-order case in standard form,

$$\frac{d^2y}{dt^2} + 2\zeta\omega_0\frac{dy}{dt} + \omega_0^2 y = \omega_0^2 x, \qquad \omega_0 = \frac{1}{\sqrt{LC}}, \quad \zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$$

The equation defines a system only with conditions attached

A differential equation by itself does not define a unique input–output map: it has a family of solutions differing by the homogeneous solution. Two conventions remove the ambiguity. The system is said to be at **initial rest** if $x(t) = 0$ for $t < t_0$ implies $y(t) = 0$ for $t < t_0$. Under initial rest, a linear constant-coefficient differential equation defines a system that is linear, time-invariant and causal. With nonzero initial conditions the system is incrementally linear (Chapter 16), and the response splits as in Section 17.5.

From the equation to $h(t)$

The impulse response is the solution under initial rest with $x = \delta(t)$. For the RC circuit, integrating the equation across $t = 0$ shows that the impulse charges the capacitor instantly to $y(0^+) = 1/\tau$, after which it discharges freely:

$$h(t) = \frac{1}{\tau} e^{-t/\tau} u(t)$$

For the second-order system with $\zeta < 1$ (underdamped),

$$h(t) = \frac{\omega_0}{\sqrt{1-\zeta^2}}\, e^{-\zeta\omega_0 t} \sin(\omega_d t)\, u(t), \qquad \omega_d = \omega_0\sqrt{1-\zeta^2}$$

In general, $h(t)$ is a sum of terms $e^{p_i t}$, one for each root p_i of the characteristic equation; these roots are the poles of the transfer function (Chapter 11). Every property of h — its duration, its oscillation, whether it decays — is set by those roots.

Worked example — step response of an RLC sensor filter. A passive filter in a vibration-sensor front end uses $L = 10$ mH, $C = 1$ µF and $R = 50\ \Omega$ in series, with output across C. Then $\omega_0 = 1/\sqrt{10^{-2}\times 10^{-6}} = 10^4$ rad/s ($f_0 = 1.59$ kHz) and $\zeta = \frac{50}{2}\sqrt{10^{-6}/10^{-2}} = 25 \times 0.01 = 0.25$. The damped frequency is $\omega_d = 10^4\sqrt{1-0.0625} = 9682$ rad/s. The unit step response peaks at $t_p = \pi/\omega_d = 0.324$ ms, with overshoot $e^{-\pi\zeta/\sqrt{1-\zeta^2}} = e^{-0.811} = 0.444$, that is, 44 %. The envelope decays as $e^{-\zeta\omega_0 t} = e^{-2500t}$; the usual estimate of 2 % settling time, $4/(\zeta\omega_0)$, gives 1.6 ms (an exact calculation gives 1.4 ms). A 1 V step from the sensor would briefly drive the following ADC to 1.44 V — if the ADC's full scale is set just above the expected signal, this overshoot clips. Raising R to about 141 Ω would give $\zeta \approx 0.707$ and an overshoot of only 4.3 %.

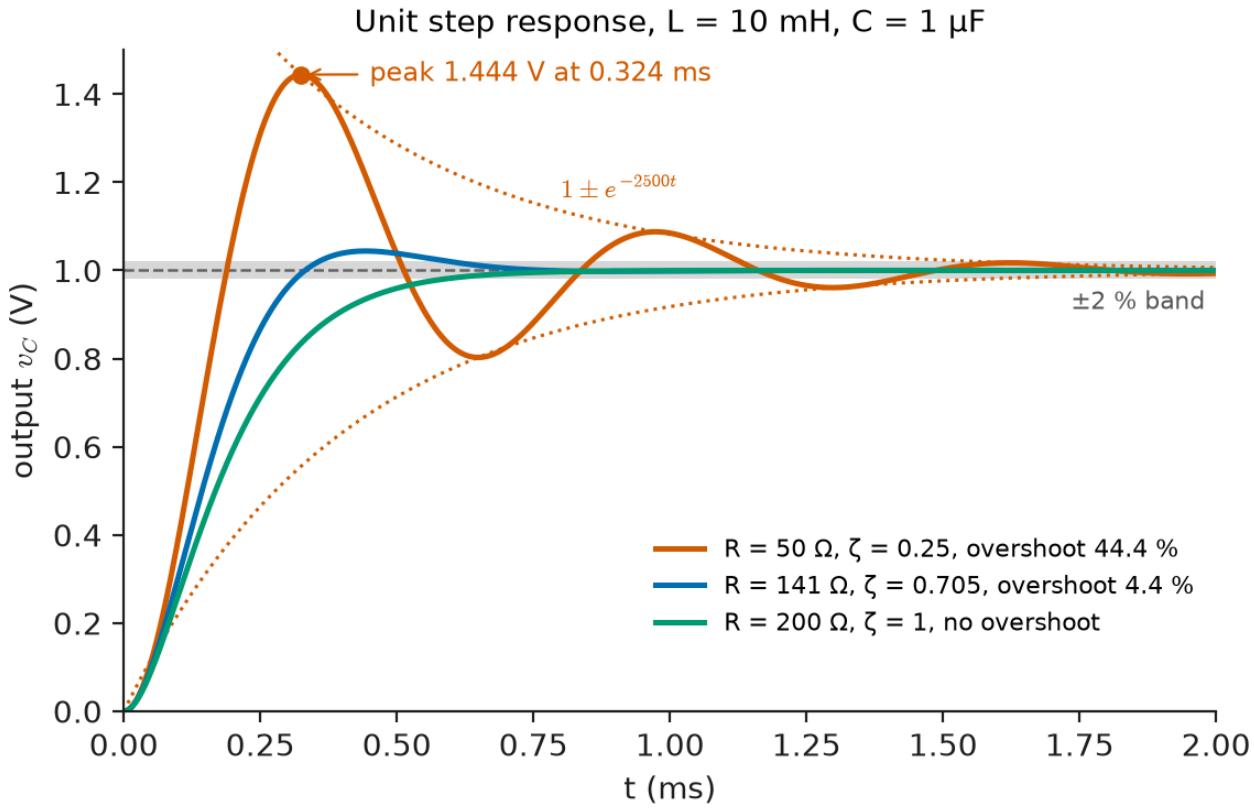

Figure 17.3 — Step response of a series RLC low-pass with L = 10 mH, C = 1 uF for three damping values.

Figure 17.3 compares three damping values. The figure also illustrates a point used repeatedly in Chapter 18: the same system that rings for a step has a resonant peak in its frequency response.

When the input is differentiated

If derivatives of x appear on the right-hand side of the equation, the impulse response can contain an impulse itself. The RC high-pass filter used as an AC-coupling stage — output across R — obeys $\tau\, dy/dt + y = \tau\, dx/dt$, and its impulse response is

$$h(t) = \delta(t) - \frac{1}{\tau} e^{-t/\tau} u(t)$$

The $\delta(t)$ term says that a fast edge passes straight through the coupling capacitor; the negative exponential then pulls the output back towards zero. The step response, $s(t) = e^{-t/\tau}u(t)$, is the familiar "tilt" or droop of a pulse passed through AC coupling. For the 0.05 Hz high-pass filter of a diagnostic ECG amplifier, $\tau = 3.18$ s, so a 1 mV calibration step droops by $1 - e^{-0.2/3.18} \approx 6$ % over 200 ms — the reason the ST segment of an ECG, which lasts a few hundred milliseconds, is sensitive to the choice of low-frequency cut-off. The integral of this h is zero: a high-pass filter has zero DC gain.

17.4 Difference equations

Recursive systems

The discrete-time counterpart of a differential equation is a linear constant-coefficient difference equation:

$$y[n] = -\sum_{k=1}^{N} a_k\, y[n-k] + \sum_{k=0}^{M} b_k\, x[n-k]$$

Written this way it is an algorithm: each new output is a weighted combination of the current input, a few past inputs and a few past outputs. It needs memory for M past inputs and N past outputs and $M + N + 1$ multiplications per sample.

Two families result:

- **FIR (finite impulse response)**: $N = 0$, no feedback. The impulse response is just the coefficient list, $h[n] = b_n$ for $0 \le n \le M$, zero otherwise. The moving average is the standard example.
- **IIR (infinite impulse response)**: at least one $a_k \neq 0$. Feedback makes the impulse response infinitely long. A handful of coefficients produce a long, possibly very selective, response — and also the possibility of instability.

As with differential equations, the difference equation defines an LTI causal system under initial rest.

The exponential smoother

The simplest IIR system, used everywhere from temperature sensors to network-traffic estimators and power meters, is the **exponential smoother** (first-order recursive low-pass):

$$y[n] = a\, y[n-1] + (1-a)\, x[n], \qquad 0 < a < 1$$

Setting $x = \delta$ and iterating from rest gives $h[0] = 1 - a$, $h[1] = a(1-a)$, $h[2] = a^2(1-a)$, ..., that is,

$$h[n] = (1-a)\, a^n\, u[n]$$

which sums to $(1-a)/(1-a) = 1$: unity DC gain. It is the sampled cousin of the RC filter.

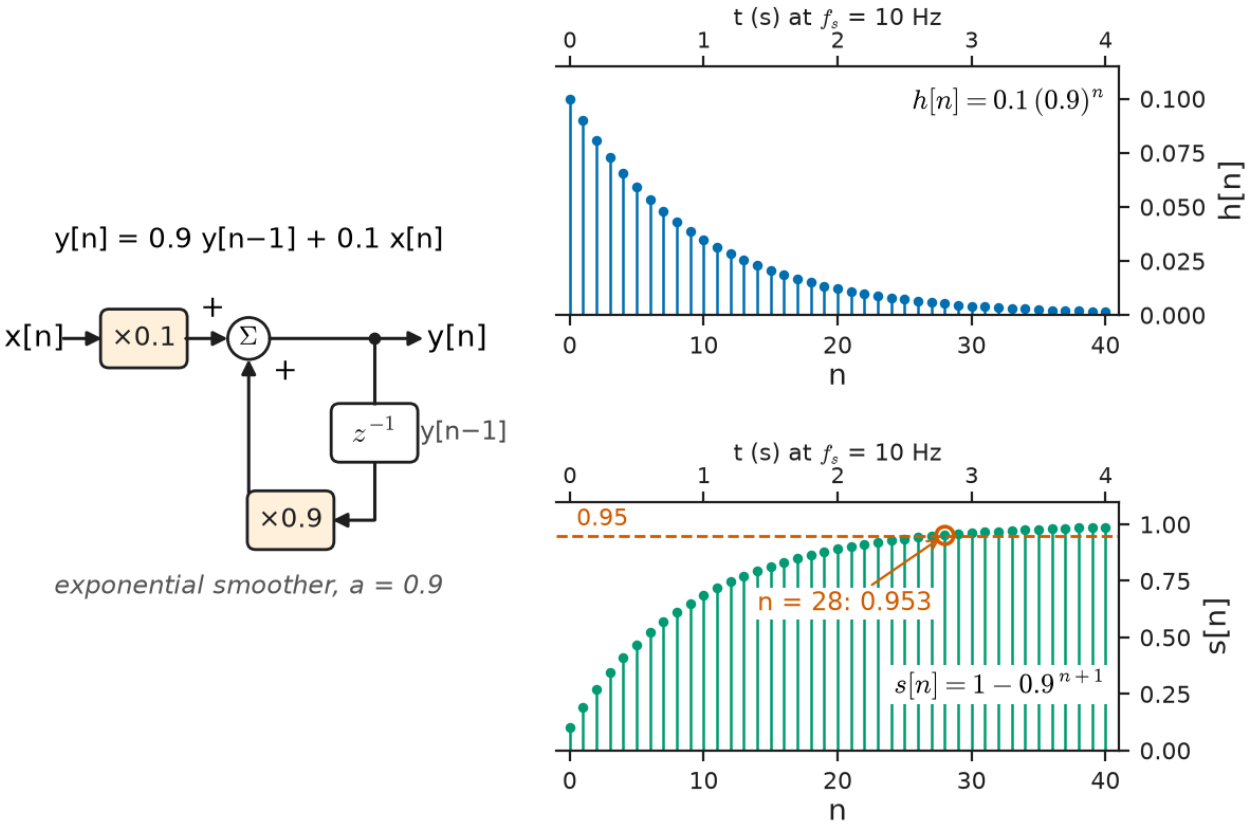

Figure 17.4 — Difference-equation realisation of the exponential smoother and its impulse and step responses.

Worked example — smoothing a temperature reading. A battery-pack temperature sensor is sampled at $f_s = 10$ Hz and smoothed with $a = 0.9$ (Figure 17.4). The impulse response is $h[n] = 0.1 \times 0.9^n$. Its equivalent time constant is $-T/\ln a = -0.1/\ln 0.9 = 0.949$ s, close to the 1 s an RC filter would need. The step response is $s[n] = \sum_{k=0}^{n} h[k] = 1 - 0.9^{n+1}$: after 10 samples (at 0.9 s), $s[9] = 0.651$; after 29 samples, $s[28] = 1 - 0.9^{29} = 0.953$, so a sudden 10 °C rise is reported to within 0.5 °C after 2.8 s. Each update costs two multiplications and one addition, and the smoother needs to remember only one number, $y[n-1]$, regardless of how long its memory effectively is. A moving average with comparable smoothing would need about 20 stored samples.

Solving difference equations

Difference equations can be solved like differential equations, with geometric sequences r^n playing the role of exponentials: the homogeneous solution is a combination of r_i^n for the roots r_i of the characteristic polynomial $r^N + a_1 r^{N-1} + \cdots + a_N = 0$, and a particular solution is added for the input. For the smoother the single root is $r = a$. Roots inside the unit circle, $|r_i| < 1$, give decaying terms; a root outside gives growth. The z-transform of Chapter 20 turns all of this into algebra, exactly as the Laplace transform does for differential equations in Chapter 11.

Difference equations as models of sampled continuous systems

A difference equation is often a model of a continuous system observed at intervals of T. Sampling the RC impulse response $e^{-t/\tau}$ at $t = nT$ gives $\left(e^{-T/\tau}\right)^n$, a geometric sequence with ratio $a = e^{-T/\tau}$. Every continuous mode e^{pt} maps in the same way to a discrete mode r^n with

$$r = e^{pT}$$

so left-half-plane poles ($\operatorname{Re} p < 0$) map to roots inside the unit circle ($|r| < 1$). The smoother with $a = 0.9$ at $T = 0.1$ s is the sampled version of a mode with $p = \ln(0.9)/0.1 = -1.054\ \text{s}^{-1}$, the time constant of 0.949 s found above. The correspondence $r = e^{pT}$ reappears as the mapping between the s- and z-planes in Chapter 20 and as the impulse-invariant design method in Chapter 23. It also shows why fast sampling of a slow system produces coefficients close to $1 - a = e^{-0.001} = 0.999$ for a 1 s time constant sampled at 1 kHz — which is a source of the finite-precision problems discussed in Chapter 21.

> **Boundary.**
> The general solution theory of linear differential and difference equations, and the Laplace and z-transforms, are developed in *The Mathematics Universe*. This chapter uses the results and focuses on what they mean for engineering systems.

17.5 System response

Zero-input and zero-state

A system with stored energy at $t = 0$ and an input applied from $t = 0$ has a total response that splits into two parts:

$$y(t) = y_{zi}(t) + y_{zs}(t)$$

The **zero-input response** y_{zi} is what the stored energy produces with the input switched off; it depends on initial conditions only and consists of the system's natural modes. The **zero-state response** y_{zs} is what the input produces from rest; it is $y_{zs} = x * h$ for $t \geq 0$. The split is exact, follows from linearity, and lets each part be computed separately — the zero-state part by convolution or transforms, the zero-input part from the initial conditions.

Natural and forced, transient and steady state

A second, independent split groups the response by *form* rather than by cause:

- The **natural response** contains the system's own modes $e^{p_i t}$ — the homogeneous solution.
- The **forced response** has the form of the input — a constant for a step, a sinusoid at the input frequency for a sinusoid.

For a stable system the natural modes decay, so the natural response is also the **transient** and the forced response for a persistent input is the **steady state**. Both splits are used: zero-input/zero-state when initial conditions are given, natural/forced when the steady state is the goal. Note that the zero-state response itself contains natural-mode terms — switching on an input excites the system's modes even from rest.

Worked example — switching a sinusoid onto an RC filter. The RC filter with $\tau = 1$ ms is at rest when a 100 Hz signal $x(t) = 10\cos(2\pi \cdot 100\,t)$ V is switched on at $t = 0$. Section 17.6 shows that the

steady-state response is the input scaled by $|H| = 1/\sqrt{1+(\omega\tau)^2}$ and shifted by $-\arctan\omega\tau$; with $\omega\tau = 0.628$ this gives amplitude 8.47 V and phase $-32.1°$. That is the forced response:

$$y_f(t) = 8.47\cos(2\pi\cdot 100\,t - 32.1°)\ \text{V}$$

At $t = 0^+$ the capacitor voltage must still be zero, but $y_f(0) = 8.47\cos(-32.1°) = 7.17$ V. The natural response supplies the difference: $y_n(t) = -7.17\,e^{-t/\tau}$ V. The complete response is $y = y_f + y_n$; the transient has fallen below 1 % of its initial value after $5\tau = 5$ ms, half a cycle of the input. Had the input been switched on at a zero crossing of the forced response instead, there would have been no transient at all. The same phenomenon, in a second-order and much larger form, is the inrush and DC-offset current when a circuit breaker closes onto a power line (Chapter 7 and *Electrical Energy & Machines*).

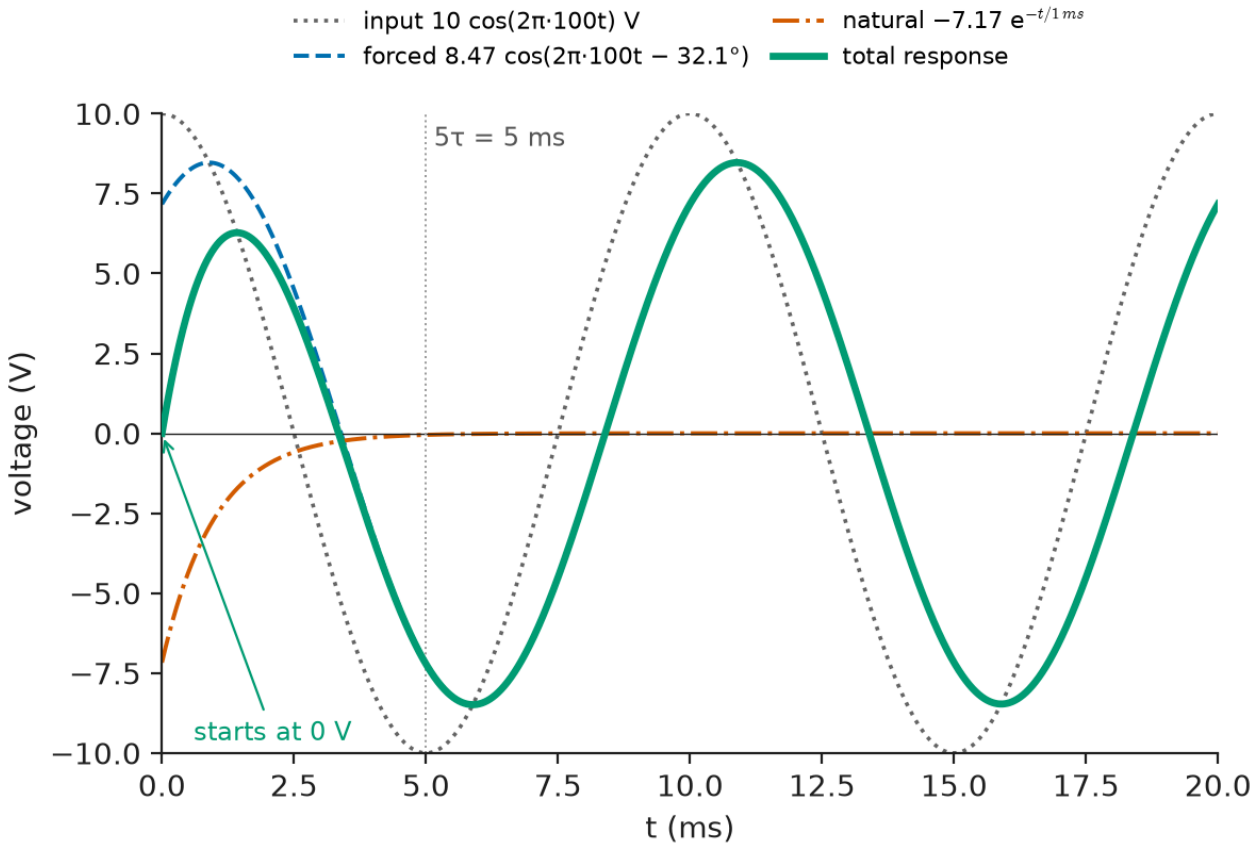

Figure 17.5 — Switching a 100 Hz sinusoid onto an RC filter with tau = 1 ms: forced response, natural response and their sum.

Figure 17.5 shows the three components.

17.6 Eigenfunctions

Complex exponentials pass through unchanged

Apply the complex exponential $x(t) = e^{st}$, with s any complex number, to an LTI system with impulse response h:

$$y(t) = \int_{-\infty}^{\infty} h(\lambda)\, e^{s(t-\lambda)}\, d\lambda = e^{st} \underbrace{\int_{-\infty}^{\infty} h(\lambda)\, e^{-s\lambda}\, d\lambda}_{H(s)}$$

The output is the input multiplied by a complex constant $H(s)$, provided the integral converges. In linear-algebra language, e^{st} is an **eigenfunction** of every LTI system and $H(s)$ is the corresponding **eigenvalue**. $H(s)$ as a function of s is the **transfer function** — the Laplace transform of h (Chapter 11). In discrete time, z^n is an eigenfunction with eigenvalue $H(z) = \sum_n h[n] z^{-n}$, the z-transform (Chapter 20).

> **Status of the principle: Eigenfunction property of LTI systems**
> *Derived*. It follows directly from the convolution integral and the factorisation $e^{s(t-\lambda)} = e^{st}e^{-s\lambda}$; it holds for every LTI system for those s at which the integral defining $H(s)$ converges.

Sinusoids

With $s = j\omega$ the eigenfunction is $e^{j\omega t}$, and the eigenvalue $H(j\omega)$ is the **frequency response**. For a real system (h real), $H(-j\omega) = H^*(j\omega)$, and since $\cos \omega t = \tfrac{1}{2}(e^{j\omega t} + e^{-j\omega t})$,

$$A\cos(\omega t + \theta) \;\longrightarrow\; A\,|H(j\omega)|\cos\big(\omega t + \theta + \angle H(j\omega)\big)$$

A sinusoid goes in, a sinusoid of the *same frequency* comes out, scaled in amplitude and shifted in phase. This is the theoretical basis of the phasor method of Chapter 8, of every Bode plot, and of the claim in Chapter 16 that an LTI system cannot create new frequencies.

Worked example — what an RC filter does to two tones. For the RC filter with $\tau = 1$ ms, $H(j\omega) = 1/(1 + j\omega\tau)$. At 100 Hz, $\omega\tau = 0.628$, so $|H| = 0.847$ and $\angle H = -32.1°$. At 1 kHz, $\omega\tau = 6.28$, so $|H| = 0.157$ (-16.1 dB) and $\angle H = -81.0°$. An input of 10 V at 100 Hz plus 10 V at 1 kHz emerges as 8.47 V at 100 Hz and 1.57 V at 1 kHz — no other frequencies. The phase shift of $-32.1°$ at 100 Hz corresponds to a time delay of $32.1/360 \times 10$ ms $= 0.893$ ms; the $-81.0°$ at 1 kHz corresponds to only 0.225 ms. The two components are delayed by different amounts, a fact that matters for pulse shapes and is quantified as group delay in Chapter 18.

Why eigenfunctions matter

The eigenfunction property turns convolution into multiplication. If an input can be written as a sum (or integral) of complex exponentials — which is what Fourier series and transforms do — then each component is simply multiplied by H at its own frequency, and the outputs are added. A difficult integral in time becomes a product in frequency. Chapter 18 builds on this; it is also why modulation (multiplication by a carrier) cannot be performed by an LTI system and requires a time-varying or nonlinear one.

Input	Output of LTI system
e^{st}	$H(s)\,e^{st}$
$e^{j\omega t}$	$H(j\omega)\,e^{j\omega t}$
$\cos(\omega t)$, real h	$\lvert H(j\omega)\rvert\cos(\omega t + \angle H(j\omega))$
constant c	$H(0)\,c$ (DC gain)
z^n	$H(z)\,z^n$
$\cos(\Omega n)$, real h	$\lvert H(e^{j\Omega})\rvert\cos(\Omega n + \angle H(e^{j\Omega}))$

Worked example — smoothing a 1 Hz oscillation. The temperature smoother of Section 17.4 ($a = 0.9$, $f_s = 10$ Hz) has $H(e^{j\Omega}) = 0.1/(1 - 0.9e^{-j\Omega})$. A 1 Hz oscillation corresponds to $\Omega = 2\pi \cdot 1/10 = 0.628$ rad/sample. Evaluating, $\lvert H\rvert = 0.168$ and $\angle H = -62.8°$. A 1 Hz ripple of ±2 °C (for example, electrical pickup from a cooling-fan controller) is reduced to ±0.34 °C. The smoother's −3 dB frequency is about 0.17 Hz, so slow genuine temperature changes pass with little attenuation.

17.7 Causal and stable systems

Causality from the impulse response

For an LTI system, the convolution integral $y(t) = \int h(\lambda)x(t-\lambda)\,d\lambda$ involves future inputs $x(t-\lambda)$ with $\lambda < 0$ only through values of $h(\lambda)$ for negative λ. Hence:

$$\text{LTI system causal} \iff h(t) = 0 \text{ for } t < 0 \qquad (h[n] = 0 \text{ for } n < 0)$$

The convolution then becomes $y(t) = \int_0^\infty h(\lambda)x(t-\lambda)\,d\lambda$: only the past and present input matter. The impulse response of a real-time system starts at or after the impulse. The ideal "brick-wall" low-pass filter of Chapter 18 has a sinc impulse response that extends to $t = -\infty$; it is non-causal and therefore can only be approximated, with delay, in real time.

Stability from the impulse response

For an LTI system,

$$\text{BIBO stable} \iff \int_{-\infty}^{\infty} \lvert h(t)\rvert\,dt < \infty \qquad \Big(\sum_n \lvert h[n]\rvert < \infty\Big)$$

Sufficiency: if $\lvert x\rvert \le B_x$ then $\lvert y(t)\rvert \le \int \lvert h(\lambda)\rvert\,\lvert x(t-\lambda)\rvert\,d\lambda \le B_x \int \lvert h\rvert$, which is finite. *Necessity:* if $\int\lvert h\rvert$ is infinite, choose the bounded input $x(t-\lambda) = \operatorname{sgn} h(\lambda)$ at a chosen time t; the output at that time is $\int\lvert h\rvert$, which is unbounded.

> **Status of the principle: Absolute-integrability test for BIBO stability**
> *Derived*. A two-line consequence of the convolution integral; it applies to LTI systems only, and for systems with rational transfer functions it is equivalent to all poles lying in the open left half-plane (continuous time) or strictly inside the unit circle (discrete time).

Applying the test:

Impulse response	$\int \lvert h \rvert$ or $\sum \lvert h \rvert$	Stable?
$\frac{1}{\tau} e^{-t/\tau} u(t)$ (RC)	1	yes
$(1-a)a^n u[n]$, $\lvert a \rvert < 1$ (smoother)	1 for $0 < a < 1$	yes
any FIR $h[n]$	finite sum	yes
$u(t)$ (integrator)	infinite	no
$\sin(\omega_0 t) u(t)$ (lossless LC)	infinite	no
$e^{\sigma t} u(t)$, $\sigma > 0$	infinite	no
$e^{-\zeta \omega_0 t} \sin(\omega_d t) u(t)$, $\zeta > 0$	finite	yes

Poles and the shape of h

For a system described by a differential equation, $h(t)$ is a sum of modes $e^{p_i t}$. A real pole $p = -\sigma$ gives a decaying exponential with time constant $1/\sigma$; a complex pair $p = -\sigma \pm j\omega_d$ gives a decaying oscillation. A pole in the right half-plane gives growth; a pole on the imaginary axis gives a constant or a sustained oscillation (marginal). Stability is decided by the sign of the real parts, and the *degree* of stability by their size — the most slowly decaying pole, the **dominant pole**, sets how long the system takes to settle.

Pole location (s-plane)	Mode in $h(t)$	Example
-1000	e^{-1000t}, 1 ms time constant	RC filter, $\tau = 1$ ms
$-2500 \pm j9682$	decaying oscillation at 1.54 kHz	RLC filter of Section 17.3
0	constant (step)	integrator, charge amplifier without reset
$\pm j\omega_0$	sustained sinusoid	lossless LC tank, ideal oscillator
$+\sigma$	growing exponential	latch-up, positive-feedback runaway, thermal runaway

Causal and stable together

A system that must run in real time and remain safe must be both causal and stable. For a rational transfer function, this means all poles strictly in the left half-plane (or inside the unit circle). The

requirement has a sharp consequence. If we want to invert a system — undo a channel's distortion — the inverse has the original's zeros as its poles. If the original has a zero in the right half-plane (a **non-minimum-phase** system), its causal inverse is unstable and its stable inverse is non-causal. Loudspeaker equalisation, channel equalisation in modems, and control of systems such as boost converters and aircraft pitch all run into this limit (Chapter 34).

17.8 Interconnection of systems

Series (cascade)

When system 1 feeds system 2, the overall impulse response is

$$h(t) = h_1(t) * h_2(t)$$

and by commutativity the order does not matter — for LTI systems. In practice the order matters for other reasons: a high-gain stage placed first sets the noise figure (Chapter 29), and a stage placed first can saturate on an out-of-band signal that a filter placed first would have removed. The cascade formula also assumes that connecting the stages does not change either one; with passive circuits this is false, because the second stage loads the first. A buffer amplifier between stages, or two-port analysis (Chapter 12), resolves this.

Worked example — two buffered RC stages. Two RC low-pass stages with time constants $\tau_1 = 1$ ms and $\tau_2 = 0.1$ ms are separated by a unity-gain buffer. Convolving the two exponential impulse responses gives

$$h(t) = \frac{e^{-t/\tau_1} - e^{-t/\tau_2}}{\tau_1 - \tau_2}\,u(t)$$

which starts at zero (a cascade of two first-order systems cannot respond instantly to an impulse), rises to a peak at $t_p = \frac{\tau_1\tau_2}{\tau_1-\tau_2}\ln\frac{\tau_1}{\tau_2} = 0.256$ ms with height $774\ \text{s}^{-1}$, and then decays essentially with the slower 1 ms time constant. The integral of h is 1 — the product of the two unit DC gains. Without the buffer, the second stage would load the first and the result would differ.

Parallel

When the same input drives two systems whose outputs are added,

$$h(t) = h_1(t) + h_2(t)$$

Multipath radio propagation is a parallel connection of delayed, attenuated paths, $h(t) = \sum_i \alpha_i\,\delta(t - \tau_i)$. A two-way loudspeaker crossover splits the signal into parallel low- and high-frequency paths whose outputs add acoustically. Parallel FIR filter banks underlie sub-band audio coding.

Feedback

When the output is fed back and subtracted from the input,

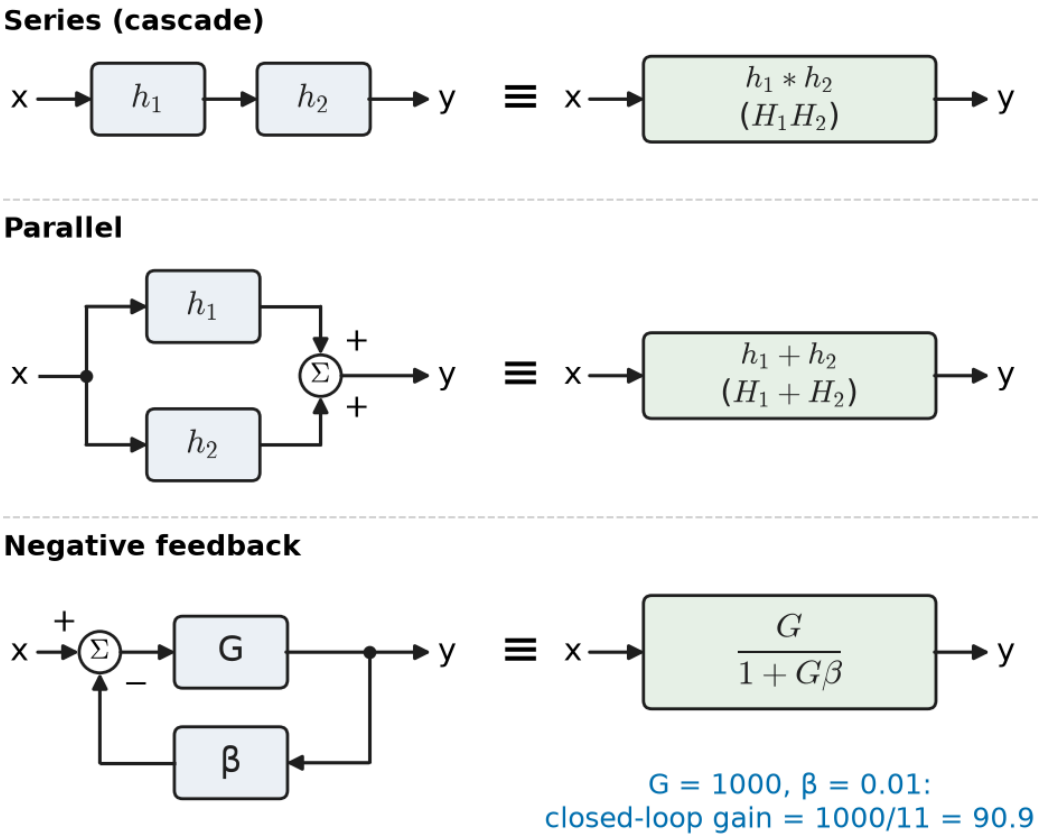

Figure 17.6 — The three basic interconnections of LTI systems and their equivalent single systems.

the relationship cannot be written as a simple convolution in the time domain, because the output depends on itself. In the transfer-function domain it becomes algebraic. With forward transfer function G and feedback β,

$$\frac{Y}{X} = \frac{G}{1 + G\beta}$$

This formula, applied at each frequency, is the basis of the feedback amplifier, the phase-locked loop and the control loop. Figure 17.6 collects the three interconnections.

Worked example — desensitising an amplifier with feedback. An amplifier has forward gain $G = 1000$, which varies by ±50 % with temperature and from unit to unit. A resistive feedback network returns $\beta = 0.01$ of the output. The closed-loop gain is $1000/(1 + 10) = 90.9$. If G falls to 500, the closed-loop gain becomes $500/(1 + 5) = 83.3$: a 50 % drop in G gives only an 8.3 % drop in closed-loop gain. As $G\beta$ grows, the gain approaches $1/\beta = 100$ and depends only on the stable resistive network. The price is that gain has been traded for accuracy (from 1000 to 91), and that feedback can turn a stable system unstable if G has enough phase lag — the subject of Chapter 34. Feedback also reduces nonlinear distortion of the forward path by roughly the factor $1 + G\beta$, which is why the second-harmonic distortion of Chapter 16 can be pushed below 0.01 % in practice.

Inverse systems and echo cancellation

An inverse system h_{inv} satisfies $h * h_{\text{inv}} = \delta$: cascading it after h restores the input. Inverses are built as feedback systems, and they are only useful when they are causal and stable.

Worked example — removing a single echo. A voice path at $f_s = 8$ kHz contains a direct signal and one echo of half the amplitude delayed by 50 ms ($D = 400$ samples): $y[n] = x[n] + 0.5\,x[n-400]$. The inverse system must satisfy $w[n] + 0.5\,w[n-400] = y[n]$, which is implemented recursively as

$$w[n] = y[n] - 0.5\,w[n-400]$$

Its impulse response is $1, -0.5, +0.25, -0.125, \ldots$ at multiples of 400 samples — an infinite train, because each correction itself produces an echo that must be corrected. The coefficients decay geometrically because $|-0.5| < 1$, so the inverse is stable; after ten terms (0.5 s) they have fallen to $0.5^{10} \approx 0.001$, or −60 dB. If the echo had been *stronger* than the direct signal (gain greater than 1, as can happen when the direct path is shadowed in a radio channel), the same recursion would grow without bound: the causal inverse would be unstable, the non-minimum-phase situation of Section 17.7. Practical echo cancellers in telephony and speakerphones do not invert the path directly; they estimate the echo path adaptively and subtract a replica of the echo (Chapter 24).

Deconvolution in practice

Inverse filtering is fragile even when the inverse is stable. Where $|H(j\omega)|$ is small, the inverse has a large gain and amplifies whatever noise is present at those frequencies. Sharpening a blurred photograph, equalising a long telephone line, and recovering the reflectivity series from a seismogram all face this trade-off between resolution and noise amplification, which the Wiener filter of Chapter 24 resolves optimally.

What remains open

Open question	Where it is answered
How is a signal written as a sum of complex exponentials, so that convolution becomes multiplication?	Chapter 18 (Fourier series and transform)
How do magnitude and phase of $H(j\omega)$ translate into filtering, bandwidth, group delay and distortion?	Chapter 18
How are transfer functions and initial conditions handled algebraically in the s-domain?	Chapter 11 (Laplace-domain circuit analysis)
How is the z-transform used to analyse difference equations and pole locations in discrete time?	Chapter 20
How is convolution computed efficiently for long signals?	Chapter 22 (DFT and FFT), Chapter 23
How are FIR and IIR filters designed to meet a specification?	Chapter 23
How do adaptive filters estimate and cancel echoes and invert channels in the presence of noise?	Chapter 24 (adaptive and Wiener filtering)
How does multipath convolution limit symbol rates, and how do OFDM and equalisation cope?	Chapter 27, Chapter 30
When does feedback make a system unstable, and how is that predicted?	Chapter 34

Next: Chapter 18 uses the eigenfunction property to move the whole analysis into the frequency domain, where filtering, bandwidth, delay and distortion become properties of a single function, $H(j\omega)$.

Chapter 18 — Frequency-Domain System Analysis

Part IV · Signals and Systems

A plant engineer at a water-treatment works is chasing two problems at once. The first is in the power system: since variable-speed drives were fitted to the pumps, transformers run hot and a capacitor bank for power-factor correction has failed twice. A power-quality analyser clipped onto the supply shows a current waveform that looks nothing like a sinusoid — two humps per half-cycle — and reports "THD 27 %". The second is in the instrumentation: a new digital filter in the flow-meter chain was supposed to remove 50 Hz pickup, and it does, but now the step response to a valve closing rings for a quarter of a second before settling, and the control loop that uses the measurement has started to hunt. Neither problem can be understood well in the time domain. Both become clear the moment the signals are described as sums of sinusoids and the systems as functions of frequency: the drive current contains 250 Hz and 350 Hz components that the capacitor bank resonates with, and the notch filter delays frequencies near 50 Hz by tens of milliseconds while passing others almost instantly.

Chapter 17 showed that complex exponentials pass through LTI systems unchanged except for a complex factor H. This chapter exploits that fact. It writes signals as sums or integrals of complex exponentials — the Fourier series and the Fourier transform — and systems as their frequency responses, then develops the vocabulary engineers use to specify and judge systems in the frequency domain: magnitude and phase, filtering, bandwidth, group delay and distortion, and the reciprocal relationship between duration and bandwidth that limits every pulse, radar and communication link.

> **How can any practical signal be built from sinusoids, and why does that make LTI analysis simple?**
> **What do magnitude, phase, bandwidth and group delay of a system actually do to a signal? What is distortion, which kinds can be undone, and why can a signal not be both short and narrowband?**

Concept	Engineering destination
Fourier series	Harmonics in power systems, PWM spectra, waveform synthesis, THD measurement
Fourier transform	Pulse spectra, radar, spectrum occupancy, EMC
Frequency response	Amplifier, filter, sensor and channel specifications
Transfer function	Filter design, pole-zero placement, feedback analysis
Magnitude and phase	Bode plots, loudspeaker and microphone responses, equalisation
Filtering	Anti-aliasing, hum removal, channel selection, noise reduction
Bandwidth	Data rate, rise time, noise power, spectrum allocation
Group delay	Pulse fidelity, data-channel equalisation, audio crossovers, control-loop delay
Distortion	Audio and RF linearity, power quality, intersymbol interference
Time–frequency relationships	Radar range resolution, pulse shaping, uncertainty in spectral measurements

Roadmap. Periodic signals as Fourier series (18.1) → aperiodic signals and the Fourier transform (18.2) → the frequency response as the system's gain at each frequency (18.3) → transfer functions, poles and zeros (18.4) → magnitude and phase (18.5) → filtering (18.6) → bandwidth (18.7) → group delay (18.8) → distortion (18.9) → time–frequency relationships (18.10).

18.1 Fourier-series representation

Periodic signals as sums of harmonics

A periodic signal with period T_0 and fundamental angular frequency $\omega_0 = 2\pi/T_0$ can be written as a sum of complex exponentials at integer multiples of ω_0:

$$x(t) = \sum_{k=-\infty}^{\infty} c_k \, e^{jk\omega_0 t}, \qquad c_k = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jk\omega_0 t}\, dt$$

The coefficient c_k is the complex amplitude of the kth **harmonic**; c_0 is the DC (average) value. For a real signal $c_{-k} = c_k^*$, and the pair $k, -k$ combines into a real sinusoid, giving the equivalent **trigonometric form**

$$x(t) = c_0 + \sum_{k=1}^{\infty} A_k \cos(k\omega_0 t + \theta_k), \qquad A_k = 2|c_k|,\ \theta_k = \angle c_k$$

The plot of A_k (or $|c_k|$) against frequency is the signal's **line spectrum**. Power-quality analysers, audio analysers and spectrum analysers display exactly this.

The series converges to $x(t)$ wherever x is continuous, for every signal an engineer will meet: the Dirichlet conditions (absolutely integrable over a period, finitely many maxima, minima and discontinuities per period) are satisfied by any physically generated waveform. At a jump, the series converges to the midpoint, and partial sums overshoot by about 9 % of the jump on either side however many terms are taken — the **Gibbs phenomenon**, visible as ringing on sharply filtered square waves.

> **Status of the principle: Fourier-series representation**
> *Derived.* A theorem of analysis: any periodic signal satisfying the Dirichlet conditions equals the sum of its harmonic series except at discontinuities; physical waveforms always satisfy those conditions.

Symmetry shortcuts

Signal symmetry	Consequence for coefficients
Even, $x(-t) = x(t)$	cosine terms only, c_k real
Odd, $x(-t) = -x(t)$	sine terms only, c_k imaginary
Half-wave symmetric, $x(t + T_0/2) = -x(t)$	odd harmonics only
Zero average	$c_0 = 0$

Half-wave symmetry matters enormously in power engineering. Any distortion that treats positive and negative half-cycles identically — magnetising current of a transformer, a six-pulse rectifier, fluorescent lamp ballasts — produces only odd harmonics, which is why 150, 250, 350 Hz (on a 50 Hz system) dominate harmonic surveys while even harmonics are usually small.

Parseval's relation and harmonic distortion

The average power of a periodic signal equals the sum of the powers of its harmonics:

$$P = \frac{1}{T_0}\int_{T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |c_k|^2 = c_0^2 + \sum_{k=1}^{\infty} \frac{A_k^2}{2}$$

This makes it possible to say how much of a signal's power lies in each harmonic, and leads to the **total harmonic distortion**,

$$\mathrm{THD} = \frac{\sqrt{\sum_{k\geq 2} A_k^2}}{A_1}$$

the RMS of all harmonics relative to the fundamental.

Worked example — the square wave. A symmetric square wave of amplitude ±1 V at 1 kHz is odd and half-wave symmetric, so it contains only odd sine harmonics, with amplitudes $A_k = 4/(\pi k)$: 1.273 V at 1 kHz, 0.424 V at 3 kHz, 0.255 V at 5 kHz, 0.182 V at 7 kHz, and so on. Its power is 1 W into 1 Ω. The fundamental carries $A_1^2/2 = 0.811$ W, 81 % of the total; the first three odd harmonics together carry 93 %. By Parseval, the harmonics above the fundamental carry $1 - 8/\pi^2 = 0.189$ W, so

$$\mathrm{THD} = \sqrt{\frac{0.189}{0.811}} = \sqrt{\frac{\pi^2}{8} - 1} = 0.483 = 48.3\ \%$$

A square wave is the extreme case of a clipped sine. An overdriven guitar amplifier approaches it deliberately; an inverter producing a "modified sine wave" is a stepped approximation whose harmonics can overheat motors and mains transformers.

Worked example — harmonic current of a drive. The pump drives of the opening use six-pulse diode rectifiers, whose input current ideally contains harmonics of order $6m \pm 1$ (5, 7, 11, 13, …). Measured magnitudes relative to the fundamental are typical of such drives: 20 % (5th, 250 Hz), 14 % (7th, 350 Hz), 9 % (11th, 550 Hz), 8 % (13th, 650 Hz). The current THD is

$$\mathrm{THD}_I = \sqrt{0.20^2 + 0.14^2 + 0.09^2 + 0.08^2} = \sqrt{0.0741} = 27.2\ \%$$

which matches the analyser reading. The RMS current is $\sqrt{1 + 0.0741} = 1.036$ times the fundamental, so conductors and transformers carry 3.6 % more current — and, because eddy-current losses grow roughly with the square of frequency, considerably more than 3.6 % extra loss. The capacitor bank, whose impedance falls with frequency, forms a parallel resonance with the supply inductance; if that resonance lies near 250 or 350 Hz, the harmonic currents are magnified, which is how the bank failed. Chapter 9 treats the power-factor side of the same problem.

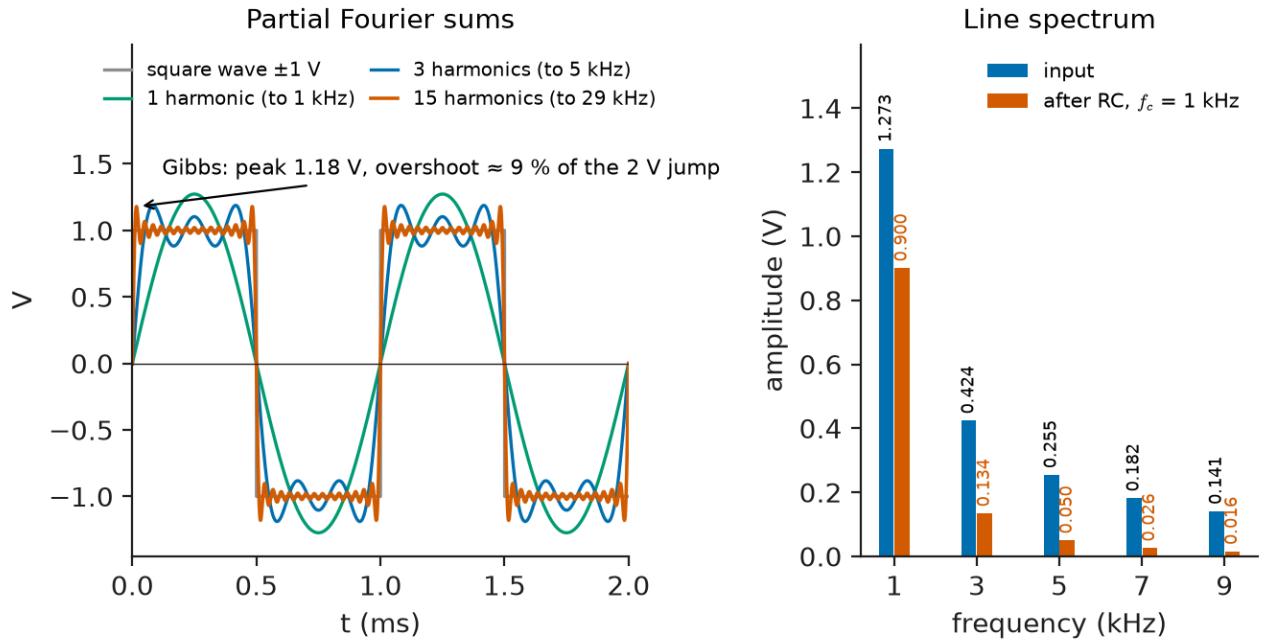

Figure 18.1 — A 1 kHz square wave as a sum of odd harmonics, and the same harmonics after an RC low-pass with cut-off 1 kHz.

Periodic signals through LTI systems

Because each harmonic is an eigenfunction, a periodic input through an LTI system gives a periodic output with the same period and coefficients

$$c_k^{(y)} = H(jk\omega_0)\, c_k^{(x)}$$

Worked example — rounding a square wave. The 1 kHz square wave is passed through an RC low-pass filter with cut-off $f_c = 1$ kHz, so $H(jk\omega_0) = 1/(1+jk)$. The harmonic amplitudes become (Figure 18.1):

Harmonic k	Input A_k (V)	$\lvert H\rvert$	$\angle H$	Output (V)
1	1.273	0.707	−45.0°	0.900
3	0.424	0.316	−71.6°	0.134
5	0.255	0.196	−78.7°	0.050
7	0.182	0.141	−81.9°	0.026
9	0.141	0.110	−83.7°	0.016

The output is dominated by the fundamental; the square wave has become a rounded, nearly triangular-sinusoidal wave. Summing $\lvert c_k^{(y)}\rvert^2$ over all harmonics gives an output power of 0.416 W, of which 0.405 W is in the fundamental. This is how an RC network converts a PWM output from a microcontroller into a smooth analog voltage, and why it needs a cut-off far below the PWM frequency to do so well.

18.2 Fourier transforms

From series to transform

An aperiodic signal — a single pulse, a transient, a burst of speech — can be viewed as a periodic signal whose period has grown without limit. The harmonics crowd together and the line spectrum becomes a continuous function, the **Fourier transform**:

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt, \qquad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

Engineers often prefer ordinary frequency f in hertz, with $X(f) = \int x(t) e^{-j2\pi f t}\, dt$ and $x(t) = \int X(f) e^{j2\pi f t}\, df$; this avoids factors of 2π and is used for spectra below when convenient. X has units of signal × seconds (V/Hz for a voltage): it is a spectral *density*. Energy is recovered by Parseval's relation for the transform,

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df$$

so $|X(f)|^2$ is the **energy spectral density**, in J/Hz into 1 Ω. Periodic signals are included by allowing impulses in X: a sinusoid at f_0 has a transform consisting of two impulses at $\pm f_0$.

> **Boundary.**
> Existence conditions, the inversion theorem and the full catalogue of properties of the Fourier transform are developed in *The Mathematics Universe*. This section collects the results used throughout this book.

Key pairs and properties

Signal $x(t)$	**Transform** $X(f)$
$\delta(t)$	1
1	$\delta(f)$
$\cos(2\pi f_0 t)$	$\tfrac{1}{2}[\delta(f-f_0)+\delta(f+f_0)]$
Rectangular pulse, height A, width τ, centred at 0	$A\tau\,\mathrm{sinc}(f\tau)$, with $\mathrm{sinc}\,u = \sin(\pi u)/(\pi u)$
$e^{-t/\tau}u(t)$	$\tau/(1+j2\pi f\tau)$
Gaussian $e^{-\pi t^2/T^2}$	$T\,e^{-\pi f^2 T^2}$
Ideal low-pass impulse response $2B\,\mathrm{sinc}(2Bt)$	rectangle of height 1 for $\lvert f\rvert < B$

Property	**Time domain**	**Frequency domain**
Linearity	$ax_1 + bx_2$	$aX_1 + bX_2$
Time shift	$x(t-t_0)$	$e^{-j2\pi f t_0}X(f)$
Frequency shift (modulation)	$x(t)e^{j2\pi f_c t}$	$X(f-f_c)$
Time scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X(f/a)$
Convolution	$x * h$	$X\,H$
Multiplication	$x\,w$	$X * W$
Differentiation	dx/dt	$j2\pi f\,X(f)$
Duality	$X(t)$	$x(-f)$

Two rows carry most of the engineering weight. The **convolution property** says that filtering in time is multiplication in frequency — this is why LTI analysis is easy in the frequency domain. The **modulation property** says that multiplying by a carrier slides the spectrum to the carrier frequency — the basis of every radio system (Chapter 26), and the reason a mixer must be time-varying (Chapter 16).

The rectangular pulse

The pulse of height A and width τ has the transform $A\tau\,\mathrm{sinc}(f\tau)$, whose main lobe extends to the first nulls at $f = \pm 1/\tau$ and whose sidelobes decay only as $1/f$ because of the sharp edges. About 90 % of the pulse energy lies within the main lobe. Shorter pulses have wider spectra: halving τ doubles the null-to-null bandwidth. This trade-off is formalised in Section 18.10.

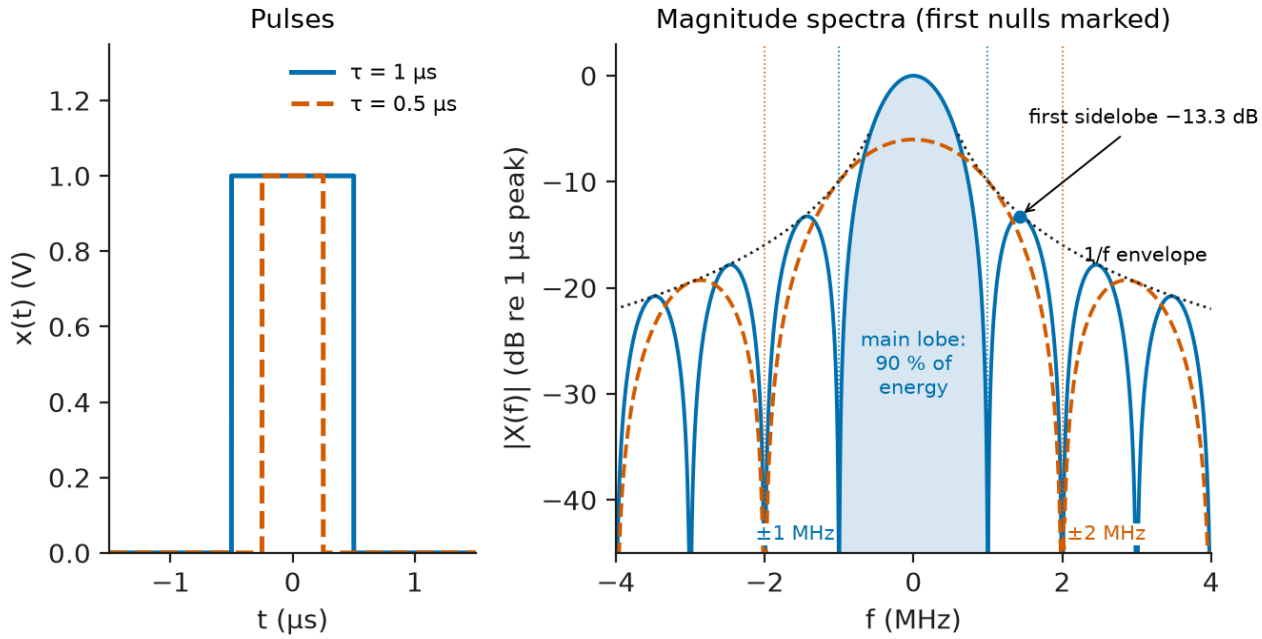

Figure 18.2 — Rectangular pulses of 1 us and 0.5 us and their sinc spectra: halving the duration doubles the spectral width.

Figure 18.2 shows this for two radar-type pulses. A rectangular switching edge in a power converter or a digital clock produces exactly this slowly decaying sidelobe structure, which is why EMC problems extend to frequencies far above the switching rate; slowing the edges (trapezoidal pulses) makes the spectrum fall as $1/f^2$ above $1/(\pi t_r)$.

18.3 Frequency response

Definition and meaning

The **frequency response** of an LTI system is the Fourier transform of its impulse response:

$$H(j\omega) = \int_{-\infty}^{\infty} h(t)\, e^{-j\omega t}\, dt$$

It is the eigenvalue of Chapter 17: the complex gain that multiplies a complex exponential at frequency ω. By the convolution property, the output spectrum of any input is

$$Y(j\omega) = H(j\omega)\, X(j\omega)$$

so the system acts independently on each frequency, changing its amplitude by $|H|$ and its phase by $\angle H$. The frequency response exists (as an ordinary function) for every BIBO-stable system, since $|H(j\omega)| \le \int |h|\, dt < \infty$.

Obtaining $H(j\omega)$

There are three practical routes:

1. **From the circuit**, using impedances $j\omega L$ and $1/(j\omega C)$ (Chapter 8). For the RC low-pass, $H = 1/(1 + j\omega RC)$.
2. **From the differential equation**, replacing each d/dt by $j\omega$ (the differentiation property). For $\sum a_k y^{(k)} = \sum b_k x^{(k)}$,

$$H(j\omega) = \frac{\sum_k b_k (j\omega)^k}{\sum_k a_k (j\omega)^k}$$

3. **By measurement**: a swept sine (network analyser, frequency-response analyser), or a broadband excitation — an impulse, a step, a chirp or noise — with $H = Y/X$ computed from the measured spectra (Chapter 22).

System	Typical frequency-response specification
Hi-fi power amplifier	20 Hz – 20 kHz within ±0.5 dB
Measurement microphone	20 Hz – 20 kHz within ±1 dB, class 1
Diagnostic ECG amplifier	0.05 – 150 Hz (−3 dB)
Telephone voice channel	300 – 3400 Hz
Current transformer for protection	flat at 50/60 Hz; bandwidth a few kHz
100 MHz oscilloscope	−3 dB at 100 MHz
Piezoelectric accelerometer	flat within 5 % up to about 1/3 of mounted resonance

The frequency response is the natural specification for amplifiers, sensors and channels because it answers directly the question a user asks: which frequencies get through, and how altered?

18.4 Transfer functions

From $H(s)$ to $H(j\omega)$

The transfer function $H(s)$ of Chapter 17 — the eigenvalue for e^{st}, equivalently the Laplace transform of h — is defined for complex $s = \sigma + j\omega$. The frequency response is its value on the imaginary axis:

$$H(j\omega) = H(s)\big|_{s=j\omega}$$

valid whenever the imaginary axis lies in the region where $H(s)$ converges, which for a causal system is precisely when the system is stable. For lumped systems $H(s)$ is a ratio of polynomials,

$$H(s) = K\frac{(s - z_1)(s - z_2)\cdots(s - z_M)}{(s - p_1)(s - p_2)\cdots(s - p_N)}$$

with zeros z_i and poles p_i. Chapter 11 develops $H(s)$ from circuits; Chapter 13 develops Bode plots from it. Here its use is geometric.

Reading the frequency response from poles and zeros

At $s = j\omega$, each factor $(j\omega - z_i)$ is a vector from the zero z_i to the point $j\omega$ on the imaginary axis, and similarly for poles. Hence

$$|H(j\omega)| = |K| \frac{\prod |j\omega - z_i|}{\prod |j\omega - p_i|}, \qquad \angle H(j\omega) = \angle K + \sum \angle(j\omega - z_i) - \sum \angle(j\omega - p_i)$$

As ω moves up the axis, the gain falls where it passes close to a zero (that zero's vector becomes short in the numerator) and rises where it passes close to a pole (that pole's vector becomes short in the denominator). Two rules summarise the geometry:

- A **zero on the imaginary axis** at $\pm j\omega_z$ makes $|H| = 0$ at ω_z: a perfect notch.
- A **pole close to the imaginary axis** at $-\sigma \pm j\omega_d$ with small σ makes a sharp peak near ω_d: a resonance, whose −3 dB width is about 2σ rad/s.

Worked example — a 50 Hz notch filter. The second-order notch

$$H(s) = \frac{s^2 + \omega_0^2}{s^2 + (\omega_0/Q)\, s + \omega_0^2}, \qquad \omega_0 = 2\pi \cdot 50 \text{ rad/s}, \quad Q = 10$$

has zeros exactly on the axis at $\pm j\omega_0$ and poles at $-\omega_0/(2Q) \pm j\omega_0\sqrt{1 - 1/(4Q^2)} = -15.7 \pm j313.8$ rad/s, just to the left of the zeros. Far from 50 Hz the pole and zero vectors have nearly equal lengths and $|H| \approx 1$; at 50 Hz the zero vector vanishes. Evaluating $H(j2\pi f)$:

f (Hz)	$\lvert H\rvert$	dB	$\angle H$
10	0.9998	−0.002	−1.2°
45	0.904	−0.88	−25.3°
47.56	0.707	−3.0	−45°
49	0.375	−8.5	−68.0°
50	0	−∞	jumps from −90° to +90°
52.56	0.707	−3.0	+45°
60	0.965	−0.31	+15.3°
100	0.998	−0.02	+3.8°

The −3 dB stop band runs from 47.56 to 52.56 Hz, a width of $f_0/Q = 5$ Hz. The notch removes mains pickup from an ECG or a strain-gauge signal while leaving most of the spectrum untouched — in magnitude. The phase column shows the cost, examined in Section 18.8.

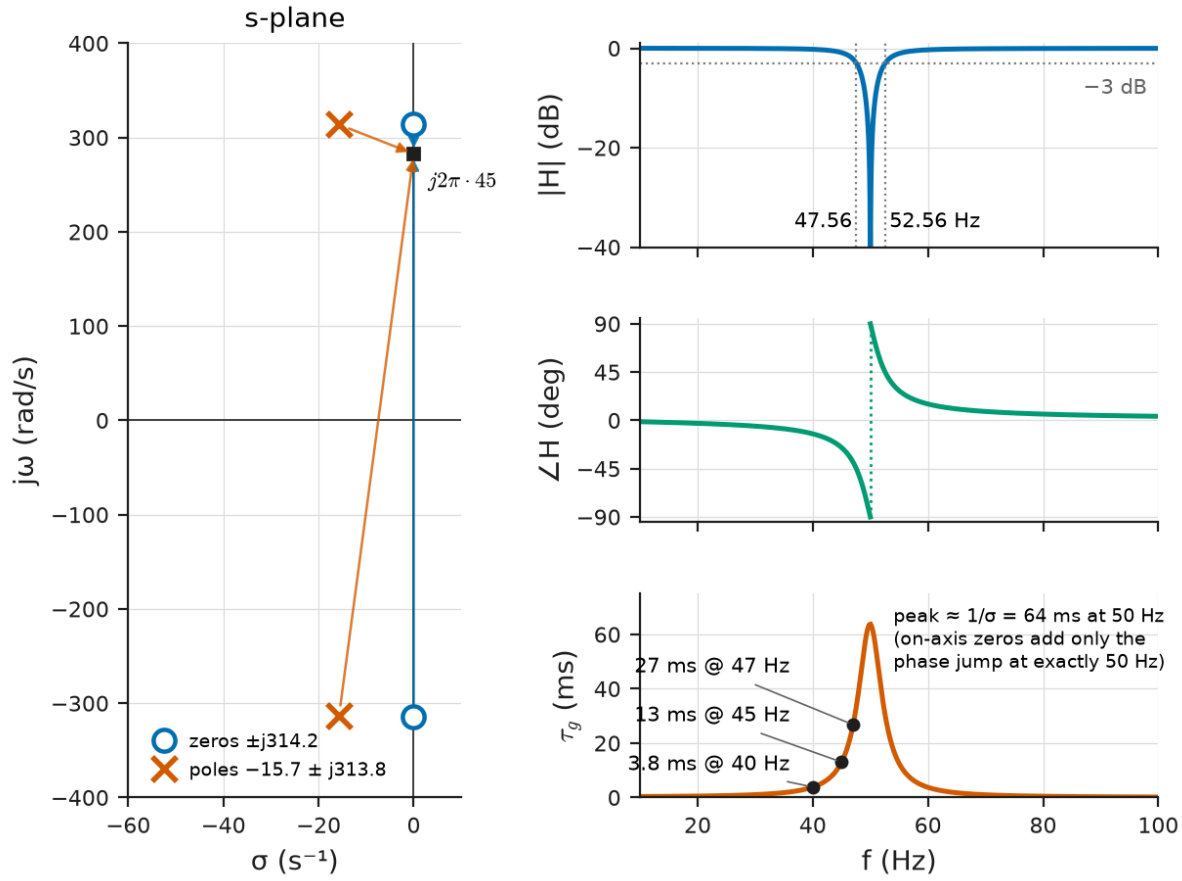

Figure 18.3 — A 50 Hz notch filter with Q = 10: pole-zero plot and the resulting magnitude, phase and group delay.

Figure 18.3 shows the pole-zero geometry and the resulting response.

18.5 Magnitude and phase

Decibels and logarithmic axes

Magnitude response is usually plotted in decibels, $20\log_{10}|H|$, against frequency on a logarithmic axis — the Bode plot of Chapter 13. The logarithm makes cascaded gains add, makes the slopes of poles and zeros straight lines (−20 dB/decade per pole, +20 dB/decade per zero), and fits the six or more decades of interest in audio, instrumentation and RF onto one page.

Ratio $\lvert H\rvert$	dB
10	+20
2	+6.0
$\sqrt{2}$	+3.0
1	0
$1/\sqrt{2} = 0.707$	−3.0
0.5	−6.0
0.1	−20
0.001	−60

What phase does

Magnitude says how much of each frequency survives; phase says how the surviving components line up in time. A constant phase shift is rarely what is wanted. The phase response that preserves waveform shape is a straight line through the origin, $\angle H = -\omega t_0$, which corresponds to a pure delay t_0 (time-shift property). Any other phase response shifts different frequencies by different amounts of time. The ear is relatively tolerant of phase changes in steady tones, but transients, pulses, digital data, ECG complexes and control signals are all sensitive to them.

Magnitude and phase are not independent for causal minimum-phase systems

For a causal, stable system with no zeros in the right half-plane — a **minimum-phase** system — the phase is determined by the magnitude (through the Hilbert-transform relations; Bode's gain-phase relation is the engineering form). Roughly, a magnitude slope of $-20n$ dB/decade, sustained over a wide band, is accompanied by a phase lag of about $n \times 90°$. This is why one cannot design a filter with an arbitrarily steep magnitude cut-off and a small phase lag at the same time, and why it is enough to check gain slope near crossover in many feedback designs (Chapter 34). Systems with right-half-plane zeros, or pure delays, have *more* phase lag than their magnitude implies: they are non-minimum-phase.

> **Status of the principle: Gain–phase relation for minimum-phase systems**
> *Derived*. For causal, stable systems with no right-half-plane zeros, the phase response is fixed by the magnitude response through the Hilbert-transform relations; the "$n \times 90°$ per $-20n$ dB/decade" rule is its approximate form for slopes held over about a decade or more.

Measuring magnitude and phase

Instrument	What it measures	Typical range
Frequency-response analyser	$\lvert H\rvert$, $\angle H$ by swept sine with correlation	about 10 µHz – 30 MHz
Audio analyser	gain, phase, THD+N	about 10 Hz – 100 kHz
Vector network analyser	S-parameters (magnitude and phase)	kHz to tens or hundreds of GHz
Dynamic signal analyser	$H = Y/X$ from FFTs of broadband excitation	DC – about 100 kHz
Power-quality analyser	harmonic magnitudes and phase angles	up to the 50th harmonic (2.5 kHz at 50 Hz)

18.6 Filtering

Ideal filters

A filter is a system designed to pass some frequencies and reject others. The ideal versions have $\lvert H\rvert = 1$ in the **passband**, $\lvert H\rvert = 0$ in the **stopband**, and linear phase throughout:

Type	Passband	Typical use
Low-pass	$\lvert f\rvert < f_c$	anti-aliasing, smoothing, PWM reconstruction
High-pass	$\lvert f\rvert > f_c$	AC coupling, ECG baseline removal, rumble filtering
Band-pass	$f_1 < \lvert f\rvert < f_2$	channel selection in radio, vibration bands, EEG rhythms
Band-stop (notch)	all except $f_1 < \lvert f\rvert < f_2$	mains hum, interfering carrier
All-pass	all frequencies, $\lvert H\rvert = 1$	phase/delay equalisation

The ideal low-pass filter with cut-off B has impulse response $h(t) = 2B\,\mathrm{sinc}(2Bt)$, which extends to $t = -\infty$. It is non-causal (Chapter 17) and cannot be built. Truncating and delaying the sinc gives a realisable approximation with a delay, a transition band and ripple — the window method of FIR design in Chapter 23. Every real filter therefore has a **transition band** between passband and stopband, a **passband ripple**, and a finite **stopband attenuation**.

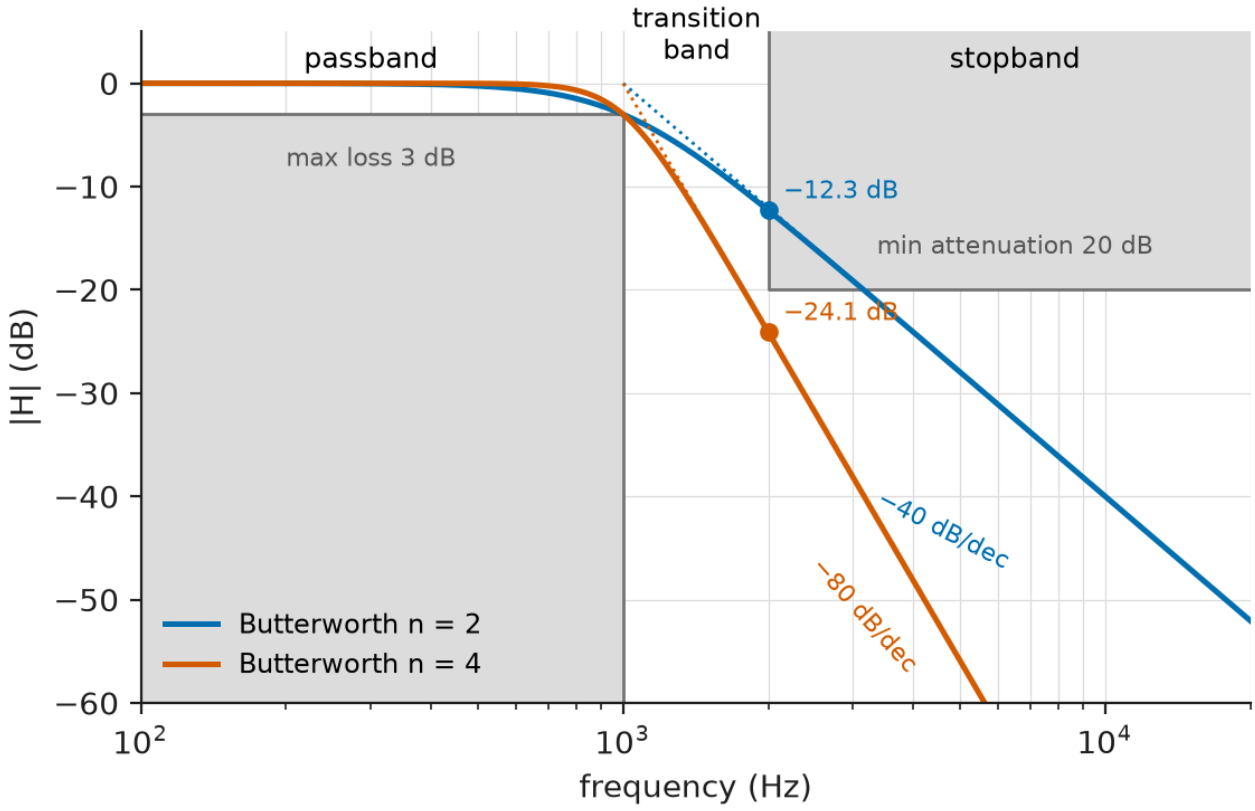

Figure 18.4 — Filter specification template and second- and fourth-order Butterworth low-pass responses with cut-off 1 kHz.

Approximating the ideal

Practical analog filters are built from standard rational approximations. The **Butterworth** response is maximally flat in the passband:

$$|H(j\omega)|^2 = \frac{1}{1 + (\omega/\omega_c)^{2n}}$$

for order n. At $\omega = \omega_c$ the loss is 3 dB for any n; one octave above, at $2\omega_c$, it is $10\log_{10}(1 + 2^{2n})$: 12.3 dB for $n = 2$ and 24.1 dB for $n = 4$ (Figure 18.4). The **Chebyshev** response trades passband ripple for a steeper transition; the **elliptic** response adds stopband ripple for the steepest transition of all; the **Bessel** response gives up selectivity for the most nearly linear phase. The choice among them is a choice about what matters more — selectivity, flatness, or waveform fidelity — and is made in Chapter 23 for digital filters and in *Electronics & Computing* for active analog realisations.

> **Status of the principle: Realisable filters cannot be ideal**
> *Derived*. A causal filter cannot have zero gain over a finite band (Paley–Wiener condition) and cannot have a discontinuous magnitude response, so every real filter has a transition band and finite stopband attenuation.

Filtering examples across EEE

Application	Filter	Typical numbers
Audio ADC anti-aliasing	low-pass	passband to 20 kHz, ≥ 80–100 dB rejection above 24 kHz (mostly digital, after oversampling)
ECG baseline wander removal	high-pass	0.05 Hz (diagnostic), 0.5 Hz (monitoring)
Mains interference removal	notch	50 or 60 Hz, Q about 5–30
FM broadcast receiver IF	band-pass	10.7 MHz centre, about 150–250 kHz wide
Protection relay	band-pass around fundamental	rejects DC offset and harmonics
Switched-mode power supply output	LC low-pass	corner well below the switching frequency
Radar receiver	matched filter	bandwidth ≈ 1/pulse length (Chapter 27)

18.7 Bandwidth

"Bandwidth" is used for both signals and systems, and it has several precise definitions. Confusing them is a common source of factors of two.

Definitions for systems

- **−3 dB (half-power) bandwidth**: the range over which $|H| \geq |H|_{\max}/\sqrt{2}$. For a low-pass system it is the cut-off frequency; for a band-pass system it is $f_2 - f_1$, and $Q = f_0/(f_2 - f_1)$.
- **Noise-equivalent bandwidth**: the width of an ideal rectangular filter, with the same peak gain, that passes the same white-noise power,

$$B_N = \frac{1}{|H|^2_{\max}} \int_0^\infty |H(f)|^2 df$$

  For a single RC pole, $B_N = \frac{\pi}{2} f_c$; for higher-order filters it approaches f_c. It is the bandwidth that sets the noise power kTB_N in Chapter 29.
- **Passband (ripple) bandwidth**: the range over which the response stays within a stated tolerance (±0.1 dB, ±1 dB). Audio and instrumentation specifications use this.

Definitions for signals

- **Null-to-null bandwidth**: the width of the main spectral lobe, $2/\tau$ for a rectangular pulse (two-sided) or $1/\tau$ one-sided.
- **Fractional-power bandwidth**: the band containing a stated fraction (99 %) of the power. Spectrum regulators use this as "occupied bandwidth".

- **RMS bandwidth**: the standard deviation of $|X(f)|^2$ treated as a distribution, used in the uncertainty relation (Section 18.10).

Bandwidth and rise time

A low-pass system cannot follow a fast edge. For a single-pole system with cut-off f_c, the 10–90 % rise time of the step response is $2.2\tau = 2.2/(2\pi f_c)$:

$$t_r \approx \frac{0.35}{f_c}$$

The constant varies slightly with the response shape (about 0.35–0.45 for typical well-damped systems), and the rule is widely used to relate bandwidth to edge speed.

> **Status of the principle: Rise time – bandwidth rule**
> *Approximate.* Exact for a single-pole low-pass ($t_r = 0.35/f_c$); within roughly ±25 % for other well-damped low-pass responses; meaningless for strongly resonant systems, whose step response rings.

Worked example — what an oscilloscope can show. A 100 MHz oscilloscope with an approximately Gaussian response has its own rise time $t_{r,\text{scope}} \approx 0.35/100\ \text{MHz} = 3.5$ ns. Cascaded rise times of well-damped systems add approximately in root-sum-square. A logic edge whose true rise time is 5 ns is displayed with

$$t_{r,\text{displayed}} \approx \sqrt{5^2 + 3.5^2} = 6.1\ \text{ns}$$

a 22 % error. To keep the error below about 2 %, the instrument must be roughly five times faster than the signal — here a 500 MHz oscilloscope. The same rule sizes the bandwidth of current probes for switching-converter measurements and of the amplifiers in a pulse-oximeter or ultrasound front end.

Signal or channel	Typical bandwidth
ECG (diagnostic)	0.05 – 150 Hz
EEG	about 0.5 – 70 Hz
Telephone speech	300 – 3400 Hz
Wideband (HD) voice	50 – 7000 Hz
High-fidelity audio	20 Hz – 20 kHz
AM broadcast channel	9 kHz (Region 1) or 10 kHz (Americas)
FM broadcast channel	about 200 kHz
Wi-Fi channel	20, 40, 80 or 160 MHz
5G NR carrier (below 6 GHz)	up to 100 MHz

18.8 Group delay

Phase delay and group delay

Write the frequency response as $H(j\omega) = |H(j\omega)|e^{j\phi(\omega)}$. Two delays can be defined from the phase:

$$\tau_p(\omega) = -\frac{\phi(\omega)}{\omega} \qquad \text{(phase delay)}, \qquad \tau_g(\omega) = -\frac{d\phi(\omega)}{d\omega} \qquad \text{(group delay)}$$

The phase delay is the time shift of a single steady sinusoid. The group delay is the time shift of the **envelope** of a narrowband signal centred on ω — a modulated carrier, a tone burst, a wave packet. To see why, consider two closely spaced tones at ω_1 and ω_2, whose sum has an envelope at the difference frequency. After the system, the envelope is shifted by $-(\phi_2 - \phi_1)/(\omega_2 - \omega_1)$, which tends to $-d\phi/d\omega$ as the tones approach each other. Since information is carried by envelopes, group delay is the delay that matters for pulses, data and modulated signals.

> **Status of the principle: Envelope travels at the group delay**
> *Approximate.* Exact for a pair of tones; for a general narrowband signal it holds when the phase is close to linear and the magnitude close to constant across the signal bandwidth — the narrowband assumption.

Constant group delay means no phase distortion

If $\phi(\omega) = -\omega t_0$ (linear phase), then $\tau_p = \tau_g = t_0$ at all frequencies and every component is delayed equally: the waveform is preserved. If the group delay varies across the band of interest, different frequency components of a pulse arrive at different times and the pulse spreads. This is the frequency-domain description of the dispersion of Chapter 14 (transmission lines) and Chapter 31 (optical fibre).

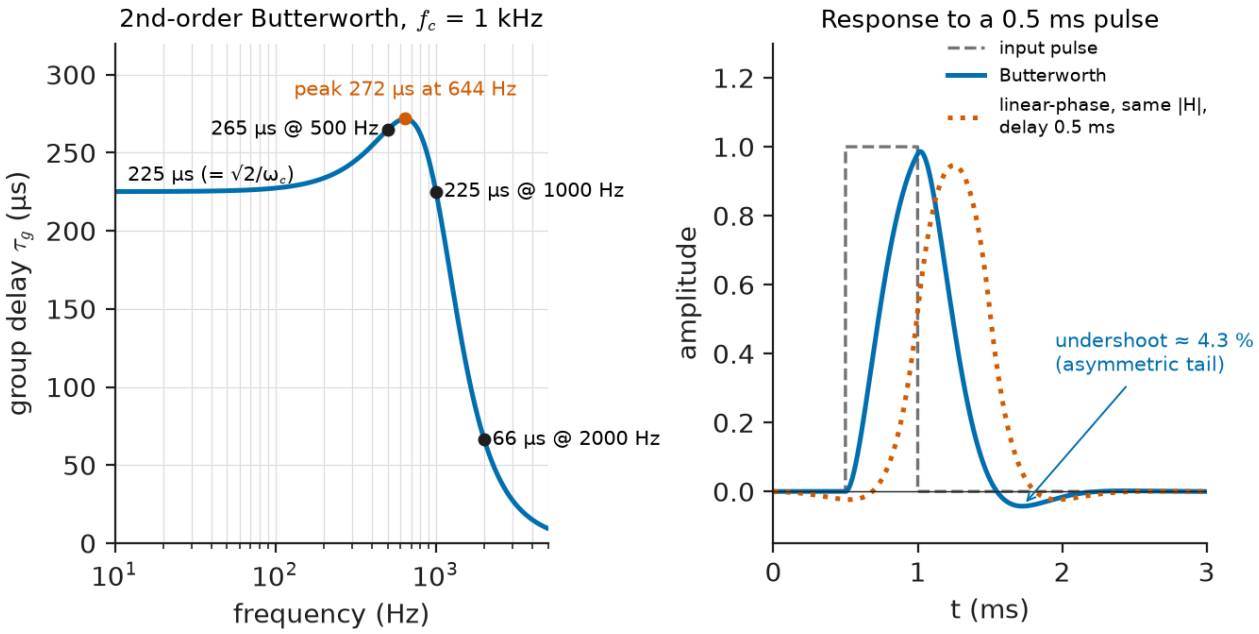

Figure 18.5 — Group delay of a 2nd-order Butterworth low-pass with fc = 1 kHz, and the effect of nonconstant delay on a pulse.

Worked example — group delay of a Butterworth filter. A second-order Butterworth low-pass with $f_c = 1$ kHz has $H(s) = \omega_c^2/(s^2 + \sqrt{2}\,\omega_c s + \omega_c^2)$, $\omega_c = 6283$ rad/s. At low frequency the group delay is $\sqrt{2}/\omega_c = 225$ µs. Computing $-d\phi/d\omega$ numerically: 265 µs at 500 Hz, a peak of 272 µs near 644 Hz, 225 µs at 1 kHz, and 66 µs at 2 kHz (Figure 18.5). The delay varies by about 21 % across the passband, peaking just below cut-off. A pulse whose spectrum spans the passband therefore emerges slightly asymmetric, with a small undershoot after it (about 4 %); the same phase distortion gives the step response its 4.3 % overshoot. For an audio crossover these differences of tens of microseconds are inaudible; for a 10 kbit/s data signal filtered by the same circuit, they distort the bit shapes and create intersymbol interference. The Bessel filter, designed for maximally flat group delay, reduces this variation at the cost of a gentler cut-off.

Worked example — the notch filter's hidden delay. The 50 Hz notch of Section 18.4 has group delay 3.8 ms at 40 Hz, 13 ms at 45 Hz and 27 ms at 47 Hz, compared with almost zero far from the notch. Components of a step or of an ECG complex lying near 50 Hz are delayed by tens of milliseconds relative to the rest; after a sharp transient they emerge late, as a decaying 50 Hz ringing of duration about $Q/(\pi f_0) \approx 64$ ms per time constant. This explains the flow-meter ringing in the opening: a notch is a sharp resonance in reverse, and sharpness in frequency always costs duration in time (Section 18.10). Lowering Q shortens the ringing but widens the notch; replacing the notch by a moving average over exactly one mains cycle (Chapter 16) trades the ringing for a fixed 10 ms delay and broader smoothing.

Group delay in practice

System	Typical group delay or its variation
Coaxial cable, velocity factor 0.66	5.05 ns/m, constant (linear phase)
Linear-phase FIR filter of N taps	$(N-1)/2$ samples, constant
Second-order Butterworth, $f_c = 1$ kHz	225 µs at DC, 272 µs peak
Narrow band-pass or notch, $Q = 10$ at 50 Hz	tens of ms near the centre
Decimation filter of a delta-sigma audio ADC at 48 kHz	typically about 0.2–1 ms, linear phase

---

18.9 Distortion

Distortionless transmission

A system is **distortionless** if the output is a scaled, delayed copy of the input,

$$y(t) = K\,x(t - t_0) \quad \Longleftrightarrow \quad H(j\omega) = K\,e^{-j\omega t_0}$$

over the band occupied by the signal: constant magnitude and linear phase. Any departure is **distortion**. It is useful to separate two kinds, because their causes, signatures and cures differ completely.

> **Status of the principle: Condition for distortionless transmission**
> *Derived*. From the time-shift property of the Fourier transform: constant gain and linear phase across the signal's band are necessary and sufficient for the output to be a scaled, delayed replica.

Linear distortion

Linear distortion arises in an LTI system that is not distortionless. It changes the relative amplitudes and timings of frequency components that are already present, but creates no new ones.

- **Amplitude (attenuation) distortion**: $|H|$ not constant across the band. The rounded square wave of Section 18.1, the muffled sound of a telephone channel, the loss of high-frequency detail through a long cable.
- **Phase (delay) distortion**: group delay not constant. The asymmetric pulse of Section 18.8, the ringing of the notch filter, the dispersion of a data pulse on a long line.

Because it is linear, this kind of distortion can in principle be undone by an inverse system, an **equaliser**, with response $1/H$ over the band (subject to the stability and noise limits of Chapter 17). Modems, DSL, optical receivers and loudspeaker processors all equalise. Offline, phase distortion can also be removed completely by forward–backward (zero-phase) filtering, a non-causal operation (Chapter 16).

Nonlinear distortion

Nonlinear distortion arises from a nonlinear system. It creates new frequencies (Chapter 16): **harmonics** kf of each input tone and **intermodulation products** $mf_1 \pm nf_2$ of pairs of tones. With a cubic term a_3x^3, two tones at f_1 and f_2 produce third-order products at $2f_1 - f_2$ and $2f_2 - f_1$, which lie close to the original tones and cannot be filtered out — the reason radio receivers specify a third-order intercept point (Chapter 32) and why intermodulation between adjacent carriers is a major constraint in cellular base stations.

Nonlinear distortion cannot be removed by an LTI equaliser. Its cures are to stay in the linear region (back off the signal level), linearise the device with feedback (Chapter 17) or predistortion, or accept and filter harmonics that fall outside the band of interest.

Measure	Definition	Typical values
THD	RMS of harmonics / fundamental	0.001–0.05 % good audio amplifier; IEEE 519 voltage limit 5–8 % on power systems
THD+N	RMS of everything except the fundamental / total	audio analyser standard
IMD	intermodulation products relative to tones	SMPTE and CCIF audio tests; −30 to −50 dBc in RF power amplifiers
Third-order intercept (IP3)	extrapolated input or output level where third-order products equal fundamentals	about +10 to +40 dBm for RF amplifiers
Adjacent-channel leakage ratio	power leaking into the next channel	about −45 dBc required in LTE

Telling them apart

A simple experiment distinguishes the two kinds. Apply a single pure tone and examine the output spectrum. If new frequencies appear, the distortion is nonlinear. If only the input frequency appears but its amplitude or phase depends on frequency, the distortion is linear. Applying the tone at two different amplitudes gives a second check: linear distortion is independent of level; nonlinear distortion grows with it.

Worked example — identifying distortion on a telemetry link. A pulse-code telemetry signal from a wind-turbine sensor is received with rounded, smeared bits. A test with a single 2 kHz tone at the transmitter shows at the receiver only 2 kHz, 6 dB down, with a 30° phase lag; at 500 Hz the loss is 0.5 dB and the lag 8°. No harmonics appear above −60 dBc. The distortion is therefore linear — amplitude and phase distortion from the cable and line filters — and an equaliser with about +5.5 dB of gain lift at 2 kHz relative to 500 Hz, together with phase correction, can restore the pulse shape. Had a 4 kHz harmonic appeared at −20 dBc, the cure would instead have been to reduce the drive level or replace a saturating line driver.

18.10 Time–frequency relationships

Scaling: short in time means wide in frequency

The scaling property $x(at) \leftrightarrow \frac{1}{|a|}X(f/a)$ says that compressing a signal in time by a factor a stretches its spectrum by the same factor. Playing a tape at double speed doubles every frequency; a pulse half as long needs twice the bandwidth. For any pulse shape, the product of a suitably defined duration Δt and bandwidth Δf is fixed by the shape, not the scale.

The uncertainty relation

With duration and bandwidth defined as RMS widths of $|x(t)|^2$ and $|X(f)|^2$,

$$\Delta t \, \Delta f \;\geq\; \frac{1}{4\pi}$$

with equality only for the Gaussian pulse. No signal can be arbitrarily concentrated in both time and frequency. Practical consequences appear everywhere:

- A spectrum analyser that must resolve two tones 1 Hz apart must observe for at least about 1 s (Chapter 22).
- A narrow filter (small bandwidth) has a long impulse response and rings: the 5 Hz-wide notch of Section 18.4 rings for tens of milliseconds.
- A short-time spectrum of speech or vibration must trade time resolution against frequency resolution (the spectrogram, Chapter 24).
- A data symbol of duration T_s needs a bandwidth of order $1/T_s$: doubling the symbol rate doubles the bandwidth (Chapter 27).

> **Status of the principle: Time–bandwidth uncertainty**
> *Derived*. A mathematical theorem about Fourier pairs (the Gabor limit), valid for any signal whatever its physical origin; the constant depends on how duration and bandwidth are defined.

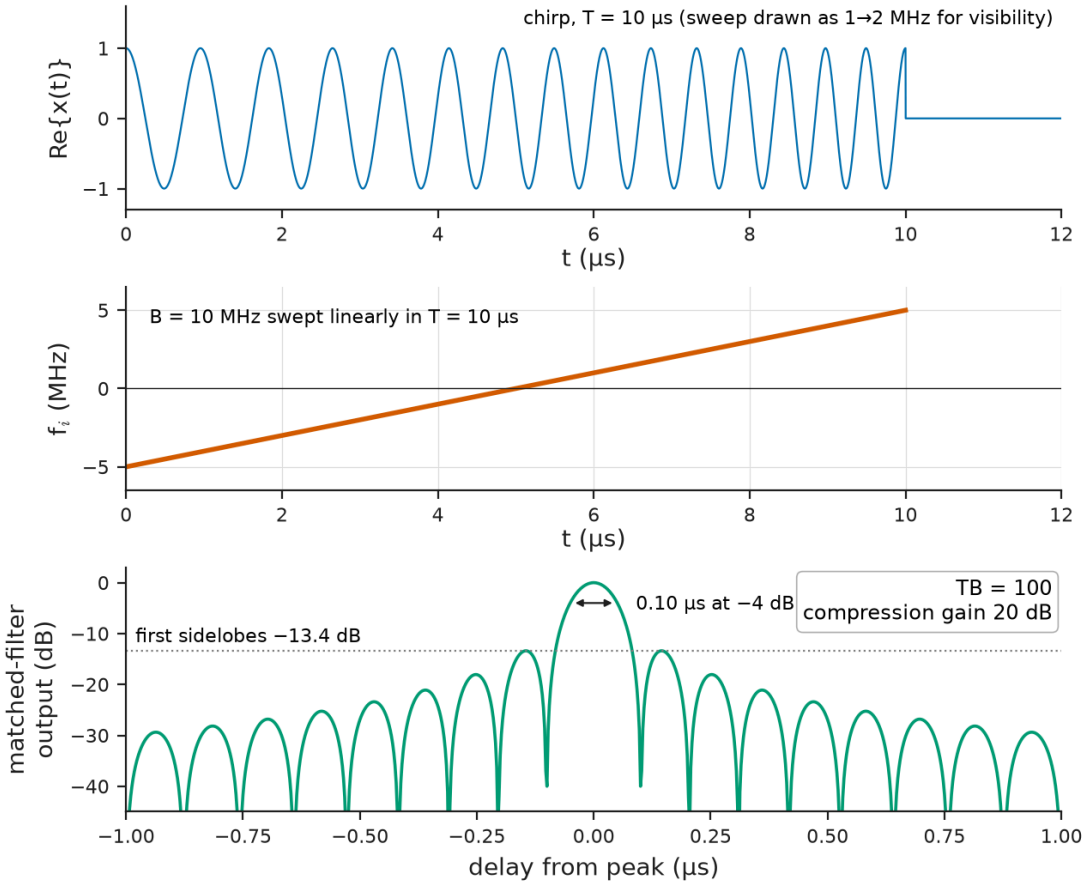

Figure 18.6 — A linear-FM chirp lasts 10 us yet occupies 10 MHz; matched filtering compresses it to a peak about 0.1 us wide.

Worked example — radar range resolution and pulse compression. A radar distinguishes two targets in range only if their echoes do not overlap. A simple rectangular pulse of duration τ gives range

resolution $\Delta R = c\tau/2$: for $\tau = 1$ µs, $\Delta R = 3 \times 10^8 \times 10^{-6}/2 = 150$ m. Shortening the pulse improves resolution but reduces the energy on target for a given peak transmitter power, and with it the detection range. The time–frequency view breaks the deadlock. Resolution actually depends on bandwidth, $\Delta R \approx c/(2B)$, not on duration as such; a simple pulse merely ties the two together through $B \approx 1/\tau$. A **chirp** — a pulse of duration $T = 10$ µs whose frequency sweeps linearly over $B = 10$ MHz — has the energy of a 10 µs pulse and the bandwidth of a 0.1 µs pulse. After a matched filter (Chapter 27), its echo is compressed to a peak about $1/B = 0.1$ µs wide, giving $\Delta R = c/(2B) = 15$ m. The time–bandwidth product $TB = 100$ is the compression ratio, a 20 dB gain in peak signal-to-noise ratio over an uncompressed pulse of the same peak power and resolution (Figure 18.6). Medical ultrasound, sonar and some automotive radars use the same principle.

Time–frequency relationships in system terms

The same reciprocity applies to systems. A system with bandwidth B has an impulse response of duration of order $1/B$, a rise time of about $0.35/B$, and — if it is selective, with a narrow band-pass of width Δf — a build-up and decay time of order $1/(\pi\Delta f)$. The table collects these links:

Frequency-domain feature	Time-domain consequence
Wide bandwidth B	fast rise time, $t_r \approx 0.35/B$
Narrow resonance, bandwidth Δf	long ringing, decay time constant $\approx 1/(\pi\Delta f)$
Sharp cut-off (steep transition)	long, oscillating impulse response; Gibbs ringing
Linear phase	symmetric impulse response, constant delay
Nonconstant group delay	dispersion, asymmetric pulse distortion
Smooth spectrum (no sharp features)	short, compact pulse
Discontinuity in time (sharp edge)	spectrum decaying as $1/f$

These are the relationships an engineer uses to estimate, in seconds, what a frequency-domain specification will do to a waveform — and to spot specifications that cannot both be met, such as a filter required to have a very steep cut-off and also to settle within a few cycles.

What remains open

Open question	Where it is answered
What happens to a signal's spectrum when it is sampled, and how is the original recovered?	Chapter 19 (sampling and reconstruction)
What is the discrete-time frequency response, and how does the z-transform relate to it?	Chapter 20
How are spectra computed from finite records of data, and what are leakage and windowing?	Chapter 22 (DFT)
How are FIR and IIR filters designed to meet passband, stopband and phase specifications?	Chapter 23
How are time and frequency resolved jointly for non-stationary signals?	Chapter 24 (time-frequency analysis, wavelets)
How is bandwidth traded against data rate and noise in communication links?	Chapters 25, 27 and 28
How is noise bandwidth used to compute noise power and SNR?	Chapter 29
How do dispersion in cables and fibres arise physically?	Chapter 14 (transmission lines), Chapter 31 (optical fibre)
How do magnitude and phase of a loop gain decide closed-loop stability?	Chapter 34 (Bode and Nyquist methods)
How are Bode plots built from poles and zeros in detail?	Chapter 13

Next: Chapter 19 applies the frequency-domain view to the bridge between continuous and discrete signals, showing when a sampled signal determines the original exactly and what goes wrong when it does not.

Chapter 19 — Sampling and Reconstruction

Part IV · Signals and Systems

A power-quality analyser is clipped onto the feeder of a factory that runs large six-pulse variable-speed drives. The instrument samples each voltage and current channel 32 times per 50 Hz cycle, 1600 samples per second, and reports a harmonic spectrum. The report shows a strong 15th harmonic at 750 Hz, and the plant engineer orders a tuned filter for it. The filter is installed, resonates, and does nothing useful, because there was very little 15th harmonic on the feeder. What the drives actually produced was a large 17th harmonic at 850 Hz, and the analyser, with no analogue filter worth the name in front of its converter, had folded 850 Hz down onto 750 Hz. Every sample the instrument took was correct; the conclusion drawn from them was wrong. The same thing happens when a cheap audio interface turns an ultrasonic bat call into an audible chirp, when a 500 Hz ECG recorder shows a slow ghost wave that is really interference from a switching supply, and when a spinning fan in a video appears to stand still.

This chapter is about the bridge between the continuous-time world of Chapters 15–18 and the discrete-time world of Chapter 20 and Part V. Sampling replaces a function of time by a list of numbers; reconstruction turns a list of numbers back into a voltage. The question is when that round trip loses nothing, what it costs when it does, and how real hardware — finite-width sampling pulses, jittery clocks, staircase outputs, finite word lengths and imperfect filters — differs from the ideal.

> **When does a sequence of samples contain everything there is to know about a continuous-time signal, and how can the signal be rebuilt from it?**
> **What happens to frequencies the sampler cannot represent, and how do real samplers, hold circuits and filters change the ideal answer?**

Concept	Engineering destination
Ideal (impulse) sampling	The mathematical model behind every ADC, digital scope and sampled-value relay
Sampling theorem	Choosing sample rates for audio, biomedical, power and radio systems
Nyquist frequency and Nyquist rate	Datasheet sample rates, oscilloscope bandwidth-to-sample-rate ratios
Aliasing	Harmonic-analyser errors, stroboscopic effects, bandpass (IF) sampling in radios
Practical sampling (finite aperture, jitter)	Track-and-hold design limits, SNR of high-frequency ADCs
Reconstruction	DAC outputs, audio playback, arbitrary waveform generators
Sample-and-hold behaviour	Sinc droop in DACs, delay in digital control loops
Quantization interface	ADC word length, dynamic range of digitised measurements
Anti-aliasing filters	Front ends of data-acquisition systems, ECG and audio converters

Roadmap. Sampling as multiplication by an impulse train (19.1) → the spectrum of a sampled signal and the sampling theorem (19.2) → Nyquist frequency and Nyquist rate (19.3) → aliasing, including its deliberate use (19.4) → practical sampling with finite pulses, apertures and jitter (19.5) → ideal and practical reconstruction (19.6) → sample-and-hold behaviour (19.7) → the quantization interface to Chapter 21 (19.8) → anti-aliasing filters (19.9).

19.1 Ideal sampling

Sampling as a model

A sampler looks at a continuous-time signal $x(t)$ at regularly spaced instants $t = nT$ and records the values

$$x[n] = x(nT), \qquad n \in \mathbb{Z}$$

where T is the **sampling interval** and $f_s = 1/T$ the **sampling rate** (samples per second, quoted in Sa/s or simply Hz). The square brackets mark a discrete-time sequence, as in Chapter 15. The angular sampling frequency is $\omega_s = 2\pi f_s$.

The sequence $x[n]$ is a list of numbers, not a function of continuous time, and that makes it awkward to compare with $x(t)$ in the frequency domain. The standard device is to represent the sampled signal as a continuous-time signal that carries the sample values as the areas of impulses:

$$x_s(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT) = \sum_{n=-\infty}^{\infty} x(nT)\,\delta(t - nT)$$

The second form uses the sifting property of the impulse (Chapter 15): multiplying a continuous function by $\delta(t - nT)$ keeps only its value at nT. The impulse train

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

is called the **sampling function** or Dirac comb. Ideal sampling is therefore modulation: the signal multiplies a periodic carrier made of impulses.

No physical circuit produces impulses. A real sampler closes a switch for a short but finite time, or a track-and-hold amplifier follows the input and then freezes it. Ideal sampling is the limiting model in which the switch closes for an infinitesimal time; Section 19.5 restores the finite width. The model is used because it keeps the information content of the samples exactly and lets the Fourier transform of Chapter 18 do the rest.

Why the impulse model gives the right answer

One might worry that $x_s(t)$ is a mathematical fiction — infinite spikes no instrument could produce. What makes it legitimate is that its Fourier transform depends only on the numbers $x(nT)$:

$$X_s(f) = \int_{-\infty}^{\infty} x_s(t)\, e^{-j2\pi f t}\, dt = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j2\pi f n T}$$

The right-hand side is precisely the discrete-time Fourier transform of the sequence $x[n]$ evaluated at the digital frequency $\Omega = 2\pi f T$ (Chapter 20 and Chapter 22). The impulse-train representation is therefore not an extra physical assumption; it is a bookkeeping device that lets the spectrum of a sequence be drawn on the same frequency axis, in hertz, as the spectrum of the original analogue signal.

> **Status of the principle: Impulse-train model of sampling**
> *Derived*. An exact mathematical representation of the sample values; it describes a physical sampler only in the limit of vanishing aperture time and zero clock jitter, which Section 19.5 relaxes.

19.2 The spectrum of a sampled signal and the sampling theorem

Periodic replication

The comb $p(t)$ is periodic with period T, so it has a Fourier series. All its coefficients are equal to $1/T$ (Chapter 18):

$$p(t) = \frac{1}{T} \sum_{k=-\infty}^{\infty} e^{j2\pi k f_s t} \qquad \Longleftrightarrow \qquad P(f) = \frac{1}{T} \sum_{k=-\infty}^{\infty} \delta(f - k f_s)$$

A comb in time is a comb in frequency, with spacing f_s. Multiplication in time becomes convolution in frequency, and convolving $X(f)$ with a shifted impulse simply shifts it, so

$$X_s(f) = X(f) * P(f) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(f - k f_s)$$

This one line is the whole theory of sampling. The spectrum of the sampled signal is the original spectrum, scaled by $1/T$, plus copies of it centred on every multiple of the sampling rate: $\pm f_s$, $\pm 2f_s$, and so on. These copies are called **images** or **spectral replicas**. The sampled spectrum is periodic in f with period f_s, which is the frequency-domain statement that sampled data cannot distinguish f from $f + f_s$.

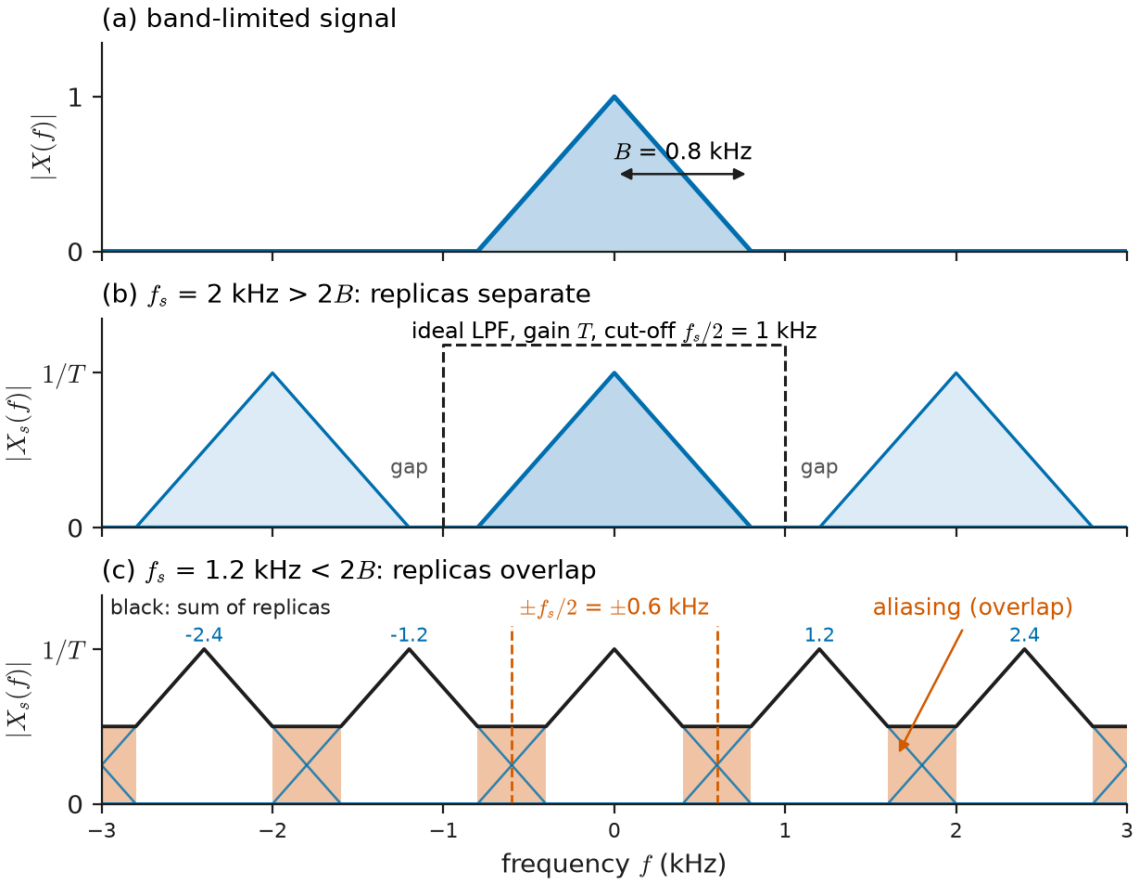

Figure 19.1 — Spectrum of a band-limited signal before and after ideal sampling, for adequate and inadequate sampling rates.

Band-limited signals and the condition for no overlap

Suppose $x(t)$ is **band-limited** to B hertz: $X(f) = 0$ for $|f| \geq B$. The baseband copy of $X_s(f)$ then occupies $-B < f < B$, the first replica occupies $f_s - B < f < f_s + B$, and the two do not overlap if

$$f_s - B \geq B \qquad \Longleftrightarrow \qquad f_s \geq 2B$$

with strict inequality needed in the general case for reasons shown in Section 19.3. When the copies do not overlap (Figure 19.1, middle), an ideal low-pass filter with cut-off anywhere between B and $f_s - B$ and gain T extracts the baseband copy exactly and returns $X(f)$. When they overlap (Figure 19.1, bottom), the tail of one replica is added to the baseband copy, and no filter can separate them afterwards: the sum is all that the samples know.

The sampling theorem

Sampling theorem.
If $x(t)$ has no spectral content at or above B hertz and is sampled at $f_s > 2B$, then $x(t)$ is uniquely determined by its samples and can be recovered exactly by

$$x(t) = \sum_{n=-\infty}^{\infty} x[n]\ \mathrm{sinc}\!\left(\frac{t - nT}{T}\right), \qquad \mathrm{sinc}(u) = \frac{\sin \pi u}{\pi u}$$

The interpolation formula follows from the filtering picture. The ideal reconstruction filter with gain T and cut-off $f_s/2$ has impulse response $h_r(t) = \mathrm{sinc}(t/T)$; passing $x_s(t)$ through it replaces each impulse $x[n]\,\delta(t - nT)$ by a sinc pulse $x[n]\,\mathrm{sinc}((t - nT)/T)$. Each sinc equals 1 at its own sampling instant and 0 at every other one, so the sum passes exactly through all the samples and, between them, fills in the one band-limited curve consistent with them.

The theorem is associated with Whittaker (1915), Kotelnikov (1933), Nyquist (whose 1928 telegraph analysis gave the $2B$ pulse rate) and Shannon (1949), who stated it in its communication-theory form. It is the reason a CD can store an orchestra as 44 100 numbers per second per channel, and the reason a protection relay can compute the impedance of a transmission line from a few dozen samples per cycle.

Status of the principle: Sampling theorem
Derived. A mathematical consequence of the Fourier transform and the periodic replication of the sampled spectrum; exact for strictly band-limited signals with uniform, jitter-free sampling and an ideal (non-causal, infinitely long) reconstruction filter.

What the theorem does not say

Three limits matter in practice and recur through the chapter.

1. **No real signal is strictly band-limited.** A signal that lasts a finite time has a spectrum that extends to infinite frequency (Chapter 18, time–frequency relationships). Every real signal therefore aliases a little. The engineering version of the theorem is "if the energy above $f_s/2$ is small enough, the aliasing error is small enough", and the anti-aliasing filter of Section 19.9 is what makes it small enough.
2. **The reconstruction filter is non-causal.** The sinc pulse extends to $t = -\infty$; it responds before its input arrives. Real reconstruction truncates, windows or approximates it, at a cost of delay and accuracy (Section 19.6).
3. **Uniformity is assumed.** If the sampling instants wander (jitter), or samples are missing, the replicas acquire noise and the formula no longer recovers $x(t)$ exactly (Section 19.5).

The theorem is also not the only sufficient condition. A signal confined to a band $f_L < |f| < f_H$ can be sampled at a rate tied to its bandwidth $f_H - f_L$ rather than to f_H (Section 19.4), and signals with other kinds of structure (sparse spectra, finite rate of innovation) can be recovered from fewer samples with nonlinear methods. Those extensions use the same replication picture.

19.3 Nyquist frequency and Nyquist rate

Two names for two different things

The literature uses two closely related terms, and confusing them causes real specification errors.

- The **Nyquist frequency** is a property of the *sampler*: $f_N = f_s/2$. It is the highest frequency that a sampled system can represent unambiguously. The band $0 \le f \le f_s/2$ is called the first Nyquist zone; $f_s/2 \le f \le f_s$ is the second zone, and so on.
- The **Nyquist rate** is a property of the *signal*: $2B$, twice its highest frequency. It is the minimum sampling rate the theorem allows for that signal.

A system is correctly sampled if its Nyquist frequency exceeds the signal's highest frequency, or equivalently if its sampling rate exceeds the signal's Nyquist rate. In the digital domain the Nyquist frequency corresponds to the digital frequency $\Omega = \pi$ rad/sample, or normalised frequency 0.5 cycles/sample, and it is the right-hand edge of every frequency-response plot in Chapters 20–23.

Why the inequality is strict

Consider a sinusoid exactly at the Nyquist frequency, $x(t) = A\sin(2\pi f_0 t + \phi)$ with $f_s = 2f_0$. Its samples are

$$x[n] = A\sin(\pi n + \phi) = A(-1)^n \sin\phi$$

The samples depend on $A\sin\phi$ only. A sine with $\phi = 0$ produces all zeros; a cosine produces $\pm A$; any phase in between produces $\pm A\sin\phi$, which is indistinguishable from a cosine of smaller amplitude. Amplitude and phase cannot be separated, so a component at exactly $f_s/2$ is not recoverable. For signals with continuous spectra this single frequency carries no energy and the distinction is academic, but for line spectra — mains harmonics, a test tone, a clock spur — it matters: a sampler must be strictly faster than twice the highest line, and in practice comfortably faster.

Representative sampling rates

Table 19.1 shows how the theorem is used across EEE. The ratio $f_s/(2B)$ — the margin over the Nyquist rate — is set by how difficult the anti-aliasing filter is allowed to be and by what the processing needs; it is rarely below 1.1.

Table 19.1 — Representative sampling rates (typical values).

Application	Signal bandwidth B	Sampling rate f_s	$f_s/(2B)$
Seismic monitoring (broadband station)	up to ~40 Hz	100 Sa/s	~1.25
Diagnostic ECG	0.05–150 Hz	500–1000 Sa/s	1.7–3.3
Clinical EEG	up to ~70–100 Hz	256–1000 Sa/s	1.3–7
Narrowband telephone speech	300–3400 Hz	8 kSa/s	1.18
Wideband (HD) speech	50–7000 Hz	16 kSa/s	1.14
CD audio	20 Hz–20 kHz	44.1 kSa/s	1.10
Professional audio and video sound	20 kHz (or more)	48, 96, 192 kSa/s	1.2–4.8
Protection relay (50 Hz system)	up to ~1 kHz after filtering	1–4 kSa/s (20–80 samples/cycle)	—
Sampled values, metering class (IEC 61850-9-2 LE, 50 Hz)	up to ~5 kHz harmonics	12.8 kSa/s (256 samples/cycle)	~1.3
Machine-vibration analysis	up to 10 kHz	25.6 kSa/s	1.28
LTE baseband, 20 MHz channel	±9 MHz complex	30.72 MSa/s	~1.7
Digital oscilloscope, 1 GHz bandwidth	1 GHz	2.5–10 GSa/s	1.25–5

The CD figure is a historical compromise. 44.1 kHz was chosen so that audio could be recorded on modified video tape recorders (three samples per video line: 245 usable lines per field at 60 fields/s, or 294 at 50 fields/s, both give 44 100), and it leaves only a 2.05 kHz transition band between the top of the audio band and the Nyquist frequency — the reason early CD players needed brutally sharp analogue filters (Section 19.9).

19.4 Aliasing

Folding

When a component at frequency f_0 lies above the Nyquist frequency, its samples are identical to those of a component at a lower frequency. For a cosine,

$$\cos(2\pi f_0 n T) = \cos(2\pi (f_0 - k f_s) n T)$$

for any integer k, because the difference in phase per sample is $2\pi k$. Since cos is even, $f_0 - k f_s$ and its negative are also indistinguishable. The frequency the sampled data appear to contain, the **alias frequency**, is therefore

$$f_a = \left| f_0 - k f_s \right|, \qquad k = \operatorname{round}\left(\frac{f_0}{f_s} \right)$$

which always lies in the first Nyquist zone, $0 \le f_a \le f_s/2$. Geometrically, the frequency axis is folded like a paper fan at $f_s/2$, f_s, $3f_s/2$, ..., and every frequency lands on the first zone. Components in odd zones land with their spectrum upright; components in even zones land mirrored (spectrally inverted), because they come from a negative-frequency image. The Nyquist frequency is sometimes called the **folding frequency** for this reason.

Aliasing is not distortion in the usual sense. The aliased component keeps its amplitude; it simply appears at the wrong frequency, and with sampled data alone there is no way to tell that it is wrong. A tone at 1 kHz, a tone at 7 kHz and a tone at 9 kHz sampled at 8 kSa/s produce exactly the same sequence of numbers.

> **Status of the principle: Alias frequency rule**
> *Derived*. Follows from the 2π periodicity of complex exponentials in the sample index; exact for any sampled sinusoid regardless of amplitude.

Familiar faces of aliasing

- **Stroboscopic effects.** A film camera samples at 24 frames per second. A wagon wheel with 12 spokes turning at 2.1 revolutions per second presents a spoke-passing frequency of 25.2 Hz; its alias is $|25.2 - 24| = 1.2$ Hz, and the wheel appears to turn slowly. The direction it appears to turn depends on which side of the fold the true frequency lies. A stroboscope used to "freeze" a rotating shaft or a vibrating blade exploits exactly this.
- **Audio.** A 30 kHz ultrasonic tone (a cleaning bath, a motion sensor, a bat) sampled at 44.1 kSa/s aliases to $44.1 - 30 = 14.1$ kHz, squarely in the audible band. A 25 kHz tone aliases to 19.1 kHz.
- **Digital oscilloscopes.** With the sample rate reduced on a long time-base setting, a 10 MHz clock can appear as a slowly drifting sine of a few kilohertz. The trace looks clean and stable — that is what makes aliasing dangerous on instruments.
- **Images.** Moiré patterns on a photographed striped shirt or a scanned halftone are two-dimensional aliasing of spatial frequencies above the sensor's pixel Nyquist frequency.

Worked example — Harmonic aliasing in a power-quality analyser. The analyser of the opening samples at $f_s = 32 \times 50 = 1600$ Sa/s, so its Nyquist frequency is 800 Hz, which is the 16th harmonic. The odd harmonics of a six-pulse drive appear at orders $6k \pm 1$: 5, 7, 11, 13, 17, 19, 23, 25, ... Harmonics up to the 15th (750 Hz) lie in the first zone and are reported correctly. Above it:

- 17th, 850 Hz: $k = \text{round}(850/1600) = 1$, $f_a = |850 - 1600| = 750$ Hz — reported as 15th harmonic.
- 19th, 950 Hz: $f_a = 650$ Hz — reported as 13th harmonic.
- 31st, 1550 Hz: $f_a = 50$ Hz — added to the fundamental.
- 33rd, 1650 Hz: $f_a = |1650 - 1600| = 50$ Hz — also on the fundamental.

In general, with M samples per cycle, harmonic h is reported as harmonic $|h - kM|$. With $M = 32$, the pairs (15, 17), (13, 19), (1, 31), (1, 33) collide. Doubling to $M = 64$ (3.2 kSa/s) moves the fold to the 32nd harmonic; then the 31st and 33rd both land at 1550 Hz and only harmonics around the 63rd and 65th land on the fundamental. Because the aliased harmonic adds as a phasor to the true one, its effect depends on relative phase: a 17th harmonic of 4 % folded onto a true 15th of 0.5 % produces a reading

anywhere from 3.5 % to 4.5 %, almost all of it fictitious. This is why IEC 61000-4-7 class instruments specify both the analysis window and a mandatory anti-aliasing filter, and why a scope-based "FFT" of a mains waveform at a low sample rate must be read with suspicion. Figure 19.2 shows the folding.

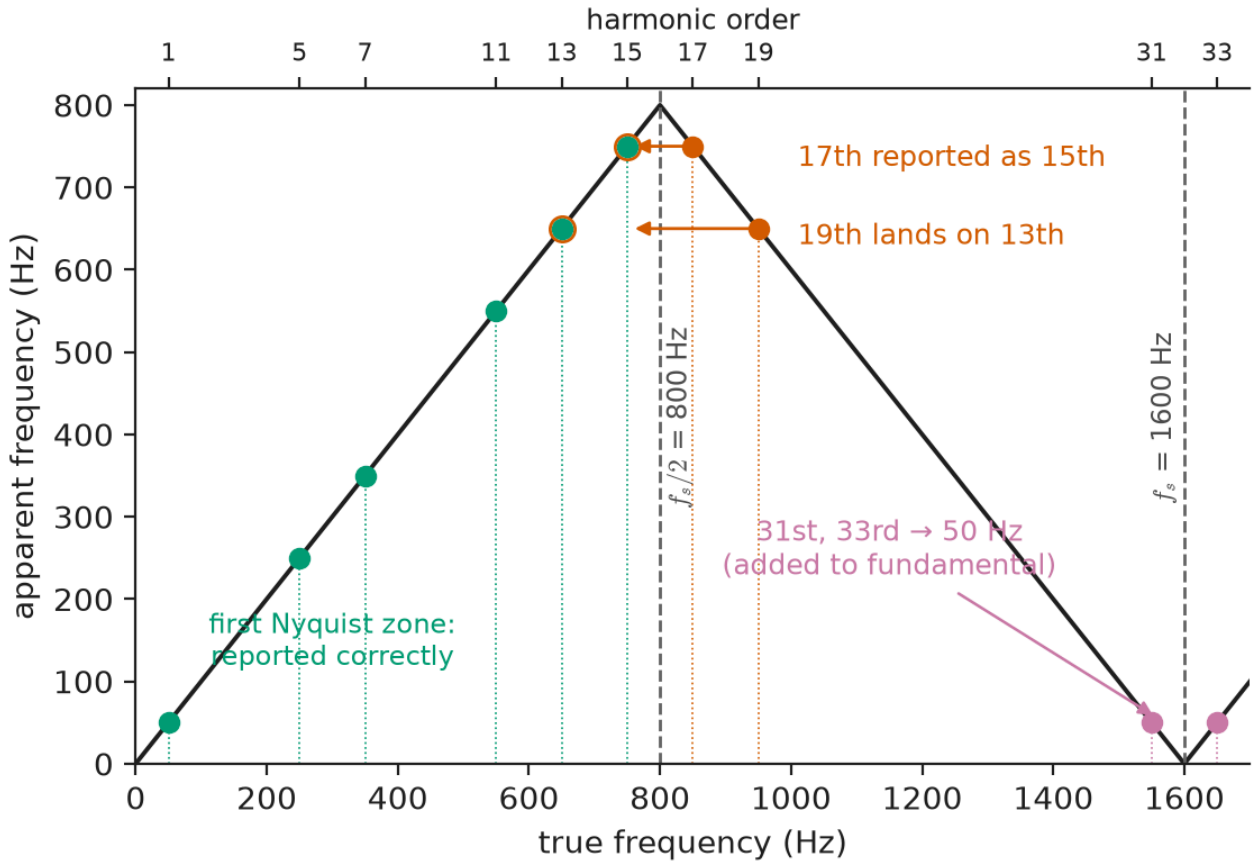

Figure 19.2 — Folding of mains harmonics into the first Nyquist zone for a sampler at 32 samples per 50 Hz cycle.

Aliasing as a tool: bandpass (undersampling) sampling

Aliasing is harmful only when a signal folds onto another signal. If a signal is confined to a band that lies entirely inside one Nyquist zone, folding moves it to baseband without overlapping itself, and nothing is lost. This is **bandpass sampling** or **undersampling**, and it is how many radio and radar receivers digitise an intermediate frequency (IF) directly with a converter running far below the IF.

For a real signal occupying $f_L \le |f| \le f_H$ with bandwidth $B = f_H - f_L$, the replicas do not overlap if

$$\frac{2f_H}{n} \le f_s \le \frac{2f_L}{n-1}, \qquad n = 1, 2, \dots, \left\lfloor \frac{f_H}{B} \right\rfloor$$

where n is the index of the Nyquist zone into which the band falls. The case $n = 1$ is ordinary low-pass sampling. The minimum possible rate is $2B$, reached only when f_H is an integer multiple of B, and only with zero margin.

> **Status of the principle: Bandpass sampling condition**
> *Derived*. The same no-overlap requirement as the sampling theorem, applied to the two sidebands of a bandpass spectrum; exact for strictly band-limited signals, but the allowed f_s windows narrow quickly as n grows and become sensitive to clock tolerance.

Worked example — Direct IF sampling. A radar receiver delivers a 20 MHz-wide signal centred on a 70 MHz IF: $f_L = 60$ MHz, $f_H = 80$ MHz, $B = 20$ MHz. Then $\lfloor f_H/B \rfloor = 4$ and the allowed windows are:

Zone n	$2f_H/n$ (MHz)	$2f_L/(n-1)$ (MHz)	Allowed f_s (MHz)
1	160	—	≥ 160
2	80	120	80–120
3	53.3	60	53.3–60
4	40	40	exactly 40 (no margin)

Choosing $f_s = 56$ MHz, near the middle of the $n = 3$ window, the band lies in the third Nyquist zone (56–84 MHz), which is odd, so it maps upright: 60 MHz → 4 MHz, 70 MHz → 14 MHz, 80 MHz → 24 MHz. The digitised signal occupies 4–24 MHz inside the 0–28 MHz first zone, with 4 MHz guard bands at each end. A converter at 56 MSa/s replaces one that would otherwise need more than 160 MSa/s, at the price of two strict requirements: an analogue bandpass filter that removes everything outside 60–80 MHz before the converter (noise from every other zone folds in too), and a sampler whose analogue input bandwidth and aperture jitter are good at 80 MHz, not at 28 MHz (Section 19.5).

19.5 Practical sampling

Natural sampling

The simplest physical sampler is a switch that connects the signal to the output for a time τ every T seconds. The output is

$$x_p(t) = x(t)\, p_\tau(t)$$

where $p_\tau(t)$ is a train of rectangular pulses of width τ and height 1. The pulse tops follow the signal, which is why this is called **natural sampling**. The pulse train is periodic, so it has Fourier coefficients $c_k = (\tau/T)\,\mathrm{sinc}(k\tau/T)$ (Chapter 18), and

$$X_p(f) = \sum_{k=-\infty}^{\infty} \frac{\tau}{T}\,\mathrm{sinc}\!\left(\frac{k\tau}{T}\right) X(f - k f_s)$$

The replicas are still exact copies of $X(f)$; only their weights change, falling off along a sinc envelope. The baseband copy is undistorted, scaled by the duty cycle τ/T. Natural sampling is how time-division

multiplexed analogue telephony (pulse-amplitude modulation) was carried, and it is the model for a chopper amplifier's switching stage.

Flat-top sampling

Most converters need a constant voltage while they work, so practical samplers capture the value at the sampling instant and hold it flat for a time τ. This **flat-top sampling** is the ideal sampled signal convolved with a rectangular pulse:

$$x_{ft}(t) = \sum_{n} x(nT)\,\Pi\!\left(\frac{t - nT - \tau/2}{\tau}\right) = x_s(t) * h_\tau(t)$$

with $h_\tau(t)$ a unit-height pulse of width τ starting at $t = 0$. Its spectrum is the replicated spectrum multiplied by the pulse transform:

$$X_{ft}(f) = \underbrace{\tau\,\mathrm{sinc}(f\tau)\,e^{-j\pi f\tau}}_{H_\tau(f)} \cdot \frac{1}{T}\sum_{k} X(f - kf_s)$$

The difference from natural sampling is decisive: here the *whole* spectrum, baseband copy included, is shaped by $\mathrm{sinc}(f\tau)$. Frequencies near the top of the band are attenuated relative to low frequencies — the **aperture effect** — and delayed by $\tau/2$. For $\tau \ll T$ the droop is negligible; for $\tau = T$ (a hold for the whole sample interval) it is the zero-order-hold droop of Section 19.7.

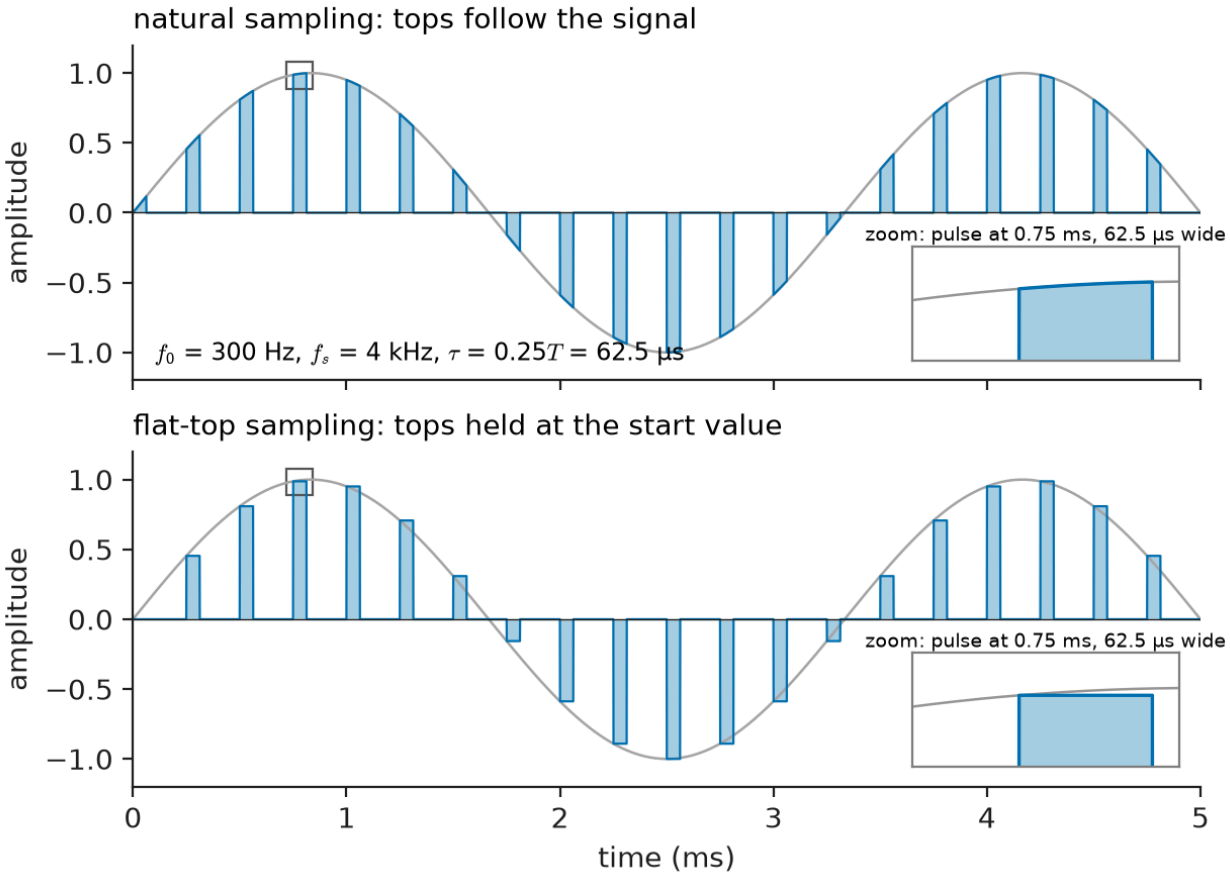

Figure 19.3 — Natural versus flat-top sampling of a 300 Hz sine with fs = 4 kHz and pulse width tau = 0.25T.

The track-and-hold and the aperture

In a modern ADC the sampling switch is followed by a hold capacitor, together forming a **track-and-hold** (T/H). During the track phase the capacitor voltage follows the input through the switch on-resistance R_{on}, a first-order RC low-pass with bandwidth $1/(2\pi R_{on} C_H)$ that must comfortably exceed the highest input frequency. At the hold command the switch opens; the instant of opening is the effective sampling instant. Three non-idealities limit how closely this approximates impulse sampling:

- **Finite tracking bandwidth.** The RC response attenuates and phase-shifts high input frequencies. Datasheets quote a "full-power analogue input bandwidth", often several times $f_s/2$ in converters designed for undersampling.
- **Aperture delay.** The fixed time between the hold command and the actual opening. It shifts all samples equally and matters only when several channels must be simultaneous, as in a three-phase power meter or a phased-array receiver.
- **Aperture uncertainty (jitter).** The random variation of the effective sampling instant from sample to sample, combining clock jitter and switch-timing noise.

> **Boundary.**
> The transistor-level design of track-and-hold circuits, converter architectures and clock oscillators belongs to *Electronics & Computing*. Here they appear only through their effect on the sampled signal.

Jitter

If sample n is taken at $nT + \varepsilon_n$ instead of nT, the value recorded is

$$x(nT + \varepsilon_n) \approx x(nT) + \varepsilon_n \, \dot{x}(nT)$$

The timing error turns into an amplitude error proportional to the slope of the signal. For a full-scale sine $A\sin(2\pi f_0 t)$ the slope has RMS value $2\pi f_0 A/\sqrt{2}$, and if the jitter ε_n is random with RMS value σ_j and uncorrelated with the signal, the error power is $(2\pi f_0 \sigma_j)^2 A^2/2$. The signal power is $A^2/2$, so the jitter-limited signal-to-noise ratio is

$$\mathrm{SNR}_j = -20\log_{10}(2\pi f_0\,\sigma_j) \ \text{dB}$$

> **Status of the principle: Jitter-limited SNR**
> *Approximate.* First-order Taylor expansion of the signal about the nominal sampling instant; valid when $2\pi f_0 \sigma_j \ll 1$ and the jitter is random and uncorrelated with the signal. Periodic jitter instead produces discrete spurs.

The formula depends on the *input* frequency, not on the sampling rate. That is why jitter is a minor issue for audio and a dominant one for IF sampling.

Worked example — How good must the clock be?

Input frequency	RMS jitter	SNR_j	Context
1 kHz	1 ns	104.0 dB	Instrumentation: jitter irrelevant
20 kHz	1 ns	78.0 dB	Audio: a cheap 1 ns clock limits a "24-bit" converter to about 13 bits
10 MHz	1 ps	84.0 dB	Baseband radio ADC
70 MHz	0.5 ps	73.2 dB	IF sampling with a good crystal oscillator
70 MHz	0.1 ps	87.1 dB	IF sampling with a low-jitter clock synthesiser
100 MHz	100 fs	84.0 dB	Wideband receiver

For the 70 MHz IF of the previous example, the receiver designer who wants about 74 dB (roughly 12 effective bits; Section 19.8) must hold total aperture jitter near 0.45 ps RMS — a requirement on the clock source and its distribution on the board, not on the converter's sample rate. Running the same converter at the same 56 MSa/s on a 4 MHz baseband signal would relax the jitter requirement by the ratio of input frequencies, $70/4 = 17.5$, almost 25 dB.

Non-uniform and missing samples

Telemetry systems drop packets; event-driven sensors report only when something changes; distributed measurement units are synchronised by GPS with microsecond rather than picosecond timing. When sample times are known but irregular, the signal can often still be recovered by least-squares fitting of a band-limited model, provided the average rate exceeds the Nyquist rate and the gaps are not too long. When sample times are irregular and *unknown*, the error behaves like jitter. Phasor measurement units on transmission grids are a clear case: an error of 1 µs in the time stamp is a phase error of $360^\circ \times 50 \times 10^{-6} = 0.018^\circ$ at 50 Hz, which is why PMU standards tie total vector error to timing accuracy.

19.6 Reconstruction

The ideal reconstruction filter

Reconstruction is the inverse problem: given $x[n]$, produce a continuous-time signal. The theory of Section 19.2 gives the ideal answer. Form the impulse train $x_s(t)$ and pass it through a low-pass filter of gain T and cut-off $f_s/2$:

$$H_r(f) = \begin{cases} T, & |f| < f_s/2 \\ 0, & |f| > f_s/2 \end{cases} \qquad \Longleftrightarrow \qquad h_r(t) = \operatorname{sinc}\!\left(\frac{t}{T}\right)$$

The output is the sinc interpolation formula. Figure 19.4 shows it working: each sample launches a sinc pulse that passes through zero at every other sampling instant, and their sum is a smooth curve through all the samples.

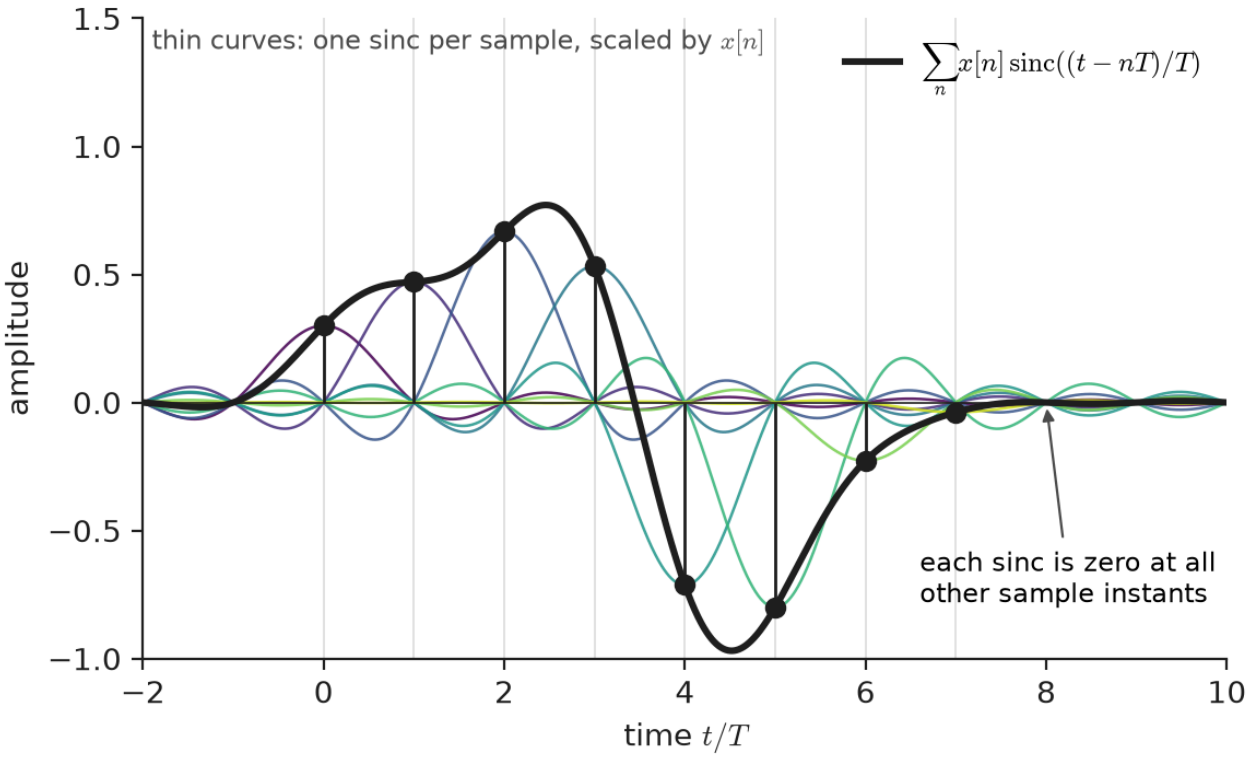

Figure 19.4 — Ideal reconstruction as a sum of sinc pulses, one per sample.

The ideal filter cannot be built. Its impulse response begins at $t = -\infty$, so it is non-causal, and it decays only as $1/t$, so truncating it to a finite number of terms leaves errors that die out slowly. Practical reconstruction makes three compromises:

1. **Delay.** A causal approximation uses a delayed and truncated sinc, accepting a fixed latency. A digital interpolator that uses 32 samples on each side of the output point introduces a delay of 32 samples.
2. **Windowing.** Multiplying the truncated sinc by a smooth window (Chapter 22) reduces the ripple caused by abrupt truncation. Windowed-sinc interpolators are standard in audio sample-rate converters.
3. **Transition band.** A realisable analogue filter needs a band between the top of the signal and the start of the first image. This is why practical systems sample faster than the Nyquist rate: the margin in Table 19.1 is the transition band of the reconstruction (and anti-aliasing) filter.

> **Status of the principle: Sinc interpolation**
> *Derived.* Exact reconstruction for a strictly band-limited signal sampled above its Nyquist rate; any finite, causal implementation is an approximation whose error is set by truncation length, window and transition band.

Simpler interpolators

When the sample rate is high compared with the signal bandwidth, crude interpolators work well, and they are often what is actually used:

- **Zero-order hold** (staircase): each sample is held until the next one. This is what most DACs output (Section 19.7).
- **First-order hold** (linear interpolation): straight lines join the samples. Its impulse response is a triangle of base $2T$, and its frequency response is $T\,\mathrm{sinc}^2(fT)$: images are suppressed more strongly than with a hold (12 dB per octave far from the band rather than 6), but the in-band droop is twice as large in decibels. Digital oscilloscopes use it, or a sinc interpolator, to draw traces between sample points; a scope that shows a sine sampled at 2.5 samples per cycle with linear interpolation looks jagged, while sinc interpolation shows the true sine.
- **Higher-order and spline interpolators** sit between linear interpolation and the windowed sinc in accuracy and cost.

Each of these is just a different reconstruction filter $h_r(t)$, and its quality can be judged by how flat its response is below $f_s/2$ and how strongly it attenuates the images at $f_s \pm f$.

The DAC and its reconstruction (image-rejection) filter

A digital-to-analogue converter does not output impulses; it outputs a held level per sample. The analogue reconstruction chain is therefore:

$$x[n] \;\longrightarrow\; \text{DAC (zero-order hold)} \;\longrightarrow\; \text{analogue low-pass (image-rejection) filter} \;\longrightarrow\; x_r(t)$$

The image-rejection filter has the same specification problem as the anti-aliasing filter (Section 19.9), mirrored: it must pass up to the signal's top frequency and suppress the first image starting at $f_s - B$. In audio this filter used to be a 9th- to 11th-order analogue design. Modern audio DACs avoid it by **oversampling**: a digital interpolation filter raises the sample rate by 8 or more before the DAC, so the first image moves to hundreds of kilohertz and a 2nd- or 3rd-order analogue filter suffices. The sharp filtering is done digitally, where it is exact and cheap (Chapters 23–24).

19.7 Sample-and-hold behaviour

The zero-order hold

A **zero-order hold** (ZOH) keeps each sample value constant for one full interval T. It is the flat-top sampler of Section 19.5 with $\tau = T$, and its impulse response is a unit pulse of width T:

$$h_0(t) = \begin{cases} 1, & 0 \le t < T \\ 0, & \text{otherwise} \end{cases} \qquad \Longleftrightarrow \qquad H_0(f) = T\,\mathrm{sinc}(fT)\,e^{-j\pi fT}$$

Equivalently, in the Laplace domain of Chapter 11, $H_0(s) = (1 - e^{-sT})/s$: a step at $t = 0$ minus a delayed step at $t = T$. Figure 19.5 shows the staircase output and the magnitude response.

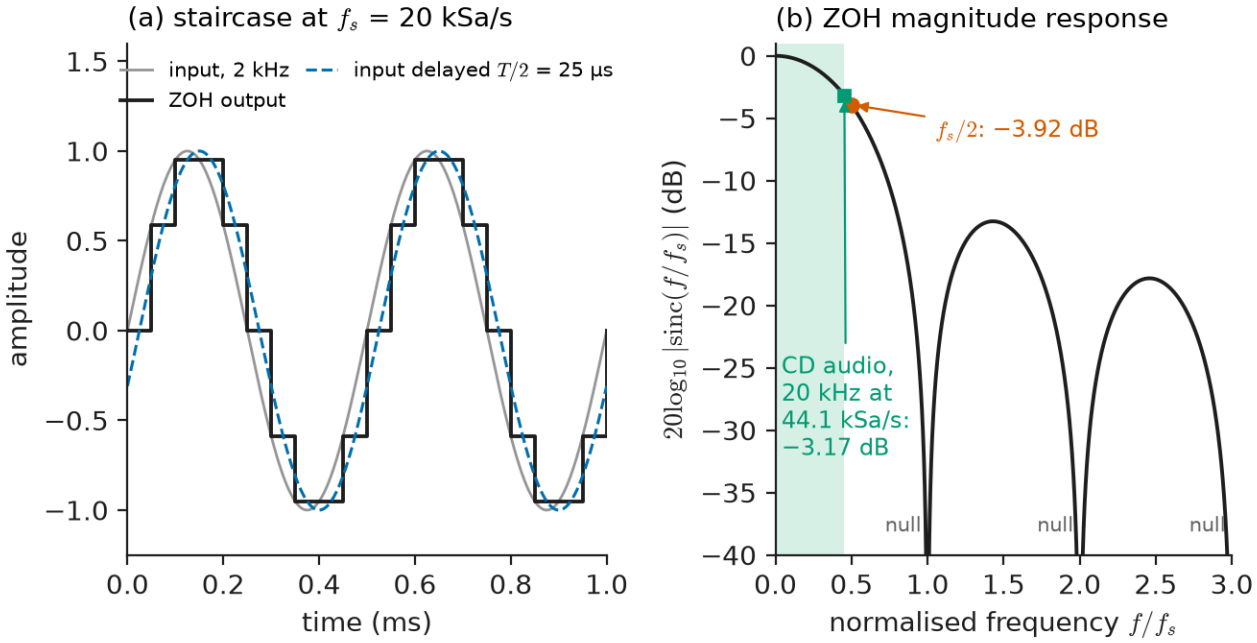

Figure 19.5 — Zero-order hold: staircase output and its sinc-shaped frequency response.

Three consequences

Amplitude droop. The magnitude $|\mathrm{sinc}(f/f_s)|$ falls from 1 at DC to $2/\pi = 0.637$ (-3.92 dB) at the Nyquist frequency. The hold acts as a gentle low-pass inside the signal band.

Partial image rejection. $H_0(f)$ has nulls at f_s, $2f_s$, ..., so it attenuates the images near those frequencies, but only weakly at their edges; an image at $f_s - f$ for small f is attenuated by only $\mathrm{sinc}(1 - f/f_s) \approx f/f_s$. The staircase still contains strong high-frequency content (its steps), which is why the image-rejection filter cannot be omitted unless the load (a loudspeaker, a motor) filters for itself.

Delay. The linear phase $e^{-j\pi fT}$ is a pure delay of $T/2$: the staircase is, on average, half a sample late. In digital control loops (Chapter 35) this delay costs phase margin: at a crossover frequency f_c the hold adds a lag of $180^\circ f_c/f_s$, which is why control loops are sampled 10–30 times faster than their crossover frequency, much faster than the sampling theorem alone would demand.

> **Status of the principle: Zero-order-hold response**
> *Derived*. Exact for an ideal DAC whose output steps instantaneously and stays flat; real DACs add settling transients and glitches at each code change, which add to the image energy.

Worked example — Sinc droop in audio and its correction. For a CD player whose DAC is a plain zero-order hold at $f_s = 44.1$ kSa/s, the response at 20 kHz is

$$|H_0| = \mathrm{sinc}\left(\frac{20}{44.1}\right) = \frac{\sin(0.4535\pi)}{0.4535\pi} = 0.694 \quad (-3.17\ \mathrm{dB})$$

against 0.99915 (-0.007 dB) at 1 kHz: a 3 dB treble loss, plainly audible. The half-sample delay is $0.5/44\,100 = 11.3$ µs. Three remedies are used:

Remedy	Response at 20 kHz	Notes
Plain ZOH at 44.1 kSa/s	-3.17 dB	Unacceptable for hi-fi
ZOH at 48 kSa/s	-2.64 dB	Higher rate helps a little
ZOH at 96 kSa/s	-0.63 dB	Higher rate helps a lot
ZOH after 8× digital oversampling (352.8 kSa/s)	-0.05 dB	Standard practice
Inverse-sinc ($x/\sin x$) digital pre-emphasis	≈ 0 dB (flat by design)	Used in video and RF DACs

The 8× entry follows from $\operatorname{sinc}(20/352.8) = \operatorname{sinc}(0.0567) = 0.9947$. Oversampling solves both the droop and the image-filter problem at once, which is why nearly every audio DAC does it. In RF transmitters, where the DAC must run close to its Nyquist frequency, an inverse-sinc digital filter is placed before the DAC to boost the upper band by up to 3.9 dB.

Sample-and-hold in front of an ADC

The hold appears on the input side too. A successive-approximation converter takes N clock cycles to resolve N bits, and the input must not change by more than half an LSB during conversion. Without a hold, a full-scale sine of frequency f_0 changes at up to $2\pi f_0 A$ per second; for a 12-bit converter (LSB $= 2A/4096$) and a 10 µs conversion, the limit would be $f_0 < 2A/(2 \times 4096)\,/\,(2\pi A \times 10^{-5}) \approx 3.9$ Hz. With a track-and-hold, the limit is set instead by aperture jitter (Section 19.5), raising the usable input frequency by orders of magnitude. The sample-and-hold is what makes the ideal sampling model apply to a converter that takes a finite time to think.

19.8 The quantization interface

Two discretisations

Sampling discretises time; **quantization** discretises amplitude. An ADC does both: after the track-and-hold freezes a value, the converter maps it to the nearest of 2^N levels. For a converter with full-scale range V_{FS} (peak to peak) the step, or LSB size, is

$$\Delta = \frac{V_{FS}}{2^N}$$

A 12-bit converter with a ± 5 V range ($V_{FS} = 10$ V) has $\Delta = 2.44$ mV. Unlike sampling, quantization is irreversible: the theorem of Section 19.2 restores the exact signal from its samples, but nothing restores the exact amplitude from its rounded value. The output is the input plus an error $e[n] = x_q[n] - x[n]$ that is confined to $\pm\Delta/2$ for an input inside the range.

The additive-noise model

When the input is busy — large compared with Δ, not locked in frequency to the sampling clock — the error behaves like a random variable uniformly distributed on $[-\Delta/2, \Delta/2]$, independent from sample to sample and of the signal. Its variance is

$$\sigma_e^2 = \frac{\Delta^2}{12}$$

and its power is spread evenly from 0 to $f_s/2$. For a full-scale sine of amplitude $V_{FS}/2$, the ratio of signal power to quantization-noise power is

$$\mathrm{SQNR} = 6.02N + 1.76 \text{ dB}$$

> **Status of the principle: Uniform quantization-noise model**
> *Approximate*. Valid for signals that span many quantization steps and are not correlated with the sampling clock; it fails for small, slow or periodic signals locked to the clock, where the error becomes deterministic harmonic distortion — the case that dither (Chapter 21) repairs.

This rule — roughly 6 dB per bit — sets the vocabulary of every data-acquisition specification. A 12-bit converter can at best reach 74.0 dB, a 16-bit one 98.1 dB and a 24-bit one 146.2 dB. Real converters fall short because of thermal noise, nonlinearity and the jitter of Section 19.5, and their performance is quoted as an **effective number of bits**, $\mathrm{ENOB} = (\mathrm{SINAD} - 1.76)/6.02$, measured with a full-scale sine.

Where sampling and quantization meet

Two interactions between the two discretisations matter for the rest of this chapter.

1. **Aliasing defines the noise floor of the anti-aliasing filter.** An interfering component that aliases into the band is harmless only if it lands below the quantization noise. That turns the anti-aliasing specification into a number: the stopband attenuation should be roughly the SQNR, about $6N$ dB, for components that can reach full scale (Section 19.9).
2. **Oversampling spreads the noise.** Because the quantization noise is spread over 0–$f_s/2$, sampling faster than needed and then filtering digitally to the signal band B removes a fraction of the noise. The in-band noise falls by $10\log_{10}(f_s/(2B))$ dB — about 3 dB, or half a bit, per doubling of f_s. Noise-shaping converters (sigma–delta) do much better by pushing the noise out of band before filtering.

Chapter 21 develops quantization fully — the quantizer as a nonlinearity, dither, headroom and crest factor, oversampling gain and the finite-precision arithmetic that follows the converter. Chapter 27 uses the same mathematics for pulse-code modulation, where the quantizer is the source encoder of a digital link.

> **Boundary.**
> Converter architectures (successive approximation, pipeline, flash, sigma–delta), their nonlinearity and their interface circuits are the subject of *Electronics & Computing*. This book uses their input–output model only.

19.9 Anti-aliasing filters

What the filter must do

The anti-aliasing filter is an analogue low-pass placed before the sampler to make the input band-limited enough. Its specification follows directly from the folding picture of Section 19.4. Let the useful signal extend to f_p (the passband edge) and let the sampling rate be f_s.

- Components between f_p and $f_s - f_p$ fold into the band from f_p to $f_s/2$, which contains no useful signal. They can be removed later by a sharp digital filter, so the analogue filter need not suppress them strongly.
- Components above $f_s - f_p$ fold *into the passband* 0–f_p, where nothing can remove them. They must be attenuated, before sampling, to below the acceptable error floor — usually the quantization noise level.

So the stopband of the anti-aliasing filter begins at $f_{stop} = f_s - f_p$, not at $f_s/2$ as is often assumed, and the required attenuation A is set by how strong out-of-band interference can be and by the converter resolution. Figure 19.6 shows the arrangement.

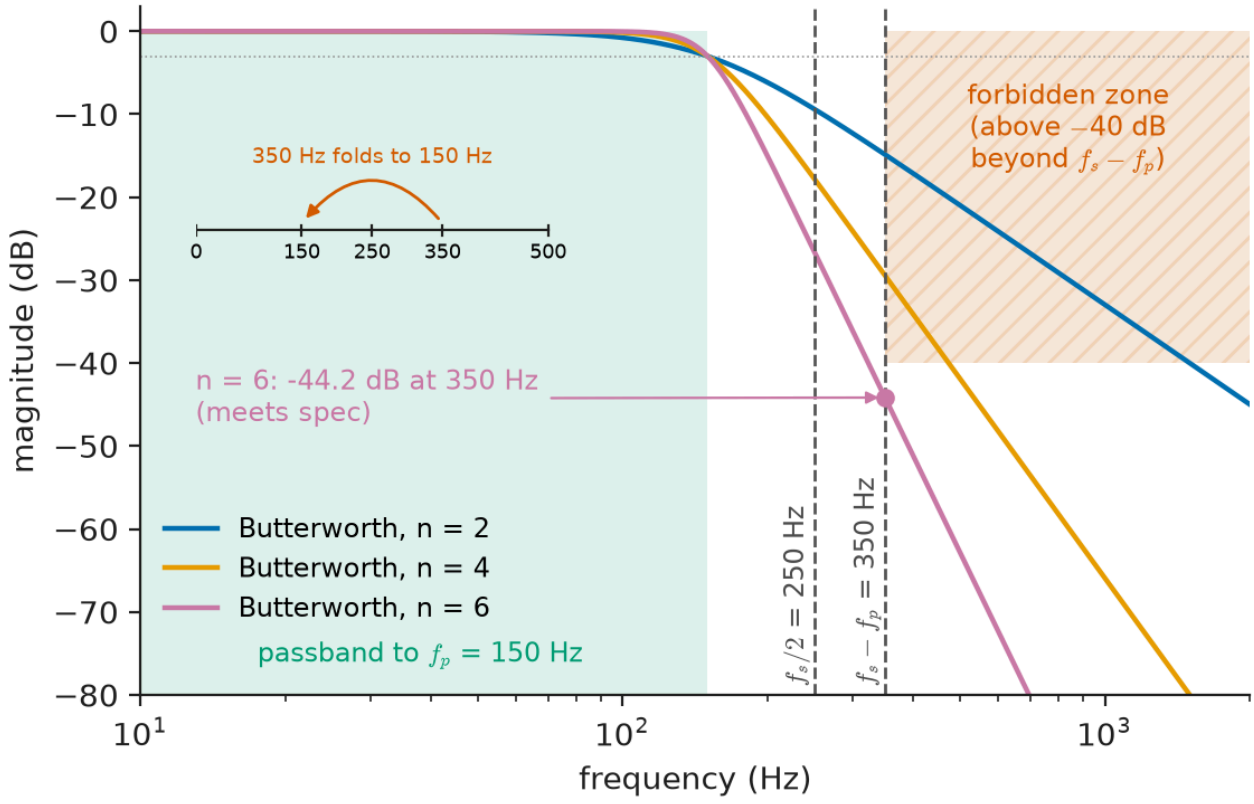

Figure 19.6 — Anti-aliasing filter specification for an ECG front end sampled at 500 Sa/s.

Filter order

For a Butterworth low-pass with -3 dB cut-off f_c and order n, the attenuation at frequency f is $10\log_{10}\left(1 + (f/f_c)^{2n}\right)$ dB. Requiring at least A dB at f_{stop} gives

$$n \;\geq\; \frac{\log_{10}\!\left(10^{A/10}-1\right)}{2\log_{10}(f_{stop}/f_c)}$$

Chebyshev and elliptic designs (Chapter 13 and Chapter 23) reach the same attenuation with lower order, at the cost of passband ripple and worse group delay — and group delay distortion matters for waveforms whose shape carries meaning, such as an ECG or a radar pulse. Bessel filters keep the waveform best and need the highest order.

> **Status of the principle: Anti-aliasing stopband rule**
> *Derived*. The stopband edge $f_s - f_p$ follows from the folding geometry; the required attenuation is an engineering choice tied to converter resolution and the expected interference level.

Worked example — ECG front end. A diagnostic ECG must pass 0.05–150 Hz and is sampled at 500 Sa/s with a 16-bit converter. Folding makes 350–500 Hz land on 150–0 Hz, so the stopband begins at $f_{stop} = 500 - 150 = 350$ Hz. The main out-of-band threats are muscle noise, electrosurgical and switching-supply interference; suppose 40 dB of rejection at 350 Hz is required. With a Butterworth response and $f_c = 150$ Hz,

$$n \geq \frac{\log_{10}(10^4-1)}{2\log_{10}(350/150)} = \frac{4.000}{2\times 0.3680} = 5.44 \;\Rightarrow\; n = 6$$

A 6th-order Butterworth gives $10\log_{10}(1+(350/150)^{12}) = 44.2$ dB at 350 Hz. Asking for 60 dB would need $n \geq 8.15$, i.e. 9th order — and an analogue 9th-order filter with matched components across many channels is expensive. The modern alternative is to sample at, say, 4 kSa/s with a simple 2nd-order analogue filter, then filter and decimate digitally to 500 Sa/s (Chapter 24). The electrode's own impedance and the input protection network often provide the first pole for free.

Worked example — Why audio converters oversample. A CD-rate system passes 20 kHz and samples at 44.1 kSa/s, so $f_{stop} = 44.1 - 20 = 24.1$ kHz. For 16-bit performance the filter should give about 96 dB there. The Butterworth order is

$$n \geq \frac{\log_{10}(10^{9.6}-1)}{2\log_{10}(24.1/20)} = \frac{9.6}{2\times 0.0810} \approx 59.3$$

A 60th-order analogue filter is absurd; even an elliptic design (0.1 dB ripple) needs 11th order with tightly matched components and severe phase distortion near 20 kHz. Now let the converter sample at $64 \times 44.1 = 2.8224$ MSa/s. The stopband moves to $2.8224 - 0.02 = 2.8024$ MHz, and

$$n \geq \frac{9.6}{2\log_{10}(2.8024/0.02)} = \frac{9.6}{2\times 2.1465} = 2.24 \;\Rightarrow\; n = 3$$

A 3rd-order filter with cut-off 50 kHz gives 104.9 dB at 2.8024 MHz, and costs only 0.018 dB of droop at 20 kHz. The sharp filtering moves into a digital decimation filter, which is exact, linear-phase and identical in every unit. This trade — analogue simplicity bought with a higher sample rate and digital processing — is the central design pattern of Part V.

Placement and practicalities

- **Before the sampler, always.** No digital processing after the converter can undo aliasing. A digital filter can only remove what is still distinguishable.
- **Every channel, matched.** In a three-phase power meter, a mismatch of filter phase between voltage and current channels becomes a power-factor error; at 50 Hz a 0.1° mismatch gives an error of 0.3 % in active power at a power factor of 0.5.
- **Noise also aliases.** Broadband thermal noise above $f_s/2$ folds into the band just as tones do. Without a filter, an amplifier with 1 MHz of noise bandwidth feeding a 10 kSa/s converter folds about 200 Nyquist zones of noise into the 5 kHz band — a 23 dB noise penalty.
- **Bandpass sampling needs bandpass filters.** In the IF example of Section 19.4 the filter must reject everything outside 60–80 MHz, including the noise in other zones.
- **The sampler's own bandwidth.** A converter's analogue input is wider than $f_s/2$; it will happily alias anything the filter lets through.

Table 19.2 collects typical anti-aliasing approaches.

Table 19.2 — Typical anti-aliasing arrangements.

System	f_s	Analogue filter	Digital stage
Diagnostic ECG (oversampled design)	2–8 kSa/s at the ADC, 500 Sa/s output	1st–2nd-order RC	FIR decimation
Power-quality analyser (IEC 61000-4-7)	10–20 kSa/s	4th–8th-order, cut-off near 3–5 kHz	Synchronised FFT window
Protection relay	1–4 kSa/s	2nd–3rd-order, cut-off a few hundred hertz	Fourier or cosine filter per cycle
Studio audio ADC (sigma–delta)	64× to 256× of 48 kSa/s	1st–2nd-order RC	Multi-stage FIR decimation
Vibration monitoring	25.6 kSa/s	6th–8th-order elliptic or switched-capacitor	Decimation for lower bands
Radio IF sampling	50–250 MSa/s	SAW or LC bandpass	Digital down-conversion

What remains open

Open question	Where it is answered
How are sequences, discrete-time convolution and the Z-transform used to describe sampled systems?	Chapter 20
How do quantization, dither, headroom and finite word length behave inside a processor?	Chapter 21
How is the spectrum of a finite record of samples computed, and what do leakage and windowing do?	Chapter 22
How are the sharp digital filters that replace analogue anti-aliasing and image filters designed?	Chapter 23
How do decimation and interpolation change sample rates inside a system?	Chapter 24
How does sampling become pulse-code modulation in a digital link?	Chapter 27
How does the hold delay of Section 19.7 affect a digital control loop?	Chapter 35
How do track-and-hold and ADC circuits actually work at transistor level?	*Electronics & Computing*
Where does the Fourier theory of the Dirac comb come from rigorously?	*The Mathematics Universe*

Next: Chapter 20 leaves the analogue world behind and studies the sampled sequence as a signal in its own right, with discrete-time convolution, difference equations and the Z-transform.

Chapter 20 — Discrete-Time Signals and Systems

Part IV · Signals and Systems

A biomedical engineer is handed an ECG recorder that samples at 500 Sa/s and asked to remove the 50 Hz mains hum that rides on every trace recorded in the ward. The analogue anti-aliasing filter of Chapter 19 is already in place and the converter is fine; what remains is a stream of numbers with an unwanted sinusoid in it. In the analogue world the cure would be a twin-T notch with precision resistors and capacitors that drift with temperature. In the sampled world it is three lines of code: each output sample is a weighted sum of the last three input samples and the last two output samples, with five coefficients computed once. That filter never drifts, is identical in every recorder, and can be retuned to 60 Hz for export by changing two numbers. To design it with confidence, however, the engineer needs to answer questions the analogue theory of Chapters 11–18 answered for circuits: what is the response to an arbitrary input, where are the poles, is the filter stable, and what does it do at every frequency?

This chapter builds that theory for sequences. Its central objects mirror those of continuous time — the impulse response, convolution, difference equations instead of differential equations, the Z-transform instead of the Laplace transform, poles and zeros in the z-plane instead of the s-plane — but several behaviours are new: frequency is periodic, the stability boundary is a circle, and every system can be implemented exactly with additions, multiplications and memory.

> **How are discrete-time signals and systems described, and how is the output of a discrete-time LTI system computed from its input?**
> **What does the Z-transform reveal about a digital system — its frequency response, its poles and zeros, its stability — and what makes a digital filter work?**

Concept	Engineering destination
Sequences and discrete frequency	Representation of every sampled measurement, audio stream and baseband radio signal
Discrete-time convolution	FIR filtering, multipath channel models, smoothing of sensor data
Difference equations	Recursive filters, digital controllers, numerical models of circuits
Z-transform	Solving difference equations, analysing digital filters and sampled-data control
System functions $H(z)$	Frequency response of digital filters, equalisers and controllers
Poles and zeros	Notch and resonator design, echo cancellers, predicting transient behaviour
Discrete-time stability	Safe recursive filters, digital control loops, adaptive algorithms
Digital filter fundamentals	Mains-hum removal, smoothing, DC blocking, the starting point for Chapter 23

Roadmap. Sequences and discrete frequency (20.1) → convolution sum (20.2) → difference equations (20.3) → the Z-transform (20.4) → system functions and frequency response (20.5) → poles and zeros in the z-plane (20.6) → stability and the unit circle (20.7) → first digital filters (20.8).

20.1 Sequences

Signals indexed by integers

A discrete-time signal is a **sequence** $x[n]$, a number for every integer n. Sequences arise from sampling ($x[n] = x_a(nT)$, Chapter 19), from inherently discrete processes (daily energy consumption, bits from a source, pixels along a line), and from computation. Once inside a processor, a sequence has no attached time unit: n is a sample index and the sample interval T is a separate piece of information that must travel with the data. Forgetting it is a common source of error when filter coefficients designed for one sampling rate are reused at another.

The elementary sequences of Chapter 15 carry over directly, and Figure 20.1 shows them.

- **Unit impulse** (Kronecker delta): $\delta[n] = 1$ for $n = 0$, and 0 otherwise. Unlike the Dirac impulse it is an ordinary, finite sequence, so the discrete theory has none of the generalised-function subtleties of continuous time.
- **Unit step**: $u[n] = 1$ for $n \geq 0$, and 0 for $n < 0$. The two are related by a first difference and a running sum: $\delta[n] = u[n] - u[n-1]$ and $u[n] = \sum_{k=-\infty}^{n} \delta[k]$.
- **Exponential**: $x[n] = a^n u[n]$. For $0 < a < 1$ it decays; for $a > 1$ it grows; for $-1 < a < 0$ it alternates in sign while decaying. Every real number a gives a valid sequence, so a decaying alternating exponential — impossible as a real continuous-time exponential — is perfectly ordinary here.
- **Sinusoid**: $x[n] = A\cos(\Omega_0 n + \phi)$, where Ω_0 is the **digital frequency** in radians per sample.
- **Complex exponential**: $x[n] = z_0^n$ with complex $z_0 = re^{j\Omega_0}$, which contains the real exponential ($\Omega_0 = 0$) and the sinusoid ($r = 1$) as special cases.

Any sequence can be written as a weighted sum of shifted impulses,

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

which is trivial to read and is the foundation of convolution in Section 20.2.

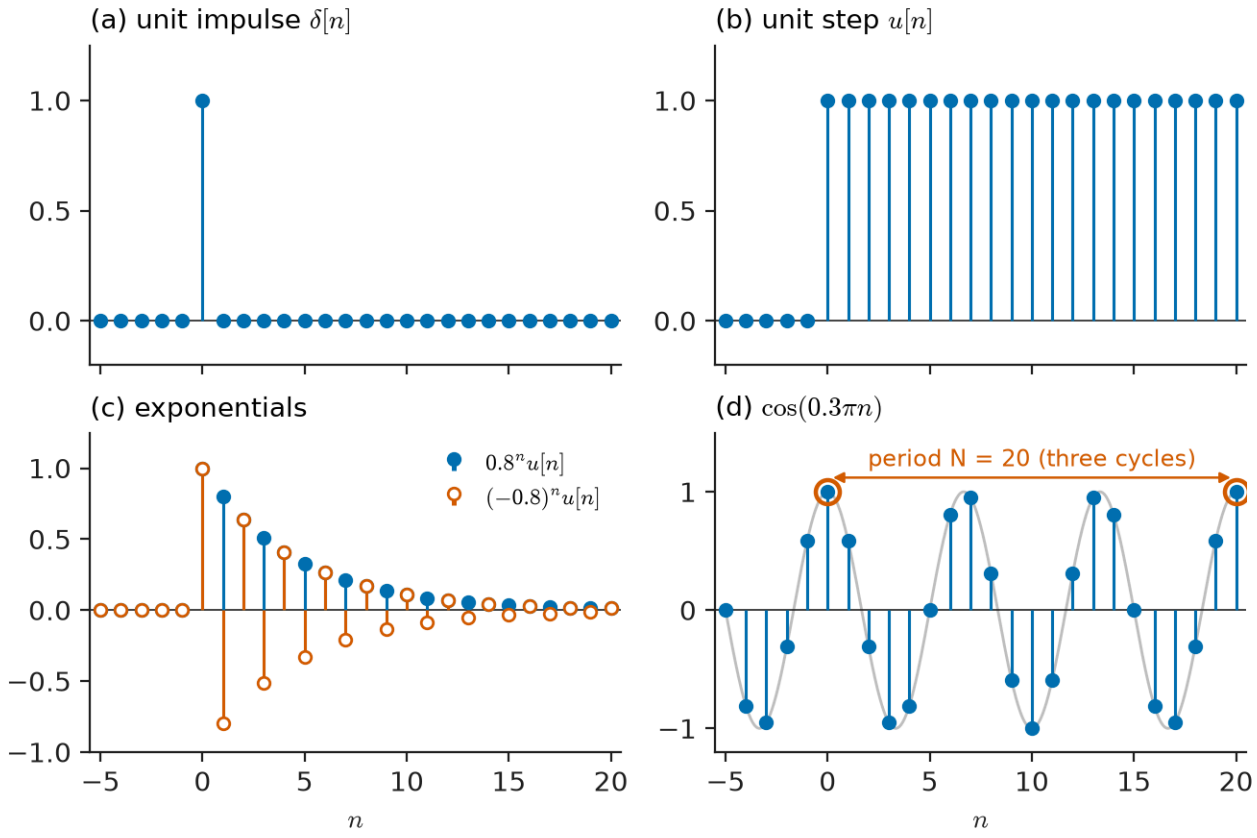

Figure 20.1 — Elementary discrete-time sequences shown as stem plots.

Discrete frequency is periodic

If the sequence came from sampling a continuous sinusoid of frequency f_0 at rate f_s, then $\Omega_0 = 2\pi f_0 / f_s$. The Nyquist frequency $f_s/2$ maps to $\Omega = \pi$. Two properties distinguish discrete from continuous sinusoids, and both follow from the integer index:

1. **Frequency is only defined modulo 2π.** $\cos((\Omega_0 + 2\pi k)n) = \cos(\Omega_0 n)$ for every integer k, because $2\pi k n$ is a multiple of 2π. This is the aliasing of Chapter 19 seen from inside the sequence: all frequency information lives in $-\pi < \Omega \le \pi$, and "high frequency" means Ω near π, where the sequence alternates in sign every sample. The fastest possible discrete oscillation is $(-1)^n = \cos(\pi n)$.
2. **A discrete sinusoid need not be periodic.** $\cos(\Omega_0 n)$ repeats after N samples only if $\Omega_0 N = 2\pi m$ for integers m and N, that is, only if $\Omega_0 / 2\pi = m/N$ is rational. The fundamental period is the smallest such N. For $\Omega_0 = 0.3\pi$, $\Omega_0/2\pi = 3/20$ and the period is 20 samples (spanning three cycles of the underlying cosine). For $\Omega_0 = 1$ rad/sample, $1/2\pi$ is irrational and the sequence never repeats exactly, although it is perfectly well behaved.

The second property is why a sampled 50 Hz waveform repeats every cycle only if f_s is an integer multiple of 50 Hz — and why power instruments lock their sampling clock to the measured mains frequency with a phase-locked loop, so that each analysis window contains exactly an integer number of cycles (a matter of spectral leakage, Chapter 22).

Operations on sequences

The operations of Chapter 15 apply with one restriction: time arguments must remain integers.

- **Shift**: $x[n - n_0]$ delays by n_0 samples. Delay is the fundamental hardware operation of DSP: a register, a memory location or a circular-buffer pointer.
- **Reversal**: $x[-n]$.
- **Down-sampling** by M: $x[Mn]$ keeps every M-th sample and discards the rest. This is not reversible in general and can alias (Chapter 24).
- **Up-sampling** by L: inserting $L - 1$ zeros between samples. Scaling time by a non-integer factor is not defined directly; it requires interpolation (Chapter 24).
- **Even–odd decomposition**, **energy** $\sum |x[n]|^2$ and **average power** $\lim_{N\to\infty} \frac{1}{2N+1} \sum_{-N}^{N} |x[n]|^2$ are defined as in Chapter 15 with sums in place of integrals.

20.2 Discrete-time convolution

The convolution sum

A discrete-time system maps an input sequence to an output sequence. Linearity and time invariance (Chapter 16) mean exactly what they did for continuous systems, with shifts by integers. For an LTI system, let $h[n]$ be the response to $\delta[n]$ — the **impulse response**. Because $x[n]$ is a sum of weighted, shifted impulses, linearity and time invariance give

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k] \equiv (x * h)[n]$$

> **Status of the principle: Convolution sum**
> *Derived*. Follows directly from linearity, time invariance and the decomposition of a sequence into impulses; exact for every discrete-time LTI system, with no regularity conditions beyond convergence of the sum.

The derivation is the same as for the continuous convolution integral of Chapter 17, but it is simpler and more immediately useful: it is an algorithm. Each output sample is a finite or infinite weighted sum of input samples, and if $h[n]$ has finite length it can be computed exactly with a known number of multiplications. Convolution is commutative, associative and distributive over addition, so cascaded systems combine by convolving their impulse responses and parallel systems by adding them (Chapter 17).

Computing a convolution

The second form, $y[n] = \sum_k h[k]x[n-k]$, has a direct reading: the output is the current input times $h[0]$, plus the previous input times $h[1]$, and so on. Graphically, $x[n-k]$ as a function of k is the input flipped and slid so that its sample at time n sits at $k = 0$; the output is the sum of the products where the two overlap (Figure 20.2). If x has length L and h length M, the result has length $L + M - 1$.

Worked example — A two-path radio channel. A narrowband telemetry link arrives by a direct path and by two reflections, and at the receiver's sampling rate the channel is modelled as an impulse response $h = [1,\ 0.5,\ 0.25]$: the first echo arrives one sample later with half the amplitude, the second two samples later with a quarter. A transmitted burst $x = [3,\ 1,\ 2]$ (starting at $n = 0$) produces

$$
\begin{aligned}
y[0] &= h[0]x[0] = 3 \\
y[1] &= h[0]x[1] + h[1]x[0] = 1 + 1.5 = 2.5 \\
y[2] &= h[0]x[2] + h[1]x[1] + h[2]x[0] = 2 + 0.5 + 0.75 = 3.25 \\
y[3] &= h[1]x[2] + h[2]x[1] = 1 + 0.25 = 1.25 \\
y[4] &= h[2]x[2] = 0.5
\end{aligned}
$$

so $y = [3,\ 2.5,\ 3.25,\ 1.25,\ 0.5]$, of length $3 + 3 - 1 = 5$. The received samples no longer resemble the transmitted ones: each symbol is smeared into its neighbours. This **intersymbol interference** is what an equaliser in a modem or a mobile receiver must undo (Chapters 24, 27 and 30). As a check, the sum of the outputs equals the product of the sums of the two sequences, $6 \times 1.75 = 10.5$, a property that follows from evaluating the Z-transforms of Section 20.4 at $z = 1$.

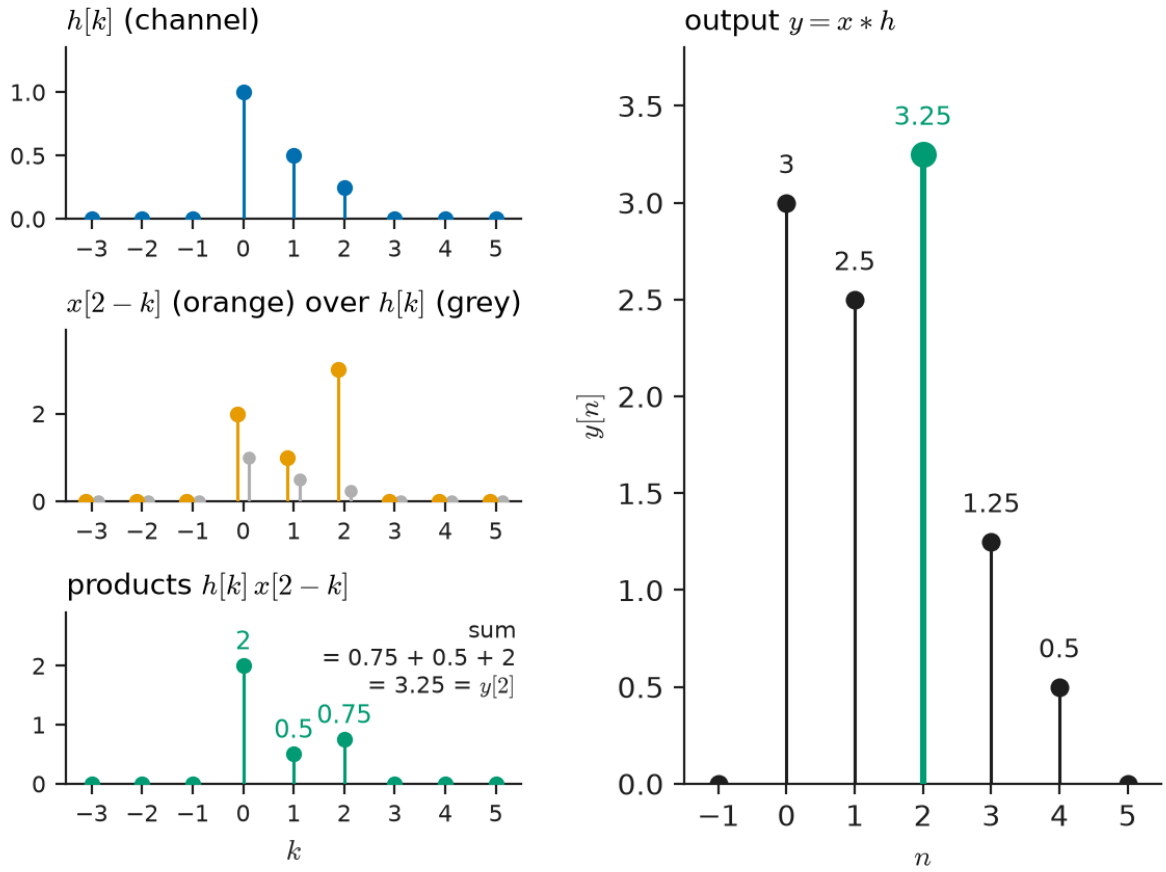

Figure 20.2 — Graphical evaluation of the convolution of x = [3, 1, 2] with the channel h = [1, 0.5, 0.25].

Finite and infinite impulse responses

The structure of $h[n]$ divides discrete-time LTI systems into two classes that run through the rest of Part V.

- **FIR (finite impulse response)** systems have $h[n] \neq 0$ only for $0 \le n \le M-1$. The convolution sum is finite: $y[n] = \sum_{k=0}^{M-1} h[k]x[n-k]$. It is computed directly, needs M multiply–accumulate operations per sample, and is always stable.
- **IIR (infinite impulse response)** systems have impulse responses that continue forever, such as $h[n] = a^n u[n]$. The convolution sum cannot be evaluated directly, but such systems can often be computed recursively from a few past outputs (Section 20.3) at a fixed, small cost per sample.

Causality and memory

A discrete-time LTI system is **causal** if $h[n] = 0$ for $n < 0$: the output depends only on present and past inputs. Real-time systems — a hearing aid, a protection relay, a digital controller — must be causal. Off-line systems processing a stored record — an image, a recorded seismogram, a post-processed ECG — need not be; a non-causal, symmetric smoother that uses future samples gives zero phase distortion, something no causal filter can achieve. A system is **memoryless** if $h[n] = c\,\delta[n]$, a pure gain.

20.3 Difference equations

Linear constant-coefficient difference equations

The discrete counterpart of a linear differential equation with constant coefficients is the **linear constant-coefficient difference equation** (LCCDE):

$$\sum_{k=0}^{N} a_k\, y[n-k] = \sum_{k=0}^{M} b_k\, x[n-k], \qquad a_0 = 1$$

Solving for the current output gives the recursion that a processor executes every sample:

$$y[n] = \sum_{k=0}^{M} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k]$$

This book uses this sign convention throughout Part V: the denominator coefficients a_k appear with a minus sign in the recursion and with a plus sign in the system function of Section 20.5, matching common DSP software. When all a_k for $k \ge 1$ are zero the equation is **non-recursive**, and the b_k are simply the FIR impulse response. When any $a_k \neq 0$ the equation is **recursive** and, in general, IIR.

Difference equations arise in three ways in EEE:

1. **As algorithms** for digital filters and controllers, where they are the design itself.
2. **As discretisations** of differential equations. Approximating dy/dt by $(y[n] - y[n-1])/T$ turns $\tau\, dy/dt + y = x$ into $y[n] = \frac{\tau}{\tau+T} y[n-1] + \frac{T}{\tau+T} x[n]$. Circuit simulators such as SPICE replace every capacitor and inductor by such a discrete model at each time step.
3. **As exact models** of inherently sampled processes: a bank balance compounded monthly, a population counted yearly, a buffer that fills and empties once per frame.

Solving by recursion

A difference equation can always be solved by running it forward from initial conditions. This is the main practical advantage over differential equations: no approximation is involved, and the solution for any input is obtained sample by sample.

Worked example — Recursive solution with an initial condition. The system $y[n] - 0.5\,y[n-1] = x[n]$ has initial condition $y[-1] = 1$ and input $x[n] = u[n]$. Recursion gives

$$y[0] = 0.5(1) + 1 = 1.5, \quad y[1] = 0.5(1.5) + 1 = 1.75, \quad y[2] = 0.5(1.75) + 1 = 1.875, \ \ldots$$

The values approach 2, the steady state at which $y = 0.5y + 1$. A closed form, found in Section 20.4, is $y[n] = 2 - 0.5(0.5)^n$ for $n \geq 0$, which reproduces 1.5, 1.75, 1.875.

Homogeneous and particular solutions

The classical method mirrors Chapter 7. The homogeneous equation $\sum a_k y[n-k] = 0$ has solutions of the form $y[n] = \lambda^n$; substituting gives the **characteristic equation**

$$\lambda^N + a_1 \lambda^{N-1} + \cdots + a_N = 0$$

whose roots λ_i are the **natural modes**. With distinct roots the natural response is $\sum C_i \lambda_i^n$; a repeated root λ of multiplicity m contributes $\lambda^n, n\lambda^n, \ldots, n^{m-1}\lambda^n$. A particular solution matching the form of the input is added, and the constants are fixed by the initial conditions. In the example above, $\lambda = 0.5$, the particular solution for a constant input is the constant 2, and $y[-1] = 1$ fixes $C = -0.5$, consistent with $C\lambda^{-1} = -1$.

Complex roots $\lambda = re^{\pm j\theta}$ give natural responses $r^n \cos(\theta n + \phi)$: a sampled damped oscillation, decaying if $r < 1$. Their magnitude sets the decay per sample and their angle sets the oscillation frequency in radians per sample — the same information as the pole positions of Section 20.6, because they *are* the poles.

Zero-input and zero-state responses

The response separates into the **zero-input response** (initial conditions, zero input) and the **zero-state response** (input applied to a system initially at rest). For the example, the zero-state response to $u[n]$ is $2 - (0.5)^n$ and the zero-input response from $y[-1] = 1$ is $0.5(0.5)^n$; their sum is the total. Only the zero-state response is described by convolution with $h[n]$. The distinction matters in practice: a digital filter that is started with stale data in its delay registers, or a controller switched in with a non-zero integrator state, shows a zero-input transient superimposed on its intended behaviour. The cure is to initialise the state deliberately, a practice called **bumpless transfer** in control.

20.4 The Z-transform

Definition

The Laplace transform turned differential equations into algebra by weighting signals with e^{-st}. The **Z-transform** does the same for sequences with powers of a complex variable z:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}$$

It is a power series in z^{-1} whose coefficients are the samples. The sample $x[n]$ is the coefficient of z^{-n}, so z^{-1} marks one sample of delay. For a sequence obtained by sampling, $z = e^{sT}$ connects it to the Laplace variable: the impulse-train representation of Chapter 19 has Laplace transform $\sum x[n]e^{-snT}$, which is $X(z)$ with $z = e^{sT}$.

> **Boundary.**
> The Z-transform as a mathematical object — Laurent series, analyticity, the contour-integral inverse — is developed in *The Mathematics Universe*. This section recaps the results needed for system analysis and uses them.

Region of convergence

The sum converges only for some values of z, and the set where it does, the **region of convergence** (ROC), is part of the transform. For the right-sided exponential $x[n] = a^n u[n]$,

$$X(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}}, \qquad |z| > |a|$$

a geometric series that converges when $|az^{-1}| < 1$. For the left-sided sequence $x[n] = -a^n u[-n-1]$ the same algebraic expression results, but with ROC $|z| < |a|$. The formula alone does not determine the sequence; the ROC does. The general rules follow from this example:

- The ROC is an annulus centred on the origin; it contains no poles.
- A **right-sided** sequence (zero before some n_0), which includes every causal sequence, has an ROC outside the outermost pole.
- A **left-sided** sequence has an ROC inside the innermost pole.
- A **two-sided** sequence has an annular ROC between two poles, or none at all.
- A **finite-length** sequence converges everywhere except possibly at $z = 0$ or $z = \infty$.

For causal systems — the usual case in real-time DSP — the ROC is always "outside the outermost pole", and the notation often leaves it implicit.

A short table

Table 20.1 — Common Z-transform pairs.

Sequence $x[n]$	$X(z)$	**ROC**
$\delta[n]$	1	all z
$\delta[n-k],\ k>0$	z^{-k}	$z \neq 0$
$u[n]$	$\dfrac{1}{1-z^{-1}}$	$\lvert z\rvert > 1$
$a^n u[n]$	$\dfrac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u[-n-1]$	$\dfrac{1}{1-az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$
$n a^n u[n]$	$\dfrac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
$r^n \cos(\Omega_0 n)\, u[n]$	$\dfrac{1 - r\cos\Omega_0\, z^{-1}}{1 - 2r\cos\Omega_0\, z^{-1} + r^2 z^{-2}}$	$\lvert z\rvert > r$
$r^n \sin(\Omega_0 n)\, u[n]$	$\dfrac{r\sin\Omega_0\, z^{-1}}{1 - 2r\cos\Omega_0\, z^{-1} + r^2 z^{-2}}$	$\lvert z\rvert > r$

Properties

The properties used constantly in system analysis are:

Property	**Sequence**	**Transform**
Linearity	$\alpha x_1[n] + \beta x_2[n]$	$\alpha X_1(z) + \beta X_2(z)$
Time shift	$x[n-k]$	$z^{-k}X(z)$
Convolution	$x[n] * h[n]$	$X(z)H(z)$
Multiplication by a^n	$a^n x[n]$	$X(z/a)$
Time reversal	$x[-n]$	$X(1/z)$
Differentiation in z	$n x[n]$	$-z\,\dfrac{dX}{dz}$
Initial value (causal x)	$x[0]$	$\lim_{z\to\infty} X(z)$
Final value (if it exists)	$\lim_{n\to\infty} x[n]$	$\lim_{z\to 1}(1-z^{-1})X(z)$

The convolution property is the reason for the whole apparatus: a convolution sum in time becomes a product in z, exactly as the convolution integral became a product in s. The final-value theorem holds only if all poles of $(1-z^{-1})X(z)$ lie strictly inside the unit circle.

> **Status of the principle: Convolution property of the Z-transform**
> *Derived*. A consequence of the definition and the time-shift property; holds wherever the ROCs of $X(z)$ and $H(z)$ overlap.

Inverse Z-transform by partial fractions

For the rational transforms of LTI systems, the inverse is found as in Chapter 11: expand in partial fractions in z^{-1} and invert each term from the table. Long division of the power series gives the first few samples directly and is a useful check.

Worked example — Impulse response of a two-pole system. A sensor-conditioning filter has

$$H(z) = \frac{1}{(1 - 0.5z^{-1})(1 - 0.25z^{-1})} = \frac{1}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

and is causal. Write $H(z) = \dfrac{A}{1 - 0.5z^{-1}} + \dfrac{B}{1 - 0.25z^{-1}}$. Setting $z^{-1} = 2$ (the root of the first factor) in $A = \left.H(z)(1 - 0.5z^{-1})\right|$ gives $A = 1/(1 - 0.25 \cdot 2) = 2$; setting $z^{-1} = 4$ gives $B = 1/(1 - 0.5 \cdot 4) = -1$. Therefore

$$h[n] = \left[2(0.5)^n - (0.25)^n\right] u[n]$$

which gives $h[0] = 1$, $h[1] = 0.75$, $h[2] = 0.4375$, $h[3] = 0.234375$, $h[4] = 0.12109$. The same values follow from running the recursion $y[n] = 0.75y[n-1] - 0.125y[n-2] + x[n]$ with $x = \delta$, and from the initial-value theorem ($H(\infty) = 1$). The DC gain is $H(1) = 1/(1 - 0.75 + 0.125) = 2.667$, which equals $\sum h[n] = 2 \times 2 - 4/3 = 2.667$. The slower pole at 0.5 dominates the tail: after 10 samples the 0.25 term has fallen below 10^{-6} while the 0.5 term is still about 2×10^{-3}.

The unilateral Z-transform and initial conditions

To handle initial conditions, the **unilateral** transform $\mathcal{Z}_u\{x\} = \sum_{n=0}^{\infty} x[n] z^{-n}$ is used, in which the shift property picks up the initial values:

$$\mathcal{Z}_u\{y[n-1]\} = z^{-1}Y(z) + y[-1]$$

Applied to the earlier example $y[n] - 0.5y[n-1] = u[n]$ with $y[-1] = 1$:

$$Y(z)\left(1 - 0.5z^{-1}\right) - 0.5 = \frac{1}{1 - z^{-1}} \quad \Rightarrow \quad Y(z) = \frac{1}{(1 - z^{-1})(1 - 0.5z^{-1})} + \frac{0.5}{1 - 0.5z^{-1}}$$

Partial fractions of the first term give $\dfrac{2}{1 - z^{-1}} - \dfrac{1}{1 - 0.5z^{-1}}$, so

$$y[n] = \underbrace{2 - (0.5)^n}_{\text{zero-state}} + \underbrace{0.5(0.5)^n}_{\text{zero-input}} = 2 - 0.5(0.5)^n, \qquad n \geq 0$$

which is the closed form quoted in Section 20.3. The role is exactly that of the one-sided Laplace transform with initial capacitor voltages in Chapter 11.

20.5 Discrete-time system functions

From difference equation to $H(z)$

Taking the Z-transform of the LCCDE of Section 20.3 with zero initial conditions and using the shift property,

$$Y(z)\sum_{k=0}^{N} a_k z^{-k} = X(z)\sum_{k=0}^{M} b_k z^{-k} \quad \Rightarrow \quad H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \cdots + b_M z^{-M}}{1 + a_1 z^{-1} + \cdots + a_N z^{-N}}$$

The **system function** $H(z)$ is the Z-transform of the impulse response, the ratio of output to input transforms, and the rational function whose coefficients are those of the difference equation — three descriptions of the same object. An FIR filter has denominator 1; an IIR filter has a non-trivial denominator. Reading coefficients off $H(z)$ gives the implementation directly, which makes the discrete transfer function more immediately practical than its analogue counterpart: there is no component-selection step between the equation and the hardware.

Frequency response

Complex exponentials are eigenfunctions of discrete-time LTI systems (Chapter 17): the input $x[n] = z_0^n$ produces

$$y[n] = \sum_k h[k] z_0^{n-k} = z_0^n \sum_k h[k] z_0^{-k} = H(z_0)\, z_0^n$$

For a sinusoidal input, $z_0 = e^{j\Omega}$ lies on the **unit circle**, and the multiplier is the **frequency response**

$$H(e^{j\Omega}) = H(z)\big|_{z=e^{j\Omega}} = \sum_n h[n] e^{-j\Omega n}$$

which is the discrete-time Fourier transform of $h[n]$. A sinusoid $\cos(\Omega n)$ emerges as $|H(e^{j\Omega})|\cos(\Omega n + \angle H(e^{j\Omega}))$, with gain and phase given by the value of $H(z)$ at the point $e^{j\Omega}$. Three facts follow at once:

- $H(e^{j\Omega})$ is periodic in Ω with period 2π, so only $-\pi < \Omega \le \pi$ needs to be plotted; for real $h[n]$ the magnitude is even and the phase odd, so $0 \le \Omega \le \pi$ suffices.
- $\Omega = 0$ (DC) corresponds to $z = 1$ and $\Omega = \pi$ (the Nyquist frequency) to $z = -1$. The DC gain is $H(1) = \sum h[n]$, the Nyquist gain $H(-1) = \sum (-1)^n h[n]$.
- In physical units, $\Omega = 2\pi f / f_s$. The same coefficients give a filter whose frequency features scale with f_s: a notch at $\Omega = 0.2\pi$ sits at 50 Hz if $f_s = 500$ Sa/s and at 60 Hz if $f_s = 600$ Sa/s.

The frequency response exists (the unit circle lies in the ROC) only for stable systems, which anticipates Section 20.7.

Worked example — An exponential smoother. A temperature transmitter sampled at $f_s = 1$ kSa/s smooths its readings with

$$y[n] = 0.9\, y[n-1] + 0.1\, x[n], \qquad H(z) = \frac{0.1}{1 - 0.9 z^{-1}}$$

The DC gain is $0.1/(1-0.9)=1$, so a steady reading passes unchanged. The step response is $1-0.9^{n+1}$: after 10 samples (10 ms) it has reached $1-0.9^{11}\approx 0.686$, and it takes $\ln 0.05/\ln 0.9 = 28.4$ samples to reach 95 %. Matching $0.9^{n}=e^{-nT/\tau}$ gives an equivalent analogue time constant $\tau = -T/\ln 0.9 = 9.49$ ms. The magnitude response is

$$|H(e^{j\Omega})| = \frac{0.1}{\sqrt{1 - 1.8\cos\Omega + 0.81}}$$

Setting this to $1/\sqrt{2}$ gives $\cos\Omega_c = 1-(0.1)^2/(2\times 0.9)$, $\Omega_c = 0.1055$ rad/sample, or $f_c = \Omega_c f_s/2\pi = 16.8$ Hz. An analogue RC filter with $\tau = 9.49$ ms has $f_c = 1/(2\pi\tau) = 16.8$ Hz too: the two agree well while $f_c \ll f_s$. They part company near Nyquist: the digital filter's gain at $\Omega=\pi$ is $0.1/1.9 = 0.0526$ (-25.6 dB) and then *rises again* as frequency increases further (the response is periodic), whereas the RC filter keeps falling at 20 dB/decade. This is why digital smoothing never replaces the analogue anti-aliasing filter of Chapter 19.

Block diagrams and realisations

A system function can be implemented by many different arrangements of adders, multipliers and unit delays (z^{-1} blocks), called **realisations** or **structures**. The **direct form I** implements the numerator (feed-forward) and denominator (feedback) sections with separate delay lines; the **direct form II** shares one delay line of length $\max(M,N)$ and is canonical (minimum delays). **Cascade** realisations factor $H(z)$ into second-order sections, and **parallel** realisations use partial fractions. All are equivalent in exact arithmetic; they differ greatly in finite precision, which is the subject of Chapter 21 and Chapter 23. Figure 20.3 shows the direct form II of a second-order section, the "biquad" that is the building block of almost every IIR filter.

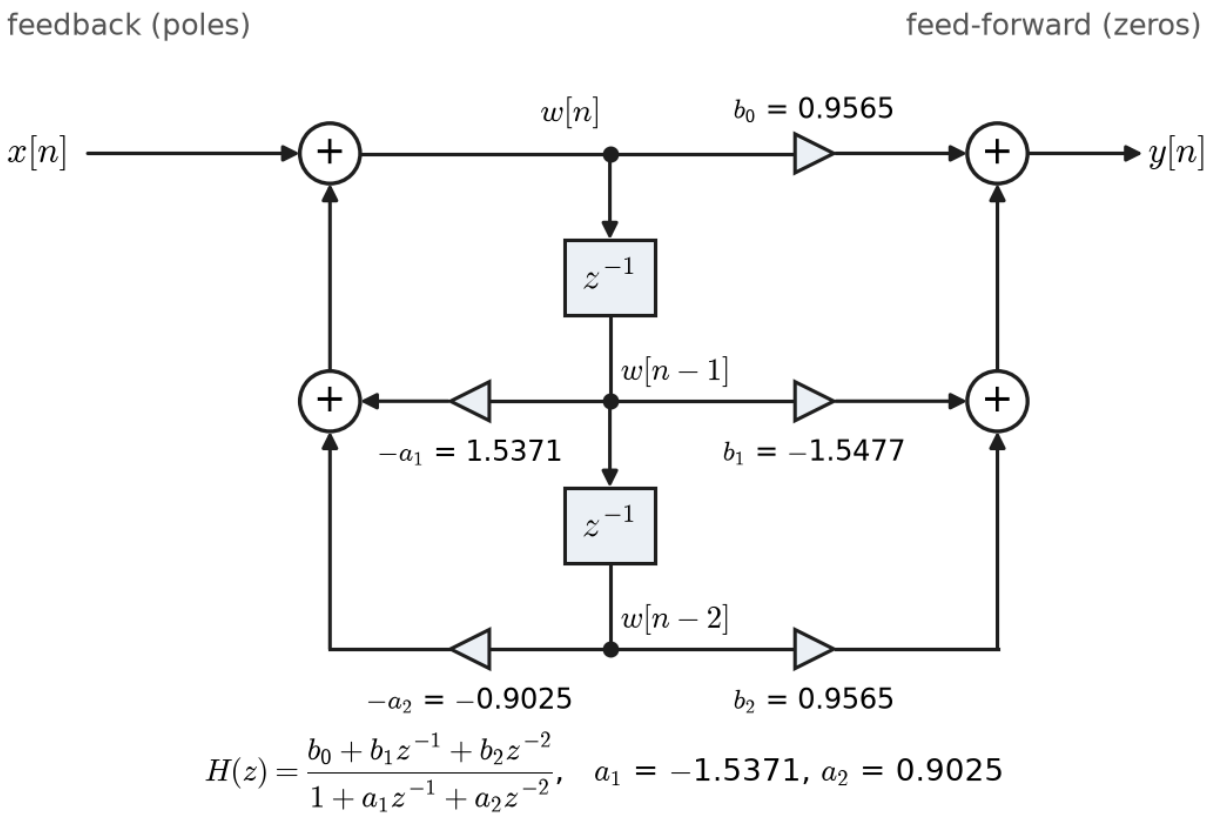

Figure 20.3 — Direct form II realisation of a second-order section (biquad).

20.6 Poles and zeros

Factorising the system function

Multiplying numerator and denominator of $H(z)$ by $z^{\max(M,N)}$ and factorising,

$$H(z) = b_0 \frac{\prod_{k=1}^{M}(1 - q_k z^{-1})}{\prod_{k=1}^{N}(1 - p_k z^{-1})}$$

The q_k are the **zeros** and the p_k the **poles**. For real coefficients they are real or occur in complex-conjugate pairs. An FIR filter has all its poles at $z = 0$, which contribute only delay. As in the s-plane (Chapters 11 and 13), the pole–zero pattern plus a gain constant determines the system completely.

Pole position and natural response

A causal pole at $p = re^{j\theta}$ contributes a mode $r^n \cos(\theta n + \phi)$ to the impulse response. The **radius** r sets the decay per sample and the **angle** θ the oscillation frequency in radians per sample:

- poles near the origin give responses that die out in a few samples;
- poles near the unit circle give long, slowly decaying responses — the time constant in samples is $-1/\ln r \approx 1/(1-r)$ for r close to 1;

- a pole on the positive real axis gives a non-oscillating decay; on the negative real axis a response that alternates in sign every sample (oscillation at the Nyquist frequency);
- a pole on the unit circle gives a sustained oscillation, outside it a growing one.

Figure 20.5 later in the chapter shows these cases side by side.

The mapping $z = e^{sT}$

When a discrete system is designed to mimic a continuous one, or a continuous signal is sampled, an s-plane pole at $s = \sigma + j\omega$ appears in the z-plane at

$$z = e^{sT} = e^{\sigma T} e^{j\omega T}$$

so $r = e^{\sigma T}$ and $\theta = \omega T$. The left half of the s-plane maps inside the unit circle, the $j\omega$ axis onto the unit circle (with ω from $-\pi/T$ to π/T covering it once — the aliasing of Chapter 19 again), and the right half-plane outside. Lines of constant damping in s become circles of constant radius; lines of constant frequency become rays from the origin. Figure 20.4 shows the mapping.

> **Status of the principle: Pole mapping $z = e^{sT}$**
> *Derived*. Exact for the poles of a sampled continuous-time impulse response (impulse invariance); other discretisations used in filter design, such as the bilinear transform of Chapter 23, map poles differently.

Worked example — Sampling a damped oscillation. A power-system oscillation mode has poles at $s = -100 \pm j2\pi(50)$ s^{-1} — a 50 Hz oscillation decaying with a 10 ms time constant. A relay samples at $f_s = 1$ kSa/s ($T = 1$ ms). The z-plane poles are at

$$r = e^{-100 \times 0.001} = e^{-0.1} = 0.9048, \qquad \theta = 2\pi \times 50 \times 0.001 = 0.3142 \text{ rad} = 18^\circ$$

that is, $z = 0.8606 \pm j0.2796$. In the sampled record the oscillation shrinks by the factor 0.9048 each sample and completes one cycle every $2\pi/0.3142 = 20$ samples, exactly as it should. At $f_s = 200$ Sa/s the angle would be 0.5π — still unambiguous; at $f_s = 100$ Sa/s it would be π, on the negative real axis, and the 50 Hz oscillation would become indistinguishable from a Nyquist-rate alternation.

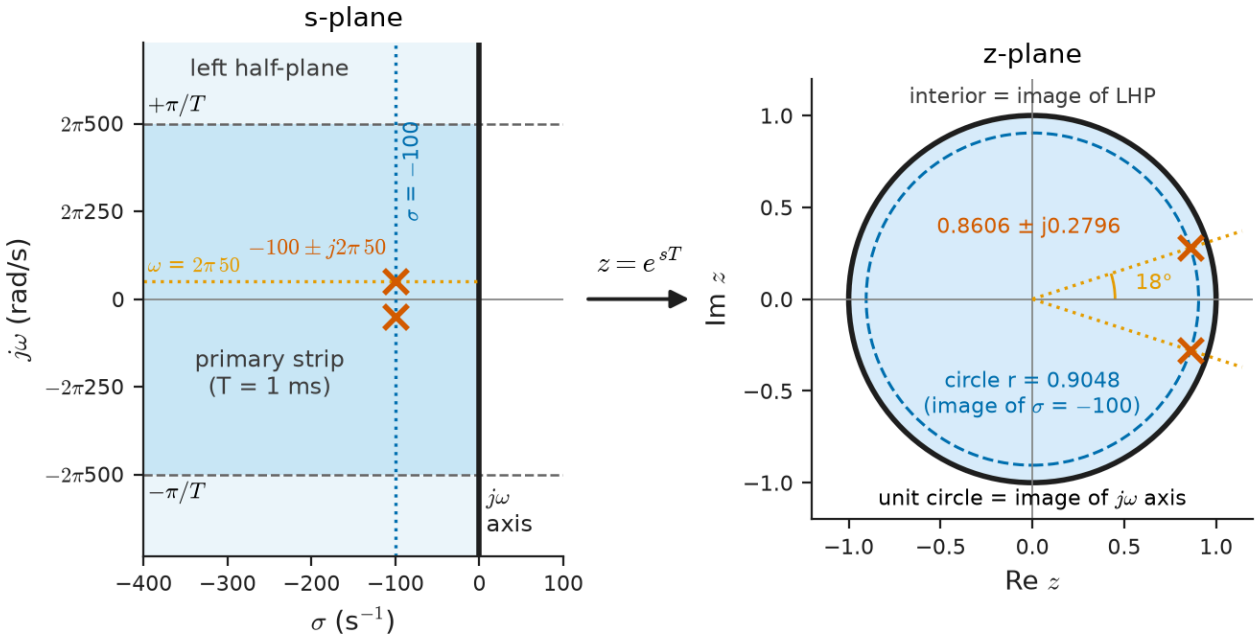

Figure 20.4 — Mapping of the s-plane to the z-plane by z = exp(sT).

Geometric evaluation of the frequency response

Write the factored $H(z)$ at a point $z = e^{j\Omega}$ on the unit circle. Each factor $(e^{j\Omega} - q_k)$ is the vector from the zero q_k to that point, and each $(e^{j\Omega} - p_k)$ the vector from a pole. Hence

$$|H(e^{j\Omega})| = |b_0| \frac{\prod_k \text{distance from zero } q_k}{\prod_k \text{distance from pole } p_k}, \qquad \angle H = \sum \angle(\text{zero vectors}) - \sum \angle(\text{pole vectors}) + \text{linea}$$

As the evaluation point walks around the circle from $\Omega = 0$ to π:

- passing close to a **zero** makes the response small; a zero *on* the circle gives an exact null;
- passing close to a **pole** makes the response large; the closer the pole to the circle, the sharper and higher the peak;
- a pole and a zero at the same angle, with the pole just inside the zero, give a narrow notch: far from them the two distances are nearly equal and cancel; near them, the zero distance shrinks to nothing while the pole distance stays finite.

This picture is the single most useful design intuition in digital filtering, and it leads straight to the notch filter of Section 20.8.

20.7 Discrete-time stability

BIBO stability and the impulse response

A discrete-time system is **bounded-input bounded-output (BIBO) stable** if every bounded input produces a bounded output (Chapter 16). For an LTI system, this holds if and only if the impulse response is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Sufficiency follows from $|y[n]| \le \sum |h[k]|\,|x[n-k]| \le B_x \sum |h[k]|$. Necessity follows by choosing the bounded input $x[n-k] = \operatorname{sgn}(h[k])$, which makes $y[n]$ equal to $\sum |h[k]|$. Every FIR filter is stable, since its sum is finite.

The unit-circle criterion

For a system with rational $H(z)$, absolute summability is equivalent to the ROC including the unit circle. For a **causal** system, whose ROC lies outside the outermost pole, this gives the working rule:

> A causal LTI system with rational system function is BIBO stable if and only if all its poles lie strictly inside the unit circle, $|p_k| < 1$.

> **Status of the principle: Unit-circle stability criterion**
> *Derived*. Follows from the absolute-summability condition and the geometric decay of each mode p_k^n; exact in infinite-precision arithmetic. In fixed-point hardware, poles very close to the circle can move onto or outside it when coefficients are rounded (Chapter 21), and nonlinear limit cycles can appear even for stable poles.

A pole exactly on the circle is **marginally stable**: the impulse response neither decays nor grows, but a bounded input at the pole's frequency produces an unbounded output (resonance in the discrete sense). The accumulator $y[n] = y[n-1] + x[n]$, with a pole at $z = 1$, is the classic example: a constant input makes the output ramp forever. Digital integrators in controllers and phase accumulators in frequency synthesisers have exactly this pole, and their implementation must guard against the consequent growth (anti-windup, modular arithmetic).

The unit-circle criterion is the image of the left-half-plane criterion of Chapter 34 under $z = e^{sT}$.

Testing a second-order denominator

For a second-order denominator $1 + a_1 z^{-1} + a_2 z^{-2}$ the poles lie inside the unit circle if and only if

$$|a_2| < 1 \qquad \text{and} \qquad |a_1| < 1 + a_2$$

These inequalities define the **stability triangle** in the (a_1, a_2) plane, with vertices at $(-2, 1)$, $(2, 1)$ and $(0, -1)$. Complex poles occur when $a_1^2 < 4a_2$, inside a parabola within the triangle. For higher orders, the Jury test (the discrete counterpart of Routh–Hurwitz, Chapter 34) or direct numerical root-finding is used; in practice filters are implemented as cascades of second-order sections precisely so that each section can be checked with this simple test.

Worked example — Is this resonator stable? A vibration-analysis rig sampling at $f_s = 1$ kSa/s uses the resonator

$$y[n] = 1.8\,y[n-1] - 0.9\,y[n-2] + x[n], \qquad A(z) = 1 - 1.8z^{-1} + 0.9z^{-2}$$

so $a_1 = -1.8$, $a_2 = 0.9$. The test: $|0.9| < 1$ and $1.8 < 1.9$ — stable. The poles are the roots of $z^2 - 1.8z + 0.9 = 0$, $z = 0.9 \pm j0.3$, with radius $\sqrt{0.9} = 0.9487$ and angle $\arctan(0.3/0.9) = 0.3218$ rad. The impulse response is a decaying oscillation at $0.3218 f_s/2\pi = 51.2$ Hz, shrinking by 0.9487 per sample (time constant about $-1/\ln 0.9487 = 19$ samples). The DC gain is $1/(1 - 1.8 + 0.9) = 10$.

Now let a coefficient error change a_2 to 1.01, i.e. $y[n] = 1.8y[n-1] - 1.01y[n-2] + x[n]$. Then $|a_2| = 1.01 > 1$ and the system is unstable: the poles move to $0.9 \pm j0.4472$, radius $\sqrt{1.01} = 1.005$, and the oscillation grows by 0.5 % per sample — a factor of about 150 per second at this sample rate. A third case, $y[n] = y[n-1] + 0.5y[n-2] + x[n]$, has $A(z) = 1 - z^{-1} - 0.5z^{-2}$: $|a_2| = 0.5 < 1$ but $|a_1| = 1 > 1 + a_2 = 0.5$, so it fails, and indeed it has a real pole at 1.366. Figure 20.5 shows the three impulse responses.

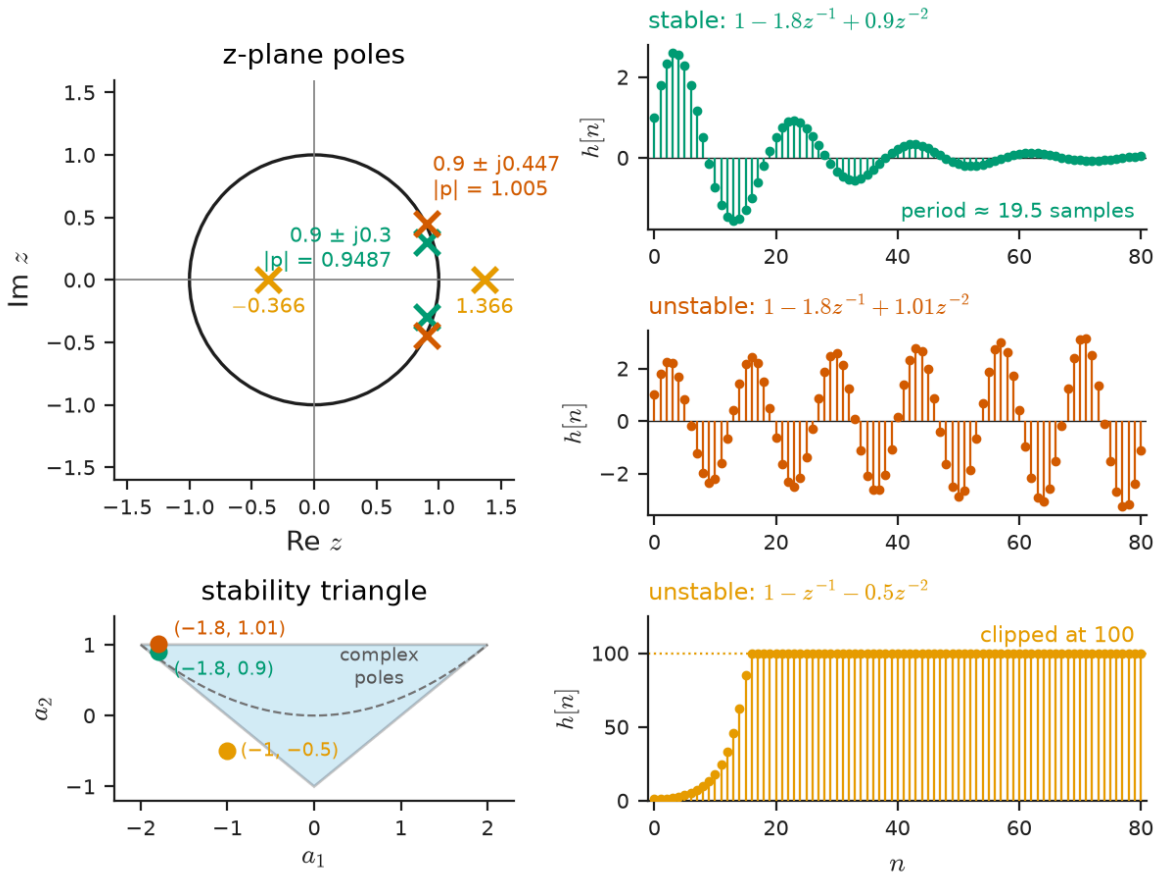

Figure 20.5 — Pole locations in the z-plane and the corresponding impulse responses.

Stability of what, exactly?

BIBO stability of the transfer function is not the whole story. A system can have a pole outside the unit circle that is cancelled by a zero, so that $H(z)$ looks stable while an internal state grows. In exact arithmetic the cancellation hides the growth; in a processor, rounding excites the hidden mode and the internal variable eventually overflows. The rule is the same as in continuous time (Chapter 35): never rely on cancelling an unstable pole. Adaptive filters (Chapter 24) and digital control loops (Chapter 35) add another layer — their stability depends on step sizes and loop gains, not only on pole positions at one instant.

20.8 Digital filter fundamentals

What a digital filter is

A digital filter is a discrete-time LTI system designed to shape the spectrum of a sequence: to pass some frequencies, attenuate others, or alter phase. Everything in Sections 20.2–20.7 applies, and the vocabulary of Chapter 13 — passband, stopband, transition band, ripple, attenuation, group delay — carries over with frequencies measured in Ω or as a fraction of f_s. Chapter 23 develops systematic design; this section collects the elementary filters that every engineer meets and that illustrate the pole–zero intuition.

FIR and IIR compared

Property	FIR	IIR
Impulse response	finite, length M	infinite
Poles	all at $z = 0$	anywhere inside the unit circle
Stability	always	must be ensured by design and implementation
Exact linear phase	possible (symmetric coefficients)	not possible with a causal filter
Order for a sharp transition	high (tens to hundreds of taps)	low (4 to 10)
Cost per output sample	M multiply–accumulates	about $2N + 1$ multiply–accumulates
Sensitivity to coefficient rounding	low	can be high for poles near the unit circle
Typical uses	audio crossovers, data converters, communications, imaging	notch and resonator filters, low-cost smoothing, emulation of analogue filters

An FIR filter with symmetric coefficients, $h[n] = h[M-1-n]$, has **linear phase**: its frequency response is $H(e^{j\Omega}) = A(\Omega)e^{-j\Omega(M-1)/2}$ with A real, so every frequency is delayed by the same $(M-1)/2$ samples and the waveform shape is preserved. That property is why FIR filters dominate where waveform fidelity matters — ECG morphology, data-communication pulses, audio crossovers.

The moving average

The N-point moving average $y[n] = \frac{1}{N}\sum_{k=0}^{N-1} x[n-k]$ is the simplest FIR low-pass. Summing the geometric series,

$$H(z) = \frac{1}{N}\,\frac{1 - z^{-N}}{1 - z^{-1}}, \qquad |H(e^{j\Omega})| = \left|\frac{\sin(N\Omega/2)}{N\sin(\Omega/2)}\right|$$

Its zeros are the N-th roots of unity except $z = 1$ (cancelled by the pole there), placing exact nulls at $f = kf_s/N$, $k = 1, \dots, N-1$. That is a useful property: a 10-point moving average at $f_s = 500$ Sa/s has

nulls at 50, 100, 150, 200 and 250 Hz, so it removes mains interference *and all its harmonics* completely — which is why integrating ADCs in digital multimeters integrate over exactly one or several mains periods (20 ms at 50 Hz) to reject hum. Its group delay is $(N-1)/2 = 4.5$ samples (9 ms). The cost is a poor low-pass shape: the first sidelobe is only about 13 dB down. The recursive form $y[n] = y[n-1] + \frac{1}{N}(x[n] - x[n-N])$ computes the same output with two additions per sample, at the price of a pole on the unit circle that relies on exact cancellation — fine in integer arithmetic, risky in floating point (Chapter 21).

The first-order IIR smoother

The exponential smoother of Section 20.5, $y[n] = a\,y[n-1] + (1-a)x[n]$, has a single pole at $z = a$ and unit DC gain. It is the digital RC filter: one multiply, one state variable, a time constant of about $1/(1-a)$ samples. It is used everywhere a slowly varying estimate is wanted from noisy samples — battery state of charge, RMS meters, automatic gain control, the "exponentially weighted moving average" of network traffic monitors.

The DC blocker

The filter $H(z) = \dfrac{1 - z^{-1}}{1 - p z^{-1}}$, with p slightly less than 1, has a zero exactly at $z = 1$ (complete rejection of DC) and a pole just inside it. Far from DC the pole and zero distances are nearly equal and the gain is close to 1 (at Nyquist it is $2/(1+p)$). With $p = 0.995$ at $f_s = 1$ kSa/s the -3 dB frequency is about $(1-p)f_s/2\pi = 0.80$ Hz: it removes electrode offset from a biosignal or the DC bias from a microphone channel while leaving everything above a few hertz untouched.

The notch filter

The pole–zero picture of Section 20.6 builds a notch in two steps. To reject a frequency Ω_0 completely, place a conjugate pair of zeros on the unit circle at $e^{\pm j\Omega_0}$. Alone, they would also attenuate a wide band around Ω_0. To make the notch narrow, place a conjugate pair of poles at the same angles at radius r just inside the circle. The result is

$$H(z) = g\,\frac{1 - 2\cos\Omega_0\, z^{-1} + z^{-2}}{1 - 2r\cos\Omega_0\, z^{-1} + r^2 z^{-2}}$$

with g chosen for unit gain at DC. For r close to 1, the -3 dB width of the notch is approximately $\Delta\Omega \approx 2(1-r)$ rad/sample, or $\Delta f \approx (1-r)f_s/\pi$.

> **Status of the principle: Notch-width approximation**
> *Approximate.* Obtained by linearising the pole distance near the notch; accurate for $1 - r \ll 1$ and notch frequencies not close to 0 or $f_s/2$.

Worked example — Removing 50 Hz hum from an ECG. With $f_s = 500$ Sa/s and $f_0 = 50$ Hz, $\Omega_0 = 2\pi \times 50/500 = 0.2\pi = 0.6283$ rad/sample and $2\cos\Omega_0 = 1.6180$. Choose $r = 0.95$. Then

$$H(z) = g\,\frac{1 - 1.6180 z^{-1} + z^{-2}}{1 - 1.5371 z^{-1} + 0.9025 z^{-2}}$$

with $2r\cos\Omega_0 = 1.5371$ and $r^2 = 0.9025$. At $z = 1$ the numerator is 0.3820 and the denominator 0.3654, so $g = 0.3654/0.3820 = 0.9565$, and the final coefficients are $b = [0.9565,\ -1.5477,\ 0.9565]$, $a = [1,\ -1.5371,\ 0.9025]$. The implementation is

$$y[n] = 0.9565x[n] - 1.5477x[n-1] + 0.9565x[n-2] + 1.5371y[n-1] - 0.9025y[n-2]$$

Its response, computed numerically:

Frequency (Hz)	**0**	**40**	**45**	**48**	**49**	**50**	**55**	**60**	**100**	**150**
Gain (dB)	0.00	-0.62	-2.17	-7.08	-12.4	$-\infty$	-2.17	-0.62	$+0.03$	$+0.05$

The -3 dB band runs from 45.96 Hz to 54.03 Hz, a width of 8.07 Hz, against the estimate $(1 - 0.95) \times 500/\pi = 7.96$ Hz. The poles have radius 0.95, so the filter's own transient — the time it takes to lock on to a hum that suddenly appears — decays with a time constant of about $1/(1 - 0.95) = 20$ samples, or 40 ms. Moving r to 0.99 would narrow the notch to about 1.6 Hz, protecting more of the ECG spectrum, but would lengthen the transient to about 200 ms and make the filter ring visibly after sharp QRS complexes, and it would be more sensitive to the mains frequency drifting by a few tenths of a hertz. The trade between selectivity and transient length is a direct consequence of the time–frequency relationship of Chapter 18. Figure 20.6 shows the pole–zero plot and the response.

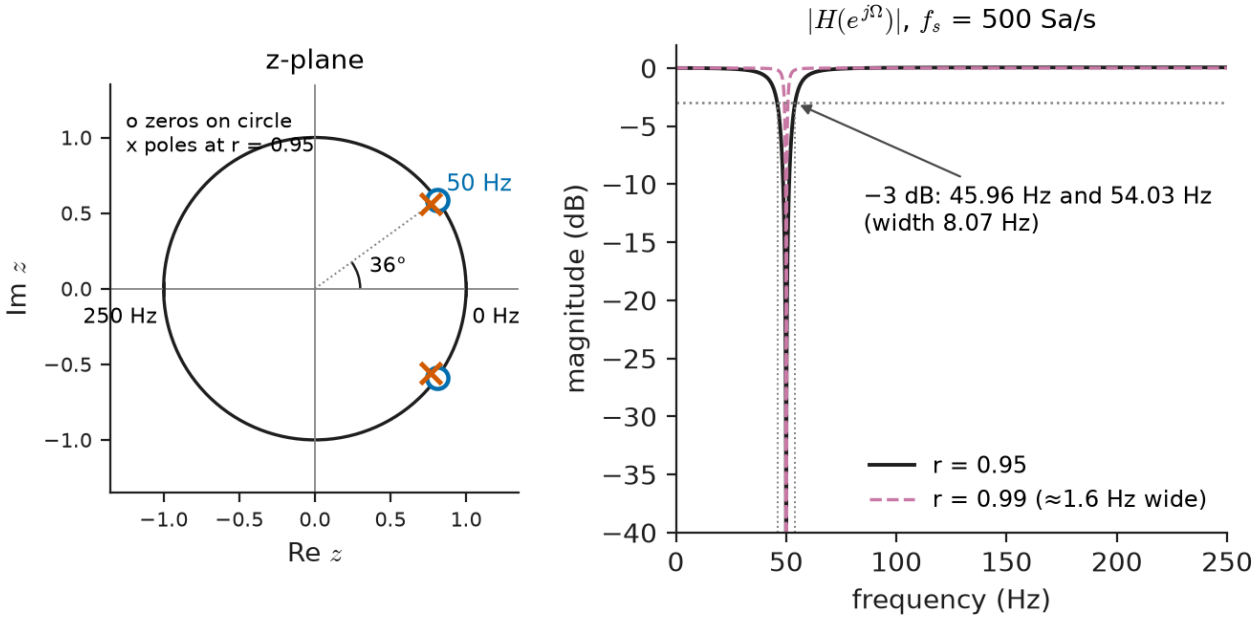

Figure 20.6 — Pole-zero plot and magnitude response of the 50 Hz notch filter for an ECG sampled at 500 Sa/s.

Resonators and combs

Swapping the roles — poles near the circle with no compensating zeros — gives a **resonator**, a narrow band-pass whose centre frequency is the pole angle. The stable example of Section 20.7 is one; tone detectors in telephony (the Goertzel algorithm for DTMF, Chapter 22) and the vibration-tracking filters of rotating machinery use the same idea. Replacing z^{-1} by z^{-D} in any filter replicates its response D times around the circle, producing a **comb filter**: $1 - z^{-D}$ places nulls at every multiple of f_s/D and is

used to remove a fundamental and all its harmonics at once, or, with feedback, to create the echoes of audio effects and the reverberation of room models.

Specifications and what comes next

A practical filter is designed from a specification: passband edge and ripple, stopband edge and attenuation, phase or group-delay requirements, and the cost of computing each sample. The simple filters above meet loose specifications by placing a few poles and zeros by hand. Chapter 23 shows how to meet tight ones — an anti-aliasing decimator flat to 0.01 dB with 100 dB of stopband rejection, say — using windowed and equiripple FIR designs and IIR designs transformed from analogue prototypes. Chapter 21 first establishes what happens when the coefficients and arithmetic of these filters are not exact.

What remains open

Open question	Where it is answered
How do finite word length, rounding and overflow change the behaviour of these filters in real hardware?	Chapter 21
How is the frequency content of a finite sequence computed efficiently?	Chapter 22
How are FIR and IIR filters designed to meet tight specifications, and how are analogue designs transformed to digital?	Chapter 23
How do down-sampling and up-sampling, and filters that adapt their own coefficients, work?	Chapter 24
How do equalisers undo the intersymbol interference of the multipath channel of Section 20.2?	Chapters 27 and 30
How is the Jury test related to Routh–Hurwitz, and how are discrete-time feedback loops analysed?	Chapter 34
How are digital controllers designed and the hold delay accounted for?	Chapter 35
What is the rigorous theory of the Z-transform and its inversion contour integral?	*The Mathematics Universe*

Next: Chapter 21 opens Part V by asking what happens when these exact sequences and system functions are carried out by real processors with finite word lengths and limited arithmetic.

Part V — Digital Signal Processing

Part IV ended with sampled signals described exactly: sequences, convolution sums, difference equations and the Z-transform. Part V turns that mathematics into working processing. A hearing aid that removes feedback whistle, a protection relay that measures fault impedance within one mains cycle, a radar that pulls a weak echo out of clutter, an ECG monitor that suppresses hum and baseline wander, a software-defined radio that demodulates a channel from a stream of complex samples — all of them read numbers from a converter, compute, and write numbers out, on time, with finite word lengths. This part asks how that is done well. It treats the processor as a signal-processing system in its own right: how signals and systems are represented in a machine and how finite precision changes them; how a finite block of samples is turned into a spectrum; how filters are specified, designed and implemented; and how sample rates are changed, filters made to adapt, and signals analysed in time and frequency together.

The emphasis throughout is on design that can be verified with numbers. Every method is taken from a specification (passband ripple, stopband attenuation, frequency resolution, SNR, latency, operations per second) to an implementation whose performance can be predicted before it is built and measured after.

By the end of this part you should understand:

- how sampling and quantization, fixed-point and floating-point arithmetic, and the real-time processing chain set the accuracy and cost of any DSP system, and how numerical error is predicted and budgeted;
- what the discrete Fourier transform computes, why the FFT makes it cheap, and how leakage, windowing and zero padding affect what a measured spectrum really shows;
- how FIR and IIR digital filters are specified, designed and implemented, and how their stability and finite-precision behaviour are ensured;
- how multirate processing changes sample rates efficiently, how adaptive filters learn their coefficients, and how spectral, time–frequency and wavelet methods analyse signals whose content changes with time;
- how these tools combine in practical audio, image, biomedical, power-system, radar and communication processing.

Chapter	Central question	What it delivers
Chapter 21 — DSP Fundamentals	What happens when the exact sequences and systems of Part IV are carried out by real converters and processors?	Discrete-time representations, quantization and SQNR, the DSP workflow and real-time budgets, fixed- and floating-point arithmetic, finite-precision effects and numerical error budgets
Chapter 22 — Discrete Fourier Transform	How is the spectrum of a finite record of samples computed and interpreted?	DFT and inverse DFT, spectral interpretation, FFT algorithms, leakage, windowing, zero padding and basic spectral estimation
Chapter 23 — Digital Filters	How is a digital filter designed to meet a specification and implemented reliably?	FIR and IIR filters, difference-equation structures, specifications, frequency transformations, FIR and IIR design methods, stability, quantization effects and implementation constraints
Chapter 24 — Advanced Signal Processing	How are sample rates changed, filters made adaptive, and non-stationary signals analysed?	Multirate processing, decimation and interpolation, Wiener and LMS adaptive filtering, spectral estimation, time–frequency analysis, wavelets, and fundamentals of image and audio processing

Part V sits between the signal theory that precedes it and the engineering destinations that follow. It depends directly on Chapter 19 (sampling, aliasing and the quantization interface) and Chapter 20 (convolution, the Z-transform, poles, zeros and stability), and it reuses the filter vocabulary of Chapter 13 and the frequency-domain ideas of Chapter 18. In return, it supplies the working tools of the rest of the book. Communication systems in Part VI are now largely digital: pulse-code modulation in Chapter 27 is quantization applied to a link, matched filters and equalisers are digital filters, OFDM in Chapter 30 is built on the FFT, and the noise analysis of Chapter 29 relies on the spectral-estimation ideas introduced here. Control systems in Part VII are implemented as difference equations running on processors, and the digital control fundamentals of Chapter 35 inherit the sampling delays, word-length limits and stability checks of this part. The mathematics of the Fourier and Z-transforms and of random processes is recapped where it is used and developed fully in *The Mathematics Universe*; converters, processors and their circuits belong to *Electronics & Computing*.

Chapter 21 — DSP Fundamentals

Part V · Digital Signal Processing

A team ports a vibration-monitoring algorithm from a laptop to a small fixed-point processor mounted on a pump. On the laptop, written in double-precision floating point, the algorithm tracks a 5 Hz structural resonance perfectly. On the target, with 16-bit coefficients, the resonance tracker drifts to the wrong frequency, then locks up at a constant output and never recovers. Elsewhere in the same plant, a data logger built on a different board timestamps its samples by adding $1/48\,000$ s to a single-precision floating-point clock at each sample; after eight minutes its clock stops advancing altogether. A third engineer computes the variance of a high-voltage test supply's ripple, a few millivolts riding on ten kilovolts, and the code reports a negative number. None of these failures is a bug in the usual sense. Each program does exactly what the mathematics of Chapter 20 says — but with numbers that have finite length.

Part IV treated sequences and systems as exact mathematical objects. Part V turns them into working processing: programs and hardware that read samples from a converter, compute, and write samples out, on time, every time. This chapter sets up the vocabulary for the whole part. It describes how discrete-time signals are represented in a machine, how sampling and quantization turn an analogue quantity into numbers, how a DSP system is organised and designed, and how finite precision — in the converter, in the coefficients and in every arithmetic operation — changes the ideal answer.

> **How is a continuous-world signal turned into numbers a processor can work on, and what is the processing chain around it?**
> **What do finite word length, fixed-point and floating-point arithmetic do to the results, and how is numerical error predicted and kept under control?**

Concept	Engineering destination
Discrete-time representations	Sample buffers, frames, I/Q streams, file formats for audio, ECG and fault records
Sampling and quantization	Choosing ADC resolution and rate, headroom, dither, oversampling converters
DSP workflow	Real-time budgets on microcontrollers, DSP chips and FPGAs; design-to-deployment flow
Finite-precision effects	Filter noise floors, limit cycles, overflow, coefficient sensitivity
Fixed-point arithmetic	Low-power audio codecs, hearing aids, motor drives, protection relays, FPGA filters
Floating-point arithmetic	Audio workstations, radar and sonar processing, scientific instruments
Numerical error in DSP	Error budgets, scaling, structure choice, robust accumulators and statistics

Roadmap. How signals and systems are represented in a machine (21.1) → sampling and quantization as one interface (21.2) → the end-to-end processing chain and design flow (21.3) → the catalogue of finite-precision effects (21.4) → fixed-point and floating-point arithmetic (21.5) → predicting and controlling numerical error (21.6).

21.1 Discrete-time representations

A sequence in memory

In Chapter 20 a sequence $x[n]$ was defined for all integers n. In a machine it is a finite array of numbers, each held in a fixed number of bits, together with the information needed to interpret them. A complete discrete-time representation therefore has several parts, all of which must travel with the data:

- **Sample values**, in a stated number format (Section 21.5): 16-bit integers, 24-bit integers packed into 32 bits, 32-bit floats.
- **Sample rate** f_s, which turns the index n into time nT and the digital frequency Ω into hertz.
- **Scale**, the physical quantity represented by full scale or by one count: volts, amperes, pascals, millivolts of ECG, g of acceleration.
- **Time reference**: the absolute time of a chosen sample, essential when signals from different recorders must be aligned (fault records from both ends of a transmission line, EEG and video).
- **Channel layout**: how several channels are stored — interleaved (L, R, L, R, … in audio files) or in separate blocks.

Table 21.1 shows how real systems package these parts.

Table 21.1 — Representative discrete-time data representations.

Domain	Typical rate and format	Scale and metadata
Consumer audio (WAV, CD)	44.1 or 48 kSa/s, 16-bit signed integer, interleaved stereo	Full scale = 0 dBFS; rate and channel count in file header
Studio audio	48–192 kSa/s, 24-bit integer or 32-bit float	Float: ± 1.0 = full scale, values above allowed internally
Diagnostic ECG	500–1000 Sa/s, 12–16 bits per lead, 12 leads	Gain in µV per count, lead names, patient time stamp
Power-system fault records (COMTRADE, IEEE C37.111)	1–20 kSa/s, 16- or 32-bit integers	Per-channel multiplier and offset to primary amperes or volts; trigger time
Software-defined radio	1–100 MSa/s, complex I/Q pairs of 12–16-bit integers	Centre frequency, gain, rate
Seismic (miniSEED)	1–200 Sa/s, 24-bit or 32-bit integer, compressed	Instrument response, station and channel code, UTC time
Machine-vibration monitoring	25.6 kSa/s, 24-bit	Sensitivity in mV per g, shaft speed reference

Real and complex sequences

Most measured signals are real, but much of DSP works with **complex** sequences. A radio receiver mixes its input with a cosine and a sine to produce in-phase and quadrature components, $x[n] = x_I[n] + jx_Q[n]$, a complex baseband signal whose spectrum need not be symmetric about zero (Chapters 26 and 27). Complex data make positive and negative frequencies distinguishable, so a complex sampler at rate f_s covers a band of width f_s rather than $f_s/2$: the LTE example in Chapter 19's Table 19.1 samples a 20 MHz channel with 30.72 MSa/s complex. Each complex sample costs two real numbers, and a complex multiplication costs four real multiplications and two additions (or three and five with a rearrangement).

Sample-by-sample and block processing

Real-time systems process data in one of two ways.

- **Sample-by-sample**: each new sample is processed immediately and an output sample produced before the next input arrives. Latency is one sample plus the converters' delay. Digital control loops, protection relays and active noise cancellation work this way.
- **Block (frame) processing**: samples are collected into blocks of L samples, typically using two alternating buffers (ping-pong buffering) filled by direct memory access while the processor works on the other. Overheads are paid once per block, and block algorithms such as the FFT (Chapter 22) become available. The price is latency: at least one block, and in practice about two. With $L = 512$ at 48 kSa/s, one block is 10.7 ms.

Representing systems

A discrete-time system is represented in a machine by three things: its **coefficients** (the b_k and a_k of Chapter 20, stored in some number format), its **structure** (the order of operations, Section 20.5) and its **state** (the contents of the delay elements). The mathematics of Chapter 20 is indifferent to structure, but finite precision is not: two structures with the same $H(z)$ can behave very differently once coefficients and state are rounded, which is the subject of Sections 21.4–21.6. A signal-flow graph or block diagram is the standard drawing of a structure; it is also, nearly literally, the data-flow graph that an FPGA synthesis tool turns into hardware.

21.2 Sampling and quantization

Chapter 19 treated sampling and introduced quantization at its interface. Here the converter is viewed from the processor's side: what the numbers mean, how good they are and how to get the most out of a given word length.

The uniform quantizer

A uniform quantizer with N bits maps an input x in a range of width V_{FS} to one of 2^N levels spaced $\Delta = V_{FS}/2^N$ apart. In a **mid-tread** quantizer zero is an output level (inputs within $\pm\Delta/2$ of zero give exactly zero); in a **mid-rise** quantizer zero is a decision threshold and the levels sit at

$\pm\Delta/2, \pm 3\Delta/2, \ldots$ Most ADCs are mid-tread with two's-complement output codes from -2^{N-1} to $2^{N-1}-1$. Rounding to the nearest level gives an error $e = Q(x) - x$ in $[-\Delta/2, \Delta/2]$ inside the range. Outside the range the output **clips** at the end levels and the error grows without limit — **overload**. Figure 21.1 shows the staircase and both kinds of error.

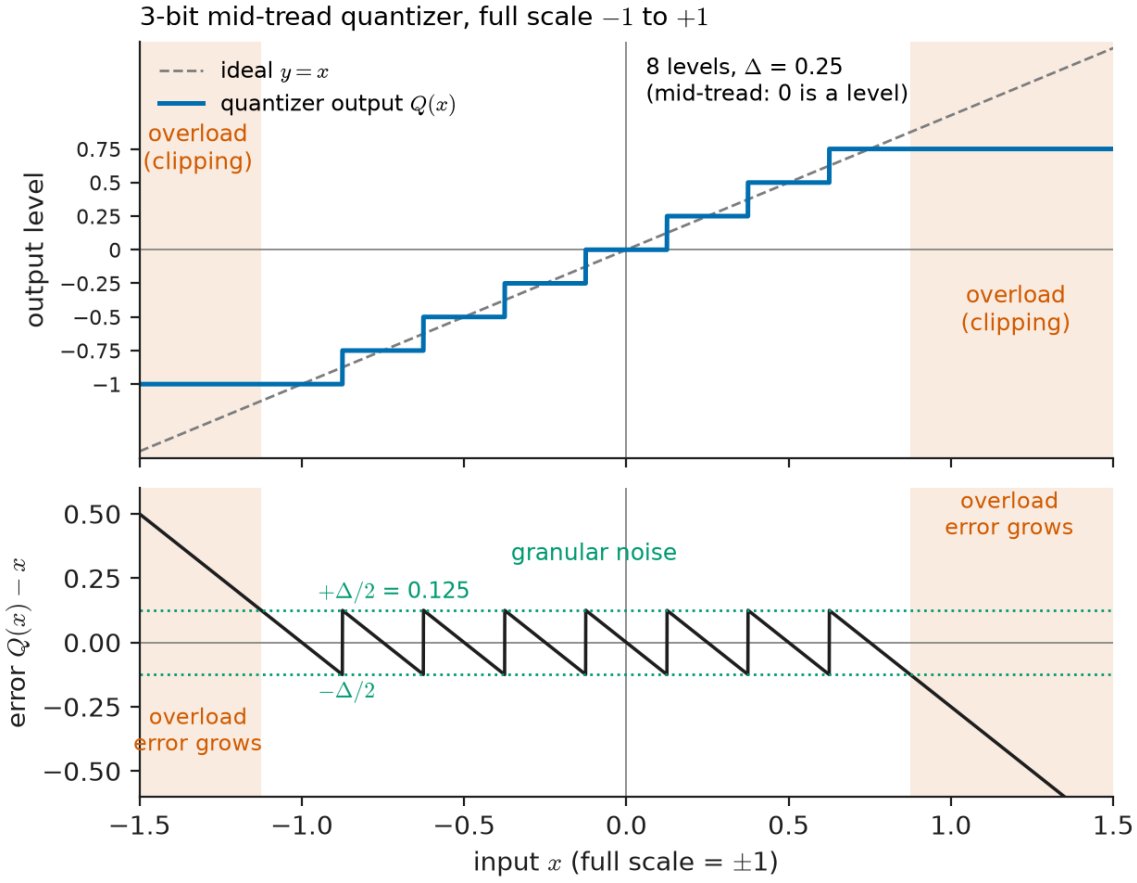

Figure 21.1 — Transfer characteristic and error of a 3-bit mid-tread quantizer, showing granular and overload regions.

The quantization-noise model and SQNR

The additive-noise model of Chapter 19 treats the granular error as white noise, uniform on $[-\Delta/2, \Delta/2]$, uncorrelated with the signal, with variance

$$\sigma_e^2 = \frac{1}{\Delta}\int_{-\Delta/2}^{\Delta/2} e^2\, de = \frac{\Delta^2}{12}$$

For a sine of amplitude A, the signal power is $A^2/2$. With A at full scale, $A = V_{FS}/2 = 2^{N-1}\Delta$, the signal-to-quantization-noise ratio is

$$\mathrm{SQNR} = 10\log_{10}\frac{A^2/2}{\Delta^2/12} = 10\log_{10}\left(\frac{3}{2}\,2^{2N}\right) = 6.02N + 1.76 \text{ dB}$$

and for a sine L dB below full scale (level $-L$ dBFS) it is $6.02N + 1.76 - L$ dB. Every bit is worth about 6 dB, every halving of signal level costs 6 dB.

> **Status of the principle: SQNR $= 6.02N + 1.76$ dB**
> *Approximate.* Exact consequence of the uniform-noise model for a full-scale sine; the model holds for busy signals spanning many levels and not synchronised with the sampling clock, and fails for small, slow or periodic signals and in overload.

Worked example — Audio word length and level. A 16-bit audio channel carrying a full-scale sine has SQNR $= 6.02 \times 16 + 1.76 = 98.1$ dB. A simulation of one million samples of a near-full-scale sine (amplitude 0.999 of full scale, at a frequency not related to f_s) rounded to 16 bits gives 98.09 dB — the model is excellent here. But music is rarely at full scale. A passage at -20 dBFS has SQNR 78.1 dB, and a quiet passage at -60 dBFS only 38.1 dB, by which point the error is no longer noise-like (see dither, below). Studio recording at 24 bits is therefore not about the final listener's dynamic range but about **headroom**: recording at -20 dBFS to avoid any risk of clipping still leaves a theoretical 126 dB, far below the converter's analogue noise floor.

Loading, crest factor and headroom

Signals other than sines have different ratios of peak to RMS value, the **crest factor** $C = x_{peak}/x_{rms}$. A sine has $C = \sqrt{2}$ (3.01 dB); speech and music have 12–20 dB; Gaussian noise has no strict peak, but values beyond 4σ occur with probability 6×10^{-5}, so $C \approx 4$ (12.0 dB) is a common loading choice; an OFDM radio signal (Chapter 30) behaves like Gaussian noise. If the quantizer range is set to $\pm C\sigma_x$, then $\Delta = 2C\sigma_x/2^N$ and

$$\mathrm{SQNR} = 6.02N + 4.77 - 20\log_{10} C \ \mathrm{dB}$$

Worked example — Loading a 16-bit converter with noise-like signals. For a Gaussian signal with $C = 4$ the formula gives SQNR $= 6.02 \times 16 + 4.77 - 12.04 = 89.0$ dB — but the formula counts granular noise only. A Gaussian signal exceeds 4σ about 0.006 % of the time, and each clipped sample carries an error of many LSBs; at 16 bits that overload noise dominates, and a simulation including it gives only 52 dB. Lower loading (larger C) reduces overload but raises granular noise; higher loading does the reverse. The optimum balances the two and moves with N: for 8 bits it lies near $C = 4$ (40.6 dB, about 9 dB below the sine figure), for 16 bits near $C = 6$ (85.5 dB, about 12.6 dB below the full-scale sine figure of 98.1 dB). The more bits a converter has, the more headroom a noise-like signal needs. Figure 21.2 shows SQNR against input level: a 6 dB-per-bit straight line in the granular region, a peak, and a cliff when overload sets in. Running a converter "hot" to gain a few decibels is a bad bargain, because clipping produces distortion spread across the whole band and, in a radio, spectral regrowth into neighbouring channels.

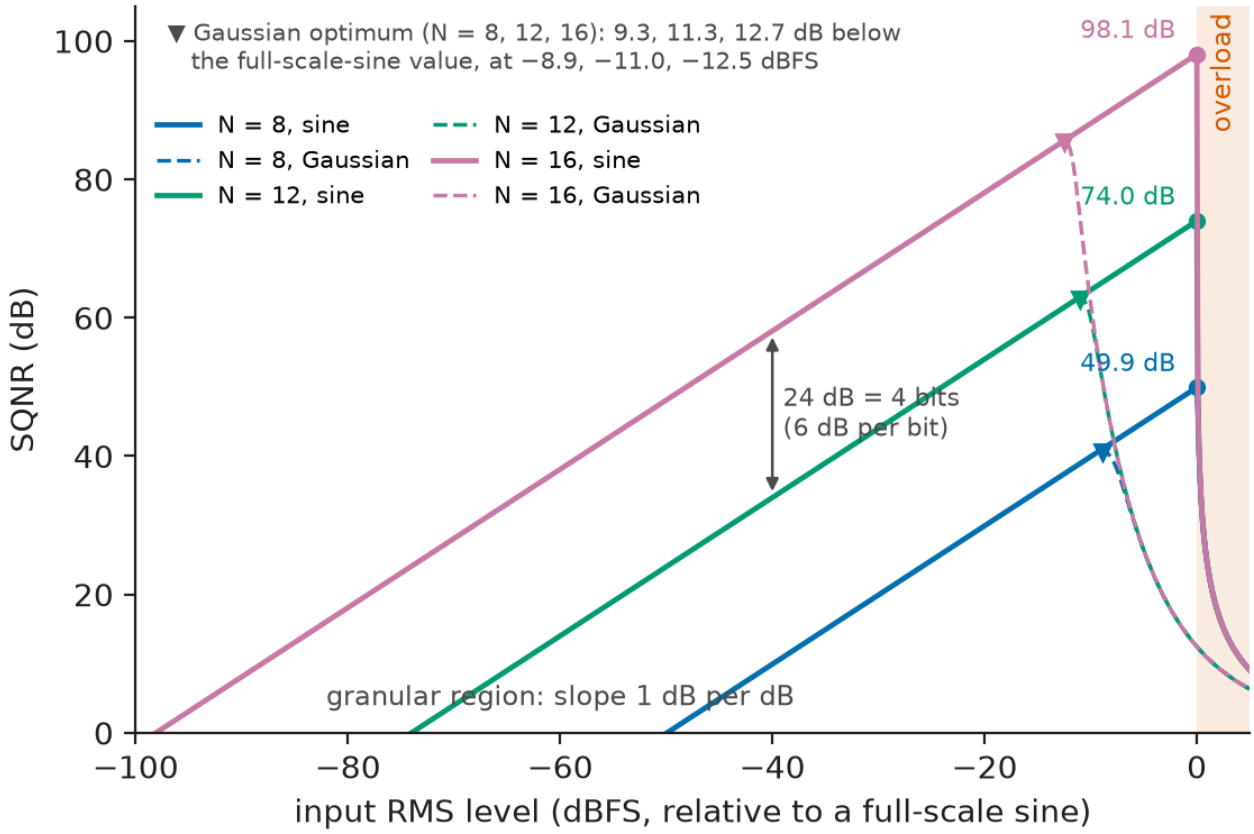

Figure 21.2 — SQNR versus input level for 8-, 12- and 16-bit uniform quantizers with sine and Gaussian inputs.

Dither

The uniform-noise model fails for small signals. A sine of amplitude 1 LSB is quantized to a sequence taking only the values -1, 0 and $+1$: the "error" is a deterministic, periodic waveform — harmonic distortion correlated with the signal, audible in the fade-out of a piano note as a gritty buzz. A 1 kHz tone at -90 dBFS in a 16-bit system has an amplitude of $32768 \times 10^{-90/20} = 1.04$ LSB, exactly in this regime. Even worse, a signal smaller than $\Delta/2$ vanishes entirely.

Dither is a small random noise added *before* quantization. With **triangular-PDF** (TPDF) dither of ± 1 LSB peak, the mean and variance of the total error become independent of the signal: distortion is replaced by a steady, signal-independent hiss, and signals far below 1 LSB remain audible or measurable after averaging. The cost is an increase in noise power from $\Delta^2/12$ to $3\Delta^2/12$, a 4.77 dB rise in the noise floor. Dither is standard when audio is reduced from 24 to 16 bits for distribution, and in measurement instruments that average many readings to resolve below one LSB.

> **Status of the principle: Dither linearises the quantizer**
> *Derived*. With TPDF dither of 2 LSB peak-to-peak, the first two moments of the total error are provably independent of the input; the noise-floor cost of 4.77 dB is exact for this dither.

Oversampling and noise shaping

Quantization noise is spread uniformly over 0 to $f_s/2$. If the signal occupies only 0 to B, a digital low-pass filter after the converter removes the noise between B and $f_s/2$, and the in-band noise falls by the **oversampling ratio** $\mathrm{OSR} = f_s/(2B)$:

$$\mathrm{SQNR} = 6.02N + 1.76 + 10\log_{10}\mathrm{OSR}\ \mathrm{dB}$$

Worked example — Trading rate for resolution. A 12-bit ADC running at 1 MSa/s digitises a sensor whose useful bandwidth is 10 kHz. $\mathrm{OSR} = 10^6/(2\times10^4) = 50$, worth $10\log_{10}50 = 17.0$ dB, or $17.0/6.02 = 2.8$ extra bits: the theoretical SQNR rises from 74.0 dB to 91.0 dB after filtering and decimating to, say, 25 kSa/s. The gain is real only if the quantization error is truly noise-like (the analogue noise of the sensor or added dither usually ensures this) and if the decimation filter is long enough. Plain oversampling is expensive: each extra bit costs a factor of 4 in rate.

Noise shaping does much better. A sigma–delta modulator places the quantizer inside a feedback loop that pushes quantization noise towards high frequencies, where the decimation filter removes it. An L-th-order loop improves SQNR by about $(2L+1)\times 3$ dB per doubling of OSR, rather than 3 dB. This is how a 1-bit or few-bit quantizer running at a few megahertz delivers 24-bit audio, and why modern audio, precision measurement and many radio converters are sigma–delta designs.

> **Boundary.**
> The circuit design of ADCs, DACs and sigma–delta modulators belongs to *Electronics & Computing*. This chapter uses only their input–output behaviour: sample rate, resolution, noise and delay.

Table 21.2 — Typical converter resolutions and their use.

Application	Resolution (bits)	Sample rate	Ideal SQNR (dB)	Notes
Video, fast oscilloscopes	8	0.1–10 GSa/s	49.9	ENOB often 6–7 at high frequency
Microcontroller ADC, motor-drive current sensing	10–12	0.1–5 MSa/s	62.0–74.0	Synchronised to PWM
Protection relays, energy meters	16–24	1–20 kSa/s	98–146	Sigma–delta common in meters
Software-defined radio	12–16	10–250 MSa/s	74–98	Jitter-limited at high IF
CD and consumer audio	16	44.1–48 kSa/s	98.1	Distribution format
Studio audio, ECG/EEG, seismic, vibration	24	0.1–192 kSa/s	146.2 (theory)	Real ENOB 18–21 bits, analogue-noise limited

21.3 The digital signal processing workflow

The processing chain

Almost every DSP system has the same skeleton (Figure 21.3):

sensor → conditioning → anti-aliasing filter → S/H + ADC → processor → DAC → reconstruction fi

The **sensor** converts a physical quantity to an electrical one: a microphone, a current transformer, an ECG electrode, an antenna, an accelerometer. **Conditioning** amplifies, level-shifts and isolates it to fit the converter's range — the scaling that determines how much of the converter's resolution is actually used. The **anti-aliasing filter** and **ADC** are Chapter 19's subject. The **processor** runs the algorithm: filtering (Chapter 23), spectral analysis (Chapter 22), detection, estimation, adaptive processing (Chapter 24), demodulation (Part VI) or control (Part VII). The output path may lead back to the analogue world through a **DAC** and a reconstruction filter — a loudspeaker, a motor drive, a transmitter — or may end in a decision, a display or a stored record, with no analogue output at all: a relay that trips a breaker, a heart-rate number on a monitor, a packet passed up a protocol stack.

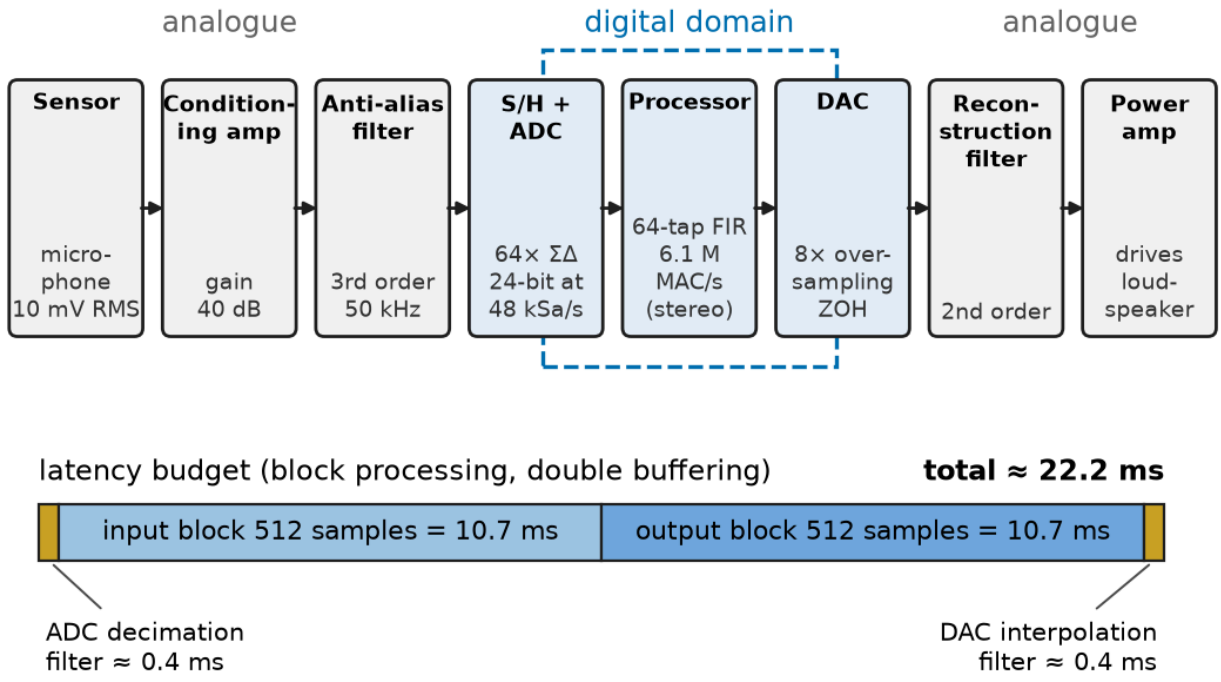

Figure 21.3 — The generic DSP processing chain with representative values for a 48 kSa/s audio system.

Real-time constraints

A real-time DSP system must finish processing each sample, or each block, before the next one arrives. Two numbers govern every design.

Throughput. At a sample rate f_s the processor has $1/f_s$ seconds per sample: 20.8 µs at 48 kSa/s, which on a 200 MHz processor is about 4170 clock cycles. The workload is usually counted in

multiply–accumulate operations (MACs), because filtering is dominated by $y = \sum h_k x_{n-k}$ and DSP processors are built to perform one or more MACs per cycle.

Latency. The delay from an input event to the corresponding output. It accumulates from the converters' internal filters, from buffering, and from the algorithm's own group delay (an FIR filter of length M with linear phase adds $(M-1)/2$ samples). Tolerances vary enormously: a hearing aid must keep total latency below about 10 ms or the wearer hears an echo of their own voice; active noise cancellation needs tens of microseconds; a protection relay must decide within one or two mains cycles (20–40 ms at 50 Hz), including the time its filters need to settle; an off-line seismic analysis has no latency constraint at all.

Worked example — A processing budget. A stereo audio crossover runs a 64-tap FIR filter on each channel at 48 kSa/s: $64 \times 48\,000 \times 2 = 6.14$ million MACs per second. A protection relay applies a 101-tap FIR to six channels — three currents and three voltages — at 10 kSa/s: $101 \times 10\,000 \times 6 = 6.06$ million MACs per second. Both fit comfortably on a 200 MHz processor capable of one MAC per cycle, using about 3 % of its capacity, leaving room for everything else the device must do. A 4096-tap room-correction filter on eight channels at 96 kSa/s needs 3.1×10^9 MACs per second by direct convolution and pushes the design towards FFT-based fast convolution (Chapter 22) or an FPGA. In block processing with $L = 512$, the audio path pays $512/48\,000 = 10.7$ ms per block of buffering; with double buffering about 21 ms — fine for a crossover in a home system, too long for a hearing aid, which uses blocks of 16–64 samples instead.

Implementation platforms

Platform	Typical arithmetic	Strengths	Typical applications
General microcontroller (with DSP instructions)	16/32-bit fixed point, often single-precision float	Low cost, low power, integrated peripherals	Motor control, sensor nodes, energy meters
Dedicated DSP processor	16/32-bit fixed point or 32-bit float, multiple MACs per cycle	Deterministic timing, efficient filtering	Audio codecs, hearing aids, telecom, relays
FPGA	Arbitrary fixed-point word lengths	Massive parallelism, very low latency	Radar, SDR front ends, high-speed test equipment
Application processor or PC	32/64-bit float, vector units	Easy development, large memory	Audio workstations, off-line analysis, instrumentation
GPU	32-bit (and 16-bit) float, thousands of cores	Very high throughput on large blocks	Radio astronomy, medical imaging, sonar beamforming

> **Boundary.**
> Processor architectures, memory hierarchies and embedded-software practice belong to *Electronics & Computing*. Here only their numerical consequences matter.

The design flow

A disciplined DSP design moves from mathematics to hardware in stages, each verified against the previous one.

1. **Specification.** What must the system achieve, measured how? Passband ripple, stopband attenuation, SNR, latency, update rate, power, cost. A specification without measurable numbers cannot be verified.
2. **Algorithm design in high precision.** Design and simulate in double-precision floating point (Python, MATLAB, Julia) using recorded or synthetic test signals. This establishes what the algorithm can achieve when arithmetic is not a concern.
3. **Finite-precision model.** Choose number formats, scaling, structure and word lengths, and simulate them **bit-exactly**: the model must reproduce every rounding, saturation and overflow of the target. Compare against the double-precision reference; the difference is the implementation error, and it must fit inside the specification's error budget.
4. **Implementation** in C, assembly or a hardware description language, often generated automatically from the bit-exact model.
5. **Verification** with test vectors: the target's output, sample for sample, must equal the bit-exact model's output, and the system's measured performance (with real converters and real signals) must meet the specification.

Most of the failures in the opening paragraph come from skipping step 3: an algorithm that was correct in double precision was implemented with the target's arithmetic without first predicting what that arithmetic would do.

21.4 Finite-precision effects

A digital system with infinite-precision numbers is exactly the LTI system of Chapter 20. Real systems differ from that ideal in four distinct ways, which are analysed separately because they behave differently and are cured differently.

1. Input quantization

The ADC's quantization error (Section 21.2) is noise added at the input. It is filtered by the system along with the signal, so its effect at the output depends on the system's frequency response over the noise band. It sets the floor below which no amount of processing can see — except by averaging or by filtering out-of-band noise, as oversampling does.

2. Coefficient quantization

The coefficients b_k, a_k are stored in finite precision, so the implemented system is not the designed one: its zeros and poles are slightly moved. This is a *deterministic* error — the filter is a different but still linear, time-invariant filter — and can be evaluated exactly by computing the response with the rounded coefficients. For FIR filters the effect is usually benign: stopband attenuation is limited to roughly $6B$ dB for B-bit coefficients, and linear phase is preserved if the rounded coefficients remain symmetric.

For IIR filters it can be severe: poles near the unit circle, which narrowband and low-frequency filters always have, move a long way for small coefficient changes, and can cross the circle (Section 21.6).

3. Arithmetic roundoff

The product of two B-bit numbers has $2B$ bits. Storing it back into B bits requires rounding or truncation, which injects a small error at every multiplication (or at every accumulator store). Under conditions similar to those for ADC quantization, each rounding point behaves like a white-noise source of variance $\Delta^2/12$, and the noise it produces at the output depends on the transfer function from that point to the output. In recursive filters, where rounded values are fed back, this noise is amplified by the feedback loop (Section 21.6).

4. Overflow

A fixed-point register can hold only a limited range. A sum that exceeds it either **wraps around** (in two's complement, a large positive value becomes a large negative one) or **saturates** at the largest representable value, if saturation logic is used. Either is a large nonlinear error. It is prevented by **scaling** — dividing the input or the coefficients so that no internal node can exceed its range for any allowed input — at the cost of a lower signal level relative to roundoff noise. Scaling is thus a trade-off between overflow and noise, exactly the loading trade-off of the ADC in Section 21.2.

Limit cycles

Rounding inside a feedback loop is a nonlinearity, and nonlinear feedback systems can oscillate on their own. A recursive filter with zero input should decay to zero; with rounding it may instead settle into a constant or a small periodic output, a **zero-input limit cycle**.

Worked example — A dead band. The first-order filter $y[n] = Q\{0.9\,y[n-1]\}$, with Q rounding to the nearest integer (in LSB units) and starting from $y = 10$, produces $10, 9, 8, 7, 6, 5, 5, 5, \ldots$ because $0.9 \times 6 = 5.4$ rounds to 5, and then $0.9 \times 5 = 4.5$ rounds back to 5. The output sticks at 5 LSB forever instead of decaying. With the coefficient -0.9 it alternates $+5, -5, +5, \ldots$ — a limit cycle at the Nyquist frequency. In general a first-order loop with coefficient a cannot decay below the **dead band** $|y| \le 0.5/(1-|a|)$ LSB, here exactly 5. For $a = 0.99$ the dead band is 50 LSB; for $a = 0.999$, 500 LSB. In audio, a low-level limit cycle in a reverberation filter is an audible idle tone; in a controller, a small persistent oscillation (hunting) of the actuator.

> **Status of the principle: Dead-band bound for first-order limit cycles**
> *Derived.* Follows from requiring that rounding can hold y constant, $|y - |a|y| \le 1/2$; exact for rounding to nearest in a first-order loop. Higher-order loops have more complicated limit-cycle behaviour, and overflow can cause far larger oscillations.

Overflow inside a feedback loop is worse: with wrap-around arithmetic, a second-order filter can enter a full-scale **overflow oscillation** that never ends. Saturation arithmetic prevents this in second-order sections, which is one reason DSP processors provide it in hardware.

The practical remedies for limit cycles are wider internal state (more bits behind the binary point, which shrinks the dead band in signal units), magnitude truncation instead of rounding in some structures, small amounts of dither, and structures that are proven free of limit cycles.

21.5 Fixed-point and floating-point arithmetic

Two's-complement integers and fractional formats

Nearly all fixed-point DSP uses **two's complement**. A B-bit word represents integers from -2^{B-1} to $2^{B-1}-1$; the most significant bit has weight -2^{B-1} and the others positive powers of two. Addition and subtraction work identically for positive and negative numbers, and an intermediate overflow does no harm if the final result is in range — a useful property in long sums.

DSP usually interprets the integer as a **fraction** by placing an implied binary point. In the **Q**m**.**n notation, m bits are integer bits (excluding the sign, in the convention used here) and n are fractional; the value is the integer divided by 2^n. The most common format is **Q15** (also written Q0.15 or Q1.15): a 16-bit word with 15 fractional bits, representing values in $[-1,\ 1-2^{-15}] = [-1,\ 0.999969]$ with resolution $2^{-15} = 3.05\times10^{-5}$. Q31 does the same in 32 bits with resolution 4.66×10^{-10}. Fractions are used because the product of two numbers less than 1 in magnitude is also less than 1 — multiplication cannot overflow, except for the single case $(-1)\times(-1)$. Figure 21.4 shows the bit layouts.

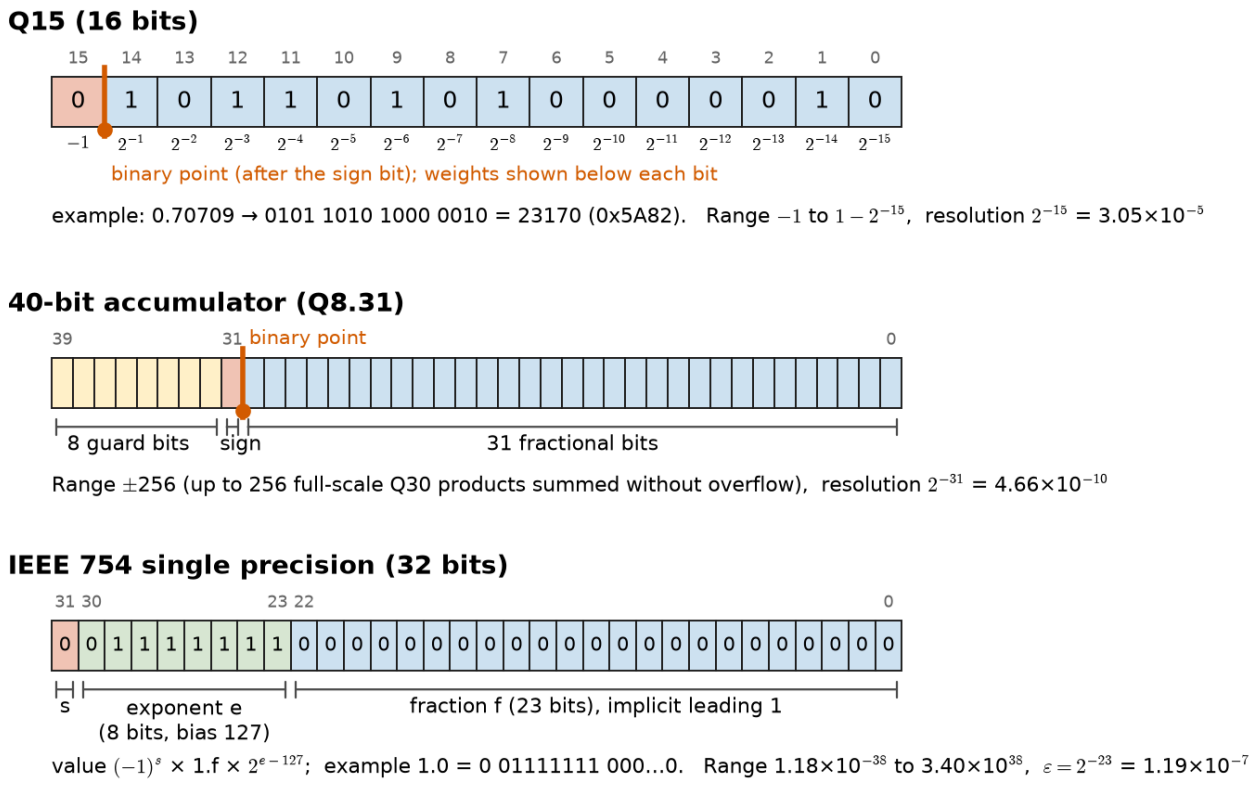

Figure 21.4 — Bit layouts of Q15 fixed-point, a 40-bit accumulator and IEEE 754 single-precision floating point.

Worked example — Q15 coefficients and products. The coefficient 0.7071 ($1/\sqrt{2}$, common in filters and FFT twiddle factors) becomes $\operatorname{round}(0.7071 \times 32768) = 23170$, which represents $23170/32768 = 0.707092$, an error of -7.7×10^{-6}, within the maximum rounding error $2^{-16} = 1.53 \times 10^{-5}$. Multiplying the Q15 numbers 0.75 (24576) and -0.5 (-16384) gives the integer $-402\,653\,184$, which is a **Q30** number: $-402\,653\,184/2^{30} = -0.375$. To store it back as Q15, shift right by 15 bits (with rounding): -12288, i.e. -0.375. Now add 0.75 and 0.5 in Q15: $24576 + 16384 = 40960$, beyond the largest Q15 value 32767. With wrap-around this becomes $40960 - 65536 = -24576$, or -0.75 — a full-scale sign error. With saturation it becomes 32767, or 0.99997, an error of 0.25 but with the correct sign. Saturation is almost always the right choice for signals; wrap-around is the right choice for phase accumulators and modular counters.

Accumulators and guard bits

A FIR filter computes $\sum_{k=0}^{M-1} h_k x_{n-k}$. If each product is kept to full Q30 precision and added in a wide **accumulator**, there is only one rounding — at the end — rather than one per product. DSP processors provide accumulators of 40 bits or more for 16-bit data: 32 bits for the full product plus 8 **guard bits** above the sign, allowing up to $2^8 = 256$ worst-case full-scale products to be summed without overflow. In general summing M terms needs $\lceil \log_2 M \rceil$ guard bits. With the accumulator protected, intermediate overflow is impossible and only the final result must be scaled and saturated. This single design feature — double-width products and guard bits — is why fixed-point FIR filters can achieve noise floors close to the ideal set by their output word length.

Rounding modes

Truncation (dropping the low bits, i.e. rounding towards $-\infty$ in two's complement) is cheap but biased: its error lies in $(-\Delta, 0]$ with mean $-\Delta/2$. A DC offset of half an LSB at every multiplication can accumulate in a recursive filter to a significant offset. **Round-to-nearest** has zero mean in the noise model. **Convergent rounding** (round half to even) removes the slight bias of ordinary rounding in the exact-half case. Most processors offer rounding as an option in the store instruction; the bit-exact model must use the same mode.

Floating point

A **floating-point** number stores a sign s, an exponent e and a significand (mantissa) m, representing $(-1)^s \times 1.m \times 2^{e - \text{bias}}$. The IEEE 754 standard defines the formats used everywhere:

Table 21.3 — Number formats used in DSP.

Format	Bits (sign/exponent/fraction)	Smallest step relative to value	Range (normal numbers)	Approximate dynamic range	Full-scale-sine SQNR equivalent
Q15 fixed point	16 total	absolute step 3.05×10^{-5}	-1 to 0.99997	90 dB	98.1 dB at full scale, falling 1 dB per dB of level
Q31 fixed point	32 total	absolute step 4.66×10^{-10}	-1 to ≈ 1	187 dB	194.4 dB at full scale
IEEE half (float16)	1/5/10	$\varepsilon = 9.77 \times 10^{-4}$	6.1×10^{-5} to 65 504	181 dB	about 68 dB at any level
IEEE single (float32)	1/8/23	$\varepsilon = 1.19 \times 10^{-7}$	1.18×10^{-38} to 3.40×10^{38}	1529 dB	about 146 dB at any level
IEEE double (float64)	1/11/52	$\varepsilon = 2.22 \times 10^{-16}$	2.2×10^{-308} to 1.8×10^{308}	12 318 dB	about 320 dB at any level

The key difference from fixed point is that the error of floating point is *relative*. Rounding a number x to the nearest representable value gives $\mathrm{fl}(x) = x(1+\delta)$ with $|\delta| \le \varepsilon/2$, where ε (**machine epsilon**) is the spacing between 1 and the next larger number: 2^{-23} for single and 2^{-52} for double precision. The signal-to-roundoff ratio of a single operation is therefore roughly constant regardless of signal level — about 146 dB for single precision (a 24-bit significand) — instead of falling 1 dB for every decibel of level as in fixed point. Overflow practically never happens, so scaling largely disappears as a design concern.

> **Status of the principle: Relative error model of floating point**
> *Derived.* IEEE 754 guarantees that each basic operation is correctly rounded, so $\mathrm{fl}(a \circ b) = (a \circ b)(1+\delta)$ with $|\delta| \le \varepsilon/2$ for normal results; the bound fails for subnormal numbers near underflow and says nothing about how errors combine across many operations.

Floating point is not free of problems. The relative error is constant, but the *absolute* error grows with the size of the number, so adding a small number to a large one loses the small number's low bits — or all of it (Section 21.6). **Subnormal** (denormal) numbers near underflow are slow on many processors: a recursive audio filter whose state decays towards zero after the music stops can drop into subnormals and multiply its CPU load several times, which is why audio software often flushes subnormals to zero or adds a tiny offset. And floating-point results depend on the order of operations, because addition is not associative; two correct implementations of the same filter can differ in their last bits, which matters when bit-exact reproducibility is required.

Block floating point and choosing a format

Between the two lies **block floating point**: a block of fixed-point values shares one exponent, chosen so that the largest value in the block nearly fills the word. FFT processors (Chapter 22) often rescale between stages this way, gaining most of floating point's dynamic range at fixed-point cost.

The choice of format is an engineering one. Fixed point uses less silicon area and power and gives deterministic timing; it dominates in hearing aids, battery-powered sensors, high-volume consumer codecs and FPGAs. Floating point shortens development, removes scaling, and dominates in audio workstations, instrumentation and radar processing on general processors. Many products mix them: a fixed-point FPGA front end at the full sample rate feeding a floating-point processor at a decimated rate.

21.6 Numerical error in DSP

The previous sections listed the sources of error. This section predicts their size at the output of a system and shows the most common ways error accumulates.

Noise gain of a roundoff source

If a white roundoff noise $e[n]$ of variance σ_e^2 is injected at some node, and $g[n]$ is the impulse response from that node to the output, the output noise variance is

$$\sigma_{out}^2 = \sigma_e^2 \sum_{n=0}^{\infty} g[n]^2 = \sigma_e^2 \, \frac{1}{2\pi} \int_{-\pi}^{\pi} |G(e^{j\Omega})|^2 \, d\Omega$$

The sum $\sum g[n]^2$ is the **noise gain** of that path (the second form is Parseval's relation). Total output noise is the sum over all rounding points, since the sources are assumed independent.

> **Status of the principle: Roundoff noise propagation**
> *Approximate.* Exact for an LTI system driven by white noise; the approximation lies in treating rounding as independent white noise, which requires busy signals and fails in limit cycles and for constant or very small inputs.

Worked example — Roundoff in an exponential smoother. An RMS meter in a motor drive smooths its squared current with $y[n] = a\,y[n-1] + (1-a)x[n]$ in Q15, with products accumulated at double width and rounded once when $y[n]$ is stored. The rounding error enters the loop at the same point as the stored output, so $g[n] = a^n u[n]$ and the noise gain is

$$\sum_{n=0}^{\infty} a^{2n} = \frac{1}{1-a^2}$$

For $a = 0.9$ the noise gain is 5.26 (7.2 dB); for $a = 0.99$ it is 50.3 (17.0 dB); for $a = 0.999$ it is 500 (27.0 dB). With $\Delta = 2^{-15}$, the rounding noise has RMS value $\Delta/\sqrt{12} = 8.81 \times 10^{-6}$ (-101.1 dB relative to full scale). For $a = 0.99$ the output noise is $8.81 \times 10^{-6} \times \sqrt{50.3} = 6.25 \times 10^{-5}$, or -84.1 dB: the filter's

own arithmetic has made its output about 17 dB noisier than the 16-bit data it processes, costing almost three bits. The lesson generalises: **the narrower the filter (the closer its poles to the unit circle), the larger its internal noise gain.** The remedy is to keep the state y in a wider format — Q31 state reduces the noise by 2^{16}, or 96 dB — or to use error-feedback structures that subtract the previous rounding error. Figure 21.5 shows the noise model.

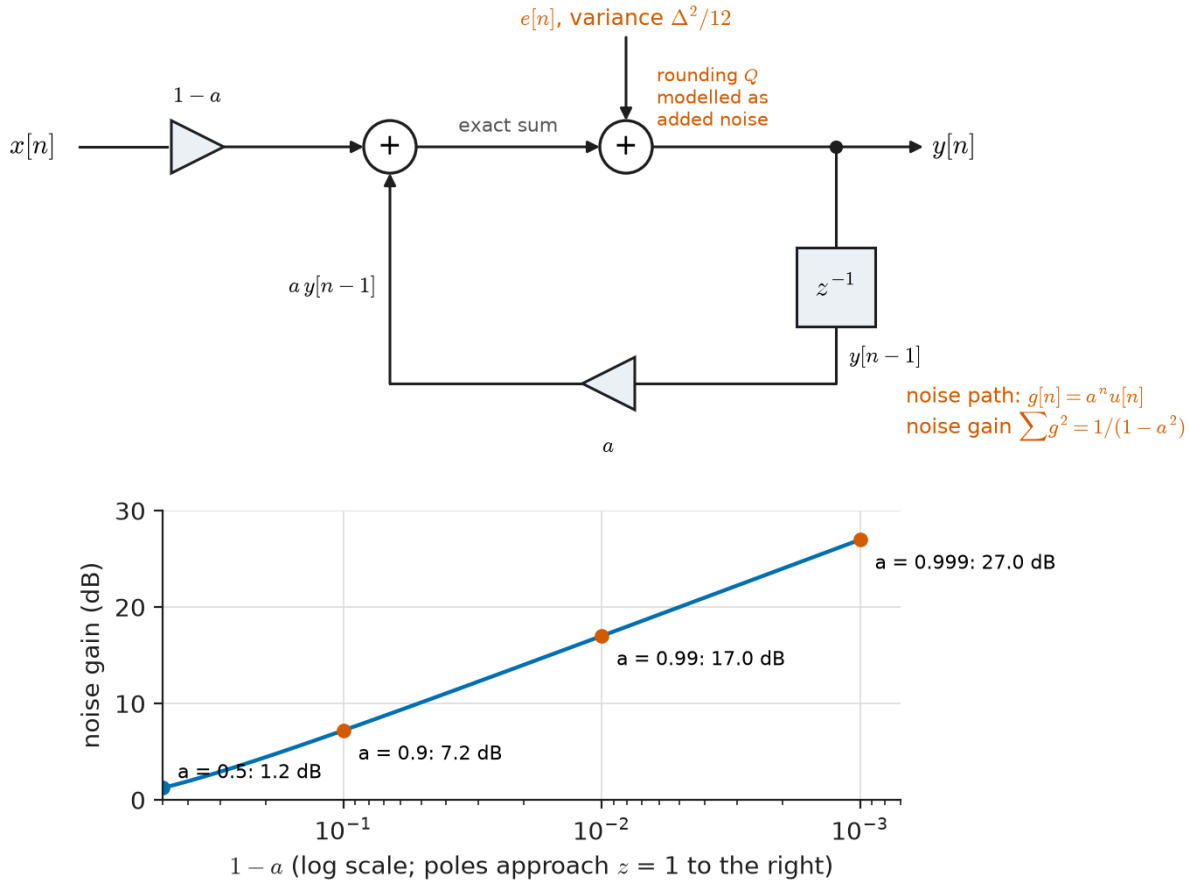

Figure 21.5 — Linear noise model of a first-order recursive filter with rounding at the state store.

Coefficient sensitivity

The poles of a direct-form second-order section are determined by $a_1 = -2r\cos\theta$ and $a_2 = r^2$. Near $\theta = 0$ (low-frequency resonances relative to f_s), $\cos\theta \approx 1 - \theta^2/2$ changes very little with θ, so a tiny change in a_1 produces a large change in pole angle; and r near 1 means a tiny change in a_2 can push the pole onto the circle. Narrowband, low-frequency, high-sample-rate filters are the worst case — which is precisely the situation of the vibration tracker in the opening paragraph.

Worked example — Quantizing a resonator. The tracker uses a resonator with poles at $r = 0.995$, $\theta = 2\pi \times 5/1000$ (5 Hz at $f_s = 1$ kSa/s). In the recursion $y[n] = c_1 y[n-1] + c_2 y[n-2] + x[n]$ the coefficients are $c_1 = 2r\cos\theta = 1.989018$ and $c_2 = -r^2 = -0.990025$. Because $|c_1| > 1$ they are stored with one integer bit and B fractional bits. Rounding to B bits and recomputing the poles:

Fractional bits B	c_1 stored	c_2 stored	Pole radius	Pole frequency
exact	1.989018	-0.990025	0.99500	5.000 Hz
15	1.989014	-0.990021	0.99500	5.000 Hz
12	1.989014	-0.989990	0.99498	4.92 Hz
8	1.988281	-0.988281	real poles at 1.000 and 0.988	none (no resonance)

At 12 bits the resonance has moved by 1.6 %. At 8 bits the complex pair has collapsed onto the real axis and one pole sits exactly on the unit circle: the "resonator" has become a marginally stable integrator whose output, once disturbed, never returns — the lock-up reported in the opening. The cure is not simply more bits but a better structure: a cascade of sections, a coupled-form (Gold–Rader) resonator whose coefficients are $r\cos\theta$ and $r\sin\theta$ directly and whose pole grid is uniform near $z = 1$, or reducing the sample rate before the resonator so that 5 Hz is no longer a tiny angle (Chapter 24). Chapter 23 returns to structure choice for IIR filters.

Accumulated error in long computations

Some errors do not average out but grow with the length of the computation.

Worked example — The clock that stopped. The data logger of the opening keeps time by executing `t = t + dt` once per sample in single precision, with $dt = 1/48\,000 = 2.083 \times 10^{-5}$ s. Each addition rounds the result to the float32 grid, whose spacing is εt — it grows with t. A bit-exact simulation shows:

Real elapsed time	Logged time	Error
1 s	1.000645 s	$+0.65$ ms
10 s	10.0357 s	$+35.7$ ms
60 s	57.809 s	-2.19 s
256 s	223.94 s	-32.1 s
474.5 s	512.000 s	$+37.5$ s, and frozen from here on

The error changes sign because the rounding of each increment depends on where t sits on the grid. When t reaches 512 s the grid spacing is $2^{-14} = 6.1 \times 10^{-5}$ s, so the increment 2.08×10^{-5} s is less than half a spacing and every addition rounds back to the same value: the clock has stopped. In double precision the same loop is off by only 16 µs after one hour. The robust solution uses neither: count samples in an integer and compute $t = n/f_s$ when needed, so that no error accumulates at all. The same mechanism afflicts phase accumulators in oscillators (keep phase in a wrapping integer), running sums in recursive moving averages (Chapter 20), and long integrations of sensor data, as in inertial navigation. Figure 21.6 plots the error.

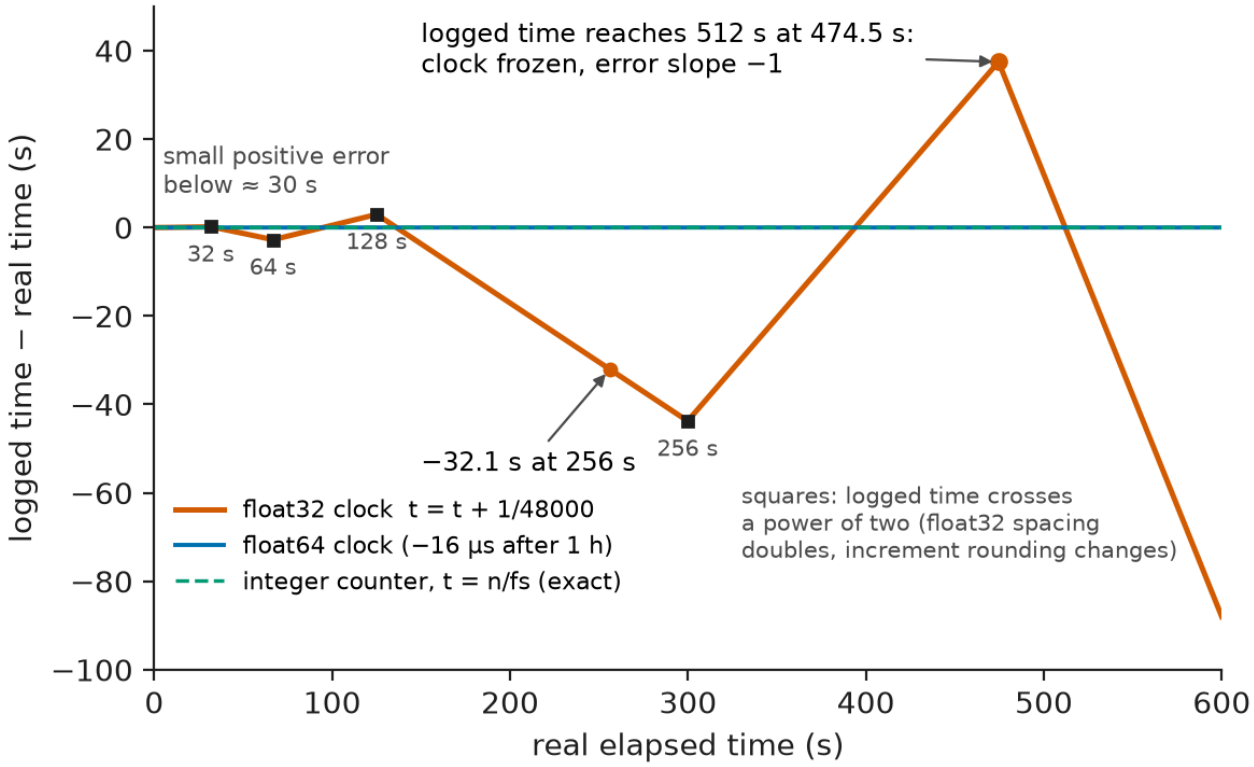

Figure 21.6 — Error of a single-precision clock incremented by 1/48000 s per sample, compared with double precision and an integer sample counter.

Catastrophic cancellation

Subtracting two nearly equal floating-point numbers leaves only their difference, but the rounding errors of the two operands — relative to their large size — remain. The relative error of the result can then be enormous.

Worked example — A negative variance. A test engineer records a 10 kV DC supply in volts and wants the RMS ripple, about 10 mV. The textbook formula $\sigma^2 = \overline{x^2} - \bar{x}^2$ is evaluated in float32 over 10 000 samples. Both $\overline{x^2}$ and $\bar{x}^2$ are about 10^8 V², and float32 resolves them only to about $10^8 \times 1.2 \times 10^{-7} \approx 12$ V² — far more than the true variance of 10^{-4} V². The computed result in one run was -8 V², a physically impossible negative variance; the true value is 9.98×10^{-5} V². Subtracting the mean first (the **two-pass** method, $\sigma^2 = \overline{(x - \bar{x})^2}$) gives 9.98×10^{-5} V² in the same float32 arithmetic, correct to four digits. Streaming algorithms that must not store the data use Welford's update, which achieves the same accuracy in one pass. The principle — remove large common components before squaring, multiplying or subtracting — applies equally to DC offsets in AC power measurements, to carrier components in demodulators and to computing small differences between large phasors in differential protection.

Error budgets and scaling

A DSP design accounts for all of these errors explicitly, just as an analogue design accounts for resistor tolerances and op-amp noise:

1. **Start from the specification** — for example, an output SNR of 90 dB for an audio processor, or an amplitude accuracy of 0.1 % for a revenue meter.
2. **Allocate the error budget** among input quantization, coefficient error, arithmetic roundoff and algorithm approximation (finite filter length, window effects).
3. **Choose scaling** so that no node overflows for the worst-case input, using the ℓ_1 norm $\sum |g[n]|$ (which guarantees no overflow), the ℓ_2 norm $\sqrt{\sum g[n]^2}$ (which bounds overflow probability for noise-like signals) or the peak of $|G(e^{j\Omega})|$ (for sinusoids), from the input to each node.
4. **Choose structure and word lengths** so that the roundoff noise computed with the noise-gain formula, after scaling, meets its share of the budget, with margin.
5. **Verify** with a bit-exact simulation, including worst-case and pathological inputs: full-scale sinusoids at the filter's peak-gain frequency, steps, silence (for limit cycles), and long runs (for accumulation).

> **Status of the principle: Linear error-budget method**
> *Approximate.* Treats each error source as independent additive noise and adds variances; adequate for most filtering and transform designs with busy signals, but it must be supplemented by deterministic checks for overflow, limit cycles, coefficient-induced instability and accumulation, which the noise model does not capture.

Table 21.4 summarises the error sources, their character and their typical cures.

Table 21.4 — Finite-precision error sources and remedies.

Source	Character	Typical size	Main remedies
ADC quantization	Additive noise (with dither or busy signals)	$\Delta^2/12$; 98 dB SQNR at 16 bits	More bits, oversampling, proper gain staging
Coefficient quantization	Deterministic change of $H(z)$	Pole shifts large near $z = 1$	More coefficient bits, second-order-section cascades, better structures, lower internal sample rate
Product roundoff	Additive noise shaped by noise gain	$\Delta^2/12$ times $\sum g^2$	Wide accumulators, wider state, error feedback
Overflow	Large nonlinear error	Up to full scale	Scaling, guard bits, saturation arithmetic
Limit cycles	Self-sustained small oscillations	Dead band of a few to hundreds of LSB	Wider state, dither, limit-cycle-free structures
Floating-point accumulation	Growing absolute error, stalls	Grows with operand size and count	Integer counters, compensated (Kahan) summation, double precision
Catastrophic cancellation	Loss of significant digits	Relative error up to 100 % or more	Remove common components first, two-pass or Welford algorithms

What remains open

Open question	Where it is answered
Where did the sampling theorem, aliasing and the quantization interface come from?	Chapter 19
What are the exact sequences, difference equations and system functions that finite precision approximates?	Chapter 20
How is the spectrum of a finite record computed, and how do FFT word lengths and block floating point behave?	Chapter 22
How are FIR and IIR filters designed, structured and quantized to meet a specification?	Chapter 23
How do decimation, interpolation and adaptive filters behave, and how is the sample rate changed before a narrowband filter?	Chapter 24
How does quantization become pulse-code modulation, with companding and line coding?	Chapter 27
How do sampling and finite precision affect digital controllers?	Chapter 35
How are ADCs, DACs and DSP processors built?	*Electronics & Computing*
What is the general theory of floating-point error analysis?	*The Mathematics Universe*

Next: Chapter 22 introduces the discrete Fourier transform and the FFT, the tools that turn a finite block of samples into a spectrum.

Chapter 22 — Discrete Fourier Transform

Part V · Digital Signal Processing

A power-quality analyser is clipped onto the incoming feeder of a factory that has just installed a bank of variable-speed drives. The utility has complained about harmonic distortion, and the analyser must report the 5th, 7th and 11th harmonic voltages to a fraction of a percent. Inside the instrument there is no bank of tuned filters and no swept spectrum analyser: there is an ADC, a block of memory holding 2048 samples, and an algorithm that turns those samples into 2048 complex numbers in under a millisecond. When the mains frequency sits at exactly 50.00 Hz the numbers are perfect. When the grid sags to 49.8 Hz during an evening peak, the same algorithm under-reports the 11th harmonic by a quarter, and a careless engineer would conclude that the drives had become cleaner. Nothing in the hardware changed. What changed was the relation between the signal and the finite, sampled window through which the algorithm sees it.

That algorithm is the discrete Fourier transform (DFT), almost always computed by a fast Fourier transform (FFT). It is the most used numerical tool in electrical engineering: it sits inside spectrum analysers, vibration monitors, OFDM modems, radar Doppler processors, audio codecs, MRI scanners and every oscilloscope's "FFT" button. This chapter defines it, inverts it, shows how to read its output as a spectrum, explains why it can be computed fast, and then deals honestly with the three things that make its output differ from the "true" spectrum of Chapter 18: leakage, the grid of bins, and random fluctuation.

> **What exactly does the DFT compute, and how is it related to the continuous Fourier transform of the signal that was sampled?**
> **Why do tones leak into neighbouring bins, what does a window buy, and why does zero padding not improve resolution? How is a reliable power spectrum estimated from a finite, noisy record?**

Concept	Engineering destination
DFT definition and bins	FFT spectrum analysers, oscilloscope FFT mode, harmonic analysers
Inverse DFT	OFDM transmitters (Wi-Fi, LTE, 5G NR, DVB-T), synthesis of test waveforms
Circular convolution	Fast FIR filtering, cyclic prefix in OFDM, correlation receivers
Spectral interpretation and scaling	Harmonic measurement (IEC 61000-4-7), audio THD+N, ADC testing
FFT algorithms	Real-time spectrum analysis, radar Doppler banks, software-defined radio
Spectral leakage	Vibration analysis of rotating machines, power-system harmonics, sonar
Windowing	Detecting weak tones beside strong ones, amplitude-accurate calibration
Zero padding	Peak interpolation, frequency estimation, display smoothing
Spectral estimation	Noise-floor measurement, EEG band power, PSD of sensor noise

Roadmap. Definition of the DFT as samples of the DTFT (22.1) → its inverse and the circular properties that follow (22.2) → reading bins as physical frequencies and amplitudes (22.3) → the FFT and why it is fast (22.4) → leakage when the record is not periodic (22.5) → windows as the cure and their costs (22.6) → zero padding and what it does not do (22.7) → estimating power spectra of random signals (22.8).

22.1 DFT definition

From the DTFT to a computable transform

Chapter 20 gave the discrete-time Fourier transform (DTFT) of a sequence $x[n]$,

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n},$$

a continuous, 2π-periodic function of the normalised angular frequency ω (radians per sample). Two features make it unsuitable for a processor: it needs infinitely many samples, and it produces a function on a continuum of frequencies. A real instrument holds a finite record of N samples, $x[0], \dots, x[N-1]$, and can store only a finite list of numbers. The natural compromise is to evaluate the DTFT of the finite record at N equally spaced frequencies,

$$\omega_k = \frac{2\pi k}{N}, \qquad k = 0, 1, \dots, N-1.$$

This defines the **discrete Fourier transform**:

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N}, \qquad k = 0, 1, \dots, N-1.$$

It is customary to write $W_N = e^{-j2\pi/N}$, the principal N-th root of unity, so that $X[k] = \sum_n x[n] W_N^{kn}$. Each output $X[k]$ is a correlation of the record with a complex exponential that completes exactly k cycles in N samples. The index k is called the **bin** number, and the set of N outputs is the spectrum on a grid.

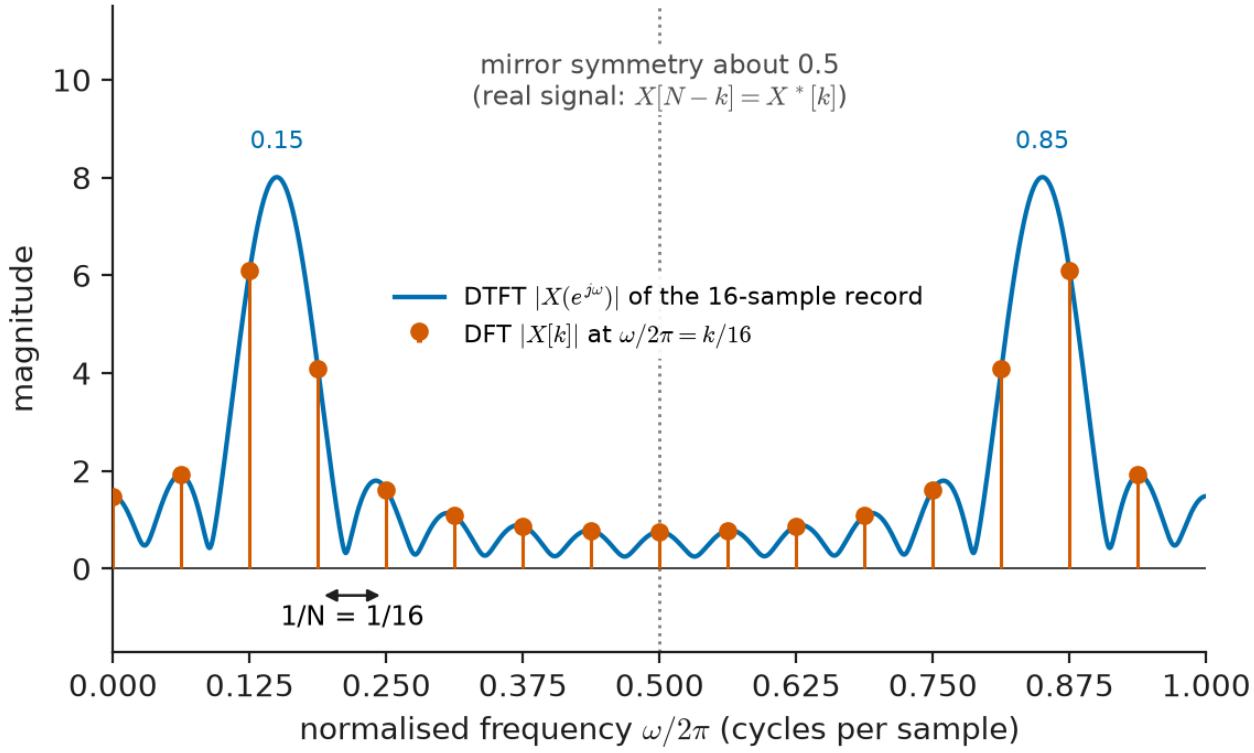

Figure 22.1 — The DFT samples the DTFT of a finite record at N equally spaced frequencies.

Figure 22.1 shows the relation for a 16-sample record of a cosine: the DTFT of the finite record is a smooth curve, and the DFT is that curve read off at 16 points. Everything that follows in this chapter is a consequence of this picture: the curve is not the spectrum of the infinitely long cosine (which would be two impulses), because the record was truncated; and the 16 points may or may not land on the curve's peaks.

Matrix form

Because the DFT is a linear map from N numbers to N numbers, it can be written as a matrix–vector product, $\mathbf{X} = \mathbf{F}_N \mathbf{x}$, with $(\mathbf{F}_N)_{kn} = W_N^{kn}$. For $N = 4$, $W_4 = e^{-j\pi/2} = -j$ and

$$\mathbf{F}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}.$$

Worked example — a four-point DFT by hand. Take $x = [1,\ 2,\ 0,\ -1]$. Then

$$X[0] = 1 + 2 + 0 - 1 = 2, \qquad X[1] = 1 + 2(-j) + 0 + (-1)(j) = 1 - 3j,$$
$$X[2] = 1 - 2 + 0 + 1 = 0, \qquad X[3] = 1 + 2(j) + 0 + (-1)(-j) = 1 + 3j.$$

$X[0]$ is the sum of the samples, N times the mean (the DC term). $X[3]$ is the complex conjugate of $X[1]$, as it must be for a real input (Section 22.3). $X[2]$, the bin at half the sampling rate, is real. Direct evaluation needed $N^2 = 16$ complex multiplications (most of them trivial here); Section 22.4 shows how to cut that cost. The energy check of the next section also works: $\sum |x[n]|^2 = 1 + 4 + 0 + 1 = 6$ and $\frac{1}{4}\sum |X[k]|^2 = \frac{1}{4}(4 + 10 + 0 + 10) = 6$.

Periodicity built into the definition

Because W_N^{kn} is periodic in both k and n with period N, the formula, if evaluated outside $0 \le k \le N-1$, simply repeats: $X[k+N] = X[k]$. More importantly, the inverse (next section) returns a sequence that is also periodic in n. The DFT therefore does not really describe a finite record; it describes **one period of a periodic sequence** whose period is the record. Every peculiarity of the DFT, from circular convolution to leakage, follows from this hidden periodic extension.

> **Status of the principle: DFT as samples of the DTFT**
> *Derived.* For a sequence that is zero outside $0 \le n \le N-1$, $X[k] = X(e^{j\omega})|_{\omega = 2\pi k/N}$ exactly; this follows directly from the definitions and needs no approximation.

22.2 Inverse DFT

The inversion formula

The N complex exponentials $e^{j2\pi kn/N}$, $k = 0, \dots, N-1$, are orthogonal over N samples:

$$\sum_{n=0}^{N-1} e^{j2\pi (k-m)n/N} = \begin{cases} N, & k = m \pmod N \\ 0, & \text{otherwise,} \end{cases}$$

because the sum of a geometric series of N equally spaced points on the unit circle is zero unless every point is 1. Multiplying the DFT by $e^{j2\pi km/N}$ and summing over k therefore isolates $x[m]$, giving the **inverse DFT**:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]\, e^{j2\pi kn/N}, \qquad n = 0, 1, \dots, N-1.$$

In matrix form $\mathbf{F}_N^{-1} = \frac{1}{N}\mathbf{F}_N^{*}$: the inverse is the complex-conjugate transpose divided by N. The scaled matrix $\frac{1}{\sqrt{N}}\mathbf{F}_N$ is unitary. The placement of the factor $1/N$ is a convention; numerical libraries (NumPy, MATLAB, FFTW) put it on the inverse as above, some textbooks split it as $1/\sqrt{N}$ on each side, and instrument firmware often applies its own amplitude scaling (Section 22.3). Mixing conventions is the commonest source of a wrong factor of N or 2 in a measured spectrum.

The inverse can be read as a synthesis equation: any N-sample block is a sum of N harmonically related complex exponentials with complex weights $X[k]/N$. This is how an OFDM transmitter works. In LTE or Wi-Fi, each subcarrier's data symbol (a QAM point, Chapter 27) is written into one bin $X[k]$, and an

inverse FFT produces the time-domain block that is sent; the receiver's forward FFT recovers the symbols. Chapter 30 develops the scheme.

> **Status of the principle: DFT invertibility**
> *Derived.* Follows from the orthogonality of the N discrete complex exponentials; the DFT pair is an exact, lossless change of basis for any finite sequence, with no band-limiting or periodicity assumption on the underlying analog signal.

Parseval's relation

Orthogonality also gives energy conservation:

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2 .$$

This is the basis of every power reading taken from an FFT: the mean-square value of the record equals the sum of $|X[k]|^2/N^2$ over all bins. A total-harmonic-distortion meter, an RMS reading from an FFT analyser and the noise-power integration of Section 22.8 all rest on it.

Properties and the meaning of "circular"

The DFT inherits most DTFT properties, with one systematic change: every shift is a **circular** shift, taken modulo N, because of the hidden periodic extension. Table 22.1 collects the properties used in practice.

Property	Time domain (N-point)	DFT domain
Linearity	$a\,x[n] + b\,y[n]$	$a\,X[k] + b\,Y[k]$
Circular time shift	$x[(n-m) \bmod N]$	$W_N^{km} X[k]$
Modulation (frequency shift)	$W_N^{-ln} x[n]$	$X[(k-l) \bmod N]$
Conjugate symmetry (real x)	$x[n]$ real	$X[N-k] = X^*[k]$
Time reversal	$x[(-n) \bmod N]$	$X[(-k) \bmod N]$
Circular convolution	$\sum_m x[m]\,h[(n-m) \bmod N]$	$X[k]\,H[k]$
Multiplication	$x[n]\,y[n]$	$\frac{1}{N}\sum_l X[l]\,Y[(k-l) \bmod N]$
Parseval	$\sum \lvert x[n]\rvert^2$	$\frac{1}{N}\sum \lvert X[k]\rvert^2$

Table 22.1 — DFT properties. All index arithmetic is modulo N.

Circular versus linear convolution

The convolution property is the reason the DFT is used for filtering, and also the reason it is easy to misuse. Multiplying two DFTs and inverting gives the **circular** convolution of the two records, in which samples that run off the end of the block wrap around to the beginning. A linear filter (Chapter 17)

produces the **linear** convolution, of length $L + M - 1$ for an input of length L and an impulse response of length M.

The two agree if both sequences are zero-padded to a length $N \geq L + M - 1$ before the DFT, because then nothing wraps. This is the principle of **fast convolution**: pad, transform, multiply, inverse-transform. For long signals the input is split into blocks and either the overlapping output tails are added (overlap-add) or the wrapped-around first $M - 1$ outputs of each block are discarded (overlap-save). Section 22.4 shows when this is cheaper than direct convolution.

OFDM exploits circularity on purpose. Each transmitted block is preceded by a **cyclic prefix**, a copy of its last few samples. A multipath channel with a delay spread shorter than the prefix then acts on the block as a circular convolution, so the receiver's DFT sees each subcarrier multiplied by a single complex channel gain $H[k]$, and equalisation reduces to one complex division per bin.

> **Status of the principle: Circular convolution theorem**
> *Derived*. Exact consequence of the DFT's orthogonality and its modulo-N indexing; it equals linear convolution only when the transform length is at least $L + M - 1$ (or a cyclic prefix makes the channel act circularly).

22.3 Spectral interpretation

Bins as physical frequencies

When $x[n] = x_a(nT_s)$ are samples of an analog signal taken at $f_s = 1/T_s$ (Chapter 19), bin k corresponds to the physical frequency

$$f_k = k\,\frac{f_s}{N} = \frac{k}{NT_s} = \frac{k}{T_{\text{rec}}},$$

where $T_{\text{rec}} = NT_s$ is the record duration. The bin spacing

$$\Delta f = \frac{f_s}{N} = \frac{1}{T_{\text{rec}}}$$

depends only on how long the record lasts, not on the sampling rate. A spectrum analyser that needs 1 Hz bins must observe for 1 s, whatever its ADC speed. Bins $k = 0, \dots, N/2$ cover 0 to $f_s/2$; bins $k = N/2 + 1, \dots, N - 1$ represent negative frequencies $-(N - k)\Delta f$, because $e^{-j2\pi kn/N} = e^{+j2\pi(N-k)n/N}$. Numerical libraries provide a "shift" function that reorders the output to run from $-f_s/2$ to $+f_s/2$.

Table 22.2 gives representative record parameters from several fields.

Application	f_s	N	T_{rec}	Δf
Power-quality harmonics (IEC 61000-4-7, 50 Hz)	10.24 kHz	2048	200 ms (10 cycles)	5 Hz
Audio analyser, THD+N	48 kHz	65 536	1.365 s	0.73 Hz
Machine vibration monitoring	25.6 kHz	4096	160 ms	6.25 Hz
EEG band-power analysis	256 Hz	512	2 s	0.5 Hz
LTE 20 MHz OFDM symbol	30.72 MHz	2048	66.7 µs	15 kHz
Wi-Fi 802.11a/g, 20 MHz channel	20 MHz	64	3.2 µs	312.5 kHz
Pulse-Doppler radar, 64 pulses	PRF 2 kHz	64	32 ms	31.25 Hz

Table 22.2 — Typical DFT record parameters. $\Delta f = 1/T_{\text{rec}}$ in every row.

Symmetry for real signals

For a real input, $X[N-k] = X^*[k]$: the negative-frequency half is the mirror image of the positive half, magnitudes symmetric and phases antisymmetric about $N/2$. Only $N/2+1$ bins carry independent information (bins 0 and $N/2$ are real). Instruments therefore display a **single-sided** spectrum from 0 to $f_s/2$, folding the negative-frequency energy onto the positive side.

Amplitude scaling

A real sinusoid $x[n] = A\cos(2\pi k_0 n/N + \phi)$ with an integer number of cycles k_0 in the record puts all of its energy in bins k_0 and $N-k_0$, each with magnitude

$$|X[k_0]| = \frac{NA}{2}.$$

So the peak amplitude is read as $A = 2|X[k_0]|/N$, the RMS amplitude as $\sqrt{2}|X[k_0]|/N$, and the phase as $\phi = \arg X[k_0]$ (relative to the first sample). The DC bin is the exception: $X[0] = N\bar{x}$ with no factor of 2, and so is the Nyquist bin $X[N/2]$. When a window $w[n]$ multiplies the record (Section 22.6), the tone magnitude becomes $\frac{A}{2}\sum_n w[n]$, and the reading must be divided by the window's **coherent gain** $\text{CG} = \frac{1}{N}\sum w[n]$.

Worked example — harmonic analysis of a factory feeder. A power-quality analyser samples a 230 V, 50 Hz phase voltage at $f_s = 10.24$ kHz and transforms $N = 2048$ samples (200 ms, ten cycles, as IEC 61000-4-7 prescribes for 50 Hz systems). The voltage contains a 5th harmonic of 5.0 %, a 7th of 3.5 % and an 11th of 1.5 % of the fundamental.

Bins. $\Delta f = 10\,240/2048 = 5$ Hz. The fundamental falls in bin $50/5 = 10$, the 5th harmonic (250 Hz) in bin 50, the 7th (350 Hz) in bin 70 and the 11th (550 Hz) in bin 110. Each harmonic of order h sits in bin $10h$, with nine empty "interharmonic" bins between successive harmonics.

Magnitudes. The fundamental has peak amplitude $230\sqrt{2} = 325.27$ V, so $|X[10]| = 2048 \times 325.27/2 = 3.331 \times 10^5$ (in volts, with no window). Converting back, $\sqrt{2}\,|X[10]|/N = 230.0$ V RMS. Likewise bin

50 gives 11.50 V, bin 70 gives 8.05 V and bin 110 gives 3.45 V RMS, and every other bin is zero to rounding error.

Distortion. The total harmonic distortion is

$$\mathrm{THD} = \frac{\sqrt{11.50^2 + 8.05^2 + 3.45^2}}{230.0} = \sqrt{0.05^2 + 0.035^2 + 0.015^2} = 6.28\ \%.$$

The 49.8 Hz evening. If the grid drifts to 49.8 Hz while the analyser keeps $f_s = 10.24$ kHz, the 200 ms record holds 9.96 cycles instead of 10. Now the fundamental reads 228.85 V in bin 10 (−0.5 %), with 10.13 V and 8.30 V spilling into bins 9 and 11. The harmonics suffer more, because a harmonic of order h is offset by $0.04h$ bins from its nominal bin: the 5th reads 10.75 V (−6.6 %), the 7th 7.20 V (−10.6 %) and the 11th 2.56 V (−25.7 %). The computed THD drops to 5.77 %. This is spectral leakage (Section 22.5). The standard's remedy is to lock the sampling clock to the measured mains frequency with a phase-locked loop, so that the record always contains exactly ten cycles; it also prescribes summing the energy of the neighbouring bins into "harmonic groups", which recovers most of the leaked power (the RMS of bins 9–11 is 229.2 V).

What a bin is not

A DFT bin is not a measurement of the signal's spectrum at one frequency. It is the output of a bandpass filter, centred on f_k, whose frequency response is the transform of the (window-weighted) record shape. For an unwindowed record this filter has the response

$$|W_R(f)| = \left| \frac{\sin(\pi N f / f_s)}{N \sin(\pi f / f_s)} \right|,$$

the **Dirichlet kernel** (periodic sinc), with nulls at every other bin centre, a main lobe two bins wide between nulls and side lobes falling only slowly. Sections 22.5 to 22.7 are all about the shape of this filter.

> **Boundary.**
> The DFT analyses sampled data; it cannot recover what sampling destroyed. Components above $f_s/2$ that reached the ADC appear in the DFT at alias frequencies (Chapter 19), indistinguishable from genuine in-band components. The anti-aliasing filter and converter hardware are treated in *Electronics & Computing*.

22.4 FFT algorithms

The cost of the definition

Computing every $X[k]$ directly from the definition needs N complex multiplications and $N-1$ complex additions per bin, about N^2 complex multiply-accumulates in total. For $N = 2048$ that is 4.2 million; for the $N = 65\,536$ audio record of Table 22.2 it is 4.3 billion. The **fast Fourier transform** is a family of algorithms that produce exactly the same numbers with work proportional to $N \log_2 N$. The modern form is due to Cooley and Tukey (1965), though Gauss used the idea in 1805 for asteroid orbits.

Radix-2 decimation in time

Let N be a power of 2. Split the sum into even- and odd-indexed samples:

$$X[k] = \sum_{m=0}^{N/2-1} x[2m]\, W_N^{2mk} + W_N^k \sum_{m=0}^{N/2-1} x[2m+1]\, W_N^{2mk}.$$

Since $W_N^2 = W_{N/2}$, each sum is an $N/2$-point DFT: $G[k]$ of the even samples and $H[k]$ of the odd samples. Both are periodic in k with period $N/2$, and $W_N^{k+N/2} = -W_N^k$, so

$$X[k] = G[k] + W_N^k H[k], \qquad X[k+N/2] = G[k] - W_N^k H[k], \qquad k = 0, \ldots, \tfrac{N}{2} - 1.$$

One complex multiplication by the **twiddle factor** W_N^k produces two outputs. This two-input, two-output operation is the **butterfly**. Applying the split recursively, an N-point DFT becomes $\log_2 N$ stages of $N/2$ butterflies each. The total is

$$\tfrac{N}{2} \log_2 N \text{ complex multiplications}, \qquad N \log_2 N \text{ complex additions}.$$

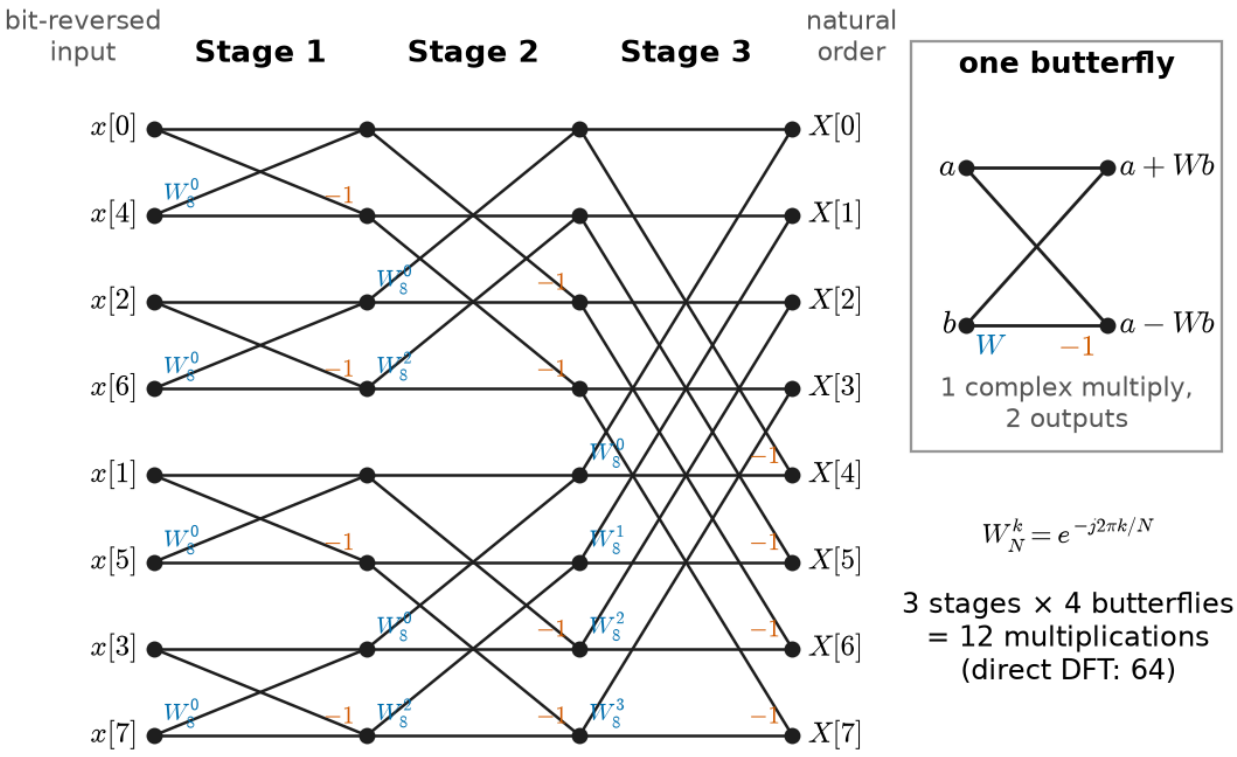

Figure 22.2 — Eight-point radix-2 decimation-in-time FFT: three stages of four butterflies, with bit-reversed input order.

Figure 22.2 shows the eight-point flow graph. Because the recursion repeatedly separates even and odd indices, the inputs enter in **bit-reversed** order (index 1 = 001 binary goes to position 100 = 4), and the outputs emerge in natural order. Each stage can overwrite its inputs with its outputs, so the algorithm runs **in place** with no extra memory beyond the N complex values and a table of $N/2$ twiddle factors.

> **Status of the principle: FFT complexity**
> *Derived*. The $O(N \log N)$ count is an exact algebraic consequence of the factorisation of W_N^{kn} for composite N; the FFT returns the DFT exactly, apart from floating- or fixed-point rounding.

Other members of the family

- **Decimation in frequency** (Sande–Tukey) splits the output index into even and odd instead; the butterfly multiplies after the subtraction, inputs are in natural order and outputs are bit-reversed. The cost is identical.
- **Radix-4 and split-radix** algorithms use the fact that multiplication by $\pm j$ is free (swap and negate), cutting the number of real multiplications by about 25 % and 33 % relative to radix 2.
- **Mixed-radix** FFTs handle $N = 2^a 3^b 5^c$; LTE uses 1536-point transforms ($2^9 \times 3$) for 15 MHz channels. **Prime-factor** and **Bluestein (chirp-z)** algorithms extend $O(N \log N)$ to any N, including primes.
- **Real-input FFTs** exploit $X[N-k] = X^*[k]$: an N-point real transform can be computed as one $N/2$-point complex FFT plus a post-processing pass, halving the work.
- The **Goertzel algorithm** computes a single bin with a second-order recursive filter, about N real multiplications per bin. When only a few bins are needed (the eight DTMF tones of a telephone keypad, or one harmonic in a protection relay) it is cheaper than a full FFT.

Table 22.3 compares the costs.

N	Direct DFT, N^2 complex mult.	Radix-2 FFT, $\frac{N}{2}\log_2 N$	Ratio
64	4096	192	21
1024	1.05×10^6	5120	205
2048	4.19×10^6	11 264	372
65 536	4.29×10^9	5.24×10^5	8192

Table 22.3 — Multiplication counts. Trivial multiplications by ± 1, $\pm j$ are not removed, so practical FFTs are somewhat faster still.

Worked example — the FFT budget of an LTE receiver. A 20 MHz LTE carrier is sampled at 30.72 MHz and each OFDM symbol is demodulated with a 2048-point FFT, giving 2048 bins spaced 15 kHz apart (of which 1200 carry data). The normal cyclic-prefix configuration sends 14 symbols per millisecond, so the receiver must perform 14 000 FFTs per second per antenna.

Direct DFT. $2048^2 = 4.194 \times 10^6$ complex multiplications per symbol, i.e. $4.194 \times 10^6 \times 14\,000 = 5.87 \times 10^{10}$ complex multiplications per second, roughly 2.3×10^{11} real multiplications per second. That is beyond a handset's power budget.

Radix-2 FFT. $\frac{2048}{2} \times \log_2 2048 = 1024 \times 11 = 11\,264$ complex multiplications per symbol, or $11\,264 \times 14\,000 = 1.58 \times 10^8$ per second, about 6.3×10^8 real multiplications per second. That is a small fraction of a modern baseband processor.

Ratio. $4.194 \times 10^6 / 11\,264 = 372$. The FFT is not a speed-up of a feasible computation; it is what makes OFDM feasible at all. With four receive antennas (Chapter 30) the FFT load still stays below 3×10^9 real multiplications per second.

Fast convolution

Filtering a long signal through an M-tap FIR filter costs M multiply-accumulates per output sample when done directly. With overlap-save using an N-point FFT, each block yields $N - M + 1$ new outputs for one forward FFT, N complex products and one inverse FFT, roughly $2 \cdot \frac{N}{2}\log_2 N + N$ complex multiplications, or about four times that in real multiplications. For $M = 1024$ and $N = 4096$, the direct method needs 1024 real multiplications per output, while the FFT method needs about $4(2 \cdot 2048 \cdot 12 + 4096)/3073 \approx 69$. The crossover usually lies between 32 and 128 taps, depending on the processor. Long room-reverberation filters in audio, acoustic echo cancellers and matched filters for long radar pulses (pulse compression) are implemented this way.

22.5 Spectral leakage

The periodic extension seen by the DFT

Section 22.1 noted that the DFT treats the record as one period of a periodic sequence. If a sinusoid completes an integer number of cycles in the record, the periodic extension is seamless and the DFT shows exactly one bin per frequency (the 50.00 Hz case above). If it does not, the extension has a jump where the end of one copy meets the start of the next. That discontinuity has energy at all frequencies, and it appears in the DFT as **spectral leakage**: the tone's power spreads into every bin, falling off only slowly with distance.

The frequency-domain view is equivalent and more useful. Truncating to N samples multiplies the infinite sequence by a rectangular window $w_R[n]$; multiplication in time is convolution in frequency, so the DTFT of the record is the true line spectrum convolved with $W_R(e^{j\omega})$, the Dirichlet kernel. A tone at frequency f_0 becomes a copy of the kernel centred on f_0. The DFT samples that copy at the bin centres. If f_0 is on a bin centre, all other samples fall on the kernel's nulls and the leakage is invisible; otherwise they sample its side lobes.

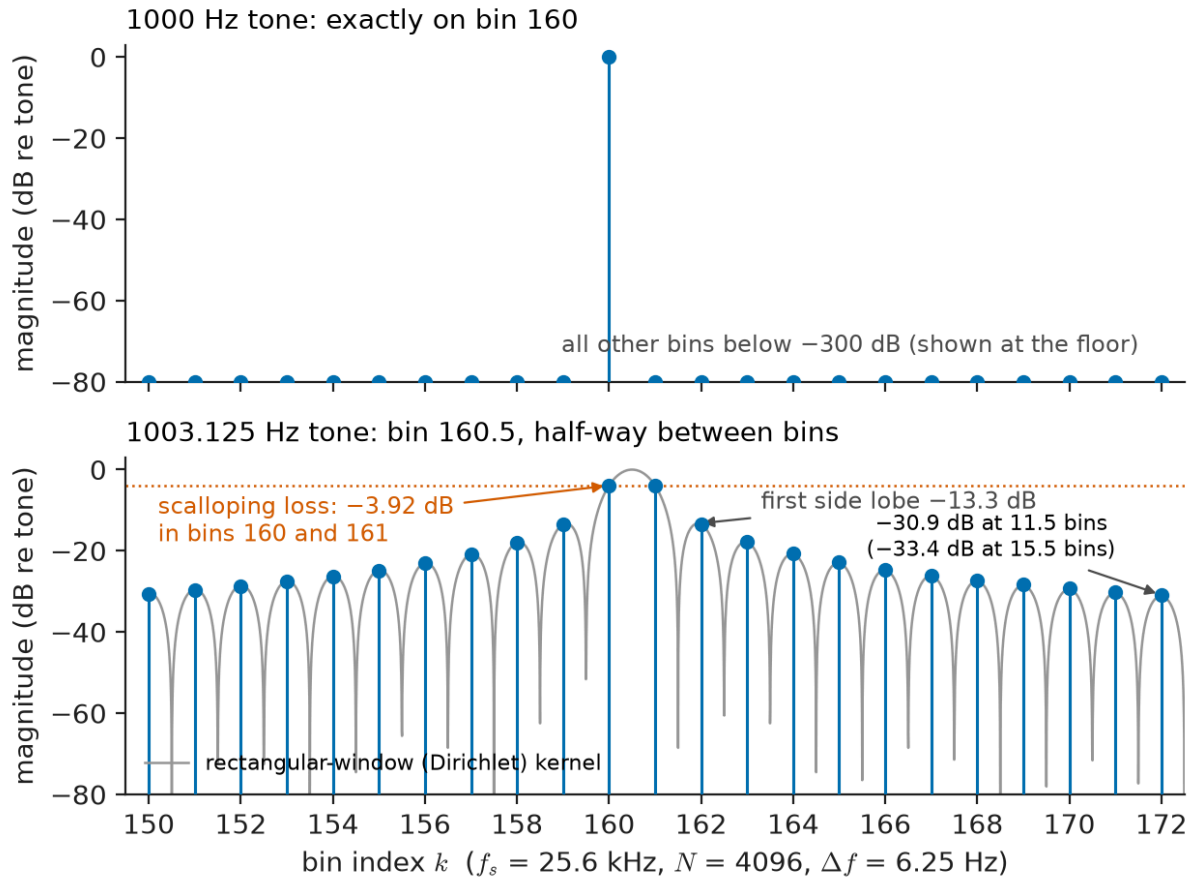

Figure 22.3 — Spectral leakage: a tone on a bin centre occupies one bin, while a tone half-way between bins spreads into every bin through the side lobes of the rectangular-window kernel.

Two consequences: scalloping and masking

Figure 22.3 shows both effects.

Scalloping loss. A tone half-way between bins ($f_0 = (k + \frac{1}{2})\Delta f$) is attenuated in the nearest bin by the kernel's value half a bin from its centre, $1/(N \sin(\pi/2N)) \approx 2/\pi$, i.e. 3.92 dB. An unwindowed FFT can therefore under-read a tone's amplitude by up to 36 %, depending only on where the tone happens to fall relative to the bin grid.

Masking by side lobes. The rectangular kernel's highest side lobe is −13.3 dB, and its side lobes decay as $1/(\pi\,\Delta k)$, i.e. only 6 dB per octave of bin offset. At 15.5 bins from an off-bin tone the leakage is still about −33 dB. Any weaker component further below than that is buried.

These are not numerical errors; they are exact properties of the finite observation. More bits, more processing or a faster FFT will not remove them. The only levers are the record length (which sets the kernel width) and the record shape, the window.

> **Status of the principle: Leakage as convolution with the window transform**
> *Derived.* Follows from the multiplication property of the Fourier transform applied to truncation; it holds exactly for any finite record, and it disappears only when every component is periodic in the record length.

22.6 Windowing

Tapering the record

A **window** $w[n]$ multiplies the record before the DFT, tapering it smoothly to (near) zero at both ends. The periodic extension then has no jump, and the window's transform has much lower side lobes than the Dirichlet kernel. The price is a wider main lobe: tapering effectively shortens the record, so two tones must be further apart to be resolved, and the noise in each bin rises.

The most used windows belong to the generalised cosine family,

$$w[n] = \sum_{m=0}^{M} (-1)^m a_m \cos\!\left(\frac{2\pi m n}{N}\right), \qquad n = 0, \dots, N-1,$$

with $a_0 = a_1 = 0.5$ for the **Hann** window, $a_0 = 0.54$, $a_1 = 0.46$ for **Hamming**, $(0.42, 0.5, 0.08)$ for **Blackman** and $(0.35875, 0.48829, 0.14128, 0.01168)$ for the four-term **Blackman–Harris**. The **Kaiser** window uses a modified Bessel function with a shape parameter β that trades main-lobe width continuously against side-lobe level. **Flat-top** windows are designed for a flat main-lobe peak, so that amplitude is read correctly wherever the tone falls.

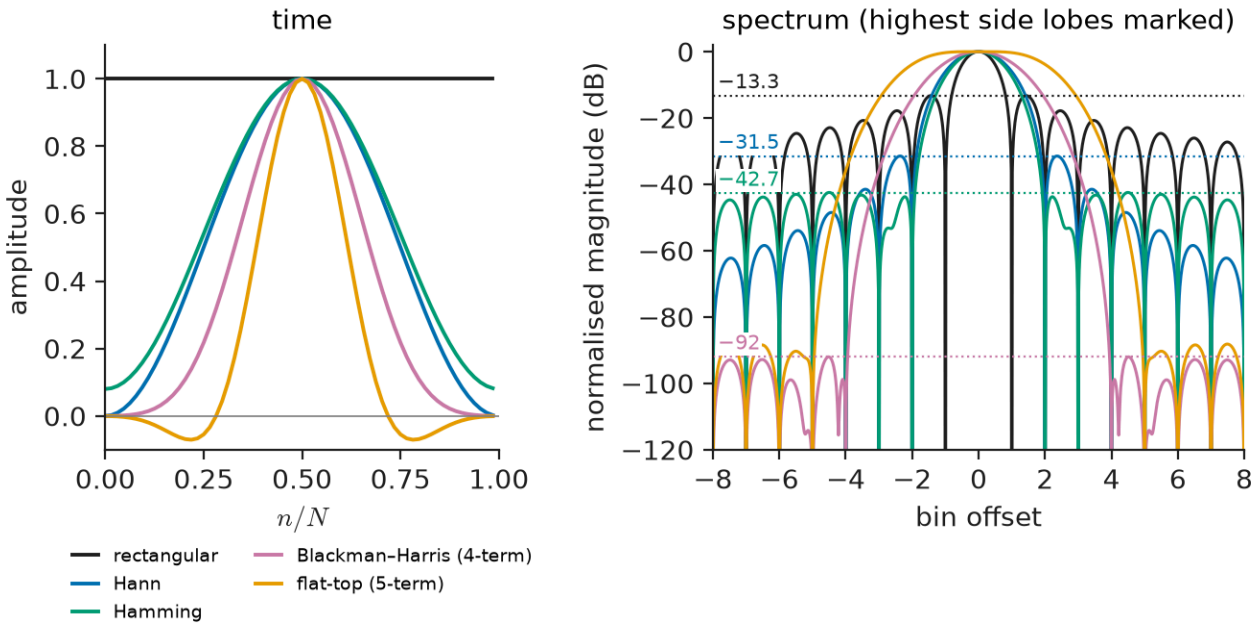

Figure 22.4 — Common windows in time (left) and their spectra in dB versus bin offset (right): narrower main lobes come with higher side lobes.

Figures of merit

Figure 22.4 shows the shapes and their spectra. Four numbers characterise a window for spectrum analysis:

- **Coherent gain** $\mathrm{CG} = \frac{1}{N}\sum w[n]$: the factor by which a bin-centred tone's DFT magnitude is reduced, used to correct amplitude readings.
- **Equivalent noise bandwidth**, $\mathrm{ENBW} = N\sum w^2[n] / \left(\sum w[n]\right)^2$ bins: the width of the rectangular filter that would pass the same white-noise power as the window's bin filter. It sets the noise floor per bin (Section 22.8).
- **Highest side lobe** and its **roll-off rate**: how far away, and how far down, a strong tone stops masking weaker ones.
- **Scalloping loss**: the worst-case amplitude error for a tone half-way between bins.

Table 22.4 gives these values, computed for $N = 4096$; they are practically independent of N for $N \geq 64$.

Window	CG	ENBW (bins)	3 dB bandwidth (bins)	Main lobe, null to centre (bins)	Highest side lobe (dB)	Roll-off (dB/octave)	Scalloping loss (dB)
Rectangular	1.00	1.00	0.89	1	−13.3	6	3.92
Hann	0.50	1.50	1.44	2	−31.5	18	1.42
Hamming	0.54	1.36	1.30	2	−42.7	6	1.75
Blackman	0.42	1.73	1.64	3	−58.1	18	1.10
Kaiser, $\beta = 9$	0.41	1.76	1.67	3.0	−66	6	1.07
Blackman–Harris (4-term)	0.36	2.00	1.90	4	−92.0	6	0.83
Flat-top (5-term)	0.22	3.77	3.73	5	−93	6	0.01

Table 22.4 — Window figures of merit (after F. J. Harris, 1978; values recomputed).

The table shows the trade-off clearly. Moving down the list, side lobes fall from −13 dB to −93 dB and scalloping falls from 3.9 dB to 0.01 dB, but the noise bandwidth grows from one bin to almost four, and the main lobe widens from ±1 to ±5 bins. No window is best for every job:

Measurement task	Usual choice	Reason
Transient or impact fully inside the record (hammer test, single pulse)	Rectangular	No leakage problem; tapering would discard signal
General-purpose spectrum, random signals	Hann	Good side-lobe roll-off, moderate ENBW
Two close tones of similar level	Hamming or Hann	Narrow main lobe
Weak tone near a strong one (ADC spurs, radar clutter, sonar)	Blackman–Harris or Kaiser with large β	Side lobes below the dynamic range of interest
Accurate amplitude calibration of tones	Flat-top	Scalloping below 0.01 dB

> **Status of the principle: Window trade-off**
> *Derived*. A window cannot have both a narrow main lobe and low side lobes, because a shorter effective duration widens the transform (the time–bandwidth relation of Chapter 18); the table values follow exactly from the window formulas.

Worked example — finding a bearing defect beside a strong shaft tone. A vibration monitor on a motor-driven pump samples an accelerometer at $f_s = 25.6$ kHz and transforms $N = 4096$ samples, giving $\Delta f = 6.25$ Hz. A strong tone from blade-pass excitation lies at 1003.125 Hz, which is bin 160.5, the worst case for scalloping. An incipient bearing defect produces a tone 70 dB weaker at 1100 Hz, in bin 176, 15.5 bins away. Can the defect be seen?

Rectangular window. The strong tone reads 3.91 dB low in bins 160 and 161. Its leakage into bin 176 is $\approx 1/(\pi \times 15.5)$, computed exactly as −33.4 dB relative to its true level. The defect at −70 dB is buried 37 dB beneath the leakage.

Hann window. Scalloping falls to 1.42 dB. Because the Hann side lobes fall at 18 dB per octave, the leakage at 15.5 bins is −81.3 dB, so the defect tone at −70 dB now stands about 11 dB above the leakage skirt.

Blackman–Harris window. Scalloping is 0.83 dB and the leakage at bin 176 is −112.6 dB. The defect tone now stands more than 40 dB clear, and its level could be tracked as it grows over months of monitoring.

The record, the FFT and the sensor are identical in the three cases; the window alone decides whether the defect is visible. The cost of the Blackman–Harris choice is a noise floor 3 dB higher than the rectangular case (ENBW 2.00 instead of 1.00) and a main lobe ±4 bins wide, which would merge tones closer than about 25 Hz.

22.7 Zero padding

What padding does

Zero padding appends zeros to an N-sample record to make it $N_{\text{pad}} > N$ samples before the DFT. Because the extra samples are zero, the DTFT of the record is unchanged; the longer DFT simply samples the same curve at N_{pad} points instead of N. The bin spacing becomes f_s/N_{pad}, finer than $1/T_{\text{rec}}$, and the plot looks smoother.

Padding is used for four practical reasons:

1. to reach a convenient FFT length (pad 1000 samples to 1024);
2. to make linear convolution out of circular convolution (Section 22.2);
3. to **interpolate** the spectrum, so that the peak of a tone between bins can be located more accurately and its amplitude read with less scalloping;
4. to produce smoother displays.

What padding does not do

Padding adds no information about the signal. The **resolution**, the ability to separate two nearby components, is set by the main-lobe width of the window transform, which depends on the record duration T_{rec} and the window shape, not on the number of DFT points. Two tones closer than roughly one main-lobe width (about $1/T_{\text{rec}}$ for a rectangular window, $2/T_{\text{rec}}$ for Hann) merge into one peak however finely that peak is sampled.

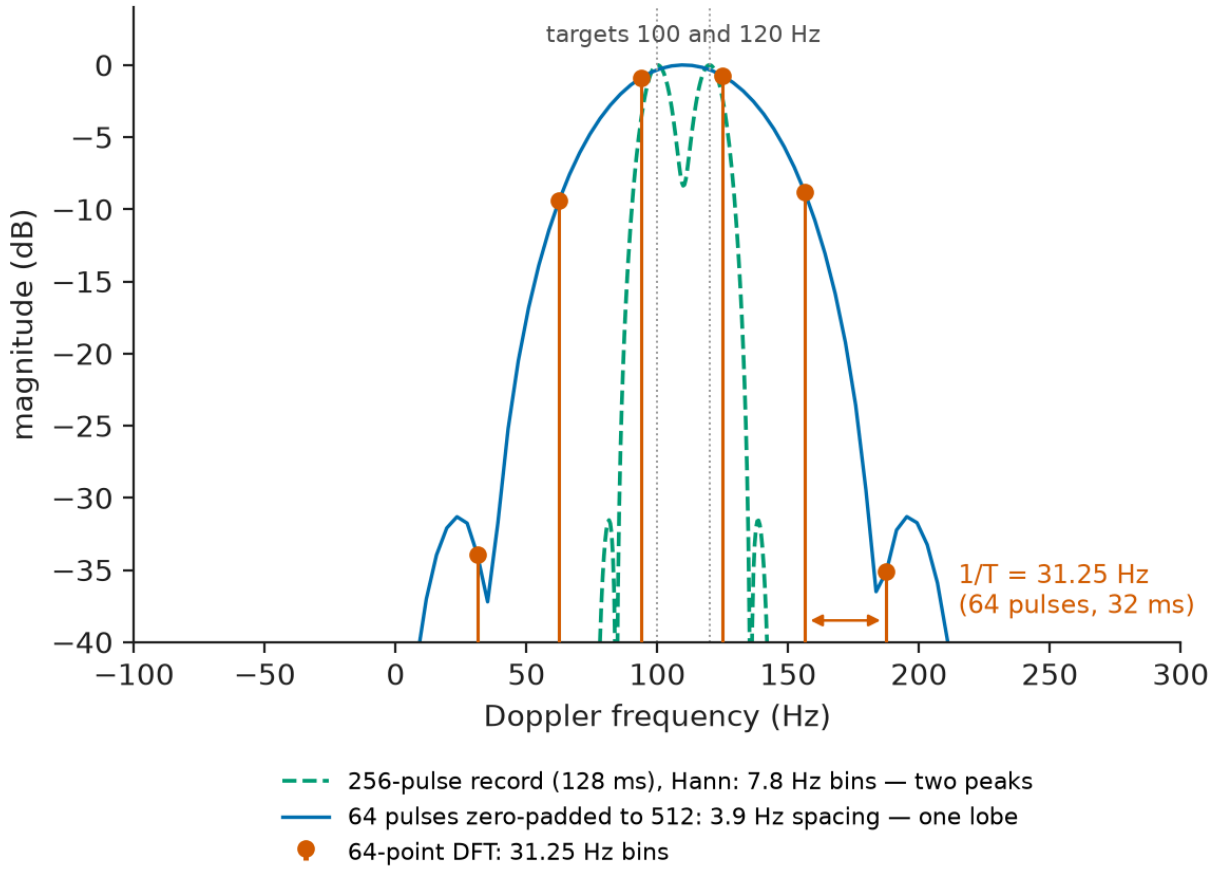

Figure 22.5 — Zero padding samples the same spectrum more densely but cannot separate two Doppler tones closer than the resolution 1/T.

Worked example — Doppler resolution in a pulse radar. An X-band pulse-Doppler radar at 10 GHz ($\lambda = c/f = 0.03$ m) transmits pulses at a PRF of 2 kHz and forms a Doppler spectrum from $N = 64$ pulses in each range cell. The radial velocity of a target is $v = f_D\lambda/2$.

Bin spacing and resolution. $\Delta f = 2000/64 = 31.25$ Hz, corresponding to $\Delta v = 31.25 \times 0.03/2 = 0.469$ m/s. The unambiguous Doppler span ± 1 kHz corresponds to ± 15 m/s.

Zero padding to 512. The bin spacing becomes $2000/512 = 3.91$ Hz, or 0.059 m/s. For a single target this helps: the interpolated peak locates its velocity to a small fraction of 0.469 m/s, limited by noise rather than by the grid, and the scalloping loss of up to 1.4 dB (Hann) disappears.

Two targets. Two aircraft at 1.50 m/s and 1.80 m/s produce Doppler tones at 100 Hz and 120 Hz, 20 Hz apart. This is less than one bin of the 64-pulse record, and with the Hann window's 3 dB width of $1.44 \times 31.25 = 45$ Hz the two tones produce a single lobe. Padding to 512 or to 8192 points draws that single lobe in finer detail; it never splits it (Figure 22.5). Separating them requires a dwell long enough that $1/T_{\text{rec}}$ is well below 20 Hz: with Hann, more than about $2000 \times 1.44/20 \approx 144$ pulses, i.e. 72 ms on target. Parametric methods (Chapter 24) can do better from short records, at the price of a signal model.

> **Status of the principle: Resolution set by record length, not by DFT length**
> *Derived*. Zero padding samples the DTFT of the windowed record more finely but leaves that DTFT unchanged, so the main-lobe width, and hence classical resolution, is fixed by T_{rec} and the window.

22.8 Spectral estimation

From deterministic tones to random signals

So far the signals were tones whose amplitudes the DFT should measure. Much of engineering practice deals instead with **random** signals: the noise floor of an ADC, the background EEG, thermal noise in a receiver, wind loading on a structure, a vibration that is broadband rather than tonal. Each record of such a signal is different, and its DFT is itself random. What is wanted is the **power spectral density** (PSD) $S_x(f)$, the distribution of the signal's mean power over frequency, defined for a wide-sense stationary process as the Fourier transform of its autocorrelation (the Wiener–Khinchin theorem). The probability theory of random processes belongs to *The Mathematics Universe*; Chapter 29 applies it to communication noise. Here the task is practical: how to estimate $S_x(f)$ from finite data.

The periodogram

The most direct estimate is the **periodogram**,

$$\hat{S}_x(f_k) = \frac{1}{f_s \sum_n w^2[n]} \left| \sum_{n=0}^{N-1} w[n]\, x[n]\, e^{-j2\pi kn/N} \right|^2 ,$$

in units of (signal unit)^2/Hz, two-sided. Doubling the bins from 1 to $N/2 - 1$ gives the one-sided PSD that instruments display. The normalisation by $\sum w^2$ makes the estimate unbiased for white noise whatever the window, and it implies that integrating $\hat{S}_x$ over frequency returns the mean-square value of the record, which is Parseval again.

Two defects limit the periodogram:

- **Bias** through leakage: the expected periodogram is the true PSD convolved with $|W(f)|^2$. Steep spectra (a 1/f noise floor, a mains line above a quiet background) are smeared unless a low-side-lobe window is used.
- **Variance** that does not decrease with record length. For Gaussian noise, each periodogram bin is approximately the true PSD times an exponentially distributed random variable with two degrees of freedom (chi-squared, $\chi^2_2/2$), so its standard deviation equals its mean. A longer record gives more bins, finer resolution, but the same 100 % fluctuation per bin. That is why a single-FFT noise spectrum looks like grass.

> **Status of the principle: Periodogram variance**
> *Approximate.* For a stationary Gaussian process and bins away from 0 and $f_s/2$, the periodogram ordinates are asymptotically independent with standard deviation equal to their mean; non-Gaussian signals and very short records depart from this.

Averaging: Bartlett and Welch

The fix is to trade resolution for stability by averaging. **Bartlett's method** divides the record into K non-overlapping segments, computes a periodogram of each and averages them; the variance falls by $1/K$ and the resolution coarsens to $1/T_{\text{seg}}$. **Welch's method** adds a window to each segment and overlaps the segments, typically by 50 % for Hann, which recovers data that the window would otherwise down-weight. For Hann segments with 50 % overlap the correlation between neighbouring segment estimates is small, and the variance is

$$\operatorname{var}\left[\hat{S}_x\right] \approx \frac{S_x^2}{K}\left(1 + 2c^2\right), \qquad c = \frac{\sum_n w[n]\, w[n + L/2]}{\sum_n w^2[n]} = \tfrac{1}{6} \text{ for Hann},$$

only 6 % worse than for independent segments. Welch's method is the default in every analysis package and in most spectrum and dynamic-signal analysers ("RMS averaging").

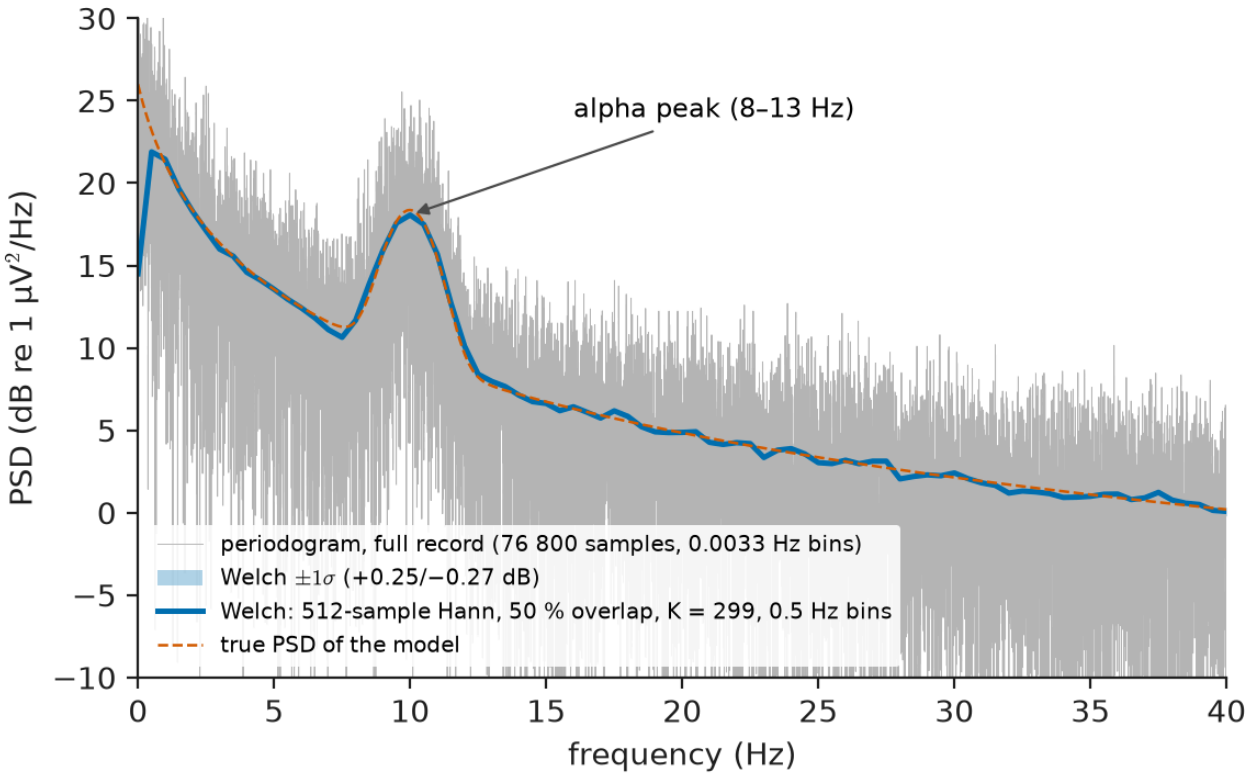

Figure 22.6 — A single periodogram of a 5-minute EEG record fluctuates by 100 percent per bin; Welch averaging of 299 two-second Hann segments gives a smooth estimate with a visible alpha peak.

Worked example — EEG band power by Welch's method. A clinical EEG channel is sampled at 256 Hz for 5 minutes, giving 76 800 samples. The neurologist wants the power in the alpha band (8–13 Hz).

Segmenting. With 2 s Hann segments ($L = 512$, 0.5 Hz bins) and 50 % overlap (hop 256 samples), the number of segments is $K = \lfloor (76\,800 - 512)/256 \rfloor + 1 = 299$. The effective resolution is the window's 3 dB bandwidth, $1.44 \times 0.5 \approx 0.72$ Hz (or its ENBW, $1.5 \times 0.5 = 0.75$ Hz), ample to separate the alpha and beta bands.

Stability. With $c = 1/6$, the relative variance is $(1 + 2/36)/299 = 3.53 \times 10^{-3}$, a standard deviation of 5.9 % of the PSD in each bin, which is $+0.25/-0.27$ dB. A single periodogram of the whole record would have finer bins (0.0033 Hz), which are useless here, and a 100 % fluctuation in every bin (Figure 22.6).

Band power. The alpha power is the PSD summed over the bins from 8 to 13 Hz (11 bins) times the bin width. Because the sum spans many bins, its relative scatter is smaller still. Comparing the alpha power with the eyes open and closed is one of the most robust measurements in clinical neurophysiology.

Noise floor and processing gain

When a tone sits on white noise, the DFT concentrates the tone in about one bin while the noise spreads over all $N/2$ bins of the one-sided spectrum. The tone-to-noise ratio in its bin therefore exceeds the wideband signal-to-noise ratio by the **FFT processing gain**

$$G_{\mathrm{FFT}} = 10 \log_{10} \frac{N}{2\,\mathrm{ENBW}} \ \mathrm{dB}.$$

The displayed noise floor of an FFT analyser drops 3 dB every time N doubles, while the tone stays put. That is why an FFT display can show spurs far below the noise level that a wideband meter would report, and why the noise floor in an FFT plot must always be quoted together with its bin width.

Worked example — measuring the noise floor of a 16-bit ADC. An ideal 16-bit converter driven by a full-scale sine has quantisation-limited SNR $= 6.02 \times 16 + 1.76 = 98.08$ dB (Chapter 21). An engineer tests it with a coherently sampled full-scale tone and a Hann-windowed FFT.

N = 8192. The processing gain is $10 \log_{10}(8192/2) = 36.12$ dB, less $10 \log_{10} 1.5 = 1.76$ dB for the Hann ENBW. The noise floor per bin is therefore $-98.08 - 36.12 + 1.76 = -132.4$ dBFS (referred to the full-scale tone).

N = 65 536. The processing gain rises to 45.15 dB and the floor falls to -141.5 dBFS. A harmonic spur at −120 dBFS, invisible in a wideband SNR measurement, now stands more than 20 dB above the floor.

Reading it back. To recover the SNR from the plot, the engineer must sum the noise power over all bins (excluding the tone's main lobe and its harmonics) and divide by the ENBW, not read the floor level. Mistaking the per-bin floor for the SNR would overstate the converter by 34–44 dB. IEEE Std 1241 for ADC testing specifies this procedure.

Representative noise and spectrum-analysis figures

Quantity	Typical value
Ideal ADC SNR, 12 / 16 / 24 bit	74.0 / 98.1 / 146.2 dB
Real 24-bit audio ADC SNR (A-weighted)	110–125 dB
Thermal noise density at 290 K	−174 dBm/Hz
Hann ENBW / Blackman–Harris ENBW	1.50 / 2.00 bins
Welch segments needed for ±0.5 dB (1σ) per bin (Hann, 50 %)	about 71
Mains interference in EEG/ECG before filtering	10–1000 µV at 50 or 60 Hz

Table 22.5 — Values used when interpreting FFT-based spectra.

Beyond the classical estimators

Periodogram-based methods make no assumption about the signal and their resolution is limited to about $1/T_{\mathrm{rec}}$. When the record is short (a few radar pulses, a brief speech frame) or when the spectrum is known to consist of a few sharp peaks, **parametric** estimators that fit an autoregressive model, and **subspace** methods such as MUSIC that fit a sum of sinusoids, can resolve components closer than $1/T_{\mathrm{rec}}$. **Multitaper** methods average several orthogonal windows over one record to reduce variance without segmenting. These, together with the time-varying spectrum of non-stationary signals, are the subject of Chapter 24.

> **Status of the principle: Welch averaging**
> *Approximate.* The $1/K$ variance reduction assumes stationarity over the whole record and nearly independent segments; non-stationary signals (a changing machine speed, sleep-stage changes in EEG) violate the first assumption and call for time-frequency methods.

What remains open

Open question	Where it is answered
How quantisation and finite word length affect FFT accuracy and the ADC noise floor	Chapter 21 (finite-precision effects); converter hardware in *Electronics & Computing*
How to design filters whose frequency response is specified on the DFT grid, and how to implement them by fast convolution	Chapter 23
How to resolve components closer than $1/T_{\text{rec}}$, and how to analyse spectra that change with time	Chapter 24 (parametric estimation, spectrograms, wavelets)
How the inverse FFT and cyclic prefix build an OFDM transmitter, and how the receiver equalises per bin	Chapter 30
How noise PSD, noise bandwidth and noise figure determine detection performance	Chapter 29
The probability theory behind PSD estimates and their confidence intervals	*The Mathematics Universe* (random processes)
Why the sampling rate and anti-aliasing filter limit what the DFT can see	Chapter 19 (back reference)

Next: Chapter 23 uses the frequency-domain view built here to specify, design and implement digital filters, FIR and IIR, and to examine what finite word length does to them.

Chapter 23 — Digital Filters

Part V · Digital Signal Processing

A portable ECG monitor is being certified. Its 50 Hz mains-rejection filter worked perfectly in the floating-point simulation on the designer's laptop: a sharp notch, 3 Hz wide, with more than 50 dB of rejection. On the production hardware, an 8-bit microcontroller storing filter coefficients as single bytes, the notch has moved to 48.9 Hz and rejects only 3 dB at 50 Hz. The trace is covered in hum. A second team member proposes replacing the notch with a 16th-order Butterworth lowpass written as one long difference equation; in 16-bit arithmetic that filter oscillates on its own, with no input at all. Neither failure is visible in a textbook transfer function. Both come from the step between a filter as mathematics and a filter as a program running on finite hardware.

A digital filter is a difference equation. Choosing its coefficients so that its frequency response meets a specification is **design**; arranging its arithmetic so that the specification survives finite word length and a real-time budget is **implementation**. This chapter treats both, for the two families that cover nearly all practice: finite-impulse-response (FIR) and infinite-impulse-response (IIR) filters.

> **When should a filter be FIR and when IIR, and how is each designed from a specification?**
> **Why do filters that are stable and correct on paper fail in fixed-point hardware, and how is the arithmetic arranged to prevent it?**

Concept	Engineering destination
FIR filters, linear phase	ECG and EEG front ends, audio crossovers, data-converter decimation, channel filters in software radio
IIR filters	Audio equalisers, anti-hum notches, protection-relay filters, sensor smoothing in control loops
Difference-equation structures	Firmware on DSPs and microcontrollers, FPGA filter cores
Filter specifications	Requirements documents, standards (IEC 61000-4-7, audio codecs, 3GPP spectral masks)
Frequency transformations	Deriving highpass, bandpass and bandstop filters from one lowpass prototype
FIR design (window, Kaiser, Parks–McClellan)	Linear-phase channel and decimation filters
IIR design (bilinear transform)	Digital versions of Butterworth, Chebyshev and elliptic analog filters
Stability	Recursive filters, adaptive filters, digital controllers
Quantisation effects	Fixed-point DSP, low-cost microcontrollers, FPGA word-length choice
Implementation constraints	MAC budgets, latency limits in hearing aids and control loops, memory

Roadmap. FIR filters (23.1) and IIR filters (23.2) as two kinds of difference equation → how each is implemented as a structure (23.3) → what a specification states (23.4) → FIR design (23.5) → IIR design via analog prototypes and the bilinear transform (23.6) → frequency transformations that turn a lowpass into any other type (23.7) → stability (23.8) → what quantisation does (23.9) → the engineering constraints that finally choose the design (23.10).

23.1 FIR filters

Definition

A **finite-impulse-response** filter computes each output as a weighted sum of the present and M past inputs:

$$y[n] = \sum_{k=0}^{M} b_k \, x[n-k].$$

Its impulse response is the coefficient list itself, $h[n] = b_n$ for $0 \le n \le M$ and zero elsewhere, so it has $N = M + 1$ **taps** and **order** M. Its transfer function (Chapter 20) is a polynomial in z^{-1},

$$H(z) = \sum_{k=0}^{M} b_k z^{-k},$$

with M zeros and all M poles at the origin. Three properties follow immediately.

1. **Always stable.** A finite sum of bounded inputs is bounded; there is no feedback, so no coefficient choice can make the filter oscillate.
2. **Exact linear phase is possible.** If the coefficients are symmetric, $b_k = b_{M-k}$, or antisymmetric, $b_k = -b_{M-k}$, the frequency response factorises as $H(e^{j\omega}) = A(\omega)\, e^{-j\omega M/2}$ (times j for antisymmetric), where $A(\omega)$ is real. Every frequency is then delayed by the same $M/2$ samples: the group delay (Chapter 18) is constant and waveform shape is preserved.
3. **Selectivity costs taps.** Because there are no poles away from the origin to create sharp resonances, a narrow transition band needs a long impulse response. Section 23.5 quantifies this: the number of taps grows inversely with the transition width.

The four linear-phase types

Symmetry and the parity of the length give four types with different built-in zeros:

Type	Symmetry	Length N	Forced zero at	Usable for
I	Symmetric	Odd	none	Any response (lowpass, highpass, bandpass, bandstop)
II	Symmetric	Even	$z=-1$ ($f_s/2$)	Lowpass, bandpass; not highpass
III	Antisymmetric	Odd	$z=1$ and $z=-1$	Differentiators, Hilbert transformers, bandpass
IV	Antisymmetric	Even	$z=1$ (DC)	Differentiators, Hilbert transformers, highpass

Table 23.1 — Linear-phase FIR types.

Linear phase matters wherever waveform shape carries information. In an ECG, a filter with non-constant group delay distorts the ST segment, which clinicians read to diagnose ischaemia; in a digital modem, it causes intersymbol interference; in a loudspeaker crossover, it smears transients. The price is latency: the 187-tap linear-phase ECG filter of Section 23.5, running at 1 kHz, delays everything by 93 ms.

> **Status of the principle: Linear phase from coefficient symmetry**
> *Derived.* Symmetric or antisymmetric real coefficients make $H(e^{j\omega})e^{j\omega M/2}$ purely real or purely imaginary, which is exactly constant group delay $M/2$; it holds for any coefficient values, before and after quantisation, because quantisation preserves symmetry.

23.2 IIR filters

Definition

An **infinite-impulse-response** filter also feeds back past outputs:

$$y[n]=\sum_{k=0}^{M} b_k x[n-k]-\sum_{k=1}^{P} a_k y[n-k], \qquad H(z)=\frac{\sum_{k=0}^{M} b_k z^{-k}}{1+\sum_{k=1}^{P} a_k z^{-k}}.$$

The denominator places poles anywhere in the z-plane. A pole close to the unit circle at angle ω_0 creates a sharp resonance near ω_0; a zero on the unit circle creates an exact null. With a handful of coefficients an IIR filter achieves selectivity that would take an FIR filter tens or hundreds of taps. The impulse response decays geometrically but never ends, hence the name.

The properties are the mirror image of FIR:

1. **Efficient.** The 50 Hz notch of the opening needs five coefficients; an FIR filter with a comparable 3 Hz notch at 1 kHz sampling would need several hundred taps.
2. **Stability is a condition, not a guarantee.** All poles must lie strictly inside the unit circle (Section 23.8).

3. **Phase is nonlinear.** A causal IIR filter cannot have exactly linear phase (that would need its poles mirrored outside the unit circle). Group delay peaks near the band edges, where the poles are closest to the circle.
4. **Finite word length matters more**, because errors recirculate through the feedback (Section 23.9).

Most IIR filters are digital versions of classical analog filters, which is why the analog-filter knowledge of Chapter 13 carries over directly (Section 23.6).

A common IIR example: the notch. Placing zeros on the unit circle at $\pm\omega_0$ and poles just inside at radius r gives

$$H(z) = \frac{1 - 2\cos\omega_0\, z^{-1} + z^{-2}}{1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2}}.$$

The zeros null ω_0 exactly; the poles cancel the zeros' effect away from ω_0, so the response is flat elsewhere. The 3 dB width of the notch is approximately $2(1-r)$ rad/sample, or $(1-r)f_s/\pi$ in hertz. For $f_0 = 50$ Hz at $f_s = 1$ kHz with $r = 0.99$: $\omega_0 = 2\pi \times 0.05$, the denominator coefficients are $a_1 = -2 \times 0.99\cos(0.1\pi) = -1.88309$ and $a_2 = 0.9801$, and the notch is about 3.2 Hz wide. At 49 Hz the attenuation is already only 5.4 dB, and at DC the gain is +0.08 dB. This is the filter whose 8-bit failure opened the chapter.

23.3 Difference-equation implementations

The transfer function fixes what a filter does; the **structure**, the arrangement of delays, multipliers and adders, fixes how the arithmetic is done. In exact arithmetic all structures with the same $H(z)$ are identical; in finite arithmetic they differ greatly.

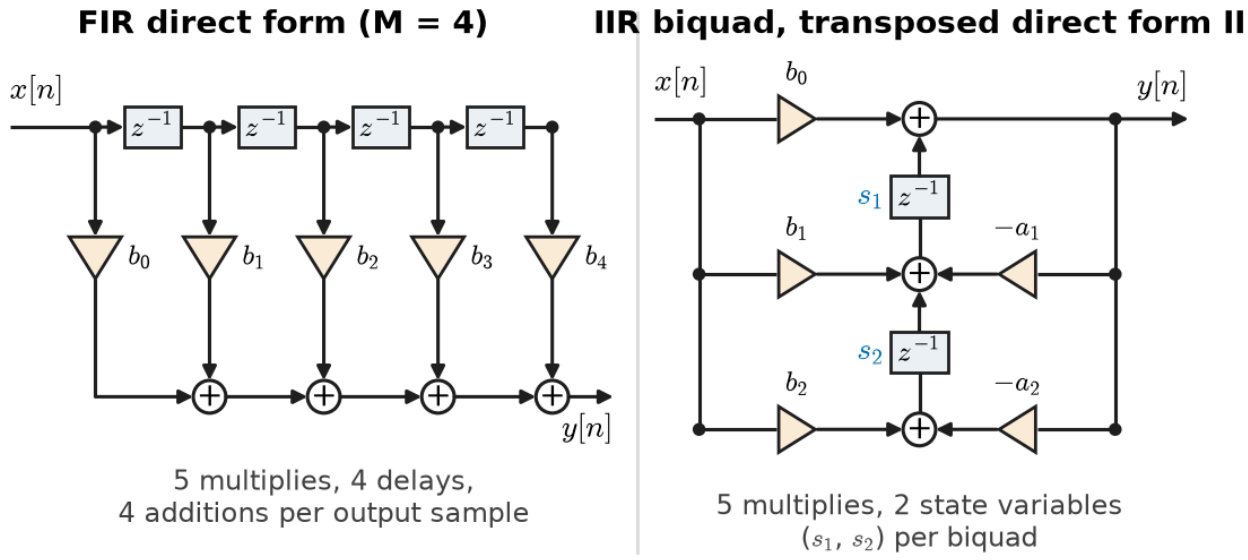

Figure 23.1 — Two basic structures: an FIR tapped delay line and a second-order IIR section in transposed direct form II.

FIR structures

The **direct form** (tapped delay line, Figure 23.1, left) is the definition drawn as hardware: M delays, $M+1$ multipliers, one accumulator. Digital signal processors are built around it: a single-cycle multiply–accumulate (MAC) unit, a circular buffer for the delay line and a zero-overhead loop. For a linear-phase filter, the symmetric structure adds $x[n-k]+x[n-M+k]$ before multiplying, halving the multiplications. The **transposed** form moves the delays to the output side, which suits FPGAs because every adder is followed by a register and the critical path is one multiply and one add, whatever the length. **Polyphase** decomposition, which splits the coefficients into interleaved sub-filters for decimation and interpolation, is developed in Chapter 24, and **fast convolution** via the FFT in Chapter 22.

IIR structures

- **Direct form I** implements the numerator and denominator as two tapped delay lines, $M+P$ delays in total. Its single accumulator sees every product, which makes it robust in fixed point when the accumulator is wide.
- **Direct form II** reverses the order (poles first, then zeros) so that the two delay lines merge into one of length $\max(M,P)$: the canonical form with the fewest delays.
- **Transposed direct form II** (Figure 23.1, right) is the usual choice in floating-point code, because it adds numbers of similar size and has good numerical behaviour.
- **Cascade (second-order sections)**: factor $H(z)$ into biquads, $H(z)=g\prod_i H_i(z)$, each with two poles and two zeros, and run them one after another. This is the standard for any IIR filter above second order.
- **Parallel form**: expand $H(z)$ by partial fractions (Chapter 11) into a sum of biquads, run them side by side and add the outputs.
- **Lattice and wave-digital** structures use reflection coefficients and have very low coefficient sensitivity; they appear in speech coding (linear prediction) and in high-end audio.

The rule that matters in practice: **never implement an IIR filter above second order as a single direct-form difference equation.** The poles of a high-order polynomial are extremely sensitive to its coefficients (Section 23.9). Factor into second-order sections, pair each pole pair with the nearest zero pair, and order the sections from the lowest-Q to the highest-Q pole pair, or in the reverse order if the concern is noise rather than overflow.

23.4 Filter specifications

The tolerance scheme

A filter requirement is not a single curve but a set of bounds (Figure 23.2):

- **passband** edge f_p and ripple: the gain must stay within $1\pm\delta_p$, often given as a peak-to-peak ripple $A_p=20\log_{10}\frac{1+\delta_p}{1-\delta_p}$ dB;
- **stopband** edge f_{st} and attenuation: the gain must stay below δ_s, given as $A_s=-20\log_{10}\delta_s$ dB;
- the **transition band** between them, where the response is unconstrained;

- the **sampling rate** f_s, because a digital filter's response is defined on normalised frequency f/f_s ;
- **phase** requirements: linear phase, a maximum group-delay variation, or a maximum total latency.

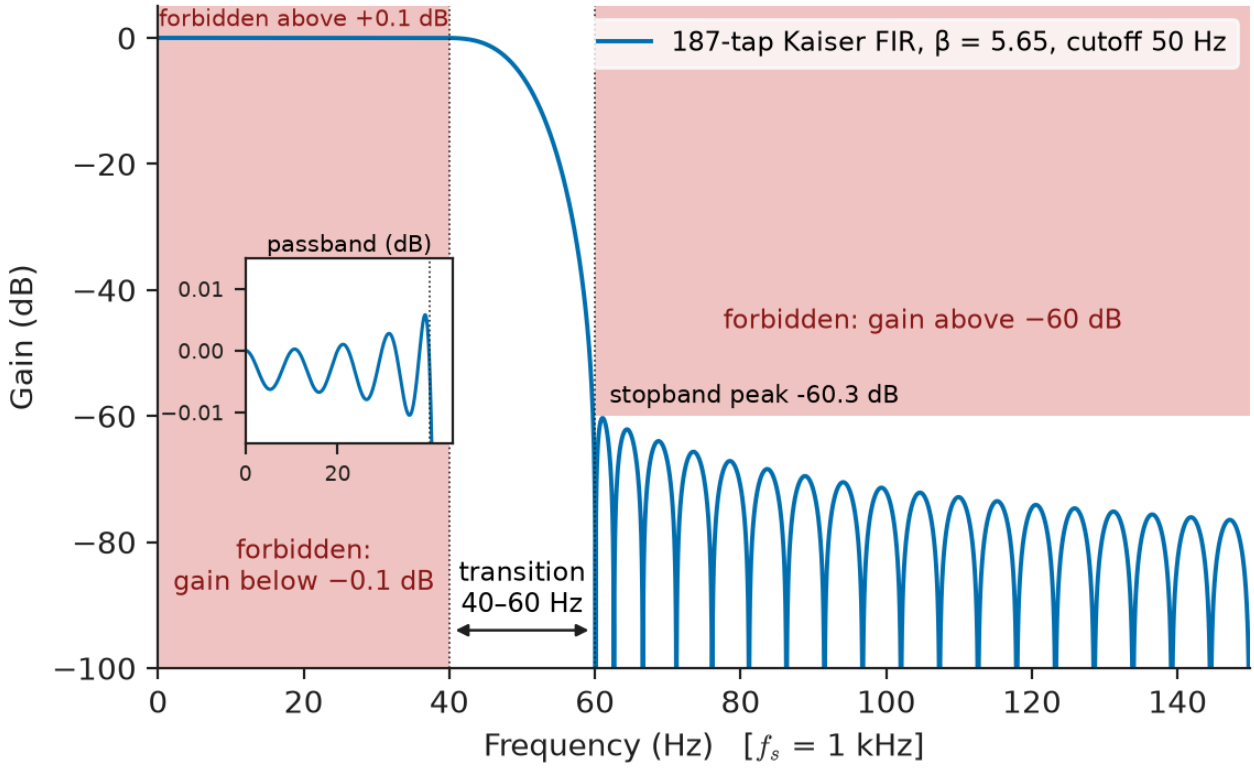

Figure 23.2 — Lowpass tolerance scheme for the ECG filter of the worked examples, with the Kaiser-window FIR response threading the allowed region.

A digital filter is a periodic function of frequency with period f_s, so a lowpass specification up to $f_s/2$ fully defines it. Table 23.2 lists representative specifications from practice.

Application	f_s	Passband	Stopband	Ripple / attenuation	Phase
Diagnostic ECG lowpass	1 kHz	0–40 Hz	> 60 Hz	±0.1 dB / 60 dB	Linear
Audio ADC decimation filter	48 kHz out	0–20 kHz	> 24 kHz	±0.005 dB / 100 dB	Linear
Telephone-band channel filter	8 kHz	300–3400 Hz	< 200 Hz, > 4 kHz	0.5 dB / 40 dB	Not critical
Power-system protection relay (fundamental extraction)	1–4 kHz	50 or 60 Hz	DC offset and harmonics	1 dB / 40–60 dB	Fixed delay within one cycle
Audio graphic-equaliser band	48 kHz	Octave around centre	none (bell shape)	±12 dB gain	Minimum phase acceptable
Seismic and vibration anti-alias decimation	100 Hz–25.6 kHz	0.4 f_s	0.5 f_s	0.1 dB / 120 dB	Linear

Table 23.2 — Representative digital-filter specifications. Values are typical, not normative.

Converting between units

The conversions are worth having at hand. A peak deviation $\delta_p = 0.0115$ is a ripple of ± 0.099 dB, or 0.20 dB peak to peak. A 60 dB stopband means $\delta_s = 0.001$; 100 dB means 10^{-5}. A 1 dB peak-to-peak passband ripple corresponds to $\delta_p = (10^{1/20} - 1)/(10^{1/20} + 1) = 0.0575$. A digital filter designer works with δ values, and a requirements document is written in decibels.

What a specification implies

Three rules of thumb guide the choice before any design starts. The **transition width** relative to f_s dominates cost: halving it doubles an FIR filter's length and adds about a third to an IIR filter's order. **Stopband attenuation** adds cost roughly linearly in decibels. And a **linear-phase** requirement effectively decides for FIR, unless the latency budget forbids it.

23.5 FIR design

The ideal filter and the window method

The ideal lowpass with cutoff ω_c has the impulse response

$$h_d[n] = \frac{\sin(\omega_c n)}{\pi n}, \qquad -\infty < n < \infty,$$

which is infinitely long and non-causal. The **window method** truncates it to $N = M + 1$ samples centred on $M/2$, multiplies by a window $w[n]$, and delays by $M/2$ to make it causal:

$$h[n] = w[n]\,\frac{\sin\!\big(\omega_c (n - M/2)\big)}{\pi (n - M/2)}, \qquad n = 0, \dots, M.$$

The frequency response is the ideal brick wall convolved with the window's transform (the same mechanism as leakage in Chapter 22). The window's main-lobe width sets the transition width; its side lobes set the passband ripple and stopband attenuation, which come out approximately equal ($\delta_p \approx \delta_s$). Rectangular truncation gives only 21 dB of stopband attenuation no matter how long the filter, because of the Gibbs phenomenon; Hann gives about 44 dB, Hamming 53 dB and Blackman 74 dB.

The Kaiser window method

Kaiser's empirical formulas turn a specification straight into a design. With $A = -20\log_{10}\min(\delta_p, \delta_s)$ and transition width $\Delta\omega = 2\pi(f_{st} - f_p)/f_s$,

$$\beta = \begin{cases} 0.1102(A - 8.7), & A > 50 \\ 0.5842(A-21)^{0.4} + 0.07886(A-21), & 21 \le A \le 50 \\ 0, & A < 21 \end{cases} \qquad M \approx \frac{A - 8}{2.285\,\Delta\omega}.$$

> **Status of the principle: Kaiser design formulas**
> *Empirical.* Fitted by Kaiser (1974) to many computed designs; the order estimate is usually within a few taps, and the result must always be checked against the specification and lengthened if it falls short.

Worked example — a linear-phase ECG lowpass. A diagnostic ECG sampled at $f_s = 1$ kHz must pass 0–40 Hz with negligible ripple, reject everything above 60 Hz (which includes the mains interference and most muscle noise) by 60 dB, and keep exactly linear phase so that the ST segment is not distorted.

Parameters. $A = 60$ dB, $\delta = 10^{-3}$. $\Delta\omega = 2\pi \times 20/1000 = 0.12566$ rad/sample. $\beta = 0.1102 \times (60 - 8.7) = 5.653$.

Length. $M \approx (60 - 8)/(2.285 \times 0.12566) = 181.1$, so $M = 182$ and $N = 183$ taps (odd, Type I). The cutoff is placed mid-transition, at 50 Hz.

Check. Evaluating the designed response shows the passband deviates by at most 0.011 dB and the stopband peaks at −59.4 dB, just short of the target, which is typical of the formula. Lengthening to $N = 187$ taps gives −60.3 dB. Figure 23.2 shows the result.

Cost. With the symmetric structure the filter needs 94 multiplications per output sample, 94 000 per second, a trivial load. Its group delay is $186/2 = 93$ samples = 93 ms. For a bedside monitor that delay is invisible; in a closed-loop pacing or defibrillation-timing application it might not be acceptable.

Optimal equiripple design (Parks–McClellan)

The window method wastes taps: its error is largest near the band edges and needlessly small elsewhere, and it cannot set δ_p and δ_s independently. The **Parks–McClellan** algorithm (the Remez exchange) finds the filter of given length that minimises the maximum weighted error,

$$\min_{h} \max_{\omega \in \text{bands}} W(\omega)\,\bigl|H(e^{j\omega}) - D(\omega)\bigr|,$$

where D is the desired response and W the weighting. The result is **equiripple**: the error oscillates with equal amplitude across each band. Weighting the stopband by δ_p/δ_s lets the designer ask for a loose passband and a tight stopband. Kaiser's estimate of the required length is

$$N \approx \frac{-20\log_{10}\sqrt{\delta_p\delta_s} - 13}{14.6\,(f_{st} - f_p)/f_s} + 1.$$

For the ECG specification with a ±0.1 dB passband ($\delta_p = 0.0115$) and $\delta_s = 10^{-3}$, the estimate is 126 taps, and the actual minimum found by the algorithm is 135 taps, about 28 % fewer than the 187-tap Kaiser design. Parks–McClellan is the default in every filter-design package.

Other FIR methods

- **Frequency sampling** specifies H at the N DFT frequencies and takes the inverse DFT; transition samples are optimised to lower the ripple. It suits filters with many narrow bands and implementations that already work in the DFT domain.
- **Least-squares** design minimises the integrated squared error instead of the peak error; the stopband energy is lower but the peak ripple is higher. Audio and image filters often prefer it.
- **Half-band** filters (cutoff at $f_s/4$, symmetric ripple) have almost every second coefficient exactly zero, which halves their cost; they are the workhorse of factor-of-2 decimation (Chapter 24).

23.6 IIR design

Starting from analog prototypes

Decades of analog filter theory (Chapter 13) have produced closed-form lowpass prototypes. IIR design reuses them: design an analog filter $H_a(s)$ that meets a (warped) specification, then map it to $H(z)$. The four classical prototypes differ in where they put their error:

Prototype	Passband	Stopband	Roll-off for given order	Phase / group delay
Butterworth	Maximally flat, monotonic	Monotonic	Slowest	Moderate group-delay peak
Chebyshev type I	Equiripple	Monotonic	Faster	Larger peak near the edge
Chebyshev type II (inverse)	Monotonic, flat	Equiripple	Faster	Moderate
Elliptic (Cauer)	Equiripple	Equiripple	Fastest (minimum order)	Largest group-delay variation
Bessel (analog)	Monotonic	Monotonic	Slow	Near-constant delay (not preserved by bilinear mapping)

Table 23.3 — Classical analog prototypes used for IIR design.

The bilinear transform

The standard mapping replaces s by

$$s = \frac{2}{T}\,\frac{1 - z^{-1}}{1 + z^{-1}}, \qquad T = \frac{1}{f_s}.$$

It maps the whole left half of the s-plane into the interior of the unit circle and the $j\Omega$ axis onto the unit circle, one to one. Two consequences follow. First, **stability is preserved**: a stable analog prototype always gives a stable digital filter. Second, **there is no aliasing**, since the infinite analog frequency axis is squeezed onto $0 \le \omega < \pi$. The squeeze is nonlinear: analog frequency Ω and digital frequency ω are related by

$$\Omega = \frac{2}{T}\tan\frac{\omega}{2}.$$

This **frequency warping** is harmless for piecewise-constant magnitude specifications if the critical frequencies are **pre-warped**: compute the analog edge frequencies from the desired digital ones with this formula, design the analog prototype at those frequencies, then transform.

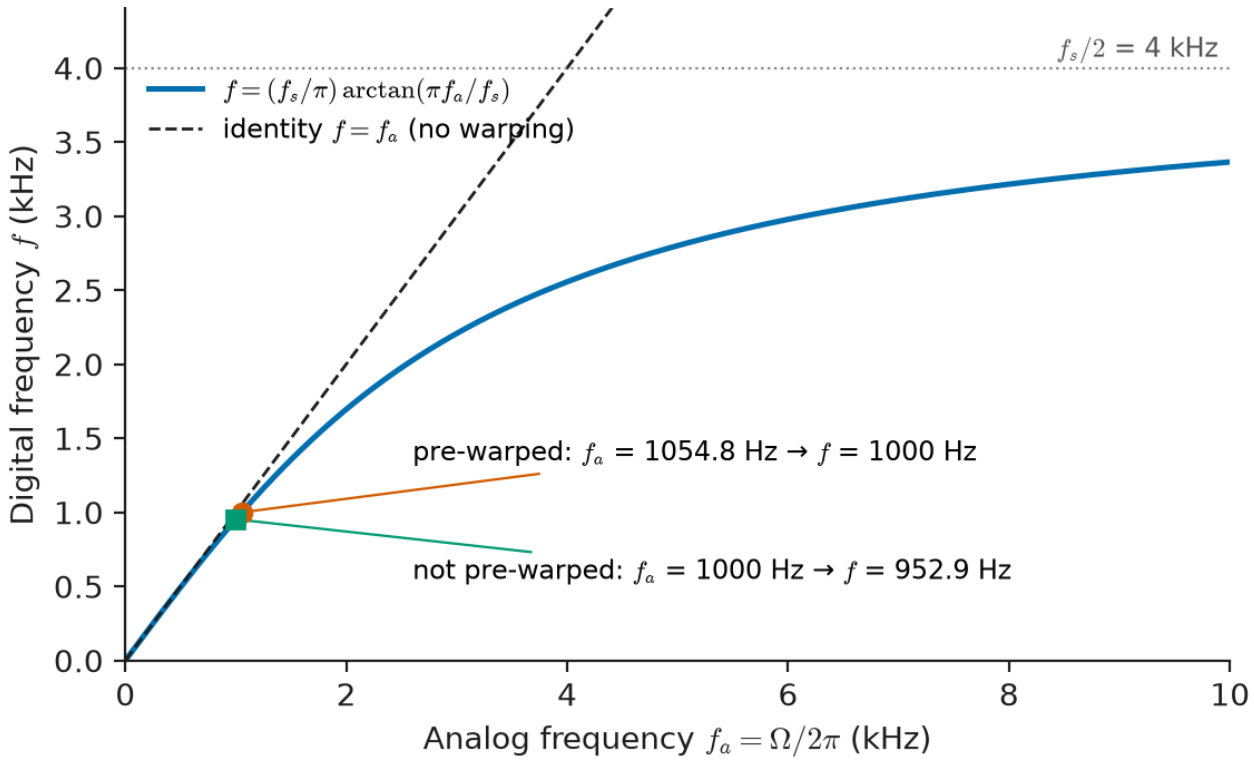

Figure 23.3 — Frequency warping of the bilinear transform at fs = 8 kHz: digital frequencies are compressed toward fs/2, so band edges must be pre-warped.

> **Status of the principle: Bilinear transform**
> *Derived.* An exact conformal map from the s-plane to the z-plane; it preserves stability, order and the magnitude values of the analog response, but relocates every frequency according to $\Omega = (2/T)\tan(\omega/2)$, so it preserves shapes only on a warped frequency axis and does not preserve the analog filter's phase or group-delay characteristics.

Worked example — a second-order Butterworth lowpass by bilinear transform. A protection relay samples current at $f_s = 8$ kHz and needs a second-order Butterworth lowpass with a 1 kHz cutoff to remove switching noise before its phasor estimator.

Pre-warp. $K = \tan(\pi f_c/f_s) = \tan(\pi/8) = 0.41421$. The analog prototype is therefore designed at $\Omega_c = 2f_sK = 6627.4$ rad/s (1054.8 Hz) rather than $2\pi \times 1000 = 6283.2$ rad/s.

Transform. For a second-order lowpass $H_a(s) = \Omega_c^2/(s^2 + (\Omega_c/Q)s + \Omega_c^2)$ with $Q = 1/\sqrt{2}$, substituting the bilinear transform and dividing through gives, with $D = 1 + K/Q + K^2 = 1 + 0.58579 + 0.17157 = 1.75736$,

$$b_0 = b_2 = \frac{K^2}{D} = 0.097631, \quad b_1 = 2b_0 = 0.195262, \quad a_1 = \frac{2(K^2-1)}{D} = -0.942809, \quad a_2 = \frac{1 - K/Q +}{D}$$

Check. The poles are at $0.47140 \pm j0.33333$, radius 0.5774, well inside the unit circle. The DC gain is $(b_0 + b_1 + b_2)/(1 + a_1 + a_2) = 0.390524/0.390524 = 1$, and the gain at 1 kHz is exactly −3.01 dB. The double zero at $z = -1$ gives infinite attenuation at 4 kHz, the image of $\Omega = \infty$.

Without pre-warping. Had the analog prototype been designed at 1000 Hz and transformed directly, the digital cutoff would have landed at $(f_s/\pi)\arctan(\pi \times 1000/8000) = 952.9$ Hz, a 4.7 % error (Figure 23.3).

Choosing the prototype and order

For a given specification, the four prototypes need very different orders. Figure 23.4 compares them for a lowpass at $f_s = 8$ kHz with 1 dB passband ripple to 1 kHz and 60 dB attenuation from 1.5 kHz.

Design	Order or length	Multiplications per sample (approx.)	Group delay in passband
Butterworth IIR	16	40 (8 biquads)	Large peak near 1 kHz
Chebyshev I IIR	8	20 (4 biquads)	Larger peak
Chebyshev II IIR	8	20 (4 biquads)	Moderate
Elliptic IIR	6	15 (3 biquads)	4.2 to 39 samples
Parks–McClellan FIR	36 taps	18 (symmetric)	17.5 samples, constant

Table 23.4 — Order needed by each design method for the same specification ($f_s = 8$ kHz, 1 dB / 60 dB, 1.0–1.5 kHz transition).

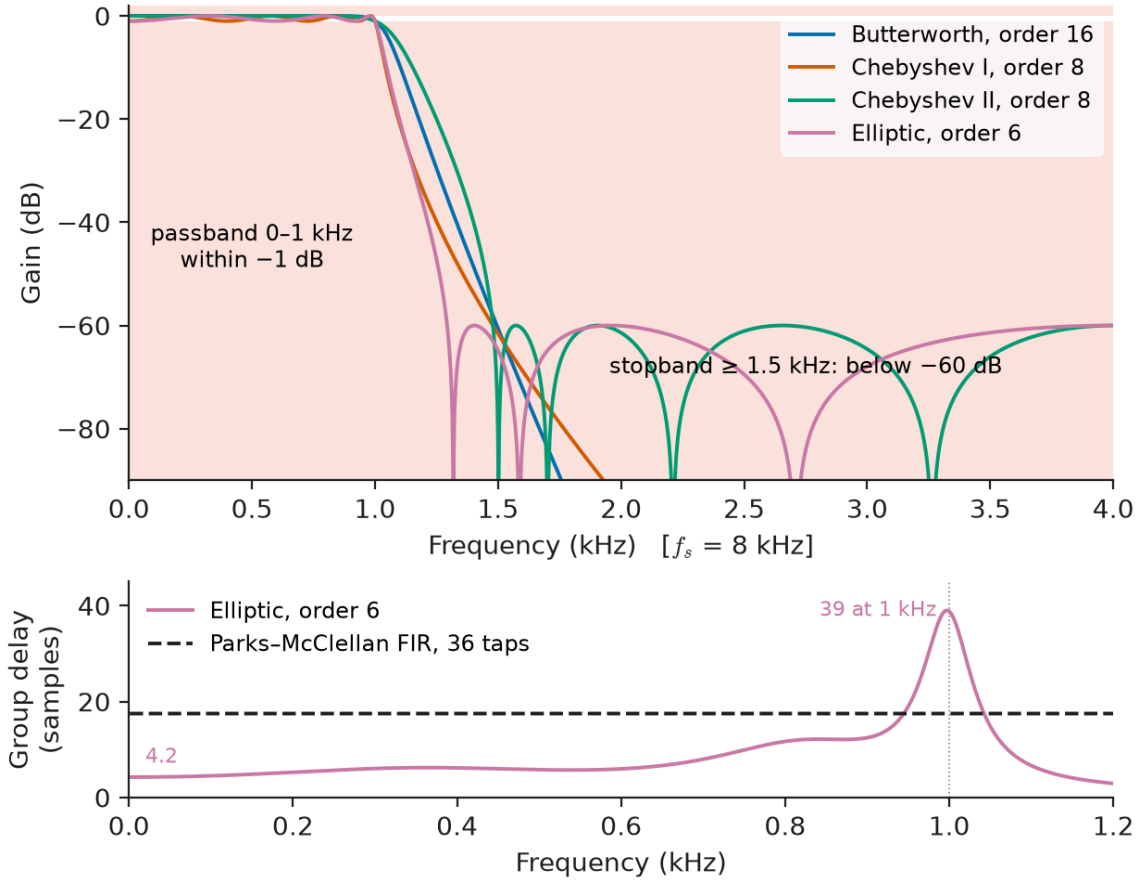

Figure 23.4 — Magnitude responses of order-16 Butterworth, order-8 Chebyshev I, order-8 Chebyshev II and order-6 elliptic filters meeting the same specification.

The elliptic filter is the cheapest in arithmetic, but its group delay varies by a factor of nine across the passband. The FIR filter with the same magnitude specification is only slightly more expensive here because the transition is relatively wide; for the narrow transition of the ECG example the gap is far larger.

Impulse invariance and other mappings

Impulse invariance samples the analog impulse response, $h[n] = T\,h_a(nT)$, which maps each analog pole s_i to $z_i = e^{s_i T}$. It preserves the time-domain shape exactly (useful when modelling a physical system's impulse response, for example a room or a sensor), but the frequency response is the aliased sum of the analog response, so it is unusable for highpass and bandstop filters and poor for lowpass filters unless the analog response is well attenuated by $f_s/2$. For a first-order RC lowpass with a 1 kHz cutoff at $f_s = 8$ kHz, the impulse-invariant filter has an attenuation at 4 kHz of only 8.5 dB relative to DC, while the analog filter at 4 kHz has 12.3 dB: aliasing has filled in the stopband. The **matched-z** transform also maps poles and zeros by $z = e^{sT}$; it is simple but has similar aliasing errors. Direct **computer optimisation** of IIR coefficients against an arbitrary response is also possible, with stability enforced as a constraint.

23.7 Frequency transformations

A designer rarely starts from scratch for a highpass, bandpass or bandstop filter. A normalised lowpass prototype (cutoff 1 rad/s) is designed once and transformed. There are two routes.

Analog transformations before the bilinear map

Substituting for s in the lowpass prototype $H_{LP}(s)$:

Target	Substitution	Order
Lowpass, cutoff Ω_c	$s \to s/\Omega_c$	n
Highpass, cutoff Ω_c	$s \to \Omega_c/s$	n
Bandpass, centre Ω_0, width B	$s \to \dfrac{s^2+\Omega_0^2}{Bs}$	$2n$
Bandstop, centre Ω_0, width B	$s \to \dfrac{Bs}{s^2+\Omega_0^2}$	$2n$

Table 23.5 — Analog frequency transformations. $\Omega_0 = \sqrt{\Omega_1\Omega_2}$ (geometric centre) and $B = \Omega_2 - \Omega_1$ for edges Ω_1, Ω_2.

Each band edge must be pre-warped first and the transformed analog filter then mapped by the bilinear transform. Bandpass and bandstop transformations double the order: a second-order prototype becomes a fourth-order (two-biquad) bandpass.

Digital (Constantinides) transformations

Alternatively, a digital lowpass with cutoff θ_p can be transformed directly by replacing z^{-1} with an all-pass function of z^{-1}. For lowpass to lowpass with new cutoff ω_p,

$$z^{-1} \to \frac{z^{-1}-\alpha}{1-\alpha z^{-1}}, \qquad \alpha = \frac{\sin\big((\theta_p-\omega_p)/2\big)}{\sin\big((\theta_p+\omega_p)/2\big)};$$

lowpass to highpass uses $z^{-1} \to -(z^{-1}+\alpha)/(1+\alpha z^{-1})$ with a cosine ratio for α, and lowpass to bandpass and bandstop use second-order all-pass substitutions. Because the substitution is all-pass, the unit circle maps onto itself and stability is preserved; the magnitude values (ripple, attenuation) are preserved and only the frequency axis is redrawn.

The practical value is **tunability**. An audio equaliser, a tracking filter that follows a machine's rotation speed, or a variable-cutoff synthesiser filter can recompute a single parameter α rather than redesigning the filter. The widely used "audio EQ cookbook" biquads (peaking, shelving, notch) are bilinear transforms of second-order analog prototypes with closed-form coefficients of exactly this kind.

> **Status of the principle: Frequency transformations**
> *Derived*. Analog substitutions are rational maps of the $j\Omega$ axis onto itself and Constantinides' maps are all-pass maps of the unit circle onto itself; both preserve the magnitude tolerance values and stability exactly, while bandpass and bandstop maps double the order.

23.8 Stability

The pole condition

A causal LTI filter is bounded-input bounded-output stable if and only if its impulse response is absolutely summable (Chapter 17). For a rational $H(z)$ this is equivalent to:

$$|p_i| < 1 \quad \text{for every pole } p_i.$$

FIR filters satisfy it trivially. For IIR filters it must be checked, and the bilinear transform guarantees it for any stable analog prototype. For a second-order section $1 + a_1 z^{-1} + a_2 z^{-2}$ the condition reduces to the **stability triangle**

$$|a_2| < 1, \qquad |a_1| < 1 + a_2,$$

which is easy to verify for each biquad in a cascade. The notch of Section 23.2 ($a_1 = -1.88309$, $a_2 = 0.9801$) lies inside the triangle with a margin of only 0.097 on the second inequality; its poles sit at radius 0.99, and a narrow notch or a sharp resonance always lives close to the edge.

Where stability is lost in practice

A filter designed stable can become unstable in implementation in three ways.

1. **Coefficient quantisation** moves poles. In a high-order direct form the movement can carry a pole across the unit circle (Section 23.9).
2. **Nonlinear arithmetic**: overflow in two's complement wraps a large positive number to a large negative one, which can sustain large-amplitude oscillations; rounding can sustain small **limit cycles**. These are not instabilities of the linear model but behaviours of the nonlinear real system.
3. **Time variation**: adaptive filters (Chapter 24) and filters whose coefficients are retuned while running can go unstable even when every frozen-time version is stable. Smooth coefficient interpolation and lattice or state-variable structures mitigate this.

For the filters inside a control loop, stability of the filter itself is only part of the story; the stability of the closed loop that contains it is the subject of Chapters 34 and 35.

Status of the principle: Pole-radius stability criterion
Derived. Follows from partial-fraction expansion of a rational $H(z)$: each pole contributes a term p_i^n to the impulse response, which is summable only if $|p_i| < 1$; it applies to the linear model with exact arithmetic.

23.9 Quantization effects

A digital filter running in B-bit fixed point differs from its design in four ways, each of which can be analysed separately.

Coefficient quantisation

Coefficients stored with a quantisation step q are rounded by up to $q/2$. The filter that runs is a slightly different, but still linear, filter. Poles and zeros move, and the magnitude response changes. The sensitivity is greatest where roots are clustered: a pole pair at small angle ω_0 near the unit circle, or many poles of a high-order filter near the passband edge.

Worked example — the 50 Hz notch in 16, 12 and 8 bits. Take the notch of Section 23.2 ($f_0 =$ 50 Hz, $f_s = 1$ kHz, $r = 0.99$). Its coefficients lie between −2 and +2, so a fixed-point format with two integer bits is used, and a B-bit word has step $q = 2^{-(B-2)}$. The numerator coefficients 1 and 1 are exact, so the zeros stay on the unit circle; only their angle can move.

Word length	b_1 (ideal −1.902113)	a_1 (ideal −1.883092)	Notch (zero) frequency	Pole radius	Attenuation at 50 Hz
16 bits ($q = 2^{-14}$)	−1.902100	−1.883118	50.003 Hz	0.99000	53 dB
12 bits ($q = 2^{-10}$)	−1.902344	−1.882813	49.94 Hz	0.99019	28 dB
8 bits ($q = 2^{-6}$)	−1.906250	−1.890625	48.92 Hz	0.99216	3.1 dB

Table 23.6 — Effect of coefficient word length on a narrow 50 Hz notch (computed by rounding the coefficients and re-solving for roots).

Figure 23.5 shows the root movements. At 8 bits the notch moves 1.08 Hz, which is a third of its own 3 Hz width, and the 50 Hz hum passes almost untouched: the opening failure. The poles also move to 49.10 Hz with a radius of 0.99216, narrowing the notch to 2.5 Hz. Sixteen-bit coefficients put the null within 0.003 Hz of 50 Hz. The structural lesson is that low-frequency, narrow features at a high sampling rate are hard in fixed point. Remedies are longer coefficients, a lower sampling rate for the filter (decimate first), or structures in which coefficients map more uniformly to pole positions (the coupled or Gold–Rader form, the lattice).

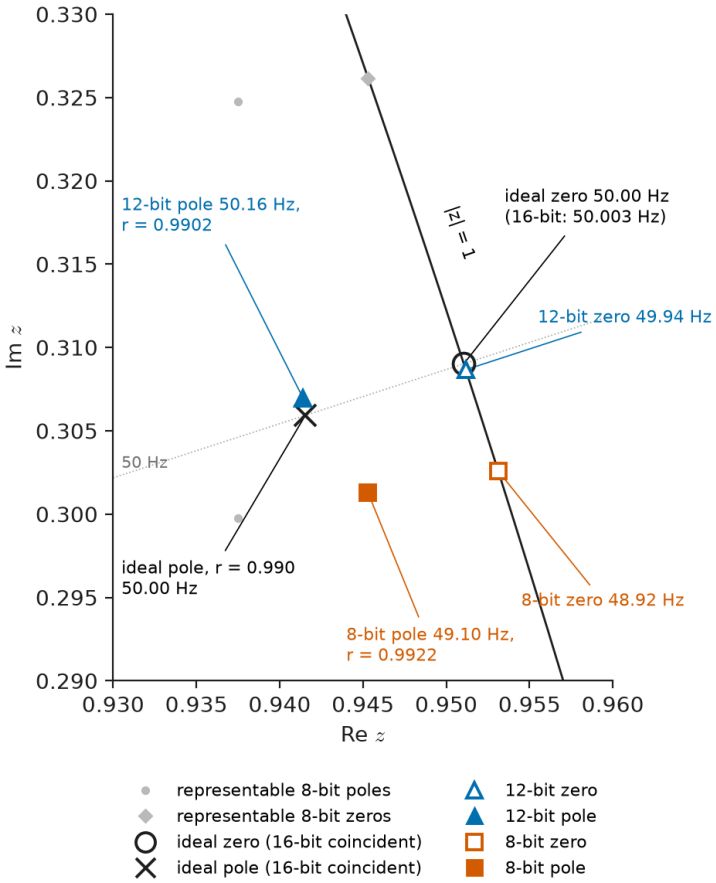

Figure 23.5 — Pole and zero positions of the 50 Hz notch near z = 1 with 16-bit, 12-bit and 8-bit coefficients, showing the 8-bit null displaced to 48.9 Hz.

High-order direct form. The same mechanism ruins high-order filters. The 16th-order Butterworth of Table 23.4 has a largest pole radius of 0.931. Written as a single direct-form denominator, its coefficients reach magnitude 168; rounded to 24-bit words (scaled to that range) the largest pole radius becomes about 0.93–0.95 (the exact value depends on the scaling convention), already off design, and at 16 bits it moves outside the unit circle (about 1.1–1.25): the filter is unstable and oscillates without input. Implemented as eight second-order sections with 16-bit coefficients, every pole stays within 0.01 % of its designed radius. This is why the cascade rule of Section 23.3 is not a matter of style.

> **Status of the principle: Root sensitivity of high-order polynomials**
> *Derived*. The derivative of a root with respect to a polynomial coefficient is inversely proportional to the product of its distances to all other roots, so clustered roots of a high-order polynomial are extremely sensitive; factoring into low-order sections removes the cluster from each polynomial.

Round-off noise

Every product of two B-bit numbers has $2B$ bits; storing it back in B bits rounds it, with an error uniformly distributed over $\pm q/2$ under the usual assumptions (Chapter 21), of variance

$$\sigma_e^2 = \frac{q^2}{12}.$$

For a Q15 word ($q = 2^{-15}$) this is 7.76×10^{-11}, or −101.1 dB relative to a full-scale value of 1. In an FIR filter each such error reaches the output once. In an IIR filter it is injected inside the feedback loop and amplified by the **noise gain** of the path from the injection point to the output, $\sum_n h_e^2[n]$. For the notch's recursive part $1/A(z)$ that gain is 265.5 (24.2 dB), because poles at radius 0.99 ring for hundreds of samples. If two products are rounded separately before the feedback, the output noise is about $2 \times 265.5 \times 7.76 \times 10^{-11} = 4.1 \times 10^{-8}$, or −73.8 dB: 27 dB, or four and a half bits, worse than the converter. The fix is the standard DSP practice of accumulating all products in a double-width (32- or 40-bit) accumulator and rounding once, and, for very narrow filters, using **error feedback** (noise shaping) or a higher-precision state.

Overflow and scaling

Fixed-point numbers have a limited range, and internal nodes of a filter can exceed the input's range even when the output does not: a resonant section has internal gain approaching $1/(1-r)$. Overflow must be prevented by **scaling** the input to each section. The L_1 norm of the impulse response to a node guarantees no overflow for any input; the L_∞ norm of the frequency response guarantees it for sinusoids; the L_2 norm controls it in an average sense for random inputs. Each choice trades overflow probability against signal-to-noise ratio. Where overflow cannot be excluded, **saturation arithmetic** (clamping at full scale) replaces two's-complement wraparound, which would otherwise produce large-amplitude oscillations.

Limit cycles

Rounding inside a feedback loop is nonlinear, and it can sustain small oscillations with zero input. For a first-order section $y[n] = Q(a\,y[n-1])$, rounding cannot shrink the output once $|y| \le q/\big(2(1-|a|)\big)$, the **deadband**. With $a = 0.9$ the deadband is $\pm 5q$: the output can stick at a nonzero value (or alternate in sign if $a < 0$) forever. In audio such a limit cycle is an audible idle tone in silence; in a control system it is a small steady oscillation of an actuator. Remedies are magnitude truncation instead of rounding, dither, or a wider internal word.

23.10 Implementation constraints

The final choice of filter is made against the hardware, not only the specification. Table 23.7 summarises the classical trade-off.

Criterion	FIR	IIR
Arithmetic for sharp transitions	High (tens to thousands of taps)	Low (a few biquads)
Phase	Exactly linear possible	Nonlinear; group delay peaks at band edges
Latency	Half the length for linear phase	A few samples (minimum phase)
Stability	Always	Must be checked; can be lost in quantisation
Sensitivity to word length	Low	High for narrow or high-order designs
Round-off noise	Low, no recirculation	Amplified by the noise gain
Adaptation (Chapter 24)	Simple, well-behaved	Difficult (stability during update)
Multirate efficiency	Polyphase saves a factor M or L	No comparable saving
Memory	N coefficients and N states	5 coefficients and 2 states per biquad

Table 23.7 — FIR versus IIR in implementation.

Arithmetic budget

The cost measure is multiply–accumulates per second: taps (or 5× biquads) times sampling rate times channels. Representative capabilities are given in Table 23.8.

Platform	Typical sustained MAC rate	Typical word length
8-bit microcontroller (software multiply)	0.5–2 M MAC/s	8/16-bit
32-bit Cortex-M4/M7 microcontroller	50–500 M MAC/s	16/32-bit fixed, 32-bit float
Audio DSP (SHARC, HiFi class)	1–10 G MAC/s	32/40-bit float or fixed
Mid-range FPGA (hundreds of DSP slices)	50–500 G MAC/s	18×27-bit multipliers
Hearing-aid DSP	10–100 M MAC/s at milliwatt power	16–24-bit fixed

Table 23.8 — Indicative processing capability. Actual figures depend on clock, memory bandwidth and code efficiency.

Worked example — choosing a crossover filter for an active loudspeaker. An active two-way loudspeaker processes stereo audio at 48 kHz on a Cortex-M7 microcontroller budgeted for 60 M MAC/s for filtering. The woofer–tweeter crossover needs a steep lowpass and highpass at 2.5 kHz.

FIR option. A linear-phase design with 255 taps per filter gives good steepness and perfect phase alignment between drivers. Cost: $255 \times 48\,000 = 12.24$ M MAC/s per filter; two filters (low and high) per channel and two channels give 49 M MAC/s, which fits, using 82 % of the budget. Its latency is $127/48\,000 = 2.6$ ms, acceptable for playback but not for a live monitor where the performer hears both acoustic and amplified sound.

IIR option. A fourth-order Linkwitz–Riley crossover (two cascaded second-order Butterworth sections per output) costs $2 \times 5 = 10$ MACs per sample per output, so $10 \times 48\,000 \times 4 = 1.92$ M MAC/s in total, with less than 1 ms of delay. Its phase is not linear, but the low- and high-pass outputs remain in phase with each other at all frequencies, so the drivers sum flat in amplitude.

Decision. The IIR crossover uses 3 % of the budget and leaves room for room-correction filters; the FIR option is reserved for products that advertise phase linearity. Both meet the magnitude specification; the constraints decide.

Latency, memory and word length

- **Latency** limits are tight in hearing aids (a few milliseconds before the delayed sound and direct sound comb-filter audibly), in active noise control (the filter must act before the sound arrives), in protection relays (a fault decision within one cycle) and in control loops (Chapter 35), where every sample of delay costs phase margin.
- **Memory** for a long FIR filter is N coefficients plus N delay-line samples per channel; on small microcontrollers this, not arithmetic, can be the binding constraint.
- **Word length** follows from the specification: stopband attenuation of A dB needs coefficients of roughly $A/6 + 2$ bits for an FIR filter (a 100 dB decimation filter needs about 18-bit coefficients), and narrow IIR filters need more, as the notch example showed.
- **Floating point** removes most scaling and overflow concerns at the cost of power and silicon; single precision (24-bit mantissa) is adequate for most audio and instrumentation filters, but very low-frequency IIR sections at high sampling rates can still need double precision.

> **Boundary.**
> The processor architectures, MAC units, FPGA DSP slices and data converters on which filters run are treated in *Electronics & Computing*. This chapter treats the filter as an algorithm and its arithmetic.

What remains open

Open question	Where it is answered
How to change the sampling rate efficiently, and how polyphase filters exploit decimation and interpolation	Chapter 24
How a filter can learn its own coefficients from data (adaptive and Wiener filters)	Chapter 24
How matched filters and pulse-shaping filters (raised cosine) are designed for digital communication	Chapter 27
How a digital filter inside a feedback loop affects closed-loop stability and phase margin	Chapters 34 and 35
The underlying quantisation model and fixed-point number formats	Chapter 21 (back reference)
The analog prototypes and their pole positions	Chapter 13 (back reference)
Hardware realisation of filters in DSPs and FPGAs	*Electronics & Computing*

Next: Chapter 24 lets filters change the sampling rate, adapt their coefficients to the data and resolve how a spectrum changes in time, and closes Part V with the fundamentals of image and audio processing.

Chapter 24 — Advanced Signal Processing

Part V · Digital Signal Processing

A wearable ECG patch has three problems its designers did not have on the bench. Its delta-sigma ADC produces a one-bit stream at several megahertz that must become clean 500 Hz samples without burning the battery. The 50 Hz interference it picks up changes in amplitude and phase every time the wearer walks past a different appliance, so no fixed notch is quite right. And the cardiologist does not want one spectrum of a 24-hour recording; she wants to see how the heart's rhythm, and the noise, change from minute to minute. None of these is solved by the fixed filter and single FFT of Chapters 22 and 23. The first needs filters that change the sampling rate; the second needs a filter that learns its own coefficients; the third needs a spectrum that is a function of time.

This chapter collects those techniques: multirate processing, adaptive and optimal (Wiener) filtering, model-based spectral estimation, time–frequency analysis and wavelets. It ends with the fundamentals of the two signal types that most people meet every day, images and audio, where all the tools of Part V come together.

> **How can the sampling rate be changed by any ratio without aliasing, and cheaply? What is the best linear filter for separating a signal from noise, and how can a filter find it from data alone? How can a spectrum be estimated from very short records, and how is a spectrum that changes in time represented?**

Concept	Engineering destination
Multirate processing, polyphase filters	Audio sample-rate converters, delta-sigma ADC decimators, software-defined radio channelisers
Decimation and interpolation	Oversampling DACs, digital down-conversion in receivers, data reduction in instrumentation
Adaptive filters	Echo cancellers in telephony and conferencing, channel equalisers, active noise control
Wiener filtering	Noise reduction in speech and biomedical signals, image restoration, optimal smoothing of sensor data
LMS algorithm	Adaptive mains-interference cancellation, beamforming, hearing aids
Parametric spectral estimation	Short-record Doppler radar, EEG rhythm analysis, speech coding (linear prediction)
Time–frequency analysis	Speech spectrograms, machine run-up analysis, radar micro-Doppler, bat and whale acoustics
Wavelets	ECG QRS detection, signal denoising, JPEG 2000, transient detection in power systems
Image and audio fundamentals	Cameras, medical imaging, codecs (JPEG, MP3, AAC), audio effects

Roadmap. Multirate systems (24.1) → decimation, interpolation and rational rate change (24.2) → filters that adapt (24.3) → the optimal Wiener filter they aim at (24.4) → the LMS algorithm that gets there from data (24.5) → spectral estimation beyond the periodogram (24.6) → time–frequency analysis (24.7) → wavelets (24.8) → images and audio (24.9).

24.1 Multirate processing

Why change the sampling rate

A **multirate** system contains more than one sampling rate. The reasons are practical:

- **Interfacing standards.** Professional audio runs at 48 kHz, CD audio at 44.1 kHz, broadcast at 32 kHz, and telephone speech at 8 or 16 kHz. Every converter between them is a multirate system.
- **Oversampled converters.** Delta-sigma ADCs sample at 64–256 times the output rate to push quantisation noise out of band (*Electronics & Computing*); a digital decimator then removes the out-of-band noise and lowers the rate. Oversampling DACs do the reverse.
- **Efficiency.** A narrowband signal sampled at a high rate wastes arithmetic. A software radio that digitises 100 MHz of spectrum but needs one 200 kHz channel shifts the channel to baseband and reduces its rate by a factor of hundreds before doing anything else.

- **Multiresolution analysis.** Filter banks and wavelets (Section 24.8) split signals into bands at different rates.

The two basic operations

Downsampling by an integer M keeps every M-th sample: $y[n] = x[nM]$. **Upsampling** by an integer L inserts $L-1$ zeros between samples: $y[n] = x[n/L]$ when n is a multiple of L, and zero otherwise. In the z-domain,

$$Y_{\downarrow M}(e^{j\omega}) = \frac{1}{M}\sum_{k=0}^{M-1} X\left(e^{j(\omega-2\pi k)/M}\right), \qquad Y_{\uparrow L}(e^{j\omega}) = X(e^{j\omega L}).$$

Downsampling stretches the spectrum by M and overlays M shifted copies: anything above the new Nyquist frequency π/M (in the old normalised units) aliases, exactly as in analog sampling (Chapter 19). Upsampling compresses the spectrum by L and creates $L-1$ **images** of the baseband between 0 and the new $f_s/2$. Figure 24.1 shows both effects. Both operations are linear but **time-varying**: shifting the input by one sample does not simply shift the output. That is why multirate systems need their own rules.

Noble identities and polyphase decomposition

Two identities let filters be moved across rate changers:

$$H(z^M) \text{ followed by } \downarrow M \;\equiv\; \downarrow M \text{ followed by } H(z), \qquad \uparrow L \text{ followed by } H(z^L) \;\equiv\; H(z) \text{ followed by } \uparrow L.$$

They are the key to efficiency. Any FIR filter can be split into M **polyphase components**, each containing every M-th coefficient:

$$H(z) = \sum_{m=0}^{M-1} z^{-m} E_m(z^M), \qquad e_m[n] = h[nM+m].$$

Combining this with the noble identity, a filter followed by a downsampler becomes a commutator that deals input samples in turn to M short sub-filters $E_m(z)$, all running at the low output rate (Figure 24.2). The arithmetic is the same as computing only the outputs that will be kept, which is a factor M cheaper than filtering at the high rate and discarding $M-1$ of every M outputs. The dual structure serves interpolation: the zeros inserted by upsampling are never multiplied.

> **Status of the principle: Noble identities and polyphase efficiency**
> *Derived*. Exact algebraic identities for LTI filters combined with integer rate changers; the factor-of-M (or L) saving holds for FIR filters, and does not carry over to recursive IIR filters, whose feedback needs every high-rate output.

24.2 Decimation and interpolation

Decimation

Decimation by M is lowpass filtering followed by downsampling. The filter, the **anti-aliasing** filter of the digital domain, must remove everything above π/M that would otherwise fold into the band of interest. Only the part of the spectrum that folds onto the wanted band needs strong attenuation; components that fold into a band that will later be discarded are harmless, which relaxes the filter in multistage designs.

Interpolation

Interpolation by L is upsampling followed by a lowpass **anti-imaging** filter with cutoff π/L and gain L. The filter fills in the zeros with values consistent with a band-limited signal; with an ideal filter the result is exactly the sequence that would have been obtained by sampling the original analog signal at L times the rate. Practical interpolators use polyphase FIR filters; simple ones use linear interpolation or a zero-order hold, which leave residual images (the sinc droop and imaging of Chapter 19).

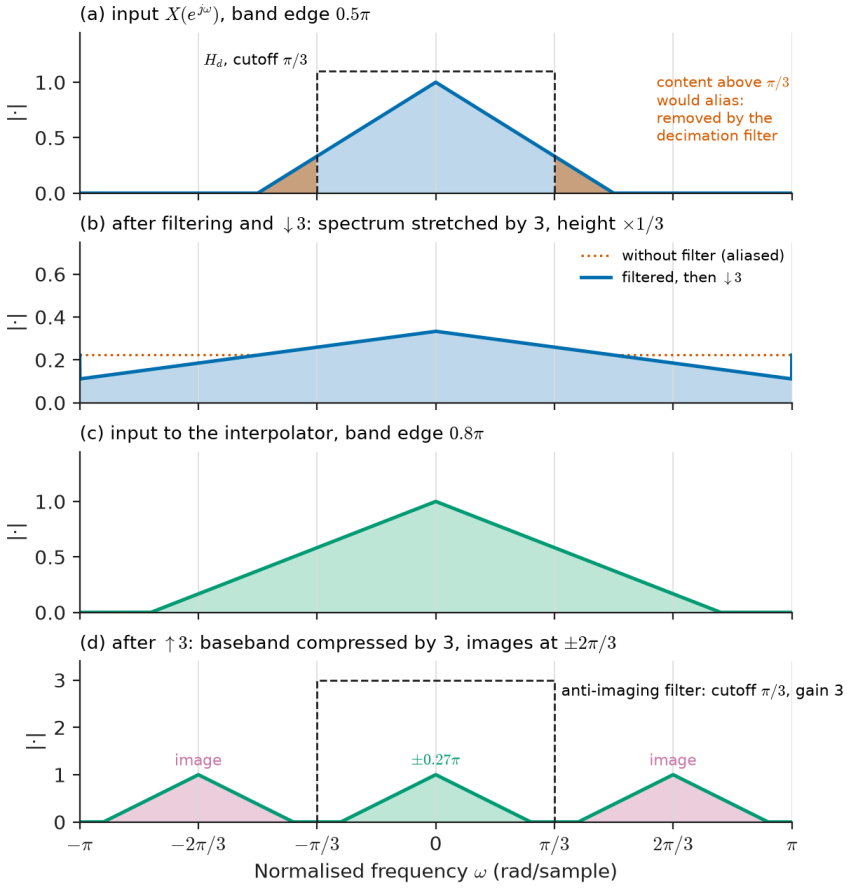

Figure 24.1 — Spectra in decimation by 3 and interpolation by 3: aliasing if the input is not band-limited to pi/3, and images that the interpolation filter must remove.

Rational sample-rate conversion

A change by a rational factor L/M is interpolation by L followed by decimation by M, with one lowpass filter whose cutoff is $\min(\pi/L, \pi/M)$ at the intermediate rate. Converting CD audio at 44.1 kHz to studio audio at 48 kHz needs

$$\frac{48\,000}{44\,100} = \frac{160}{147},$$

so $L = 160$, $M = 147$ and an intermediate rate of $44\,100 \times 160 = 7.056$ MHz. Nobody computes at that rate. In polyphase form, each output sample needs only one of the 160 polyphase branches, selected by a cycling index; if the prototype filter has $160 \times 32 = 5120$ taps, each output costs 32 multiply–accumulates, or 1.54 M MAC/s per channel at 48 kHz. Converters between unrelated or drifting clocks (asynchronous sample-rate converters in digital audio receivers) compute the branch coefficients on the fly from an estimated ratio.

Multistage decimation and CIC filters

A large factor M is best done in stages. The early stages run at the highest rate but only need to protect the narrow band that will survive, so their transition bands are wide and their filters short. The final stage runs at the lowest rate and carries the sharp transition. Filters with no multiplications at all are used at the highest rate.

The **cascaded integrator–comb (CIC)** filter is the standard first stage. N integrators at the input rate, a downsampler by R, and N comb differentiators at the output rate implement

$$H(z) = \left(\frac{1 - z^{-R}}{1 - z^{-1}} \right)^{N}, \qquad |H(f)| = \left| \frac{\sin(\pi f R / f_{s,\text{in}})}{R \sin(\pi f / f_{s,\text{in}})} \right|^{N}$$

(normalised to unit DC gain). It needs only adders and registers, so it runs at hundreds of megahertz in an FPGA or ASIC. Its nulls fall exactly on the frequencies that alias onto DC after downsampling. Its drawbacks are passband droop, which a later compensation filter corrects, and a gain of R^N, which requires $N \log_2 R$ extra bits in the integrators.

Worked example — the decimator of a 24-bit audio ADC. A delta-sigma audio ADC modulates at $f_{s,\text{in}} = 64 \times 48$ kHz $= 3.072$ MHz. The output must be 48 kHz with a 0–20 kHz passband and 100 dB rejection of everything that would alias into it.

Single stage. A single FIR lowpass at 3.072 MHz with a 20–24 kHz transition needs, by Kaiser's formula (Chapter 23), $N \approx (100 - 8)/(2.285 \times 2\pi \times 4000/3.072 \times 10^6) = 4921$ taps. Even in polyphase form, computing only the outputs at 48 kHz, it costs $4921 \times 48\,000 = 236$ M MAC/s.

Stage 1: CIC, $R = 16$, $N = 4$. This takes the rate from 3.072 MHz to 192 kHz with no multiplications. The bands that fold onto 0–20 kHz are centred on multiples of 192 kHz; the nearest edge is $192 - 20 = 172$ kHz, where the CIC attenuates by 75.2 dB. The droop at 20 kHz is 0.62 dB, to be equalised later, and the integrators need 16 extra bits. If 75 dB is not enough, a fifth-order CIC gives 94.0 dB at 172 kHz with 0.77 dB of droop.

Stage 2: FIR, 192 kHz to 96 kHz. The band 76–116 kHz folds onto 0–20 kHz, so the transition can be 20 to 76 kHz: $N \approx 92/(2.285 \times 2\pi \times 56/192) \approx 22$ taps, run at 96 kHz output: 2.1 M MAC/s (about half that as a half-band filter).

Stage 3: FIR, 96 kHz to 48 kHz. This carries the sharp 20–24 kHz transition: $N \approx 92/(2.285 \times 2\pi \times 4/96) \approx 154$ taps at 48 kHz output: 7.4 M MAC/s, and it can include the CIC droop correction.

Total. About 9.5 M MAC/s against 236 M MAC/s, a factor of 25, with most of the hardware at the highest rate made of adders. Every audio ADC and every digital down-converter in a radio receiver is built this way.

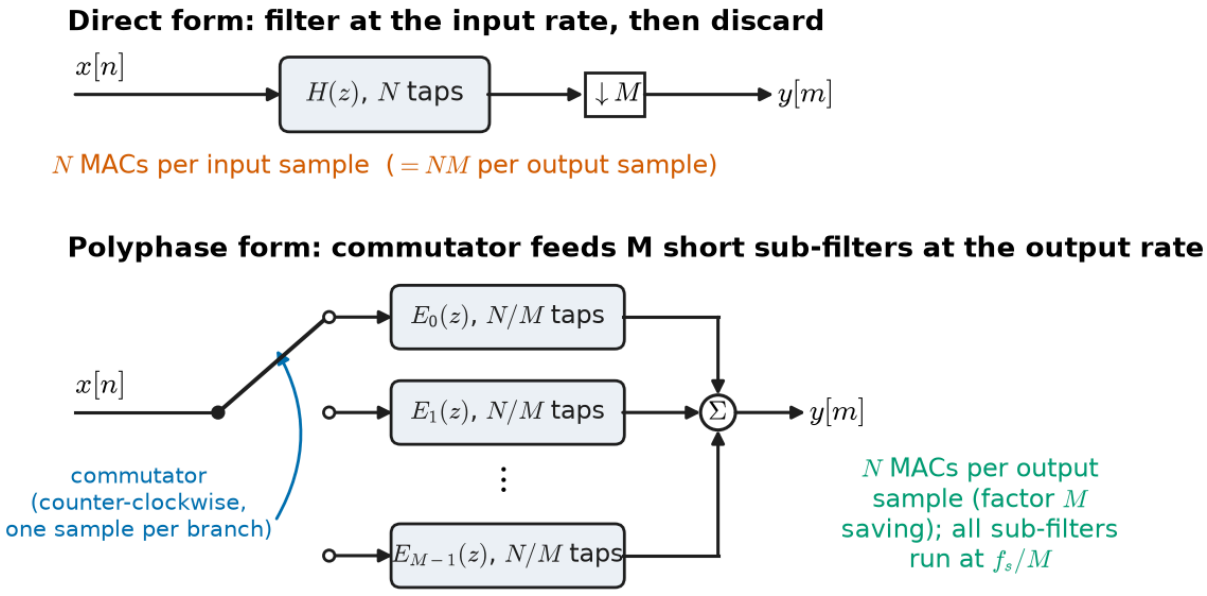

Figure 24.2 — Decimation by M: direct form (filter at the high rate, then discard samples) and the equivalent polyphase form (commutator feeding M short sub-filters at the low rate).

> **Status of the principle: Multistage decimation saving**
> *Approximate.* The cost estimates use Kaiser's empirical length formula and assume the stopband need only protect the band that finally survives; the order-of-magnitude saving is robust, the exact stage split is an optimisation that depends on the hardware.

24.3 Adaptive filters

When fixed filters are not enough

The filters of Chapter 23 are designed once from a specification. Many signals violate the assumption that the specification is known in advance: the echo path from a loudspeaker to a microphone changes whenever someone moves in the room; the frequency response of a telephone line or a radio channel depends on the call; mains interference on a patient changes with posture and nearby equipment. An **adaptive filter** adjusts its coefficients continuously to minimise an error signal, tracking such changes.

The structure is always the same. An FIR filter with weight vector $\mathbf{w}[n] = [w_0, \dots, w_{M-1}]^{\mathsf{T}}$ processes an input vector $\mathbf{x}[n] = [x[n], \dots, x[n-M+1]]^{\mathsf{T}}$ to give $y[n] = \mathbf{w}^{\mathsf{T}}[n]\,\mathbf{x}[n]$. A **desired** signal $d[n]$ is available, the **error** is $e[n] = d[n] - y[n]$, and an algorithm updates $\mathbf{w}$ to make e small in the mean-square sense. FIR (transversal) filters are used almost always, because their error surface is a simple quadratic bowl and they cannot become unstable through the filter itself.

Four configurations

What counts as the input and desired signal defines the application (Figure 24.3):

1. **System identification.** The same input drives the unknown system and the adaptive filter; the desired signal is the system's output. When the error is minimised, the filter is a model of the system. *Example:* echo cancellation. The far-end speech drives both the echo path and the filter; the microphone signal is $d[n]$; the filter's output is the predicted echo, and the error is the near-end speech with the echo removed. A network echo canceller (ITU-T G.168) covers 64–128 ms of echo tail at 8 kHz, i.e. 512–1024 taps; an acoustic echo canceller for a conference room at 16 kHz covers 100–250 ms with 2048–4096 taps.
2. **Inverse modelling (equalisation).** The filter is placed after an unknown channel and trained to reproduce the transmitted signal (a known training sequence, or the receiver's own decisions). *Example:* channel equalisers in digital modems and wireline transceivers (Chapter 27).
3. **Prediction.** The input is a delayed version of the signal and the desired signal is the signal itself; the filter learns to predict the next sample. *Example:* linear-predictive speech coding in mobile phones, where only the prediction error is transmitted.
4. **Interference cancellation.** A primary sensor receives signal plus interference; a reference sensor receives interference correlated with that in the primary but not with the signal. The filter shapes the reference to match the interference, and the error is the cleaned signal. *Examples:* removing mains hum from ECG using a reference derived from the mains, removing maternal ECG from foetal ECG recorded on the abdomen, active noise-cancelling headphones.

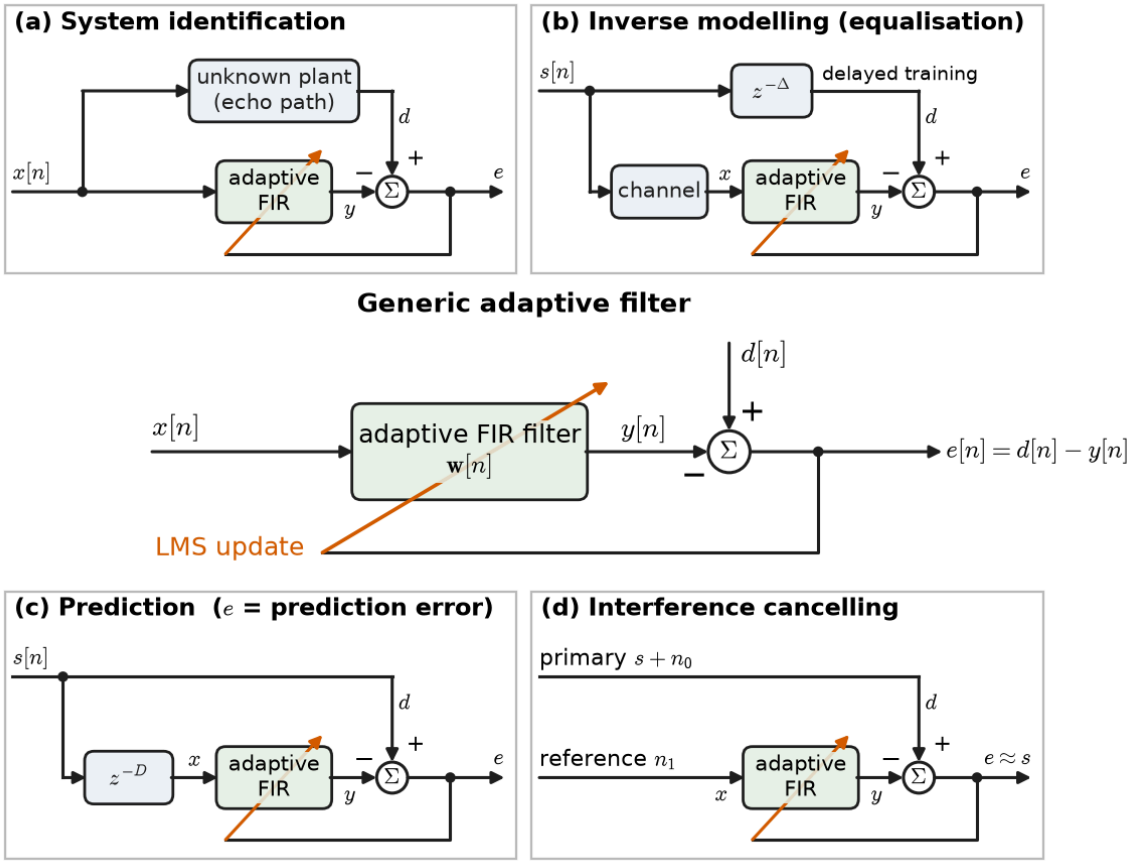

Figure 24.3 — The adaptive filter and its four standard configurations: system identification, inverse modelling, prediction and interference cancellation.

What the error converges to, and how fast, are separate questions: the first is answered by the Wiener filter, the second by the adaptation algorithm.

24.4 Wiener filtering

The optimal linear filter

Suppose $x[n]$ and $d[n]$ are jointly wide-sense stationary with known second-order statistics. Which fixed FIR filter $\mathbf{w}$ minimises the mean-square error $J(\mathbf{w}) = E\{e^2[n]\}$? Expanding,

$$J(\mathbf{w}) = \sigma_d^2 - 2\,\mathbf{p}^{\mathsf{T}}\mathbf{w} + \mathbf{w}^{\mathsf{T}}\mathbf{R}\,\mathbf{w},$$

where $\mathbf{R} = E\{\mathbf{x}\mathbf{x}^{\mathsf{T}}\}$ is the $M \times M$ input autocorrelation matrix (Toeplitz, entries $r_{xx}[i-j]$) and $\mathbf{p} = E\{d\,\mathbf{x}\}$ is the cross-correlation vector. J is a quadratic bowl with a single minimum, found by setting the gradient to zero:

$$\mathbf{R}\,\mathbf{w}_o = \mathbf{p} \qquad \text{(Wiener–Hopf equations)}, \qquad J_{\min} = \sigma_d^2 - \mathbf{p}^{\mathsf{T}}\mathbf{w}_o.$$

At the optimum the error is uncorrelated with every input sample used by the filter (the **orthogonality principle**): any remaining correlation could be used to reduce the error further.

The frequency-domain Wiener filter

If the filter may be infinitely long and non-causal (offline processing of a recorded signal), the solution is simplest in the frequency domain. For a signal s observed in additive noise v uncorrelated with it, $x = s + v$ and $d = s$:

$$H_o(e^{j\omega}) = \frac{S_s(e^{j\omega})}{S_s(e^{j\omega}) + S_v(e^{j\omega})}, \qquad J_{\min} = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{S_s S_v}{S_s + S_v}\, d\omega.$$

The interpretation is immediate: pass frequencies where the signal dominates (gain near 1), suppress those where noise dominates (gain near 0), and in between weight by the local signal fraction. Speech-enhancement algorithms in phones and hearing aids apply this gain per bin of a short-time FFT (Section 24.7), with S_v estimated during speech pauses ("spectral subtraction" and its refinements). Image deblurring uses the same formula with the blur's frequency response included.

> **Status of the principle: Wiener filter**
> *Derived.* The Wiener–Hopf equations follow exactly from minimising mean-square error over linear filters for wide-sense stationary signals with known second-order statistics; the filter is optimal among all estimators only if the signals are jointly Gaussian, and it is only as good as the statistics assumed.

Worked example — smoothing a noisy pressure sensor. A pressure transducer on a hydraulic line produces a slowly varying signal modelled as a first-order autoregressive process with unit variance and correlation $\rho = 0.9$ between successive samples ($r_{ss}[k] = 0.9^{|k|}$). The ADC adds white noise of unit variance, so the input SNR is 0 dB. What is gained by Wiener filtering?

One tap. $\mathbf{R} = [2]$, $\mathbf{p} = [1]$, $w = 0.5$, $J_{\min} = 1 - 0.5 = 0.5$: a 3.0 dB improvement, by simple shrinkage.

Two taps. $\mathbf{R} = \begin{bmatrix} 2 & 0.9 \\ 0.9 & 2 \end{bmatrix}$, $\mathbf{p} = \begin{bmatrix} 1 \\ 0.9 \end{bmatrix}$. Solving, $\mathbf{w}_o = [0.3730,\ 0.2821]^{\mathsf{T}}$ and $J_{\min} = 1 - (0.3730 + 0.9 \times 0.2821) = 0.3730$: 4.28 dB.

More taps. With 5 taps $J_{\min} = 0.3075$, with 10 taps 0.3036, and with 20 taps 0.3036 (5.18 dB). The causal filter has converged; a longer past does not help, because an AR(1) signal's dependence on the past decays as 0.9^k.

Non-causal. Allowing the filter to use future samples as well (offline smoothing, using the frequency-domain formula) gives $J_{\min} = 0.218$, a 6.62 dB improvement. The extra 1.4 dB is what a real-time controller gives up by having to act on the latest sample; a post-processing analysis of the same record can have it.

24.5 LMS algorithms

Steepest descent without statistics

In practice $\mathbf{R}$ and $\mathbf{p}$ are unknown and may change. Widrow and Hoff (1960) proposed replacing the true gradient of J, $-2(\mathbf{p}-\mathbf{R}\mathbf{w})$, by its instantaneous estimate $-2e[n]\mathbf{x}[n]$, and stepping downhill:

$$\mathbf{w}[n+1] = \mathbf{w}[n] + \mu\, e[n]\,\mathbf{x}[n], \qquad e[n] = d[n] - \mathbf{w}^{\mathsf{T}}[n]\mathbf{x}[n].$$

(The factor 2 is absorbed into the step size μ; some texts keep it, which halves every μ below.) This is the **least-mean-squares (LMS)** algorithm: $2M$ multiplications per sample, no matrix inversion, no statistics. Its simplicity is why it, and its variants, run inside most echo cancellers, equalisers and noise cancellers.

Convergence, step size and misadjustment

Averaged over the data, the weights follow the steepest-descent path toward $\mathbf{w}_o$. Decomposing along the eigenvectors of $\mathbf{R}$ with eigenvalues λ_i, each mode decays as $(1-\mu\lambda_i)^n$. Hence:

- **Stability in the mean** requires $0 < \mu < 2/\lambda_{\max}$. Since $\lambda_{\max}$ is rarely known, the safe practical bound uses $\lambda_{\max} \le \operatorname{tr}\mathbf{R} = M\sigma_x^2$: $0 < \mu < 2/(M\sigma_x^2)$, and for good behaviour of the mean-square error, μ is chosen an order of magnitude smaller.
- **Time constants**: mode i converges with $\tau_i \approx 1/(\mu\lambda_i)$ samples. The slowest mode, set by $\lambda_{\min}$, dominates; the ratio $\lambda_{\max}/\lambda_{\min}$ (the eigenvalue spread, large for coloured inputs such as speech) is the main weakness of LMS.
- **Misadjustment**: because the gradient estimate is noisy, the weights jitter about $\mathbf{w}_o$ and the steady-state error exceeds $J_{\min}$ by the fraction

$$\mathcal{M} \approx \frac{\mu \operatorname{tr}\mathbf{R}}{2}.$$

Small μ gives low misadjustment but slow convergence and tracking; large μ the reverse. Every LMS design is a choice on this trade-off.

> **Status of the principle: LMS convergence and misadjustment formulas**
> *Approximate.* Derived under the independence assumption (successive input vectors uncorrelated) and small μ; they predict real behaviour well for small step sizes and stationary inputs, and less well near the stability limit or for strongly correlated tapped-delay-line inputs.

Variants

- **Normalised LMS (NLMS)** divides the step by the input power, $\mathbf{w} \leftarrow \mathbf{w} + \tilde{\mu}\, e\,\mathbf{x}/(\epsilon + \|\mathbf{x}\|^2)$, with $0 < \tilde{\mu} < 2$. It is insensitive to the input level, which matters for speech whose level varies by 40 dB, and is the standard in echo cancellation.
- **Sign–error and sign–sign LMS** replace e or $\mathbf{x}$ by their signs, removing multiplications for very-high-rate hardware (serial-link equalisers).

- **Frequency-domain (block) LMS** updates in the FFT domain, reducing the cost of very long filters and partly whitening the input.
- **Recursive least squares (RLS)** minimises an exponentially weighted sum of squared errors exactly at each step. It converges in about $2M$ samples regardless of eigenvalue spread, but costs $O(M^2)$ operations per sample and needs care with numerical stability; it is used for short filters where fast convergence is essential (fast-fading radio channels, Chapter 30).

Worked example — an adaptive mains canceller for ECG. An ECG is sampled at $f_s = 500$ Hz. A reference is derived from the mains: $x_1[n] = \cos(2\pi \cdot 50\,n/f_s)$ and $x_2[n] = \sin(2\pi \cdot 50\,n/f_s)$, amplitude $C = 1$. A two-weight LMS filter with $\mu = 0.01$ shapes the reference to match the interference in the ECG, and its error output is the cleaned ECG (Widrow's adaptive noise canceller).

Correlation matrix. The two reference inputs are orthogonal with power $C^2/2 = 0.5$ each: $\mathbf{R} = \operatorname{diag}(0.5, 0.5)$, $\operatorname{tr}\mathbf{R} = 1$, $\lambda_{\max} = \lambda_{\min} = 0.5$. The mean stability bound is $\mu < 4$; no eigenvalue spread exists, so LMS behaves ideally.

Convergence. $\tau = 1/(\mu\lambda) = 1/(0.01 \times 0.5) = 200$ samples $= 0.4$ s. If the hum amplitude or phase changes (a patient turns over), the canceller re-converges in about $3\tau = 1.2$ s. A direct simulation confirms that the weight error falls to 0.37 of its initial value after 200 samples.

Equivalent filter. Analysing the loop shows that, from primary input to error output, the canceller is a linear time-invariant notch,

$$H(z) = \frac{1 - 2\cos\omega_0\, z^{-1} + z^{-2}}{1 - 2\big(1 - \tfrac{\mu C^2}{2}\big)\cos\omega_0\, z^{-1} + (1 - \mu C^2) z^{-2}},$$

with zeros exactly at 50 Hz and poles at radius $\sqrt{1 - \mu C^2} = 0.99499$. Its 3 dB bandwidth is $\mu C^2 f_s/(2\pi) = 0.80$ Hz: far narrower than the fixed notch of Chapter 23, and centred exactly on the reference frequency by construction, so the coefficient-quantisation failure of that chapter cannot occur. If the mains drifts to 49.8 Hz, the reference (taken from the mains itself) drifts with it.

Misadjustment. $\mathcal{M} = \mu \operatorname{tr}\mathbf{R}/2 = 0.005$: the residual excess error is 0.5 % of the minimum, negligible. Halving μ would halve the notch width and double the convergence time; the clinical requirement (fast recovery after movement, no distortion of the ECG's low-frequency content) sets the choice.

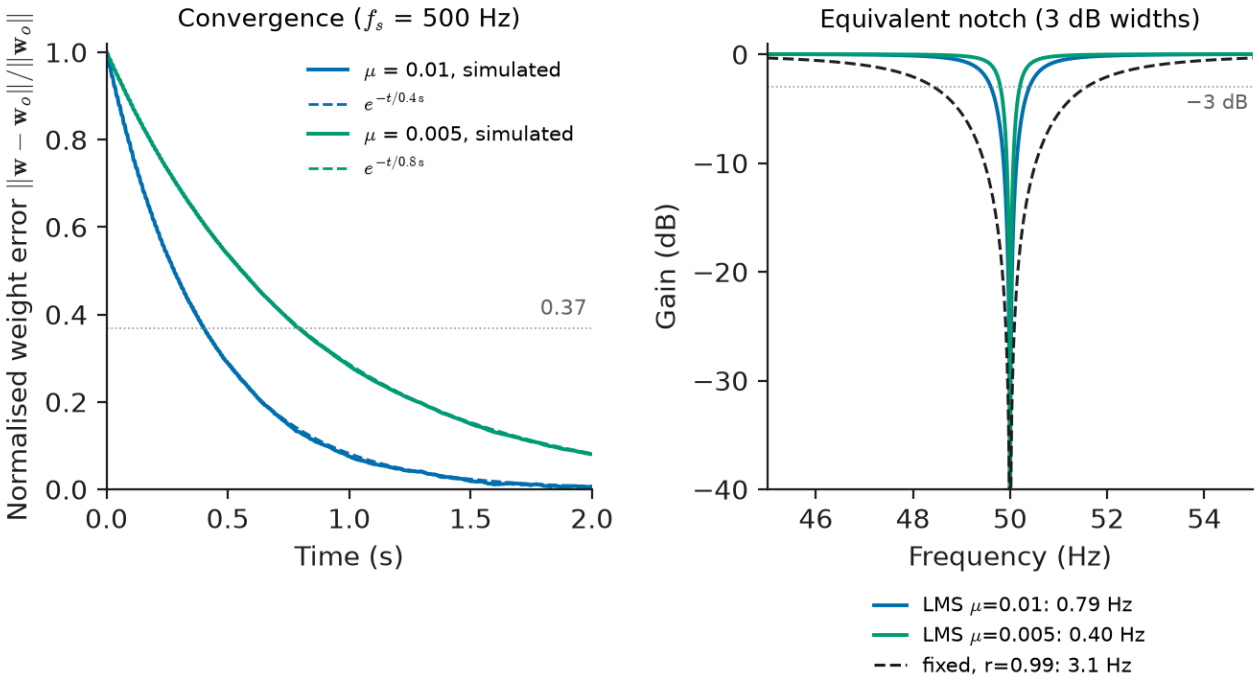

Figure 24.4 — Adaptive mains cancellation in ECG: weight convergence with time constant 0.4 s and the equivalent 0.8 Hz-wide notch at 50 Hz.

Figure 24.4 shows both the convergence and the equivalent notch.

24.6 Spectral estimation

Limits of the classical methods

Chapter 22 estimated spectra non-parametrically: windowed periodograms averaged by Welch's method. Such estimators make no assumption about the signal, but their resolution cannot be better than about $1/T_{\text{rec}}$, and averaging for low variance costs further resolution. When records are short (a radar dwell of a few pulses, a 1 s EEG epoch before the brain state changes, a speech frame of 20 ms), the classical spectrum is either blurred or noisy. **Parametric** methods assume a model with few parameters, estimate the parameters, and compute the spectrum from the model. If the model fits, resolution and smoothness come from the model rather than from the record length.

Autoregressive (AR) models

The most used model is the **autoregressive** process of order p: white noise $\varepsilon[n]$ of variance σ^2 driving an all-pole filter,

$$x[n] = -\sum_{k=1}^{p} a_k x[n-k] + \varepsilon[n], \qquad S_x(e^{j\omega}) = \frac{\sigma^2}{\left|1 + \sum_{k=1}^{p} a_k e^{-j\omega k}\right|^2}.$$

Poles near the unit circle produce sharp spectral peaks, which suits resonant phenomena: vocal-tract formants, mechanical resonances, brain rhythms, clutter in radar. Multiplying the model equation by

$x[n-m]$ and taking expectations gives the **Yule–Walker equations**,

$$\sum_{k=1}^{p} a_k\, r[m-k] = -r[m], \quad m = 1, \dots, p, \qquad \sigma^2 = r[0] + \sum_{k=1}^{p} a_k r[k],$$

a Toeplitz system solved efficiently by the **Levinson–Durbin recursion** in $O(p^2)$ operations. The recursion's intermediate results, the **reflection coefficients**, are what speech codecs quantise and transmit (they guarantee a stable synthesis filter when each has magnitude below 1). **Burg's method** estimates the reflection coefficients directly from the data and behaves better on short records. Choosing p is part of the problem: too low smooths away peaks, too high creates spurious ones; information criteria (AIC, MDL) automate the choice.

Worked example — alpha rhythm from a one-second EEG epoch. An EEG channel sampled at 128 Hz shows a dominant alpha rhythm. From a 1 s epoch (128 samples) the normalised autocorrelation estimates are $r[0]=1$, $r[1]=0.881$, $r[2]=0.573$. Fit an AR(2) model.

Yule–Walker.

$$\begin{bmatrix} 1 & 0.881 \\ 0.881 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = - \begin{bmatrix} 0.881 \\ 0.573 \end{bmatrix} \;\Rightarrow\; a_1 = -1.6806, \quad a_2 = 0.9076.$$

The driving-noise variance is $\sigma^2 = 1 + (-1.6806)(0.881) + (0.9076)(0.573) = 0.0394$.

Poles and peak. The poles of $1 + a_1 z^{-1} + a_2 z^{-2}$ have radius $\sqrt{a_2} = 0.9527$ and angle $\arccos\!\left(-a_1/(2\sqrt{a_2})\right) = 0.4906$ rad, i.e. $0.4906 \times 128/(2\pi) = 10.0$ Hz. The model spectrum peaks at 9.95 Hz with a half-power width of about 2.0 Hz.

Comparison. A Hann-windowed periodogram of the same 1 s epoch has a 3 dB resolution of $1.44 \times 1 = 1.44$ Hz and a 100 % fluctuation per bin; the AR model gives a smooth spectrum and a peak frequency that can be tracked epoch by epoch, which is what sleep-staging and anaesthesia-depth monitors do. The price is model dependence: if the real spectrum has two close peaks (alpha and mu rhythms), an AR(2) model will merge them and a higher order is needed.

Subspace and multitaper methods

When the signal is a sum of a few sinusoids in white noise (Doppler returns from several targets, multiple sources arriving at a sensor array), **subspace** methods such as **MUSIC** and **ESPRIT** split the eigenvectors of the estimated autocorrelation matrix into a signal subspace and a noise subspace and locate frequencies where the steering vector is orthogonal to the noise subspace. They resolve sinusoids much closer than $1/T_{\text{rec}}$ at moderate SNR, and are standard in direction-of-arrival estimation for antenna arrays and automotive radar.

Multitaper estimation (Thomson) stays non-parametric but replaces the single window by several orthogonal **Slepian** tapers, each concentrating its energy in a band $\pm W$; averaging the resulting periodograms reduces variance without segmenting the record. It is widely used in neuroscience and geophysics.

Method	Assumption	Resolution	Variance	Typical use
Periodogram / Welch	None (stationarity)	About $1/T$ (per segment)	Reduced by averaging	General-purpose, calibrated PSD
Multitaper	None (stationarity)	About $2W$	Reduced by taper count	EEG, geophysics, climate
AR (Yule–Walker, Burg)	All-pole spectrum	Model-dependent, can beat $1/T$	Low (few parameters)	Speech, EEG, radar clutter, vibration
MUSIC / ESPRIT	Few sinusoids in white noise	Well below $1/T$ at good SNR	Low, if model holds	Array direction finding, Doppler, frequency estimation

Table 24.1 — Spectral estimators compared.

> **Status of the principle: Model-based super-resolution**
> *Approximate.* Resolution beyond $1/T_{\text{rec}}$ comes from the assumed model, not from the data; it holds when the model order and structure are right and the SNR is adequate, and fails (spurious or merged peaks) when they are not.

24.7 Time-frequency analysis

Spectra that change

A Fourier spectrum averages over the whole record and discards when each component occurred. Speech changes its spectrum every 10–20 ms; a motor run-up sweeps its vibration frequencies through the structure's resonances; a bat's echolocation call is a chirp lasting milliseconds; a fault in a power system produces a transient lasting a few cycles; a helicopter's rotor blades modulate a radar echo (micro-Doppler). These are **non-stationary** signals, and they need representations in time and frequency together.

The short-time Fourier transform

The **short-time Fourier transform (STFT)** slides a window $w[n]$ along the signal and takes a DFT at each position m, spaced by a hop H:

$$X[m,k] = \sum_{n=0}^{L-1} x[n+mH]\, w[n]\, e^{-j2\pi kn/L}.$$

Its squared magnitude, displayed as an image of time against frequency, is the **spectrogram**. Each column is a windowed DFT (Chapter 22), so everything there applies: bin spacing f_s/L, the window's main-lobe width and side lobes, ENBW. With a hop of $L/2$ or $L/4$ and a suitable window (Hann at 50 % overlap sums to a constant), the STFT is invertible by **overlap-add**, which is how STFT-domain processing such as Wiener-gain noise suppression and time-stretching is done.

The uncertainty trade-off

The window length fixes both resolutions at once: time resolution is about the window duration $T_w = L/f_s$, frequency resolution about $1/T_w$ times a window factor. They cannot both be improved. For any window, the root-mean-square durations in time and frequency satisfy

$$\Delta t \, \Delta f \;\geq\; \frac{1}{4\pi},$$

with equality only for a Gaussian window (the Gabor limit, the signal-processing form of the uncertainty relation; its proof belongs to Fourier analysis in *The Mathematics Universe*).

> **Status of the principle: Time–frequency uncertainty**
> *Derived*. A theorem of Fourier analysis (Cauchy–Schwarz applied to a function and its derivative); it limits every linear time–frequency representation built from one window, not the measurement equipment.

Worked example — choosing the window for a speech spectrogram. Speech is sampled at 16 kHz. A male speaker's voice has a pitch of about 120 Hz (harmonics every 120 Hz, glottal pulses every 8.3 ms), and formants that move over tens of milliseconds.

Narrowband spectrogram, 25 ms Hann window ($L = 400$). Bin spacing $16\,000/400 = 40$ Hz; 3 dB width $1.44 \times 40 = 57.6$ Hz; ENBW 60 Hz. The 120 Hz harmonics, three bins apart and separated by more than twice the 3 dB width, appear as separate horizontal striations. Time resolution is about 25 ms, so individual glottal pulses are not seen. With a 10 ms hop (160 samples) there are 100 frames per second; this is the standard front end for speech recognition, whose mel-frequency features are computed from these frames.

Wideband spectrogram, 5 ms Hann window ($L = 80$). Bin spacing 200 Hz and 3 dB width 288 Hz: the harmonics merge into broad formant bands, but each glottal pulse, 8.3 ms apart, appears as a separate vertical striation.

Neither picture is "right"; they answer different questions, and phoneticians routinely use both (Figure 24.5).

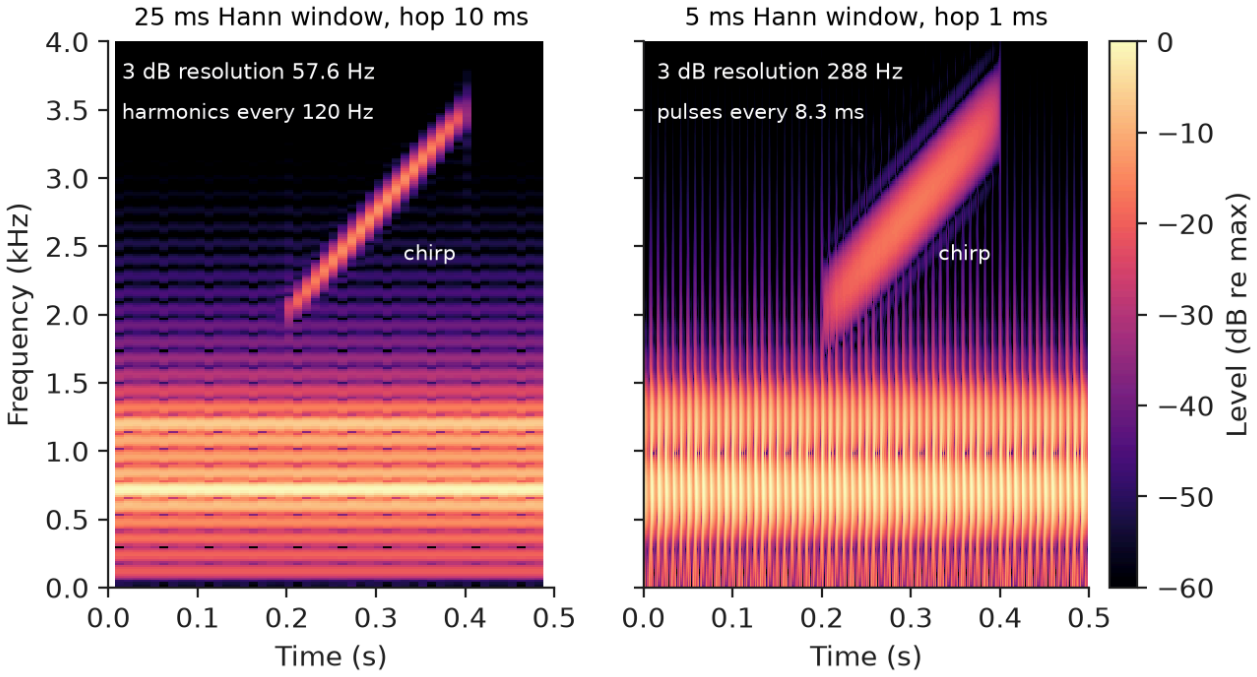

Figure 24.5 — Spectrograms of the same signal with a 25 ms and a 5 ms Hann window: long windows resolve harmonics, short windows resolve pulses in time.

Other time–frequency representations

The **Wigner–Ville distribution** achieves better joint concentration than the spectrogram for a single chirp, but produces strong cross-terms between components; smoothed versions (Cohen's class) trade cross-terms against resolution. **Reassignment** sharpens a spectrogram by moving energy to its local centre of gravity. The **constant-Q transform**, whose bins are spaced logarithmically with bandwidth proportional to frequency, matches musical scales and human hearing. The most important alternative, however, is the wavelet transform, which builds the varying resolution into its basis.

24.8 Wavelet fundamentals

Scaling instead of modulating

The STFT uses one window at all frequencies: the same time resolution for a 50 Hz rhythm and a 5 kHz click. Many signals combine long, low-frequency components with short, high-frequency transients: an ECG's slow T wave and sharp QRS complex, a power system's 50 Hz waveform and switching transients, an image's smooth regions and edges. Good analysis of such signals uses long windows at low frequencies and short windows at high frequencies. The **wavelet transform** does this by correlating the signal with scaled and shifted copies of one prototype, the **mother wavelet** $\psi(t)$:

$$W_x(a,b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} x(t)\, \psi^*\!\left(\frac{t-b}{a}\right) dt.$$

Small scale a means a compressed wavelet, good time resolution and a high centre frequency; large a means a stretched wavelet and good frequency resolution at low frequency. The relative bandwidth

$\Delta f/f$ is constant: wavelet analysis is constant-Q analysis. The wavelet must have zero mean (it is a bandpass function) and finite energy; common choices are the Morlet wavelet (a Gaussian-windowed sinusoid) for continuous analysis and the Daubechies family for discrete analysis.

The discrete wavelet transform as a filter bank

For computation, scales are restricted to powers of two, $a = 2^j$, and shifts to multiples of the scale. The resulting **discrete wavelet transform (DWT)** is implemented exactly by an iterated two-channel filter bank (Mallat's algorithm): a lowpass filter h and a highpass filter g split the signal into halves of the band, each output is decimated by 2, and the lowpass branch is split again (Figure 24.6). The highpass outputs at each level are the **detail** coefficients d_j; the final lowpass output is the **approximation** a_J. With **orthogonal** wavelets, the filters satisfy perfect-reconstruction conditions, the total number of coefficients equals the number of samples, and energy is preserved. The cost is $O(N)$, even cheaper than an FFT.

For a signal sampled at f_s, level j details occupy roughly the band $f_s/2^{j+1}$ to $f_s/2^j$. For an ECG sampled at 360 Hz (the MIT-BIH arrhythmia database standard):

Coefficients	Approximate band	ECG content
d_1	90–180 Hz	Mostly noise, EMG
d_2	45–90 Hz	Mains interference (50/60 Hz), EMG
d_3	22.5–45 Hz	QRS complex energy
d_4	11.25–22.5 Hz	QRS complex energy
d_5	5.6–11.25 Hz	P and T waves, QRS
a_5	0–5.6 Hz	P and T waves, baseline wander

Table 24.2 — Five-level DWT bands at $f_s = 360$ Hz.

QRS detectors look for coincident maxima in d_3 and d_4; baseline wander is removed by suppressing the deepest approximation.

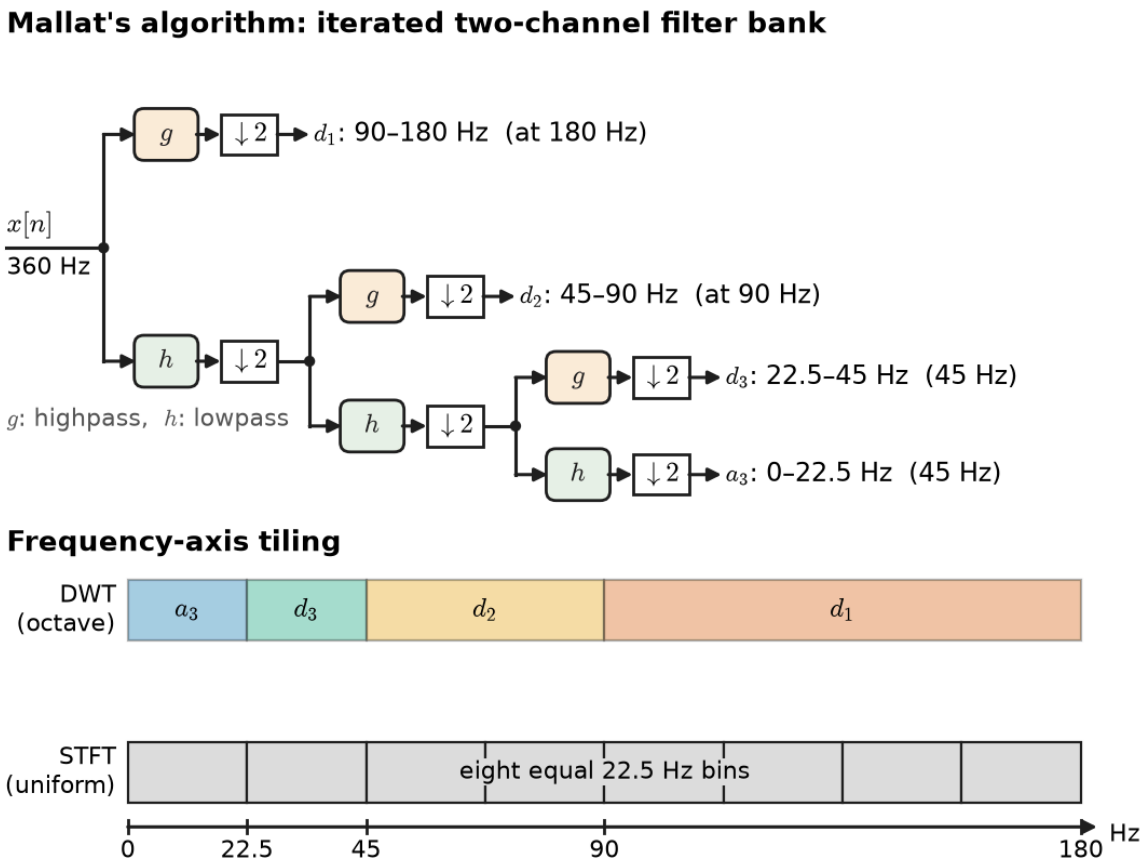

Figure 24.6 — Three-level discrete wavelet transform as an iterated two-channel filter bank, with the resulting octave-band tiling of the frequency axis for an ECG sampled at 360 Hz.

A Haar example

The simplest orthogonal wavelet, the **Haar** wavelet, has filters $h = \frac{1}{\sqrt{2}}[1,\ 1]$ and $g = \frac{1}{\sqrt{2}}[1,\ -1]$: scaled pairwise sums and differences.

Worked example — a three-level Haar transform. Take eight samples of a slowly rising and falling signal, $x = [4, 6, 10, 12, 8, 6, 5, 5]$, with energy $\sum x^2 = 446$.

Level 1. Pairwise sums and differences divided by $\sqrt{2}$: approximation $a_1 = [7.071,\ 15.556,\ 9.899,\ 7.071]$ and detail $d_1 = [-1.414,\ -1.414,\ 1.414,\ 0]$.

Level 2. Applied to a_1: $a_2 = [16,\ 12]$, $d_2 = [-6,\ 2]$.

Level 3. Applied to a_2: $a_3 = 19.799$, $d_3 = 2.828$.

Result. The transform is $[a_3; d_3; d_2; d_1] = [19.799;\ 2.828;\ -6,\ 2;\ -1.414,\ -1.414,\ 1.414,\ 0]$. Its energy is $19.799^2 + 2.828^2 + 36 + 4 + 3 \times 2 + 0 = 392 + 8 + 40 + 6 = 446$, unchanged as orthogonality requires. Most of the energy (88 %) sits in the single approximation coefficient, and the finest details are small: the sparsity that wavelet compression exploits. Setting d_1 to zero and inverting reproduces the pairwise means $[5, 5, 11, 11, 7, 7, 5, 5]$, a smoothed signal from which only 6 units of energy (1.3 %) have been removed.

Wavelet denoising

That example contains the idea of wavelet **denoising**. A smooth signal with occasional sharp features is represented by a few large coefficients; white noise spreads evenly over all coefficients. Shrinking small coefficients toward zero (soft thresholding) removes mostly noise while keeping edges sharp, which a linear lowpass filter cannot do. Donoho's **universal threshold** is $\lambda = \sigma\sqrt{2\ln N}$, where σ is estimated robustly from the finest details as $\text{median}(|d_1|)/0.6745$. For a 10 s ECG at 360 Hz ($N = 3600$), $\lambda \approx 4.05\sigma$. Wavelet coding is also the basis of JPEG 2000, used in digital cinema and medical imaging (DICOM).

> **Status of the principle: Perfect reconstruction of the orthogonal DWT**
> *Derived*. With filters satisfying the orthogonality (perfect-reconstruction) conditions, the iterated two-channel filter bank is an orthogonal change of basis, exactly invertible and energy-preserving; the band-to-level mapping in Table 24.2 is only approximate, because practical wavelet filters have finite transition bands.

24.9 Digital image and audio processing fundamentals

Images as two-dimensional signals

A digital image is a two-dimensional sequence $x[m, n]$: a grid of samples (pixels) of a light-intensity field, with one value per pixel for grey-scale and three (red, green, blue) for colour, typically 8 bits each, and 10–16 bits in medical and scientific imaging. Every one-dimensional idea of Part IV and V extends:

- **2-D sampling and aliasing.** Spatial frequency is measured in cycles per pixel or per millimetre. Patterns finer than half the pixel pitch alias into moiré fringes, which is why cameras put an optical low-pass filter in front of the sensor.
- **2-D convolution.** A linear shift-invariant image filter is $y[m, n] = \sum_k \sum_l h[k, l]\, x[m - k, n - l]$, with a small **kernel** h. A 3×3 or 5×5 averaging or Gaussian kernel blurs and reduces noise; the **Sobel** kernels approximate horizontal and vertical derivatives and detect edges; a Laplacian kernel sharpens. A kernel that factorises as $h[k, l] = h_1[k]h_2[l]$ is **separable** and can be applied as a row pass and a column pass.
- **2-D DFT.** Computed by 1-D FFTs of all rows and then all columns. Its magnitude shows dominant orientations and periodic patterns; filtering by multiplication in the 2-D frequency domain removes periodic noise such as scan lines.
- **Nonlinear filters.** The **median** filter replaces each pixel by the median of its neighbourhood; it removes impulsive ("salt-and-pepper") noise while preserving edges, which no linear filter can do.

Transform coding: the DCT and JPEG

Natural images are highly correlated between neighbouring pixels, so their energy concentrates in few low-frequency transform coefficients. The **discrete cosine transform (DCT)**, a real relative of the DFT that implicitly extends a block symmetrically and so avoids the edge discontinuities that cause

leakage, compacts energy close to optimally for such signals. **JPEG** divides the image into 8×8 blocks, applies the 2-D DCT, quantises the 64 coefficients with steps that grow with spatial frequency (because the eye is less sensitive to fine detail), and entropy-codes the result (Chapter 28). Compression ratios of 10:1 are visually lossless for photographs. The same DCT block coding, with motion compensation between frames, underlies video standards from MPEG-2 to H.265.

Worked example — the processing load of real-time video filtering. A machine-vision camera produces 1920×1080 colour frames at 60 frames per second, 24 bits per pixel.

Raw rate. One frame is $1920 \times 1080 \times 24 = 49.77$ Mbit, and the stream is $49.77 \times 60 = 2.99$ Gbit/s. This is why video is almost never stored or transmitted uncompressed.

Edge detection. Two 3×3 Sobel kernels on the luminance channel cost $2 \times 9 = 18$ multiply–accumulates per pixel: $1920 \times 1080 \times 18 \times 60 = 2.24 \times 10^9$ MAC/s.

Smoothing. A 5×5 Gaussian kernel applied directly costs 25 MAC per pixel, 3.11×10^9 MAC/s; applied separably (a 5-tap row pass and a 5-tap column pass), it costs 10 MAC per pixel, 1.24×10^9 MAC/s, a 60 % saving.

Platform. Loads of a few giga-MAC per second point to an FPGA or a GPU rather than a general-purpose microcontroller (Table 23.8), and they explain why camera pipelines use separable and integer kernels wherever possible.

Audio fundamentals

Audio processing works with signals defined by human hearing, roughly 20 Hz to 20 kHz and a dynamic range of about 120 dB. Sound pressure level is measured in dB SPL relative to 20 μPa; the threshold of hearing is near 0 dB SPL at 1–4 kHz, conversation is about 60 dB SPL, and the threshold of pain is 120–130 dB SPL. **Weighting** filters (A-weighting for low-level noise measurements) approximate the ear's frequency-dependent sensitivity.

Format	Sampling rate	Word length	Bit rate (stereo)	Typical use
Narrowband telephony (G.711)	8 kHz	8-bit companded	64 kbit/s (mono)	Fixed-line voice
Wideband speech (AMR-WB)	16 kHz	coded	6.6–23.85 kbit/s (mono)	Mobile voice
CD audio	44.1 kHz	16-bit	1411.2 kbit/s	Music distribution
Studio / broadcast	48 or 96 kHz	24-bit	2304–4608 kbit/s	Production
Perceptual codec (MP3, AAC)	44.1/48 kHz	coded	96–320 kbit/s	Streaming

Table 24.3 — Common audio formats.

The core audio operations are the tools of this part:

- **Equalisation and crossovers** with biquad IIR filters (Chapter 23).
- **Dynamic-range processing**: compressors and limiters measure the signal envelope and apply a time-varying gain, which is a nonlinear, level-dependent system, and must be designed so that the gain changes do not themselves create audible distortion.
- **Reverberation and spatial audio**: long FIR filters (measured room impulse responses, seconds long) applied by fast convolution (Chapter 22), and head-related transfer functions for headphones.
- **Sample-rate conversion** (Section 24.2) between 44.1 and 48 kHz domains.
- **Noise reduction**: STFT-domain Wiener gains (Sections 24.4 and 24.7) and adaptive noise cancelling (Section 24.5).
- **Perceptual coding**: MP3 and AAC split the signal into about 32 subbands or a 1024-line modified DCT, estimate from a psychoacoustic model which components are masked by louder neighbours in time and frequency, and spend bits only on what is audible. A 128 kbit/s MP3 stream is 11 times smaller than CD audio ($1411.2/128 = 11.0$), and at that rate most listeners cannot reliably distinguish it from the original on most material.

> **Status of the principle: Perceptual masking models**
> *Empirical.* The masking thresholds used by audio codecs come from listening experiments and are fitted models of human hearing; they hold for typical listeners and material, and codec artefacts appear when signals (sharp transients, pure tones, unusual listeners) fall outside the conditions the models were fitted to.

> **Boundary.**
> Image sensors, optics, microphones, loudspeakers and the converters that connect them to processors belong to *Engineering Physics* and *Electronics & Computing*. This section covers the signal-processing operations performed on the resulting samples.

What remains open

Open question	Where it is answered
How a delta-sigma converter shapes quantisation noise so that decimation can remove it	Chapter 21 (back reference); converter circuits in *Electronics & Computing*
How adaptive equalisers and matched filters recover symbols from a dispersive channel	Chapter 27
How entropy coding compresses the quantised DCT and wavelet coefficients of images and audio	Chapter 28
The statistics of noise and random processes on which Wiener and LMS theory rest	Chapter 29; the probability theory itself in *The Mathematics Universe*
How OFDM, MIMO and array processing use multirate filter banks, adaptive filters and subspace methods	Chapter 30
How digital filters, sample rates and delays enter the design of digital controllers	Chapter 35
The windowed DFT and Welch estimator that the spectrogram and the classical estimators build on	Chapter 22 (back reference)

Next: Chapter 25 opens Part VI by placing these signal-processing tools inside a complete communication system of source, transmitter, channel, noise and receiver.

Part VI — Communication Theory

Parts I–V built the tools to describe how circuits and systems shape signals: networks and transmission lines, transforms and transfer functions, sampling, the DFT and digital filters. Part VI turns those tools to the purpose for which much of electrical engineering exists — moving information from one place, or one time, to another. A communication link is a chain of signal-processing blocks between a source and a destination, with a channel in the middle that attenuates, distorts, delays and adds noise and interference, all within a power and bandwidth fixed by physics, batteries and regulators. This part asks how such a chain should be designed and how well it can possibly work, and answers in numbers: link budgets, signal-to-noise ratios, energy per bit, error probabilities and the Shannon capacity that bounds them.

By the end of this part you should understand:

- the generic model of a communication system and the roles of source, transmitter, channel, receiver and destination;
- how thermal and other noise, interference, bandwidth and signal-to-noise ratio set the performance of a link, and how to compute a link budget in decibels;
- how amplitude, frequency and phase modulation place an analog message on a carrier, what each costs in bandwidth and power, and how the superheterodyne receiver recovers it;
- how analog messages are digitised (PCM, delta modulation), how bits are put on a line (line coding) or a carrier (ASK, FSK, PSK, QPSK, QAM), and why the matched filter is the optimum receiver;
- how bit-error probability depends on E_b/N_0 for each modulation, and how the power–bandwidth trade-off is bounded by channel capacity;
- what entropy, mutual information and channel capacity mean, and how source and channel codes approach the theoretical limits;
- how random-process models describe communication noise, noise figure and detection performance;
- how radio propagation, fading and multipath shape wireless systems, and how cellular architecture, multiple access, OFDM and MIMO respond to them;
- how fibre attenuation and dispersion limit optical links, and how S-parameters and antennas describe the RF front end.

Chapter	Central question	What it delivers
25. Communication-System Fundamentals	What is a communication system made of, and which quantities decide how well it works?	System model, sources, transmitters and receivers, channel models, kTB noise, bandwidth definitions, SNR and E_b/N_0, performance measures and the Shannon bound
26. Analog Communication	How is an analog message carried on a carrier, and what does each method cost and buy?	AM, DSB-SC, SSB, VSB, FM and PM; modulation indices and Bessel spectra; Carson's rule; modulators, demodulators and PLLs; the superheterodyne; noise figures of merit and threshold
27. Digital Communication	How are bits turned into waveforms, and how often does the receiver decide wrongly?	PCM and companding, delta modulation, line codes, Nyquist pulse shaping, constellations, ASK/FSK/PSK, QPSK, QAM, matched filtering, BER and SER in AWGN
28. Information Theory and Coding	How much information can a channel carry, and how do codes approach that limit?	Entropy, mutual information, channel capacity, source coding and compression, error detection, block and convolutional codes, modern coding methods
29. Random Signals and Communication Noise	How is noise modelled as a random process, and how does it set detection performance?	Gaussian and white-noise models, noise bandwidth, noise figure and cascades, detection in noise, BER and SER, performance trade-offs
30. Wireless and Mobile Communications	How do real radio paths behave, and how do mobile systems cope with them?	Path loss, fading, multipath, diversity, cellular architecture, multiple access, OFDM, MIMO, link budgets and system performance
31. Optical Communications	What limits transmission over optical fibre, and how are fibre links designed?	Fibre attenuation and dispersion, optical transmitters and receivers, WDM, optical link budgets and system limits
32. RF and Microwave Fundamentals	How do circuits and antennas behave at radio and microwave frequencies?	RF signal behaviour, matching, S-parameters, RF components and resonators, microwave networks, antennas, radiation patterns, gain and directivity, RF measurements

Part VI draws on almost everything before it. Fourier analysis and distortion (Chapter 18) describe channels; sampling and quantization (Chapters 19, 21) underlie PCM; the DFT (Chapter 22) is the engine of OFDM; digital and adaptive filters (Chapters 23–24) implement pulse shaping and equalisation; resonant circuits and transmission lines (Chapters 13–14) return as preselectors, matching networks and the Smith chart. Part VII reuses its feedback ideas: phase-locked loops and automatic gain control are control systems, analysed with the stability tools of Chapter 34. Probability and random processes are recapped, not derived (*The Mathematics Universe*); wave and antenna physics belong to *Engineering Physics*; the transistor circuits of amplifiers, mixers and converters to *Electronics & Computing*.

Chapter 25 — Communication-System Fundamentals

Part VI · Communication Theory

A hospital installs wireless telemetry so that cardiac patients can walk the corridors while their ECG is watched from the nursing station. On the bench the system is flawless. On the ward, three things go wrong. Near the lift shaft the trace drops out for a few seconds at a time. Beside a bank of infusion pumps it shows a fuzz of spikes that the arrhythmia software reads as ventricular beats. And when a second ward installs the same system, both begin to lose packets whenever the two transmitters happen to be on adjacent channels. Nothing is broken. Each failure is a different face of the same problem: a message of a few hundred hertz of bandwidth has to cross a channel that attenuates it, adds noise, admits interference from other users and other machines, and does all of this at a power level and in a bandwidth that a regulator and a battery have fixed in advance.

This chapter builds the vocabulary for that problem. It treats a communication system as a chain of signal-processing blocks, each of which the earlier parts of this book already know how to analyse, and introduces the quantities used to decide whether the chain works: bandwidth, signal-to-noise ratio, energy per bit, error probability, spectral and power efficiency.

> **What is a communication system made of, and which properties of the source, the channel and the noise decide how well it can work?**
> **How are bandwidth, signal power and noise traded against one another, and what single numbers summarise a link's performance?**

Concept	Engineering destination
Block model of a communication system	Architecture of radio, cable, fibre, telemetry and storage links
Information sources and their bandwidths	Codec choice, sampling rates, channel sizing for voice, audio, video, biosignals
Transmitter and receiver functions	RF front ends, modems, software-defined radios
Channel models	Link design for twisted pair, coax, fibre, radio and power-line carriers
Thermal noise kTB	Receiver sensitivity, radar range, instrumentation amplifier noise floors
Interference	Frequency planning, EMC, co-channel reuse in cellular networks
Bandwidth definitions	Spectrum licensing, filter specifications, regulatory masks
SNR, E_b/N_0, C/N_0	Link budgets, modem fallback rates, satellite and deep-space links
Performance measures and the Shannon limit	Comparing modulation and coding schemes, setting design targets

Roadmap. The generic system model (25.1) → what the source delivers (25.2) → what transmitters and receivers do (25.3) → what the channel does (25.4) → noise and interference (25.5) → bandwidth (25.6) → signal-to-noise ratio and its variants (25.7) → the performance measures that tie them together, bounded by Shannon's capacity formula (25.8).

25.1 Communication-system models

The generic chain

Every communication system, from a 19th-century telegraph to a 5G handset, can be drawn as the same chain (Figure 25.1). An **information source** produces a message: a voltage from a microphone, an ECG electrode or a strain gauge; a stream of characters; the samples of a video frame. A **transmitter** converts the message into a signal suited to the channel. The **channel** is the physical medium — a pair of copper wires, a coaxial cable, an optical fibre, the radio path between two antennas, the magnetic surface of a disk — together with everything that happens to the signal on the way. The **receiver** processes what arrives and produces an estimate of the original message for the **destination**: a loudspeaker, a display, a protection relay, a computer.

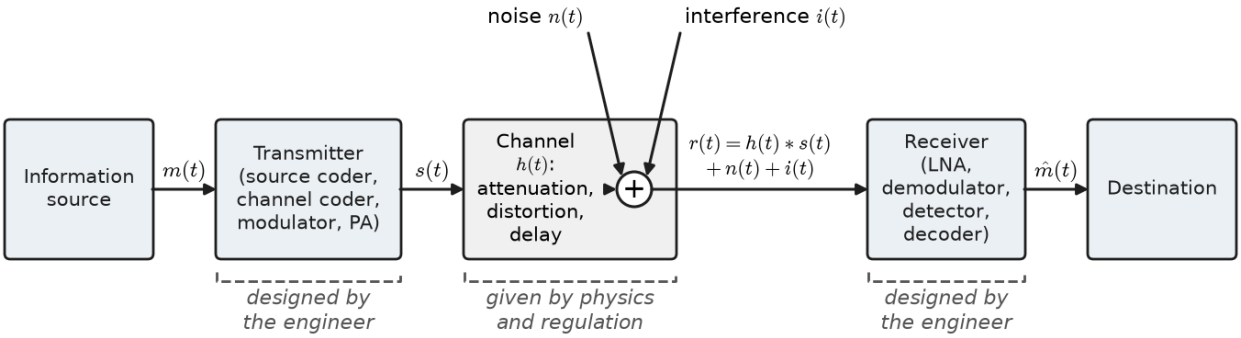

Figure 25.1 — The generic communication-system model: source, transmitter, channel with additive noise and interference, receiver and destination.

The division in Figure 25.1 between what the engineer designs (transmitter and receiver) and what is given (the channel) is the organising idea of communication engineering. The channel is not chosen freely: the physics of propagation, the noise temperature of the environment, the transmit power permitted by regulation or by a battery, and the bandwidth allocated by a licence are all constraints. The transmitter and receiver are then designed to deliver the best fidelity possible within those constraints.

Mathematically the chain is a sequence of systems in the sense of Chapter 16. Most blocks are linear and time-invariant (filters, cables, amplifiers in their linear range); some are linear but time-varying (mixers, modulators — Chapter 26 shows that multiplying by a carrier is linear but not time-invariant); a few are deliberately nonlinear (quantizers, detectors, limiters). The analysis tools of Parts III–V — impulse response and convolution (Chapter 17), Fourier transforms and transfer functions (Chapter 18), sampling (Chapter 19) and the DFT (Chapter 22) — therefore apply block by block.

The simplest useful channel model

The channel in Figure 25.1 is described by the received signal

$$r(t) = \int_{-\infty}^{\infty} h(\tau)\, s(t-\tau)\, d\tau + n(t) + i(t)$$

where $h(t)$ is the channel's impulse response (attenuation, delay and linear distortion), $n(t)$ is random noise and $i(t)$ is interference from other transmitters or machines. Three special cases recur through Part VI:

1. **Additive white Gaussian noise (AWGN) channel**: $h(t) = \alpha\,\delta(t - t_0)$, a pure gain and delay, plus white Gaussian noise. It models a short cable, a line-of-sight microwave link or a satellite link well, and it is the reference against which every modulation scheme is compared.
2. **Linear filter channel**: $h(t)$ has memory, so different frequencies are attenuated and delayed differently. It models telephone loops, long coaxial cables, DSL and backplanes; the result is intersymbol interference (Chapter 27).
3. **Time-varying (fading) channel**: $h(t)$ changes with time because the transmitter, receiver or scatterers move. It models mobile radio (Chapter 30).

> **Status of the principle: The additive-noise linear channel model**
> *Approximate.* Valid when the medium is linear at the signal levels used and noise is independent of the signal; it fails for power amplifiers driven into compression, for fibre at high launch power (Chapter 31), and for quantization noise that is correlated with the signal.

Classifications of systems

Several classifications help locate a system before analysing it.

- **Analog and digital.** An analog system delivers a continuous waveform whose fidelity is judged by SNR and distortion (Chapter 26). A digital system delivers symbols from a finite alphabet whose fidelity is judged by error probability (Chapter 27). Many modern systems carry analog messages digitally: the voice is sampled and coded, transmitted as bits, and reconstructed.
- **Baseband and passband.** A baseband signal occupies frequencies from near zero up to some maximum (an Ethernet line code, the signal on an ECG lead wire). A passband signal is centred on a carrier frequency f_c far above its bandwidth (an FM broadcast at 98.1 MHz, Wi-Fi at 2.437 GHz). Radio requires passband signals because antennas are efficient only when their size is comparable to a wavelength and because the spectrum must be shared by frequency.
- **Simplex, half-duplex and full-duplex.** Broadcast radio is simplex. A walkie-talkie is half-duplex (push to talk). A telephone is full-duplex, achieved either by separate frequencies (frequency-division duplex) or alternating time slots (time-division duplex).
- **Point-to-point, broadcast and multiple access.** A microwave backhaul link joins two points; a TV transmitter serves an unlimited audience; a cellular cell serves many users who share the channel by frequency, time, code or space (Chapter 30).

Layering

Practical networks divide the chain of Figure 25.1 into layers. The **physical layer** — the subject of this part of the book — is responsible for turning bits into waveforms and back. Above it, link-layer protocols frame bits into packets and retransmit lost ones, and network and transport layers route and deliver them end to end. The layered view explains why the telemetry system of the opening paragraph "loses packets": the physical layer delivered errors, and the link layer discarded the corrupted frames. Here we stay at the physical layer, where the questions are about waveforms, spectra, noise and probability.

25.2 Information sources

Analog and digital sources

The source sets what the system must carry. An **analog source** produces a waveform $m(t)$ characterised by its amplitude range, its spectrum (usually summarised as a bandwidth W), and the fidelity the destination needs. A **digital source** produces a sequence of symbols drawn from a finite alphabet at some rate; it is characterised by its symbol rate and by the statistics of the symbols. An

analog source becomes digital after sampling at a rate of at least $2W$ (Chapter 19) and quantizing each sample to n bits (Chapter 27), giving a raw bit rate

$$R_b = n\, f_s \geq 2nW$$

Table 25.1 collects the sources that appear most often in electrical engineering.

Table 25.1 — Representative information sources (typical values).

Source	Baseband bandwidth	Typical digital form	Bit rate
Telephone speech	300–3400 Hz	8 kHz × 8 bit companded PCM (G.711)	64 kb/s
Speech, mobile codec	300–3400 Hz	AMR, EVS speech codecs	4.75–24.4 kb/s
Wideband speech	50–7000 Hz	16 kHz sampling	16–64 kb/s (coded)
High-fidelity audio	20 Hz–20 kHz	44.1 kHz × 16 bit × 2 channels (CD)	1.411 Mb/s raw; 128–320 kb/s (AAC/MP3)
Diagnostic ECG	0.05–150 Hz	500 Hz × 12 bit per lead	6 kb/s per lead
EEG	0.5–100 Hz	256–512 Hz × 16–24 bit per channel	4–12 kb/s per channel
Surface EMG	10–500 Hz	1–2 kHz × 16 bit	16–32 kb/s
Power-system sampled values (IEC 61850-9-2)	up to ~2 kHz	80 samples per 50 Hz cycle, 8 channels	~5 Mb/s per merging unit (with framing)
Analog composite video (historic)	4.2 MHz (525-line), 5–6 MHz (625-line)	—	—
HD video 1080p60	~30 MHz luminance equivalent	Uncompressed 4:2:2, 10 bit (3G-SDI)	2.97 Gb/s raw; 3–10 Mb/s (H.265/H.264)
Telemetry from slow sensors	< 1 Hz	1 sample/s × 16 bit	tens of b/s

Two lessons follow from the table. First, bandwidths span nine orders of magnitude, from sensor data at a fraction of a hertz to video at tens of megahertz, so there is no single "right" system. Second, the raw digital rate is usually far larger than what is actually sent: speech at 64 kb/s is routinely compressed to 12 kb/s, and HD video by a factor of several hundred, because real sources are redundant. How far a source can be compressed without loss is governed by its **entropy**, the subject of Chapter 28. For now it is enough to know that the relevant measure of a source is the rate at which it produces *information*, not the rate at which it produces samples.

Source statistics that matter to the transmitter

Beyond bandwidth, three statistical properties of an analog source affect the design of the transmitter.

- **Peak-to-average ratio (crest factor).** Speech has a crest factor of 15–20 dB; a sinusoidal test tone only 3 dB. A modulator or quantizer must accommodate the peaks without clipping, which means the average power is far below the maximum the hardware can handle. Companding in telephone PCM (Chapter 27) and pre-emphasis in FM broadcasting (Chapter 26) are both responses to this.
- **Spectral shape.** Most of the power of speech lies below 1 kHz, while intelligibility depends on 1–3 kHz. Audio and video spectra fall steeply with frequency. Systems that add flat noise therefore damage the high-frequency content most, which is why emphasis networks exist.
- **DC and low-frequency content.** An ECG must preserve frequencies down to 0.05 Hz to show the ST segment faithfully; a voice channel need not pass anything below 300 Hz. A channel that cannot pass DC (transformer-coupled lines, AC-coupled amplifiers) forces the use of modulation or of DC-free line codes (Chapter 27).

Worked example — the ECG telemetry stream. A 12-lead diagnostic ECG recorder samples each lead at 500 Hz with 12-bit resolution. The raw payload rate is

$$R_b = 12 \text{ leads} \times 500 \text{ s}^{-1} \times 12 \text{ bit} = 72 \text{ kb/s}.$$

With 25 % protocol overhead (headers, sequence numbers, a cyclic redundancy check), the link must carry about 90 kb/s. The ward telemetry radio in the opening paragraph uses a channel of 25 kHz; to carry 90 kb/s in 25 kHz requires a spectral efficiency of $90/25 = 3.6$ b/s/Hz. Section 25.8 shows that this demands an SNR of at least $2^{3.6} - 1 \approx 11$, or 10.4 dB, even for an ideal system — and practical modems need several decibels more. The design team therefore has three options: a wider channel, lossless compression of the ECG (typically 2–3:1, since successive samples are highly correlated), or sending fewer leads. The source analysis has already shaped the system before any modulation is chosen.

25.3 Transmitters and receivers

The transmitter

A digital transmitter performs, in order, the operations shown in the top half of Figure 25.2.

1. **Source coding** removes redundancy: speech, audio and video codecs; lossless compression of data (Chapter 28).
2. **Channel coding** adds controlled redundancy so that the receiver can detect or correct errors (Chapter 28).
3. **Modulation** maps bits to waveforms: line codes on a cable, phase or amplitude states on a carrier (Chapter 27). Analog transmitters skip the coding steps and impress the message directly on the carrier by amplitude, frequency or phase modulation (Chapter 26).
4. **Pulse shaping and filtering** confine the spectrum to the allocated channel.
5. **Up-conversion** moves the signal from baseband or a low intermediate frequency to the carrier frequency.
6. **Power amplification** raises the signal to the transmitted power, which ranges from microwatts (a capsule endoscope) through 0.2 W (a mobile handset), 1–40 W (a cellular base-station sector)

and 50 kW (a medium-wave broadcaster) to megawatts of peak power in a radar.

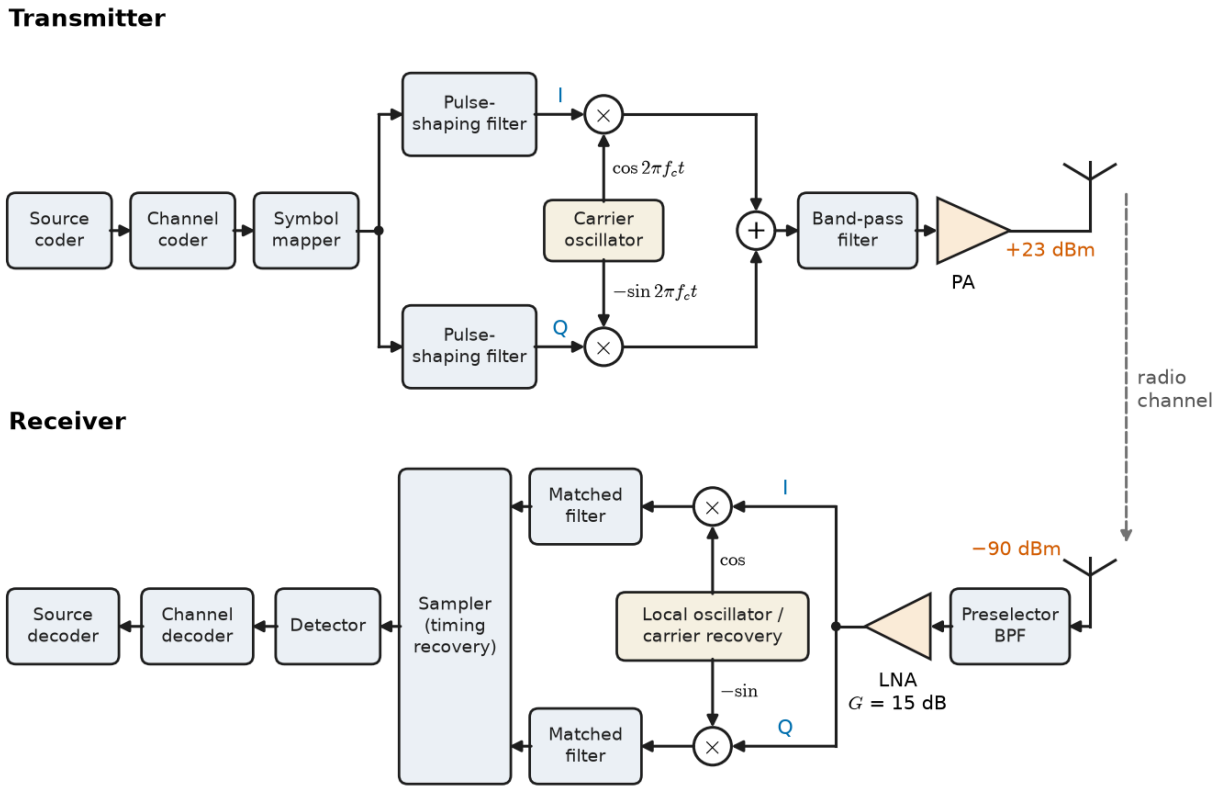

Figure 25.2 — Generic digital transmitter and superheterodyne-style receiver, showing the coding, modulation and frequency-conversion stages and the I/Q representation.

Complex-baseband representation

The I/Q branches in Figure 25.2 reflect the standard mathematical representation of a passband signal. Any real signal confined to a band around f_c can be written as

$$s(t) = s_I(t)\cos(2\pi f_c t) - s_Q(t)\sin(2\pi f_c t) = \operatorname{Re}\left\{\tilde{s}(t)\, e^{j2\pi f_c t}\right\}, \qquad \tilde{s}(t) = s_I(t) + j\, s_Q(t)$$

where s_I and s_Q are real baseband signals — the **in-phase** and **quadrature** components — and $\tilde{s}(t)$ is the **complex envelope**. The amplitude and phase of the carrier are $|\tilde{s}(t)|$ and $\arg \tilde{s}(t)$. The representation is exact for any bandpass signal whose bandwidth is less than $2f_c$, and it is how every modern modem and software-defined radio works internally: the carrier is applied or removed in a single mixing stage, and all the signal processing happens on the complex envelope at a sample rate set by the bandwidth, not by the carrier. A 20 MHz Wi-Fi channel at 5.2 GHz is processed with complex samples at a few tens of megasamples per second, not at the 10 GS/s that direct sampling of the carrier would require.

Status of the principle: Complex-envelope (I/Q) representation
Derived. A mathematical identity for real bandpass signals, following from the Fourier-transform symmetry of real signals (Chapter 18); it requires only that the signal's spectrum vanish near zero frequency, which holds whenever the bandwidth is less than twice the carrier.

The receiver

The receiver must undo the transmitter's operations, but it faces three problems the transmitter does not.

- **The signal is weak.** A transmitter operates at watts; the receiver input of a cellular phone at the cell edge is around −100 dBm (0.1 pW), and that of a GPS receiver about −130 dBm, below the thermal noise in its bandwidth. The first stage must therefore be a **low-noise amplifier** (LNA), because noise added later is amplified less than noise added first (the cascade formula is derived in Chapter 29).
- **The wanted signal is surrounded by unwanted ones.** A receiver tuned to one FM station has dozens of others, some 80 dB stronger, within a few megahertz. **Selectivity** — the ability to reject adjacent and distant signals — is provided by filters, and the superheterodyne architecture of Chapter 26 was invented precisely to make selectivity independent of tuning.
- **The receiver does not know the carrier phase, frequency or symbol timing exactly.** It must estimate them from the received signal (**synchronization**): carrier recovery loops, phase-locked loops and timing-recovery loops. A 1 ppm error in a 2.4 GHz crystal oscillator is a 2.4 kHz frequency offset, which rotates a QPSK constellation through a full turn in 0.4 ms.

After amplification, filtering, down-conversion and synchronization, the receiver applies a **detector**. For analog signals this is a demodulator (envelope detector, frequency discriminator) whose output is the estimated message. For digital signals it is a matched filter followed by a decision device that chooses the most likely transmitted symbol (Chapter 27).

Boundary.
The circuits that implement LNAs, mixers, oscillators and power amplifiers — transistor biasing, matching networks, linearity of real devices — belong to *Electronics & Computing*. This chapter treats them as blocks with a gain, a noise contribution and a frequency response.

25.4 Channels

Physical media

The channel is where the signal suffers. Table 25.2 compares the guided media used in practice; Chapter 14 analysed them as transmission lines, and Chapter 31 treats fibre in detail.

Table 25.2 — Attenuation of common guided media (typical values).

Medium	Frequency	Attenuation	Typical use
Telephone twisted pair, 0.5 mm (24 AWG)	1 kHz	~0.6 dB/km (with loading) to 1.5 dB/km	Voice-band local loop
Same pair	1 MHz	~20 dB/km	ADSL
Cat 5e/6 UTP	1 MHz / 100 MHz	2 / 22 dB per 100 m	Ethernet
RG-58 coax	100 MHz / 1 GHz	~16 / ~55 dB per 100 m	Lab and short RF runs
RG-6 coax	100 MHz / 1 GHz	~6.5 / ~20 dB per 100 m	Cable TV drops
7/8-inch foam coax	2 GHz	~6 dB per 100 m	Base-station feeders
Single-mode fibre	1310 nm / 1550 nm	0.33 / 0.20 dB/km	Long-haul and access networks
Power-line carrier (MV line)	50–500 kHz	0.1–1 dB/km plus coupling losses	Utility protection signalling

Attenuation in decibels grows linearly with distance on a guided medium: 10 km of fibre at 1550 nm loses 2 dB, 100 km loses 20 dB. In free space, by contrast, the power spreads over a sphere and the loss grows only with the square of distance. Between isotropic antennas,

$$\frac{P_R}{P_T} = G_T G_R \left(\frac{\lambda}{4\pi d}\right)^2, \qquad L_{fs} = 20\log_{10}\frac{4\pi d}{\lambda}\ \text{dB}$$

the **Friis transmission formula**, where G_T and G_R are the antenna gains (Chapter 32). Each doubling of distance adds 6 dB of loss, and each factor of ten adds 20 dB. This difference is why a few hundred metres of coax is enough to lose a 2 GHz signal completely, whereas a radio link reaches a geostationary satellite 36 000 km away: at long distances, logarithmic growth beats linear growth. Terrestrial radio paths deviate from free space because of ground reflection, obstruction and multipath; Chapter 30 replaces the exponent 2 by empirical values between about 2 and 5.

> **Status of the principle: Friis free-space transmission formula**
> *Derived*. Follows from energy conservation over a spherical wavefront and the definition of antenna gain and effective aperture; valid in the far field of both antennas with an unobstructed path and no ground reflection.

The decibel bookkeeping

Because the chain of Figure 25.1 multiplies gains and losses, link calculations are done in decibels, where they add. Power levels are expressed in dBm (decibels relative to 1 mW) or dBW. The conversions to remember are: 0 dBm = 1 mW, 30 dBm = 1 W, −30 dBm = 1 µW, −100 dBm = 0.1 pW; and 3 dB is a factor 2, 10 dB a factor 10.

Worked example — a 2.4 GHz link budget. An industrial monitoring link transmits $P_T = 1$ W (30 dBm) at 2.4 GHz between two antennas of 2 dBi gain, 100 m apart in a clear line of sight. The wavelength is $\lambda = c/f = 0.125$ m, so

$$L_{fs} = 20\log_{10}\frac{4\pi \times 100}{0.125} = 20\log_{10}(10\,053) = 80.05 \text{ dB}.$$

The received power is

$$P_R = 30 + 2 + 2 - 80.05 = -46.05 \text{ dBm} \approx 25 \text{ nW}.$$

At 1 km the free-space loss rises by exactly 20 dB to 100.05 dB, and $P_R = -66.05$ dBm. Whether either level is "enough" cannot be decided yet: it depends on the noise in the receiver's bandwidth, which Section 25.5 supplies, and the example continues in Section 25.7.

Linear and nonlinear distortion

Attenuation alone is harmless: an amplifier can restore the level. What damages the signal is distortion. A channel is **distortionless** if its output is a scaled, delayed copy of its input, which requires (Chapter 18) a flat magnitude response and a linear phase response — constant group delay — across the signal's band. Real channels violate both:

- A telephone loop attenuates 3 kHz more than 300 Hz and has a group delay that rises sharply near the band edges; voice survives, but a fast data signal smears into its neighbours (intersymbol interference).
- A long coax has loss that grows as $\sqrt{f}$ because of the skin effect, so a fast digital edge arrives with a slow tail. Cable equalizers invert this response.
- A radio path with two routes of different lengths has a frequency response with periodic notches (Chapter 30).

Nonlinear distortion occurs when the channel includes a device whose output is not proportional to its input: a power amplifier near saturation, a fibre at high launch power, a mixer driven too hard. A nonlinearity creates new frequencies. Two tones at f_1 and f_2 passing through a cubic nonlinearity produce third-order intermodulation products at $2f_1 - f_2$ and $2f_2 - f_1$, which fall close to the original tones and cannot be filtered out. This is why multi-carrier transmitters must be backed off from saturation by several decibels, sacrificing efficiency for linearity.

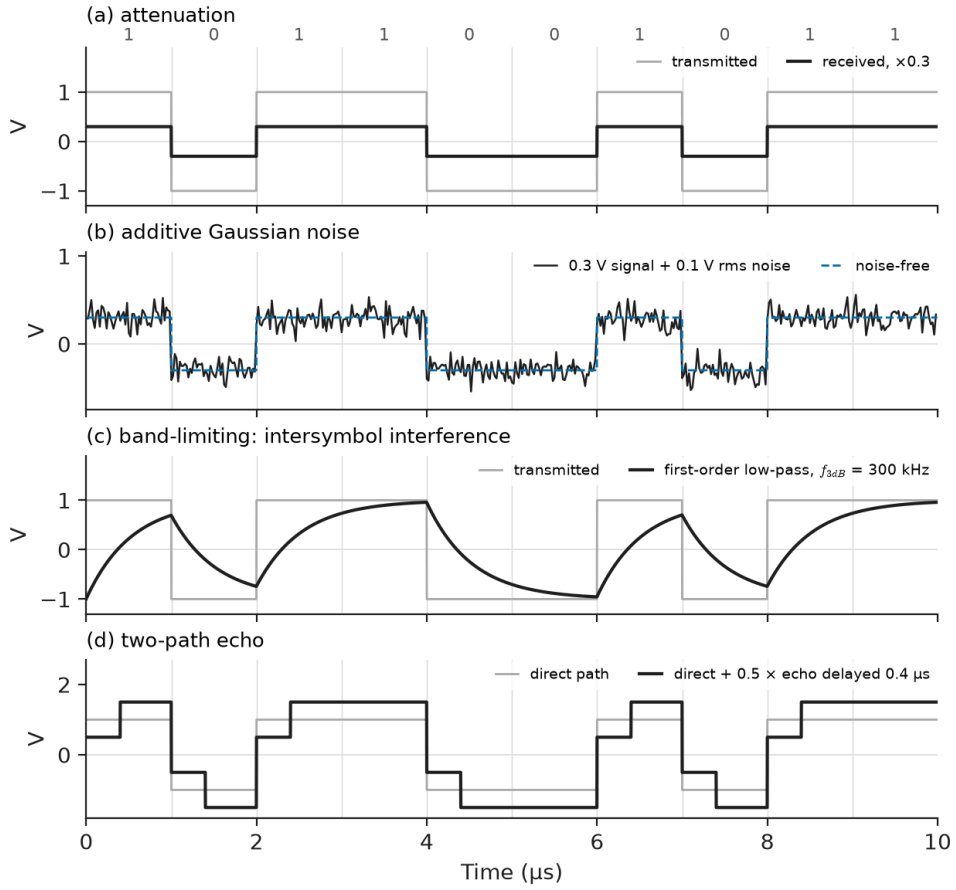

Figure 25.3 — Four channel impairments acting on the same 1 Mb/s NRZ bit pattern: attenuation, additive noise, band-limiting (intersymbol interference) and a two-path echo.

Figure 25.3 shows the four impairments side by side. Attenuation (a) alone would be harmless; noise (b) and band-limiting (c) both make the received waveform ambiguous, and an echo (d) is a band-limiting filter in disguise, since its frequency response has notches.

Storage as a channel

A hard disk, a flash memory or an optical disc is a channel in time rather than space: data written now is read later, after degradation. The same models apply. A flash cell's stored charge leaks and is disturbed by reads of neighbouring cells, giving a noisy channel whose error rate rises with age; a disk read head sees intersymbol interference from neighbouring magnetic transitions. Error-correcting codes (Chapter 28) are as essential in storage as in radio.

25.5 Noise and interference

Thermal noise

Every resistor at absolute temperature T generates a random voltage from the thermal agitation of its electrons. Over a bandwidth B small compared with kT/h (about 6 THz at room temperature), the available noise power — the power delivered into a matched load — is

$$N = kTB$$

where $k = 1.380\,649 \times 10^{-23}$ J/K is Boltzmann's constant. The power is independent of the resistance and proportional to bandwidth: thermal noise is **white**, with a one-sided power spectral density $N_0 = kT$. At the standard reference temperature $T_0 = 290$ K,

$$kT_0 = 4.00 \times 10^{-21}\ \text{W/Hz} = -174.0\ \text{dBm/Hz}.$$

This single number underlies every link budget. Table 25.3 gives the thermal noise floor for common channel bandwidths.

Table 25.3 — Thermal noise power kT_0B at 290 K.

Bandwidth	System	kT_0B
1 Hz	reference	−174.0 dBm
3.1 kHz	telephone channel	−139.1 dBm
25 kHz	narrowband telemetry/PMR	−130.0 dBm
200 kHz	GSM carrier, FM broadcast	−121.0 dBm
1 MHz	—	−114.0 dBm
6 MHz	cable TV channel	−106.2 dBm
20 MHz	Wi-Fi channel, LTE carrier	−101.0 dBm
400 MHz	5G mmWave carrier	−88.0 dBm

> **Status of the principle: Thermal noise power kTB (Johnson–Nyquist)**
> *Fundamental.* Follows from statistical thermodynamics (equipartition of energy among the modes of a transmission line in equilibrium); the classical form kTB is accurate when $hf \ll kT$, which holds up to hundreds of gigahertz at room temperature but fails for cryogenic receivers at millimetre-wave frequencies.

A real receiver adds noise of its own on top of the thermal noise of its source. This is summarised by its **noise figure** NF: the ratio, in decibels, of the SNR at its input to the SNR at its output, when the input is at 290 K. A receiver with $NF = 7$ dB behaves as if its input noise were 7 dB above kT_0B. Typical values: 0.5–1.5 dB for a satellite or GPS LNA, 3–5 dB for a Wi-Fi or LTE receiver, 6–9 dB for a low-cost receiver chip. The theory of noise figure and of noise in cascades is the subject of Chapter 29.

Worked example — receiver sensitivity. A GSM-type receiver with a 200 kHz channel has a noise figure of 7 dB and needs an SNR of 9 dB at the detector to meet its error-rate target. Its noise floor referred to the input is

$$N = -174.0 + 10\log_{10}(2 \times 10^5) + 7 = -174.0 + 53.0 + 7 = -114.0\ \text{dBm}.$$

The **sensitivity**, the smallest input that meets the target, is $-114.0 + 9 = -105.0$ dBm, or about 32 fW. Every dB of noise figure removed and every dB of required SNR saved through better modulation or

coding lowers this figure one-for-one, which, through the path-loss law, extends range. With a free-space exponent of 2, 6 dB doubles the range; with the exponent of 3.5 typical of urban radio (Chapter 30), 6 dB extends range by a factor $10^{6/35} = 1.48$.

Other internal noise sources

Thermal noise is not the only noise generated inside the receiver.

- **Shot noise** arises because current in a diode, transistor or photodetector consists of discrete charges crossing a barrier at random times. Its power spectral density is $2qI$ (A²/Hz), white up to very high frequencies. It dominates optical receivers (Chapter 31) and some radiation detectors.
- **Flicker (1/f) noise** has a power spectral density inversely proportional to frequency. It matters below about 1–10 kHz in semiconductor devices and is the reason precision instrumentation amplifiers and ECG front ends use chopper stabilisation, and why direct-conversion radio receivers struggle with signals near DC.
- **Phase noise** of oscillators spreads a pure carrier into a skirt of noise, which mixes strong adjacent signals into the wanted channel (reciprocal mixing) and rotates constellation points randomly.
- **Quantization noise** is introduced deliberately by analog-to-digital conversion (Chapters 19, 21, 27).

External noise

The antenna collects noise from its surroundings as well as signal. Below about 30 MHz, atmospheric noise from lightning worldwide dominates and can exceed thermal noise by 40–60 dB; at VHF and low UHF, man-made noise from ignition systems, switch-mode power supplies, LED drivers and power-line corona is the limit in urban areas; between roughly 1 and 10 GHz the sky is quiet, with a background of only a few kelvin, which is why satellite navigation, satellite communication and radio astronomy cluster there. Above 10 GHz, absorption by water vapour and oxygen adds noise and attenuation. Figure 25.4 summarises the frequency dependence.

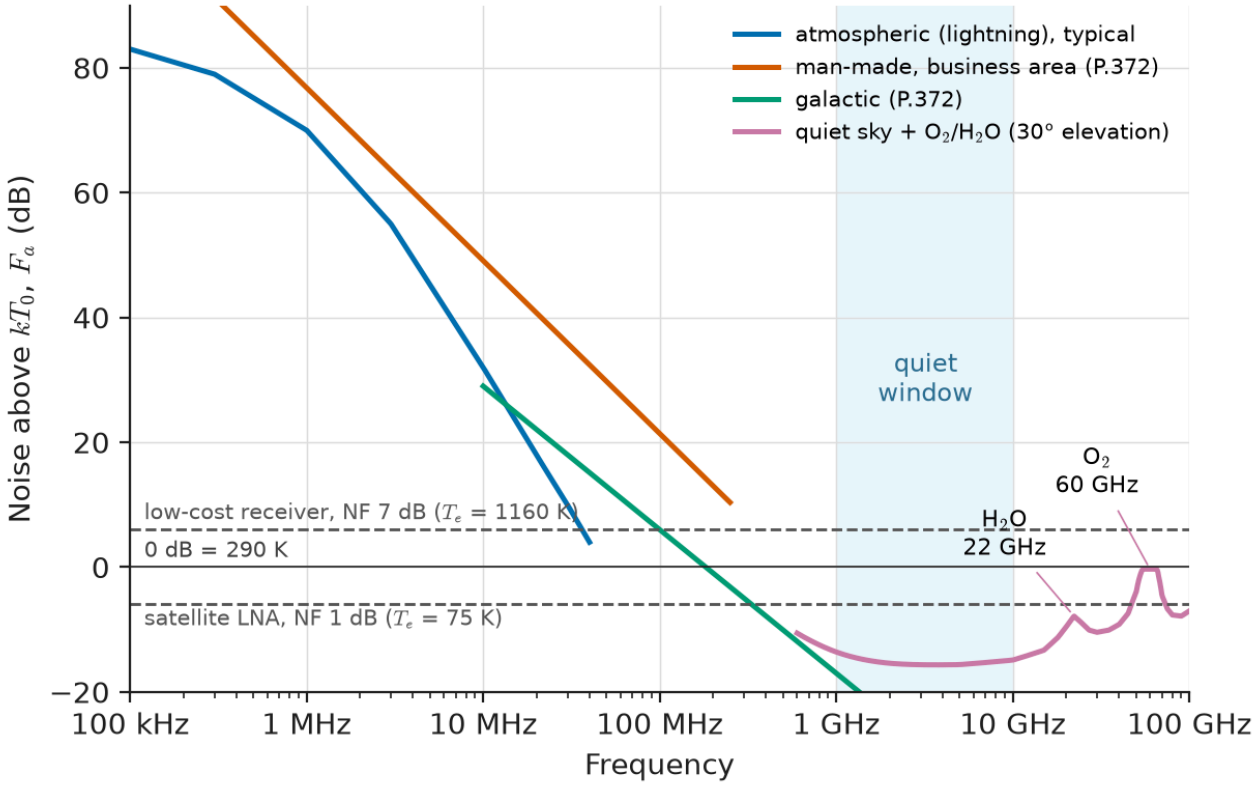

Figure 25.4 — Typical external noise levels relative to kT₀B versus frequency, compared with receiver noise figures.

The consequence for design is that a low receiver noise figure is wasted at HF, where the antenna noise is already far above thermal, but is decisive at microwave frequencies, where the antenna sees cold sky. A medium-wave AM receiver with a 10 dB noise figure is perfectly adequate; a GPS receiver with the same noise figure would lose nearly 10 dB of margin.

Interference

Interference is unwanted signal with structure — another transmitter, a harmonic, a switching transient — rather than random noise. Its main forms are:

- **Co-channel interference**: another transmitter on the same frequency. Cellular systems reuse frequencies in distant cells, so co-channel interference, not noise, usually limits them; their figure of merit is the signal-to-interference ratio (SIR) or SINR.
- **Adjacent-channel interference**: energy spilling from a neighbouring channel because of imperfect transmitter spectra (sidebands, spectral regrowth from nonlinear amplifiers) or imperfect receiver selectivity. This is the failure in the ward-telemetry story.
- **Intermodulation**: products of two or more strong signals mixing in a nonlinear receiver stage, appearing in the wanted channel.
- **Crosstalk**: coupling between adjacent pairs in a cable. In DSL, near-end and far-end crosstalk from other subscribers' pairs in the same binder limit the data rate more than thermal noise does.
- **Power-line and switching interference**: 50/60 Hz and its harmonics coupled into biomedical and instrumentation leads; broadband impulses from switch-mode converters, motor drives and relays. A typical ECG signal of 1 mV can be accompanied by 50 Hz common-mode

pickup of tens of millivolts; it is the differential amplifier's common-mode rejection (often 100 dB or more) and the right-leg drive circuit, not filtering alone, that make the ECG readable.

Noise and interference differ in what can be done about them. Noise is unpredictable and can only be overcome by more signal energy or by averaging over more time. Interference has structure that can be exploited: it may be filtered (if spectrally separate), cancelled (if it can be measured, as with adaptive mains-hum cancellation, Chapter 24), spatially rejected by directional antennas, or avoided by frequency planning.

25.6 Bandwidth

Why there is more than one definition

Strictly time-limited signals have spectra that extend to infinite frequency (Chapter 18), so "bandwidth" always means "the frequency range that contains most of what matters". Several definitions are in use, and they give different numbers for the same signal (Figure 25.5).

- **Absolute bandwidth**: the range outside which the spectrum is exactly zero. Only strictly band-limited signals have one; it is a theoretical idealisation.
- **3-dB (half-power) bandwidth**: the range over which the power spectral density stays within 3 dB of its peak. Standard for filters and amplifiers.
- **Null-to-null bandwidth**: the width of the main lobe of the spectrum, convenient for rectangular-pulse modulation.
- **Equivalent noise bandwidth**: the width of a rectangular filter with the same peak gain that passes the same white-noise power (Chapter 29). It is the B that belongs in kTB.
- **Occupied (power-containment) bandwidth**: the range containing a specified fraction of the power, usually 99 %. This is the regulatory definition.
- **Bounded-spectrum (mask) bandwidth**: the range outside which the spectrum is below a specified level, such as -50 dB relative to the peak. Regulators publish emission masks of this type.

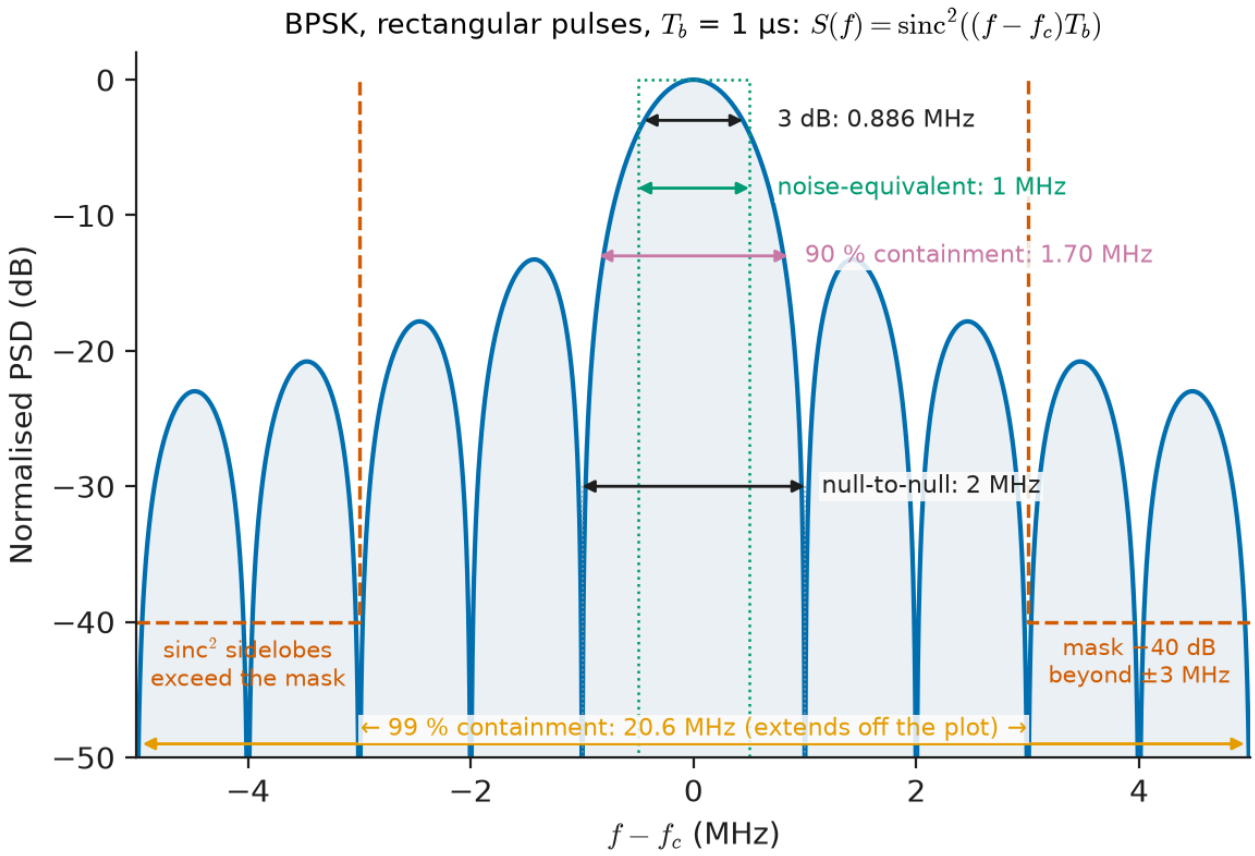

Figure 25.5 — Different bandwidth definitions applied to the power spectral density of a rectangular-pulse BPSK signal at 1 Mb/s.

For the rectangular-pulse BPSK spectrum in Figure 25.5 the definitions give widely different answers: about 0.89 MHz (3 dB), 1 MHz (noise-equivalent), 2 MHz (null-to-null), and roughly 20 MHz for 99 % containment, because the sinc^2 sidelobes decay only as $1/f^2$. This large ratio is exactly why practical transmitters shape their pulses (Chapter 27): a raised-cosine pulse with 35 % excess bandwidth confines essentially all the power to 1.35 MHz for the same 1 Mb/s BPSK signal.

Bandwidth of the message and of the transmitted signal

Two bandwidths must be distinguished. The **message bandwidth** W belongs to the source (3.4 kHz for telephone speech, 15 kHz for FM broadcast audio). The **transmission bandwidth** B_T is what the modulated signal occupies on the channel. Their ratio depends on the modulation: single-sideband AM gives $B_T = W$, conventional AM $B_T = 2W$, and broadcast FM $B_T \approx 12W$ (Chapter 26). For digital signals the relevant ratio is between the bit rate R_b and B_T, the **spectral efficiency** $\eta = R_b/B_T$ in b/s/Hz (Section 25.8).

Time–bandwidth duality

Bandwidth and time are reciprocal (Chapter 18): a pulse of duration T occupies a bandwidth of order $1/T$. A system that sends R_s symbols per second therefore needs bandwidth proportional to R_s. Nyquist showed (Chapter 27) that the minimum baseband bandwidth for R_s independent pulses without intersymbol interference is $R_s/2$; passband transmission of the same complex symbols needs

R_s. Radar is governed by the same reciprocity: a pulse compressed to 1 µs resolves targets 150 m apart in range, and resolving 15 m requires a bandwidth of about 10 MHz.

Bandwidth as a regulated resource

Radio spectrum is allocated by national regulators under the international framework of the ITU Radio Regulations. Table 25.4 gives representative channel widths; they set the B that every subsequent calculation in a given service must use.

Table 25.4 — Representative channel bandwidths (typical; regional variations apply).

Service	Band	Channel width
AM broadcast (medium wave)	531–1602 kHz (ITU Regions 1, 3); 530–1700 kHz (Region 2)	9 kHz; 10 kHz
FM broadcast	87.5–108 MHz	200 kHz (Region 2 spacing); 100 kHz raster elsewhere
Narrowband private mobile radio	VHF/UHF	12.5 kHz, 6.25 kHz
Medical telemetry (WMTS, USA)	608–614, 1395–1432 MHz	up to 6 MHz blocks
GSM	900/1800 MHz	200 kHz
LTE	0.45–3.8 GHz	1.4–20 MHz
5G NR	FR1 / FR2	up to 100 MHz / 400 MHz
Wi-Fi	2.4, 5, 6 GHz	20–320 MHz
Analog/digital TV	VHF/UHF	6, 7 or 8 MHz
Cable TV (DOCSIS 3.0 downstream)	54–1002 MHz	6 MHz (USA), 8 MHz (Europe)

25.7 Signal-to-noise ratio

Definition

The **signal-to-noise ratio** is the ratio of average signal power to average noise power at a specified point in the system:

$$\mathrm{SNR} = \frac{S}{N}, \qquad \mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10} \frac{S}{N}$$

Both powers must be measured at the same point and in the same bandwidth. Three points matter in practice. The **input SNR** (or **carrier-to-noise ratio**, CNR) is measured at the receiver input or IF, in the transmission bandwidth B_T. The **detection SNR** is measured at the input to the demodulator. The

output SNR is measured at the destination, in the message bandwidth W. For analog systems the ratio between output SNR and a reference input SNR is the figure of merit of the modulation (Chapter 26).

Table 25.5 lists SNR values that mark useful quality levels.

Table 25.5 — Output SNR for acceptable quality (typical).

Application	Output SNR
Intelligible speech, emergency radio	10–15 dB
Commercial-quality telephone speech	25–40 dB
AM broadcast (good reception)	~40 dB
FM broadcast (stereo, good reception)	50–70 dB
Analog television picture (good)	40–50 dB (weighted)
CD audio (16-bit)	~96–98 dB dynamic range
Diagnostic ECG (1 mV R wave)	30–40 dB above input-referred noise of a few μV

Energy per bit and C/N_0

For digital systems, the SNR in a bandwidth is less fundamental than the energy available per bit. If the received power is S and the bit rate R_b, each bit carries energy $E_b = S/R_b$. With noise spectral density N_0, the ratio

$$\frac{E_b}{N_0} = \frac{S}{N_0 R_b} = \frac{S}{N}\cdot\frac{B}{R_b}$$

is independent of the bandwidth chosen to measure the noise. It is the quantity in which error-probability formulas are written (Chapter 27), and it makes systems of different rates and bandwidths comparable. Its companion, the **carrier-to-noise-density ratio** $C/N_0 = S/N_0$ (in dB-Hz), describes the link alone, before a bit rate is chosen: a GPS receiver typically sees C/N_0 of 35–50 dB-Hz.

Worked example — the 2.4 GHz link, completed. Continue the link of Section 25.4 with $P_R = -46.05$ dBm at 100 m, a receiver noise figure of 6 dB and a 20 MHz channel. The noise floor is

$$N = -174.0 + 10\log_{10}(2\times10^{7}) + 6 = -174.0 + 73.0 + 6 = -95.0 \text{ dBm},$$

so SNR $= -46.05 - (-94.96) = 48.9$ dB. At 1 km, $P_R = -66.05$ dBm and SNR $= 28.9$ dB. Suppose the modem transmits $R_b = 40$ Mb/s in the 20 MHz channel. At 1 km,

$$\frac{E_b}{N_0} = 28.9 + 10\log_{10}\frac{20\times10^{6}}{40\times10^{6}} = 28.9 - 3.0 = 25.9 \text{ dB}.$$

Chapter 27 shows that QPSK needs about 10.5 dB and 64-QAM about 19 dB of E_b/N_0 for a bit-error rate of 10^{-6}, so the link has a comfortable margin at 1 km in free space — but Chapter 30 will show that a realistic multipath path loses 20–30 dB more and that fading margins of 10–20 dB are needed.

SINR

When interference matters, it is added to the noise power in the denominator:

$$\mathrm{SINR} = \frac{S}{N + I}$$

The approximation that treats interference as additional Gaussian noise is reasonable when there are many interferers or when the interferer is itself a wideband digitally modulated signal; it is pessimistic for a single narrowband interferer that a receiver could notch out. In a well-designed cellular network, $I \gg N$ over most of the cell, and increasing every transmitter's power improves nothing: the system is **interference-limited**.

How SNR degrades along a chain

Every lossy element or noisy amplifier lowers the SNR, and the SNR can never be increased by processing that only filters or amplifies within the signal band. What can be done is to remove out-of-band noise by filtering (raising SNR measured in a wide bandwidth to SNR measured in the signal bandwidth), to integrate over a longer time (averaging), or to exploit structure in the signal (correlation, matched filtering, coding). An ECG averaged over 64 heartbeats synchronised to the R wave improves SNR by $10 \log_{10} 64 = 18$ dB for uncorrelated noise; a radar that integrates 100 pulses coherently gains 20 dB. These are examples of the general principle, made precise in Chapters 27 and 29, that SNR at a decision is set by signal *energy* relative to noise density, and energy grows with observation time.

25.8 Communication performance measures

Measures of fidelity

The first question about any link is how faithfully the message arrives.

- For analog systems: **output SNR**, **distortion** (total harmonic distortion, intermodulation) and frequency response flatness over the message band.
- For digital systems: the **bit-error rate** (BER, or bit-error probability P_b), the **symbol-error rate** (SER), and at higher layers the **packet** or **frame-error rate** and the residual error rate after decoding. A BER of 10^{-3} is tolerable for coded speech, 10^{-6} is a common uncoded design point, 10^{-12} is required of optical transport networks, and 10^{-15} of storage after error correction.
- For both: **availability**, the fraction of time a target fidelity is met. A microwave backhaul link may be specified for 99.999 % availability (about 5 minutes of outage per year), which is set by rain fading, not by average SNR.

Measures of resource use

The second question is what the link costs in the two resources the channel limits.

- **Spectral (bandwidth) efficiency** $\eta = R_b / B$ in b/s/Hz: how many bits each hertz carries.

- **Power (energy) efficiency**: the E_b/N_0 needed to reach a target BER. A scheme that reaches 10^{-6} at 10.5 dB is 4 dB more power-efficient than one that needs 14.5 dB, meaning it can use 2.5 times less power, or reach 1.58 times farther in free space.
- **Latency**: the delay from source to destination. It includes propagation (5 μs/km in fibre, about 240 ms up and down through a geostationary satellite at the subsatellite point), processing, buffering and coding delay. Power-system protection signalling must deliver a trip command in a few milliseconds; a phone call tolerates about 150 ms one way.
- **Complexity, cost and power consumption** of the transmitter and receiver: a capsule endoscope or a coin-cell sensor cannot afford the processing of a base station.

Table 25.6 gives representative spectral efficiencies.

Table 25.6 — Representative spectral efficiencies of deployed systems (typical, including overheads where noted).

System	Modulation	Approximate η (b/s/Hz)
Bluetooth basic rate	GFSK, 1 Mb/s in 1 MHz	1
GSM	GMSK, 270.8 kb/s in 200 kHz	1.35
DVB-S2 satellite TV	QPSK–32APSK with LDPC	1–4.5
V.34 telephone modem	up to 33.6 kb/s in ~3.4 kHz	~10
DOCSIS 3.0 downstream	256-QAM, 42.88 Mb/s in 6 MHz	7.1
LTE, 20 MHz, 2×2 MIMO (peak)	64-QAM	~7.5
Wi-Fi 6, 80 MHz, 1 stream	1024-QAM, rate 5/6	~7.5

The power–bandwidth trade-off and the Shannon limit

The two resources can be traded. FM broadcasting spends bandwidth to save power (Chapter 26). M-ary FSK does the same for digital data; QAM spends power to save bandwidth (Chapter 27). The limit of what any trade can achieve was found by Shannon in 1948. For a channel of bandwidth B with additive white Gaussian noise and average received SNR S/N, no system can transmit error-free faster than

$$C = B \log_2\left(1 + \frac{S}{N}\right) \ \text{b/s}$$

and for any rate below C there exist codes that make the error probability as small as desired. This is the **Shannon–Hartley capacity formula**; its derivation from mutual information belongs to Chapter 28. Here we use it as a ceiling.

> **Status of the principle: Shannon–Hartley channel capacity**
> *Derived*. A theorem of information theory for the band-limited AWGN channel with an average power constraint; it bounds every possible modulation and code but does not say how to achieve the bound, and it does not apply directly to fading, interference-limited or non-Gaussian channels without modification.

Worked example — the telephone channel. A voice-band telephone channel passes roughly 300–3400 Hz, $B \approx 3.1$ kHz, and a good end-to-end connection has an SNR of about 35 dB ($S/N = 3162$). Its capacity is

$$C = 3100 \log_2(1 + 3162) = 3100 \times 11.63 \approx 36 \text{ kb/s}.$$

The V.34 modem standard of 1994 reached 33.6 kb/s, within 7 % of this limit, which is why dial-up modems stopped improving there. The later 56 kb/s V.90 modems did not beat Shannon: they exploited the fact that in the downstream direction the "channel" was a digital 64 kb/s PCM link with only the final analog loop, whose SNR is much higher than 35 dB.

Writing $S = E_b R_b$ and $N = N_0 B$ and setting $R_b = C$ gives a relation between spectral efficiency $\eta = C/B$ and the minimum E_b/N_0:

$$\frac{E_b}{N_0} \geq \frac{2^{\eta} - 1}{\eta}$$

Figure 25.6 plots this bound. As $\eta \to 0$ (unlimited bandwidth) the right-hand side tends to $\ln 2 = 0.693$, or -1.59 dB: no system, however much bandwidth it uses, can communicate reliably with less energy per bit than this. At $\eta = 1$ the bound is 0 dB; at $\eta = 4$ it is $15/4 = 3.75$, or 5.7 dB; at $\eta = 8$ it is $255/8$, or 15.0 dB. Each additional b/s/Hz at high efficiency costs about 3 dB more energy per bit.

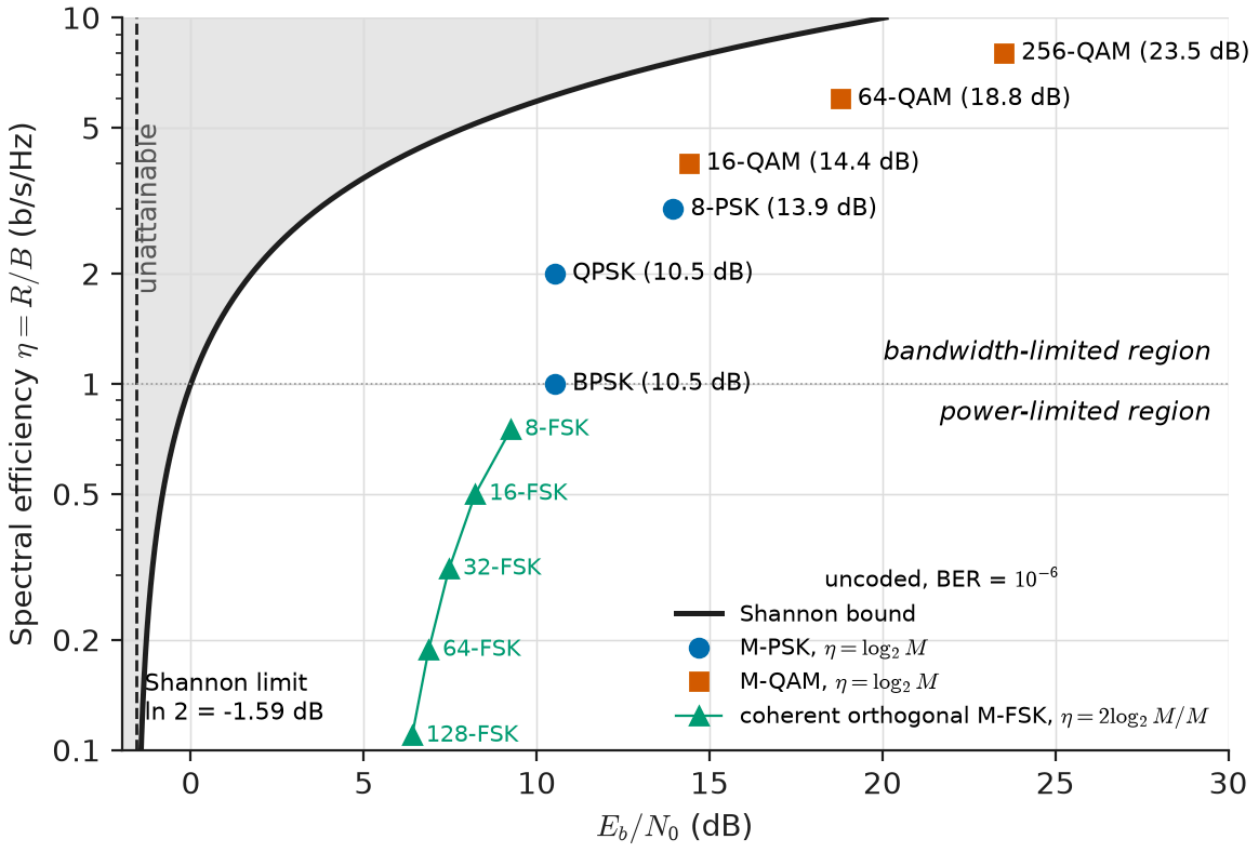

Figure 25.6 — The Shannon bound on spectral efficiency versus Eb/No, with uncoded modulation schemes at a bit-error rate of 1e-6 for comparison.

The gap between a marker in Figure 25.6 and the curve is the room left for improvement by coding. Uncoded QPSK at 10^{-6} sits about 9 dB to the right of the bound at $\eta = 2$ (1.76 dB); modern LDPC and

turbo codes close most of that gap, operating within about 1 dB of capacity (Chapter 28).

Two regimes

The bound divides practical systems into two families.

- **Power-limited systems** have bandwidth to spare but little received energy: deep-space probes, GPS, satellite telemetry, low-power sensor networks, implanted medical devices. They operate at $\eta < 1$, use robust modulations (BPSK, orthogonal FSK, spread spectrum) and strong codes, and are judged by E_b/N_0.
- **Bandwidth-limited systems** have expensive spectrum but adequate SNR: cable modems, microwave backhaul, telephone modems, Wi-Fi close to the access point. They operate at η from 4 to 10, use dense QAM, and are judged by b/s/Hz.

A single system can move between regimes as conditions change. Wi-Fi, LTE and DSL all measure SNR continuously and adapt their modulation and code rate — this is the "modem that falls back to a lower rate" of everyday experience, and the telemetry radio that should have done so beside the infusion pumps.

Choosing a performance measure

No single number captures a link. A practical specification states the target fidelity (output SNR, or BER after decoding) at a specified availability, the data rate, the allowed bandwidth and transmit power, the latency limit, and the environment (noise, interference, fading). The rest of Part VI supplies the tools to meet such a specification: Chapters 26 and 27 relate modulation to bandwidth and SNR, Chapter 28 relates coding to capacity, Chapter 29 sharpens the noise models, and Chapters 30–32 supply the channels.

> **Status of the principle: The power–bandwidth trade-off**
> *Derived*. A direct consequence of the Shannon–Hartley formula; practical modulation schemes realise it only partially, and in interference-limited or fading channels the effective trade-off is set by SINR statistics rather than by thermal noise.

What remains open

Open question	Where it is answered
How are AM, FM and PM signals generated, demodulated and compared in noise?	Chapter 26
How are bits mapped to waveforms, and what BER does each modulation achieve at a given E_b/N_0?	Chapter 27
What is information, and why is the Shannon–Hartley formula the capacity? How do codes approach it?	Chapter 28
How are random noise processes, noise bandwidth and cascaded noise figure described?	Chapter 29
How do real radio paths differ from free space (path-loss exponents, fading, multipath)?	Chapter 30
How do fibre attenuation and dispersion limit optical links?	Chapter 31
Where do antenna gain and impedance matching come from?	Chapters 14 and 32
How are Fourier transforms, power spectral densities and the statistics of noise defined mathematically?	Chapter 18; *The Mathematics Universe*

Next: Chapter 26 puts a message onto a carrier by varying its amplitude, frequency or phase, and asks what each choice costs in bandwidth and buys in noise immunity.

Chapter 26 — Analog Communication

Part VI · Communication Theory

On a summer evening a listener in a farmhouse tunes a medium-wave AM station 300 km away. The voice is clear but carries a steady hiss and, every few seconds, the crackle of a distant thunderstorm. Switching to an FM station only 60 km away, the music is clean and quiet — until the car drives behind a hill, when the FM signal does not fade gently like the AM one did but collapses abruptly into bursts of clicks and then into roaring noise. In the control tower of the nearby airport, meanwhile, two pilots transmit at the same instant on the same VHF channel, and the controller hears both voices, garbled together, with a whistling heterodyne between them; had the channel used FM, the controller would have heard only the stronger pilot and been unaware of the other.

Every one of these effects — the hiss, the crackle, the clean FM sound, the abrupt FM collapse, the capture of one signal by another — follows from how the message is impressed on the carrier. This chapter develops the three ways of doing so for an analog message: varying the carrier's amplitude, its frequency or its phase.

> **How is an analog message placed on a high-frequency carrier, and how is it recovered?**
> **What does each method cost in bandwidth and transmitter power, and how well does each resist noise?**

Concept	Engineering destination
Conventional AM, envelope detection	Medium- and short-wave broadcasting, aviation VHF voice
DSB-SC and coherent detection	LVDT and strain-bridge carrier amplifiers, lock-in amplifiers, FM stereo subcarrier
SSB and VSB	HF marine and amateur voice, historic carrier telephony, analog TV video
Frequency modulation	FM broadcasting, two-way radio, analog instrumentation and biotelemetry
Phase modulation	Indirect FM transmitters, phase-modulated sensors, the bridge to digital PSK
Modulation index, Bessel spectra, Carson's rule	Channel spacing, transmitter mask compliance, deviation measurement
Superheterodyne receiver	Almost every radio receiver, spectrum analysers, radar receivers
Noise performance, threshold, pre-emphasis	Broadcast coverage planning, choice between AM and FM

Roadmap. Amplitude modulation and its variants (26.1) → frequency modulation (26.2) and phase modulation (26.3) → the modulation index and the spectra it produces (26.4) → how modulators and demodulators work (26.5) → the superheterodyne receiver (26.6) → noise performance of each scheme (26.7) → complete analog systems in use (26.8).

26.1 Amplitude modulation

Why modulate at all

A message such as speech occupies 300–3400 Hz. Sent directly by radio it would need an antenna comparable to its wavelength of about 100 km, and every transmitter in a region would interfere with every other. **Modulation** solves both problems by shifting the message to a band around a **carrier** frequency f_c chosen to suit the channel and to separate users. The carrier is

$$c(t) = A_c \cos(2\pi f_c t + \phi_c)$$

and the message $m(t)$, of bandwidth $W \ll f_c$, may vary its amplitude, its frequency or its phase. Modulation also permits **frequency-division multiplexing**: many messages share one channel on different carriers.

Conventional AM

In conventional amplitude modulation (AM, also "double-sideband with carrier", DSB-LC) the envelope of the carrier follows the message:

$$s(t) = A_c \left[1 + k_a m(t)\right] \cos(2\pi f_c t)$$

where k_a is the amplitude sensitivity of the modulator. For a single tone $m(t) = A_m \cos(2\pi f_m t)$ the **modulation index** (or modulation depth) is $\mu = k_a A_m$, and

$$s(t) = A_c \cos(2\pi f_c t) + \frac{\mu A_c}{2} \cos\big(2\pi (f_c + f_m) t\big) + \frac{\mu A_c}{2} \cos\big(2\pi (f_c - f_m) t\big)$$

The spectrum consists of the carrier and two **sidebands**, an upper and a lower, each a frequency-shifted copy of the message spectrum. For a general message of bandwidth W the transmission bandwidth is

$$B_T = 2W$$

so a 5 kHz audio channel needs 10 kHz; medium-wave broadcasting uses 9 kHz or 10 kHz channel spacing and band-limits audio accordingly.

The modulation index can be read from the envelope on an oscilloscope (Figure 26.1). If the envelope swings between $A_{\max}$ and $A_{\min}$,

$$\mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}$$

An envelope swinging between 15 V and 5 V corresponds to $\mu = 10/20 = 0.5$. For $\mu \le 1$ the envelope $A_c[1 + \mu \cos 2\pi f_m t]$ is never negative and is an exact copy of the message plus a constant. For $\mu > 1$ (**overmodulation**) the bracket changes sign, the envelope folds at zero, an envelope detector produces a distorted output, and the clipped transmitter spreads splatter into adjacent channels. Broadcasters limit negative peaks to μ slightly below 1 for this reason.

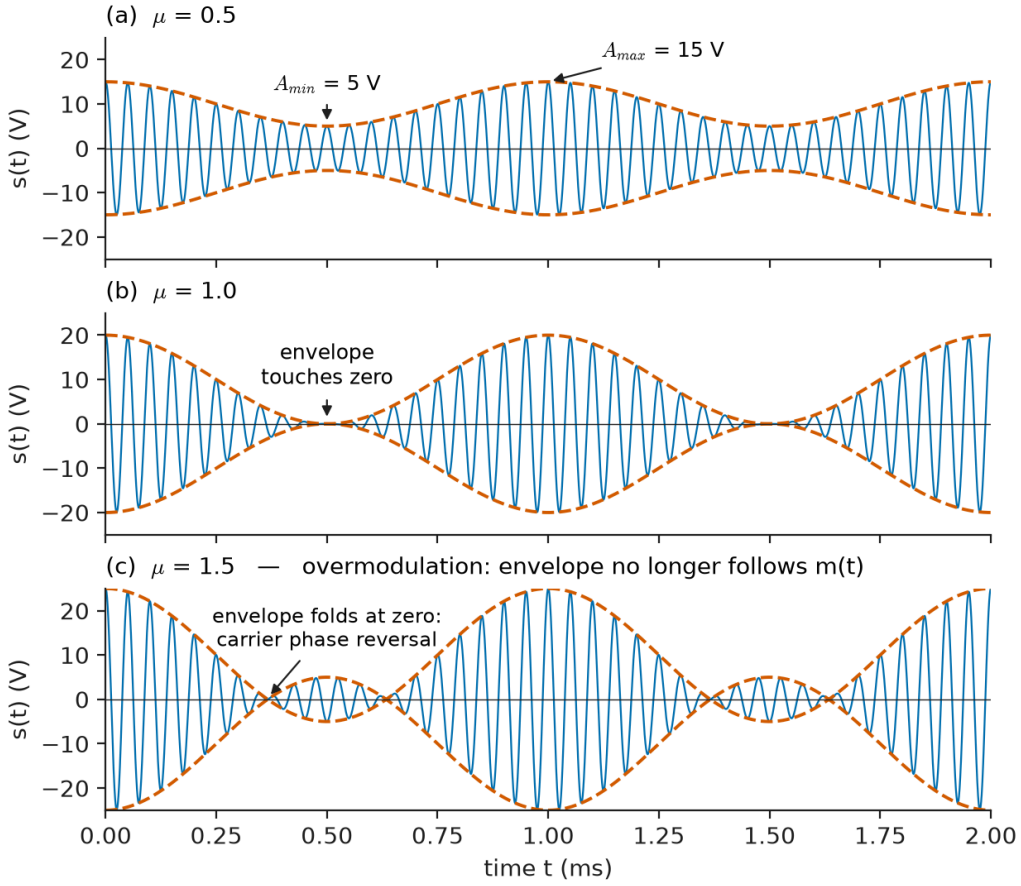

Figure 26.1 — Conventional AM waveforms for modulation indices 0.5, 1.0 and 1.5, with the envelope drawn dashed.

Power and efficiency

The average power of the tone-modulated AM signal into a resistance R is the sum of the carrier and sideband powers:

$$P_t = P_c\left(1 + \frac{\mu^2}{2}\right), \qquad P_c = \frac{A_c^2}{2R}$$

The carrier carries no information; only the sidebands do. The **power efficiency** is

$$\eta_{AM} = \frac{\mu^2/2}{1 + \mu^2/2} = \frac{\mu^2}{2 + \mu^2}$$

which is at most 1/3 (at $\mu = 1$). Real programme material has a crest factor of 10 dB or more, so the average modulation depth is around 0.3 and the efficiency is about 4 %.

Worked example — a 10 kW AM transmitter. A medium-wave transmitter delivers a carrier power of 10 kW into a 50 Ω antenna feed and is modulated to $\mu = 0.8$ by a test tone. The total power is

$$P_t = 10\left(1 + \frac{0.64}{2}\right) = 13.2\ \text{kW},$$

of which the two sidebands carry 3.2 kW (1.6 kW each), an efficiency of $3.2/13.2 = 24.2$ %. The RMS feed current rises from $\sqrt{10\,000/50} = 14.14$ A unmodulated to $\sqrt{13\,200/50} = 16.25$ A modulated — a 15 % rise that broadcast engineers use as a quick check of modulation depth. With ordinary programme at an average $\mu \approx 0.3$, only $0.09/2.09 = 4.3$ % of the radiated power is in the sidebands. The attraction that compensates for this waste is the receiver: an envelope detector (Section 26.5) costs a diode, a resistor and a capacitor.

> **Status of the principle: AM spectrum and power relations**
> *Derived*. Follow from the product of a carrier with $1 + k_a m(t)$ and the modulation theorem of the Fourier transform (Chapter 18); they assume an ideal linear modulator and an unmodulated carrier of constant amplitude.

Double-sideband suppressed carrier (DSB-SC)

Removing the carrier gives

$$s(t) = A_c\, m(t) \cos(2\pi f_c t)$$

All transmitted power is now in the sidebands, but the envelope is $|m(t)|$, not $m(t)$: an envelope detector produces a full-wave-rectified message. Recovery requires **coherent (synchronous) detection**: multiplying by a local carrier of the same frequency and phase and low-pass filtering,

$$s(t)\cos(2\pi f_c t + \phi) = \tfrac{1}{2}A_c m(t)\cos\phi + \tfrac{1}{2}A_c m(t)\cos(4\pi f_c t + \phi) \xrightarrow{\text{LPF}} \tfrac{1}{2}A_c m(t)\cos\phi$$

A phase error ϕ attenuates the output by $\cos\phi$ and a frequency error Δf multiplies it by a beat $\cos(2\pi \Delta f\, t)$, which is unacceptable. The receiver must therefore recover the carrier, for example with a Costas loop or by squaring the signal and dividing by two (Section 26.5).

DSB-SC is the natural modulation of instrumentation. An LVDT position sensor or a strain-gauge bridge excited at 5 kHz produces an output that is the excitation multiplied by the displacement or strain — DSB-SC with the excitation as carrier. Because the excitation is available at the receiver, coherent detection is trivial, and the sign of the displacement is preserved by the phase. A lock-in amplifier applies the same principle to dig a signal 60 dB below the noise out of an optical or biomedical measurement.

Single sideband (SSB)

Both sidebands of AM or DSB-SC carry the same information. **Single-sideband** modulation transmits only one, halving the bandwidth to

$$B_T = W$$

An SSB signal can be written using the Hilbert transform $\hat{m}(t)$ of the message (a 90° phase shift of every frequency component):

$$s(t) = \frac{A_c}{2}\left[m(t)\cos(2\pi f_c t) \mp \hat{m}(t)\sin(2\pi f_c t)\right]$$

with the minus sign for the upper and the plus sign for the lower sideband. SSB is generated by filtering one sideband from a DSB-SC signal (feasible when the message has no energy near DC, as speech has none below 300 Hz) or by the phasing method suggested by the equation above. It is demodulated coherently; a small frequency error does not cancel the output as in DSB-SC but shifts every audio frequency by Δf, which for speech is tolerable up to about 50 Hz (a "Donald Duck" voice). SSB is the standard voice mode of HF marine, military and amateur radio, and was the basis of analog carrier telephony, in which 4 kHz SSB channels were stacked into groups of 12 between 60 and 108 kHz.

Vestigial sideband (VSB)

Signals with significant low-frequency content, such as television video, cannot be split into sidebands by a realisable filter. **Vestigial sideband** transmits one sideband in full and a gradual vestige of the other. If the VSB filter's response has odd symmetry about the carrier, the vestige exactly compensates for the partial suppression of the wanted sideband near f_c, and coherent detection returns the message undistorted. Analog television used VSB for its 4.2–5 MHz video in a 6–8 MHz channel, and the ATSC digital television standard uses 8-level VSB.

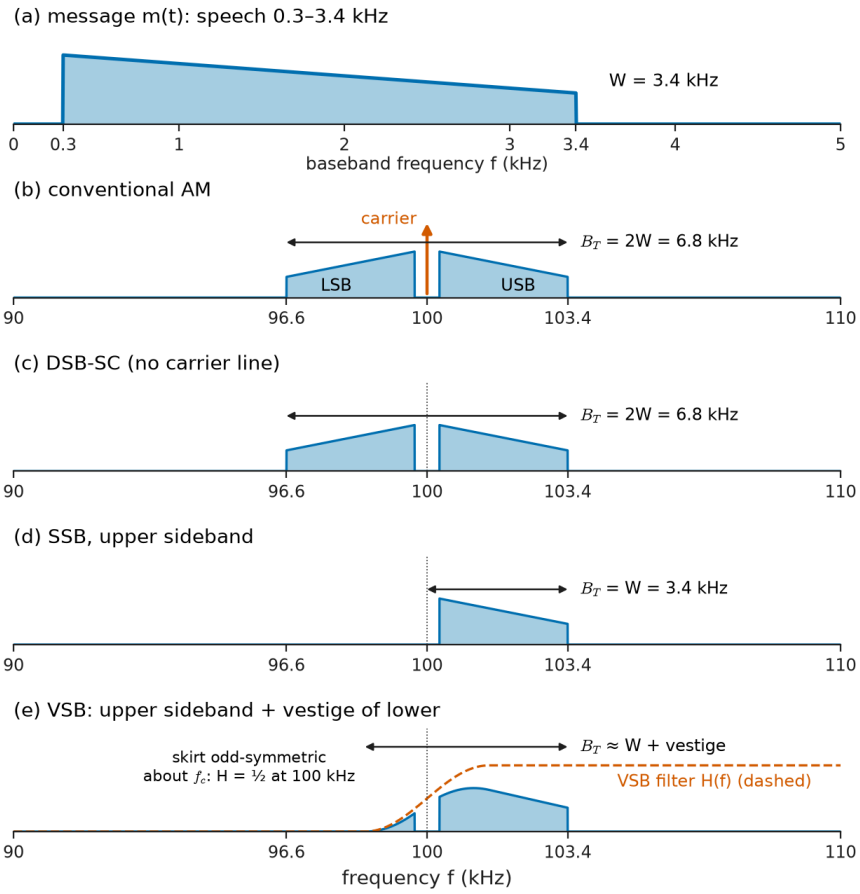

Figure 26.2 — Spectra of a 300-3400 Hz speech message and of the corresponding AM, DSB-SC, SSB (upper sideband) and VSB signals on a 100 kHz carrier.

Figure 26.2 summarises the family. Conventional AM spends power on the carrier to make the receiver cheap; DSB-SC removes the carrier at the price of a coherent receiver; SSB halves the bandwidth at the

price of a coherent receiver and a sharper filter; VSB is the compromise for signals with low-frequency content.

26.2 Frequency modulation

Instantaneous frequency

AM keeps the carrier's angle fixed and varies its amplitude. **Angle modulation** keeps the amplitude fixed and varies the angle:

$$s(t) = A_c \cos \theta(t), \qquad f_i(t) = \frac{1}{2\pi}\frac{d\theta}{dt}$$

where $f_i(t)$ is the **instantaneous frequency**. In **frequency modulation** (FM) the instantaneous frequency follows the message:

$$f_i(t) = f_c + k_f\, m(t)$$

with k_f the frequency sensitivity in Hz/V. Integrating gives the phase and hence the signal:

$$s(t) = A_c \cos\left[2\pi f_c t + 2\pi k_f \int_0^t m(\tau)\, d\tau\right]$$

The **peak frequency deviation** is $\Delta f = k_f \max|m(t)|$. In FM broadcasting $\Delta f = 75$ kHz at 100 % modulation; in narrowband two-way radio on 12.5 kHz channels, $\Delta f = 2.5$ kHz.

Properties that distinguish FM from AM

Three properties of the FM signal follow immediately from the constant amplitude, and they explain much of the opening story.

1. **Constant envelope.** The transmitted power $A_c^2/(2R)$ does not depend on the message. The power amplifier can run in saturation (class C), with efficiencies of 60–80 %, since amplitude nonlinearity does not distort a constant-envelope signal. This is why FM dominated mobile radio, where battery life and heat matter.
2. **Amplitude noise is irrelevant.** The information is in the zero crossings, not the amplitude, so a receiver can clip (limit) the signal and remove amplitude noise and fading before demodulating.
3. **Capture effect.** When two FM signals share a channel, the demodulator output is dominated by the stronger one once the ratio exceeds a few decibels; the weaker one is suppressed almost completely. Aviation VHF voice deliberately uses AM so that simultaneous transmissions remain audible and are not silently lost.

The price is bandwidth. FM is a nonlinear modulation: the spectrum of an FM signal is not a shifted copy of the message spectrum, and it is in general much wider (Section 26.4).

> **Status of the principle: Instantaneous frequency of an angle-modulated signal**
> *Derived*. A definition consistent with the Fourier spectrum only when the frequency varies slowly compared with the carrier ($W \ll f_c$); the quasi-static picture fails when the modulating frequency approaches the deviation, where the Bessel-function spectrum of Section 26.4 is needed.

26.3 Phase modulation

In **phase modulation** (PM) the phase, rather than the frequency, follows the message:

$$s(t) = A_c \cos\left[2\pi f_c t + k_p\, m(t)\right]$$

where k_p is the phase sensitivity in rad/V and the **peak phase deviation** is $\Delta\phi = k_p \max|m(t)|$. Its instantaneous frequency is

$$f_i(t) = f_c + \frac{k_p}{2\pi}\frac{dm}{dt}$$

So PM is FM driven by the derivative of the message, and FM is PM driven by the integral of the message. A phase modulator preceded by an integrator is a frequency modulator; this "indirect" method (Armstrong's) produces FM from a crystal-stable carrier and was used in early broadcast transmitters. Conversely, an FM signal can be demodulated as PM followed by differentiation.

The distinction is visible in the dependence of deviation on modulating frequency. For a tone $m(t) = A_m \cos 2\pi f_m t$:

- FM: $\Delta f = k_f A_m$, independent of f_m; phase deviation $\Delta f / f_m$ falls with f_m.
- PM: $\Delta\phi = k_p A_m$, independent of f_m; frequency deviation $k_p A_m f_m$ rises with f_m.

Worked example — PM versus FM deviation. A phase modulator has $k_p = 0.8$ rad/V and is driven by a 1 V tone. At $f_m = 1$ kHz the peak phase deviation is 0.8 rad and the peak frequency deviation is 0.8×1 kHz $= 0.8$ kHz. At $f_m = 3$ kHz the phase deviation is still 0.8 rad, but the frequency deviation is 2.4 kHz. A PM transmitter fed with speech therefore deviates most on the high-frequency components, which is equivalent to FM with a 6 dB/octave pre-emphasis — precisely the property exploited in Section 26.7. In narrowband two-way radio the transmitted signal is in practice somewhere between the two: FM with pre-emphasis, which behaves like PM over the speech band.

Phase modulation is also the analog ancestor of the digital phase-shift keying of Chapter 27: when $m(t)$ takes only discrete values, PM becomes PSK. Analog PM appears in its own right in phase-modulated fibre sensors, in some deep-space telemetry carriers (where a residual carrier is kept for tracking) and in the measurement of oscillator phase noise.

> **Status of the principle: FM–PM equivalence**
> *Derived*. Exact identity following from the definition of instantaneous frequency: FM of $m(t)$ equals PM of $2\pi k_f \int m\,dt$; it holds for any message, so every result about one transfers to the other through an integrator or differentiator.

26.4 Modulation index

Three indices, one idea

Each modulation has a dimensionless index that sets both its spectrum and its noise performance.

- **AM:** $\mu = k_a A_m$, the fractional change of the envelope. It must stay at or below 1 for envelope detection.
- **FM:** $\beta = \Delta f / f_m$ for a tone; more generally the **deviation ratio** $D = \Delta f / W$, the peak deviation divided by the highest message frequency.
- **PM:** $\beta_p = \Delta\phi = k_p A_m$, the peak phase deviation in radians.

For a tone, the FM and PM signals have the same form, $A_c \cos(2\pi f_c t + \beta \sin 2\pi f_m t)$ with β the peak phase deviation, so their spectra are identical for equal β.

The Bessel spectrum

Expanding the tone-modulated angle signal in a Fourier series (the Jacobi–Anger expansion) gives

$$s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos\big(2\pi (f_c + n f_m) t\big)$$

where J_n is the Bessel function of the first kind of order n, with $J_{-n} = (-1)^n J_n$. The spectrum is a carrier line of amplitude $A_c J_0(\beta)$ and an infinite set of sidebands at $f_c \pm n f_m$ with amplitudes $A_c |J_n(\beta)|$. Since $\sum J_n^2 = 1$, the total power is always $A_c^2/(2R)$: modulation does not add power, it redistributes it from the carrier into the sidebands. Table 26.1 lists the values.

Table 26.1 — Bessel coefficients $J_n(\beta)$ (amplitudes relative to the unmodulated carrier).

β	J_0	J_1	J_2	J_3	J_4	J_5	J_6	J_7	J_8
0.25	0.984	0.124	0.008	—	—	—	—	—	—
0.5	0.938	0.242	0.031	0.003	—	—	—	—	—
1.0	0.765	0.440	0.115	0.020	0.002	—	—	—	—
2.0	0.224	0.577	0.353	0.129	0.034	0.007	0.001	—	—
2.405	0.000	0.519	0.432	0.199	0.065	0.016	0.003	0.001	—
5.0	−0.178	−0.328	0.047	0.365	0.391	0.261	0.131	0.053	0.018

Two regimes emerge. For $\beta \ll 1$ (**narrowband FM**, NBFM) only $J_0 \approx 1$ and $J_1 \approx \beta/2$ matter: the spectrum looks like AM, with bandwidth $2f_m$, except that one sideband is inverted so that the sidebands add in quadrature to the carrier — they modulate its phase, not its amplitude. For $\beta \gg 1$ (**wideband FM**) many sidebands are significant, and the spectrum spreads over roughly $\pm\Delta f$ about the carrier. The carrier line vanishes at $\beta = 2.405$, 5.520, 8.654, …, the zeros of J_0; measuring the drive level at

which the carrier disappears on a spectrum analyser is the classic method of calibrating deviation (the **Bessel-null method**).

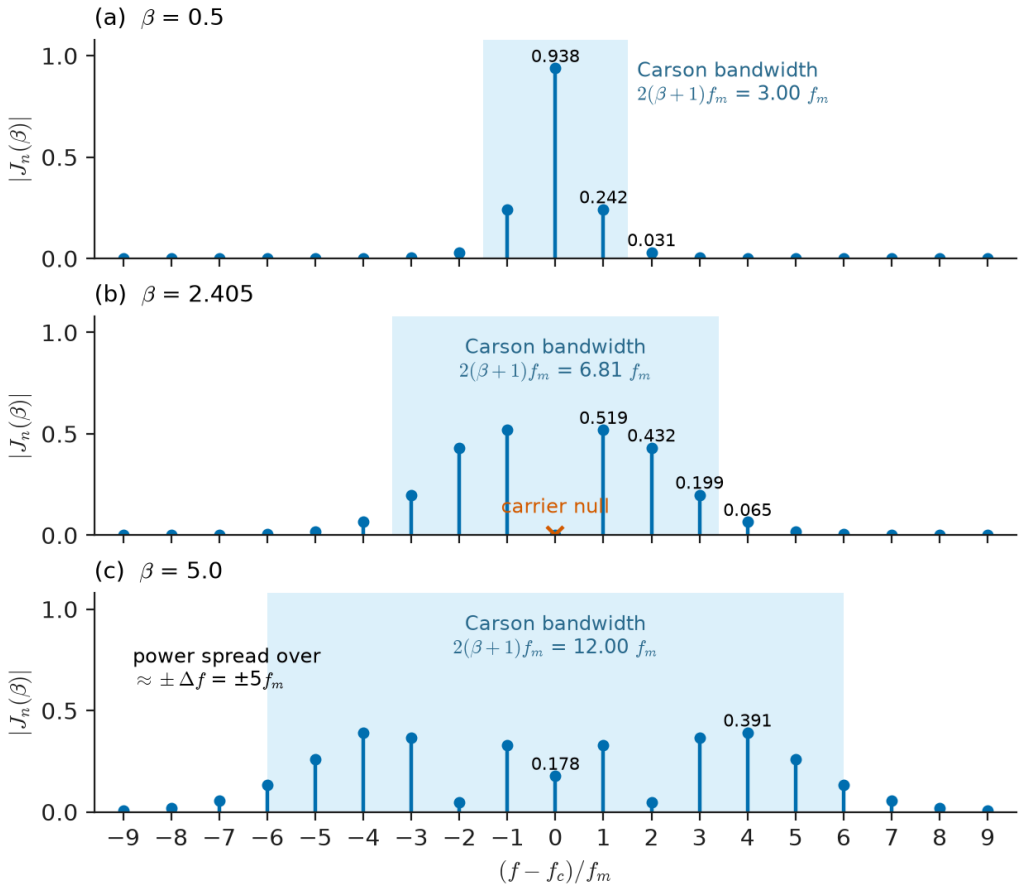

Figure 26.3 — Line spectra of a tone-modulated FM signal for modulation indices 0.5, 2.405 and 5, showing the redistribution of power from the carrier into sidebands.

Carson's rule

Figure 26.3 shows that the sidebands become negligible beyond $n \approx \beta + 1$. Keeping these gives **Carson's rule** for the transmission bandwidth:

$$B_T \approx 2(\Delta f + W) = 2W(D + 1)$$

It reduces correctly to $2W$ for narrowband FM and to $2\Delta f$ for very wideband FM. For a tone, the sidebands up to $n = \beta + 1$ contain about 98 % of the power or more. A more conservative rule, used where adjacent-channel interference is critical, counts sidebands down to 1 % of the carrier amplitude and gives roughly $2(\Delta f + 2W)$.

> **Status of the principle: Carson's rule**
> *Empirical.* A rule of thumb based on the observed decay of Bessel coefficients beyond $n = \beta + 1$; it captures about 98 % of the power for tone modulation and is somewhat generous for real programme material, but it underestimates the bandwidth needed to keep distortion very low or to meet strict adjacent-channel masks.

Worked example — FM broadcast bandwidth. A broadcast FM station transmits a 15 kHz test tone at full deviation $\Delta f = 75$ kHz. The modulation index is $\beta = 75/15 = 5$. Carson's rule gives

$$B_T \approx 2(75 + 15) = 180 \text{ kHz},$$

which fits the 200 kHz channel spacing used in the Americas. From Table 26.1, the carrier now carries $J_0^2(5) = 0.178^2 = 3.2$ % of the power, and the carrier plus the six pairs of sidebands inside ± 90 kHz carry

$$J_0^2 + 2\sum_{n=1}^{6} J_n^2(5) = 0.9936,$$

that is, 99.4 % of the total. For comparison, a narrowband two-way radio with $\Delta f = 2.5$ kHz and $W = 3$ kHz has $D = 0.83$ and $B_T \approx 2(2.5 + 3) = 11$ kHz, which fits a 12.5 kHz channel. A stereo FM broadcast whose baseband (Section 26.8) extends to 53 kHz needs, by Carson, $2(75 + 53) = 256$ kHz — wider than the channel. In practice the deviation contributed by the stereo subcarrier is smaller than the total and the real occupied bandwidth stays near 200 kHz, which illustrates the limited precision of the rule.

26.5 Modulator and demodulator concepts

Generating AM and DSB-SC

All amplitude modulators multiply the carrier by something. The ways of doing so fall into three families.

- **Product (multiplier) modulators**: an analog multiplier, or a Gilbert-cell mixer, forms $m(t)\cos(2\pi f_c t)$ directly — DSB-SC. Adding a DC offset to $m(t)$ gives conventional AM.
- **Switching modulators**: the carrier switches the message on and off (multiplying it by a square wave at f_c), and a band-pass filter at f_c keeps the fundamental. The **diode ring (double-balanced) modulator** switches the polarity of the message, giving DSB-SC with high carrier suppression (typically 30–50 dB). In high-power broadcast AM the final stage itself is the switch: its supply voltage is varied by the audio (plate or collector modulation), and modern transmitters use pulse-duration or digital modulation of a switching supply, reaching overall efficiencies above 85 %.
- **Nonlinear (square-law) modulators**: a device with $i = a_1 v + a_2 v^2$, driven by carrier plus message, produces a term $2a_2 m(t)\cos(2\pi f_c t)$ which a band-pass filter extracts. Simple but inefficient; useful for understanding where mixer products come from.

Envelope detection

The conventional AM receiver uses an **envelope detector** (Figure 26.4): a diode charges a capacitor to the peaks of the carrier, and a resistor lets it discharge slowly enough to bridge the carrier cycles but fast enough to follow the envelope. The time constant must satisfy

$$\frac{1}{f_c} \ll RC \ll \frac{1}{W}$$

More precisely, to avoid **diagonal clipping** — the capacitor failing to discharge as fast as the envelope falls — for a tone of frequency W and depth μ,

$$RC \le \frac{\sqrt{1-\mu^2}}{2\pi W \mu}$$

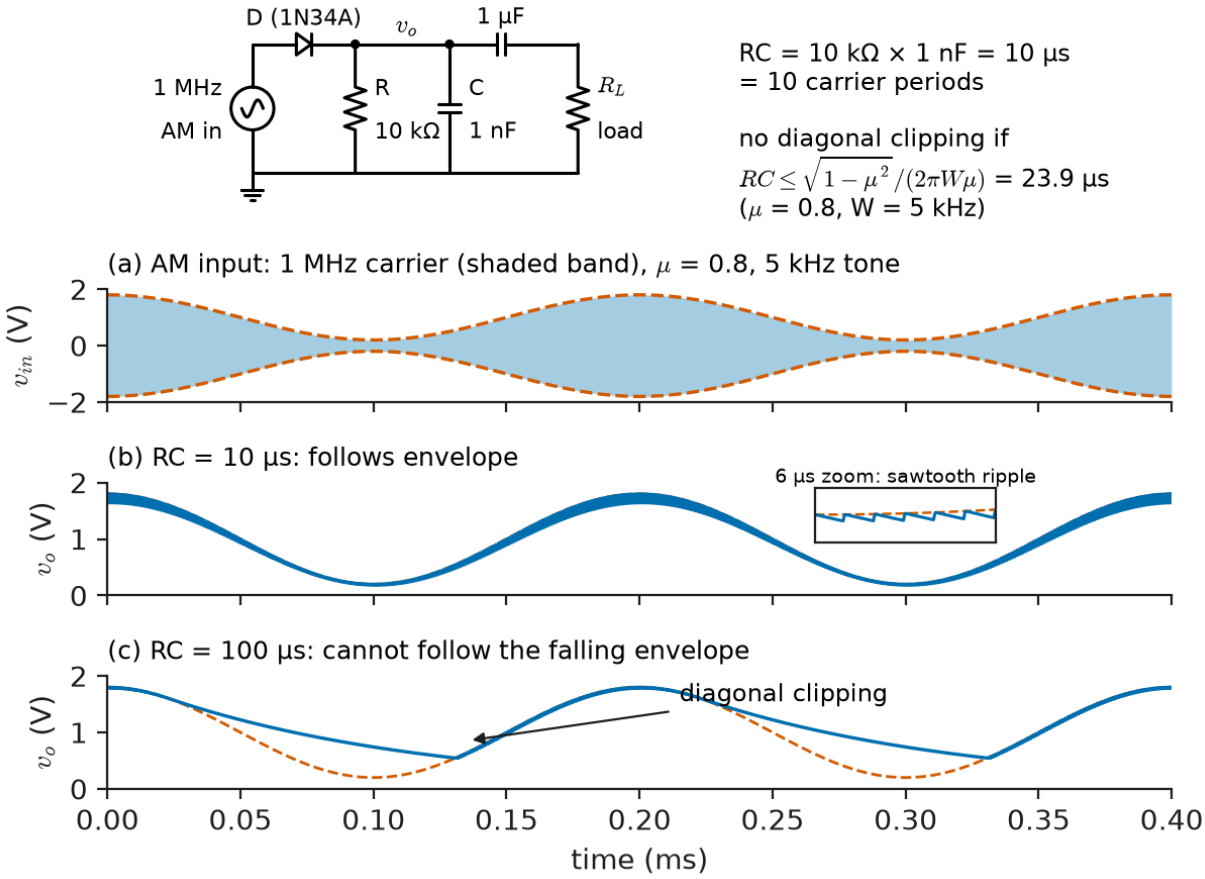

Figure 26.4 — Diode envelope detector for AM with its input and output waveforms, showing carrier ripple and diagonal clipping when RC is too large.

Worked example — envelope-detector time constant. An AM receiver's detector sees a 1 MHz carrier modulated to $\mu = 0.8$ at audio frequencies up to 5 kHz. The upper limit is

$$RC \le \frac{\sqrt{1-0.64}}{2\pi \times 5000 \times 0.8} = \frac{0.6}{25\,133} = 23.9\ \mu\text{s},$$

and the lower limit requires RC to be many carrier periods of 1 µs. Choosing $R = 10$ kΩ and $C = 1$ nF gives $RC = 10$ µs: ten carrier periods, so the carrier ripple is about $1\ \mu s/10\ \mu s = 10$ % of the peak and is removed by the band-limited audio stage that follows, and the choice sits comfortably inside the clipping limit (Figure 26.4b). With $C = 10$ nF ($RC = 100$ µs) the output cannot follow the troughs of the envelope (Figure 26.4c). In a superheterodyne receiver the detector works at the intermediate frequency of 455 kHz, where $1/455\ \text{kHz} = 2.2$ µs: the same $RC = 10$ µs still satisfies both limits, though it spans only about 4.5 IF periods and the ripple roughly doubles.

The envelope detector works only if the carrier is strong compared with the noise. When noise dominates, the envelope is mostly noise, and the message is lost not gradually but abruptly — the **threshold effect** of AM, discussed in Section 26.7.

Coherent detection and carrier recovery

DSB-SC, SSB and VSB need a local carrier synchronised to the transmitted one. Three ways to obtain it are standard.

- **Transmitted pilot**: send a small carrier or a tone related to it. FM stereo sends a 19 kHz pilot from which the receiver doubles the frequency to regenerate the 38 kHz subcarrier.
- **Squaring loop**: squaring a DSB-SC signal produces $m^2(t)\cos^2(2\pi f_c t)$, which contains a component at $2f_c$ because m^2 has a nonzero mean. A phase-locked loop at $2f_c$ followed by a divide-by-two recovers the carrier (with a 180° ambiguity).
- **Costas loop**: two mixers driven by the local oscillator in phase and in quadrature; the product of their outputs is proportional to $\sin 2\phi$, the phase error, which drives the oscillator to lock. The Costas loop is also the standard carrier-recovery circuit for BPSK (Chapter 27).

The phase-locked loop

A **phase-locked loop** (PLL) consists of a phase detector, a loop filter and a voltage-controlled oscillator (VCO), connected in a feedback loop that forces the VCO to track the phase of the input. Linearised around lock, it is a feedback control system (Chapter 34) whose closed-loop transfer function from input phase to VCO phase is low-pass. When the input is FM, the VCO control voltage needed to keep lock is proportional to the instantaneous frequency deviation — so the control voltage *is* the demodulated message. PLL demodulators also lower the FM threshold by a few decibels, because the loop bandwidth is narrower than the Carson bandwidth.

FM demodulators

Every FM demodulator converts frequency variations into amplitude variations and then detects them.

- **Slope detector and discriminator**: a tuned circuit operated on its skirt converts frequency into amplitude; the balanced discriminator (Foster–Seeley, ratio detector) uses two such circuits for linearity.
- **Differentiator plus envelope detector**: differentiating $A_c\cos\theta(t)$ gives an amplitude $A_c\,d\theta/dt = 2\pi A_c f_i(t)$, an AM signal whose envelope is the instantaneous frequency. A limiter before the differentiator removes amplitude variations.
- **Quadrature detector**: the signal is multiplied by a copy of itself phase-shifted by a network whose phase depends on frequency; common in FM receiver integrated circuits.
- **PLL demodulator**, as above.
- **Digital demodulator**: in a software-defined receiver the complex envelope $\tilde{s} = s_I + j s_Q$ is sampled and the instantaneous frequency is computed as the rate of change of $\arg \tilde{s}$, for example from $\operatorname{Im}\{\tilde{s}[n]\tilde{s}^*[n-1]\}$.

> **Boundary.**
> The transistor-level design of mixers, VCOs, limiters and detectors belongs to *Electronics & Computing*; the PLL as a control loop is analysed in Chapters 33–34. Here they are treated through their input–output functions.

26.6 Superheterodyne receivers

The tuning problem

An early **tuned-radio-frequency** receiver amplified the signal at its own frequency with a chain of tuned stages, all of which had to be retuned together. Its selectivity changes with frequency: a tuned circuit with a fixed Q has bandwidth f/Q, so a Q that gives 10 kHz bandwidth at 540 kHz gives 32 kHz at 1700 kHz. Armstrong's **superheterodyne** (1918) solved this by converting every incoming signal to a fixed **intermediate frequency** (IF), where all the selectivity and most of the gain can be provided by fixed filters (Figure 26.5).

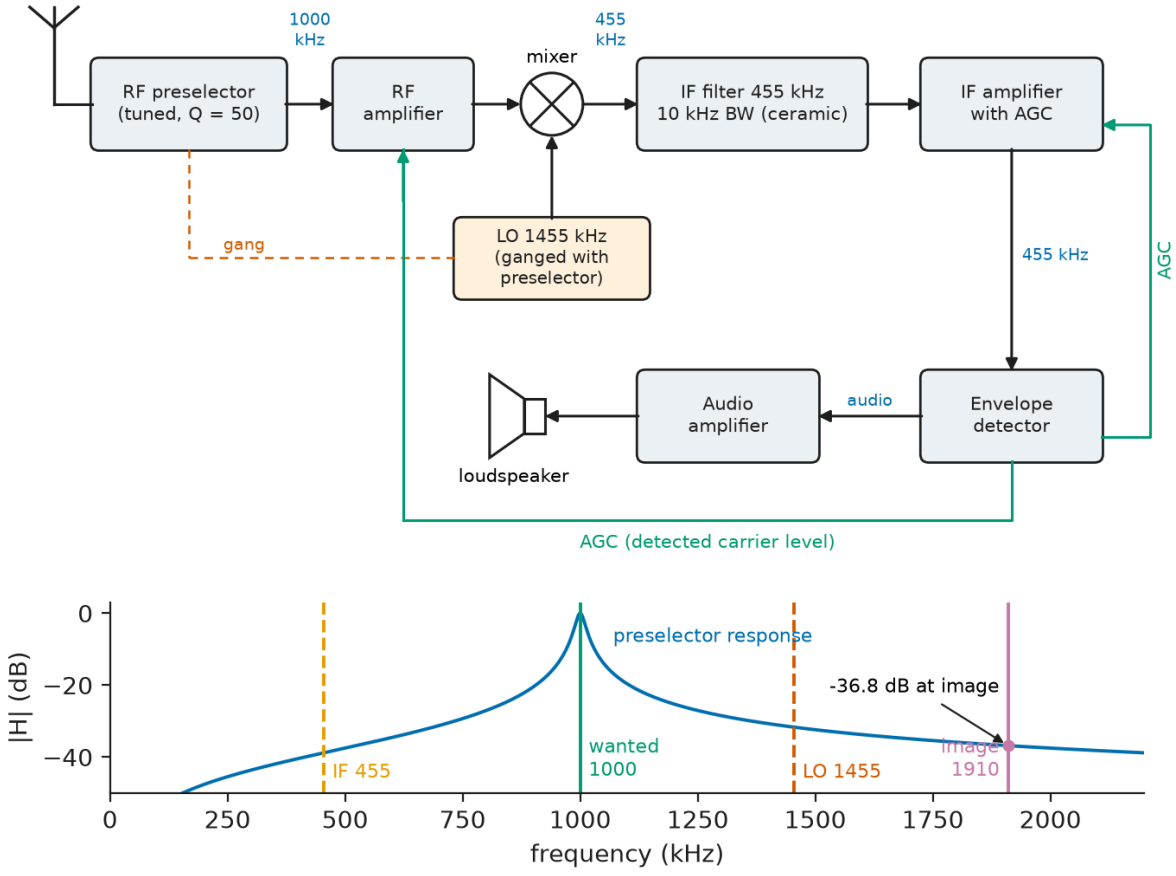

Figure 26.5 — Superheterodyne AM broadcast receiver tuned to 1000 kHz, with the frequencies at each point and the image frequency.

Frequency conversion and the image

A mixer multiplies the incoming RF signal at f_{RF} by a local oscillator (LO) at f_{LO}, producing sum and difference frequencies. The IF filter selects the difference:

$$f_{IF} = |f_{LO} - f_{RF}|$$

With high-side injection, $f_{LO} = f_{RF} + f_{IF}$. But a second input frequency also produces the same difference: the **image frequency**

$$f_{im} = f_{LO} + f_{IF} = f_{RF} + 2f_{IF}$$

Once a signal at the image has reached the mixer, it is indistinguishable from the wanted one. It must be removed before the mixer by the **preselector**, a tuned RF filter. A single tuned circuit of loaded quality factor Q tuned to f_{RF} attenuates the image by

$$\alpha = \sqrt{1 + Q^2\rho^2}, \qquad \rho = \frac{f_{im}}{f_{RF}} - \frac{f_{RF}}{f_{im}}$$

the result from the frequency response of a resonant circuit (Chapter 13).

Worked example — image rejection in AM and FM receivers. An AM broadcast receiver with $f_{IF} = 455$ kHz is tuned to 1000 kHz. The LO runs at 1455 kHz and the image is at $1000 + 910 = 1910$ kHz, inside the broadcast band. With a preselector of $Q = 50$,

$$\rho = \frac{1910}{1000} - \frac{1000}{1910} = 1.910 - 0.524 = 1.386, \qquad \alpha = \sqrt{1 + (50 \times 1.386)^2} = 69.3,$$

an image rejection of $20\log_{10} 69.3 = 36.8$ dB, adequate for broadcast. An FM receiver with the standard $f_{IF} = 10.7$ MHz tuned to 100 MHz has its image at 121.4 MHz, outside the FM band (it falls in the aeronautical band). There, $\rho = 1.214 - 0.824 = 0.390$, and a single $Q = 50$ circuit gives only $20\log_{10}\sqrt{1 + 19.5^2} = 25.8$ dB; FM tuners therefore use two or three tuned RF stages. The general lesson is that the image rejection depends on the ratio f_{IF}/f_{RF}: a high IF eases image rejection, a low IF eases adjacent-channel selectivity. Receivers that need both use **double conversion** — a high first IF for image rejection followed by a low second IF for selectivity. Communication receivers commonly use first IFs of 45–70 MHz and a second IF of 455 kHz; spectrum analysers convert up to a first IF of several gigahertz.

> **Status of the principle: Superheterodyne frequency conversion and the image response**
> *Derived.* Follows from the product of two sinusoids generating sum and difference frequencies; it assumes an ideal multiplier, and real mixers add spurious responses at $|mf_{LO} \pm nf_{RF}| = f_{IF}$ for small integers m, n that must also be filtered.

Tracking, AGC and other functions

The LO and the preselector are tuned together (ganged). For an AM receiver covering 530–1700 kHz with high-side injection, the LO covers 985–2155 kHz, a frequency ratio of 2.19 against 3.21 for the signal, so the capacitance ratios needed are $2.19^2 = 4.8$ and $3.21^2 = 10.3$. Low-side injection would

require the LO to span 75–1245 kHz, a ratio of 16.6 — impractical with a variable capacitor, which is why AM receivers use high-side injection. Modern receivers tune with a synthesiser, and the preselector is replaced by switched band filters.

The IF stage also provides **automatic gain control** (AGC): the detected carrier level reduces the gain of the RF and IF amplifiers, so that a signal varying over 80–100 dB at the antenna produces a nearly constant level at the detector. AGC is what makes the farmhouse AM listener's distant station fade only gradually.

Beyond the superheterodyne

Two alternatives have become common with integrated circuits. The **direct-conversion (zero-IF)** receiver mixes straight to baseband with I and Q mixers; it has no image (the image is the signal itself, separated by I/Q processing) but suffers from DC offsets, flicker noise and LO leakage. The **low-IF** receiver uses an IF of a few hundred kilohertz with I/Q image rejection. Both depend on the complex-envelope representation of Chapter 25, and both are now the standard architectures of mobile handsets and Wi-Fi chips. The superheterodyne remains the reference architecture for high-performance receivers, spectrum analysers and radar.

26.7 Noise performance

The comparison framework

To compare modulations fairly, all are given the same received signal power S_R, the same noise spectral density N_0 at the receiver input, and the same message bandwidth W. The reference is the **baseband SNR**

$$\gamma = \frac{S_R}{N_0 W}$$

which is what a baseband system sending the message directly would achieve. Each modulation then yields an output SNR SNR_o, and the ratio SNR_o/γ is its **figure of merit**. The detailed derivations, which treat the bandpass noise as $n_I(t)\cos 2\pi f_c t - n_Q(t)\sin 2\pi f_c t$ with independent Gaussian components, belong to Chapter 29; the results are summarised in Table 26.2, with $m(t)$ normalised so that $|m(t)| \le 1$ and P_m its average power.

Table 26.2 — Noise performance of analog modulations (above threshold).

Modulation	B_T	SNR_o/γ	Tone value	Detector
Baseband	W	1	1	—
DSB-SC	$2W$	1	1	Coherent
SSB	W	1	1	Coherent
AM, envelope	$2W$	$\dfrac{\mu^2 P_m}{1+\mu^2 P_m}$	$\dfrac{\mu^2}{2+\mu^2} \le \frac{1}{3}$	Envelope (above threshold)
PM	$\approx 2(\beta_p+1)W$	$\beta_p^2 P_m$	$\beta_p^2/2$	Phase
FM	$\approx 2(D+1)W$	$3D^2 P_m$	$\frac{3}{2}\beta^2$	Discriminator (above threshold)

The table carries three messages.

1. **Linear modulations cannot beat baseband.** DSB-SC and SSB both achieve $\mathrm{SNR}_o = \gamma$. DSB-SC uses twice the bandwidth of SSB and thus admits twice the noise, but the coherent detector adds the two sidebands' signal voltages coherently and their noise powers incoherently, which exactly compensates.
2. **Conventional AM wastes at least 4.8 dB** (factor 3) at full tone modulation and about 14 dB at the typical programme depth of $\mu = 0.3$, because the carrier carries most of the power.
3. **FM trades bandwidth for SNR.** The figure of merit grows as the square of the deviation ratio, while the bandwidth grows only linearly. Doubling the deviation quadruples the output SNR (+6 dB).

Why FM noise has a triangular spectrum

The FM advantage has a physical explanation. Additive noise of amplitude n on a carrier of amplitude A_c perturbs the phase by about n/A_c; a phase perturbation at frequency f from the carrier produces a frequency perturbation proportional to f. The noise power spectral density at the output of an FM discriminator therefore rises as f^2 — the "triangular" noise spectrum (triangular in amplitude) — while the signal is unaffected. The large deviation scales the signal up; the strong carrier scales the noise down. For PM the output noise spectrum is flat.

The rising noise spectrum suggests a refinement. Since the message spectrum of audio falls at high frequencies while the FM noise rises, the transmitter boosts high frequencies before modulation (**pre-emphasis**) and the receiver cuts them correspondingly after detection (**de-emphasis**), attenuating the noise where it is strongest. Broadcast FM uses a first-order network with a time constant of 75 µs (Americas, Korea) or 50 µs (Europe and elsewhere), corner frequencies of 2.12 kHz and 3.18 kHz. For a 15 kHz audio band, the improvement in output SNR for white message spectra is

$$I = \frac{(W/f_1)^3}{3\left[W/f_1 - \arctan(W/f_1)\right]}$$

which evaluates to 13.2 dB for 75 µs and 10.2 dB for 50 µs. The Dolby noise-reduction systems of analog tape and the RIAA equalisation of vinyl records are the same idea applied to other channels.

The threshold effect

Table 26.2 holds only when the carrier is much stronger than the noise in the transmission bandwidth. When it is not, both the envelope detector and the FM discriminator fail abruptly. For FM, the noise occasionally drives the resultant phasor around the origin, adding a full 2π to the phase; each such event produces an impulse — a click — at the discriminator output. The click rate rises steeply as the carrier-to-noise ratio in B_T falls below about 10 dB, and the output SNR collapses (Figure 26.6). This is the car-behind-the-hill experience of the opening paragraph.

The **FM threshold** occurs approximately when

$$\frac{S_R}{N_0 B_T} \approx 10 \ (10 \text{ dB}), \qquad \text{i.e.} \qquad \gamma_{th} \approx 20(D+1)$$

so wideband FM, with its larger B_T, reaches threshold at a higher γ. This is the hidden cost of the FM trade: above threshold the advantage grows with deviation; below it, the system fails at higher signal levels the more deviation it uses. PLL and feedback demodulators lower the threshold by 2–5 dB by tracking the signal with a bandwidth narrower than B_T.

> **Status of the principle: Above-threshold noise formulas for AM and FM**
> *Approximate.* Derived under a high carrier-to-noise assumption (noise small compared with the carrier inside B_T); they fail below a carrier-to-noise ratio of roughly 10 dB, where envelope detectors and discriminators exhibit threshold and the output SNR falls much faster than the input.

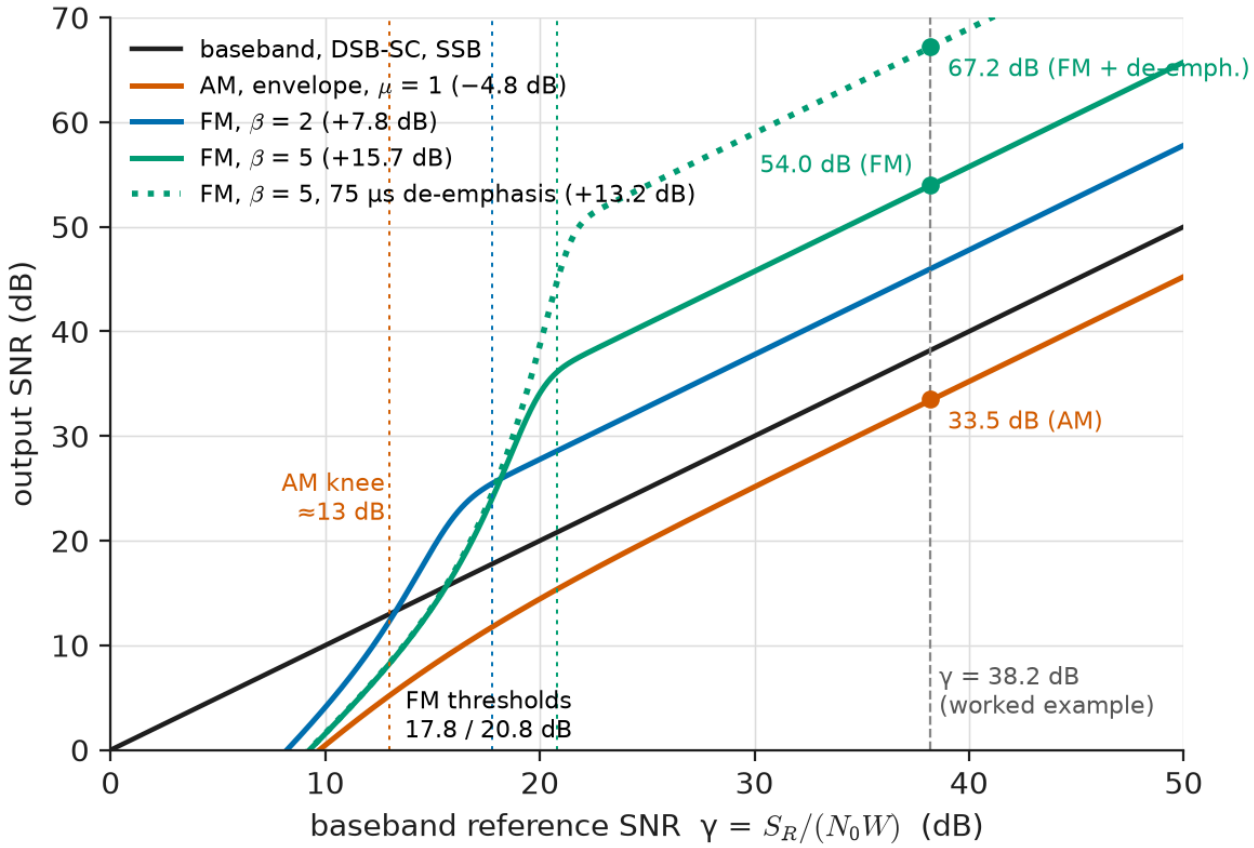

Figure 26.6 — Output SNR versus baseband reference SNR gamma for baseband/DSB-SC/SSB, AM with mu = 1, and FM with beta = 2 and beta = 5, showing the FM threshold.

Worked example — AM versus FM at the same received power. A broadcast receiver with a noise figure of 4 dB ($N_0 = -174 + 4 = -170$ dBm/Hz) receives $S_R = -90$ dBm, and the audio bandwidth is $W = 15$ kHz. The reference SNR is

$$\gamma = -90 - (-170 + 10\log_{10} 15\,000) = -90 - (-170 + 41.76) = 38.2 \text{ dB}.$$

- **AM**, full tone modulation $\mu = 1$: $\text{SNR}_o = 38.2 + 10\log_{10}(1/3) = 38.2 - 4.8 = 33.5$ dB. With typical programme modulation $\mu = 0.3$: $38.2 + 10\log_{10}(0.09/2.09) = 24.6$ dB — audible hiss.
- **FM**, $\beta = 5$: $\text{SNR}_o = 38.2 + 10\log_{10}(1.5 \times 25) = 38.2 + 15.7 = 54.0$ dB. With 75 μs de-emphasis, about $54.0 + 13.2 = 67.2$ dB.
- **Threshold check**: the carrier-to-noise ratio in the Carson bandwidth of 180 kHz is $-90 - (-170 + 52.55) = 27.4$ dB, well above 10 dB. The threshold value of γ is $10\log_{10}(20 \times 6) = 20.8$ dB, so the signal can fade by about 17 dB before the FM output collapses.

The FM system gives 20.5 dB more output SNR than full-modulation AM, and more than 40 dB more than AM with real programme and de-emphasis — at the cost of 18 times the bandwidth (180 kHz against 10 kHz for 5 kHz AM audio). The same calculation for narrowband FM with $D = 0.83$ gives a figure of merit of only $1.5 \times 0.83^2 = 1.04$ (0.2 dB): narrowband two-way radio chooses FM for its constant envelope, its immunity to amplitude fading and its capture property, not for an SNR gain.

Interference performance

The capture effect is the interference-domain counterpart of the FM noise advantage. An interfering carrier of relative amplitude $\rho < 1$ at offset f_d produces an output at frequency f_d with amplitude proportional to ρf_d in FM, compared with ρ in AM. Interference close to the wanted carrier is strongly suppressed, and as ρ approaches 1 the receiver switches abruptly to whichever signal is stronger. FM broadcast planning requires a co-channel protection ratio of roughly 37–45 dB for high-quality stereo, but mobile FM voice systems work with co-channel ratios of 12–18 dB, far less than AM would need for comparable intelligibility.

26.8 Analog communication systems

Although most new systems are digital, analog modulation remains in service where its simplicity, robustness or installed base matters, and it lives on inside digital systems as the carrier-handling stage. Table 26.3 surveys the main systems.

Table 26.3 — Representative analog communication systems (typical parameters).

System	Modulation	Carrier range	Message bandwidth	Channel	Notes
Long/medium-wave broadcast	AM, $\mu \approx 0.3$ average	148–283 kHz, 526.5–1606.5 kHz (Region 1); 530–1700 kHz (Region 2)	4.5–5 kHz	9 or 10 kHz	Ground-wave by day, sky-wave by night
Short-wave broadcast	AM	3.9–26.1 MHz	4.5 kHz	5 kHz raster	Ionospheric, worldwide
HF marine, military, amateur voice	SSB	2–30 MHz	300–2700 Hz	3 kHz	Power-efficient long range
Aviation VHF voice	AM	118–137 MHz	300–3000 Hz	25 or 8.33 kHz	No capture: overlapping calls are heard
FM broadcast	WBFM, $\Delta f =$ 75 kHz	87.5–108 MHz	15 kHz per channel (stereo baseband to 53 kHz, RDS at 57 kHz)	200 kHz	Pre-emphasis 50 or 75 µs
Land mobile and PMR	NBFM, $\Delta f =$ 2.5–5 kHz	30–960 MHz	300–3000 Hz	12.5–25 kHz	Being replaced by digital (DMR, TETRA, P25)
Analog TV (legacy)	VSB video, FM sound	VHF/UHF	4.2–6 MHz video	6–8 MHz	Replaced by digital in most countries
Instrumentation telemetry	FM/FM subcarriers (IRIG 106)	VHF, L/S band	tens of Hz to tens of kHz per subcarrier	0.5–several MHz	Flight testing, historical biotelemetry
LVDT and bridge carrier amplifiers	DSB-SC, coherent	1–20 kHz excitation	DC to ~1/10 of carrier	—	Position, force and pressure sensors
Power-line carrier (utility)	AM/SSB, FSK	30–500 kHz	Voice and protection tones	4 kHz	Teleprotection over HV lines

FM stereo: a multiplex inside a modulation

FM stereo is a compact demonstration of most of this chapter. The left and right audio channels L and R, each limited to 15 kHz, are combined into a composite baseband:

- L + R from 30 Hz to 15 kHz (heard by mono receivers);
- a 19 kHz pilot tone at about 10 % of deviation;

- L − R as DSB-SC on a 38 kHz subcarrier, occupying 23–53 kHz;
- RDS data at 57 kHz (3 × 19 kHz) on many stations.

This composite signal then frequency-modulates the carrier. A stereo receiver demodulates the FM, doubles the 19 kHz pilot to recover the 38 kHz carrier, demodulates the L − R subcarrier coherently, and forms L and R by sum and difference. Because the L − R channel lies at the top of the composite baseband where FM noise is highest, stereo reception is noisier than mono by roughly 20 dB, and receivers blend to mono when the signal is weak.

Where analog modulation survives in digital systems

The analog modulation concepts of this chapter are not only historical. Every digital radio uses I/Q modulators that are DSB-SC modulators on quadrature carriers; QAM (Chapter 27) is two DSB-SC signals sharing one channel. GSM's GMSK and Bluetooth's GFSK are constant-envelope frequency modulations chosen for amplifier efficiency, exactly the FM argument of Section 26.2. The superheterodyne and PLL are in every receiver. And the analog figures of merit introduced here — power efficiency, bandwidth expansion, threshold — reappear in Chapter 27 as E_b/N_0, spectral efficiency and the cliff-like BER curve of a digital link.

What remains open

Open question	Where it is answered
How are the noise formulas of Table 26.2 derived from a bandpass Gaussian noise model?	Chapter 29
How does a phase-locked loop behave as a feedback system, and when does it lose lock?	Chapters 33 and 34
How does modulation of a carrier by discrete symbols (PSK, FSK, QAM) differ, and how is it detected?	Chapter 27
How do fading and multipath affect AM and FM reception on mobile paths?	Chapter 30
How are resonant preselectors, IF filters and their Q factors designed?	Chapter 13
How are the mixers, oscillators and limiters built from transistors?	*Electronics & Computing*
What is the ultimate limit on the SNR–bandwidth trade-off that FM only partially exploits?	Chapters 25 and 28

Next: Chapter 27 replaces the continuous message with a stream of bits, and asks how to map them onto pulses and carriers, how many can be sent per hertz, and how often the receiver will guess wrong.

Chapter 27 — Digital Communication

Part VI · Communication Theory

A cable-modem customer complains that downloads are fast in the morning and slow every evening. The technician's meter tells the story in three numbers. In the morning the downstream channel shows a modulation error ratio of 36 dB, and the modem runs 256-QAM. In the evening, as the temperature of the street amplifiers rises and a neighbour's faulty connector lets in ingress from a nearby FM transmitter, the ratio falls to 29 dB, the uncorrected codeword errors climb, and the system falls back to 64-QAM — losing a quarter of its throughput. Nothing in the network has failed; the link has simply crossed the boundary at which one modulation stops being reliable and a more robust one must take over.

An analog link degrades gracefully: a little more noise gives a little more hiss. A digital link behaves differently. Its error rate is negligible over a wide range of conditions and then rises by orders of magnitude over a few decibels. Understanding where that cliff lies — for each way of turning bits into waveforms — is the central task of this chapter.

> **How are analog messages turned into bits, and how are bits turned into waveforms for a cable or a radio channel?**
> **How many bits can each symbol carry, how should the receiver filter and decide, and how often will it decide wrongly?**

Concept	Engineering destination
Pulse-code modulation, companding	Digital telephony (E1/T1), CD audio, ECG and instrumentation data acquisition
Delta and sigma-delta modulation	Low-cost voice links, audio and precision ADCs
Line coding	Ethernet, USB, CAN, E1/T1 trunks, optical links
Bit and symbol rates, Nyquist pulse shaping	Channel sizing, roll-off filters, eye-diagram testing
ASK, FSK, PSK	RFID, garage-door and tyre-pressure sensors, Bluetooth, satellite telemetry
QPSK and QAM	Satellite TV, cable modems, Wi-Fi, LTE and 5G, microwave backhaul
Constellation diagrams	Transmitter and receiver testing with vector signal analysers (EVM, MER)
Matched filtering	Every digital receiver, radar pulse compression, ultrasound
Error probability versus E_b/N_0	Link budgets, adaptive modulation thresholds

Roadmap. Digitising the message by PCM (27.1) and delta modulation (27.2) → putting bits on a wire by line coding (27.3) → symbol and bit rates and the bandwidth they need (27.4) → the geometric picture of constellations (27.5) → carrier modulations ASK, FSK, PSK (27.6), QPSK (27.7) and QAM (27.8) → the optimum receiver filter (27.9) → detection and error probability (27.10).

27.1 Pulse-code modulation

Sample, quantize, encode

Pulse-code modulation (PCM) turns an analog message into bits in three steps.

1. **Sampling** at $f_s \geq 2W$ after an anti-aliasing filter (Chapter 19). Telephone speech, band-limited to 3.4 kHz, is sampled at 8 kHz; audio at 44.1 or 48 kHz; a diagnostic ECG at 500 Hz to 1 kHz.
2. **Quantization**: each sample is rounded to one of $L = 2^n$ levels.
3. **Encoding**: each level is represented by an n-bit binary word.

The bit rate is $R_b = n f_s$: $8 \times 8000 = 64$ kb/s for a telephone channel, $16 \times 44\,100 \times 2 = 1.4112$ Mb/s for stereo CD audio.

Quantization noise

A uniform quantizer with step Δ over a full-scale range $\pm V$ has $\Delta = 2V/2^n$. If the signal is busy enough that the rounding error e is uniformly distributed over $\pm\Delta/2$ and uncorrelated with the signal, its mean-square value is

$$\overline{e^2} = \frac{\Delta^2}{12}$$

For a full-scale sinusoid of power $V^2/2$, the **signal-to-quantization-noise ratio** is

$$\mathrm{SQNR} = \frac{V^2/2}{\Delta^2/12} = \frac{3}{2}\,2^{2n}, \qquad \mathrm{SQNR}_{\mathrm{dB}} = 6.02n + 1.76$$

Each extra bit buys 6 dB. Table 27.1 shows the consequences.

Table 27.1 — PCM parameters of common systems (typical).

Application	f_s	n (bits)	Bit rate per channel	Full-scale SQNR (sine)
Telephone, G.711	8 kHz	8 (companded)	64 kb/s	~38 dB over a 40 dB range
Diagnostic ECG	500 Hz–1 kHz	12–16	6–16 kb/s	74–98 dB
Power-quality and protection recorders	4–20 kHz	16	64–320 kb/s	98 dB
CD audio	44.1 kHz	16	705.6 kb/s	98.1 dB
Studio audio	48–192 kHz	24	1.15–4.6 Mb/s	~146 dB theoretical, 110–120 dB achieved
Seismic recorders	100–1000 Hz	24	2.4–24 kb/s	limited to ~120–140 dB by analog noise

> **Status of the principle: The 6 dB-per-bit quantization rule**
> *Approximate.* Rests on modelling quantization error as uniform, white and independent of the signal; accurate for busy signals spanning many levels, but it fails for small or slowly varying signals (where the error is correlated and appears as distortion) and is corrected in practice by dither.

Companding

The full-scale SQNR is reached only by a signal that fills the quantizer. Speech varies over 40 dB between quiet and loud talkers, and a uniform 8-bit quantizer that gives 50 dB for a loud talker gives only 20 dB for a talker 30 dB quieter. Telephony instead uses **companding**: compressing the signal logarithmically before uniform quantization and expanding it after decoding. The **μ-law** characteristic (North America, Japan) is

$$y = \operatorname{sgn}(x)\,\frac{\ln(1+\mu|x|)}{\ln(1+\mu)}, \qquad |x| \le 1,\ \mu = 255$$

and the **A-law** characteristic (Europe and elsewhere, $A = 87.6$) is similar. For small inputs the slope of the μ-law curve is $\mu/\ln(1+\mu) = 46$, so quiet signals are amplified by 33 dB before quantization; for large inputs the steps are large. The SQNR becomes almost independent of signal level,

$$\text{SQNR} \approx 6.02n + 4.77 - 20\log_{10}\ln(1+\mu)\ \text{dB} = 38\ \text{dB for } n = 8,$$

over a range of about 40 dB (Figure 27.1). A uniform quantizer would need about 13 bits to match this over the same range.

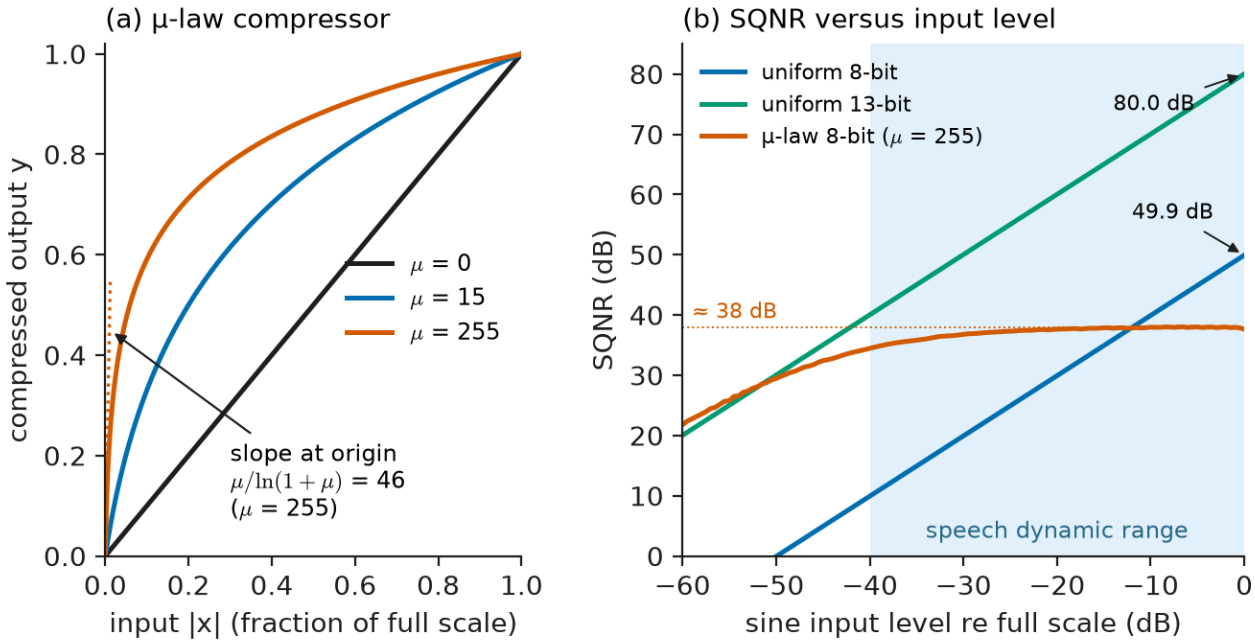

Figure 27.1 — Mu-law companding: compression characteristic and the resulting SQNR versus input level compared with uniform 8-bit and 13-bit quantization.

PCM in telephony and instrumentation

PCM channels are multiplexed in time. The European **E1** carrier interleaves 32 time slots of 64 kb/s (30 voice channels plus framing and signalling) into 2.048 Mb/s; the North American **T1** carries 24 channels plus 8 kb/s of framing at 1.544 Mb/s. The same structure underlies the IEC 61850 sampled-value streams in digital substations, where merging units sample current and voltage transformers at 80 samples per cycle and send them as packets to protection relays.

PCM's decisive advantage over analog transmission is **regeneration**. An analog repeater amplifies noise along with signal, so noise accumulates over every hop. A digital repeater decides each bit afresh and transmits a clean pulse; as long as each hop's error probability is small, errors accumulate only additively (k hops at P_b each give about kP_b), while the noise does not accumulate at all. Transcontinental digital links therefore deliver the same quality as local ones.

Worked example — PCM for a multichannel biomedical recorder. A sleep laboratory records 8 EEG channels and 2 ECG leads, each sampled at 500 Hz with 16-bit resolution, plus 3 auxiliary channels (respiration, oximetry, body position) at 50 Hz and 12 bits. The bit rate is

$$R_b = 10 \times 500 \times 16 + 3 \times 50 \times 12 = 80\,000 + 1800 = 81.8\ \text{kb/s}.$$

The full-scale SQNR of the 16-bit channels is $6.02 \times 16 + 1.76 = 98.1$ dB. An EEG signal of 50 μV peak recorded on an input range of ±5 mV sits 40 dB below full scale, so its SQNR is about 58 dB — still well above the analog noise floor of a good EEG amplifier (about 0.5–1 μV rms), which means 16 bits is enough and the analog front end, not the quantizer, limits the recording. A 12-bit converter would give only 34 dB on the same EEG signal, comparable to the analog noise; this is why biopotential ADCs moved from 12 to 16–24 bits.

27.2 Delta modulation

One bit per sample

Delta modulation (DM) quantizes not the sample but the change from a running estimate, using one bit. The encoder keeps a staircase approximation $\hat{m}(t)$; at each sampling instant it sends 1 if $m > \hat{m}$ (and steps the staircase up by Δ) or 0 otherwise (and steps it down). The decoder rebuilds the same staircase and low-pass filters it. The hardware is a comparator, a flip-flop and an integrator — far simpler than a multi-bit ADC. In exchange, DM must sample many times faster than the Nyquist rate so that successive samples are strongly correlated and a single step can follow the signal.

Slope overload and granular noise

DM has two distinct error mechanisms (Figure 27.2).

- **Slope overload**: the staircase can rise at most Δf_s per second. If the message rises faster, the staircase falls behind. For a tone $A\sin(2\pi f_m t)$, whose maximum slope is $2\pi f_m A$, overload is avoided if

$$\Delta f_s \geq 2\pi f_m A$$

- **Granular noise**: when the signal is flat, the staircase hunts up and down by $\pm\Delta$ around it. This error behaves like quantization noise of power about $\Delta^2/3$ spread over 0 to f_s, of which the fraction W/f_s survives the output filter.

Large steps reduce overload but increase granular noise; small steps do the reverse. The optimum step lies at the overload limit. For a tone at that limit the SQNR is

$$\mathrm{SQNR}_{DM} \approx \frac{3 f_s^3}{8\pi^2 f_m^2 W}$$

growing as f_s^3: 9 dB per doubling of sampling rate, compared with 6 dB per bit (per doubling of L) in PCM.

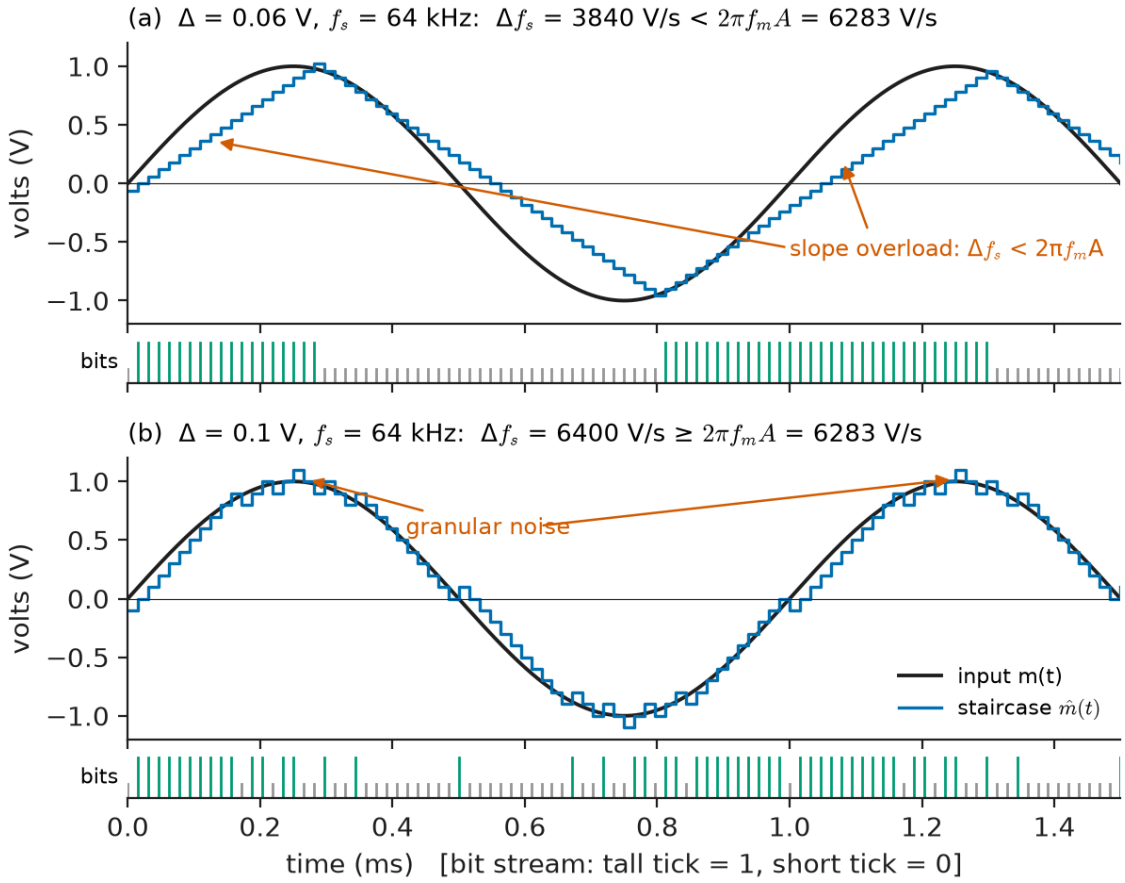

Figure 27.2 — Delta modulation of a 1 kHz tone showing the staircase approximation, slope overload where the step is too small, and granular noise on the flat portions.

Worked example — designing a delta modulator for speech. A delta modulator for a voice channel ($W = 4$ kHz) runs at $f_s = 64$ kb/s, the same bit rate as PCM telephony, and must handle a 1 kHz tone of 1 V peak without overload. The minimum step is

$$\Delta \geq \frac{2\pi \times 1000 \times 1}{64\,000} = 98.2 \text{ mV}.$$

At this step, the SQNR is

$$\text{SQNR} = \frac{3 \times (64\,000)^3}{8\pi^2 \times 1000^2 \times 4000} = 2490 \;\; (34.0 \text{ dB}).$$

Doubling the rate to 128 kb/s raises it to 43.0 dB. At equal bit rate, 8-bit uniform PCM gives 49.9 dB for the same full-scale tone, so plain DM is inferior for a constant-level signal. Its practical forms make up for this. **Adaptive delta modulation** (continuously variable slope delta modulation, CVSD) enlarges the step during runs of identical bits and shrinks it otherwise, extending the dynamic range; CVSD at 16–64 kb/s was used in military and Bluetooth voice links because it degrades gracefully under bit errors. **Differential PCM** generalises DM to multi-bit quantization of a prediction error, and underlies ADPCM speech coding at 32 kb/s.

Sigma-delta conversion

Placing the integrator before the quantizer, inside the loop, gives the **sigma-delta** (ΔΣ) modulator. Its quantization noise is pushed toward high frequencies (noise shaping), and a digital decimation filter (Chapter 24) removes it. A second-order sigma-delta converter oversampling by 64 reaches 16 bits from a 1-bit quantizer. Nearly all audio ADCs and DACs, and many precision instrumentation ADCs for strain gauges, thermocouples and ECG, work this way.

> **Boundary.**
> The circuit design of ADCs, DACs and sigma-delta modulators belongs to *Electronics & Computing*; this chapter uses their input–output behaviour.

27.3 Line coding

What a line code must do

To send bits over a cable or fibre without a carrier, each bit (or group of bits) is represented by a pulse shape. The choice — the **line code** — must satisfy several requirements at once:

- **Spectrum** matched to the channel: little or no DC content for transformer- or capacitor-coupled lines; little energy at high frequencies where cable loss and crosstalk are large.
- **Timing content**: enough transitions for the receiver to recover the clock, even in long runs of identical bits.
- **Error monitoring**: some codes have redundancy that lets violations be detected without decoding.
- **Efficiency**: bandwidth needed per bit, and transmit power needed for a given error rate.

The common codes

Figure 27.3 shows the main binary line codes for the same bit sequence, and Table 27.2 compares their properties.

- **Unipolar NRZ (on–off)**: 1 = $+V$ for the whole bit, 0 = 0 V. Simple; has a DC component and a spectral line at DC; long runs of 1s or 0s carry no timing.
- **Polar NRZ**: 1 = $+V$, 0 = $-V$. No DC on average for balanced data, best noise performance for a given peak voltage, but still no timing in long runs. Used inside chips, on RS-232 and (as differential NRZ with bit stuffing) on CAN and USB.
- **Return-to-zero (RZ)**: the pulse occupies only half the bit. Doubles the bandwidth but gives a transition in every 1.
- **Bipolar or alternate mark inversion (AMI)**: 0 = 0 V, successive 1s alternate between $+V$ and $-V$. No DC; a violation of the alternation reveals an error. Long runs of 0s lose timing, which **HDB3** (E1) and **B8ZS** (T1) cure by substituting deliberate violation patterns for runs of four or eight zeros.

- **Manchester (biphase)**: each bit has a transition in the middle (by the IEEE 802.3 convention, low-to-high for 1). No DC, guaranteed timing, but twice the bandwidth of NRZ. Used by 10 Mb/s Ethernet and many RFID and industrial links.
- **Block codes**: **4B5B** (100 Mb/s Ethernet, with MLT-3 three-level signalling), **8b/10b** (Gigabit Ethernet, Fibre Channel, PCIe 1–2) and **64b/66b** or **128b/130b** (10 Gb/s Ethernet, PCIe 3+) map groups of bits to longer code words with guaranteed transitions and bounded running disparity (DC balance), at an overhead of 25 %, 25 %, 3.1 % and 1.6 % respectively.
- **Multilevel codes**: **2B1Q** (ISDN), **PAM-4** (25/50 Gb/s per lane links, 400G Ethernet) and **PAM-5** (1000BASE-T) send more than one bit per symbol to reduce the symbol rate.

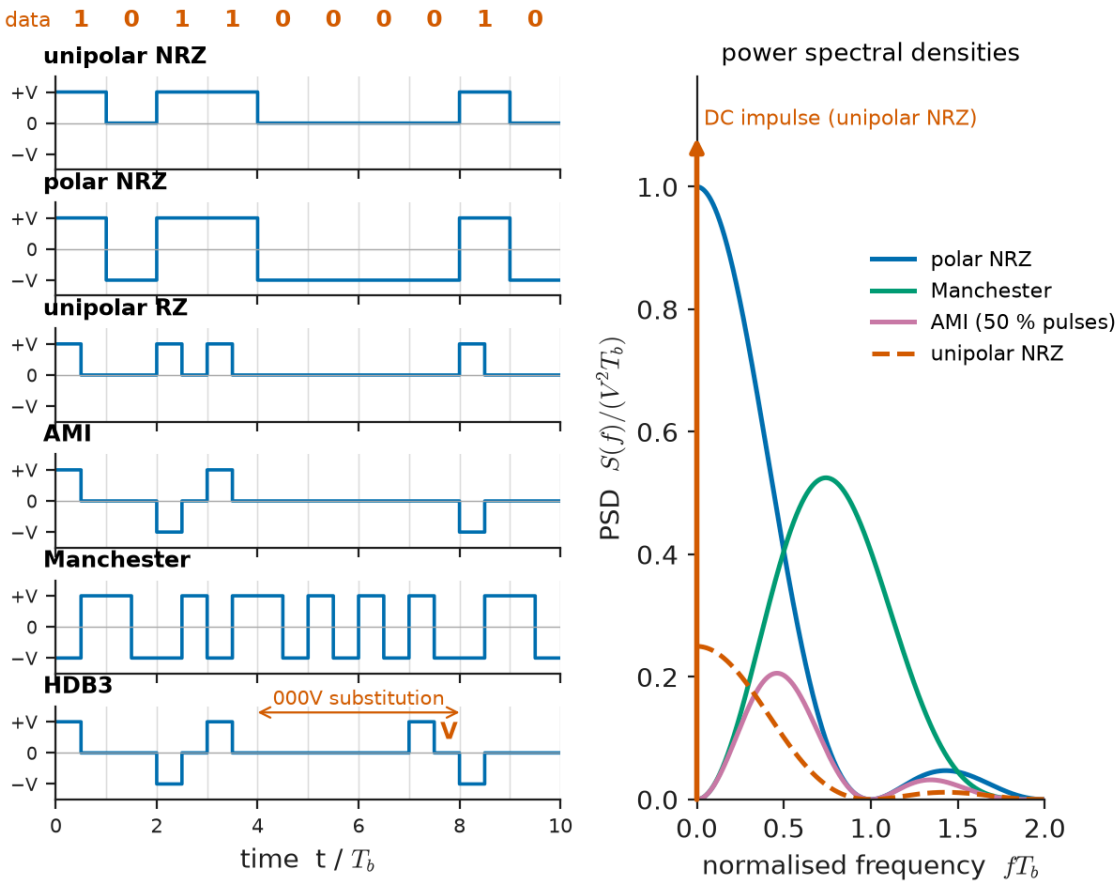

Figure 27.3 — Common binary line codes for the bit sequence 1 0 1 1 0 0 0 0 1 0 and their power spectral densities.

Table 27.2 — Properties of binary line codes (data rate R_b, equiprobable bits).

Code	DC component	First spectral null	Timing in long runs	Error monitoring	Typical use
Unipolar NRZ	Yes	R_b	No	No	Logic-level links, optical OOK
Polar NRZ	No (on average)	R_b	No	No	RS-232, CAN, PCIe lanes after block coding
Unipolar RZ	Yes	$2R_b$	For 1s only	No	Some optical and IR links
AMI / HDB3 / B8ZS	No	R_b	Yes with HDB3/B8ZS	Yes (bipolar violations)	E1, T1 trunks
Manchester	No	$2R_b$	Yes	Limited	10BASE-T Ethernet, RFID, MIL-STD-1553
8b/10b	No (bounded disparity)	$1.25R_b$	Yes	Yes (invalid words)	Gigabit Ethernet, SATA, Fibre Channel

> **Status of the principle: Line-code power spectral densities**
> *Derived*. Obtained from the autocorrelation of the coded symbol sequence and the Fourier transform of the pulse shape; valid for equiprobable, independent data bits, and altered by scramblers, framing patterns or long-run statistics of real traffic.

27.4 Symbol and bit rates

Definitions

A digital modulator sends **symbols**, each chosen from an alphabet of M possibilities, at a **symbol rate** R_s (in baud, symbols per second). Each symbol carries $k = \log_2 M$ bits, so the **bit rate** is

$$R_b = R_s \log_2 M$$

A modem sending 2400 symbols per second with 16 possible symbols delivers 9600 b/s; a cable modem sending 5.36 Msymbol/s with 256-QAM delivers $5.36 \times 8 = 42.9$ Mb/s. Correspondingly, the **energy per symbol** is $E_s = kE_b$, and the symbol duration is $T_s = 1/R_s = kT_b$. The distinction matters because the bandwidth depends on the symbol rate, while the information delivered depends on the bit rate: raising M increases the bits per hertz without increasing the bandwidth — but, as later sections show, at a cost in power.

Intersymbol interference and the Nyquist criterion

Rectangular pulses have sinc spectra with slowly decaying sidelobes, which waste bandwidth and violate spectral masks (Chapter 25, Figure 25.5). If the pulses are band-limited instead, they spread in time and overlap their neighbours, causing **intersymbol interference** (ISI). Nyquist showed that a pulse $p(t)$ causes no ISI at the sampling instants if

$$p(nT_s) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} \quad \Longleftrightarrow \quad \sum_{k=-\infty}^{\infty} P\left(f - \frac{k}{T_s}\right) = T_s$$

The narrowest spectrum satisfying this is a rectangle of width $R_s/2$ (the sinc pulse), so the **minimum baseband bandwidth** for R_s symbols per second is

$$B_{\min} = \frac{R_s}{2}$$

> **Status of the principle: Nyquist's zero-ISI criterion**
> *Derived*. Follows from sampling the pulse train and applying the Poisson summation formula (Chapters 18–19); it guarantees zero ISI only at ideal sampling instants on a channel whose overall response (transmit filter, channel, receive filter) has the specified form, and it says nothing about noise or timing jitter.

Raised-cosine pulses

The ideal sinc pulse needs a brick-wall filter and decays as $1/t$, so a small timing error causes large ISI. The practical family is the **raised-cosine** spectrum, whose transition band is a half-cosine of width set by the **roll-off factor** α ($0 \le \alpha \le 1$):

$$B = \frac{R_s}{2}(1+\alpha) \ \text{(baseband)}, \qquad B_T = R_s(1+\alpha) \ \text{(passband)}$$

Its pulse decays as $1/t^3$, so tails are tamed. Typical values: $\alpha = 0.22$ in UMTS, 0.35 in DVB-S, 0.20–0.05 in DVB-S2X, 0.12–0.18 in DOCSIS. In practice the filtering is split equally between transmitter and receiver as a pair of **root-raised-cosine** filters, which together form a raised cosine and — as Section 27.9 shows — make the receive filter matched to the transmitted pulse.

The quality of the pulse shaping is checked with the **eye diagram**: the received baseband waveform is displayed on an oscilloscope triggered by the symbol clock, so that all symbol periods are overlaid (Figure 27.4). The vertical opening of the eye at the sampling instant is the noise margin; the horizontal opening is the tolerance to timing error; the thickness of the traces at the crossing is jitter.

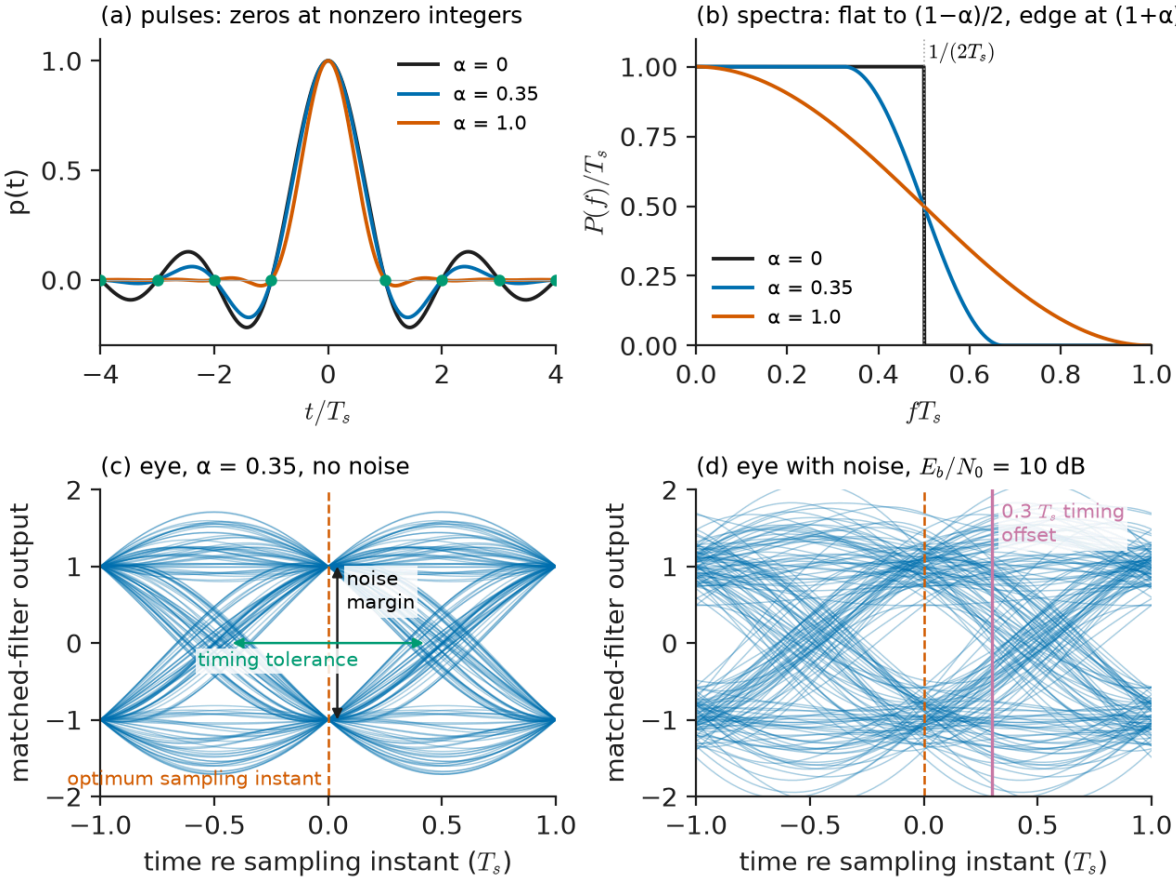

Figure 27.4 — Raised-cosine pulse shaping: pulse shapes and spectra for roll-off 0, 0.35 and 1, and eye diagrams of polar binary data with and without noise.

Spectral efficiency of digital modulation

With raised-cosine shaping, a passband signal of M-ary symbols achieves

$$\eta = \frac{R_b}{B_T} = \frac{\log_2 M}{1+\alpha} \ \text{b/s/Hz}$$

Worked example — a QPSK satellite carrier and a QAM cable channel. A satellite modem must send $R_b = 2$ Mb/s using QPSK ($M = 4$, $k = 2$) with roll-off $\alpha = 0.35$. The symbol rate is $R_s = 2/2 = 1$ Msymbol/s and the occupied bandwidth $B_T = 1 \times 1.35 = 1.35$ MHz, a spectral efficiency of 1.48 b/s/Hz. A DOCSIS 3.0 cable channel of 6 MHz runs 64-QAM at 5.057 Msymbol/s with $\alpha = 0.18$, so $B_T = 5.057 \times 1.18 = 5.97$ MHz and $R_b = 5.057 \times 6 = 30.34$ Mb/s; with 256-QAM it runs 5.361 Msymbol/s at $\alpha = 0.12$, occupying 6.00 MHz and delivering $5.361 \times 8 = 42.88$ Mb/s. The step from 64- to 256-QAM gains 41 % in throughput in the same bandwidth — and, as Section 27.10 shows, costs about 6 dB in required SNR, which is exactly the margin the evening customer of the opening paragraph lost.

27.5 Constellation diagrams

Signal space

Most digital modulations can be described with two orthonormal basis functions over a symbol period,

$$\phi_1(t) = \sqrt{\frac{2}{T_s}}\cos(2\pi f_c t), \qquad \phi_2(t) = -\sqrt{\frac{2}{T_s}}\sin(2\pi f_c t), \qquad 0 \le t < T_s$$

(with pulse shaping, the rectangular window is replaced by the shaping pulse). Each transmitted symbol is $s_i(t) = a_i\phi_1(t) + b_i\phi_2(t)$ and is fully described by the point (a_i, b_i) in a plane — its **constellation point**. The energy of a symbol is the squared distance of its point from the origin, $E_i = a_i^2 + b_i^2$, and the energy of the difference between two symbols is the squared distance between their points. The plane is exactly the complex-envelope plane of Chapter 25: a_i is the in-phase and b_i the quadrature component.

This geometric picture turns the design of a modulation into a packing problem. Additive white Gaussian noise adds a circularly symmetric Gaussian cloud of variance $N_0/2$ per dimension to each point. Errors occur when the cloud around one point reaches into the decision region of another, so the error probability is governed by the **minimum distance** $d_{\min}$ between points relative to $\sqrt{N_0}$, while the transmitted power is governed by the **average energy** E_s. A good constellation places its points as far apart as possible for a given average energy.

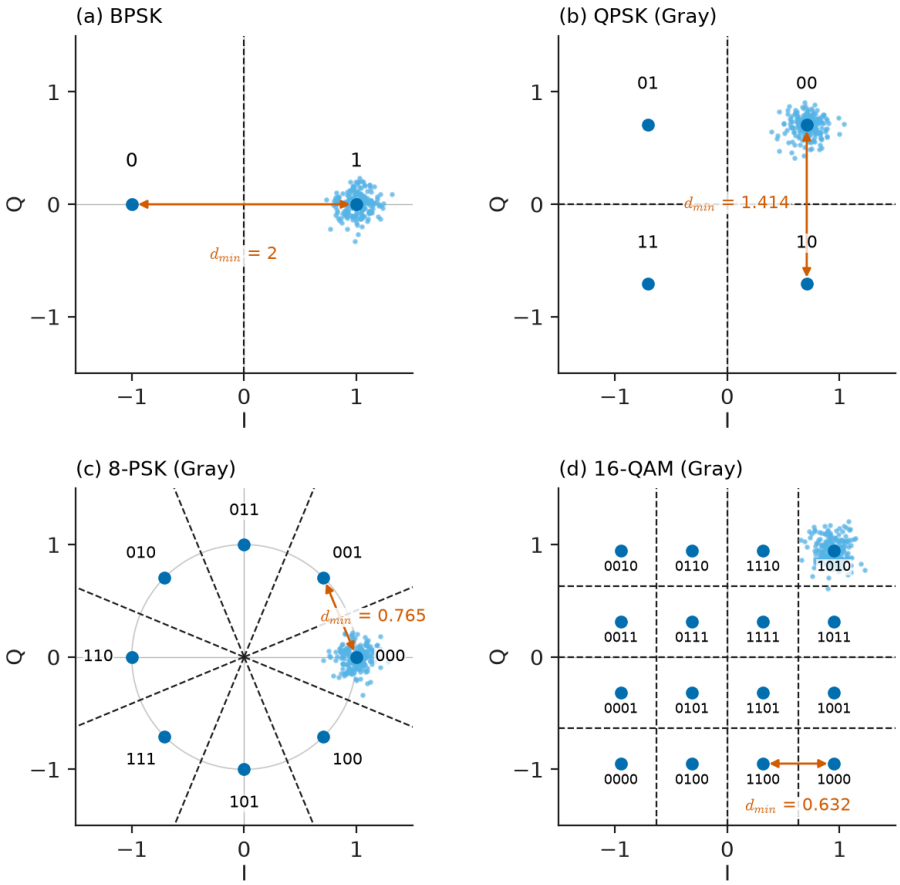

Figure 27.5 — Gray-coded constellations of BPSK, QPSK, 8-PSK and 16-QAM, all normalised to unit average symbol energy, with decision boundaries and minimum distances.

Reading a constellation on an instrument

A vector signal analyser demodulates a real signal and plots the received points. Each impairment has a recognisable signature: additive noise gives circular clouds; phase noise smears points along arcs; a carrier-frequency offset rotates the whole constellation; I/Q gain imbalance stretches it into a rectangle; quadrature error skews it into a parallelogram; amplifier compression pulls the outer points inward. The spread is summarised by the **error-vector magnitude** (EVM), the rms distance between received and ideal points relative to the rms ideal amplitude, or equivalently by the **modulation error ratio** $\mathrm{MER} = -20\log_{10}(\mathrm{EVM})$. Standards specify maximum EVM per modulation: roughly 17.5 % for QPSK, 8 % for 64-QAM and 3.5 % for 256-QAM in LTE base stations, and about 1.3 % (-38 dB) for 4096-QAM in Wi-Fi 7. The cable technician's "MER of 36 dB" in the opening paragraph is this measurement.

> **Status of the principle: Signal-space (constellation) representation**
> *Derived.* Exact for any set of finite-energy signals by Gram–Schmidt orthogonalisation; the two-dimensional form covers all linear carrier modulations, and its use for error analysis assumes additive white Gaussian noise and a receiver that projects onto the same basis.

27.6 ASK, FSK and PSK

The three binary carrier modulations vary the amplitude, frequency or phase of the carrier between two values per bit, exactly as the analog modulations of Chapter 26 vary them continuously.

Amplitude-shift keying

Binary ASK, in its usual form **on–off keying** (OOK), transmits the carrier for a 1 and nothing for a 0:

$$s(t) = \begin{cases} A_c \cos(2\pi f_c t), & \text{bit } 1 \\ 0, & \text{bit } 0 \end{cases}$$

It is the passband version of unipolar NRZ: its constellation is two points on the I axis at 0 and $\sqrt{2E_b}$ (average energy E_b). OOK can be detected with an envelope detector, so receivers are cheap; it is used in garage-door and car-key fobs, tyre-pressure sensors at 315/433 MHz, RFID backscatter, infrared remote controls and, as intensity modulation, in most short-reach optical links (Chapter 31). Its drawbacks are poor power efficiency, sensitivity to amplitude fading, and a decision threshold that depends on the received signal level.

Frequency-shift keying

Binary FSK sends one of two frequencies:

$$s_i(t) = A_c \cos(2\pi f_i t), \qquad i \in \{0, 1\}$$

If the tones are separated by a multiple of $1/(2T_b)$ for coherent detection, or of $1/T_b$ for noncoherent detection, the two signals are **orthogonal**: their constellation points lie on two different axes, $\sqrt{E_b}$ apart from the origin in perpendicular directions. **Minimum-shift keying** (MSK) uses the minimum

spacing $1/(2T_b)$ with continuous phase; filtered by a Gaussian low-pass filter it becomes **GMSK**, the modulation of GSM ($BT = 0.3$) and, as **GFSK**, of Bluetooth and many industrial telemetry radios. FSK can be detected noncoherently with a pair of filters and envelope detectors, or with an FM discriminator, and has a constant envelope, so the transmitter amplifier can run saturated. The occupied bandwidth is roughly $B_T \approx |f_1 - f_0| + 2R_b$ for rectangular pulses, much less with Gaussian shaping. The classic Bell 103 modem used 1070/1270 Hz tones at 300 b/s; HART instrumentation signalling still superimposes 1200/2200 Hz FSK at 1200 b/s on 4–20 mA loops.

M-ary orthogonal FSK uses M tones. Its bandwidth grows in proportion to $M/\log_2 M$ while its required E_b/N_0 falls — it is the power-limited end of the Shannon trade-off (Chapter 25), used in deep-space links, LoRa-type chirp systems and paging.

Phase-shift keying

Binary PSK (BPSK) transmits the carrier with phase 0 or π:

$$s(t) = \pm A_c \cos(2\pi f_c t) = \pm\sqrt{\frac{2E_b}{T_b}}\cos(2\pi f_c t)$$

Its constellation is two points at $\pm\sqrt{E_b}$: antipodal signals, as far apart as possible for a given energy. BPSK is the passband version of polar NRZ and has the best error performance of any binary modulation. It requires coherent detection with a recovered carrier (Costas or squaring loop, Chapter 26), which has an inherent 180° ambiguity; **differential PSK** (DPSK) resolves this by encoding each bit as a change, or no change, of phase relative to the previous symbol, and allows a simple receiver that compares consecutive symbols. BPSK is used in GPS (on the C/A code), deep-space telemetry, RFID and as the most robust mode of Wi-Fi and LTE.

The error performance of each of the three binary schemes is derived in Section 27.10; in summary, for the same E_b/N_0, coherent BPSK is 3 dB better than coherent FSK or OOK (on an average-energy basis), and noncoherent detection costs about 1 dB more at useful error rates.

27.7 QPSK

Two BPSK signals in quadrature

Quadrature PSK sends two bits per symbol by choosing one of four carrier phases. Equivalently — and this is how it is built — it sends two independent BPSK signals on the cosine and the sine carriers:

$$s(t) = a_I\sqrt{\frac{E_s}{T_s}}\cos(2\pi f_c t) - a_Q\sqrt{\frac{E_s}{T_s}}\sin(2\pi f_c t), \qquad a_I, a_Q \in \{\pm 1\}$$

so that $E_s = 2E_b$. Because the cosine and sine are orthogonal, the receiver separates them with the I and Q mixers of Figure 25.2 and detects each independently. Each branch is a BPSK link with energy E_b per bit; therefore QPSK has **the same bit-error probability as BPSK at the same** E_b/N_0, while sending twice as many bits in the same bandwidth. This "free" doubling is why QPSK, not BPSK, is the workhorse of satellite TV (DVB-S), early cellular data, and the control channels of LTE and 5G.

Gray coding

The four points are labelled so that neighbours differ in only one bit — **Gray coding** (00, 01, 11, 10 around the circle; Figure 27.5). The most likely error, to an adjacent point, then produces one bit error rather than two, and the bit-error probability is close to the symbol-error probability divided by $\log_2 M$. All practical PSK and QAM constellations are Gray-coded.

Variants for amplifier efficiency

When the data on both branches change at once, the phase jumps by 180° and, after pulse shaping, the envelope passes through zero. A nonlinear amplifier then spreads the spectrum. **Offset QPSK** (OQPSK) delays the Q branch by half a symbol, so only one branch changes at a time and phase jumps are limited to ±90°; it was used in IS-95 CDMA uplinks and is used in IEEE 802.15.4 (Zigbee). **π/4-QPSK** alternates between two QPSK constellations rotated by 45°, limiting jumps to ±135° and allowing differential detection; it was used in North American TDMA and TETRA. These variants have the same BER as QPSK in AWGN and differ only in their envelope behaviour.

> **Status of the principle: QPSK as two independent BPSK channels**
> *Derived*. Follows from the orthogonality of cosine and sine carriers over a symbol interval; it requires accurate carrier phase and I/Q balance, and a phase error ϕ couples the branches, reducing each branch's useful amplitude by $\cos\phi$ and adding crosstalk proportional to $\sin\phi$.

27.8 QAM

Amplitude and phase together

Quadrature amplitude modulation extends QPSK by letting each branch take more than two levels. With $\sqrt{M}$ levels per branch in a square grid, M-QAM carries $\log_2 M$ bits per symbol: 16-QAM carries 4, 64-QAM 6, 256-QAM 8, 1024-QAM 10 and 4096-QAM 12. With levels $\pm 1, \pm 3, \ldots, \pm(\sqrt{M}-1)$ times a spacing parameter $d/2$, the minimum distance is d and the average symbol energy is

$$E_s = \frac{(M-1)d^2}{6}$$

so for a fixed E_s the distance between points shrinks as $1/\sqrt{M-1}$. Each quadrupling of M (two more bits per symbol) halves the distance and costs about 6 dB of SNR at the same symbol-error rate — the cost the cable modem of the opening paragraph pays for 256- versus 64-QAM.

QAM is far more efficient than M-PSK for $M > 4$. In M-PSK all points lie on a circle, and their spacing $2\sqrt{E_s}\sin(\pi/M)$ shrinks almost as $1/M$; QAM fills the plane. At 10^{-6}, 16-QAM needs about 14.4 dB of E_b/N_0, while 16-PSK needs about 18.4 dB. Only 8-PSK survives in practice (EDGE, DVB-S2), where its constant amplitude suits saturated satellite amplifiers; DVB-S2 also uses amplitude-and-phase rings (16- and 32-APSK) as a compromise.

Where QAM is used

Table 27.3 lists representative uses and the SNR each needs. The pattern is consistent across media: QAM order is chosen to match the available SNR, and adaptive systems change it continuously.

Table 27.3 — QAM and PSK orders in deployed systems (typical uncoded requirements at a BER of 10^{-6} in AWGN).

Modulation	Bits/symbol	E_b/N_0 required	E_s/N_0 required	Representative use
BPSK	1	10.5 dB	10.5 dB	GPS, deep space, most robust Wi-Fi/LTE modes
QPSK	2	10.5 dB	13.5 dB	DVB-S, LTE/5G control channels, satellite telemetry
8-PSK	3	14.0 dB	18.7 dB	EDGE, DVB-S2
16-QAM	4	14.4 dB	20.4 dB	LTE, Wi-Fi, microwave backhaul
64-QAM	6	18.8 dB	26.6 dB	DOCSIS, DVB-C, LTE, Wi-Fi
256-QAM	8	23.5 dB	32.5 dB	DOCSIS 3.0, 5G, Wi-Fi 5, backhaul
1024–4096-QAM	10–12	~28.5–33.7 dB	~38.5–44.5 dB	DOCSIS 3.1, Wi-Fi 6/7, high-capacity backhaul

Coding (Chapter 28) reduces these requirements by several decibels, and practical receivers lose 1–3 dB to implementation effects (phase noise, finite-precision arithmetic, imperfect equalisation), which is why EVM limits are set tighter than the table alone would suggest.

> **Status of the principle: The 6-dB-per-two-bits rule for square QAM**
> *Derived*. Follows from $E_s = (M-1)d^2/6$ for square constellations; exact for the minimum-distance relation and approximate for error rates, and it assumes a linear channel and amplifier — compression of the outer points makes dense QAM degrade faster than the rule predicts.

27.9 Matched filtering

The problem

A known pulse $p(t)$ of duration T arrives in white Gaussian noise of two-sided spectral density $N_0/2$. The receiver passes the signal through a linear filter $h(t)$ and samples the output at $t = T$ to decide. Which filter makes the decision most reliable? Since the error probability falls as the SNR at the sampling instant rises, the question is which $h(t)$ maximises

$$\mathrm{SNR}_{out} = \frac{\left|\int_{-\infty}^{\infty} H(f)P(f)e^{j2\pi fT}\,df\right|^2}{\frac{N_0}{2}\int_{-\infty}^{\infty}|H(f)|^2\,df}$$

The solution

By the Cauchy–Schwarz inequality the numerator is at most $\int|H|^2df\cdot\int|P|^2df$, with equality when $H(f)$ is proportional to $P^*(f)e^{-j2\pi fT}$. In the time domain,

$$h(t) = p(T-t)$$

the **matched filter** is the time-reversed, delayed copy of the pulse it expects. The maximum output SNR is

$$\mathrm{SNR}_{\max} = \frac{2E}{N_0}, \qquad E = \int_0^T p^2(t)\,dt$$

Two facts stand out. The best achievable SNR depends only on the pulse *energy*, not on its shape, bandwidth or peak amplitude. And the matched filter is equivalent to a **correlator**: multiplying the received signal by $p(t)$ and integrating over the symbol, $\int_0^T r(t)p(t)\,dt$, produces the same sample at $t = T$. For a rectangular pulse the correlator is an **integrate-and-dump** circuit.

> **Status of the principle: The matched-filter theorem**
> *Derived*. A consequence of the Cauchy–Schwarz inequality for a known signal in additive white noise; it maximises the SNR at one sampling instant, and for coloured noise the optimum filter becomes a whitening filter followed by a matched filter.

Consequences across EEE

- **Digital receivers**: the root-raised-cosine receive filter is matched to the root-raised-cosine transmit pulse, so the standard Nyquist pair gives both zero ISI and maximum SNR.
- **Radar**: a long transmitted chirp of 100 µs with a bandwidth of 10 MHz has the energy of a long pulse; its matched filter compresses it to the 0.1 µs resolution of a short pulse (pulse compression), a gain of $100\times10 = 1000$ (30 dB) in time–bandwidth product.
- **Ultrasound, sonar and seismic exploration** use the same pulse compression with coded excitations.
- **Biomedical signal detection**: QRS detectors in ECG monitors correlate the signal with a QRS template; GPS receivers correlate with the satellite's spreading code and recover signals 20 dB below the thermal noise in the receiver bandwidth.

Worked example — matched versus simple filtering. A baseband polar link sends rectangular pulses of amplitude ±0.3 mV and duration $T = 1$ µs into a receiver whose input noise has one-sided density $N_0 = 10^{-14}$ V²/Hz (referred to a 1 Ω normalisation). The pulse energy is $E_b = A^2T = (0.3\times10^{-3})^2\times10^{-6} = 9\times10^{-14}$ V²s, so

$$\frac{E_b}{N_0} = 9\ (9.54\ \mathrm{dB}), \qquad \mathrm{SNR}_{\max} = \frac{2E_b}{N_0} = 18\ (12.55\ \mathrm{dB}).$$

An integrate-and-dump receiver deciding by sign gives (Section 27.10) $P_b = Q(\sqrt{18}) = Q(4.24) = 1.1 \times 10^{-5}$. If the integrator is replaced by a simple RC low-pass filter, the best time constant ($RC \approx 0.8T$) gives an output SNR only 0.81 times the matched value — a loss of 0.89 dB — and the error probability rises to $Q(\sqrt{18 \times 0.81}) = 6.7 \times 10^{-5}$, about six times worse. The matched filter is not a refinement; it is worth almost a decibel of transmitter power for free.

27.10 Detection and error probability

The decision rule

After the matched filter, the receiver holds one number per dimension per symbol and must decide which symbol was sent. The **maximum a posteriori** (MAP) rule picks the symbol with the largest posterior probability given the observation; with equally likely symbols it becomes the **maximum-likelihood** (ML) rule, and in additive white Gaussian noise ML is the **minimum-distance** rule: choose the constellation point nearest to the received point. The decision boundaries are the perpendicular bisectors between neighbouring points (the dashed lines in Figure 27.5).

> **Status of the principle: Minimum-distance detection in AWGN**
> *Derived*. The ML rule for equiprobable symbols in additive white Gaussian noise, since the likelihood decreases monotonically with Euclidean distance; it is not optimal for unequal priors, non-Gaussian (impulsive) noise or channels with ISI, which need weighted thresholds or sequence detection.

Binary signals in Gaussian noise

For two signals separated by distance d in signal space, the noise component along the line joining them is Gaussian with variance $N_0/2$, and an error occurs when it exceeds $d/2$. The error probability is

$$P_b = Q\left(\frac{d}{\sqrt{2N_0}}\right), \qquad Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-u^2/2}\, du$$

where Q is the tail of the standard Gaussian distribution (its properties are developed in *The Mathematics Universe*). Applying this to the three binary schemes:

- **Antipodal (BPSK, polar NRZ)**: $d = 2\sqrt{E_b}$, so
$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$
- **Orthogonal (coherent FSK)**: $d = \sqrt{2E_b}$, giving $P_b = Q\left(\sqrt{E_b/N_0}\right)$, 3 dB worse.
- **On–off (coherent OOK, unipolar NRZ)** with average energy E_b: points at 0 and $\sqrt{2E_b}$, also $d = \sqrt{2E_b}$, so $P_b = Q\left(\sqrt{E_b/N_0}\right)$ — 3 dB worse than BPSK on average energy, 6 dB worse on peak energy.

Noncoherent receivers cannot exploit the carrier phase and pay a penalty:

- **Noncoherent FSK**: $P_b = \tfrac{1}{2} e^{-E_b/(2N_0)}$
- **DPSK**: $P_b = \tfrac{1}{2} e^{-E_b/N_0}$

Table 27.4 gives representative values, and Figure 27.6 plots the curves.

Table 27.4 — Bit-error probability of binary modulations in AWGN.

E_b/N_0	BPSK / QPSK	DPSK	Coherent FSK, OOK	Noncoherent FSK
0 dB	7.9×10^{-2}	1.8×10^{-1}	1.6×10^{-1}	3.0×10^{-1}
4 dB	1.3×10^{-2}	4.1×10^{-2}	5.6×10^{-2}	1.4×10^{-1}
6 dB	2.4×10^{-3}	9.3×10^{-3}	2.3×10^{-2}	6.8×10^{-2}
8 dB	1.9×10^{-4}	9.1×10^{-4}	6.0×10^{-3}	2.1×10^{-2}
10 dB	3.9×10^{-6}	2.3×10^{-5}	7.8×10^{-4}	3.4×10^{-3}
12 dB	9.0×10^{-9}	6.5×10^{-8}	3.4×10^{-5}	1.8×10^{-4}
Required for 10^{-6}	10.5 dB	11.2 dB	13.5 dB	14.2 dB

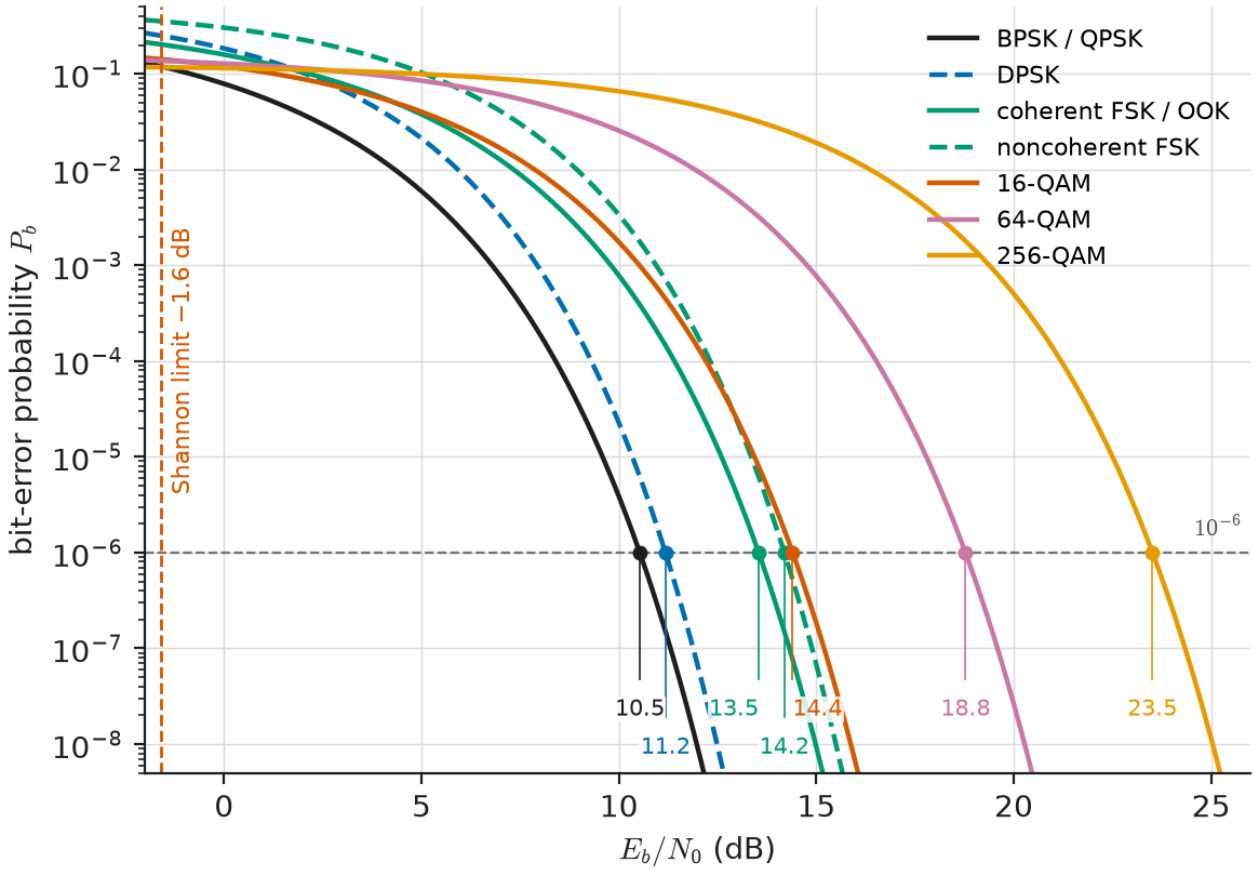

Figure 27.6 — Bit-error probability versus Eb/No in AWGN for binary and M-ary modulations, with the Shannon limit for reference.

> **Status of the principle: BER formulas for binary modulation in AWGN**
> *Derived*. Exact for ideal coherent (or ideal noncoherent) detection of equiprobable binary signals in additive white Gaussian noise with perfect timing and carrier recovery; they ignore ISI, phase noise, fading and interference, each of which raises the error rate and can create an error floor that no increase in E_b/N_0 removes.

M-ary error probability

For M-ary constellations the symbol-error probability is dominated by errors to nearest neighbours. A useful **nearest-neighbour approximation** is

$$P_s \approx N_{nn}\, Q\!\left(\frac{d_{\min}}{\sqrt{2N_0}}\right)$$

where N_{nn} is the average number of nearest neighbours. For square M-QAM, treating each branch as a $\sqrt{M}$-level PAM signal,

$$P_s = 1-(1-p)^2, \qquad p = 2\left(1-\frac{1}{\sqrt{M}}\right) Q\!\left(\sqrt{\frac{3E_s}{(M-1)N_0}}\right)$$

With Gray coding, $P_b \approx P_s/\log_2 M$.

Worked example — a QPSK link budget and a 16-QAM upgrade. A point-to-point telemetry link for a remote substation sends $R_b = 2$ Mb/s with QPSK. The receiver has a noise figure of 3 dB, so $N_0 = -174 + 3 = -171$ dBm/Hz. At a received power of $P_R = -100$ dBm,

$$\frac{E_b}{N_0} = P_R - N_0 - 10\log_{10} R_b = -100 + 171 - 63.0 = 8.0 \text{ dB},$$

giving $P_b = Q(\sqrt{2\times 10^{0.8}}) = 1.9\times 10^{-4}$ — about 380 errors per second, too many for protection signalling. For 10^{-6} the link needs 10.5 dB, that is $P_R \ge -171 + 63.0 + 10.5 = -97.5$ dBm, 2.5 dB more than it has. If instead the received power falls by 5 dB in a rain fade to -105 dBm, $E_b/N_0 = 3.0$ dB and $P_b = 2.3\times 10^{-2}$: the link is useless. The 5 dB fade has increased the error rate by a factor of 120, which is the "cliff" of digital communication.

Now suppose the operator wants to double the rate to 4 Mb/s in the same 1.35 MHz bandwidth by changing to 16-QAM at the same symbol rate of 1 Msymbol/s, and improves the antennas so that $P_R = -90$ dBm. Then $E_s/N_0 = P_R - N_0 - 10\log_{10} R_s = -90 + 171 - 60.0 = 21.0$ dB. The QAM formula above gives a symbol-error rate of 7.8×10^{-7} and a bit-error rate of about 2.0×10^{-7} (at 20 dB the figures would be 1.2×10^{-5} and 2.9×10^{-6}, showing again how steep the curve is). The 16-QAM link therefore needs roughly 7 dB more received power than the 2 Mb/s QPSK link (20.4 dB against 13.5 dB of E_s/N_0 for 10^{-6}, Table 27.3) to double the throughput — the power-for-bandwidth trade of Chapter 25 in concrete form.

Power-limited and bandwidth-limited choices

Putting the results together, Figure 27.6 and Table 27.3 show the two directions in which a designer can move from QPSK:

- toward **more bits per symbol** (16-, 64-, 256-QAM): more b/s/Hz, at about 3 dB more E_b/N_0 per extra bit at high order;
- toward **orthogonal signalling** (M-FSK) or **coding** (Chapter 28): less E_b/N_0 needed, at the cost of bandwidth or complexity.

Both trades are bounded by the Shannon curve of Figure 25.6, and uncoded modulations at 10^{-6} all sit about 8.5–10.5 dB to its right. Closing that gap is the job of error-control coding, the subject of the next chapter.

Bit-error rate in practice

Error probabilities are measured with a **bit-error-rate tester** that sends a known pseudo-random sequence and counts errors. To measure P_b with reasonable confidence, about 100 errors must be observed, so verifying 10^{-12} on a 10 Gb/s optical link takes $100/(10^{10}\times 10^{-12}) = 10^4$ s, nearly three hours; engineers therefore measure at deliberately degraded SNR and extrapolate along the theoretical curve, or use the Q-factor of the eye diagram. In fading channels (Chapter 30) the average BER falls only inversely with average E_b/N_0 rather than exponentially, and diversity and coding are needed to restore the steep curves of this section.

What remains open

Open question	Where it is answered
How much can a source be compressed, and how do error-correcting codes approach the Shannon limit?	Chapter 28
How are bandpass Gaussian noise, noise bandwidth, BER and SER treated as random-process problems?	Chapter 29
How do fading, multipath and Doppler change error probability, and how do OFDM and MIMO restore it?	Chapter 30
How are OOK and coherent QAM used on optical fibre, and what limits them?	Chapter 31
How are equalisers designed to remove ISI on dispersive channels?	Chapters 23 and 24 (adaptive filters)
What sets the quantization and finite-precision limits of the ADCs that feed PCM systems?	Chapters 19 and 21; *Electronics & Computing*
What are the properties of the Gaussian distribution and the Q-function used in the error formulas?	*The Mathematics Universe*

Next: Chapter 28 asks what information actually is, how much of it a channel can carry, and how codes let a link approach that limit.

Chapter 28 — Information Theory and Coding

Part VI · Communication Theory

Voyager 1 is more than 24 billion kilometres from Earth. Its transmitter radiates about 23 W at 8.4 GHz from a 3.7 m dish; by the time the signal reaches a 70 m antenna of the Deep Space Network, the received power is below an attowatt (10^{-18} W), far beneath the thermal noise in any bandwidth wide enough to carry the signal comfortably. Yet engineering telemetry still arrives with bit errors so rare that the mission does not notice them. Nothing in Chapter 27 explains how this is possible. The link is operated at an energy per bit only a few decibels above the noise density, where uncoded BPSK would make several errors in every hundred bits. What saves the link is *coding*: a convolutional code of constraint length 7, decoded by the Viterbi algorithm, wrapped in a Reed–Solomon code that cleans up what the Viterbi decoder misses. The redundancy costs bandwidth, but it buys a factor of hundreds in required signal power.

At the other end of the scale, a smartphone downloads a 30 MB software update that would occupy 90 MB uncompressed, over a 5G link whose data channel is protected by a low-density parity-check code that operates within about a decibel of a theoretical limit computed in 1948. Both stories rest on the same two ideas. Information can be *measured*, and a source can be compressed down to that measure and no further. A noisy channel has a *capacity*, and below that rate errors can be made arbitrarily rare by coding, while above it they cannot.

> **How much information does a source produce, and how far can it be compressed without loss?**
> **What is the largest rate a noisy channel can carry reliably, and which codes approach it in practice?**

Concept	Engineering destination
Entropy	Compression limits for audio, images, telemetry and logged sensor data
Mutual information	Rate a given channel and modulation can actually deliver; feature selection in measurement
Channel capacity (Shannon–Hartley)	Ceiling for modem, Wi-Fi, cellular and satellite throughput; link planning
Source coding, Huffman, arithmetic coding	JPEG, MP3/AAC entropy stages, fax, H.264/H.265
Lossless compression (LZ, DEFLATE)	ZIP, PNG, FLAC, firmware images, smart-meter logs
Error detection (parity, CRC)	Ethernet, CAN bus, USB, Modbus, storage sectors, ARQ protocols
Block codes (Hamming, BCH, Reed–Solomon)	ECC memory, QR codes, CDs and DVDs, DVB, deep-space links
Convolutional codes, Viterbi decoding	GSM, satellite modems, legacy Wi-Fi, space telemetry
Turbo, LDPC and polar codes	3G/4G, 5G NR, DVB-S2, Wi-Fi 6, 10GBASE-T, flash memory

Roadmap. Information and entropy (28.1) → mutual information (28.2) → channel capacity (28.3) → source coding and lossless compression (28.4–28.5) → error detection (28.6) → error correction in general (28.7) → block codes (28.8) → convolutional codes (28.9) → modern near-capacity codes (28.10).

28.1 Information and entropy

Measuring surprise

The engineering question behind information theory is concrete: how many binary digits must be sent, on average, to tell a receiver which of several possible messages a source produced? If a pressure sensor on a gas pipeline reports "normal" 999 times in 1000, a message saying "normal" tells the control room almost nothing it did not already expect, while "alarm" tells it a great deal. Information must therefore depend on probability, and it should add for independent events: learning the outcomes of two independent sensors should give the sum of the two amounts. The only function with both properties (up to the choice of base) is the logarithm. The **self-information** of an outcome x with probability $p(x)$ is

$$I(x) = -\log_2 p(x) \quad \text{bits}.$$

A fair coin flip carries 1 bit; one of 256 equally likely byte values carries 8 bits; the pipeline "alarm" with probability 10^{-3} carries $\log_2 1000 \approx 9.97$ bits, while "normal" carries only $-\log_2 0.999 \approx 0.0014$ bits.

With the natural logarithm the unit is the *nat* (1 nat $= 1.443$ bits).

Entropy of a discrete source

A **discrete memoryless source** emits symbols from an alphabet $\{x_1, \dots, x_M\}$, independently, with probabilities p_i. Its **entropy** is the average self-information per symbol:

$$H(X) = -\sum_{i=1}^{M} p_i \log_2 p_i \quad \text{bits/symbol}.$$

The entropy obeys

$$0 \le H(X) \le \log_2 M,$$

with $H = 0$ only when one symbol is certain, and $H = \log_2 M$ only when all symbols are equally likely. Entropy is a property of the probability distribution, not of the symbols' names or values: a source emitting $\{-5\ \text{V}, +5\ \text{V}\}$ equiprobably and one emitting $\{\text{red}, \text{green}\}$ equiprobably both have 1 bit per symbol.

For a binary source with $P(1) = p$ the entropy is the **binary entropy function**

$$H_b(p) = -p \log_2 p - (1-p)\log_2 (1-p),$$

which rises from 0 at $p = 0$ to 1 bit at $p = \tfrac{1}{2}$ and falls symmetrically (Figure 28.1). It reappears throughout the chapter, because the capacity of the binary symmetric channel and the limits of binary compression are both written with it. A few values are worth remembering: $H_b(0.1) = 0.469$, $H_b(0.11) = 0.500$, $H_b(0.01) = 0.081$, $H_b(10^{-3}) = 0.0114$.

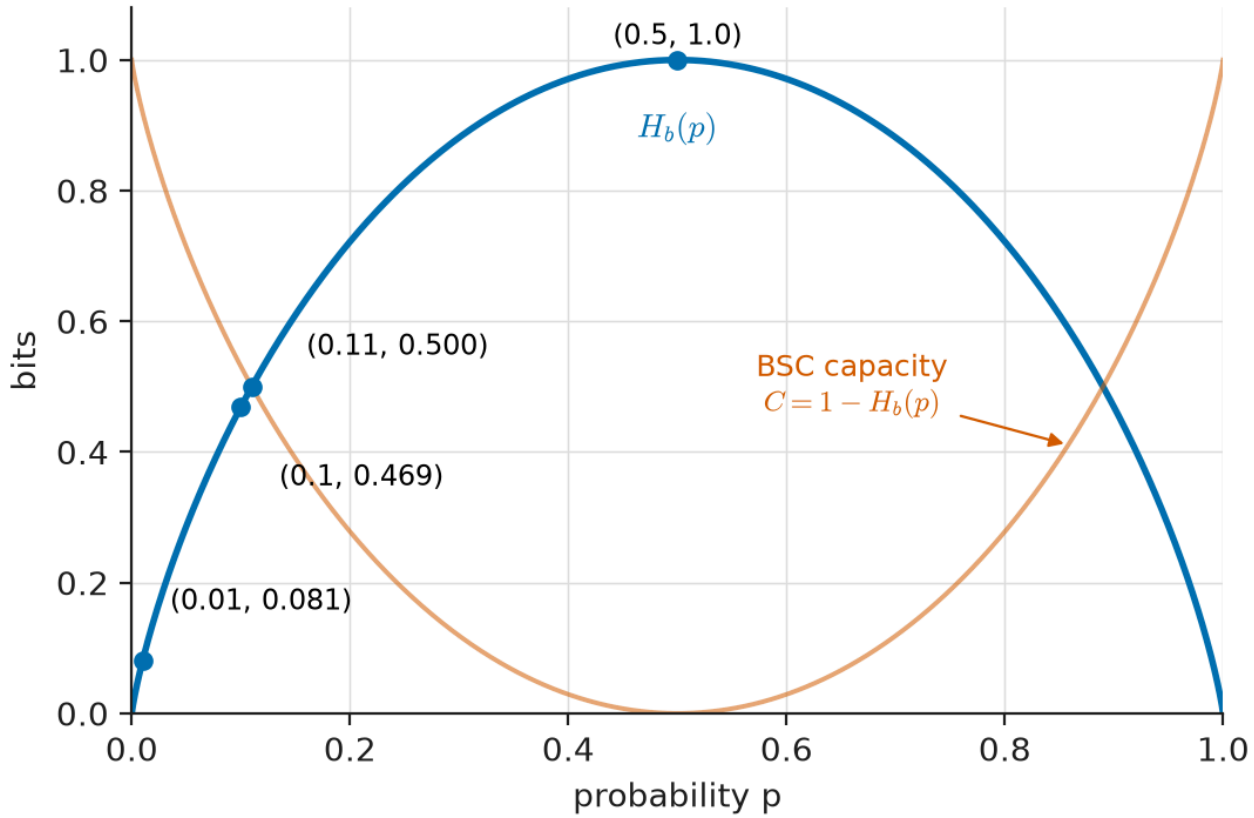

Figure 28.1 — The binary entropy function: a biased binary source carries less than one bit per symbol.

Joint entropy and sources with memory

For two random variables the **joint entropy** is $H(X,Y) = -\sum_{x,y} p(x,y)\log_2 p(x,y)$, and the **conditional entropy** $H(Y\mid X) = -\sum_{x,y} p(x,y)\log_2 p(y\mid x)$ is the average uncertainty that remains about Y once X is known. They are linked by the chain rule

$$H(X,Y) = H(X) + H(Y\mid X) \le H(X) + H(Y),$$

with equality only for independent variables. Real sources have memory: in English text "q" is almost always followed by "u", successive ECG samples are strongly correlated, and neighbouring pixels in a photograph resemble each other. The relevant measure for such a source is the **entropy rate**

$$\mathcal{H} = \lim_{n\to\infty} \frac{1}{n} H(X_1, X_2, \dots, X_n),$$

which is never larger than the single-symbol entropy and is often much smaller. Exploiting memory, by prediction, transforms or dictionaries, is what turns a modest single-symbol compressor into a powerful one.

Table 28.1 — Representative entropies and information rates (typical values).

Source	Raw representation	Approximate information content
Fair die	3 bits/symbol (fixed-length)	$\log_2 6 = 2.585$ bits/symbol
English text, single letters (27 symbols)	8 bits/character ASCII	about 4.1 bits/character
English text, long context (Shannon's estimates)	8 bits/character	roughly 1–1.5 bits/character
Telephone speech, 8 kHz × 8 bit A-law	64 kb/s	coded at 8–13 kb/s by speech codecs with good quality (lossy)
CD audio, 44.1 kHz × 16 bit × 2	1411 kb/s	FLAC lossless typically 700–1000 kb/s
12-lead ECG, 500 Hz × 12 bit	72 kb/s	lossless typically 2–3:1 reduction
Greyscale photograph, 8 bit/pixel	8 bits/pixel	lossless typically 3.5–5 bits/pixel (PNG, JPEG-LS)

The lossy entries are included only for scale: they discard information deliberately (rate–distortion theory, outside this chapter's scope), whereas the lossless entries are bounded below by the entropy rate.

> **Status of the principle: Entropy as the measure of information**
> *Derived*. Entropy is the unique (up to scale) measure satisfying continuity, monotonic growth with the number of equiprobable outcomes, and additivity for independent events; its operational meaning comes from the source coding theorem of Section 28.4.

Worked example — Entropy of a telemetry status word. A substation monitoring unit reports one of five breaker states every second: *closed–normal* (probability 0.4), *open–normal* (0.2), *closed–alarm* (0.2), *open–alarm* (0.1) and *fault* (0.1). Its entropy is

$$H = -\left[0.4\log_2 0.4 + 2(0.2\log_2 0.2) + 2(0.1\log_2 0.1)\right] = 0.529 + 0.929 + 0.664 = 2.122 \text{ bits/symbol}.$$

A fixed-length code needs $\lceil \log_2 5 \rceil = 3$ bits per report. The entropy says that, on average, no lossless code can use fewer than 2.122 bits, so up to 29 % of the fixed-length bit budget is redundant. Section 28.4 shows how to recover most of it.

28.2 Mutual information

What the output tells about the input

A channel maps an input X to an output Y according to a set of transition probabilities $p(y \mid x)$. Before seeing Y, the receiver's uncertainty about X is $H(X)$; after seeing it, the remaining uncertainty is $H(X \mid Y)$, called the **equivocation**. The reduction is the **mutual information**

$$I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X) = \sum_{x,y} p(x,y) \log_2 \frac{p(x,y)}{p(x)\,p(y)}.$$

Mutual information is symmetric, non-negative, and zero only when X and Y are independent. It is the number of bits per channel use that actually get through: $H(X)$ bits are sent, $H(X \mid Y)$ of them are lost in the noise.

The second form, $H(Y) - H(Y \mid X)$, is usually the easier to compute, because $H(Y \mid X)$ is fixed by the channel: it is the entropy the noise adds, independent of the message.

The binary symmetric and erasure channels

Two discrete channels serve as standard models for a hard-decision digital link.

- **Binary symmetric channel (BSC).** Each bit is flipped with crossover probability p, independently. It models a BPSK demodulator followed by a hard decision, with $p = Q\left(\sqrt{2E_b/N_0}\right)$ from Chapter 27. Here $H(Y \mid X) = H_b(p)$ for any input distribution.
- **Binary erasure channel (BEC).** Each bit is either received correctly or marked as erased (lost, with known position) with probability ε. It models packet loss in a network where a CRC flags corrupted packets, and lost symbols in a storage array.

Worked example — Mutual information of a biased link. A remote-terminal unit sends a binary "valve open" flag with $P(X=0) = 0.7$ over a BSC with $p = 0.1$. The output probability is $P(Y=0) = 0.7(0.9) + 0.3(0.1) = 0.66$, so

$$H(Y) = H_b(0.66) = 0.925 \text{ bit}, \qquad H(Y \mid X) = H_b(0.1) = 0.469 \text{ bit},$$

$$I(X;Y) = 0.925 - 0.469 = 0.456 \text{ bit per channel use.}$$

The source entropy is $H(X) = H_b(0.7) = 0.881$ bit, so the equivocation is $0.881 - 0.456 = 0.425$ bit: almost half of what is sent is lost. If the input were equiprobable, $I(X;Y) = 1 - H_b(0.1) = 0.531$ bit, higher. The input distribution matters, and maximizing over it is exactly what defines capacity.

Mutual information beyond communications

Mutual information measures statistical dependence in general. It quantifies coupling between EEG channels where correlation would miss non-linear relationships, and medical image registration (aligning an MRI with a CT scan) chooses the alignment that maximizes the mutual information between pixel intensities.

> **Status of the principle: Data-processing inequality**
> *Derived.* For any chain $X \to Y \to Z$ in which Z depends on X only through Y, $I(X;Z) \le I(X;Y)$: no processing of a received signal, however clever, can create information about the transmitted message. Hard decisions, quantization and filtering can only lose it, which is why soft-decision decoding (Section 28.7) outperforms hard-decision decoding.

28.3 Channel capacity

Definition and the coding theorem

The **capacity** of a channel is the largest mutual information obtainable by choosing the input distribution:

$$C = \max_{p(x)} I(X;Y) \quad \text{bits per channel use.}$$

Its meaning is given by Shannon's **noisy-channel coding theorem**: for every rate $R < C$ there exist codes of sufficiently long block length whose probability of decoding error is as small as desired; for $R > C$ the error probability is bounded away from zero for every code. Capacity is therefore a sharp boundary between possible and impossible, not a soft guideline. The theorem is non-constructive: Shannon proved that randomly chosen long codes work on average, without saying how to build or decode one efficiently. The 60-year search for practical codes that approach capacity is the subject of Sections 28.7–28.10.

> **Status of the principle: Noisy-channel coding theorem**
> *Derived*. A mathematical theorem about memoryless channels with known statistics; it assumes unbounded block length (hence unbounded delay and complexity) and gives no construction. Practical codes approach the limit within fractions of a decibel only at block lengths of thousands of bits.

Capacity of the discrete channels

For the BSC the uniform input is optimal and

$$C_{\text{BSC}} = 1 - H_b(p).$$

At $p = 0.01$, $C = 1 - 0.081 = 0.919$ bit per channel use: a rate-0.9 code can, in principle, make a link with 1 % raw bit errors essentially error-free. At $p = 0.11$, $C \approx 0.5$: a rate-1/2 code is the most that can be supported. At $p = 0.5$, $C = 0$: the output is independent of the input. For the BEC, $C = 1 - \varepsilon$, an intuitively satisfying result: the fraction of bits not erased is the fraction of capacity left.

The AWGN channel and the Shannon–Hartley law

For a band-limited channel of bandwidth B with additive white Gaussian noise of one-sided power spectral density N_0 (Chapter 29) and average received signal power S, the capacity is

$$C = B \log_2\!\left(1 + \frac{S}{N}\right) = B \log_2\!\left(1 + \frac{S}{N_0 B}\right) \quad \text{bit/s.}$$

It is achieved by a Gaussian-distributed input. Two regimes follow directly:

- **Bandwidth-limited regime** ($S/N \gg 1$): $C \approx B \log_2(S/N)$. Every doubling of signal power (+3 dB) adds only 1 bit/s/Hz. Doubling bandwidth nearly doubles capacity.

- **Power-limited regime** ($S/N \ll 1$): $C \approx (S/N_0)\log_2 e$, independent of bandwidth. Adding bandwidth no longer helps; only power does.

Writing $S = E_b R$ and the spectral efficiency $\eta = R/B$, operation at capacity ($R = C$) requires

$$\frac{E_b}{N_0} \geq \frac{2^{\eta} - 1}{\eta}.$$

As $\eta \to 0$ this tends to $\ln 2 = 0.693$, or -1.59 dB, the **Shannon limit**: no communication system of any kind can work reliably with less energy per bit than this, relative to the noise density. At $\eta = 1$ the bound is 0 dB, at $\eta = 2$ it is 1.76 dB, at $\eta = 4$ it is 5.74 dB and at $\eta = 6$ it is 10.2 dB. Figure 28.2 places practical modulation schemes on this plane. Uncoded BPSK needs 9.6 dB for a bit error rate of 10^{-5} at $\eta = 1$, about 9.6 dB above the bound for that spectral efficiency; this gap is what coding exists to close.

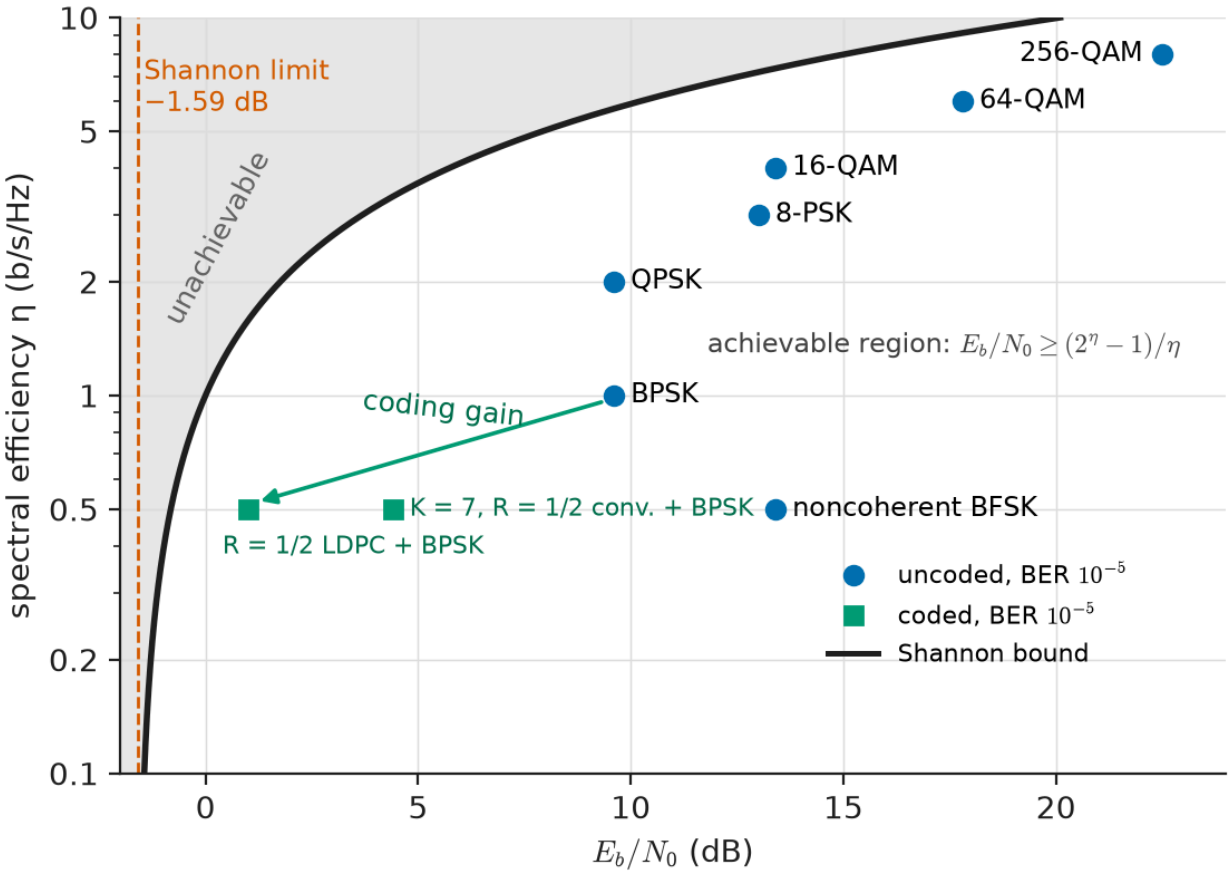

Figure 28.2 — The bandwidth-efficiency plane: the Shannon bound separates achievable from impossible operating points.

> **Status of the principle: Shannon–Hartley capacity**
> *Derived.* Follows from the coding theorem applied to a band-limited channel with additive white Gaussian noise and an average power constraint. It does not hold as stated for fading channels, interference that is not Gaussian, peak-power-limited transmitters, or channels with memory; Chapter 30 extends it to fading and multi-antenna channels.

Worked example — Capacity of a telephone line and of a cellular carrier. A voice-band telephone channel passes roughly 300–3400 Hz, so $B \approx 3.1$ kHz, with a signal-to-noise ratio around 35 dB ($S/N = 3162$). Then

$$C = 3100 \log_2(1 + 3162) = 3100 \times 11.63 = 36.0 \text{ kb/s}.$$

This is why V.34 analog modems topped out at 33.6 kb/s: they ran at about 93 % of capacity. (V.90's 56 kb/s downstream exploited the fact that the downstream path was digital up to the subscriber's line card, removing most of the quantization noise; it does not violate the bound.) A 20 MHz cellular carrier at a cell-centre SNR of 20 dB has $C = 20 \times 10^6 \log_2 101 = 133$ Mb/s for one antenna; a single-antenna LTE carrier peaks near 75–100 Mb/s after overheads, again within sight of the limit. Higher rates require more bandwidth or more antennas (Chapter 30), not cleverer modulation alone.

28.4 Source coding

The task

Source coding maps source symbols to binary codewords so that the average number of bits per symbol is as small as possible, while still allowing exact recovery. It is the theoretical core of compression. Fixed-length codes waste bits whenever the symbol probabilities are unequal; **variable-length codes** give frequent symbols short codewords and rare symbols long ones, as Morse code did in the 1840s (E is a single dot, Q is dash-dash-dot-dash).

A variable-length code must be **uniquely decodable**: a concatenated bit stream must parse into codewords in only one way. The most useful subclass is **prefix codes** (also called instantaneous codes), in which no codeword is the beginning of another. A prefix code can be decoded bit by bit, emitting a symbol as soon as a codeword is complete, which is exactly what a hardware decoder or a firmware routine wants.

The Kraft inequality

A prefix code with codeword lengths $\ell_1, \dots, \ell_M$ exists if and only if

$$\sum_{i=1}^{M} 2^{-\ell_i} \le 1.$$

The inequality expresses that short codewords are a scarce resource: a codeword of length 1 uses up half of all possible binary strings as prefixes. McMillan showed the same inequality holds for every uniquely decodable code, so nothing is lost by restricting attention to prefix codes.

The source coding theorem

Minimizing the average length $\bar{L} = \sum p_i \ell_i$ subject to the Kraft inequality gives the ideal lengths $\ell_i = -\log_2 p_i$, whose average is exactly $H(X)$. Because lengths must be integers, the optimum satisfies

$$H(X) \le \bar{L} < H(X) + 1.$$

Coding blocks of n symbols at a time spreads the "+1" over n symbols, so $\bar{L}/n \to H(X)$ as n grows. This is Shannon's **source coding theorem**: a source can be compressed losslessly to its entropy rate, and no further. The efficiency of a code is $H/\bar{L}$; its redundancy is $\bar{L} - H$.

> **Status of the principle: Source coding theorem**
> *Derived.* A theorem about sources with known, stationary statistics. Practical compressors must estimate the statistics (or adapt to them), and they meet the bound only asymptotically; for non-stationary sources such as mixed text and binary files the "entropy" itself is model-dependent.

Huffman coding

Huffman's 1952 algorithm builds an optimal prefix code (minimum $\bar{L}$ among all symbol-by-symbol prefix codes):

1. List the symbols with their probabilities.
2. Merge the two least probable entries into a node whose probability is their sum; label the two branches 0 and 1.
3. Repeat until one node (the root) remains.
4. Read each codeword from the root to its leaf.

Worked example — Huffman code for the status word. For the five-state breaker monitor of Section 28.1 ($0.4, 0.2, 0.2, 0.1, 0.1$), the algorithm proceeds:

- merge *open–alarm* (0.1) and *fault* (0.1) → node 0.2;
- merge *open–normal* (0.2) and *closed–alarm* (0.2) → node 0.4;
- merge the 0.2 node from step 1 with *closed–normal* (0.4) → node 0.6;
- merge 0.4 and 0.6 → root.

One valid set of codewords is: *closed–normal* 11, *open–normal* 00, *closed–alarm* 01, *open–alarm* 100, *fault* 101 (Figure 28.3). The Kraft sum is $3 \times 2^{-2} + 2 \times 2^{-3} = 1$, a complete code. The average length is

$$\bar{L} = 0.4(2) + 0.2(2) + 0.2(2) + 0.1(3) + 0.1(3) = 2.2 \text{ bits/symbol},$$

against an entropy of 2.122 bits: an efficiency of 96.5 %, and a saving of 27 % relative to the 3-bit fixed-length code. Across thousands of monitors on a narrowband utility radio network, that saving is significant.

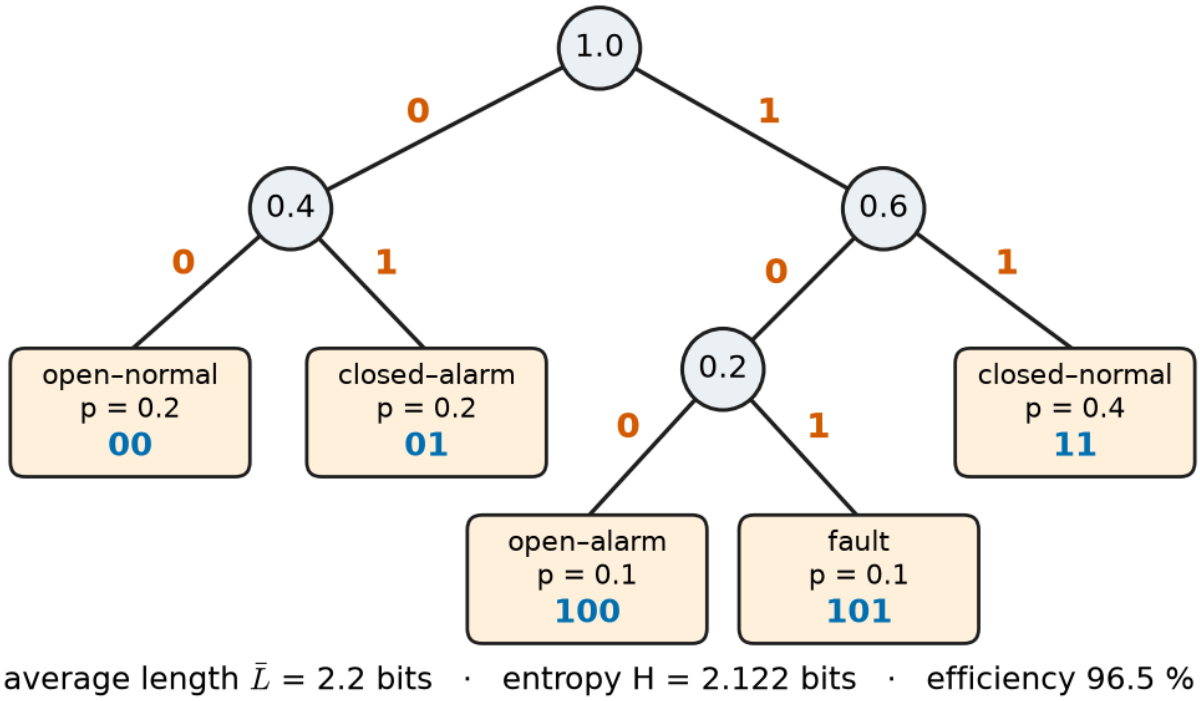

Figure 28.3 — Huffman code tree for the five-state breaker monitor.

Huffman codes have two weaknesses: a very probable symbol ($p = 0.95$, information 0.074 bit) still costs at least one bit, and the probability model must be known in advance. The next section addresses both.

28.5 Lossless compression

Lossless versus lossy

Lossless compression reconstructs the input bit for bit. It is mandatory for executable code, financial records, protection-relay settings, medical images used for diagnosis in many jurisdictions, and any data that will be further processed numerically. **Lossy compression** (JPEG, MP3, AAC, H.265, speech codecs) discards information the human eye or ear does not miss and achieves far higher ratios; its theory is rate–distortion theory, and its practice belongs to Chapter 24's audio and image processing. Every lossy codec, however, ends with a lossless entropy-coding stage, so the techniques below are inside every media file.

Two principles organize lossless compressors: **modelling**, which predicts the next symbol and assigns it a probability, and **coding**, which turns probabilities into bits as close to $-\log_2 p$ as possible. Good compressors separate the two.

Run-length coding

The simplest model: runs of identical symbols are sent as (symbol, count) pairs. Fax machines (ITU-T T.4) Huffman-code the run lengths of white and black pixels along each scan line and achieve 10:1 or more on typed pages; JPEG run-length codes the long runs of zero coefficients after quantization.

Arithmetic coding

Arithmetic coding represents an entire message by a single number in $[0, 1)$. Starting with the unit interval, each symbol narrows the current interval in proportion to its probability; the final interval has width equal to the product of all symbol probabilities, $\prod p(x_k)$, and a binary fraction identifying a point inside it needs about $-\sum \log_2 p(x_k)$ bits, plus at most two. The overhead is therefore two bits per *message*, not per symbol, and a symbol with probability 0.95 costs close to its true 0.074 bit. Arithmetic coding also separates model and coder cleanly: the probabilities can change for every symbol, driven by an adaptive context model. The context-adaptive binary arithmetic coder (CABAC) of H.264 and H.265 video saves roughly 10–15 % of bit rate over the Huffman-style alternative for the same quality. Range coding and asymmetric numeral systems (ANS, used in Zstandard) are engineering variants with the same efficiency and faster integer implementations.

Dictionary methods: Lempel–Ziv

When the statistics are unknown, a compressor can learn them from the data itself. **LZ77** (Lempel and Ziv, 1977) replaces a repeated string by a pointer (distance back, length) into a sliding window of recently seen data; **LZ78** and its variant **LZW** build an explicit dictionary of phrases. Lempel–Ziv coding is **universal**: for any stationary ergodic source, its compression ratio approaches the entropy rate as the input grows, without the compressor knowing the source statistics in advance. DEFLATE (ZIP, gzip, PNG) combines LZ77 with Huffman coding of the pointers and literals; LZW is used in GIF and TIFF.

Prediction and transforms for signals

For sampled signals, the best lossless coders first remove the predictable part. FLAC fits a linear predictor (up to order 32) to each audio block and entropy-codes the small-variance residual with Rice codes; JPEG-LS predicts each pixel from its neighbours; lossless ECG compressors use prediction or integer wavelets (Chapter 24). The ratio achieved depends on predictability: white noise does not compress at all.

Table 28.2 — Typical lossless compression ratios.

Data	Method	Typical ratio
English text, source code	gzip (DEFLATE) / Zstandard / xz	2.5–4 : 1
SCADA or smart-meter logs (repetitive)	DEFLATE, Zstandard	5–20 : 1
CD-quality music	FLAC	1.4–2 : 1
ECG, 12 bit samples	Linear prediction + entropy coding	2–3 : 1
Typed fax page	T.4 modified Huffman run-length	10–20 : 1
Photographs	PNG / JPEG-LS	1.5–2.5 : 1
Encrypted or already-compressed data	Any	about 1 : 1 (may expand slightly)

The last row is a direct consequence of the source coding theorem: well-encrypted data is designed to look like independent, equiprobable bits, whose entropy equals their length. A counting argument shows that no lossless compressor can shorten *every* input; any that shortens some must lengthen others.

28.6 Error detection

Detection and retransmission

Many links do not need to *correct* errors, only to *detect* them: if a corrupted packet is recognized, the receiver discards it and the protocol asks for a retransmission (**automatic repeat request**, ARQ). Ethernet, Wi-Fi, USB, CAN bus and TCP all rely on detection plus retransmission at some layer. Detection needs far less redundancy than correction, and an undetected error is usually much more harmful than a detected one: a frame flagged as bad costs a few milliseconds; a corrupted circuit-breaker trip command accepted as valid can black out a feeder.

Parity and checksums

A single **parity bit** makes the number of 1s in a word even (or odd). It detects every odd number of bit errors and no even number, so it catches every single error. On a BSC with $p = 10^{-4}$ and 8-bit bytes plus parity, the probability of an undetected error is dominated by double errors, $\binom{9}{2} p^2 \approx 3.6 \times 10^{-7}$ per byte. Parity is still used in UART framing and in memory buses where errors are rare and independent.

Checksums add the data words (in one's-complement arithmetic for IP, TCP and UDP, or with Fletcher and Adler variants) and append the sum. They are cheap in software but weak against some error patterns, such as reordered words, which leave a simple sum unchanged.

Cyclic redundancy checks

The workhorse of error detection is the **cyclic redundancy check (CRC)**. The k message bits are treated as the coefficients of a polynomial $m(x)$ over the binary field (where addition is XOR), and an r-bit check is the remainder

$$c(x) = \left[x^{r} m(x) \right] \bmod g(x),$$

where $g(x)$ is a generator polynomial of degree r. The transmitted word $x^{r}m(x) + c(x)$ is exactly divisible by $g(x)$; the receiver divides what it receives and declares an error if the remainder is non-zero. In hardware the division is a shift register with XOR feedback taps at the positions of the non-zero coefficients of $g(x)$, processing one bit per clock cycle; table-driven software versions process a byte at a time.

A well-chosen degree-r generator guarantees detection of all single and double errors (within a design length), all odd numbers of errors if $(x+1)$ divides $g(x)$, all burst errors of length $\le r$, and all but a fraction of about 2^{-r} of longer random error patterns.

Worked example — Computing a CRC-4. Take the generator $g(x) = x^4 + x + 1$ (binary 10011) and the 10-bit message 1101011011. Append four zeros and divide by 10011 using XOR long division:

$$11010110110000 \ \bmod \ 10011 = 1110.$$

The transmitted word is 1101011011 1110. At the receiver, dividing the full 14-bit word by 10011 leaves remainder 0000, so it is accepted. If a burst corrupts any 4 or fewer consecutive bits, the remainder is non-zero and the error is detected, guaranteed; a random multi-bit corruption escapes detection with probability of roughly $2^{-4} = 1/16$, which is why real protocols use 16 or 32 check bits.

Table 28.3 — CRCs in common use.

CRC	Check bits	Used in	Undetected fraction of random errors
CRC-8 (various)	8	SMBus, 1-Wire, automotive sensors	$\approx 4 \times 10^{-3}$
CRC-15-CAN	15	CAN bus frames	$\approx 3 \times 10^{-5}$
CRC-16-CCITT	16	HDLC, Bluetooth, X.25	$\approx 1.5 \times 10^{-5}$
CRC-16 (Modbus)	16	Modbus RTU in industrial control	$\approx 1.5 \times 10^{-5}$
CRC-24A	24	LTE and 5G NR transport blocks	$\approx 6 \times 10^{-8}$
CRC-32	32	Ethernet, ZIP, PNG, SATA	$\approx 2.3 \times 10^{-10}$

> **Status of the principle: CRC detection guarantees**
> *Derived*. The burst-detection and odd-weight guarantees follow from polynomial algebra over GF(2); the 2^{-r} figure for random patterns is an average that assumes errors are "random enough" and the frame is within the design length of the polynomial.

ARQ in brief

Stop-and-wait ARQ idles the channel during each round trip (about 0.5 s on a geostationary hop); **go-back-N** and **selective-repeat** ARQ keep a window of frames in flight. Modern cellular systems use **hybrid ARQ** (HARQ): each transmission is protected by a forward error-correcting code, and if decoding fails the receiver keeps the soft information and combines it with the retransmission, which may carry additional parity bits (incremental redundancy). HARQ adapts the effective code rate to the channel actually experienced, one of the reasons LTE and 5G throughput degrades gracefully rather than collapsing at the cell edge.

28.7 Error-correcting codes

Forward error correction

When retransmission is impossible or too slow, the receiver must correct errors itself: **forward error correction (FEC)**. Deep-space probes cannot wait hours for a repeat; a broadcast receiver has no return channel; a storage medium may be read years after it was written; a voice call cannot tolerate the delay. FEC adds $n - k$ redundant bits to every k information bits, giving a **code rate**

$$R = \frac{k}{n} < 1.$$

The cost is bandwidth expansion by $1/R$ at a fixed information rate (or a lower information rate at fixed bandwidth), plus decoder complexity and latency.

Hamming distance and the power of a code

The **Hamming distance** $d(\mathbf{u}, \mathbf{v})$ between two binary words is the number of positions in which they differ; the **weight** $w(\mathbf{u})$ is the number of 1s. The **minimum distance** $d_{\min}$ of a code is the smallest distance between any two distinct codewords. A code with minimum distance $d_{\min}$ can

- detect any pattern of up to $d_{\min} - 1$ errors, and
- correct any pattern of up to $t = \lfloor (d_{\min} - 1)/2 \rfloor$ errors,

because spheres of radius t around the codewords do not overlap; a received word inside a sphere is decoded to its centre. Some of the distance can be traded between the two jobs: a code with $d_{\min} = 4$ can correct one error and simultaneously detect two.

Coding gain, hard and soft decisions

The benefit of a code is quoted as **coding gain**: the reduction in E_b/N_0 (energy per *information* bit) needed to reach a target bit error rate, compared with uncoded transmission. Each coded bit carries energy $E_c = RE_b$, so the raw channel bits are *noisier* than uncoded bits would be; the code must recover more than it loses.

A **hard-decision** decoder receives the demodulator's 0/1 decisions, turning the channel into a BSC. A **soft-decision** decoder receives the unquantized (or finely quantized, typically 3–6 bits) matched-filter outputs, which carry the reliability of each bit. By the data-processing inequality hard decisions throw away information; for the AWGN channel the loss is about 2 dB at low rates. At high SNR the asymptotic coding gain is approximately

$$G_{\text{soft}} \approx 10\log_{10}(R\,d_{\min})\ \text{dB}, \qquad G_{\text{hard}} \approx 10\log_{10}(R\,(t+1))\ \text{dB}.$$

Bounds on codes

No code can have every desirable property at once. The **Hamming (sphere-packing) bound** limits how many correctable patterns fit in the space:

$$2^{n-k} \geq \sum_{i=0}^{t} \binom{n}{i}.$$

Codes that meet it with equality are **perfect**; the Hamming codes and the binary Golay (23,12) code are the only non-trivial binary examples. The **Singleton bound** $d_{\min} \leq n - k + 1$ is met with equality by Reed–Solomon codes, which are therefore called maximum-distance-separable.

> **Status of the principle: Minimum-distance correction rule**
> *Derived*. Correction of $t = \lfloor (d_{\min}-1)/2 \rfloor$ errors is guaranteed by geometry for bounded-distance decoding; maximum-likelihood decoders often correct many patterns beyond t, so $d_{\min}$ sets a worst-case, not an average, performance.

28.8 Block codes

Linear block codes

An (n, k) **linear block code** is a k-dimensional subspace of the space of binary n-tuples: the sum (XOR) of any two codewords is a codeword. Linearity makes encoding and decoding matrix operations over GF(2). Encoding is multiplication of the message row vector $\mathbf{m}$ by a $k \times n$ **generator matrix**:

$$\mathbf{c} = \mathbf{m}\mathbf{G}, \qquad \mathbf{G} = [\mathbf{I}_k \;\; \mathbf{P}]$$

in **systematic** form, where the first k code bits are the message itself and the last $n-k$ are parity bits. The $(n-k) \times n$ **parity-check matrix** $\mathbf{H} = [\mathbf{P}^T \;\; \mathbf{I}_{n-k}]$ satisfies $\mathbf{c}\mathbf{H}^T = \mathbf{0}$ for every codeword. For a linear code, $d_{\min}$ equals the smallest weight of a non-zero codeword, and also equals the smallest number of columns of $\mathbf{H}$ that sum to zero.

Syndrome decoding

If $\mathbf{r} = \mathbf{c} + \mathbf{e}$ is received, with $\mathbf{e}$ the error pattern, the **syndrome**

$$\mathbf{s} = \mathbf{r}\mathbf{H}^{T} = \mathbf{e}\mathbf{H}^{T}$$

depends only on the error, not on the codeword. A decoder precomputes a table from each syndrome to the most likely (lowest-weight) error pattern, adds that pattern to $\mathbf{r}$, and recovers $\mathbf{c}$. For a single error in position i, the syndrome is simply the i-th column of $\mathbf{H}$.

Hamming codes

For any $m \geq 2$, the **Hamming code** has $n = 2^{m} - 1$, $k = n - m$, and a parity-check matrix whose columns are all the distinct non-zero m-bit patterns. Since no two columns are equal, no two columns sum to zero, so $d_{\min} = 3$ and $t = 1$. The syndrome of a single error *is* the column of $\mathbf{H}$ at the error position, so correction is a direct lookup.

Worked example — Hamming (7,4) encoding and correction. Take

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}.$$

For the message $\mathbf{m} = (1,0,1,1)$, the parity bits are $\mathbf{m}\mathbf{P} = (1\oplus 0\oplus 1\oplus 1,\ 1\oplus 0\oplus 1\oplus 0,\ 0\oplus 0\oplus 1\oplus 1) = (1,0,0)$, so $\mathbf{c} = (1,0,1,1,1,0,0)$. Suppose bit 3 is flipped in a noisy memory cell: $\mathbf{r} = (1,0,0,1,1,0,0)$. The syndrome is

$$s_1 = r_1 \oplus r_3 \oplus r_4 \oplus r_5 = 1, \qquad s_2 = r_1 \oplus r_2 \oplus r_3 \oplus r_6 = 1, \qquad s_3 = r_2 \oplus r_3 \oplus r_4 \oplus r_7 = 1,$$

so $\mathbf{s} = \mathbf{r}\mathbf{H}^{T} = (1,1,1)$, which matches the third column of $\mathbf{H}$. Flipping bit 3 restores $\mathbf{c}$, and the message is read from the first four bits.

On a BSC with $p = 10^{-3}$, an uncoded 4-bit word is wrong with probability $1-(1-p)^{4} = 3.99\times 10^{-3}$. The (7,4) code fails only if two or more of its seven bits are wrong:

$$P_{\text{word}} = 1 - (1-p)^{7} - 7p(1-p)^{6} = 2.09 \times 10^{-5},$$

a 190-fold improvement for 75 % more bits. At $p = 10^{-2}$ the improvement shrinks to about 19-fold (3.94×10^{-2} to 2.03×10^{-3}): the advantage of a code grows rapidly as the raw channel improves. But on an AWGN channel at equal energy per information bit, the coded bits are 2.4 dB noisier ($R = 4/7$), and the net hard-decision coding gain of the (7,4) code at a bit error rate of 10^{-5} is only about 0.4 dB. Short codes help little against Gaussian noise; they are valuable where errors are rare and isolated.

That is exactly the situation in computer memory. **SEC-DED** (single-error-correcting, double-error-detecting) ECC memory uses an extended Hamming code, (72,64), with $d_{\min} = 4$: eight check bits per 64-bit word correct any single-bit upset caused by an alpha particle or cosmic-ray neutron, and flag any double error.

Cyclic codes, BCH and Reed–Solomon codes

A **cyclic code** is a linear code in which every cyclic shift of a codeword is a codeword. Cyclic codes are generated by a polynomial $g(x)$, exactly like the CRC of Section 28.6, and encoded by the same shift-register hardware; a CRC is simply a cyclic code used for detection only.

BCH codes (Bose, Chaudhuri, Hocquenghem) are cyclic codes designed for a chosen error-correcting capability t, with $n = 2^m - 1$ and $n - k \le mt$. Algebraic decoders (Berlekamp–Massey, then Chien search) correct up to t errors efficiently. A (255,239) BCH code corrects $t = 2$; the outer code of DVB-S2 is a BCH code correcting 8–12 errors in frames of up to about 58,000 bits. BCH codes also protect NAND flash pages, where tens of bit errors per kilobyte must be corrected as the cells wear.

Reed–Solomon (RS) codes are non-binary BCH codes over GF(2^m): each symbol is an m-bit byte, and an (n, k) RS code with $n \le 2^m - 1$ corrects any $t = (n - k)/2$ symbol errors. Because one symbol error covers up to m consecutive bit errors, RS codes are excellent against **bursts**. The (255,223) code used in deep space and the CCSDS standards corrects 16 byte errors, or any burst up to 121 bits. CDs use two interleaved shortened RS codes (cross-interleaved Reed–Solomon coding), which together with interleaving correct bursts of about 4000 bits, a scratch of 2.5 mm along the track. QR codes use RS codes with four selectable levels, recovering from about 7 %, 15 %, 25 % or 30 % of damaged codewords, which is why a QR code with a logo printed over its centre still scans.

Table 28.4 — Block codes and their parameters.

Code	(n, k)	Rate	$d_{\min}$	Corrects	Typical use
Single parity	$(k+1, k)$	$k/(k+1)$	2	detection only	UART, buses
Hamming	(7,4)	0.571	3	1 bit	Teaching, small links
Extended Hamming	(72,64)	0.889	4	1 bit, detects 2	ECC DRAM
Golay	(23,12)	0.522	7	3 bits	Early space links, radio paging
BCH	(255,239)	0.937	5	2 bits	Optical and storage links
Reed–Solomon	(255,223) bytes	0.875	33 symbols	16 bytes	CCSDS deep space, DVB
Reed–Solomon	(204,188) bytes	0.922	17 symbols	8 bytes	DVB-T, DVB-C
Reed–Solomon (CD)	(32,28) and (28,24)	0.75 overall	5 symbols each	2 bytes each, plus erasures	Compact Disc

Interleaving

Real channels produce errors in bursts: a fade on a radio link, a scratch on a disc, an impulsive noise spike on a power-line communication link from a switching motor. Most codes are designed for

scattered errors. An **interleaver** permutes the coded bits before transmission (for example, writing into a matrix by rows and reading out by columns) and the deinterleaver restores the order, so a burst of b channel errors is spread across b different codewords, each of which sees only one or two. The price is latency: an interleaver spanning a fade of 20 ms delays every bit by at least that much, a real constraint for voice and control loops.

28.9 Convolutional codes

The encoder

A **convolutional code** does not cut the data into independent blocks. Instead, each group of input bits produces output bits that depend on the current input and on the previous $K-1$ inputs held in a shift register, where K is the **constraint length**. A rate-$1/n$ encoder has n modulo-2 adders, each tapping a subset of the register stages; the tap patterns are written as **generator polynomials**, usually in octal.

The standard teaching example has $R = 1/2$, $K = 3$ and generators $(7, 5)_8$, that is, $g_1 = 111$ and $g_2 = 101$:

$$c_k^{(1)} = u_k \oplus u_{k-1} \oplus u_{k-2}, \qquad c_k^{(2)} = u_k \oplus u_{k-2}.$$

Each output bit is a discrete-time convolution, over GF(2), of the input with a generator sequence, which is where the name comes from (compare the convolution sum of Chapter 20). The encoder is a finite-state machine with $2^{K-1} = 4$ states, the contents of the two memory cells (Figure 28.4).

Worked example — Encoding with the (7,5) code. Encode the message 1011, followed by two zero "tail" bits that return the encoder to the all-zero state. Starting from state 00:

Step	Input u_k	State (u_{k-1}, u_{k-2}) before	$c^{(1)} = u_k \oplus u_{k-1} \oplus u_{k-2}$	$c^{(2)} = u_k \oplus u_{k-2}$	Output
1	1	00	1	1	11
2	0	10	1	0	10
3	1	01	0	0	00
4	1	10	0	1	01
5	0 (tail)	11	0	1	01
6	0 (tail)	01	1	1	11

The transmitted sequence is 11 10 00 01 01 11: twelve channel bits for four information bits. The tail makes the effective rate $4/12$ here; for realistic frames of hundreds or thousands of bits, the two tail bits are negligible and $R \approx 1/2$.

The trellis and free distance

Unrolling the state diagram in time gives the **trellis**: four state nodes at each time step, two branches leaving each node (input 0 or 1), each branch labelled with its two output bits. Every codeword is a path through the trellis. The relevant distance measure is the **free distance** d_{free}, the minimum Hamming weight of any path that leaves the all-zero state and later returns to it. For the (7,5) code, $d_{\text{free}} = 5$ (the path driven by a single 1 produces 11 10 11, weight 5). The asymptotic soft-decision coding gain is about $10\log_{10}(R\,d_{\text{free}}) = 10\log_{10} 2.5 = 4.0$ dB.

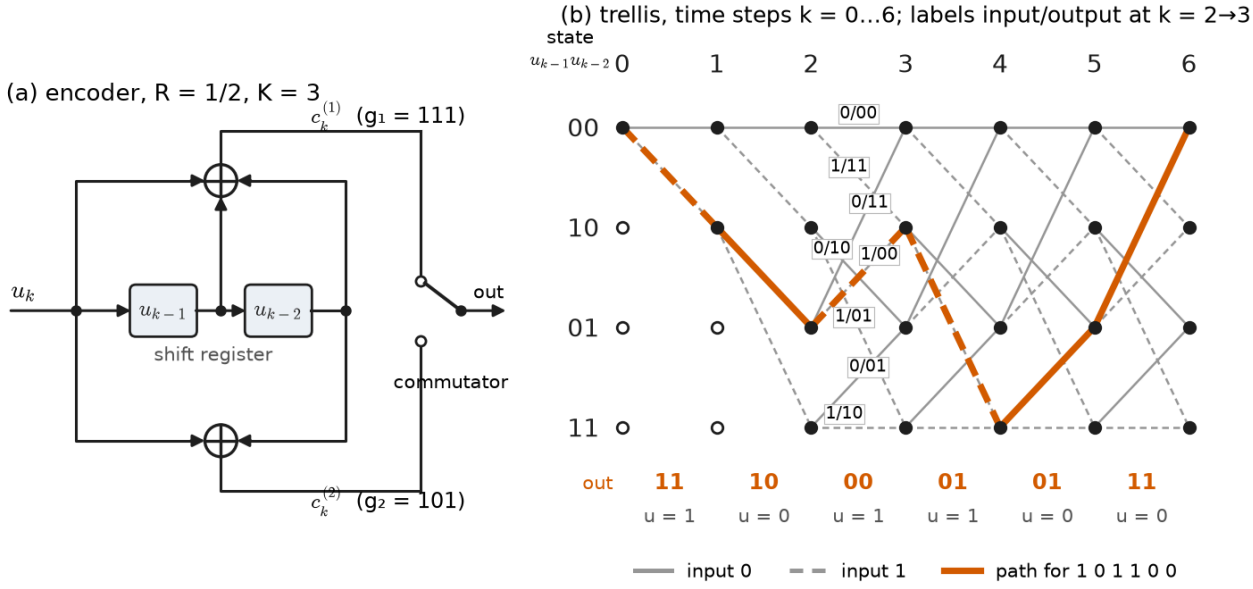

Figure 28.4 — The rate-1/2, K = 3, (7,5) convolutional encoder and its trellis.

Viterbi decoding

Maximum-likelihood decoding chooses the trellis path closest to the received sequence: in Hamming distance for hard decisions, in Euclidean distance for soft decisions. The **Viterbi algorithm** (1967) does this with effort that grows linearly with the message length rather than exponentially. At each time step, for each state, it keeps only the best path (the *survivor*) entering that state, because any continuation will add the same future metric to all paths that meet there. Per step the work is 2^{K-1} add–compare–select operations. After a decoding depth of about 5–7 constraint lengths the survivors almost always share a common history, which is output. With $K = 7$ (64 states), the decoder is a few tens of thousands of gates and runs at hundreds of megabits per second.

The same trellis search, an instance of dynamic programming, reappears in maximum-likelihood equalization of intersymbol interference and in hard-disk read channels.

The industry-standard code and puncturing

The $K = 7$, $R = 1/2$ code with generators $(171, 133)_8$ has $d_{\text{free}} = 10$ and an asymptotic gain of $10\log_{10} 5 = 7.0$ dB. In practice, with 3-bit soft decisions, it reaches a bit error rate of 10^{-5} at about 4.4 dB, a coding gain of roughly 5.2 dB over uncoded BPSK. It has been used on Voyager and many later space missions, GSM speech channels (with $K = 5$), 802.11a/g Wi-Fi and countless satellite modems.

Higher rates are obtained by **puncturing**: periodically deleting some coded bits according to a known pattern. Deleting one of every four bits from the rate-1/2 code gives rate 2/3; the decoder inserts neutral (zero-confidence) values at the deleted positions and runs the same Viterbi algorithm. 802.11a/g uses this to derive rates 2/3 and 3/4 from the single $K = 7$ mother code, adapting protection to channel quality without new hardware.

> **Status of the principle: Viterbi decoding is maximum-likelihood**
> *Derived*. The Viterbi algorithm finds the maximum-likelihood path exactly for a finite-state encoder in memoryless noise, given unlimited decoding depth. The truncated depth and finite soft-decision quantization used in practice cost typically 0.1–0.3 dB.

28.10 Fundamentals of modern coding methods

The gap that remained

By 1990 the best practical systems, such as the $K = 7$ convolutional code concatenated with a (255,223) Reed–Solomon outer code, reached a bit error rate of 10^{-5} at about 2.3–2.5 dB for an overall rate near 0.44. The Shannon limit for that rate with binary inputs is close to 0 dB: a gap of about 2.5 dB remained. Many coding theorists considered it practically unclosable. Three ideas closed it within a decade: long codes built from simple components, iterative decoding that exchanges soft information, and decoders that operate on a sparse graph.

Concatenated codes

Concatenation (Forney, 1966) puts two codes in series: an *inner* code close to the channel cleans up most errors, and an *outer* code removes the residual errors, which arrive in bursts from the inner decoder and are therefore well suited to a Reed–Solomon outer code with an interleaver between the two. This was the deep-space standard for decades and is still used in DVB-S and DVB-T.

Turbo codes

Turbo codes (Berrou, Glavieux and Thitimajshima, 1993) encode the data with two simple recursive convolutional encoders in parallel, the second fed through a long pseudo-random interleaver, and transmit the systematic bits plus both parity streams. The decoder runs two soft-input soft-output component decoders (using the BCJR, or MAP, algorithm, which computes a posterior probability for every bit rather than one best path). Each decoder passes its **extrinsic information**, the new information it has inferred about each bit, to the other as prior knowledge, and the process iterates 6–10 times. With block lengths of several thousand bits, a rate-1/2 turbo code reaches 10^{-5} within about 0.5–0.7 dB of the Shannon limit. Turbo codes were adopted for 3G (UMTS, CDMA2000), 4G LTE data channels and the CCSDS deep-space standard.

Low-density parity-check codes

LDPC codes, invented by Gallager in 1962 and largely forgotten until rediscovered in the mid-1990s, are linear block codes whose parity-check matrix **H** is very sparse: each column and each row contains only a few 1s, regardless of a block length of thousands of bits. The code is represented by a bipartite **Tanner graph** with one *variable node* per code bit and one *check node* per parity equation, connected wherever **H** has a 1 (Figure 28.5). Decoding is **belief propagation** (also called message passing or the sum–product algorithm): variable nodes send their current bit probabilities to the checks they participate in; each check node computes, for each connected bit, what the other bits imply about it; and the process repeats, typically 10–50 iterations, until all parity checks are satisfied. Because every node does a small local computation, the decoder parallelizes naturally, giving throughputs of tens of gigabits per second in hardware.

Well-designed irregular LDPC codes come within 0.1 dB of capacity at very long block lengths, and within about 0.5–1 dB at practical lengths. They are used in DVB-S2 (satellite television), 802.11n/ac/ax Wi-Fi, 10GBASE-T Ethernet, the data channels of 5G NR, and the read channels of hard disks and solid-state drives.

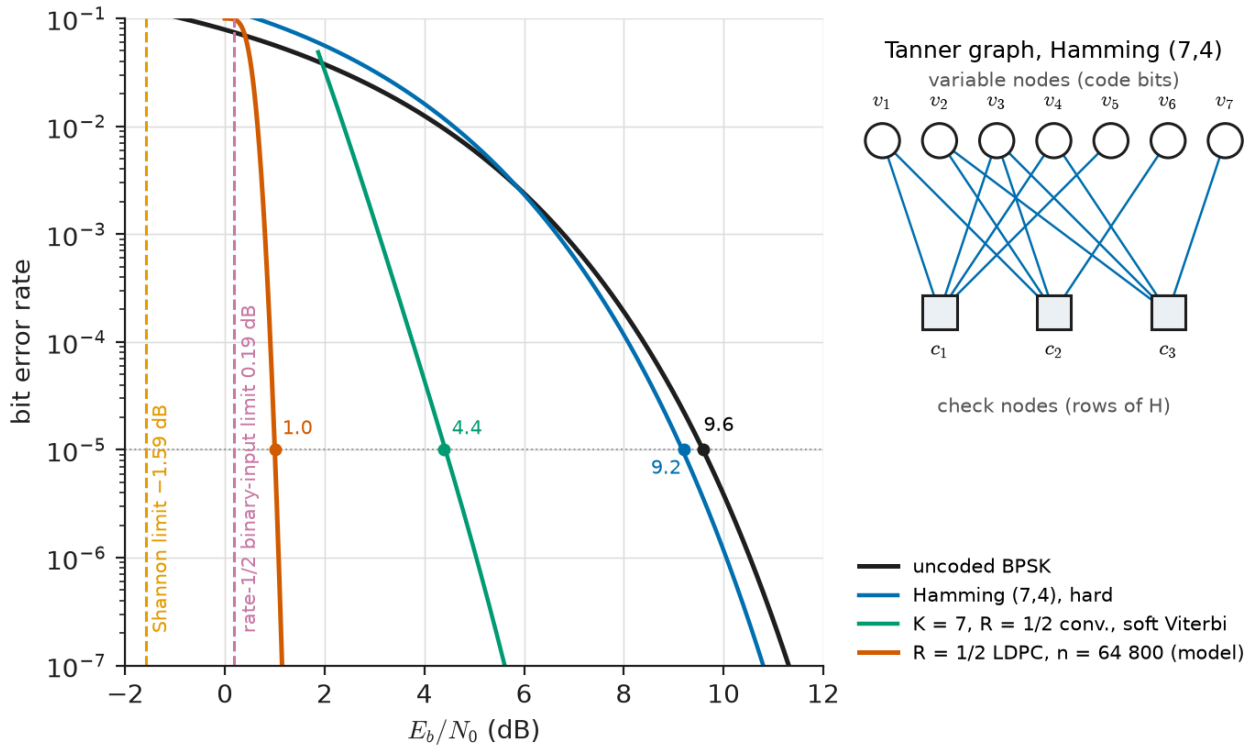

Figure 28.5 — Coded performance on the AWGN channel, with the Tanner graph of a small parity-check code.

Polar codes

Polar codes (Arıkan, 2009) are the first codes *proved* to achieve the capacity of binary-input symmetric channels with an explicit construction and low-complexity encoding and decoding ($O(n \log n)$). A recursive transform, built from the 2×2 kernel $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, converts n uses of the channel into synthetic bit-channels that *polarize*: as n grows, a fraction C of them become nearly perfect and the rest nearly useless. Information is sent on the good bit-channels; the bad ones are "frozen" to known

values. With successive-cancellation list decoding and an embedded CRC, polar codes perform very well at short block lengths, which is why 5G NR uses them for its control channels while LDPC protects the data channels.

Table 28.5 — Channel codes in current standards.

System	Data channel code	Control or other code	Note
GSM (2G)	Convolutional, $K = 5$, $R = 1/2$	Fire code, convolutional	Viterbi decoding
UMTS / LTE (3G/4G)	Turbo, $R = 1/3$ mother code	Convolutional (tail-biting)	HARQ with incremental redundancy
5G NR	LDPC (two base graphs)	Polar with CRC	Rates from about 1/5 to 0.95
802.11a/g Wi-Fi	Convolutional $K = 7$, punctured	—	Rates 1/2, 2/3, 3/4
802.11n/ac/ax Wi-Fi	LDPC (optional in n, standard later) or convolutional	—	Codeword lengths 648–1944 bits
DVB-S2	LDPC inner, BCH outer	—	Within about 0.7–1 dB of capacity
CCSDS deep space	Turbo, LDPC, or conv + RS	—	Frame lengths up to thousands of bits
ECC DRAM	Extended Hamming (72,64)	—	SEC-DED
NAND flash SSD	LDPC or BCH	—	Soft reads for LDPC as cells wear

What modern codes cost

Near-capacity codes are not free. They need long blocks, and therefore latency: a 64,800-bit DVB-S2 frame at 10 Mb/s takes 6.5 ms to fill before decoding can start. Iterative decoders consume power proportional to the number of iterations. And they exhibit an **error floor**: below a certain error rate, typically 10^{-6} to 10^{-10}, the steep "waterfall" of the BER curve flattens because of low-weight codewords or graph structures, which is why DVB-S2 adds an outer BCH code. Choosing a code is therefore an engineering trade among rate, block length, latency, decoder power and the error rate the application needs, not a search for the code closest to capacity.

> **Status of the principle: Near-capacity performance of iterative codes**
> *Empirical.* Turbo and LDPC performance within a fraction of a decibel of capacity is established by simulation and density-evolution analysis rather than by closed-form proof (except for polar codes asymptotically), and holds for the long block lengths, good interleavers and AWGN-like channels for which they are designed.

> **Boundary.**
> The algebra of finite fields used by BCH and Reed–Solomon codes, and the probability theory behind entropy, are part of *The Mathematics Universe*; this chapter quotes the results it needs. Hardware implementation of encoders and decoders (shift registers, LDPC decoder ASICs, memory controllers) belongs to *Electronics & Computing*.

What remains open

Open question	Where it is answered
Where does the AWGN model, and the noise density N_0, come from physically, and how is it measured in a receiver?	Chapter 29 (random signals, white noise, noise figure)
How are bit and symbol error rates of each modulation computed, which feed the coding-gain comparisons here?	Chapter 29 (BER and SER); Chapter 27 for the modulations themselves
What is the capacity of a fading channel and of a multi-antenna channel?	Chapter 30 (fading, diversity, MIMO capacity)
How do coding and interleaving interact with fading, OFDM and link budgets in a real cellular system?	Chapter 30
How do fibre-optic links use FEC to run close to their noise limits?	Chapter 31 (optical receivers and link budgets)
How are the convolution, z-transform and trellis ideas used in the decoders tied to discrete-time systems?	Chapters 20 and 23 (discrete-time systems, digital filters)
How are prediction and transforms used in lossy audio and image compression?	Chapter 24 (digital image and audio processing fundamentals)
What are the probability theory and finite-field algebra behind these results?	*The Mathematics Universe*

Next: Chapter 29 examines the noise itself, the random processes that make every channel imperfect, and turns the AWGN model into concrete bit and symbol error rates, noise figures and detection thresholds.

Chapter 29 — Random Signals and Communication Noise

Part VI · Communication Theory

A GPS satellite 20,200 km overhead delivers about −130 dBm to a phone's antenna, 10^{-16} W. In the 2 MHz bandwidth the signal occupies, the thermal noise generated by the receiver itself is about −109 dBm, so the signal arrives 21 dB *below* the noise: on a spectrum analyser it is invisible. Yet the phone computes its position to a few metres within seconds. A weather radar listens for echoes from raindrops that return a femtowatt; an EEG amplifier must resolve microvolt brain rhythms against the thermal noise of the electrode resistance; a 5G handset must decide whether a received symbol was one of 256 constellation points while the noise jostles it. In each case the engineering limit is not the signal's absolute size but its size *relative to the noise*, and in each case the noise is a random signal that can only be described statistically.

Chapter 28 assumed an additive white Gaussian noise channel and computed its capacity. This chapter explains where that model comes from, how noise is characterized and measured in real receivers, and how it turns into the bit and symbol error rates that decide whether a link works.

> **What kind of random signal is receiver noise, and how much of it does a given bandwidth and amplifier chain admit?**
> **How should a receiver decide between possible signals in noise, and what error rate results?**

Concept	Engineering destination
Random processes, autocorrelation, PSD	Spectrum analysers, interference analysis, vibration and seismic monitoring
Gaussian noise, Q-function	Error-rate prediction, false-alarm rates, measurement uncertainty
White noise, kT	Receiver sensitivity, EEG/ECG front ends, ADC driver design
Noise bandwidth	Comparing filters, integrating detectors, specifying spectrum-analyser RBW
Noise figure and temperature, Friis	LNA placement, satellite ground stations, radio-telescope receivers
Detection in noise, matched filter, Neyman–Pearson	Radar and sonar thresholds, ECG QRS detection, digital receivers
BER and SER	Link specifications, modem rate adaptation, test equipment
Performance trade-offs	Choosing modulation, coding, bandwidth and power for a system

Roadmap. Random processes in communication (29.1) → Gaussian noise (29.2) → white-noise models (29.3) → noise bandwidth (29.4) → noise figure (29.5) → detection in noise (29.6) → BER and SER (29.7) → performance trade-offs (29.8).

29.1 Random processes in communication

Why randomness is the right model

A message is unknown to the receiver in advance; if it were known, it would carry no information (Chapter 28). Noise is the superposition of an enormous number of microscopic events, the thermal agitation of electrons, the arrival of individual photons, the discharges of distant lightning, which cannot be predicted individually. Interference from other users, fading of a radio path and the jitter of a clock are random as well. A deterministic waveform model of any of these is useless; what can be modelled and measured are their statistical regularities: average values, power, spectra, and the probability of large excursions.

A **random process** $X(t)$ is a family of random variables indexed by time. Each experiment produces one **sample function** (one recorded noise waveform); the collection of all possible sample functions is the **ensemble**. At a fixed instant t_1, $X(t_1)$ is an ordinary random variable with a probability density $f_X(x;t_1)$.

Mean and autocorrelation

The two most useful descriptors are the **mean** and the **autocorrelation function**:

$$\mu_X(t) = E[X(t)], \qquad R_X(t_1, t_2) = E[X(t_1)X(t_2)].$$

The autocorrelation says how strongly the process at one instant resembles itself at another. For noise from a wide-band amplifier it falls to zero within nanoseconds; for slowly drifting baseline wander in an ECG it stays high for seconds.

Stationarity and ergodicity

A process is **wide-sense stationary (WSS)** if its mean is constant and its autocorrelation depends only on the time difference $\tau = t_2 - t_1$:

$$\mu_X(t) = \mu_X, \qquad R_X(t, t+\tau) = R_X(\tau).$$

Thermal noise in a receiver at constant temperature is stationary; the noise in a switched-mode power supply, which varies with the switching phase, is *cyclostationary* (its statistics repeat periodically); the noise floor of a radio channel over a whole day is not stationary at all. The value $R_X(0) = E[X^2]$ is the average power (in watts into 1 Ω, or in V^2), a fact used constantly.

In the laboratory there is only one sample function, recorded over a finite time. A process is **ergodic** (in the mean and autocorrelation) if time averages of a single sample function converge to the ensemble averages:

$$\lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2} x(t)\,dt = \mu_X, \qquad \lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2} x(t)x(t+\tau)\,dt = R_X(\tau).$$

Ergodicity is what justifies measuring noise with a voltmeter, an oscilloscope or a spectrum analyser. It is an assumption; it fails, for instance, for a process whose mean depends on a random but fixed initial condition, such as the DC offset of one particular amplifier drawn from a production batch.

Power spectral density and the Wiener–Khinchin theorem

For a WSS process the **power spectral density (PSD)** $S_X(f)$ is the Fourier transform of the autocorrelation:

$$S_X(f) = \int_{-\infty}^{\infty} R_X(\tau)\,e^{-j2\pi f\tau}\,d\tau, \qquad R_X(\tau) = \int_{-\infty}^{\infty} S_X(f)\,e^{j2\pi f\tau}\,df.$$

This is the **Wiener–Khinchin theorem**. $S_X(f)$ has units of $\mathrm{W/Hz}$ (or $\mathrm{V^2/Hz}$), is real, even and non-negative, and its integral is the total power $R_X(0)$. It tells how the noise power is distributed over frequency, which is what a spectrum analyser displays and what a filter acts upon. A random signal has no Fourier transform in the ordinary sense (its energy is infinite), so the PSD, not the amplitude spectrum, is the correct frequency-domain description of noise.

> **Status of the principle: Wiener–Khinchin theorem**
> *Derived.* A mathematical result for wide-sense stationary processes; it gives the PSD meaning as the Fourier transform of the autocorrelation. For non-stationary signals (speech, radar returns, fading channels) it applies only over intervals short enough to be treated as stationary, which motivates the short-time spectral methods of Chapter 24.

Random processes through LTI systems

When a WSS process with PSD $S_X(f)$ passes through a stable LTI system with frequency response $H(f)$, the output is WSS with

$$\mu_Y = \mu_X H(0), \qquad S_Y(f) = |H(f)|^2 S_X(f).$$

Phase does not matter for the output power, only the magnitude response. This one formula underlies every noise calculation in this chapter: the noise power at a filter output is the integral of $|H(f)|^2$ times the input PSD. Moreover, if the input is Gaussian, so is the output, because a linear system forms weighted sums of its input and any linear combination of jointly Gaussian variables is Gaussian.

> **Boundary.**
> The general theory of random processes, ensembles and stochastic integrals belongs to *The Mathematics Universe*. This chapter uses the second-order (mean, autocorrelation, PSD) description, which is all that linear noise analysis requires.

29.2 Gaussian noise

Why noise is Gaussian

The voltage across a resistor is the sum of tiny contributions from a vast number of electrons moving independently. By the **central limit theorem**, the sum of many independent small contributions tends to a Gaussian distribution, whatever the distribution of each. Thermal noise, shot noise at reasonably high current, and the aggregate of many weak interferers are all very nearly Gaussian. The Gaussian density with mean μ and variance σ^2 is

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\exp\!\left[-\frac{(x-\mu)^2}{2\sigma^2}\right].$$

For a zero-mean noise voltage, σ is the RMS value, and σ^2 is the average power in a 1 Ω reference.

The Q-function

Error rates are governed by the probability that Gaussian noise exceeds a threshold. The **Q-function** is the tail probability of a standard Gaussian:

$$Q(x) = \frac{1}{\sqrt{2\pi}}\int_x^\infty e^{-u^2/2}\,du = \tfrac{1}{2}\operatorname{erfc}\!\left(\frac{x}{\sqrt{2}}\right).$$

A zero-mean Gaussian of standard deviation σ exceeds a level a with probability $Q(a/\sigma)$. The tail falls extremely fast: for large x, $Q(x) \approx e^{-x^2/2}/(x\sqrt{2\pi})$, so each additional σ of margin reduces the probability by roughly an order of magnitude or more.

Table 29.1 — Values of the Gaussian tail probability.

x	$Q(x)$	Where this matters
1.0	0.159	Noise exceeds its RMS value about 16 % of the time in one direction
2.0	2.28×10^{-2}	
3.0	1.35×10^{-3}	Radar false-alarm probability 10^{-3} needs a threshold of 3.09σ
4.0	3.17×10^{-5}	
4.27	1.0×10^{-5}	BPSK at a BER of 10^{-5}
4.75	1.0×10^{-6}	Threshold for $P_{FA} = 10^{-6}$
6.0	9.9×10^{-10}	
7.0	1.3×10^{-12}	Serial links specified at BER 10^{-12}

The table explains the "crest factor" engineers use for Gaussian noise: although the peak of Gaussian noise is unbounded, a peak-to-RMS ratio of 4 to 5 is exceeded only a tiny fraction of the time, so an ADC or amplifier sized for about 4.5 times the RMS noise rarely clips on noise alone.

Narrowband noise and the Rayleigh envelope

After a bandpass filter of bandwidth B centred at $f_c \gg B$, Gaussian noise looks like a sinusoid at f_c whose amplitude and phase wander slowly, at a rate set by B. It can be written in **in-phase and quadrature** form:

$$n(t) = n_I(t)\cos 2\pi f_c t - n_Q(t)\sin 2\pi f_c t = r(t)\cos[2\pi f_c t + \phi(t)]\,.$$

The baseband components n_I and n_Q are independent, zero-mean Gaussian processes, each with the same variance σ^2 as $n(t)$ itself. The **envelope** $r = \sqrt{n_I^2 + n_Q^2}$ then has the **Rayleigh** density and the phase ϕ is uniform on $[0, 2\pi)$:

$$f_R(r) = \frac{r}{\sigma^2}\exp\!\left(-\frac{r^2}{2\sigma^2}\right), \qquad r \ge 0.$$

If a sinusoid of amplitude A is present as well, the envelope follows the **Rician** density, which approaches a Gaussian centred at A when $A \gg \sigma$. These two distributions govern every envelope (non-coherent) detector: AM receivers, radar detectors, the noncoherent FSK demodulators of Section 29.7, and, as Chapter 30 shows, the fading of radio signals, which is mathematically the same phenomenon. Figure 29.1 shows a Gaussian noise record, its histogram and the envelope distribution.

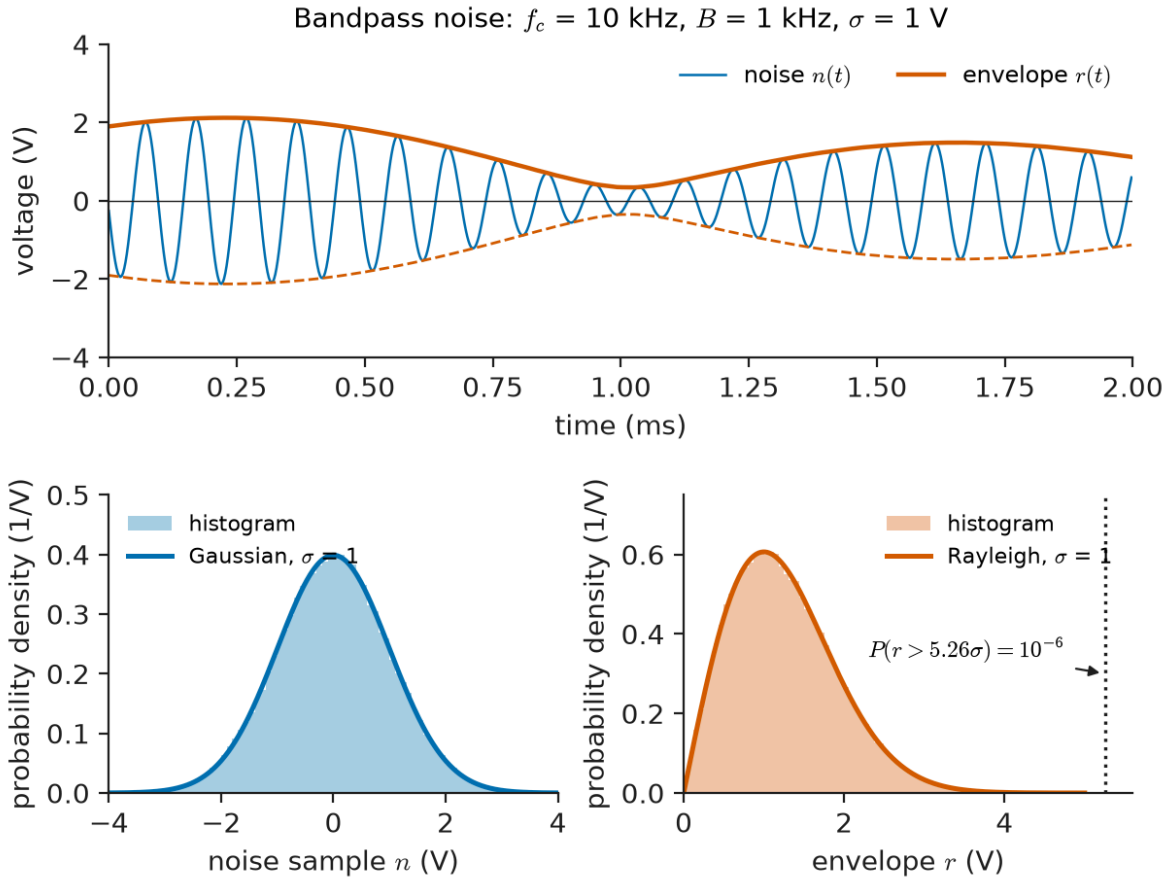

Figure 29.1 — Gaussian noise and its Rayleigh envelope.

> **Status of the principle: Gaussian model of receiver noise**
> *Approximate.* It follows from the central limit theorem applied to many independent microscopic contributions, and fits thermal and shot noise closely. It fails for impulsive noise (ignition, switching transients, power-line communication channels, lightning), for a few strong interferers, and for quantization error in a coarse ADC, whose tails are lighter or heavier than Gaussian.

29.3 White-noise models

Thermal noise

In 1928 Johnson measured, and Nyquist explained, the random voltage across any resistor in thermal equilibrium. A resistor R at absolute temperature T behaves as a noiseless resistor in series with a noise voltage source whose mean-square value in a bandwidth B is

$$\overline{v_n^2} = 4kTRB,$$

or, equivalently, a noiseless resistor in parallel with a noise current source of mean-square value $4kTB/R$, where $k = 1.381 \times 10^{-23}\ \mathrm{J/K}$ is Boltzmann's constant. The **available noise power**, delivered to a matched load, is independent of R:

$$P_n = kTB.$$

At the standard reference temperature $T_0 = 290\ \mathrm{K}$, $kT_0 = 4.00 \times 10^{-21}\ \mathrm{W/Hz}$, which is the most important number in receiver design:

$$kT_0 = -174\ \mathrm{dBm/Hz}.$$

Table 29.2 — Thermal noise in practical bandwidths ($T_0 = 290$ K).

Bandwidth	Example	Noise power kT_0B
1 Hz	Reference	-174 dBm
1 kHz	Narrowband telemetry	-144 dBm
25 kHz	Land-mobile FM voice channel	-130 dBm
200 kHz	GSM carrier	-121 dBm
1 MHz	—	-114 dBm
20 MHz	Wi-Fi or LTE carrier	-101 dBm
100 MHz	5G NR carrier (FR1)	-94 dBm

Noise voltages for representative sources:

Source	Conditions	RMS noise voltage $\sqrt{4kTRB}$
1 kΩ resistor	290 K, 20 kHz audio band	0.57 μV
50 Ω source	290 K, 1 MHz	0.89 μV
10 kΩ electrode–skin impedance	310 K, 1 kHz EEG/ECG band	0.41 μV

The last row is the floor that a biopotential amplifier cannot beat by any electronic design: EEG rhythms of 10–100 μV are well above it, but evoked potentials of 1 μV or less must be extracted by averaging hundreds of stimulus-locked responses, which reduces uncorrelated noise by the square root of the number of averages.

White noise as a model

A process with a flat PSD is called **white**, by analogy with white light. The standard model of receiver noise is **additive white Gaussian noise (AWGN)** with two-sided PSD

$$S_n(f) = \frac{N_0}{2}, \qquad -\infty < f < \infty,$$

so that the one-sided density is N_0 (W/Hz) and the noise power in a bandwidth B is $N_0 B$. The autocorrelation is an impulse,

$$R_n(\tau) = \frac{N_0}{2}\,\delta(\tau),$$

which means samples at any two distinct instants are uncorrelated. For thermal noise referred to a matched input, $N_0 = kT$.

White noise is an idealization: its total power $\int S_n\,df$ is infinite. Real thermal noise is flat only up to frequencies where hf approaches kT, about 6 THz at 290 K, where the quantum correction cuts it off. For every electronic system that is effectively infinite bandwidth, which is why the model is used without apology. What matters is that the noise is flat over the band of the system that receives it; outside that band, the receiver's own filters determine what gets through, so the shape of the noise there is irrelevant.

White noise through a filter: kT/C

Passing white noise through a filter produces **coloured** noise with PSD $|H(f)|^2 N_0/2$. A classic result follows for a resistor R in parallel with a capacitor C: the noise voltage across the capacitor has mean-square value

$$\overline{v_C^2} = \frac{kT}{C},$$

independent of R. A larger resistor produces more noise density but a narrower bandwidth, and the two effects cancel exactly. At 300 K, a 1 pF sampling capacitor carries 64 μV RMS of noise and a 10 pF capacitor 20 μV. This "kT/C noise" sets the minimum capacitor size in the sample-and-hold of a high-resolution ADC and in CMOS image-sensor pixels, independent of how clever the switch design is.

Other noise mechanisms

- **Shot noise.** Current made of discrete charges crossing a barrier (a pn junction, a photodiode, a vacuum tube) fluctuates with mean-square value $\overline{i_n^2} = 2qI_{DC}B$. It is white up to very high frequencies and Gaussian at practical currents. A 1 mA diode current in 1 MHz carries 17.9 nA RMS of shot noise; a photodiode current of 10 µA in 1 GHz carries 57 nA. Shot noise sets the quantum limit of optical receivers (Chapter 31).
- **Flicker ($1/f$) noise.** A PSD rising as $1/f^{\alpha}$ (with $\alpha \approx 1$) at low frequencies, found in all active devices and in carbon resistors. Below the **corner frequency**, from a few hertz for precision bipolar op-amps to tens of kilohertz or more for small MOSFETs, it dominates white noise. It limits DC-coupled instrumentation such as strain-gauge bridges and thermocouple amplifiers, and is the reason chopper-stabilized amplifiers exist.
- **Quantization noise.** The error of an ideal N-bit ADC with a full-scale sine input behaves, under suitable conditions, like white noise with an SNR of $6.02N + 1.76$ dB (74 dB for 12 bits); see Chapter 21.
- **External noise.** Atmospheric noise (dominant below about 30 MHz), man-made noise from vehicles and switching equipment (strong in urban areas up to several hundred megahertz), galactic noise and the sky temperature seen by an antenna (Section 29.5).

> **Status of the principle: Johnson–Nyquist thermal noise, $\overline{v_n^2} = 4kTRB$**
> *Fundamental.* Follows from thermodynamic equilibrium (the fluctuation–dissipation theorem) and holds for any dissipative element at temperature T. It is white only for $hf \ll kT$, and it applies to the resistive part of an impedance; ideal reactances generate no noise.

29.4 Noise bandwidth

Definition

Filters do not have rectangular responses, so "the noise in bandwidth B" needs a precise meaning. The **equivalent noise bandwidth** B_N of a filter is the width of a rectangular filter, with the same peak power gain $|H_{\max}|^2$, that passes the same white-noise power:

$$B_N = \frac{1}{|H_{\max}|^2} \int_0^{\infty} |H(f)|^2 \, df.$$

The output noise power from white noise of one-sided density N_0 is then exactly $N_0 |H_{\max}|^2 B_N$. The noise bandwidth is always larger than the -3 dB bandwidth, because real skirts pass extra noise.

First-order RC filter

For $H(f) = 1/(1 + jf/f_c)$, with $f_c = 1/(2\pi RC)$,

$$B_N = \int_0^{\infty} \frac{df}{1 + (f/f_c)^2} = \frac{\pi}{2} f_c = \frac{1}{4RC}.$$

The noise bandwidth is 57 % wider than the 3 dB bandwidth (Figure 29.2). Steeper filters approach their 3 dB bandwidth: the ratio B_N/f_c is 1.571 for first-order, 1.111 for second-order, 1.047 for third-order, 1.026 for fourth-order and 1.006 for eighth-order Butterworth low-pass filters. For a bandpass response, the same ratios apply to the half-bandwidth on each side of centre.

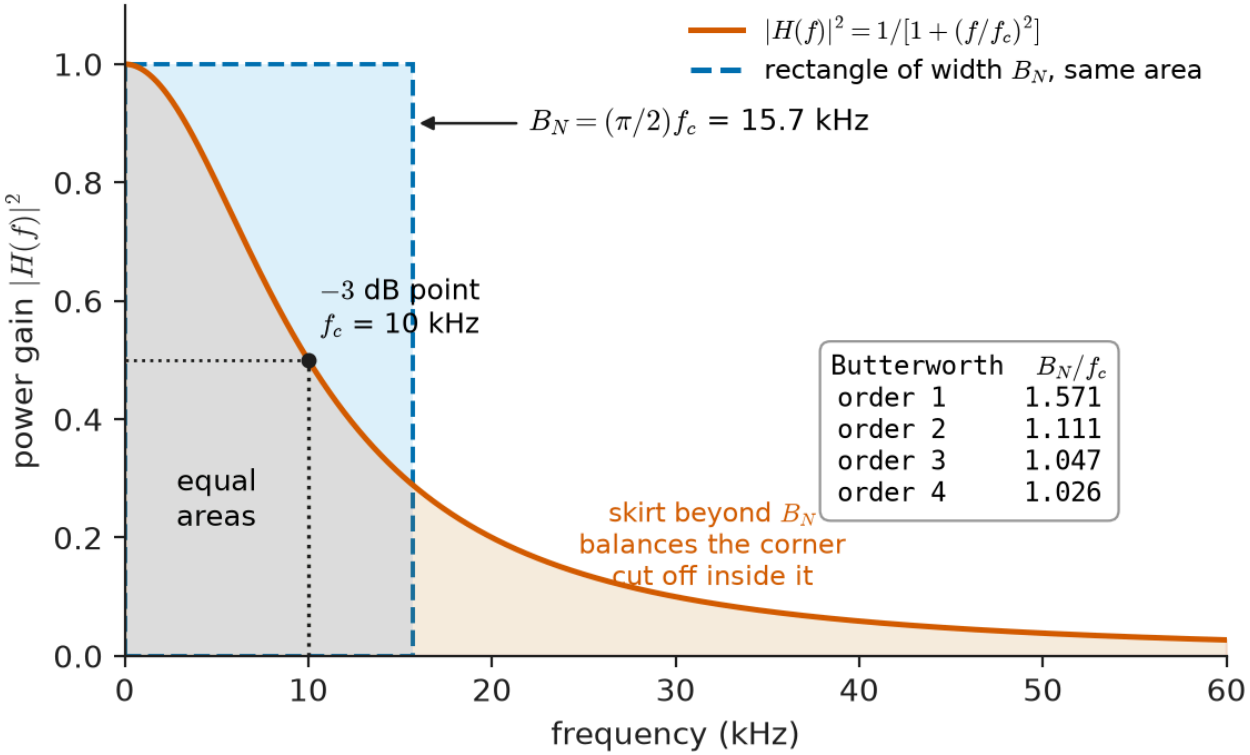

Figure 29.2 — Equivalent noise bandwidth of a first-order RC low-pass filter.

Integrators and averaging

A detector that averages its input over a time T (a moving-average or integrate-and-dump filter) has $|H(f)| = |\mathrm{sinc}(fT)|$ and a noise bandwidth $B_N = 1/(2T)$. Averaging for 1 s therefore gives a noise bandwidth of 0.5 Hz: the basis of lock-in amplifiers that recover nanovolt signals buried in noise, and of the long coherent integrations in GPS and radio astronomy. Halving the noise bandwidth halves the noise power, gaining 3 dB, at the cost of doubling the measurement time.

In communication receivers, the noise bandwidth of the receive filter is what turns a noise density N_0 into a noise power N. For a matched filter with a root-raised-cosine response at symbol rate R_s, $B_N = R_s$ exactly, independent of the roll-off factor: this is why E_s/N_0 and the SNR in the "symbol-rate bandwidth" are numerically equal.

Worked example — Noise of an instrumentation front end. A vibration-monitoring accelerometer on a gas turbine has a 1 kΩ source resistance and feeds an amplifier followed by a first-order anti-aliasing filter with $f_c = 10$ kHz. The amplifier's own input voltage noise is $4\ \mathrm{nV}/\sqrt{\mathrm{Hz}}$ (flat above its 1/f corner). The noise bandwidth is

$$B_N = \frac{\pi}{2} \times 10\ \mathrm{kHz} = 15.7\ \mathrm{kHz}.$$

Thermal noise of the source: $\sqrt{4kTR} = \sqrt{4(1.381\times10^{-23})(290)(1000)} = 4.00\ \text{nV}/\sqrt{\text{Hz}}$, the same density as the amplifier. Uncorrelated noises add in power, so the total input-referred density is $\sqrt{4.00^2 + 4^2} = 5.66\ \text{nV}/\sqrt{\text{Hz}}$ and the RMS input noise is

$$v_n = 5.66\ \text{nV}/\sqrt{\text{Hz}} \times \sqrt{15\,700\ \text{Hz}} = 0.71\ \mu\text{V}.$$

Using the 3 dB bandwidth instead of B_N would have underestimated this as 0.57 µV. With a 100 mV/g sensor, the noise floor corresponds to about 7 µg RMS, comfortably below the vibration levels of interest; if it were not, the remedy would be a narrower filter, not a quieter amplifier, since the amplifier contributes only half the noise power.

29.5 Noise figure

Noise factor and noise figure

Every amplifier, mixer and cable adds noise to what it receives. The **noise factor** F measures how much a two-port degrades the signal-to-noise ratio when its input is driven from a source at the reference temperature $T_0 = 290\ \text{K}$:

$$F = \left.\frac{(S/N)_{\text{in}}}{(S/N)_{\text{out}}}\right|_{T_{\text{source}} = 290\ \text{K}}, \qquad \text{NF} = 10\log_{10} F\ \ \text{dB}.$$

A noiseless device has $F = 1$ (NF $= 0$ dB). Equivalently, if the device has available gain G and adds noise power N_a at its output, $F = 1 + N_a/(GkT_0B)$.

A passive lossy element at temperature T_0, such as a cable or filter with loss L (power ratio, $L > 1$), has $F = L$: a 2 dB cable loss before an amplifier is a 2 dB noise figure.

Noise temperature

For low-noise systems the noise factor is an awkward scale (0.5 dB and 0.6 dB look nearly equal but differ by 8 K). The **effective input noise temperature** T_e is the temperature of a source resistor that would generate the device's added noise:

$$T_e = (F - 1)T_0, \qquad F = 1 + \frac{T_e}{T_0}.$$

Table 29.3 — Typical noise figures and temperatures.

Device	NF (dB)	T_e (K)
Cryogenic HEMT LNA (radio astronomy, deep space)	0.05–0.1	3–7
Satellite-TV LNB at 11–12 GHz	0.5–0.8	35–58
GaAs/GaN pHEMT LNA, 1–6 GHz	0.5–1.2	35–92
Smartphone receiver front end (after switches and filters)	3–6	290–865
Wi-Fi access point receiver	4–7	440–1160
Passive diode mixer	6–9 (≈ conversion loss)	865–2010
Spectrum analyser without preamplifier	20–30	2.9×10^{4}–2.9×10^{5}

Cascades: the Friis formula

For stages in cascade, with noise factors $F_1, F_2, \ldots$ and available power gains $G_1, G_2, \ldots$ (as ratios), the overall noise factor is

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots, \qquad T_e = T_{e1} + \frac{T_{e2}}{G_1} + \frac{T_{e3}}{G_1 G_2} + \cdots .$$

The noise of each stage is divided by the total gain ahead of it. The first stage therefore dominates, provided its gain is reasonable, which is the whole design principle of receivers: put a **low-noise amplifier (LNA)** with enough gain as close to the antenna as possible, and avoid loss in front of it.

> **Status of the principle: Friis noise cascade formula**
> *Derived*. Follows from adding independent noise powers referred to the input of a chain of linear stages, each specified by available gain and noise factor. It assumes linear operation, uncorrelated stage noise, and gains defined for the actual source and load impedances; mismatch between stages changes both G and F.

Worked example — Receiver noise figure and sensitivity. A GSM-band base-station receiver consists of an LNA (NF 1.0 dB, gain 15 dB), a passive mixer (NF 8 dB, gain -7 dB) and an IF amplifier (NF 4 dB, gain 30 dB), as in Figure 29.3. As ratios: $F_1 = 1.259$, $G_1 = 31.62$; $F_2 = 6.310$, $G_2 = 0.1995$; $F_3 = 2.512$. Then

$$F = 1.259 + \frac{6.310 - 1}{31.62} + \frac{2.512 - 1}{31.62 \times 0.1995} = 1.259 + 0.168 + 0.240 = 1.666,$$

so NF $= 2.22$ dB. Placing the mixer first instead gives $F = 7.85$, NF $= 8.95$ dB: 6.7 dB worse for the same components. Inserting a 2 dB lossy cable between the antenna and the LNA (as when the LNA is at the bottom of the mast instead of the top) adds exactly 2 dB, giving 4.22 dB; this is why tower-mounted amplifiers exist.

The **sensitivity** is the input level that produces the minimum SNR the demodulator needs. In dBm,

$$P_{\text{sens}} = -174 + 10\log_{10} B + \text{NF} + \text{SNR}_{\text{min}}.$$

For a 200 kHz GSM carrier, NF = 2.22 dB and a required SNR of 10 dB: $P_{\text{sens}} = -174 + 53.0 + 2.2 + 10 = -108.8$ dBm. Each decibel of noise figure saved is a decibel of sensitivity gained, worth about 12 % of range in free space and more in terms of cell coverage area.

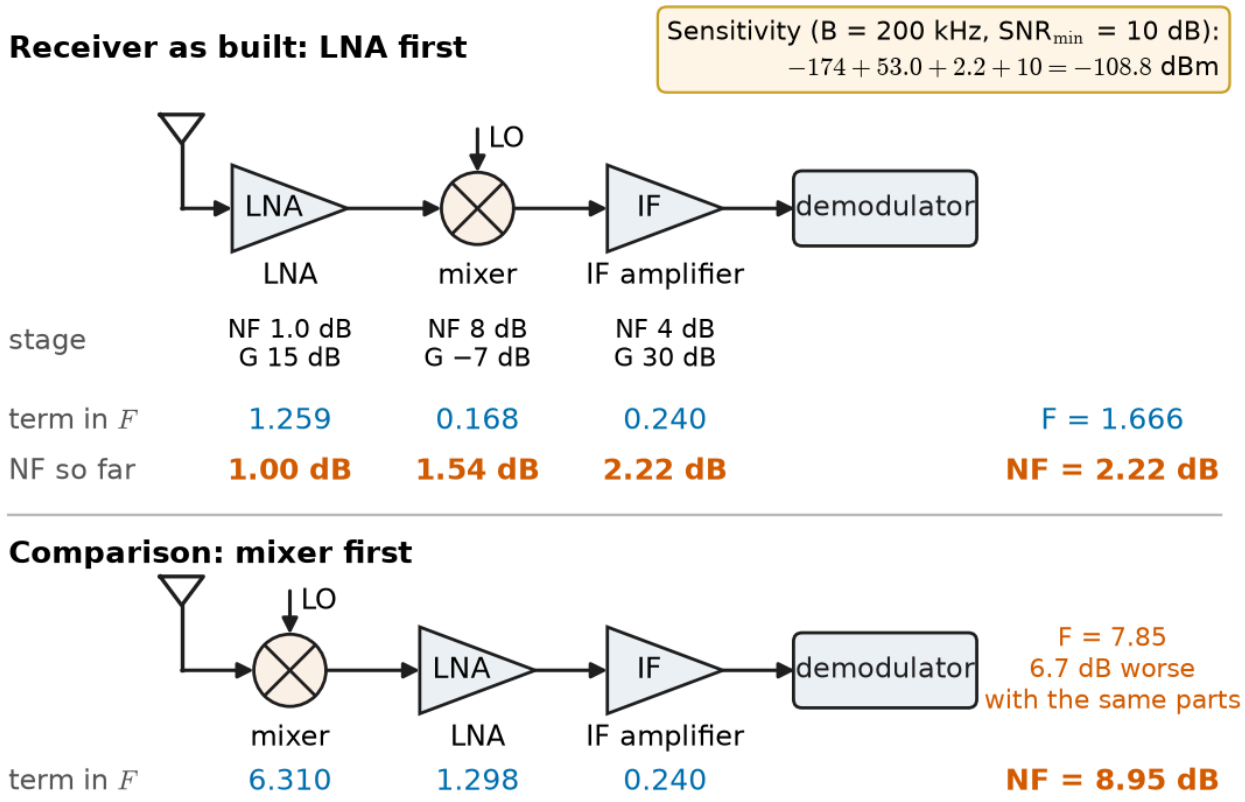

Figure 29.3 — Noise-figure budget of a three-stage receiver.

System noise temperature and G/T

At microwave frequencies an antenna pointed at the cold sky picks up little noise: its **antenna temperature** T_A may be 20–60 K for a satellite ground antenna at high elevation, rising toward 290 K when it looks at the ground or into rain. The **system noise temperature** is $T_{\text{sys}} = T_A + T_e$, and the noise density is $N_0 = kT_{\text{sys}}$. A satellite ground station is rated by its **figure of merit** G/T, the antenna gain divided by the system temperature.

For example, a 60 cm satellite-TV dish (gain about 36 dBi at 12 GHz) with $T_A = 50$ K and an LNB of NF 0.7 dB ($T_e = 50.7$ K) has $T_{\text{sys}} = 100.7$ K and $G/T = 36 - 10\log_{10} 100.7 = 16.0$ dB/K. If the LNB were replaced by one with NF 2 dB ($T_e = 170$ K), T_{sys} would rise to 220 K and G/T would fall by 3.4 dB, roughly halving the margin against rain fade. For cold-sky receivers the noise temperature, not the noise figure, is the natural language.

> **Status of the principle: Noise factor referred to $T_0 = 290$ K**
> *Approximate.* The noise factor is defined with a 290 K source; so it predicts the SNR degradation exactly only when the actual source is at 290 K. For a cold antenna the degradation is larger than NF suggests (an amplifier with $T_e = 50$ K doubles the noise of a 50 K sky, a 3 dB loss, although its NF is only 0.7 dB); use $T_{\text{sys}} = T_A + T_e$ instead.

29.6 Detection in noise

The decision problem

A receiver observes a noisy signal and must decide which of several hypotheses is true: was a 0 or a 1 sent; is there a target in this range cell or only noise; is this deflection in an ECG a QRS complex or a muscle artefact? In the simplest case there are two hypotheses, and after the receiver's front-end processing (Chapter 27's matched filter) the evidence is reduced to one number y:

$$H_0 : \; y = n, \qquad H_1 : \; y = s + n,$$

with n Gaussian of variance σ^2. The two conditional densities $f(y \mid H_0)$ and $f(y \mid H_1)$ are Gaussian curves centred at 0 and s (Figure 29.4). Any decision rule amounts to choosing a threshold γ: decide H_1 if $y > \gamma$.

Likelihood-ratio tests

All optimal detectors compare the **likelihood ratio** with a threshold:

$$\Lambda(y) = \frac{f(y \mid H_1)}{f(y \mid H_0)} \underset{H_0}{\overset{H_1}{\gtrless}} \eta.$$

Different design criteria differ only in the choice of η:

- **Maximum a posteriori (MAP)**, which minimizes the probability of error, uses $\eta = P(H_0)/P(H_1)$.
- **Maximum likelihood (ML)** uses $\eta = 1$; it equals MAP for equiprobable hypotheses, the normal case for data bits after scrambling or source coding.
- **Bayes** detection with unequal costs of the two errors scales η by the cost ratio.
- **Neyman–Pearson** detection, used when prior probabilities are meaningless (a radar does not know how likely a target is), maximizes the detection probability subject to a fixed false-alarm probability.

For Gaussian noise the log-likelihood ratio is linear in y, so every one of these rules is a threshold on the matched-filter output; only the threshold changes.

> **Status of the principle: Likelihood-ratio detection (Neyman–Pearson lemma)**
> *Derived*. The likelihood-ratio test is optimal for each of these criteria, a theorem of statistical decision theory. Its practical performance depends on knowing the noise statistics; if the noise level drifts, a fixed threshold no longer gives the designed false-alarm rate, which is why radars use constant-false-alarm-rate (CFAR) processors that estimate the noise from neighbouring cells.

The matched filter as the optimum front end

For a known pulse $s(t)$ of energy E in white noise of density $N_0/2$, the matched filter (impulse response $s(T-t)$) maximizes the output SNR at the sampling instant, giving

$$\left(\frac{S}{N}\right)_{\max} = \frac{2E}{N_0},$$

independent of pulse shape (Chapter 27). For coherent detection of a known signal, the normalized separation between the two hypotheses, the **deflection**, is therefore $d = \sqrt{2E/N_0}$. Everything in the rest of this chapter depends on the signal only through its energy, which is why pulse shape can be chosen for spectral reasons without any penalty in noise performance.

False alarm and detection probabilities

With the matched-filter output normalized so that the noise has unit variance, a threshold γ gives

$$P_{FA} = Q(\gamma), \qquad P_D = Q(\gamma - d).$$

Raising the threshold reduces false alarms and detections together; the trade-off is shown by the **receiver operating characteristic (ROC)**, a plot of P_D against P_{FA} for fixed d. Only more signal energy, less noise or longer integration moves the ROC curve towards the ideal corner.

For noncoherent detection, where the carrier phase is unknown (most radars, on–off keyed links), the detector forms the envelope. Under H_0 the envelope is Rayleigh, so $P_{FA} = \exp(-\gamma^2/2\sigma^2)$, and for $P_{FA} = 10^{-6}$ the threshold is $\gamma = \sqrt{2\ln 10^6}\,\sigma = 5.26\sigma$. Noncoherent detection costs typically 0.5–1 dB of SNR relative to coherent detection at radar operating points.

Worked example — Radar detection threshold. A surveillance radar must keep false alarms rare: with about 10^6 range–azimuth cells examined per second, $P_{FA} = 10^{-6}$ gives about one false alarm per second, which a tracker can tolerate. For coherent detection, the threshold is $\gamma = Q^{-1}(10^{-6}) = 4.75$ (in units of the noise RMS). Then:

- With $2E/N_0 = 13$ dB, $d = \sqrt{19.95} = 4.47$, so $P_D = Q(4.75 - 4.47) = Q(0.29) = 0.39$.
- With $2E/N_0 = 15$ dB, $d = 5.62$, so $P_D = Q(-0.87) = 0.81$.
- To reach $P_D = 0.9$, $d = 4.75 + 1.28 = 6.03$, so $2E/N_0 = 36.4$, or 15.6 dB, that is, $E/N_0 = 12.6$ dB.

If the operator relaxed the false-alarm requirement to 10^{-3}, the threshold would fall to 3.09 and the 13 dB echo would be detected with $P_D = Q(3.09 - 4.47) = 0.92$. The example shows how steep the trade is: 2 dB of extra SNR doubles the detection probability at a fixed false-alarm rate. It also shows why radar designers integrate many pulses: coherently integrating n pulses multiplies E by n.

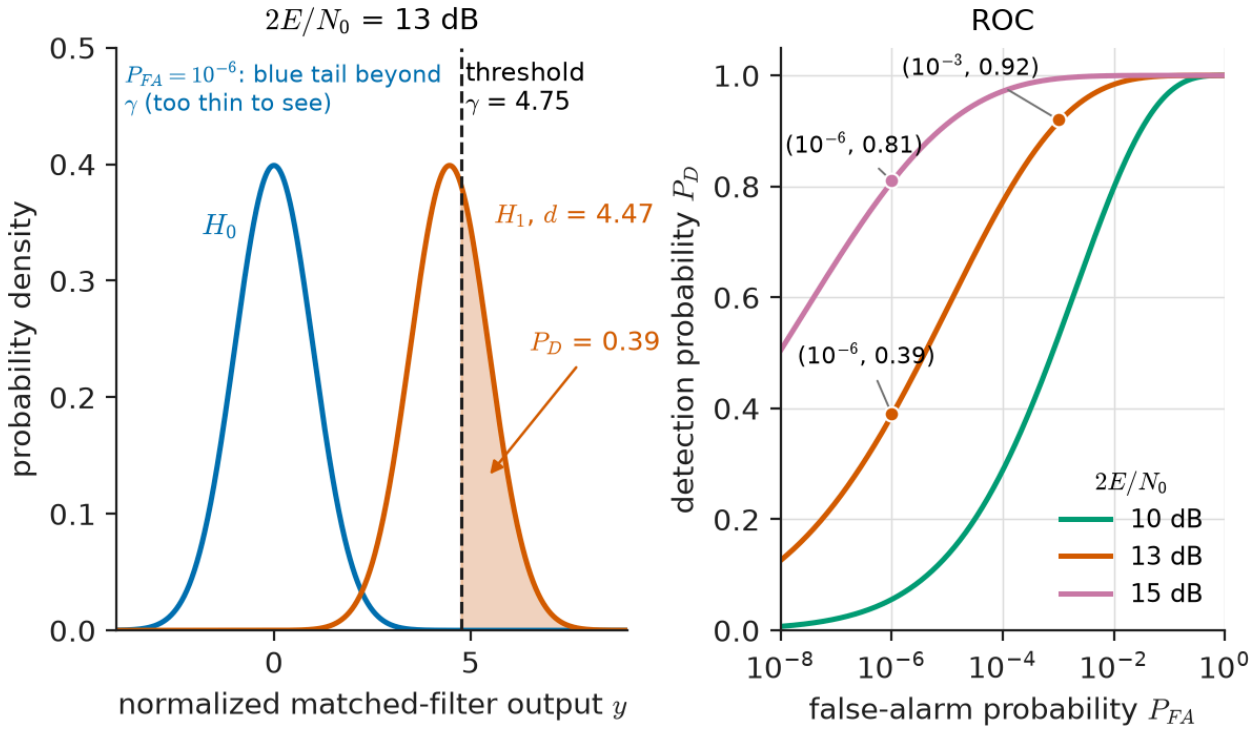

Figure 29.4 — Binary detection in Gaussian noise: threshold, false alarm and detection probabilities, and the ROC.

Detection beyond communications

The same structure appears throughout signal processing. An ECG QRS detector band-pass filters the signal to the 5–15 Hz range where QRS energy concentrates (an approximate matched filter), then thresholds it, with an adaptive threshold that tracks the noise level, a CFAR idea in medical guise. A GPS receiver correlates the received signal against a local replica of each satellite's code: the despreading gain of the 1023-chip code is $10\log_{10}1023 = 30$ dB. At a carrier-to-noise density of $C/N_0 = -130 - (-174 + 2) = 42$ dB-Hz (receiver NF 2 dB), integrating for 1 ms gives an SNR of $C/N_0 - 10\log_{10}1000 = 42 - 30 = 12$ dB in the 1 kHz post-correlation bandwidth (the despreading gain is what collapses the 1 MHz spread signal into that 1 kHz bandwidth), enough to detect the satellite; longer integration in weak-signal receivers adds roughly 3 dB per doubling. This is how the signal 21 dB below the noise at the antenna becomes detectable.

29.7 BER and SER

Definitions

The **bit error rate (BER)**, or bit error probability P_b, is the probability that a delivered bit differs from the transmitted bit. The **symbol error rate (SER)** P_s is the probability that a received M-ary symbol is decided wrongly. They are related through the mapping of bits to symbols. With **Gray coding**, adjacent constellation points differ in one bit, and since at reasonable SNR almost all symbol errors are to a nearest neighbour,

$$P_b \approx \frac{P_s}{\log_2 M}.$$

Performance is compared at equal **energy per bit** E_b relative to N_0, so that schemes carrying different numbers of bits per symbol are compared fairly: $E_s = E_b \log_2 M$. In terms of measured SNR in the noise bandwidth B_N, $\mathrm{SNR} = (E_b/N_0)(R_b/B_N)$.

Binary schemes

For coherent binary signalling with equiprobable symbols and ML detection, the error probability depends only on the Euclidean distance d_{12} between the two signal points: $P_b = Q\!\left(\sqrt{d_{12}^2/2N_0}\right)$. This gives the standard results:

$$P_{b,\mathrm{BPSK}} = Q\!\left(\sqrt{\frac{2E_b}{N_0}}\right), \qquad P_{b,\text{coh. BFSK}} = Q\!\left(\sqrt{\frac{E_b}{N_0}}\right).$$

Antipodal BPSK has $d_{12}^2 = 4E_b$; orthogonal FSK has $d_{12}^2 = 2E_b$, a 3 dB penalty. On–off keying with the same *average* energy also performs like orthogonal signalling. Noncoherent schemes, which need no carrier-phase reference, give exponential rather than Q-function forms:

$$P_{b,\mathrm{DPSK}} = \tfrac{1}{2} e^{-E_b/N_0}, \qquad P_{b,\text{nc. BFSK}} = \tfrac{1}{2} e^{-E_b/2N_0}.$$

M-ary schemes

QPSK is two independent BPSK channels in quadrature, so its *bit* error rate equals BPSK's at the same E_b/N_0 while using half the bandwidth; its symbol error rate is $P_s = 1 - (1 - P_b)^2 \approx 2P_b$.

For **M-PSK** with $M \geq 4$, the nearest-neighbour approximation gives

$$P_s \approx 2Q\!\left(\sqrt{\frac{2E_s}{N_0}}\sin\frac{\pi}{M}\right).$$

For square **M-QAM** with $\sqrt{M}$ levels on each axis,

$$P_s = 1 - \left[1 - 2\left(1 - \frac{1}{\sqrt{M}}\right) Q\!\left(\sqrt{\frac{3E_s}{(M-1)N_0}}\right)\right]^2.$$

Every additional bit per symbol beyond QPSK costs roughly 2–3 dB more E_b/N_0 for square QAM and 3.5–5 dB for PSK, because the points are packed closer together for the same average energy.

Table 29.4 — Required E_b/N_0 (dB) for given bit error rates in AWGN (uncoded, Gray-coded).

Scheme	b/s/Hz (Nyquist)	$P_b = 10^{-3}$	$P_b = 10^{-5}$	$P_b = 10^{-6}$
BPSK	1	6.8	9.6	10.5
QPSK	2	6.8	9.6	10.5
DPSK (binary)	1	7.9	10.3	11.2
Coherent BFSK	0.5–1	9.8	12.6	13.5
Noncoherent BFSK	0.5	10.9	13.4	14.2
8-PSK	3	10.0	13.0	14.0
16-QAM	4	10.5	13.4	14.4
64-QAM	6	14.8	17.8	18.8
256-QAM	8	19.4	22.5	23.5

Figure 29.5 plots the corresponding curves. Their characteristic "waterfall" shape is the Q-function: at high SNR, each decibel buys about an order of magnitude in BER.

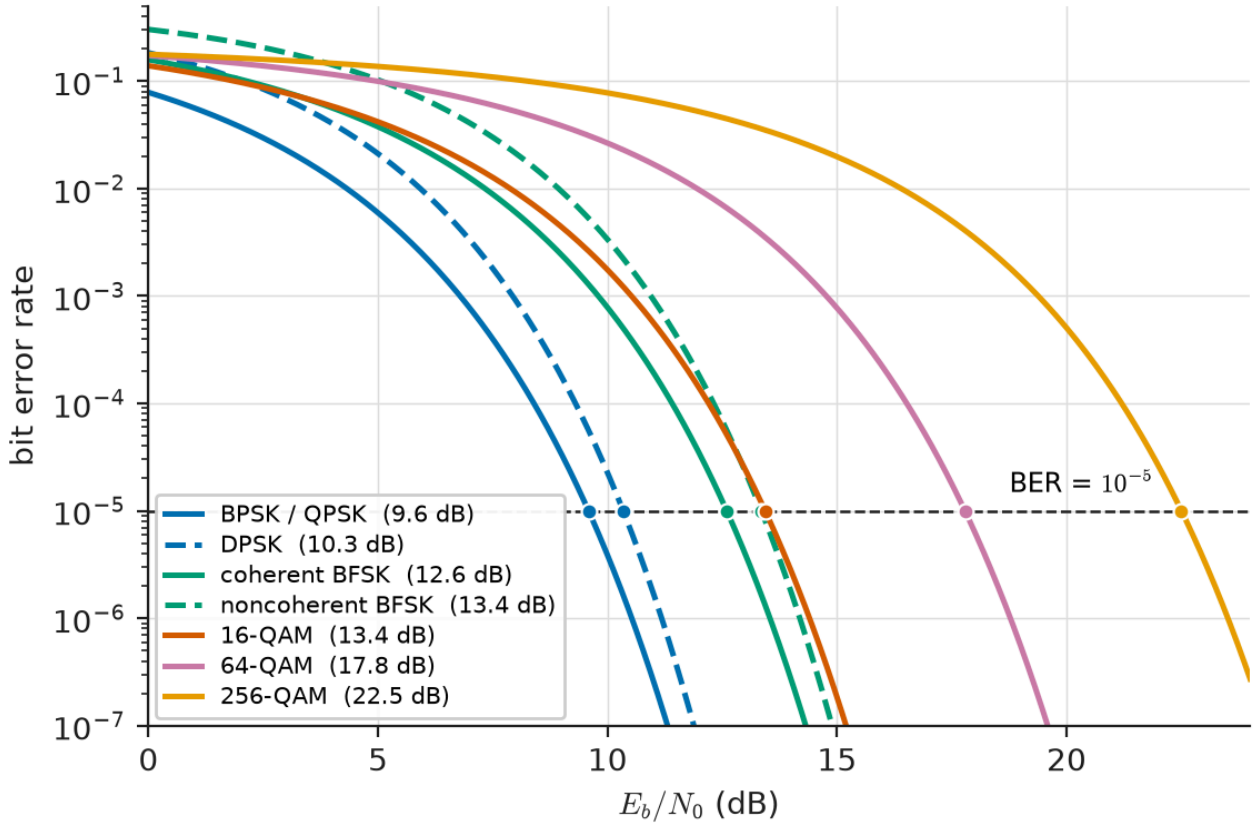

Figure 29.5 — Bit error rate versus Eb/N0 for common modulations in AWGN.

> **Status of the principle: Error-rate formulas for AWGN**
> *Derived*. Exact or tightly approximate consequences of Gaussian noise statistics and ML detection with perfect synchronization. Real receivers lose typically 0.5–2 dB to imperfect carrier and timing recovery, phase noise, amplifier non-linearity and finite ADC resolution, which are quoted as "implementation loss" in link budgets.

Worked example — 16-QAM on a microwave backhaul link. A point-to-point microwave link uses 16-QAM at 50 Mbaud, giving 200 Mb/s before coding. The demodulator measures $E_s/N_0 = 18$ dB, so $E_s/N_0 = 63.1$ and

$$\sqrt{\frac{3E_s}{15N_0}} = \sqrt{12.62} = 3.55, \qquad Q(3.55) = 1.92 \times 10^{-4},$$

$$P_s = 1 - \left[1 - 2(0.75)(1.92 \times 10^{-4})\right]^2 = 5.73 \times 10^{-4}, \qquad P_b \approx \frac{P_s}{4} = 1.43 \times 10^{-4}.$$

The corresponding E_b/N_0 is $18 - 10\log_{10} 4 = 12.0$ dB. At 200 Mb/s that is about 29,000 bit errors per second before error correction, but well within the correcting power of the link's LDPC code, which for a rate-0.9 code tolerates a raw BER of a few times 10^{-3}. If rain fade reduced E_s/N_0 by 2 dB to 16 dB, the raw BER would rise to 1.8×10^{-3}, still correctable; at around 14 dB the code fails, and adaptive modulation would step down to QPSK. This is why modern links quote throughput as a function of SNR rather than a single BER.

The union bound and error floors

For constellations without a closed-form result, the **union bound** sums pairwise error probabilities:

$$P_s \le \frac{1}{M}\sum_{i}\sum_{j \neq i} Q\!\left(\frac{d_{ij}}{\sqrt{2N_0}}\right),$$

which at high SNR is dominated by the nearest neighbours, giving the familiar form $P_s \approx N_{\min} Q\big(d_{\min}/\sqrt{2N_0}\big)$, where $N_{\min}$ is the average number of nearest neighbours. When a non-random impairment such as phase noise or intersymbol interference is present, the BER curve flattens into an **error floor** that no increase of transmit power removes; the cure is to remove the impairment, not to add power.

29.8 Performance trade-offs

The design variables

A communication link is specified by a handful of quantities that cannot all be optimized together:

- **Power** (or E_b/N_0): limited by battery life, regulations, amplifier cost, interference to others and, in satellites, solar-panel area.
- **Bandwidth**: licensed spectrum is expensive and finite; for a given data rate, the spectral efficiency $\eta = R_b/B$ is the measure.

- **Error rate**: the reliability the application needs, from 10^{-3} for voice to 10^{-12} or better for computer interconnects.
- **Complexity and power consumption of the receiver**: coherent detection, equalizers and iterative decoders consume energy and silicon.
- **Latency**: interleavers, long codes and retransmissions add delay that voice, control loops and protection signalling cannot tolerate.

Power-limited and bandwidth-limited regimes

The Shannon bound $E_b/N_0 \geq (2^{\eta}-1)/\eta$ (Chapter 28) divides systems into two families. **Power-limited** systems (deep space, GNSS, low-power IoT, sensor networks) have bandwidth to spare and use low-order modulation, low code rates, spread spectrum or orthogonal signalling: M-ary FSK even improves with increasing M in E_b/N_0 terms, at the cost of bandwidth. **Bandwidth-limited** systems (cellular, cable modems, microwave backhaul, Wi-Fi near the access point) have SNR to spare and use dense constellations with high-rate codes.

Table 29.5 — Typical operating points of real systems.

System	Modulation and coding	Spectral efficiency (b/s/Hz)	Operating SNR or E_b/N_0	Regime
Deep-space telemetry	BPSK, turbo or LDPC, $R \approx 1/6$–$1/2$	0.1–0.5	$E_b/N_0 \approx 0.5$–2 dB	Power-limited
GPS L1 C/A	BPSK, DSSS 1.023 Mchip/s, 50 b/s	2.5×10^{-5}	$C/N_0 \approx 35$–50 dB-Hz	Extremely power-limited
LoRa (IoT)	Chirp spread spectrum	0.01–0.3	SNR down to -20 dB	Power-limited
DVB-S2 satellite TV	QPSK to 32APSK, LDPC	1–4.5	SNR about 1–16 dB	Intermediate
LTE / 5G at cell edge	QPSK, low-rate LDPC/turbo	0.1–1	SINR -5 to 5 dB	Interference-limited
LTE / 5G at cell centre	64- to 256-QAM, high-rate code	4–7.5 per layer	SINR 20–30 dB	Bandwidth-limited
Microwave backhaul	Up to 4096-QAM, LDPC	8–11	SNR 25–40 dB	Bandwidth-limited
DOCSIS 3.1 cable	Up to 4096-QAM OFDM, LDPC	8–10	SNR 35–41 dB	Bandwidth-limited

Trading power for bandwidth and complexity

Three instructive comparisons follow from the formulas of Section 29.7 and Chapter 28.

1. **QPSK versus BPSK.** Same E_b/N_0 for the same BER, half the bandwidth: a free lunch made possible by using both quadrature dimensions. It is why BPSK survives only where simplicity or robustness to phase error matters.
2. **16-QAM versus QPSK.** Twice the spectral efficiency for about 3.8 dB more E_b/N_0 at $P_b = 10^{-5}$ (13.4 versus 9.6 dB). At high SNR this is a good trade; near the noise floor it is not.
3. **Coded versus uncoded.** A rate-1/2 LDPC code with QPSK needs about 1 dB for $P_b = 10^{-5}$ at 1 b/s/Hz, versus 9.6 dB for uncoded BPSK at the same spectral efficiency. The 8.6 dB saved is paid for in decoder complexity and latency, now cheap enough that almost no modern link runs uncoded.

Adding coherent detection saves roughly 1 dB over differential detection but requires carrier recovery; noncoherent FSK is still chosen for the cheapest radios (key fobs, many sub-GHz sensors) because a 3–4 dB penalty is acceptable when it saves a phase-locked loop. Equalization, diversity and adaptive modulation, discussed in Chapter 30, extend the same trade-offs to channels that are not simply additive noise.

Adaptive operation

Because the channel SNR changes, modern links select their operating point continuously. Wi-Fi rate adaptation, LTE and 5G link adaptation (driven by channel-quality reports from the terminal), DOCSIS profile management and adaptive microwave radios all pick the modulation and code rate that maximize throughput while keeping the block error rate near a target (typically 10 % before HARQ in cellular systems). The result is a throughput curve that tracks the Shannon bound with a gap of a few decibels across a wide range of SNR, rather than a single fixed operating point that is either wasteful or fragile.

> **Status of the principle: Power–bandwidth trade-off**
> *Derived*. The ultimate trade-off is set by the Shannon bound for the AWGN channel; the positions of practical schemes relative to it come from the error-rate formulas above. In fading and interference-limited channels the trade-off shifts, and diversity and multiple antennas become additional design variables.

What remains open

Open question	Where it is answered
How do multipath fading and Doppler change the error-rate curves derived here for AWGN?	Chapter 30 (fading, diversity)
How are noise figure, path loss and fade margins combined into a complete link budget?	Chapter 30 (link budgets)
What limits optical receivers, where shot noise and amplifier noise compete?	Chapter 31 (optical receivers)
How is noise figure measured at RF and microwave frequencies, and how do impedance mismatches alter it?	Chapter 32 (RF measurements)
How do quantization and finite word length add noise in digital receivers?	Chapters 21 and 23
How are the PSDs of non-stationary signals estimated in practice?	Chapters 22 and 24 (spectral estimation, time–frequency analysis)
How do coding and capacity move the operating points in Table 29.4 toward the Shannon bound?	Chapter 28
How are low-noise amplifiers and their transistors designed?	*Electronics & Computing*
What is the probability and random-process theory behind the results quoted here?	*The Mathematics Universe*

Next: Chapter 30 takes these noise and detection results onto real radio channels, where path loss, multipath fading, Doppler and interference from other users dominate, and shows how diversity, OFDM, MIMO and cellular design overcome them.

Chapter 30 — Wireless and Mobile Communications

Part VI · Communication Theory

A passenger on a train at 120 km/h is on a video call. The phone is 2 km from the nearest base station, behind the steel and glass of the carriage, and the radio signal arriving at it is not one wave but dozens: reflections from buildings, scattered from trees and lamp posts, each with its own delay and Doppler shift. Every few centimetres of travel the reflections add up differently, so the received power plunges by 20 or 30 dB several hundred times a second and recovers just as fast. The average signal reaching it is more than a million million times weaker than what the base station transmitted. And the base station is serving hundreds of other users on the same frequencies, while neighbouring cells reuse those frequencies a kilometre away. The call still works.

The AWGN channel of Chapters 28 and 29 is a gentle idealization of this. Real radio channels lose power with distance far faster than free space predicts, fade randomly in time and frequency, smear symbols over one another, and are shared by many users who interfere with one another. This chapter builds the models for these effects and the techniques (diversity, cellular reuse, multiple access, OFDM, MIMO) that turned them from obstacles into resources.

> **How does a radio signal lose power and fluctuate between transmitter and receiver, and how much margin must a link carry?**
> **How do cellular systems, OFDM and multiple antennas deliver high, reliable throughput to many moving users over such channels?**

Concept	Engineering destination
Radio propagation, Fresnel zones	Siting microwave links, antenna mast heights, indoor coverage
Path loss models (free space, two-ray, Hata)	Cell planning, Wi-Fi coverage, IoT gateway range, drone control links
Multipath, delay spread, Doppler	Equalizer and OFDM design, high-speed-rail and vehicle communications
Fading statistics, outage	Fade margins, reliability of SCADA radio and emergency services
Diversity, MRC	Base-station receive antennas, Wi-Fi, satellite radio, hearing-aid links
Cellular architecture, reuse, Erlang B	Network dimensioning, capacity planning
Multiple access (FDMA, TDMA, CDMA, OFDMA)	2G–5G standards, Wi-Fi, satellite networks
OFDM	LTE, 5G NR, Wi-Fi, DVB-T, DAB, DSL, power-line communication
MIMO	4G/5G throughput, Wi-Fi 5/6/7, massive-MIMO base stations, beamforming
Link budgets	Every radio link: cellular, satellite, telemetry, radar, IoT

Roadmap. Radio propagation (30.1) → path loss (30.2) → multipath (30.3) → fading (30.4) → diversity (30.5) → cellular architecture (30.6) → multiple access (30.7) → OFDM (30.8) → MIMO (30.9) → link budgets (30.10) → wireless-system performance (30.11).

30.1 Radio propagation

Mechanisms

Between two antennas, a radio wave reaches the receiver by four basic mechanisms, each the subject of field theory in *Engineering Physics* and modelled here by its consequences:

- **Free-space propagation** along a clear line of sight: power density falls with the square of distance because the same power spreads over a sphere of growing area.
- **Reflection** from surfaces large compared with the wavelength: the ground, building walls, the sea, the ionosphere at HF. The reflected wave has a magnitude and phase set by the surface's permittivity, conductivity and the angle of incidence.
- **Diffraction** around edges and over obstacles (rooftops, hilltops, the curvature of the earth), which lets signals reach shadowed regions with additional loss, more at higher frequency.
- **Scattering** from objects comparable to or smaller than the wavelength (foliage, lamp posts, rough surfaces, rain), which redistributes energy in many directions.

The relative importance of these mechanisms depends strongly on frequency. Table 30.1 shows the bands and their typical uses.

Table 30.1 — Radio bands and their propagation character.

Band	Frequency	Wavelength	Dominant propagation	Typical uses
LF/MF	30 kHz–3 MHz	10 km–100 m	Ground wave, diffraction over earth	Time signals, AM broadcasting, navigation
HF	3–30 MHz	100–10 m	Ionospheric reflection (sky wave)	Long-distance broadcasting, aviation, amateur radio
VHF	30–300 MHz	10–1 m	Line of sight with diffraction	FM, DAB, aircraft, land-mobile radio
UHF	300 MHz–3 GHz	1 m–10 cm	Line of sight, reflection, building penetration	Cellular (700 MHz–2.6 GHz), GNSS, Wi-Fi 2.4 GHz, TV
SHF (lower)	3–6 GHz	10–5 cm	Line of sight, strong reflection, weaker penetration	5G n77/n78, Wi-Fi 5/6 GHz, point-to-point links
SHF (upper)	6–30 GHz	5–1 cm	Line of sight, rain attenuation above ~10 GHz	Satellite TV (Ku), backhaul, radar
EHF (mm-wave)	30–300 GHz	10–1 mm	Short-range LOS, blocked by bodies, oxygen absorption at 60 GHz	5G FR2 (24–48 GHz), automotive radar 77 GHz, 60 GHz WiGig

Line of sight and Fresnel zones

A radio "line of sight" is not a ray but a volume. The **first Fresnel zone** is the ellipsoid around the direct path within which reflected or diffracted paths differ in length by less than half a wavelength. At a point that is d_1 from one end and d_2 from the other, its radius is

$$r_1 = \sqrt{\frac{\lambda d_1 d_2}{d_1 + d_2}}.$$

Obstacles intruding into this zone cause measurable extra loss even if the direct optical path is clear. The usual planning rule keeps at least 60 % of r_1 clear. For a 10 km link at 5.8 GHz ($\lambda = 5.17$ cm), the radius at mid-path is $\sqrt{0.0517 \times 5000 \times 5000/10\,000} = 11.4$ m, so 6.8 m of clearance is needed above trees and buildings at the midpoint, in addition to the earth's bulge. This, not optical visibility, sets microwave mast heights.

30.2 Path loss

Free space

The **Friis transmission equation** gives the received power between two antennas in free space, separated by d in the far field:

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2 .$$

The **free-space path loss** is the factor $(4\pi d/\lambda)^2$; in decibels, with d in kilometres and f in megahertz,

$$\mathrm{FSPL} = 20\log_{10} d_{\mathrm{km}} + 20\log_{10} f_{\mathrm{MHz}} + 32.44 \ \ \mathrm{dB}.$$

Loss grows by 20 dB per decade of distance and 20 dB per decade of frequency. The frequency dependence is a statement about antennas, not about space: an isotropic antenna's effective aperture $\lambda^2/4\pi$ shrinks with frequency. At fixed antenna *size*, gain rises as f^2 and the frequency dependence reverses, which is why high-frequency links use dishes and why 5G mm-wave equipment uses arrays with many elements.

Representative values: at 2.4 GHz, 1 m gives 40.0 dB and 100 m gives 80.0 dB; a 10 km link at 5.8 GHz has 127.7 dB; a geostationary satellite downlink at 12 GHz over 35,786 km has 205.1 dB.

> **Status of the principle: Friis free-space equation**
> *Derived*. Follows from energy conservation for a spherical wave and the reciprocity relation between antenna gain and effective aperture. It holds in the far field ($d \gg 2D^2/\lambda$ for an antenna of size D) with an unobstructed first Fresnel zone and no reflections; it is accurate for satellite and many microwave links, and optimistic for almost every terrestrial mobile link.

The two-ray ground-reflection model

Over flat ground the receiver sees the direct wave plus a ground-reflected wave with nearly 180° phase reversal at grazing incidence. When the path difference becomes small compared with λ, the two nearly cancel, and beyond the **breakpoint distance**

$$d_b \approx \frac{4 h_t h_r}{\lambda}$$

the received power falls as d^{-4} and becomes independent of frequency:

$$P_r \approx P_t G_t G_r \frac{h_t^2 h_r^2}{d^4}, \qquad L \approx 40\log_{10} d - 20\log_{10}(h_t h_r) \ \ \mathrm{dB}.$$

For a 30 m base-station antenna and a 1.5 m handset at 900 MHz, $d_b = 540$ m; at 5 km the two-ray loss is 114.9 dB, against 105.5 dB in free space. The model explains why raising an antenna extends coverage so effectively (+6 dB per doubling of either height) and why mobile path loss falls faster than free space.

Log-distance models and shadowing

Real environments are summarized by a **path-loss exponent** n and a reference loss at distance d_0:

$$L(d) = L(d_0) + 10n \log_{10} \frac{d}{d_0} + X_\sigma,$$

where X_σ is a zero-mean Gaussian random variable (in dB) with standard deviation σ: **log-normal shadowing**, caused by buildings, hills and foliage that differ from one location to another at the same distance. Its correlation distance is tens of metres outdoors.

Table 30.2 — Path-loss exponents and shadowing (typical values).

Environment	Exponent n	Shadowing σ (dB)
Free space	2	0
Rural macrocell, flat terrain	3–3.5	6–8
Urban macrocell	3.5–4	8–10
Dense urban, shadowed	4–5	8–12
Indoor, line of sight (corridor)	1.6–1.8	3–5
Indoor office, obstructed	3–4	6–10
Factory floor, obstructed	2–3	6–8

To guarantee coverage at the cell edge with 90 % probability despite shadowing, the link needs a **shadow margin** of 1.28σ: 10.3 dB for $\sigma = 8$ dB.

Empirical models: Okumura–Hata

For macrocell planning, path loss is predicted by models fitted to extensive measurements. The **Hata model**, a curve fit to Okumura's 1960s Tokyo measurements, gives the median urban loss for $150 \le f \le 1500$ MHz, base-station height $30 \le h_b \le 200$ m, mobile height $1 \le h_m \le 10$ m and $1 \le d \le 20$ km:

$$L_{\text{urban}} = 69.55 + 26.16 \log_{10} f - 13.82 \log_{10} h_b - a(h_m) + (44.9 - 6.55 \log_{10} h_b) \log_{10} d,$$

with f in MHz, d in km, and, for a small or medium city, $a(h_m) = (1.1 \log_{10} f - 0.7) h_m - (1.56 \log_{10} f - 0.8)$. The COST-231 extension covers 1500–2000 MHz; suburban and rural corrections subtract several decibels to tens of decibels. Figure 30.1 compares the three models.

> **Status of the principle: Okumura–Hata path-loss model**
> *Empirical.* A regression fit to measurements in a particular city and frequency range; it predicts the median loss with a typical error of several decibels and must not be extrapolated outside its stated parameter ranges, to microcells with antennas below rooftop height, or to indoor coverage.

Worked example — Macrocell coverage at 900 MHz. A base station radiates 43 dBm (20 W) into a 15 dBi antenna at 30 m height through 3 dB of feeder loss. A handset at 1.5 m height, 2 km away in an urban area, has a 0 dBi antenna and 3 dB body loss. The Hata correction is $a(1.5) = 0.016$ dB, the distance slope is $44.9 - 6.55\log_{10} 30 = 35.2$ dB/decade (exponent $n = 3.52$), and

$$L = 69.55 + 26.16(2.954) - 13.82(1.477) - 0.016 + 35.2(0.301) = 137.0 \text{ dB},$$

against 97.6 dB in free space at the same distance. The median received power is

$$P_r = 43 + 15 - 3 - 137.0 + 0 - 3 = -85.0 \text{ dBm}.$$

With $\sigma = 8$ dB shadowing, coverage at 90 % of edge locations requires planning for $-85.0 - 10.3 = -95.3$ dBm. A handset with NF 7 dB in a 10 MHz carrier has a noise floor of $-174 + 70 + 7 = -97$ dBm, leaving an SNR of only 1.7 dB at that 90th-percentile point, before multipath fading or interference are considered. At 5 km the median loss rises to 151.0 dB: the cell radius is limited, and to cover more area the operator must add sites, raise antennas or lower the frequency.

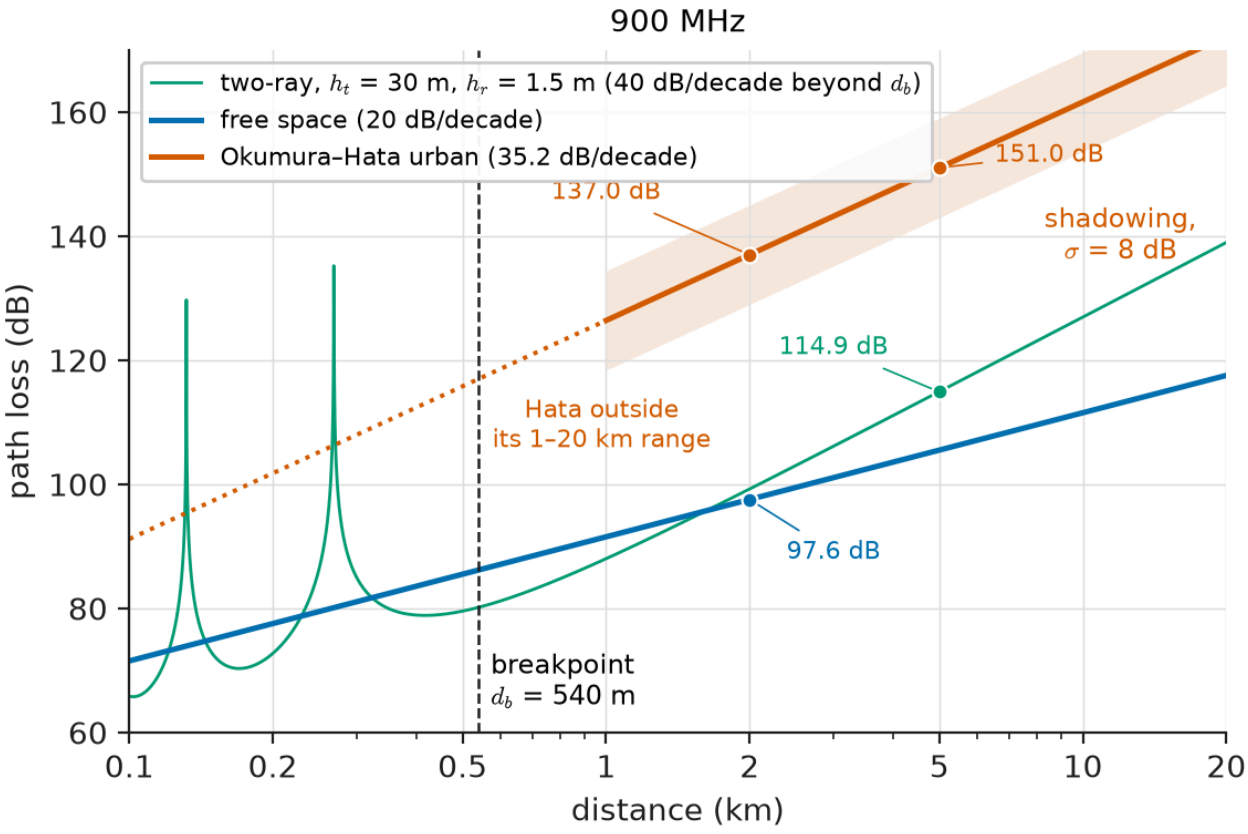

Figure 30.1 — Path loss versus distance at 900 MHz for free-space, two-ray and Okumura-Hata urban models.

30.3 Multipath

Delay spread and coherence bandwidth

In a typical mobile channel, the signal arrives over many paths of different lengths. The channel's impulse response is a set of delayed, attenuated and phase-rotated copies:

$$h(t,\tau) = \sum_i a_i(t)\, e^{-j\phi_i(t)}\, \delta(\tau - \tau_i).$$

The spread of the delays is measured by the **RMS delay spread** σ_τ, the standard deviation of the delays weighted by path power. Its frequency-domain counterpart is the **coherence bandwidth** B_c, the range of frequencies over which the channel response is strongly correlated. A common rule of thumb is

$$B_c \approx \frac{1}{5\sigma_\tau}$$

(for correlation 0.5; $1/(50\sigma_\tau)$ for correlation 0.9).

Table 30.3 — Typical RMS delay spreads.

Environment	σ_τ	Coherence bandwidth $1/(5\sigma_\tau)$
Indoor office or home	10–50 ns	4–20 MHz
Factory hall (metal)	50–200 ns	1–4 MHz
Urban microcell	0.1–0.5 µs	0.4–2 MHz
Urban macrocell	0.5–3 µs	70–400 kHz
Hilly terrain	3–10 µs	20–70 kHz

The comparison between signal bandwidth B_s (roughly the symbol rate) and B_c classifies the channel:

- **Flat fading** ($B_s \ll B_c$, symbol period $\gg \sigma_\tau$): all frequency components fade together; the channel is a single complex gain. A 200 kHz GSM carrier in a 1 µs macrocell is close to this boundary.
- **Frequency-selective fading** ($B_s > B_c$): different parts of the band fade differently and, in the time domain, symbols overlap (intersymbol interference, ISI). A single-carrier 20 MHz signal in a 1 µs channel spans about 20 symbols of ISI and needs a long equalizer, or OFDM (Section 30.8).

Doppler spread and coherence time

A receiver moving at speed v sees each path shifted in frequency by up to the **maximum Doppler shift**

$$f_D = \frac{v}{\lambda} = \frac{v f_c}{c}.$$

Paths arriving from different directions have different shifts between $-f_D$ and $+f_D$, so the received spectrum of a pure tone is spread over $2f_D$. For uniformly distributed arrival angles this gives the classic U-shaped **Jakes (Clarke) Doppler spectrum**. The time-domain counterpart is the **coherence time** T_c, over which the channel stays roughly constant; one common definition is

$$T_c \approx \frac{0.423}{f_D}.$$

At 120 km/h and 2 GHz, $f_D = 33.3 \times 2 \times 10^9/(3 \times 10^8) = 222$ Hz and $T_c \approx 1.9$ ms. A pedestrian at 1.4 m/s at 2.4 GHz gives $f_D = 11$ Hz and $T_c \approx 38$ ms; a high-speed train at 300 km/h at 2.6 GHz gives 723 Hz and 0.59 ms. If T_c is much longer than a symbol the fading is **slow** (the channel can be estimated and tracked); if comparable, it is **fast**, and channel estimates go stale before they can be used.

> **Status of the principle: Coherence bandwidth and coherence time rules**
> *Approximate.* $B_c \approx 1/(5\sigma_\tau)$ and $T_c \approx 0.423/f_D$ are rules of thumb tied to a chosen correlation level and to the shapes of the delay and Doppler profiles (exponential delay profile, uniform scattering in azimuth). They give order-of-magnitude guidance for system design, not exact values.

30.4 Fading

Rayleigh fading

When there is no dominant path and many scattered components of comparable size arrive with independent random phases, the in-phase and quadrature components of the received signal are sums of many random terms. By the central limit theorem they are Gaussian, exactly as for narrowband noise (Chapter 29), and the received envelope is **Rayleigh** distributed. The instantaneous received power γ (or SNR) is then exponentially distributed about its mean $\bar{\gamma}$:

$$f(\gamma) = \frac{1}{\bar{\gamma}} e^{-\gamma/\bar{\gamma}}, \qquad P(\gamma < \gamma_0) = 1 - e^{-\gamma_0/\bar{\gamma}}.$$

The probability that power falls 10 dB below the mean is $1 - e^{-0.1} = 9.5\ \%$; 20 dB below, about 1 %; 30 dB below, about 0.1 %. Deep fades are short: their duration scales with $1/f_D$, typically a few milliseconds for vehicles. Successive fades occur about every half wavelength, which is 7.5 cm at 2 GHz (Figure 30.2).

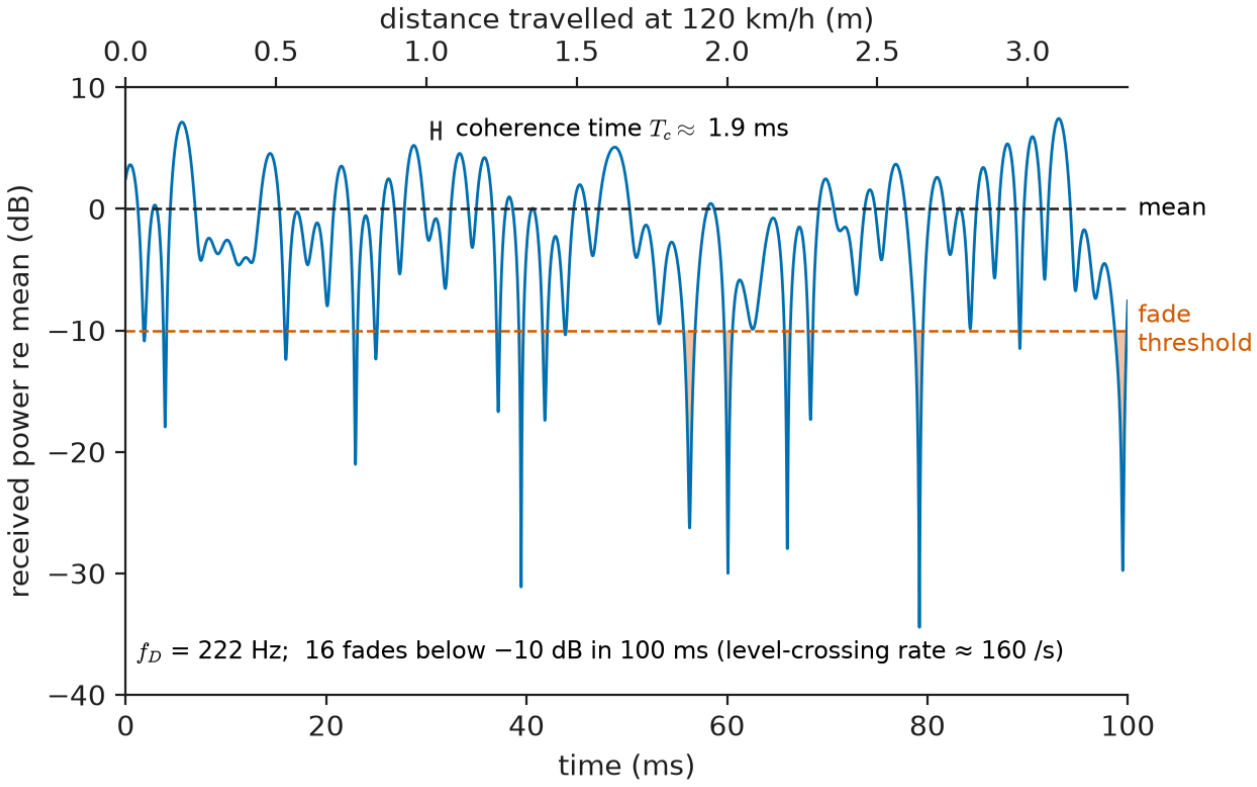

Figure 30.2 — Simulated Rayleigh fading envelope for a vehicle at 120 km/h at 2 GHz.

Rician fading

When a line-of-sight or dominant specular path is present, the envelope follows the **Rician** distribution, characterized by the **K-factor**, the ratio of power in the dominant component to the power in the scattered components. $K = 0$ is Rayleigh; $K \to \infty$ is no fading. Typical values are 3–10 dB for line-of-sight microcells and indoor links, higher for fixed links with good clearance. **Nakagami-m** fading is a flexible alternative fitted to measurements.

The three scales of variation

A received signal level therefore varies on three scales, which are modelled separately and combined in link planning:

1. **Path loss**, the median versus distance (Section 30.2), varying over kilometres.
2. **Shadowing**, log-normal around the median, varying over tens of metres.
3. **Multipath fading** (small-scale), Rayleigh or Rician, varying over fractions of a wavelength.

Error rate in fading

Fading changes the error-rate behaviour dramatically. Averaging the BPSK error probability $Q(\sqrt{2\gamma})$ over the exponential distribution of γ gives

$$\bar{P}_b = \frac{1}{2}\left(1 - \sqrt{\frac{\bar{\gamma}}{1+\bar{\gamma}}}\right) \approx \frac{1}{4\bar{\gamma}} \quad (\bar{\gamma} \gg 1).$$

In AWGN, BER falls *exponentially* with SNR; in Rayleigh fading only *inversely*. At a mean E_b/N_0 of 10 dB, AWGN gives 3.9×10^{-6} while Rayleigh fading gives 2.3×10^{-2}. To reach 10^{-3}, AWGN needs 6.8 dB and Rayleigh fading 24.0 dB: a penalty of 17 dB. The errors come almost entirely from the deep fades, not from the average conditions. Adding power is a very expensive cure; the efficient cures are diversity and coding with interleaving, which break the dependence on any single fade.

> **Status of the principle: Rayleigh and Rician fading models**
> *Approximate*. They follow from the central limit theorem when many independent scattered paths of comparable strength are present (Rayleigh), or such paths plus one fixed component (Rician). They fit urban mobile measurements well, but fail for channels dominated by a few strong paths (sparse mm-wave channels), and the independence of successive fades assumed in analysis depends on motion and bandwidth.

Outage and fade margin

For a link that must work most of the time rather than achieve an average BER, the relevant measure is the **outage probability**: the fraction of time (or locations) where the SNR drops below the minimum γ_0 the receiver needs. With Rayleigh fading, $P_{\text{out}} = 1 - e^{-\gamma_0/\bar{\gamma}}$; to keep outage at P_{out} the mean SNR must exceed γ_0 by a **fade margin** of $-10\log_{10}[-\ln(1-P_{\text{out}})]$: 9.8 dB for 10 %, 20.0 dB for 1 %, and 30.0 dB for 0.1 %. Each additional "nine" of availability costs 10 dB, a factor of ten in power.

Worked example — Outage of a telemetry radio. A water-utility SCADA radio needs an instantaneous SNR of 5 dB to decode a packet. The mean SNR at a remote pumping station is 20 dB, and the channel is Rayleigh. The outage probability is

$$P_{\text{out}} = 1 - \exp\left(-\frac{10^{0.5}}{10^{2}}\right) = 1 - e^{-0.0316} = 3.1\ \%.$$

About one packet in 32 is lost to fades. With two receive antennas spaced far enough apart for independent fading, and selection of the stronger, outage requires both to fade: $P_{\text{out}} = (0.0311)^2 = 9.7 \times 10^{-4}$, a 32-fold improvement without any extra transmitted power. To reach the same 0.1 % outage with one antenna, the mean SNR would have to rise from 20 dB to $5 + 30.0 = 35.0$ dB: 15 dB more power. This is the case for diversity.

30.5 Diversity

The principle

Diversity provides the receiver with several replicas of the same information that fade independently. The probability that all of them fade at once is the product of the individual probabilities, so deep fades become rare. The number of independent branches, L, is the **diversity order**; at high SNR the error probability falls as $\bar{\gamma}^{-L}$ instead of $\bar{\gamma}^{-1}$.

Sources of diversity

- **Space diversity**: several antennas separated enough for independent fading. At a handset surrounded by scatterers, about $\lambda/2$ suffices (7.5 cm at 2 GHz); at an elevated base station, which sees scatterers in a narrow angle, 10–20 wavelengths are needed.
- **Polarization diversity**: two orthogonally polarized antennas in one housing, the standard for cellular base-station panels.
- **Time diversity**: repetition or coding plus interleaving over an interval longer than the coherence time. It fails for slow users: a stationary handset has an effectively infinite T_c, which is why interleaving alone does not protect pedestrians.
- **Frequency diversity**: transmitting over frequencies separated by more than B_c, as in frequency hopping, or by coding across the subcarriers of OFDM.
- **Multipath diversity**: a RAKE receiver (CDMA) resolves paths separated by more than one chip and combines them, turning multipath into diversity.

Combining methods

- **Selection combining (SC)**: use the branch with the highest SNR. Simple, needs only one receiver chain plus switching.
- **Maximal-ratio combining (MRC)**: co-phase the branches and weight each by its amplitude (divided by its noise power), then add. The output SNR is the sum of the branch SNRs, $\gamma_{\text{MRC}} = \sum_{i} \gamma_i$; this is optimal for independent noise.
- **Equal-gain combining (EGC)**: co-phase and add with equal weights; within about 1 dB of MRC.

For BPSK with L-branch MRC in independent Rayleigh fading, with mean SNR $\bar{\gamma}$ per branch and $\mu = \sqrt{\bar{\gamma}/(1+\bar{\gamma})}$,

$$\bar{P}_b = \left(\frac{1-\mu}{2}\right)^{L} \sum_{k=0}^{L-1} \binom{L-1+k}{k} \left(\frac{1+\mu}{2}\right)^{k} \approx \binom{2L-1}{L} \frac{1}{(4\bar{\gamma})^{L}}.$$

At $\bar{\gamma} = 10$ dB per branch, the BER is 2.3×10^{-2} for $L=1$, 1.6×10^{-3} for $L=2$, and 9.7×10^{-6} for $L=4$. The SNR needed for 10^{-3} falls from 24.0 dB ($L=1$) to 11.1 dB ($L=2$) and 4.0 dB ($L=4$) per branch, approaching the AWGN value of 6.8 dB in total combined SNR (Figure 30.3).

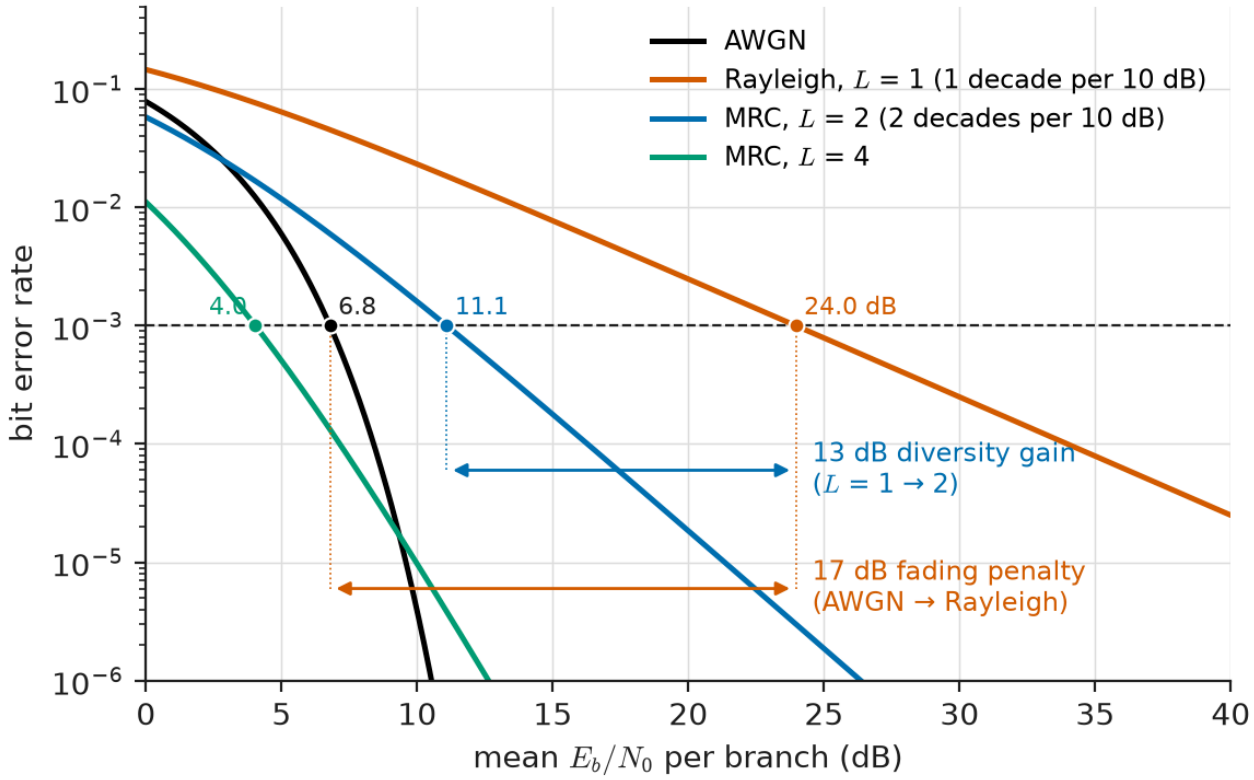

Figure 30.3 — BPSK bit error rate in AWGN and in Rayleigh fading with maximal-ratio diversity.

Transmit diversity: the Alamouti code

Handsets are small, so it is often easier to add antennas at the base station and transmit from them. Transmitting the same symbol from two antennas simply produces another fading sum, with no diversity. The **Alamouti scheme** (1998) sends two symbols s_1, s_2 over two symbol periods from two antennas as

	antenna 1	antenna 2
time 1	s_1	s_2
time 2	$-s_2^*$	s_1^*

With a single receive antenna and channel gains h_1, h_2 constant over the two periods, simple linear processing separates the symbols and gives each the SNR $(|h_1|^2 + |h_2|^2)E_s/(2N_0)$, the full second-order diversity of two-branch MRC, at full rate, with a 3 dB power penalty because the transmit power is split. It is used in LTE's transmit-diversity mode and in Wi-Fi.

> **Status of the principle: Diversity gain**
> *Derived*. The $\bar{\gamma}^{-L}$ behaviour follows from the probability that L independent fades coincide, and MRC's SNR summation from Cauchy–Schwarz. The full gain requires independent branches with accurate channel estimates; correlated branches (closely spaced antennas in a narrow angular spread) yield less, and estimation errors at low SNR erode MRC's advantage.

30.6 Cellular architecture

Frequency reuse

Spectrum is scarce, and a single high-power transmitter covering a city can serve only as many simultaneous users as it has channels. The **cellular concept** (Bell Labs, 1947, deployed from the late 1970s) divides the coverage area into cells, each served by a base station of modest power, and reuses the same frequencies in cells far enough apart that their mutual interference is tolerable. Capacity then grows with the number of cells, not with spectrum.

In the hexagonal model, cells are grouped into **clusters** of N cells, each using a different set of channels; possible cluster sizes satisfy $N = i^2 + ij + j^2$ for integers i, j (1, 3, 4, 7, 9, 12, …). The distance between co-channel cells, relative to the cell radius R, is the **reuse ratio**

$$\frac{D}{R} = \sqrt{3N}.$$

With six first-tier co-channel interferers at distance D and path-loss exponent n, a mobile at the cell edge experiences a signal-to-interference ratio of approximately

$$\frac{S}{I} \approx \frac{(D/R)^n}{6} = \frac{(\sqrt{3N})^n}{6}.$$

Table 30.4 — Cluster size and edge SIR (six interferers).

N	D/R	SIR, $n=3$	SIR, $n=4$
3	3.00	6.5 dB	11.3 dB
4	3.46	8.4 dB	13.8 dB
7	4.58	12.1 dB	18.7 dB
12	6.00	15.6 dB	23.3 dB

First-generation analog FM (AMPS) needed about 18 dB, hence $N = 7$; GSM tolerates about 9 dB and used $N = 3$ or 4 with sectoring and frequency hopping. CDMA, LTE and 5G use **universal frequency reuse** ($N = 1$), accepting low SINR at the cell edge and managing interference by coding, power control, scheduling and coordination between cells. **Sectoring**, dividing each site into three 120° sectors with directional antennas, cuts the number of first-tier interferers from six to about two, raising SIR by roughly 4.8 dB. Figure 30.4 shows a seven-cell reuse pattern.

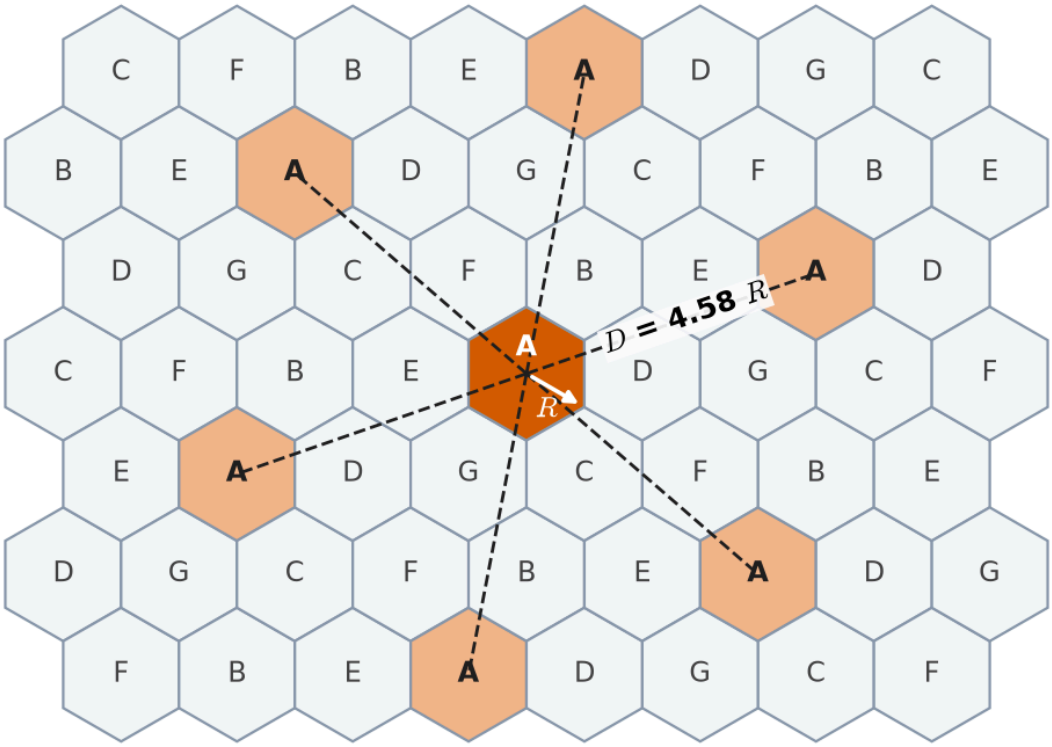

Figure 30.4 — Hexagonal cell layout with a cluster size of 7.

Cell types and the network

Real networks are heterogeneous: **macrocells** with antennas above rooftops covering 1–30 km, **microcells** below rooftop level covering a few hundred metres, and **small cells** (pico and femto) indoors or on street furniture covering tens of metres, all overlaid. A mobile network also has a core: base stations connect over backhaul (fibre or microwave) to the core network, which handles authentication, mobility management, routing to the internet and the telephone network. As a user moves, the network performs **handover** from one cell to the next, deciding on the basis of measured signal strengths with hysteresis to prevent ping-ponging at a boundary, and the core tracks idle users by location area so that they can be paged.

Traffic and trunking: Erlang B

The number of channels a cell needs depends on the offered traffic in **erlangs** (average number of simultaneous calls: 1 erlang is one channel continuously occupied). For blocked-calls-cleared systems with random arrivals, the **Erlang B formula** gives the blocking probability with C channels and offered traffic A:

$$P_B = \frac{A^C / C!}{\sum_{k=0}^{C} A^k / k!}.$$

At 2 % blocking, 10 channels carry 5.08 erlangs (51 % utilization), 30 channels carry 21.9 erlangs (73 %) and 100 channels carry 88.0 erlangs (88 %). This **trunking efficiency** is why large pools of channels are more efficient than small ones, and one reason why sectoring, which splits the pool, has a capacity cost as well as an interference benefit. Packet-switched data networks replace Erlang B with queueing and scheduling models, but the principle of statistical multiplexing is the same.

> **Status of the principle: Hexagonal reuse and SIR formula**
> *Approximate*. It assumes identical hexagonal cells, a single path-loss exponent, equal transmit powers and only first-tier interferers, and ignores shadowing. Real cell layouts follow terrain and demand; the formula guides reuse choice but planning tools use measured or ray-traced propagation.

30.7 Multiple access

Multiple access shares a channel among many users; **duplexing** separates the two directions of one link, either by frequency (FDD, paired bands) or by time (TDD, one band alternating).

FDMA and TDMA

Frequency-division multiple access (FDMA) gives each user a separate carrier: first-generation analog systems used 25–30 kHz channels. It needs guard bands and a separate radio chain per carrier at the base station, but each user's narrowband signal sees flat fading.

Time-division multiple access (TDMA) gives each user a time slot on a shared carrier. GSM combines both: 200 kHz carriers (FDMA), each carrying eight time slots of 577 µs (TDMA), with frequency hopping for diversity. TDMA requires accurate timing, including **timing advance** to compensate for propagation delay (3.3 µs per km each way), and the bursty transmission causes the familiar 217 Hz audio buzz in nearby loudspeakers.

CDMA

Code-division multiple access (CDMA) lets all users transmit simultaneously over the same wide band, each spreading its data with a distinct high-rate code. The receiver correlates with the desired user's code: that signal is despread back to its narrow bandwidth, while the others remain spread and appear as noise. The **processing gain** is the ratio of chip rate to data rate; for UMTS voice, 3.84 Mchip/s over 12.2 kb/s gives $10\log_{10}(3.84\times10^{6}/12\,200) = 25.0$ dB. CDMA naturally supports $N = 1$ reuse, soft handover (connection to several base stations at once) and RAKE multipath diversity, but needs fast, accurate power control (1500 updates per second in UMTS) to solve the **near–far problem**: an uncontrolled nearby user could swamp distant ones by tens of decibels.

OFDMA, SC-FDMA and random access

Orthogonal frequency-division multiple access (OFDMA), used in LTE and 5G downlinks and in Wi-Fi 6, assigns users different sets of OFDM subcarriers and time slots (resource blocks of 12

subcarriers × 1 slot in LTE), and the scheduler can give each user the subcarriers where its channel is currently strong, turning frequency-selective fading into a source of gain (**multi-user diversity**). The LTE uplink uses **SC-FDMA** (DFT-spread OFDM), which keeps the subcarrier allocation flexibility but lowers the peak-to-average power ratio of the handset's transmitter. Unlicensed systems such as classical Wi-Fi use **carrier-sense multiple access with collision avoidance (CSMA/CA)**: listen, wait a random back-off, and transmit if the channel is free.

Table 30.5 — Multiple access in major systems.

System	Multiple access	Duplex	Channel bandwidth
AMPS (1G)	FDMA	FDD	30 kHz
GSM (2G)	TDMA/FDMA, frequency hopping	FDD	200 kHz, 8 slots
IS-95 (2G)	CDMA, 1.2288 Mchip/s	FDD	1.25 MHz
UMTS (3G)	Wideband CDMA, 3.84 Mchip/s	FDD (TDD option)	5 MHz
LTE (4G)	OFDMA downlink, SC-FDMA uplink	FDD or TDD	1.4–20 MHz per carrier
5G NR	OFDMA (DFT-s-OFDM option on uplink)	Mostly TDD above 3 GHz	Up to 100 MHz (FR1), 400 MHz (FR2)
Wi-Fi 6 (802.11ax)	OFDMA plus CSMA/CA	TDD	20–160 MHz

30.8 OFDM fundamentals

Why many slow carriers beat one fast one

A single carrier at 20 Msymbol/s in a channel with 1–5 μs delay spread suffers ISI across 20–100 symbols; equalizing it in the time domain is costly and sensitive. **Orthogonal frequency-division multiplexing (OFDM)** divides the data into N parallel streams on closely spaced subcarriers, each at $1/N$ of the rate. Each subcarrier is so narrow that it sees flat fading, and each symbol is so long that the delay spread is only a small fraction of it.

Orthogonality and the FFT

Subcarriers spaced by $\Delta f = 1/T_u$, where T_u is the useful symbol duration, are **orthogonal** over T_u: the integral of the product of any two distinct subcarriers over one symbol is zero. Their spectra (sinc-shaped) overlap, but at the centre of each subcarrier all others pass through zero, so no guard bands are needed between them. The baseband OFDM symbol

$$x(t) = \sum_{k=0}^{N-1} X_k \, e^{j 2\pi k \Delta f t}, \qquad 0 \le t < T_u,$$

sampled at N points, is exactly an inverse DFT of the complex data symbols X_k. Modulation is therefore an **IFFT** and demodulation an **FFT** (Chapter 22), which is what made OFDM practical in cheap silicon (Figure 30.5).

The cyclic prefix

Multipath still smears each symbol into the next. OFDM prepends a **cyclic prefix (CP)**, a copy of the last T_{CP} of the symbol, to its beginning. If T_{CP} exceeds the channel's maximum excess delay:

1. ISI from the previous symbol falls entirely within the prefix, which the receiver discards;
2. the channel's linear convolution with the symbol becomes a **circular** convolution over the FFT window, so in the frequency domain each subcarrier is simply multiplied by the channel's frequency response H_k:

$$Y_k = H_k X_k + N_k.$$

Equalization reduces to one complex division per subcarrier ($\hat{X}_k = Y_k / H_k$), using pilot subcarriers to estimate H_k. The cost is overhead: the CP carries no new data.

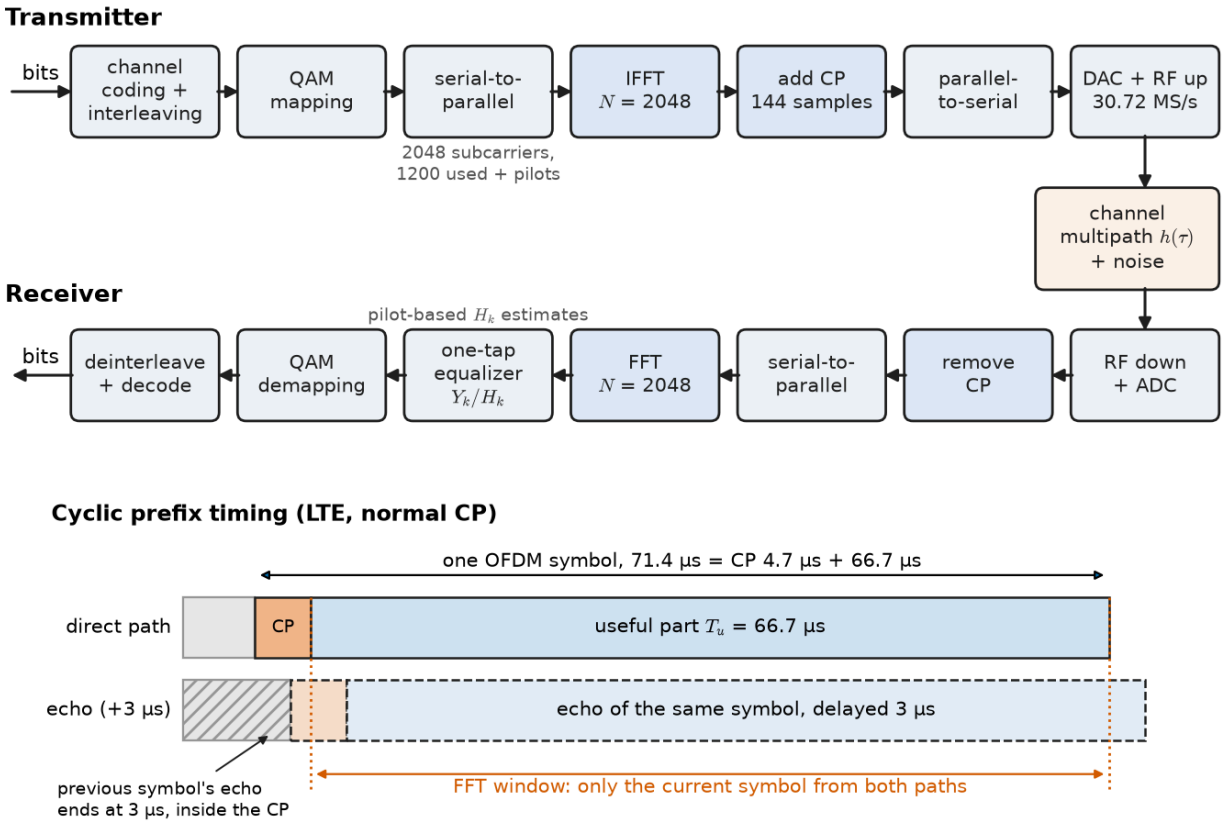

Figure 30.5 — OFDM transmitter and receiver with cyclic prefix.

Numbers from real systems

LTE (20 MHz carrier). Subcarrier spacing 15 kHz, so $T_u = 1/15\,000 = 66.7$ µs. 1200 subcarriers are used (100 resource blocks of 12) within a 2048-point FFT sampled at 30.72 MHz. The normal CP is 144 samples, $144/30.72 \times 10^6 = 4.69$ µs (160 samples in the first symbol of each slot), enough for typical

urban delay spreads, at an overhead of $144/(2048+144) = 6.6$ %. Fourteen symbols fit in each 1 ms subframe; with 64-QAM, the raw resource-element rate is $1200 \times 14 \times 6/10^{-3} = 100.8$ Mb/s per antenna layer, before control, pilot and coding overheads.

Wi-Fi (802.11a/g, 20 MHz). A 64-point FFT with 312.5 kHz spacing gives $T_u = 3.2$ µs; a 0.8 µs guard interval makes a 4 µs symbol. Of 52 used subcarriers, 48 carry data and 4 are pilots. With 64-QAM and rate-3/4 coding, the data rate is $48 \times 6 \times 0.75/4\ \mu\text{s} = 54$ Mb/s, the familiar top rate of 802.11a/g. Indoor delay spreads of tens of nanoseconds fit comfortably within 0.8 µs.

5G NR uses scalable numerology: spacings of $15 \times 2^{\mu}$ kHz (15, 30, 60, 120 kHz), shortening symbols and CPs in proportion, so that wide mm-wave carriers with large phase noise use wide spacings and short symbols, while sub-6 GHz macrocells use 15 or 30 kHz.

OFDM's weaknesses

- **Peak-to-average power ratio (PAPR).** The sum of many independent subcarriers is nearly Gaussian; its peaks reach 10–12 dB above the average with non-negligible probability. The power amplifier must back off to stay linear, wasting efficiency, which matters most in battery-powered handsets, hence SC-FDMA on the LTE uplink.
- **Frequency-offset and phase-noise sensitivity.** A carrier frequency error of a fraction of Δf, from oscillator tolerance or Doppler, destroys orthogonality and causes inter-carrier interference. A 15 kHz spacing tolerates offsets of a few hundred hertz.
- **Faded subcarriers.** Subcarriers in a spectral null are lost; OFDM relies on coding and interleaving across subcarriers (*coded* OFDM) to recover them, which converts frequency selectivity into frequency diversity.

> **Status of the principle: OFDM with cyclic prefix diagonalizes the channel**
> *Derived*. If the CP is longer than the channel impulse response and the channel is constant over one symbol, the DFT exactly converts linear convolution into per-subcarrier multiplication. Delay spreads exceeding the CP, or Doppler shifts that are a significant fraction of the subcarrier spacing, break the result and cause inter-symbol and inter-carrier interference.

30.9 MIMO fundamentals

The channel matrix

With N_t transmit and N_r receive antennas, the narrowband channel is described by an $N_r \times N_t$ complex matrix $\mathbf{H}$, whose element h_{ij} is the gain from transmit antenna j to receive antenna i:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}.$$

In an OFDM system there is one such matrix per subcarrier. In a rich scattering environment the entries are roughly independent complex Gaussian variables; with line of sight and closely spaced antennas they are strongly correlated.

Three uses of multiple antennas

1. **Diversity** (Section 30.5): sending or receiving the same information over independently fading paths, improving reliability. The maximum diversity order is $N_t N_r$.
2. **Beamforming (array gain)**: weighting the antennas so their signals add coherently in the desired direction, increasing average SNR by up to a factor $N_t N_r$ and reducing interference to and from other directions. Massive-MIMO base stations with 64 or more elements use it to serve several users simultaneously in different directions.
3. **Spatial multiplexing**: transmitting independent data streams from different antennas at the same frequency and time, and separating them at the receiver using their different spatial signatures. Up to $\min(N_t, N_r)$ streams (layers) can be supported.

MIMO capacity

With equal power on each transmit antenna and the channel known at the receiver, the capacity of a fixed channel at total SNR ρ is

$$C = \log_2 \det\left(\mathbf{I}_{N_r} + \frac{\rho}{N_t}\mathbf{H}\mathbf{H}^H\right) = \sum_{i=1}^{r} \log_2\left(1 + \frac{\rho}{N_t}\lambda_i\right) \ \text{b/s/Hz},$$

where λ_i are the eigenvalues of $\mathbf{H}\mathbf{H}^H$ (the squared singular values of $\mathbf{H}$) and r is its rank. The channel behaves as r parallel independent channels, the **eigenmodes**, each with its own gain. At high SNR in rich scattering, capacity grows as $\min(N_t, N_r)\log_2 \rho$: linearly with the number of antennas, not logarithmically with power. This is why every recent generation of Wi-Fi and cellular systems has multiplied its antenna count.

> **Status of the principle: MIMO capacity formula**
> *Derived*. Follows from Shannon capacity applied to a vector Gaussian channel with receiver channel knowledge and equal power allocation (with transmitter knowledge, water-filling over eigenmodes does slightly better). The linear growth with $\min(N_t, N_r)$ requires a well-conditioned channel matrix, which needs rich scattering or adequate antenna spacing; a line-of-sight channel between compact arrays may have effective rank 1.

Worked example — Capacity of a 2×2 link. A 2×2 Wi-Fi link at a total SNR of 20 dB ($\rho = 100$) has the measured channel

$$\mathbf{H} = \begin{bmatrix} 1.0 & 0.6 - 0.3j \\ 0.4 + 0.5j & 0.9 \end{bmatrix}.$$

The eigenvalues of $\mathbf{H}\mathbf{H}^H$ are $\lambda_1 = 2.556$ and $\lambda_2 = 0.114$. With $\rho/N_t = 50$:

$$C = \log_2(1 + 50 \times 2.556) + \log_2(1 + 50 \times 0.114) = 7.01 + 2.75 = 9.76 \ \text{b/s/Hz}.$$

A single-antenna link with unit gain at the same SNR would give $\log_2 101 = 6.66$ b/s/Hz. The second eigenmode is 13.5 dB weaker than the first, so it contributes less, but the total is still 47 % above the single-antenna value. Averaged over independent Rayleigh-fading 2×2 channels at 20 dB, simulation gives an ergodic capacity of about 11.3 b/s/Hz, against 5.9 b/s/Hz for a single Rayleigh-faded antenna pair: close to the factor of two that $\min(N_t, N_r) = 2$ promises.

Practical MIMO

Receivers separate spatial streams with linear detectors (zero-forcing or MMSE) or near-ML detectors; transmitters with channel knowledge (from feedback in FDD, or from reciprocity in TDD) **precode** onto the eigenmodes. LTE supports up to 4 layers to a handset (8 in later releases), 5G NR up to 8 on the downlink, Wi-Fi 6 up to 8 spatial streams. **Multi-user MIMO** sends different streams to different single- or dual-antenna users from a many-antenna base station, so that spatial multiplexing is limited by the number of base-station antennas rather than by the handset.

30.10 Link budgets

The bookkeeping

A **link budget** tallies all gains and losses between transmitter and receiver, in decibels, to find the received power, the SNR, and the margin left over for fading and uncertainty:

$$P_r\ (\text{dBm}) = P_t + G_t - L_t - L_p + G_r - L_r - L_{\text{misc}},$$

where L_t and L_r are feeder losses, L_p is the path loss and L_{misc} covers polarization mismatch, body loss, building penetration, rain and atmospheric absorption. The noise power is $N = -174 + 10\log_{10} B +$ NF dBm, the SNR is $P_r - N$, and the **link margin** is the excess over the SNR the modem requires. The margin must cover the fading and shadowing allowances of Sections 30.2–30.4 plus implementation losses.

Worked example — A 5.8 GHz point-to-point backhaul link. A wireless internet provider links a hilltop tower to a village 10 km away using two 28 dBi dishes. The radio transmits 20 dBm into 1 dB of cable loss at each end; the receiver has NF 5 dB in a 20 MHz channel, and its 256-QAM mode with rate-5/6 coding needs an SNR of about 25 dB. Fresnel clearance has been checked (Section 30.1). The budget, shown in Table 30.6 and Figure 30.6:

Table 30.6 — Link budget for a 10 km, 5.8 GHz link.

Item	Value
Transmit power	+20.0 dBm
Transmit cable loss	−1.0 dB
Transmit antenna gain	+28.0 dBi
EIRP	+47.0 dBm
Free-space path loss, 10 km at 5.8 GHz	−127.7 dB
Receive antenna gain	+28.0 dBi
Receive cable loss	−1.0 dB
Received power	**−53.7 dBm**
Noise floor: $-174 + 10\log_{10}(2 \times 10^7) + 5$	−96.0 dBm
SNR	**42.3 dB**
Required SNR, 256-QAM rate 5/6	25.0 dB
Link margin	**17.3 dB**

The EIRP of 47 dBm (50 W) must also be checked against the regulatory limit for the band, which varies by country. The 17.3 dB margin protects against multipath fading on the link (which over a clear, fixed path is Rician with high K-factor, so the margin needed is much less than the Rayleigh values) and some antenna misalignment caused by wind. At 5.8 GHz rain attenuation over 10 km is small, roughly 1–3 dB in heavy rain; on a 23 GHz or 38 GHz link the same budget would be dominated by a rain allowance of 10–30 dB. When the margin is eaten into, adaptive modulation steps down to 64-QAM or 16-QAM, trading throughput for continued operation (Chapter 29).

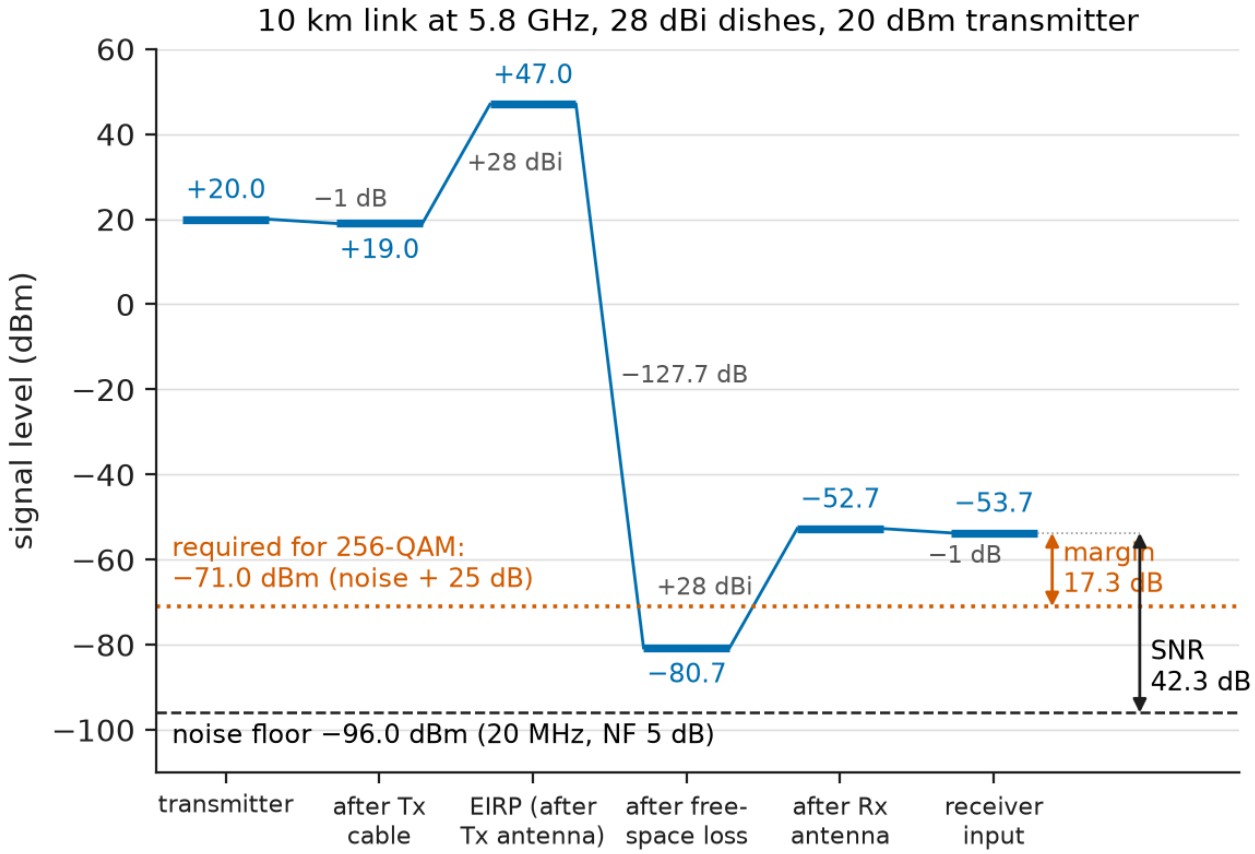

Figure 30.6 — Signal-level diagram for the 10 km, 5.8 GHz link budget.

Cellular and satellite budgets

A mobile link budget differs mainly in the allowances. The macrocell example of Section 30.2 left an SNR of only 1.7 dB at the 90 % shadowing point before multipath; a real budget adds penetration loss (10–20 dB into buildings, 5–15 dB into vehicles), interference margin (3–6 dB in loaded networks), and diversity and coding gains on the other side of the ledger. The cell edge is set where the remaining SINR just supports the lowest-rate modulation and coding scheme, typically around −5 dB with QPSK and low-rate codes plus HARQ repetition. Satellite budgets, by contrast, have enormous path losses (205 dB for GEO at 12 GHz) offset by high antenna gains and cold-sky noise temperatures, use G/T as the receiving figure of merit (Chapter 29), and reserve their margins for rain.

> **Status of the principle: Link budget**
> *Derived.* A link budget is energy conservation and noise accounting in decibels; it is exact for its inputs. Its accuracy is limited entirely by the propagation model and margins chosen, which are approximate or empirical, so budgets are always quoted with an availability or coverage percentage.

30.11 Wireless-system performance

Measures of performance

A wireless system is judged by several measures at once:

- **Throughput** per user and per cell, in Mb/s, and **spectral efficiency** in b/s/Hz per cell or per user; for an area, **area spectral efficiency** in b/s/Hz/km².
- **Coverage and outage**: the fraction of locations and time with acceptable service, often expressed through the cell-edge (5th-percentile) user throughput.
- **Reliability and latency**: block error rate after HARQ and delay; 5G ultra-reliable low-latency service targets a packet error rate of 10^{-5} within 1 ms for industrial control and protection signalling.
- **Capacity**: the number of users or erlangs per cell, or the aggregate traffic served.
- **Energy efficiency**: bits per joule at the base station and battery life at the terminal, critical for IoT sensors that must run for ten years on a coin cell.

Fading and capacity

Combining Shannon's formula with fading gives the **ergodic capacity** $E[\log_2(1+\gamma)]$, the long-term average achievable rate when coding spans many fades, and the **outage capacity**, the rate supported except for a given fraction of time when coding cannot span fades. For a single-antenna Rayleigh channel at a mean SNR of 20 dB, the ergodic capacity is about 5.9 b/s/Hz against 6.66 in AWGN, a modest loss once coding and interleaving average over the fades; the outage capacity at 1 % is far lower, because a 1 % outage requires backing off by about 20 dB. Diversity and MIMO raise both.

How the techniques combine

A modern cellular link uses every technique of this chapter at once: OFDM to convert the frequency-selective channel into flat subchannels; a scheduler that gives each user its best resource blocks (multi-user diversity); MIMO for spatial multiplexing at the cell centre and diversity or beamforming at the edge; adaptive modulation and coding that tracks SINR from QPSK at rate 0.1 to 256-QAM at rate 0.93; HARQ that absorbs residual errors; and $N = 1$ reuse with interference coordination. Their combined effect is a throughput that tracks the Shannon bound with a gap of roughly 2–4 dB across the usable SINR range.

Table 30.7 — Evolution of cellular performance (typical peak and practical figures).

Generation	Key technologies	Peak downlink (per carrier or configuration)	Typical user throughput	Cell spectral efficiency
2G GSM/EDGE	TDMA, GMSK/8-PSK	0.2–0.5 Mb/s (EDGE)	0.05–0.2 Mb/s	≈0.2 b/s/Hz
3G HSPA	WCDMA, 16/64-QAM, HARQ	14–42 Mb/s	1–5 Mb/s	≈0.5–1 b/s/Hz
4G LTE-A	OFDMA, 2×2–4×4 MIMO, carrier aggregation	150 Mb/s–1 Gb/s	10–50 Mb/s	≈1.5–3 b/s/Hz
5G NR (FR1)	OFDMA, massive MIMO, 100 MHz carriers	1–4 Gb/s	100–500 Mb/s	≈5–10 b/s/Hz
5G NR (FR2 mm-wave)	400 MHz carriers, beamforming arrays	up to ~10 Gb/s (aggregated)	0.5–2 Gb/s, short range	depends on beam management

Most of the improvement in throughput per site has come from three sources: more spectrum (wider carriers, aggregation), more antennas (MIMO, beamforming), and more sites (densification). Improvements in modulation and coding, which brought systems from about 10 dB from the Shannon bound to 1–2 dB, are nearly exhausted; the remaining capacity lies in space and spectrum, consistent with the capacity formulas of Chapter 28 and Section 30.9.

Limits of the models

Everything in this chapter treats the channel as a random process with known statistics. Real deployments add effects these models handle poorly: correlated shadowing between nearby cells, non-Gaussian interference from bursty users, blockage of mm-wave beams by a hand or a passing bus, and hardware impairments (phase noise, amplifier non-linearity, ADC resolution) that put an error floor under high-order QAM. Field measurement and system-level simulation with standardized channel models (3GPP and ITU) remain essential complements to the formulas.

> **Boundary.**
> Electromagnetic propagation physics (reflection coefficients, diffraction theory, atmospheric absorption) belongs to *Engineering Physics*; RF power amplifiers, low-noise amplifiers and transceiver integrated circuits belong to *Electronics & Computing*. Antenna gain and patterns are introduced in Chapter 32 of this book.

What remains open

Open question	Where it is answered
How are antenna gain, directivity and radiation patterns, used throughout the link budgets here, defined and measured?	Chapter 32 (antenna fundamentals, gain and directivity)
How are RF components such as filters, mixers and matching networks in a radio front end characterized?	Chapter 32 (S-parameters, RF components)
How do fibre-optic links, which carry the backhaul behind every cell, budget their power and dispersion?	Chapter 31 (optical link budgets)
How are the FFTs and filters inside OFDM transceivers implemented digitally?	Chapters 22 and 23
How do adaptive equalizers and channel estimators track a time-varying channel?	Chapter 24 (adaptive filters, LMS)
Where do the error-rate and noise-figure results used in link budgets come from?	Chapter 29
How close can coded systems get to the capacity limits quoted here?	Chapter 28
How are power control and other feedback loops in a radio network analysed for stability?	Chapters 34 and 35
What is the electromagnetic physics of reflection, diffraction and scattering?	*Engineering Physics*

Next: Chapter 31 moves from radio to light, where the channel is a glass fibre whose attenuation and dispersion, rather than fading and interference, set the limits of the links that carry most of the world's traffic.

Chapter 31 — Optical Communications

Part VI · Communication Theory

A regional operator lights a new 80 km fibre between two exchanges. The transceivers are standard 10 Gb/s plug-in modules, the fibre is new, the connectors have been cleaned and inspected, and the link comes up error-free on the first day. Three months later a second operator, using the same fibre type and the same distance, swaps in cheaper modules with directly modulated lasers. The received power is identical, well above the receiver's quoted sensitivity, yet the link runs at a bit-error ratio near 10^{-6} and the forward-error-correction counters never stop climbing. Nothing is broken. The light arrives in sufficient quantity; it simply arrives *smeared*, because each wavelength component of a chirped laser pulse travels the 80 km at a slightly different speed. The second link is dispersion-limited.

This chapter treats optical fibre as a communication channel in the sense of Chapter 25. The mathematics is that of a cable or radio link; what changes are the numbers — carriers near 200 THz, losses of 0.2 dB/km, bandwidths of terahertz — and the physics that sets them.

> **Why can a hair-thin glass fibre carry tens of terabits per second over thousands of kilometres, when copper cable struggles past a few hundred metres at 10 Gb/s? What limits an optical link — the light that is lost, the shape that is smeared, or the noise added on the way — and how does an engineer tell which one?**

Concept	Engineering destination
Total internal reflection, numerical aperture, modes	Choosing single-mode or multimode fibre, connector and splice tolerances
Fibre attenuation spectrum	Choosing 1310 or 1550 nm, span lengths between amplifiers, PON reach
Modal, chromatic and polarization dispersion	Data-centre reach limits, 10–400 Gb/s long-haul design, dispersion compensation
Laser and LED transmitters	Transceiver choice: VCSEL, Fabry–Perot, DFB, externally modulated laser
PIN and APD receivers, Q factor	Receiver sensitivity, BER specifications, FEC thresholds
Wavelength-division multiplexing and EDFAs	Backbone and submarine capacity, OSNR budgets
Optical link budget	Acceptance testing of a new route, margin for repairs and ageing
Nonlinearity and system limits	Launch-power optimisation, capacity of a fibre pair

Roadmap. Why light and how a fibre guides it (31.1) → the families of fibre and their parameters (31.2) → loss (31.3) and delay spread (31.4) as the two channel impairments → the transmitter (31.5) and

receiver (31.6) at the ends → multiplying capacity with many wavelengths (31.7) → putting the numbers together in a link budget (31.8) → what finally limits a fibre system (31.9).

31.1 Optical transmission fundamentals

Why an optical carrier

The capacity of any channel scales with its usable bandwidth (Shannon, Chapter 28). A carrier at frequency f_c can, in practice, be modulated over a bandwidth that is some fraction of f_c before the components that generate, guide and detect it stop behaving. At 2 GHz that fraction gives tens of megahertz; at the optical carrier frequency

$$f_c = \frac{c}{\lambda} = \frac{2.998 \times 10^8 \text{ m/s}}{1550 \text{ nm}} \approx 193.4 \text{ THz}$$

even a fraction of a percent is tens of terahertz. The low-loss window of silica fibre around 1550 nm is about 35 nm wide in the conventional amplifier band alone, which converts to

$$\Delta f = \frac{c\,\Delta\lambda}{\lambda^2} = \frac{2.998 \times 10^8 \times 35 \times 10^{-9}}{(1547.5 \times 10^{-9})^2} \approx 4.4 \text{ THz}$$

— more than all the radio spectrum below 100 GHz put together, in a single strand of glass. The second reason is loss. A good coaxial cable loses about 20 dB per 100 m at 1 GHz (Chapter 14), and the loss rises as $\sqrt{f}$ because of the skin effect. A modern single-mode fibre loses about 0.2 dB per *kilometre*, and that loss is essentially independent of the modulation rate. The third reason is immunity: glass picks up no 50 Hz field, forms no ground loops, and can run inside the earth wire of a transmission line (OPGW cable).

Photons and power

Light comes in quanta of energy $E = hc/\lambda$: at 1550 nm one photon carries 1.28×10^{-19} J (0.80 eV), so 1 µW is about 7.8×10^{12} photons per second. The granularity is irrelevant for power budgets but sets the ultimate floor on receiver noise (Section 31.6). Optical power is quoted in dBm as in RF work (Chapter 25): a data-centre VCSEL launches −2 to +2 dBm, a long-haul DFB 0 to +5 dBm per channel, and a 10 Gb/s receiver needs roughly −18 to −28 dBm.

Guiding light: total internal reflection and numerical aperture

A step-index fibre is a glass core of refractive index n_1 surrounded by a cladding of slightly lower index n_2, both usually silica, with the core doped (typically with germanium dioxide) to raise its index by a fraction of a percent. The relative index difference

$$\Delta = \frac{n_1 - n_2}{n_1}$$

is about 0.35 % for standard single-mode fibre and about 1 % for multimode fibre (Figure 31.1). A ray inside the core striking the core–cladding interface at an angle (from the normal) greater than the

critical angle $\theta_c = \arcsin(n_2/n_1)$ is totally reflected and stays in the core. Tracing that condition back through the end face of the fibre gives the largest launch angle θ_a (in air) that is guided:

$$\mathrm{NA} = \sin\theta_a = \sqrt{n_1^2 - n_2^2} \approx n_1\sqrt{2\Delta}$$

For $n_1 = 1.48$ and $\Delta = 0.01$ the numerical aperture is 0.21 and the acceptance half-angle about 12°; for single-mode fibre the NA is about 0.12 (7°). The NA sets how easily light can be coupled in: a large-area, wide-angle source such as an LED couples only to large-NA multimode fibre, whereas single-mode fibre needs a laser whose beam is focused onto a spot only a few micrometres across.

> **Status of the principle: Ray guiding by total internal reflection**
> *Approximate.* Geometrical optics is valid when the core is many wavelengths across (multimode fibre, core 50–62.5 µm); for single-mode cores of 8–10 µm the guided field must be found from Maxwell's equations as a waveguide mode, and the ray picture gives only qualitative insight.

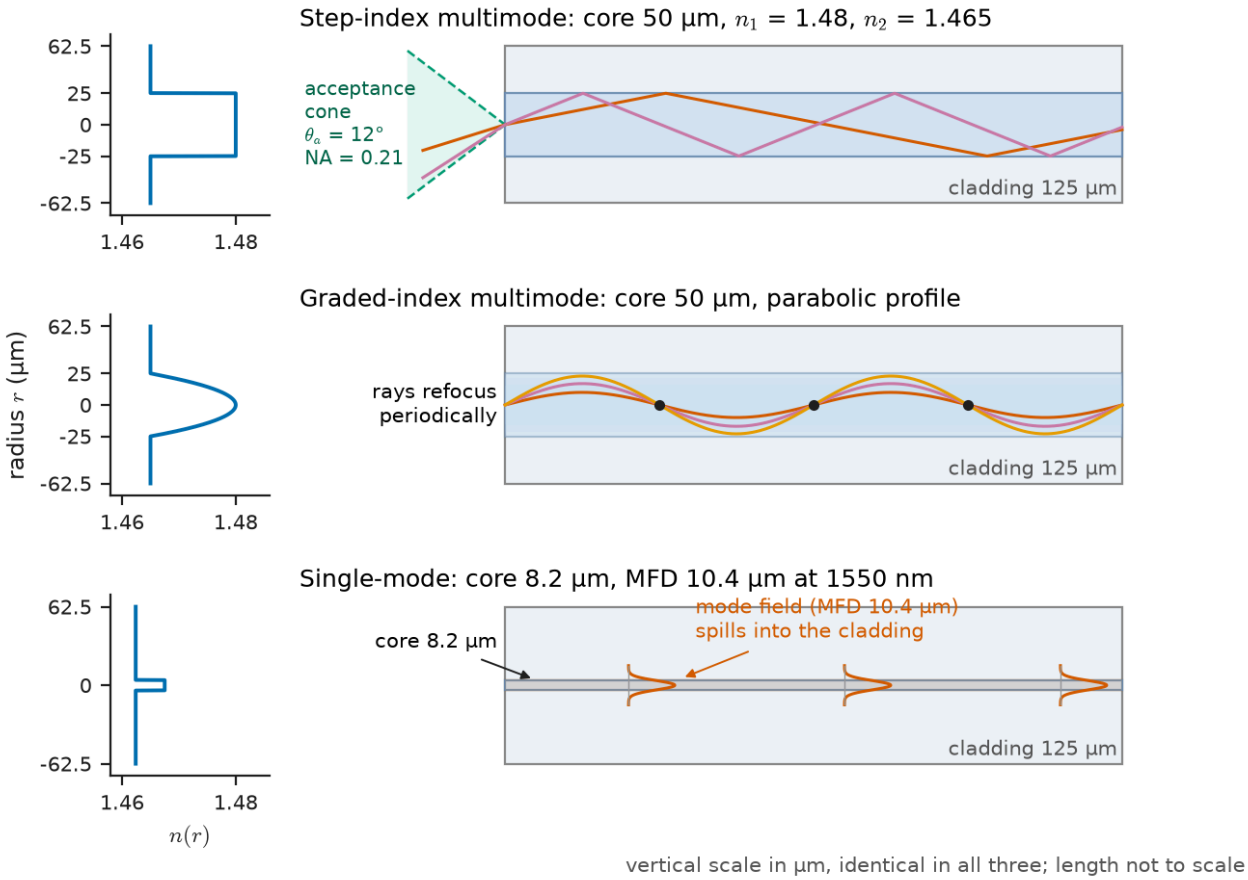

Figure 31.1 — Light guiding in step-index multimode, graded-index multimode and single-mode fibre, with the acceptance cone set by the numerical aperture.

Modes and the V number

Because the core is a waveguide, only discrete field patterns — modes — propagate, each with its own propagation constant, exactly as in the rectangular waveguides of Chapter 32. How many modes a step-index fibre supports is set by the normalised frequency

$$V = \frac{2\pi a}{\lambda}\,\mathrm{NA}$$

where a is the core radius. For $V < 2.405$ (the first zero of the Bessel function J_0) only the fundamental mode, LP_{01}, propagates, and the fibre is **single-mode**. For large V the number of modes is approximately $V^2/2$. With $a = 4.1\ \mu$m and $\mathrm{NA} = 0.12$, $V = 2.36$ at 1310 nm and 1.99 at 1550 nm: single-mode at both, with cutoff wavelength

$$\lambda_c = \frac{2\pi a\,\mathrm{NA}}{2.405} \approx 1285\ \mathrm{nm}$$

A multimode fibre with a 50 μm core and NA 0.2 has $V \approx 37$ at 850 nm and so carries several hundred modes.

> **Status of the principle: Single-mode condition** $V < 2.405$
> *Derived*. Follows from solving the weakly guiding wave equation in a step-index cylinder; exact for the ideal step profile, while real profiles and cabled bends shift the effective cutoff by tens of nanometres, which is why standards specify a *cable* cutoff wavelength below 1260 nm.

The fundamental mode is roughly Gaussian and extends into the cladding, with a **mode-field diameter** (MFD) of about 9.2 μm at 1310 nm and 10.4 μm at 1550 nm; the MFD sets splice loss and laser coupling, and is why single-mode connectors align fibres to about a micrometre.

31.2 Fiber characteristics

The fibre families

Practically all telecommunications fibre is silica with a 125 μm cladding and a 250 μm (or 200 μm) polymer coating. What distinguishes the families is the core design and therefore the modal and dispersion behaviour. Table 31.1 collects the types an engineer meets.

Table 31.1 — Common fibre types (typical specified values).

Fibre	Core / MFD	Main window	Loss (dB/km)	Dispersion / bandwidth	Typical use
ITU-T G.652.D standard single-mode	8.2 µm core; MFD 9.2 µm at 1310 nm	1310 and 1550 nm	≤0.35 at 1310, ≤0.22 at 1550 (typ. 0.32 and 0.19)	Zero dispersion 1300–1324 nm; about 17 ps/(nm·km) at 1550 nm	Access, metro, long haul — the bulk of installed fibre
G.657 bend-insensitive single-mode	As G.652, trench-assisted	1310–1625 nm	As G.652	As G.652	Fibre-to-the-home drops, tight cabinets
G.655 non-zero dispersion-shifted	MFD 8–9 µm at 1550 nm	1530–1625 nm	≤0.22 at 1550	2–6 ps/(nm·km) at 1550 nm	Older DWDM long haul (limits four-wave mixing)
G.654 cut-off-shifted, large effective area	Aeff 110–125 µm²	1550 nm	0.15–0.17	≈20–21 ps/(nm·km)	Submarine and coherent terrestrial long haul
OM3 / OM4 laser-optimised multimode	50 µm graded index	850 nm	≈2.5–3.0 at 850, ≈0.7–1.0 at 1300	Effective modal bandwidth 2000 / 4700 MHz·km at 850 nm	Data-centre links up to 100–400 m
OM1 legacy multimode	62.5 µm graded index	850 / 1300 nm	≈3.0 / 1.0	≈200 / 500 MHz·km	Older building backbones
Polymer optical fibre (PMMA)	≈1 mm step index	650 nm	≈150–200	Tens of MHz over 50 m	Automotive (MOST), short industrial links

Parameters that matter to a link designer

For a link designer a fibre is described by five numbers: the attenuation α in dB/km (Section 31.3); the chromatic dispersion D in ps/(nm·km) and its slope $S = dD/d\lambda$ (Section 31.4); the modal bandwidth of multimode fibre in MHz·km; the PMD coefficient in ps/√km; and the effective area A_{eff}, about 80 µm² for G.652 at 1550 nm, which governs nonlinearity (Section 31.9). Bend radius, proof strain and ageing enter the link calculation only as extra attenuation.

> **Boundary.**
> The manufacture of preforms, the doping chemistry that sets n_1 and n_2, and the glass physics behind Rayleigh scattering belong to *The Materials Core*; the electromagnetic solution for guided modes belongs to *Engineering Physics*. Here the fibre is a channel described by α, D, bandwidth and A_{eff}.

Fibre as a linear channel

For powers below a few milliwatts per channel, a single-mode fibre is a linear, time-invariant system for the optical field (Chapter 17). Its transfer function at an offset ω from the carrier is, to second order,

$$H(\omega) = e^{-\alpha_n L/2}\, e^{-j\beta(\omega) L}, \qquad \beta(\omega) \approx \beta_0 + \beta_1 \omega + \tfrac{1}{2}\beta_2 \omega^2$$

where α_n is the power attenuation in nepers per unit length. The three terms in β have direct meanings: β_0 is a phase that does not matter, $\beta_1 = 1/v_g$ is the group delay per unit length (about 4.9 µs/km for a group index of 1.468), and β_2 is **group-velocity dispersion**, the frequency dependence of group delay that Chapter 18 identified as the cause of distortion. Loss and dispersion are therefore exactly the magnitude and phase of an all-pass-with-loss filter. The difference from a copper line is that the loss is almost flat over the signal bandwidth and the phase is almost purely quadratic.

31.3 Attenuation

Loss in decibels per kilometre

Power decays exponentially along the fibre:

$$P(L) = P(0)\, 10^{-\alpha L/10}$$

with α in dB/km. At 0.2 dB/km a 100 km span loses 20 dB — a factor of 100 — which is why unrepeated links of 80–120 km are routine and why amplifiers in long-haul systems are spaced 50–100 km apart.

> **Status of the principle: Exponential attenuation with constant α**
> *Approximate.* Holds for a uniform fibre in the linear regime; splices, connectors and bends add discrete losses that are budgeted separately, and α itself is wavelength dependent and rises slowly with hydrogen exposure and radiation.

The mechanisms

Rayleigh scattering dominates in the low-loss windows. Glass is frozen liquid; its density fluctuates on a scale far below a wavelength, and each fluctuation scatters a little light in all directions. The scattering coefficient scales as λ^{-4}, so moving from 1310 to 1550 nm reduces it by $(1550/1310)^4 \approx 1.96$ — almost exactly the ratio of the specified losses (0.32 against 0.17–0.19 dB/km). Rayleigh scattering also returns a tiny fraction of the light backwards, which is what an optical time-domain reflectometer (OTDR) measures to map loss along an installed fibre.

Infrared absorption by the vibrations of Si–O bonds rises steeply beyond about 1600 nm and sets the long-wavelength edge of the window. **Ultraviolet absorption** by electronic transitions is negligible at telecom wavelengths but contributes at 850 nm and below. **Impurity absorption** used to be dominated by hydroxyl (OH) ions, with an overtone peak near 1383 nm; "low water peak" fibre (G.652.D) has removed it to below 0.35 dB/km, opening the whole range from 1260 to 1625 nm for coarse WDM. **Bending loss** occurs when a fibre is curved tightly enough that part of the mode would

have to travel faster than light in the cladding; it is radiated. Macrobends of radius below about 15 mm cause measurable loss in G.652 fibre, especially at 1550 and 1625 nm, which is why technicians test at the longest wavelength to find bends; G.657 fibre tolerates radii down to 5–7.5 mm.

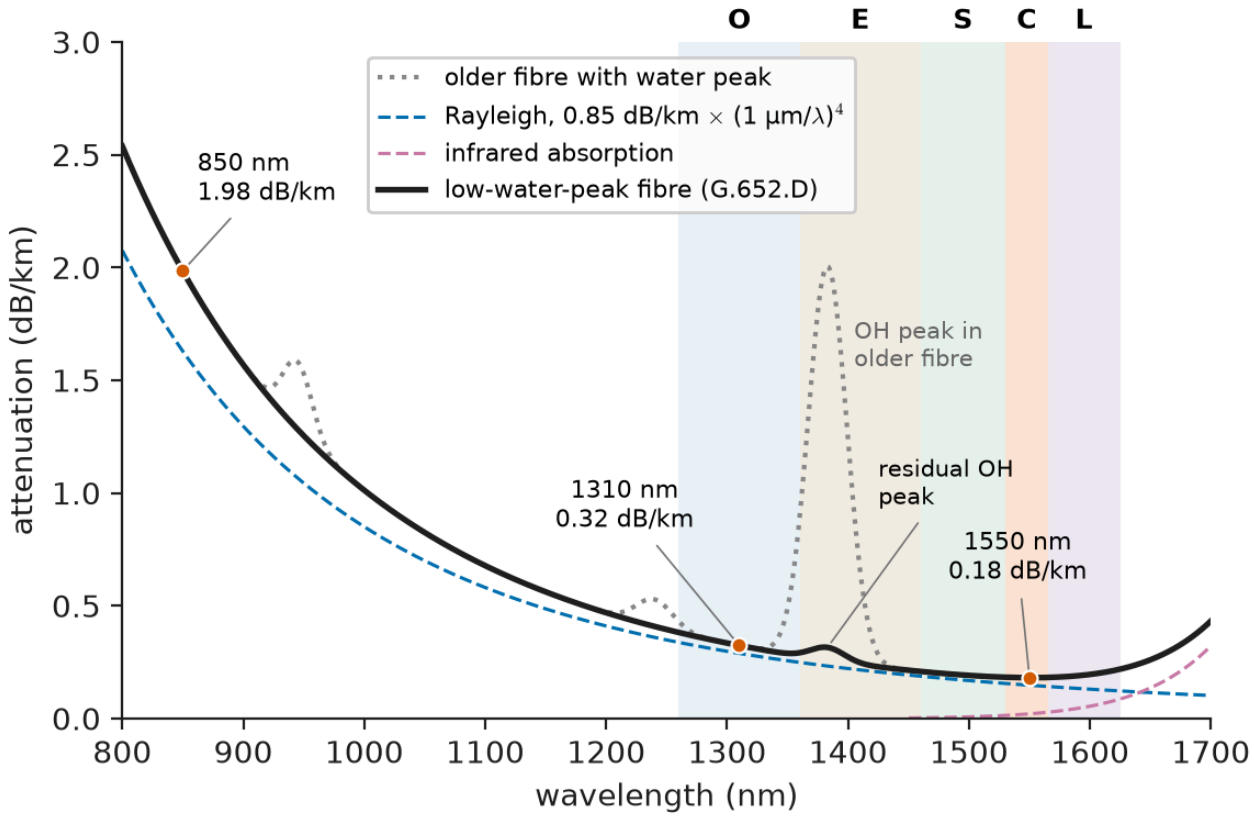

Figure 31.2 — Attenuation spectrum of low-water-peak single-mode silica fibre with the Rayleigh and infrared contributions and the conventional transmission bands.

Figure 31.2 shows the result: three historical windows at 850, 1310 and 1550 nm, now merged into a continuous low-loss region labelled by ITU bands O, E, S, C and L. Table 31.2 gives representative values for the discrete losses that are added to the distributed loss.

Table 31.2 — Representative attenuation contributions.

Item	Typical loss	Notes
G.652.D fibre at 1310 nm	0.32–0.35 dB/km	Cabled values about 0.03 dB/km higher
G.652.D fibre at 1550 nm	0.18–0.22 dB/km	Design value 0.22–0.25 dB/km cabled
Ultra-low-loss pure-silica-core fibre, 1550 nm	0.15–0.16 dB/km	Submarine and premium long haul
OM3/OM4 multimode at 850 nm	2.3–3.0 dB/km	Short reach only
Fusion splice	0.02–0.1 dB	Mean about 0.05 dB with modern splicers
Mechanical splice	0.1–0.3 dB	Field repairs
Connector pair (single-mode, PC/APC)	0.2–0.5 dB	Budget 0.5 dB; 0.75 dB worst case in some standards
1:2 passive splitter	3.5–3.8 dB	3.01 dB ideal plus excess loss
1:32 passive splitter (PON)	16–17.5 dB	$10\log_{10}32 = 15.05$ dB ideal
Repair margin on a long route	2–3 dB	For future splices after cable cuts

Worked example — span loss of an 80 km route. A route of 80 km uses cabled G.652.D fibre with a design attenuation of 0.22 dB/km at 1550 nm. The cable is delivered on 4 km drums, so there are $80/4 - 1 = 19$ fusion splices at 0.05 dB each, and a connector at each end at 0.5 dB. The span loss is

$$L_{\text{span}} = 80 \times 0.22 + 19 \times 0.05 + 2 \times 0.5 = 17.6 + 0.95 + 1.0 = 19.55 \text{ dB}$$

The fibre itself accounts for 90 % of the loss. At 1310 nm, with 0.35 dB/km, the same route would lose $28 + 0.95 + 1.0 \approx 30$ dB — beyond the reach of most 10 Gb/s receivers — which is why long unamplified links operate at 1550 nm even though dispersion is worse there.

31.4 Dispersion

Attenuation reduces the energy of a pulse; dispersion spreads it in time. When a pulse spreads into its neighbours' bit slots, it causes intersymbol interference (Chapter 27) that no amount of extra power can cure. Three mechanisms spread a pulse in fibre.

Modal dispersion

In a multimode fibre each mode — each ray angle in the geometrical picture — travels a different path length. In a step-index fibre the axial ray and the ray at the critical angle differ in transit time by

$$\Delta\tau_{\text{mod}} \approx \frac{L\, n_1 \Delta}{c}$$

For $n_1 = 1.48$, $\Delta = 0.01$: $\Delta\tau/L \approx 49$ ns/km. Keeping the spread below about a quarter of a bit period limits the bit rate–distance product to roughly 5 Mb/s·km — fine for an industrial sensor link, useless for data. A **graded-index** profile, with the index falling parabolically from the axis, makes the oblique rays travel faster in the lower-index outer region so that they arrive almost together; the ideal profile reduces the spread to

$$\Delta\tau_{\mathrm{GI}} \approx \frac{L\, n_1 \Delta^2}{8c} \approx 62 \text{ ps/km}$$

an improvement of $8/\Delta = 800$. Real profiles never achieve the ideal, and the industry instead specifies an **effective modal bandwidth** measured with standardised laser launches: 2000 MHz·km for OM3 and 4700 MHz·km for OM4 at 850 nm. That is why a 10 Gb/s short-reach link (10GBASE-SR) is specified to 300 m on OM3 and 400 m on OM4, and 100 Gb/s links over four 25 Gb/s lanes to 70–100 m.

> **Status of the principle: Modal-dispersion estimates**
> *Approximate.* The ray formulas give the scale and the improvement from grading; real multimode bandwidth depends on profile imperfections and on which modes the laser excites, so link design uses measured effective modal bandwidth, not the formula.

Chromatic dispersion

Single-mode fibre removes modal dispersion, but the group velocity of the single mode still depends on wavelength, for two reasons. **Material dispersion** arises because the refractive index of silica varies with wavelength (the same effect that makes a prism separate colours); **waveguide dispersion** arises because the fraction of the mode in the cladding, and hence its effective index, changes with wavelength. Their sum is described by the dispersion parameter

$$D = \frac{1}{L}\frac{d\tau_g}{d\lambda} = -\frac{2\pi c}{\lambda^2}\beta_2 \qquad [\mathrm{ps}/(\mathrm{nm}\cdot\mathrm{km})]$$

A source of spectral width $\Delta\lambda$ produces a delay spread

$$\Delta\tau_{\mathrm{chr}} = |D|\, L\, \Delta\lambda$$

For G.652 fibre the material and waveguide contributions cancel near 1310 nm (zero-dispersion wavelength λ_0 between 1300 and 1324 nm), and D rises to about +17 ps/(nm·km) at 1550 nm with a slope of about 0.056 ps/(nm²·km) there. Dispersion-shifted fibres move λ_0 by enlarging the waveguide contribution with a more complex index profile; non-zero dispersion-shifted fibres (G.655) deliberately leave D at 2–6 ps/(nm·km) at 1550 nm, because zero dispersion lets WDM channels stay in phase and generate strong four-wave-mixing products (Section 31.9).

> **Status of the principle: Linear delay-spread formula $\Delta\tau = |D|L\Delta\lambda$**
> *Approximate.* Valid when the source spectrum is wider than the modulation bandwidth (LEDs, Fabry–Perot lasers) and λ is far from λ_0; for narrow-linewidth lasers the signal's own modulation spectrum sets the spreading, and near λ_0 the dispersion slope (third-order term) takes over.

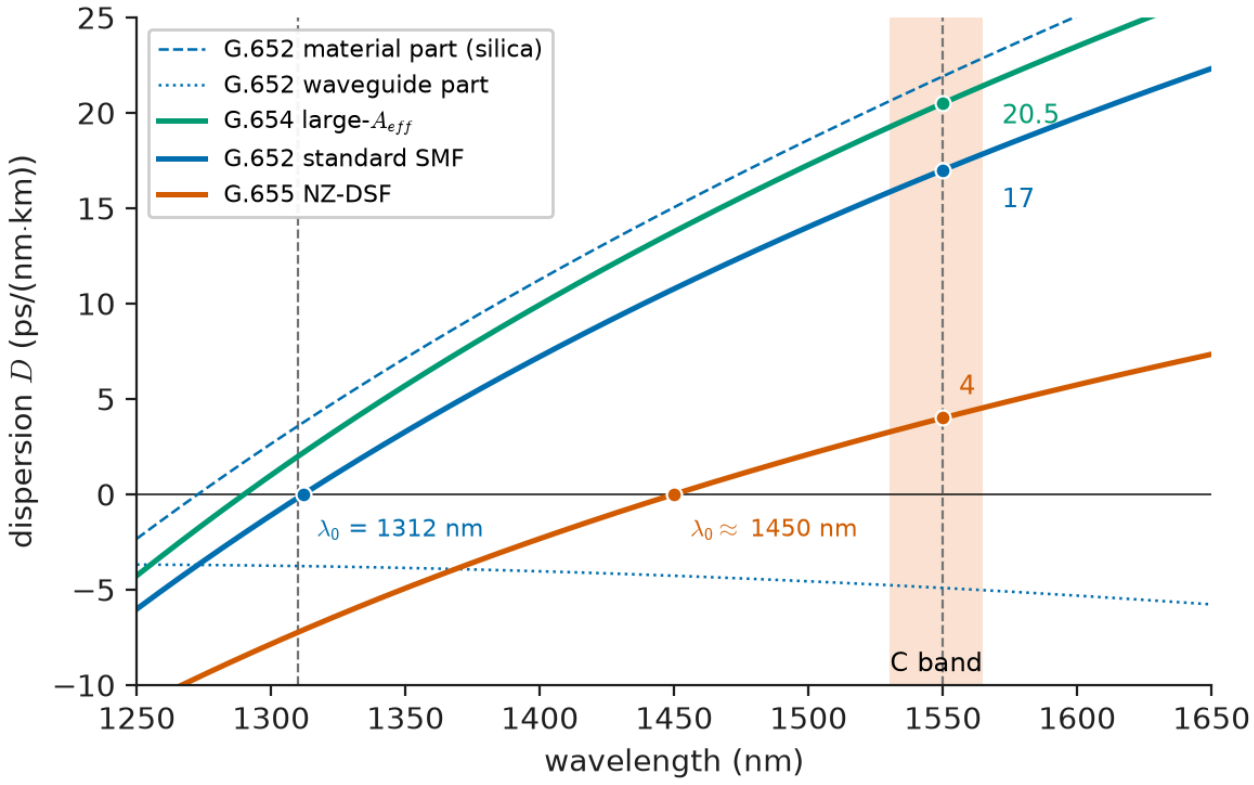

Figure 31.3 — Chromatic dispersion versus wavelength for standard, non-zero dispersion-shifted and large-effective-area single-mode fibres.

Figure 31.3 shows how the three fibre designs place their dispersion relative to the C band.

Worked example — the bit-rate limit of a Fabry–Perot laser at 1550 nm. A low-cost Fabry–Perot (FP) laser emits several longitudinal modes with an rms spectral width of 2 nm. Used at 1550 nm over 40 km of G.652 fibre ($D = 17$ ps/(nm·km)), it produces a spread

$$\Delta\tau = 17 \times 40 \times 2 = 1360 \text{ ps}$$

With the usual rule that the rms spread must stay below a quarter of a bit period, $B < 1/(4\Delta\tau) = 184$ Mb/s. The same laser at 1310 nm, near λ_0, would be limited by the dispersion slope instead and can run at 1–2.5 Gb/s over tens of kilometres — which is why early single-mode systems and today's short-reach O-band transceivers use FP lasers at 1310 nm, and why 1550 nm systems need single-frequency DFB lasers.

Dispersion with single-frequency lasers

When the laser linewidth is far narrower than the modulation bandwidth, the spectral width of the signal is set by the bit rate itself: a 10 Gb/s NRZ signal occupies about $\Delta f \approx 10$ GHz, equivalent to

$$\Delta\lambda = \frac{\lambda^2 \Delta f}{c} = \frac{(1550 \times 10^{-9})^2 \times 10^{10}}{2.998 \times 10^{8}} \approx 0.080 \text{ nm}$$

The spread now grows as $DL\Delta\lambda \propto DLB$, and it is spread relative to a bit period $1/B$ that matters, so the reach scales as $1/B^2$. Using $\beta_2 = -D\lambda^2/(2\pi c) \approx -21.7$ ps²/km and the criterion $B\sqrt{|\beta_2| L} < 1/4$ gives:

Table 31.3 — Approximate chromatic-dispersion reach of chirp-free NRZ on G.652 fibre at 1550 nm.

Bit rate	$B\sqrt{\lvert\beta_2\rvert L} < 1/4$ estimate	Industry dispersion tolerance (≈1–2 dB penalty)	Equivalent length at 17 ps/(nm·km)
2.5 Gb/s	≈460 km	≈16 000 ps/nm	≈900 km
10 Gb/s	≈29 km	≈1000 ps/nm (up to 1600 with negative-chirp EML)	≈60 km (≈95 km)
40 Gb/s	≈1.8 km	≈60 ps/nm	≈4 km

The two columns differ because the criterion is conservative and receivers tolerate some intersymbol interference at a power penalty; the scaling with $1/B^2$ is the robust result. The factor 16 between 10 and 40 Gb/s is why the move to 40 Gb/s in the 2000s was so difficult with direct detection, and why coherent receivers (Section 31.6) took over.

Chirp makes matters worse. A directly modulated laser shifts its frequency by several gigahertz as the drive current changes, broadening the spectrum and, on standard fibre at 1550 nm, compressing or expanding the pulse edges in a way that closes the eye after 20–40 km at 10 Gb/s. That is the failure in this chapter's opening: the cheap modules were directly modulated and the 80 km route accumulated $80 \times 17 = 1360$ ps/nm, beyond what they tolerate.

Dispersion compensation

Because chromatic dispersion is a linear, deterministic all-pass phase $e^{-j\beta_2\omega^2 L/2}$, it can be undone by its inverse: with **dispersion-compensating fibre** ($D \approx -100$ ps/(nm·km); about 13 km compensates 80 km of G.652 at 6–8 dB extra loss), with chirped **fibre Bragg gratings**, or **electronically** — in a coherent receiver an FIR filter (Chapter 23) applies the inverse quadratic phase to the sampled field and removes tens of thousands of ps/nm.

Polarization-mode dispersion

A real fibre is not perfectly circular, and stresses make it slightly birefringent, so the two polarisations of the "single" mode travel at slightly different speeds. The birefringence axes change randomly along the fibre and with temperature, so the differential group delay (DGD) is a random variable whose mean grows as $\sqrt{L}$:

$$\langle \Delta\tau_{\mathrm{PMD}} \rangle = \mathrm{PMD}_{\mathrm{Q}}\sqrt{L}$$

Modern fibre has a link-design value $\mathrm{PMD}_{\mathrm{Q}} \le 0.1$ ps/√km (typically 0.02–0.05), giving 1 ps over 100 km and 3 ps over 1000 km — harmless at 10 Gb/s (bit period 100 ps) but a real constraint on older fibre with 0.5–1 ps/√km at 40 Gb/s. Because PMD is statistical and time-varying, it is budgeted as an outage probability rather than a fixed penalty — the same logic as fading margins in Chapter 30.

> **Status of the principle: $\sqrt{L}$ growth of PMD**
> *Empirical.* Follows from modelling the fibre as many randomly oriented birefringent sections and is confirmed by field measurements; the PMD coefficient itself is a measured property of each cable, and the instantaneous DGD follows an approximately Maxwellian distribution.

31.5 Optical transmitters

An optical transmitter converts an electrical data signal into a modulated optical power (or, for coherent systems, a modulated optical field). Four source types cover nearly all practice.

Light-emitting diodes

An LED emits by spontaneous recombination in a forward-biased junction. The light is incoherent, spectrally broad (30–150 nm) and spread over a wide angle, so only microwatts couple into multimode fibre and essentially nothing into single-mode fibre; carrier lifetime limits modulation to roughly 50–200 MHz. LEDs survive in plastic-fibre links and optocouplers because they are cheap and robust.

Laser diodes: threshold and slope efficiency

A semiconductor laser adds optical feedback (cleaved facets or gratings) so that stimulated emission dominates once the gain overcomes the losses. Its light–current (L–I) characteristic, shown in Figure 31.4, has two regimes: below the **threshold current** I_{th} the output is weak spontaneous emission; above it the output power rises linearly,

$$P_{\mathrm{out}} \approx \eta_s \, (I - I_{\mathrm{th}})$$

with slope efficiency η_s in W/A. Typical values for a 1550 nm DFB laser are $I_{\mathrm{th}} = 8$–15 mA and $\eta_s = 0.1$–0.3 W/A at 25 °C. The threshold rises roughly exponentially with temperature (characteristic temperature T_0 of 50–100 K for InGaAsP lasers), and the slope efficiency falls, so laser drivers either regulate temperature with a thermoelectric cooler or use automatic power control from a monitor photodiode. The **extinction ratio** $r_e = P_1/P_0$ between "1" and "0" levels, typically 6–10 dB (4–10 linear), is a design compromise: biasing well above threshold for the "0" level gives speed and low chirp but wastes power in the zero, which costs receiver sensitivity.

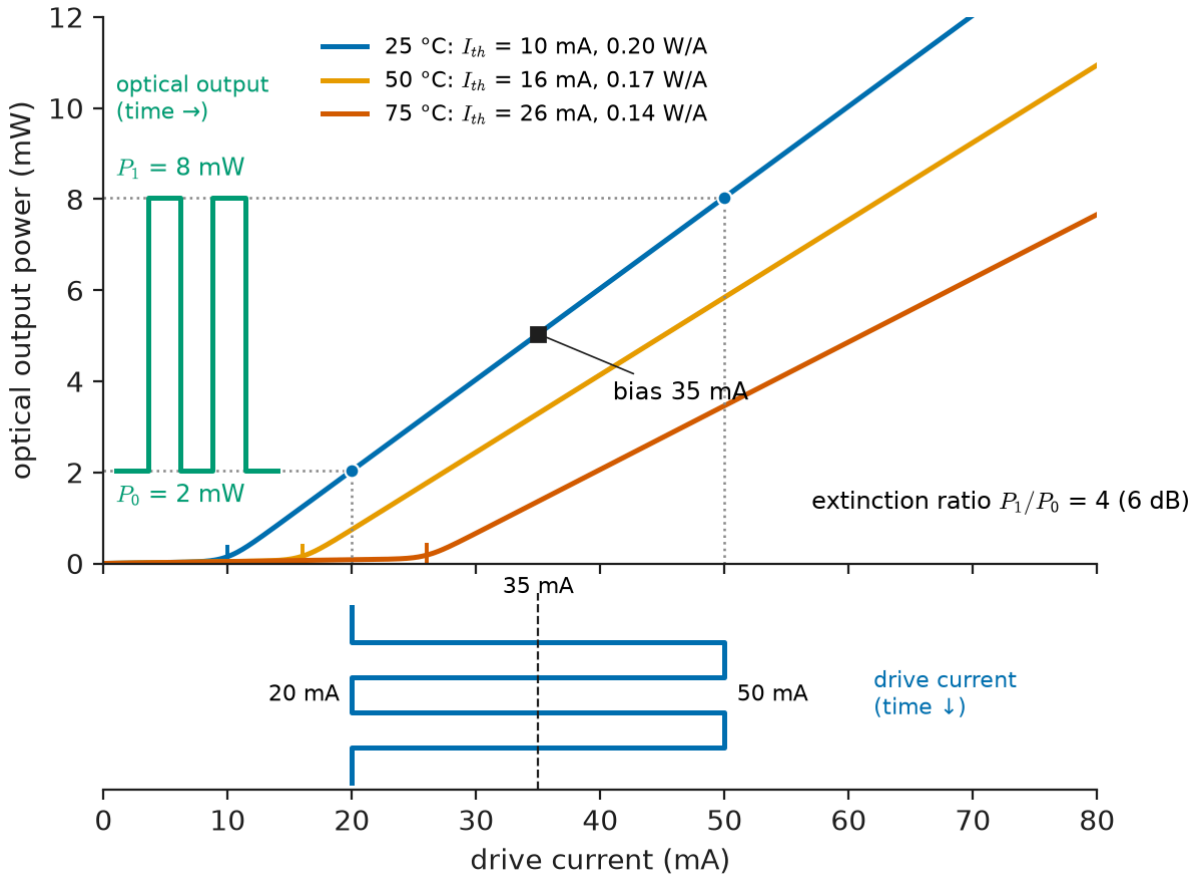

Figure 31.4 — Light–current characteristic of a 1550 nm DFB laser at three temperatures, with the bias point and modulation swing for on-off keying.

Laser structures

Fabry–Perot (FP) lasers use the cleaved facets as mirrors and oscillate on several longitudinal modes; their 1–3 nm spectral width restricts them to 1310 nm or to short distances at 1550 nm (Section 31.4). **Distributed-feedback (DFB) lasers** etch a Bragg grating along the active region so that only one mode oscillates (side-mode suppression 35–50 dB, linewidth 1–10 MHz); they are the workhorse at 1310 and 1550 nm and the source for DWDM. **VCSELs** emit perpendicular to the wafer, with sub-milliampere thresholds and circular beams that couple well into multimode fibre; 850 nm VCSELs over OM3/OM4 carry almost every short data-centre link. Coherent systems need **tunable external-cavity lasers** with 10–300 kHz linewidth so that phase noise does not swamp the symbols.

Table 31.4 — Representative optical sources.

Source	Wavelength	Spectral width	Coupled power	Modulation	Typical application
LED	650, 850, 1300 nm	30–150 nm	−30 to −15 dBm (MMF/POF)	≤200 MHz	POF, industrial, optocouplers
VCSEL	850 nm (also 940, 1310)	<0.45 nm rms	−3 to +2 dBm	Direct, 10–50 Gb/s	Data-centre short reach over OM3/OM4
Fabry–Perot laser	1310 nm	1–3 nm rms	−3 to +3 dBm	Direct, ≤10 Gb/s	Short single-mode links, PON upstream
DFB laser	1270–1610 nm	1–10 MHz linewidth, 35–50 dB SMSR	0 to +10 dBm	Direct ≤25 Gb/s or external	Metro and long haul, CWDM, DWDM
Electro-absorption modulated laser (EML)	1310 / 1550 nm	DFB plus small chirp	−2 to +3 dBm	10–100 Gb/s	10G ZR, 100G/400G PAM-4
Tunable laser + Mach–Zehnder IQ modulator	C and L band	10–300 kHz linewidth	0 to +3 dBm per channel	30–130 GBd QAM	Coherent 100G–1.2T

Direct and external modulation

Direct current modulation is limited by the laser's relaxation-oscillation bandwidth and by chirp. **External modulation** keeps the laser running continuously and modulates its output afterwards, either with an **electro-absorption modulator** integrated with the DFB (the EML) or with a **Mach–Zehnder modulator** (MZM), which splits the light into two arms whose phases are shifted by an applied voltage (via the electro-optic effect in lithium niobate, indium phosphide or silicon) and recombines them; the output field is proportional to $\cos(\pi V / 2V_\pi)$, where V_π is the voltage for a π phase shift (typically 2–5 V). Biased at the null and driven between $\pm V_\pi$, the MZM produces a chirp-free field that takes both positive and negative values — the optical equivalent of a balanced modulator (Chapter 26). Two MZMs in quadrature, one for each of I and Q, and two such pairs on orthogonal polarisations, form the dual-polarisation IQ modulator that generates the QPSK and QAM constellations of Chapter 27 at optical frequencies.

> **Boundary.**
> Laser-driver and modulator-driver circuits, their transistor stages and the thermoelectric control loop are the subject of *Electronics & Computing*; the semiconductor physics of gain, recombination and the electro-optic effect belongs to *Engineering Physics* and *The Materials Core*. This chapter uses their measured characteristics.

Intensity modulation and noise

Most short-reach links use **intensity modulation with direct detection** (IM/DD): the transmitted *power* carries the information, as NRZ on-off keying or four-level PAM-4 at 50–100 Gb/s per lane. The laser adds **relative intensity noise** (RIN, typically −140 to −155 dB/Hz for a DFB), which matters only at high received power. Reflections into the laser worsen RIN and linewidth, which is why transmitters contain isolators and PON and video systems use angled (APC) connectors.

31.6 Optical receivers

Photodetectors and responsivity

A photodiode absorbs photons and generates electron–hole pairs; reverse bias sweeps them out as a photocurrent proportional to the optical power,

$$I_p = \mathcal{R} P, \qquad \mathcal{R} = \frac{\eta q}{h f} = \frac{\eta\, \lambda[\mu\mathrm{m}]}{1.24}\ \mathrm{A/W}$$

where η is the quantum efficiency. Silicon absorbs well below about 1000 nm and is used at 850 nm ($\mathcal{R} \approx 0.5$ A/W). Indium gallium arsenide (InGaAs) covers 1000–1650 nm; with $\eta = 0.8$ at 1550 nm, $\mathcal{R} = 0.8 \times 1.55/1.24 = 1.0$ A/W. A **PIN photodiode** inserts an intrinsic region between the p and n layers so that absorption happens in a depleted zone; its bandwidth is set by transit time and capacitance (tens of gigahertz for small-area devices). An **avalanche photodiode** (APD) adds a high-field region in which carriers gain enough energy to ionise further pairs, multiplying the photocurrent by a mean gain M, typically 5–20 for InGaAs APDs.

The photocurrent drives a **transimpedance amplifier** (TIA) that converts microamperes into millivolts with low noise, followed by a limiting amplifier or an ADC, a clock-and-data-recovery circuit and a decision circuit.

> **Status of the principle: Linear photodetection $I_p = \mathcal{R}P$**
> *Fundamental.* Each absorbed photon releases at most one electron–hole pair (before avalanche gain), so current is proportional to photon flux; it is a square-law detector of the optical *field*, which is why intensity-modulated systems lose the optical phase and need coherent detection to recover it.

Noise sources

Three noise currents compete with the signal in a direct-detection receiver.

Shot noise arises because photoelectrons arrive at random (Poisson statistics). Its variance in a noise bandwidth B_n is

$$\sigma_{\mathrm{shot}}^2 = 2q\, I_p\, B_n$$

Thermal (circuit) noise is the Johnson noise of the load or feedback resistor plus the amplifier's own noise (Chapter 29). For a simple load resistor R_L it is $4kTB_n/R_L$; for a TIA it is specified as an input-referred noise current density i_n in pA/√Hz, giving $\sigma_{\text{th}}^2 = i_n^2 B_n$.

APD excess noise. Avalanche multiplication is itself random, so the shot noise is multiplied by M^2 and by an excess-noise factor

$$F_A = k_A M + (1 - k_A)\left(2 - \frac{1}{M}\right)$$

where k_A is the ratio of hole to electron ionisation coefficients (about 0.02–0.05 for silicon, 0.4–0.5 for InGaAs/InP). The signal grows as M, the shot noise as $M\sqrt{F_A}$, the thermal noise not at all: there is an optimum gain at which the APD's growing shot noise starts to outweigh its advantage over the fixed thermal noise.

> **Status of the principle: Shot-noise formula $\sigma^2 = 2qIB$**
> *Derived*. Follows from Poisson photon (or electron) statistics in the limit of many events per bit period; with fewer than a few hundred photons per bit the Gaussian approximation degrades and exact Poisson statistics are needed.

Q factor and BER

For on-off keying with Gaussian noise of standard deviation σ_1 on the "1" level and σ_0 on the "0" level, and the decision threshold placed optimally, the bit-error ratio is (Chapter 29)

$$\text{BER} = \frac{1}{2}\operatorname{erfc}\!\left(\frac{Q}{\sqrt{2}}\right), \qquad Q = \frac{I_1 - I_0}{\sigma_1 + \sigma_0}$$

Q = 6 gives BER ≈ 1.0×10^{-9} and Q = 7 gives 1.3×10^{-12}. Optical engineers often quote Q in decibels as $20\log_{10} Q$: 15.6 dB and 16.9 dB for those two values. Modern links rarely need such low raw error rates because forward error correction (Chapter 28) cleans up the stream: a hard-decision FEC threshold of 3.8×10^{-3} corresponds to $Q = 2.67$ (8.5 dB), and soft-decision FECs operate at pre-FEC BERs of 10^{-2} or higher. The receiver **sensitivity** is the minimum average received power for a specified BER.

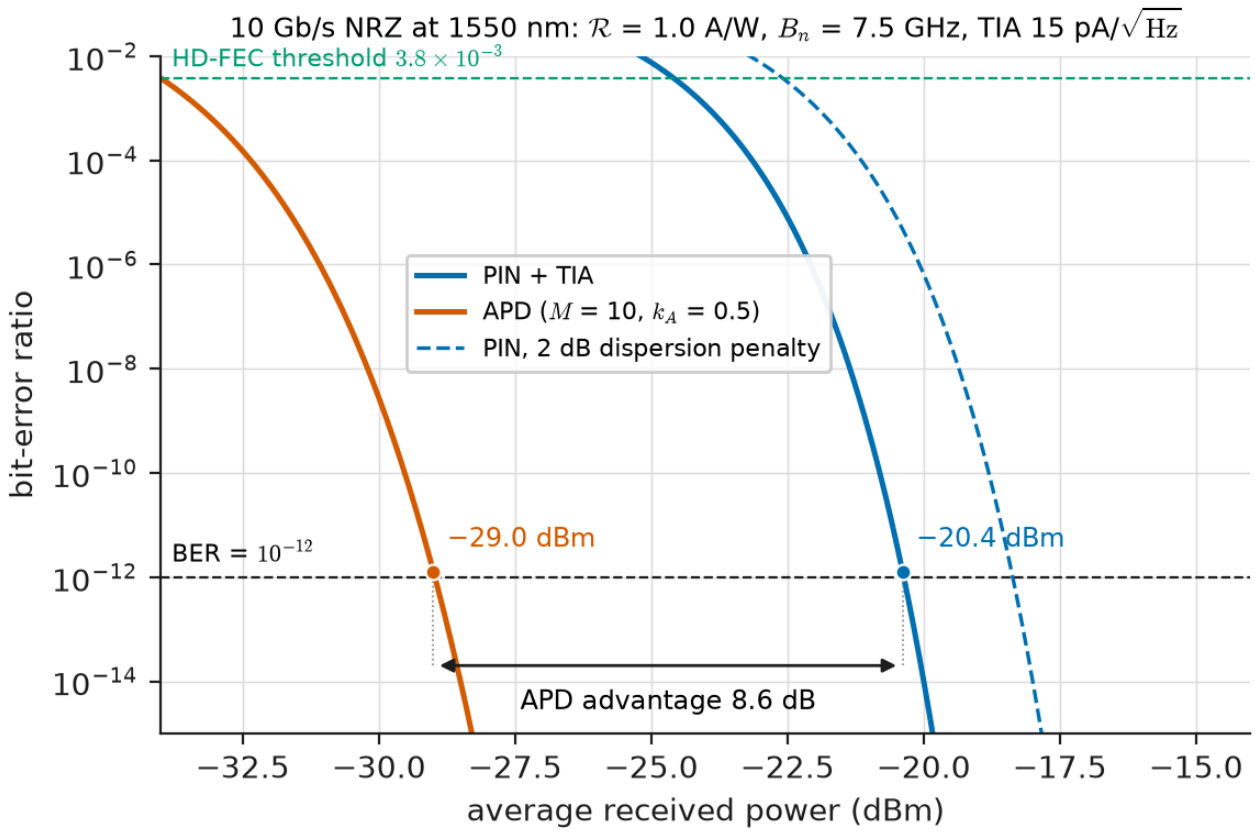

Figure 31.5 — Bit-error ratio versus average received power for a 10 Gb/s InGaAs PIN–TIA receiver and an APD receiver with gain 10, at 1550 nm.

Worked example — sensitivity of a 10 Gb/s PIN and APD receiver. A 10 Gb/s NRZ receiver at 1550 nm uses an InGaAs photodiode with $\mathcal{R} = 1.0$ A/W and a TIA with input noise density 15 pA/√Hz; the noise bandwidth is $B_n = 7.5$ GHz. The received average power is -20 dBm $= 10\ \mu$W with infinite extinction ratio, so $P_1 = 20\ \mu$W and $P_0 = 0$.

Thermal noise: $\sigma_{\text{th}} = 15 \times 10^{-12}\sqrt{7.5 \times 10^{9}} = 1.30\ \mu$A.

Signal: $I_1 = \mathcal{R}P_1 = 20\ \mu$A, $I_0 = 0$.

Shot noise on the "1": $\sigma_{\text{shot}} = \sqrt{2 \times 1.602 \times 10^{-19} \times 20 \times 10^{-6} \times 7.5 \times 10^{9}} = 0.22\ \mu$A, so $\sigma_1 = \sqrt{1.30^2 + 0.22^2} = 1.32\ \mu$A and $\sigma_0 = 1.30\ \mu$A.

$$Q = \frac{20}{1.32 + 1.30} = 7.64, \qquad \text{BER} = \tfrac{1}{2}\operatorname{erfc}(7.64/\sqrt{2}) \approx 1 \times 10^{-14}$$

The receiver is thermal-noise limited: shot noise adds only 1.4 % to σ_1. Solving for Q = 7 gives a sensitivity of -20.4 dBm. Replacing the PIN with an APD of gain $M = 10$ and $k_A = 0.5$ ($F_A = 0.5 \times 10 + 0.5 \times 1.9 = 5.95$) multiplies the signal current by ten while the APD shot noise grows only as $M\sqrt{F_A}$; repeating the calculation gives a Q = 7 sensitivity of -29.0 dBm, an improvement of 8.6 dB (Figure 31.5). Raising the gain further helps little: $M = 15$ gives -29.6 dBm, $M = 20$ gives -29.6 dBm. Commercial modules quote about -18 dBm (PIN) and -26 dBm (APD) at 10 Gb/s, 2–3 dB worse than these ideal figures, because of finite extinction ratio, intersymbol interference in the receiver filter and design margin.

The quantum limit for ideal on-off keying, about 10 photons per bit for BER 10^{-9}, is 12.8 nW (−49 dBm) at 10 Gb/s; practical direct-detection receivers sit 20–30 dB above it because the electronics, not the photons, set the noise.

Coherent detection

A **coherent receiver** mixes the incoming signal with a local-oscillator laser in a 90° optical hybrid and detects the beat products on balanced photodiodes, giving photocurrents proportional to the in-phase and quadrature components of the received *field* in each polarisation — the baseband I/Q of a radio receiver (Chapter 26). ADCs sample them and a DSP chip compensates chromatic dispersion, separates polarisations with an adaptive 2×2 MIMO equaliser (Chapter 24), recovers carrier phase and decodes the FEC. The strong local oscillator lifts the signal above circuit noise, phase recovery permits QPSK and QAM, and linear impairments become digital problems. Long-haul links above 100 Gb/s per wavelength are now coherent.

Table 31.5 — Representative receiver figures (1550 nm unless stated).

Receiver	Responsivity	Multiplication	Typical sensitivity	Notes
Si PIN, 850 nm	≈0.5 A/W	1	−12 to −10 dBm at 10–25 Gb/s	VCSEL short reach (stressed sensitivity)
InGaAs PIN + TIA	0.8–1.0 A/W	1	≈−18 dBm at 10 Gb/s	10GBASE-LR, metro
InGaAs APD + TIA	0.8–1.0 A/W (unity gain)	5–15 ($k_A \approx 0.4$–0.5)	≈−26 to −28 dBm at 10 Gb/s	10G ER/ZR, PON
EDFA pre-amplified PIN	1.0 A/W	Optical gain 20–30 dB	≈−30 to −35 dBm at 10 Gb/s	Long unrepeated links
Coherent DP-QPSK 100 Gb/s	Balanced PIN pairs	LO gain	OSNR-limited: ≈11–12 dB required OSNR (0.1 nm)	Metro / long haul

31.7 Wavelength-division multiplexing

The idea

Each laser–receiver pair uses only a small part of the fibre's 4–10 THz of low-loss bandwidth. **Wavelength-division multiplexing** (WDM) sends many independently modulated carriers at different wavelengths down the same fibre — frequency-division multiplexing (Chapter 25) with optical carriers. A passive multiplexer combines them (arrayed-waveguide gratings or thin-film filters) and a demultiplexer separates them at the far end. Because the fibre is linear at moderate powers, the channels do not interact and each behaves as an independent link.

> **Status of the principle: Independence of WDM channels**
> *Approximate.* Holds while the total power is low enough that fibre nonlinearity (cross-phase modulation, four-wave mixing, stimulated Raman scattering) is negligible and filter crosstalk is below about −25 dB; at high channel counts and launch powers the channels do interact, which sets the optimum launch power (Section 31.9).

Grids and families

Coarse WDM (CWDM, ITU-T G.694.2) uses 18 channels at 20 nm spacing from 1271 to 1611 nm, wide enough for uncooled lasers to drift without leaving their channel; it serves metro access and mobile front-haul but cannot be amplified across its whole range.

Dense WDM (DWDM, ITU-T G.694.1) places channels on a frequency grid anchored at 193.1 THz with spacings of 100, 50 or 25 GHz, or on a flexible grid of 12.5 GHz slots for coherent super-channels. A spacing of 100 GHz corresponds to 0.80 nm at 1550 nm, 50 GHz to 0.40 nm. Lasers must be temperature-stabilised to a few gigahertz.

Optical amplifiers

What made DWDM dominant is the **erbium-doped fibre amplifier** (EDFA): a few metres of fibre doped with erbium ions, pumped by a 980 or 1480 nm laser, which amplifies all wavelengths across the C band (1530–1565 nm) — or the L band (1565–1625 nm) with a longer doped fibre — simultaneously and without converting them to electrical form. One amplifier replaces dozens of electronic regenerators. Typical gain is 20–30 dB, output power +17 to +23 dBm total, and noise figure 4.5–6 dB (the quantum limit for a high-gain phase-insensitive amplifier is 3 dB). **Raman amplification**, which uses the fibre itself as the gain medium pumped about 100 nm below the signal, provides distributed gain with lower effective noise figure and is used on long spans.

Capacity

Worked example — DWDM channel count and the OSNR budget. The C band from 1530 to 1565 nm covers 4.38 THz (Section 31.1): about 44 channels at 100 GHz spacing and 88 at 50 GHz. At 10 Gb/s per channel the 50 GHz grid carries about 0.9 Tb/s; with 200 Gb/s coherent channels it carries 12–17 Tb/s, and C+L-band systems with 800 Gb/s wavelengths reach 40–60 Tb/s per fibre pair.

Every EDFA adds amplified spontaneous emission (ASE) noise, and the ratio of signal to ASE power in a reference bandwidth of 0.1 nm (12.5 GHz) is the **optical signal-to-noise ratio** (OSNR). For a chain of N identical spans, each with loss L_{span} exactly compensated by an amplifier of noise figure NF, a widely used design rule is

$$\mathrm{OSNR}_{0.1\,\mathrm{nm}}\,[\mathrm{dB}] \approx 58 + P_{\mathrm{ch}}\,[\mathrm{dBm}] - L_{\mathrm{span}}\,[\mathrm{dB}] - NF\,[\mathrm{dB}] - 10\log_{10} N$$

where 58 dB is $-10\log_{10}(h f\, B_{\mathrm{ref}})$ in dBm at 1550 nm. With $P_{\mathrm{ch}} = 0$ dBm, $L_{\mathrm{span}} = 20$ dB, $NF = 5$ dB and $N = 10$ spans (about 800 km of 80 km spans):

$$\mathrm{OSNR} \approx 58 + 0 - 20 - 5 - 10 = 23\ \mathrm{dB}$$

A 100 Gb/s DP-QPSK transceiver needs about 11–12 dB at its FEC threshold, so this link has some 11 dB of margin; a 400 Gb/s DP-16QAM channel needs about 20–22 dB and would barely close. The OSNR rule exposes the levers: every 3 dB more launch power or 3 dB less span loss buys 3 dB of OSNR, and every doubling of the number of spans costs 3 dB. Shorter spans and lower-loss fibre are therefore worth more than better amplifiers, whose noise figure cannot go below 3 dB.

> **Status of the principle: The "58 dB" OSNR rule**
> *Approximate.* Assumes identical spans with loss fully compensated, ASE-limited noise, a 0.1 nm reference bandwidth at 1550 nm and no nonlinear noise; it overestimates OSNR at high launch powers where nonlinear interference adds noise of its own.

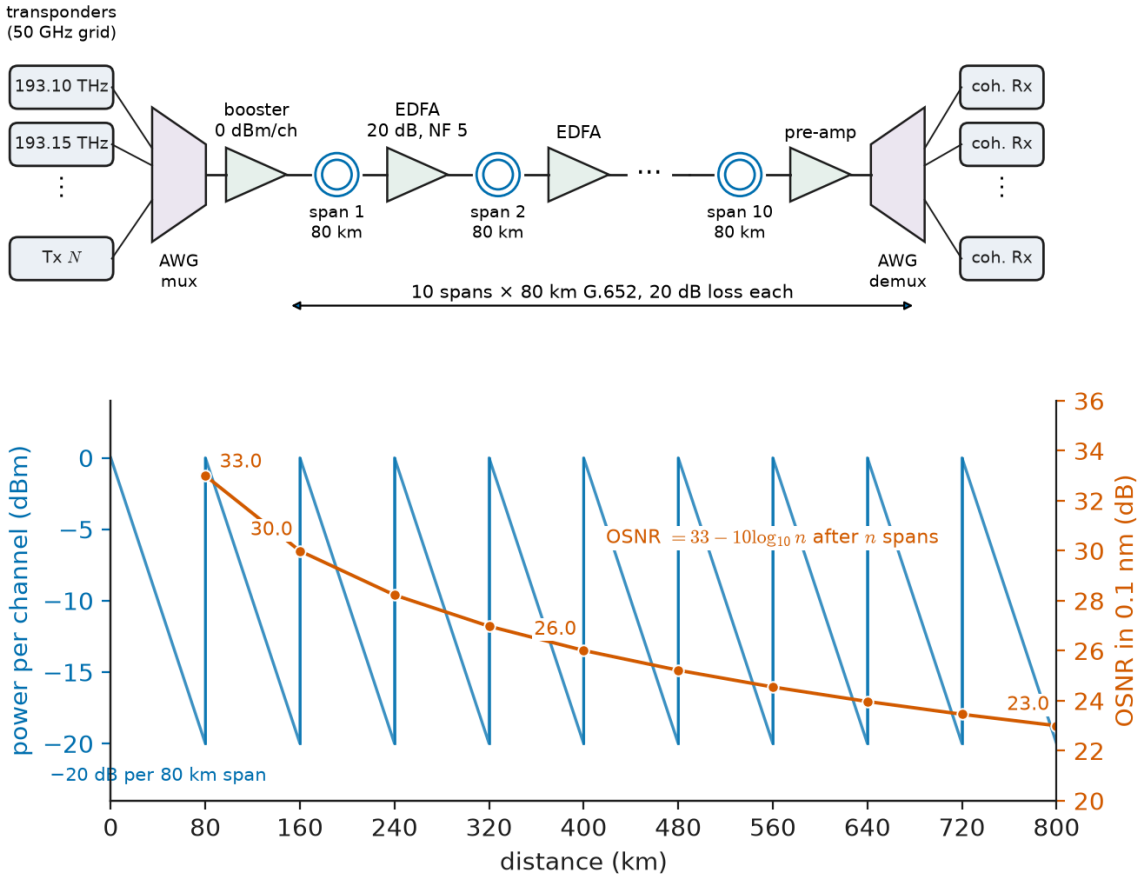

Figure 31.6 — A DWDM line system: transponders, multiplexer, booster amplifier, fibre spans with in-line EDFAs, pre-amplifier, demultiplexer and receivers, with the power and OSNR evolution along the route.

Figure 31.6 shows the saw-tooth of signal power along such a line and the steady decline of OSNR as ASE accumulates.

31.8 Optical link budgets

The power budget

An optical link budget is the same bookkeeping as the radio link budget of Chapter 30, with simpler propagation and different margins:

$$P_{\mathrm{rx}} = P_{\mathrm{tx}} - \alpha L - n_s\, a_s - n_c\, a_c - L_{\mathrm{other}}$$

$$M = P_{\mathrm{rx}} - S_{\mathrm{rx}} - P_{\mathrm{penalty}}$$

where S_{rx} is the receiver sensitivity, $n_s a_s$ and $n_c a_c$ the splice and connector losses, L_{other} splitters or passive filters, and P_{penalty} the power penalties from dispersion, extinction ratio, reflections and ageing. A margin M of 3 dB is a common minimum for a new link; operators of long routes reserve more for future cable repairs. Very short links may instead exceed the receiver's **overload** level and need an attenuator.

The rise-time (dispersion) budget

The second budget asks whether the link is fast enough. The 10–90 % rise times of transmitter, fibre and receiver add approximately in quadrature,

$$t_{\mathrm{sys}} = \sqrt{t_{\mathrm{tx}}^2 + t_{\mathrm{mod}}^2 + t_{\mathrm{chr}}^2 + t_{\mathrm{rx}}^2}$$

and must stay below about 70 % of the bit period for NRZ (35 % for RZ). In modern single-mode systems the rise-time budget has been replaced by the dispersion-tolerance specification of the transceiver (in ps/nm) and a power penalty, but the logic is the same: a link must close in *both* power and bandwidth.

> **Status of the principle: Quadrature addition of rise times**
> *Approximate.* Exact for cascaded Gaussian impulse responses; acceptable for most practical responses, but it misrepresents strongly chirped or non-Gaussian dispersion, which is why modern specifications use measured dispersion penalties instead.

Worked example — closing an 80 km, 10 Gb/s link. Take the 80 km span of Section 31.3 (19.55 dB loss) and a transmitter launching +2 dBm.

$$P_{\mathrm{rx}} = 2 - 19.55 = -17.55 \text{ dBm}$$

Option A: PIN receiver, datasheet sensitivity −18 dBm. The raw margin is only 0.45 dB before any dispersion penalty. The route accumulates $80 \times 17 = 1360$ ps/nm, beyond the ≈1000 ps/nm tolerance of a chirp-free 10 Gb/s receiver, so the link fails on dispersion as well as on power — the opening scenario.

Option B: EML transmitter and APD receiver, datasheet sensitivity −26 dBm, dispersion tolerance 1600 ps/nm with a 2 dB penalty budgeted.

$$M = -17.55 - (-26) - 2 = 6.45 \text{ dB}$$

The link closes with more than 3 dB to spare. This is essentially the 10GBASE-ZR "80 km" module class: an externally modulated laser that tolerates the dispersion and an APD that supplies the sensitivity.

Option C: 1310 nm, where dispersion is near zero. The loss at 0.35 dB/km is about 30 dB; with a +2 dBm source and a −26 dBm APD the margin is $2 - 30 - (-26) = -2$ dB. The link fails on power. At 80 km, 1550 nm with an EML is the only direct-detection option that closes.

Table 31.6 — Link-budget template for the 80 km example (Option B).

Item	Value	Running total
Transmitter launch power	+2.0 dBm	+2.0 dBm
Fibre, 80 km × 0.22 dB/km	−17.6 dB	−15.6 dBm
19 fusion splices × 0.05 dB	−0.95 dB	−16.55 dBm
2 connectors × 0.5 dB	−1.0 dB	−17.55 dBm
Received power		−17.55 dBm
Receiver sensitivity (APD, BER 10^{-12})	−26.0 dBm	
Dispersion penalty (1360 ps/nm)	2.0 dB	
Margin	**6.45 dB**	

Passive optical networks

Fibre-to-the-home uses **passive optical networks** (PONs): one exchange transceiver serves 32 or 64 homes through a passive splitter whose loss is $10\log_{10} N$ plus excess — 15.05 dB ideal, about 17 dB in practice for 1:32 — so a 20 km PON budget is dominated by the split, not the fibre. GPON Class B+ allows 28 dB between transceivers. Downstream data is broadcast and encrypted; upstream uses time-division multiple access (Chapter 30).

31.9 System limitations

A designer who has closed the power and dispersion budgets still faces limits that are not captured by either. They divide naturally by the regime in which they bite.

Power-limited versus dispersion-limited regimes

The loss limit $L_\alpha = (P_{\text{tx}} - S_{\text{rx}} - M)/\alpha$ falls only slowly with bit rate (thermal-limited sensitivity worsens by roughly 1.5 dB per doubling), whereas the dispersion limit falls as $1/B$ for broad sources and $1/B^2$ for narrow ones. Low-rate links are therefore **power-limited** and high-rate links **dispersion-limited**; for G.652 at 1550 nm with a DFB laser the crossover lies around 2.5–10 Gb/s over 50–100 km. Amplifiers remove the power limit and compensation the dispersion limit; what remains is noise and nonlinearity.

Fibre nonlinearity

The refractive index of silica depends slightly on optical intensity, $n = n_0 + n_2 I$, with $n_2 \approx 2.6 \times 10^{-20}$ m²/W. The resulting nonlinear coefficient is

$$\gamma = \frac{2\pi n_2}{\lambda A_{\text{eff}}} = \frac{2\pi \times 2.6 \times 10^{-20}}{1.55 \times 10^{-6} \times 80 \times 10^{-12}} \approx 1.3 \ \text{W}^{-1}\text{km}^{-1}$$

Because the power decays along the span, nonlinear effects accumulate over an **effective length** $L_{\text{eff}} = (1 - e^{-\alpha_n L})/\alpha_n$, about 21 km for an 80 km span at 0.2 dB/km. A channel at 10 mW acquires a nonlinear phase $\gamma P L_{\text{eff}} \approx 1.3 \times 0.01 \times 21.2 \approx 0.28$ rad per span — small in one span, significant after twenty. The resulting effects are:

- **Self-phase modulation (SPM):** each channel modulates its own phase in proportion to its own power, broadening the spectrum and interacting with dispersion.
- **Cross-phase modulation (XPM):** the power of neighbouring WDM channels modulates a channel's phase.
- **Four-wave mixing (FWM):** three channels beat to produce new frequencies $f_i + f_j - f_k$ that fall on other channels of an equally spaced grid. FWM is efficient only when the channels stay in phase, i.e. near zero dispersion — the reason G.655 fibre retains a small non-zero D.
- **Stimulated Brillouin scattering (SBS):** a narrow-linewidth carrier above a threshold of roughly +5 to +10 dBm is reflected back by an acoustic wave it generates; it limits the power of unmodulated or analogue-video carriers and is countered by dithering the laser frequency.
- **Stimulated Raman scattering (SRS):** transfers power from short-wavelength to long-wavelength channels across tens of terahertz, tilting the spectrum of wideband C+L systems.

> **Status of the principle: Kerr nonlinearity of silica**
> *Approximate.* The intensity-dependent index is the leading term of the third-order nonlinear susceptibility, accurate at the powers used in telecommunications (milliwatts to around a watt total); its effects on a system are captured by perturbation models (the Gaussian-noise model) that treat nonlinear interference as an extra additive noise growing as the cube of launch power.

The optimum launch power and the capacity of a fibre

The interplay between ASE and nonlinearity gives every long-haul system an **optimum launch power**. Raising P_{ch} improves OSNR one-for-one (Section 31.7) but the nonlinear interference power grows as P_{ch}^3; the total signal-to-noise ratio peaks where the nonlinear noise is half the ASE noise, typically at 0 to +3 dBm per 50 GHz channel on G.652 fibre. Beyond that, adding power makes the link *worse*. This means that, unlike the additive-white-Gaussian-noise channel of Chapter 28, the capacity of a fibre does not grow without limit as power increases — the "nonlinear Shannon limit", estimated at roughly 6–8 b/s/Hz per polarisation over transoceanic distances and higher over shorter links. Commercial coherent systems now operate within 1–2 dB of it.

Other limits

Laser phase noise limits constellation density in coherent systems; connector reflections cause multipath interference in IM/DD links; PMD and polarisation-dependent loss become outage limits on

old routes; and the bandwidth of modulators and of the ADCs and DACs in coherent DSP sets the symbol rate (presently 130–200 GBd).

> **Boundary.**
> Fibre amplifiers, nonlinear propagation and the full Gaussian-noise model are treated in specialised optical-networking texts; the DSP algorithms of coherent receivers use the equalisation and adaptive-filtering methods of Chapters 23 and 24, and the coding of Chapter 28.

What remains open

Open question	Where it is answered
How are the QPSK and QAM constellations used on coherent wavelengths generated and detected, and what is their error probability?	Chapter 27 (digital modulation, matched filtering, detection)
How close can a coded system get to Shannon capacity, and how do soft-decision FECs work?	Chapter 28 (information theory and coding)
Where do the Gaussian noise and Q-factor results come from?	Chapter 29 (random signals and communication noise)
How are the adaptive equalisers in a coherent DSP designed?	Chapters 23 and 24 (digital filters, LMS adaptive filtering)
How does a waveguide support discrete modes, and how are high-frequency electrical interconnects in a transceiver designed?	Chapter 32 (RF and microwave fundamentals); Chapter 14 (transmission lines)
How are laser drivers, TIAs and photodiode bias circuits built?	*Electronics & Computing*
Why does silica scatter as λ^{-4}, and how is a low-loss preform made?	*The Materials Core*; *Engineering Physics*
How does a laser's automatic power control or a wavelength locker keep its output stable?	Chapters 34–35 (feedback and control design)

Next: Chapter 32 returns from optical frequencies to the radio and microwave bands, where circuits become distributed, impedance matching and S-parameters replace Ohm's law, and antennas launch signals into free space.

Chapter 32 — RF and Microwave Fundamentals

Part VI · Communication Theory

A team builds a 2.4 GHz sensor board. The power amplifier, taken from a reference design, should deliver +20 dBm into the antenna. On the bench, with a power meter on the output connector, it delivers +14 dBm and runs hot. Every component value matches the schematic. The difference is the layout: the matching network was moved 8 mm further from the amplifier pin to make room for a connector, the 1 nF "DC block" capacitor is a size that resonates with its own lead inductance at 190 MHz and looks like a small inductor at 2.4 GHz, and the ground return of the shunt capacitor runs through a via whose inductance is a significant fraction of the capacitor's reactance. At 2.4 GHz, 8 mm of microstrip on FR-4 is more than a tenth of a wavelength; a capacitor is a resonator; a via is an inductor. The circuit on the schematic and the circuit on the board are different circuits.

This chapter is about the frequency range — roughly 30 MHz to 300 GHz — where that happens. The laws of Parts I–III still hold, but the lumped-element assumption of Chapter 1 fails, power flows as waves along transmission lines (Chapter 14), and voltages and currents become hard to define and harder to measure. Engineers respond by changing their variables: from voltages and currents to incident and reflected waves, from impedance matrices to scattering matrices, and from wires to antennas that launch the waves into space.

> **Why does a circuit that works at 10 MHz stop working at 2 GHz with the same component values?**
> **How do RF engineers describe, match, measure and radiate signals when voltage and current are no longer the natural variables?**

Concept	Engineering destination
Electrical length, skin effect, parasitics	PCB layout of radios, choosing capacitors and inductors, high-speed digital interconnect
Impedance matching	Power-amplifier output networks, LNA inputs, antenna feeds
S-parameters	Datasheets of every RF part, network-analyser measurements, circuit simulation
RF components	Receiver and transmitter front ends: filters, LNAs, mixers, couplers, isolators
Resonators and Q	Oscillators, filters, frequency references, MRI coils, particle accelerators
Microwave networks and waveguides	Radar feeds, satellite ground stations, high-power transmitters
Antennas, radiation patterns, gain	Wi-Fi and cellular links, radar, satellite dishes, RFID
RF measurements	Production test, EMC pre-compliance, link troubleshooting

Roadmap. What changes at radio frequencies (32.1) → matching source to load (32.2) → describing networks by waves: S-parameters (32.3) → the components of an RF chain (32.4) and the resonators inside them (32.5) → waveguides and networks of S-parameter blocks (32.6) → leaving the circuit: antennas (32.7), their patterns (32.8) and gain (32.9) → how all of this is measured (32.10).

32.1 RF signal behavior

The bands

Radio frequencies are conventionally divided into decades (ITU) and, above 1 GHz, into the letter bands inherited from radar. Table 32.1 lists them with representative uses and wavelengths.

Table 32.1 — Radio-frequency bands and representative applications.

Band	Frequency	Free-space wavelength	Representative uses
HF	3–30 MHz	100–10 m	Shortwave broadcast, over-the-horizon radar, 13.56 MHz RFID/NFC
VHF	30–300 MHz	10–1 m	FM broadcast (88–108 MHz), air-band, MRI at 64 and 128 MHz
UHF	300 MHz–1 GHz	1 m–30 cm	TV, 433/868/915 MHz ISM, cellular 700–900 MHz
L band	1–2 GHz	30–15 cm	GNSS (1.575 GHz), cellular 1.8 GHz, L-band radar
S band	2–4 GHz	15–7.5 cm	Wi-Fi and Bluetooth 2.4 GHz, 5G 3.5 GHz, weather radar
C band	4–8 GHz	7.5–3.75 cm	Wi-Fi 5–6 GHz, satellite downlinks
X band	8–12 GHz	3.75–2.5 cm	Marine and military radar, satellite links
Ku / K / Ka	12–40 GHz	25–7.5 mm	Satellite TV (10.7–12.75 GHz), 5G mm-wave 26–28 GHz
V / W	40–110 GHz	7.5–2.7 mm	60 GHz short-range links, 77 GHz automotive radar

Electrical length and the end of the lumped model

The lumped-element model treats every connection as instantaneous. It is accurate when every dimension of the circuit is small compared with the wavelength in the medium. On FR-4 microstrip the effective permittivity is about 3.3, so the guided wavelength is $\lambda_g = c/(f\sqrt{\varepsilon_{\text{eff}}})$; a common rule says a trace longer than $\lambda_g/10$ must be treated as a transmission line.

Table 32.2 — When an FR-4 microstrip trace stops being a wire ($\varepsilon_{\text{eff}} \approx 3.3$).

Frequency	Free-space λ_0	$\lambda_g/10$ on FR-4
100 MHz	3.0 m	165 mm
1 GHz	300 mm	16.5 mm
2.4 GHz	125 mm	6.9 mm
10 GHz	30 mm	1.65 mm
28 GHz	10.7 mm	0.59 mm

At 2.4 GHz the 8 mm move in the opening story crosses this threshold: the trace adds a phase shift of $360^\circ \times 8/68.8 \approx 42^\circ$, which rotates the impedance seen by the amplifier around the Smith chart (Chapter 14) and wrecks the match. For digital engineers the same table applies with f replaced by the knee frequency $0.35/t_r$ of an edge with rise time t_r: a 100 ps edge behaves like a 3.5 GHz signal.

> **Status of the principle: The lumped-element approximation**
> *Approximate.* Kirchhoff's laws hold when the circuit is electrically small (largest dimension below about $\lambda/10$) so that propagation delay and radiation can be neglected; beyond that the distributed model of Chapter 14, ultimately Maxwell's equations, must be used.

Skin effect

At high frequency the current in a conductor crowds into a thin layer at the surface, of thickness

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

For copper ($\sigma = 5.8 \times 10^7$ S/m) this is 9.3 mm at 50 Hz, 66 μm at 1 MHz, 6.6 μm at 100 MHz, 2.1 μm at 1 GHz and 0.66 μm at 10 GHz. The resistance of a conductor of perimeter p therefore rises as $\sqrt{f}$ — the surface resistance is $R_s = 1/(\sigma\delta)$, 26 mΩ per square for copper at 10 GHz — and surface finish matters: copper roughness of 1–2 μm, normal for PCB foil, raises loss noticeably above a few gigahertz, and the plating (nickel is magnetic and lossy) can dominate. Silver plating of waveguide interiors and smooth "low-profile" copper for 28 GHz boards are both responses to the skin effect.

Components are not what they seem

Every real component has parasitics that become important once their reactances approach those of the intended element. A capacitor has an equivalent series inductance (ESL) of 0.3–1 nH for small surface-mount sizes, plus mounting inductance, and an equivalent series resistance (ESR) of tens to hundreds of milliohms. It is a series RLC circuit with a **self-resonant frequency**

$$f_{\mathrm{SRF}} = \frac{1}{2\pi\sqrt{L_{\mathrm{ESL}} C}}$$

A 1 nF capacitor with 0.7 nH has $f_{\mathrm{SRF}} = 190$ MHz; above it the "capacitor" is an inductor. A 10 pF part with the same ESL resonates at 1.9 GHz — which is why RF designers pick DC-blocking capacitors whose SRF lies near, or just below, the operating frequency, where the series impedance is lowest. Inductors have inter-winding capacitance and a parallel self-resonance; resistors have lead inductance and shunt capacitance. A via through a 1.6 mm board has roughly 0.5–1 nH, which at 2.4 GHz is 8–15 Ω of reactance — comparable to the impedances being matched.

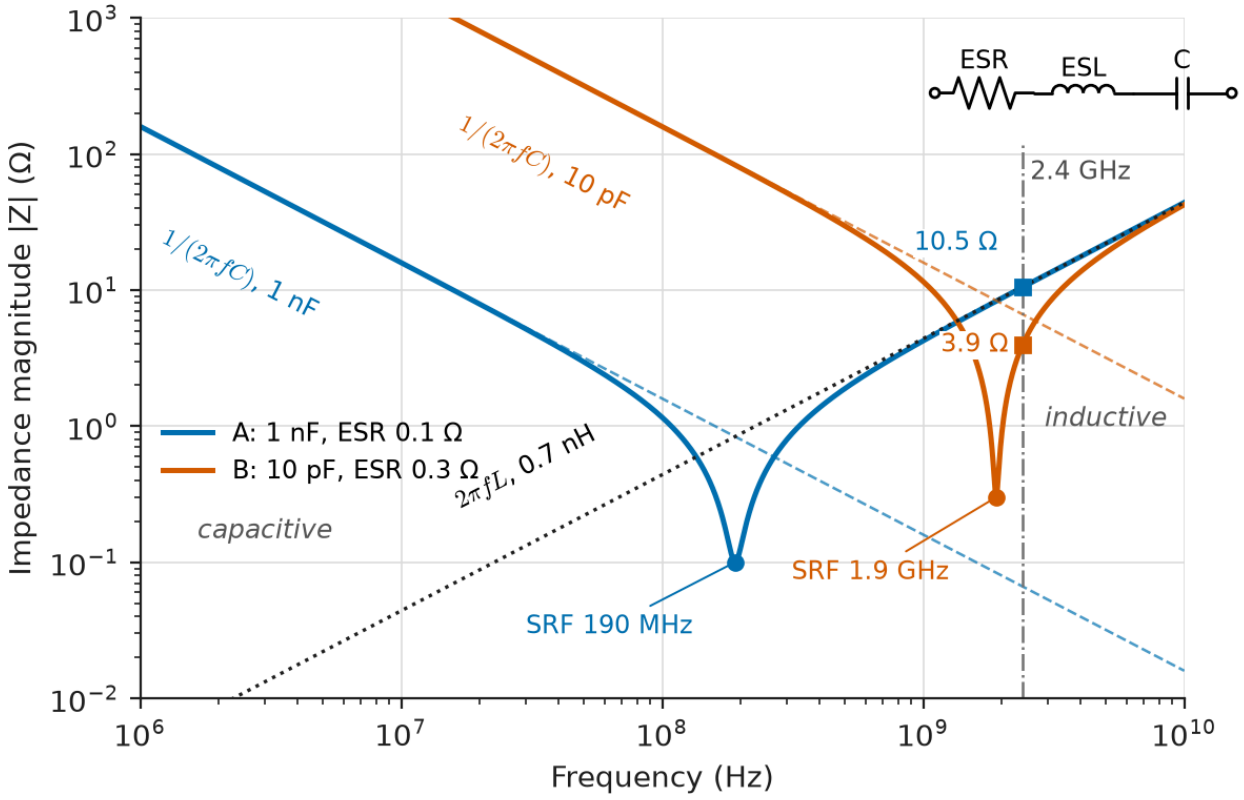

Figure 32.1 — Impedance magnitude of real surface-mount capacitors versus frequency, showing capacitive, self-resonant and inductive regions.

Figure 32.1 shows the result: below SRF the impedance follows $1/\omega C$, above it ωL, with a minimum equal to the ESR at resonance.

Power, decibels and the 50 Ω world

At RF, power is the robust quantity: voltage depends on where along a line it is measured, but power flowing in a matched line does not. Levels are quoted in dBm, gains in dB, and the reference impedance is almost always 50 Ω (75 Ω in video and cable TV). 0 dBm into 50 Ω is 224 mV rms or 0.63 V peak-to-peak. The choice of 50 Ω is a compromise for air-dielectric coaxial line between minimum loss (about 77 Ω) and maximum power handling (about 30 Ω).

32.2 Impedance matching

Why match

Matching serves three different purposes, which do not always agree:

1. **Maximum power transfer.** A source of impedance Z_S delivers the most power to a load $Z_L = Z_S^*$ (Chapter 4). For a power amplifier the "source impedance" is replaced by the optimum load impedance found from load-pull measurements, which is chosen for output power or efficiency rather than conjugate match.

2. **No reflections.** A line terminated in its characteristic impedance carries no reflected wave (Chapter 14); a mismatched load reflects a fraction $|\Gamma|^2$ of the incident power back to the source, where it may cause instability, heating or ripple in frequency response.
3. **Minimum noise.** An LNA has an optimum source impedance Z_{opt} that minimises its noise figure (Chapter 29); it generally differs from the conjugate match, so LNA designers trade gain against noise.

The reflection coefficient, return loss, VSWR and mismatch loss are four ways of expressing the same thing:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}, \quad \text{RL} = -20\log_{10}|\Gamma|, \quad \text{VSWR} = \frac{1+|\Gamma|}{1-|\Gamma|}, \quad \text{ML} = -10\log_{10}(1-|\Gamma|^2)$$

Table 32.3 — Return loss, reflection coefficient, VSWR and mismatch loss.

Return loss	$\lvert\Gamma\rvert$	VSWR	Reflected power	Mismatch loss	Typical meaning
6 dB	0.50	3.0	25 %	1.26 dB	Poor antenna at band edge
10 dB	0.32	1.92	10 %	0.46 dB	Common minimum for antennas
14 dB	0.20	1.50	4 %	0.18 dB	Good amplifier port
20 dB	0.10	1.22	1 %	0.04 dB	Good cable assembly, attenuator
30 dB	0.032	1.07	0.1 %	0.004 dB	Calibration-grade load

Table 32.3 makes a useful point: a 10 dB return loss, which looks bad, wastes only 0.46 dB of power. Matching is often more about stability, filter response and repeatability than about lost power.

The L-network

The simplest lossless matching network uses two reactances: one in series with the smaller resistance and one in shunt with the larger. To match a load resistance R_L to a larger source resistance R_0, the series reactance raises the load to $R_L(1+Q^2)$ in parallel form, which must equal R_0:

$$Q = \sqrt{\frac{R_0}{R_L} - 1}, \qquad X_s = Q R_L, \qquad X_p = \frac{R_0}{Q}$$

The **Q** of the network is fixed by the resistance ratio and sets its bandwidth: a large ratio forces a high Q and a narrow match. When more bandwidth is needed, two or more L-sections in cascade step through intermediate resistances; when a specific Q is wanted (for harmonic filtering), a π or T network adds a third element.

Status of the principle: L-network matching equations
Derived. Exact at the design frequency for ideal lossless reactances, from equating real and imaginary parts of the transformed impedance; component Q, parasitics and layout inductance shift the match, so practical networks are tuned on the board.

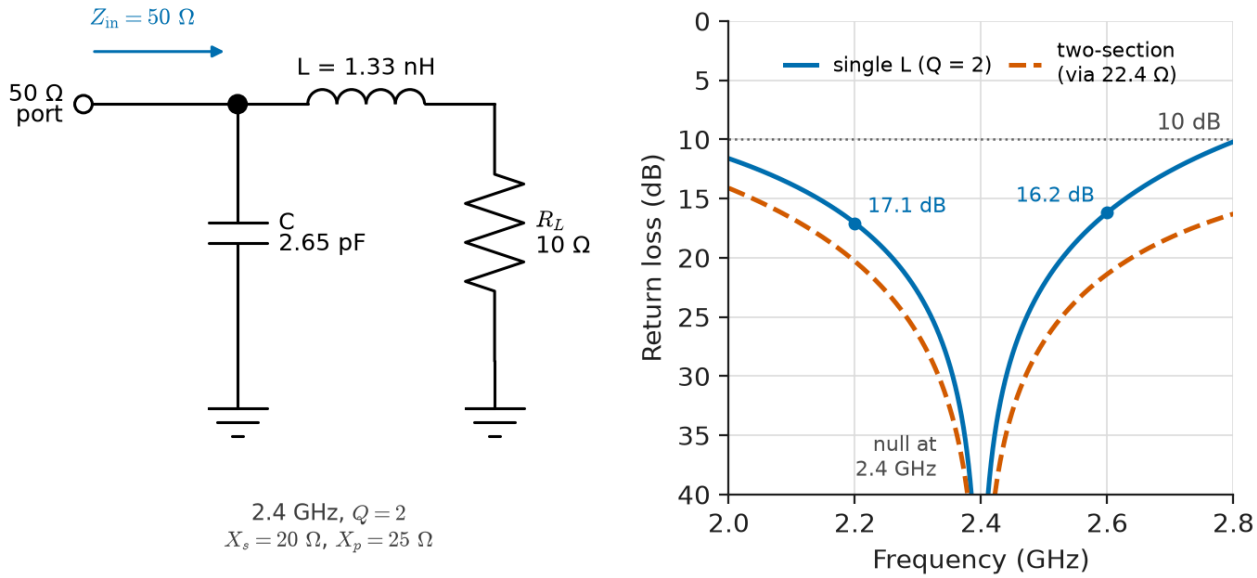

Figure 32.2 — L-network matching a 10 ohm amplifier output to a 50 ohm antenna feed at 2.4 GHz, and the resulting return loss versus frequency.

Worked example — L-network for a 2.4 GHz power amplifier. A power-amplifier stage presents an optimum load of 10 Ω (real) at 2.4 GHz and must drive a 50 Ω antenna feed. Then

$$Q = \sqrt{\frac{50}{10} - 1} = 2, \qquad X_s = 2 \times 10 = 20\ \Omega, \qquad X_p = \frac{50}{2} = 25\ \Omega$$

Choosing a series inductor and shunt capacitor (a low-pass form that also attenuates harmonics), with $\omega = 2\pi \times 2.4 \times 10^9 = 1.508 \times 10^{10}$ rad/s:

$$L = \frac{20}{1.508 \times 10^{10}} = 1.33\ \text{nH}, \qquad C = \frac{1}{1.508 \times 10^{10} \times 25} = 2.65\ \text{pF}$$

Check: the series branch is $10 + j20\ \Omega$; in parallel with $-j25\ \Omega$,

$$Z_{\text{in}} = \frac{(10 + j20)(-j25)}{10 + j20 - j25} = \frac{500 - j250}{10 - j5} = 50\ \Omega$$

Because $Q = 2$ the match is broad: computed return loss is better than 16 dB from 2.2 to 2.6 GHz (Figure 32.2). Two lessons from the opening story follow directly. A via of 0.8 nH in the capacitor's ground return adds $+j12\ \Omega$ in series with $-j25\ \Omega$, nearly halving the shunt reactance's magnitude and moving the match far from 50 Ω. And 1.33 nH is less than the inductance of 2 mm of thin trace, so at these values the "inductor" is often a printed line.

Transmission-line matching

At microwave frequencies, line sections replace lumped elements. A **quarter-wave transformer** of characteristic impedance

$$Z_T = \sqrt{Z_0 R_L}$$

matches a real load R_L to Z_0 at the frequency where it is a quarter wavelength long (Chapter 14). For the example above, $Z_T = \sqrt{50 \times 10} = 22.4\ \Omega$, a wide microstrip about 17 mm long on FR-4 at 2.4 GHz. A **single-stub** match uses a length of line to rotate the load onto the unit-conductance circle of the Smith chart and a short- or open-circuited stub in shunt to cancel the remaining susceptance. Multi-section binomial or Chebyshev transformers trade length for bandwidth, exactly as multi-section filters trade order for flatness (Chapter 23).

> **Status of the principle: Bode–Fano limit on matching bandwidth**
> *Derived*. For a load with reactive storage (for example a parallel RC), the integral of $\ln(1/|\Gamma|)$ over frequency is bounded by $\pi/(RC)$, so no lossless network can match it perfectly over a wide band; a matching network can only redistribute the reflections in frequency.

32.3 S-parameters

Why waves instead of voltages

The Z, Y, h and ABCD parameters of Chapter 12 are defined by open-circuit or short-circuit terminations. At microwave frequencies those terminations are hard to realise (an "open" radiates and has fringing capacitance), and many active devices oscillate when presented with them. Moreover, voltage and current are not uniquely defined in a waveguide. **Scattering parameters** avoid all three problems: they are defined with every port terminated in the reference impedance Z_0 (usually 50 Ω), which is easy to realise and keeps devices stable, and they relate power waves that a directional coupler can separate and measure.

At each port n define the incident and reflected waves, normalised so that $|a_n|^2$ and $|b_n|^2$ are powers:

$$a_n = \frac{V_n + Z_0 I_n}{2\sqrt{Z_0}}, \qquad b_n = \frac{V_n - Z_0 I_n}{2\sqrt{Z_0}}$$

For a two-port,

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

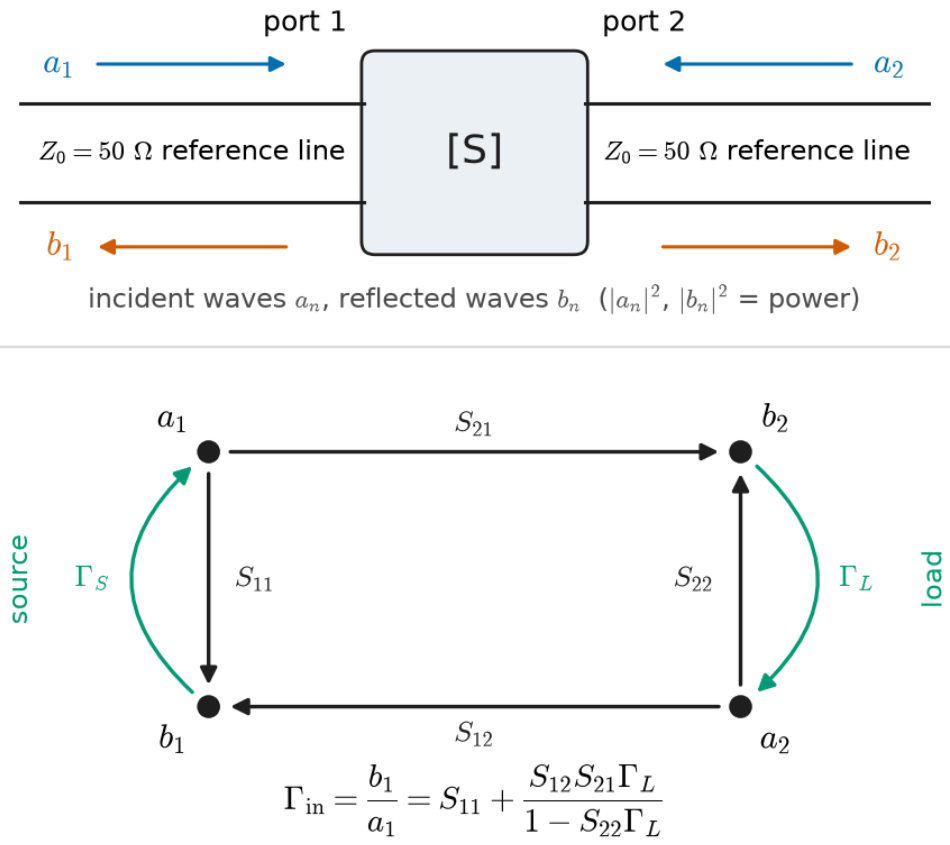

Figure 32.3 — Incident and reflected waves at a two-port and the signal-flow graph of S-parameters with source and load reflections.

Each parameter has a measurable meaning when the other port is matched ($a_2 = 0$ or $a_1 = 0$), as Figure 32.3 shows:

- $S_{11} = b_1/a_1$ — input reflection coefficient; $-20\log_{10}|S_{11}|$ is the input return loss.
- $S_{21} = b_2/a_1$ — forward transmission; $20\log_{10}|S_{21}|$ is the gain of an amplifier, or minus the insertion loss of a passive part.
- $S_{12} = b_1/a_2$ — reverse transmission (isolation of an amplifier).
- $S_{22} = b_2/a_2$ — output reflection coefficient.

Each is a complex number at each frequency, so a datasheet gives tables or Touchstone (.s2p) files of magnitude and phase from, say, 100 MHz to 20 GHz. S-parameters are linear: they describe small-signal behaviour, and for a power amplifier they say nothing about compression or harmonics.

> **Status of the principle: The scattering-matrix description**
> *Derived.* An exact, complete description of any linear time-invariant N-port at each frequency, equivalent to its impedance matrix through $\mathbf{S} = (\mathbf{Z} - Z_0\mathbf{I})(\mathbf{Z} + Z_0\mathbf{I})^{-1}$; it does not describe nonlinear or time-varying behaviour.

Properties

Reciprocity: a network of linear, isotropic materials (no ferrites, no active devices) has $\mathbf{S} = \mathbf{S}^T$, i.e. $S_{12} = S_{21}$. **Losslessness:** a lossless network conserves power, so $\mathbf{S}$ is unitary, $\mathbf{S}^H\mathbf{S} = \mathbf{I}$; for a two-port this means $|S_{11}|^2 + |S_{21}|^2 = 1$. **Symmetry:** a physically symmetric two-port has $S_{11} = S_{22}$. These properties are quick checks on measurements and on simulator output: a passive part whose measured $|S_{11}|^2 + |S_{21}|^2$ exceeds 1 indicates a calibration error.

Two simple networks

Worked example — S-parameters of a series element. A series impedance Z between two 50 Ω ports has, by direct circuit analysis with port 2 terminated in Z_0,

$$S_{11} = S_{22} = \frac{Z}{Z + 2Z_0}, \qquad S_{21} = S_{12} = \frac{2Z_0}{Z + 2Z_0}$$

(a) Series 50 Ω resistor: $S_{11} = 50/150 = 0.333$ (return loss 9.5 dB) and $S_{21} = 100/150 = 0.667$ (insertion loss 3.52 dB). Then $|S_{11}|^2 + |S_{21}|^2 = 0.111 + 0.444 = 0.556$: 44 % of the incident power is dissipated in the resistor, as expected for a lossy network.

(b) Series 1 pF capacitor at 2.4 GHz: $Z = 1/(j\omega C) = -j66.3\ \Omega$. Then

$$S_{11} = \frac{-j66.3}{100 - j66.3} = 0.306 - j0.461 = 0.553\angle -56.4^\circ, \qquad S_{21} = \frac{100}{100 - j66.3} = 0.695 + j0.461 = 0.83$$

The insertion loss is $-20\log_{10} 0.833 = 1.58$ dB, and $|S_{11}|^2 + |S_{21}|^2 = 0.306 + 0.694 = 1.000$: the capacitor is lossless, so all the "lost" power was reflected, not absorbed. A 1 pF DC block is far too small at 2.4 GHz — a reflection of 31 % of the power — which is the arithmetic behind choosing DC blocks of tens of picofarads.

Connecting S-parameters to the rest of the system

When port 2 is terminated in a load with reflection coefficient Γ_L instead of Z_0, the signal-flow graph of Figure 32.3 (solved with Mason's rule, Chapter 33) gives the input reflection

$$\Gamma_{\text{in}} = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L}$$

and, with a matched source, the transducer power gain

$$G_T = \frac{|S_{21}|^2(1 - |\Gamma_L|^2)}{|1 - S_{22}\Gamma_L|^2}$$

Worked example — an amplifier driving a mismatched filter. An LNA has $S_{11} = 0.2$, $S_{21} = 5.62$ (15 dB), $S_{12} = 0.056$ (−25 dB) and $S_{22} = 0.25$ at the frequency of interest (phases taken as zero for simplicity). It drives a filter whose input has $\Gamma_L = 0.2$. Then

$$\Gamma_{\text{in}} = 0.2 + \frac{0.056 \times 5.62 \times 0.2}{1 - 0.25 \times 0.2} = 0.2 + \frac{0.0630}{0.95} = 0.267$$

$$G_T = \frac{31.6 \times (1 - 0.04)}{(0.95)^2} = 33.6 \quad (15.27\ \text{dB})$$

The load mismatch has changed the input return loss from 14.0 dB to 11.5 dB through the amplifier's reverse transmission, and changed the gain by +0.27 dB. With other phases the gain change could just as easily be negative; this is the origin of gain ripple in cascaded RF chains and of the **mismatch uncertainty** in measurements (Section 32.10). An amplifier with $S_{12} = 0$ — a **unilateral** one — would isolate its input from its load completely.

32.4 RF components

An RF chain is assembled from a small vocabulary of building blocks, each specified by S-parameters plus a few nonlinear and noise figures.

Passive components

Attenuators are resistive networks matched at both ports; they set levels and, because they improve the return loss seen through them by twice their attenuation, are used as "pads" to tame mismatches. **Power dividers** split a signal; the **Wilkinson divider** — two quarter-wave lines of $\sqrt{2}Z_0$ with a $2Z_0$ resistor between the outputs — splits equally with all ports matched and 20–30 dB isolation between outputs. **Directional couplers** sample a fixed fraction (e.g. −10 or −20 dB) of the wave travelling in one direction only; their **directivity** (typically 20–40 dB) measures how well they reject the opposite direction, and they are the heart of every network analyser and power-amplifier output monitor. **Hybrids** (90° branch-line, 180° rat-race) split into two equal outputs with defined phase and underlie balanced amplifiers and IQ modulators. **Circulators** use magnetised ferrite to route power 1→2→3→1; they are non-reciprocal, so $S_{21} \neq S_{12}$. Terminating one port gives an **isolator**, which protects a power amplifier or laser from reflected power. **Filters** — lumped, microstrip, cavity, ceramic, SAW and BAW — select bands; their synthesis follows Chapters 13 and 23, with resonators (Section 32.5) as building blocks.

Active components

A **low-noise amplifier** (LNA) sets the receiver noise figure (Chapter 29) and must be the first gain stage. A **power amplifier** (PA) delivers output power efficiently and is characterised by output power at 1 dB compression ($P_{1\text{dB}}$), saturated power and power-added efficiency. A **mixer** multiplies the signal by a local oscillator to translate it in frequency (Chapter 26), with conversion loss, LO-to-RF isolation and image rejection as key figures. An **oscillator** — a resonator plus a gain element — generates the LO; its phase noise (dBc/Hz at an offset) limits adjacent-channel selectivity and constellation density.

Nonlinearity is specified by two figures. The **1 dB compression point** is the input or output power at which gain has fallen 1 dB below its small-signal value. The **third-order intercept point** (IP3) is the extrapolated power at which the third-order intermodulation products of a two-tone test would equal the fundamental; because those products grow at 3 dB per dB of input, the intermodulation level falls by 2 dB for every 1 dB reduction in drive. For many amplifiers the output IP3 lies roughly 10 dB above $P_{1\text{dB}}$.

Status of the principle: OIP3 ≈ $P_{1\text{dB}}$ + 10 dB
Empirical. Follows from a pure cubic nonlinearity, which predicts 9.6 dB, and is observed within a few decibels for many class-A amplifiers; it fails for devices with higher-order or memory nonlinearity and for linearised or class-AB power amplifiers.

Table 32.4 — Representative RF component specifications (1–6 GHz, typical).

Component	Key figures	Typical values
LNA (GaAs pHEMT or SiGe)	Gain; NF; OIP3	15–20 dB; 0.5–1.5 dB; +20 to +35 dBm
Driver amplifier	Gain; $P_{1\text{dB}}$	15–25 dB; +20 to +27 dBm
Cellular base-station PA (GaN)	Saturated output; efficiency	40–50 dBm; 40–60 % (Doherty)
Passive double-balanced mixer	Conversion loss; LO drive; IIP3	6–8 dB; +7 to +17 dBm; +15 to +25 dBm
Wilkinson divider	Split loss; isolation	3.0 dB + 0.2–0.5 dB; 20–30 dB
Directional coupler	Coupling; directivity	10 or 20 dB; 20–40 dB
Ferrite isolator	Insertion loss; isolation	0.2–0.5 dB; 20–25 dB
SAW/BAW band filter	Insertion loss; rejection	1–3 dB; 30–50 dB
Coaxial cable, RG-58 / low-loss ½-inch	Loss at 1 GHz	≈0.6 dB/m / ≈0.07 dB/m
SMA connector	Upper frequency; return loss	18–26.5 GHz; >20 dB

Boundary.
Transistor-level LNA, PA and oscillator design — bias networks, stability circles, load-pull, device models — belongs to *Electronics & Computing*. This chapter treats components through their S-parameters and figures of merit.

32.5 Resonators

Lumped resonators and Q

A resonator stores energy that oscillates between two forms — electric and magnetic in an LC circuit (Chapter 7) — at a natural frequency f_0. Its quality factor

$$Q = 2\pi \frac{\text{energy stored}}{\text{energy lost per cycle}} = \frac{f_0}{\Delta f_{3\,\text{dB}}}$$

measures both how slowly it rings down and how narrow its response is (Chapter 13). For a series RLC, $Q = \omega_0 L/R$. A 10 nH inductor and 1 pF capacitor resonate at 1.59 GHz; if the inductor has $Q = 50$ its loss resistance is $\omega_0 L/Q = 2\ \Omega$, and the circuit's Q can be no higher than 50. At RF, lumped inductors are the weak point: their Q is limited to 30–100 by skin-effect resistance and substrate loss, which is why high-Q microwave resonators abandon lumped elements altogether.

Unloaded and loaded Q

A resonator in a circuit is coupled to external ports, which also remove energy. The **unloaded Q** Q_0 counts only internal losses; the **external Q** Q_e counts the energy leaving through the coupling; the **loaded Q** combines them,

$$\frac{1}{Q_L} = \frac{1}{Q_0} + \frac{1}{Q_e}, \qquad \beta = \frac{Q_0}{Q_e}$$

where β is the coupling coefficient. At **critical coupling** ($\beta = 1$) the resonator is matched to the feed and $Q_L = Q_0/2$. A filter designer sets the external Q to obtain the desired bandwidth; an oscillator designer keeps Q_L high to minimise phase noise.

Distributed resonators

A transmission line short-circuited at one end and a quarter wavelength long, or open-circuited and a half wavelength long, behaves near resonance like a parallel LC circuit (Chapter 14). **Microstrip resonators** reach Q_0 of 100–300, limited by conductor and dielectric loss and radiation. **Coaxial resonators** with ceramic filling reach 500–2000 at 1–3 GHz and form the duplexers in cellular base stations. **Cavity resonators** — closed metal boxes — and **dielectric resonators** — pucks of high-permittivity, low-loss ceramic — reach thousands to tens of thousands. Quartz crystals exploit mechanical resonance coupled through piezoelectricity and reach 10^4–10^6.

Table 32.5 — Representative unloaded Q of resonators.

Resonator	Frequency	Unloaded Q_0	Typical use
Lumped chip LC	0.1–3 GHz	30–100	Matching networks, simple filters
Microstrip line	1–30 GHz	100–300	PCB filters, VCOs
Ceramic coaxial	0.8–6 GHz	500–2000	Handset and base-station duplexers
Copper waveguide cavity	10 GHz	5000–12 000	Radar filters, frequency discriminators
Dielectric resonator	2–30 GHz	10 000–40 000 (at about 5 GHz)	Low-phase-noise oscillators, satellite filters
SAW / BAW resonator	0.1–6 GHz	500–3000	Mobile-phone band filters
Quartz crystal	1–100 MHz	10^4–10^6	Clock and frequency references
Superconducting niobium cavity	1.3 GHz	10^9–10^{10}	Particle accelerators

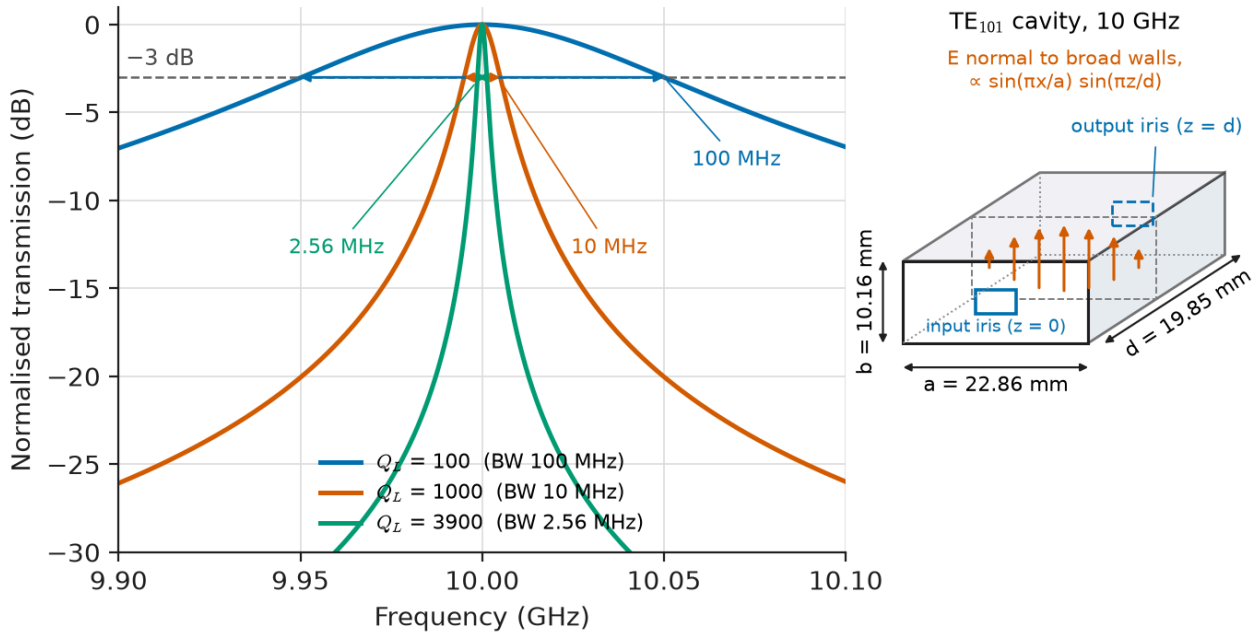

Figure 32.4 — Transmission response of resonators with different loaded Q around a 10 GHz centre frequency.

Worked example — a 10 GHz waveguide cavity. A section of WR-90 waveguide (internal dimensions $a = 22.86$ mm, $b = 10.16$ mm) closed at both ends forms a rectangular cavity. Its lowest mode, TE_{101}, resonates at

$$f_{101} = \frac{c}{2}\sqrt{\frac{1}{a^2} + \frac{1}{d^2}}$$

For $f = 10$ GHz the required length is $d = \left[(2f/c)^2 - 1/a^2\right]^{-1/2} = 19.85$ mm. The copper walls have surface resistance $R_s = \sqrt{\pi f \mu_0 / \sigma} = 26.1$ mΩ at 10 GHz, and the standard cavity-loss formula then gives an unloaded Q of about 7800. Critically coupled, $Q_L = 3900$ and the 3 dB bandwidth is

10 GHz/3900 = 2.6 MHz (Figure 32.4). In practice surface roughness and joint losses lower Q_0 by 10–30 %, and silver plating raises it by about 3 %.

32.6 Microwave networks

Waveguides

Above about 3 GHz, and at powers beyond what coaxial cable can handle, hollow metal **waveguides** carry microwave energy. A rectangular guide supports transverse-electric (TE) and transverse-magnetic (TM) modes, each with a **cutoff frequency** below which it does not propagate:

$$f_{c,mn} = \frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

The dominant mode is TE_{10} with $f_c = c/2a$. For WR-90 ($a = 22.86$ mm, $b = 10.16$ mm) this is 6.56 GHz; the next modes are TE_{20} at 13.1 GHz and TE_{01} at 14.8 GHz, so the recommended single-mode band is 8.2–12.4 GHz (X band). Above cutoff the guide is dispersive, with guided wavelength

$$\lambda_g = \frac{\lambda_0}{\sqrt{1 - (f_c/f)^2}}$$

At 10 GHz, $\lambda_0 = 30.0$ mm and $\lambda_g = 39.7$ mm: the guide wavelength is longer than in free space, and the group velocity is $c\sqrt{1-(f_c/f)^2} = 0.755c$. This group-delay dispersion is the microwave counterpart of chromatic dispersion in fibre (Chapter 31). Waveguide loss is low — about 0.1 dB/m for copper WR-90 — and power handling is hundreds of kilowatts peak, which is why radar transmitters and satellite ground-station feeds use it.

> **Status of the principle: Waveguide cutoff**
> *Fundamental.* Follows from Maxwell's equations with the boundary condition of zero tangential electric field on perfectly conducting walls; finite wall conductivity adds loss but does not shift the cutoff appreciably.

Planar lines

Most microwave circuits are printed. **Microstrip** (a strip over a ground plane) is cheap and easy to mount components on, but its field is partly in air, so it is slightly dispersive and radiates at discontinuities. **Stripline** (a strip between two grounds) is non-radiating and non-dispersive. **Coplanar waveguide** puts the grounds beside the strip on the same layer, which suits wafer probing and millimetre-wave integrated circuits. Their characteristic impedance and effective permittivity follow from geometry (Chapter 14); a 50 Ω microstrip on 1.6 mm FR-4 is about 3 mm wide, on 0.25 mm high-frequency laminate about 0.55 mm.

Networks of networks

A microwave system is a network of S-parameter blocks. For a chain of two-ports the scattering parameters do not multiply directly, but the **transfer (T) parameters**, which relate the waves at port 1 to those at port 2, do; equivalently each block can be converted to ABCD parameters (Chapter 12), cascaded by matrix multiplication and converted back. Circuit simulators do this at every frequency point. For arbitrary topologies — a balanced amplifier with two hybrids, a diplexer with a common port — the waves at every connection form a signal-flow graph, and Mason's gain formula (Chapter 33) or direct matrix solution gives any transfer of interest. The same graphs show how errors propagate, which is the basis of network-analyser calibration (Section 32.10).

Worked example — a two-way radar feed. An X-band radar transmitter (10 GHz, 50 kW peak) feeds an antenna through 6 m of WR-90 (0.11 dB/m), a ferrite circulator (0.3 dB insertion loss, 23 dB isolation) and a rotary joint (0.2 dB). The forward loss is $6 \times 0.11 + 0.3 + 0.2 = 1.16$ dB, so the antenna receives $50 \times 10^{-0.116} = 38.3$ kW. If the antenna presents a 14 dB return loss, $10^{-1.4} = 4\%$ of that power — about 1.5 kW — is reflected; after returning through the joint and guide (another 0.86 dB) the remaining 1.25 kW reaches the circulator, which routes it to the receiver arm, where a limiter protects the LNA; only $1.25\,\text{kW} \times 10^{-2.3} \approx 6.3$ W leaks back toward the transmitter. The same network diagram, with echoes arriving at microwatt levels, carries the received echo; the radar equation of Section 32.9 takes over from there.

32.7 Antenna fundamentals

From guided to radiated waves

An antenna is a transducer between a guided wave on a transmission line and a free-space wave. Any accelerated charge radiates; a structure becomes an efficient antenna when its size is comparable to a wavelength and its current distribution does not cancel its own field at a distance. Seen from its feed terminals, an antenna is a load impedance

$$Z_A = R_{\text{rad}} + R_{\text{loss}} + jX_A$$

where the **radiation resistance** R_{rad} represents power carried away to space, R_{loss} represents ohmic and dielectric loss, and X_A is stored reactive energy. The **radiation efficiency** is $\eta_r = R_{\text{rad}}/(R_{\text{rad}} + R_{\text{loss}})$.

> **Boundary.**
> The derivation of radiated fields from currents (retarded potentials, the Hertzian dipole) belongs to *Engineering Physics*. Here an antenna is characterised by its impedance, pattern, directivity and gain, and treated as a component in a link.

Reciprocity

For antennas made of linear, isotropic materials, the transmitting and receiving patterns, impedances and gains are identical. This **reciprocity** is the antenna form of the network reciprocity of Section 32.3

and Chapter 4, and it allows an antenna to be measured in whichever direction is more convenient.

> **Status of the principle: Antenna reciprocity**
> *Derived.* A consequence of the Lorentz reciprocity theorem for Maxwell's equations in linear, isotropic, time-invariant media; it fails for antennas containing ferrites, plasmas or active circuits.

Basic antenna types

Short dipole. A dipole much shorter than a wavelength, of length l, has $R_{\text{rad}} = 80\pi^2 (l/\lambda)^2\ \Omega$; for $l = \lambda/10$ this is only 7.9 Ω, against a large capacitive reactance. Electrically small antennas — car-radio whips at AM frequencies, implant antennas, 433 MHz key fobs — are therefore inefficient and narrowband: a few ohms of conductor and matching loss compete with a few ohms of radiation resistance.

Half-wave dipole. A centre-fed conductor $\lambda/2$ long — 1.5 m at 100 MHz, 62 mm at 2.4 GHz — has $R_{\text{rad}} \approx 73\ \Omega$ and, cut slightly short (about 0.47λ), no reactance; it matches 50–75 Ω cable well and has a directivity of 1.64 (2.15 dBi).

Monopole over a ground plane. By image theory a $\lambda/4$ whip over a large ground behaves as half a dipole: $R_{\text{rad}} \approx 36.5\ \Omega$ and twice the dipole's directivity (5.15 dBi over an infinite ground). The mobile-phone and vehicle antenna is a monopole whose "ground plane" is the phone's circuit board or the car roof.

Patch antenna. A rectangular conductor about $\lambda_g/2$ long over a ground plane, printed on a laminate; at 2.45 GHz on FR-4 ($\varepsilon_{\text{eff}} \approx 4.1$) the length is about 30 mm. Patches radiate broadside with 6–8 dBi gain, are thin and cheap, and form the elements of phased arrays — but their bandwidth is only a few percent.

Aperture antennas. Horns and parabolic reflectors radiate from an aperture many wavelengths across; their gain is set by aperture area (Section 32.9).

Arrays. Several elements fed with controlled amplitudes and phases form an array whose pattern is the element pattern multiplied by an **array factor**. Steering the phases steers the beam electronically: the basis of phased-array radar, 5G base stations with 64 elements and the MIMO systems of Chapter 30.

Near and far field

Close to an antenna the reactive near field dominates; far away the fields decay as $1/r$, are transverse, and have a fixed angular shape. For an antenna of largest dimension D the far field begins at roughly the **Fraunhofer distance**

$$r_{\text{ff}} = \frac{2D^2}{\lambda}$$

For a 1.2 m dish at 12 GHz ($\lambda = 25$ mm) this is 115 m, which is why large antennas are measured on long outdoor ranges or with near-field scanners whose data are transformed mathematically to the far field.

32.8 Radiation patterns

Describing a pattern

The **radiation pattern** gives the radiated power density (or field) as a function of direction (θ, ϕ) at a fixed far-field distance, usually normalised to its maximum and plotted in decibels. Two principal-plane cuts are used: the **E-plane**, containing the electric field and the direction of maximum radiation, and the **H-plane**, perpendicular to it. A pattern is characterised by:

- the **main lobe** and its **half-power beamwidth** (HPBW), the angle between the −3 dB points;
- **sidelobes**, especially the first sidelobe level relative to the main lobe (−13.2 dB for a uniformly illuminated aperture, −20 to −30 dB for tapered illumination);
- **nulls** between lobes;
- the **front-to-back ratio** for directional antennas;
- **polarisation** — the orientation of the electric field (linear vertical or horizontal, circular right- or left-handed) — and the cross-polar level.

A half-wave dipole's pattern is a doughnut: $\cos(\frac{\pi}{2}\cos\theta)/\sin\theta$ in field, with nulls along the wire axis, omnidirectional in the plane perpendicular to it, and an E-plane HPBW of 78°. A parabolic dish produces a pencil beam whose HPBW in degrees is approximately

$$\theta_{3\,\mathrm{dB}} \approx 70\frac{\lambda}{D}$$

for typical edge-tapered illumination.

> **Status of the principle:** $\theta_{3\,\mathrm{dB}} \approx 70\lambda/D$
> *Empirical.* A rule fitted to typical reflector illuminations (the uniform aperture gives about $51\lambda/D$, heavily tapered ones up to $80\lambda/D$); it is used for first estimates of beamwidth and pointing tolerance.

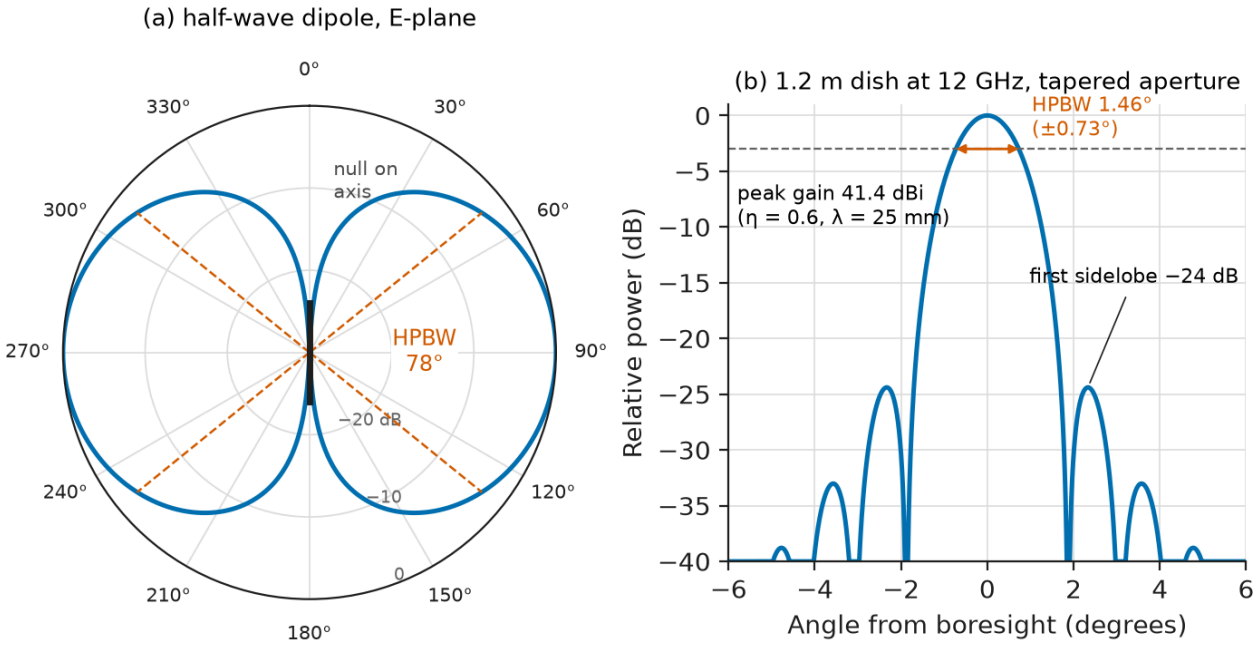

Figure 32.5 — Radiation patterns of a half-wave dipole and of a 1.2 m parabolic dish at 12 GHz.

Figure 32.5 compares the broad dipole pattern with the pencil beam of a dish.

Patterns and the link

Patterns matter in three ways. They set **coverage**: a sector antenna on a cellular mast with 65° horizontal HPBW serves a third of the circle. They set **interference**: sidelobes of an earth-station antenna pointing at one satellite illuminate its neighbours spaced 2° away along the geostationary arc, so regulators specify a sidelobe envelope ($29 - 25\log_{10}\theta$ dBi). And they set **pointing tolerance**: a 1.46° beam must be aimed within about a fifth of its beamwidth to lose less than 0.5 dB. Polarisation also matters: a linearly polarised antenna receiving a circularly polarised wave loses 3 dB, and two orthogonal linear polarisations can carry independent signals — the frequency-reuse trick of satellite TV and of dual-polarised MIMO.

32.9 Gain and directivity

Definitions

The **directivity** of an antenna is the ratio of the radiation intensity in a given direction to the intensity averaged over all directions:

$$D(\theta,\phi) = \frac{4\pi\, U(\theta,\phi)}{P_{\text{rad}}}$$

Its maximum value is "the directivity". An isotropic radiator has $D = 1$ (0 dBi). The **gain** includes the radiation efficiency, $G = \eta_r D$, and is what a link uses; realized gain additionally includes the mismatch loss $1 - |\Gamma|^2$. For a pencil beam, directivity can be estimated from the two principal-plane beamwidths in degrees,

$$D \approx \frac{41\,253}{\theta_E\,\theta_H}$$

where 41 253 is the number of square degrees in a sphere.

Effective aperture

A receiving antenna collects power as if it presented an area A_e to the incoming wave: $P_r = S\,A_e$ for incident power density S. Gain and effective aperture are linked by a universal relation:

$$G = \frac{4\pi A_e}{\lambda^2}$$

For aperture antennas $A_e = \eta_a A_{\text{phys}}$, with aperture efficiency η_a of 0.5–0.75 for reflectors and horns. Hence the gain of a dish of diameter D:

$$G = \eta_a \left(\frac{\pi D}{\lambda}\right)^2$$

> **Status of the principle:** $G = 4\pi A_e/\lambda^2$
> *Derived*. Follows from reciprocity and thermodynamic equilibrium of an antenna with a matched load in a black-body enclosure; it holds for every lossless, matched antenna regardless of type.

Friis transmission and EIRP

Combining the gain of a transmitting antenna with the effective aperture of a receiving one gives the Friis equation for a free-space link of length R,

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2$$

or in decibels $P_r = P_t + G_t + G_r - \text{FSPL}$ with $\text{FSPL} = 20\log_{10}(4\pi R/\lambda)$, the free-space path loss used in the link budgets of Chapter 30. The product $P_t G_t$ is the **effective isotropic radiated power** (EIRP), the quantity regulators limit — 20 dBm (100 mW) EIRP for 2.4 GHz Wi-Fi in Europe, for example.

A monostatic radar applies Friis twice — out to a target of radar cross-section σ, which reradiates, and back to the same antenna:

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$$

The R^4 law is brutal: an X-band radar with $P_t = 1$ kW, $G = 30$ dBi and $\lambda = 3$ cm receives only -103 dBm from a 1 m² target at 10 km, and doubling the range costs 12 dB.

Table 32.6 — Representative antenna gains.

Antenna	Gain	Beamwidth	Application
Isotropic (reference)	0 dBi	—	Link-budget reference
Short dipole	1.76 dBi	90°	Small-antenna limit
Half-wave dipole	2.15 dBi	78° (E-plane)	FM, reference antenna
Quarter-wave monopole on large ground	5.15 dBi	39° elevation	Vehicles, base whips
Phone or IoT chip antenna	−5 to +2 dBi	Near omnidirectional	Handsets, sensors
Microstrip patch	6–8 dBi	60–80°	GNSS, Wi-Fi access points, array elements
Yagi–Uda (6–15 elements)	9–15 dBi	30–50°	TV reception, point-to-point UHF
Cellular sector panel	15–18 dBi	65° horizontal	Base stations
Standard-gain horn	10–25 dBi	10–50°	Calibration, feeds
1.2 m dish at 12 GHz	≈41 dBi	≈1.5°	Satellite earth station
70 m deep-space dish at 8.4 GHz	≈74 dBi	≈0.04°	Deep-space network

Worked example — a satellite dish and a Wi-Fi link. *(a) Dish.* A 1.2 m reflector at 12 GHz ($\lambda = 2.998 \times 10^8 / 12 \times 10^9 = 25.0$ mm) with aperture efficiency 0.6:

$$G = 0.6 \left(\frac{\pi \times 1.2}{0.0250} \right)^2 = 0.6 \times (150.9)^2 = 1.37 \times 10^4 \quad (41.4 \text{ dBi})$$

The beamwidth is $70 \times 0.025 / 1.2 = 1.46^\circ$, and the beamwidth formula for directivity gives $41\,253 / 1.46^2 = 1.9 \times 10^4$ (42.9 dBi) — about 1.5 dB above the gain, as expected, since the beamwidth estimate ignores sidelobe power and efficiency. The effective aperture is $0.6 \times \pi \times 0.6^2 = 0.68$ m². The free-space loss to a geostationary satellite at a slant range of 38 000 km is $20 \log_{10}(4\pi \times 3.8 \times 10^7 / 0.025) = 205.6$ dB — the 41 dB of dish gain is what makes a satellite signal receivable at all.

(b) Wi-Fi. An access point at 2.4 GHz transmits 20 dBm into a 2 dBi antenna; a laptop 50 m away in open space also has 2 dBi. With $\lambda = 0.125$ m,

$$\text{FSPL} = 20 \log_{10} \frac{4\pi \times 50}{0.125} = 74.0 \text{ dB}, \qquad P_r = 20 + 2 + 2 - 74.0 = -50.0 \text{ dBm}$$

A receiver needing −70 dBm for its highest rates has 20 dB of margin in free space — margin that walls, bodies and multipath fading (Chapter 30) consume indoors.

32.10 RF measurements

The vector network analyser

The **vector network analyser** (VNA, Figure 32.6) measures S-parameters directly. A swept source drives one port of the device under test (DUT); directional couplers at each port separate incident and reflected waves; receivers down-convert them and measure amplitude and phase ratios. Modern VNAs cover 10 MHz to 67 GHz or beyond in coaxial form and reach 1 THz with waveguide extenders, with dynamic range of 100–140 dB.

A VNA measures not the DUT alone but the DUT embedded in imperfect couplers, cables and connectors. **Calibration** removes these systematic errors. In the 12-term error model for a two-port measurement, each port has directivity, source-match and reflection-tracking errors, plus load-match, transmission-tracking and isolation terms — all obtained by measuring known standards. **SOLT** calibration uses short, open, load and through standards with known (modelled) responses; **TRL** uses a through, a reflect and a line of known length and is preferred on printed circuits and wafers where good loads are hard to make. After calibration the **reference plane** sits at the DUT's terminals, and **de-embedding** can move it further to remove test fixtures.

> **Status of the principle: Error-model calibration**
> *Derived*. Because the test set is linear, its errors form a linear network whose S-parameters can be solved from measurements of known standards; the accuracy is then limited by how well the standards are known and by drift and connector repeatability.

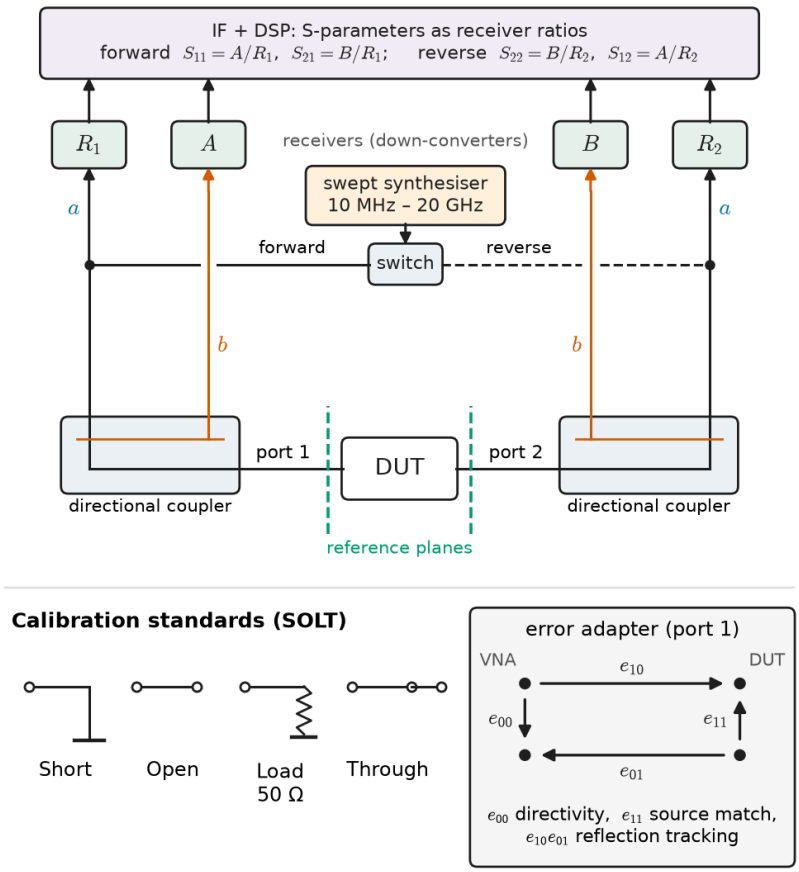

Figure 32.6 — Two-port vector network analyser with its directional couplers, receivers and calibration standards.

Mismatch uncertainty

Even after calibration, residual source and load mismatches create uncertainty. With a source match Γ_S and DUT reflection Γ_L, the multiple reflection between them gives a power uncertainty of

$$\Delta P_{\text{dB}} = 20\log_{10}(1 \pm |\Gamma_S||\Gamma_L|)$$

A power sensor with 20 dB return loss ($|\Gamma| = 0.1$) measuring a source with 10 dB return loss (0.316) has uncertainty of +0.27/−0.28 dB from this term alone — often the largest item in an RF power-measurement budget, and the reason attenuator pads are placed at measurement ports.

Spectrum analysis and power measurement

The **spectrum analyser** displays power versus frequency, by a swept superheterodyne (Chapter 26) or an FFT of digitised IF (Chapter 22). Its **resolution bandwidth** (RBW) sets frequency resolution and noise floor: the displayed average noise level is

$$\text{DANL} \approx -174\ \text{dBm/Hz} + 10\log_{10}(\text{RBW}) + NF$$

so an analyser with a 24 dB noise figure at 1 kHz RBW shows a floor of about −120 dBm; reducing RBW to 10 Hz lowers it by 20 dB at the cost of a 100× slower sweep. Analysers measure harmonics, spurious emissions, adjacent-channel power, phase noise and EMC emissions. **Power meters** (thermistor, thermocouple or diode sensors) give the most accurate absolute power, ±0.1–0.2 dB; **noise-figure**

analysers use the Y-factor method with a calibrated noise source switched between two temperatures (Chapter 29). **Time-domain reflectometry** — sending a fast edge and observing reflections — locates impedance discontinuities in cables and PCB traces, and VNAs offer the same view by inverse-transforming $S_{11}(f)$.

Antenna measurements

Antenna gain and pattern are measured in **anechoic chambers**, lined with absorber, on far-field ranges longer than $2D^2/\lambda$, in compact ranges that use a reflector to create a plane wave, or with **near-field scanners** that sample the field close to the antenna and transform to the far field. Gain is obtained by comparison with a calibrated standard-gain horn or by the three-antenna method, which uses Friis' equation for three pairs to solve for three unknown gains. Over-the-air (OTA) testing of phones and base stations extends these methods to complete radios with integrated antennas.

Table 32.7 — Representative RF measurement instruments.

Instrument	Measures	Typical performance
Vector network analyser	S-parameters vs frequency	100–140 dB dynamic range; ±0.05–0.1 dB, ±0.5° after calibration
Spectrum / signal analyser	Spectrum, ACP, EVM, phase noise	DANL −150 to −165 dBm/Hz with preamplifier; ±0.3–0.5 dB amplitude
Thermocouple / diode power sensor	Average or peak power	−70 to +20 dBm; ±0.1–0.2 dB
Noise-figure analyser	NF and gain	±0.05–0.2 dB NF uncertainty
TDR / TDT	Impedance vs distance	15–35 ps rise time; mm resolution
Anechoic chamber / near-field scanner	Pattern, gain, EIRP	±0.3–0.5 dB gain uncertainty

What remains open

Open question	Where it is answered
How do reflection coefficient, standing waves and the Smith chart follow from the telegrapher's equations?	Chapter 14 (transmission lines)
How are Z, Y, h and ABCD parameters defined and converted?	Chapter 12 (two-port networks)
How do noise figure and cascaded noise determine receiver sensitivity?	Chapter 29 (random signals and communication noise)
How are mixers and superheterodyne receivers used in analog modulation?	Chapter 26 (analog communication)
How do path loss, fading and MIMO arrays enter a wireless link budget?	Chapter 30 (wireless and mobile communications)
How do waveguide modes appear at optical frequencies in fibre?	Chapter 31 (optical communications)
How is Mason's gain formula, used here for wave-flow graphs, derived?	Chapter 33 (dynamic-system modeling)
How do oscillators and power amplifiers stay stable, and how is automatic gain or power control designed?	Chapters 34–35 (feedback, stability and control design); *Electronics & Computing*
How do Maxwell's equations produce radiation, and why does a waveguide have cutoff?	*Engineering Physics*

Next: Chapter 33 opens Part VII by turning the same mathematics — differential equations, transfer functions and flow graphs — toward modelling physical systems that are to be controlled.

Part VII — Control-System Fundamentals

The earlier parts of this book analysed circuits and signals as given: a network is built, a signal arrives, and the task is to predict the response. Control turns the question around. A motor must hold a position against a changing load, a generator must keep its speed when demand jumps, an oven must reach and hold a temperature, a laser must keep its output power constant as it ages, a drone gimbal must stay level in a gust. In each case a measurement of the output is fed back to adjust the input, and the closed loop behaves very differently from its parts — faster, more accurate and less sensitive to disturbances when the design is right, oscillating or unstable when it is not. This part develops the theory behind that behaviour. It uses only mathematics the book has already built — differential equations, the Laplace and Z-transforms, transfer functions, poles and zeros, frequency response — and applies it to systems from every energy domain, so that the same tools describe an RLC circuit, a vehicle suspension, a power-system oscillation and a position servo.

By the end of this part you should understand

- How electrical, mechanical, thermal and fluid systems are reduced to lumped models, and why their equations share the same few forms.
- How nonlinear systems are linearised about an operating point, and where the linear model stops being trustworthy.
- How transfer functions, block diagrams and signal-flow graphs represent interconnected systems, and how loading invalidates naive block multiplication.
- How a model is simplified by dominant poles, lumped delays and step-test fits without losing the dynamics that matter for control.
- What feedback does to poles, sensitivity, disturbances and steady-state error, and why it can destabilise a system that is stable on its own.
- How stability is decided algebraically (Routh–Hurwitz), graphically in the s-plane (root locus) and from the loop's frequency response (Bode and Nyquist), and what gain and phase margins mean physically.
- How PID controllers are structured and tuned, how transient specifications translate into pole locations and margins, and how state-space, controllability, observability and digital

implementation extend those ideas.

Chapters

Chapter	Central question	What it delivers
Chapter 33 — Dynamic-System Modeling	How do we turn a physical device into equations that predict its response, and how far can those equations be simplified?	Lumped models and analogies across domains, linearisation, transfer functions, block diagrams, signal-flow graphs and Mason's rule, interconnection and loading effects, model reduction
Chapter 34 — Feedback and Stability	What does closing a loop do to a system, and how do we know whether the closed loop is stable?	Open- and closed-loop behaviour, characteristic equations, Routh–Hurwitz, root locus, steady-state error, Bode and Nyquist stability criteria
Chapter 35 — Control Design Fundamentals	How do we choose a controller that meets speed, accuracy and robustness specifications, in continuous or digital form?	PID control and tuning, transient-response specifications, gain and phase margins as design targets, state-space models, controllability, observability and digital control

How this part connects to the rest of the book

Part VII is the last destination of the book's backbone, circuit laws → network analysis → transients and AC → network theory → signals and systems → DSP → communication → control, and it draws on nearly all of it. The first- and second-order responses of Chapters 6–7 become the step-response specifications of control design; the s-domain analysis and pole–zero methods of Chapter 11 become the language of transfer functions and root loci; the Bode plots of Chapter 13 become the loop-shaping and margin plots of Chapter 34; the LTI-system and convolution results of Chapters 16–18 justify block-diagram algebra; and the Z-transform and digital filters of Chapters 20 and 23 return as discrete-time controllers in Chapter 35. Communication systems supply many of the loops: automatic gain control, phase-locked loops, power control in cellular networks and laser bias control in optical transmitters (Chapters 26–31) are all feedback systems. Signal-flow graphs, introduced here for control, already served for wave analysis in Chapter 32. The part stops at the theory: motor and converter hardware belong to *Electrical Energy & Machines*, controller circuits and microcontrollers to *Electronics & Computing*, and practical robot and industrial controllers to the robotics and automation books.

Chapter 33 — Dynamic-System Modeling

Part VII · Control-System Fundamentals

A camera gimbal on a small drone holds the horizon steady while the airframe pitches and rolls. Its designer raises the gain of the pitch loop to make it stiffer against wind gusts, and at a certain gain the gimbal starts to buzz at about 40 Hz. Raising the gain further makes the buzz louder until the camera image is useless. Nothing is broken: the motor, the sensor and the controller each do exactly what their datasheets say. The buzz comes from how they interact — the motor's electrical lag, the gimbal's inertia, the flexible mount, the filter on the gyro signal and the controller's sampling delay add up to a phase lag that turns negative feedback into positive feedback at one particular frequency. To predict that frequency before building the hardware, and to choose a gain that avoids it, the designer needs a **model**: a set of equations that describes how each part turns its inputs into outputs over time, and how the parts are connected.

Part VII is about such models and what feedback does to them. This first chapter builds the models. It uses no new mathematics: the differential equations of Chapters 6–7, the Laplace transform of Chapter 11 and the LTI-system ideas of Chapters 16–17 are applied to motors, tanks, ovens, vehicles and generators as well as circuits. What is new is the point of view — the model exists to be *controlled*, so it must be simple enough to design with and accurate enough, in the right frequency range, to be trusted.

> **How do we turn a physical device — electrical, mechanical, thermal, fluid — into equations that predict its response to an input?**
> **How are the models of separate parts combined into one system model, and how far can that model be simplified before it stops being useful?**

Concept	Engineering destination
Lumped physical models and analogies	Motor drives, suspensions, thermal management, process tanks
Differential equations and linearisation	Operating-point models of generators, valves, converters and aircraft
Transfer functions	Controller design, frequency-response analysis, simulation
Block diagrams	Control-system documentation, Simulink-type simulation, drive specifications
Signal-flow graphs and Mason's rule	Closed-form transfer functions of multi-loop systems, RF wave graphs
System interconnections and loading	Cascaded filters, converter–filter interaction, sensor and actuator chains
Model simplification	Dominant-pole approximations, controller tuning from step tests, real-time models

Roadmap. Physical system models and their analogies (33.1) → the differential equations they produce, and linearisation (33.2) → transfer functions (33.3) → block diagrams (33.4) and signal-flow graphs (33.5) as two pictures of the same algebra → how interconnected parts behave (33.6) → how and when a model can be simplified (33.7).

33.1 Physical system models

What a model is for

A model is a mathematical description of a system that predicts its outputs from its inputs, accurate enough for a stated purpose. The same motor may be described by a static torque–speed curve (for sizing), a second-order transfer function (for speed-loop design), a nonlinear model with saturation and friction (for simulation of large moves), or a finite-element thermal model (for overload rating). None of these is "the" model; each is good for one question. Control engineers usually want the simplest model that reproduces the dynamics in the frequency band where the controller acts.

Models are classified along a few axes, all of which recur in this Part:

- **Static or dynamic:** whether outputs depend only on present inputs, or also on stored energy (Chapter 16).
- **Linear or nonlinear:** whether superposition holds. Almost every real system is nonlinear; almost every design method in Chapters 34–35 assumes linearity, so linearisation (Section 33.2) is central.
- **Lumped or distributed:** whether a finite number of state variables suffices (ordinary differential equations) or the state varies continuously in space (partial differential equations — a transmission line, a heated slab, a flexible beam).
- **Time-invariant or time-varying**, **continuous or discrete-time**, **deterministic or stochastic.**
- **White-box, grey-box or black-box:** derived from physical laws, physical structure with fitted parameters, or fitted entirely from input–output data.

Lumped elements in every energy domain

A lumped model divides a system into elements that each store, dissipate or supply energy, connected at points where a single value of each variable is meaningful. In every physical domain two kinds of variable appear. An **across variable** is measured between two points: voltage, velocity, angular velocity, pressure, temperature. A **through variable** flows through an element: current, force, torque, volumetric flow, heat flow. Their product is power (except in the thermal domain, where heat flow is already a power). Each domain has elements that store energy in one of two ways, and elements that dissipate it, which produces the **analogies** of Table 33.1.

Table 33.1 — Analogous lumped elements (force–current analogy).

Domain	Across variable	Through variable	Dissipative element	Through-storage (inertia-like)	Across-storage (compliance-like)
Electrical	Voltage v (V)	Current i (A)	Resistor: $v = Ri$	Inductor: $v = L\,di/dt$	Capacitor: $i = C\,dv/dt$
Mechanical translation	Velocity u (m/s)	Force f (N)	Damper: $f = c\,u$	Spring: $f = k\int u\,dt$	Mass: $f = m\,du/dt$
Mechanical rotation	Angular velocity ω (rad/s)	Torque τ (N·m)	Rotary damper: $\tau = b\,\omega$	Torsional spring: $\tau = k_\theta \int \omega\,dt$	Inertia: $\tau = J\,d\omega/dt$
Fluid	Pressure p (Pa)	Flow q (m³/s)	Restriction: $p = R_f q$	Inertance: $p = I_f\,dq/dt$	Tank: $q = C_f\,dp/dt$
Thermal	Temperature T (K)	Heat flow $\dot{Q}$ (W)	Thermal resistance: $T = R_\theta \dot{Q}$	— (none)	Heat capacity: $\dot{Q} = C_\theta\,dT/dt$

In this "mobility" or force–current analogy, a mass is the analogue of a capacitor (both store energy proportional to the square of an across variable: $\tfrac{1}{2}mu^2$, $\tfrac{1}{2}Cv^2$) and a spring is the analogue of an inductor. The older force–voltage analogy pairs mass with inductance instead; both give correct equations, but the force–current form preserves the topology — elements in parallel mechanically appear in parallel electrically. The thermal domain has no inductance analogue: heat does not have inertia, which is why purely thermal systems never oscillate on their own. Figure 33.1 sets three analogous systems side by side.

> **Status of the principle: Lumped-parameter modelling**
> *Approximate.* Valid when the system's spatial extent is small compared with the wavelength (or diffusion length) of the fastest dynamics of interest; a shaft becomes a distributed torsional line at its resonant frequencies, a heated slab a diffusion PDE at short times, exactly as a circuit becomes a transmission line at high frequency (Chapter 14).

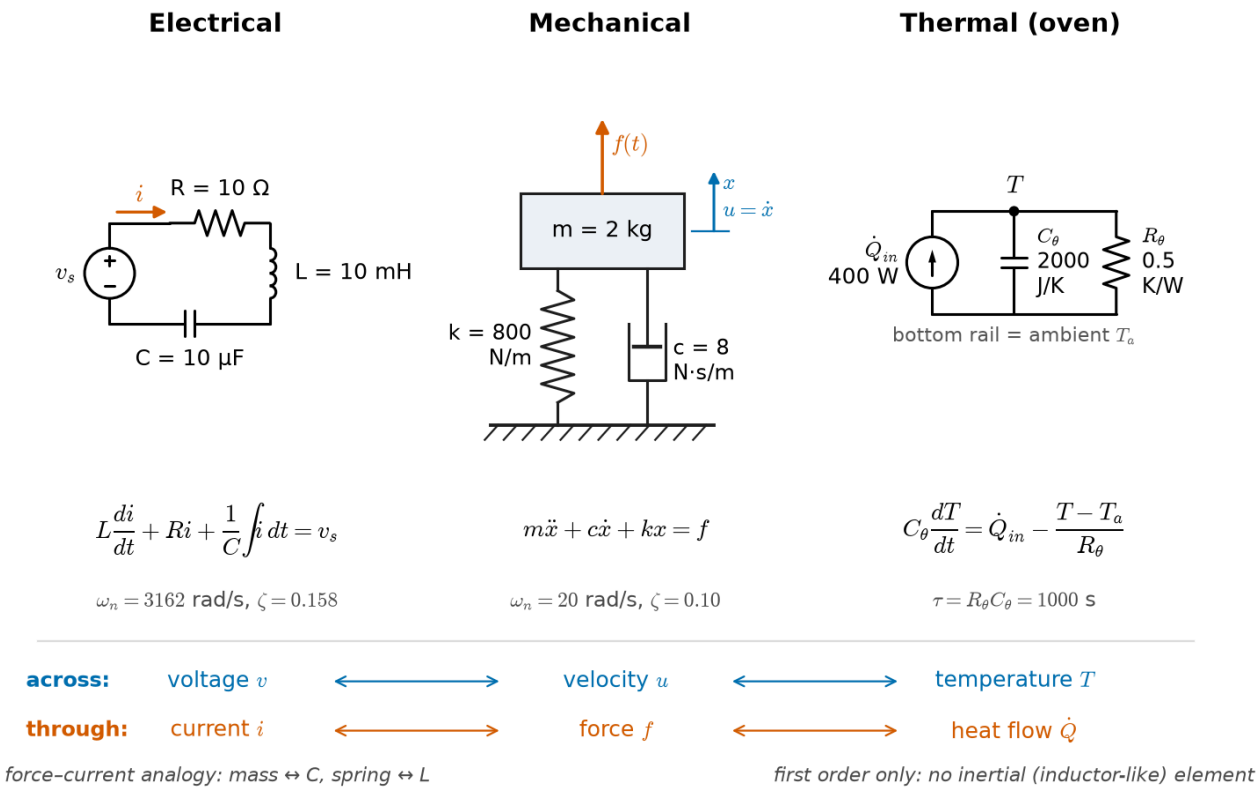

Figure 33.1 — Three analogous lumped systems: a series RLC circuit, a mass–spring–damper and a first-order thermal RC network, with the variables that correspond.

Representative parameter values

Table 33.2 — Representative parameters of dynamic systems met in EEE.

System	Parameter	Typical value
Small permanent-magnet servo motor (100–200 W)	Rotor inertia J; torque constant K_t; armature R, L	2×10^{-5}–10^{-4} kg·m²; 0.03–0.1 N·m/A; 0.5–3 Ω, 0.5–3 mH
Quarter-car suspension	Sprung mass; spring rate; damping ratio	250–450 kg; 15–35 kN/m; 0.2–0.4
Hard-disk or optical pickup actuator	Moving mass; resonance	Grams; 1–10 kHz structural modes
Synchronous generator	Inertia constant H	2–9 s (thermal and hydro units)
Electromechanical oscillation of a power system	Frequency	0.1–0.8 Hz inter-area; 0.8–2.5 Hz local mode
Thermocouple in a sheath	Time constant	0.1–10 s
Domestic oven or heated enclosure	Time constant	10–60 min
Building heating zone	Time constant	Hours
Process tank (1–10 m³)	Level time constant	Minutes to tens of minutes
MEMS accelerometer	Proof-mass resonance	1–30 kHz

The spread of time constants — microseconds in an armature, hours in a building — already says something about modelling: in any given loop, many dynamics are either too fast to matter or too slow to change during the response, and Section 33.7 will exploit that.

33.2 Differential equations

Writing the equations

The equations of a lumped model come from two kinds of law. **Element laws** relate across and through variables of each element (Table 33.1). **Interconnection laws** express conservation at nodes and compatibility around loops: Kirchhoff's current and voltage laws in circuits (Chapter 2), Newton's second law and d'Alembert's principle in mechanics, mass and energy balances in fluid and thermal systems. Combining them yields one differential equation per energy-storing element.

> **Status of the principle: Conservation laws as interconnection rules**
> *Fundamental.* Newton's second law, conservation of charge, mass and energy are physical laws; their lumped forms (KCL at a node, force balance on a rigid body, energy balance on a well-mixed volume) inherit the lumped approximation above.

Series RLC circuit. KVL around the loop with source v_s:

$$L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt = v_s \quad \Longleftrightarrow \quad LC\frac{d^2v_C}{dt^2} + RC\frac{dv_C}{dt} + v_C = v_s$$

With $R = 10\ \Omega$, $L = 10$ mH, $C = 10\ \mu$F: $\omega_n = 1/\sqrt{LC} = 3162$ rad/s (503 Hz) and $\zeta = \frac{R}{2}\sqrt{C/L} = 0.158$, an underdamped response (Chapter 7).

Mass–spring–damper. Newton's law on a mass m with spring k, damper c and applied force f:

$$m\ddot{x} + c\dot{x} + kx = f$$

For $m = 2$ kg, $k = 800$ N/m, $c = 8$ N·s/m: $\omega_n = \sqrt{k/m} = 20$ rad/s (3.18 Hz), $\zeta = c/(2\sqrt{km}) = 0.10$, and the static deflection is $1/k = 1.25$ mm per newton. Both systems have the standard form $\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2 y = \omega_n^2 u$, so everything learned about the RLC circuit in Chapter 7 — overshoot, ringing frequency, settling — transfers directly to vehicle suspensions, machine mounts and seismometers.

Thermal oven. An oven of heat capacity $C_\theta = 2000$ J/K, losing heat to ambient T_a through $R_\theta = 0.5$ K/W, heated by power P:

$$C_\theta \frac{dT}{dt} = P - \frac{T - T_a}{R_\theta} \quad\Longrightarrow\quad \tau\frac{dT}{dt} + (T - T_a) = R_\theta P, \qquad \tau = R_\theta C_\theta = 1000\ \text{s}$$

A 400 W step raises the steady temperature by $R_\theta P = 200$ K with a time constant of about 17 minutes — first order, like the RC circuit of Chapter 6.

An electromechanical example: the DC motor

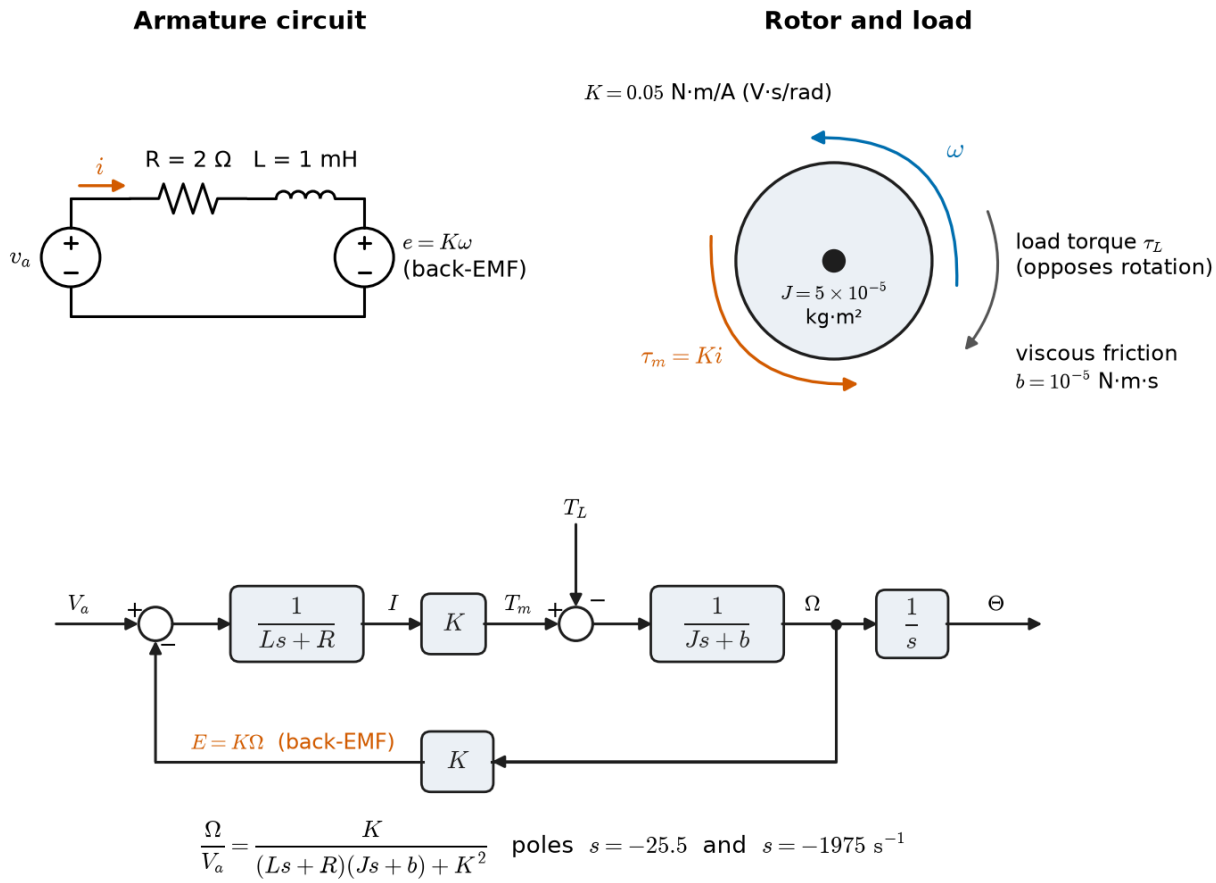

Figure 33.2 — Permanent-magnet DC motor: armature circuit, mechanical load and the resulting block diagram with back-EMF feedback.

Worked example — transfer function of a servo motor. A permanent-magnet DC motor has armature resistance $R = 2\ \Omega$, inductance $L = 1$ mH, torque and back-EMF constant $K = 0.05$ N·m/A (= V·s/rad), rotor-plus-load inertia $J = 5 \times 10^{-5}$ kg·m² and viscous friction $b = 1 \times 10^{-5}$ N·m·s. Its equations couple the electrical and mechanical domains (Figure 33.2):

$$L\frac{di}{dt} + Ri = v_a - K\omega, \qquad J\frac{d\omega}{dt} + b\omega = Ki - \tau_L$$

Taking Laplace transforms with zero initial conditions and $\tau_L = 0$, eliminating $I(s)$:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K}{(Ls+R)(Js+b) + K^2} = \frac{0.05}{5\times10^{-8}s^2 + 1.0001\times10^{-4}s + 2.52\times10^{-3}}$$

The denominator roots are $s = -25.5$ and $s = -1975\ \text{s}^{-1}$: a slow **mechanical** pole (time constant 39 ms) and a fast **electrical** pole (0.51 ms). The DC gain is $0.05/2.52\times10^{-3} = 19.8$ rad/s per volt (189 rpm/V), so 12 V gives about 238 rad/s (2270 rpm) at no load. Note how the back-EMF term K^2 couples the domains: without it the poles would be at $-R/L = -2000$ and $-b/J = -0.2$; the electrical damping of the back-EMF moves the mechanical pole from -0.2 to -25.5, which is why a motor driven from a voltage source stops quickly and one driven from a current source coasts.

> **Boundary.**
> The construction, magnetic design and power electronics of motors belong to *Electrical Energy & Machines* and *Electronics & Computing*; practical drive controllers to the robotics and automation books. Here the motor is a two-state linear model with measured constants.

State-variable form

The same equations can be written as first-order equations in the energy-storage variables. With state $\mathbf{x} = [i\ \ \omega]^T$:

$$\dot{\mathbf{x}} = \begin{bmatrix} -R/L & -K/L \\ K/J & -b/J \end{bmatrix}\mathbf{x} + \begin{bmatrix} 1/L \\ 0 \end{bmatrix} v_a$$

Every lumped linear model can be written as $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$, $y = \mathbf{C}\mathbf{x} + \mathbf{D}u$, with one state per independent energy store; the eigenvalues of $\mathbf{A}$ are the poles. This form is the starting point for computer simulation and for the state-space methods of Chapter 35.

Linearisation about an operating point

Real systems are nonlinear: valve flow depends on the square root of pressure drop, aerodynamic drag on the square of speed, generator power on the sine of rotor angle. Linear methods remain usable because control systems normally hold a system near an **operating point** (an equilibrium), and small deviations from it obey linear equations. For $\dot{x} = f(x,u)$ with equilibrium $f(x_0,u_0) = 0$, a first-order Taylor expansion in the deviations $\delta x = x - x_0$, $\delta u = u - u_0$ gives

$$\dot{\delta x} \approx \left.\frac{\partial f}{\partial x}\right|_0 \delta x + \left.\frac{\partial f}{\partial u}\right|_0 \delta u$$

Status of the principle: Small-signal linearisation
Approximate. Accurate when deviations are small enough that second-order Taylor terms are negligible; it fails at large excursions, near hard nonlinearities (saturation, backlash, Coulomb friction, switching), and where the Jacobian vanishes — the same small-signal idea as transistor models in electronics.

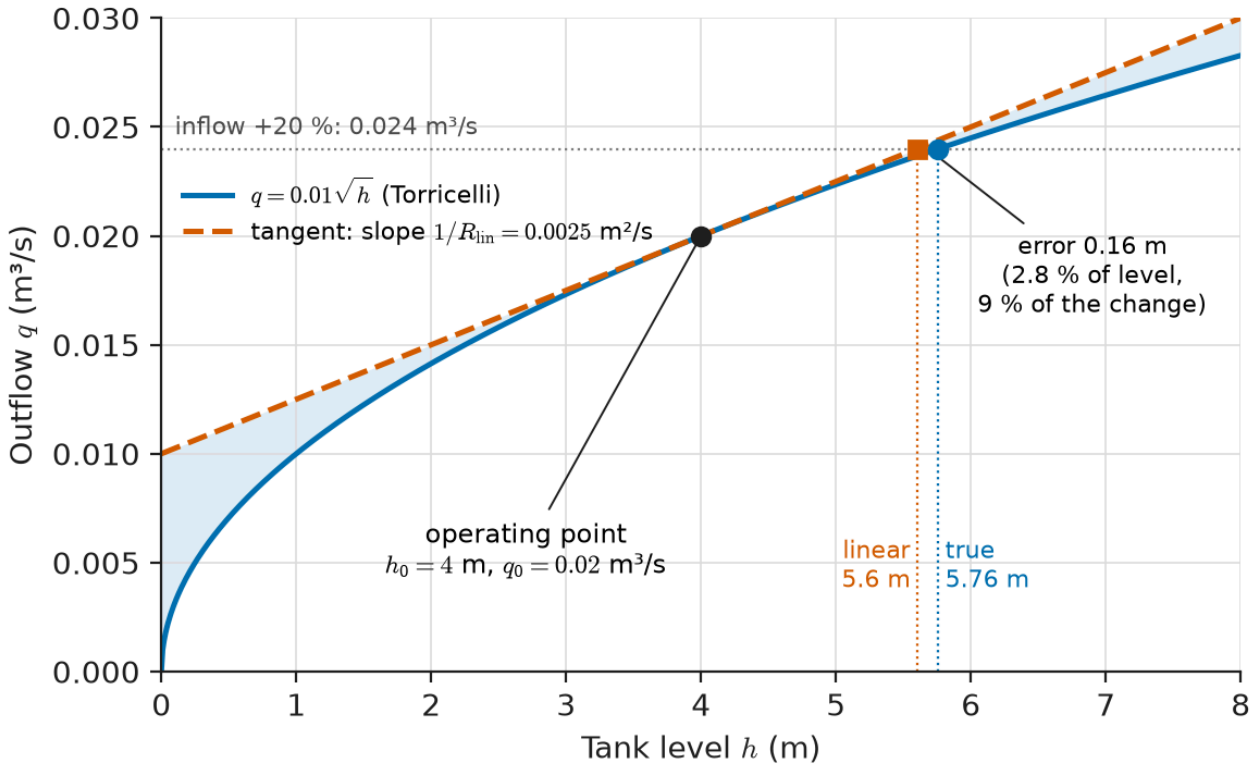

Figure 33.3 — Nonlinear outflow of a gravity-drained tank and its tangent-line linearisation about a 4 m operating level.

Worked example — linearising a draining tank. A tank of cross-section $A = 2$ m² is fed with inflow q_{in} and drains through an orifice with outflow $q_{\text{out}} = k\sqrt{h}$ (Torricelli's law). At the operating point $h_0 = 4$ m, $q_0 = 0.02$ m³/s, so $k = q_0/\sqrt{h_0} = 0.01$ m$^{2.5}$/s. The mass balance is

$$A\frac{dh}{dt} = q_{\text{in}} - k\sqrt{h}$$

Linearising $k\sqrt{h}$ about h_0: $\partial q_{\text{out}}/\partial h = k/(2\sqrt{h_0}) = 0.0025$ m²/s, i.e. a linearised outlet resistance $R_{\text{lin}} = 2\sqrt{h_0}/k = 400$ s/m². Hence

$$A\frac{d\,\delta h}{dt} = \delta q_{\text{in}} - \frac{\delta h}{R_{\text{lin}}} \quad\Longrightarrow\quad \frac{\delta H(s)}{\delta Q_{\text{in}}(s)} = \frac{R_{\text{lin}}}{AR_{\text{lin}}s + 1} = \frac{400}{800s + 1}$$

a first-order lag with time constant 800 s (13 minutes). *Check of validity:* a 20 % increase of inflow ($\delta q = 0.004$ m³/s) gives a linear prediction of $\delta h = 400 \times 0.004 = 1.6$ m, i.e. $h = 5.6$ m; the true new equilibrium is $h = (0.024/0.01)^2 = 5.76$ m. The 0.16 m discrepancy, 9 % of the true change, is the price of linearity at a 20 % excursion (Figure 33.3); for a 5 % step it falls to about 2.5 %. Note also that the

time constant depends on the operating point ($\tau = 2A\sqrt{h_0}/k$ grows with level), so a controller tuned at 4 m behaves differently at 1 m.

Worked example — the swing equation of a generator. A synchronous generator connected to a large grid obeys, in per unit, the nonlinear swing equation

$$\frac{2H}{\omega_s}\frac{d^2\delta}{dt^2} = P_m - P_{\max}\sin\delta - D\frac{d\delta}{dt}$$

where δ is the rotor angle relative to the grid, H the inertia constant, $\omega_s = 2\pi \times 50$ rad/s, and $P_{\max}\sin\delta$ the electrical power. Linearising about δ_0 gives the **synchronising power coefficient** $P_s = P_{\max}\cos\delta_0$ and a second-order model

$$M\,\Delta\ddot{\delta} + D\,\Delta\dot{\delta} + P_s\,\Delta\delta = \Delta P_m, \qquad M = \frac{2H}{\omega_s}$$

With $H = 5$ s, $M = 10/314.16 = 0.0318$ s²/rad; with $P_s = 1.5$ pu/rad the natural frequency is $\omega_n = \sqrt{P_s/M} = 6.86$ rad/s, or 1.09 Hz — the "local mode" of Table 33.2. The inherent damping is small (with $D = 0.02$ pu·s/rad, $\zeta = D/(2\sqrt{P_s M}) = 0.046$), which is why generators carry power-system stabilisers. The linear model is valid for small disturbances; a fault that swings δ past 90° needs the nonlinear equation and leads to the transient-stability problem studied in power-system texts.

33.3 Transfer functions

Definition

For a linear time-invariant system described by

$$a_n\frac{d^n y}{dt^n} + \cdots + a_1\frac{dy}{dt} + a_0 y = b_m\frac{d^m u}{dt^m} + \cdots + b_0 u$$

the Laplace transform with **all initial conditions zero** gives the **transfer function**

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + \cdots + b_0}{a_n s^n + \cdots + a_0} = K\frac{\prod_i (s - z_i)}{\prod_k (s - p_k)}$$

It is the Laplace transform of the impulse response (Chapter 17) and, evaluated at $s = j\omega$, the frequency response (Chapter 18). The roots of the denominator are the **poles** — the natural modes $e^{p_k t}$ that appear in every response — and the roots of the numerator are the **zeros**, which shape how strongly each mode is excited. The denominator set to zero is the **characteristic equation**. Physical systems are **proper** ($m \le n$): no physical system has gain rising without limit at high frequency.

> **Status of the principle: Transfer-function representation**
> *Derived.* Follows from the linearity and differentiation property of the Laplace transform applied to a linear constant-coefficient ODE; it describes only the zero-state input–output behaviour, so initial conditions must be added separately and any modes cancelled between numerator and denominator are hidden (Chapter 35, controllability and observability).

Standard forms and their parameters

Control engineers read transfer functions through a handful of standard forms (Table 33.3), each tied to a measurable property of the step response (Chapters 6–7).

Table 33.3 — Standard transfer-function forms.

Form	$G(s)$	Parameters	Example
Gain	K	Static sensitivity	Tachogenerator, 0.05 V·s/rad
Integrator	K/s	Rate of rise per unit input	Motor angle from speed; tank level from net flow
First-order lag	$K/(\tau s+1)$	DC gain K, time constant τ	Oven ($\tau = 1000$ s), RC filter, thermocouple
Second order	$K\omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$	ω_n, ζ	Suspension, RLC, generator swing
Lead / lag	$K(\tau_z s+1)/(\tau_p s+1)$	Zero and pole time constants	Compensators, sensor networks
Dead time	e^{-sT_d}	Delay T_d	Conveyor transport, network latency, ADC/DSP latency
First order plus dead time	$Ke^{-sT_d}/(\tau s+1)$	K, τ, T_d	Process plant fitted from step tests

The dead time e^{-sT_d} is not a rational function: it has unit magnitude and phase $-\omega T_d$ that grows without bound with frequency. It is the model of transport delay (fluid in a pipe, material on a belt) and of computation and sampling delay in digital controllers (Chapter 35), and it is one of the main reasons the drone gimbal in the opening story oscillated.

Poles, zeros and the shape of the response

Pole locations translate directly into time-domain behaviour. A real pole at $-a$ contributes e^{-at}; a complex pair $-\sigma \pm j\omega_d$ contributes a decaying oscillation $e^{-\sigma t}\cos(\omega_d t + \phi)$; a pole in the right half-plane grows without bound (Chapter 34). Zeros do not create modes but change the weighting: a left-half-plane zero near the dominant pole adds overshoot; a **right-half-plane zero** makes the step response start in the wrong direction — the "inverse response" of a boiler drum level when feedwater is increased, or the initial dip in output voltage of a boost converter when its duty cycle is increased. Such non-minimum-phase zeros limit achievable control bandwidth no matter how the controller is designed.

Multiple inputs and outputs

Most systems have more than one input — a motor has voltage and load torque; an oven has heater power and ambient temperature. By superposition, the output is the sum of each input's contribution through its own transfer function. For the motor,

$$\Omega(s) = \frac{K}{(Ls+R)(Js+b)+K^2}V_a(s) - \frac{Ls+R}{(Ls+R)(Js+b)+K^2}T_L(s)$$

Both transfer functions share the denominator — the poles are properties of the system, not of the input — while the numerators differ. With several outputs as well, the transfer functions form a **transfer matrix** $\mathbf{G}(s)$, with $\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$ in state-space terms.

33.4 Block diagrams

Elements and rules

A **block diagram** draws a system as blocks (each a transfer function from its input arrow to its output arrow), **summing junctions** (which add or subtract signals) and **pick-off points** (which copy a signal to several destinations). Unlike a circuit diagram, a block diagram shows signal flow, not physical connection: each block is unidirectional, and connecting two blocks assumes the second does not load the first (Section 33.6). Block diagrams are the lingua franca of control: drive manufacturers specify their speed and current loops with them, and simulation tools are built on them.

Three connection rules generate all the others.

Table 33.4 — Block-diagram algebra.

Connection	Equivalent single block
Series (cascade) G_1 then G_2	$G_1 G_2$
Parallel G_1 and G_2 summed	$G_1 \pm G_2$
Negative feedback: forward G, feedback H	$\dfrac{G}{1+GH}$
Positive feedback	$\dfrac{G}{1-GH}$
Moving a summing junction ahead of a block G	Divide the moved input by G
Moving a pick-off point behind a block G	Multiply the picked-off branch by $1/G$
Moving a pick-off point ahead of a block G	Multiply the picked-off branch by G

> **Status of the principle: Block-diagram reduction rules**
> *Derived*. Each rule is an algebraic identity for linear operators in the Laplace domain (the feedback formula follows from $E = R - HY$, $Y = GE$); they apply to LTI blocks with no loading between them, and not to nonlinear blocks such as saturations.

The closed-loop formula $G/(1+GH)$ is the most important single result of Part VII; Chapter 34 is largely a study of what it implies. The term GH is the **loop gain** (open-loop transfer function), and $1 + GH = 0$ is the closed-loop characteristic equation.

Reducing a multi-loop diagram

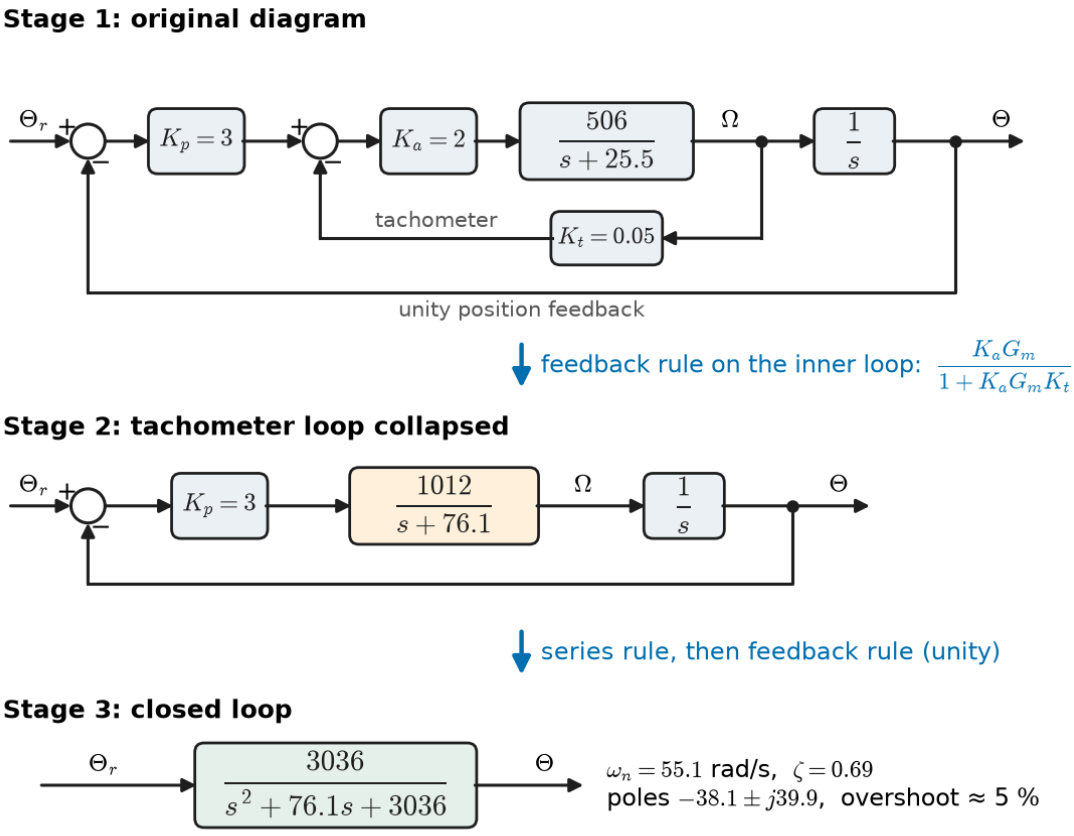

Figure 33.4 — Reduction of a cascaded position servo with an inner tachometer loop to a single closed-loop transfer function.

Worked example — reducing a position servo. The motor of Section 33.2, simplified to its dominant pole (Section 33.7), has speed response $G_m(s) = 506/(s + 25.5)$ rad/s per volt. It is driven by a power amplifier of gain $K_a = 2$, with a tachometer of $K_t = 0.05$ V·s/rad feeding speed back to the amplifier input, an integrator $1/s$ from speed to angle, and an outer position loop with proportional gain $K_p = 3$ and unity position feedback (Figure 33.4).

Step 1 — inner loop. By the feedback rule,

$$G_{\text{in}}(s) = \frac{K_a G_m}{1 + K_a G_m K_t} = \frac{1012/(s + 25.5)}{1 + 50.6/(s + 25.5)} = \frac{1012}{s + 76.1}$$

The tachometer loop has moved the speed pole from -25.5 to -76.1 — three times faster — while reducing the DC speed gain from 39.7 to 13.3 rad/s per volt.

Step 2 — series. The outer forward path is $K_p G_{\text{in}}(s)/s = 3036/[s(s + 76.1)]$.

Step 3 — outer loop. With unity feedback,

$$\frac{\Theta(s)}{\Theta_r(s)} = \frac{3036}{s^2 + 76.1s + 3036}$$

which is the standard second-order form with $\omega_n = \sqrt{3036} = 55.1$ rad/s and $\zeta = 76.1/(2 \times 55.1) = 0.69$. The closed-loop poles are $-38.1 \pm j39.9$, and the step response overshoots by about 5 % (Chapter 7). Reduction has turned a diagram with two loops into one transfer function whose parameters can be read off directly — and shows that the inner tachometer loop supplies the damping that the outer loop needs.

33.5 Signal-flow graphs

Nodes and branches

A **signal-flow graph** carries the same information as a block diagram in a sparser form. Each **node** is a signal (a variable); each directed **branch** carries a gain (a transfer function) from one node to another; the value of a node is the sum of all signals entering it. Summing junctions disappear into nodes, and blocks become branches. The vocabulary:

- A **path** is a connected sequence of branches in the arrow direction that visits no node twice; a **forward path** runs from the input node to the output node.
- A **loop** is a closed path returning to its starting node; its **loop gain** is the product of its branch gains.
- Two loops are **non-touching** if they share no node.

Signal-flow graphs come from S. J. Mason (1953), who used them for electronic feedback circuits; they reappear as wave-flow graphs for S-parameters in microwave networks (Chapter 32).

Mason's gain formula

The transfer function from an input node to an output node is

$$T = \frac{1}{\Delta} \sum_k P_k \Delta_k$$

where P_k is the gain of the k-th forward path, and the **graph determinant** is

$$\Delta = 1 - \sum L_i + \sum L_i L_j - \sum L_i L_j L_k + \cdots$$

with $\sum L_i$ the sum of all loop gains, $\sum L_i L_j$ the sum of products of all pairs of non-touching loops, and so on. Δ_k is Δ evaluated with all loops that touch the k-th forward path removed.

> **Status of the principle: Mason's gain formula**
> *Derived.* A closed-form solution, by Cramer's rule, of the linear simultaneous equations that the graph represents; it is exact for any linear graph but becomes error-prone by hand when there are many non-touching loop combinations, where direct matrix solution is safer.

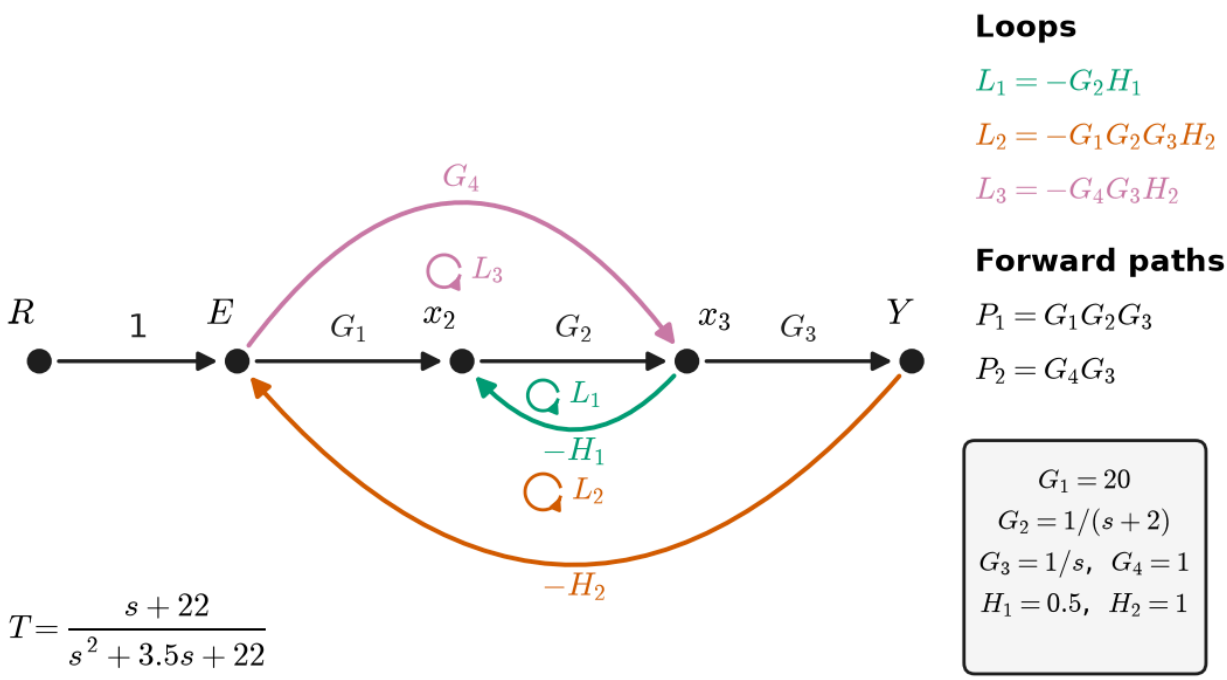

Figure 33.5 — Signal-flow graph with two forward paths and three loops used to illustrate Mason's gain formula.

Worked example — a two-path, three-loop system. Figure 33.5 shows a system with nodes R, E, x_2, x_3, Y: forward branches G_1 ($E \to x_2$), G_2 ($x_2 \to x_3$), G_3 ($x_3 \to Y$), a feedforward branch G_4 ($E \to x_3$), a local feedback $-H_1$ ($x_3 \to x_2$) and an outer feedback $-H_2$ ($Y \to E$).

Forward paths: $P_1 = G_1G_2G_3$ and $P_2 = G_4G_3$.

Loops: $L_1 = -G_2H_1$ (between x_2 and x_3), $L_2 = -G_1G_2G_3H_2$, $L_3 = -G_4G_3H_2$. Every pair shares a node (L_1 and L_3 share x_3), so there are no non-touching pairs.

Determinants: $\Delta = 1 + G_2H_1 + G_1G_2G_3H_2 + G_4G_3H_2$. Both forward paths touch all three loops, so $\Delta_1 = \Delta_2 = 1$.

$$T = \frac{G_1G_2G_3 + G_4G_3}{1 + G_2H_1 + G_1G_2G_3H_2 + G_4G_3H_2}$$

With $G_1 = 20$, $G_2 = 1/(s+2)$, $G_3 = 1/s$, $G_4 = 1$, $H_1 = 0.5$, $H_2 = 1$, multiply numerator and denominator by $s(s+2)$:

$$T(s) = \frac{20 + (s+2)}{s(s+2) + 0.5s + 20 + (s+2)} = \frac{s+22}{s^2 + 3.5s + 22}$$

The poles are $-1.75 \pm j4.35$ ($\omega_n = 4.69$ rad/s, $\zeta = 0.37$), the DC gain is $22/22 = 1$, and the feedforward branch has introduced a zero at -22. Solving the four node equations directly ($E = R - H_2Y$, $x_2 = G_1E - H_1x_3$, $x_3 = G_2x_2 + G_4E$, $Y = G_3x_3$) gives the same result, as it must.

When to use which picture

Block diagrams are better for communicating structure — they mirror the hardware and show where sensors and actuators sit. Signal-flow graphs are better for computing a transfer function by hand in a

multi-loop system, and for wave problems where each node is a travelling-wave amplitude. Software tools do neither: they assemble the state-space matrices of all components and solve numerically.

33.6 System interconnections

The three basic connections

Every interconnection of LTI systems is built from **series**, **parallel** and **feedback** connections (Table 33.4), and every one of them produces a new LTI system whose poles are determined by the interconnection: series and parallel connections keep the poles of the parts (unless cancelled), whereas feedback moves them. That single fact is why feedback is so powerful and so dangerous — it can make slow systems fast and unstable systems stable, or the reverse (Chapter 34).

> **Status of the principle: Poles of interconnected systems**
> *Derived*. For series and parallel connections the denominator is the product of the component denominators; for feedback it is $D_G D_H + N_G N_H$, so the closed-loop poles are new roots that depend on the loop gain — a direct consequence of the algebra in Table 33.4.

Loading: when blocks are not independent

Block-diagram multiplication assumes that connecting block 2 to block 1 does not change block 1's behaviour — that block 1 has zero output impedance or block 2 infinite input impedance. Physical subsystems rarely satisfy this, and ignoring it is one of the commonest modelling errors.

Worked example — two RC sections in cascade. One RC low-pass section with $R = 10\ \text{k}\Omega$ and $C = 10\ \mu\text{F}$ has $G(s) = 1/(\tau s + 1)$ with $\tau = 0.1$ s. Two identical sections connected through an ideal buffer amplifier give $1/(\tau s + 1)^2$: a double pole at $-10\ \text{s}^{-1}$. Connected directly, the second section draws current from the first capacitor, and nodal analysis (Chapter 3) gives

$$\frac{V_2(s)}{V_1(s)} = \frac{1}{\tau^2 s^2 + 3\tau s + 1} = \frac{1}{0.01 s^2 + 0.3 s + 1}$$

whose poles are -3.82 and $-26.2\ \text{s}^{-1}$. The loaded network is markedly slower: its step response reaches 50 % at 0.22 s instead of 0.17 s, and its 10–90 % rise time is 0.59 s against 0.34 s. Multiplying the individual transfer functions would have been wrong by almost a factor of two in speed. The cure is either to model the connection at the level of the underlying physics (writing the equations for the combined network) or to insert a buffer.

The same issue appears across EEE. A sensor with significant output impedance is loaded by the input of its amplifier; a mechanical gearbox reflects the load inertia back to the motor as J_L/N^2; and a **switched-mode converter with an input filter** is the classic power-electronics example. A tightly regulated converter draws constant power, so its input current falls as its input voltage rises: a small-signal input resistance of $-V^2/P$ — for a 100 W load at 48 V, $-23\ \Omega$. Connected to an LC input filter whose characteristic impedance is higher than 23 Ω, this negative resistance cancels the filter's damping and the combination oscillates, although each part is stable alone. Middlebrook's criterion — keep the

filter's output impedance well below the converter's input impedance magnitude — is an interconnection rule derived from exactly this loading analysis.

Two-port and impedance-based modelling

A systematic way to handle loading is to model each subsystem as a two-port (Chapter 12) with effort and flow variables at each port, rather than as a one-way block. Cascading then uses ABCD (transmission) matrices, which multiply correctly regardless of loading; the one-way block diagram is recovered only when one port's impedance is negligible. Bond graphs and object-oriented modelling languages (Modelica and the "acausal" component libraries of modern simulators) are built on this idea: components are connected by physical ports, and the software works out the direction of computation.

Disturbances, sensors and actuators

A model of a controlled system interconnects at least four kinds of subsystem: the **plant** to be controlled, the **actuator** (amplifier, valve, heater driver) that converts controller output into plant input, the **sensor** that measures the output (with its own dynamics, gain and noise), and the **controller**. External signals enter at defined points: the reference, **disturbances** (load torque, ambient temperature, wind gusts, grid voltage dips) and **measurement noise**. The generic arrangement, with each of these as a block, is the starting point of Chapter 34; its value lies in making explicit *where* each disturbance enters, because a disturbance entering before a slow plant pole is filtered by it, whereas one entering at the output is not.

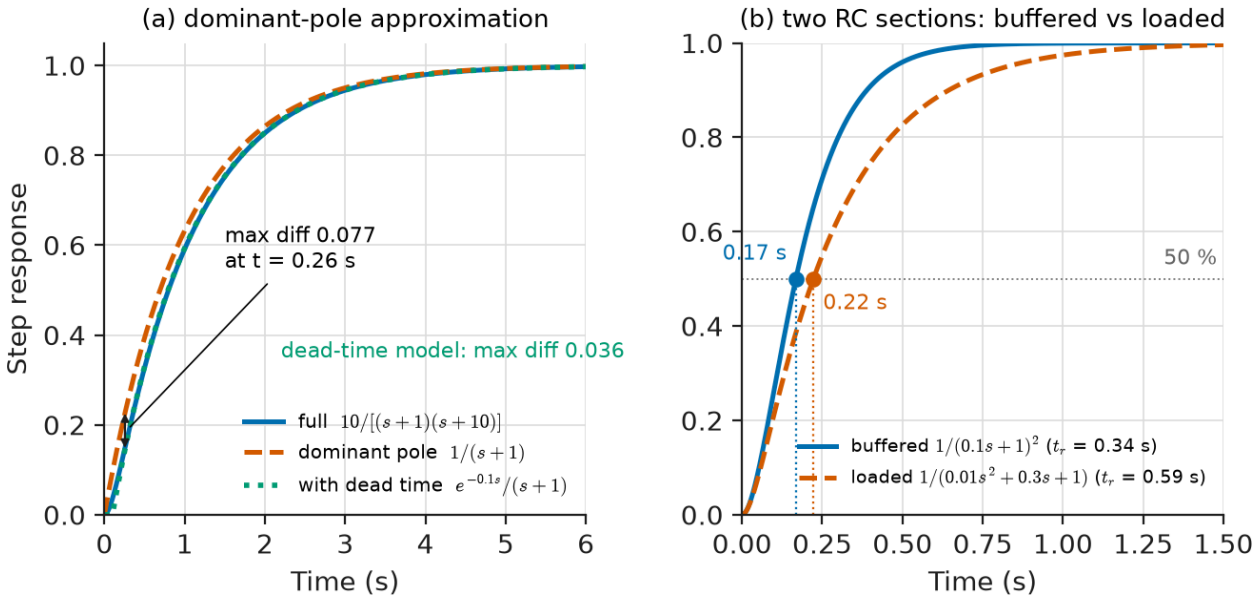

Figure 33.6 — Step responses of full and simplified models: a two-pole plant versus its dominant-pole and dead-time approximations, and loaded versus buffered RC sections.

33.7 Model simplification

Why simplify

A detailed model contains every pole the physics produces: the motor's electrical pole, the shaft's torsional mode, the sensor filter, the amplifier bandwidth, the load's structural resonances. Most of them lie far outside the frequency range in which the controller acts. Keeping them makes the model harder to understand, harder to design with, and — when parameters are uncertain — not necessarily more accurate. Simplification keeps what matters at the frequencies of interest and replaces the rest by gains or small delays.

Dominant poles

If one pole (or complex pair) is much closer to the imaginary axis than all others, its mode decays slowest and dominates the response after a short initial interval. Removing the faster poles **while preserving the DC gain** gives a lower-order model. For $G(s) = 10/[(s+1)(s+10)]$ the pole at -10 is ten times faster than the one at -1, and

$$G(s) = \frac{10}{(s+1)(s+10)} = \frac{1}{(s+1)(0.1s+1)} \approx \frac{1}{s+1}$$

The fast pole must be written in time-constant form $(0.1s+1)$ before it is dropped, so that the DC gain is kept; dropping $(s+10)$ from the first form would change the gain by a factor of ten — a common error.

> **Status of the principle: Dominant-pole approximation**
> *Approximate.* Reliable when the neglected poles are at least about five to ten times further from the imaginary axis than the dominant ones and no zeros lie near the dominant poles; it fails when feedback with high gain moves the "fast" poles toward the slow ones — the reason a neglected pole often reappears as the cause of oscillation at high gain.

Worked example — how good is the dominant-pole model? Comparing the step responses of the full model $10/[(s+1)(s+10)]$ and the approximation $1/(s+1)$ (Figure 33.6): the largest difference is 0.077 (7.7 % of the final value) at $t = 0.26$ s, where the full model lags because of its second pole. The 10–90 % rise times are 2.21 s and 2.20 s, and the 2 % settling times 4.02 s and 3.91 s. For the purpose of designing a slow outer loop, the first-order model is entirely adequate.

A better approximation replaces the neglected lag by an equal **dead time**: $1/(0.1s+1) \approx e^{-0.1s}$, since both have phase $\approx -0.1\omega$ at low frequency. The model $e^{-0.1s}/(s+1)$ reduces the largest step-response error to 0.036. This "lump the small lags into a delay" rule is the basis of the first-order-plus-dead-time models used throughout process control.

For the servo motor of Section 33.2, the same reasoning gives $G_m(s) = 19.8/(0.0392s+1) = 506/(s+25.5)$, keeping the mechanical pole and the DC gain and neglecting the electrical pole at -1975, which is 77 times faster. The step responses of the full and reduced models differ by at most 1.2 % of the final speed. The reduced model was used in Section 33.4 — legitimately, because the closed-loop poles at $-38 \pm j40$ are still more than 30 times slower than the neglected pole.

Pole–zero cancellation and near-cancellation

A zero close to a pole nearly cancels it: the mode is present but weakly excited. For $(s+2.1)/[(s+2)(s+20)]$, the residue of the pole at -2 is $0.1/18 \approx 0.006$, so the model is well approximated by $1.05/(s+20)$ — the factor $2.1/2 = 1.05$ preserving DC gain. But near-cancellation of an *unstable* or lightly damped pole is never acceptable: the mode is still there, and any disturbance entering inside the cancelled loop excites it (Chapter 35 calls this a loss of controllability or observability).

Neglecting fast dynamics and very slow ones

Two systematic simplifications follow from separating time scales:

- **Fast dynamics** (time constants much shorter than the loop's response) become static: the amplifier becomes a gain, the motor's current loop becomes a unit gain, the sensor filter a small delay. Formally, this is a singular-perturbation approximation: set $\epsilon \dot{x}_{\text{fast}} = 0$.
- **Very slow dynamics** (time constants much longer than the response) become constants or slowly varying parameters: the temperature drift of a motor's winding resistance during a 50 ms positioning move; the slow thermal time constant of a power transistor during a 1 ms current-loop transient.

Data-driven simplification: step-test models

Many plants — boilers, ovens, chemical reactors, building zones — are too complex to model from first principles but easy to test. Applying a step to the actuator and recording the output gives a **process reaction curve**, to which a first-order-plus-dead-time model $K e^{-sT_d}/(\tau s + 1)$ is fitted. A common two-point method uses the times t_{28} and t_{63} at which the response reaches 28.3 % and 63.2 % of its final change:

$$\tau = 1.5\,(t_{63} - t_{28}), \qquad T_d = t_{63} - \tau$$

> **Status of the principle: First-order-plus-dead-time step-test models**
> *Empirical.* The FOPDT form is fitted to observed reaction curves; it captures the low-frequency behaviour of many overdamped processes well enough for PID tuning (Chapter 35) but misrepresents oscillatory, integrating or inverse-response plants.

Worked example — fitting a heater from a step test. An electric heater on a test chamber is stepped from 0 to 2 kW. The chamber temperature rises from 20 °C and settles at 80 °C, so $K = 60/2 = 30$ K/kW. The response reaches 28.3 % of its change (37.0 °C) at $t_{28} = 140$ s and 63.2 % (57.9 °C) at $t_{63} = 290$ s. Then

$$\tau = 1.5 \times (290 - 140) = 225 \text{ s}, \qquad T_d = 290 - 225 = 65 \text{ s}$$

and the model is $G(s) = 30\,e^{-65s}/(225s + 1)$ K/kW. The apparent dead time of 65 s is not a true transport delay; it is the sum of the heater element's, the air's and the sensor's small lags lumped into a delay, exactly as in the dominant-pole example. The ratio $T_d/\tau \approx 0.29$ already tells a control engineer that a PID loop on this chamber can be reasonably fast without oscillating; a ratio above 1 would signal a delay-dominated process that is hard to control tightly.

Validating a model

A simplified model is accepted only after it has been compared with data the fitting did not use: a step of a different size (to check linearity), a step at a different operating point (to check parameter variation, as in the tank example), and, where possible, a frequency response measured around the expected crossover frequency (Chapter 34). The model must be good *where the controller will use it* — near the loop's crossover frequency — and may be poor elsewhere. A model that is accurate at DC and at the crossover but misses a lightly damped structural resonance at ten times crossover is still dangerous, because high controller gain there can excite it: that is precisely the drone gimbal of the opening paragraph.

Table 33.5 — Simplification methods and their limits.

Method	What it keeps	When it fails
Dominant pole	Slow modes and DC gain	High loop gain moves fast poles inward
Lumping small lags into a dead time	Low-frequency phase	Above about $1/T_d$, where the delay's phase keeps growing
Pole–zero cancellation	Input–output response	Cancelled mode is unstable or lightly damped
Singular perturbation (fast states static)	Slow subsystem	Fast and slow time scales not well separated
FOPDT step-test fit	Low-frequency gain and phase	Oscillatory, integrating or inverse-response plants
Linearisation	Small-signal behaviour at one operating point	Large excursions, saturation, friction, switching

What remains open

Open question	Where it is answered
What does feedback do to the poles of a model, and when does a closed loop become unstable?	Chapter 34 (feedback and stability)
How can stability be decided without computing the roots, and how do the poles move as gain changes?	Chapter 34 (Routh–Hurwitz, root locus)
How large is the steady-state error of a loop, and how do Bode and Nyquist plots of the loop gain predict stability?	Chapter 34 (steady-state error, frequency-domain stability)
How are PID controllers tuned from a step-test model, and how is a state-space model used for control?	Chapter 35 (control design fundamentals)
What do controllability and observability say about cancelled modes?	Chapter 35
How are continuous models turned into discrete-time models for a digital controller?	Chapter 35 (digital control); Chapter 20 (Z-transform)
Where do the Laplace-domain circuit methods and second-order step responses come from?	Chapters 6, 7 and 11
How are motors, generators and converters built, and how are robot controllers implemented?	*Electrical Energy & Machines*; *Electronics & Computing*; robotics and automation books

Next: Chapter 34 closes the loop around these models, asking what feedback does to their poles, how stability is tested, and how the Bode and Nyquist plots of the loop gain predict the behaviour of the closed-loop system.

Chapter 34 — Feedback and Stability

Part VII · Control-System Fundamentals

A small quadcopter hovers well on its factory settings, but it drifts in gusts, so its owner raises the attitude-loop gain. At twice the original gain the drone holds its angle more tightly and recovers from a push faster. At three times the gain it starts to buzz: a fast, growing wobble of the arms appears a fraction of a second after take-off, and within two seconds the craft flips. Nothing broke. The motors, propellers and sensors were exactly the same in all three flights. What changed was the *loop*: the gain around the path sensor → controller → motor → airframe → sensor. Raising the gain made the loop correct errors more forcefully, and above a certain value the corrections arrived late enough, and large enough, to feed the error instead of cancelling it.

The same story is told by a power-system operator whose generator voltage regulator hunts after a retrofit, by an audio designer whose feedback amplifier sings at 2 MHz when a long cable is connected, and by a process engineer whose reactor temperature swings $\pm 5\ ^\circ\mathrm{C}$ around its set point after a valve was replaced with a faster one. Feedback is the most powerful idea in engineering for making imprecise hardware behave precisely; it is also the easiest way to turn a stable piece of hardware into an oscillator. This chapter builds the tools that tell the two apart.

> **Why does closing a loop make a system insensitive to its own imperfections, and what does that cost?**
> **How can we decide, from a model of the open loop alone, whether the closed loop will be stable, how far it is from instability, and how accurately it will follow its command?**

Concept	Engineering destination
Open-loop and closed-loop structure	Thermostats, cruise control, voltage regulators, automatic gain control
Loop gain and sensitivity	Feedback amplifiers, generator AVRs, precision instruments
Characteristic equation	Every closed-loop design: poles of the loop decide its behaviour
Stability (BIBO, asymptotic, marginal)	Drones, power converters, phase-locked loops, grid oscillations
Routh–Hurwitz criterion	Gain limits from a polynomial, without solving for roots
Root locus	How closed-loop poles move as a gain is raised
Steady-state error and system type	Position servos, radar tracking, temperature and pressure regulation
Nyquist criterion	Stability of loops with delays, open-loop unstable plants, measured responses
Bode-plot margins	Day-to-day design of amplifiers, SMPS loops, motor drives, PLLs

Roadmap. Open and closed loops (34.1) → why feedback works and what it costs (34.2) → the characteristic equation (34.3) → what stability means (34.4) → the algebraic test: Routh–Hurwitz (34.5) → the geometric picture: root locus (34.6) → accuracy: steady-state error (34.7) → stability from the frequency response: the Nyquist criterion (34.8) → Bode and Nyquist methods in practice (34.9).

34.1 Open-loop and closed-loop systems

Two ways to make a system do what we want

An electric kettle with a timer is an **open-loop** system: the heater runs for a fixed time computed from what the designer expects the water, the kettle and the room to be. Nothing measures the result. If the kettle holds less water than assumed, it boils dry; if the mains voltage is 10 % low, the heater delivers 19 % less power ($P \propto V^2$) and the water is not hot enough. A kettle with a thermostat that cuts the heater at boiling is a **closed-loop** system: the action depends on a measurement of the result.

Chapter 33 described both structures as block diagrams. In the open loop (Figure 34.1, top) a reference r drives a controller $C(s)$, which drives the plant $G(s)$ to give the output

$$Y(s) = C(s)G(s)R(s) + G_d(s)D(s)$$

where $D(s)$ is a disturbance entering through its own path $G_d(s)$. In the closed loop (Figure 34.1, bottom) a sensor $H(s)$ measures the output, the comparator forms the error $E = R - HY$, and the controller acts on the error:

$$Y(s) = \frac{C(s)G(s)}{1 + C(s)G(s)H(s)}R(s) + \frac{G_d(s)}{1 + C(s)G(s)H(s)}D(s)$$

The product

$$L(s) = C(s)G(s)H(s)$$

is the **loop transfer function** (also called the open-loop transfer function or the return ratio): it is what a signal experiences if we break the loop at any point and follow it once round. Almost everything in this chapter is a statement about $L(s)$.

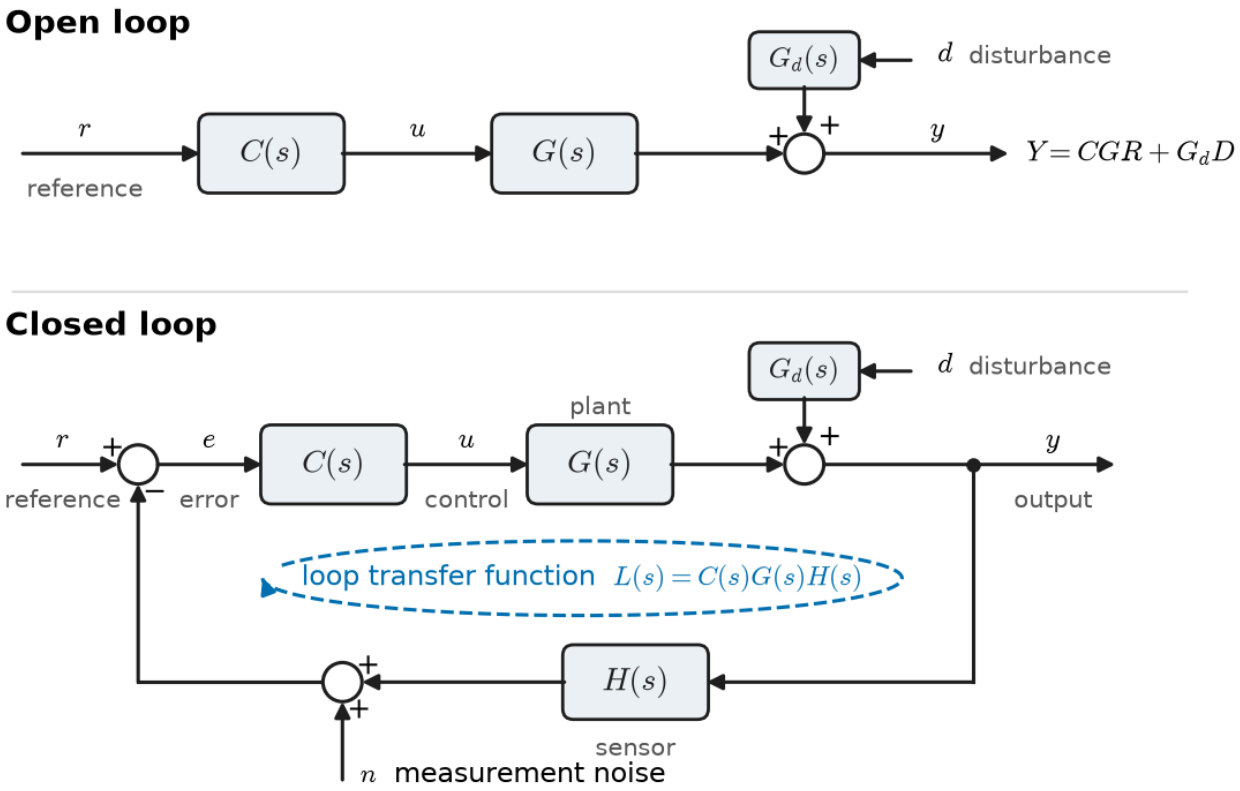

Figure 34.1 — Open-loop (top) and closed-loop (bottom) control structures with reference, controller, plant, disturbance and sensor; the closed loop defines the loop transfer function L = CGH.

Figure 34.1 also shows the measurement noise n added at the sensor. It enters the loop exactly where the reference does (with the opposite sign), which is the first hint of the price of feedback developed in Section 34.2.

Where each structure is used

Open-loop control is not obsolete. It is cheap, it cannot become unstable through the loop (a stable plant driven open loop stays stable), and it is fast because it does not wait for an error to appear. A stepper motor driven by counted pulses, a traffic light on a timer, a feedforward term that cancels a measured disturbance before it reaches the output, and the volume control of an amplifier are all open-loop. Their weakness is that every unknown, every drift and every disturbance passes straight to the output.

Closed-loop control pays for its accuracy with a sensor, with complexity, and with the possibility of instability. Table 34.1 lists typical loops across EEE and the speed at which each is designed to act; the **crossover frequency** (where $|L(j\omega)| = 1$, defined in Section 34.9) is a convenient measure of how fast a loop responds.

Table 34.1 — Representative feedback loops and typical loop crossover frequencies.

Loop	Measured variable	Actuator	Typical crossover
Op-amp non-inverting amplifier	Output voltage (divider)	Output stage	0.1–100 MHz (gain–bandwidth limited)
Phase-locked loop (RF synthesiser)	Phase error	VCO	10 kHz–1 MHz
Switched-mode supply, voltage loop	Output voltage	PWM duty cycle	1/20–1/5 of switching frequency (e.g. 5–50 kHz)
Motor drive, current loop	Phase current	Inverter voltage	0.5–5 kHz
Motor drive, speed loop	Shaft speed	Current reference	20–200 Hz
Servo position loop (robot joint, disk head)	Angle	Speed or current reference	5–2000 Hz
Generator automatic voltage regulator	Terminal voltage	Field excitation	0.5–5 rad/s (0.1–1 Hz)
Building or furnace temperature	Temperature	Heater, valve	10^{-4}–10^{-2} Hz

> **Boundary.**
> The hardware in these loops (op-amp stages, PWM modulators, inverters, excitation systems, valves) belongs to *Electronics & Computing* and *Electrical Energy & Machines*; practical embedded controllers belong to the robotics and automation books. Here they appear only as transfer functions.

34.2 Feedback principles

Loop gain buys insensitivity

Consider the feedback amplifier that H. S. Black patented in 1928 to make telephone repeaters precise. A forward amplifier of large but poorly controlled gain A is wrapped in a feedback network of accurately known attenuation β (a resistor divider). With $C = A$, $G = 1$, $H = \beta$:

$$T = \frac{Y}{R} = \frac{A}{1 + A\beta} = \frac{1}{\beta} \cdot \frac{A\beta}{1 + A\beta}$$

When the loop gain $A\beta \gg 1$, $T \approx 1/\beta$: the closed-loop gain depends on the passive divider, not on the active device. How strongly it depends on A is measured by the **sensitivity**, the ratio of relative changes:

$$S_A^T = \frac{\partial T / T}{\partial A / A} = \frac{1}{1 + A\beta}$$

The same function appears for any loop. Writing $L = CGH$ and taking the plant G as the uncertain element,

$$S(s) = \frac{1}{1 + L(s)}, \qquad T(s) = \frac{L(s)}{1 + L(s)}$$

S is the **sensitivity function** and T the **complementary sensitivity**. They obey the identity

$$S(s) + T(s) = 1$$

at every frequency.

> **Status of the principle: Feedback sensitivity reduction**
> *Derived.* $S = 1/(1+L)$ follows algebraically from the block diagram of a linear time-invariant loop; it holds for small parameter changes and at each frequency separately, so its benefit exists only where $|L(j\omega)|$ is large.

Worked example — Black's amplifier. An amplifier with open-loop gain $A = 2 \times 10^5$ is used with a divider $\beta = 0.01$. The loop gain is $A\beta = 2000$, so

$$T = \frac{2 \times 10^5}{1 + 2000} = 99.950, \qquad S = \frac{1}{2001} = 5.0 \times 10^{-4}.$$

If the open-loop gain falls by 20 % to 1.6×10^5 (a hot day, a different batch of transistors), the closed-loop gain becomes $1.6 \times 10^5 / 1601 = 99.938$, a change of -0.0125 %. The 20 % change was divided by roughly 1600, close to the loop gain, as S predicts for a large change ($20\,\%/1601$). The precise value $T \approx 1/\beta = 100$ is set by two resistors, and the 0.05 % shortfall ($1/(1+A\beta)$) is itself predictable.

Disturbance rejection

The second term of the closed-loop equation shows that a disturbance reaching the output is multiplied by S:

$$Y_d(s) = S(s)\, G_d(s) D(s)$$

A generator **automatic voltage regulator** (AVR) is a power-system example. Without regulation, the terminal voltage of a synchronous generator falls substantially between no load and full load because of the armature reaction and synchronous reactance; take a drop of 8 % for a small machine with a fixed field. With an AVR whose steady-state loop gain is 50, the remaining drop is $8\,\%/(1+50) = 0.16\,\%$. The regulator does not know why the voltage fell; it only sees the error. That is the essential virtue of feedback: it corrects disturbances that were never modelled.

Linearisation, bandwidth and speed

Feedback also linearises. If the forward gain of an audio power stage varies with signal level (crossover distortion), the closed-loop distortion is reduced by approximately the loop gain at the frequency of the distortion product. And feedback speeds up a slow plant: a first-order plant $G = K/(\tau s + 1)$ under proportional feedback $C = K_c$ gives

$$T(s) = \frac{K_c K}{\tau s + 1 + K_c K} = \frac{K_c K / (1 + K_c K)}{\dfrac{\tau}{1 + K_c K} s + 1}$$

whose time constant is shorter by the factor $1 + K_c K$. A heater with $\tau = 300$ s under a loop gain of 9 responds with a 30 s time constant, at the price of a heater that must briefly deliver ten times its final power, which it usually cannot (actuator saturation, Chapter 35).

The costs of feedback

1. **Sensor noise is passed to the output.** From Figure 34.1, the output response to measurement noise is $-T(s)N(s)$. Where $T \approx 1$ (inside the loop bandwidth) the loop follows the noise as faithfully as it follows the reference. A loop cannot be more accurate than its sensor.
2. **S and T cannot both be small.** Since $S + T = 1$, good disturbance rejection (small S) at a frequency means noise passes ($T \approx 1$) at that frequency. Designers therefore make $|L|$ large at low frequency, where disturbances and references live, and small at high frequency, where sensor noise and model errors live.
3. **Stability can be lost.** A loop that contains lags (time constants, delays, sampling) returns the correction late. If, at some frequency, the signal comes back round the loop inverted in phase (an extra 180° beyond the negative sign of the comparator) and not attenuated, it reinforces itself. This is the drone of the opening paragraph.
4. **Sensitivity cannot be pushed down everywhere.** For a stable loop whose L falls off at high frequency at least as fast as $1/s^2$ (relative degree two or more), Bode's sensitivity integral states that $\int_0^\infty \ln|S(j\omega)|\, d\omega = 0$ (more if L has right-half-plane poles): pushing $|S|$ below 1 in one band forces it above 1 somewhere else. This "waterbed effect" is why every real loop has a sensitivity peak, usually near crossover.

> **Status of the principle: Bode sensitivity integral**
> *Derived*. A consequence of complex-function theory applied to a stable, rational loop with relative degree at least two; it sets a hard limit on what any controller can achieve and is independent of the design method.

34.3 Characteristic equations

The denominator decides

Every closed-loop transfer function of the single loop in Figure 34.1 (reference to output, disturbance to output, noise to output, reference to error) has the same denominator, $1 + L(s)$. Writing $L(s) = N_L(s)/D_L(s)$ as a ratio of polynomials,

$$1 + L(s) = 0 \quad \Longleftrightarrow \quad D_L(s) + N_L(s) = 0$$

is the **characteristic equation** of the loop, and its roots are the **closed-loop poles**. Chapter 11 showed that the poles of a transfer function fix the modes $e^{p_i t}$ of its natural response; Chapter 7 showed what the two poles of an RLC circuit mean physically. Closing a loop moves the poles from the roots of

D_L (the open-loop poles) to the roots of $D_L + N_L$. The zeros of the reference-to-output transfer function, by contrast, are mostly those of C and G (plus the poles of H): feedback moves poles, not zeros.

Worked example — the running example of this chapter. A position servo (an antenna drive, a robot joint, a disk-drive actuator, a radar pedestal) is modelled by a motor with an integrator from speed to angle, a mechanical time constant of 1 s and an amplifier/current-loop lag of 0.2 s, normalised so that

$$G(s) = \frac{1}{s(s+1)(s+5)}$$

with a proportional controller $C = K$ and a unity-gain sensor $H = 1$. Then

$$L(s) = \frac{K}{s(s+1)(s+5)}, \qquad 1 + L(s) = 0 \iff s^3 + 6s^2 + 5s + K = 0.$$

For $K = 10$ the roots are $s = -5.418$ and $s = -0.291 \pm j1.327$. The closed loop is stable (all real parts negative), but the complex pair has damping ratio $\zeta = 0.291/\sqrt{0.291^2 + 1.327^2} = 0.214$, so the step response will overshoot by roughly half (48.6 % from simulation, Figure 34.6). The rest of this chapter asks how these roots move as K changes, without solving the cubic each time.

Physical origin of a characteristic equation

The characteristic equation is not a mathematical artefact of block diagrams. In a feedback amplifier it is the condition under which a nonzero output can exist with zero input, i.e. the oscillation condition (the **Barkhausen condition** $L(j\omega_0) = -1$ is the special case of a root on the imaginary axis). In a power system, the roots of the linearised machine–network–regulator equations are the electromechanical modes (0.1–2 Hz) seen in grid oscillation studies. In a biomedical context, the baroreflex that regulates blood pressure is a loop whose characteristic roots produce the slow Mayer waves (about 0.1 Hz) seen in blood-pressure records. Each is a closed loop whose natural modes are the roots of $1 + L = 0$.

34.4 Stability concepts

Three definitions

Chapter 16 introduced **bounded-input bounded-output (BIBO) stability**: every bounded input produces a bounded output. For an LTI system this is equivalent to an absolutely integrable impulse response, $\int_0^\infty |h(t)|\,dt < \infty$, which for a rational transfer function means that all poles have strictly negative real parts.

Asymptotic (internal) stability is the state-space version: with zero input, every initial condition decays to zero. It concerns the natural modes of all internal states, including any that do not appear in the input–output transfer function (Chapter 35 shows how modes can hide). For a loop with no hidden pole–zero cancellations, the two definitions coincide.

Marginal stability describes non-repeated poles on the imaginary axis and none in the right half-plane: the natural response neither decays nor grows (an undamped oscillator, an integrator). A marginally stable system is not BIBO-stable: a bounded input at the resonant frequency, or a step into an integrator, produces an unbounded output.

> **Status of the principle: Stability from pole locations**
> *Derived.* For a linear time-invariant system with a rational transfer function, BIBO stability is equivalent to all poles lying in the open left half of the s-plane; the result follows from partial-fraction expansion of the impulse response (Chapter 11).

Pole location and response

Figure 34.2 summarises the link between pole position and impulse response. A pole at $s = \sigma \pm j\omega_d$ contributes $e^{\sigma t}\cos(\omega_d t + \phi)$: the real part sets the decay (or growth) rate and the imaginary part the ringing frequency. Lines of constant damping ratio are rays from the origin at angle $\cos^{-1}\zeta$ to the negative real axis.

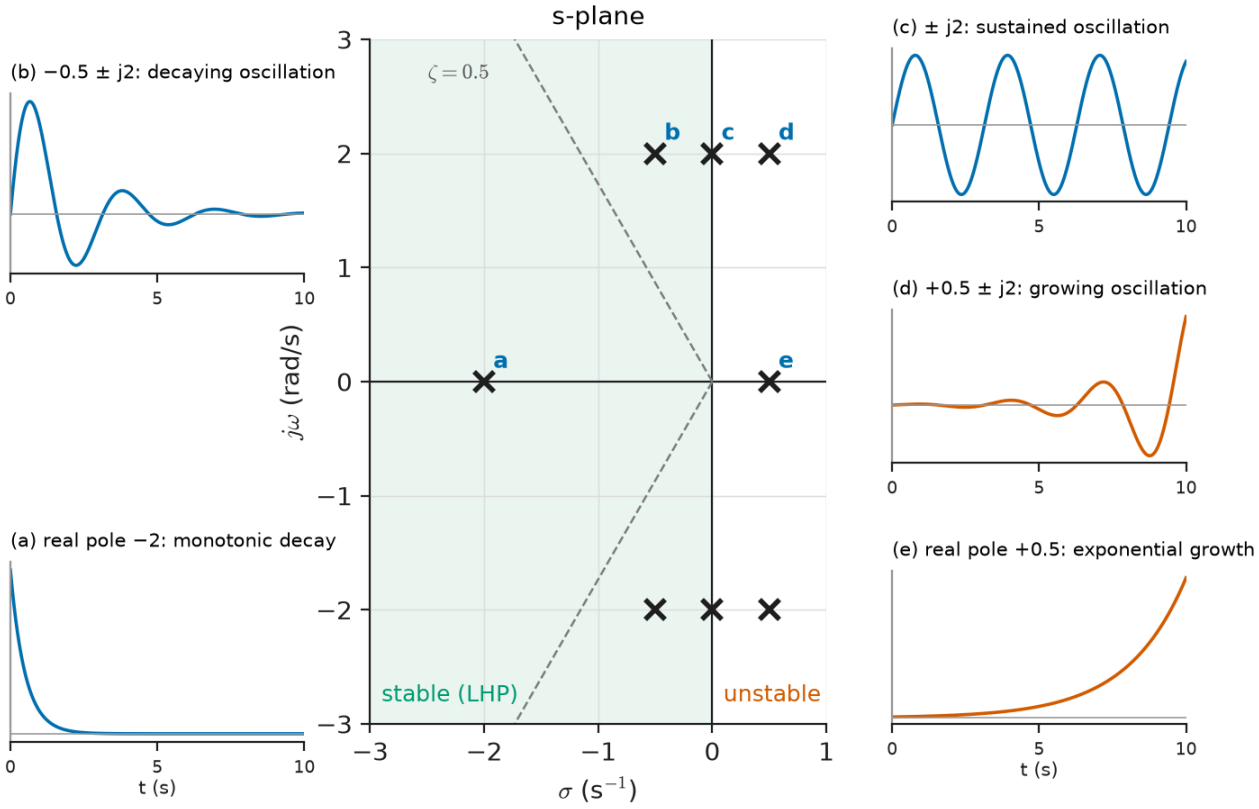

Figure 34.2 — Pole positions in the s-plane and the impulse responses they produce: decaying, oscillatory, sustained and growing modes.

Relative stability asks not only whether the poles are in the left half-plane but how far inside. Two systems that are both stable can behave very differently: the servo above with $K = 10$ has a pole pair at $\zeta = 0.21$ and rings for over 12 s; with $K = 1.13$ it barely overshoots. Engineering specifications are always about relative stability: damping ratio, decay rate, gain and phase margins.

Nonlinear and time-varying caution

Real loops saturate: an amplifier clips, a valve opens fully, an inverter reaches its current limit. The linear stability of $1 + L = 0$ is a statement about small deviations around an operating point. A loop that is linearly stable can still develop a **limit cycle** (a sustained oscillation of fixed amplitude) through saturation, backlash or relay action, and a linearly unstable loop can be held in a bounded oscillation by saturation, as in every electronic oscillator.

> **Status of the principle: Linearised stability**
> *Approximate.* Pole locations of the linearised model decide stability for small deviations about an operating point (Lyapunov's indirect method); large signals, saturation, backlash and parameter changes with operating point lie outside it.

34.5 Routh–Hurwitz criterion

Why a test without roots

Before cheap computing, solving a fifth-order polynomial for each trial gain was impractical; today it is trivial for numbers but still impractical for *symbols*. The question "for which values of K (or of a time constant, or of a line length) is the loop stable?" requires conditions on the coefficients. E. J. Routh (1877) and A. Hurwitz (1895) independently supplied them.

Necessary condition

For a polynomial

$$a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0, \qquad a_n > 0,$$

to have all roots in the open left half-plane, all coefficients must be present and positive. A missing or negative coefficient proves instability (or marginal stability) at once. The condition is sufficient only for $n \le 2$.

The Routh array

Arrange the coefficients in two rows, alternating:

$$\begin{array}{c|ccccc} s^n & a_n & a_{n-2} & a_{n-4} & \cdots \\ s^{n-1} & a_{n-1} & a_{n-3} & a_{n-5} & \cdots \\ s^{n-2} & b_1 & b_2 & b_3 & \cdots \\ s^{n-3} & c_1 & c_2 & \cdots & \\ \vdots & & & & \\ s^0 & \cdot & & & \end{array}$$

with each new entry formed from the two rows above it,

$$b_1 = \frac{a_{n-1}a_{n-2} - a_n a_{n-3}}{a_{n-1}}, \qquad b_2 = \frac{a_{n-1}a_{n-4} - a_n a_{n-5}}{a_{n-1}}, \qquad c_1 = \frac{b_1 a_{n-3} - a_{n-1}b_2}{b_1}, \ \ldots$$

The **Routh–Hurwitz criterion**: the number of roots in the right half-plane equals the number of sign changes in the first column. The polynomial is stable if and only if every first-column entry is positive.

> **Status of the principle: Routh–Hurwitz criterion**
> *Derived*. An algebraic consequence of the argument principle and Sturm-sequence theory for real polynomials; it is exact for any polynomial with real coefficients, but says nothing about how well damped the stable roots are, and it does not apply directly to time delays (which are not polynomial).

Worked example — gain limit of the servo. For $s^3 + 6s^2 + 5s + K$:

$$\begin{array}{c|cc} s^3 & 1 & 5 \\ s^2 & 6 & K \\ s^1 & \dfrac{30 - K}{6} & 0 \\ s^0 & K & \end{array}$$

Stability requires $K > 0$ and $30 - K > 0$, so $0 < K < 30$. At $K = 30$ the s^1 row vanishes entirely. A row of zeros means a pair of roots symmetric about the origin; the **auxiliary polynomial** formed from the row above, $6s^2 + 30 = 0$, gives them: $s = \pm j\sqrt{5} = \pm j2.236$. At $K = 30$ the servo oscillates at 2.236 rad/s (period 2.81 s). The current design with $K = 10$ therefore has a gain margin of $30/10 = 3$; that number returns in Section 34.9 from a Bode plot and in Chapter 35 as the starting point of Ziegler–Nichols tuning.

Worked example — counting unstable roots. For $s^4 + 2s^3 + 3s^2 + 4s + 5$ every coefficient is positive, so the necessary condition gives no warning. The array:

$$\begin{array}{c|ccc} s^4 & 1 & 3 & 5 \\ s^3 & 2 & 4 & \\ s^2 & \dfrac{2\cdot 3 - 1\cdot 4}{2} = 1 & 5 & \\ s^1 & \dfrac{1\cdot 4 - 2\cdot 5}{1} = -6 & & \\ s^0 & 5 & & \end{array}$$

The first column $1, 2, 1, -6, 5$ changes sign twice, so two roots lie in the right half-plane. Numerically the roots are $-1.288 \pm j0.858$ and $+0.288 \pm j1.416$, confirming the count.

Special cases and uses

If a first-column element is zero but its row is not, replace it by a small positive ε and continue; the signs as $\varepsilon \to 0^+$ give the count. If a whole row is zero, use the derivative of the auxiliary polynomial as the replacement row; the auxiliary roots are the symmetric (often imaginary-axis) roots.

Routh's array is the fastest way to obtain a **stability region in a parameter**. Shifting $s \to s - \sigma_0$ before forming the array counts roots to the right of $-\sigma_0$, which turns it into a test of minimum decay rate. Its limits are equally clear: it says nothing about damping, needs a polynomial (a transport delay

must be approximated, e.g. by a Padé approximant), and becomes unwieldy above about sixth order with several parameters.

34.6 Root-locus fundamentals

The idea

W. R. Evans (1948) asked a geometric question: as a single gain K in $L(s) = KF(s)$ increases from 0 to ∞, what paths do the roots of $1 + KF(s) = 0$ trace in the s-plane? The answer, the **root locus**, shows the whole design trade-off on one picture. A point s lies on the locus when

$$KF(s) = -1 \quad \Longleftrightarrow \quad \angle F(s) = \pm 180^\circ (2k+1), \qquad K = \frac{1}{|F(s)|}.$$

The **angle condition** decides *where* the locus is; the **magnitude condition** then gives the gain at any point on it.

Construction rules

With $F(s) = \prod (s - z_i) / \prod (s - p_j)$, n poles and m zeros ($n \geq m$):

1. The locus has n branches. They start ($K = 0$) at the open-loop poles and end ($K \to \infty$) at the open-loop zeros or at infinity ($n - m$ branches).
2. On the real axis, the locus occupies segments that have an odd number of real poles plus zeros to their right.
3. The $n - m$ branches that go to infinity follow straight-line asymptotes at angles $\pm 180^\circ (2k+1)/(n-m)$, radiating from the **centroid**
$$\sigma_a = \frac{\sum p_j - \sum z_i}{n - m}.$$
4. Breakaway and break-in points on the real axis are among the roots of $dK/ds = 0$, with $K = -1/F(s)$.
5. Imaginary-axis crossings follow from the Routh array (or from substituting $s = j\omega$).
6. The locus is symmetric about the real axis; departure angles from complex poles follow from the angle condition.

> **Status of the principle: Root-locus rules**
> *Derived.* Each rule is a direct consequence of the angle and magnitude conditions on $1 + KF(s) = 0$; they are exact for one real gain parameter and a rational $F(s)$, and are sketching aids rather than approximations.

Worked example — locus of the servo. For $F(s) = 1/[s(s+1)(s+5)]$ there are three poles $(0, -1, -5)$ and no zeros.

- Real-axis segments: between 0 and -1 (one pole to the right) and left of -5 (three poles to the right).

- Asymptotes: $n - m = 3$, angles $\pm 60^\circ$ and 180°, centroid $\sigma_a = (0 - 1 - 5)/3 = -2$.
- Breakaway: $K = -s(s+1)(s+5) = -(s^3 + 6s^2 + 5s)$, so $dK/ds = 0$ gives $3s^2 + 12s + 5 = 0$, with roots $s = -0.4725$ and $s = -3.528$. Only -0.4725 lies on a locus segment; there $K = 0.4725 \times 0.5275 \times 4.5275 = 1.128$.
- Imaginary-axis crossing: $K = 30$ at $s = \pm j2.236$ (Section 34.5).

The two branches that leave 0 and -1 meet at -0.4725 when $K = 1.128$ (a double real pole: critical damping of the dominant pair), split into a complex pair, and bend along the $\pm 60^\circ$ asymptotes towards the right half-plane, crossing at $K = 30$ (Figure 34.3). The third branch runs from -5 to $-\infty$.

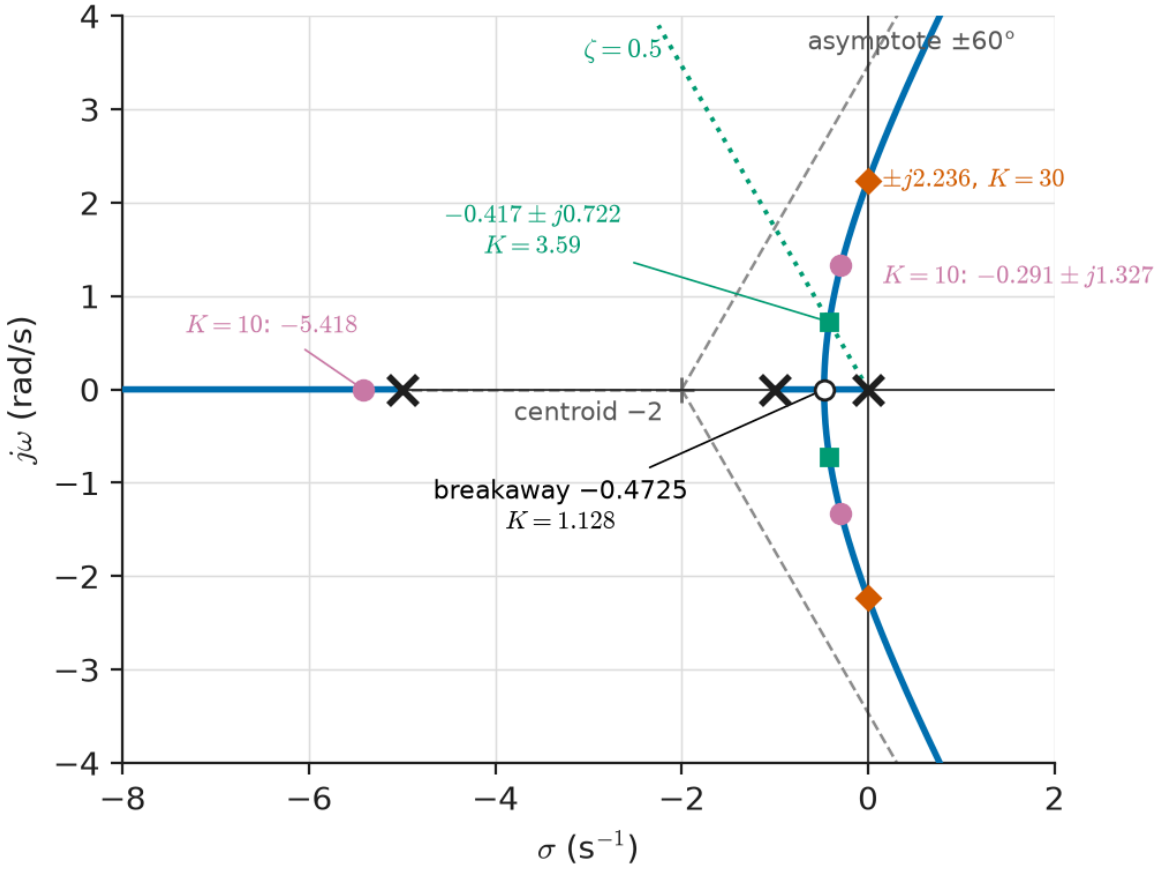

Figure 34.3 — Root locus of L(s) = K/[s(s+1)(s+5)]: breakaway at -0.47 for K = 1.13, zeta = 0.5 at K = 3.59 and imaginary-axis crossing at K = 30.

Reading a design from the locus

The locus turns a specification into a gain. Suppose we want the dominant pair at $\zeta = 0.5$ (about 16 % overshoot, Chapter 35). The $\zeta = 0.5$ ray makes 120° with the positive real axis; searching along it for the angle condition gives the intersection $s = -0.417 \pm j0.722$ (radius $5/6$), and the magnitude condition gives

$$K = |s|\,|s+1|\,|s+5| = 3.588.$$

The third pole is then at -5.167, twelve times further from the axis than the dominant pair, so the second-order approximation is good: simulation gives 16.1 % overshoot against 16.3 % for an ideal second-order system with $\zeta = 0.5$. But the locus also shows the limitation: no gain gives a dominant

pair that is both well damped and fast, because the branches bend towards the right half-plane. Making the system faster *and* well damped needs the locus itself reshaped, by adding a zero (lead compensation) or moving poles; that is Chapter 35's work.

Root loci generalise: the parameter can be a time constant, a feedback-network value or a load inertia (by rewriting the characteristic equation so that the parameter appears as a multiplier), and positive-feedback loci (0° condition) describe oscillator start-up.

34.7 Steady-state error

Error after the transient

Stability is the first requirement; the second is accuracy once transients have died. For the unity-feedback loop ($H = 1$) the error is

$$E(s) = \frac{1}{1 + L(s)} R(s) = S(s) R(s)$$

and, provided the closed loop is stable, the final-value theorem (recapped from *The Mathematics Universe*) gives

$$e_{ss} = \lim_{t \to \infty} e(t) = \lim_{s \to 0} \frac{s R(s)}{1 + L(s)}.$$

System type and error constants

Write $L(s) = K\,\tilde{L}(s)/s^{N}$ with $\tilde{L}(0) = 1$. The integer N, the number of pure integrators in the loop, is the **system type**. For the three standard test inputs:

- Step $R = A/s$: $e_{ss} = A/(1 + K_p)$ with position constant $K_p = \lim_{s\to 0} L(s)$.
- Ramp $R = A/s^2$: $e_{ss} = A/K_v$ with velocity constant $K_v = \lim_{s\to 0} sL(s)$.
- Parabola $R = A/s^3$: $e_{ss} = A/K_a$ with acceleration constant $K_a = \lim_{s\to 0} s^2 L(s)$.

Table 34.2 — Steady-state error versus system type (unity feedback, stable closed loop).

System type N	Step A/s	Ramp A/s^2	Parabola A/s^3	Typical example
0	$A/(1+K_p)$	∞	∞	Temperature loop with P control; AVR with proportional exciter
1	0	A/K_v	∞	Position servo with P control; PI-controlled speed loop
2	0	0	A/K_a	Radar or antenna tracker with PI on a position plant; type-2 PLL

> **Status of the principle: Error constants and system type**
> *Derived*. Follows from the final-value theorem applied to $S(s)R(s)$; valid only when the closed loop is stable (the theorem fails otherwise) and only for the polynomial test inputs; disturbances entering elsewhere in the loop need their own analysis.

Worked example — a tracking antenna. The servo of Section 34.3 drives a ground-station antenna that must follow a low-orbit satellite. Near the pass midpoint the required azimuth rate may be about 1°/s, which is locally a ramp. The loop is type 1 with

$$K_v = \lim_{s\to 0} s\frac{K}{s(s+1)(s+5)} = \frac{K}{5}.$$

With $K = 10$, $K_v = 2\ \mathrm{s}^{-1}$ and the tracking lag is $e_{ss} = 1/2 = 0.5^\circ$, ten times the beam-width tolerance of a narrow-beam Ka-band dish (beam widths of a few tenths of a degree are typical). With the well-damped gain $K = 3.588$ from the root locus, $K_v = 0.718\ \mathrm{s}^{-1}$ and the lag grows to 1.39°. Accuracy and damping pull in opposite directions under proportional control: the gain that gives 16 % overshoot gives a ramp error almost three times larger than the gain that gives 49 % overshoot. Chapter 35 resolves this with compensators that raise K_v without lowering the damping.

Worked example — a type-0 temperature loop. A furnace has a steady-state gain of 1.9 °C per percent of heater power and is controlled by a proportional controller of 10 % per °C. The loop gain at DC is $K_p = 19$, so a step change of set point leaves a fraction $1/(1+19) = 5\ \%$ uncorrected: asking for a 200 °C rise yields 190 °C. Doubling the gain halves the error but invites oscillation once the furnace's lags and transport delay are counted. Adding an integrator (PI control, Chapter 35) makes the loop type 1 and removes the step error entirely.

Disturbances and the internal-model idea

The same reasoning applies to disturbances: a constant load torque on the servo, or a constant heat loss in the furnace, is rejected with zero steady-state error only if an integrator sits *between the error and the point where the disturbance enters*. An integrator in the plant after the disturbance does not help. The general statement, the **internal-model principle**, is that to reject a persistent signal with zero error the loop must contain a model of its generator: an integrator for steps, a double integrator for ramps, a resonator $1/(s^2+\omega_0^2)$ for a sinusoid. Grid-connected inverters use exactly such resonant controllers at 50 or 60 Hz to track sinusoidal current references with zero error.

34.8 Frequency-domain stability

Why another criterion

Routh and the root locus need the characteristic polynomial. Three common situations deny us one:

1. **Time delay.** A transport lag $e^{-s\tau}$ (a conveyor, a long pipe, a network link, the computation delay of a digital controller) makes the characteristic equation transcendental, with infinitely many roots.

2. **Measured plants.** A loop may be known only from a measured frequency response: a network-analyser sweep of a power-supply plant, a frequency-response test on a turbine-governor.
3. **High order.** Distributed effects (a transmission line inside the loop, flexible structures) produce models of very high order whose coefficients are ill-conditioned.

All three are handled by working with $L(j\omega)$, which is available for each.

The Nyquist criterion

The closed-loop poles are the zeros of $1 + L(s)$. The **argument principle** (recapped from complex analysis in *The Mathematics Universe*) states that as s traverses a closed contour clockwise, the image $F(s)$ encircles the origin clockwise $Z - P$ times, where Z and P are the numbers of zeros and poles of F inside the contour. Take the contour to be the whole right half-plane (the **Nyquist contour**: up the imaginary axis, round a large semicircle, with small indentations around any imaginary-axis poles of L), and $F = 1 + L$. The poles of $1 + L$ are the open-loop poles of L; its zeros are the closed-loop poles. Encircling the origin by $1 + L$ is the same as encircling the point -1 by L. Hence:

$$Z = N + P$$

where Z is the number of closed-loop poles in the right half-plane, P the number of open-loop poles of L in the right half-plane, and N the number of clockwise encirclements of -1 by the **Nyquist plot** of $L(j\omega)$, ω from $-\infty$ to ∞. For stability $Z = 0$, so $N = -P$: the plot must encircle -1 counter-clockwise once for every unstable open-loop pole. In the commonest case, an open-loop-stable L ($P = 0$), **the closed loop is stable if and only if the Nyquist plot does not encircle** -1.

> **Status of the principle: Nyquist stability criterion**
> *Derived.* A direct application of Cauchy's argument principle to $1 + L(s)$; it is exact for any loop transfer function analytic on the contour, including time delays and measured responses, provided the open-loop unstable pole count P is known.

Reading the plot

Because $L(-j\omega)$ is the complex conjugate of $L(j\omega)$, only positive frequencies need be drawn; the negative-frequency half is the mirror image. Figure 34.5 shows the servo with $K = 10$: the plot starts at $-j\infty$ (the integrator gives -90° at low frequency), sweeps round through the third quadrant, crosses the negative real axis at $-1/3$ when $\omega = 2.236$ rad/s, and spirals into the origin at -270°. The point -1 lies outside the curve, so $N = 0$ and, with $P = 0$, the loop is stable. Tripling the gain stretches the plot so that it passes through -1; beyond that it encircles -1 twice clockwise and $Z = 2$, exactly the two right-half-plane roots that Routh predicts for $K > 30$.

Worked example — an open-loop unstable plant. Magnetic levitation (maglev bearings, a levitated steel ball in a laboratory) and an inverted pendulum have a right-half-plane pole: without feedback they fall. Take $L(s) = K/[(s - 1)(s + 2)]$, so $P = 1$. At $\omega = 0$, $L = -K/2$; as $\omega \to \infty$, $L \to 0$ from the direction of -180°. The positive-frequency half runs from $-K/2$ round through the third quadrant to the origin, and its mirror image closes the loop through the second quadrant, so the full plot is a closed curve crossing the negative real axis only at $-K/2$. Encirclement of -1 requires $K/2 >$

1, and the direction is counter-clockwise, so $N = -1$ and $Z = N + P = 0$ for $K > 2$. Stability needs the loop gain to be *above* a minimum. The check by algebra: $1 + L = 0$ gives $s^2 + s + (K - 2) = 0$, stable exactly for $K > 2$. Such loops have a gain margin that is less than one (a *lower* gain limit), and they are fragile in ways an engineer must respect: an actuator that saturates reduces the effective gain, and the loop can be lost.

Time delay

A pure delay $e^{-j\omega\tau}$ has unit magnitude and phase $-\omega\tau$: it rotates every point of the Nyquist plot clockwise by an angle proportional to frequency, without changing its distance from the origin. If the plot crosses the unit circle at frequency ω_c with an angle ϕ_m to spare before -180°, the loop tolerates any added delay up to the **delay margin**

$$\tau_{\max} = \frac{\phi_m}{\omega_c} \quad (\phi_m \text{ in radians}).$$

For the servo with $K = 3.588$, $\omega_c = 0.609$ rad/s and $\phi_m = 51.7^\circ = 0.903$ rad, so $\tau_{\max} = 1.48$ s. A 0.3 s communication delay in a teleoperated version of the same drive costs $0.609 \times 0.3 = 0.18$ rad $= 10.5^\circ$ of phase. A digital controller adds about one and a half sample periods of delay (Chapter 35), so the same formula decides how fast it must sample.

34.9 Bode and Nyquist methods

Margins: distance from -1

The Nyquist criterion answers yes or no; engineering needs a distance. Two classical measures of how far $L(j\omega)$ stays from the critical point (Figure 34.4 and Figure 34.5):

- **Gain margin** GM: at the **phase-crossover frequency** ω_{180}, where $\angle L = -180^\circ$, $\text{GM} = 1/|L(j\omega_{180})|$, or $-20\log_{10}|L(j\omega_{180})|$ dB. It is the factor by which the gain can rise before instability.
- **Phase margin** PM: at the **gain-crossover frequency** ω_c, where $|L| = 1$, $\text{PM} = 180^\circ + \angle L(j\omega_c)$. It is the extra phase lag the loop tolerates before instability.

A third, the **modulus margin** $\min_\omega |1 + L(j\omega)| = 1/\max|S|$, is the true shortest distance from the plot to -1 and the most reliable single robustness number; Chapter 35 uses all three as design targets.

> **Status of the principle: Gain and phase margins as stability distances**
> *Approximate.* The margins are exact statements about how much pure gain or pure phase change a loop tolerates; as measures of general robustness and damping they assume a Nyquist plot that crosses the unit circle and the negative real axis once, and can mislead for loops with several crossovers or resonances.

Bode-plot reading

Chapter 13 drew Bode plots of networks; for a loop the same plots of $|L(j\omega)|$ in dB and $\angle L(j\omega)$ in degrees give the margins by inspection (Figure 34.4). The gain margin is read on the magnitude plot at the frequency where the phase crosses -180°; the phase margin is read on the phase plot at the frequency where the magnitude crosses 0 dB. Because multiplying L by a constant only shifts the magnitude curve vertically, the effect of the gain is visible at a glance.

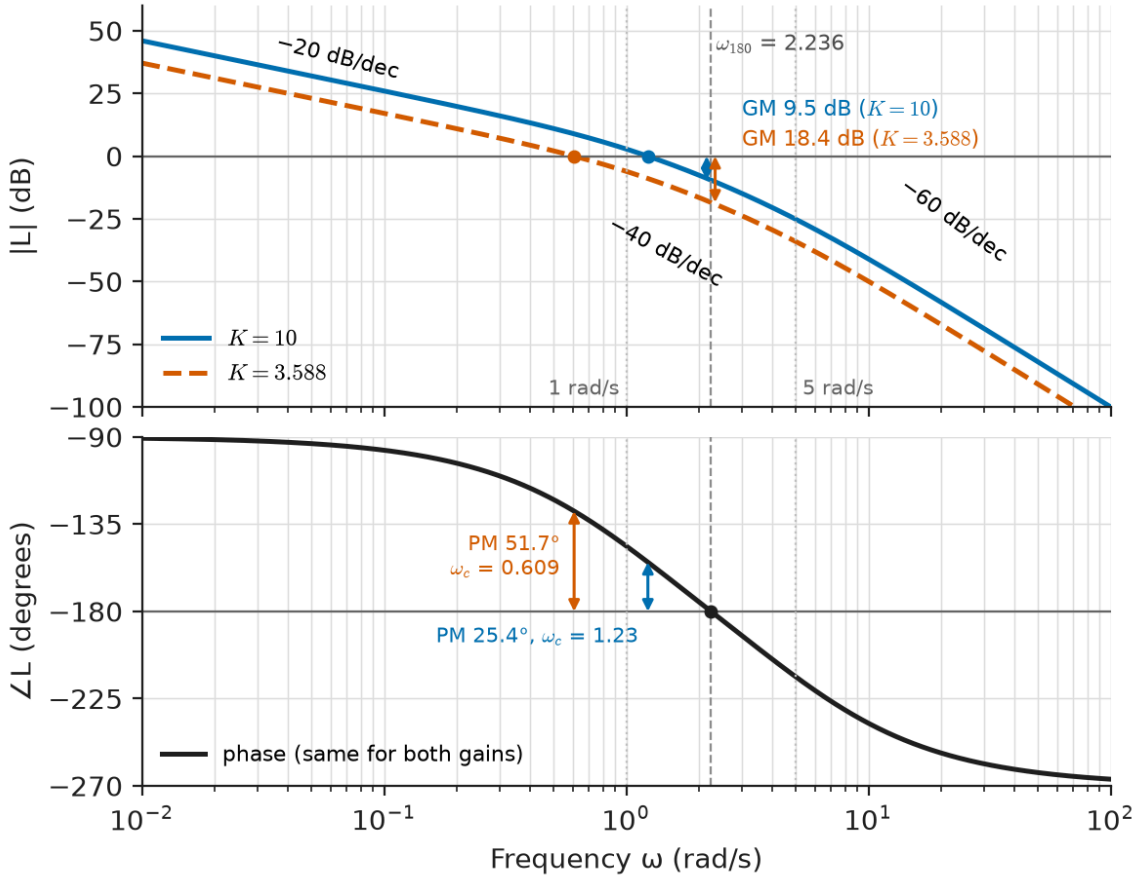

Figure 34.4 — Bode plot of the servo loop L(s) = K/[s(s+1)(s+5)] for K = 10 and K = 3.59, showing gain and phase margins.

Worked example — margins of the servo. For $L = 10/[s(s+1)(s+5)]$, the phase is $-90^\circ - \tan^{-1}\omega - \tan^{-1}(\omega/5)$, which reaches -180° when $\tan^{-1}\omega + \tan^{-1}(\omega/5) = 90^\circ$, i.e. $\omega \cdot \omega/5 = 1$, $\omega_{180} = \sqrt{5} = 2.236$ rad/s. There

$$|L| = \frac{10}{\sqrt{5}\,\sqrt{6}\,\sqrt{30}} = \frac{10}{30} = \frac{1}{3}, \qquad \text{GM} = 3 = 9.54\ \text{dB},$$

in agreement with the Routh limit $K < 30$. The gain crossover, found numerically from $|L(j\omega)| = 1$, is $\omega_c = 1.227$ rad/s, where the phase is -154.6°, so PM $= 25.4^\circ$. Repeating the calculation for $K = 3.588$ gives $\omega_c = 0.609$ rad/s, PM $= 51.7^\circ$ and GM $= 30/3.588 = 8.36 = 18.4$ dB.

Table 34.3 — The servo at four gains: one loop, four views.

K	Closed-loop poles	Dominant ζ	Step overshoot	K_v (s^{-1})	ω_c (rad/s)	PM	GM
1.128	$-0.48,\ -0.46,\ -5.06$	about 1	0 %	0.23	0.22	75.1°	28.5 dB
3.588	$-0.42 \pm j0.72,\ -5.17$	0.50	16.1 %	0.72	0.61	51.7°	18.4 dB
10	$-0.29 \pm j1.33,\ -5.42$	0.21	48.6 %	2.0	1.23	25.4°	9.5 dB
30	$\pm j2.24,\ -6$	0	sustained oscillation	6.0	2.24	0°	0 dB

Table 34.3 is the chapter in one view. The root locus (pole column), the time response (overshoot), the accuracy (velocity constant) and the Bode margins describe the same physical trade-off. Note the correlation between phase margin and damping: PM $\approx 100\zeta$ degrees holds roughly for well-separated dominant pairs (51.7° and $\zeta = 0.50$; 25.4° and $\zeta = 0.21$), a rule Chapter 35 uses to translate overshoot specifications into frequency-domain targets. The step responses are drawn in Figure 34.6.

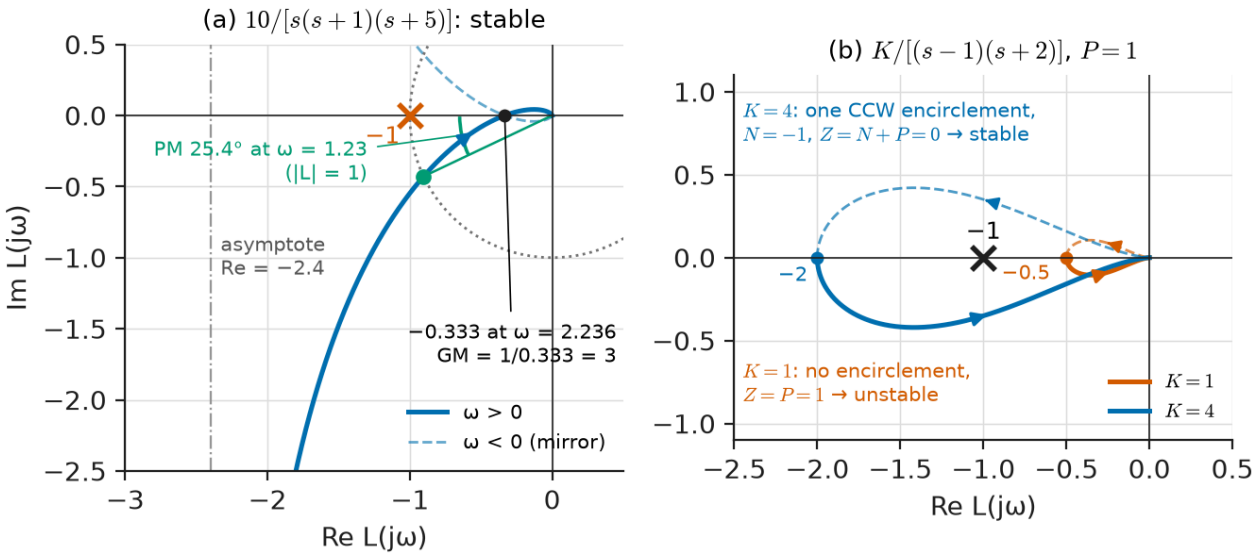

Figure 34.5 — Nyquist plots: (a) servo loop with K = 10, stable with gain and phase margins marked; (b) open-loop unstable plant K/[(s-1)(s+2)], stable only for K above 2.

Nyquist versus Bode

The Bode plot is the working tool: it is easy to sketch from asymptotes (Chapter 13), compensators add to it graphically, and margins are read directly. It quietly assumes a "simple" loop, one that is open-loop stable and crosses 0 dB and -180° once each. The Nyquist plot is the authority: it handles open-loop unstable plants, multiple crossings (a lightly damped mechanical resonance at high frequency that pokes back above 0 dB, a conditionally stable loop that becomes unstable if the gain is *reduced*), and it shows the modulus margin geometrically. A designer reads margins on Bode and checks unusual loops on Nyquist; the **Nichols chart** (magnitude in dB against phase) is a third view used in aerospace and servo work.

Where Bode and Nyquist meet hardware

Three practical cases show the methods at work.

- **Switched-mode power supplies.** Engineers inject a small signal into the feedback path and measure $L(j\omega)$ with a network analyser; a typical target is crossover at a tenth of the switching frequency, PM of 45–60° and GM of at least 6–10 dB. A buck converter's output LC filter gives a -180° phase step at its resonance, which is why its compensators add phase lead (Chapter 35).
- **Op-amp stability with capacitive loads.** A long cable adds a pole with the amplifier's output resistance, eroding phase margin; the ringing on a step and the peaking in the closed-loop response are the direct signs of lost margin (the circuit-level fixes belong to *Electronics & Computing*).
- **Phase-locked loops.** A PLL is a type-2 loop (the VCO integrates frequency into phase, the loop filter adds a second integrator); its phase margin decides reference-spur suppression and lock transients, and its crossover sets the phase-noise trade-off between reference and VCO.

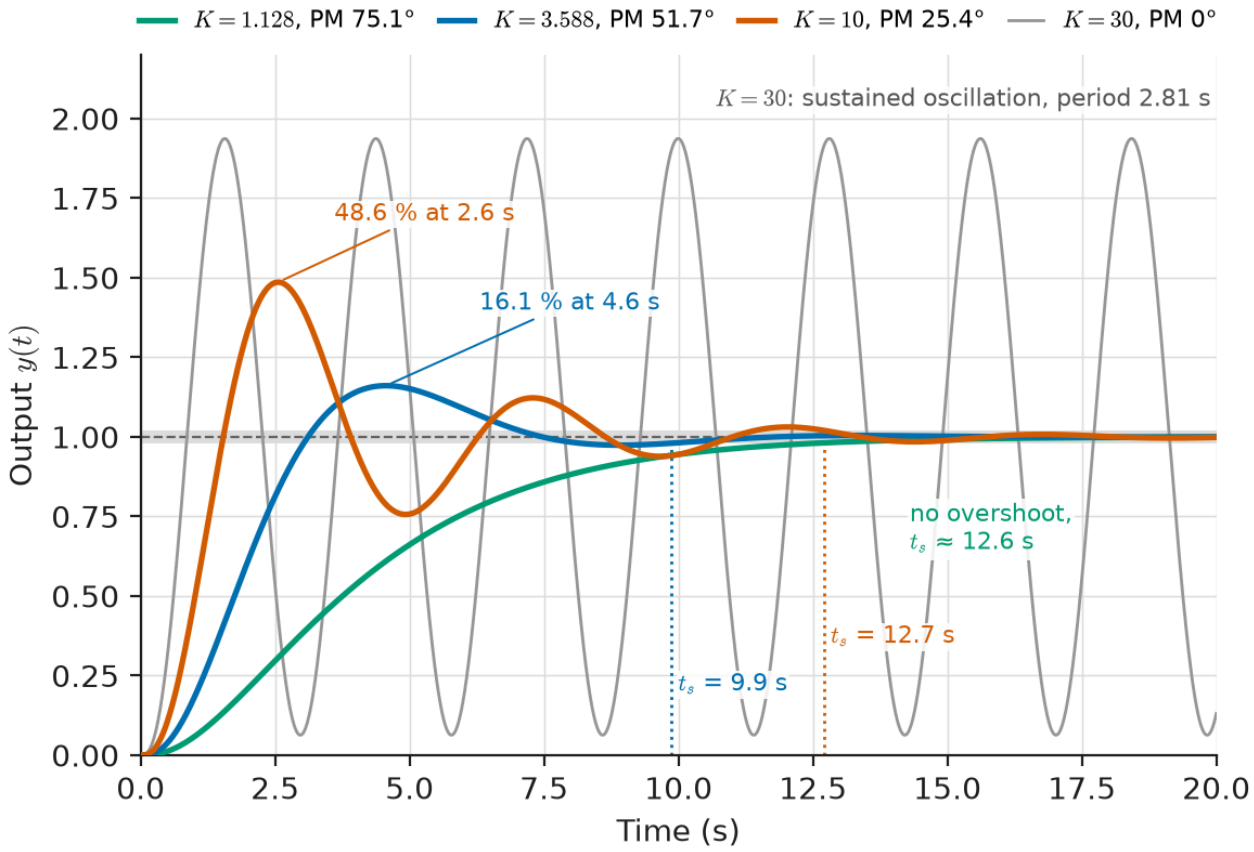

Figure 34.6 — Closed-loop step responses of the servo for K = 1.13, 3.59, 10 and 30, showing the trade-off between speed, overshoot and stability.

Figure 34.6 closes the loop on the opening story. Raising the drone's gain did what the table predicts: faster rise (rise time falls from 7.1 s at $K = 1.13$ to 0.94 s at $K = 10$ in our normalised servo), larger overshoot, longer ringing and finally oscillation at the frequency where the loop phase reaches -180°. The fix is never "less feedback" in general; it is the right feedback at the right frequencies, which is the subject of control design.

What remains open

Open question	Where it is answered
How are transient specifications (overshoot, settling time) turned into margin and pole targets?	Chapter 35 (transient-response specifications, gain and phase margins)
How can a controller raise accuracy without losing damping, or add phase where the loop needs it?	Chapter 35 (PID controllers, controller tuning)
What happens to modes that cancel in the transfer function, and how are multi-variable loops analysed?	Chapter 35 (state-space introduction, controllability, observability)
How do sampling and computation delay change the stability picture?	Chapter 35 (digital control fundamentals); z-plane stability in Chapter 20
How were the block diagrams and transfer functions of motors, heaters and circuits obtained?	Chapter 33 (dynamic-system modeling); circuit transfer functions in Chapter 11
Where do Bode asymptotes and the meaning of poles and zeros come from?	Chapters 11 and 13
Why does a feedback amplifier's own transistor circuit limit its loop gain and bandwidth?	*Electronics & Computing*
How do excitation systems and governors close loops around real generators?	*Electrical Energy & Machines*
Argument principle, final-value theorem and Laplace transform theory	*The Mathematics Universe*

Next: Chapter 35 turns analysis into design: it specifies the transient response, shapes the loop with PID and lead–lag controllers, introduces state-space models with their controllability and observability, and moves the controller into a sampled, digital computer.

Chapter 35 — Control Design Fundamentals

Part VII · Control-System Fundamentals

A commissioning engineer at a plastics plant is asked to tune the temperature controller of an extruder barrel zone. The previous settings came from a neighbouring line and produce a 12 °C overshoot on every start-up, enough to degrade the polymer. She has a choice of recipes: the 1942 Ziegler–Nichols rules printed in the controller manual, an "autotune" button that briefly relays the heater on and off, or a model-based rule that needs a step test first. Across the plant, a drives engineer is porting a well-behaved analog position loop into a microcontroller and finds that the same gains now make the axis ring; a power-electronics designer has a converter whose loop measures 25° of phase margin and must find 30° more without slowing it down. None of them is short of theory from Chapter 34: they can all tell whether a loop is stable. What they need is *design*: a specification written in numbers, a controller structure, a way to choose its parameters, and a check that the result survives the hardware it runs on.

This last chapter of the book supplies that toolkit at an introductory level. It starts from what a good response looks like, builds the PID controller and its tuning rules, uses gain and phase margins as design targets, opens the state-space view that underlies modern control, and ends with the controller as it is almost always implemented today: a program running on a sampled-data computer.

> **How are "fast", "well damped" and "accurate" turned into numbers a controller can be designed to meet?**
> **How do the PID controller, loop shaping and state feedback meet those numbers, and what changes when the controller becomes a sampled digital program?**

Concept	Engineering destination
Transient-response specifications	Servo and drive commissioning, process start-up, power-supply transient tests
PID controllers	More than 90 % of industrial loops: temperature, pressure, flow, level, speed
Controller tuning rules	Field commissioning, autotuning in process and HVAC controllers
Gain and phase margins as design targets	SMPS compensation, PLL loop filters, motor drives, flight control
Lead and lag compensation	Adding phase where needed, raising low-frequency accuracy
State-space models	Multivariable plants, aircraft, power-electronic converters, estimation
Controllability and observability	Choosing actuators and sensors, deciding what feedback can achieve
Digital control	Every microcontroller, DSP and PLC loop; networked control

Roadmap. Specifications (35.1) → the PID controller (35.2) and its tuning (35.3) → margins as design targets, with lead and lag compensation (35.4) → state-space models (35.5) → what an actuator can

steer: controllability (35.6) → what a sensor can see: observability (35.7) → the controller as a sampled program (35.8).

35.1 Transient-response specifications

The step response as a contract

A customer rarely writes "place the poles at $-1.8 \pm j2.3$". They write "the antenna shall settle within 0.05° of a new pointing command in 3 s, with less than 10 % overshoot, and track 1°/s with a lag below 0.5°". The standard way to make such statements precise is the response to a unit step of the reference, with the measures defined in Figure 35.1:

- **Rise time** t_r: time to go from 10 % to 90 % of the final value (sometimes 0–100 % for underdamped responses).
- **Peak time** t_p: time of the first maximum.
- **Percent overshoot** M_p: $(y_{\max} - y_\infty)/y_\infty \times 100\ \%$.
- **Settling time** t_s: the time after which the response stays within a band, usually $\pm 2\ \%$ (sometimes $\pm 5\ \%$ or $\pm 1\ \%$) of the final value.
- **Steady-state error** e_{ss}: the remaining difference, set by the system type (Section 34.7).

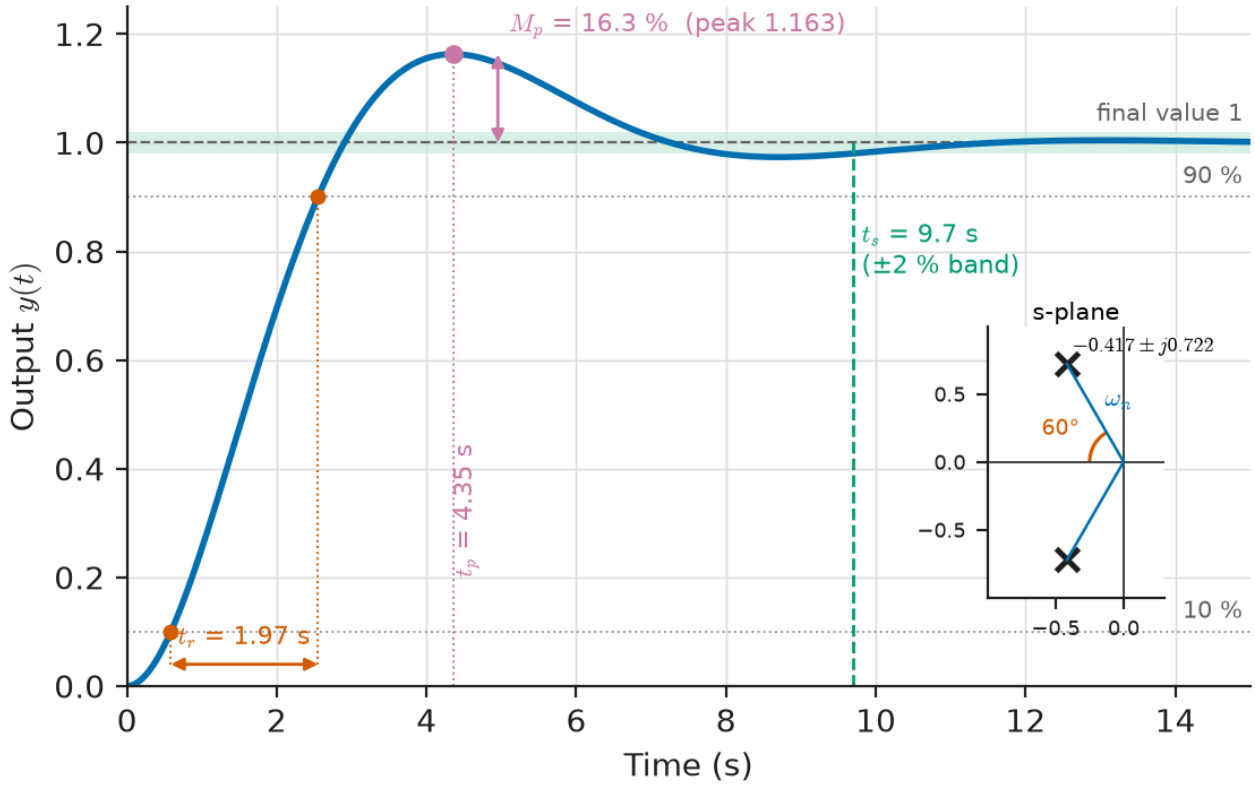

Figure 35.1 — Step-response specifications: rise time, peak time, overshoot, settling band and steady-state value, drawn on a second-order response with zeta = 0.5 and natural frequency 0.833 rad/s.

The second-order template

Most specifications are translated through the standard second-order system of Chapter 7,

$$T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \qquad s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2},$$

for which the step-response measures have closed forms (exact for M_p and t_p, approximate for t_r and t_s):

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}, \qquad t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}, \qquad t_s \approx \frac{4}{\zeta\omega_n}\ (2\ \%), \qquad t_r \approx \frac{1.8}{\omega_n}\ (\zeta \approx 0.5).$$

Overshoot depends only on ζ, the angle of the poles; settling time depends on $\zeta\omega_n$, their distance from the imaginary axis; rise and peak time scale with $1/\omega_n$, their distance from the origin. A specification therefore defines a permitted region of the s-plane: to the left of a vertical line $\sigma = -4/t_s$ and inside a wedge of half-angle $\cos^{-1}\zeta_{\min}$.

Table 35.1 — Second-order damping ratio and its time- and frequency-domain equivalents. (PM is the phase margin of the unity-feedback loop $L = \omega_n^2/[s(s+2\zeta\omega_n)]$ that produces this T; M_r is the closed-loop resonant peak.)

ζ	**Overshoot** M_p	**Phase margin**	**Resonant peak** M_r	**Typical use**
0.3	37.2 %	33.3°	1.75 (4.9 dB)	Too oscillatory for most loops
0.4	25.4 %	43.1°	1.36 (2.7 dB)	Fast mechanical servos tolerating overshoot
0.5	16.3 %	51.8°	1.15 (1.2 dB)	Common general compromise
0.6	9.5 %	59.2°	1.04 (0.35 dB)	Drives, power supplies
0.7	4.6 %	65.2°	1.00 (0 dB)	Process loops, "Butterworth" response
0.8	1.5 %	69.9°	none	Loops where overshoot is costly
1.0	0 %	76.3°	none	Critically damped: no overshoot, slower

The phase-margin column shows the rule of thumb already noticed in Chapter 34, PM $\approx 100\zeta$ degrees for $\zeta \le 0.6$.

> **Status of the principle: Dominant second-order approximation**
> *Approximate.* Specifications translate exactly only for a two-pole system with no zeros; for higher-order loops they hold when the other poles are at least about five times further from the imaginary axis than the dominant pair and no zeros lie near it. Closed-loop zeros (from PI or PD action) increase overshoot beyond the formula.

Worked example — translating a servo specification. The antenna of Section 34.7 must meet: (i) $M_p \le 10\ \%$; (ii) $t_s \le 3$ s (2 %); (iii) ramp error $\le 0.5°$ at 1°/s, i.e. $K_v \ge 2\ \text{s}^{-1}$. From (i), $\zeta \ge$

$\sqrt{(\ln 0.1)^2/(\pi^2+(\ln 0.1)^2)} = 0.591$. From (ii), $\zeta\omega_n \geq 4/3 = 1.33\ \text{s}^{-1}$. From (iii), with the plant $G = 1/[s(s+1)(s+5)]$ and a gain K, $K \geq 10$. Chapter 34 showed that proportional control cannot meet these together: $K = 10$ gives $\zeta = 0.21$ and 48.6 % overshoot; the gain giving $\zeta = 0.5$ gives $\zeta\omega_n = 0.417\ \text{s}^{-1}$ (settling about 9.6 s by the formula, 9.9 s simulated) and $K_v = 0.72$. Section 35.4 meets all three with a lead compensator.

Other specifications

Step specifications are not the whole contract. **Disturbance specifications** (the peak deviation and recovery time after a load step) matter more than set-point response in most process and power loops, where the set point rarely changes. **Integral criteria** compress a response into one number: integral of absolute error (IAE), of squared error (ISE) or of time-weighted absolute error (ITAE). **Robustness specifications** bound the peak sensitivity $M_s = \max|S(j\omega)|$ (Section 35.4), and **actuator specifications** bound control effort: a valve stroke rate, an inverter current limit, the voltage of a battery. A design that looks excellent on the step plot but demands a 10 kA pulse from a 50 A drive has not met its specification.

35.2 PID controllers

Three actions on the error

The **proportional–integral–derivative (PID)** controller acts on the present error, its accumulated past and its predicted future:

$$u(t) = K_p\left[e(t) + \frac{1}{T_i}\int_0^t e(\tau)\,d\tau + T_d\frac{de(t)}{dt}\right], \qquad C(s) = K_p\left(1 + \frac{1}{T_i s} + T_d s\right)$$

This is the **standard (ISA, ideal) form**, with integral time T_i and derivative time T_d. The **parallel form** $C = K_p + K_i/s + K_d s$ uses $K_i = K_p/T_i$, $K_d = K_p T_d$; the **series (interacting) form** $K_p'(1 + 1/T_i' s)(1 + T_d' s)$ was natural in pneumatic and early analog hardware and survives in some industrial controllers. Tuning values must be converted when moving between forms.

Table 35.2 — Effect of each PID action (typical, for a stable plant).

Action	What it reacts to	Main benefit	Main cost
Proportional K_p	Present error	Speed; reduces error by factor $1 + K_p G(0)$	Steady-state error remains (type 0); too much gives oscillation
Integral K_p/T_i	Accumulated error	Zero steady-state error to steps; disturbance rejection	Phase lag ($-90°$ at low frequency); overshoot; windup
Derivative $K_p T_d$	Rate of change	Phase lead near crossover; damping; anticipation	Amplifies sensor noise; "derivative kick" on set-point steps

The frequency-domain view is the most useful one for design. PI control is a *lag* element: it adds infinite gain at DC and costs phase below the frequency $1/T_i$. PD control is a *lead* element: it adds phase (up to $+90^\circ$) above $1/T_d$ at the price of rising gain. A PID controller places the integral corner well below crossover and the derivative corner near it.

What a practical PID contains

An ideal derivative has a gain that rises without bound, so measurement noise at high frequency is amplified enormously. Practical controllers filter it:

$$C(s) = K_p\left(1 + \frac{1}{T_i s} + \frac{T_d s}{1 + T_d s/N}\right), \qquad N \approx 5\text{–}20,$$

limiting the high-frequency derivative gain to N times the proportional gain. Figure 35.2 shows the other standard additions.

- **Derivative on measurement.** Differentiating $-y$ instead of e avoids the impulse ("derivative kick") that a set-point step would produce.
- **Set-point weighting.** Using $bR - Y$ ($0 \le b \le 1$) in the proportional term reduces overshoot to set-point changes without changing disturbance response.
- **Anti-windup.** When the actuator saturates (heater at 100 %, inverter at current limit, valve fully open), the loop is effectively open, but the integrator keeps integrating the error. When the error finally reverses, the large stored integral holds the actuator saturated and causes a large overshoot. Back-calculation feeds the difference between the computed and the saturated control signal back to the integrator through a gain $1/T_t$ (with T_t typically between T_d and T_i), or the integrator is simply frozen while the output is saturated (conditional integration).
- **Bumpless transfer.** Initialising the integrator when switching from manual to automatic so the output does not jump.

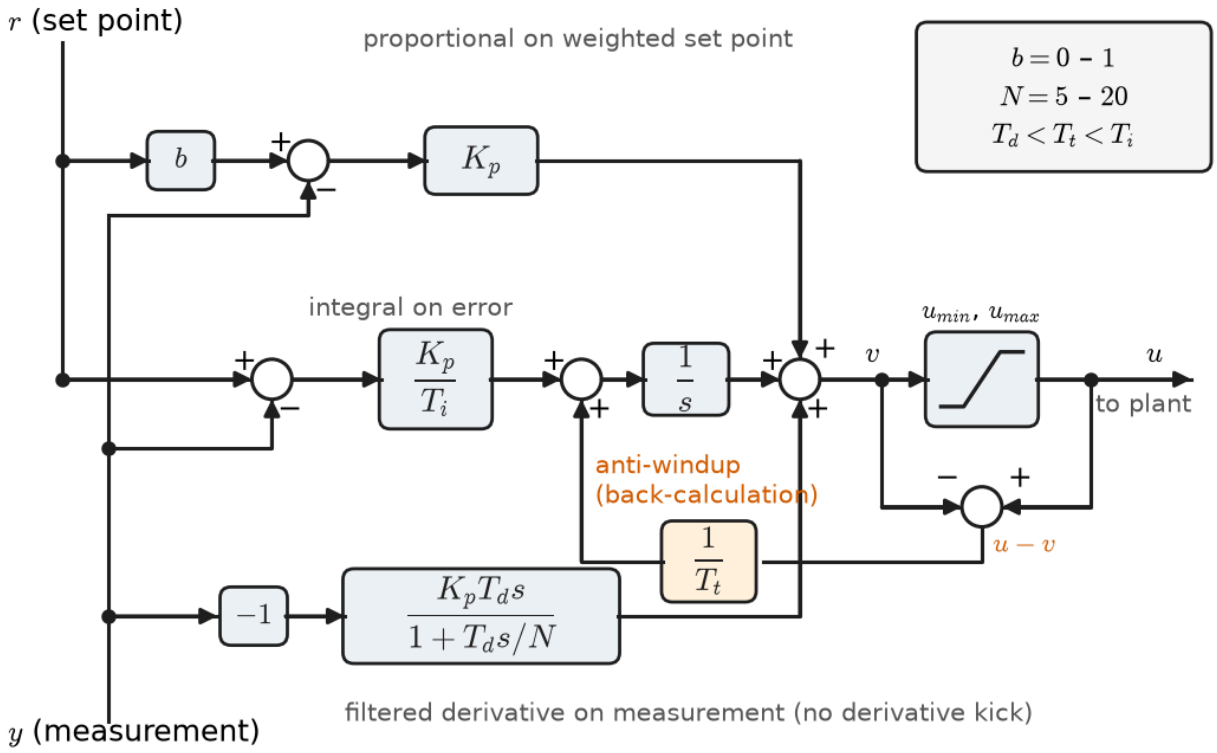

Figure 35.2 — A practical PID controller: proportional action on weighted set point, integral on error with back-calculation anti-windup, filtered derivative on measurement, and actuator saturation.

> **Status of the principle: PID control as a design structure**
> *Approximate.* A PID controller can place at most three closed-loop poles or shape the loop around one crossover; it is sufficient for plants dominated by one or two time constants and a modest delay, and insufficient for lightly damped resonances, long dead times relative to the time constants, or strongly coupled multivariable plants.

Where PID sits in hardware

PID is the dominant controller in process industry, HVAC, motor drives (where the current and speed loops of a field-oriented drive are PI controllers), power-electronic converters (type II and type III compensators are PI and PID with extra high-frequency poles), and in instrumentation (laser temperature control, oven-controlled crystal oscillators). In biomedical devices, infusion pumps and closed-loop anaesthesia systems use PID-type laws with heavy safety limits. The analog implementations with op-amps belong to *Electronics & Computing*; here the PID is a transfer function and, in Section 35.8, a difference equation.

35.3 Controller tuning

The tuning problem

Tuning means choosing K_p, T_i and T_d for a particular plant. Three families of method are used: rules based on a measured oscillation (ultimate-cycle and relay methods), rules based on a measured step response (reaction-curve, IMC/SIMC), and model-based loop shaping (Section 35.4). The first two are fast to apply in the field and require only a few numbers.

Ziegler–Nichols ultimate-cycle method

J. G. Ziegler and N. B. Nichols (1942) proposed: with integral and derivative action off, raise the proportional gain until the loop oscillates with constant amplitude. Record the **ultimate gain** K_u and the **ultimate period** T_u. Then set the controller from Table 35.3.

Table 35.3 — Classical tuning rules. (K = process gain, τ = time constant, θ = dead time of a first-order-plus-dead-time model.)

Method	Controller	K_p	T_i	T_d
Ziegler–Nichols ultimate cycle	P	$0.5K_u$	—	—
	PI	$0.45K_u$	$T_u/1.2$	—
	PID	$0.6K_u$	$T_u/2$	$T_u/8$
Ziegler–Nichols reaction curve	PI	$0.9\tau/(K\theta)$	3.33θ	—
	PID	$1.2\tau/(K\theta)$	2θ	0.5θ
SIMC (Skogestad), closed-loop time constant τ_c	PI	$\tau/[K(\tau_c+\theta)]$	$\min[\tau,\ 4(\tau_c+\theta)]$	—

The ultimate gain and period are exactly what Routh's array delivers from a model: for the servo of Chapter 34, $K_u = 30$ and $T_u = 2\pi/\sqrt{5} = 2.81$ s. The method was designed for **quarter-amplitude decay** (each overshoot a quarter of the previous one, $\zeta \approx 0.2$), which is far more aggressive than most loops tolerate today.

> **Status of the principle: Ziegler–Nichols tuning rules**
> *Empirical.* The constants were fitted to simulations and plant tests of typical process loops in the 1940s; they give a workable but oscillatory starting point (phase margins near 20–35°) and should be refined, not trusted, especially on plants with long dead time or an integrator.

Worked example — ultimate-cycle PID on a three-lag process. A process with three equal lags (a jacketed reactor with heater, wall and contents; a cascade of three thermal masses) is modelled as $G(s) = 1/(s+1)^3$ with time in minutes. The characteristic equation under P control is $s^3 + 3s^2 + 3s + 1 + K = 0$; Routh gives the limit $9 > 1 + K$, so $K_u = 8$, and the auxiliary polynomial $3s^2 + 9$ gives $\omega_u = \sqrt{3} = 1.732$ rad/min, $T_u = 3.63$ min. The Ziegler–Nichols PID settings are

$$K_p = 0.6 \times 8 = 4.8, \qquad T_i = 3.63/2 = 1.81 \text{ min}, \qquad T_d = 3.63/8 = 0.453 \text{ min}.$$

The resulting loop has a crossover at 1.38 rad/min, a phase margin of 30.6° and a sensitivity peak $M_s = 2.13$; the step response overshoots by 40.6 % and settles (2 %) in 9.4 min. For comparison, P control at $K_p = 0.5K_u = 4$ gives 54 % overshoot and leaves a 20 % steady-state error ($1/(1+4)$). The PID result is usable but aggressive; practitioners commonly reduce K_p by a factor of 1.5–2 or apply set-point weighting to cut the overshoot.

Relay autotuning

Driving a plant into sustained oscillation by hand is risky. K. J. Åström and T. Hägglund (1984) replaced the proportional gain with a relay of amplitude $\pm d$: most plants then settle into a limit cycle at very nearly the ultimate frequency, with amplitude a. The describing-function approximation of the relay gives

$$K_u \approx \frac{4d}{\pi a}, \qquad T_u \approx \text{period of the limit cycle.}$$

A relay of ± 5 % heater power that produces a temperature oscillation of ± 0.8 °C gives $K_u \approx 4 \times 5/(\pi \times 0.8) = 7.96$ % per °C. The oscillation amplitude is controlled by d and stays small, which is why this is the "autotune" button on most industrial temperature controllers.

> **Status of the principle: Relay (describing-function) estimate of the ultimate point**
> *Approximate.* The describing function treats the relay as a gain acting on the fundamental of a sinusoidal input; it is accurate to a few percent when the plant filters harmonics well (several lags), and poorer for plants dominated by a single lag or dead time.

Step-response (model-based) rules

Many process plants can be described, after an open-loop step test, by a **first-order-plus-dead-time (FOPDT)** model

$$G(s) = \frac{K\, e^{-\theta s}}{\tau s + 1}$$

obtained by drawing the tangent of steepest slope (the reaction curve) or by fitting the 28 % and 63 % points. Table 35.4 lists representative values.

Table 35.4 — Representative FOPDT parameters (order of magnitude; plants vary widely).

Process	Gain K	Time constant τ	Dead time θ
Liquid flow through a control valve	0.5–2 (flow %/valve %)	1–5 s	0.2–2 s
Small electric furnace or extruder zone	1–5 °C per % power	2–20 min	0.2–2 min
Building zone heating	0.1–0.5 °C per %	0.5–3 h	5–30 min
Liquid level in a tank (integrating)	— (integrator)	—	1–30 s
Distillation column composition	0.1–2	10–60 min	2–20 min

The **SIMC rule** (Skogestad's simple internal-model-control rule, 2003) derives from cancelling the plant lag with the integral term and choosing the closed-loop time constant τ_c; the recommended default $\tau_c = \theta$ gives a robust loop.

> **Status of the principle: IMC/SIMC tuning**
> *Approximate.* Derived analytically from an FOPDT model by direct synthesis with a first-order Taylor approximation of the delay; the resulting margins (with $\tau_c = \theta$: GM about 3, PM about 60°, $M_s \approx 1.6$) are reliable as long as the FOPDT model is a fair description of the plant.

Worked example — tuning the extruder zone. A step test of 10 % heater power on the extruder zone of the opening produces a 20 °C rise after a 20 s dead time, with a time constant of 120 s. The model is $K = 2$ °C/%, $\tau = 120$ s, $\theta = 20$ s.

SIMC PI with $\tau_c = \theta = 20$ s:

$$K_p = \frac{120}{2\,(20 + 20)} = 1.5\ \%/^\circ\mathrm{C}, \qquad T_i = \min(120,\ 4 \times 40) = 120\ \mathrm{s}.$$

Ziegler–Nichols reaction-curve PI:

$$K_p = \frac{0.9 \times 120}{2 \times 20} = 2.7\ \%/^\circ\mathrm{C}, \qquad T_i = 3.33 \times 20 = 66.6\ \mathrm{s}.$$

Simulation of both, with the delay modelled exactly (Figure 35.3), gives:

Tuning	Set-point overshoot	2 % settling	PM	GM	M_s	IAE after 1 % load step
SIMC	4.1 %	121 s	61.4°	3.14 (9.9 dB)	1.59	80 °C·s
Ziegler–Nichols	59.6 %	240 s	28.9°	1.63 (4.2 dB)	3.08	25 °C·s

The Ziegler–Nichols setting rejects a load disturbance (a cold-material surge, modelled as a 1 % step of heater-equivalent power, which would raise the temperature 2 °C open loop) three times better in IAE terms, but at the cost of 60 % set-point overshoot and a gain margin of only 1.6: a 60 % rise in process gain (a change of polymer grade) would make it oscillate. The SIMC loop is calm and robust but slow to remove disturbances, because with $\tau \gg \theta$ the integral time equals the long plant time constant. Skogestad's own refinement for such lag-dominant plants, and the common field practice, is to shorten τ_c or T_i until a compromise (for example PM $\approx 45^\circ$) is reached. Tuning rules are starting points; the trade-off between set-point response, disturbance rejection and robustness is the actual design.

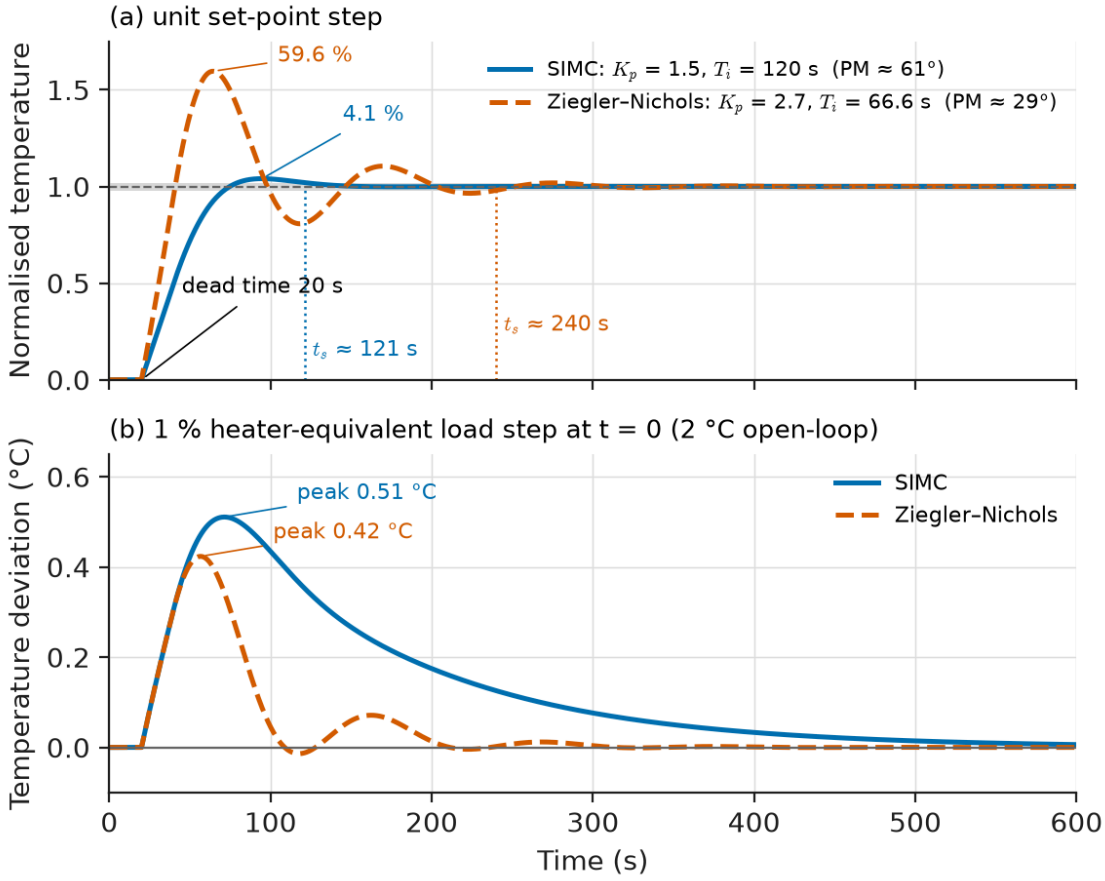

Figure 35.3 — Extruder-zone temperature loop (K = 2, tau = 120 s, theta = 20 s) under SIMC and Ziegler-Nichols PI tuning: set-point step (top) and 1 percent load-disturbance step (bottom).

Tuning on other plants

The servo of Chapter 34 already contains an integrator (from speed to angle), so integral action is rarely needed for set-point tracking and can be harmful: the Ziegler–Nichols PI setting ($K_p = 13.5$, $T_i = 2.34$ s) puts its closed-loop poles at $-0.016 \pm j1.50$, practically on the stability boundary, because the added integrator pushes the loop to type 2 with -180° of phase at low frequency. The lesson generalises: integral action belongs where the loop needs it (to reject a load torque, for instance) and with an integral time long compared with the crossover period. For integrating processes (tank level, servo position) dedicated rules exist, and for plants with resonances or long dead times model-based design is the better tool.

35.4 Gain and phase margins

Margins as design targets

Chapter 34 defined the gain margin (GM), phase margin (PM) and peak sensitivity M_s as distances from instability. In design they become targets, because they protect against what the model leaves out: gain variations of plant and actuator, unmodelled lags and delays, sampling. Table 35.5 collects typical values.

Table 35.5 — Typical margin specifications.

Application	Phase margin	Gain margin	M_s
General process control	45–65°	2–4 (6–12 dB)	1.4–2.0
Switched-mode power supply voltage loop	45–70°	at least 6–10 dB	about 1.5
Motor-drive speed and position loops	40–60°	6–12 dB	1.3–2.0
Phase-locked loop	45–65°	at least 6 dB	—
Aircraft flight control (per loop, traditional)	at least 45°	at least 6 dB	—

The peak sensitivity is the tightest of the three, since $1/M_s$ is the actual shortest distance from the Nyquist plot to -1. A loop with $M_s \le 2$ is guaranteed GM ≥ 2 and PM $\ge 29^\circ$; with $M_s \le 1.4$, GM ≥ 3.5 and PM $\ge 42^\circ$ (from GM $\ge M_s/(M_s-1)$ and PM $\ge 2\sin^{-1}[1/(2M_s)]$).

> **Status of the principle: Margin specifications**
> *Empirical.* The numerical targets come from decades of practice relating margins to acceptable overshoot and tolerance to plant variation; the bounds linking M_s to GM and PM are exact geometry, but the choice of a target value is engineering judgement.

Frequency-domain correlations

Three approximate translations connect the margins to the time domain (for a loop with a dominant crossover and a slope near -20 dB/decade there):

- Damping: $\zeta \approx \mathrm{PM}/100$ (Table 35.1).
- Speed: closed-loop bandwidth $\omega_b \approx (1\text{–}2)\,\omega_c$, and rise time roughly $t_r \approx (1.5\text{–}2.5)/\omega_c$.
- Delay tolerance: $\tau_{\max} = \mathrm{PM}/\omega_c$ (Chapter 34).

Loop shaping with lead and lag

Loop shaping modifies $L(j\omega) = C(j\omega)G(j\omega)$ directly on the Bode plot: high gain at low frequency for accuracy, a crossover at the desired speed with a slope of about -20 dB/decade and adequate phase, and low gain at high frequency for noise and robustness. Two elementary compensators do most of the work:

$$C_{\text{lead}}(s) = K_c\frac{1+s/z}{1+s/p}\ (p>z), \qquad C_{\text{lag}}(s) = K_c\frac{s+z}{s+p}\ (p<z).$$

The lead network adds a phase bump with maximum

$$\phi_{\max} = \sin^{-1}\frac{p/z-1}{p/z+1} \quad \text{at} \quad \omega_m = \sqrt{zp},$$

(for $p/z = 10$, $\phi_{\max} = 54.9^\circ$), and is the frequency-domain twin of PD control. The lag network raises low-frequency gain by z/p while leaving the crossover nearly untouched, and is the twin of PI control. Figure 35.4 compares them on the servo.

Worked example — lead compensation of the antenna servo. Specification from Section 35.1: $K_v \ge 2$, $M_p \le 10\ \%$, $t_s \le 3$ s. Proportional control with $K = 10$ meets K_v but has PM $= 25.4^\circ$ (Chapter 34). Place the lead zero on the plant pole at -1 (cancelling the slow mechanical lag) and the lead pole ten times higher:

$$C(s) = 10\,\frac{s+1}{0.1s+1}, \qquad L(s) = \frac{10}{s(s+5)(0.1s+1)} = \frac{100}{s(s+5)(s+10)}.$$

At DC the compensator gain is 10, so $K_v = 10/5 = 2$ is unchanged. The new loop has $\omega_c = 1.85$ rad/s and PM $= 59.3^\circ$. The phase crossover moves to $\sqrt{50} = 7.07$ rad/s (from Routh on $s^3 + 15s^2 + 50s + 100k$, the limit is $k < 7.5$), so GM $= 7.5 = 17.5$ dB, and $M_s = 1.45$. The closed-loop poles are $-1.81 \pm j2.35$ ($\zeta = 0.61$) and -11.4. In simulation the step response overshoots 8.5 % and settles in 2.1 s: all three specifications are met. The price is visible in the compensator: its high-frequency gain is $10 \times 10 = 100$, ten times that of the proportional controller, so encoder quantisation noise and resonances above 10 rad/s are amplified tenfold. The delay margin is $1.035/1.85 = 0.56$ s.

Worked example — lag compensation, for contrast. Keep the well-damped gain $K = 3.588$ (PM 51.7°, $K_v = 0.718$) and raise the low-frequency gain by $\beta = 2/0.718 = 2.79$ with a lag whose zero sits a decade below crossover:

$$C(s) = 3.588\,\frac{s+0.06}{s+0.0215}.$$

Now $K_v = 3.588 \times (0.06/0.0215)/5 = 2.0$, $\omega_c = 0.61$ rad/s, PM $= 48.0^\circ$, GM $= 7.97$ (18.0 dB). The accuracy specification is met with almost the original damping, but the speed is not: overshoot is 22.6 % and the 2 % settling time 17.3 s, lengthened by a slow closed-loop pole–zero pair near -0.06 that leaves a long tail. Lead buys speed and phase; lag buys low-frequency gain. Lead–lag compensators, and PID, combine both.

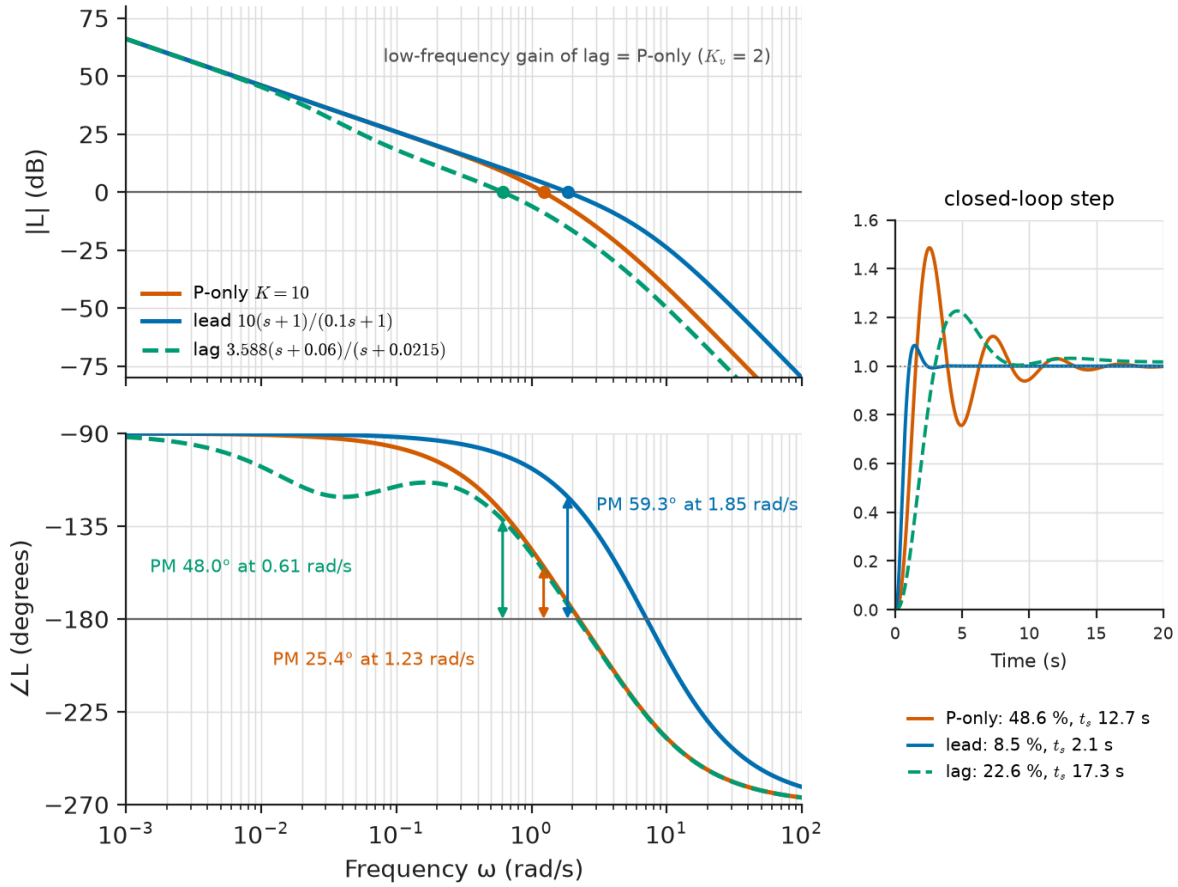

Figure 35.4 — Bode plots of the antenna-servo loop with proportional (K = 10), lead and lag compensation, with crossovers and phase margins marked.

Limits no compensator removes

Some plant features cap the achievable crossover regardless of design skill. A dead time θ limits ω_c to roughly $1/\theta$ or less (it consumes 57° of phase at $\omega = 1/\theta$). A right-half-plane zero at z (a boost converter's output in continuous conduction, the initial wrong-way response of a steam drum level) limits ω_c to about $z/2$. An unstable pole at p requires ω_c above about $2p$. Lightly damped resonances must be kept well above or carefully inside the crossover. These limits, consequences of the Bode integral of Chapter 34, are why a sensible specification is written only after looking at the plant.

35.5 State-space introduction

Why another description

Transfer functions describe single-input single-output (SISO) LTI systems from the outside. Three needs push beyond them: plants with several inputs and outputs (an aircraft, a multi-machine power system, a three-phase converter in dq coordinates); internal behaviour that may not appear at the output (Section 35.7); and design by placing every closed-loop pole, not just a dominant pair. The **state-space** model writes an nth-order system as n first-order differential equations:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \qquad \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

with state vector $\mathbf{x}$ ($n \times 1$), input $\mathbf{u}$, output $\mathbf{y}$, and constant matrices $\mathbf{A}$ (dynamics), $\mathbf{B}$ (input), $\mathbf{C}$ (output) and $\mathbf{D}$ (direct feedthrough). In circuits the natural states are capacitor voltages and inductor currents (the energy-storing variables of Chapter 5); in mechanics, positions and velocities.

Link to transfer functions

Taking Laplace transforms with zero initial state:

$$\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}, \qquad \det(s\mathbf{I} - \mathbf{A}) = 0 \text{ gives the poles.}$$

The eigenvalues of $\mathbf{A}$ are the poles of the system (unless some cancel, Section 35.7). The time-domain solution uses the matrix exponential (a recap from the linear-algebra and ODE chapters of *The Mathematics Universe*):

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\,d\tau.$$

The state description is not unique: any invertible change of variables $\mathbf{x} = \mathbf{T}\mathbf{z}$ gives $(\mathbf{T}^{-1}\mathbf{A}\mathbf{T},\ \mathbf{T}^{-1}\mathbf{B},\ \mathbf{C}\mathbf{T},\ \mathbf{D})$ with the same transfer function. Standard choices include the **controllable canonical form**, read directly from the transfer-function coefficients: $G = (b_1 s + b_0)/(s^2 + a_1 s + a_0)$ has

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} b_0 & b_1 \end{bmatrix}.$$

Worked example — a DC servo motor in state space. A small permanent-magnet servo motor driven by a voltage amplifier, with negligible armature inductance, has speed dynamics $\dot{\omega} = -\omega/\tau_m + (K_m/\tau_m)u$ with mechanical time constant $\tau_m = 0.5$ s and gain $K_m = 10$ rad/s per volt. With states $x_1 = \theta$ (shaft angle) and $x_2 = \omega$ (speed) and a position encoder as output:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 20 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad \mathbf{D} = 0.$$

Then $(s\mathbf{I} - \mathbf{A})^{-1} = \dfrac{1}{s(s+2)}\begin{bmatrix} s+2 & 1 \\ 0 & s \end{bmatrix}$ and

$$G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} = \frac{20}{s(s+2)},$$

with poles at the eigenvalues 0 and -2 of $\mathbf{A}$. The same motor with armature inductance included would gain a third state, the current, and the model of a buck converter has inductor current and capacitor voltage as states; state space handles all of them with the same notation.

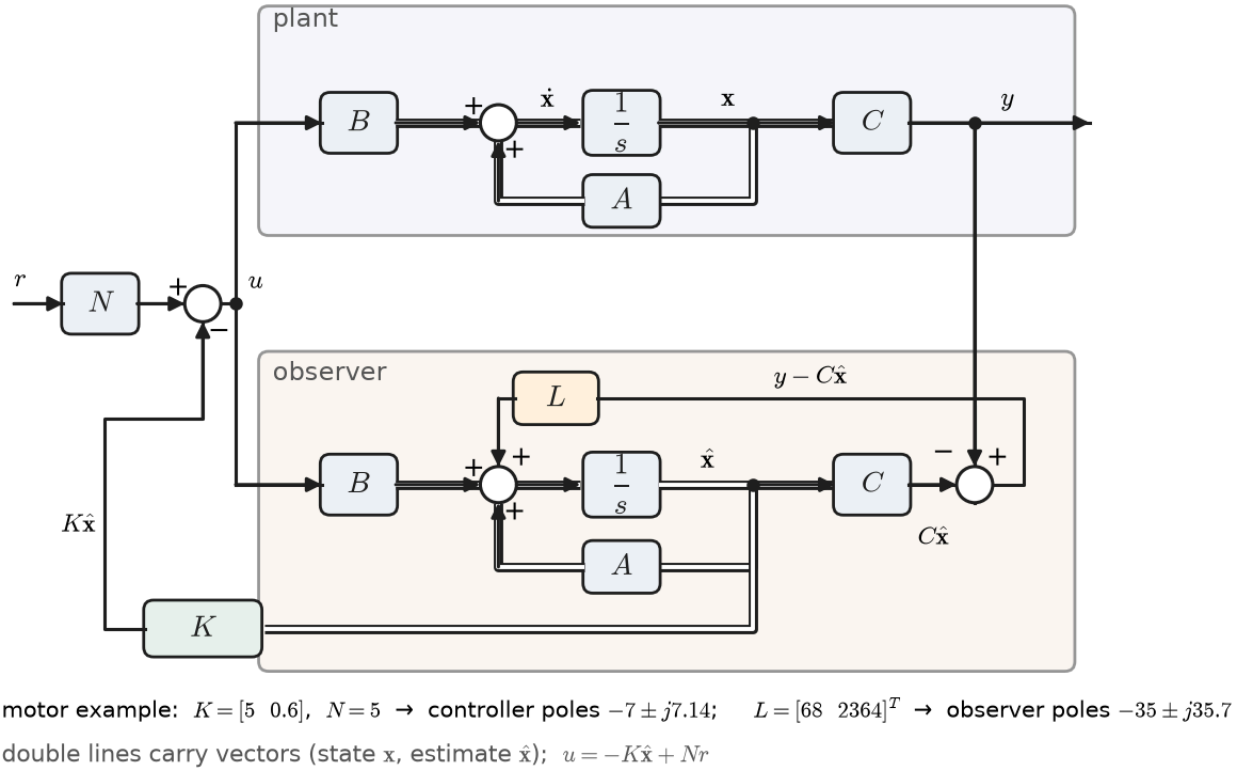

Figure 35.5 — State-space plant with state feedback u = -Kx + Nr and a Luenberger observer that reconstructs the state from the measured output.

> **Status of the principle: State-space representation**
> *Derived.* Any finite-dimensional linear time-invariant system described by ordinary differential equations can be written in state-space form, and its transfer function follows exactly; the model is only as good as the lumped, linear, time-invariant assumptions behind the differential equations.

35.6 Controllability

Can the input steer every state?

A system is **controllable** if, for any initial state and any target state, there is an input that moves the state from one to the other in finite time. Controllability asks whether the actuators reach every mode of the plant. Kalman's test (1960): an nth-order system is controllable if and only if the **controllability matrix**

$$\mathcal{C} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B} & \cdots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix}$$

has rank n.

> **Status of the principle: Kalman controllability rank test**
> *Derived.* Follows from the Cayley–Hamilton theorem applied to the solution $e^{\mathbf{A}t}$; exact for LTI models, though a matrix that is full rank but nearly singular indicates a mode that can be steered only with very large inputs.

For the motor, $\mathbf{AB} = [20,\ -40]^T$ and

$$\mathcal{C} = \begin{bmatrix} 0 & 20 \\ 20 & -40 \end{bmatrix}, \qquad \det \mathcal{C} = -400 \neq 0,$$

so the motor is controllable from its voltage input, as intuition suggests.

Worked example — an uncontrollable circuit. Two RC branches, each a resistor in series with a capacitor, are connected in parallel across the same source u. Let both have the same time constant, $R_1C_1 = R_2C_2 = \tau$ (for example 10 kΩ with 100 nF and 1 kΩ with 1 µF, both $\tau = 1$ ms). With the capacitor voltages as states,

$$\mathbf{A} = \begin{bmatrix} -1/\tau & 0 \\ 0 & -1/\tau \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1/\tau \\ 1/\tau \end{bmatrix}, \quad \mathcal{C} = \frac{1}{\tau}\begin{bmatrix} 1 & -1/\tau \\ 1 & -1/\tau \end{bmatrix},$$

which has rank 1. Whatever the source does, the difference $v_1 - v_2$ obeys $\frac{d}{dt}(v_1 - v_2) = -(v_1 - v_2)/\tau$: it decays on its own and cannot be driven. Physically, the two branches respond identically to every input, so no input can separate them. The same structure appears in a balanced bridge (the balanced condition makes the detector mode uncontrollable from the supply) and in symmetrical power networks, where the "differential" mode between identical generators on a common bus cannot be excited by a common input. If the time constants differ ($\tau_1 \neq \tau_2$), the rank becomes 2 and the system is controllable, though nearly uncontrollable when $\tau_1 \approx \tau_2$.

Pole placement by state feedback

The payoff of controllability is that full **state feedback**

$$u = -\mathbf{K}\mathbf{x} + Nr$$

gives the closed-loop matrix $\mathbf{A} - \mathbf{BK}$, whose eigenvalues can be placed *anywhere* (in complex-conjugate pairs) if and only if $(\mathbf{A}, \mathbf{B})$ is controllable. Uncontrollable modes stay where they are, whatever $\mathbf{K}$; if such a mode is unstable, no controller can stabilise the plant (it is not **stabilisable**).

Worked example — placing the motor poles. Choose closed-loop poles with $\zeta = 0.7$ and $\omega_n = 10$ rad/s, i.e. the characteristic polynomial $s^2 + 14s + 100$. With $\mathbf{K} = [k_1 \ \ k_2]$,

$$\mathbf{A} - \mathbf{BK} = \begin{bmatrix} 0 & 1 \\ -20k_1 & -2 - 20k_2 \end{bmatrix}, \qquad \det(s\mathbf{I} - \mathbf{A} + \mathbf{BK}) = s^2 + (2 + 20k_2)s + 20k_1.$$

Matching coefficients: $20k_1 = 100$, $k_1 = 5$ V/rad; $2 + 20k_2 = 14$, $k_2 = 0.6$ V·s/rad. The poles are $-7 \pm j7.14$. The reference gain that gives unit steady-state gain is $N = -1/[\mathbf{C}(\mathbf{A} - \mathbf{BK})^{-1}\mathbf{B}] = 5$ (here equal to k_1, because the plant contains an integrator). The step response overshoots 4.6 % and settles in 0.60 s, exactly as Table 35.1 predicts for $\zeta = 0.7$ with $4/(\zeta\omega_n) = 0.57$ s. The controller is a PD law in disguise: a gain on angle plus a gain on speed. For higher-order plants the gains are found by

Ackermann's formula or numerically; the **linear-quadratic regulator** (LQR), which chooses $\mathbf{K}$ to minimise a weighted sum of state error and control effort, is the systematic route taken in advanced texts.

35.7 Observability

Can the sensors see every state?

State feedback needs every state, but sensors are expensive, noisy or impossible (rotor flux in an induction motor, the temperature inside a battery cell, the core temperature of a patient). A system is **observable** if the initial state can be determined from the output (and the known input) over a finite interval. The test is dual to controllability: the **observability matrix**

$$\mathcal{O} = \begin{bmatrix} \mathbf{C} \\ \mathbf{C}\mathbf{A} \\ \vdots \\ \mathbf{C}\mathbf{A}^{n-1} \end{bmatrix}$$

must have rank n.

> **Status of the principle: Observability and duality**
> *Derived*. The rank test follows from expanding the output in terms of $e^{\mathbf{A}t}\mathbf{x}(0)$; $(\mathbf{A}, \mathbf{C})$ is observable exactly when $(\mathbf{A}^T, \mathbf{C}^T)$ is controllable, so every controllability result has an observability twin.

Worked example — encoder versus tachometer. For the motor with a position encoder, $\mathbf{C} = [1 \;\; 0]$, $\mathbf{C}\mathbf{A} = [0 \;\; 1]$, so $\mathcal{O} = \mathbf{I}$, rank 2: the speed can be reconstructed from position (by differentiation, or better by an observer). With only a tachometer, $\mathbf{C} = [0 \;\; 1]$, $\mathbf{C}\mathbf{A} = [0 \;\; -2]$, so

$$\mathcal{O} = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}, \qquad \text{rank } 1.$$

The angle is unobservable from speed: two shafts that turn identically but started at different angles produce identical tachometer signals. Integrating the speed gives the angle only up to an unknown constant, and any offset or drift in the integration accumulates without correction. This is why inertial navigation, which integrates accelerations, needs periodic position fixes (GNSS) to stay bounded.

Hidden modes

Controllability and observability explain an effect that transfer functions conceal. Kalman's decomposition splits any system into four parts: controllable and observable, controllable but unobservable, uncontrollable but observable, and neither. **Only the controllable-and-observable part appears in the transfer function.** A pole–zero cancellation in $G(s)$ is the visible sign of a mode that is uncontrollable, unobservable or both. Cancelling an unstable plant pole with a controller zero is therefore forbidden: the transfer function from reference to output looks stable, but an internal

signal grows without bound (the loop is not *internally stable*). The lead design of Section 35.4 cancelled the *stable* pole at -1, which is acceptable: the hidden mode decays as e^{-t}, visible only in the response to a disturbance entering at the plant input.

Observers

An **observer** (Luenberger, 1964) is a software copy of the plant driven by the same input and corrected by the measured output:

$$\dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}\mathbf{u} + \mathbf{L}(\mathbf{y} - \mathbf{C}\hat{\mathbf{x}}).$$

The estimation error $\mathbf{e} = \mathbf{x} - \hat{\mathbf{x}}$ obeys $\dot{\mathbf{e}} = (\mathbf{A} - \mathbf{L}\mathbf{C})\mathbf{e}$, so its eigenvalues can be placed arbitrarily if the system is observable. For the motor with an encoder, $\mathbf{A} - \mathbf{L}\mathbf{C} = \begin{bmatrix} -l_1 & 1 \\ -l_2 & -2 \end{bmatrix}$ has characteristic polynomial $s^2 + (l_1 + 2)s + (2l_1 + l_2)$. Placing the observer poles five times faster than the controller, at $\zeta = 0.7$, $\omega_n = 50$ rad/s ($s^2 + 70s + 2500$), gives $l_1 = 68$ and $l_2 = 2500 - 136 = 2364$, observer poles $-35 \pm j35.7$. Figure 35.5 shows the complete structure.

The **separation principle** states that, for an LTI plant, the controller gain $\mathbf{K}$ and observer gain $\mathbf{L}$ can be designed independently: the closed loop has the controller poles plus the observer poles. When measurement and process noise are significant, the optimal choice of $\mathbf{L}$ is the **Kalman filter**, used in everything from GNSS receivers to battery state-of-charge estimation. Observers put an obligation on the engineer as well: the observer is only as good as its model, and a fast observer amplifies sensor noise just as derivative action does.

35.8 Digital control fundamentals

The sampled loop

Almost every controller built today is a program. Figure 35.6 shows the structure: an anti-aliasing filter and ADC sample the measurement every T seconds (Chapter 19), the processor computes $u[k]$ from current and past samples, and a DAC or PWM unit holds that value constant until the next update, the **zero-order hold** (ZOH). The plant remains continuous-time; the controller sees it only at the sampling instants.

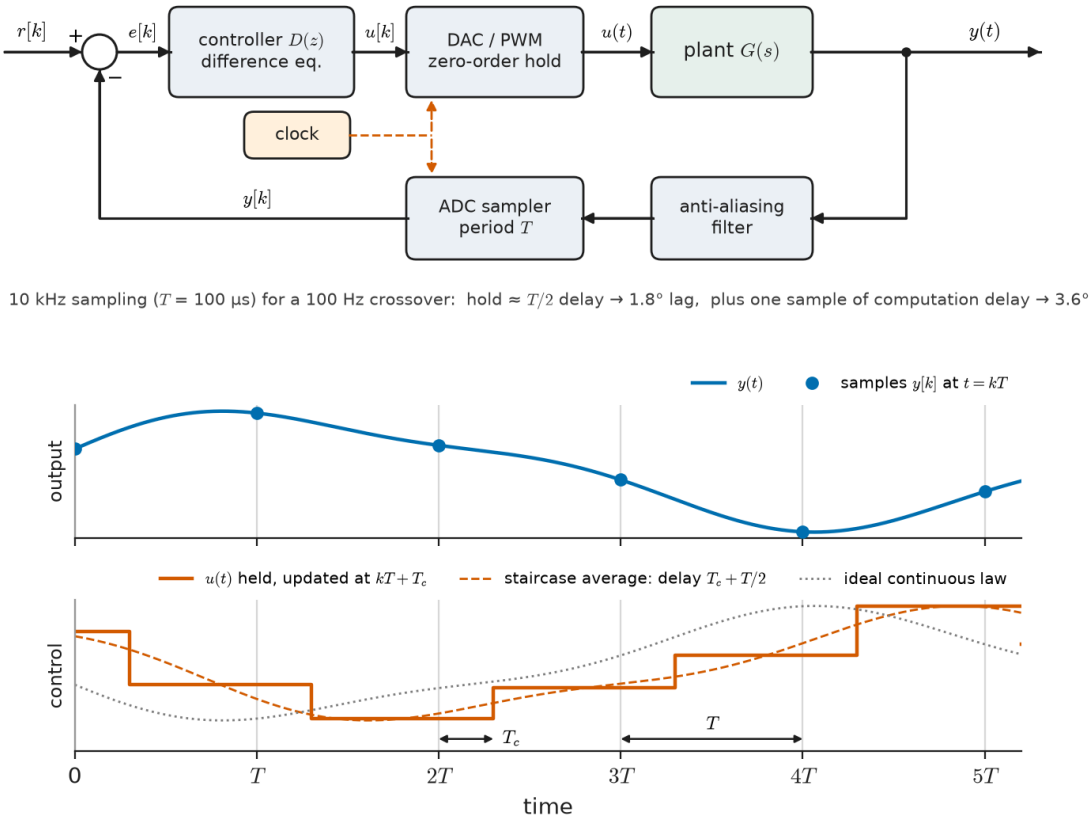

Figure 35.6 — A digital control loop: anti-aliasing filter, ADC sampling at period T, discrete controller difference equation, zero-order hold and continuous plant, with a timing diagram showing sampling, computation delay and the staircase control signal.

Sampling rate and the cost of the hold

The ZOH output is a staircase whose average lags the continuous signal by half a sample. In the frequency domain the hold behaves approximately as a delay of $T/2$, adding phase lag

$$\Delta\phi_{\text{ZOH}} \approx \frac{\omega T}{2} \text{ rad}$$

at frequency ω. Most implementations also compute $u[k]$ from $y[k]$ and apply it at the next sampling instant, adding one more sample of delay. With a total effective delay τ_d, the phase lost at crossover is $\omega_c \tau_d$.

> **Status of the principle: Zero-order hold as a half-sample delay**
> *Approximate.* The exact hold response is $(1 - e^{-sT})/s$; replacing it by $e^{-sT/2}$ is accurate for frequencies well below the Nyquist frequency π/T (within a few percent below about a fifth of it) and fails near Nyquist, where the exact discrete-time analysis in the z-domain is required.

Worked example — how fast must a drive loop sample? A motor-drive speed loop is designed in continuous time for crossover at 100 Hz ($\omega_c = 628$ rad/s) with PM $= 55^\circ$.

- At $f_s = 10$ kHz ($T = 100$ µs): the hold costs $628 \times 50\ \mu\text{s} = 0.0314$ rad $= 1.8^\circ$ and a one-sample computation delay a further 3.6°; total 5.4°, leaving PM $\approx 49.6^\circ$. The analog design transfers almost unchanged.
- At $f_s = 1$ kHz ($T = 1$ ms): the hold costs 18° and the computation delay 36°, a total of 54°, leaving about 1° of margin. The loop will ring violently or oscillate, which is exactly what the drives engineer of the opening found.

The common rule of thumb follows: sample at 10–30 times the closed-loop bandwidth (about 20–60 times the crossover in hertz for tight loops), or redesign the controller in discrete time to account for the delay explicitly.

Table 35.6 — Representative sampling rates in digital control.

Loop	Typical sampling rate
Switched-mode power supply (digital controller)	50 kHz–1 MHz (often once per switching period)
Motor-drive current loop	8–20 kHz (PWM frequency)
Motor-drive speed and position loops	1–10 kHz
Quadcopter attitude loop	0.5–8 kHz (IMU-rate dependent)
Grid-converter current control	5–20 kHz
Generator AVR / power-system stabiliser (digital)	100 Hz–2 kHz
Process temperature, flow, level (PLC/DCS)	0.1–10 Hz
Building HVAC	0.01–1 Hz

From s to z: discretising the controller and the plant

Chapter 20 introduced the z-transform and showed that a discrete system is stable when its poles lie inside the unit circle. A continuous pole at s maps to a discrete pole at

$$z = e^{sT},$$

so the left half s-plane maps to the inside of the unit circle and the imaginary axis to the circle itself. There are two standard routes to a discrete controller.

1. **Discretise the continuous design** with a substitution. The **bilinear (Tustin)** transform

   $$s \leftarrow \frac{2}{T}\,\frac{z-1}{z+1}$$

   maps the stable region exactly onto the unit disc and preserves stability; backward Euler $s \leftarrow (1 - z^{-1})/T$ is simpler and also stable but more distorting. Either works well when T is small compared with the loop time constants.
2. **Discretise the plant exactly** with the ZOH equivalent and design directly in the z-domain. For a state-space plant

$$\mathbf{x}[k+1] = \mathbf{A}_d \mathbf{x}[k] + \mathbf{B}_d u[k], \quad \mathbf{A}_d = e^{\mathbf{A}T}, \quad \mathbf{B}_d = \int_0^T e^{\mathbf{A}\tau}\, d\tau\, \mathbf{B}.$$

> **Status of the principle: Exact ZOH discretisation**
> *Derived*. Integrating the state equation over one sample with the input held constant gives $\mathbf{A}_d$ and $\mathbf{B}_d$ exactly at the sampling instants; it says nothing about intersample behaviour, which can hide ripple if the sampling is too slow.

Worked example — the motor in discrete time. With $T = 10$ ms, the motor of Section 35.5 gives

$$\mathbf{A}_d = \begin{bmatrix} 1 & 0.009901 \\ 0 & 0.98020 \end{bmatrix}, \qquad \mathbf{B}_d = \begin{bmatrix} 0.000993 \\ 0.19801 \end{bmatrix}.$$

The pole at $s = -2$ maps to $z = e^{-0.02} = 0.9802$, and the integrator pole at $s = 0$ to $z = 1$, on the unit circle as expected. The state-feedback design of Section 35.6 placed the poles at $s = -7 \pm j7.14$; their discrete images are $z = e^{sT} = 0.930 \pm j0.0665$, of magnitude 0.932, comfortably inside the unit circle. A discrete pole-placement design would target those z values directly.

Worked example — a PI controller as a difference equation. The SIMC controller of the extruder ($K_p = 1.5$, $T_i = 120$ s), sampled at $T = 1$ s with Tustin's rule, becomes

$$u[k] = u[k-1] + q_0 e[k] + q_1 e[k-1], \qquad q_0 = K_p\left(1 + \frac{T}{2T_i}\right) = 1.50625, \quad q_1 = -K_p\left(1 - \frac{T}{2T_i}\right) = -$$

This "velocity form" computes the *change* of control, which makes anti-windup trivial (clamp $u[k]$ to the actuator range before storing it) and transfers bumplessly from manual mode. At $T = 1$ s against a 20 s dead time and 120 s time constant, the sampling delay costs only a few degrees of phase at the 0.025 rad/s crossover ($0.025 \times 1.5 = 0.0375$ rad, about 2°).

Practical issues

- **Aliasing.** A sampled controller cannot distinguish a disturbance at frequency f from one at $f_s - f$ (Chapter 19). Mains pickup at 50 Hz sampled at 52 Hz appears as a 2 Hz disturbance that the loop will dutifully "correct". An analog anti-aliasing filter, or oversampling with digital filtering, is required, and its own phase lag must be counted in the margin budget.
- **Quantisation and word length.** ADC resolution sets a noise floor that derivative action amplifies; a 12-bit encoder count differentiated at 10 kHz produces large speed steps. Fixed-point controller coefficients such as q_0 and q_1 above, which differ by less than 1 %, must be represented precisely or the integral action is distorted (the finite-precision effects of Chapter 21 and Chapter 23).
- **Timing jitter and networked control.** Variable computation time or a network link makes the delay uncertain; the delay margin of Chapter 34 sets the tolerance.
- **Multi-rate structures.** Cascaded drives run current loops fast, speed loops slower and position loops slower still, exactly the separation of time scales of Table 34.1.

> **Boundary.**
> Microcontroller peripherals (ADCs, timers, PWM units), real-time operating systems and embedded code belong to *Electronics & Computing* and to the robotics and automation books. This section covers only the signal-and-system consequences of sampling a loop.

What remains open

Open question	Where it is answered
How are the plant models and block diagrams used here obtained from physics?	Chapter 33 (dynamic-system modeling)
Stability tests, root locus, steady-state error and the Nyquist criterion behind every design here	Chapter 34 (feedback and stability)
Second-order response, damping ratio and natural frequency of RLC circuits	Chapter 7
Laplace-domain poles and zeros, Bode asymptotes	Chapters 11 and 13
Sampling, aliasing, z-transform and digital-filter implementation of controllers	Chapters 19, 20, 21 and 23
Kalman filtering of noisy measurements and the statistics of sensor noise	Chapter 29 (random signals); probability in *The Mathematics Universe*
Optimal (LQR, LQG), robust (H_∞), nonlinear, adaptive and model-predictive control	Beyond this book: advanced control texts and the robotics and automation books
Op-amp and digital hardware that implements PID compensators	*Electronics & Computing*
Drives, excitation systems and governors that these loops command	*Electrical Energy & Machines*
Embedded controllers and real robot joints	Robotics and automation books

Closing. From Ohm's law in Chapter 1 to a sampled state-feedback controller in this chapter, the book has used one body of mathematics, linear systems described by poles, zeros and frequency responses, to analyse circuits, filter signals, carry information across noisy channels and, finally, make imprecise machines behave precisely; the engineer who can move between those views is ready for every specialised discipline built on them.